K-STABILITY OF PRIME FANO THREEFOLDS OF GENUS TEN AND TWELVE

Abstract. We prove that all smooth prime Fano threefolds of genus ten (family 1.9) are K-stable and that all smooth prime Fano threefolds of genus twelve (family 1.10) are K-semistable, and thereby solve Problem 10 of Xu–Zhuang's open problem list on K-stability of Fano varieties, which asks to show that any smooth Fano threefold of Picard number one is K-semistable. For smooth prime Fano threefolds of genus twelve, we show that K-stability is equivalent to each special line intersecting another line and to the GIT-stability of the Hilbert scheme of lines as a plane quartic, that K-polystability is equivalent to the GIT-polystability of the quartic, and that each K-polystable one is isomorphic to a member of an explicit family over an open set of binary octics whose fiber over zero is the Mukai–Umemura threefold. We also disprove the original formulation of Donaldson's conjecture, which predicts that such a threefold without a torus action is not K-polystable as soon as an anti-canonical divisor has a specific simple elliptic singularity, by an explicit K-stable example with finite automorphism group and with such a divisor, and we prove the prediction when the singular point is contained in a line.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Smooth Fano threefolds of Picard number one were classified by Iskovskikh [Isk77, Isk78]. We call *1.n* the family No. n of [IP99, Section 12.2, Table 12.2]. The families 1.9 and 1.10 consist of the smooth Fano threefolds X with $\mathrm{Pic}(X) = \mathbb{Z}[-K_X]$ and $(-K_X)^3 = 18$ and $(-K_X)^3 = 22$, respectively, that is, of the smooth prime Fano threefolds of genus 10 and of genus 12. Any smooth prime Fano threefold of genus 10 is a codimension two linear section of the adjoint variety of the group G_2, a fivefold in $\mathbb{P}^{13}$ [Muk88], [Muk92, Section 2], see also [Vir26, Theorem 2.2], and the smooth prime Fano threefolds of genus 12 are described by nets of alternating forms on $\mathbb{C}^7$ [Muk89], [Muk92, Section 3, Theorem 3], [BKM26b, Theorem 1.1] (Definition 2.17). See also [IP99, Examples 5.2.2(iv) and (vi) and Theorem 5.2.3].

By [KPS18, Theorem 1.1.2], which extends [Pro90], the automorphism group of any smooth member of family 1.9 is finite. A smooth member X of family 1.10 has infinite $\mathrm{Aut}(X)$ exactly in the following cases [KPS18, Theorem 1.1.2], [DFK25, Theorem 1] (see Definition-Theorem 2.25): the Mukai–Umemura threefold X^{MU} [MU83], with $\mathrm{Aut}(X^{\mathrm{MU}}) \cong \mathrm{PGL}_2$, a threefold X^a, unique up to isomorphism, with $\mathrm{Aut}(X^a) \cong \mathbb{G}_a \rtimes \mu_4$, and the threefolds with a faithful $\mathbb{G}_m$-action other than X^{MU}, which form a one-parameter family and have automorphism group $\mathbb{G}_m \rtimes \mu_2$ [KP18, Theorem 1.3].

K-stability was introduced by Tian [Tia97] and Donaldson [Don02], and by the Yau–Tian–Donaldson theorem [CDS15, Tia15], [Ber16, Theorem 1.1], a Fano manifold admits a Kähler–Einstein metric if and only if it is K-polystable in the sense of Definition 2.4. In this paper K-stability is tested through valuations. We use the stability threshold $\delta(X)$ of [FO18, BJ20] (Definition 2.2) and the valuative criterion recalled in Theorem 2.5. K-semistability is open in $\mathbb{Q}$-Gorenstein families [BLX22, Corollary 1.2], a K-polystable Fano variety has reductive automorphism group [ABHLX20, Theorem 1.3], and a K-stable Fano variety has finite automorphism group [BX19, Corollary 1.3]. Thus a K-polystable Fano manifold X with infinite $\mathrm{Aut}(X)$ satisfies $\delta(X) = 1$: indeed, $\delta(X) \geq 1$ since X is K-semistable [BJ20, Theorem B], and $\delta(X) > 1$ would make X uniformly K-stable [BX19, Theorem 2.9], hence K-stable with finite $\mathrm{Aut}(X)$ [BX19, Corollary 1.3].

For Picard number one, the K-stability of all families other than 1.9 and 1.10 is known.

(1) Any smooth Fano threefold X of Picard number one with $(-K_X)^3 \notin \{18, 22\}$, other than $\mathbb{P}^3$, the smooth quadric threefold, and V_5, is uniformly K-stable [AZ23, Theorem C]. Thus [AZ23, Theorem C] covers the families 1.1–1.8 and 1.11–1.14.

(2) The threefolds V_5, the smooth quadric threefold, and $\mathbb{P}^3$ of the families 1.15, 1.16, and 1.17 have infinite automorphism groups, and they are K-polystable but not K-stable [ACC+23, Section 3.2 and Example 3.2].

Xu and Zhuang ask whether any smooth Fano threefold of Picard number one is K-semistable [XZ26, Problem 10]. They point out that only the families 1.9 and 1.10 are open. For family 1.9, the conjecture in the Introduction of [ACC+23] predicts more, namely that any smooth member is K-stable.

We now explain why the method of [AZ22, AZ23] does not reach $(-K_X)^3 = 18$ and $(-K_X)^3 = 22$ directly. Let X be a smooth Fano threefold of Picard number one, let $H = -K_X$, let $x \in X$, and let $S \in |H|$ be an integral surface which is smooth at x. By [AZ23, Theorem 3.1],

$$\delta_x(X) \geq \frac{4}{(-K_X)^3}\epsilon_x(H|_S),$$

where $\delta_x(X)$ is $\delta_Z(X)$ of Definition 2.2 with $Z = \{x\}$, and $\epsilon_x(H|_S)$ is the Seshadri constant of $H|_S$ at x. Since $((H|_S)^2)_S = (-K_X)^3$, we have $\epsilon_x(H|_S) \leq \sqrt{(-K_X)^3}$ [AZ23, Section 2]. Thus the lower bound for $\delta_x(X)$ is at most $4/\sqrt{(-K_X)^3}$, which equals

$$\frac{4}{\sqrt{18}} \approx 0.943 \ \text{ if } (-K_X)^3 = 18, \qquad \frac{4}{\sqrt{22}} \approx 0.853 \ \text{ if } (-K_X)^3 = 22,$$

and the bound is smaller than 1 at any point of X. In this paper we use instead the adjunction estimate [AZ22, Theorem 3.2]: for a plt-type prime divisor F over X whose center on X contains x, it bounds $\delta_x(X)$ from below by the minimum of $A_X(F)/S_X(F)$ and the infimum of the stability thresholds, at the points of F over x, of the refinement of $|-K_X|$ by F, with the different of F as the boundary on F. We choose F according to the lines on X through x, that is, the curves $L \subset X$ with $(-K_X \cdot L) = 1$ that pass through x. To choose the extracted divisors and to treat the points and curves where the local estimates are not strict, we use Iskovskikh's double projection from a line [Isk89] in genus 10 (Theorem 2.22, [KPZ14, Theorem 2.6]) and in genus 12 (Theorem 2.23), the link from a conic in genus 12 (Theorem 2.24), the lines through a point [KP23, Lemma 5.6 and Corollary A.6], and the width of Reid's pagoda (Definition-Proposition 2.21).

For family 1.9, the general member is K-stable [ACC+23, Example 4.10], by a degeneration to a K-stable Fano threefold with one Gorenstein terminal singular point and the smoothings of such threefolds [Nam97, Theorem 11], [JR11, Theorem 1.4]. No previous result covered all smooth members of family 1.9.

Family 1.10 has a special place in the history of K-stability. Tian [Tia97] showed that some small deformations of X^{MU} without non-zero holomorphic vector fields admit no Kähler–Einstein metric, and Tian's example motivated the notion of K-stability [KLPZ26, Section 1.4.1]. The following members of family 1.10 were known to be K-polystable or K-semistable.

(1) The threefold X^{MU} is Kähler–Einstein [Don07, Theorem 1 and Corollary 1], [Don08, Theorem 5.1], hence K-polystable.
(2) Any member with a faithful $\mathbb{G}_m$-action is Kähler–Einstein, hence K-polystable, as Donaldson expected in [Don08, Section 5.3]. All but two of the members with a faithful $\mathbb{G}_m$-action are treated in [CS23, Theorem 1.6 and Corollary 1.7], and the other two in [Fuj23, Theorem 1.2]; see also [ACC+23, Example 4.11].
(3) The threefold X^a is K-semistable and not K-polystable [CS23, Example 1.4]. Indeed, $\mathrm{Aut}(X^a)$ is not reductive, so X^a is not K-polystable by [ABHLX20, Theorem 1.3].
(4) The member with a faithful action of $\mathrm{PSL}_2(\mathbb{F}_7)$, which is the variety of polar hexagons of the Klein quartic, is K-stable, hence so is the general member [ACC+23, Example 4.12].

General members together with the openness of K-semistability and of K-stability [BLX22, Corollary 1.2 and Theorem 4.5] only yield K-stability on a non-empty open subset of the members of each family. Moreover, in family 1.10 K-stability fails on a closed subset. The threefold X^{MU} and the members with a faithful $\mathbb{G}_m$-action are K-polystable with infinite automorphism groups, so $\delta = 1$ for them, and X^a is strictly K-semistable. Thus a local estimate that holds for all members of family 1.10 cannot be strict everywhere, and where equality occurs, the proof has to use the global geometry of X.

We now recall the description of family 1.10 by nets of alternating forms, which is the setting of Donaldson's conjecture. Let $V = \mathbb{C}^7$. For a net $\Pi \subset \Lambda^2 V^\vee$, that is, a three-dimensional linear subspace, let $X_\Pi \subset \mathrm{Gr}(3, V)$ be the scheme of three-dimensional subspaces of V that are isotropic for any $\omega \in \Pi$, and let Ω be the set of non-degenerate nets Π, that is, nets whose non-zero forms all have rank 6, such that X_Π is a smooth prime Fano threefold of genus 12. The precise definitions are given in Definition 2.17. Any smooth prime Fano threefold X of genus 12 is isomorphic to X_Π for some $\Pi \in \Omega$ [Muk89], [Muk92, Section 3, Theorem 3], [BKM26b, Theorem 1.1].

At X^{MU}, the Kuranishi slice is a neighbourhood of 0 in the space $\mathrm{Sym}^8 \mathbb{C}^2$ of binary octics with the action of PSL_2 [Don08, Section 5.3], and Tian's deformations are the ones given by GIT-unstable octics, that is, by octics whose orbit closure contains 0. Donaldson explains that a small deformation of X^{MU} given by a non-zero octic is Kähler–Einstein if and only if the octic is GIT-polystable, that is, the octic has no root of multiplicity at least four, or it has two distinct roots of multiplicity four, in which case the deformation carries a $\mathbb{C}^*$-action [Don18, Section 4.1]. Theorem 1.7 below is a global algebraic version of Donaldson's local picture, with the harmonic

slice, that is, the plane quartics q^2+b, where $q=xz-y^2$ and b runs through the space Harm_4 of harmonic quartics defined in Theorem 1.7, in place of the Kuranishi slice.

Donaldson also points out that the deformation argument implies nothing about nets far from the net of X^{MU} [Don18, Section 4.1, p. 217]. Thus a global description of the K-polystable members needs global invariants. In this paper we use the GIT of nets under $\mathrm{SL}(V)$ and the GIT of plane quartics under SL_3 (Theorems 1.3 and 1.5).

Donaldson's global prediction concerns the set $\Delta'\subset\Omega$ of nets Π such that some divisor in $|-K_{X_\Pi}|$ has a singular point whose germ is analytically isomorphic to $(\{y^2=x^3+t^4x\}\ni 0)$ (Definition 2.17). In Donaldson's terminology, K-stability means what is now called K-polystability [Don18, Section 1.3, Remark (1)], and we formulate his prediction in our terms.

Conjecture 1.1 (Donaldson [Don18, Section 4.1])**.** *For any $\Pi\in\Omega$, the threefold X_Π is not K-polystable if and only if X_Π admits no non-trivial $\mathbb{C}^*$-action and Π belongs to the closure of the set Δ' of Definition 2.17.*

A non-trivial $\mathbb{C}^*$-action on X_Π induces a faithful $\mathbb{G}_m$-action, since the kernel of a non-trivial action is a finite subgroup μ_n and $\mathbb{G}_m/\mu_n\cong\mathbb{G}_m$. Theorem 1.3 is formulated with faithful $\mathbb{G}_m$-actions. The net of Theorem 1.8 belongs to Δ' itself, so it does not matter in which space the closure of Δ' is taken. The book [ACC+23] contains a form of Conjecture 1.1 as [ACC+23, Conjecture 7.1], and we discuss it in Remark 1.9.

Family 1.10 is also described by plane quartics. Any smooth prime Fano threefold X of genus 12 is isomorphic to the variety $\mathrm{VSP}(f,6)$ of polar hexagons of a plane quartic f [Muk92, Section 6, Theorem 11], and, for quartics f satisfying the hypothesis of [Muk92, Section 6, Theorem 11], the projective equivalence class of f is determined by X. For any X, Theorem 1.5(3) and Proposition 16.10(2) show that the points $[f]\in\mathbb{P}(\mathbb{C}[x,y,z]_4)$ such that f has invertible middle catalecticant and $\mathrm{VSP}(f,6)\cong X$ form one PGL_3-orbit. For example, X^{MU} is the variety of polar hexagons of a double smooth conic [Muk92, Section 7, Theorem 14], [DKK15, Section 2.1]. The Hilbert scheme $\Sigma(X)$ of lines on X is another plane quartic [Muk04, Section 4]. It is related to f by the covariant quartic of f, which is built from the Aronhold invariant of the polar cubics of f [Muk04, Section 5], [DK93, (6.5) and Section 7], and which can be computed by [Ott24, Algorithm 5].

By [Sch01, Theorem 6.1], for general X the Hilbert scheme $\Sigma(X)$ is isomorphic to the covariant quartic of f, and according to [KPS18, Remark 5.4.8], a complete proof for all members of family 1.10 has not been published. For X^{MU}, X^a, and the members with a faithful $\mathbb{G}_m$-action other than X^{MU}, [KPS18, Warning 5.4.5] describes $\Sigma(X)$ without proof as a double conic, a union of two conics with one common point, and a union of two conics tangent at two points, respectively. By [KPS18, Proposition 5.4.4], $\Sigma(X)$ is singular for any member X of family 1.10 with infinite $\mathrm{Aut}(X)$, and such X is not K-stable [BX19, Corollary 1.3]. Thus a comparison of K-stability with the GIT of plane quartics has to cover the members with singular $\Sigma(X)$, and an isomorphism known only for general X is not enough.

Such a comparison is suggested in [ACC+23, Section 7.1], where Conjecture 1.1 is tested against the correspondence between the GIT of plane quartics and the K-stability of the threefolds. For the action of SL_3, a plane quartic is GIT-stable if and only if it is reduced and has only singular points of type A_1 and A_2, and it is GIT-semistable if and only if it has no point of multiplicity at least three and it is not the union of a cubic curve and a line tangent to the cubic at an inflection point. The GIT-polystable quartics that are not GIT-stable are the quartics projectively equivalent to $(xz-ay^2)(xz-by^2)$ with $(a,b)\neq(0,0)$, that is, the double smooth conics, the unions of two smooth conics tangent at two points, and the unions of a smooth conic and two lines tangent to it [MFK94, Chapter 4, Section 2], [HL10, Theorem 2], [OSS16, Lemma 5.9], [Laz06, Section 4.1.1] (see Theorem 2.29).

A close analogue is the K-moduli space of del Pezzo surfaces of degree two, which is the blow-up of the GIT quotient of plane quartics at the point of the double smooth conic, with exceptional divisor the GIT quotient of binary octics [OSS16, Theorems 1.1, 5.10 and 5.13]. A

K-moduli space also coincides with a GIT quotient for cubic threefolds [LX19], and for the pairs $(\mathbb{P}^2, cC)$, where C is a plane curve of degree at least four and $c > 0$ is smaller than a threshold depending on the degree of C [ADL24, Theorem 1.3(1)].

The boundary of the K-moduli space of family 1.10 was studied in [KLPZ26]. The boundary of the K-moduli stack, which parametrizes the singular members, is purely divisorial with four irreducible components, which correspond to the four families of one-nodal degenerations of the members of family 1.10 [Pro16], see also [KP25], and the forgetful morphism from the moduli of pairs of a K-semistable such threefold and an anti-canonical K3 surface with at worst ADE singularities to the moduli of polarized K3 surfaces of genus 12 is an open immersion [KLPZ26, Theorems 1.1 and 1.2]. A general one-nodal threefold of each of the four types is K-polystable [DK25, Theorem 1.1]. Before the K-moduli space, no compactification of the moduli of smooth members of family 1.10 with a modular meaning was known, and the GIT of the Grassmannian construction [GLT14, Section 2.1] is difficult to analyze [KLPZ26, p. 2].

The goal of this paper is to settle the K-stability of the smooth members of the families 1.9 and 1.10. We prove that any smooth member of family 1.9 is K-stable (Theorem 1.2). For family 1.10, we characterize K-stability and K-polystability through lines, through orbits of nets, and through two plane quartics (Theorems 1.3 and 1.5), we show that the harmonic slice at X^{MU} contains a representative of any K-polystable member (Theorem 1.7), and we construct an explicit counterexample to Conjecture 1.1 (Theorem 1.8).

1.1. **K-stability in genus ten and twelve.** Let X be a smooth prime Fano threefold of genus g, that is, a smooth projective threefold with $-K_X$ ample, $\mathrm{Pic}(X) = \mathbb{Z}[-K_X]$, and $(-K_X)^3 = 2g-2$ (see Definition 2.11), and write $H := -K_X$. In the numbering of [IP99, Section 12.2, Table 12.2], the smooth prime Fano threefolds of genus 10 form the family 1.9, and those of genus 12 form the family 1.10. For $g \in \{10, 12\}$ the divisor H is very ample [IP99, Proposition 4.1.11], and a *line* on X is a line in the anti-canonical embedding $X \subset \mathbb{P}^{g+1}$, that is, a closed subscheme $L \subset X$ with Hilbert polynomial $t+1$ with respect to H. A line L on X is *special* if

$$N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$$

(see Definition 2.11).

A prime divisor F over X is a prime divisor on a normal projective variety Y with a birational morphism $\mu : Y \to X$. We write $A_X(F)$ for the log discrepancy of F with respect to X, and

$$S_X(F) := \frac{1}{(-K_X)^3} \int_0^{+\infty} \mathrm{vol}_Y(-\mu^* K_X - tF)dt,$$

$$\beta_X(F) := A_X(F) - S_X(F), \qquad \delta(X) := \inf_F \frac{A_X(F)}{S_X(F)},$$

where the infimum runs through prime divisors F over X (see Definition 2.2). For K-semistability, K-polystability, K-stability, and uniform K-stability we use Definition 2.4. By [LXZ22, Theorems 2.9 and 1.6] (see Theorem 2.5), X is K-semistable if and only if $\delta(X) \geq 1$, and X is K-stable if and only if $\beta_X(F) > 0$ for every prime divisor F over X, if and only if X is uniformly K-stable, if and only if $\delta(X) > 1$.

Our first main result concerns all smooth members of the family 1.9.

Theorem 1.2. *Let X be a smooth prime Fano threefold of genus 10, that is, a smooth member of the family 1.9. Then $\beta_X(F) = A_X(F) - S_X(F) > 0$ for any prime divisor F over X. In particular, X is K-stable and uniformly K-stable, and $\delta(X) > 1$.*

For genus 12 we use the description of X by a net of alternating forms. We use the notation V, X_Π, and Ω of Definition 2.17, and we write $\mathbb{B} := \mathrm{Gr}(3, \Lambda^2 V^\vee)$, where $\dim V = 7$ and a point $\Pi \in \mathbb{B}$ is a net of alternating forms on V (the letter Π for a net follows [Don18, Section 4.1]). A net $\Pi \in \mathbb{B}$ is *non-degenerate* if every non-zero form $\omega \in \Pi$ has rank 6. The non-degenerate nets form an open subset $\mathbb{B}^{\mathrm{nd}} \subset \mathbb{B}$, and $\Omega \subset \mathbb{B}^{\mathrm{nd}}$. The group $\mathrm{SL}(V)$ acts on $\mathbb{B}$ and preserves $\mathbb{B}^{\mathrm{nd}}$ and

Ω. We denote by $\Omega^{\mathrm{st}} \subset \Omega$ the set of nets $\Pi \in \Omega$ such that the orbit $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and the stabilizer of Π in $\mathrm{SL}(V)$ is finite (see Definition 2.17).

Let X be a smooth prime Fano threefold of genus 12, and let $\mathcal{U}_X$ be the Mukai bundle of X, the exceptional vector bundle of rank 3 on X characterized in Definition-Theorem 2.20. We set $V_X := H^0(X, \mathcal{U}_X^\vee)^\vee$, so that $\dim V_X = 7$. The *canonical Mukai net* of X is

$$\Pi_X := \mathrm{Ker}\left(\Lambda^2 V_X^\vee \to H^0(X, \Lambda^2 \mathcal{U}_X^\vee)\right).$$

To view Π_X as a point of $\mathbb{B}$, we choose a linear isomorphism $V_X \cong V$. For any such isomorphism, the image of Π_X in $\mathbb{B}$ belongs to Ω and $X \cong X_{\Pi_X}$, and the images for different isomorphisms form one $\mathrm{SL}(V)$-orbit (see Definition-Theorem 2.20). In particular, the conditions $\Pi_X \in \Omega^{\mathrm{st}}$ and "$\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$" depend only on X.

Our second main result describes K-stability of all smooth members of the family 1.10 in terms of lines on X and of the canonical Mukai net.

Theorem 1.3. *Let X be a smooth prime Fano threefold of genus 12, and let $\Pi_X \in \Omega$ be the canonical Mukai net of X, viewed in $\mathbb{B}$ via a linear isomorphism $V_X \cong V$. Then X is K-semistable, and the following hold.*

(1) The following conditions are equivalent: (a) $\delta(X) > 1$, (b) X is K-stable, (c) X is uniformly K-stable, (d) $\Pi_X \in \Omega^{\mathrm{st}}$, (e) the orbit $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and the algebraic group $\mathrm{Aut}(X)$ is finite, and (f) every special line on X intersects another line on X. If (a)–(f) hold, then $A_X(F) > S_X(F)$ for any prime divisor F over X. If (a)–(f) fail, then $\delta(X) = 1$.

(2) The following conditions are equivalent: (i) there exist $D \in |-K_X|$ and a point $p \in D$ contained in a line on X such that the germ $(D \ni p)$ is analytically isomorphic to $(\{y^2 = x^3 + t^4 x\} \ni 0)$, (ii) X contains a special line which intersects no other line on X, (iii) X admits an anti-canonical test configuration with smooth central fiber on which the induced $\mathbb{G}_m$-action is non-trivial (product test configurations are allowed), (iv) the orbit $\mathrm{SL}(V) \cdot \Pi_X$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, or X admits a faithful $\mathbb{G}_m$-action, and (v) $\Pi_X \notin \Omega^{\mathrm{st}}$. Moreover, for D and p as in (i), the line through p is unique, it is special, and it intersects no other line on X.

(3) X is K-polystable if and only if $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$, if and only if X admits a faithful $\mathbb{G}_m$-action or there exists no pair (D, p) as in (2)(i). If $\mathrm{SL}(V) \cdot \Pi_X$ is closed and $\mathrm{Aut}(X)$ is infinite, then X is K-polystable, not K-stable, and $\delta(X) = 1$. If $\mathrm{SL}(V) \cdot \Pi_X$ is not closed, then X is K-semistable, not K-polystable, and $\delta(X) = 1$.

Conditions (2)(ii) and (2)(v) are the negations of conditions (1)(f) and (1)(d), so the conditions in (2) are equivalent to the failure of the conditions in (1).

The anti-canonical test configurations in (2)(iii) and the product test configurations are those of Definition 2.4. The germ $(\{y^2 = x^3 + t^4 x\} \ni 0)$ in (2)(i) is a simple elliptic singularity of type $\tilde{E}_8$ [Sai74], and the exceptional curve of the minimal resolution of the germ is isomorphic to the elliptic curve $\{y^2 = x^3 + x\}$, whose j-invariant is 1728.

Since $\mathbb{B}^{\mathrm{nd}}$ is the non-vanishing locus of an $\mathrm{SL}(V)$-invariant section $c_{\mathrm{Pf}} \in H^0(\mathbb{B}, \mathcal{A}^{14})$ of a power of the Plücker line bundle $\mathcal{A}$ (see Definition-Proposition 16.2), every point of $\mathbb{B}^{\mathrm{nd}}$ is GIT-semistable, and $\mathbb{B}^{\mathrm{nd}}$ is saturated with respect to the quotient map $\mathbb{B}^{\mathrm{ss}} \to \mathbb{B}^{\mathrm{ss}} /\!\!/ \mathrm{SL}(V)$, where $\mathbb{B}^{\mathrm{ss}} \subset \mathbb{B}$ is the GIT-semistable locus for the linearization $\mathcal{A}$. Thus the orbit $\mathrm{SL}(V) \cdot \Pi$ of a net $\Pi \in \mathbb{B}^{\mathrm{nd}}$ is closed in $\mathbb{B}^{\mathrm{nd}}$ if and only if Π is GIT-polystable, and Ω^{st} is the set of GIT-stable nets in Ω.

The condition in (2)(i) that p is contained in a line on X cannot be removed (see Theorem 1.8). Condition (2)(iii) allows product test configurations. Indeed, if X admits a faithful $\mathbb{G}_m$-action, then X satisfies (2)(iv), hence (2)(i) and (2)(ii), and the product test configuration induced by the $\mathbb{G}_m$-action has central fiber X with non-trivial induced $\mathbb{G}_m$-action, hence satisfies (2)(iii) (see also Theorem 18.1). The smoothness of the central fiber in (2)(iii) cannot be removed: for any $D \in |H|$, the filtration of $\bigoplus_{m \geq 0} H^0(X, mH)$ by the order of vanishing along D gives an

anti-canonical test configuration whose central fiber is the projective cone over D with non-trivial induced $\mathbb{G}_m$-action, so without smoothness (2)(iii) would hold for any X.

Part (3) of Theorem 1.3 is a corrected form of Conjecture 1.1. In part (3), the non-trivial $\mathbb{C}^*$-action of Conjecture 1.1 becomes a faithful $\mathbb{G}_m$-action, which is an equivalent condition, as explained after Conjecture 1.1. The condition $\Pi \in \overline{\Delta'}$ of Conjecture 1.1 becomes the existence of a pair (D, p) as in (2)(i), where p is contained in a line on X.

By Theorems 1.2 and 1.3 and the results on the other families recalled at the beginning of the introduction, the question of Xu and Zhuang [XZ26, Problem 10] has an affirmative answer.

Corollary 1.4. *Let X be a smooth Fano threefold of Picard number one. Then X is K-semistable.*

We prove Corollary 1.4 in Section 22.

1.2. **Plane quartics and the slice at the Mukai–Umemura threefold.** Recall from Definition-Theorem 2.25 the Mukai–Umemura threefold X^{MU}, the threefold X^a, and the members with a faithful $\mathbb{G}_m$-action.

Let X be a smooth prime Fano threefold of genus 12. The seven principal 6×6 Pfaffians of the alternating 7×7 matrix $(\omega \to \omega(e_i, e_j))_{0 \le i,j \le 6}$ of the canonical Mukai net Π_X, whose entries belong to $\Pi_X^\vee$, for a basis $e_0, \dots, e_6$ of V_X generate a Gorenstein ideal of codimension three of $\operatorname{Sym} \Pi_X^\vee$, whose quotient ring has socle degree four and has Macaulay inverse system in degree four spanned by a quartic $f_X \in \operatorname{Sym}^4 \Pi_X$, the *apolar quartic* of X (see Definition-Proposition 2.28). For a ternary quartic f, let $f^\perp \subset \mathbb{C}[\partial] := \mathbb{C}[\partial_x, \partial_y, \partial_z]$ be its apolar ideal, the ideal of differential operators annihilating f, and let $\mathrm{Cat}_2(f) : \mathbb{C}[\partial]_2 \to \mathbb{C}[x, y, z]_2$, $P \to P(f)$, be its middle catalecticant, whose kernel is $(f^\perp)_2$ (Definition 2.27). For a ternary quartic f, $\mathrm{VSP}(f, 6)$ is the closure in Hilb^6 of the locus of six distinct points $[\ell_1], \dots, [\ell_6]$ with $f \in \mathbb{C}\ell_1^4 + \cdots + \mathbb{C}\ell_6^4$, and $\mathrm{VPS}(f, 6) \supset \mathrm{VSP}(f, 6)$ is the scheme of length-six subschemes whose homogeneous ideal is contained in $f^\perp$ (see Definition 2.27).

The *Scorza quartic* of f is

$$\mathrm{Sc}(f)(a) = \mathfrak{A}\big(a_1 \partial_x f + a_2 \partial_y f + a_3 \partial_z f\big) \quad \text{for } a = (a_1, a_2, a_3) \in \mathbb{C}^3,$$

regarded as a quartic in (x, y, z) by renaming (a_1, a_2, a_3) as (x, y, z), where $\mathfrak{A}$ is the Aronhold invariant of plane cubics (see Definition 2.27). The map $f \to \mathrm{Sc}(f)$ is the Scorza transform [DK93, Section 7], and $\mathrm{Sc}(f)$ is the covariant quartic of f [Muk04, Section 5]. We write

$$Q_X := (\mathrm{Sc}(f_X) = 0) \subset \mathbb{P}(\Pi_X^\vee).$$

The quartics f_X and Q_X play different roles. The quartic f_X belongs to the side of nets. Indeed, every net $\Pi \in \mathbb{B}^{\mathrm{nd}}$ has an apolar quartic f_Π with invertible middle catalecticant (Definition-Proposition 2.28), and by Proposition 20.11, for $\Pi, \Pi' \in \mathbb{B}^{\mathrm{nd}}$, Π belongs to the closure of $\mathrm{SL}(V) \cdot \Pi'$ in $\mathbb{B}^{\mathrm{nd}}$ if and only if $[f_\Pi]$ belongs to the closure of $\mathrm{PGL}_3 \cdot [f_{\Pi'}]$ in the open subset U^{cat} of quartics with invertible middle catalecticant. Thus GIT of nets translates into GIT of f_X. The quartic Q_X is isomorphic to the Hilbert scheme $\Sigma(X)$ of lines by Theorem 1.5(1), an object visible in the geometry of X. The curve Q_X is defined by the Scorza quartic of f_X, and the curves $(f_X = 0)$ and Q_X need not be projectively equivalent, for instance when f_X is projectively equivalent to $F_0 = yz(2x^2 + yz)$ (Remark 1.6).

We use GIT of plane quartics with respect to the action of SL_3 on $\mathbb{P}(\mathbb{C}[x, y, z]_4)$ with the linearization $\mathcal{O}(1)$. The GIT-stable and the GIT-polystable quartics are described in Theorem 2.29 (see [MFK94, Chapter 4, Section 2] and [CDGF+25, Theorem 5.1]). In particular, every smooth plane quartic is GIT-stable.

Theorem 1.5. *Let X be a smooth prime Fano threefold of genus 12, let $f_X \in \operatorname{Sym}^4 \Pi_X$ be the apolar quartic of the canonical Mukai net Π_X, and let $Q_X := (\mathrm{Sc}(f_X) = 0) \subset \mathbb{P}(\Pi_X^\vee)$. All GIT notions for plane quartics refer to the action of SL_3 on $\mathbb{P}(\mathbb{C}[x, y, z]_4)$ with the linearization $\mathcal{O}(1)$. Then the following hold.*

(1) *The middle catalecticant of f_X is invertible, and the Hilbert scheme $\Sigma(X)$ of lines on X, with the scheme structure, is isomorphic to the plane quartic Q_X. The quartics f_X and Q_X are GIT-semistable.*

(2) *X is K-stable, that is, the equivalent conditions of Theorem 1.3(1) hold, if and only if f_X is GIT-stable, if and only if Q_X is GIT-stable. Moreover, X is K-polystable, that is, the equivalent conditions of Theorem 1.3(3) hold, if and only if f_X is GIT-polystable, if and only if Q_X is GIT-polystable. In particular, X is K-semistable but not K-polystable if and only if f_X, equivalently Q_X, is GIT-semistable but not GIT-polystable, and X is K-polystable but not K-stable if and only if f_X, equivalently Q_X, is GIT-polystable but not GIT-stable.*

(3) *Let $U^{\mathrm{sm}} \subset \mathbb{P}(\mathbb{C}[x,y,z]_4)$ be the set of $[f]$ with $(f^\perp)_2 = 0$ such that $\mathrm{VPS}(f,6)$ is smooth. For $[f] \in U^{\mathrm{sm}}$, $\mathrm{VSP}(f,6) = \mathrm{VPS}(f,6)$, and $\mathrm{VSP}(f,6)$ is a smooth prime Fano threefold of genus 12. Sending $[f]$ to $\mathrm{VPS}(f,6)$ induces a bijection from the set of PGL_3-orbits in U^{sm} onto the set of isomorphism classes of smooth prime Fano threefolds of genus 12, whose inverse sends X to the orbit of $[f_X]$, and (1) and (2) hold for $X = \mathrm{VPS}(f,6)$ with f and $\mathrm{Sc}(f)$ in place of f_X and $\mathrm{Sc}(f_X)$.*

(4) *The quartic Q_X is non-reduced if and only if Q_X is a double smooth conic, if and only if $X \cong X^{\mathrm{MU}}$. The quartic Q_X is the union of two distinct smooth conics tangent at two distinct points if and only if $X \ncong X^{\mathrm{MU}}$ and X admits a faithful $\mathbb{G}_m$-action. The double smooth conics and the unions of two smooth conics tangent at two points are exactly the GIT-polystable but not GIT-stable quartics Q_X. The quartic Q_X has a singular point of type A_7 if and only if $X \cong X^a$, if and only if Q_X is the union of two smooth conics with one common point, where the intersection multiplicity is four. Every reduced Q_X has only singular points of type A_k with $k \le 7$. Neither a smooth conic together with two distinct tangent lines nor a cubic together with an inflectional tangent line is of the form Q_X.*

(5) *Up to projective equivalence, the quartics $F_h = \frac{h}{6}x^4 + 2x^2yz + y^2z^2$ of Definition 2.27, $h \in \mathbb{C}\backslash\{1,2\}$, are the apolar quartics of the smooth prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action, and $h = 6$ corresponds to X^{MU}.*

Theorem 19.1 in Section 19 adds that the projective Scorza map is étale at $[f_X]$ when Q_X is not GIT-stable, computes the Scorza quartics of F_h and of an apolar quartic of X^a, and describes the GIT-polystable quartics in the orbit closures of $[f_X]$ and $[\mathrm{Sc}(f_X)]$. The equivalences in Theorem 1.5 between the quartics f_X, Q_X and the orbit of Π_X are geometric and do not use K-stability: GIT-stability is read off from the singularities of $\Sigma(X)$ at the special lines (Propositions 10.8 and 16.17), and K-stability enters (2) only through Theorem 1.3(1) and (3).

Remark 1.6. By Theorem 1.5(5), $F_0 = yz(2x^2+yz)$ is the apolar quartic of a smooth prime Fano threefold of genus 12. It is a smooth conic together with two tangent lines, which by Theorem 1.5(4) is not of the form Q_X. Hence, for this X, the curve $(f_X = 0)$ is not projectively equivalent to the curve Q_X, which is isomorphic to the Hilbert scheme of lines $\Sigma(X)$ by Theorem 1.5(1) (compare [ACC+23, Section 7.1]). Theorem 1.5 contains the statements of [KPS18, Remark 5.4.8] for every smooth prime Fano threefold of genus 12, and the statements of [KPS18, Warning 5.4.5] for X^{MU}, X^a, and the members with a faithful $\mathbb{G}_m$-action.

Next we describe a slice at the Mukai–Umemura threefold. Let $q := xz - y^2$. The group $\mathrm{SO}(q) \cong \mathrm{PSL}_2$ acts on $\mathbb{C}[x,y,z]_1 \cong \mathrm{Sym}^2\mathbb{C}^2$ through the conic $(x,y,z) = (u^2, uv, v^2)$, and

$$\mathbb{C}[x,y,z]_4 \cong \mathrm{Sym}^4(\mathrm{Sym}^2\mathbb{C}^2) \cong \mathrm{Sym}^8\mathbb{C}^2 \oplus \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathbb{C}$$

as SL_2-representations, the three summands being the space $\mathrm{Harm}_4 := \{b \in \mathbb{C}[x,y,z]_4 : (4\partial_x\partial_z - \partial_y^2)b = 0\}$ of harmonic quartics, q times the space of quadrics c with $(4\partial_x\partial_z - \partial_y^2)c = 0$, and $\mathbb{C}q^2$. We consider the quartics $f_b := q^2 + b$ with $b \in \mathrm{Harm}_4$.

The open subset $\mathrm{Harm}_4^\circ \subset \mathrm{Harm}_4$ below is defined through VPS rather than VSP. By Lemma 16.3(2), for every quartic f with $(f^\perp)_2 = 0$, $\mathrm{VPS}(f,6) \cong X_\Pi$ for a net $\Pi \in \mathbb{B}^{\mathrm{nd}}$ with a basis in which f_Π is a non-zero multiple of f. Thus $\mathrm{VPS}(f,6)$ is smooth if and only if $\Pi \in \Omega$

(Proposition 16.10(1)). No such f is lost in this way: if $\mathrm{VSP}(f,6)$ is a smooth prime Fano threefold of genus 12, then $\mathrm{VSP}(f,6)=\mathrm{VPS}(f,6)$ (Proposition 16.10(2)). The condition $\det \mathrm{Cat}_2(f_b)\neq 0$ guarantees that f_b is the apolar quartic of a non-degenerate net.

Scalar matrices in SL_3 act trivially on $\mathbb{P}(\mathbb{C}[x,y,z]_4)$, so the SL_3-orbits and the PGL_3-orbits in $\mathbb{P}(\mathbb{C}[x,y,z]_4)$ coincide, and the GIT quotient $\mathbb{P}(\mathbb{C}[x,y,z]_4)^{\mathrm{ss}}/\!\!/\mathrm{SL}_3$ is also a good quotient by PGL_3. We write $U^{\mathrm{sm}}/\!\!/\mathrm{PGL}_3$ for the image of U^{sm} in the GIT quotient $\mathbb{P}(\mathbb{C}[x,y,z]_4)^{\mathrm{ss}}/\!\!/\mathrm{SL}_3$. Since U^{sm} is a saturated invariant open subset of the GIT-semistable locus by Theorem 20.14(1), $U^{\mathrm{sm}}/\!\!/\mathrm{PGL}_3$ is an open subscheme of the GIT quotient and a good quotient of U^{sm} [Alp13, Remark 6.2]. Let $M^{\mathrm{K}}_{3,22}$ be the K-moduli space, the projective good moduli space of K-semistable $\mathbb{Q}$-Fano threefolds of anti-canonical volume 22 (see [KLPZ26, Theorem 2.25] and the references given before it).

Theorem 1.7. *Let $q:=xz-y^2$, let $\mathrm{Harm}_4:=\{b\in\mathbb{C}[x,y,z]_4:(4\partial_x\partial_z-\partial_y^2)b=0\}$, and let $\mathrm{SO}(q):=\{g\in\mathrm{SL}_3:q\circ g=q\}\cong\mathrm{PSL}_2$ act on Harm_4 by $b\to b\circ g^{-1}$, where restriction to the conic $(x,y,z)=(u^2,uv,v^2)$ induces an isomorphism of Harm_4 onto the space $\mathrm{Sym}^8\mathbb{C}^2$ of binary octics. For $b\in\mathrm{Harm}_4$ let $f_b:=q^2+b$, and let*

$$\mathrm{Harm}_4^\circ:=\{b\in\mathrm{Harm}_4:\det\mathrm{Cat}_2(f_b)\neq 0 \text{ and } \mathrm{VPS}(f_b,6) \text{ is smooth}\}.$$

Let $\varpi:\mathrm{Harm}_4\to\mathrm{Harm}_4/\!\!/\mathrm{SO}(q)$ be the affine quotient. Then the following hold.

1. *Harm_4° is an $\mathrm{SO}(q)$-invariant open subset containing 0 with $\mathrm{Harm}_4^\circ=\varpi^{-1}(\varpi(\mathrm{Harm}_4^\circ))$, and $\varpi(\mathrm{Harm}_4^\circ)$ is open in $\mathrm{Harm}_4/\!\!/\mathrm{SO}(q)$ and is the quotient $\mathrm{Harm}_4^\circ/\!\!/\mathrm{SO}(q)$.*
2. *The threefolds $X_b:=\mathrm{VPS}(f_b,6)=\mathrm{VSP}(f_b,6)$, $b\in\mathrm{Harm}_4^\circ$, form an $\mathrm{SO}(q)$-equivariant smooth projective family of smooth prime Fano threefolds of genus 12, polarized by the relative anti-canonical bundle, with $X_0\cong X^{\mathrm{MU}}$. The morphism $\mathrm{PGL}_3\times^{\mathrm{SO}(q)}\mathrm{Harm}_4\to\mathbb{P}(\mathbb{C}[x,y,z]_4)$ given by $[g,b]\to g\cdot[f_b]$ is étale along the zero section.*
3. *Every X_b is K-semistable. Every smooth K-polystable prime Fano threefold of genus 12 is isomorphic to X_b for some $b\in\mathrm{Harm}_4^\circ$ whose $\mathrm{SO}(q)$-orbit is closed in Harm_4.*
4. *Sending b to $[f_b]$ induces a surjective morphism $\rho:\varpi(\mathrm{Harm}_4^\circ)\to U^{\mathrm{sm}}/\!\!/\mathrm{PGL}_3$, and the closed points of $U^{\mathrm{sm}}/\!\!/\mathrm{PGL}_3$ correspond bijectively to the isomorphism classes of smooth K-polystable prime Fano threefolds of genus 12.*
5. *The family $(X_b)_{b\in\mathrm{Harm}_4^\circ}$ induces a morphism $\kappa:\varpi(\mathrm{Harm}_4^\circ)\to M^{\mathrm{K}}_{3,22}$ to the projective good moduli space of K-semistable $\mathbb{Q}$-Fano threefolds of anti-canonical volume 22. The image of κ on $\mathbb{C}$-points is the locus of smooth K-polystable prime Fano threefolds of genus 12. For $b\in\mathrm{Harm}_4^\circ$, $\kappa(\varpi(b))$ is the class of the K-polystable degeneration of X_b, namely $\mathrm{VPS}(f_{\mathrm{cl}},6)$, where $[f_{\mathrm{cl}}]\in U^{\mathrm{sm}}$ is a GIT-polystable quartic in the quotient fiber of $[f_b]$, unique up to projective equivalence. Moreover, $\kappa(\varpi(b))$ is the class of $X_{b'}$ for some $b'\in\mathrm{Harm}_4^\circ$ whose $\mathrm{SO}(q)$-orbit is closed.*

1.3. **The original formulation of Donaldson's conjecture.** The following threefold is a K-stable member of family 1.10 with finite automorphism group whose net belongs to Δ'.

Theorem 1.8. *Let $V=\mathbb{C}^7$ with basis $e_0,\dots,e_6$, let $e_{ij}:=e_i^\vee\wedge e_j^\vee$, and let*

$$\begin{aligned}\omega_1&=6e_{02}-2e_{05}+10e_{14}-47e_{23}+6e_{35},\\ \omega_2&=6e_{03}-2e_{06}+6e_{12}+8e_{15}-37e_{24}+6e_{36}+6e_{45},\\ \omega_3&=6e_{13}-2e_{16}+10e_{25}-47e_{34}+6e_{46}.\end{aligned}$$

Let $\Pi:=\mathbb{C}\omega_1+\mathbb{C}\omega_2+\mathbb{C}\omega_3$ and $X:=X_\Pi\subset\mathrm{Gr}(3,V)$. Then the following hold.

1. *$\Pi\in\mathbb{B}^{\mathrm{nd}}$, and in the basis $(x,y,z):=(\omega_1/6,\omega_2/12,\omega_3/6)$ of Π the apolar quartic f_Π is a non-zero multiple of $f_1:=(xz-y^2)^2+y(x^3+z^3)$, the member $s=1$ of the pencil $f_s:=(xz-y^2)^2+sy(x^3+z^3)$. The quartic f_1 is smooth with invertible middle catalecticant.*
2. *X is a smooth prime Fano threefold of genus 12 with $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_X\cong\mathcal{O}_X(-K_X)$, and the scheme defined by $\omega_1,\omega_2,\omega_3$ over $\mathbb{Z}_{(3)}$ is smooth and projective of relative dimension three.*

(3) *There exist* $D \in |-K_X|$ *and* $p \in D$*, given explicitly in Construction-Proposition 21.1, such that the germ* $(D \ni p)$ *is analytically isomorphic to* $(\{y^2 = x^3 + t^4x\} \ni 0)$*, a simple elliptic singularity of type* $\tilde{E}_8$ *with* j*-invariant* 1728 *and Milnor number* 10*. In particular,* $\Pi \in \Delta'$*.*

(4) X *is K-stable and uniformly K-stable,* $\delta(X) > 1$*, and the algebraic group* $\mathrm{Aut}(X)$ *is finite. In particular,* X *admits no faithful* $\mathbb{G}_m$*-action.*

Hence X *contradicts Conjecture 1.1.*

The model over $\mathbb{Z}_{(3)}$ in (2) is how the smoothness of X is proved: the locus where the model over $\mathbb{Z}_{(3)}$ is not smooth of relative dimension three is proper over $\mathbb{Z}_{(3)}$, and its fiber over $\mathbb{F}_3$ is empty (Construction-Proposition 21.1). By (1) and (2), $\Pi \in \Omega$. Since $\mathrm{Aut}(X)$ is finite, X admits no non-trivial $\mathbb{C}^*$-action. Thus $\Pi \in \Delta'$ by (3), and X_Π is K-polystable by (4), which is excluded by Conjecture 1.1. Since X is K-stable, Theorem 1.3(2) shows that p is contained in no line on X, so the condition in Theorem 1.3(2)(i) that p is contained in a line cannot be dropped. The quartic f_1 has the form of Theorem 1.7: $f_1 = q^2 + b$, where $b = y(x^3 + z^3) \in \mathrm{Harm}_4$ restricts to the binary octic $uv(u^6 + v^6)$ on the conic $(x, y, z) = (u^2, uv, v^2)$.

Remark 1.9. In [ACC+23, Conjecture 7.1], Conjecture 1.1 appears in the following form: a smooth prime Fano threefold X of genus 12 is K-polystable if and only if X admits a faithful $\mathbb{G}_m$-action or no element of $|-K_X|$ has singularities of the form $y^2 = x^3 + t^4x$ or worse. The threefold X of Theorem 1.8 also contradicts the form in [ACC+23, Conjecture 7.1]: X is K-polystable, since it is K-stable, X admits no faithful $\mathbb{G}_m$-action, and $|-K_X|$ contains a divisor with a singular point analytically isomorphic to $(\{y^2 = x^3 + t^4x\} \ni 0)$.

Sketch of the proof. Theorems 1.2 and 1.3 are proved in three steps.

(1) By Theorem 2.5, we bound $A_X(F) - S_X(F)$ for prime divisors F over X according to $\mathrm{center}_X F$, sorting closed points x by their incidence type, that is, the lengths of the local rings of the scheme $\Sigma(X, x)$ of lines through x, which takes seven values (Proposition 4.1). Known arguments treat the centers that are prime divisors or curves C with $(H \cdot C) \geq 2$ in genus 10, or $(H \cdot C) \geq 3$ in genus 12 (Lemma 4.2, and Proposition 4.6 after [AZ23, proof of Theorem 5.12]). The lines, the conics in genus 12, and all closed points need new estimates.

At each center we extract a prime divisor E of plt type depending on the lines through it, bound the refinement W of the anti-canonical series by E in the sense of [AZ22] (Definition 2.8, Theorem 2.9), and deduce a lower bound for $A_X(F)$ in terms of $S_X(F)$ on X by Theorem 2.9 or Lemma 3.1. In genus 10 the estimates at all centers are strict (Sections 4–9).

(2) In genus 12 the estimates are strict except in two configurations without local estimate: the special lines L for which the estimate of Corollary 12.8(3) does not exclude $A_X(F) = S_X(F)$ for a prime divisor F with $\mathrm{center}_X F = L$, and the points of incidence type 3 on exactly one conic (Lemma 15.3). In either case X degenerates smoothly to a K-polystable member with a faithful $\mathbb{G}_m$-action (Lemma 12.3, Proposition 15.2, Theorem 2.26), so X is K-semistable by openness (Theorem 17.1).

(3) If $\Pi_X \in \Omega^{\mathrm{st}}$, the relative Mukai net of such a degeneration stays in the closed orbit $\mathrm{SL}(V) \cdot \Pi_X$, so the central fiber is isomorphic to X and has finite automorphism group, which contradicts the faithful $\mathbb{G}_m$-action on the central fiber (Proposition 16.13, Lemma 17.3). Thus neither configuration occurs on X, and all estimates are strict.

If $\Pi_X \notin \Omega^{\mathrm{st}}$, then $\mathrm{Aut}(X)$ is infinite or a one-parameter subgroup of $\mathrm{SL}(V)$ yields a non-product test configuration with Donaldson–Futaki invariant zero, so $\delta(X) = 1$ (Proposition 17.4), which proves Theorem 17.5. For $\Pi = \Pi_X$, part (1) of Theorem 17.5 is the K-semistability of X and the equivalence of (1)(a)–(e) in Theorem 1.3, and part (3) is the first equivalence of (3) there. Theorems 18.1 and 19.1 express the condition $\Pi_X \in \Omega^{\mathrm{st}}$ through the lines on X and through plane quartics, which completes Theorem 1.3 and implies Theorem 1.5. They depend on the line

link of Theorem 2.23 (the blow-up of a line, a flop, and the blow-down to V_5), on the widths of the flopping curves of the line link, that is, the number of blow-ups in Reid's pagoda resolution of the flop (Definition-Proposition 2.21, Proposition 10.8), and on the isomorphism $\Sigma(X) \cong Q_X$ for any X (Theorem 16.11). Theorem 1.7 follows from Theorems 20.14 and 1.5, and Theorem 1.8 from Construction-Proposition 21.1, Theorem 2.29, Theorem 16.12(2) and (6), and Theorem 17.5. The conclusions hold for any smooth member, and the K-stability of a given member can be checked through one plane quartic.

Ideas (a) and (b) concern the tightest cases. We now explain four ideas of the proof.

(a) Keeping the coefficient of E. At several centers our bound on E is only $S(W;v) \le A_{E,\Delta_E}(v)$ for the invariant $S(W;v)$ of the refinement W along a prime divisor v over E (Definition 2.8) and the log discrepancy A_{E,Δ_E} of the pair (E,Δ_E), where Δ_E is the different of E (Theorem 2.9). At a point p on three lines in genus 10, the blow-up E of p has $A_X(E)/S_X(E) = 4/3$, but Theorem 6.18 bounds $S(W;v)$ only by $A_{E,\Delta_E}(v)$ for the prime divisors v over E (here $E \cong \mathbb{P}^2$ and $\Delta_E = 0$), so [AZ22, Theorem 3.2] implies only $\delta_p(X) \ge 1$.

The inequality [AZ22, (3.6)] in the proof of [AZ22, Theorem 3.2] contains the term $(A_X(E) - \lambda S_X(E))\operatorname{mult}_F E$, and the statement of [AZ22, Theorem 3.2] drops the term. Lemma 3.1 is a variant of [AZ22, (3.6)] that keeps the term for any λ with $\lambda S(W;v) \le A_{E,\Delta_E}(v)$ for the prime divisors v over E. Thus a bound on E without margin still implies $A_X(F) > S_X(F)$ whenever $A_X(E) > S_X(E)$ and $\operatorname{mult}_F E > 0$, and $A_X(E) - S_X(E) = \frac{3}{4}$ at p. Lemma 3.1 also yields the strict inequality at the points of incidence type 2 and at several other points in genus 12. At centers without margin we obtain no lower bound for $\delta_x(X)$ larger than 1, and $\delta(X) > 1$ follows from K-stability by Theorem 2.5(3)(4), without an explicit bound.

(b) Exact refined series. At points on special lines or on several lines, W has to be bounded with little or no loss. Bounding W by the sections on E that satisfy base conditions given by the lines can lose too much. In the configuration of Theorem 6.5(2), $E \cong \mathbb{F}_1$ is the exceptional divisor of the blow-up of a line through the point x of Theorem 6.5, with negative section s and fiber f_x over x, and such a bound, using only a subscheme of length two of the base scheme at the point $q = s \cap f_x$, yields only $53/48 > 1$ as an upper bound for the invariant S of q on the further refinement of W by f_x (see the paragraph before Construction 6.4). There, for the orders of vanishing at q up to a cutoff given by a piecewise linear function of (m,j), we use the dimensions $\dim W_{m,j}$ of the graded pieces of W instead, which Lemma 3.3 turns into an upper bound for S.

At several other points we compute the associated graded ring of $\bigoplus_m H^0(X,mH)$ for ord_E exactly, by a degeneration of X (Proposition 6.9) or by finding $N_m := h^0(X,mH)$ distinct leading monomials of products of explicit sections in degree m (for example Construction-Proposition 15.4). A one-parameter subgroup then degenerates W to a series spanned by eigenvectors with explicit basis type divisors, and the lower semicontinuity of log canonical thresholds [BL21, Proposition 10(2)] carries their log canonicity back to W (Lemma 3.6).

(c) Degenerations and orbits of nets. Openness [BLX22, Corollary 1.2] turns the degenerations of step (2) only into K-semistability. The new input is that the Mukai bundle of the central fiber is exceptional, so over $\operatorname{Spec}\mathbb{C}[[\tau]]$ it extends over the degeneration and defines a relative net. The relative net stays in $\mathrm{SL}(V)\cdot\Pi_X$ if $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ (Proposition 16.13), which is what step (3) uses.

In the other direction, the CM line bundle of the universal family over $\mathbb{B}^{(3)}$, extended to $\mathbb{B}$ through the Koszul complex, is $\mathrm{SL}(V)$-equivariantly isomorphic to $\mathcal{A}^{175}$ for the Plücker line bundle $\mathcal{A}$ (Proposition 16.7). Thus the Donaldson–Futaki invariant of the test configuration given by a one-parameter subgroup is a positive multiple of the vanishing order of the invariant Pfaffian section (Definition-Proposition 16.2) along the orbit at the limit, which is zero for a non-degenerate limit net. Proposition 16.7 concerns only the test configurations given by one-parameter subgroups of $\mathrm{SL}(V)$, so it implies $\delta(X) \le 1$ for $\Pi_X \notin \Omega^{\mathrm{st}}$, not K-semistability.

(d) The harmonic slice. The étaleness of the harmonic slice $\{[q^2+b] : b \in \mathrm{Harm}_4\}$ in Theorem 1.7(2) is local near $[q^2]$. To reach a GIT-polystable quartic $[f]$ far from q^2, we need an invertible

symmetric matrix A and $t \neq 0$ with $AC_f(A) = tI_3$, where C_f is the linear map on symmetric 3×3 matrices given by the coefficients of f (Definition-Proposition 20.1). The equation $AC_f(A) = tI_3$ is complex-algebraic, unlike the moment map equation of the Kempf–Ness theorem [KN79], [MFK94, Theorem 8.3]. Properness of the projection to $\mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$ of the closed set of pairs $([f], [A])$ with $AC_f(A)$ scalar yields only a non-zero solution A, possibly singular (Definition-Proposition 20.1(4)).

We count solutions. The space of complete conics is the closure of the graph of $[A] \to [\operatorname{adj} A]$ in $\mathbb{P}^5 \times \mathbb{P}^5$ (Construction-Proposition 20.7), and its intersection number with the graph of C_f is $1+2+4+4+2+1 = 14$. For a GIT-stable f whose inverse quartic $f^\vee$ (Definition-Proposition 20.1, Lemma 20.5) defines a reduced curve, the solutions with A singular correspond injectively to singular points of $(f = 0)$ and of $(f^\vee = 0)$, and their total length is at most $6 + 7 = 13$, by the bounds for the total Milnor numbers (Lemma 20.5, Proposition 20.6). With explicit solutions in the remaining cases, any GIT-polystable orbit of non-degenerate quartics intersects the slice. The argument proves existence and does not single out a point of the intersection.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce the refinements and adjunction of Abban–Zhuang, weighted blow-ups, Mukai models and the canonical Mukai net, the width of Reid's pagoda, line links, apolar quartics and Scorza quartics, and K-polystability of members with a torus action. Section 3 collects the tools used repeatedly in the local estimates: the Abban–Zhuang inequality with the coefficient of the extracted divisor kept, weight inequalities for subspaces of plane forms, rank truncation, asymptotics of semigroups, flatness of torus degenerations of curvilinear base conditions, and torus degenerations of refinements with semicontinuity of log canonical thresholds. Section 4 treats what genus 10 and genus 12 have in common: the incidence type of a closed point (the lengths of the local rings of the scheme of lines through the point), divisorial centers, points contained in no line, the volume function of the blow-up of a point through which only non-special lines pass, and curves of degree at least 2 in genus 10 and at least 3 in genus 12.

In Section 5 we study prime divisors whose center on a threefold of genus 10 is a line. Section 6 treats points of incidence types 1, $1+1$, and $1+1+1$ in genus 10, that is, points through which one, two, or three lines pass, all of them non-special. Section 7 treats points of incidence type 2 in genus 10. Section 8 treats points of incidence type $2+1$ in genus 10. Section 9 treats points of incidence type 3 in genus 10.

In Section 10 we study line links, widths, and the local structure of the Hilbert scheme of lines in genus 12. Section 11 treats curves of degree one and two in genus 12, except one configuration of a special line, which is treated in Section 12 by a degeneration to a member with a torus action. Section 12 constructs smooth degenerations to members with a torus action from special lines, in particular from special lines which intersect no other line, and completes the treatment of curves of degree one and two in genus 12. Section 13 treats points of incidence types 1, $1+1$, and $1+1+1$ in genus 12. Section 14 treats points of incidence types 2 and $2+1$ in genus 12. Section 15 treats points of incidence type 3 in genus 12.

In Section 16 we study nets of alternating forms and GIT of plane quartics. Section 17 concerns K-semistability in genus 12 and the criterion in terms of net orbits. Section 18 relates special lines meeting no other line to the simple elliptic germ. Section 19 develops the dictionary between plane quartics, lines, and net orbits. Section 20 studies the harmonic slice. Section 21 studies the explicit net of Theorem 1.8. Section 22 contains the proofs of the main theorems.

Appendix A contains the Zariski decompositions for points on three lines and the integral tables for genus 10. Appendix B contains the integral tables for genus 12 and the Scorza differential at the quartics F_h. Appendix C contains the explicit identities used for the threefold of Theorem 1.8.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. **Valuations and K-stability.** For a real number ξ we write $\xi_+ := \max\{\xi, 0\}$.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a Fano variety with klt singularities of dimension d, let F be a prime divisor over X, and let $\mu : Y \to X$ be a birational morphism from a normal projective variety Y such that F is a prime divisor on Y. We write $\operatorname{center}_X F := \mu(F)$ and $A_X(F) := a(F, X, 0)$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on a normal variety Y' birational to X, we write $\operatorname{mult}_F D$ for the coefficient of F in the pullback of D to a common resolution of Y and Y'. The *expected vanishing order* of F is

$$S_X(F) := \frac{1}{(-K_X)^d} \int_0^{+\infty} \operatorname{vol}_Y(-\mu^* K_X - tF) dt,$$

and we set $\beta_X(F) := A_X(F) - S_X(F)$. For a divisorial valuation $v = c \operatorname{ord}_F$ with $c > 0$, let $A_X(v) := cA_X(F)$ and $S_X(v) := cS_X(F)$. The *stability threshold* of X is $\delta(X) := \inf_F A_X(F)/S_X(F)$, and for an irreducible closed subset $Z \subset X$ we set

$$\delta_Z(X) := \inf\{A_X(F)/S_X(F) : Z \subset \operatorname{center}_X F\},$$

where F runs through prime divisors over X, as in [AZ22, Section 2.2]. For a closed point $x \in X$ we write $\delta_x(X) := \delta_{\{x\}}(X)$.

Definition 2.3. For a pair (Y, B) and an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on Y, the *log canonical threshold* is $\operatorname{lct}(Y, B; D) := \sup\{t \geq 0 : (Y, B + tD) \text{ is lc}\}$, and $\operatorname{lct}(Y; D) := \operatorname{lct}(Y, 0; D)$.

Definition 2.4. Let X be a Fano variety with klt singularities. A *test configuration* of X consists of a normal variety $\mathcal{X}$ with a flat projective morphism $\pi : \mathcal{X} \to \mathbb{A}^1$, a $\mathbb{G}_m$-action on $\mathcal{X}$ covering the standard action on $\mathbb{A}^1$, and a π-ample $\mathbb{G}_m$-linearized $\mathbb{Q}$-line bundle $\mathcal{L}$, together with a $\mathbb{G}_m$-equivariant isomorphism

$$(\mathcal{X}, \mathcal{L})|_{\pi^{-1}(\mathbb{A}^1 \setminus \{0\})} \cong (X, -K_X) \times (\mathbb{A}^1 \setminus \{0\}),$$

where $\mathbb{G}_m$ acts trivially on X. It is *anti-canonical* if $K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier and $\mathcal{L} = -K_{\mathcal{X}/\mathbb{A}^1}$. It is a *product* if $(\mathcal{X}, \mathcal{L}) \cong (X \times \mathbb{A}^1, p_1^*(-K_X))$ with $\mathbb{G}_m$ acting on X through a one-parameter subgroup of $\operatorname{Aut}(X)$, and it is *trivial* if moreover $\mathbb{G}_m$ acts trivially on X. The Donaldson–Futaki invariant $\operatorname{DF}(\mathcal{X}, \mathcal{L})$ and the invariant $J^{\mathrm{NA}}(\mathcal{X}, \mathcal{L})$ are defined as in [Fuj19, Definition 2.3(4) and (2)]. The variety X is *K-semistable* if $\operatorname{DF} \geq 0$ for any test configuration, *K-polystable* if moreover $\operatorname{DF} = 0$ only for products, *K-stable* if moreover $\operatorname{DF} = 0$ only for the trivial test configuration, and *uniformly K-stable* if there exists $\epsilon > 0$ such that

$$\operatorname{DF}(\mathcal{X}, \mathcal{L}) \geq \epsilon J^{\mathrm{NA}}(\mathcal{X}, \mathcal{L})$$

for any test configuration $(\mathcal{X}, \mathcal{L})$. K-semistability, K-polystability, K-stability, and uniform K-stability coincide with the notions of [Fuj19, Definition 2.7]. A Fano variety which is K-semistable but not K-polystable is called *strictly K-semistable*.

The Donaldson–Futaki invariant is normalized as in [Fuj19, Proposition 2.12(2)(iii)]. In the definition of products and of the trivial test configuration, the isomorphism $\mathcal{X} \cong X \times \mathbb{A}^1$ is $\mathbb{G}_m$-equivariant, and $\mathcal{L} \cong p_1^*(-K_X)$ is an isomorphism of $\mathbb{Q}$-line bundles, not of $\mathbb{G}_m$-linearized ones. Since $\mathcal{O}(X \times \mathbb{A}^1)^* = \mathbb{C}^*$, two $\mathbb{G}_m$-linearizations of a $\mathbb{Q}$-line bundle on $X \times \mathbb{A}^1$ differ by a character of $\mathbb{G}_m$, that is, by replacing $\mathcal{L}$ with $\mathcal{L} + c\mathcal{X}_0$ for some $c \in \mathbb{Q}$, where $\mathcal{X}_0 = \pi^*(0)$ is a principal divisor. Hence the trivial test configuration and the products correspond to the trivial and product-type test configurations of [Fuj19, Definition 2.2(2)]. Requiring $\mathcal{X}$ to be normal in Definition 2.4 follows [LX14, Definition 6 and Section 8.2]: normalizing a test configuration does not increase the Donaldson–Futaki invariant, with equality if and only if the normalization is an isomorphism in codimension one. Hence the normality of $\mathcal{X}$ does not affect K-semistability. For K-stability and K-polystability, it excludes the non-normal test configurations that are isomorphic in codimension one to the trivial test configuration or to a product [LX14, Section 8.2, Example 4].

Theorem 2.5 ([LXZ22, Theorems 2.9 and 1.6])**.** *Let X be a Fano variety with klt singularities. Then the following hold.*

(1) X is K-semistable if and only if $A_X(F) \geq S_X(F)$ for any prime divisor F over X, if and only if $\delta(X) \geq 1$.
(2) X is K-stable if and only if $A_X(F) > S_X(F)$ for any prime divisor F over X.
(3) X is uniformly K-stable if and only if $\delta(X) > 1$.
(4) X is K-stable if and only if X is uniformly K-stable.

The valuative criteria in (1) and (2) are due to Fujita [Fuj19, Theorems 1.4 and 1.6] and Li [Li17], and (2) for all prime divisors over X, not only for the dreamy ones of [Fuj19, Theorem 1.6], is due to Blum and Xu [BX19, Corollary 4.2]. The characterizations by $\delta(X)$ in (1) and (3) are due to Fujita and Odaka [FO18] and Blum and Jonsson [BJ20, Theorems B and C] (see also [BX19, Theorem 2.9]), and (4) is [LXZ22, Theorem 1.6].

Openness of K-semistability is used in this paper only for the degenerations in Sections 12 and 15: there the central fiber of a smooth family is a K-polystable member with a faithful $\mathbb{G}_m$-action, and Lemma 2.6 implies that the general fiber is K-semistable.

Lemma 2.6 ([BLX22, Corollary 1.2])**.** *Let T be a smooth connected curve, and let $\pi : \mathcal{X} \to T$ be a smooth projective morphism whose fibers are Fano threefolds, such that $-K_{\mathcal{X}/T}$ is π-ample. Assume that the fiber $\mathcal{X}_0$ over a point $0 \in T$ is K-semistable. Then $\mathcal{X}_t$ is K-semistable for any t in a non-empty open subset of T. In particular, if $\mathcal{X} \to \mathbb{A}^1$ is a test configuration of a Fano threefold X with smooth K-semistable central fiber, then X is K-semistable.*

Proof. Since π is smooth and projective with connected fibers and $-K_{\mathcal{X}/T}$ is π-ample, $(\mathcal{X}, 0) \to T$ is a $\mathbb{Q}$-Gorenstein family of log Fano pairs over the normal base T in the sense of [BLX22, Definition 2.1]. Thus, by [BLX22, Corollary 1.2], the points $t \in T$ such that $\mathcal{X}_t$ is K-semistable form an open subset of T containing 0.

Now let $\pi : \mathcal{X} \to \mathbb{A}^1$ be a test configuration of a Fano threefold X such that $\mathcal{X}_0$ is smooth and K-semistable. Since π is flat, $\mathcal{X}_t \cong X$ for $t \neq 0$, and $\mathcal{X}_0$ is smooth, π is a smooth morphism. Since π is flat and projective and $\mathcal{X}_t \cong X$ is connected for $t \neq 0$, $\mathcal{X}_0$ is connected [Har77, Exercise III.11.4], hence irreducible. Since $\mathcal{L}$ and $-K_{\mathcal{X}/\mathbb{A}^1}$ coincide on $\mathcal{X} \backslash \mathcal{X}_0$, we have $\mathcal{L} + K_{\mathcal{X}/\mathbb{A}^1} \sim_{\mathbb{Q}} c\mathcal{X}_0$ for some $c \in \mathbb{Q}$, and $\mathcal{X}_0 = \pi^*(0) \sim 0$. Thus $\mathcal{L} \sim_{\mathbb{Q}} -K_{\mathcal{X}/\mathbb{A}^1}$, so $-K_{\mathcal{X}/\mathbb{A}^1}$ is π-ample and $\mathcal{X}_0$ is a Fano threefold. Applying the first paragraph of this proof with $T = \mathbb{A}^1$, we see that the points $t \in \mathbb{A}^1$ such that $\mathcal{X}_t$ is K-semistable form a non-empty open subset of $\mathbb{A}^1$, which contains some $t \neq 0$. Since $\mathcal{X}_t \cong X$, X is K-semistable. □

Theorem 2.7 ([ABHLX20, Theorem 1.3] and [BX19, Corollary 1.3])**.** *Let X be a Fano variety with klt singularities. Then $\mathrm{Aut}(X)$ is reductive if X is K-polystable, and $\mathrm{Aut}(X)$ is finite if X is K-stable.*

2.2. **Refinements and adjunction.** Definition 2.8 is taken from [AZ22, Sections 2.2–2.6, Definitions 2.6 and 2.9, Lemma 2.1, and Example 2.6], where the plt type divisor is denoted by F.

Definition 2.8. Let (X, Δ) be a projective klt pair and $L_1, \dots, L_r$ $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor classes on X, that is, $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors up to $\mathbb{Z}$-linear equivalence. For a $\mathbb{Q}$-divisor D on X we set $\mathcal{O}_X(D) := \mathcal{O}_X(\lfloor D \rfloor)$, so that $H^0(X, \mathcal{O}_X(D)) = \{0\} \cup \{f \in \mathbb{C}(X)^* : (f) + D \geq 0\}$, where (f) is the principal divisor of f. For $0 \neq s \in H^0(X, \mathcal{O}_X(D))$ given by $f \in \mathbb{C}(X)^*$ we write $\operatorname{div}(s) := (f) + D \geq 0$, and if D is $\mathbb{Q}$-Cartier, then for a valuation v over X we set $v(s) := v(\operatorname{div}(s))$ [AZ22, Section 2.1, Definition 2.9, and Remark 2.3]. An $\mathbb{N}^r$-graded linear series $V_\bullet$ on X associated to $L_1, \dots, L_r$ consists of subspaces

$$V_{\vec{a}} \subset H^0(X, \mathcal{O}_X(a_1 L_1 + \dots + a_r L_r)), \qquad \vec{a} = (a_1, \dots, a_r) \in \mathbb{N}^r,$$

with $V_{\vec{0}} = \mathbb{C}$ and $V_{\vec{a}} \cdot V_{\vec{b}} \subset V_{\vec{a}+\vec{b}}$. Each such series considered below has bounded support and contains an ample series in the sense of [AZ22, Section 2.5]. For $m \geq 0$ let $N_m := \sum_{\vec{a}: a_1 = m} \dim V_{\vec{a}}$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X, a non-zero subspace $U \subset H^0(X, \mathcal{O}_X(D))$, and a divisorial valuation v over X, the *jumping numbers* of v on U are the real numbers t at which $\dim\{s \in U : v(s) \geq t\}$ drops, each counted with the size of the drop. A prime divisor over X is identified with the valuation defined by the prime divisor. Let $S_m(V; v)$ be the sum of the jumping numbers of v on all $V_{\vec{a}}$ with $a_1 = m$, divided by mN_m. The *expected vanishing order* $S(V; v) := \lim_{m \to +\infty} S_m(V; v)$ exists [AZ22, Lemma 2.9]. For an irreducible closed subset $Z \subset X$ we set

$$\delta_Z(X, \Delta; V) := \inf\{A_{X,\Delta}(v)/S(V; v) : Z \subset \operatorname{center}_X v\},$$

where $A_{X,\Delta}(v) := a(v, X, \Delta)$. An *$m$-basis type divisor* of V is

$$\frac{1}{mN_m} \sum_{\vec{a}: a_1 = m} \sum_i \operatorname{div}(s_{\vec{a},i}),$$

where $(s_{\vec{a},i})_i$ is a basis of $V_{\vec{a}}$ for each $\vec{a}$. The divisor is *compatible* with v if each basis $(s_{\vec{a},i})_i$ contains a basis of each subspace $\{s \in V_{\vec{a}} : v(s) \geq t\}$. For an ample line bundle L on X, the *complete series* of L is $V_m := H^0(X, mL)$, $m \geq 0$, so that $N_m = h^0(X, mL)$. If X is a Fano variety with klt singularities, $\Delta = 0$, and $V_m := H^0(X, \mathcal{O}_X(-mK_X))$, then $S(V; F) = S_X(F)$ and $\delta_Z(X, 0; V) = \delta_Z(X)$ [AZ22, Section 2.6]. In this case we write $S_m(v) := S_m(V; v)$, and $S_m(F) := S_m(V; \operatorname{ord}_F)$ for a prime divisor F over X.

Let E be a prime divisor over X of *plt type*, that is, there exists a projective birational morphism $\pi : Y \to X$ from a normal variety Y such that E is a prime divisor on Y, $-E$ is $\mathbb{Q}$-Cartier and π-ample, and $(Y, \Delta_Y + E)$ is plt [KM98, Definition 2.34], where Δ_Y is the strict transform of Δ with the component E removed. If E is a prime divisor on X which is not a component of Δ, we allow $\pi = \mathrm{id}_X$ [AZ22, Definition 2.6]. The morphism π is uniquely determined by E [AZ22, Definition 2.6(1)]. We call π the *plt extraction* of E, and we let $\Delta_E := \operatorname{Diff}_E(\Delta_Y)$ be the different [Kol13, Definition 4.2]. For a $\mathbb{Q}$-Cartier Weil divisor D on Y we write $D|_E$ for the $\mathbb{Q}$-divisor class on E with $\mathcal{O}_Y(D)/\mathcal{O}_Y(D - E) \cong \mathcal{O}_E(D|_E)$ [AZ22, Lemma 2.1]. Assume that $\pi^*(a_1 L_1 + \dots + a_r L_r)$ is a Weil divisor for all $\vec{a} \in \mathbb{N}^r$, for instance that $L_1, \dots, L_r$ are Cartier. Then the *refinement* of $V_\bullet$ by E is the $\mathbb{N}^{r+1}$-graded linear series $W_\bullet$ on E, associated to $\pi^* L_1|_E, \dots, \pi^* L_r|_E$ and $-E|_E$, with

$$W_{\vec{a},j} := \operatorname{Im}\big(\{s \in V_{\vec{a}} : \operatorname{ord}_E(s) \geq j\} \to H^0(E, (\pi^*(a_1 L_1 + \dots + a_r L_r) - jE)|_E)\big), \qquad \vec{a} \in \mathbb{N}^r,\ j \geq 0;$$

it has bounded support and contains an ample series, as follows. Since each $\pi^*(a_1 L_1 + \dots + a_r L_r) - jE$ is a $\mathbb{Q}$-Cartier Weil divisor, by [AZ22, Lemma 2.1] we have

$$W_{\vec{a},j} \cong \{s \in V_{\vec{a}} : \operatorname{ord}_E(s) \geq j\}/\{s \in V_{\vec{a}} : \operatorname{ord}_E(s) \geq j + 1\},$$

so $\sum_{j \geq 0} \dim W_{\vec{a},j} = \dim V_{\vec{a}}$ and $\sum_{\vec{a}: a_1 = m} \sum_{j \geq 0} \dim W_{\vec{a},j} = N_m$ for $m \geq 0$. In particular $W_{\vec{a},j} \neq 0$ if and only if $\operatorname{ord}_E(s) = j$ for some $s \in V_{\vec{a}}$. For Cartier $L_1, \dots, L_r$, bounded support and the ample series are [AZ22, Example 2.6]. In general, each $\pi^* L_i$ is a Weil divisor on the normal

variety Y, hence Cartier at the generic point of E, so condition (2) of an ample series follows from [AZ22, Lemma 2.4] for the flag $Y \supset E$, and bounded support and conditions (1) and (3) follow from those of $V_{\vec{\bullet}}$, as in [AZ22, Example 2.6]. For the complete series of L we have

$$W_{m,j} = \operatorname{Im}\left(H^0(Y, m\pi^*L - jE) \to H^0(E, (m\pi^*L - jE)|_E)\right).$$

We write $A_{E,\Delta_E}(v) := a(v, E, \Delta_E)$, and $S(W; v)$ and $\delta_{Z'}(E, \Delta_E; W)$ are defined by the formulas for $S(V; v)$ and $\delta_Z(X, \Delta; V)$ with (E, Δ_E) and $W_{\vec{\bullet}}$ in place of (X, Δ) and $V_{\vec{\bullet}}$, all normalized by mN_m. For a prime divisor C over E of plt type over (E, Δ_E), the refinement of $W_{\vec{\bullet}}$ by C is denoted $W^C_{\vec{\bullet}}$, and for a closed point q of C we write $S(W^C; q) := S(W^C; \operatorname{ord}_q)$.

All local estimates in Sections 4–15 are instances of Theorem 2.9: we extract a plt type divisor E over the center, depending on the lines through the center, and bound $\delta_Z(X)$ from below by the thresholds of the refined series on E with the different. Strict inequalities come from the stronger form in Lemma 3.1.

Theorem 2.9 ([AZ22, Theorem 3.2])**.** *Let (X, Δ) be a projective klt pair, $V_{\vec{\bullet}}$ a multigraded linear series on X as in Definition 2.8, $Z \subset X$ an irreducible closed subset, and E a prime divisor over X of plt type with $Z \subset \operatorname{center}_X E$. Let $\pi : Y \to X$ be the plt extraction of E, Δ_E the different, and $W_{\vec{\bullet}}$ the refinement of $V_{\vec{\bullet}}$ by E. Then*

$$\delta_Z(X, \Delta; V) \geq \min\left\{\frac{A_{X,\Delta}(E)}{S(V; E)},\ \inf_{Z'} \delta_{Z'}(E, \Delta_E; W)\right\},$$

where Z' runs through the irreducible closed subsets of E with $\pi(Z') = Z$. In particular, if X is a Fano variety with klt singularities, $\Delta = 0$, and $V_{\vec{\bullet}}$ is the complete series of $-K_X$, then

$$\delta_Z(X) \geq \min\left\{\frac{A_X(E)}{S_X(E)},\ \inf_{Z'} \delta_{Z'}(E, \Delta_E; W)\right\}.$$

By [AZ22, Lemma 2.9], $\delta_Z(X, \Delta; V)$ is the stability threshold of $V_{\vec{\bullet}}$ defined through m-basis type divisors in [AZ22, Section 2.6], and the infimum defining it may be taken over all valuations v with $A_{X,\Delta}(v) < +\infty$ and $Z \subset \operatorname{center}_X v$. The refinements in Definition 2.8 also have bounded support and contain an ample series, so by [AZ22, Lemma 2.9] the threshold $\delta_{Z'}(E, \Delta_E; W)$ is also the stability threshold of $W_{\vec{\bullet}}$ defined through m-basis type divisors, and the infimum defining it may be taken over all valuations v over E with $A_{E,\Delta_E}(v) < +\infty$ and $Z' \subset \operatorname{center}_E v$. In [AZ22, Theorem 3.2], $L_1, \dots, L_r$ are Cartier divisors. If k is a positive integer such that $kL_1, \dots, kL_r$ are Cartier, then replacing $V_{\vec{\bullet}}$ with $V_{k\vec{\bullet}}$ multiplies $S(V; \cdot)$ by k and divides $\delta_Z(X, \Delta; V)$ by k [AZ22, Lemma 2.11]. The refinement of $V_{k\vec{\bullet}}$ by E is the series $W_{k\vec{a},j}$, $(\vec{a}, j) \in \mathbb{N}^{r+1}$. Since S_m is normalized by mN_m with $m = a_1$, for $r = 1$ Definition 2.8 shows termwise that S on E is multiplied by k. In general, the argument in the proof of [AZ22, Lemma 2.11], applied to the Okounkov bodies of $W_{\vec{\bullet}}$ and of $W_{k\vec{\bullet},\bullet}$ via $(\vec{a}, j, \nu) \to (k\vec{a}, j, \nu)$, which rescales only the coordinates of $\vec{a}$, shows that S on E is multiplied by k and $\delta_{Z'}(E, \Delta_E; W)$ is divided by k. Hence Theorem 2.9 follows from the Cartier case.

The plt type divisors in later sections are mostly exceptional divisors of weighted blow-ups.

Definition 2.10. Let X be a smooth threefold, $x \in X$ a closed point, z_1, z_2, z_3 regular parameters of $\mathcal{O}_{X,x}$, and w_1, w_2, w_3 positive integers. For $k \geq 0$ let $\mathcal{I}_k \subset \mathcal{O}_X$ be the ideal sheaf which is equal to $\mathcal{O}_X$ outside x and whose stalk at x is generated by the monomials $z_1^{a_1} z_2^{a_2} z_3^{a_3}$ with $w_1a_1 + w_2a_2 + w_3a_3 \geq k$. The *weighted blow-up* of X at x with weights (w_1, w_2, w_3) in the parameters (z_1, z_2, z_3) is the morphism

$$\operatorname{Proj}_X \bigoplus_{k \geq 0} \mathcal{I}_k \to X,$$

and the *monomial valuation* with weights (w_1, w_2, w_3) is the valuation v of $\mathbb{C}(X)$ with $v(g) = \max\{k : g \in \mathcal{I}_{k,x}\}$ for $0 \neq g \in \mathcal{O}_{X,x}$. We write $\mathbb{P}(w_1, w_2, w_3) := \operatorname{Proj} \mathbb{C}[y_1, y_2, y_3]$, where y_i has degree w_i, for the weighted projective plane.

2.3. Prime Fano threefolds of genus ten and twelve.

Definition 2.11. A *smooth prime Fano threefold of genus g* is a smooth projective threefold X with $-K_X$ ample, $\mathrm{Pic}(X) = \mathbb{Z}[-K_X]$, and $(-K_X)^3 = 2g-2$. We write $H := -K_X$. A *line* (resp. *conic*) on X is a closed subscheme with Hilbert polynomial $t+1$ (resp. $2t+1$) with respect to H. Let $\Sigma(X) := \mathrm{Hilb}^{t+1}(X)$ be the Hilbert scheme of lines, and for a closed point $x \in X$ let $\Sigma(X,x) \subset \Sigma(X)$ be the closed subscheme of lines containing x, that is, the fiber over x of the universal line. A line L is *special* if $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and *non-special* otherwise. If $\Sigma(X,x)$ is finite, the *incidence type* of x is the partition of the length of $\Sigma(X,x)$ given by the lengths of the local rings of $\Sigma(X,x)$, written as a sum such as $2+1$, and written $\emptyset$ if $\Sigma(X,x)$ is empty.

The incidence type is the basis of the classification of points in this paper. By Proposition 4.1, $\Sigma(X,x)$ is always finite and the incidence type of x is one of $\emptyset$, 1, $1+1$, $1+1+1$, 2, $2+1$, and 3. The local ring of $\Sigma(X,x)$ is non-reduced exactly at the special lines through x, and the incidence type of x determines which divisor over X we extract at x.

Lemma 2.12 ([IP99, Proposition 4.1.11, Corollary 4.1.13, Proposition 2.1.12 and Lemma 4.2.1], [KPS18, Lemma 2.1.4 and Lemma 2.2.3])**.** *Let X be a smooth prime Fano threefold of genus $g \in \{10, 12\}$. Then the following hold.*

(1) H is very ample, and the morphism defined by $|H|$ embeds X into $\mathbb{P}^{g+1}$ as an intersection of quadrics.
(2) For every integer $m \geq 0$,

$$N_m = h^0(X, mH) = \frac{(2g-2)m(m+1)(2m+1)}{12} + 2m + 1.$$

(3) Every line $L \subset X$ is isomorphic to $\mathbb{P}^1$, and $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ or $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.
(4) Every irreducible component of $\Sigma(X)$ is a curve.

Proof. Part (1) is [IP99, Proposition 4.1.11 and Corollary 4.1.13]. Part (2) follows from Riemann–Roch and the vanishing [IP99, Proposition 2.1.12]. Part (4) is [KPS18, Lemma 2.2.3]. For (3), let $L \subset X$ be a line. By (1), L is a line in $\mathbb{P}^{g+1}$, so $L \cong \mathbb{P}^1$. By [KPS18, Lemma 2.1.4], $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(-1-a)$ for some integer $a \geq 0$. Since

$$N_{L/X} \subset N_{L/\mathbb{P}^{g+1}} \cong \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus g},$$

we have $a \leq 1$. This is also [IP99, Lemma 4.2.1(i)]. □

Theorem 2.13 ([Muk88], [IP99, Theorem 5.2.3], [Vir26, Theorem 2.2])**.** *Every smooth prime Fano threefold of genus 10 is isomorphic to a codimension two linear section of the adjoint variety $\mathrm{G}_2\,\mathrm{Gr}(2,7) \subset \mathbb{P}^{13}$, with H the restriction of the hyperplane class.*

Theorem 2.13 is due to Mukai [Muk88], [Muk92, Section 2], and [Vir26, Theorem 2.2] gives it for every smooth prime Fano threefold of genus 10. In Sections 6–9 we use the coordinates on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ of [Kap11, Section 2]. The Pfaffian matrix below, and the realization of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ as a linear section of $\mathrm{Gr}(2,7)$, are in [Kap11, Section 2, Lemma 2.1], where our coordinates M and J are written m and j. For a finite-dimensional complex vector space, $\mathbb{P}(\cdot)$ denotes the projective space of one-dimensional subspaces, and $[e]$ denotes the point of a non-zero vector e.

Set-up 2.14. Let $(a, b, \dots, n)$, A, $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, $\mathrm{Pf}_{j_1j_2j_3j_4}$, $\mathcal{W}$, and $G_{\mathcal{W}}$ be as follows.

(1) $(a, b, c, d, e, f, g, h, i, J, k, l, M, n)$ are homogeneous coordinates on $\mathbb{P}^{13}$.

(2) A is the alternating matrix

$$A := \begin{pmatrix} 0 & -f & e & g & h & i & a \\ f & 0 & -d & J & k & l & b \\ -e & d & 0 & M & n & -g-k & c \\ -g & -J & -M & 0 & c & -b & d \\ -h & -k & -n & -c & 0 & a & e \\ -i & -l & g+k & b & -a & 0 & f \\ -a & -b & -c & -d & -e & -f & 0 \end{pmatrix},$$

and A_{rs} denotes the (r,s)-entry of A.

(3) $\mathrm{G}_2\,\mathrm{Gr}(2,7) \subset \mathbb{P}^{13}$ is the adjoint variety of G_2, realized as the subscheme cut out by the 4×4 Pfaffians of A.

(4) For $1 \le j_1 < j_2 < j_3 < j_4 \le 7$, $\mathrm{Pf}_{j_1j_2j_3j_4} := A_{j_1j_2}A_{j_3j_4} - A_{j_1j_3}A_{j_2j_4} + A_{j_1j_4}A_{j_2j_3}$ is the Pfaffian of the principal 4×4 submatrix of A with rows and columns j_1, j_2, j_3, j_4.

(5) $\mathcal{W} \subset \Lambda^2\mathbb{C}^7$ is the fourteen-dimensional space of the matrices A, so that $\mathbb{P}^{13} = \mathbb{P}(\mathcal{W})$, and $G_{\mathcal{W}} := \{\alpha \in \mathrm{SL}_7 : \alpha\mathcal{W}\alpha^t = \mathcal{W}\}$ acts on $\mathbb{P}^{13}$ by $A \to \alpha A\alpha^t$.

(6) $G^\circ_{\mathcal{W}}$ is the identity component of $G_{\mathcal{W}}$.

Lemma 2.15. *Notation as in Set-up 2.14, and let $p := [h]$ be the point with $h = 1$ and all other coordinates zero. Then the following hold.*

(1) The affine chart $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{h \ne 0\}$ is the graph of the polynomial map $\mathbb{A}^5 \to \mathbb{A}^8$, $(i,a,e,n,g) \to (f,k,c,b,d,J,l,M)$, given on $\{h=1\}$ by

$$\begin{aligned} f = ie - a^2, \quad k = -g - ae - in, \quad c = e^2 + an, \quad b = -ag - ic, \\ d = eg + ac, \quad J = fc + gk, \quad l = -af + ik, \quad M = -ec + gn, \end{aligned} \tag{2.1}$$

read from left to right: each coordinate on a right-hand side is expressed through (i,a,e,n,g) by an identity listed before it. In particular, $p \in \mathrm{G}_2\,\mathrm{Gr}(2,7)$, the embedded tangent space of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ at p is $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7) = \{f = c = g + k = b = d = l = M = J = 0\} \cong \mathbb{P}^5$, and (i,a,e,n,g) are coordinates on $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$.

(2) The affine chart $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{i \ne 0\}$ is the graph of the polynomial map $\mathbb{A}^5 \to \mathbb{A}^8$, $(a,f,g,h,l) \to (e,k,b,c,n,d,M,J)$, given on $\{i=1\}$ by

$$\begin{aligned} e = hf + a^2, \quad k = af + hl, \quad b = -f^2 + al, \quad c = -ag - hb, \\ n = -a^3 - 2ahf - hg - h^2l, \quad d = gf - ab, \quad M = eb - g(g+k), \quad J = gl - fb, \end{aligned} \tag{2.2}$$

read from left to right: each coordinate on a right-hand side is expressed through (a,f,g,h,l) by an identity listed before it.

(3) The scheme of lines on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ through p, embedded in $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$ by tangent directions, is the twisted cubic Γ_p defined by $g = 0$ and $a^2 = ie$, $e^2 = -an$, $ae = -in$, and the morphism $\mathbb{P}^1 \to \Gamma_p$ given by $[\sigma:\tau] \to (i:a:e:n:g) = (\sigma^3 : \sigma^2\tau : \sigma\tau^2 : -\tau^3 : 0)$ is an isomorphism. Moreover, $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is the projective cone with vertex p over Γ_p as a set, and also as a scheme outside p.

(4) The group $G_{\mathcal{W}}$ preserves $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, and the stabilizer of p in $G^\circ_{\mathcal{W}}$ acts on Γ_p through a surjection onto $\mathrm{Aut}(\Gamma_p) \cong \mathrm{PGL}_2$. In particular, the stabilizer of p in $G^\circ_{\mathcal{W}}$ acts triply transitively on Γ_p.

Proof. We first prove (1). Set $h = 1$, so that $A_{15} = 1$. For $r, s \in \{2,3,4,6,7\}$ with $r < s$, the Pfaffian with rows and columns $1, 5, r, s$ is, up to sign,

$$A_{15}A_{rs} - A_{1r}A_{5s} + A_{1s}A_{5r} = A_{rs} - A_{1r}A_{5s} + A_{1s}A_{5r}.$$

Writing out the ten Pfaffians with rows and columns $1, 5, r, s$, their vanishing is equivalent to the identities (2.1), where $(r,s) = (6,7), (3,6), (3,7)$ give the first three identities, the pairs $(2,7)$ and $(4,6)$ both give the identity for b, and the pairs $(2,3)$ and $(4,7)$ both give the identity for d, and the pairs $(2,4)$, $(2,6)$, $(3,4)$ give the identities for J, l, M, respectively. Modulo (2.1),

$A = uv^t - vu^t$, where u and v are the first and the fifth columns of A, because $A_{15} = 1$ and $A_{11} = A_{55} = 0$. Hence all 4×4 Pfaffians of A vanish modulo (2.1). Thus the chart $\{h \neq 0\}$ of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is the graph of the polynomial map $\mathbb{A}^5 \to \mathbb{A}^8$ defined by (2.1), with coordinates (i,a,e,n,g) on $\mathbb{A}^5$. In particular, $p \in \mathrm{G}_2\,\mathrm{Gr}(2,7)$.

The linear parts at p of the right-hand sides of (2.1) are 0, except the linear part $-g$ of k. Hence the embedded tangent space at p is

$$\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7) = \{f = c = g + k = b = d = l = M = J = 0\} \cong \mathbb{P}^5,$$

with homogeneous coordinates (h,i,a,e,n,g). Thus (i,a,e,n,g) are coordinates on $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$.

We prove (2) as (1), with $A_{16} = i = 1$ in place of $A_{15} = h = 1$. On $\{i = 1\}$ we have

$$\begin{aligned}
&\mathrm{Pf}_{1567} = a^2 - e + hf, \quad \mathrm{Pf}_{1256} = k - af - hl,\\
&\mathrm{Pf}_{1267} = al - b - f^2, \quad \mathrm{Pf}_{1456} = ag + hb + c,\\
&\mathrm{Pf}_{1356} = ae + hg + hk + n, \quad \mathrm{Pf}_{1467} = gf - ab - d,\\
&\mathrm{Pf}_{1346} = M - eb + g(g + k), \quad \mathrm{Pf}_{1246} = J + fb - gl.
\end{aligned}$$

For $x = e, k, b, c, n, d, M, J$, the Pfaffians Pf_{1567}, Pf_{1256}, Pf_{1267}, Pf_{1456}, Pf_{1356}, Pf_{1467}, Pf_{1346}, Pf_{1246}, respectively, are $\pm x$ plus a polynomial in (a,f,g,h,l) and the coordinates listed before x in (2.2). Substituting the identities for e and k into $n = -ae - hg - hk$ gives the identity for n. Thus the vanishing of $\mathrm{Pf}_{1567}, \dots, \mathrm{Pf}_{1246}$ on $\{i = 1\}$ is equivalent to (2.2). Modulo (2.2), $A = uv^t - vu^t$, where u and v are the first and the sixth columns of A, so all 4×4 Pfaffians of A vanish modulo (2.2). Thus the chart $\{i \neq 0\}$ of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is the graph of the polynomial map $\mathbb{A}^5 \to \mathbb{A}^8$ defined by (2.2).

Next we prove (3). Let $I \subset \mathbb{C}[i,a,e,n,g]$ be the ideal

$$I := (a^2 - ie,\ ae + in,\ e^2 + an,\ ag,\ eg,\ ig,\ ng,\ g^2).$$

On $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ we have $A_{rs} = 0$ for all $r, s \notin \{1,5\}$, since the entries A_{rs} with $r, s \notin \{1,5\}$ are $\pm f, \pm d, \pm J, \pm l, \pm b, \pm M, \pm(g+k), \pm c$. Hence a 4×4 Pfaffian whose index set does not contain both 1 and 5 restricts to zero, and the ten Pfaffians with indices $1, 5, r, s$ restrict to $\pm(A_{1r}A_{5s} - A_{1s}A_{5r})$, that is, to the quadratic parts of the identities (2.1) for f, k, c, b, d, J, l, M. On $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$, the quadratic parts are $ie - a^2$, $ae + in$, $e^2 + an$, ag, eg, g^2, ig, ng, respectively, up to sign. Thus $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is the subscheme of $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ defined by the generators of I, which do not involve h. Since $g \cdot (i,a,e,n,g) \subset I$, the saturation of I is $(g,\ a^2 - ie,\ ae + in,\ e^2 + an)$. Hence the subscheme of $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$ defined by I is

$$\Gamma_p = \{g = 0,\ a^2 = ie,\ e^2 = -an,\ ae = -in\},$$

and outside p, where one of i, a, e, n, g is invertible, $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is defined by g, $a^2 - ie$, $ae + in$, and $e^2 + an$, that is, it coincides with the projective cone with vertex p over Γ_p. In the coordinates $(i,a,e,-n)$ on the hyperplane $\{g = 0\}$, the three quadrics $ie - a^2$, $-(ae + in)$, and $-(e^2 + an)$ are the 2×2 minors of

$$\begin{pmatrix} i & a & e \\ a & e & -n \end{pmatrix},$$

and

$$\gamma : \mathbb{P}^1 \to \mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)), \quad [\sigma : \tau] \to (i : a : e : n : g) = (\sigma^3 : \sigma^2\tau : \sigma\tau^2 : -\tau^3 : 0)$$

is the cubic Veronese embedding in the coordinates $(i,a,e,-n)$. Hence Γ_p is a twisted cubic and γ is an isomorphism onto Γ_p. Since $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is cut out by quadrics, [KP23, Lemma 5.6] shows that the scheme of lines through p, embedded by tangent directions, is the subscheme of $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$ defined by the restrictions of the 4×4 Pfaffians of A to $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$, that is, by I. Thus the scheme of lines through p is Γ_p.

Finally we prove (4). Let $\epsilon_1, \dots, \epsilon_7$ be the standard basis of $\mathbb{C}^7$, and for an alternating 7×7 matrix B let $\omega_B := \sum_{r<s} B_{rs}\epsilon_r \wedge \epsilon_s$. Then $\omega_{\alpha B \alpha^t} = (\Lambda^2\alpha)\omega_B$ for $\alpha \in \mathrm{SL}_7$, and

$$\omega_B \wedge \omega_B = 2 \sum_{j_1<j_2<j_3<j_4} \big(B_{j_1j_2}B_{j_3j_4} - B_{j_1j_3}B_{j_2j_4} + B_{j_1j_4}B_{j_2j_3}\big)\epsilon_{j_1} \wedge \epsilon_{j_2} \wedge \epsilon_{j_3} \wedge \epsilon_{j_4}.$$

Hence the 4×4 Pfaffians of $\alpha A \alpha^t$ are linear combinations of the 4×4 Pfaffians of A, with the entries of $\Lambda^4\alpha$ as coefficients. For $\alpha \in G_{\mathcal{W}}$, the linear automorphisms $A \to \alpha A \alpha^t$ and $A \to \alpha^{-1} A \alpha^{-t}$ of $\mathcal{W}$ both map the ideal of 4×4 Pfaffians of A into itself. Thus $G_{\mathcal{W}}$ preserves $\mathrm{G}_2\,\mathrm{Gr}(2,7)$.

Let E_{rs} be the elementary 7×7 matrix with entry 1 at (r,s), and for $t \in \mathbb{C}$ let

$$u_+(t) := \mathbf{1}_7 + t(E_{15} - E_{24} + 2E_{67} + E_{73}) + t^2 E_{63},$$
$$u_-(t) := \mathbf{1}_7 + t(E_{51} - E_{42} + 2E_{37} + E_{76}) + t^2 E_{36}.$$

We have $u_\pm(t) = \exp(tZ_\pm)$ for the nilpotent matrices $Z_+ = E_{15} - E_{24} + 2E_{67} + E_{73}$ and $Z_- = E_{51} - E_{42} + 2E_{37} + E_{76}$, and $\det u_\pm(t) = 1$. A direct calculation shows that $u_+(t) A u_+(t)^t$ is the matrix A with the coordinates replaced by

$$\begin{gathered} a + 2te - t^2 n, \quad b - 2td + t^2 M, \quad c, \quad d - tM, \quad e - tn, \\ f + t(g+k) - t^2 c, \quad g - tc, \quad h, \quad i + 3ta + 3t^2 e - t^3 n, \quad J, \\ k - tc, \quad l + 3tb - 3t^2 d + t^3 M, \quad M, \quad n, \end{gathered}$$

and that $u_-(t) A u_-(t)^t$ is the matrix A with the coordinates replaced by

$$\begin{gathered} a + ti, \quad b + tl, \quad c - t(g+k) - t^2 f, \quad d - 2tb - t^2 l, \quad e + 2ta + t^2 i, \\ f, \quad g + tf, \quad h, \quad i, \quad J, \quad k + tf, \quad l, \\ M - 3td + 3t^2 b + t^3 l, \quad n - 3te - 3t^2 a - t^3 i, \end{gathered}$$

in the order $(a,b,c,d,e,f,g,h,i,J,k,l,M,n)$. Hence $u_\pm(t) \in G_{\mathcal{W}}$. Since $t \to u_\pm(t)$ is a morphism $\mathbb{A}^1 \to G_{\mathcal{W}}$ with $u_\pm(0) = \mathbf{1}_7$, we have $u_\pm(t) \in G^\circ_{\mathcal{W}}$. The coordinate h is fixed and does not occur in the other new coordinates, so $u_\pm(t)$ fixes p. On $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$, $u_\pm(t)$ acts by

$$u_+(t) : (h,i,a,e,n,g) \to (h,\ i + 3ta + 3t^2 e - t^3 n,\ a + 2te - t^2 n,\ e - tn,\ n,\ g),$$
$$u_-(t) : (h,i,a,e,n,g) \to (h,\ i,\ a + ti,\ e + 2ta + t^2 i,\ n - 3te - 3t^2 a - t^3 i,\ g),$$

so $u_\pm(t)$ preserves $\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$. Therefore, on the twisted cubic,

$$u_+(t)\gamma([\sigma:\tau]) = \gamma([\sigma + t\tau : \tau]), \qquad u_-(t)\gamma([\sigma:\tau]) = \gamma([\sigma : \tau + t\sigma]).$$

The stabilizer of p in $G^\circ_{\mathcal{W}}$ preserves $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ and p, hence acts on the scheme of lines through p, and the embedding by tangent directions is equivariant. Thus the stabilizer acts on Γ_p through a homomorphism to $\mathrm{Aut}(\Gamma_p) \cong \mathrm{PGL}_2$, whose image contains the images of the matrices $\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ t & 1 \end{smallmatrix}\right)$ for all $t \in \mathbb{C}$. Matrices of both forms generate SL_2, so the homomorphism is surjective. Since PGL_2 acts triply transitively on $\mathbb{P}^1$, the stabilizer of p in $G^\circ_{\mathcal{W}}$ acts triply transitively on Γ_p. □

Lemma 2.16. *Notation as in Set-up 2.14. For any two-dimensional subspace $W \subset \mathcal{W}^\vee$, the linear section $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{P}(W^\perp)$ is a Gorenstein scheme of pure dimension three, and the universal linear section over $\mathrm{Gr}(2, \mathcal{W}^\vee)$ is flat.*

Proof. The variety $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is a homogeneous space of G_2, hence smooth. By [Lat23, p. 568], $\dim \mathrm{G}_2\,\mathrm{Gr}(2,7) = 5$ and the degree of $\mathrm{G}_2\,\mathrm{Gr}(2,7) \subset \mathbb{P}(\mathcal{W})$ is 18. By Bertini's theorem [Har77, Theorem II.8.18], a general hyperplane section Z of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is smooth, and a general hyperplane section X of Z is smooth. By [Lat23, Theorem 2.1], X is a smooth prime Fano threefold of genus 10, so $\mathrm{Pic}(X) \cong \mathbb{Z}$. By [Laz04a, Example 3.1.25] for $Z \subset \mathrm{G}_2\,\mathrm{Gr}(2,7)$ and for $X \subset Z$, restriction induces an isomorphism $\mathrm{Pic}(\mathrm{G}_2\,\mathrm{Gr}(2,7)) \cong \mathrm{Pic}(X) \cong \mathbb{Z}$. If $\mathcal{O}(1)$ is k times

a generator, then k^3 divides $(\mathcal{O}_X(1)^3) = 18$, so $k = \pm 1$. Hence $\mathrm{Pic}(\mathrm{G}_2\,\mathrm{Gr}(2,7)) = \mathbb{Z}[\mathcal{O}(1)]$. By [Kap11, Section 2], $\mathbb{P}(\mathcal{W})$ is the linear span of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, so $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is not contained in a hyperplane.

Let $w \in \mathcal{W}^\vee$ be non-zero, let D be the divisor of w on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, and write $D = \sum_j m_j D_j$ with D_j prime and $m_j > 0$. Then $D_j \sim a_j\mathcal{O}(1)$ for integers a_j, and

$$a_j(\mathcal{O}(1)^5) = (D_j \cdot \mathcal{O}(1)^4) > 0$$

implies $a_j > 0$. Since $\sum_j m_j a_j = 1$, D is a prime divisor with multiplicity one, so the hyperplane section $(w = 0)$ is integral. Let $w' \in \mathcal{W}^\vee$ be linearly independent of w. If w' vanished on $(w = 0)$, then the integral hyperplane section $(w' = 0)$ would contain $(w = 0)$, hence would be equal to $(w = 0)$. Then $w' = \lambda w$ on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ for some $\lambda \in \mathbb{C}^*$, and $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ would be contained in the hyperplane $(w' - \lambda w = 0)$, a contradiction. Thus w' is a non-zero divisor on $(w = 0)$, and (w, w') is a regular sequence at any point of $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{P}(W^\perp)$, where W is spanned by w and w'. Hence $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{P}(W^\perp)$ is a Cohen–Macaulay and Gorenstein scheme of pure dimension three [BH98, Theorem 2.1.3(a) and Proposition 3.1.19(b)].

Let $\mathcal{N} := \mathrm{Ker}(\mathcal{W}^\vee \otimes \mathcal{O} \to \mathcal{O}(1))$ on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, a vector bundle of rank 13. The universal linear section $\mathcal{X}$ is the relative Grassmannian of two-dimensional subspaces of $\mathcal{N}$, so $\mathcal{X}$ is smooth and

$$\dim \mathcal{X} = 5 + 2 \cdot 11 = 27.$$

The base $\mathrm{Gr}(2, \mathcal{W}^\vee)$ is smooth with $\dim \mathrm{Gr}(2, \mathcal{W}^\vee) = 24$, and any fiber of $\mathcal{X} \to \mathrm{Gr}(2, \mathcal{W}^\vee)$ is a linear section of pure dimension $3 = 27 - 24$. Hence $\mathcal{X} \to \mathrm{Gr}(2, \mathcal{W}^\vee)$ is flat by [Mat89, Theorem 23.1]. □

Definition 2.17. Let V be a complex vector space of dimension 7 and $\mathbb{B} := \mathrm{Gr}(3, \Lambda^2 V^\vee)$. For a net $\Pi \in \mathbb{B}$, let $X_\Pi \subset \mathrm{Gr}(3, V)$ be the zero scheme of the section of $\Lambda^2\mathcal{U}^\vee \otimes \Pi^\vee$ induced by restriction of the forms of Π to the tautological subbundle $\mathcal{U}$, that is, the scheme of three-dimensional subspaces isotropic for every $\omega \in \Pi$. The net Π is *non-degenerate* if every non-zero $\omega \in \Pi$ has rank 6, and $\mathbb{B}^{\mathrm{nd}} \subset \mathbb{B}$ denotes the open subset of non-degenerate nets. Let $\Omega \subset \mathbb{B}^{\mathrm{nd}}$ be the set of Π such that X_Π is a smooth prime Fano threefold of genus 12, and let $\Omega^{\mathrm{st}} \subset \Omega$ be the set of Π whose $\mathrm{SL}(V)$-orbit is closed in $\mathbb{B}^{\mathrm{nd}}$ and whose stabilizer in $\mathrm{SL}(V)$ is finite. Let $\Delta' \subset \Omega$ be the set of Π such that some divisor in $|-K_{X_\Pi}|$ has a singular point whose germ is analytically isomorphic to $(\{y^2 = x^3 + t^4 x\} \ni 0)$.

Remark 2.18. The scheme structure on X_Π is the one of [BKM26b, (4.1), Definition 4.1, and Lemma 4.2]. In the definition of Ω^{st}, the closedness of the orbit does not imply the finiteness of the stabilizer: the orbit of the net of the Mukai–Umemura threefold X^{MU} is closed with infinite stabilizer (Theorem 2.26(1), Theorem 17.5(3), and Theorem 16.12(2)). The orbit of the net of X^a is not closed (Theorem 2.26(2) and Theorem 17.5(3)). The open subset $\mathbb{B}^{\mathrm{nd}} \subset \mathbb{B}$ is affine and $\mathrm{SL}(V)$-invariant, and all points of $\mathbb{B}^{\mathrm{nd}}$ are GIT-semistable (Definition-Proposition 16.2). Theorem 17.5 shows that Ω^{st} consists exactly of the nets $\Pi \in \Omega$ such that X_Π is K-stable.

Lemma 2.19. *Let V be a complex vector space of dimension 7, let $\mathrm{Gr}(3, V) \subset \mathbb{P}(\Lambda^3 V)$ be the Plücker embedding, and let $\mathrm{Fl}(2, 4; V)$ be the variety of flags $W_2 \subset W_4 \subset V$ with $\dim W_2 = 2$ and $\dim W_4 = 4$. Then the morphism from $\mathrm{Fl}(2, 4; V)$ to the Hilbert scheme of lines of $\mathrm{Gr}(3, V)$ sending $(W_2 \subset W_4)$ to the line $\{U \in \mathrm{Gr}(3, V) : W_2 \subset U \subset W_4\}$ is an isomorphism. In particular, for a local Artinian $\mathbb{C}$-algebra A, the families of lines of $\mathrm{Gr}(3, V)$ over $\mathrm{Spec}\, A$ correspond to the pairs $W_2 \subset W_4$ of direct summands of $V \otimes A$ of ranks 2 and 4, and for $U \in \mathrm{Gr}(3, V)$ the scheme of lines of $\mathrm{Gr}(3, V)$ through $[U]$ is $\mathbb{P}(U^\vee) \times \mathbb{P}(V/U)$, where the pair of points given by a hyperplane $W_2 \subset U$ and a line $W_4/U \subset V/U$ corresponds to the line $\{U' \in \mathrm{Gr}(3, V) : W_2 \subset U' \subset W_4\}$.*

Proof. Let F be the Hilbert scheme of lines of $\mathrm{Gr}(3, V)$. The family $\{(W_2 \subset W_4, U) : W_2 \subset U \subset W_4\}$ over $\mathrm{Fl}(2, 4; V)$ is a $\mathbb{P}^1$-bundle whose fiber over $(W_2 \subset W_4)$ is $\mathbb{P}(W_4/W_2)$, and its fibers are lines of $\mathrm{Gr}(3, V)$, since $\Lambda^3(W_2 \oplus \mathbb{C}x) = \Lambda^2 W_2 \wedge x$ is linear in $x \in W_4/W_2$. Hence it defines a morphism $\phi : \mathrm{Fl}(2, 4; V) \to F$, which is bijective on closed points by [Har92, Exercise 6.9(i)].

Let $\ell \in F$ be the line given by $W_2 \subset W_4$, so that $\ell = \mathbb{P}(W_4/W_2)$. Then $\mathcal{U}|_\ell$ is an extension of $\mathcal{O}_\ell(-1)$ by $W_2 \otimes \mathcal{O}_\ell$, and $(V/\mathcal{U})|_\ell$ is an extension of $(V/W_4) \otimes \mathcal{O}_\ell$ by $(W_4 \otimes \mathcal{O}_\ell)/\mathcal{U}|_\ell \cong \mathcal{O}_\ell(1)$. Since $H^1(\ell, \mathcal{O}_\ell(1)) = 0$, both extensions split, so

$$\mathcal{U}|_\ell \cong \mathcal{O}_\ell^{\oplus 2} \oplus \mathcal{O}_\ell(-1), \qquad (V/\mathcal{U})|_\ell \cong \mathcal{O}_\ell(1) \oplus \mathcal{O}_\ell^{\oplus 3}.$$

By [Har92, Example 16.1], $T_{\mathrm{Gr}(3,V)} = \mathcal{H}om(\mathcal{U}, V/\mathcal{U})$, so

$$T_{\mathrm{Gr}(3,V)}|_\ell \cong \mathcal{O}_\ell(2) \oplus \mathcal{O}_\ell(1)^{\oplus 5} \oplus \mathcal{O}_\ell^{\oplus 6}.$$

The inclusion $T_\ell \cong \mathcal{O}_\ell(2) \to T_{\mathrm{Gr}(3,V)}|_\ell$ is an isomorphism onto the summand $\mathcal{O}_\ell(2)$, because $\mathrm{Hom}(\mathcal{O}_\ell(2), \mathcal{O}_\ell(j)) = 0$ for $j < 2$. Hence $N_{\ell/\mathrm{Gr}(3,V)} \cong \mathcal{O}_\ell(1)^{\oplus 5} \oplus \mathcal{O}_\ell^{\oplus 6}$, so $h^1(\ell, N_{\ell/\mathrm{Gr}(3,V)}) = 0$ and

$$h^0(\ell, N_{\ell/\mathrm{Gr}(3,V)}) = 16 = \dim \mathrm{Gr}(2, V) + \dim \mathrm{Gr}(2, V/W_2) = \dim \mathrm{Fl}(2, 4; V).$$

Thus F is smooth of dimension 16 at every point [Kol96, Theorem I.2.8 and Theorem I.2.15]. Since ϕ is bijective and $\mathrm{Fl}(2, 4; V)$ is irreducible, F is a smooth irreducible variety, and the bijective morphism ϕ of smooth varieties is birational. Since $\mathrm{Fl}(2, 4; V)$ is projective, ϕ is a projective birational morphism with finite fibers onto the normal variety F. Hence ϕ is finite [Har77, Exercise III.11.2] and satisfies $\phi_* \mathcal{O}_{\mathrm{Fl}(2,4;V)} = \mathcal{O}_F$ (see the proof of [Har77, Corollary III.11.4]), so ϕ is an isomorphism.

The first statement after "In particular" is the functor of points of $\mathrm{Fl}(2, 4; V)$ on $\operatorname{Spec} A$. The scheme of lines through $[U]$ is the fiber over $[U]$ of the universal line, which under ϕ is the closed subscheme $\{(W_2 \subset W_4) : W_2 \subset U \subset W_4\} \cong \mathrm{Gr}(2, U) \times \mathrm{Gr}(1, V/U) = \mathbb{P}(U^\vee) \times \mathbb{P}(V/U)$ of $\mathrm{Fl}(2, 4; V)$, by the functor of points. □

Definition-Theorem 2.20 ([BKM26a, Theorem 1.1] and [BKM26b, Theorem 1.1 and Corollary 6.10])**.** Let X be a smooth prime Fano threefold of genus 12. There exists a unique exceptional vector bundle $\mathcal{U}_X$ of rank 3 on X with $\det \mathcal{U}_X \cong \mathcal{O}_X(K_X)$, $H^\bullet(X, \mathcal{U}_X) = 0$, $\mathcal{U}_X^\vee$ globally generated, $h^0(X, \mathcal{U}_X^\vee) = 7$, and $H^{>0}(X, \mathcal{U}_X^\vee) = 0$, the *Mukai bundle* of X. Let $V_X := H^0(X, \mathcal{U}_X^\vee)^\vee$. The *canonical Mukai net* of X is

$$\Pi_X := \mathrm{Ker}\left(\Lambda^2 V_X^\vee \to H^0(X, \Lambda^2 \mathcal{U}_X^\vee)\right).$$

Then Π_X is three-dimensional and non-degenerate, the evaluation map induces an isomorphism of X onto the scheme of three-dimensional subspaces of V_X isotropic for every form in Π_X, and H is the restriction of the Plücker class. In particular, for every linear isomorphism $V_X \cong V$ the image of Π_X in $\mathbb{B}$ belongs to Ω, and the images for different isomorphisms form one $\mathrm{SL}(V)$-orbit.

Proof. The existence and the uniqueness of $\mathcal{U}_X$ are [BKM26a, Theorem 1.1]. By [BKM26b, Theorem 1.1, Lemma 4.2, and Corollary 6.10], the evaluation map induces an isomorphism of X onto the scheme of three-dimensional subspaces of V_X isotropic for every form in a non-degenerate net $\sigma_0 \subset \Lambda^2 V_X^\vee$, under which $\mathcal{U}_X$ is the restriction of the tautological subbundle and H is the restriction of the Plücker class. Every form in σ_0 vanishes on $\mathcal{U}_X$, so $\sigma_0 \subset \Pi_X$. By [BKM26b, Lemma 6.9 and its proof], a very general surface $S \in |H|$ is a prime K3 surface of genus 12 with Mukai bundle $\mathcal{U}_S := \mathcal{U}_X|_S$, and restriction gives $H^0(X, \mathcal{U}_X^\vee) = H^0(S, \mathcal{U}_S^\vee)$. Hence Π_X is contained in the kernel of $\Lambda^2 V_X^\vee \to H^0(S, \Lambda^2 \mathcal{U}_S^\vee)$, which is three-dimensional by [BKM26b, Proposition 4.5 and its proof]. Comparing dimensions, $\Pi_X = \sigma_0$, so Π_X is three-dimensional and non-degenerate. Thus for any linear isomorphism $V_X \cong V$, the image Π of Π_X is non-degenerate and $X_\Pi \cong X$, so $\Pi \in \Omega$. Two linear isomorphisms $V_X \cong V$ differ by an element of $\mathrm{GL}(V)$. Since scalars act trivially on $\mathbb{B}$ and $\mathrm{GL}(V) = \mathbb{C}^* \cdot \mathrm{SL}(V)$, the images form one $\mathrm{SL}(V)$-orbit. □

For every $\Pi \in \mathbb{B}$, $X_{\Pi'} \cong X_\Pi$ for every Π' in the $\mathrm{SL}(V)$-orbit of Π, and the stabilizers of two points of one orbit are conjugate. Hence membership in Ω, closedness of the orbit, and finiteness of the stabilizer depend only on the $\mathrm{SL}(V)$-orbit. Since $\mathcal{U}_X$ is unique, Π_X depends only on X. Thus the isomorphism class of X corresponds to the $\mathrm{SL}(V)$-orbit of Π_X, and $\mathrm{Aut}(X)$ is the stabilizer of Π_X in $\mathrm{PGL}(V)$, while the stabilizer of Π_X in $\mathrm{SL}(V)$ is an extension of $\mathrm{Aut}(X)$ by μ_7 (Theorem 16.12).

2.4. **Links and flops.** Along a line, and in genus twelve also along a conic, the volumes and the refined series are computed on the other side of a Sarkisov link, where the linear systems are explicit: on the quadric Q for a line in genus ten (Theorem 2.22), on V_5 for a line in genus twelve (Theorem 2.23), and on the quadric Q for a conic in genus twelve (Theorem 2.24). The fixed parts of the refined series in later sections come from divisors on the other side of the link.

We use the width of a flopping curve in the sense of [Rei83, Definition (5.3), Theorem (5.4), and (5.8)]. The description of Reid's pagoda as a tower of blow-ups of curves is [BO95, Section 3].

Definition-Proposition 2.21. Let Y be a smooth threefold and $C \subset Y$ a smooth rational curve with $N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ which is contracted by a small contraction $Y \to \bar{Y}$ to an isolated singular point. The *width* $w(C)$ of C is the number of blow-ups of curves in Reid's pagoda resolution of the flop of C, where the last blow-up, whose exceptional divisor is $\mathbb{P}^1 \times \mathbb{P}^1$, is counted. Then $\bar{Y}$ is analytically locally given by $uv = z(z - t^{w(C)})$ near the image of C, and conversely, if $\bar{Y}$ is analytically locally given by $uv = z(z - t^n)$ near the image of C for an integer $n \geq 1$, then $w(C) = n$. We call C an *Atiyah flopping curve*, and the flop of C an *Atiyah flop*, if $N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. The curve C is an Atiyah flopping curve if and only if $w(C) = 1$.

Proof. Let $n := w(C)$. By [Rei83, Remark (5.13)(b)], $\bar{Y}$ is analytically locally given by $xy = z^2 - t^{2n}$ near the image of C. Replacing z by $z + t^n$ and rescaling t, we obtain the equation $uv = z(z - t^n)$, see also [DW16, Example 3.10]. Conversely, let $f_n := uv - z(z - t^n)$. The polynomial f_n is weighted homogeneous of degree $2n$ for the weights $(n, n, n, 1)$ of (u, v, z, t). Its partial derivatives are $\partial_u f_n = v$, $\partial_v f_n = u$, $\partial_z f_n = -(2z - t^n)$, and $\partial_t f_n = nzt^{n-1}$, and

$$\begin{aligned}\dim_{\mathbb{C}} \mathbb{C}\{u, v, z, t\}/(\partial_u f_n, \partial_v f_n, \partial_z f_n, \partial_t f_n) &= \dim_{\mathbb{C}} \mathbb{C}\{u, v, z, t\}/(v,\ u,\ 2z - t^n,\ nzt^{n-1}) \\ &= \dim_{\mathbb{C}} \mathbb{C}\{t\}/(t^{2n-1}) = 2n - 1.\end{aligned}$$

Since $2nf_n = nu\partial_u f_n + nv\partial_v f_n + nz\partial_z f_n + t\partial_t f_n$, f_n belongs to its Jacobian ideal, so the dimension computed above equals $\tau(f_n) := \dim_{\mathbb{C}} \mathbb{C}\{u, v, z, t\}/(f_n, \partial_u f_n, \partial_v f_n, \partial_z f_n, \partial_t f_n)$. An analytic isomorphism of hypersurface germs of order at least two is induced by an analytic change of ambient coordinates, and the local equations then differ by a unit [GLS07, Chapter I, Lemma 1.23]. The ideal $(f, \partial_u f, \ldots, \partial_t f)$ does not change when f is multiplied by a unit, and it is mapped to the corresponding ideal under a change of coordinates. Hence $\tau(f_n) = 2n - 1$ depends only on the germ $(f_n = 0)$, and the germs for different n are pairwise non-isomorphic. Thus if $\bar{Y}$ is analytically locally given by $uv = z(z - t^n)$, then $n = w(C)$. Finally, by [Rei83, Definition (5.3)] and the sentence following it, $w(C) = 1$ if and only if $N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, that is, C is an Atiyah flopping curve if and only if $w(C) = 1$. □

Theorem 2.22 ([KPZ14, Theorem 2.6, Proposition 2.2, Proposition 2.12, and Theorem 2.13])**.** *Let X be a smooth prime Fano threefold of genus 10, $L \subset X$ a line, and $\pi : Y := \mathrm{Bl}_L X \to X$ the blow-up with exceptional divisor E. Then $-K_Y$ is nef and big, and there exists a flop $Y \dashrightarrow Y^+$ followed by the blow-up $Y^+ = \mathrm{Bl}_\Gamma Q \to Q$ of a smooth curve Γ of genus 2 and degree 7 in a smooth quadric threefold $Q \subset \mathbb{P}^4$. Under strict transform, the pullback h of $\mathcal{O}_Q(1)$ and the exceptional divisor D' of $Y^+ \to Q$ satisfy $h = \pi^*H - 2E$ and $D' = 2\pi^*H - 5E$. The flopping curves are the strict transforms of the lines on X intersecting L and, if L is special, the negative section of $E \cong \mathbb{F}_3$. They are pairwise disjoint, and each of them has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ in Y. The image of E in Q is a quartic surface containing Γ, which is normal if and only if L is non-special, and whose normalization is $\mathbb{F}_0$ or $\mathbb{F}_2$ if L is special.*

The link of Theorem 2.22 is Iskovskikh's double projection from the line L [Isk89], see [KPZ14, Theorem 2.6]. By the statement of [KPZ14, Theorem 2.6], the double projection is defined by $|\pi^*H - 2E|$ and D' is the strict transform of a divisor in $|2\pi^*H - 5E|$, which implies the relations $h = \pi^*H - 2E$ and $D' = 2\pi^*H - 5E$. The divisor $-K_Y$ is nef and big, see the proof of [KPZ14, Theorem 2.6].

For a variety T and a closed subscheme $Z \subset T$, we write $\mathrm{Aut}(T; Z)$ for the subgroup of $\mathrm{Aut}(T)$ consisting of the automorphisms g with $g(Z) = Z$.

Theorem 2.23 ([KPS18, Theorem 5.2.2, Remark 5.2.4, Lemma 5.2.5 and the paragraph following it, and Remark 5.2.6])**.** *Let X be a smooth prime Fano threefold of genus 12, $L \subset X$ a line, and $\pi : Y := \mathrm{Bl}_L X \to X$ the blow-up with exceptional divisor E. Then there exists a flop $Y \dashrightarrow Y^+$ followed by the blow-up $Y^+ = \mathrm{Bl}_\Gamma V_5 \to V_5$ of a rational normal quintic curve Γ in the quintic del Pezzo threefold $V_5 \subset \mathbb{P}^6$. Under strict transform, the pullback h of the hyperplane class of V_5 and the exceptional divisor R of $Y^+ \to V_5$ satisfy $h = \pi^*H - 2E$, $R = \pi^*H - 3E$, and $\pi^*H = 3h - 2R$. The divisor R is the strict transform of the unique member of $|\pi^*H - 3E|$, and E is the strict transform of the unique hyperplane section of V_5 containing Γ. The flopping curves are pairwise disjoint. They are the strict transforms of the lines on X intersecting L and, if L is special, the negative section of $E \cong \mathbb{F}_3$, and the flopped curves are the strict transforms of the bisecant lines of Γ. Conversely, for every rational normal quintic $\Gamma \subset V_5$ the linear system $|3h - 2R|$ on $\mathrm{Bl}_\Gamma V_5$ defines the inverse link to a smooth prime Fano threefold of genus 12 with a marked line, and $\mathrm{Aut}(X; L) \cong \mathrm{Aut}(V_5; \Gamma)$.*

The link of Theorem 2.23 is the double projection from the line L [Isk89], see [KPS18, Theorem 5.2.2] and [IP99, Theorem 4.3.7].

The width enters in two places. First, for a non-special line L and a line $M \neq L$ meeting L at y, if n is the width of the strict transform of M in $\mathrm{Bl}_L X$, then the completed local ring at (M, y) of the scheme of pairs (M', y') with $M' \neq L$ a line and $y' \in M' \cap L$ is $\mathbb{C}[t]/(t^n)$ (Proposition 5.3 and Definition-Proposition 10.9). Second, in genus twelve, let L be a special line and w_L the width of the negative section of $E \cong \mathbb{F}_3$. If $w_L \leq 2$, then w_L determines the germ of $\Sigma(X)$ at $[L]$, which is of type A_1 or A_2. If $w_L = 3$, the completed local ring of $\Sigma(X)$ at $[L]$ is determined only up to the form $\mathbb{C}[[a, b]]/(a^2 + \varphi(b))$ with $\varphi \in b^4\mathbb{C}[[b]]$, where $\varphi = 0$ is allowed (Proposition 10.8).

Theorem 2.24 ([KP18, Theorem 2.2, the proof of Theorem 2.2, and Lemma 2.4])**.** *Let X be a smooth prime Fano threefold of genus 12, $C \subset X$ a smooth conic, and $\pi : Y := \mathrm{Bl}_C X \to X$ the blow-up with exceptional divisor E. Then $|-K_Y|$ is base point free and defines a morphism that contracts no divisor, and there exists a flop $Y \dashrightarrow Y^+$ followed by the blow-up $Y^+ = \mathrm{Bl}_\Gamma Q \to Q$ of a smooth rational quadratically normal sextic curve Γ in a smooth quadric threefold $Q \subset \mathbb{P}^4$. Under strict transform, the pullback h of the hyperplane class of Q and the exceptional divisor R of $Y^+ \to Q$ satisfy $h = \pi^*H - 2E$, $R = 2\pi^*H - 5E$, $\pi^*H = 5h - 2R$, and $E = 2h - R$. The divisor R is the strict transform of the unique member of $|2\pi^*H - 5E|$, and E is the strict transform of the unique member of $|2h - R|$.*

By [KP18, Theorem 2.2(i)], the map $Y \dashrightarrow Q$ of Theorem 2.24 is given by the linear system $|\pi^*H - 2E|$ and contracts the unique member of $|2\pi^*H - 5E|$. See also [Tak89] and [IP99, Theorem 4.4.11].

2.5. **Automorphisms and members with a torus action.** The smooth Fano threefolds of Picard rank one with infinite automorphism group were classified in [Pro90] and [KPS18, Theorem 1.1.2]. In genus 12, the threefold with automorphism group PGL_2 was constructed in [MU83], and the threefold with automorphism group $\mathbb{G}_a \rtimes \mu_4$ appears in [KPS18, Example 5.3.2 and Proposition 5.4.6] and [DFK25, Theorem 1].

Definition-Theorem 2.25 ([KPS18, Theorems 1.1.2 and 5.3.10, Proposition 5.4.6])**.** Let X be a smooth prime Fano threefold of genus 10 or 12. Then the following hold.

(1) If X has genus 10, then $\mathrm{Aut}(X)$ is finite.
(2) If X has genus 12, then $\mathrm{Aut}(X)$ is infinite if and only if X is isomorphic to one of the following: the unique, up to isomorphism, smooth prime Fano threefold of genus 12 with automorphism group isomorphic to PGL_2, the *Mukai–Umemura threefold* X^{MU}; the unique, up to isomorphism, smooth prime Fano threefold of genus 12 with automorphism group isomorphic to $\mathbb{G}_a \rtimes \mu_4$, denoted X^a; or a member of a one-parameter family of smooth prime Fano threefolds of genus 12 with automorphism group isomorphic to $\mathbb{G}_m \rtimes \mu_2$.

The uniqueness statements in Definition-Theorem 2.25(2) are part of the classification [KPS18, Theorem 5.3.10 and Proposition 5.4.6], in which the automorphism groups of the three classes are pairwise non-isomorphic. The smooth prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action, X^{MU} among them, are described in [KP18, Theorem 1.3], see also [DKK15, Section 2.1]. In this paper we parametrize them by the nets Π_h of Definition-Proposition 16.14, and the stabilizers of the nets Π_h are compared with $\mathrm{Aut}(X_{\Pi_h})$ in Theorem 16.12.

The members with a faithful $\mathbb{G}_m$-action are the central fibers of the smooth degenerations constructed in Sections 12 and 15.

Theorem 2.26 ([CS23, Corollary 1.7, Remark 1.8, and Example 1.4], [Fuj23, Theorem 1.2], [Ber16, Theorem 1.1])**.** *Let X be a smooth prime Fano threefold of genus 12. Then the following hold.*

(1) If X admits a faithful $\mathbb{G}_m$-action, then X is K-polystable.
(2) If $X \cong X^a$, then X is K-semistable and not K-polystable.

Proof. For (1), the smooth prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action are X^{MU} and the members of the one-parameter family in Definition-Theorem 2.25(2). Together they form the one-parameter family of [CS23, Introduction]. By [CS23, Corollary 1.7], every member of the family of [CS23, Introduction] except two admits a Kähler–Einstein metric, and X^{MU} is not an exceptional member [CS23, Introduction]. The exceptional members admit Kähler–Einstein metrics by [Fuj23, Theorem 1.2] (see [CS23, Remark 1.8]). A Fano manifold with a Kähler–Einstein metric is K-polystable by [Ber16, Theorem 1.1]. For X^{MU}, a Kähler–Einstein metric was also obtained by Donaldson [Don07, Theorem 1 and Corollary 1], see also [Don08, Theorem 5.1 and the paragraph after it].

For (2), the threefold X^a is K-semistable by [CS23, Example 1.4]. The identity component of $\mathrm{Aut}(X^a) \cong \mathbb{G}_a \rtimes \mu_4$ is $\mathbb{G}_a$, so $\mathrm{Aut}(X^a)$ is not reductive. Thus X^a is not K-polystable by Theorem 2.7. □

2.6. **Plane quartics.**

Definition 2.27. Let $\mathbb{C}[\partial] := \mathbb{C}[\partial_x, \partial_y, \partial_z]$ act on $\mathbb{C}[x, y, z]$ by differentiation, and let $f \in \mathbb{C}[x, y, z]_4$ be non-zero. The *apolar ideal* of f is

$$f^\perp := \{P \in \mathbb{C}[\partial] \mid P(f) = 0\}.$$

The *middle catalecticant* of f is the linear map $\mathrm{Cat}_2(f) : \mathbb{C}[\partial]_2 \to \mathbb{C}[x, y, z]_2$, $P \to P(f)$. The quartic f is *non-degenerate* if $\mathrm{Cat}_2(f)$ is invertible. Since $\operatorname{Ker} \mathrm{Cat}_2(f) = (f^\perp)_2$ and $\dim \mathbb{C}[\partial]_2 = \dim \mathbb{C}[x, y, z]_2 = 6$, f is non-degenerate if and only if $(f^\perp)_2 = 0$.

For a basis $(P_1, \dots, P_6)$ of $\mathbb{C}[\partial]_2$, the *determinant of* $\mathrm{Cat}_2(f)$ *in the basis* $(P_1, \dots, P_6)$ is

$$\det \mathrm{Cat}_2(f) := \det \big((P_iP_j)(f)\big)_{1 \le i,j \le 6},$$

that is, the determinant of the symmetric bilinear form $(P, P') \to (PP')(f)$ on $\mathbb{C}[\partial]_2$ with respect to $(P_1, \dots, P_6)$. The bilinear form $(P, P') \to (PP')(f)$ is the composition of $\mathrm{Cat}_2(f)$ with the perfect pairing $\mathbb{C}[\partial]_2 \times \mathbb{C}[x, y, z]_2 \to \mathbb{C}$, $(P, q) \to P(q)$. Hence $\det \mathrm{Cat}_2(f) \neq 0$ if and only if $\mathrm{Cat}_2(f)$ is invertible.

Let $\mathrm{VPS}(f, 6) \subset \mathrm{Hilb}^6(\mathbb{P}(\mathbb{C}[x, y, z]_1))$ be the closed subscheme of length-six subschemes whose homogeneous ideal, viewed in $\mathbb{C}[\partial]$, is contained in $f^\perp$. Let $\mathrm{VSP}(f, 6) \subset \mathrm{VPS}(f, 6)$ be the closure of the locus of six distinct points $[\ell_1], \dots, [\ell_6]$ with $f \in \mathbb{C}\ell_1^4 + \cdots + \mathbb{C}\ell_6^4$. By the apolarity lemma [IK99, Lemma 1.15], the homogeneous ideal of six distinct points $[\ell_1], \dots, [\ell_6]$ is contained in $f^\perp$ if and only if $f \in \mathbb{C}\ell_1^4 + \cdots + \mathbb{C}\ell_6^4$, so $\mathrm{VSP}(f, 6) \subset \mathrm{VPS}(f, 6)$.

Let $\mathfrak{A}$ be the Aronhold invariant, the equation of the closure of the SL_3-orbit of $[x^3 + y^3 + z^3]$ in $\mathbb{P}(\mathbb{C}[x, y, z]_3)$, which is an SL_3-invariant of degree four determined up to a non-zero scalar [Ott24, Section 3]. The *Scorza quartic* of f is

$$\mathrm{Sc}(f)(a) := \mathfrak{A}\big(a_1\partial_x f + a_2\partial_y f + a_3\partial_z f\big) \quad \text{for } a = (a_1, a_2, a_3) \in \mathbb{C}^3,$$

regarded as an element of $\mathbb{C}[x,y,z]_4$ by renaming (a_1,a_2,a_3) as (x,y,z). The *projective Scorza map* is the rational map $\mathbb{P}(\mathbb{C}[x,y,z]_4) \dashrightarrow \mathbb{P}(\mathbb{C}[x,y,z]_4)$, $[f] \to [\mathrm{Sc}(f)]$, defined where $\mathrm{Sc}(f) \neq 0$.

Let $U^{\mathrm{cat}} \subset \mathbb{P}(\mathbb{C}[x,y,z]_4)$ be the open subset of non-degenerate quartics, and let $U^{\mathrm{sm}} \subset U^{\mathrm{cat}}$ be the subset of $[f]$ such that $\mathrm{VPS}(f,6)$ is smooth. For $h \in \mathbb{C}\backslash\{1\}$, let

$$F_h := \frac{h}{6}x^4 + 2x^2yz + y^2z^2.$$

For the Scorza quartic we refer to [DK93, Section 7] and [Ott24, Algorithm 5]. The value $h = 1$ is excluded in Definition 2.27 because $\det \mathrm{Cat}_2(F_h) = 4096(h-1)$ in the basis of Proposition 16.16(3), so F_1 is degenerate. The quartic F_2 is non-degenerate, but the scheme X_{Π_2} attached to the net Π_2 with apolar quartic F_2 is singular by Definition-Proposition 16.14(1)(4)(5), and for this reason Theorem 1.5(5) also excludes $h = 2$. The correspondence between U^{cat} and $\mathbb{B}^{\mathrm{nd}}$ is Proposition 20.11, and the points of U^{cat} are GIT-semistable by Proposition 16.16.

Definition-Proposition 2.28. Let $\Pi \in \mathbb{B}^{\mathrm{nd}}$. After the choice of a basis x,y,z of Π, we identify $\mathrm{Sym}\,\Pi$ with $\mathbb{C}[x,y,z]$ and $\mathrm{Sym}\,\Pi^\vee$ with $\mathbb{C}[\partial]$, and we apply Definition 2.27 through such an identification. Let $e_0,\dots,e_6$ be a basis of V. The seven principal 6×6 Pfaffians of the alternating 7×7 matrix $(\omega \to \omega(e_i,e_j))_{0\le i,j\le 6}$ of Π, whose entries belong to $\Pi^\vee$, generate a Gorenstein ideal of codimension three of $\mathrm{Sym}\,\Pi^\vee$ whose quotient has socle degree four, and the ideal does not depend on the basis $e_0,\dots,e_6$. The Macaulay inverse system of the quotient in degree four is spanned by a quartic $f_\Pi \in \mathrm{Sym}^4\Pi$, determined up to a non-zero scalar, the *apolar quartic* of Π. The apolar ideal of f_Π is the Pfaffian ideal, and $\mathrm{Cat}_2(f_\Pi)$ is invertible, that is, f_Π is non-degenerate in the sense of Definition 2.27. The subschemes $(f_\Pi = 0)$ and $(\mathrm{Sc}(f_\Pi) = 0)$ of $\mathbb{P}(\Pi^\vee)$ do not depend on the choice of the basis x,y,z. For a smooth prime Fano threefold X of genus 12, let $f_X := f_{\Pi_X}$ and $Q_X := (\mathrm{Sc}(f_X) = 0) \subset \mathbb{P}(\Pi_X^\vee)$.

Proof. Let $S := \mathrm{Sym}\,\Pi^\vee$, let $M := (\omega \to \omega(e_i,e_j))_{0\le i,j\le 6}$, and let $I \subset S$ be the ideal generated by the seven principal 6×6 Pfaffians of M. Let $\omega_\Pi \in \Lambda^2V^\vee \otimes S_1$ be the element corresponding to the inclusion $\Pi \subset \Lambda^2V^\vee$. The element ω_Π does not depend on any basis. In the basis $e_0,\dots,e_6$ we have $\omega_\Pi = \sum_{i<j} M_{ij}e_i^\vee \wedge e_j^\vee$, so the coefficients of $\omega_\Pi^{\wedge 3}/3! \in \Lambda^6V^\vee \otimes S_3$ in the basis $(e_0^\vee \wedge \cdots \wedge \widehat{e_k^\vee} \wedge \cdots \wedge e_6^\vee)_k$ are the principal 6×6 Pfaffians of M. Hence I_3 is the image of the contraction $\Lambda^6V \to S_3$ with $\omega_\Pi^{\wedge 3}$, and I does not depend on the basis $e_0,\dots,e_6$.

For $0 \neq \omega \in \Pi$, the value $M(\omega)$ of M at ω is the matrix $(\omega(e_i,e_j))_{i,j}$ of ω, of rank 6 since Π is non-degenerate. An alternating matrix of rank 6 has an invertible principal 6×6 submatrix: indeed, let v span the kernel of $M(\omega)$, and choose k with $v_k \neq 0$. If w is in the kernel of the principal submatrix omitting the index k, and $\tilde{w}$ is w extended by 0 at k, then $M(\omega)\tilde{w} = ce_k$ with $cv_k = v^tM(\omega)\tilde{w} = -(M(\omega)v)^t\tilde{w} = 0$, so $c = 0$, $\tilde{w} \in \mathbb{C}v$, and $\tilde{w}_k = 0$ implies $w = 0$. Thus some principal 6×6 Pfaffian of M does not vanish at ω. Thus the zero locus of I in Π is $\{0\}$, and I is primary to the ideal $\mathfrak{m} := \Pi^\vee S$ by the Nullstellensatz. Hence I has codimension three in the three-dimensional polynomial ring S, so the grade of I is three. The entries of M are linear and the principal 6×6 Pfaffians of M are cubic. By [BE77, Theorem 2.1], S/I is Gorenstein with self-dual minimal free resolution

$$0 \to S(-7) \to S(-4)^7 \xrightarrow{M} S(-3)^7 \to S \to S/I \to 0.$$

The resolution gives the Hilbert series

$$\frac{1-7t^3+7t^4-t^7}{(1-t)^3} = 1+3t+6t^2+3t^3+t^4$$

of S/I. Thus S/I is an Artinian Gorenstein algebra of socle degree four. By Macaulay duality [IK99, Lemma 2.12], [Sch01, Theorem 2.1], the inverse system of S/I in degree four is spanned by a quartic $f_\Pi \in \mathrm{Sym}^4\Pi$, determined up to a non-zero scalar, and $f_\Pi^\perp = I$. Since I is generated by cubics, $(f_\Pi^\perp)_2 = I_2 = 0$, so $\mathrm{Cat}_2(f_\Pi)$ is invertible.

The subscheme $(f_\Pi = 0) \subset \mathbb{P}(\Pi^\vee)$ is the zero scheme of $f_\Pi \in \operatorname{Sym}^4 \Pi$, and so does not depend on the basis x, y, z. Under the identification of $\operatorname{Sym} \Pi^\vee$ with $\mathbb{C}[\partial]$, $\xi \in \Pi^\vee$ acts on $\operatorname{Sym} \Pi$ as a derivation, and $\operatorname{Sc}(f_\Pi)$ is the function $\xi \to \mathfrak{A}(\xi(f_\Pi))$ on $\Pi^\vee$. Let x, y, z and x', y', z' be bases of Π, and let $\mathfrak{A}_{x,y,z}$ and $\mathfrak{A}_{x',y',z'}$ be the functions on $\operatorname{Sym}^3 \Pi$ given by $\mathfrak{A}$ through the identifications $\operatorname{Sym}^3 \Pi \cong \mathbb{C}[x, y, z]_3$ defined by x, y, z and by x', y', z'. The bases x, y, z and x', y', z' differ by an element of $\mathrm{GL}_3(\mathbb{C}) = \mathbb{C}^* \cdot \mathrm{SL}_3(\mathbb{C})$. Since $\mathfrak{A}$ is an SL_3-invariant of degree four, $\mathfrak{A}_{x',y',z'} = c\mathfrak{A}_{x,y,z}$ for some $c \in \mathbb{C}^*$. Hence $\operatorname{Sc}(f_\Pi) \in \operatorname{Sym}^4 \Pi$ changes by a non-zero scalar, and $(\operatorname{Sc}(f_\Pi) = 0)$ does not depend on the basis.

Finally, the definition of f_Π uses only a seven-dimensional vector space V and a non-degenerate net Π on V. By Definition-Theorem 2.20, Π_X is a non-degenerate net on V_X with $\dim V_X = 7$, so f_X and Q_X are defined. □

The definition of f_Π through the Pfaffian ideal follows [Muk92, Section 6, Lemma 12] and [Sch01, Theorems 2.3 and 2.6]. By Theorem 16.11, Q_X is isomorphic to $\Sigma(X)$. The curves $(f_X = 0)$ and Q_X need not be projectively equivalent (Remark 1.6).

We use the GIT-semistable points of [MFK94, Definition 1.7], and we call the properly stable points of [MFK94, Definition 1.8] *GIT-stable*, that is, a GIT-semistable point is GIT-stable if its orbit is closed in the GIT-semistable locus and its stabilizer is finite [MFK94, Amplification 1.11]. A point is *GIT-unstable* if it is not GIT-semistable. We call a GIT-semistable point p *GIT-polystable* if the orbit of p is closed in the GIT-semistable locus. We let GL_3 act on $\mathbb{C}[x, y, z]$ by $(g \cdot f)(v) = f(g^{-1}v)$, and on $\mathbb{C}[\partial]$ by the contragredient action on $\mathbb{C}[\partial]_1$, so that $g \cdot (P(f)) = (g \cdot P)(g \cdot f)$ for any $g \in \mathrm{GL}_3$, $P \in \mathbb{C}[\partial]$, and $f \in \mathbb{C}[x, y, z]$. We use the Hilbert–Mumford criterion [MFK94, Theorem 2.1 and Proposition 2.3] in the following form: for non-zero $f \in \mathbb{C}[x, y, z]_4$, the point $[f]$ is not GIT-stable if and only if there exists a non-trivial one-parameter subgroup λ of SL_3 such that $\lim_{t\to 0} \lambda(t) \cdot f$ exists in $\mathbb{C}[x, y, z]_4$.

Theorem 2.29 ([MFK94, Chapter 4, Section 2] and [CDGF+25, Theorem 5.1])**.** *Let $f \in \mathbb{C}[x, y, z]_4$ be non-zero, and consider the action of SL_3 on $\mathbb{P}(\mathbb{C}[x, y, z]_4)$ with the linearization $\mathcal{O}(1)$. Then the following hold.*

(1) $[f]$ is GIT-stable if and only if $(f = 0)$ is reduced and has only singular points of type A_1 and A_2.

(2) $[f]$ is GIT-unstable if and only if $(f = 0)$ has a point of multiplicity at least three, or $(f = 0)$ is the union of a cubic curve and a line tangent to the cubic at an inflection point.

(3) $[f]$ is GIT-polystable but not GIT-stable if and only if f is projectively equivalent to $(xz - ay^2)(xz - by^2)$ for some $(a, b) \in \mathbb{C}^2 \setminus \{(0, 0)\}$. Such a quartic is a double smooth conic if $a = b$, the union of two smooth conics tangent at two points if $ab \neq 0$ and $a \neq b$, and the union of a smooth conic and two lines tangent to the conic if $ab = 0$.

In particular, every smooth plane quartic is GIT-stable.

For Theorem 2.29(3) see [CDGF+25, Theorem 5.1 and Theorem 1.12(2)–(4)]. The GIT-semistable closed orbits are also listed in [HL10, Theorem 2] and [Laz06, Section 4.1.1]. The quartics of Theorem 2.29(3) that belong to U^{cat} are determined in Proposition 16.16, and their relation with the members with a faithful $\mathbb{G}_m$-action is given by Definition-Proposition 16.14.

3. Tools for local estimates

In this section, we prove six lemmas used in Sections 4–15.

Lemma 3.1. *Let (X, Δ) be a projective klt pair, $V_\bullet^\bullet$ a multigraded linear series on X as in Definition 2.8, associated to $L_1, \dots, L_r$, E a prime divisor over X of plt type with plt extraction $\pi : Y \to X$ and different Δ_E, and $\lambda > 0$ a real number. If $\pi^*(a_1L_1 + \cdots + a_rL_r)$ is a Weil divisor for all $\vec{a} \in \mathbb{N}^r$, let $W_\bullet^\bullet$ be the refinement of $V_\bullet^\bullet$ by E. Otherwise, fix an integer $k \geq 1$ such that $kL_1, \dots, kL_r$ are Cartier, let $W_\bullet^\bullet$ be the refinement by E of the series $(V_{k\vec{a}})_{\vec{a}}$ associated*

to $kL_1, \dots, kL_r$, and let $S(W; v)$ denote k^{-1} times the expected vanishing order of v on $W_\bullet$ in Definition 2.8. Let $G \subset E$ be either a closed subset or the generic point of an irreducible closed subset of E. Assume that $\lambda S(W; v) \le A_{E,\Delta_E}(v)$ for any prime divisor v over E with $\operatorname{center}_E v \cap G \neq \emptyset$. Then

$$A_{X,\Delta}(F) \ge \lambda S(V; F) + \big(A_{X,\Delta}(E) - \lambda S(V; E)\big) \operatorname{mult}_F E$$

for any prime divisor F over X with $\operatorname{center}_Y F \cap G \neq \emptyset$. In particular, if X is a Fano variety with klt singularities, $\Delta = 0$, and $V_\bullet$ is the complete series of $-K_X$, then $A_X(F) \ge \lambda S_X(F) + (A_X(E) - \lambda S_X(E)) \operatorname{mult}_F E$ for these F.

Proof. We first reduce to the case where $\pi^*(a_1L_1 + \cdots + a_rL_r)$ is a Weil divisor for all $\vec{a}$. Suppose that $\pi^*(a_1L_1 + \cdots + a_rL_r)$ is not a Weil divisor for some $\vec{a}$, and let k and $W_\bullet$ be as in the statement. Let $V'_{\vec{a}} := V_{k\vec{a}}$. Then $V'_\bullet$ is a multigraded linear series associated to the Cartier divisors $kL_1, \dots, kL_r$, $\pi^*(ka_1L_1 + \cdots + ka_rL_r)$ is Cartier for all $\vec{a}$, and $W_\bullet$ is the refinement of $V'_\bullet$ by E. By [AZ22, Lemma 2.11(1)], $S(V'; v) = kS(V; v)$ for any valuation v over X. The assumption of the lemma says that $(\lambda/k)S'(W; v) \le A_{E,\Delta_E}(v)$, where $S'(W; v) = kS(W; v)$ is the expected vanishing order in Definition 2.8. Thus the lemma for $V'_\bullet$ and λ/k gives, for any prime divisor F over X with $\operatorname{center}_Y F \cap G \neq \emptyset$,

$$\begin{aligned} A_{X,\Delta}(F) &\ge \frac{\lambda}{k} S(V'; F) + \Big(A_{X,\Delta}(E) - \frac{\lambda}{k} S(V'; E)\Big) \operatorname{mult}_F E \\ &= \lambda S(V; F) + \big(A_{X,\Delta}(E) - \lambda S(V; E)\big) \operatorname{mult}_F E. \end{aligned}$$

Possibly replacing $V_\bullet$ with $V'_\bullet$ and λ with λ/k, we may assume that $\pi^*(a_1L_1 + \cdots + a_rL_r)$ is a Weil divisor for all $\vec{a}$. Then for a non-zero $s \in V_{\vec{a}}$, the $\mathbb{Q}$-divisor $\pi^* \operatorname{div}(s)$ differs from $\pi^*(a_1L_1 + \cdots + a_rL_r)$ by a principal divisor, so $\operatorname{ord}_E(s)$ is an integer, and $\pi^* \operatorname{div}(s) - jE$ is a $\mathbb{Q}$-Cartier Weil divisor for any integer j.

Fix a prime divisor F over X with $\operatorname{center}_Y F \cap G \neq \emptyset$, and let z be a point of $\operatorname{center}_Y F \cap G$. If G is the generic point of an irreducible closed subset of E, then z is that generic point. Let $Z' \subset E$ be the closure of z. For any prime divisor v over E with $Z' \subset \operatorname{center}_E v$, we have $z \in \operatorname{center}_E v \cap G$, hence $\lambda S(W; v) \le A_{E,\Delta_E}(v)$. Thus

$$\delta_{Z'}(E, \Delta_E; W) \ge \lambda.$$

Let λ' be a real number with $0 < \lambda' < \lambda$. Since $(Y, \Delta_Y + E)$ is plt, the pair (E, Δ_E) is klt by adjunction [Kol13, Lemma 4.8]. The series $W_\bullet$ has bounded support and contains an ample series (Definition 2.8). Following [AZ22, Section 2.6], for $m \ge 1$ let $\delta_{Z',m}(E, \Delta_E; W)$ be the infimum, over all m-basis type divisors D of $W_\bullet$, of the largest real number c such that $(E, \Delta_E + cD)$ is lc at the generic point of Z'. By [AZ22, Lemma 2.9], $\delta_{Z',m}(E, \Delta_E; W)$ converges to $\delta_{Z'}(E, \Delta_E; W) \ge \lambda > \lambda'$. Thus there exists an integer m_0 such that, for any $m \ge m_0$ and any m-basis type divisor D of $W_\bullet$, the pair $(E, \Delta_E + \lambda' D)$ is lc in a neighborhood of z.

Fix $m \ge m_0$. By [AZ22, Lemma 3.1], for any $\vec{a}$ with $a_1 = m$ there exists a basis $(s_{\vec{a},i})_i$ of $V_{\vec{a}}$ compatible with both ord_E and ord_F. Let D_m be the m-basis type divisor of $V_\bullet$ defined by the bases $(s_{\vec{a},i})_i$. Then $\operatorname{mult}_E D_m = S_m(V; E)$, $\operatorname{mult}_F D_m = S_m(V; F)$, and

$$\pi^* D_m = S_m(V; E)E + \Gamma_m,$$

where $\Gamma_m \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y which does not contain E. For a basis element $s \in V_{\vec{a}}$ with $\operatorname{ord}_E(s) = j$, the restriction of $\pi^* \operatorname{div}(s) - jE$ to E is the divisor of the image of s in $W_{\vec{a},j}$. Since the bases are compatible with ord_E, the images of the basis elements s with $\operatorname{ord}_E(s) = j$ form a basis of $W_{\vec{a},j}$. Hence $\Gamma_m|_E$ is an m-basis type divisor of $W_\bullet$, and $(E, \Delta_E + \lambda'\Gamma_m|_E)$ is lc in a neighborhood of z. Since Γ_m is $\mathbb{Q}$-Cartier, $\operatorname{Diff}_E(\Delta_Y + \lambda'\Gamma_m) = \Delta_E + \lambda'\Gamma_m|_E$. Let $Y^\circ \subset Y$ be an open neighborhood of z such that $(E \cap Y^\circ, \Delta_E + \lambda'\Gamma_m|_E)$ is lc. By inversion of adjunction [Kaw07, Theorem] applied to Y°, the pair $(Y, \Delta_Y + E + \lambda'\Gamma_m)$ is lc in a neighborhood of z. Since $z \in \operatorname{center}_Y F$, we obtain

$$a(F, Y, \Delta_Y + E + \lambda'\Gamma_m) \ge 0.$$

On the other hand, $K_Y + \Delta_Y + E = \pi^*(K_X + \Delta) + A_{X,\Delta}(E)E$, so

$$K_Y + \Delta_Y + E + \lambda'\Gamma_m = \pi^*(K_X + \Delta + \lambda' D_m) + \big(A_{X,\Delta}(E) - \lambda' S_m(V;E)\big)E.$$

Therefore

$$\begin{aligned} 0 &\le a(F, X, \Delta + \lambda' D_m) - \big(A_{X,\Delta}(E) - \lambda' S_m(V;E)\big)\operatorname{mult}_F E \\ &= A_{X,\Delta}(F) - \lambda' S_m(V;F) - \big(A_{X,\Delta}(E) - \lambda' S_m(V;E)\big)\operatorname{mult}_F E. \end{aligned}$$

By [AZ22, Lemma 2.9], $S_m(V;F) \to S(V;F)$ and $S_m(V;E) \to S(V;E)$ as $m \to +\infty$. Letting $m \to +\infty$ and then $\lambda' \to \lambda$, we obtain the inequality of the lemma. The second conclusion follows, because $S(V;F) = S_X(F)$ and $S(V;E) = S_X(E)$ for the complete series of $-K_X$ (Definition 2.8). □

Lemma 3.1 is a variant of [AZ22, (3.6)], proved by the argument for [AZ22, Theorem 3.2]. In [AZ22, (3.6)], λ is the right-hand side of [AZ22, (3.2)], so $\lambda \le A_{X,\Delta}(E)/S(V;E)$, whereas in Lemma 3.1, λ may exceed $A_{X,\Delta}(E)/S(V;E)$ and the assumption concerns only the prime divisors v over E with $\operatorname{center}_E v \cap G \ne \emptyset$. As in [AZ22, (3.6)], and unlike the statement of [AZ22, Theorem 3.2], the term $\big(A_{X,\Delta}(E) - \lambda S(V;E)\big)\operatorname{mult}_F E$ is kept. With $\lambda = 1$, it gives $A_{X,\Delta}(F) > S(V;F)$ whenever $A_{X,\Delta}(E) > S(V;E)$ and $\operatorname{mult}_F E > 0$.

At a point not on any line and at a point on exactly one line, the spaces $W_{m,j}$ of the refinement by the exceptional divisor of the blow-up of the point are known only through their dimensions, which come from the projection from the point or from the link of the line. Lemma 3.2 bounds the total weight of any divisorial valuation on a subspace of plane forms in terms of the dimension of the subspace alone, and depends on the concavity of the square root of the volume [BJ20, Proposition 2.3]. We do not know a reference for Lemma 3.2.

Lemma 3.2. *Let v be a divisorial valuation over $\mathbb{P}^2$ and $a := A_{\mathbb{P}^2}(v)$, and let $S_{\mathcal{O}(1)}(v) := S(V;v)$, where $V_\bullet$ is the complete series of $\mathcal{O}_{\mathbb{P}^2}(1)$. For $j \ge 0$ and a linear subspace $U \subset H^0(\mathbb{P}^2, \mathcal{O}(j))$ let $w_v(U)$ be the sum of the jumping numbers of the filtration induced by v on U, and for $j \ge 1$ let $h_j := \dim H^0(\mathbb{P}^2, \mathcal{O}(j)) = \binom{j+2}{2}$. Then the following hold.*

(1) $S_{\mathcal{O}(1)}(v) \le a/3$, and for any $\rho \in [0,1]$ the sum of the largest $\lfloor \rho h_j \rfloor$ jumping numbers of v on $H^0(\mathbb{P}^2, \mathcal{O}(j))$ is at most $jh_j S_{\mathcal{O}(1)}(v)(3\rho - 2\rho^{3/2}) + o(j^3)$, uniformly in ρ.

(2) For any r with $0 \le r \le h_j$ and any subspace $U \subset H^0(\mathbb{P}^2, \mathcal{O}(j))$ of dimension r,

$$w_v(U) \le a\Big(jr - \frac{2\sqrt{2}}{3} r^{3/2}\Big) + o(j^3),$$

where the error term depends on v but not on r or U.

Proof. Let $V_j := H^0(\mathbb{P}^2, \mathcal{O}(j))$, and for $t \ge 0$ let $\mathcal{F}^t V_j := \{s \in V_j : v(s) \ge t\}$. Let $\tau_{j,1} \ge \cdots \ge \tau_{j,h_j} \ge 0$ be the jumping numbers of v on V_j, so that $\dim \mathcal{F}^t V_j = \#\{i : \tau_{j,i} \ge t\}$. Let

$$\nu_j := \frac{1}{h_j} \sum_{i=1}^{h_j} \delta_{\tau_{j,i}/j} \quad \text{and} \quad f_j(t) := \nu_j([t, +\infty)) = \frac{\dim \mathcal{F}^{jt} V_j}{h_j}.$$

Write $v = c \operatorname{ord}_F$, where $c > 0$ and F is a prime divisor on a smooth projective surface Z with a birational morphism $\mu : Z \to \mathbb{P}^2$. Let A_Z be an ample divisor on Z and $\ell \subset \mathbb{P}^2$ a line. For a non-zero $s \in V_j$, we have $\mu^* \operatorname{div}(s) - \operatorname{ord}_F(s) F \ge 0$, so $j(\mu^*\ell \cdot A_Z) - \operatorname{ord}_F(s)(F \cdot A_Z) \ge 0$. Thus $v(s) \le T' j$, where $T' := c(\mu^*\ell \cdot A_Z)/(F \cdot A_Z)$, and all ν_j are supported in $[0, T']$.

For $t \ge 0$ let $V_\bullet^t$ be the graded linear series with $V_j^t := \mathcal{F}^{jt} V_j$, and let $T := \sup\{t \ge 0 : \operatorname{vol}(V_\bullet^t) > 0\}$. Since $\mathcal{F}^{jt} V_j = 0$ for $t > T'$, we have $T \le T'$. By [BC11, Theorem 1.11] (see also [BJ20, Theorem 2.8]) and [BJ20, Theorem 3.3], ν_j converges weakly to a probability measure ν on $[0, T']$, and $f(t) := \nu([t, +\infty))$ satisfies

$$f(t) = \lim_j f_j(t) = \operatorname{vol}(V_\bullet^t)/\operatorname{vol}(\mathcal{O}_{\mathbb{P}^2}(1)) = \operatorname{vol}(V_\bullet^t) \quad \text{for } 0 \le t < T.$$

By [BJ20, Proposition 2.3], the function $t \to \mathrm{vol}(V_\bullet^t)^{1/2}$ is concave on $[0,T)$. Thus $f^{1/2}$ is non-increasing and concave on $[0,T)$, and $f(0)=1$. Being concave and non-increasing with value 1 at 0, the function $f^{1/2}$ is continuous on $[0,T)$. Let $\varphi : [0,T] \to [0,1]$ be the continuous extension of $f^{1/2}|_{[0,T)}$. Then φ is non-increasing and concave, and $\varphi^2 = f$ almost everywhere on $[0,+\infty)$, where $\varphi := 0$ on $(T,+\infty)$. Since the supports are bounded, $S_{\mathcal{O}(1)}(v) = \lim_j \int x d\nu_j = \int x d\nu$, and

$$S_0 := S_{\mathcal{O}(1)}(v) = \int_0^T f(t)dt.$$

For $s \in [0,1]$ let $g(s) := \max\{t \in [0,T] : \varphi(t) \geq s\}$. Since φ is continuous and non-increasing, $\{t \in [0,T] : \varphi(t) \geq s\} = [0, g(s)]$ and $\varphi(g(s)) \geq s$. The function g is concave: for $s_1, s_2 \in [0,1]$ and $0 \leq \theta \leq 1$, the concavity of φ implies

$$\varphi\big(\theta g(s_1) + (1-\theta)g(s_2)\big) \geq \theta\varphi(g(s_1)) + (1-\theta)\varphi(g(s_2)) \geq \theta s_1 + (1-\theta)s_2,$$

hence $g(\theta s_1 + (1-\theta)s_2) \geq \theta g(s_1) + (1-\theta)g(s_2)$. Let $R(s) := g(s)/(1-s)$ for $0 \leq s < 1$. Then R is non-decreasing: if $0 \leq s < s' < 1$, then s' is a convex combination of s and 1, and the concavity of g together with $g(1) \geq 0$ implies

$$g(s') \geq \frac{1-s'}{1-s} g(s) + \frac{s'-s}{1-s} g(1) \geq \frac{1-s'}{1-s} g(s).$$

For $\rho \in [0,1]$ let $C(\rho) := \int_0^{T'} \min\{\rho, f(t)\}dt$. Since the Lebesgue measure of $\{t \in [0,T] : \varphi(t)^2 \geq u\}$ is $g(\sqrt{u})$ for $u \in (0,1]$, Fubini's theorem implies

$$C(\rho) = \int_0^\rho g(\sqrt{u})du = \int_0^{\sqrt{\rho}} 2sg(s)ds = \int_0^{\sqrt{\rho}} R(s)\psi(s)ds, \qquad \psi(s) := 2s(1-s).$$

In particular, $S_0 = C(1) = \int_0^1 R(s)\psi(s)ds$. Let $I(\sigma) := \int_0^\sigma \psi(s)ds = \sigma^2 - \frac{2}{3}\sigma^3$. For $0 < \sigma < 1$, since R is non-decreasing, we have

$$\int_0^\sigma R\psi ds \leq R(\sigma)I(\sigma) \quad \text{and} \quad \int_\sigma^1 R\psi ds \geq R(\sigma)\big(I(1) - I(\sigma)\big),$$

hence

$$I(1)\int_0^\sigma R\psi ds \leq I(\sigma)\int_0^\sigma R\psi ds + R(\sigma)I(\sigma)\big(I(1)-I(\sigma)\big) \leq I(\sigma)\int_0^1 R\psi ds.$$

Taking $\sigma = \sqrt{\rho}$ and using $I(1) = 1/3$, we obtain

$$C(\rho) \leq S_0 \frac{I(\sqrt{\rho})}{I(1)} = S_0\big(3\rho - 2\rho^{3/2}\big) \quad \text{for any } \rho \in [0,1].$$

We bound S_0. By [BJ20, Corollary 7.16], formula (7.3), $\delta(\mathbb{P}^2) = 1$, because the barycenter of the polytope of $-K_{\mathbb{P}^2}$ is the origin. By [BJ20, Theorem C], $\delta(\mathbb{P}^2)$ is the infimum of $A_{\mathbb{P}^2}(v')/S_{\mathbb{P}^2}(v')$ over divisorial valuations v' over $\mathbb{P}^2$. Thus $S_{\mathbb{P}^2}(v) \leq a$. Since $-K_{\mathbb{P}^2} \sim \mathcal{O}(3)$, $S_{\mathbb{P}^2}(v) = 3S_0$, hence $S_0 \leq a/3$.

We pass to finite levels. For $k \leq h_j$, the sum of the largest k jumping numbers of v on V_j is

$$\sum_{i=1}^k \tau_{j,i} = \int_0^{+\infty} \min\{k, \dim \mathcal{F}^t V_j\}dt = jh_j \int_0^{T'} \min\Big\{\frac{k}{h_j}, f_j(t)\Big\}dt.$$

Let $C_j(\rho) := \int_0^{T'} \min\{\rho, f_j(t)\}dt$. By weak convergence, $f_j(t) \to f(t)$ for any t with $\nu(\{t\}) = 0$, that is, for all t outside a countable set. By dominated convergence, $C_j(\rho) \to C(\rho)$ for any $\rho \in [0,1]$. Each function $\rho \to C_j(\rho)$ is T'-Lipschitz, so the convergence is uniform on $[0,1]$. Moreover, $|\min\{\lfloor \rho h_j \rfloor / h_j, y\} - \min\{\rho, y\}| \leq 1/h_j$ for any y. Therefore, uniformly in $\rho \in [0,1]$,

$$\sum_{i=1}^{\lfloor \rho h_j \rfloor} \tau_{j,i} = jh_j C(\rho) + o(jh_j) \leq jh_j S_0\big(3\rho - 2\rho^{3/2}\big) + o(j^3).$$

This proves (1).

For (2), let $U \subset V_j$ be a subspace with $\dim U = r$, and let $\mathcal{F}^t U := U \cap \mathcal{F}^t V_j$. For any $t \geq 0$,

$$\dim \mathcal{F}^t U \leq \min\{r, \dim \mathcal{F}^t V_j\}.$$

Integrating over $t \in [0, +\infty)$, we obtain that $w_v(U)$ is at most the sum of the largest r jumping numbers of v on V_j. Since $3\rho - 2\rho^{3/2} \geq 0$ on $[0,1]$ and $S_0 \leq a/3$, (1) with $\rho = r/h_j$ implies

$$w_v(U) \leq ajh_j\Big(\frac{r}{h_j} - \frac{2}{3}\Big(\frac{r}{h_j}\Big)^{3/2}\Big) + o(j^3) = a\Big(jr - \frac{2}{3}\cdot\frac{j}{\sqrt{h_j}}r^{3/2}\Big) + o(j^3),$$

where the error term depends only on v. Since $h_j = (j+1)(j+2)/2$, we have $\sqrt{2} - 2\sqrt{2}/(j+2) \leq j/\sqrt{h_j} \leq \sqrt{2}$. As $r^{3/2} \leq h_j^{3/2} = O(j^3)$, replacing $j/\sqrt{h_j}$ by $\sqrt{2}$ changes the right-hand side by $O(j^2)$, uniformly in r and U. This proves (2). □

When the base conditions only show $W_{\vec{a},j} \subset U$ for a space U that is too large, Lemma 3.3, applied with $U' = W_{\vec{a},j}$, bounds the first c steps of the filtration of $W_{\vec{a},j}$ by $\dim W_{\vec{a},j}$ and the remaining steps by those of U. Lemma 3.3 is used in Theorem 6.5 and, through Lemma 8.5(2), in Proposition 8.8.

Lemma 3.3. *Let $U' \subset U$ be vector spaces with $\dim U < +\infty$, let $(\mathcal{F}^h U)_{h\geq 0}$ be a decreasing filtration of U by subspaces with $\mathcal{F}^0 U = U$ and $\mathcal{F}^h U = 0$ for $h \gg 0$, and let $\mathcal{F}^h U' := U' \cap \mathcal{F}^h U$. Then for any integer $c \geq 0$,*

$$\sum_{h\geq 1} \dim \mathcal{F}^h U' \leq c \dim U' + \sum_{h>c} \dim \mathcal{F}^h U.$$

Proof. Both sides are finite sums. For $1 \leq h \leq c$ we have $\dim \mathcal{F}^h U' \leq \dim U'$, and for $h > c$ we have $\dim \mathcal{F}^h U' \leq \dim \mathcal{F}^h U$. Summing over $h \geq 1$ implies the inequality. □

The next lemma computes expected vanishing orders of series spanned by monomials. The proof uses a theorem of Khovanskii on finitely generated semigroups [KK12, Part I, Section 1], recalled in the second paragraph of the proof.

Lemma 3.4. *Let $\Gamma \subset \mathbb{Z}_{\geq 0} \times \mathbb{Z}^n$ be a finitely generated semigroup with $\Gamma \cap (\{0\} \times \mathbb{Z}^n) = \{0\}$, let $\Gamma_m := \{u \in \mathbb{Z}^n : (m,u) \in \Gamma\}$, and assume that Γ generates $\mathbb{Z}^{n+1}$ as a group. Let $\Delta \subset \mathbb{R}^n$ be the closure of $\bigcup_{m\geq 1} m^{-1}\Gamma_m$. Then Δ is a convex polytope, $\#\Gamma_m = \mathrm{vol}(\Delta)m^n + O(m^{n-1})$, and for any function $\phi : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ which is Lipschitz and positively homogeneous of degree one, for instance a continuous piecewise linear function which is positively homogeneous of degree one,*

$$\sum_{u\in\Gamma_m} \phi(m,u) = m^{n+1}\int_\Delta \phi(1,x)dx + O(m^n).$$

Moreover, if Γ is generated by $(m_1,x_1),\dots,(m_r,x_r)$ with $m_1,\dots,m_r \geq 1$, then Δ is the convex hull of $x_1/m_1,\dots,x_r/m_r$.

Proof. Let $(m_1,u_1),\dots,(m_k,u_k)$ be the non-zero elements of a finite set of generators of Γ. Since $\Gamma \cap (\{0\} \times \mathbb{Z}^n) = \{0\}$, we have $m_i \geq 1$ for any i. Let $\Sigma \subset \mathbb{R}^{n+1}$ be the closed convex cone generated by Γ, and let $\Delta' := \mathrm{Conv}(u_1/m_1,\dots,u_k/m_k)$. A point $(s,u) = \sum_i c_i(m_i,u_i)$ with $c_i \geq 0$ and $s > 0$ satisfies $u/s = \sum_i (c_i m_i/s)(u_i/m_i)$ with $\sum_i c_i m_i/s = 1$. Thus

$$\Sigma \cap (\{s\} \times \mathbb{R}^n) = \{s\} \times s\Delta' \quad \text{for any } s > 0.$$

In particular, $m^{-1}\Gamma_m \subset \Delta'$ for any $m \geq 1$, so $\Delta \subset \Delta'$. Conversely, let $p_1,\dots,p_k$ and M be positive integers with $\sum_i p_i = M$, and let $N := \mathrm{lcm}(m_1,\dots,m_k)$. Then

$$\Big(MN, \sum_i p_i \frac{N}{m_i}u_i\Big) = \sum_i p_i \frac{N}{m_i}(m_i,u_i) \in \Gamma,$$

so $\sum_i (p_i/M)(u_i/m_i) \in (MN)^{-1}\Gamma_{MN}$. Hence all rational convex combinations of the u_i/m_i belong to $\bigcup_m m^{-1}\Gamma_m$, and $\Delta' \subset \Delta$. Thus $\Delta = \Delta'$ is a convex polytope. Since Γ generates $\mathbb{Z}^{n+1}$, the cone Σ has non-empty interior, hence so does Δ.

By a theorem of Khovanskii [KK12, Part I, Section 1], since Γ generates $\mathbb{Z}^{n+1}$, there exists $\gamma_0 = (m_0, u_0) \in \Gamma$ such that $(\gamma_0 + \Sigma) \cap \mathbb{Z}^{n+1} \subset \Gamma$. By the description of the slices of Σ, for any $m \geq m_0$ we obtain

$$\big(u_0 + (m - m_0)\Delta\big) \cap \mathbb{Z}^n \subset \Gamma_m \subset m\Delta \cap \mathbb{Z}^n.$$

For $u \in \mathbb{Z}^n$ let $Q_u := u + [0,1)^n$. Let $P \subset \mathbb{R}^n$ be a convex polytope with non-empty interior. For $y \in \mathbb{R}^n$ and $s \geq 1$, the boundary of $y + sP$ is the union of finitely many convex subsets of hyperplanes of diameter $O(s)$, so at most $O(s^{n-1})$ cubes Q_u intersect the boundary of $y + sP$, uniformly in y. A lattice point $u \in y + sP$ with $Q_u \not\subset y + sP$, and a cube Q_u which intersects $y + sP$ with $u \notin y + sP$, both determine cubes intersecting the boundary. Since

$$\#\{u : Q_u \subset y + sP\} \leq \mathrm{vol}(sP) \leq \#\{u : Q_u \cap (y + sP) \neq \emptyset\},$$

we obtain

$$\#\big((y + sP) \cap \mathbb{Z}^n\big) = \mathrm{vol}(P)s^n + O(s^{n-1}),$$

uniformly in y. Applying the estimate to $P = \Delta$ with $(y, s) = (u_0, m - m_0)$ and $(y, s) = (0, m)$, we obtain $\#\Gamma_m = \mathrm{vol}(\Delta)m^n + O(m^{n-1})$ and

$$\#\big((m\Delta \cap \mathbb{Z}^n)\backslash\Gamma_m\big) = O(m^{n-1}).$$

Let c_ϕ be a Lipschitz constant of ϕ. By homogeneity $\phi(0) = 0$, so $|\phi(m, y)| \leq c_\phi \|(m, y)\| = O(m)$ for $y \in m\Delta$, since Δ is bounded. Thus

$$\sum_{u \in \Gamma_m} \phi(m, u) = \sum_{u \in m\Delta \cap \mathbb{Z}^n} \phi(m, u) + O(m^n).$$

For any u with $Q_u \subset m\Delta$, we have $|\phi(m, u) - \int_{Q_u} \phi(m, y)dy| \leq c_\phi\sqrt{n}$, and there are at most $\mathrm{vol}(m\Delta) = O(m^n)$ such u. The remaining lattice points of $m\Delta$ and the remaining part of $m\Delta$ are contained in the $O(m^{n-1})$ cubes Q_u intersecting the boundary of $m\Delta$, on which $|\phi(m, \cdot)| = O(m)$. Hence

$$\begin{aligned}\sum_{u \in m\Delta \cap \mathbb{Z}^n} \phi(m, u) &= \int_{m\Delta} \phi(m, y)dy + O(m^n) \\ &= m^n \int_\Delta \phi(m, mx)dx + O(m^n) \\ &= m^{n+1} \int_\Delta \phi(1, x)dx + O(m^n),\end{aligned}$$

where the last equality follows from the homogeneity of ϕ. Finally, a continuous piecewise linear function has finitely many linear pieces, hence is Lipschitz.

To prove the last conclusion, assume that Γ is generated by $(m_1, x_1), \ldots, (m_r, x_r)$ with $m_1, \ldots, m_r \geq 1$. An element $u \in \Gamma_m$ is of the form $u = \sum_i n_i x_i$ with $n_i \in \mathbb{Z}_{\geq 0}$ and $\sum_i n_i m_i = m$, so $u/m = \sum_i (n_i m_i/m)(x_i/m_i)$ is a convex combination of the x_i/m_i. Thus $\Delta \subset \mathrm{Conv}(x_1/m_1, \ldots, x_r/m_r)$, since the convex hull is closed. Conversely, $x_i/m_i \in m_i^{-1}\Gamma_{m_i} \subset \Delta$ for any i, and Δ is convex, so $\mathrm{Conv}(x_1/m_1, \ldots, x_r/m_r) \subset \Delta$. □

Lemma 3.5 is used in Sections 6 and 8 for torus degenerations of curvilinear base conditions: since the family of ideals below is flat, the dimension of a space of sections vanishing along the corresponding subschemes is upper semicontinuous in the parameter [Har77, Theorem III.12.8].

Lemma 3.5. *Let $P_\lambda(b), Q_\lambda(b) \in \mathbb{C}[\lambda][b]$, where Q_λ is monic of degree $\ell \geq 1$ in b and $\deg_b P_\lambda < \ell$, and let $I_\lambda := (a - P_\lambda(b), Q_\lambda(b)) \subset \mathbb{C}[\lambda][a, b]$. Then for any integer $w \geq 1$ the monomials $a^i b^{i'}$ with $0 \leq i < w$ and $0 \leq i' < \ell(w - i)$ form a basis of $\mathbb{C}[\lambda][a, b]/I_\lambda^w$ over $\mathbb{C}[\lambda]$. In particular the family of ideals $(I_\lambda^w)_{\lambda \in \mathbb{A}^1}$ is flat over $\mathbb{A}^1$.*

Proof. Let $R := \mathbb{C}[\lambda][a,b]$ and $a' := a - P_\lambda(b)$. Then $R = \mathbb{C}[\lambda][a',b]$, $I_\lambda = (a', Q_\lambda)$, and I_λ^w is generated by $a'^k Q_\lambda^{w-k}$, $0 \le k \le w$.

We first show that the elements $a'^i b^{i'}$ with $0 \le i < w$ and $0 \le i' < \ell(w-i)$ form a basis of R/I_λ^w. Any element of R is of the form $\sum_i a'^i c_i(b)$ with $c_i \in \mathbb{C}[\lambda][b]$. The terms with $i \ge w$ belong to I_λ^w. For $i < w$, since Q_λ^{w-i} is monic of degree $\ell(w-i)$, we may write $c_i = q_i Q_\lambda^{w-i} + r_i$ with $\deg_b r_i < \ell(w-i)$, and $a'^i q_i Q_\lambda^{w-i} \in I_\lambda^w$. Thus the elements span. For linear independence, the coefficient of a'^i in an element $\sum_k a'^k Q_\lambda^{w-k} g_k$ of I_λ^w is a sum of terms divisible by Q_λ^{w-k} with $k \le i$, hence is divisible by Q_λ^{w-i}. If $\sum_{i<w} a'^i r_i(b) \in I_\lambda^w$ with $\deg_b r_i < \ell(w-i)$, then Q_λ^{w-i} divides r_i, and since Q_λ^{w-i} is monic, $r_i = 0$.

Now

$$a^i b^{i'} = (a' + P_\lambda(b))^i b^{i'} = a'^i b^{i'} + \sum_{k<i} \binom{i}{k} a'^k P_\lambda(b)^{i-k} b^{i'}.$$

If $i' < \ell(w-i)$, then for $k < i$,

$$\deg_b \left(P_\lambda^{i-k} b^{i'}\right) \le (i-k)(\ell-1) + \ell(w-i) - 1 = \ell(w-k) - (i-k) - 1 < \ell(w-k).$$

Thus $a'^k P_\lambda^{i-k} b^{i'}$ is a $\mathbb{C}[\lambda]$-linear combination of the elements $a'^k b^{i''}$ with $i'' < \ell(w-k)$. Ordering both sets by the exponent i, the transition matrix from $\{a'^i b^{i'}\}$ to $\{a^i b^{i'}\}$ is unitriangular. Hence the monomials $a^i b^{i'}$ form a basis of R/I_λ^w over $\mathbb{C}[\lambda]$. In particular R/I_λ^w is a free $\mathbb{C}[\lambda]$-module, hence flat over $\mathbb{A}^1$. □

Lemma 3.6 reduces log canonicity for all basis type divisors of a refinement, and hence an upper bound on $S(W;v)$ for all v, to log canonicity for the divisors of eigenvectors spanning the limit of the refinement under a one-parameter subgroup, through the lower semicontinuity of log canonical thresholds [BL21, Proposition 10(2)]. In applications, the limit is determined by exhibiting N_m linearly independent monomials in it, and Lemma 3.4 computes the expected vanishing orders as integrals of piecewise linear functions over a polytope.

Lemma 3.6. *Let (E, Δ_E) be a projective klt pair such that Δ_E is a $\mathbb{Q}$-divisor, and let $W_\bullet$ be a multigraded linear series on E as in Definition 2.8, associated to $\mathbb{Q}$-Cartier Weil divisors $L_1, \dots, L_r$, with numbers N_m. For each m let $\lambda_m : \mathbb{G}_m \to \mathrm{Aut}(E)$ be a one-parameter subgroup with $\lambda_m(t)_* \Delta_E = \Delta_E$ such that the reflexive sheaves $\mathcal{O}_E(a_1 L_1 + \cdots + a_r L_r)$ with $a_1 = m$ carry λ_m-linearizations, and for $\vec{a}$ with $a_1 = m$ let $\mathrm{in}_{\lambda_m} W_{\vec{a}}$ be the limit of $\lambda_m(t) \cdot W_{\vec{a}}$ as $t \to 0$ in the Grassmannian of $H^0(E, \mathcal{O}_E(a_1 L_1 + \cdots + a_r L_r))$. Call an* eigen-basis divisor *at level m a $\mathbb{Q}$-divisor*

$$\frac{1}{m N_m} \sum_{\vec{a}: a_1 = m} \sum_i \mathrm{div}(s_{\vec{a},i}),$$

where $(s_{\vec{a},i})_i$ is a basis of $\mathrm{in}_{\lambda_m} W_{\vec{a}}$ consisting of λ_m-eigenvectors for each $\vec{a}$. Then the following hold.

1. *Assume that there exist real numbers $c_m > 0$ with $\lim_m c_m = c > 0$ such that $(E, \Delta_E + c_m D)$ is log canonical for any $m \gg 0$ and any eigen-basis divisor D at level m. Then $cS(W;v) \le A_{E,\Delta_E}(v)$ for any prime divisor v over E, and $(E, \Delta_E + c_m D)$ is log canonical for any $m \gg 0$ and any m-basis type divisor D of $W_\bullet$.*
2. *Let $q \in E$ be a closed point fixed by every λ_m. Assume that there exist real numbers $c_m > 0$ with $\lim_m c_m = c > 0$ such that $(E, \Delta_E + c_m D)$ is klt at q for any $m \gg 0$ and any eigen-basis divisor D at level m. Then $cS(W;v) \le A_{E,\Delta_E}(v)$ for any prime divisor v over E with $q \in \mathrm{center}_E v$.*

Proof. For a non-zero ideal sheaf $\mathfrak{b} \subset \mathcal{O}_E$, let $\mathrm{lct}(E, \Delta_E; \mathfrak{b})$ be the infimum of $A_{E,\Delta_E}(F)/\mathrm{mult}_F \mathfrak{b}$ over the prime divisors F over E with $\mathrm{mult}_F \mathfrak{b} > 0$. If $D' \ge 0$ is a $\mathbb{Q}$-divisor on E and $M'D'$ is Cartier for a positive integer M', then $\mathrm{lct}(E, \Delta_E; \mathcal{O}_E(-M'D')) = \mathrm{lct}(E, \Delta_E; D')/M'$ (Definition 2.3).

Step 1. In this step we construct, for a fixed m and a fixed prime divisor v over E, a family of divisors on $E \times \mathbb{A}^1$ which degenerates a v-compatible m-basis type divisor of $W_{\bullet,\bullet}$ to an eigen-basis divisor.

For $\vec{a}$ with $a_1 = m$, let $L_{\vec{a}} := a_1 L_1 + \cdots + a_r L_r$, and let $H^0(E, \mathcal{O}_E(L_{\vec{a}})) = \bigoplus_{k \in \mathbb{Z}} H_{\vec{a},k}$ be the weight decomposition, where $\lambda_m(t)$ acts on sections through the λ_m-linearization, so that $\operatorname{div}(\lambda_m(t) \cdot s) = \lambda_m(t)(\operatorname{div} s)$, and $\lambda_m(t) \cdot s = t^k s$ for $s \in H_{\vec{a},k}$. For a non-zero section $s = \sum_k s_k$ with $s_k \in H_{\vec{a},k}$, let $k(s) := \min\{k : s_k \neq 0\}$ and $\bar{s} := s_{k(s)}$. For $k \in \mathbb{Z}$, let $\mathcal{G}^k W_{\vec{a}} \subset W_{\vec{a}}$ be the subspace of sections s with $s = 0$ or $k(s) \geq k$. By [AZ22, Lemma 3.1], there exists a basis $(s_{\vec{a},i})_i$ of $W_{\vec{a}}$ compatible with both the filtration of $W_{\vec{a}}$ induced by v and the filtration $\mathcal{G}^\bullet W_{\vec{a}}$.

We claim that $(\bar{s}_{\vec{a},i})_i$ is a basis of $\operatorname{in}_{\lambda_m} W_{\vec{a}}$ consisting of λ_m-eigenvectors. Suppose that $\sum_i e_i \bar{s}_{\vec{a},i} = 0$ for complex numbers e_i, not all zero. Let k be the minimum of $k(s_{\vec{a},i})$ over i with $e_i \neq 0$. Since $\bar{s}_{\vec{a},i} \in H_{\vec{a},k(s_{\vec{a},i})}$, we have $\sum_{i:k(s_{\vec{a},i})=k} e_i \bar{s}_{\vec{a},i} = 0$, so $s := \sum_{i:k(s_{\vec{a},i})=k} e_i s_{\vec{a},i}$ belongs to $\mathcal{G}^{k+1} W_{\vec{a}}$. Since the basis is compatible with $\mathcal{G}^\bullet W_{\vec{a}}$, the elements $s_{\vec{a},i}$ with $k(s_{\vec{a},i}) \geq k+1$ form a basis of $\mathcal{G}^{k+1} W_{\vec{a}}$, which contradicts the linear independence of $(s_{\vec{a},i})_i$. Thus the $\bar{s}_{\vec{a},i}$ are linearly independent. For $t \neq 0$, the subspace $\lambda_m(t) \cdot W_{\vec{a}}$ has the basis

$$\sigma_{\vec{a},i}(t) := t^{-k(s_{\vec{a},i})} \lambda_m(t) \cdot s_{\vec{a},i} = \bar{s}_{\vec{a},i} + \sum_{k > k(s_{\vec{a},i})} t^{k-k(s_{\vec{a},i})} s_{\vec{a},i,k},$$

where $s_{\vec{a},i,k}$ is the weight-k component of $s_{\vec{a},i}$. As $t \to 0$, the Plücker vector of the basis $(\sigma_{\vec{a},i}(t))_i$ converges to the Plücker vector of $(\bar{s}_{\vec{a},i})_i$, which is non-zero. This proves the claim.

The right-hand side above defines $\sigma_{\vec{a},i} \in H^0(E, \mathcal{O}_E(L_{\vec{a}})) \otimes \mathbb{C}[t]$, a section of the reflexive pullback of $\mathcal{O}_E(L_{\vec{a}})$ to $E \times \mathbb{A}^1$, with $\sigma_{\vec{a},i}(0) = \bar{s}_{\vec{a},i}$ and $\sigma_{\vec{a},i}(1) = s_{\vec{a},i}$. Let

$$\mathcal{D} := \frac{1}{mN_m} \sum_{\vec{a}: a_1 = m} \sum_i \operatorname{div}(\sigma_{\vec{a},i}),$$

and let D_t be the restriction of $\mathcal{D}$ to $E \times \{t\}$, which is defined because $\sigma_{\vec{a},i}(t) \neq 0$ for any t. Then $D := D_1$ is an m-basis type divisor of $W_{\bullet,\bullet}$ compatible with v, so $v(D) = S_m(W; v)$, and D_0 is an eigen-basis divisor at level m. For $t \neq 0$, $D_t = \lambda_m(t)(D)$ is the image of D under the automorphism $\lambda_m(t)$. Since $\lambda_m(t)_* \Delta_E = \Delta_E$, the pair $(E, \Delta_E + c' D_t)$ is isomorphic to $(E, \Delta_E + c' D)$ for any $c' \geq 0$ and $t \neq 0$.

Let M be a positive integer such that $ML_1, \ldots, ML_r$ are Cartier. Then $MmN_m\mathcal{D}$ is the divisor of the section $\prod_{\vec{a},i} \sigma_{\vec{a},i}^M$ of an invertible sheaf, hence is Cartier. Let $\mathfrak{a} := \mathcal{O}_{E \times \mathbb{A}^1}(-MmN_m\mathcal{D})$. Since $\mathfrak{a}$ is invertible and $E \times \mathbb{A}^1$ is normal, $\mathcal{O}_{E \times \mathbb{A}^1}/\mathfrak{a}$ has no embedded associated points, so its associated points are the generic points of the components of $\mathcal{D}$. No component of $\mathcal{D}$ is contained in a fiber of $E \times \mathbb{A}^1 \to \mathbb{A}^1$, hence $\mathcal{O}_{E \times \mathbb{A}^1}/\mathfrak{a}$ is flat over $\mathbb{A}^1$ [Har77, Proposition III.9.7]. The restriction of $\mathfrak{a}$ to $E \times \{t\}$ is $\mathfrak{a}_t := \mathcal{O}_E(-MmN_m D_t)$. The zero set of $\mathfrak{a}$ is proper over $\mathbb{A}^1$, because E is projective. The family $(E \times \mathbb{A}^1, \Delta_E \times \mathbb{A}^1) \to \mathbb{A}^1$ is a $\mathbb{Q}$-Gorenstein flat family of klt pairs in the sense of [BL21, Definition 7].

Step 2. In this step we prove (1).

Let m_0 be an integer such that the assumption of (1) holds for any $m \geq m_0$, and fix $m \geq m_0$ and a prime divisor v over E. Let $\mathcal{D}$, D_t, and $\mathfrak{a}$ be as in **Step 1**. By [BL21, Proposition 10(2)], the function $t \to \operatorname{lct}(E, \Delta_E; \mathfrak{a}_t)$ is lower semicontinuous on $\mathbb{A}^1$ with respect to the Zariski topology. By **Step 1**, the function is constant on $\mathbb{A}^1 \backslash \{0\}$. For any $\epsilon > 0$, the set of t with $\operatorname{lct}(E, \Delta_E; \mathfrak{a}_t) > \operatorname{lct}(E, \Delta_E; \mathfrak{a}_0) - \epsilon$ is open and contains 0, hence contains a non-zero t. Thus

$$\operatorname{lct}(E, \Delta_E; D) = MmN_m \operatorname{lct}(E, \Delta_E; \mathfrak{a}_1) \geq MmN_m \operatorname{lct}(E, \Delta_E; \mathfrak{a}_0) = \operatorname{lct}(E, \Delta_E; D_0) \geq c_m,$$

where the last inequality is the assumption of (1). Hence $(E, \Delta_E + c_m D)$ is lc, and

$$c_m S_m(W; v) = c_m v(D) \leq A_{E, \Delta_E}(v).$$

Since m_0 does not depend on v, the inequality holds for any $m \geq m_0$ and any prime divisor v over E. By [AZ22, Lemma 2.9], $S_m(W; v) \to S(W; v)$, so $cS(W; v) \leq A_{E, \Delta_E}(v)$.

Let D' be any m-basis type divisor of $W_{\bullet}$ with $m \geq m_0$, defined by bases $(s'_{\vec{a},i})_i$. For any prime divisor v over E and any $\vec{a}$, if $v(s'_{\vec{a},1}) \geq v(s'_{\vec{a},2}) \geq \dots$, then $s'_{\vec{a},1}, \dots, s'_{\vec{a},i}$ span an i-dimensional subspace of $\{s \in W_{\vec{a}} : v(s) \geq v(s'_{\vec{a},i})\}$, so $v(s'_{\vec{a},i})$ is at most the i-th largest jumping number of v on $W_{\vec{a}}$. Thus $v(D') \leq S_m(W; v)$, and

$$a(v, E, \Delta_E + c_m D') = A_{E,\Delta_E}(v) - c_m v(D') \geq A_{E,\Delta_E}(v) - c_m S_m(W; v) \geq 0.$$

Hence $(E, \Delta_E + c_m D')$ is lc.

Step 3. In this step we prove (2).

We say that $(E, \Delta_E + c'D')$ is lc at q if $a(F, E, \Delta_E + c'D') \geq 0$ for any prime divisor F over E with $q \in \operatorname{center}_E F$. Let $\mathfrak{m}_q$ be the ideal sheaf of q. We first show that, for a $\mathbb{Q}$-divisor $D' \geq 0$ on E with $M'D'$ Cartier and a real number $c' > 0$, the pair $(E, \Delta_E + c'D')$ is lc at q if and only if

$$\operatorname{lct}\left(E, \Delta_E; \mathcal{O}_E(-M'D') + \mathfrak{m}_q^{\kappa}\right) \geq c'/M' \quad \text{for any integer } \kappa \geq 1.$$

Let F be a prime divisor over E. If $q \notin \operatorname{center}_E F$, then $\operatorname{mult}_F\left(\mathcal{O}_E(-M'D') + \mathfrak{m}_q^{\kappa}\right) = 0$. If $q \in \operatorname{center}_E F$, then $\operatorname{mult}_F \mathfrak{m}_q > 0$ and

$$\operatorname{mult}_F\left(\mathcal{O}_E(-M'D') + \mathfrak{m}_q^{\kappa}\right) = \min\{M' \operatorname{mult}_F D', \kappa \operatorname{mult}_F \mathfrak{m}_q\}.$$

If $(E, \Delta_E + c'D')$ is lc at q, then

$$A_{E,\Delta_E}(F) \geq c' \operatorname{mult}_F D' \geq (c'/M') \min\{M' \operatorname{mult}_F D', \kappa \operatorname{mult}_F \mathfrak{m}_q\}$$

for any such F and κ. Conversely, for F with $q \in \operatorname{center}_E F$, choose κ with $\kappa \operatorname{mult}_F \mathfrak{m}_q \geq M' \operatorname{mult}_F D'$. Then the inequality on the threshold implies $A_{E,\Delta_E}(F) \geq c' \operatorname{mult}_F D'$.

Let m_0 be an integer such that the assumption of (2) holds for any $m \geq m_0$, and fix $m \geq m_0$ and a prime divisor v over E with $q \in \operatorname{center}_E v$. Let $\mathcal{D}$, D_t, and $\mathfrak{a}$ be as in **Step 1**, and for $\kappa \geq 1$ let

$$\mathfrak{b}^{(\kappa)} := \mathfrak{a} + I_{q\times\mathbb{A}^1}^{\kappa} \subset \mathcal{O}_{E\times\mathbb{A}^1},$$

where $I_{q\times\mathbb{A}^1}$ is the ideal sheaf of $q \times \mathbb{A}^1$. The zero set of $\mathfrak{b}^{(\kappa)}$ is contained in $q \times \mathbb{A}^1$, hence is proper over $\mathbb{A}^1$, and the restriction of $\mathfrak{b}^{(\kappa)}$ to $E \times \{t\}$ is $\mathfrak{a}_t + \mathfrak{m}_q^{\kappa}$. Since $\lambda_m(t)$ fixes q, it maps $\mathfrak{a}_1 + \mathfrak{m}_q^{\kappa}$ to $\mathfrak{a}_t + \mathfrak{m}_q^{\kappa}$ for $t \neq 0$, so $t \to \operatorname{lct}(E, \Delta_E; \mathfrak{a}_t + \mathfrak{m}_q^{\kappa})$ is constant on $\mathbb{A}^1\backslash\{0\}$. The pair $(E, \Delta_E + c_m D_0)$ is klt at q by assumption, hence lc at q, so

$$\operatorname{lct}(E, \Delta_E; \mathfrak{a}_0 + \mathfrak{m}_q^{\kappa}) \geq \frac{c_m}{MmN_m} \quad \text{for any } \kappa \geq 1.$$

By [BL21, Proposition 10(2)] applied to $\mathfrak{b}^{(\kappa)}$, the argument of **Step 2** implies

$$\operatorname{lct}(E, \Delta_E; \mathfrak{a}_1 + \mathfrak{m}_q^{\kappa}) \geq \frac{c_m}{MmN_m} \quad \text{for any } \kappa \geq 1.$$

Thus $(E, \Delta_E + c_m D)$ is lc at q. Since $q \in \operatorname{center}_E v$,

$$c_m S_m(W; v) = c_m v(D) \leq A_{E,\Delta_E}(v).$$

Letting $m \to +\infty$ and using $S_m(W; v) \to S(W; v)$ [AZ22, Lemma 2.9], we obtain

$$cS(W; v) \leq A_{E,\Delta_E}(v). \quad \square$$

In Lemma 3.6, the valuation v is not assumed to be invariant under λ_m, and the one-parameter subgroup λ_m may depend on m.

4. Points and curves in genus ten and twelve

In this section, we prove the estimates for the centers whose proofs apply without change to smooth prime Fano threefolds of genus ten and of genus twelve. From this section on, we prove $A_X(F) \geq S_X(F)$ according to the center $\operatorname{center}_X F$, and the proofs of Theorems 1.2 and 17.1 are organized by the center of F. Here we treat prime divisors on X, the scheme of lines through a closed point, which divides the closed points of X into seven types, the points on no line, and the curves of degree at least two in genus ten and at least three in genus twelve, and we compute the volume function of the blow-up of a point through which only non-special lines pass (Proposition 4.5), which is used in Sections 6 and 13.

Proposition 4.1. *Let X be a smooth prime Fano threefold of genus $g \in \{10, 12\}$ and $x \in X$ a closed point. Then $\Sigma(X, x)$ is finite of length at most three, and for any line $L \subset X$ with $x \in L$, the Zariski tangent space of $\Sigma(X, x)$ at $[L]$ is $H^0(L, N_{L/X}(-x))$. Each local ring of $\Sigma(X, x)$ is isomorphic to $\mathbb{C}$, $\mathbb{C}[t]/(t^2)$, or $\mathbb{C}[t]/(t^3)$. A local ring isomorphic to $\mathbb{C}$ corresponds to a non-special line, and a local ring of length two or three corresponds to a special line. Thus the incidence type of x is one of $\emptyset$, 1, $1+1$, $1+1+1$, 2, $2+1$, and 3. Moreover, for the embedding $\Sigma(X, x) \subset \mathbb{P}(T_x X)$ given by the tangent directions, no subscheme of $\Sigma(X, x)$ of length three is contained in a line.*

Proof. Since $g \geq 7$, [KP23, Corollary A.6] implies that $\Sigma(X, x)$ is finite of length at most three.

Let $L \subset X$ be a line with $x \in L$, and let R be the local ring of $\Sigma(X, x)$ at the point $[L]$, with maximal ideal $\mathfrak{m}$. Then R is an Artinian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$. By Lemma 2.12(3), $L \cong \mathbb{P}^1$. By [Kol96, Theorem I.2.8] and its proof, the Zariski tangent space of $\Sigma(X)$ at $[L]$ is $H^0(L, N_{L/X})$: a first-order deformation $L_\epsilon \subset X \times \operatorname{Spec} \mathbb{C}[\epsilon]$ of L corresponds to a homomorphism $f : I_L/I_L^2 \to \mathcal{O}_L$, and L_ϵ is defined by the functions $\phi + \epsilon \tilde{f}(\phi)$, where ϕ runs through the local sections of I_L and $\tilde{f}(\phi)$ is a local section of $\mathcal{O}_X$ lifting $f(\phi)$. The deformation L_ϵ contains $x \times \operatorname{Spec} \mathbb{C}[\epsilon]$ if and only if f vanishes at x. Since $N_{L/X}$ is locally free, the sections of $N_{L/X}$ vanishing at x form $H^0(L, N_{L/X}(-x))$. Since $\Sigma(X, x)$ is the closed subscheme of $\Sigma(X)$ of lines containing x, we obtain

$$(\mathfrak{m}/\mathfrak{m}^2)^\vee \cong H^0(L, N_{L/X}(-x)),$$

see also [Kol96, Theorem II.1.7]. By Lemma 2.12(3),

$$h^0(L, N_{L/X}(-x)) = h^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)\big) = 0$$

if L is non-special, and

$$h^0(L, N_{L/X}(-x)) = h^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-3)\big) = 1$$

if L is special.

If $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 = 0$, then $\mathfrak{m} = \mathfrak{m}^2$, so $\mathfrak{m} = 0$ by Nakayama's lemma and $R \cong \mathbb{C}$. If $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 = 1$, pick $t \in \mathfrak{m}$ whose class generates $\mathfrak{m}/\mathfrak{m}^2$. Nakayama's lemma implies $\mathfrak{m} = (t)$, hence $R \cong \mathbb{C}[t]/(t^k)$, where k is the length of R. Since $\mathfrak{m} \neq 0$ and the length of $\Sigma(X, x)$ is at most three, $k \in \{2, 3\}$. Thus a local ring isomorphic to $\mathbb{C}$ corresponds to a non-special line, and a local ring of length two or three corresponds to a special line. Since the lengths of the local rings of $\Sigma(X, x)$ are positive integers with sum at most three, the incidence type of x is a partition of 0, 1, 2, or 3, that is, one of the seven listed partitions.

It remains to show that no subscheme of $\Sigma(X, x)$ of length three is contained in a line. By Lemma 2.12(1), we regard X as a closed subscheme of $\mathbb{P}^{g+1}$, and there exist quadratic forms $Q_1, \dots, Q_n$ on $\mathbb{C}^{g+2}$ such that $X = \bigcap_i \{Q_i = 0\}$ as schemes. Let $\hat{x} \in \mathbb{C}^{g+2}$ be a non-zero vector representing x, and let B_i be the symmetric bilinear form with $B_i(w, w) = Q_i(w)$. Since X is cut out by the Q_i as a scheme, the linear subspace

$$\hat{T} := \{w \in \mathbb{C}^{g+2} : B_i(\hat{x}, w) = 0 \text{ for } 1 \leq i \leq n\}$$

is the affine cone over the embedded tangent space of X at x, and $\mathbb{P}(T_xX)$ is the projective space of one-dimensional subspaces of $\hat{T}/\mathbb{C}\hat{x}$. By [KP23, Lemma 5.6], there is a natural embedding $\Sigma(X,x) \subset \mathbb{P}(T_xX)$. We describe the equations of $\Sigma(X,x)$ in $\mathbb{P}(T_xX)$. For a $\mathbb{C}$-algebra R' and $w \in R'^{g+2}$, we have

$$Q_i(s\hat{x} + s'w) = 2ss'B_i(\hat{x}, w) + s'^2Q_i(w)$$

in $R'[s,s']$. Comparing coefficients, the line through x in the direction of w is contained in $\{Q_i = 0\}$ if and only if $B_i(\hat{x},w) = 0$ and $Q_i(w) = 0$. For $w \in \hat{T}$ and $c \in \mathbb{C}$, we have $Q_i(w + c\hat{x}) = Q_i(w)$, so Q_i induces a quadratic form q_i on $\hat{T}/\mathbb{C}\hat{x}$. Thus $\Sigma(X,x)$ is the closed subscheme of $\mathbb{P}(T_xX) \cong \mathbb{P}^2$ defined by $q_1, \dots, q_n$.

Suppose that a subscheme of $\Sigma(X,x)$ of length three is contained in a line $\Lambda \subset \mathbb{P}(T_xX)$. For each i, the restriction $q_i|_\Lambda \in H^0(\Lambda, \mathcal{O}_\Lambda(2))$ vanishes on a subscheme of length three, so $q_i|_\Lambda = 0$. Thus $\Lambda \subset \Sigma(X,x)$, which contradicts the finiteness of $\Sigma(X,x)$. □

Lemma 4.2. *Let X be a smooth prime Fano threefold of genus g and $F \subset X$ a prime divisor. Then $F \sim aH$ for an integer $a \geq 1$, $A_X(F) = 1$, $S_X(F) = \frac{1}{4a}$, and $A_X(F)/S_X(F) = 4a \geq 4$.*

Proof. Since $\mathrm{Pic}(X) = \mathbb{Z}[H]$, we have $F \sim aH$ for some $a \in \mathbb{Z}$, and $a(2g-2) = (F \cdot H^2) > 0$ implies $a \geq 1$. Since X is smooth and F is a prime divisor on X, $A_X(F) = 1$. For $0 \leq t \leq 1/a$, the class $H - tF \equiv (1-at)H$ is nef, and for $t > 1/a$, $((H-tF)\cdot H^2) = (1-at)(2g-2) < 0$, so $H - tF$ is not pseudo-effective. By [BJ20, Corollary 3.9],

$$S_X(F) = \frac{1}{2g-2}\int_0^{1/a} (2g-2)(1-at)^3 dt = \frac{1}{4a}.$$

□

By Lemma 2.12(4), $\Sigma(X)$ is one-dimensional, so the points of X contained in some line form the image in X of the universal line, which is a closed subset of dimension at most two. Thus the points on no line form a dense open subset of X. In particular a general point of X is contained in no line. Among the seven incidence types of Proposition 4.1, the type $\emptyset$ is the only one whose treatment does not use lines. The Sarkisov link of [Tak89], see also [KP23], centered at a point of incidence type $\emptyset$ determines only the dimensions of the pieces of the refinement by the exceptional divisor of the blow-up of the point, and Lemma 3.2(2) together with Jensen's inequality turns the dimensions into the bound 33/32. The following lemma collects the volumes and the dimensions.

Lemma 4.3. *Let X be a smooth prime Fano threefold of genus $g \in \{10, 12\}$, let $x \in X$ be a closed point with $\Sigma(X,x) = \emptyset$, and let $\pi : Y \to X$ be the blow-up of x, with exceptional divisor E. For integers $m \geq 1$ and $j \geq 0$, let $W_{m,j}$ be the image of the restriction map $H^0(Y, m\pi^*H - jE) \to H^0(E, \mathcal{O}_E(j))$, and let $r_{m,j} := \dim W_{m,j}$. Then the following hold.*

(1) $\mathrm{vol}_Y(\pi^*H - tE) = 2g - 2 - t^3$ *for* $0 \leq t \leq 2$.

(2) If $g = 10$, *then, with* $u := t - 2$,

$$\mathrm{vol}_Y(\pi^*H - tE) = \begin{cases} 10 - 12u - 6u^2 + 8u^3, & 2 \leq t \leq 3, \\ 0, & t \geq 3. \end{cases}$$

(3) If $g = 12$, *then, with* $u := t - 2$,

$$\mathrm{vol}_Y(\pi^*H - tE) = \begin{cases} 14 - 12u - 6u^2 + 5u^3, & 2 \leq t \leq 3, \\ (10-3t)^3, & 3 \leq t \leq 10/3, \\ 0, & t \geq 10/3. \end{cases}$$

(4) $S_X(E) = 2$ *if* $g = 10$, *and* $S_X(E) = 71/33$ *if* $g = 12$.

(5) $r_{m,j} = h^0(Y, m\pi^*H - jE) - h^0(Y, m\pi^*H - (j+1)E)$, *and* $\sum_{j\geq 0} r_{m,j} = N_m$.

(6) $r_{m,j} = \binom{j+2}{2}$ *for* $0 \leq j \leq 2m-1$. *Moreover,* $r_{m,j} = 0$ *for* $j > 3m$ *if* $g = 10$, *and for* $j > 10m/3$ *if* $g = 12$.

(7) As $m \to +\infty$,

$$\sum_{j=2m}^{3m-1} r_{m,j} = \begin{cases} \frac{5}{3}m^3 + o(m^3), & g = 10, \\ \frac{13}{6}m^3 + o(m^3), & g = 12, \end{cases}$$

and if $g = 12$, then $\sum_{j \geq 3m} r_{m,j} = \frac{1}{6}m^3 + O(m^2)$.

Proof. The proof is long, so we divide it into several steps. We write H for π^*H.

Step 1. In this step we recall the Sarkisov link centered at x and compare intersection numbers on Y and on the flop of Y.

Since $\mathrm{Pic}(X) = \mathbb{Z}[K_X]$ and $g \in \{10, 12\}$ (Definition 2.11), X satisfies the standing assumptions (5.0.1) and (5.1.3) of [KP23, Section 5]: condition (5.0.1) is that $\mathrm{Pic}(X) = \mathbb{Z} \cdot K_X$ and $g \in \{7, 9, 10, 12\}$, and condition (5.1.3) is that $\mathrm{Pic}(X) = \mathbb{Z} \cdot K_X$ and $g \geq 6$. Since $K_Y = \pi^*K_X + 2E$ [KM98, Lemma 2.29], we have $-K_Y = H - 2E$. Since $E \cong \mathbb{P}^2$ and $\mathcal{O}_E(E) \cong \mathcal{O}_E(-1)$,

$$H^3 = 2g - 2, \qquad H^2 \cdot E = H \cdot E^2 = 0, \qquad E^3 = 1,$$

hence

$$\begin{aligned} &(-K_Y)^3 = 2g - 10, \quad (-K_Y)^2 \cdot E = 4, \quad (-K_Y) \cdot E^2 = -2, \\ &(-K_Y)^2 \cdot H = (-K_Y) \cdot H^2 = 2g - 2, \quad (-K_Y) \cdot H \cdot E = 0. \end{aligned} \tag{4.1}$$

By [KP23, Lemma 5.7], $|-K_Y|$ is base point free. Since $(-K_Y)^3 = 2g - 10 > 0$, the divisor $-K_Y$ is nef and big.

By [KP23, Definition 5.1, Remark 5.2, Lemma 5.7, and Theorem 5.17(iii)(iv)], there exists a Sarkisov link with center x, that is, a diagram

$$Y \xrightarrow{\varphi} \bar{Y} \xleftarrow{\varphi_+} Y^+ \xrightarrow{\sigma_+} X^+,$$

where $\bar{Y}$ is a normal projective variety, φ and φ_+ are small crepant contractions, $\psi := \varphi_+^{-1} \circ \varphi : Y \dashrightarrow Y^+$ is a flop, σ_+ is an extremal contraction, and Y^+ is smooth. Moreover, if $g = 10$, then $X^+ \cong \mathbb{P}^1$ and σ_+ is a del Pezzo fibration of degree six, and if $g = 12$, then $X^+ \cong \mathbb{P}^3$ and σ_+ is the blow-up of $\mathbb{P}^3$ along a smooth rational curve Γ of degree six.

Since ψ is an isomorphism in codimension one between smooth projective varieties, taking strict transforms induces an isomorphism $\mathrm{Pic}(Y) \cong \mathrm{Pic}(Y^+)$ and isomorphisms $H^0(Y, D) \cong H^0(Y^+, D^+)$ for any divisor D on Y with strict transform D^+. We denote D^+ again by D. In particular $-K_Y$ corresponds to $-K_{Y^+}$, $\mathrm{Pic}(Y^+) = \mathbb{Z}H \oplus \mathbb{Z}E$, and $\mathrm{vol}_Y(D) = \mathrm{vol}_{Y^+}(D)$ for any $\mathbb{R}$-divisor D on Y.

We show that $-K_{Y^+}$ is nef and that

$$(-K_Y \cdot D_1 \cdot D_2)_Y = (-K_{Y^+} \cdot D_1 \cdot D_2)_{Y^+} \tag{4.2}$$

for any divisors D_1, D_2 on Y. Since φ is crepant, $(-K_Y \cdot C) = 0$ for any curve C contracted by φ, and $\mathrm{Exc}(\varphi)$ is a finite union of such curves since φ is small. Since $|-K_Y|$ is base point free, a general member $S \in |-K_Y|$ contains no irreducible component C of $\mathrm{Exc}(\varphi)$, and since $(S \cdot C) = (-K_Y \cdot C) = 0$, we have $S \cap \mathrm{Exc}(\varphi) = \emptyset$. Since ψ restricts to an isomorphism $Y \setminus \mathrm{Exc}(\varphi) \to Y^+ \setminus \mathrm{Exc}(\varphi_+)$ and S is proper, $\psi(S)$ is a closed surface in $Y^+ \setminus \mathrm{Exc}(\varphi_+)$, isomorphic to S, and $\psi(S) \in |-K_{Y^+}|$. Thus

$$(-K_Y \cdot D_1 \cdot D_2)_Y = (D_1|_S \cdot D_2|_S)_S = (D_1|_{\psi(S)} \cdot D_2|_{\psi(S)})_{\psi(S)} = (-K_{Y^+} \cdot D_1 \cdot D_2)_{Y^+}.$$

Moreover, the strict transforms of the members of $|-K_Y|$ have no common point outside $\mathrm{Exc}(\varphi_+)$, so $(-K_{Y^+} \cdot C) \geq 0$ for any curve $C \not\subset \mathrm{Exc}(\varphi_+)$, and $(-K_{Y^+} \cdot C) = 0$ for any curve $C \subset \mathrm{Exc}(\varphi_+)$ since φ_+ is crepant. Thus $-K_{Y^+}$ is nef, and it is big since $\mathrm{vol}_{Y^+}(-K_{Y^+}) = \mathrm{vol}_Y(-K_Y) > 0$.

For a nef $\mathbb{R}$-divisor D on Y or on Y^+ we have $\mathrm{vol}(D) = D^3$: for nef $\mathbb{Q}$-divisors this is [Laz04a, Corollary 1.4.41], and the general case follows since a nef class is a limit of ample $\mathbb{Q}$-classes and the volume is continuous [Laz04a, Theorem 2.2.44 and Corollary 2.2.45].

Step 2. In this step we prove (1) and (2), and we show that $h^0(Y, mH - jE) = 0$ for $j > 3m$ if $g = 10$.

For $0 \le t \le 2$, the $\mathbb{R}$-divisor $H - tE = (1 - \frac{t}{2})H + \frac{t}{2}(-K_Y)$ is nef, so **Step 1** implies

$$\mathrm{vol}_Y(H - tE) = (H - tE)^3 = 2g - 2 - t^3.$$

This proves (1).

Assume that $g = 10$. By [KP23, Theorem 5.17(iii)], the map $\sigma_+ \circ \psi$ is given by the linear system $|H - 3E|$. Hence $H - 3E = \sigma_+^* \mathcal{O}_{\mathbb{P}^1}(1)$ on Y^+, a general fiber S of σ_+ is a member of $|H - 3E|$, and $H - 3E$ is nef on Y^+. The surface S is a smooth del Pezzo surface of degree six, and $\mathcal{O}_S(S) \cong \mathcal{O}_S$, so $-K_{Y^+}|_S = -K_S$. By (4.1) and (4.2), and since two general fibers of σ_+ are disjoint, we have

$$(-K_{Y^+})^3 = 10, \qquad (-K_{Y^+})^2 \cdot (H-3E) = (K_S^2)_S = 6, \qquad (-K_{Y^+}) \cdot (H-3E)^2 = (H-3E)^3 = 0.$$

For $2 \le t \le 3$ and $u = t - 2$, we have $H - tE = (1 - u)(-K_{Y^+}) + u(H - 3E)$, which is nef on Y^+. By **Step 1**,

$$\mathrm{vol}_Y(H - tE) = 10(1 - u)^3 + 18(1 - u)^2 u = 10 - 12u - 6u^2 + 8u^3.$$

At $t = 3$ the volume is 0. Since $E \ge 0$, $t \to \mathrm{vol}_Y(H - tE)$ is non-increasing, so $\mathrm{vol}_Y(H - tE) = 0$ for $t \ge 3$. This proves (2).

For integers $m \ge 1$ and $j > 3m$, we have

$$mH - jE = (3m - j)(-K_{Y^+}) + (j - 2m)(H - 3E).$$

Suppose that $s \in H^0(Y^+, mH - jE)$ is non-zero. A general fiber S of σ_+ is not contained in the divisor $\mathrm{div}(s)$, so $(\mathrm{div}(s)|_S \cdot (-K_S))_S \ge 0$. However,

$$(\mathrm{div}(s)|_S \cdot (-K_S))_S = (3m - j)(K_S^2)_S = 6(3m - j) < 0,$$

a contradiction. Thus $h^0(Y, mH - jE) = h^0(Y^+, mH - jE) = 0$.

Step 3. In this step we prove (3), and we compute $h^0(Y, mH - jE)$ for $j \ge 3m$ if $g = 12$.

Assume that $g = 12$, and write $f := \sigma_+ : Y^+ \to \mathbb{P}^3$. Let $M := f^* \mathcal{O}_{\mathbb{P}^3}(1)$ and let D be the exceptional divisor of f. Since Γ is a smooth rational curve of degree six, $\deg N_{\Gamma/\mathbb{P}^3} = 4 \cdot 6 - 2 = 22$, and [IP99, Lemma 2.2.14] implies

$$M^3 = 1, \qquad M^2 \cdot D = 0, \qquad M \cdot D^2 = -6, \qquad D^3 = -\deg N_{\Gamma/\mathbb{P}^3} = -22.$$

Since $K_{Y^+} = f^* K_{\mathbb{P}^3} + D$ [KM98, Lemma 2.29], we have $-K_{Y^+} = 4M - D$, hence

$$(-K_{Y^+})^3 = 14, \qquad (-K_{Y^+})^2 \cdot M = 10, \qquad (-K_{Y^+})^2 \cdot D = 26, \qquad (-K_{Y^+}) \cdot M^2 = 4.$$

We compute the class of E on Y^+. Since Y^+ is the blow-up of $\mathbb{P}^3$ along Γ, $\mathrm{Pic}(Y^+) = \mathbb{Z}M \oplus \mathbb{Z}D$ [Har77, Exercise II.8.5(a)]. Since $\mathrm{Pic}(Y) = \mathbb{Z}H \oplus \mathbb{Z}E$ and $-K_Y = H - 2E$, the classes $-K_{Y^+}$ and E also form a basis of $\mathrm{Pic}(Y^+)$. Write $E = \alpha M - \beta D$ with $\alpha, \beta \in \mathbb{Z}$. Since $-K_{Y^+} = 4M - D$, the change of basis has determinant ± 1, that is, $\alpha - 4\beta = \pm 1$. By (4.1) and (4.2),

$$4 = (-K_{Y^+})^2 \cdot E = 10\alpha - 26\beta.$$

Substituting $\alpha = 4\beta \pm 1$ yields $14\beta = 4 \mp 10$. Only the sign $\alpha - 4\beta = -1$ leads to an integer, namely $\beta = 1$ and $\alpha = 3$. Thus

$$E = 3M - D, \qquad M = H - 3E, \qquad D = 3H - 10E, \qquad H = 10M - 3D. \tag{4.3}$$

For $2 \le t \le 3$ and $u = t - 2$, (4.3) implies $H - tE = (1 - u)(-K_{Y^+}) + uM$, which is nef on Y^+. By **Step 1**,

$$\mathrm{vol}_Y(H - tE) = 14(1 - u)^3 + 30(1 - u)^2 u + 12(1 - u)u^2 + u^3 = 14 - 12u - 6u^2 + 5u^3.$$

For $3 \le t \le 10/3$, (4.3) implies

$$H - tE = (10 - 3t)M + (t - 3)D.$$

Since D is f-exceptional and $\mathbb{P}^3$ is normal, $f_*\mathcal{O}_{Y^+}(kD) = \mathcal{O}_{\mathbb{P}^3}$ for any integer $k \geq 0$: a rational function on an open subset $U \subset \mathbb{P}^3$ which is regular outside the codimension two subset $U \cap \Gamma$ is regular on U. By the projection formula,

$$H^0(Y^+, cM + kD) \cong H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(c)) \tag{4.4}$$

for any integers c and $k \geq 0$. For rational $t \in [3, 10/3]$, (4.4) applied to the multiples of $H - tE$ with integer coefficients implies $\mathrm{vol}_Y(H - tE) = (10 - 3t)^3$, and the continuity of the volume function [Laz04a, Corollary 2.2.45] extends the formula to real $t \in [3, 10/3]$. At $t = 10/3$ the volume is 0, and since $t \to \mathrm{vol}_Y(H - tE)$ is non-increasing, $\mathrm{vol}_Y(H - tE) = 0$ for $t \geq 10/3$. This proves (3).

For integers $m \geq 1$ and $j \geq 3m$, (4.3) and (4.4) imply

$$h^0(Y, mH - jE) = h^0\big(Y^+, (10m - 3j)M + (j - 3m)D\big) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(10m - 3j)).$$

In particular $h^0(Y, mH - 3mE) = \binom{m+3}{3}$, and $h^0(Y, mH - jE) = 0$ for $j > 10m/3$.

Step 4. In this step we prove (4).

By Definition 2.2, $S_X(E) = \frac{1}{2g-2}\int_0^{+\infty} \mathrm{vol}_Y(H - tE)dt$. If $g = 10$, then (1) and (2) imply

$$\int_0^2 (18 - t^3)dt = 32, \qquad \int_0^1 (10 - 12u - 6u^2 + 8u^3)du = 4,$$

so $S_X(E) = 36/18 = 2$. If $g = 12$, then (1) and (3) imply

$$\int_0^2 (22 - t^3)dt = 40, \qquad \int_0^1 (14 - 12u - 6u^2 + 5u^3)du = \frac{29}{4}, \qquad \int_3^{10/3} (10 - 3t)^3 dt = \frac{1}{12},$$

so

$$S_X(E) = \frac{1}{22}\Big(40 + \frac{29}{4} + \frac{1}{12}\Big) = \frac{1}{22} \cdot \frac{142}{3} = \frac{71}{33}.$$

Step 5. In this step we prove (5), (6), and (7).

Since $\mathcal{O}_E(-E) \cong \mathcal{O}_E(1)$ and $\pi^*H|_E$ is trivial, there is an exact sequence

$$0 \to \mathcal{O}_Y(mH - (j+1)E) \to \mathcal{O}_Y(mH - jE) \to \mathcal{O}_E(j) \to 0.$$

The kernel of the restriction map $H^0(Y, mH - jE) \to H^0(E, \mathcal{O}_E(j))$ is $H^0(Y, mH - (j+1)E)$, which proves the formula for $r_{m,j}$ in (5). By **Steps 2 and 3**, $h^0(Y, mH - jE) = 0$ for $j > 3m$ if $g = 10$ and for $j > 10m/3$ if $g = 12$, so the sum of the $r_{m,j}$ telescopes to $h^0(Y, mH) = h^0(X, mH) = N_m$. This proves (5).

Let $0 \leq j \leq 2m - 1$. Then

$$\big(mH - (j+1)E\big) - K_Y = (m+1)H - (j+3)E = \Big(m + 1 - \frac{j+3}{2}\Big)H + \frac{j+3}{2}(-K_Y)$$

with $0 < j + 3 \leq 2(m+1)$. Since H and $-K_Y$ are nef and big, $(m+1)H - (j+3)E$ is nef and big, and Kawamata–Viehweg vanishing [KM98, Theorem 2.64] implies $H^1(Y, mH - (j+1)E) = 0$. Thus the restriction map is surjective and $r_{m,j} = h^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(j)) = \binom{j+2}{2}$. The vanishing of $r_{m,j}$ in (6) follows from (5) and the vanishing of $h^0(Y, mH - jE)$ in **Steps 2 and 3**. This proves (6).

By (5),

$$\sum_{j=2m}^{3m-1} r_{m,j} = h^0(Y, -mK_Y) - h^0(Y, m(H - 3E)).$$

Since $-K_Y = \pi^*H - 2E$, (1) gives $\mathrm{vol}_Y(-K_Y) = 2g - 10$, so $h^0(Y, -mK_Y) = \frac{2g-10}{6}m^3 + o(m^3)$. If $g = 10$, then $\mathrm{vol}_Y(H - 3E) = 0$ by (2), so $h^0(Y, m(H - 3E)) = o(m^3)$, and the sum is $\frac{5}{3}m^3 + o(m^3)$. If $g = 12$, then $h^0(Y, m(H - 3E)) = \binom{m+3}{3} = \frac{1}{6}m^3 + O(m^2)$ by **Step 3**, and the sum is $\frac{14-1}{6}m^3 + o(m^3) = \frac{13}{6}m^3 + o(m^3)$. Finally, if $g = 12$, then (5) and **Step 3** imply $\sum_{j \geq 3m} r_{m,j} = h^0(Y, m(H - 3E)) = \frac{1}{6}m^3 + O(m^2)$. This proves (7). □

Proposition 4.4. *Let X be a smooth prime Fano threefold of genus $g \in \{10, 12\}$ and $x \in X$ a closed point with $\Sigma(X, x) = \emptyset$. Let $\pi : Y := \mathrm{Bl}_x X \to X$ be the blow-up of x, with exceptional divisor $E \cong \mathbb{P}^2$, and let $W_\bullet$ be the refinement of $(H^0(X, mH))_m$ by E (Definition 2.8). Then $S_X(E) = 2$ if $g = 10$ and $S_X(E) = 71/33$ if $g = 12$, and*

$$S(W; v) \le c_g A_{E,\Delta_E}(v)$$

for any prime divisor v over E, where $c_{10} := (126 - 10\sqrt{30})/81$ and $c_{12} := (174 - 13\sqrt{39})/99$. Both constants are smaller than $32/33$. In particular $\delta_x(X) \ge 33/32$.

Proof. The varieties Y and $E \cong \mathbb{P}^2$ are smooth, $-E$ is π-ample since $\mathcal{O}_E(-E) \cong \mathcal{O}_E(1)$, and (Y, E) is plt. Thus E is of plt type, π is the plt extraction of E, $\Delta_E = 0$, and $A_{E,\Delta_E}(v) = A_E(v)$ for any prime divisor v over E. Since $K_Y = \pi^* K_X + 2E$ [KM98, Lemma 2.29], we have $A_X(E) = 3$. Since $(m\pi^* H - jE)|_E \cong \mathcal{O}_E(j)$, Definition 2.8 implies that $W_{m,j}$ is the space $W_{m,j}$ of Lemma 4.3, and we let $r_{m,j}$ be as there. The values of $S_X(E)$ follow from Lemma 4.3(4).

Fix a prime divisor v over E, and let w_v be as in Lemma 3.2. By Lemma 3.2(2) with $U = W_{m,j}$, for $j \ge 1$,

$$w_v(W_{m,j}) \le A_{E,\Delta_E}(v)\Big(j r_{m,j} - \frac{2\sqrt{2}}{3} r_{m,j}^{3/2}\Big) + o(j^3),$$

where the error term depends on v but not on m. Moreover $w_v(W_{m,0}) = 0$, since $W_{m,0} = H^0(E, \mathcal{O}_E)$ consists of constants, and $r_{m,0} = 1$. By Lemma 4.3(6), $r_{m,j} = 0$ for $j > 3m$ if $g = 10$ and for $j > 10m/3$ if $g = 12$. In particular $r_{m,j} = 0$ for $j > 4m$ in both genera. Since the sum of the error terms over $1 \le j \le 4m$ is $o(m^4)$, and the grade $j = 0$ changes the right-hand side by $O(1)$, we obtain

$$\sum_{j \ge 0} w_v(W_{m,j}) \le A_{E,\Delta_E}(v)\Big(\sum_{j \ge 0} j r_{m,j} - \frac{2\sqrt{2}}{3} \sum_{j \ge 0} r_{m,j}^{3/2}\Big) + o(m^4). \tag{4.5}$$

We bound the sums on the right-hand side of (4.5). For the first sum, $\{s \in H^0(X, mH) : \mathrm{ord}_E(s) \ge j\} = H^0(Y, m\pi^* H - jE)$, so Lemma 4.3(5) implies that the jumping numbers of ord_E on $H^0(X, mH)$ are the integers $j \ge 0$, each counted $r_{m,j}$ times. Thus $\frac{1}{mN_m} \sum_j j r_{m,j} = S_m(V; E)$, where V is the complete series of H, and

$$\lim_{m \to +\infty} \frac{1}{mN_m} \sum_{j \ge 0} j r_{m,j} = S(V; E) = S_X(E)$$

by [AZ22, Lemma 2.9] and Definition 2.8.

For the second sum, we use Jensen's inequality for the convex function $s \to s^{3/2}$: for non-negative real numbers $\rho_1, \dots, \rho_k$,

$$\sum_{i=1}^{k} \rho_i^{3/2} \ge \frac{1}{\sqrt{k}} \Big(\sum_{i=1}^{k} \rho_i\Big)^{3/2}. \tag{4.6}$$

We split the grades into three blocks, using Lemma 4.3(6)(7).

- For $0 \le j \le 2m - 1$, $r_{m,j} = \binom{j+2}{2} \ge j^2/2$, so
$$\sum_{j=0}^{2m-1} r_{m,j}^{3/2} \ge \frac{1}{2\sqrt{2}} \sum_{j=0}^{2m-1} j^3 = \sqrt{2} m^4 + O(m^3).$$
- The block $2m \le j \le 3m - 1$ consists of m grades, so (4.6) implies
$$\sum_{j=2m}^{3m-1} r_{m,j}^{3/2} \ge \frac{1}{\sqrt{m}} \Big(\sum_{j=2m}^{3m-1} r_{m,j}\Big)^{3/2} = \begin{cases} (5/3)^{3/2} m^4 + o(m^4), & g = 10, \\ (13/6)^{3/2} m^4 + o(m^4), & g = 12. \end{cases}$$

- If $g = 12$, the block $3m \le j \le 10m/3$ consists of at most $m/3+1$ grades, so (4.6) implies

$$\sum_{j \ge 3m} r_{m,j}^{3/2} \ge \frac{1}{\sqrt{m/3+1}} \Big(\frac{1}{6} m^3 + O(m^2) \Big)^{3/2} = \frac{1}{6\sqrt{2}} m^4 + o(m^4).$$

If $g = 10$, we bound the grades $j \ge 3m$ from below by 0.

Thus

$$\liminf_{m \to +\infty} \frac{1}{m^4} \sum_{j \ge 0} r_{m,j}^{3/2} \ge \begin{cases} \sqrt{2} + (5/3)^{3/2}, & g = 10, \\ \sqrt{2} + (13/6)^{3/2} + \frac{1}{6\sqrt{2}}, & g = 12. \end{cases} \tag{4.7}$$

By Lemma 2.12(2), $mN_m = \frac{g-1}{3} m^4 + O(m^3)$. Since $S(W; v) = \lim_m S_m(W; v)$ [AZ22, Lemma 2.9], dividing (4.5) by mN_m and letting $m \to +\infty$ implies

$$S(W; v) \le A_{E, \Delta_E}(v) \Big(S_X(E) - \frac{2\sqrt{2}}{g-1} \liminf_{m \to +\infty} \frac{1}{m^4} \sum_{j \ge 0} r_{m,j}^{3/2} \Big).$$

If $g = 10$, then (4.7) implies that the factor in parentheses is at most

$$2 - \frac{2\sqrt{2}}{9} \Big(\sqrt{2} + \Big(\frac{5}{3} \Big)^{3/2} \Big) = 2 - \frac{4}{9} - \frac{10\sqrt{30}}{81} = \frac{126 - 10\sqrt{30}}{81} = c_{10}.$$

If $g = 12$, then (4.7) implies that the factor in parentheses is at most

$$\frac{71}{33} - \frac{2\sqrt{2}}{11} \Big(\sqrt{2} + \Big(\frac{13}{6} \Big)^{3/2} + \frac{1}{6\sqrt{2}} \Big) = \frac{71}{33} - \frac{4}{11} - \frac{13\sqrt{39}}{99} - \frac{1}{33} = \frac{174 - 13\sqrt{39}}{99} = c_{12}.$$

Thus $S(W; v) \le c_g A_{E, \Delta_E}(v)$. Since $\sqrt{30} > 5$ and $\sqrt{39} > 6$,

$$c_{10} < \frac{126 - 50}{81} = \frac{76}{81} < \frac{32}{33}, \qquad c_{12} < \frac{174 - 78}{99} = \frac{32}{33}.$$

Finally, by Theorem 2.9 with $Z = \{x\}$, where Z' runs through the irreducible closed subsets of E,

$$\delta_x(X) \ge \min \Big\{ \frac{A_X(E)}{S_X(E)}, \inf_{Z'} \delta_{Z'}(E, 0; W) \Big\}.$$

For any Z', $\delta_{Z'}(E, 0; W) \ge \inf_v A_E(v)/S(W; v) \ge 1/c_g > 33/32$, where v runs through the prime divisors over E. Moreover $A_X(E)/S_X(E) = 3/2$ if $g = 10$ and $A_X(E)/S_X(E) = 99/71$ if $g = 12$, and both numbers are larger than $33/32$. Thus $\delta_x(X) \ge 33/32$. □

Let x be a closed point such that $\Sigma(X, x)$ is reduced and non-empty. The strict transforms of the lines through x have negative degree against $\pi^* H - tE$ past $t = 1$, so the volume is computed on the threefold Z^+ of Proposition 4.5(3), obtained by blowing up the strict transforms of the lines and flopping the negative sections of the exceptional divisors.

Proposition 4.5. *Let X be a smooth prime Fano threefold of genus $g \in \{10, 12\}$, let $d := 2g - 2$, and let $x \in X$ be a closed point such that $\Sigma(X, x)$ is reduced of length $k \in \{1, 2, 3\}$. Let $\pi : Y := \mathrm{Bl}_x X \to X$ be the blow-up of x with exceptional divisor E, let W be the refinement of $(H^0(X, mH))_m$ by E (Definition 2.8), and let $r_{m,j} := \dim W_{m,j}$. Let $\tilde{L}_1, \dots, \tilde{L}_k \subset Y$ be the strict transforms of the k lines through x, and let $q : Z \to Y$ be the blow-up of $\tilde{L}_1 \cup \cdots \cup \tilde{L}_k$. Then the following hold.*

(1) We have

$$\mathrm{vol}_Y(\pi^* H - tE) = \begin{cases} d - t^3 & 0 \le t \le 1, \\ d - t^3 + \frac{k}{2}(t-1)^3 & 1 \le t \le 2. \end{cases}$$

(2) *$W_{m,j} = H^0(E, \mathcal{O}_E(j))$ for $0 \le j \le m-2$, and $r_{m,j} = \frac{j^2}{2} - \frac{k}{4}(j-m)^2 + O(m)$ uniformly for integers $m \le j < 2m$. In particular,*

$$\lim_{m\to+\infty} m^{-4} \sum_{0\le j<2m} r_{m,j}^{3/2} = \int_0^1 \left(\frac{t^2}{2}\right)^{3/2} dt + \int_1^2 \left(\frac{t^2}{2} - \frac{k}{4}(t-1)^2\right)^{3/2} dt.$$

(3) *The curves $\tilde{L}_1, \dots, \tilde{L}_k$ are pairwise disjoint, and the exceptional divisor of q is $F := F_1 + \cdots + F_k$ with $q(F_i) = \tilde{L}_i$ and $F_i \cong \mathbb{F}_1$. The negative sections of $F_1, \dots, F_k$ can be flopped simultaneously to a smooth projective threefold Z^+ on which the strict transform of each F_i is isomorphic to $\mathbb{P}^2$ with normal bundle $\mathcal{O}_{\mathbb{P}^2}(-2)$.*

(4) *For $1 \le t \le 2$, the strict transform on Z^+ of $\pi^*H - tE - \frac{t-1}{2}F$ is nef. Moreover, for integers $m \le j \le 2m$, the strict transform $P_{m,j}$ on Z^+ of the divisor $m\pi^*H - jE - \lceil \frac{j-m}{2} \rceil F$ is nef, and*

$$H^0(Y, m\pi^*H - jE) = H^0(Z^+, P_{m,j})$$

via strict transform and multiplication by the canonical section of the strict transform of $\lceil \frac{j-m}{2} \rceil F$.

Proof. The proof is long, so we divide it into several steps. Let $L_1, \dots, L_k$ be the lines through x, so that $\tilde{L}_i$ is the strict transform of L_i.

Step 1. In this step we compute the normal bundles of $\tilde{L}_i$ and of the negative section of F_i.

Since $\Sigma(X, x)$ is reduced, each local ring of $\Sigma(X, x)$ is isomorphic to $\mathbb{C}$. Thus each L_i is non-special (Proposition 4.1), and hence $N_{L_i/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ (Lemma 2.12(3)). By [KM98, Lemma 2.29], $-K_Y = \pi^*H - 2E$. Since L_i is smooth at x and $(H \cdot L_i) = 1$, we have $(\pi^*H \cdot \tilde{L}_i) = (E \cdot \tilde{L}_i) = 1$. Let $y_i := \tilde{L}_i \cap E$. Since $\pi|_{\tilde{L}_i} : \tilde{L}_i \to L_i$ is an isomorphism, the differential $d\pi : T_Y|_{\tilde{L}_i} \to T_X|_{L_i}$ induces a morphism $\psi_i : N_{\tilde{L}_i/Y} \to N_{L_i/X}$, which is an isomorphism away from y_i. At y_i the image of $d\pi_{y_i}$ is the tangent line $T_x L_i$, so ψ_i vanishes at y_i and factors through an injective morphism $\psi_i' : N_{\tilde{L}_i/Y} \to N_{L_i/X}(-x)$ of rank two vector bundles on $\tilde{L}_i \cong \mathbb{P}^1$. Both bundles have degree -3:

$$\deg N_{\tilde{L}_i/Y} = -2 - (K_Y \cdot \tilde{L}_i) = -2 + (\pi^*H \cdot \tilde{L}_i) - 2(E \cdot \tilde{L}_i) = -3, \qquad \deg N_{L_i/X}(-x) = -1 - 2 = -3.$$

Thus $\det \psi_i'$ is a non-zero section of a line bundle of degree 0, so ψ_i' is an isomorphism and

$$N_{\tilde{L}_i/Y} \cong N_{L_i/X}(-x) \cong \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2).$$

Since $\Sigma(X, x)$ is reduced, [KP23, Lemma 5.11(i)] holds at x, so the curves $\tilde{L}_1, \dots, \tilde{L}_k$ are pairwise disjoint and intersect E at k distinct points, the base locus of $|\pi^*H - 2E|$ is $\tilde{L}_1 \sqcup \cdots \sqcup \tilde{L}_k$, and $|-K_Z|$ is base point free. In particular, the exceptional divisor of q is $F = F_1 + \cdots + F_k$ with $q(F_i) = \tilde{L}_i$, and $F_i \cong \mathbb{P}(N_{\tilde{L}_i/Y}) \cong \mathbb{F}_1$ [Har77, Theorem II.8.24]. Since $\tilde{L}_i \not\subset E$, q^*E is the strict transform of E. By [KM98, Lemma 2.29] again,

$$-K_Z = \pi^*H - 2E - F.$$

Let $s_i \subset F_i$ be the negative section and f_i a fiber of $F_i \to \tilde{L}_i$, so that $(s_i^2) = -1$, $(s_i \cdot f_i) = 1$, and $(f_i^2) = 0$. For a divisor D on Y we have $(q^*D)|_{F_i} \equiv (D \cdot \tilde{L}_i) f_i$. Since $((F_i|_{F_i}) \cdot f_i) = -1$ and

$$((F_i|_{F_i})^2) = (F_i^3) = -\deg N_{\tilde{L}_i/Y} = 3$$

[IP99, Lemma 2.2.14], we have $(-F_i)|_{F_i} \equiv s_i + 2f_i$. Let $M := \pi^*H - E$. Since $(\pi^*H \cdot \tilde{L}_i) = (E \cdot \tilde{L}_i) = 1$, we obtain

$$M|_{F_i} \equiv 0, \qquad (-K_Z)|_{F_i} \equiv -f_i + s_i + 2f_i = s_i + f_i, \qquad (F_i \cdot s_i) = -((s_i + 2f_i) \cdot s_i) = -1.$$

In the exact sequence

$$0 \to N_{s_i/F_i} \to N_{s_i/Z} \to N_{F_i/Z}|_{s_i} \to 0$$

both outer terms are isomorphic to $\mathcal{O}_{\mathbb{P}^1}(-1)$, since $(s_i^2) = -1$ and $(F_i \cdot s_i) = -1$. As $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$, the sequence splits, so

$$N_{s_i/Z} \cong \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1).$$

Step 2. In this step we construct the simultaneous flop $Z \dashrightarrow Z^+$ of $s_1, \dots, s_k$ and prove (3).

By Lemma 2.12(1), $|H|$ embeds X into $\mathbb{P}^{g+1}$, and the morphism defined by $|M|$ on Y is the composite of $Y \to \mathrm{Bl}_x \mathbb{P}^{g+1}$ and the projection from x. In particular, $|M|$ is base point free. Let $C' \subset Y$ be a curve with $(M \cdot C') = 0$. Since $M|_E \cong \mathcal{O}_{\mathbb{P}^2}(1)$, $C' \not\subset E$. Thus $\pi(C')$ is contained in a line of $\mathbb{P}^{g+1}$ through x, so $\pi(C')$ is a line on X through x and C' is one of $\tilde{L}_1, \dots, \tilde{L}_k$. Hence any curve $C' \subset Z$ with $(M \cdot C') = 0$ is contained in $F_1 \cup \cdots \cup F_k$. Since $s_i + f_i$ is the pullback of a line under the contraction $F_i \to \mathbb{P}^2$ of s_i, the only curve $C' \subset F_i$ with $((s_i + f_i) \cdot C') = 0$ is s_i. Therefore the curves $C' \subset Z$ with $((-K_Z + M) \cdot C') = 0$ are exactly $s_1, \dots, s_k$.

The divisor $-K_Z + M$ is base point free, and it is big since $(M^3) = d - 1 > 0$. Let $h : Z \to \bar{Z}$ be the Stein factorization of the morphism defined by $|-K_Z + M|$. Then h is a small birational morphism with exceptional locus $s_1 \cup \cdots \cup s_k$, $\bar{Z}$ is projective, and the points $p_i := h(s_i)$ are pairwise distinct because the fibers of h are connected. The morphisms defined by $|-K_Z|$ and $|M|$ contract $s_1, \dots, s_k$. Since $h_* \mathcal{O}_Z = \mathcal{O}_{\bar{Z}}$, [Kol96, Proposition II.5.3] gives a factorization through h over a neighborhood of each point of $\bar{Z}$, and the factorizations over different neighborhoods glue because h is surjective and $\bar{Z}$ is reduced. Thus $K_{\bar{Z}}$ is Cartier, $K_Z = h^* K_{\bar{Z}}$, and $M = h^* \bar{M}$ for a base point free divisor $\bar{M}$ on $\bar{Z}$. Since h is small and Z is smooth, $\bar{Z}$ has terminal singularities [KM98, Lemma 2.30 and Corollary 2.31(2)]. Since $s_1, \dots, s_k$ are the only curves contracted by h and $(-F \cdot s_i) = 1$, $-F$ is h-ample by [KM98, Theorem 1.42].

For each i let $U_i := \bar{Z} \backslash \{p_j : j \neq i\}$, so that the exceptional locus of h over U_i is s_i. By [Kol89, Theorem 2.4] (see also [KM98, Theorems 6.14 and 6.15]), applied to $h^{-1}(U_i) \to U_i$ and the Cartier divisor F, the F-flop $h_i^+ : Z_i^+ \to U_i$ exists and Z_i^+ is smooth. By [KM98, Definition 6.10], h_i^+ is small and the strict transform of F is h_i^+-ample. For $i \neq j$, h_i^+ and h_j^+ are isomorphisms over $U_i \cap U_j$, so they glue to a morphism $h^+ : Z^+ \to \bar{Z}$. The strict transform of F on Z^+ is h^+-ample, since relative ampleness is local on $\bar{Z}$. As $\bar{Z}$ is projective, Z^+ is a smooth projective threefold, and $Z \dashrightarrow Z^+$ is an isomorphism in codimension one. Moreover, $K_{Z^+} = h^{+*} K_{\bar{Z}}$ and $M = h^{+*} \bar{M}$ on Z^+, so $-K_{Z^+}$ and M are base point free on Z^+.

To identify the strict transforms of the F_i on Z^+ and to compare intersection numbers on Z and Z^+, we describe the flop $Z \dashrightarrow Z^+$ through the blow-up $\tau : Z' \to Z$ of $s_1 \cup \cdots \cup s_k$. Let $G_i := \tau^{-1}(s_i)$. Since $N_{s_i/Z} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, each G_i is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$ with $G_i|_{G_i} \cong \mathcal{O}(-1, -1)$, and the ruling of G_i other than the fibers of $\tau|_{G_i}$ can be contracted, giving a smooth threefold [Rei83, (5.5)]. Let $\gamma_i \subset G_i$ be a fiber of $\tau|_{G_i}$, let $\sigma_i \subset G_i$ be a curve of the other ruling, which τ maps isomorphically onto s_i, and let $\tau^+ : Z' \to Z''$ be the contraction of the rulings of $G_1, \dots, G_k$ containing $\sigma_1, \dots, \sigma_k$. Since $h \circ \tau$ contracts each σ_i, it factors through a morphism $Z'' \to \bar{Z}$ [Kol96, Proposition II.5.3], and the curves contracted by $Z'' \to \bar{Z}$ are $\tau^+(G_1), \dots, \tau^+(G_k)$. Since $s_i \subset F_i$ and F_i is smooth, $\tau_*^{-1} F = \tau^* F - \sum_i G_i$. As $(\tau^* F \cdot \gamma_i) = 0$, $(\tau^* F \cdot \sigma_i) = (F \cdot s_i) = -1$, and $(G_i \cdot \gamma_i) = (G_i \cdot \sigma_i) = -1$, we obtain

$$(\tau_*^{-1} F \cdot \gamma_i) = 1, \qquad (\tau_*^{-1} F \cdot \sigma_i) = 0.$$

Let F'' be the strict transform of F on Z''. Then $\tau_*^{-1} F - \tau^{+*} F'' = \sum_i b_i G_i$ for some $b_i \in \mathbb{Z}$, and $0 = ((\tau_*^{-1} F - \tau^{+*} F'') \cdot \sigma_i) = -b_i$, so $\tau_*^{-1} F = \tau^{+*} F''$. Since τ^+ maps γ_i isomorphically onto $\tau^+(G_i)$,

$$(F'' \cdot \tau^+(G_i)) = (\tau_*^{-1} F \cdot \gamma_i) = 1.$$

Thus $Z'' \to \bar{Z}$ is small and F'' is ample over $\bar{Z}$ [KM98, Theorem 1.42]. By the uniqueness of the F-flop [KM98, Definition 6.10 and Corollary 6.4(3)], $Z'' = Z^+$ over $\bar{Z}$, so $\tau^+ : Z' \to Z^+$ is a birational morphism contracting the fibers σ_i of each G_i.

Let F_i' be the strict transform of F_i on Z'. Since s_i is a Cartier divisor on the smooth surface F_i, $\tau|_{F_i'} : F_i' \to F_i$ is an isomorphism [Har77, Corollary II.7.15 and Proposition II.7.14]. The curve $F_i' \cap G_i$ is the section $\mathbb{P}(N_{s_i/F_i})$ of $G_i \to s_i$. Since $N_{s_i/F_i} \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ is a sub-line-bundle of

$N_{s_i/Z} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ given by a constant vector, this section is a fiber σ_i of $\tau^+|_{G_i}$. As $F_i \cap F_j = \emptyset$ for $i \neq j$, we have $F_i' \cap G_j = \emptyset$, so τ^+ contracts exactly the curve $F_i' \cap G_i \cong s_i$ of F_i'. Thus the strict transform $\tau^+(F_i')$ of F_i on Z^+ is a Cartier divisor on the smooth threefold Z^+ which is smooth outside one point, and hence it is normal by Serre's criterion. The birational morphism $\mathbb{F}_1 \cong F_i' \to \tau^+(F_i')$ contracts only the negative section, so $\tau^+(F_i') \cong \mathbb{P}^2$ by Zariski's Main Theorem applied to the contraction $\mathbb{F}_1 \to \mathbb{P}^2$. Since $F_1', \dots, F_k'$ are pairwise disjoint and $F_i' \cap G_j = \emptyset$ for $i \neq j$, the strict transforms of $F_1, \dots, F_k$ on Z^+ are pairwise disjoint.

Let D be an $\mathbb{R}$-divisor on Z with strict transform D^+ on Z^+, and let $c_i := (D \cdot s_i)$. We claim that

$$\tau^{+*} D^+ = \tau^* D + \sum_i c_i G_i, \qquad (D^{+3}) = (D^3) - \sum_i c_i^3.$$

Since τ and τ^+ are isomorphisms outside $G_1 \cup \dots \cup G_k$, the $\mathbb{R}$-divisor $\tau^{+*} D^+ - \tau^* D$ is supported on $G_1 \cup \dots \cup G_k$. Then $(\tau^{+*} D^+ \cdot \sigma_i) = 0$, $(\tau^* D \cdot \sigma_i) = c_i$, and $(G_i \cdot \sigma_i) = -1$, which implies the first equality. Moreover, $((\tau^* D)^2 \cdot G_i) = 0$, $(\tau^* D \cdot G_i^2) = -c_i$, and $(G_i^3) = ((G_i|_{G_i})^2) = 2$, so

$$(D^{+3}) = (D^3) + \sum_i \big(3c_i^2(-c_i) + 2c_i^3\big) = (D^3) - \sum_i c_i^3.$$

Since $(-K_Z \cdot s_i) = (M \cdot s_i) = 0$, we have $\tau^{+*}(-K_{Z^+}) = \tau^*(-K_Z)$ and $\tau^{+*} M = \tau^* M$. Restricting to F_i', **Step 1** implies that $-K_{Z^+}|_{F_i}$ and $M|_{F_i}$ pull back to $s_i + f_i$ and 0 under the contraction $F_i \to \mathbb{P}^2$ of s_i. Thus, on Z^+,

$$-K_{Z^+}|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}(1), \qquad M|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}.$$

By adjunction,

$$\mathcal{O}_{\mathbb{P}^2}(-3) \cong \mathcal{O}_{F_i}(K_{Z^+} + F_i) \cong \mathcal{O}_{\mathbb{P}^2}(-1) \otimes \mathcal{O}_{F_i}(F_i),$$

so $F_i|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}(-2)$. This proves (3).

Step 3. In this step we prove (4).

On Z we have $\pi^* H - tE = (2-t)M + (t-1)(-K_Z + F)$, since $-K_Z = \pi^* H - 2E - F$. As $Z \dashrightarrow Z^+$ is an isomorphism in codimension one, the strict transforms on Z^+ satisfy

$$\pi^* H - tE - \frac{t-1}{2} F = (2-t)M + (t-1)\Big(- K_{Z^+} + \frac{1}{2} F\Big).$$

By **Step 2**,

$$\Big(- K_{Z^+} + \frac{1}{2} F\Big)\Big|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}(1) \otimes \mathcal{O}_{\mathbb{P}^2}(-1) \cong \mathcal{O}_{\mathbb{P}^2},$$

and any curve $C' \subset Z^+$ not contained in $\operatorname{Supp} F$ satisfies $(F \cdot C') \geq 0$ and $(-K_{Z^+} \cdot C') \geq 0$. Thus $-K_{Z^+} + \frac{1}{2} F$ is nef, and hence $\pi^* H - tE - \frac{t-1}{2} F$ is nef for $1 \leq t \leq 2$.

Fix integers $m \leq j \leq 2m$, and let $\ell := j - m$ and $a := \lceil \ell/2 \rceil$. Since $m\pi^* H - jE = (m-\ell)M + \ell(-K_{Z^+} + F)$ on Z^+, **Step 2** implies

$$(m\pi^* H - jE - bF)|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}(2b - \ell)$$

for any integer b. For $0 \leq b < a$ we have $2b - \ell < 0$, so $H^0(F_i, \mathcal{O}_{\mathbb{P}^2}(2b-\ell)) = 0$. Since $F_1, \dots, F_k$ are pairwise disjoint, the exact sequence

$$0 \to \mathcal{O}_{Z^+}(m\pi^* H - jE - (b+1)F) \to \mathcal{O}_{Z^+}(m\pi^* H - jE - bF) \to \bigoplus_i \mathcal{O}_{F_i}(2b - \ell) \to 0$$

implies that multiplication by the canonical section of F is an isomorphism

$$H^0(Z^+, m\pi^* H - jE - (b+1)F) \to H^0(Z^+, m\pi^* H - jE - bF).$$

By induction on b,

$$H^0(Z^+, m\pi^* H - jE) = H^0(Z^+, P_{m,j}), \qquad P_{m,j} := m\pi^* H - jE - aF.$$

Since $q_*\mathcal{O}_Z = \mathcal{O}_Y$ and $Z \dashrightarrow Z^+$ is an isomorphism in codimension one between normal varieties, $H^0(Y, m\pi^*H - jE) = H^0(Z^+, m\pi^*H - jE)$ via strict transform. Finally,

$$P_{m,j} = \begin{cases} (m-\ell)M + \ell\big(-K_{Z^+} + \frac{1}{2}F\big) & \ell \text{ even}, \\ (m-\ell)M + (\ell-1)\big(-K_{Z^+} + \frac{1}{2}F\big) - K_{Z^+} & \ell \text{ odd}, \end{cases}$$

and $m - \ell \geq 0$, so $P_{m,j}$ is nef by **Step 2** and the previous paragraph. This proves (4).

Step 4. In this step we prove (1).

For $0 \leq t \leq 1$, $\pi^*H - tE = (1-t)\pi^*H + tM$ is nef on Y. Since $((\pi^*H)^3) = d$, $((\pi^*H)^2 \cdot E) = (\pi^*H \cdot E^2) = 0$, and $(E^3) = 1$, we obtain $\mathrm{vol}_Y(\pi^*H - tE) = ((\pi^*H - tE)^3) = d - t^3$.

Let $1 \leq t \leq 2$ be rational, let $u := t - 1$, and let $P_t := \pi^*H - tE - \frac{u}{2}F$ on Z^+. For any positive integer m with $mt \in \mathbb{Z}$ and $mu \in 2\mathbb{Z}$, **Step 3** with $j = mt$ implies $H^0(Y, m(\pi^*H - tE)) \cong H^0(Z^+, mP_t)$. Since P_t is nef, $\mathrm{vol}_Y(\pi^*H - tE) = \mathrm{vol}_{Z^+}(P_t) = (P_t^3)$. To compute (P_t^3), let $D_t := \pi^*H - tE - \frac{u}{2}F$ on Z. Since $((\pi^*H - tE) \cdot \tilde{L}_i) = 1 - t = -u$, **Step 1** implies

$$((\pi^*H - tE)^2 \cdot F_i) = 0, \qquad ((\pi^*H - tE) \cdot F_i^2) = u, \qquad (F_i^3) = 3$$

on Z, and $(F_i \cdot F_j) = 0$ for $i \neq j$. Thus

$$(D_t^3) = d - t^3 + 3k\Big(\frac{u}{2}\Big)^2 u - 3k\Big(\frac{u}{2}\Big)^3 = d - t^3 + \frac{3k}{8}u^3.$$

Moreover, $(D_t \cdot s_i) = -u - \frac{u}{2}(F_i \cdot s_i) = -\frac{u}{2}$. By the formula of **Step 2**,

$$(P_t^3) = d - t^3 + \frac{3k}{8}u^3 + \frac{k}{8}u^3 = d - t^3 + \frac{k}{2}(t-1)^3.$$

By the continuity of volume [Laz04a, Theorem 2.2.44 and Corollary 2.2.45], the formula for $\mathrm{vol}_Y(\pi^*H - tE)$ holds for any real $t \in [1, 2]$. This proves (1).

Step 5. In this step we prove (2).

For any integers $m, j \geq 0$, the exact sequence

$$0 \to \mathcal{O}_Y(m\pi^*H - (j+1)E) \to \mathcal{O}_Y(m\pi^*H - jE) \to \mathcal{O}_E(j) \to 0$$

implies $r_{m,j} = h^0(Y, m\pi^*H - jE) - h^0(Y, m\pi^*H - (j+1)E)$. For $0 \leq j \leq m-2$,

$$(m\pi^*H - (j+1)E) - K_Y = (m-j-2)\pi^*H + (j+3)M$$

is nef and big, so $H^1(Y, m\pi^*H - (j+1)E) = 0$ by the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64]. Thus $W_{m,j} = H^0(E, \mathcal{O}_E(j))$.

Now let $m \leq j \leq 2m$, and keep the notation ℓ, a, and $P_{m,j}$ of **Step 3**. Let $\epsilon := a - \frac{\ell}{2} \in \{0, \frac{1}{2}\}$ and

$$P'_{m,j} := (m-\ell)M + \ell\Big(-K_{Z^+} + \frac{1}{2}F\Big) = (2m-j)M + (j-m)\Big(-K_{Z^+} + \frac{1}{2}F\Big),$$

so that $P_{m,j} = P'_{m,j} - \epsilon F$. Since $P'_{m,j}|_{F_i} \cong \mathcal{O}_{\mathbb{P}^2}$, we have

$$(P'^2_{m,j} \cdot F_i) = (P'_{m,j} \cdot F_i^2) = (P'_{m,j} \cdot (-K_{Z^+}) \cdot F_i) = 0.$$

By **Step 2**, $(F_i^3) = 4$ and $(-K_{Z^+} \cdot F_i^2) = -2$ on Z^+. Thus

$$(P_{m,j}^3) = (P'^3_{m,j}) - 4k\epsilon^3, \qquad (P_{m,j}^2 \cdot (-K_{Z^+})) = (P'^2_{m,j} \cdot (-K_{Z^+})) - 2k\epsilon^2.$$

By **Step 1** and the formula of **Step 2**, $((-K_{Z^+})^3) = ((-K_Z)^3) = ((\pi^*H - 2E)^3) + 3k - 3k = d - 8 > 0$, so $-K_{Z^+}$ is nef and big. Hence $P_{m,j} - K_{Z^+}$ is nef and big, and $H^i(Z^+, P_{m,j}) = 0$ for $i > 0$ by the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64]. By the Hirzebruch–Riemann–Roch theorem [Har77, Appendix A, Theorem 4.1] and **Step 3**,

$$h^0(Y, m\pi^*H - jE) = \chi(Z^+, P_{m,j}) = R(m,j) - \frac{2}{3}k\epsilon^3 - \frac{1}{2}k\epsilon^2 - \frac{\epsilon}{12}\big(F \cdot (K_{Z^+}^2 + c_2(Z^+))\big),$$

where

$$R(m,j) := \frac{(P'^3_{m,j})}{6} + \frac{(P'^2_{m,j}\cdot(-K_{Z^+}))}{4} + \frac{(P'_{m,j}\cdot(K^2_{Z^+}+c_2(Z^+)))}{12} + \chi(Z^+,\mathcal{O}_{Z^+})$$

is a polynomial in (m,j) of degree at most three. For $m \le j \le 2m$ we have $P'_{m,j} = mP_{j/m}$, so **Step 4** implies

$$\frac{(P'^3_{m,j})}{6} = \frac{1}{6}\Big(dm^3 - j^3 + \frac{k}{2}(j-m)^3\Big) =: R_3(m,j).$$

As both sides are polynomials, R_3 is the homogeneous cubic part of R. For $m \le j < 2m$ we obtain

$$r_{m,j} = R(m,j) - R(m,j+1) + O(1) = -\frac{\partial R_3}{\partial j}(m,j) + O(m) = \frac{j^2}{2} - \frac{k}{4}(j-m)^2 + O(m),$$

uniformly in j, since the second derivatives of R_3 and the first derivatives of $R - R_3$ are $O(m)$ for $0 \le j \le 2m$.

Let $\phi(t) := \frac{t^2}{2}$ for $0 \le t \le 1$ and $\phi(t) := \frac{t^2}{2} - \frac{k}{4}(t-1)^2$ for $1 \le t \le 2$. Then $\phi > 0$ on $(0,2]$ as $k \le 3$. By the first paragraph, $r_{m,j} = \frac{(j+1)(j+2)}{2}$ for $0 \le j \le m-2$, and $0 \le r_{m,m-1} \le h^0(E, \mathcal{O}_E(m-1)) = O(m^2)$. Thus $r_{m,j}m^{-2} = \phi(j/m) + O(1/m)$ uniformly for $0 \le j < 2m$ with $j \ne m-1$, and the term $j = m-1$ contributes $O(1/m)$ to $m^{-4}\sum_j r^{3/2}_{m,j}$. Since $t \to t^{3/2}$ is Lipschitz on bounded subsets of $[0,+\infty)$, $m^{-4}\sum_{0\le j<2m} r^{3/2}_{m,j}$ is a Riemann sum of $\phi^{3/2}$ on $[0,2]$ up to $O(1/m)$, and hence converges to $\int_0^2 \phi(t)^{3/2}dt$. This proves (2). □

The next proposition covers every integral curve on X except the lines in both genera and the conics in genus twelve, which are treated in Sections 5, 11, and 12.

Proposition 4.6. *There exists a real number $\epsilon_0 > 0$ with the following property. Let X be a smooth prime Fano threefold of genus g, let $d := 2g-2$, and let $C \subset X$ be an integral curve. Assume that H is very ample, $d \ge 6$, and $\lfloor (2d+2)/(H\cdot C)\rfloor \le 20$. Then $\delta_C(X) \ge 1+\epsilon_0$. In particular, $A_X(F) > S_X(F)$ for any prime divisor F over X with $C \subset \operatorname{center}_X F$.*

Moreover, the assumptions on X and C hold for any integral curve C with $(H\cdot C) \ge 2$ on a smooth prime Fano threefold of genus 10, and for any integral curve C with $(H\cdot C) \ge 3$ on a smooth prime Fano threefold of genus 12.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we define the constant ϵ_0.

For a smooth surface T, a closed point $p \in T$, and a prime divisor F' over T with $p \in \operatorname{center}_T F'$, let $A_T(F') := a(F',T,0)$. For an ideal $\mathfrak{a} \subset \mathcal{O}_{T,p}$ let $\operatorname{mult}_{F'}\mathfrak{a} := \min_{f\in\mathfrak{a}} \operatorname{ord}_{F'}(f)$, and for an $\mathbb{R}$-divisor $B \ge 0$ on T let

$$\operatorname{lct}(T \ni p; \mathfrak{a}) := \inf_{F'} \frac{A_T(F')}{\operatorname{mult}_{F'}\mathfrak{a}}, \qquad \operatorname{lct}(T \ni p; B) := \inf_{F'} \frac{A_T(F')}{\operatorname{mult}_{F'} B},$$

where F' runs through the prime divisors over T with $p \in \operatorname{center}_T F'$. The invariant $\operatorname{lct}(T \ni p; B)$ is the local version of Definition 2.3. We show that there exists a real number α with $\alpha > \frac{1}{3}$ such that

$$\operatorname{lct}(T \ni p; \mathfrak{a}) \ge \alpha$$

for any smooth surface T, any closed point $p \in T$, and any ideal $\mathfrak{a} \subset \mathcal{O}_{T,p}$ with $\ell(\mathcal{O}_{T,p}/\mathfrak{a}) \le 20$.

Let $\mathfrak{a}$ be such an ideal and $\ell := \ell(\mathcal{O}_{T,p}/\mathfrak{a})$. If $\ell = 0$, then $\operatorname{lct}(T \ni p; \mathfrak{a}) = +\infty$. Assume that $\ell \ge 1$. In the notation of [Zhu20, Definition 3.2], [Zhu20, Lemma 3.4] bounds from below the minimal colength of an ideal $\mathfrak{c} \subset \mathcal{O}_{T,p}$ such that $(T, \frac{1}{3}\cdot\mathfrak{c})$ is not klt at p, by a constant which is at least $3\cdot(2\cdot 3+1) = 21$ by [Zhu20, Lemma 5.2]. As $\ell \le 20$, $(T, \frac{1}{3}\cdot\mathfrak{a})$ is klt near p. Since $\operatorname{lct}(T \ni p; \mathfrak{a})$ is computed by a divisor on a log resolution of $\mathfrak{a}$, we have $\operatorname{lct}(T \ni p; \mathfrak{a}) > \frac{1}{3}$.

Let $\mathfrak{m}_p$ be the maximal ideal of $\mathcal{O}_{T,p}$. The ideals $\mathfrak{a}+\mathfrak{m}_p^i$ strictly decrease in i until they are equal to $\mathfrak{a}$ by Nakayama's lemma, so $\mathfrak{m}_p^{20}\subset\mathfrak{m}_p^{\ell}\subset\mathfrak{a}$. Choose local coordinates u,v at p. Then $\mathfrak{a}$ is the preimage of an ideal $\bar{\mathfrak{a}}$ of $\mathcal{O}_{T,p}/\mathfrak{m}_p^{20}\cong\mathbb{C}[u,v]/(u,v)^{20}$, and we let $\mathfrak{a}'\subset\mathbb{C}[u,v]$ be the preimage of $\bar{\mathfrak{a}}$. Since $\mathfrak{a}$ and $\mathfrak{a}'$ have isomorphic completions, $\operatorname{lct}(T\ni p;\mathfrak{a})=\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{a}')$. Fix a monomial order, and let $\mathfrak{b}$ be the initial ideal of $\mathfrak{a}'$. Then $(u,v)^{20}\subset\mathfrak{b}$, and $\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{b})\leq\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{a}')$ by the lower semicontinuity of log canonical thresholds [DK01, Main Theorem 0.2], applied to the flat degeneration of $\mathfrak{a}'$ to $\mathfrak{b}$ [Eis95, Proposition 15.16 and Theorem 15.17]. We also have $\ell(\mathbb{C}[u,v]/\mathfrak{b})=\ell(\mathbb{C}[u,v]/\mathfrak{a}')=\ell$, since by [Eis95, Theorem 15.3] the monomials not in $\mathfrak{b}$ form a basis of both $\mathbb{C}[u,v]/\mathfrak{a}'$ and $\mathbb{C}[u,v]/\mathfrak{b}$. A monomial ideal containing $(u,v)^{20}$ is determined by the monomials $u^{i_1}v^{i_2}$ with $i_1+i_2<20$ that it contains, so the set $\mathcal{M}$ of monomial ideals $\mathfrak{d}\subset\mathbb{C}[u,v]$ with $(u,v)^{20}\subset\mathfrak{d}$ and $1\leq\ell(\mathbb{C}[u,v]/\mathfrak{d})\leq 20$ is finite, and $\mathcal{M}$ is non-empty since $(u,v)\in\mathcal{M}$. Let

$$\alpha:=\min_{\mathfrak{d}\in\mathcal{M}}\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{d}).$$

By the previous paragraph, applied to $(\mathbb{A}^2,0)$, each $\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{d})$ with $\mathfrak{d}\in\mathcal{M}$ is larger than $\frac{1}{3}$. Thus $\alpha>\frac{1}{3}$, and since $\mathfrak{b}\in\mathcal{M}$,

$$\operatorname{lct}(T\ni p;\mathfrak{a})\geq\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{b})\geq\alpha.$$

Let $\kappa:=\frac{4\alpha}{1+\alpha}$ and $\epsilon_0:=\kappa-1$. Since $\alpha>\frac{1}{3}$, we have $\kappa>1$, and $\epsilon_0>0$ does not depend on X or C.

Step 2. In this step we choose a hyperplane section T and a curve $G\subset T$.

Let $e:=(H\cdot C)$, and fix a closed point $y\in X\backslash C$. We take T through y because [AZ23, Theorem 5.8] concerns hyperplane sections through a given point: a very general member T of $|H|$ containing y satisfies $\rho(T)=1$. Since H is very ample, a general member of $|H|$ containing y is smooth, also at y, and since $y\notin C$, the members of $|H|$ containing y have no base point on C, so a general such member intersects C transversally at e distinct smooth points of C [Har77, Corollary III.10.9 and Remark III.10.9.2]. We choose $T\in|H|$ smooth with $\rho(T)=1$ and intersecting C transversally at e distinct smooth points of C, and let $Z:=T\cap C=\{p_1,\dots,p_e\}$ and $L:=H|_T$, so that $(L^2)=d$.

By adjunction $K_T=(K_X+T)|_T\sim 0$. Moreover, $H^1(X,\mathcal{O}_X)=H^1(X,K_X+H)=0$ by the Kodaira vanishing theorem [Laz04a, Theorem 4.2.1] and $H^2(X,\mathcal{O}_X(-T))=H^2(X,K_X)=0$ by Serre duality, so the exact sequence $0\to\mathcal{O}_X(-T)\to\mathcal{O}_X\to\mathcal{O}_T\to 0$ implies $H^1(T,\mathcal{O}_T)=0$. Thus T is a K3 surface. By [Laz04a, Theorem 3.1.17 and Example 3.1.18], the restriction $H^2(X,\mathbb{Z})\to H^2(T,\mathbb{Z})$ is injective with torsion-free cokernel. Since $H^2(X,\mathbb{Z})=\operatorname{Pic}(X)=\mathbb{Z}[H]$ and $\operatorname{Pic}(T)\subset H^2(T,\mathbb{Z})$ as $H^1(T,\mathcal{O}_T)=0$, the subgroup $\mathbb{Z}[L]$ is saturated in $H^2(T,\mathbb{Z})$. As $\rho(T)=1$, we obtain $\operatorname{Pic}(T)=\mathbb{Z}[L]$. Let $h_n:=h^0(T,nL)$. By the Riemann–Roch theorem and the Kodaira vanishing theorem on T [Har77, Theorem V.1.6 and Remark III.7.15],

$$h_0=1,\qquad h_n=\frac{d}{2}n^2+2\quad\text{for } n\geq 1.$$

In particular, h_n is nondecreasing in $n\geq 0$ and $h^0(T,2L)=2d+2$.

Let $G\in|L|$ be a general member. Since L is very ample and Z is finite, G is a smooth irreducible curve with $G\cap Z=\emptyset$ [Har77, Corollary III.10.9 and Remark III.10.9.1]. The flag $X\supset T\supset G$ is admissible, and each member is a Cartier divisor in the previous one.

Step 3. In this step we describe the basis type divisors on T.

Let W be the refinement of $(H^0(X,mH))_m$ by T (Definition 2.8, with plt extraction id_X), and let $\mathcal{F}$ be the filtration of W by ord_G, that is, the filtration induced by the flag $X\supset T\supset G$. Since the sections of mH vanishing to order at least j along T are the products of the j-th power of the canonical section of T with $H^0(X,(m-j)H)$, $W_{m,j}$ is the image of $H^0(X,(m-j)H)\to H^0(T,(m-j)L)$ for $0\leq j\leq m$, and $W_{m,j}=0$ for $j>m$. For

$n \geq 1$, $H^1(X,(n-1)H) = H^1(X,K_X+nH) = 0$ by the Kodaira vanishing theorem [Laz04a, Theorem 4.2.1], and $H^0(X,\mathcal{O}_X) \to H^0(T,\mathcal{O}_T)$ is an isomorphism. Thus

$$W_{m,j} = H^0(T,(m-j)L) \quad (0 \leq j \leq m), \qquad N_m = \sum_{n=0}^{m} h_n.$$

Following [AZ22, Section 2.1 and Definition 2.12], for an $\mathbb{R}$-divisor $D \geq 0$ on T let $\operatorname{lct}_Z(T;D)$ be the largest $\lambda \geq 0$ with $Z \not\subset \operatorname{Nlc}(T,\lambda D)$. As $Z = \{p_1,\dots,p_e\}$,

$$\operatorname{lct}_Z(T;D) = \max_i \operatorname{lct}(T \ni p_i; D).$$

Let $\delta_{Z,m}(T;W,\mathcal{F})$ be the infimum of $\operatorname{lct}_Z(T;D)$, where D runs through the m-basis type divisors of W compatible with ord_G (Definition 2.8), and let $\delta_Z(T;W,\mathcal{F}) := \limsup_{m\to+\infty} \delta_{Z,m}(T;W,\mathcal{F})$.

Let D be an m-basis type divisor of W compatible with ord_G. Since a section of $H^0(T,nL)$ defines a divisor linearly equivalent to nL,

$$D \sim_{\mathbb{Q}} c_m L, \qquad c_m := \frac{1}{mN_m} \sum_{n=0}^{m} n h_n.$$

In $H^0(T,nL)$, the sections vanishing to order at least i along G form a subspace of dimension h_{n-i} for $1 \leq i \leq n$, and 0 for $i > n$. As the basis of each $W_{m,j}$ is compatible with ord_G,

$$\operatorname{mult}_G D = a_m := \frac{1}{mN_m} \sum_{n=0}^{m} \sum_{i=1}^{n} h_{n-i} = \frac{1}{mN_m} \sum_{l=0}^{m-1} (m-l) h_l.$$

Since $N_m = \frac{d}{6}m^3 + O(m^2)$, $\sum_{n=0}^{m} n h_n = \frac{d}{8}m^4 + O(m^3)$, and $\sum_{l=0}^{m-1}(m-l)h_l = \frac{d}{24}m^4 + O(m^3)$, we have $c_m \to \frac{3}{4}$ and $a_m \to \frac{1}{4}$. Write

$$D = a_m G + \Gamma, \qquad \Gamma \geq 0, \qquad G \not\subset \operatorname{Supp}\Gamma.$$

Then $\Gamma \sim_{\mathbb{Q}} \lambda_m L$, where $\lambda_m := c_m - a_m \to \frac{1}{2}$. The numbers c_m, a_m, and λ_m do not depend on D.

Step 4. In this step we bound the coefficients of Γ and define the boundary B.

Let $C' \subset T$ be an integral curve. We show that $\operatorname{mult}_{C'} \Gamma \leq a_m$. If $C' = G$, then $\operatorname{mult}_{C'} \Gamma = 0$. Otherwise $\operatorname{mult}_{C'} \Gamma = \operatorname{mult}_{C'} D$. Since $\operatorname{Pic}(T) = \mathbb{Z}[L]$ and $(L \cdot C') > 0$, $C' \sim rL$ for an integer $r \geq 1$. In $H^0(T,nL)$, the sections vanishing to order at least i along C' form a subspace of dimension $h^0(T,(n-ir)L)$, which is h_{n-ir} if $ir \leq n$ and 0 otherwise. For any basis $(\sigma_1,\dots,\sigma_{h_n})$ of $H^0(T,nL)$, at most $\dim\{\sigma : \operatorname{ord}_{C'}(\sigma) \geq i\}$ of the σ_l satisfy $\operatorname{ord}_{C'}(\sigma_l) \geq i$, so

$$\sum_l \operatorname{ord}_{C'}(\sigma_l) = \sum_{i\geq 1} \#\{l : \operatorname{ord}_{C'}(\sigma_l) \geq i\} \leq \sum_{i=1}^{\lfloor n/r \rfloor} h_{n-ir} \leq \sum_{i=1}^{n} h_{n-i},$$

where the last inequality holds since $n \to h_n$ is nondecreasing. Summing over n and dividing by mN_m, we obtain $\operatorname{mult}_{C'} D \leq a_m$. The bound $\operatorname{mult}_{C'} D \leq a_m$ holds for any m-basis type divisor D of W and any integral curve $C' \subset T$.

For $m \geq 2$ large enough that $a_m > 0$ and $\lambda_m > 0$, let

$$\eta_m := \Big(1 - \frac{1}{m}\Big) \min\Big\{1, \frac{1}{4a_m}, \frac{1}{2\lambda_m}\Big\}.$$

Then $0 < \eta_m < 1$, $\eta_m \to 1$, $4\eta_m a_m < 1$, and $\eta_m \lambda_m < \frac{1}{2}$. Let $B := 4\eta_m \Gamma$. By the previous paragraph, $\operatorname{mult}_{C'} B \leq 4\eta_m a_m < 1$ for any integral curve $C' \subset T$, so (T,B) is klt outside a finite set. Moreover,

$$2L - (K_T + B) \sim_{\mathbb{Q}} (2 - 4\eta_m\lambda_m)L$$

is ample.

Step 5. In this step we show that $\delta_{Z,m}(T;W,\mathcal{F}) \geq \kappa\eta_m$.

Let D and B be as in **Steps 3 and 4**, and let $\mathcal{J} := \mathcal{J}(T, B)$ be the multiplier ideal [Laz04b, Definition 9.2.1]. By **Step 4**, $\mathcal{O}_T/\mathcal{J}$ has finite support. By the Nadel vanishing theorem [Laz04b, Theorem 9.4.8], $H^1(T, \mathcal{O}_T(2L) \otimes \mathcal{J}) = 0$, so the restriction

$$H^0(T, 2L) \to H^0(T, \mathcal{O}_T(2L) \otimes \mathcal{O}_T/\mathcal{J})$$

is surjective. Thus $\ell(\mathcal{O}_T/\mathcal{J}) \leq h^0(T, 2L) = 2d + 2$. Since Z consists of e distinct points, there exists $p \in Z$ with

$$\ell(\mathcal{O}_{T,p}/\mathcal{J}_p) \leq \left\lfloor \frac{2d+2}{e} \right\rfloor \leq 20.$$

By **Step 1**, $\operatorname{lct}(T \ni p; \mathcal{J}_p) \geq \alpha$.

Let F' be a prime divisor over T with $p \in \operatorname{center}_T F'$. Computing $\mathcal{J}$ on a log resolution of (T, B) on which F' is a divisor, we have $\operatorname{mult}_{F'} \mathcal{J}_p + A_T(F') > \operatorname{mult}_{F'} B$. As $A_T(F') \geq \alpha \operatorname{mult}_{F'} \mathcal{J}_p$, we obtain

$$\operatorname{mult}_{F'} B < \left(1 + \frac{1}{\alpha}\right) A_T(F').$$

Thus $\operatorname{lct}(T \ni p; B) \geq \frac{\alpha}{1+\alpha}$, which is the local form of [Zhu20, Theorem 1.6 and Remark 3.1]. Since $B = 4\eta_m \Gamma$, we have $\operatorname{lct}(T \ni p; \Gamma) \geq \frac{4\eta_m \alpha}{1+\alpha} = \kappa \eta_m$. Since $p \notin G$, $D = \Gamma$ near p, and hence

$$\operatorname{lct}_Z(T; D) \geq \operatorname{lct}(T \ni p; D) \geq \kappa \eta_m.$$

The point p depends on D, but the bound does not. Taking the infimum over D, we obtain $\delta_{Z,m}(T; W, \mathcal{F}) \geq \kappa \eta_m$ for any large m. As $\eta_m \to 1$, we have $\delta_Z(T; W, \mathcal{F}) \geq \kappa$.

Step 6. In this step we conclude the proof.

Since $T \sim H$ is a prime divisor on X, Lemma 4.2 with $a = 1$ gives $A_X(T)/S_X(T) = 4$. Moreover, $\kappa = 4 - \frac{4}{1+\alpha} < 4$. By [AZ22, Theorem 3.4], applied to the complete series of H, the admissible flag $X \supset T \supset G$ with the filtration $\mathcal{F}$, and the curve C, whose intersection Z with T is allowed to be reducible, we have

$$\delta_C(X) \geq \min\left\{\frac{A_X(T)}{S_X(T)}, \delta_Z(T; W, \mathcal{F})\right\} \geq \min\{4, \kappa\} = \kappa = 1 + \epsilon_0.$$

Let F be a prime divisor over X with $C \subset \operatorname{center}_X F$. Since H is ample, $S_X(F) > 0$, and hence $A_X(F) \geq \kappa S_X(F) > S_X(F)$.

Finally, let X be a smooth prime Fano threefold of genus 10 or 12. Then H is very ample (Lemma 2.12(1)), and $d = 18$ or $d = 22$. If $d = 18$ and $(H \cdot C) \geq 2$, then $\lfloor 38/(H \cdot C) \rfloor \leq \lfloor 38/2 \rfloor = 19 \leq 20$. If $d = 22$ and $(H \cdot C) \geq 3$, then $\lfloor 46/(H \cdot C) \rfloor \leq \lfloor 46/3 \rfloor = 15 \leq 20$. □

Remark 4.7. The argument for Proposition 4.6 follows the curve estimate for curves C with $(H \cdot C) \geq 2$ in [AZ23, proof of Theorem 5.12], with the colength bound for the relevant multiplier ideals taken to be at most 20, so that the bound $\operatorname{lct} > 1/3$ of [Zhu20, Lemmas 3.4 and 5.2] can be used. The constant ϵ_0 is not explicit, and only $\epsilon_0 > 0$ is used.

5. Lines in genus ten

In this section, we prove that $A_X(F) > S_X(F)$ for any prime divisor F over a smooth prime Fano threefold X of genus 10 such that $\operatorname{center}_X F$ is a line (Corollary 5.10). We treat non-special lines in Theorem 5.5 and special lines in Propositions 5.7 and 5.9, and we prove Corollary 5.10 at the end of this section. The line link and the incidence scheme introduced below are also used for points on lines in Sections 6 and 8.

Set-up 5.1. Let X be a smooth prime Fano threefold of genus 10 and $L \subset X$ a line. Let $\pi : Y := \operatorname{Bl}_L X \to X$ be the blow-up of L with exceptional divisor E, write H also for $\pi^* H$, and let $Y \dashrightarrow Y^+ = \operatorname{Bl}_\Gamma Q$, $h = H - 2E$, and $D' = 2H - 5E$ be given by the line link of Theorem 2.22.

(1) If L is non-special, then $E \cong \mathbb{F}_1$. In this case, let s be the negative section and f the class of a fiber of $E \to L$, for $x \in L$ let f_x be the fiber over x, let
$$U := \operatorname{Im}\big(H^0(Y, H-2E) \to H^0(E, \mathcal{O}_E(2s+3f))\big)$$
with base scheme $Z_U \subset E$, let D be the unique member of $|2H-5E|$, which is the strict transform of D' (Lemma 5.2 with $m=2$ and $j=5$), and let
$$W_{m,j} := \operatorname{Im}\big(H^0(Y, mH-jE) \to H^0(E, \mathcal{O}_E(js+(m+j)f))\big).$$

(2) If L is special, then $E \cong \mathbb{F}_3$, s denotes the negative section of E, and P denotes the exceptional divisor of $\operatorname{Bl}_s Y \to Y$, viewed as a prime divisor over X.

The line link of Set-up 5.1 is the diagram
$$X \xleftarrow{\pi} Y \dashrightarrow Y^+ \to Q,$$
in which the last arrow is the blow-up of Γ. The descriptions of E in (1) and (2) follow from Lemma 2.12(3): E is the projectivization of $N_{L/X}$ over $L \cong \mathbb{P}^1$, and the projectivizations of $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ and $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ are $\mathbb{F}_1$ and $\mathbb{F}_3$ [Har77, Proposition V.2.8]. If L is special, let f also denote the class of a fiber of $E \to L$. In both cases, $H|_E \sim (H \cdot L)f = f$, $(E|_E \cdot f) = -1$, and $((E|_E)^2) = (E^3) = -\deg N_{L/X} = 1$ [IP99, Lemma 2.2.14], so $-E|_E \sim s + bf$ with $((s+bf)^2) = (s^2)+2b = 1$ [Har77, Propositions V.2.3 and V.2.9]. Together with $D \sim 2H-5E$ (Lemma 5.2), we obtain

$$\begin{aligned}
&L \text{ non-special:} && (s^2) = -1, \quad -E|_E \sim s+f, \quad (mH-jE)|_E \sim js + (m+j)f,\\
&&& (H-2E)|_E \sim 2s+3f, \quad D|_E \sim 5s+7f,\\
&L \text{ special:} && (s^2) = -3, \quad -E|_E \sim s+2f, \quad (mH-jE)|_E \sim js + (m+2j)f,\\
&&& (H-2E)|_E \sim 2s+5f, \quad (2H-5E)|_E \sim 5s+12f.
\end{aligned} \tag{5.1}$$

By Theorem 2.22 and [KPZ14, Proposition 2.12(3)], near each flopping curve of $Y \dashrightarrow Y^+$ the flopping contraction is Reid's pagoda, so any flopping curve satisfies the conditions of Definition-Proposition 2.21 and has a width.

Lemma 5.2. *Notation as in Set-up 5.1, with L special or non-special. Then the strict transform of D' on Y is a prime divisor different from E and linearly equivalent to $2H-5E$, $A_X(E) = 2$, and*
$$\operatorname{vol}(H-tE) = \begin{cases} 18-3t^2-t^3 & 0 \le t \le 1,\\ 18-3t^2-t^3+4(t-1)^3 & 1 \le t \le 2,\\ 2(5-2t)^3 & 2 \le t \le 5/2,\\ 0 & t \ge 5/2, \end{cases}$$
so that $S_X(E) = 101/72$. The image E_Q of E in Q is a member of $|\mathcal{O}_Q(2)|$, and the strict transform of E on Y^+ maps birationally onto E_Q. For integers $m \ge 1$ and $2m \le j \le 5m/2$, multiplication by the $(j-2m)$-th power of a section defining the strict transform of D' is an isomorphism
$$H^0(Y, (5m-2j)h) \to H^0(Y, mH-jE),$$
and $h^0(Y, qh) = h^0(Q, \mathcal{O}_Q(q))$ for any integer $q \ge 0$. Moreover $H^0(Y, mH-jE) = 0$ for $j > 5m/2$. Finally, $N_m = 3m^3 + O(m^2)$, and with $t := j/m$,
$$h^0(Y, mH-jE) - h^0(Y, mH-(j+1)E) = \frac{m^2}{2}w(t) + O(m)$$
uniformly in j on each of the intervals $0 \le t < 1$, $1 < t < 2$, and $2 < t < 5/2$, where $w(t) = t^2+2t$, $w(t) = -3t^2+10t-4$, and $w(t) = 4(5-2t)^2$, respectively. In particular
$$\lim_{m \to +\infty} \frac{1}{mN_m} \sum_{j > 2m} (j-2m)\big(h^0(Y, mH-jE) - h^0(Y, mH-(j+1)E)\big) = \frac{1}{72}.$$

Proof. The proof is long, so we divide it into several steps. Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one, we identify divisors on Y and on Y^+ by strict transform. Let $\pi^+ : Y^+ \to Q$ be the blow-up of Γ.

Step 1. In this step we compute the intersection numbers and the canonical classes, and we describe the image of E in Q.

Since $L \cong \mathbb{P}^1$ and $(K_X \cdot L) = -1$, we have $\deg N_{L/X} = -1$. Since Γ is a smooth curve of genus 2 and degree 7 and $-K_Q$ is three times a hyperplane class, we have $\deg N_{\Gamma/Q} = 2\cdot 2 - 2 + 3\cdot 7 = 23$. By [IP99, Lemma 2.2.14],

$$(H^3) = 18, \quad (H^2 \cdot E) = 0, \quad (H \cdot E^2) = -1, \quad (E^3) = -\deg N_{L/X} = 1$$

on Y, and

$$(h^3) = 2, \quad (h^2 \cdot D') = 0, \quad (h \cdot D'^2) = -7, \quad (D'^3) = -\deg N_{\Gamma/Q} = -23$$

on Y^+ (see also [KPZ14, (2.6.3)–(2.6.4)]). The relations $h = H - 2E$ and $D' = 2H - 5E$ are equivalent to

$$H = 5h - 2D', \qquad E = 2h - D'.$$

Since $K_Y = \pi^* K_X + E$ [KM98, Lemma 2.29], we have $-K_Y = H - E$ and $A_X(E) = 2$. Since $K_{Y^+} = \pi^{+*} K_Q + D'$ [KM98, Lemma 2.29] and $\pi^{+*}(-K_Q) \sim 3h$, we have $-K_{Y^+} = 3h - D'$, which is the strict transform of $H - E$.

The divisor D' is prime, since it is the exceptional divisor of the blow-up of the smooth irreducible curve Γ. Thus the strict transform of D' on Y is a prime divisor linearly equivalent to $2H - 5E$. It is different from E, since $\pi^+(D') = \Gamma$, while π^+ maps the strict transform of E onto a surface (Theorem 2.22). Let σ_D be a section defining it. Finally, for any divisor M on Y we have $H^0(Y, M) = H^0(Y^+, M)$ as subspaces of the function field, since Y and Y^+ are normal and isomorphic in codimension one [Har77, Proposition II.6.3A]. In particular, $\mathrm{vol}_Y(M) = \mathrm{vol}_{Y^+}(M)$.

Let E^+ be the strict transform of E on Y^+, and let $E_Q := \pi^+(E^+)$. Since $E = 2h - D'$ on Y^+ (Theorem 2.22) and $\pi^+_* D' = 0$, the divisor $\pi^+_* E^+ = \deg(E^+ \to E_Q) \cdot E_Q$ is a member of $|\mathcal{O}_Q(2)|$. Since E_Q is a quartic surface (Theorem 2.22) and the members of $|\mathcal{O}_Q(2)|$ are surfaces of degree 4 in $\mathbb{P}^4$, we obtain $\deg(E^+ \to E_Q) = 1$. Thus $E_Q \in |\mathcal{O}_Q(2)|$, and $E^+ \to E_Q$ is birational.

Step 2. In this step we compute $H^0(Y^+, qh + kD')$ for integers q and $k \geq 0$.

Since D' is π^+-exceptional and Q is normal, $\pi^+_* \mathcal{O}_{Y^+}(kD') = \mathcal{O}_Q$ for any integer $k \geq 0$: a rational function on an open subset $V \subset Q$ which is regular outside the codimension two subset $V \cap \Gamma$ is regular on V [Har77, Proposition II.6.3A]. By the projection formula, multiplication by σ_D^k is an isomorphism $\pi^{+*} H^0(Q, \mathcal{O}_Q(q)) \to H^0(Y^+, qh + kD')$ for any integers q and $k \geq 0$. In particular, $H^0(Y^+, qh + kD') = 0$ if $q < 0$. For $q \geq 0$, the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^4}(q-2) \to \mathcal{O}_{\mathbb{P}^4}(q) \to \mathcal{O}_Q(q) \to 0$$

implies

$$h^0(Q, \mathcal{O}_Q(q)) = \binom{q+4}{4} - \binom{q+2}{4} = \frac{(q+1)(q+2)(2q+3)}{6} =: \rho(q).$$

Let $m \geq 1$ and j be integers. By **Step 1**, $mH - jE = (5m - 2j)h + (j - 2m)D'$. If $2m \leq j \leq 5m/2$, then multiplication by σ_D^{j-2m} is an isomorphism $H^0(Y, (5m-2j)h) \to H^0(Y, mH - jE)$, and $h^0(Y, qh) = h^0(Q, \mathcal{O}_Q(q))$ for $q \geq 0$. If $j > 5m/2$, then $5m - 2j < 0$ and $j - 2m > 0$, so $H^0(Y, mH - jE) = 0$. Applied to multiples of $\mathbb{Q}$-divisors, the isomorphism given by multiplication by σ_D^k shows that $\mathrm{vol}((5-2t)h + (t-2)D') = \mathrm{vol}((5-2t)h) = 2(5-2t)^3$ for $2 \leq t \leq 5/2$, since h is nef with $(h^3) = 2$, and that $\mathrm{vol}(H - tE) = 0$ for $t \geq 5/2$.

Step 3. In this step we compute the volume function and $S_X(E)$.

For $0 \leq t \leq 1$, the $\mathbb{R}$-divisor $H - tE = (1-t)H + t(H - E)$ is nef on Y, since $-K_Y$ is nef (Theorem 2.22). Thus $\mathrm{vol}(H - tE) = ((H - tE)^3) = 18 - 3t^2 - t^3$.

For $1 \le t \le 2$, by **Step 1**,

$$H - tE = (2-t)(3h - D') + (t-1)h,$$

which is nef on Y^+, since h is nef and $-K_{Y^+}$ is nef [KPZ14, Lemma 2.7]. Writing $H - tE = (5-2t)h + (t-2)D'$, **Step 1** implies

$$\mathrm{vol}(H - tE) = 2(5-2t)^3 - 21(5-2t)(t-2)^2 - 23(t-2)^3 = 18 - 3t^2 - t^3 + 4(t-1)^3,$$

where the last equality is a direct calculation. For $t \ge 2$, $\mathrm{vol}(H - tE)$ is given by **Step 2**. Therefore

$$S_X(E) = \frac{1}{18}\int_0^{5/2} \mathrm{vol}(H - tE)dt = \frac{1}{18}\Big(\frac{67}{4} + \frac{33}{4} + \frac{1}{4}\Big) = \frac{101}{72}.$$

Step 4. In this step we prove the rank asymptotics and compute the limit.

By Lemma 2.12(2), $N_m = \frac{18m(m+1)(2m+1)}{12} + 2m + 1 = 3m^3 + O(m^2)$.

Let $m \ge 1$ be an integer. For $0 \le j \le m$, the divisor $(mH - jE) - K_Y = (m+1)H - (j+1)E$ is nef and big on Y by **Step 3**. For $m \le j \le 2m$, the divisor $(m+1)H - (j+1)E$ is nef and big on Y^+, since $1 \le \frac{j+1}{m+1} < 2$. By the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64] and **Step 1**, $h^0(Y, mH - jE)$ equals $\chi(Y, mH - jE)$ for $0 \le j \le m$ and $\chi(Y^+, mH - jE)$ for $m \le j \le 2m$. By the Riemann–Roch theorem [Har77, Appendix A, Exercise 6.7], each of $\chi(Y, mH - jE)$ and $\chi(Y^+, mH - jE)$ is a polynomial of degree three in (m, j), whose homogeneous part of degree three is $\frac{m^3}{6}v(j/m)$, where $v(t)$ is the first, respectively the second, polynomial in the volume formula. Thus, for $0 \le j < m$, respectively $m < j < 2m$,

$$h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E) = -\frac{m^2}{6}v'(t) + O(m)$$

uniformly in j, and $-\frac{1}{3}v'(t)$ equals $t^2 + 2t$, respectively $-3t^2 + 10t - 4$. For $2m < j < 5m/2$, let $q := 5m - 2j \ge 1$. By **Step 2**, $h^0(Y, mH - jE) = \rho(q)$, and $h^0(Y, mH - (j+1)E) = \rho(q-2)$, also when $j + 1 > 5m/2$, since $\rho(-1) = 0$. Hence

$$\begin{aligned} h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E) &= \rho(q) - \rho(q-2) \\ &= 2q^2 + 2q + 1 = \frac{m^2}{2}\cdot 4(5-2t)^2 + O(m). \end{aligned}$$

This proves the rank asymptotics, with errors uniform in j since $j = O(m)$.

Let $a_j := h^0(Y, mH - jE)$ and $J_m := \lfloor 5m/2 \rfloor$. Since $a_j = 0$ for $j > J_m$, summation by parts implies

$$\sum_{j>2m} (j - 2m)(a_j - a_{j+1}) = \sum_{j=2m+1}^{J_m} a_j.$$

By **Step 2**, $a_j = \rho(5m - 2j) = \frac{m^3}{6}\cdot 2(5 - 2j/m)^3 + O(m^2)$ uniformly for $2m < j \le J_m$. There are $O(m)$ such j, and the Riemann sum of the monotone function $2(5-2t)^3$ with mesh $1/m$ differs from the integral by $O(1/m)$. Hence

$$\sum_{j=2m+1}^{J_m} a_j = \frac{m^4}{6}\int_2^{5/2} 2(5-2t)^3 dt + O(m^3).$$

Dividing by $mN_m = 3m^4 + O(m^3)$, we obtain

$$\lim_{m\to+\infty} \frac{1}{mN_m}\sum_{j>2m}(j-2m)(a_j - a_{j+1}) = \frac{1}{18}\int_2^{5/2} 2(5-2t)^3 dt = \frac{1}{18}\cdot\frac{1}{4} = \frac{1}{72}. \qquad \square$$

By Proposition 5.3 below, for a point x of a non-special line L, the lines through x other than L correspond to the points of $Z_U \cap f_x$ on $E \cong \mathbb{F}_1$, the length of Z_U at any point of $Z_U \cap f_x$ is the width of the corresponding line, and the widths of the lines intersecting L sum to four. The sum four restricts the configurations of lines through a point of L in Sections 6 and 8. Here, for

a composite of point blow-ups resolving the base points of a linear system on a smooth surface, a center is a simple free point if it has multiplicity one in the base scheme and is either a point of the surface or a point of the last exceptional curve and of no other exceptional curve [Cas00, Section 3.6].

Proposition 5.3. *Notation and conditions as in Set-up 5.1, with L non-special. Let Inc_L be the preimage of L in the universal line over $\Sigma(X)\backslash\{[L]\}$, so that the closed points of Inc_L are the pairs (M, y), where $M \neq L$ is a line and $y \in M \cap L$. Then there exists an isomorphism $\mathrm{Inc}_L \cong Z_U$ commuting with the maps to L. At a pair (M, y), if n is the width of the strict transform of M in Y, both completed local rings are isomorphic to $\mathbb{C}[t]/(t^n)$. The widths of the lines intersecting L sum to 4, and the base points of U are resolved by exactly four blow-ups at simple free points, proper or infinitely near, counted over the distinct points of $\operatorname{Supp} Z_U$. For any $x \in L$,*

$$\Sigma(X, x)\backslash\{[L]\} \cong Z_U \cap f_x,$$

and if $y \in Z_U \cap f_x$ corresponds to a line of width n and $\psi(t)$ is the image of a local parameter of L at x under $\mathcal{O}_{L,x} \to \widehat{\mathcal{O}}_{Z_U,y} \cong \mathbb{C}[t]/(t^n)$, then the local ring of $Z_U \cap f_x$ at y is $\mathbb{C}[t]/(t^n, \psi(t))$. In particular, if $\Sigma(X, x)$ consists of $[L]$ and a curvilinear double point at a line M, then the width of M is 2, 3, or 4, and ψ has order exactly two if the width is 3 or 4.

Proof. The proof is long, so we divide it into several steps. Let $\gamma : Y \to \bar{Y}$ and $\gamma^+ : Y^+ \to \bar{Y}$ be the flopping contractions of $Y \dashrightarrow Y^+$, let $\pi^+ : Y^+ \to Q$ be the blow-up of Γ, and let E^+ be the strict transform of E on Y^+. We write h also for the strict transform of h on Y^+, which is $\pi^{+*}\mathcal{O}_Q(1)$. By Theorem 2.22, the flopping curves are the strict transforms of the lines $M \neq L$ intersecting L.

Step 1. In this step we fix a line $M \neq L$ intersecting L, and we compute the base ideal of $|h|$ near the strict transform C_M of M in Y.

By Lemma 2.12(1), L and M are lines in $\mathbb{P}^{11}$, so $M \cap L$ is a reduced point y. Thus $(E \cdot C_M) = 1$, and C_M intersects E transversally in one point y_M. Let n be the width of C_M. By Definition-Proposition 2.21, the completion of the local ring of $\bar{Y}$ at $\gamma(C_M)$ is

$$R := \mathbb{C}[[u, v, z, t]]/(uv - z(z - t^n)).$$

By [Rei83, Remark (5.13)(b)] (see also [DW16, Example 3.10]), the small resolutions of $\operatorname{Spec} R$ are the blow-ups of the ideals (u, z) and $(u, z - t^n)$. The automorphism of R which sends z to $t^n - z$ exchanges (u, z) and $(u, z - t^n)$, so, possibly replacing z with $t^n - z$, we may assume that γ is the blow-up of (u, z) near C_M, and then γ^+ is the blow-up of $(u, z - t^n)$ near the flopped curve C_M^+. Thus the formal completion $\widehat{Y}$ of Y along C_M is covered by the formal spectra of $\mathbb{C}[\lambda][[a, t]]$ and $\mathbb{C}[\mu][[b, t]]$, glued by

$$\mu = \lambda^{-1}, \qquad b = \lambda^2 a - \lambda t^n,$$

with $C_M = \{a = t = 0\} = \{b = t = 0\}$, and γ is given by $(u, v, z) = (a, b, \lambda a)$ in the first chart and by $(u, v, z) = (\mu(\mu b + t^n), b, \mu b + t^n)$ in the second chart.

Let $J := (u, z - t^n) \subset R$, and let $S_\infty := \{\mu = 0\} \subset \widehat{Y}$. In the first chart $J\mathcal{O}_{\widehat{Y}} = (a, \lambda a - t^n) = (a, t^n)$, and in the second chart $J\mathcal{O}_{\widehat{Y}} = \mu \cdot (\mu b + t^n, b) = \mu \cdot (b, t^n)$. Since $(a, t^n) = (b, t^n)$ where λ is invertible,

$$J\mathcal{O}_{\widehat{Y}} = I_\Xi \cdot \mathcal{O}_{\widehat{Y}}(-S_\infty),$$

where $\Xi \subset \widehat{Y}$ is the closed formal subscheme defined by (a, t^n) and by (b, t^n). The ring $R/J \cong \mathbb{C}[[v, t]]$ is a regular domain of dimension two, so J is prime.

The kernels of the restriction maps between the Picard groups of the infinitesimal neighborhoods of C_M are quotients of the groups $H^1(C_M, \operatorname{Sym}^k N^\vee_{C_M/Y})$, $k \geq 1$, which vanish since $N^\vee_{C_M/Y} \cong \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$ or $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2)$. Hence a line bundle on $\widehat{Y}$ is determined by the degree of the restriction to C_M. Since $(h \cdot C_M) = 1 - 2 = -1 = -(S_\infty \cdot C_M)$, we obtain $\mathcal{O}_Y(h)|_{\widehat{Y}} \cong \mathcal{O}_{\widehat{Y}}(-S_\infty)$. Since

$\gamma_*\mathcal{O}_Y = \mathcal{O}_{\bar{Y}}$, the theorem on formal functions [Har77, Theorem III.11.1] implies $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = R$, so $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}(-S_\infty))$ is the ideal of the functions in R vanishing on $\gamma(S_\infty) = \{u = z - t^n = 0\}$, which is J. Since $\gamma_*\mathcal{O}_Y(h) = \gamma_*^+\mathcal{O}_{Y^+}(h)$, J is also the module of sections of $\mathcal{O}_{Y^+}(h)$ over the formal completion of Y^+ along C_M^+.

Since the automorphism $z \to t^n - z$ exchanges Y and Y^+ near $\gamma(C_M)$, line bundles on the formal completion of Y^+ along C_M^+ are also determined by their degree on C_M^+. The curve C_M^+ is mapped by π^+ isomorphically onto a line [KPZ14, Proposition 2.12(4)], so $(h \cdot C_M^+) = 1$, which is the degree of the invertible sheaf $J\mathcal{O}_{Y^+}$ on the exceptional curve C_M^+ of the blow-up of J. Thus $\mathcal{O}_{Y^+}(h) \cong J\mathcal{O}_{Y^+}$ near C_M^+. The restrictions of u and $z - t^n$ to C_M^+ are two sections of $\mathcal{O}_{\mathbb{P}^1}(1)$ without common zero, so the restriction map $J/\mathfrak{m}J \to H^0(C_M^+, \mathcal{O}_{Y^+}(h)|_{C_M^+})$ is an isomorphism of two-dimensional spaces. Since h is base point free on Y^+, the image of $H^0(Y^+, h)$ in $H^0(C_M^+, \mathcal{O}_{Y^+}(h)|_{C_M^+}) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1))$ has no base point, hence is all of $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1))$. Thus $H^0(Y, h) = H^0(Y^+, h)$ generates $J/\mathfrak{m}J$, and by Nakayama's lemma it generates J. Therefore the base ideal of $|h|$ on $\widehat{Y}$ is I_Ξ.

Step 2. In this step we compute the embedded deformations of C_M in Y and the base ideal of U at y_M.

Since $I_\Xi = (a, t^n)$ in the first chart and $I_\Xi = (b, t^n)$ in the second chart, possibly replacing (λ, a) with (μ, b), we may assume that y_M is contained in the first chart. Since E is transversal to C_M at y_M, there exist $\lambda_0 \in \mathbb{C}$ and a power series $\vartheta(a, t)$ with $\vartheta(0, 0) = 0$ such that E is defined by $\lambda = \lambda_0 + \vartheta(a, t)$ near y_M. Thus (a, t) are local coordinates of E at y_M, and by **Step 1** the base ideal of U at y_M is (a, t^n). In particular $\widehat{\mathcal{O}}_{Z_U, y_M} \cong \mathbb{C}[t]/(t^n)$.

Let A be a local Artinian $\mathbb{C}$-algebra with residue field $\mathbb{C}$, and let $C_A \subset \widehat{Y} \times \operatorname{Spec} A$ be a flat deformation of C_M. The projection $\lambda : C_A \to \mathbb{P}^1_A$ is projective, since C_A is projective over $\operatorname{Spec} A$ as an infinitesimal deformation of $C_M \cong \mathbb{P}^1$ (cf. [Har77, Exercise II.4.9]), and quasi-finite, hence finite [Har77, Exercise III.11.2]. On each affine chart the corresponding map of algebras is surjective by Nakayama's lemma, since it is surjective on the closed fiber, and then injective by the flatness of C_A over A and Nakayama's lemma. Thus $C_A \cong \mathbb{P}^1_A$. The function t on C_A is then a constant $\tau \in \mathfrak{m}_A$, and C_A is the union of the graphs $a = a(\lambda)$ and $b = b(\mu)$ with $a(\lambda) \in \mathfrak{m}_A[\lambda]$ and $b(\mu) \in \mathfrak{m}_A[\mu]$. The gluing condition reads

$$b(\lambda^{-1}) = \lambda^2 a(\lambda) - \lambda\tau^n.$$

The left side has only non-positive powers of λ, and the right side has only positive powers of λ. Hence $a = b = 0$ and $\tau^n = 0$. Conversely, any $\tau \in \mathfrak{m}_A$ with $\tau^n = 0$ defines the flat deformation $\{a = 0, t = \tau\} \cup \{b = 0, t = \tau\}$. Thus the completed local ring of the Hilbert scheme of Y at $[C_M]$ is $\mathbb{C}[\tau]/(\tau^n)$. The universal curve $\{a = 0, t = \tau\}$ intersects E in the point $(a, t, \lambda) = (0, \tau, \lambda_0 + \vartheta(0, \tau))$, which is a point of Z_U since $a = 0$ and $\tau^n = 0$. The induced map $\widehat{\mathcal{O}}_{Z_U, y_M} = \mathbb{C}[t]/(t^n) \to \mathbb{C}[\tau]/(\tau^n)$ sends t to τ, so it is an isomorphism.

Step 3. In this step we identify the deformations of C_M in Y with the deformations of (M, y) in Inc_L.

Let A be as in **Step 2**, $M_A \subset X \times \operatorname{Spec} A$ a flat family of lines with closed fiber M, and y_A a section of $M_A \to \operatorname{Spec} A$ with $y_A \subset L \times \operatorname{Spec} A$ and closed point y. Let $\ell_1, \dots, \ell_r$ be linear forms defining L in $\mathbb{P}^{11}$, and let $\ell_y \in H^0(M_A, \mathcal{O}_{M_A}(1))$ define y_A on $M_A \cong \mathbb{P}^1_A$. Each $\ell_i|_{M_A}$ vanishes on y_A, so $\ell_i|_{M_A} = c_i\ell_y$ for some $c_i \in A$, and some c_i is a unit since some $\ell_i|_M$ is non-zero. Thus the ideal of L restricts to the invertible ideal of y_A on M_A. By the universal property of the blow-up [Har77, Proposition II.7.14], $M_A \to X$ lifts uniquely to Y, and the lift is a deformation C_A of C_M whose intersection with $E \times \operatorname{Spec} A$ is the lift of y_A. Conversely, let C_A be a flat deformation of C_M in Y. By **Step 2**, $C_A \cong \mathbb{P}^1_A$, and $\pi^*H|_{C_A} \cong \mathcal{O}_{\mathbb{P}^1_A}(1)$. The restrictions of the linear forms on $\mathbb{P}^{11}$ span $H^0(C_M, \mathcal{O}_{C_M}(1))$, hence span $H^0(C_A, \mathcal{O}_{C_A}(1))$ by Nakayama's lemma. Thus π maps C_A isomorphically onto a flat family of lines M_A, and $\pi(C_A \cap (E \times \operatorname{Spec} A))$ is a section y_A of

M_A over $\operatorname{Spec} A$ contained in $L \times \operatorname{Spec} A$. By the universal property of the blow-up, the map $(M_A, y_A) \to C_A$ and the map $C_A \to (\pi(C_A), \pi(C_A \cap (E \times \operatorname{Spec} A)))$ are mutually inverse. Thus the completed local ring of Inc_L at (M, y) is isomorphic to the completed local ring of the Hilbert scheme of Y at $[C_M]$, compatibly with the maps to L.

Step 4. In this step we prove the isomorphism $\operatorname{Inc}_L \cong Z_U$ and the description of the fibers.

Outside the flopping curves, $Y \dashrightarrow Y^+$ is an isomorphism and h is base point free. Thus $\operatorname{Supp} Z_U$ is the set of the points y_M, one for each line $M \neq L$ intersecting L, and the closed points of Inc_L are the pairs (M, y) with $y = M \cap L$. Applied to the universal family of lines over Inc_L, the map $(M_A, y_A) \to C_A$ of **Step 3** followed by the intersection with E defines a morphism $\operatorname{Inc}_L \to E$ commuting with the maps to L. By **Steps 2 and 3**, the morphism $\operatorname{Inc}_L \to E$ is bijective onto Z_U on closed points and induces isomorphisms of completed local rings. Hence $\operatorname{Inc}_L \cong Z_U$, and both completed local rings at (M, y) are $\mathbb{C}[t]/(t^n)$.

Let $x \in L$. Since L is non-special, Proposition 4.1 implies that the local ring of $\Sigma(X, x)$ at $[L]$ is $\mathbb{C}$, so $[L]$ is an isolated reduced point of $\Sigma(X, x)$, and $\Sigma(X, x) \backslash \{[L]\}$ is the fiber of Inc_L over x. The fiber of Z_U over x is $Z_U \cap f_x$. Hence $\Sigma(X, x) \backslash \{[L]\} \cong Z_U \cap f_x$. If $y \in Z_U \cap f_x$ corresponds to a line of width n, then f_x is defined near y by the pullback of a local parameter of L at x, so the local ring of $Z_U \cap f_x$ at y is $\mathbb{C}[t]/(t^n, \psi(t))$.

Step 5. In this step we prove that the widths sum to four.

Let $y_M \in \operatorname{Supp} Z_U$, and let (a, t) be the coordinates of **Step 2**, so that the base ideal of U at y_M is (a, t^n). In the chart $a = a_1 t$ of the blow-up of y_M, the pullback of (a, t^n) is $t \cdot (a_1, t^{n-1})$. By induction on n, the ideal (a, t^n) is resolved by n blow-ups. The first is the blow-up of y_M, and for $1 \leq k < n$ the $(k+1)$-st is the blow-up of the origin of the chart with coordinates (a_k, t), $a = a_k t^k$, in which the pullback of (a, t^n) is $t^k(a_k, t^{n-k})$. The center of the $(k+1)$-st blow-up is a point of the exceptional curve of the k-th blow-up and of no other exceptional curve. Each center has multiplicity one in the base scheme, and the pullback of (a, t^n) is invertible. Let w_{tot} be the sum of the widths of the lines $M \neq L$ intersecting L, let $p_T : T \to E$ be the composite of the w_{tot} blow-ups resolving the base points of U, with total exceptional classes $e_1, \dots, e_{w_{\mathrm{tot}}}$, and let $\Lambda_T := p_T^*(2s + 3f) - \sum_{i=1}^{w_{\mathrm{tot}}} e_i$. Then $p_T^* U$ has fixed part $\sum_i e_i$ and a base point free moving part contained in $|\Lambda_T|$, and

$$(\Lambda_T^2) = ((2s + 3f)^2) - w_{\mathrm{tot}} = 8 - w_{\mathrm{tot}}.$$

On the other hand, $H^0(Y, h) = \pi^{+*} H^0(Q, \mathcal{O}_Q(1))$ by Lemma 5.2. Thus the morphism $\phi_T : T \to \mathbb{P}^4$ defined by the moving part of $p_T^* U$ is the composite of p_T, the birational map $E \dashrightarrow E^+$, π^+, and the embedding $Q \subset \mathbb{P}^4$, and the image of ϕ_T is $E_Q := \pi^+(E^+)$. By Lemma 5.2, $E^+ \to E_Q$ is birational, so ϕ_T is birational, and E_Q is a quartic surface (Theorem 2.22). Hence [Har77, Proposition V.3.2 and Appendix A, A4]

$$(\Lambda_T^2) = \deg E_Q = 4.$$

Thus $w_{\mathrm{tot}} = 4$, the widths of the lines intersecting L sum to 4, and the base points of U are resolved by exactly four blow-ups at simple free points.

Finally, assume that $\Sigma(X, x)$ consists of $[L]$ and a curvilinear double point at a line M of width n. By **Step 4**, the length of $\mathbb{C}[t]/(t^n, \psi(t))$, which is $\min\{n, \operatorname{ord} \psi\}$, is two. Thus $n \geq 2$, and $n \leq 4$ since the widths sum to four. If $n \geq 3$, then $\operatorname{ord} \psi = 2$. □

Construction-Proposition 5.4. Notation and conditions as in Set-up 5.1, with L non-special. Let $p_T : T \to E$ be the four blow-ups resolving Z_U (Proposition 5.3), with total exceptional classes $e_1, \dots, e_4$, let $\Lambda_T := p_T^*(2s + 3f) - \sum_i e_i$, and for integers $r \geq 0$ let

$$U_r := \operatorname{Im}\left(H^0(Y, rh) \to H^0(E, \mathcal{O}_E(r(2s + 3f)))\right).$$

Then the following hold.

(1) (a) Λ_T is nef with $\Lambda_T^2 = 4$.

(b) The morphism defined by $|\Lambda_T|$ maps T birationally onto the image of E in Q, an integral quartic surface containing Γ.
(c) At most one of the four centers is a point of the strict transform of s.

(2) (a) $D|_E \sim 5s + 7f$.
(b) $D_T := p_T^*(D|_E) - 3\sum_i e_i$ satisfies $D_T \geq 0$ and $(\Lambda_T \cdot D_T) = 7$.
(c) Exactly one irreducible component C of D_T has positive Λ_T-degree.
(d) The coefficient of C in D_T is one.
(e) The morphism defined by $|\Lambda_T|$ maps C birationally onto Γ.

(3) For integers $m \geq 1$, $W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m+j)f) \otimes I_{Z_U}^{j-m})$ for $m \leq j \leq 2m$, and $W_{m,j} = d_E^{j-2m} \cdot U_{5m-2j}$ for $2m \leq j \leq \lfloor 5m/2 \rfloor$, where d_E is the restriction to E of a section defining D; moreover $U_r \subset H^0(E, \mathcal{O}_E(r(2s+3f)) \otimes I_{Z_U}^r)$ for any $r \geq 0$.

(4) Let $x \in L$ be a closed point such that $Z_U \cap f_x$ is one reduced point p. Then f_x is not a component of $D|_E$, and the divisor $(D|_E)|_{f_x}$ on f_x has degree 5 and contains $3p$.

Proof. The proof is long, so we divide it into several steps. Let $\pi^+ : Y^+ \to Q$ be the blow-up of Γ, let E^+ be the strict transform of E on Y^+, and let $E_Q := \pi^+(E^+)$ be the image of E in Q. We write h also for $\pi^{+*}\mathcal{O}_Q(1)$.

Step 1. In this step we describe the blow-ups over a point of $\operatorname{Supp} Z_U$ and prove (1).

Let $y \in \operatorname{Supp} Z_U$ and let n be the width of the corresponding line. By Proposition 5.3, $\widehat{\mathcal{O}}_{Z_U,y} \cong \mathbb{C}[t]/(t^n)$. The ring $\mathbb{C}[t]/(t^n)$ has embedding dimension at most one, so I_{Z_U} contains an element a of $\widehat{\mathcal{O}}_{E,y}$ which is part of a system of coordinates (a,t), and then $I_{Z_U}\widehat{\mathcal{O}}_{E,y} = (a, t^n)$. Over y, the morphism p_T is the composite of the blow-ups of the origins of the charts with coordinates (a_l, t), $a = a_l t^l$, $0 \leq l \leq n-1$, and the strict transform G_l of the l-th exceptional curve defines the monomial valuation v_l with $v_l(a) = l$ and $v_l(t) = 1$. The total exceptional classes over y sum to $\sum_{l=1}^n lG_l$. Since $v_l(\varphi)$ is the minimum of $l\alpha + \beta$ over the monomials $a^\alpha t^\beta$ of φ, and a monomial satisfies $l\alpha + \beta \geq kl$ for $1 \leq l \leq n$ if and only if it belongs to $(a, t^n)^k$, we obtain, for any integer $k \geq 0$ and any $\varphi \in \widehat{\mathcal{O}}_{E,y}$,

$$p_T^* \operatorname{div} \varphi - k\sum_i e_i \geq 0 \text{ over } y \iff v_l(\varphi) \geq kl \text{ for } 1 \leq l \leq n \iff \varphi \in I_{Z_U}^k \widehat{\mathcal{O}}_{E,y}. \tag{5.2}$$

In particular $I_{Z_U}\mathcal{O}_T = \mathcal{O}_T(-\sum_i e_i)$, so p_T^*U has fixed part $\sum_i e_i$ and a base point free moving part contained in $|\Lambda_T|$. Thus Λ_T is nef and $(\Lambda_T^2) = 8 - 4 = 4$.

Since $K_E = -2s - 3f$ [Har77, Corollary V.2.11] and $K_T = p_T^*K_E + \sum_i e_i$ [Har77, Proposition V.3.3], we have $\Lambda_T = -K_T$. Since $-2K_T$ is nef and big, the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64] and the Riemann–Roch theorem on the rational surface T [Har77, Theorem V.1.6] imply $h^0(T, \Lambda_T) = 1 + (\Lambda_T^2) = 5$. On the other hand, $h^0(Y, h) = h^0(Q, \mathcal{O}_Q(1)) = 5$ and $H^0(Y, h - E) = H^0(Y, H - 3E) = 0$ by Lemma 5.2, so $\dim U = 5$. Thus $|\Lambda_T|$ is the moving part of p_T^*U. Let $\phi_T : T \to \mathbb{P}^4$ be the morphism defined by $|\Lambda_T|$. As U is the restriction of $\pi^{+*}H^0(Q, \mathcal{O}_Q(1))$, the morphism ϕ_T is the composite of p_T, the birational map $E \dashrightarrow E^+$, π^+, and the embedding $Q \subset \mathbb{P}^4$. By Lemma 5.2, $E^+ \to E_Q$ is birational, and E_Q is a quartic surface containing Γ (Theorem 2.22). Hence E_Q is integral as the image of E, and ϕ_T maps T birationally onto E_Q.

Let k_s be the number of centers of p_T contained in the strict transform of s. Since s is smooth, the strict transform s_T of s on T is $p_T^*s - \sum_{i \in I_s} e_i$, where I_s is the set of indices of the k_s centers. As $(\Lambda_T \cdot p_T^*s) = ((2s+3f) \cdot s) = 1$ and $(\Lambda_T \cdot e_i) = 1$,

$$0 \leq (\Lambda_T \cdot s_T) = 1 - k_s.$$

Thus $k_s \leq 1$, which completes the proof of (1).

Step 2. In this step we compare $|h|$ on Y and on Y^+ by a common resolution.

Let $\gamma : Y \to \bar{Y}$ and $\gamma^+ : Y^+ \to \bar{Y}$ be the flopping contractions of $Y \dashrightarrow Y^+$, and let $g : \widetilde{Y} \to Y$ and $g^+ : \widetilde{Y} \to Y^+$ be birational morphisms from a smooth projective threefold

$\widetilde{Y}$ with $\gamma \circ g = \gamma^+ \circ g^+$. For a divisor M_0 on Y with strict transform M_0^+ on Y^+, we have $H^0(Y, \mathcal{O}_Y(M_0)) = H^0(Y^+, \mathcal{O}_{Y^+}(M_0^+))$ as spaces of rational functions, since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one between normal varieties.

Fix $h_0 \in |h|$ on Y, and let $\Delta := g^* h_0 - g^{+*} h_0^+$. For any rational function φ with $\operatorname{div} \varphi + h_0 \geq 0$,

$$\operatorname{div} \varphi + g^* h_0 = (\operatorname{div} \varphi + g^{+*} h_0^+) + \Delta,$$

and the divisors $\operatorname{div} \varphi + g^{+*} h_0^+$ form the base point free linear system $g^{+*}|h|$. Thus $\Delta \geq 0$, and Δ is the fixed part of $g^*|h|$. Moreover $g_* \Delta = 0$, since $g_* g^{+*} h_0^+ = h_0$.

The variety $\bar{Y}$ has hypersurface singularities (Definition-Proposition 2.21), so $K_{\bar{Y}}$ is Cartier, and $K_Y = \gamma^* K_{\bar{Y}}$ and $K_{Y^+} = \gamma^{+*} K_{\bar{Y}}$ since γ and γ^+ are small. Let $B_0 \in |-K_{\bar{Y}}|$. Then $\gamma^* B_0 \in |H - E|$, $\gamma^{+*} B_0 \in |3h - D'|$, and $g^* \gamma^* B_0 = g^{+*} \gamma^{+*} B_0$. For integers $a, b \geq 0$, let $M_0 := a\gamma^* B_0 + b h_0$, which is linearly equivalent to $a(H - E) + bh$. Then $g^* M_0 = g^{+*} M_0^+ + b\Delta$, so for any section ξ of $a(H - E) + bh$ on Y, given by a rational function φ with $\operatorname{div} \varphi + M_0 \geq 0$,

$$g^*(\operatorname{div} \varphi + M_0) = (\operatorname{div} \varphi + g^{+*} M_0^+) + b\Delta \geq b\Delta. \tag{5.3}$$

Since $D \sim 3h - (H - E)$ on Y and $\gamma^{+*} B_0 \sim 3h - D'$ on Y^+,

$$g^* D \sim 3 g^{+*} h_0^+ + 3\Delta - g^{+*} \gamma^{+*} B_0 \sim g^{+*} D' + 3\Delta.$$

Both $g^* D \geq 0$ and $g^{+*} D' + 3\Delta \geq 0$ hold, and $h^0(\widetilde{Y}, g^* D) = h^0(Y, D) = 1$, so $g^* D = g^{+*} D' + 3\Delta$.

Let $E_{\widetilde{Y}}$ be the strict transform of E on $\widetilde{Y}$, let E'' be a resolution of the irreducible component of $E_{\widetilde{Y}} \times_E T$ dominating E, and let $\alpha : E'' \to \widetilde{Y}$ and $r_T : E'' \to T$ be the induced morphisms, so that $g \circ \alpha = p_T \circ r_T$ and r_T is birational. Since Δ is g-exceptional and $g^+(E_{\widetilde{Y}}) = E^+ \not\subset D'$, neither Δ nor $g^{+*} D'$ contains $E_{\widetilde{Y}}$. Since $H^0(Y, h - E) = 0$, the restriction map $H^0(Y, h) \to U$ is an isomorphism, so $\alpha^* g^* |h| = r_T^* p_T^* U$. Computing through g^+, the linear system $\alpha^* g^* |h|$ has fixed part $\alpha^* \Delta$ and base point free moving part $\alpha^* g^{+*} |h|$. Computing through **Step 1**, the linear system $r_T^* p_T^* U$ has fixed part $r_T^* \sum_i e_i$ and base point free moving part $r_T^* |\Lambda_T|$. Comparing the fixed parts and the moving parts, we obtain

$$\alpha^* \Delta = r_T^* \sum_i e_i, \qquad r_T^* |\Lambda_T| = \alpha^* g^{+*} |h|. \tag{5.4}$$

Step 3. In this step we prove (2).

By (5.1), $D|_E \sim 5s + 7f$. By $g^* D = g^{+*} D' + 3\Delta$ and (5.4),

$$r_T^* p_T^* (D|_E) = \alpha^* g^* D = \alpha^* g^{+*} D' + 3 r_T^* \sum_i e_i,$$

so $r_T^* D_T = \alpha^* g^{+*} D' \geq 0$, and $D_T = r_{T*} r_T^* D_T \geq 0$. Since $(\Lambda_T \cdot p_T^*(5s + 7f)) = ((2s + 3f) \cdot (5s + 7f)) = 19$ and $(\Lambda_T \cdot e_i) = 1$,

$$(\Lambda_T \cdot D_T) = 19 - 3 \cdot 4 = 7.$$

By (5.4), $\phi_T \circ r_T = \pi^+ \circ g^+ \circ \alpha$ up to the embedding $Q \subset \mathbb{P}^4$. Any component of D_T is $r_T(C'')$ for a component C'' of $\alpha^* g^{+*} D'$ which is not contracted by r_T, and $\pi^+(g^+(\alpha(C''))) \subset \pi^+(D') = \Gamma$. Thus ϕ_T maps any component of D_T into Γ. A component C of D_T with $(\Lambda_T \cdot C) > 0$ is mapped onto Γ, and $(\Lambda_T \cdot C) = 7 \deg(C \to \Gamma)$ since $\deg \Gamma = 7$. Since $(\Lambda_T \cdot D_T) = 7$, exactly one component C of D_T has positive Λ_T-degree, the coefficient of C in D_T is one, and $\deg(C \to \Gamma) = 1$. This proves (2).

Step 4. In this step we prove (3).

Let $m \leq j \leq 2m$. Since $mH - jE \sim (2m - j)(H - E) + (j - m)h$, (5.3) with $a = 2m - j$ and $b = j - m$ and (5.4) imply, for any $\xi \in H^0(Y, mH - jE)$ with $\xi|_E \neq 0$,

$$r_T^* p_T^* \operatorname{div}(\xi|_E) = \alpha^* g^* \operatorname{div} \xi \geq (j - m) \alpha^* \Delta = (j - m) r_T^* \sum_i e_i.$$

Applying r_{T*}, we obtain $p_T^*\operatorname{div}(\xi|_E) \geq (j-m)\sum_i e_i$. By (5.2) and the faithful flatness of completion, $\xi|_E$ is a section of $\mathcal{O}_E(js+(m+j)f)\otimes I_{Z_U}^{j-m}$. With $a=0$ and $b=r$, (5.3), (5.4), and (5.2) imply $U_r \subset H^0(E,\mathcal{O}_E(r(2s+3f))\otimes I_{Z_U}^r)$. For $2m \leq j \leq \lfloor 5m/2 \rfloor$, Lemma 5.2 implies $H^0(Y,mH-jE) = \sigma_D^{j-2m}\cdot H^0(Y,(5m-2j)h)$, where σ_D defines D. Restricting to E, we obtain $W_{m,j} = d_E^{j-2m}\cdot U_{5m-2j}$, with $d_E = \sigma_D|_E \neq 0$ since $D \neq E$.

Step 5. In this step we prove (4).

Since $Z_U \cap f_x$ is the reduced point p, f_x contains no point of $\operatorname{Supp} Z_U$ other than p, and $I_{Z_U}\mathcal{O}_{f_x,p} = \mathfrak{m}_{f_x,p}$. Let (a,t) be the coordinates of **Step 1** at p, so that $I_{Z_U}\widehat{\mathcal{O}}_{E,p} = (a,t^n)$. If $n \geq 2$, then, since $Z_U \cap f_x$ is reduced at p, the local ring $\mathbb{C}[t]/(t^n,\psi(t))$ of Proposition 5.3 has length one, so ψ has order one. As the local equation of f_x maps to $\psi(t)$ in $\widehat{\mathcal{O}}_{Z_U,p} \cong \mathbb{C}[t]/(t^n)$, it is congruent to a unit times t modulo $(a)+\mathfrak{m}_p^2$. Hence the strict transform of f_x does not contain the second center over p, which is the tangent direction of $\{a=0\}$. Thus the strict transform f_T of f_x on T is $p_T^*f_x - e_{i_0}$, where e_{i_0} is the total transform of the blow-up of p, and

$$(\Lambda_T \cdot f_T) = ((2s+3f)\cdot f) - 1 = 1.$$

Suppose that f_x is a component of $D|_E$. Then f_T is a component of D_T, since $p_T^*(D|_E) \geq p_T^*f_x \geq f_T$ and the classes e_i are supported on p_T-exceptional curves. As $(\Lambda_T\cdot f_T) = 1 > 0$, we have $f_T = C$ by (2)(c), so ϕ_T maps f_T birationally onto Γ by (2)(e). This is impossible, since $f_T \cong \mathbb{P}^1$ and Γ has genus two. Thus f_x is not a component of $D|_E$, and $((D|_E)\cdot f_x) = ((5s+7f)\cdot f) = 5$. By (2) and (5.2), $d_E \in I_{Z_U}^3\widehat{\mathcal{O}}_{E,p}$. Since $I_{Z_U}\mathcal{O}_{f_x,p} = \mathfrak{m}_{f_x,p}$, the restriction $d_E|_{f_x}$ belongs to $\mathfrak{m}_{f_x,p}^3$, so $(D|_E)|_{f_x}$ contains $3p$. □

We now treat non-special lines. Theorem 5.5 is the non-special case of Corollary 5.10.

Theorem 5.5. *Let X be a smooth prime Fano threefold of genus 10 and $L \subset X$ a non-special line. Then $\delta_L(X) \geq 63000/62779$. In particular $A_X(F) > S_X(F)$ for any prime divisor F over X with $\operatorname{center}_X F = L$.*

Proof. The proof is long, so we divide it into several steps. We use the notation of Set-up 5.1. Since L is non-special, $E \cong \mathbb{F}_1$, with negative section s and fiber class f. For integers $m \geq 1$ and $j \geq 0$ let $r_{m,j} := \dim W_{m,j}$.

Step 1. In this step we reduce the theorem to an upper bound for $S(W;C)$, where C runs through the integral curves $C \subset E$ with $\pi(C) = L$.

The morphism π is the blow-up of the smooth curve L in the smooth threefold X. Thus Y and E are smooth, $-E$ is π-ample, and (Y,E) is plt [KM98, Corollary 2.31(3)]. Hence E is a prime divisor over X of plt type with $\operatorname{center}_X E = L$, π is the plt extraction of E, and $\operatorname{Diff}_E(0) = 0$ by adjunction. By Definition 2.8, the series $W_{\bullet,\bullet}$ of Set-up 5.1 is the refinement by E of the complete series of H. By Theorem 2.9 with $Z = L$,

$$\delta_L(X) \geq \min\left\{\frac{A_X(E)}{S_X(E)},\ \inf_{Z'}\delta_{Z'}(E,0;W)\right\},$$

where Z' runs through the irreducible closed subsets of E with $\pi(Z') = L$. By Lemma 5.2, $A_X(E) = 2$ and $S_X(E) = 101/72$, so $A_X(E)/S_X(E) = 144/101$.

An irreducible closed subset $Z' \subset E$ with $\pi(Z') = L$ is either E or an integral curve $C \subset E$ with $\pi(C) = L$. No prime divisor over E has center E, so $\delta_E(E,0;W) = +\infty$. Let $C \subset E$ be an integral curve with $\pi(C) = L$. A projective birational morphism onto the smooth surface E is an isomorphism over the complement of finitely many points, so the only prime divisor over E whose center contains C is C itself. Since $A_{E,0}(C) = 1$, we obtain $\delta_C(E,0;W) = 1/S(W;C)$, where $S(W;C) := S(W;\operatorname{ord}_C)$. Therefore

$$\delta_L(X) \geq \min\left\{\frac{144}{101},\ \inf_C \frac{1}{S(W;C)}\right\}, \tag{5.5}$$

where C runs through the integral curves $C \subset E$ with $\pi(C) = L$. For such C we have $(C \cdot f) \geq 1$, because $\pi|_C : C \to L$ is surjective.

We recall how $S_X(E)$ is computed from the dimensions of the spaces $W_{m,j}$, and $S(W;C)$ from the dimensions of their subspaces of sections vanishing to a given order along C. The kernel of the restriction map from $H^0(Y, mH - jE)$ to $H^0(E, (mH - jE)|_E)$ is $H^0(Y, mH - (j+1)E)$, and $H^0(Y, mH - jE) = 0$ for $j > 5m/2$ (Lemma 5.2). Since $H^0(Y, mH) = H^0(X, mH)$, we obtain

$$r_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E), \qquad \sum_{j \geq 0} r_{m,j} = N_m. \tag{5.6}$$

Thus the jumping numbers of ord_E on $H^0(X, mH)$ are the integers j, each counted $r_{m,j}$ times. By Definition 2.8 and Lemma 5.2,

$$\lim_{m \to +\infty} \frac{1}{mN_m} \sum_{j \geq 0} j r_{m,j} = S_X(E) = \frac{101}{72}. \tag{5.7}$$

For an integral curve $C \subset E$ and an integer $k \geq 0$ let

$$W^{\geq k}_{m,j}(C) := \{\sigma \in W_{m,j} : \mathrm{ord}_C \sigma \geq k\}.$$

The jumping numbers of ord_C on $W_{m,j}$ are non-negative integers, and their sum is

$$\sum_{k \geq 1} \dim W^{\geq k}_{m,j}(C).$$

Hence

$$S_m(W;C) = \frac{1}{mN_m} \sum_{j \geq 0} \sum_{k \geq 1} \dim W^{\geq k}_{m,j}(C), \tag{5.8}$$

and $S(W;C) = \lim_{m \to +\infty} S_m(W;C)$ [AZ22, Lemma 2.9].

Step 2. In this step we show that $S(W;C) \leq 101/144$ for any integral curve $C \subset E$ with $(C \cdot f) \geq 2$.

Let $a := (C \cdot f) \geq 2$ and let $\sigma \in W_{m,j}$ be non-zero. By (5.1), $\mathrm{div}(\sigma) \geq 0$ is a divisor of class $js + (m+j)f$. Write $\mathrm{div}(\sigma) = (\mathrm{ord}_C \sigma)C + R$ with $R \geq 0$. Since f is nef, $(R \cdot f) \geq 0$, and

$$j = (\mathrm{div}(\sigma) \cdot f) = a \,\mathrm{ord}_C \sigma + (R \cdot f) \geq a \,\mathrm{ord}_C \sigma.$$

Hence each jumping number of ord_C on $W_{m,j}$ is at most j/a. By (5.8),

$$S_m(W;C) \leq \frac{1}{a} \cdot \frac{1}{mN_m} \sum_{j \geq 0} j r_{m,j}.$$

Letting $m \to +\infty$, (5.7) implies $S(W;C) \leq 101/(72a) \leq 101/144$.

Step 3. In this step we let $C \subset E$ be an integral curve with $(C \cdot f) = 1$, and we bound the terms of (5.8) with $j < 2m$.

Write $C \sim s + bf$. If $C = s$, then $b = 0$. Otherwise $b - 1 = (C \cdot s) \geq 0$. Hence $b \geq 0$. For integers $a_0 \geq 0$ and $b_0 \geq a_0$, the pushforward of $\mathcal{O}_E(a_0 s + b_0 f)$ to $L \cong \mathbb{P}^1$ is $\bigoplus_{i=0}^{a_0} \mathcal{O}_{\mathbb{P}^1}(b_0 - i)$ [Har77, Proposition V.2.8, Exercise III.8.4(a), and Exercise II.5.1(d)], so

$$h^0(E, a_0 s + b_0 f) = \sum_{i=0}^{a_0} (b_0 - i + 1) = \frac{(a_0 + 1)(2b_0 - a_0 + 2)}{2}. \tag{5.9}$$

Let $k \geq 0$ be an integer and let $\sigma \in W^{\geq k}_{m,j}(C)$ be non-zero. Then $\mathrm{div}(\sigma) - kC \geq 0$ is a divisor of class $(j-k)s + (m+j-kb)f$. Since kbf is effective, division by the k-th power of a section defining C, followed by multiplication by a section defining a member of $|kbf|$, embeds $W^{\geq k}_{m,j}(C)$ into $H^0(E, (j-k)s + (m+j)f)$. Moreover the argument of **Step 2** with $a = 1$ implies $\mathrm{ord}_C \sigma \leq j$ for any non-zero $\sigma \in W_{m,j}$. Therefore

$$\dim W^{\geq k}_{m,j}(C) \leq \min\left\{r_{m,j}, h^0(E, (j-k)s + (m+j)f)\right\} \quad \text{for } 0 \leq k \leq j, \tag{5.10}$$

and $W^{\geq k}_{m,j}(C) = 0$ for $k > j$. Let

$$\varphi_1(t,u) := (t-u)(2+t+u) = t^2 + 2t - 2u - u^2.$$

By (5.9) with $a_0 = j-k$ and $b_0 = m+j$,

$$h^0(E,(j-k)s+(m+j)f) = \frac{(j-k+1)(2m+j+k+2)}{2} = \frac{m^2}{2}\varphi_1\Big(\frac{j}{m},\frac{k}{m}\Big) + O(m)$$

uniformly for $0 \leq k \leq j \leq 5m/2$. By Lemma 5.2, $r_{m,j} = \frac{m^2}{2}w(j/m) + O(m)$ uniformly for $m < j < 2m$, where $w(t) = -3t^2 + 10t - 4$. For $j \leq m$ we use only the second term of (5.10), and for $m < j < 2m$ we use the minimum. The corresponding integrals are

$$I_1 := \int_0^1 \int_0^t \varphi_1(t,u)dudt, \qquad I_2 := \int_1^2 \int_0^t \min\{w(t), \varphi_1(t,u)\}dudt.$$

Since $\int_0^t \varphi_1(t,u)du = t^2 + \frac{2}{3}t^3$, we have

$$I_1 = \frac{1}{2}, \qquad \int_1^2 \int_0^t \varphi_1(t,u)dudt = \frac{29}{6}. \tag{5.11}$$

For $1 \leq t \leq 2$ let $d := t-1$. Then

$$\varphi_1(t,u) - w(t) = 4d^2 - 2u - u^2,$$

and $\frac{6}{5}d^2 \leq 1 + d = t$, since $\frac{6}{5}d^2 - d - 1 < 0$ for $0 \leq d \leq 1$. We bound $\min\{w(t), \varphi_1(t,u)\}$ by $w(t)$ for $0 \leq u \leq \frac{6}{5}d^2$ and by $\varphi_1(t,u)$ for $\frac{6}{5}d^2 \leq u \leq t$. Thus

$$\begin{aligned}\int_0^t \min\{w(t), \varphi_1(t,u)\}du &\leq \int_0^t \varphi_1(t,u)du - \int_0^{6d^2/5} (4d^2 - 2u - u^2)du \\ &= \int_0^t \varphi_1(t,u)du - \frac{84}{25}d^4 + \frac{72}{125}d^6.\end{aligned}$$

Integrating over $1 \leq t \leq 2$ and using (5.11), we obtain

$$I_2 \leq \frac{29}{6} - \frac{84}{125} + \frac{72}{875} = \frac{29}{6} - \frac{516}{875}. \tag{5.12}$$

Step 4. In this step we keep C as in **Step 3**, and we bound the terms of (5.8) with $j \geq 2m$.

Let $2m \leq j \leq 5m/2$, $q := 5m - 2j$, and $k_0 := j - 2m$. By Construction-Proposition 5.4(3), $W_{m,j} = d_E^{k_0} \cdot U_q$, where U_q is the space U_r of that statement for $r = q$ and d_E is the restriction to E of a section σ_D defining D. Since D is a prime divisor different from E (Lemma 5.2), $d_E \neq 0$, and multiplication by $d_E^{k_0}$ is injective on sections of line bundles on the integral surface E. Hence $\dim U_q = \dim W_{m,j} = r_{m,j}$. For $\tau \in U_q$ we have $\operatorname{ord}_C\big(d_E^{k_0}\tau\big) = k_0 \operatorname{mult}_C(D|_E) + \operatorname{ord}_C \tau$. By (5.1), $D|_E \geq 0$ is a divisor of class $5s + 7f$. Since f is nef and $(C \cdot f) = 1$, we obtain $\operatorname{mult}_C(D|_E) \leq (D|_E \cdot f) = 5$. Let $U_q^{\geq k}(C) := \{\tau \in U_q : \operatorname{ord}_C \tau \geq k\}$. Then

$$\sum_{k\geq 1} \dim W^{\geq k}_{m,j}(C) = k_0 \operatorname{mult}_C(D|_E) r_{m,j} + \sum_{k\geq 1} \dim U_q^{\geq k}(C). \tag{5.13}$$

The argument of **Step 3**, with $js + (m+j)f$ replaced by $2qs + 3qf$, implies

$$\dim U_q^{\geq k}(C) \leq \min\big\{r_{m,j}, h^0(E,(2q-k)s+3qf)\big\} \quad \text{for } 0 \leq k \leq 2q,$$

and $U_q^{\geq k}(C) = 0$ for $k > 2q$. Let

$$\varphi_2(c,u) := (2c-u)(4c+u) = 8c^2 - 2cu - u^2.$$

By (5.9) with $a_0 = 2q - k$ and $b_0 = 3q$,

$$h^0(E,(2q-k)s+3qf) = \frac{(2q-k+1)(4q+k+2)}{2} = \frac{m^2}{2}\varphi_2\Big(\frac{q}{m},\frac{k}{m}\Big) + O(m)$$

uniformly for $0 \le k \le 2q$. By Lemma 5.2, $r_{m,j} = \frac{m^2}{2} \cdot 4c^2 + O(m)$ uniformly for $2m < j < 5m/2$, where $c := q/m = 5 - 2t$ and $t := j/m$. The corresponding integral is

$$I_3 := \int_2^{5/2} \int_0^{2(5-2t)} \min\left\{4(5-2t)^2, \varphi_2(5-2t, u)\right\} du dt.$$

For $c \ge 0$ we have $\int_0^{2c} \varphi_2(c,u)du = \frac{28}{3}c^3$ and $\frac{6}{5}c \le 2c$. Bounding the minimum by $4c^2$ for $0 \le u \le \frac{6}{5}c$ and by $\varphi_2(c,u)$ for $\frac{6}{5}c \le u \le 2c$, we obtain

$$\int_0^{2c} \min\{4c^2, \varphi_2(c,u)\}du \le \frac{28}{3}c^3 - \int_0^{6c/5} (4c^2 - 2cu - u^2)du = \Big(\frac{28}{3} - \frac{348}{125}\Big)c^3.$$

Since $\int_2^{5/2}(5-2t)^3 dt = \frac{1}{8}$,

$$I_3 \le \frac{1}{8}\Big(\frac{28}{3} - \frac{348}{125}\Big) = \frac{7}{6} - \frac{87}{250}. \tag{5.14}$$

For the first term of (5.13), Lemma 5.2 implies

$$\lim_{m \to +\infty} \frac{1}{mN_m} \sum_{j > 2m} (j - 2m) \operatorname{mult}_C(D|_E) r_{m,j} = \frac{\operatorname{mult}_C(D|_E)}{72} \le \frac{5}{72}. \tag{5.15}$$

Step 5. In this step we conclude the proof.

Let $C \subset E$ be an integral curve with $(C \cdot f) = 1$. In **Steps 3 and 4**, each bound for a summand of (5.8) or of the last sum in (5.13) has the form $\frac{m^2}{2}G(j/m, k/m) + O(m)$ with G continuous and piecewise polynomial, and there exist $O(m^2)$ pairs (j,k). The indices $j \in \{m, 2m, 5m/2\}$, which are excluded from the uniform rank estimates, contribute $O(m^3)$, because $r_{m,j} \le h^0(E, js + (m+j)f) = O(m^2)$ and $k \le j \le 5m/2$. Since $mN_m = 3m^4 + O(m^3)$ by Lemma 5.2, replacing the double sums by Riemann integrals changes $S_m(W;C)$ by $O(1/m)$, and the factor $\frac{m^2}{2} \cdot m^2/(mN_m)$ tends to $\frac{1}{6}$. By (5.8), (5.13), and (5.15), we obtain

$$S(W;C) \le \frac{1}{6}(I_1 + I_2 + I_3) + \frac{\operatorname{mult}_C(D|_E)}{72}.$$

By (5.11), (5.12), and (5.14),

$$I_1 + I_2 + I_3 \le \frac{1}{2} + \frac{29}{6} - \frac{516}{875} + \frac{7}{6} - \frac{87}{250} = \frac{4867}{875}.$$

Since $\operatorname{mult}_C(D|_E) \le 5$,

$$S(W;C) \le \frac{4867}{5250} + \frac{5}{72} = \frac{62779}{63000}. \tag{5.16}$$

The bound (5.16) does not depend on b in $C \sim s + bf$.

By **Step 2** and (5.16), $S(W;C) \le \max\{101/144, 62779/63000\} = 62779/63000$ for any integral curve $C \subset E$ with $\pi(C) = L$. By (5.5),

$$\delta_L(X) \ge \min\left\{\frac{144}{101}, \frac{63000}{62779}\right\} = \frac{63000}{62779}.$$

Finally, let F be a prime divisor over X with $\operatorname{center}_X F = L$. Then $A_X(F)/S_X(F) \ge \delta_L(X) \ge 63000/62779 > 1$, so $A_X(F) > S_X(F)$. □

We now turn to special lines. For a special line L, the refinement of the anti-canonical series by E alone does not suffice for the prime divisors over X whose center on Y is the negative section s of E: along s it only bounds the relevant coefficient by a number larger than one (see the proof of Proposition 5.9). For the prime divisors over X with center s on Y, we use the exceptional divisor P of $\operatorname{Bl}_s Y \to Y$ (Set-up 5.1). Proposition 5.9 gives $\delta_L(X) \ge 1$ if $S_X(P) \le 173/72$ and $\delta_L(X) \ge 72/71$ if $S_X(P) \le 43/18$, so we compute $S_X(P) = 43/18$ exactly in Proposition 5.7.

For $0 \le k \le 2$, the volume of $H - kP$ is computed on the blow-up of X along the non-reduced conic supported on L. For $2 \le k \le 4$, we reverse the order of the blow-ups on the quadric side of

the link: we first blow up the singular line of the image of E in Q, and then the strict transform of Γ. The following construction describes both blow-ups.

Construction-Proposition 5.6. Notation and conditions as in Set-up 5.1, with L special. Then the image E_Q of E in Q is a non-normal quartic surface containing Γ whose singular locus is a line $\Lambda \subset Q$, and the strict transform of Λ in Y^+ is the flopped curve of s. Let $Q' := \mathrm{Bl}_\Lambda Q$ with exceptional divisor $P_{Q'}$, write h also for the pullback of $\mathcal{O}_Q(1)$, and let $l := h - P_{Q'}$. Then the following hold.

(1) (a) The projection from Λ induces a $\mathbb{P}^1$-bundle $p : Q' \to \mathbb{P}^2$, and $l = p^*\mathcal{O}_{\mathbb{P}^2}(1)$.
(b) The surface $P_{Q'}$ is isomorphic to $\mathbb{F}_1$, with negative section g, and η is the class of a fiber of the ruling $\varrho : P_{Q'} \to \Lambda$.
(c) We have $h|_{P_{Q'}} = \eta$ and $P_{Q'}|_{P_{Q'}} = -g$.
(d) The morphism p contracts g to a point.

(2) (a) The strict transform $\bar{E}_Q$ of E_Q in Q' is $p^{-1}(\kappa)$ for a smooth conic $\kappa \subset \mathbb{P}^2$.
(b) The morphism $\bar{E}_Q \to E_Q$ is the normalization.
(c) Let $\bar{f}$ be the class of a fiber of $\bar{E}_Q \to \kappa$. Then exactly one of the following holds.
(i) $\bar{E}_Q \cong \mathbb{F}_0$, $\bar{\sigma}$ is the class of the other ruling, $h|_{\bar{E}_Q} = \bar{\sigma} + 2\bar{f}$, and $P_{Q'}|_{\bar{E}_Q} = \bar{\sigma}$.
(ii) $\bar{E}_Q \cong \mathbb{F}_2$, $\bar{\sigma}$ is the class of the negative section, $h|_{\bar{E}_Q} = \bar{\sigma} + 3\bar{f}$, and $P_{Q'}|_{\bar{E}_Q} = \bar{\sigma} + \bar{f}$.

(3) (a) The strict transform Γ' of Γ in Q' is contained in $\bar{E}_Q$, and the class of Γ' is $2\bar{\sigma} + 3\bar{f}$ if $\bar{E}_Q \cong \mathbb{F}_0$ and $2\bar{\sigma} + 5\bar{f}$ if $\bar{E}_Q \cong \mathbb{F}_2$.
(b) The morphism $\Gamma' \to \Gamma$ is an isomorphism.
(c) The scheme $\Xi := \Gamma' \cap P_{Q'}$ has length three and maps isomorphically onto $\Gamma \cap \Lambda$.
(d) The scheme $\Gamma' \cap g$ has length at most two.

(4) Let $Q'' := \mathrm{Bl}_{\Gamma'} Q'$ with exceptional divisor D'', write h and l also for their pullbacks to Q'', let $\bar{E}_{Q''}$ be the strict transform of $\bar{E}_Q$, and write P for the strict transform of $P_{Q'}$. By (c), P is the divisor P of Set-up 5.1.
(a) The divisor $-K_{Q''} = 3h - P - D''$ is base point free.
(b) We have $\bar{E}_{Q''} \sim 2h - 2P - D''$.
(c) The prime divisors on Q'' that are exceptional over X are exactly $\bar{E}_{Q''}$, corresponding to E, and the strict transform of $P_{Q'}$, which as a prime divisor over X is the divisor P of Set-up 5.1. Hence $A_X(P) = 3$.
(d) With $H_{Q''} := 5h - 2D'' - P$, the strict transform of a general member of $|H|$, we have

$$\{\phi \in H^0(X, mH) : \mathrm{ord}_P(\phi) \geq j\} = H^0(Q'', mH_{Q''} - jP)$$

for all integers $m, j \geq 0$, as spaces of rational functions.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the assertions on E_Q and Λ.

By Theorem 2.22, E_Q is a quartic surface containing Γ, and E_Q is non-normal since L is special. By Lemma 5.2, $E_Q \in |\mathcal{O}_Q(2)|$. By [KPZ14, Proposition 2.2(2)(b) and (b')], the singular locus of E_Q is a line Λ, and Λ is a 3-secant line of Γ. Let $Y \to \bar{Y}$ and $Y^+ \to \bar{Y}$ be the flopping contractions, let $s^+ \subset Y^+$ be the flopped curve of s, and let Λ^+ be the strict transform of Λ in Y^+. By [KPZ14, Proposition 2.12(4) and the proof of Theorem 2.13], the image of s^+ in Q is a line along which E_Q is non-normal. The non-normal locus of E_Q is contained in the singular locus Λ of E_Q [KPZ14, Proposition 2.2(2)], so the image of s^+ in Q is Λ. Since s^+ is irreducible and maps onto $\Lambda \not\subset \Gamma$, s^+ is the strict transform of Λ, that is, $\Lambda^+ = s^+$. Thus the images of Λ^+ and s in $\bar{Y}$ are both equal to one point $\bar{y} \in \bar{Y}$. Since the flopping curves are pairwise disjoint (Theorem 2.22), the fiber of $Y \to \bar{Y}$ over $\bar{y}$ is s.

Step 2. In this step we prove (1).

Since the orthogonal group of a non-degenerate quadratic form in five variables acts transitively on the two-dimensional isotropic subspaces of $\mathbb{C}^5$, that is, on the lines in Q, there exist homogeneous coordinates $[x : y : z : w : v]$ on $\mathbb{P}^4$ such that

$$Q = (q_Q = 0), \quad q_Q := xv - yw - z^2, \qquad \Lambda = (z = w = v = 0).$$

The morphism $p : Q' \to \mathbb{P}^2$ of (1)(a) is induced by the projection $[x : y : z : w : v] \to [z : w : v]$ from Λ. The points of the plane spanned by Λ and $[0 : 0 : z_0 : w_0 : v_0]$ are $[x : y : tz_0 : tw_0 : tv_0]$, and on the plane

$$q_Q(x, y, tz_0, tw_0, tv_0) = t(v_0x - w_0y - z_0^2t).$$

Thus the fiber of p over $[z_0 : w_0 : v_0]$ is the line $(v_0x - w_0y - z_0^2t = 0)$ in the plane with coordinates $[x : y : t]$, and $(v_0, -w_0, -z_0^2) \neq 0$. Hence p is a $\mathbb{P}^1$-bundle, and each fiber of p maps isomorphically onto a line of Q. The linear system $|l| = |h - P_{Q'}|$ is induced by the hyperplanes containing Λ, so $l = p^*\mathcal{O}_{\mathbb{P}^2}(1)$. The exceptional divisor $P_{Q'}$ is $(t = 0)$. Over $[z_0 : w_0 : v_0] \neq p_0 := [1 : 0 : 0]$, the surface $P_{Q'}$ contains the single point $[w_0 : v_0 : 0]$ of the fiber, and $P_{Q'}$ contains the fiber over p_0. Thus $p|_{P_{Q'}} : P_{Q'} \to \mathbb{P}^2$ is a birational morphism which contracts exactly the curve $p^{-1}(p_0)$.

The surface $P_{Q'} = \mathbb{P}(N_{\Lambda/Q})$ is a $\mathbb{P}^1$-bundle over $\Lambda \cong \mathbb{P}^1$ with Picard number two. Since $P_{Q'}$ admits a birational morphism to $\mathbb{P}^2$ contracting one irreducible curve, $P_{Q'} \cong \mathbb{F}_1$ and $p^{-1}(p_0)$ is the negative section g. Since $\deg N_{\Lambda/Q} = -K_Q \cdot \Lambda - 2 = 1$, [IP99, Lemma 2.2.14] implies

$$h^3 = 2, \quad h^2P_{Q'} = 0, \quad hP_{Q'}^2 = -1, \quad P_{Q'}^3 = -1 \tag{5.17}$$

on Q'. The class $h|_{P_{Q'}}$ is the pullback of $\mathcal{O}_\Lambda(1)$, that is, η. Write $P_{Q'}|_{P_{Q'}} \equiv c_1g + c_2\eta$. Then $c_1 = (\eta \cdot P_{Q'}|_{P_{Q'}}) = hP_{Q'}^2 = -1$ and $-c_1^2 + 2c_1c_2 = P_{Q'}^3 = -1$, so $c_2 = 0$ and $P_{Q'}|_{P_{Q'}} = -g$. In particular $l|_{P_{Q'}} = g + \eta$ is the pullback of a line under $P_{Q'} \to \mathbb{P}^2$.

Step 3. In this step we prove (2).

Since $E_Q \in |\mathcal{O}_Q(2)|$, there exists a quadratic form q_2 with $E_Q = Q \cap (q_2 = 0)$ as schemes. Since $\Lambda \subset E_Q$, we may write $q_2 = z\ell_1 + w\ell_2 + v\ell_3$ with linear forms ℓ_1, ℓ_2, ℓ_3. Since Q is smooth and E_Q is singular at any point of Λ, the differentials dq_2 and dq_Q are linearly dependent at any point $[x : y : 0 : 0 : 0]$ of Λ. There, the partial derivatives of q_Q with respect to (x, y, z, w, v) are $(0, 0, 0, -y, x)$, and those of q_2 are $(0, 0, \ell_1, \ell_2, \ell_3)|_\Lambda$. Since $\ell_1|_\Lambda, \ell_2|_\Lambda, \ell_3|_\Lambda$ are linear forms in x, y, linear dependence for all $[x : y]$ implies

$$\ell_1|_\Lambda = 0, \qquad x \cdot \ell_2|_\Lambda + y \cdot \ell_3|_\Lambda = 0,$$

hence $\ell_2|_\Lambda = -\lambda y$ and $\ell_3|_\Lambda = \lambda x$ for a constant λ. Then $\ell_1 + \lambda z$, $\ell_2 + \lambda y$, and $\ell_3 - \lambda x$ vanish on Λ, so

$$q_\kappa := q_2 - \lambda q_Q = z(\ell_1 + \lambda z) + w(\ell_2 + \lambda y) + v(\ell_3 - \lambda x)$$

is a quadratic form in z, w, v, and $E_Q = Q \cap (q_\kappa = 0)$. Let $\kappa := (q_\kappa = 0) \subset \mathbb{P}^2$. Since z, w, v are the pullbacks under p of the coordinates of $\mathbb{P}^2$, q_κ defines $p^*\kappa \in |2l| = |2h - 2P_{Q'}|$, and the pullback of E_Q to Q' is $p^*\kappa + 2P_{Q'}$. Since $P_{Q'}$ maps onto $\mathbb{P}^2$, $p^*\kappa$ does not contain $P_{Q'}$. Thus $\bar{E}_Q = p^{-1}(\kappa) \sim 2l$. Since $\bar{E}_Q \to E_Q$ is birational and E_Q is integral, κ is an integral conic, hence smooth. Thus $\bar{E}_Q$ is a smooth $\mathbb{P}^1$-bundle over $\kappa \cong \mathbb{P}^1$.

The curves contracted by $Q' \to Q$ are the fibers of $P_{Q'} \to \Lambda$, and p maps any such fiber onto a line in $\mathbb{P}^2$ since $(l \cdot \eta) = ((g + \eta) \cdot \eta)_{P_{Q'}} = 1$. Since κ contains no line, $\bar{E}_Q = p^{-1}(\kappa)$ contains no fiber of $P_{Q'} \to \Lambda$. Thus $\bar{E}_Q \to E_Q$ is finite and birational, and $\bar{E}_Q$ is smooth, so $\bar{E}_Q \to E_Q$ is the normalization. By Theorem 2.22, $\bar{E}_Q \cong \mathbb{F}_0$ or $\bar{E}_Q \cong \mathbb{F}_2$.

Since the fibers of $\bar{E}_Q \to \kappa$ are fibers of p, we have $(h \cdot \bar{f}) = 1$ and $l|_{\bar{E}_Q} = 2\bar{f}$. Moreover

$$(h|_{\bar{E}_Q})^2 = h^2 \cdot (2h - 2P_{Q'}) = 4.$$

Write $h|_{\bar{E}_Q} \equiv \alpha\bar{\sigma} + \beta\bar{f}$. Then $\alpha = (h \cdot \bar{f}) = 1$. If $\bar{E}_Q \cong \mathbb{F}_0$, then $2\alpha\beta = 4$, so $\beta = 2$. If $\bar{E}_Q \cong \mathbb{F}_2$, then $-2\alpha^2 + 2\alpha\beta = 4$, so $\beta = 3$. In both cases $P_{Q'}|_{\bar{E}_Q} = h|_{\bar{E}_Q} - l|_{\bar{E}_Q}$, which is $\bar{\sigma}$ if $\bar{E}_Q \cong \mathbb{F}_0$ and $\bar{\sigma} + \bar{f}$ if $\bar{E}_Q \cong \mathbb{F}_2$. This proves (2).

Step 4. In this step we prove (3).

Since $\Gamma \subset E_Q$ and $\Gamma \not\subset \Lambda$, we have $\Gamma' \subset \bar{E}_Q$. Since Γ is a smooth curve not contained in Λ, the ideal $\mathcal{I}_\Lambda \cdot \mathcal{O}_\Gamma$ is invertible, so by [Har77, Proposition II.7.14] the inclusion $\Gamma \subset Q$ lifts to a morphism $\Gamma \to Q'$ whose image is Γ'. Thus $\Gamma' \cong \Gamma$ is a smooth curve of genus two with $(h \cdot \Gamma') = 7$. Write $\Gamma' \equiv a\bar{\sigma} + b\bar{f}$ on $\bar{E}_Q$. If $\bar{E}_Q \cong \mathbb{F}_0$, then $K_{\bar{E}_Q} = -2\bar{\sigma} - 2\bar{f}$, and

$$(h \cdot \Gamma') = 2a + b = 7, \qquad (\Gamma' \cdot (\Gamma' + K_{\bar{E}_Q})) = 2ab - 2a - 2b = 2$$

by the adjunction formula. Thus $(a-1)(3-a) = 1$, so $a = 2$ and $b = 3$. If $\bar{E}_Q \cong \mathbb{F}_2$, then $K_{\bar{E}_Q} = -2\bar{\sigma} - 4\bar{f}$, and

$$(h \cdot \Gamma') = a + b = 7, \qquad (\Gamma' \cdot (\Gamma' + K_{\bar{E}_Q})) = -2a^2 + 2ab - 2b = 2.$$

Thus $(a-2)^2 = 0$, so $a = 2$ and $b = 5$. This is consistent with [KPZ14, Proposition 2.2(2)(b) and (b')].

Since $\Gamma' \not\subset P_{Q'}$, the scheme $\Xi = \Gamma' \cap P_{Q'}$ is the zero scheme of a section of $\mathcal{O}_{\Gamma'}(P_{Q'})$, of length

$$(P_{Q'} \cdot \Gamma') = (P_{Q'}|_{\bar{E}_Q} \cdot \Gamma')_{\bar{E}_Q} = \begin{cases} \bar{\sigma} \cdot (2\bar{\sigma} + 3\bar{f}) = 3 & \text{if } \bar{E}_Q \cong \mathbb{F}_0, \\ (\bar{\sigma} + \bar{f}) \cdot (2\bar{\sigma} + 5\bar{f}) = -4 + 5 + 2 = 3 & \text{if } \bar{E}_Q \cong \mathbb{F}_2. \end{cases}$$

Since $\mathcal{I}_\Lambda \cdot \mathcal{O}_{Q'} = \mathcal{O}_{Q'}(-P_{Q'})$, the ideal of Ξ in $\mathcal{O}_{\Gamma'}$ is $\mathcal{I}_\Lambda \cdot \mathcal{O}_{\Gamma'}$, which corresponds to the ideal $\mathcal{I}_\Lambda \cdot \mathcal{O}_\Gamma$ of $\Gamma \cap \Lambda$ under $\Gamma' \cong \Gamma$. Thus $\Xi \to \Gamma \cap \Lambda$ is an isomorphism.

Recall that $g = p^{-1}(p_0)$. If $p_0 \notin \kappa$, then $g \cap \bar{E}_Q = \emptyset$, so $\Gamma' \cap g = \emptyset$. If $p_0 \in \kappa$, then g is a fiber of $\bar{E}_Q \to \kappa$, and $g \not\subset \Gamma'$ since Γ' is an irreducible curve of genus two. By [Har77, Proposition V.1.4], the length of $\Gamma' \cap g$ is $(\Gamma' \cdot \bar{f})_{\bar{E}_Q} = 2$.

Step 5. In this step we prove (4)(a) and (4)(b).

Let $q : Q'' \to Q'$ be the blow-up. Since Q' and Q'' are blow-ups of smooth curves in smooth threefolds, [KM98, Lemma 2.29] implies $K_{Q'} = -3h + P_{Q'}$ and $K_{Q''} = q^*K_{Q'} + D''$. Since $\Gamma' \not\subset P_{Q'}$, we have $P = q^*P_{Q'}$, and since $\bar{E}_Q$ is smooth along $\Gamma' \subset \bar{E}_Q$, we have $\bar{E}_{Q''} = q^*\bar{E}_Q - D''$. Thus

$$-K_{Q''} = 3h - P - D'', \qquad \bar{E}_{Q''} \sim 2h - 2P - D'', \qquad l = h - P. \tag{5.18}$$

Since $q^{-1}\mathcal{I}_{\Gamma'} \cdot \mathcal{O}_{Q''} = \mathcal{O}_{Q''}(-D'')$, it suffices to show that $\mathcal{F} := \mathcal{I}_{\Gamma'}(3h - P_{Q'})$ is globally generated on Q'.

We first treat the restriction to $P_{Q'}$. Recall that $\varrho : P_{Q'} \to \Lambda$ is the ruling of (1)(b). By (1)(c), $(3h - P_{Q'})|_{P_{Q'}} = g + 3\eta$, and by [Har77, Exercise III.8.4(a)] and [Har77, Propositions V.2.8 and V.2.9],

$$\varrho_*\mathcal{O}_{P_{Q'}}(g + 3\eta) \cong \mathcal{O}_\Lambda(3) \oplus \mathcal{O}_\Lambda(2),$$

where the summand $\mathcal{O}_\Lambda(3)$ is $\sigma_g \cdot \varrho_*\mathcal{O}_{P_{Q'}}(3\eta)$ for a section σ_g defining g. By (3)(c), $\varrho|_\Xi$ is a closed immersion. Near a point of $\varrho(\Xi)$, let t be a local coordinate on Λ and u an affine coordinate on the fibers of ϱ such that $u^{-1}(\infty)$ is disjoint from Ξ, which is possible since Ξ intersects each fiber of ϱ in at most one point. After shrinking the neighborhood, $\mathcal{O}_{P_{Q'}}(g + 3\eta) \cong \mathcal{O}_{P_{Q'}}(u^{-1}(\infty))$, so the local sections of $\varrho_*\mathcal{O}_{P_{Q'}}(g + 3\eta)$ are the fiber-linear forms $\alpha(t) + \beta(t)u$. Since $\Xi \to \varrho(\Xi)$ is an isomorphism, the ideal of Ξ is $(u - \phi(t), \psi(t))$ for local functions ϕ and ψ. The evaluation map

$$\varrho_*\mathcal{O}_{P_{Q'}}(g + 3\eta) \to \varrho_*\mathcal{O}_\Xi(g + 3\eta), \qquad \alpha + \beta u \to (\alpha + \beta\phi) \bmod \psi,$$

is surjective, and the kernel $\mathcal{K}_0$ is locally free with local basis $u - \phi$ and ψ. Thus $\mathcal{K}_0$ is a vector bundle of rank two and degree $3 + 2 - 3 = 2$ on Λ, so $\mathcal{K}_0 \cong \mathcal{O}_\Lambda(a) \oplus \mathcal{O}_\Lambda(2 - a)$ with $a \geq 1$, and $a \leq 3$ since $\mathcal{K}_0 \subset \mathcal{O}_\Lambda(3) \oplus \mathcal{O}_\Lambda(2)$. Suppose that $a = 3$. Since $\mathrm{Hom}(\mathcal{O}_\Lambda(3), \mathcal{O}_\Lambda(2)) = 0$, the summand $\mathcal{O}_\Lambda(3)$ of $\mathcal{K}_0$ is the summand $\sigma_g \cdot \varrho_*\mathcal{O}_{P_{Q'}}(3\eta)$. Thus σ_g vanishes on Ξ, that is, $\Xi \subset g$, which contradicts the bound on the length of $\Gamma' \cap g$ in (3)(d). Hence $\mathcal{K}_0 \cong \mathcal{O}_\Lambda(2) \oplus \mathcal{O}_\Lambda$ or $\mathcal{K}_0 \cong \mathcal{O}_\Lambda(1)^{\oplus 2}$, and $\mathcal{K}_0$ is globally generated. Over a neighborhood of a point of $\varrho(\Xi)$, the images of $u - \phi$ and ψ generate $(u - \phi, \psi)$ in the affine chart, and the image of $u - \phi$ does

not vanish at the point ∞ of the fiber. Over $\Lambda\backslash\varrho(\Xi)$, we have $\mathcal{K}_0 = \varrho_*\mathcal{O}_{P_{Q'}}(g+3\eta)$, and $\varrho^*\varrho_*\mathcal{O}_{P_{Q'}}(g+3\eta) \to \mathcal{O}_{P_{Q'}}(g+3\eta)$ is surjective. Hence the natural map $\varrho^*\mathcal{K}_0 \to \mathcal{I}_{\Xi/P_{Q'}}(g+3\eta)$ is surjective, and $\mathcal{I}_{\Xi/P_{Q'}}(g+3\eta)$ is globally generated since $\mathcal{K}_0$ is globally generated.

Next we lift sections. Since a local equation of $P_{Q'}$ is a nonzerodivisor on $\mathcal{O}_{\Gamma'}$, we have $\mathcal{I}_{\Gamma'} \otimes \mathcal{O}_{P_{Q'}} = \mathcal{I}_{\Xi/P_{Q'}}$ and an exact sequence

$$0 \to \mathcal{I}_{\Gamma'}(3h-2P_{Q'}) \to \mathcal{F} \to \mathcal{I}_{\Xi/P_{Q'}}(g+3\eta) \to 0. \tag{5.19}$$

Since $\Gamma' \subset \bar{E}_Q$ and $\bar{E}_Q \sim 2h-2P_{Q'}$, we have exact sequences

$$0 \to \mathcal{O}_{Q'}(h) \to \mathcal{I}_{\Gamma'}(3h-2P_{Q'}) \to \mathcal{O}_{\bar{E}_Q}\big((3h-2P_{Q'})|_{\bar{E}_Q} - \Gamma'\big) \to 0 \tag{5.20}$$

and

$$0 \to \mathcal{O}_{Q'}(h+P_{Q'}) \to \mathcal{F} \to \mathcal{O}_{\bar{E}_Q}\big((3h-P_{Q'})|_{\bar{E}_Q} - \Gamma'\big) \to 0. \tag{5.21}$$

By (2) and (3), $(3h-2P_{Q'})|_{\bar{E}_Q} - \Gamma'$ is $-\bar{\sigma}+3\bar{f}$ or $-\bar{\sigma}+2\bar{f}$, whose intersection number with $\bar{f}$ is $(-\bar{\sigma}\cdot\bar{f}) = -1$, and $(3h-P_{Q'})|_{\bar{E}_Q} - \Gamma' = 3\bar{f}$ in both cases. By (1), $(h+P_{Q'})|_{P_{Q'}} = \eta - g$, and $((\eta-g)\cdot\eta) = -1$. Thus each of the line bundles $\mathcal{O}_{\bar{E}_Q}((3h-2P_{Q'})|_{\bar{E}_Q}-\Gamma')$ and $\mathcal{O}_{P_{Q'}}(\eta-g)$ is $\mathcal{O}(-1)$ twisted by the pullback of a line bundle on the base of the $\mathbb{P}^1$-bundle $\bar{E}_Q \to \kappa$, resp. $\varrho : P_{Q'} \to \Lambda$, its direct images vanish [Har77, Exercises III.8.3 and III.8.4(a)], and all its cohomology groups vanish [Har77, Exercise III.8.1]. Moreover $H^1(Q',h) = H^1(Q,\mathcal{O}_Q(1)) = 0$, and we have the exact sequence

$$0 \to \mathcal{O}_{Q'}(h) \to \mathcal{O}_{Q'}(h+P_{Q'}) \to \mathcal{O}_{P_{Q'}}(\eta-g) \to 0. \tag{5.22}$$

By (5.22), $H^1(Q',h) = 0$, and the vanishing of the cohomology of $\mathcal{O}_{P_{Q'}}(\eta-g)$, we have $H^1(Q',h+P_{Q'}) = 0$. By (5.20), $H^1(Q',h) = 0$, and the vanishing of the cohomology of $\mathcal{O}_{\bar{E}_Q}((3h-2P_{Q'})|_{\bar{E}_Q}-\Gamma')$, we have

$$H^1(Q',\mathcal{I}_{\Gamma'}(3h-2P_{Q'})) = 0,$$

so (5.19) shows that $H^0(Q',\mathcal{F}) \to H^0(P_{Q'},\mathcal{I}_{\Xi/P_{Q'}}(g+3\eta))$ is surjective. By (5.21) and $H^1(Q',h+P_{Q'}) = 0$, the restriction $H^0(Q',\mathcal{F}) \to H^0(\bar{E}_Q,\mathcal{O}_{\bar{E}_Q}(3\bar{f}))$ is surjective.

Now let $\xi \in Q'$. If $\xi \in P_{Q'}$, then the global sections of $\mathcal{F}$ generate $\mathcal{F}_\xi$ modulo a local equation of $P_{Q'}$, hence generate $\mathcal{F}_\xi$ by Nakayama's lemma. If $\xi \notin P_{Q'}$, then the sections $\sigma_{P_{Q'}}\tau$ with $\tau \in H^0(Q',h)$ and $\sigma_{P_{Q'}}$ a section defining $P_{Q'}$ generate $\mathcal{O}_{Q'}(h+P_{Q'})$ at ξ, and $3\bar{f}$ is base point free on $\bar{E}_Q$. Thus (5.21) and the surjectivity onto $H^0(\bar{E}_Q,\mathcal{O}_{\bar{E}_Q}(3\bar{f}))$ imply that the global sections of $\mathcal{F}$ generate $\mathcal{F}_\xi$. Hence $\mathcal{F}$ is globally generated, and $-K_{Q''}$ is base point free.

Step 6. In this step we prove (4)(c).

Since $\Gamma \not\subset \Lambda$, $Q' \to Q$ is an isomorphism near the generic point of Γ, so $\mathrm{ord}_{D''} = \mathrm{ord}_{D'}$ as valuations of the function field. Since $\Lambda \not\subset \Gamma$, $Y^+ \to Q$ is an isomorphism near the generic point of Λ, so P is also the exceptional divisor of the blow-up of Y^+ along Λ^+, and P is not a divisor on Y^+. Thus, as valuations, the prime divisors on Q'' are P and the prime divisors on Y^+. The flop $Y \dashrightarrow Y^+$ is an isomorphism in codimension one, and the prime divisors on Y are E and the strict transforms of prime divisors on X. Since E maps onto E_Q, the strict transform of E in Q'' is $\bar{E}_{Q''}$. Hence the prime divisors on Q'' exceptional over X are exactly $\bar{E}_{Q''}$ and P.

Since $Y^+ = \mathrm{Bl}_\Gamma Q$ is smooth, $A_{Y^+}(P) = 2$. Since the flop is crepant, $A_Y(P) = A_{Y^+}(P) = 2$ [KM98, Lemma 2.30]. The center of P on Y is contained in the fiber of $Y \to \bar{Y}$ over $\bar{y}$, which is s by **Step 1**. On the smooth threefold Y, the blow-up of a closed point or a smooth curve has an exceptional divisor F_0 with $A_Y(F_0)$ equal to the codimension of the center, and any other prime divisor F over Y whose center on Y equals the center of F_0 on Y has center on the blow-up contained in F_0, so $A_Y(F) \geq 2 + \mathrm{mult}_F F_0 \geq 3$ [KM98, Lemma 2.29, Lemma 2.30, Corollary 2.31(2)]. Thus a divisor over Y whose center is a closed point has log discrepancy at least 3, and the only divisor over Y with center s and log discrepancy 2 is the exceptional

divisor of $\mathrm{Bl}_s Y \to Y$. Hence the strict transform P of $P_{Q'}$ is the divisor P of Set-up 5.1. Since $K_Y = \pi^* K_X + E$, $s \subset E$, and E is smooth, we have $\mathrm{mult}_P E = 1$ and

$$A_X(P) = A_Y(P) + \mathrm{mult}_P E = 3.$$

Since $\bar{E}_{Q''}$ is the strict transform of E, it defines the valuation ord_E, and $A_X(\bar{E}_{Q''}) = A_X(E) = 2$ since E is the exceptional divisor of the blow-up of the smooth curve L in the smooth threefold X [KM98, Lemma 2.29].

Step 7. In this step we prove (4)(d).

Let $H_0 \in |H|$ be a general member and $H_{Q''}$ the strict transform of H_0 in Q''. Since $|H|$ is base point free, $\mathrm{mult}_F H_0 = 0$ for any prime divisor F exceptional over X. Comparing $K_{Q''}$ and K_X on a common resolution and pushing forward to Q'', **Step 6** implies

$$K_{Q''} \sim -H_{Q''} + (A_X(\bar{E}_{Q''}) - 1)\bar{E}_{Q''} + (A_X(P) - 1)P = -H_{Q''} + \bar{E}_{Q''} + 2P.$$

By (5.18),

$$H_{Q''} \sim (3h - P - D'') + (2h - 2P - D'') + 2P = 5h - 2D'' - P.$$

Let ϕ be a rational function on X. Since X is normal, $\phi \in H^0(X, mH)$ if and only if $\mathrm{ord}_F(\phi) + m\,\mathrm{mult}_F H_0 \geq 0$ for any prime divisor F on X. By **Step 6**, the prime divisors on X are exactly the prime divisors on Q'' not exceptional over X, and the coefficient of F in $H_{Q''}$ is $\mathrm{mult}_F H_0$. If $\phi \in H^0(X, mH)$, then $\mathrm{ord}_F(\phi) \geq 0$ for $F = \bar{E}_{Q''}$ and $F = P$, since $\mathrm{mult}_F H_0 = 0$ and $\mathrm{div}(\phi) + mH_0 \geq 0$, so the pullback of $\mathrm{div}(\phi) + mH_0$ to Q'' is ≥ 0. Since the coefficients of $\bar{E}_{Q''}$ and P in $mH_{Q''} - jP$ are 0 and $-j$, we obtain

$$\{\phi \in H^0(X, mH) : \mathrm{ord}_P(\phi) \geq j\} = H^0(Q'', mH_{Q''} - jP)$$

for all integers $m, j \geq 0$. □

We now compute the volume function of P.

Proposition 5.7. *Notation and conditions as in Set-up 5.1. Assume that L is special. Then $A_X(P) = 3$ and*

$$\mathrm{vol}(H - kP) = \begin{cases} 18 - \frac{3}{2}k^2 & 0 \leq k \leq 2, \\ 18 - \frac{3}{2}k^2 + \frac{3}{4}(k-2)^3 & 2 \leq k \leq 4, \\ 0 & k \geq 4. \end{cases}$$

In particular $S_X(P) = 43/18$.

Proof. By Construction-Proposition 5.6(4)(c), $A_X(P) = 3$.

Step 1. In this step we prove the formula for $0 \leq k \leq 2$.

Since $N^\vee_{L/X} \cong \mathcal{O}_L(-1) \oplus \mathcal{O}_L(2)$, there exists a closed subscheme $C \subset X$ with $\mathcal{I}_L^2 \subset \mathcal{I}_C \subset \mathcal{I}_L$ such that $\mathcal{I}_L/\mathcal{I}_C \cong \mathcal{O}_L(-1)$ is the quotient of $\mathcal{I}_L/\mathcal{I}_L^2 \cong N^\vee_{L/X}$ by the summand $\mathcal{O}_L(2)$. Near any point of L there exist local coordinates x, y, z of X with $\mathcal{I}_L = (x, y)$ and $\mathcal{I}_C = (y, x^2)$. Thus C is a local complete intersection, and C is the non-reduced conic supported on L [KPS18, Lemma 2.1.1, Remarks 2.1.2 and 2.1.7]. Let $\langle C \rangle \subset \mathbb{P}^{11}$ be the plane of C in the embedding of Lemma 2.12(1).

We show that $X \cap \langle C \rangle = C$ as schemes. By Lemma 2.12(1), X is an intersection of quadrics. The restriction to $\langle C \rangle$ of any quadric containing X vanishes on C, hence is a scalar multiple of the quadratic equation of C in $\langle C \rangle$. If all restrictions vanish, then $\langle C \rangle \subset X$. Since $\mathrm{Pic}\, X = \mathbb{Z}H$, a surface in X is a member of $|aH|$ for some integer $a \geq 1$, and $(H^2 \cdot aH) = 18a \neq 1$, so X contains no plane. Thus $X \cap \langle C \rangle = C$, so $\mathcal{I}_C$ is generated by the linear forms vanishing on $\langle C \rangle$, and $\mathcal{I}_C(H)$ is globally generated.

Let $X_C := \mathrm{Bl}_C X$ with exceptional Cartier divisor B, and write H also for the pullback of H. Since $\mathcal{I}_C \cdot \mathcal{O}_{X_C} = \mathcal{O}_{X_C}(-B)$, the divisor $H - B$ is base point free. Thus

$$H - \tfrac{k}{2}B = (1 - \tfrac{k}{2})H + \tfrac{k}{2}(H - B)$$

is nef for $0 \le k \le 2$. In the local coordinates above, X_C is covered by the charts $y = x^2T$ and $x^2 = yS$. The first chart is smooth with $B = 2(x = 0)$. The second chart is a hypersurface in a smooth fourfold whose singular locus $(x = y = S = 0)$ has codimension two, so it is normal [Har77, Proposition II.8.23(b)], and $B = \operatorname{div}(y) = 2(x = y = 0)$. Hence X_C is normal and $B = 2P_C$ for a prime divisor P_C, the only prime divisor on X_C exceptional over X. Let R be the local ring of X at the generic point of L. The valuation of P_C is given by the localization of $R[y/x^2]$ at (x). On Y, the negative section s corresponds to the summand $\mathcal{O}_L(1)$ of $N_{L/X}$, so at the generic point of L the curve s is defined by $(x, y/x)$ in $R[y/x]$, and the blow-up of s has the chart $R[y/x][y/x^2] = R[y/x^2]$ with exceptional divisor (x). Thus $\operatorname{ord}_{P_C} = \operatorname{ord}_P$. Since X_C is normal and P_C is the only exceptional prime divisor, for rational k and integers m with $mk/2 \in \mathbb{Z}$, the space $H^0(X_C, m(H - \frac{k}{2}B))$ is the space of sections $\phi \in H^0(X, mH)$ with $\operatorname{ord}_P(\phi) \ge mk$. Thus $\operatorname{vol}(H - kP) = \operatorname{vol}(H - \frac{k}{2}B)$ on X_C.

Since $C \subset X$ is a regular embedding, $B = \mathbb{P}(N_{C/X})$ and $\mathcal{O}_{X_C}(B)|_B = \mathcal{O}_{\mathbb{P}(N_{C/X})}(-1)$. By the definition of Segre classes [Ful98, Appendix B.6.2–B.6.3 and Sections 3.1–3.2],

$$H^3 = 18, \qquad H^2B = 0, \qquad HB^2 = -(H \cdot [C]) = -2, \qquad B^3 = -\deg N_{C/X}.$$

Since C is a plane conic, $\omega_C = \mathcal{O}_C(-1)$, so $\det N_{C/X} = \omega_C \otimes \omega_X^{-1}|_C \cong \mathcal{O}_C$ and $\deg N_{C/X} = 0$. Since $H - \frac{k}{2}B$ is nef, [Laz04a, Corollary 1.4.41] implies

$$\operatorname{vol}(H - kP) = \left(H - \tfrac{k}{2}B\right)^3 = 18 - \tfrac{3}{2}k^2, \qquad 0 \le k \le 2. \tag{5.23}$$

Step 2. In this step we reduce the formula for $2 \le k \le 4$ to a computation of M_k^3, and prove $\operatorname{vol}(H - kP) = 0$ for $k \ge 4$.

We use the notation of Construction-Proposition 5.6. By Construction-Proposition 5.6(4)(d), $\operatorname{vol}(H - kP) = \operatorname{vol}(H_{Q''} - kP)$ on Q''. For $2 \le k \le 4$, let

$$M_k := \tfrac{4-k}{2}(-K_{Q''}) + \tfrac{k-2}{2}l.$$

By (5.18) and $H_{Q''} = 5h - 2D'' - P$,

$$H_{Q''} - kP = M_k + \tfrac{k}{2}\bar{E}_{Q''}. \tag{5.24}$$

Since $-K_{Q''}$ and l are base point free, M_k is nef. Let $\bar{f}_0$ be a general fiber of $\bar{E}_Q \to \kappa$. Since $\Gamma' \to \kappa$ has degree $(\Gamma' \cdot \bar{f}) = 2$, $\bar{f}_0$ intersects Γ' transversally in two points. Let $\bar{f}''$ be the strict transform of $\bar{f}_0$ in $\bar{E}_{Q''}$. By Construction-Proposition 5.6(2), $(P_{Q'} \cdot \bar{f}_0) = 1$ in both cases, so

$$(h \cdot \bar{f}'') = 1, \quad (P \cdot \bar{f}'') = 1, \quad (D'' \cdot \bar{f}'') = 2, \quad (\bar{E}_{Q''} \cdot \bar{f}'') = 2 - 2 - 2 = -2,$$

and $((H_{Q''} - kP) \cdot \bar{f}'') = 5 - 4 - 1 - k = -k$. Let k be rational, let m be a positive integer such that mM_k and $\frac{mk}{2}\bar{E}_{Q''}$ are integral divisors, and let $G \in |m(H_{Q''} - kP)|$. Write $G = c\bar{E}_{Q''} + G_1$ with $G_1 \ge 0$ not containing $\bar{E}_{Q''}$. Since the curves $\bar{f}''$ cover $\bar{E}_{Q''}$, we have $(G_1 \cdot \bar{f}'') \ge 0$ for general $\bar{f}''$, that is, $-mk + 2c \ge 0$. Thus $c \ge mk/2$, and multiplication by the $\frac{mk}{2}$-th power of a section defining $\bar{E}_{Q''}$ is an isomorphism

$$H^0(Q'', mM_k) \to H^0(Q'', m(H_{Q''} - kP)).$$

Since M_k is nef, [Laz04a, Corollary 1.4.41] implies

$$\operatorname{vol}(H - kP) = M_k^3, \qquad 2 \le k \le 4, \tag{5.25}$$

for rational k, and for real k by the continuity of the volume function [Laz04a, Theorem 2.2.44 and Corollary 2.2.45]. In particular $\operatorname{vol}(H - 4P) = M_4^3 = l^3 = 0$, since l is pulled back from $\mathbb{P}^2$. Since $\operatorname{vol}(H - kP)$ is non-increasing in k, $\operatorname{vol}(H - kP) = 0$ for $k \ge 4$.

Step 3. In this step we compute M_k^3 and $S_X(P)$.

On Q' we use the intersection numbers (5.17). By Construction-Proposition 5.6(3), $(h\cdot\Gamma')=7$ and $(P_{Q'}\cdot\Gamma')=3$. Thus $(-K_{Q'}\cdot\Gamma')=((3h-P_{Q'})\cdot\Gamma')=18$ and $\deg N_{\Gamma'/Q'}=18+2\cdot 2-2=20$. By [IP99, Lemma 2.2.14], on Q'' the products of D'' with two pullbacks vanish, and

$$hD''^2=-7,\qquad PD''^2=-3,\qquad D''^3=-20.$$

A direct calculation from (5.18) shows

$$(-K_{Q''})^3=12,\qquad (-K_{Q''})^2\cdot l=8,\qquad (-K_{Q''})\cdot l^2=2,\qquad l^3=0. \tag{5.26}$$

For instance, with $a:=3h-P$, we have $a^3=54-9+1=46$ and $(a\cdot\Gamma')=18$, so $(-K_{Q''})^3=a^3-3(a\cdot\Gamma')+20=12$. By (5.26), for $2\le k\le 4$,

$$\begin{aligned}M_k^3&=12\Big(\frac{4-k}{2}\Big)^3+24\Big(\frac{4-k}{2}\Big)^2\frac{k-2}{2}+6\cdot\frac{4-k}{2}\Big(\frac{k-2}{2}\Big)^2\\&=\frac{3}{2}(4-k)^3+3(4-k)^2(k-2)+\frac{3}{4}(4-k)(k-2)^2\\&=12+9k-6k^2+\frac{3}{4}k^3=18-\frac{3}{2}k^2+\frac{3}{4}(k-2)^3.\end{aligned}$$

Together with (5.23) and (5.25), this proves the formula for $\mathrm{vol}(H-kP)$. Finally,

$$\int_0^2\Big(18-\frac{3}{2}k^2\Big)dk=32,\qquad \int_2^4\Big(12+9k-6k^2+\frac{3}{4}k^3\Big)dk=11,$$

so $S_X(P)=\frac{1}{18}(32+11)=\frac{43}{18}$. □

We next bound the orders of vanishing of the restricted series along the curves of E that dominate L (Definition-Lemma 5.8), and then prove the criterion of Proposition 5.9, which bounds $\delta_L(X)$ in terms of $S_X(P)$.

Definition-Lemma 5.8. Notation and conditions as in Set-up 5.1, with L special. Let f be the class of a fiber of $E\to L$, let D be the strict transform of D' on Y, a prime divisor with $D\sim 2H-5E$ (Lemma 5.2), and for integers $m,j\ge 0$ let

$$W_{m,j}:=\mathrm{Im}\big(H^0(Y,mH-jE)\to H^0(E,\mathcal{O}_E(js+(m+2j)f))\big).$$

For a curve $C\subset E$ with $\pi(C)=L$ and an integer $k\ge 1$, let $W_{m,j}^{\ge k}(C):=\{\sigma\in W_{m,j}:\mathrm{ord}_C\,\sigma\ge k\}$. Then W is the refinement by E of the complete series of H (Definition 2.8),

$$S_m(W;C)=\frac{1}{mN_m}\sum_{j\ge 0}\sum_{k\ge 1}\dim W_{m,j}^{\ge k}(C)\quad\text{for every } m\ge 1,$$

and the following hold.

(1) $S(W;s)\le 619/432$.
(2) If $C\ne s$ and $(C\cdot f)=1$, then $S(W;C)\le 277/432$.
(3) If $(C\cdot f)\ge 2$, then $S(W;C)\le 101/(72(C\cdot f))\le 101/144$.

Proof. On $E\cong\mathbb{F}_3$ we have $(s^2)=-3$, $(s\cdot f)=1$, and $(f^2)=0$ [Har77, Propositions V.2.3 and V.2.9]. By (5.1),

$$E|_E\sim -s-2f,\quad (mH-jE)|_E\sim js+(m+2j)f,\quad (H-2E)|_E\sim 2s+5f,\quad D|_E\sim 5s+12f.$$

Since π is the blow-up of the smooth curve L in the smooth threefold X, E is of plt type with plt extraction π, so W is the refinement by E of the complete series of H (Definition 2.8). Since ord_C takes integer values, the non-zero jumping numbers of ord_C on $W_{m,j}$ are the integers $k\ge 1$, each counted $\dim W_{m,j}^{\ge k}(C)-\dim W_{m,j}^{\ge k+1}(C)$ times. Summing over k and j gives the formula for $S_m(W;C)$. Let $r_{m,j}:=\dim W_{m,j}$. Then

$$r_{m,j}=h^0(Y,mH-jE)-h^0(Y,mH-(j+1)E).$$

For real numbers α,β, the class $\alpha s+\beta f$ is nef if and only if $\beta\ge 3\alpha\ge 0$ [Har77, Proposition V.2.20]. For $0\le\beta<3\alpha$ the Zariski decomposition of $\alpha s+\beta f$ has negative

part $(\alpha-\beta/3)s$ and positive part $(\beta/3)(s+3f)$ [Laz04a, Theorem 2.3.19] (Zariski [Zar62, Theorem 7.7], extended to pseudo-effective divisors by Fujita [Fuj79], and [Bau09] for $\mathbb{R}$-divisors). Hence, by [Laz04a, Corollary 1.4.41 and Corollary 2.3.22],

$$\operatorname{vol}(\alpha s+\beta f)=\begin{cases}2\alpha\beta-3\alpha^2 & \beta\ge 3\alpha\ge 0,\\ \beta^2/3 & 0\le\beta\le 3\alpha,\\ 0 & \alpha<0 \text{ or } \beta<0.\end{cases} \tag{5.27}$$

For integers $\alpha\ge 0$ and β,

$$h^0(E,\mathcal{O}_E(\alpha s+\beta f))=\sum_{i=0}^{\alpha}\max\{\beta-3i+1,0\}$$

[Har77, Proposition V.2.8, Exercise III.8.4(a), and Exercise II.5.1(d)], and $h^0(E,\mathcal{O}_E(\alpha s+\beta f))=0$ for $\alpha<0$. The right side is a Riemann sum of $\int_0^\alpha\max\{\beta-3x,0\}dx=\operatorname{vol}(\alpha s+\beta f)/2$. Thus, for M in a bounded set of $\mathbb{R}$-classes on E and integers m with mM integral,

$$h^0(E,\mathcal{O}_E(mM))=\frac{m^2}{2}\operatorname{vol}(M)+O(m). \tag{5.28}$$

We describe $W_{m,j}$ for $j\ge 2m$. Let σ_D be a section defining D. By Lemma 5.2, $H^0(Y,mH-jE)=\sigma_D^{j-2m}\cdot H^0(Y,(5m-2j)h)$ for $2m\le j\le 5m/2$, and $H^0(Y,mH-jE)=0$ for $j>5m/2$. For integers $q\ge 0$ let U_q be the image of $H^0(Y,qh)$ in $H^0(E,\mathcal{O}_E(q(2s+5f)))$, where the class is $q(2s+5f)$, and not $q(2s+3f)$ as in Construction-Proposition 5.4, because $E\cong\mathbb{F}_3$. For $k\ge 1$ let $U_q^{\ge k}(C):=\{\rho\in U_q:\operatorname{ord}_C\rho\ge k\}$. Let $d_E:=\sigma_D|_E$. Then

$$W_{m,j}=d_E^{j-2m}\cdot U_{5m-2j}\quad\text{for } 2m\le j\le 5m/2,$$

and $W_{m,j}=0$ for $j>5m/2$. Since D is a prime divisor different from E, $E\not\subset D$, so $d_E\ne 0$ and multiplication by d_E^{j-2m} is injective.

We prove (1) and (2). Let $C'=s$ if $C=s$, and $C'=s+3f$ if $C\ne s$ and $(C\cdot f)=1$. In the second case $C\sim s+bf$ with $b=(C\cdot s)+3\ge 3$, so for any class M and any $k\ge 1$,

$$h^0(E,\mathcal{O}_E(M-kC))\le h^0(E,\mathcal{O}_E(M-kC')),$$

because $M-kC'=(M-kC)+k(b-3)f$ and $|k(b-3)f|\ne\emptyset$. Since f is nef, $D|_E-\operatorname{mult}_C(D|_E)C\ge 0$, and $(C\cdot f)=1$,

$$\operatorname{mult}_C(D|_E)=\operatorname{mult}_C(D|_E)(C\cdot f)\le(D|_E\cdot f)=5.$$

For $2m\le j\le 5m/2$, let $q:=5m-2j$. Any $w\in W_{m,j}$ is $d_E^{j-2m}\rho$ with $\rho\in U_q$, and $\operatorname{ord}_C w=(j-2m)\operatorname{mult}_C(D|_E)+\operatorname{ord}_C\rho$. Taking a basis of U_q compatible with the filtration by ord_C, we obtain

$$\sum_{k\ge 1}\dim W_{m,j}^{\ge k}(C)\le 5(j-2m)r_{m,j}+\sum_{k\ge 1}\dim U_q^{\ge k}(C).$$

Since $W_{m,j}^{\ge k}(C)$ and $U_q^{\ge k}(C)$ consist of sections of $\mathcal{O}_E(js+(m+2j)f-kC)$ and of $\mathcal{O}_E(q(2s+5f)-kC)$, respectively, summing over j gives

$$\begin{aligned}S_m(W;C)\le\frac{1}{mN_m}\Big(&\sum_{0\le j<2m}\sum_{k\ge 1}h^0\big(E,\mathcal{O}_E(js+(m+2j)f-kC')\big)\\ &+\sum_{2m\le j\le 5m/2}\sum_{k\ge 1}h^0\big(E,\mathcal{O}_E((5m-2j)(2s+5f)-kC')\big)\\ &+5\sum_{j\ge 2m}(j-2m)r_{m,j}\Big).\end{aligned} \tag{5.29}$$

For an $\mathbb{R}$-class M on E let $I_{C'}(M):=\int_0^\infty\operatorname{vol}(M-uC')du$. The indices j and k in the first two sums of (5.29) run over $O(m)$ values, and by (5.28) each term equals $\frac{m^2}{2}\operatorname{vol}(M)+O(m)$, where

mM is the class in that term. Since $N_m = 3m^3 + O(m^2)$ (Lemma 2.12(2)) and $\mathrm{vol}(M - uC')$ is a continuous piecewise polynomial function of the class M and of u, the first two sums in (5.29), divided by mN_m, are Riemann sums with mesh $1/m$ and differ by $O(1/m)$ from

$$\frac{1}{6}\int_0^2 I_{C'}\big(ts + (1+2t)f\big)dt + \frac{1}{6}\int_2^{5/2} I_{C'}\big((5-2t)(2s+5f)\big)dt.$$

Since $r_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$, the last limit in Lemma 5.2 shows that the last term in (5.29) divided by mN_m converges to $\frac{5}{72}$. A direct calculation from (5.27) shows the values of $I_{C'}$ in Table 1. For instance, for $1 \le t \le 2$ the class $ts + (1+2t)f$ has negative part $\frac{t-1}{3}s$, and the positive part has volume $(1+2t)^2/3$.

TABLE 1. The order integrals $I_{C'}$ of $ts + (1+2t)f$ for $t \le 2$ and of $c(2s+5f)$ for $t \ge 2$ on $E \cong \mathbb{F}_3$, and their integrals over t.

	$0 \le t \le 1$	$1 \le t \le 2$	$2 \le t \le 5/2$, $c = 5 - 2t$
I_s	$t^2 + t^3$	$\frac{(1+2t)^2(7t-1)}{27}$	$\frac{325c^3}{27}$
integral over t	$\frac{7}{12}$	$\frac{329}{54}$	$\frac{325}{216}$
I_{s+3f}	t^2	$\frac{(1+2t)^3}{27}$	$\frac{125c^3}{27}$
integral over t	$\frac{1}{3}$	$\frac{68}{27}$	$\frac{125}{216}$

Since $S(W;C) = \lim_{m\to+\infty} S_m(W;C)$ (Definition 2.8), it follows that

$$S(W;s) \le \frac{1}{6}\Big(\frac{7}{12} + \frac{329}{54} + \frac{325}{216}\Big) + \frac{5}{72} = \frac{619}{432},$$

and, if $C \ne s$ and $(C \cdot f) = 1$,

$$S(W;C) \le \frac{1}{6}\Big(\frac{1}{3} + \frac{68}{27} + \frac{125}{216}\Big) + \frac{5}{72} = \frac{277}{432}.$$

This proves (1) and (2).

We prove (3). Let $a := (C \cdot f) \ge 2$. Since f is nef, any non-zero $\sigma \in W_{m,j}$ satisfies $a \operatorname{ord}_C \sigma \le ((js + (m+2j)f) \cdot f) = j$, so the jumping numbers of ord_C on $W_{m,j}$ are at most j/a. Thus

$$S_m(W;C) \le \frac{1}{a} \cdot \frac{1}{mN_m} \sum_{j\ge 0} j r_{m,j}.$$

By the rank asymptotics of Lemma 5.2 and $N_m = 3m^3 + O(m^2)$, the right side converges to $\frac{1}{a} \cdot \frac{1}{6}\int_0^{5/2} tw(t)dt = \frac{101}{72a}$. Hence $S(W;C) \le 101/(72a) \le 101/144$. □

Proposition 5.9. *Notation and conditions as in Set-up 5.1. Assume that L is special. Then $\delta_L(X) \ge 1$ if $S_X(P) \le 173/72$, and $\delta_L(X) \ge 72/71$ if $S_X(P) \le 43/18$.*

Proof. We use the notation of Definition-Lemma 5.8. Let $\theta \in \{1, 72/71\}$ satisfy

$$\theta\big(S_X(P) - S_X(E)\big) \le 1. \tag{5.30}$$

Since $S_X(E) = 101/72$ (Lemma 5.2), $\theta = 1$ satisfies (5.30) if $S_X(P) \le 173/72$, and $\theta = 72/71$ satisfies it if $S_X(P) \le 43/18$, because $\frac{72}{71}\big(\frac{43}{18} - \frac{101}{72}\big) = 1$. Hence it suffices to show $\delta_L(X) \ge \theta$ for any such θ. We show that $A_X(F) \ge \theta S_X(F)$ for any prime divisor F over X with $L \subset \operatorname{center}_X F$.

Suppose that $\operatorname{center}_X F \ne L$. Then $\operatorname{center}_X F$ is a prime divisor on X, so F is a prime divisor on X and $F \sim aH$ for some integer $a \ge 1$. Then $A_X(F) = 1$ and $\mathrm{vol}(H - tF) = 18(1-at)^3$ for $0 \le t \le 1/a$, so $S_X(F) = 1/(4a)$ and $A_X(F)/S_X(F) = 4a > \theta$. Suppose that $F = E$. Then $A_X(E) = 2$ [KM98, Lemma 2.29], so $A_X(E)/S_X(E) = 144/101 > 72/71 \ge \theta$.

Suppose that $\operatorname{center}_X F = L$ and $F \ne E$. Let $C := \operatorname{center}_Y F$. Then $C \subset \pi^{-1}(L) = E$, $\pi(C) = L$, and $C \ne E$, so C is a curve with $\pi(C) = L$. Fix a rational number λ with

$72/101 < \lambda < \theta$. For $m \geq 1$, by [AZ22, Lemma 3.1] there exists a basis $\{\sigma_1, \dots, \sigma_{N_m}\}$ of $H^0(X, mH)$ compatible with the filtrations induced by F and by E. Let D_m be the m-basis type divisor associated to $\{\sigma_i\}$. Then $\operatorname{mult}_F D_m = S_m(F)$ and $\operatorname{mult}_E D_m = S_m(E)$ [AZ22, Section 2.2], so

$$\pi^* D_m = S_m(E)E + \Omega_m$$

with $\Omega_m \geq 0$ and $E \not\subset \operatorname{Supp} \Omega_m$. Let $a_m := \lambda S_m(E) - 1$ and $\Delta_m := a_m E + \lambda \Omega_m$. Since $K_Y = \pi^* K_X + E$, we have $K_Y + \Delta_m = \pi^*(K_X + \lambda D_m)$, hence

$$A_X(F) - \lambda S_m(F) = A_Y(F) - \operatorname{mult}_F \Delta_m. \tag{5.31}$$

Since $S_m(E) \to 101/72$ [AZ22, Lemma 2.9] and $1 < \lambda \cdot 101/72 < 101/71 < 2$, we have $0 < a_m < 1$ for $m \gg 0$.

We bound the coefficient of C in $\Omega_m|_E$. For $0 \leq j \leq 5m/2$, the elements σ_i with $\operatorname{mult}_E \pi^* \sigma_i = j$ restrict to a basis $\{w_i\}$ of $W_{m,j}$, and $\Omega_m|_E = \frac{1}{mN_m} \sum_i \operatorname{div}(w_i)$, the sum running over all j. For any $k \geq 1$, by linear independence, the number of basis elements w_i of $W_{m,j}$ with $\operatorname{ord}_C w_i \geq k$ is at most $\dim W_{m,j}^{\geq k}(C)$. Summing over k and j, Definition-Lemma 5.8 gives

$$\operatorname{mult}_C(\Omega_m|_E) \leq S_m(W; C). \tag{5.32}$$

Suppose that $C \neq s$. Since $277/432 < 101/144$, (5.32) and Definition-Lemma 5.8(2)(3) imply

$$\limsup_{m \to +\infty} \operatorname{mult}_C(\lambda \Omega_m|_E) \leq \frac{72}{71} \cdot \frac{101}{144} = \frac{101}{142} < 1.$$

Let $m \gg 0$, and let $C^0 \subset C$ be a non-empty smooth open subset disjoint from the other components of $\Omega_m|_E$. Then $(E, \lambda \Omega_m|_E)$ is lc in a neighborhood of C^0 [KM98, Corollary 2.31(3)]. By inversion of adjunction [Kaw07, Theorem], $(Y, E + \lambda \Omega_m)$ is lc in a neighborhood of C^0, and so is (Y, Δ_m) since $a_m \leq 1$.

Suppose that $C = s$. Since P is the exceptional divisor of the blow-up of the smooth curve s in the smooth threefold Y, we have $\operatorname{mult}_P E = 1$ and $\operatorname{mult}_P \Omega_m = \operatorname{mult}_s \Omega_m$. Since $\operatorname{mult}_P D_m \leq S_m(P)$,

$$\operatorname{mult}_s \Omega_m = \operatorname{mult}_P(\pi^* D_m) - S_m(E) \leq S_m(P) - S_m(E).$$

By [AZ22, Lemma 2.9] and (5.30),

$$\limsup_{m \to +\infty} \lambda(S_m(P) - S_m(E)) \leq \lambda/\theta < 1.$$

By (5.32) and Definition-Lemma 5.8(1),

$$\limsup_{m \to +\infty} \lambda\big(\operatorname{mult}_s(\Omega_m|_E) + S_m(E)\big) \leq \lambda\Big(\frac{619}{432} + \frac{101}{72}\Big) = \lambda \cdot \frac{1225}{432} < \frac{72}{71} \cdot \frac{1225}{432} = \frac{1225}{426} < 3.$$

Thus for $m \gg 0$,

$$\operatorname{mult}_s(\lambda \Omega_m) \leq 1 \quad \text{and} \quad \operatorname{mult}_s(\lambda \Omega_m|_E) \leq 3 - \lambda S_m(E) = 2 - a_m.$$

Let T be a general member of a very ample linear system on Y. Then T is smooth, $T \cap s$ is a finite set of general points of s, T is transversal to s at each of them, and T contains neither E nor any component of Ω_m. By [Har77, Theorem II.8.18 and Remark III.7.9.1] applied to E, the curve $E|_T$ is smooth and irreducible. Let $p \in T \cap s$. Then

$$\operatorname{mult}_p(\Omega_m|_T) = \operatorname{mult}_s \Omega_m \quad \text{and} \quad (E|_T \cdot \Omega_m|_T)_p = \operatorname{mult}_s(\Omega_m|_E).$$

Thus $\operatorname{mult}_p(\lambda \Omega_m|_T) \leq 1$ and $(E|_T \cdot \lambda \Omega_m|_T)_p \leq 2 - a_m$. Since $a_m \in \mathbb{Q} \cap [0, 1]$, [CRZ19, Proposition 3.3] implies that $(T, (a_m E + \lambda \Omega_m)|_T)$ is lc at p. By inversion of adjunction [Kaw07, Theorem], $(Y, T + a_m E + \lambda \Omega_m)$ is lc near p, and so is (Y, Δ_m). Since the non-lc locus of (Y, Δ_m) is closed and does not contain the general point p of s, (Y, Δ_m) is lc in a neighborhood of a non-empty open subset of s.

In both cases, for $m \gg 0$ the pair (Y, Δ_m) is lc in a neighborhood of a non-empty open subset of $C = \operatorname{center}_Y F$, so $A_Y(F) \geq \operatorname{mult}_F \Delta_m$. By (5.31), $A_X(F) \geq \lambda S_m(F)$ for $m \gg 0$. Since

$S_m(F) \to S_X(F)$ [AZ22, Lemma 2.9], $A_X(F) \geq \lambda S_X(F)$. Since $\lambda < \theta$ is an arbitrary rational number close to θ, $A_X(F) \geq \theta S_X(F)$. Thus $\delta_L(X) \geq \theta$. □

Corollary 5.10. *Let X be a smooth prime Fano threefold of genus 10 and $L \subset X$ a line. Then $\delta_L(X) \geq 63000/62779 > 1$. More precisely, $\delta_L(X) \geq 63000/62779$ if L is non-special, and $\delta_L(X) \geq 72/71$ if L is special. In particular $A_X(F) > S_X(F)$ for any prime divisor F over X with $\operatorname{center}_X F = L$.*

Proof. By Lemma 2.12(3), $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ or $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, that is, L is non-special or special. If L is non-special, then $\delta_L(X) \geq 63000/62779$ by Theorem 5.5. If L is special, then $S_X(P) = 43/18$ by Proposition 5.7, so $\delta_L(X) \geq 72/71$ by Proposition 5.9. Since

$$72 \cdot 62779 - 71 \cdot 63000 = 47088 > 0,$$

we have $72/71 > 63000/62779 > 1$, so $\delta_L(X) \geq 63000/62779 > 1$ in both cases. Finally, let F be a prime divisor over X with $\operatorname{center}_X F = L$. Since $S_X(F) > 0$, $A_X(F) \geq \delta_L(X) S_X(F) > S_X(F)$. □

6. Points on non-special lines in genus ten

In this section, we prove that $A_X(F) > S_X(F)$ for any smooth prime Fano threefold X of genus 10, any closed point $x \in X$ of incidence type 1, $1+1$, or $1+1+1$, and any prime divisor F over X with $x \in \operatorname{center}_X F$. By Proposition 4.1, the incidence type of x is 1, $1+1$, or $1+1+1$ if and only if $\Sigma(X, x)$ is non-empty and reduced, and then any line through x is non-special. The three types are treated in Proposition 6.1, Theorem 6.5, and Theorem 6.18, respectively.

6.1. **Points on one line.** In this subsection, we treat the points x such that $\Sigma(X, x)$ consists of one reduced point. We use the blow-up of the point, the line link of Theorem 2.22 (see [Fuj26a, Theorem 4.3(2)(3)(ii)], after Iskovskikh's double projection from a line [Isk89]), and the rank estimate of Lemma 3.2.

Proposition 6.1. *Let X be a smooth prime Fano threefold of genus 10 and $x \in X$ a closed point such that $\Sigma(X, x)$ consists of one reduced point. Let $\pi : Y := \operatorname{Bl}_x X \to X$ with exceptional divisor $E \cong \mathbb{P}^2$ and $W_{m,j} := \operatorname{Im}\big(H^0(Y, m\pi^*H - jE) \to H^0(E, \mathcal{O}_E(j))\big)$. Then $H^0(Y, m\pi^*H - jE) = 0$ for $m \geq 1$ and $j > 3m$, and*

$$S(W; v) \leq \frac{1903}{1944} A_E(v)$$

for any prime divisor v over E. In particular $\delta_x(X) \geq 1944/1903$.

Proof. The proof is long, so we divide it into several steps.

Let L be the line through x. Since $[L]$ is a reduced point of $\Sigma(X, x)$, we have $H^0(L, N_{L/X}(-x)) = 0$ and $N_{L/X} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1)$, that is, L is non-special (Proposition 4.1). Let $\pi_L : Y_L := \operatorname{Bl}_L X \to X$ with exceptional divisor E_L, let $f_x \subset E_L$ be the fiber over x, and write H also for $\pi_L^* H$. By Theorem 2.22, there exists a flop $Y_L \dashrightarrow Y_L^+ = \operatorname{Bl}_\Gamma Q$, where $Q \subset \mathbb{P}^4$ is a smooth quadric threefold and $\Gamma \subset Q$ is a smooth curve of genus 2 and degree 7. Let h be the pullback of $\mathcal{O}_Q(1)$ and D' the exceptional divisor of $Y_L^+ \to Q$. We compare divisors on Y_L and Y_L^+ by strict transform. Then Theorem 2.22 implies

$$h = H - 2E_L, \qquad D' = 2H - 5E_L, \qquad H = 5h - 2D', \qquad E_L = 2h - D' \tag{6.1}$$

in $\operatorname{Pic} Y_L = \mathbb{Z}H \oplus \mathbb{Z}E_L$. Moreover, the flopping curves of $Y_L \dashrightarrow Y_L^+$ are the strict transforms of the lines $M \neq L$ with $M \cap L \neq \emptyset$ (Theorem 2.22).

Step 1. In this step we show that f_x is mapped isomorphically onto a smooth conic $C \subset Q$ such that $\Gamma \cap C$ has length 5, and that Γ is not contained in a hyperplane of $\mathbb{P}^4$.

Let $M \neq L$ be a line with $M \cap L \neq \emptyset$. Since L is the only line through x, we have $x \notin M$, so the strict transform of M intersects E_L only over $M \cap L \not\ni x$. Thus f_x is disjoint from the

flopping curves, and the flop is an isomorphism on a neighborhood of f_x. We denote the image of f_x in Y_L^+ again by f_x. Since $(H \cdot f_x) = 0$ and $(E_L \cdot f_x) = -1$, (6.1) implies

$$(h \cdot f_x) = 2, \qquad (D' \cdot f_x) = 5.$$

The curve f_x is not contained in D', since $(h \cdot f_x) = 2$, whereas an integral curve $C_0 \subset D'$ either maps to a point of Γ, and then $(h \cdot C_0) = 0$, or maps onto Γ, and then $(h \cdot C_0)$ is a positive multiple of $(h \cdot \Gamma) = 7$. Since $Y_L^+ \to Q$ is an isomorphism off D', f_x maps birationally onto an integral curve $C \subset Q$ with $\deg C = (h \cdot f_x) = 2$. An integral curve of degree 2 in $\mathbb{P}^4$ is a smooth plane conic, so $f_x \to C$ is a finite birational morphism of smooth projective curves, hence an isomorphism. The ideal of Γ generates $\mathcal{O}_{Y_L^+}(-D')$, so the ideal of $\Gamma \cap C$ in C corresponds to $\mathcal{O}_{f_x}(-D')$ under $f_x \cong C$. Thus

$$\operatorname{length}(\Gamma \cap C) = (D' \cdot f_x) = 5,$$

where Γ and C may intersect non-transversally.

Suppose that Γ is contained in a hyperplane of $\mathbb{P}^4$. Then $h - D'$ is effective on Y_L^+, hence on Y_L, as the flop is an isomorphism in codimension one. By (6.1), $h - D' = -H + 3E_L$. Let $D_1 \geq 0$ be a divisor with $D_1 \sim -H + 3E_L$. If D_1 has a component other than E_L, then $\pi_{L*}D_1$ is a non-zero member of $|-H|$, which is impossible since H is ample. Otherwise $D_1 = cE_L$ for some $c \geq 0$, which is impossible since $cE_L \not\sim -H + 3E_L$. Thus

$$H^0(Q, \mathcal{I}_\Gamma(1)) = 0. \tag{6.2}$$

Step 2. In this step we construct a fibration of $\operatorname{Bl}_C Q$ in quadric surfaces and compute the push-forward of the hyperplane sections through the strict transform of Γ.

Let $Q_C := \operatorname{Bl}_C Q$ with exceptional divisor E_C, and write h also for the pullback of $\mathcal{O}_Q(1)$ to Q_C. Let $\langle C \rangle \subset \mathbb{P}^4$ be the plane spanned by C. Since Q contains no plane, $Q \cap \langle C \rangle$ is a conic containing C, so $C = Q \cap \langle C \rangle$ is the complete intersection of Q with two hyperplanes. Thus the pencil of hyperplanes containing $\langle C \rangle$ defines a morphism $\varpi : Q_C \to \mathbb{P}^1$ with fiber class

$$\Phi := h - E_C.$$

For $b \in \mathbb{P}^1$ let ℓ_b be a linear form defining the corresponding hyperplane. The fiber $Q_b := \varpi^{-1}(b)$ is isomorphic to the quadric surface $Q \cap (\ell_b = 0)$, since C is a Cartier divisor on $Q \cap (\ell_b = 0)$. The curve C is a hyperplane section of Q_b in $(\ell_b = 0) \cong \mathbb{P}^3$, and C is smooth, so Q_b is smooth along C. By Bertini's theorem a general member of the pencil is smooth off the base locus C. Thus a general Q_b is a smooth quadric surface. Moreover, $\mathcal{O}_{Q_C}(h)$ is very ample over $\mathbb{P}^1$, and $H^1(Q_b, \mathcal{O}_{Q_b}(h)) = 0$ for any $b \in \mathbb{P}^1$.

Let $\Gamma' \subset Q_C$ be the strict transform of Γ. Since the ideal of $\Gamma \cap C$ in Γ is invertible, $\Gamma' \to \Gamma$ is an isomorphism, so Γ' is a smooth curve of genus 2. By **Step 1**,

$$(h \cdot \Gamma') = 7, \qquad (E_C \cdot \Gamma') = 5, \qquad (\Phi \cdot \Gamma') = 2.$$

Thus $\varpi|_{\Gamma'} : \Gamma' \to \mathbb{P}^1$ is finite and flat of degree 2. Let

$$\mathcal{V} := \varpi_* \mathcal{O}_{Q_C}(h), \qquad \mathcal{U} := \varpi_* \mathcal{O}_{\Gamma'}(h), \qquad \mathcal{K} := \ker(\mathcal{V} \to \mathcal{U}).$$

For any $b \in \mathbb{P}^1$, the scheme $\Gamma' \cap Q_b$ has length 2, and the hyperplanes of $(\ell_b = 0)$ separate any subscheme of length 2, so the restriction

$$H^0(Q_b, \mathcal{O}_{Q_b}(h)) \to H^0(\Gamma' \cap Q_b, \mathcal{O}_{\Gamma' \cap Q_b}(h))$$

is surjective. Since $H^1(Q_b, \mathcal{O}_{Q_b}(h)) = 0$, cohomology and base change [Har77, Theorem III.12.11] implies that $\mathcal{V}$ is a vector bundle of rank 4 with fibers $H^0(Q_b, \mathcal{O}_{Q_b}(h))$ and that $\mathcal{V} \to \mathcal{U}$ is surjective at any point $b \in \mathbb{P}^1$, also at the points over which $\varpi|_{\Gamma'}$ is branched and at the points b with Q_b singular. Thus $\mathcal{U}$ and $\mathcal{K}$ are vector bundles of rank 2.

The restriction $H^0(\mathbb{P}^4,\mathcal{O}(1)) \to H^0(Q_b,\mathcal{O}_{Q_b}(h))$ is surjective with kernel $\mathbb{C}\ell_b$, and ℓ_b depends linearly on the coordinates of b. Thus there exists an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^1}(-1) \to \mathcal{O}_{\mathbb{P}^1}^5 \to \mathcal{V} \to 0,$$

and $\deg \mathcal{V} = 1$. By the Riemann–Roch theorem on Γ', $\chi(\mathbb{P}^1,\mathcal{U}) = \chi(\Gamma',\mathcal{O}_{\Gamma'}(h)) = 7+1-2 = 6$, so $\deg\mathcal{U} = 6-2 = 4$ and $\deg\mathcal{K} = 1-4 = -3$. Moreover, $H^0(\mathbb{P}^1,\mathcal{K})$ is the space of sections of $\mathcal{O}_Q(1)$ vanishing on Γ, which is zero by (6.2). By [Gro57, Théorème 2.1], $\mathcal{K} \cong \mathcal{O}_{\mathbb{P}^1}(a_1) \oplus \mathcal{O}_{\mathbb{P}^1}(a_2)$ with $a_1+a_2 = -3$ and $a_1, a_2 \le -1$. Thus

$$\mathcal{K} \cong \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2), \qquad \mathcal{K}(2) \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}. \tag{6.3}$$

Let $\tilde{Q} := \mathrm{Bl}_{\Gamma'} Q_C$ with exceptional divisor E_Γ, write E_C, h, and Φ also for the pullbacks to $\tilde{Q}$, and let

$$R := 3h - 2E_C - E_\Gamma = h + 2\Phi - E_\Gamma. \tag{6.4}$$

Step 3. In this step we show that

$$H^0(Y, m\pi^*H - jE) = H^0(\tilde{Q}, 5mh - 2mE_\Gamma - jE_C) \tag{6.5}$$

for all integers $m, j \ge 0$, and that the strict transform $\tilde{E}_L$ of E_L in $\tilde{Q}$ satisfies $\tilde{E}_L = 2h - E_C - E_\Gamma$.

Let $Y_L' := \mathrm{Bl}_{f_x} Y_L$ with exceptional divisor E'. Choose local coordinates (z,u,v) of X at x with $L = (u = v = 0)$. On the chart of Y_L with coordinates $(z,u,v/u)$ we have $\mathfrak{m}_x\mathcal{O}_{Y_L} = (z,u)$, the ideal of f_x, and on the chart with coordinates $(z,u/v,v)$ we have $\mathfrak{m}_x\mathcal{O}_{Y_L} = (z,v)$, again the ideal of f_x. Thus $\mathfrak{m}_x\mathcal{O}_{Y_L'} = \mathcal{O}_{Y_L'}(-E')$, and by the universal property of blowing up [Har77, Proposition II.7.14], $Y_L' \to X$ factors through a birational morphism $g : Y_L' \to Y$ with $g^*E = E'$. Since $g_*\mathcal{O}_{Y_L'} = \mathcal{O}_Y$, the projection formula [Har77, Exercise II.5.1(d)] implies

$$H^0(Y, m\pi^*H - jE) = H^0(Y_L', mg^*\pi^*H - jE').$$

Since the flop is an isomorphism near f_x, it lifts to a birational map $Y_L' \dashrightarrow \mathrm{Bl}_{f_x} Y_L^+$ which is an isomorphism in codimension one. The smooth projective varieties $\mathrm{Bl}_{f_x} Y_L^+$ and $\tilde{Q}$ are birational over Q, and on both of them the prime divisors exceptional over Q are the divisors given by the blow-ups of Γ and of C at their generic points, because $Y_L^+ \to Q$ is an isomorphism near the generic point of C and f_x maps onto C. Hence $\mathrm{Bl}_{f_x} Y_L^+ \dashrightarrow \tilde{Q}$ is an isomorphism in codimension one, and we obtain a birational map $\phi : Y_L' \dashrightarrow \tilde{Q}$ which is an isomorphism in codimension one and satisfies $\phi_*E' = E_C$. In particular, the prime divisors on $\tilde{Q}$ exceptional over X are exactly $\tilde{E}_L$ and E_C.

A general member of $|h|$ on Y_L^+ is the pullback of a hyperplane section of Q which contains neither C nor Γ, and $f_x \notin D'$ by **Step 1**. Thus the pullbacks of h and D' to Y_L' are mapped by ϕ_* to h and E_Γ, and (6.1) implies $\phi_*g^*\pi^*H = 5h - 2E_\Gamma$. Since $E_L \cong \mathbb{F}_1$ is smooth along f_x, the pullback of E_L to Y_L' is the strict transform of E_L plus E'. Thus (6.1) implies $\tilde{E}_L = 2h - E_\Gamma - E_C$. Since ϕ is an isomorphism in codimension one between normal varieties, it preserves the spaces of global sections of corresponding divisors, which implies (6.5).

Step 4. In this step we show that $|R|$ has no fixed divisor and that, for a general $b \in \mathbb{P}^1$, the scheme $\Gamma' \cap Q_b$ consists of two points contained in different rulings of Q_b.

Since $\tilde{Q} \to Q_C$ is the blow-up of the smooth curve Γ', the left exactness of ϖ_* and the projection formula [Har77, Exercise II.5.1(d)] imply $H^0(\tilde{Q}, h + 2\Phi - E_\Gamma) = H^0(\mathbb{P}^1,\mathcal{K}(2))$. By (6.3), (6.4), and **Step 3**,

$$h^0(\tilde{Q}, R) = h^0(\mathbb{P}^1,\mathcal{K}(2)) = 3, \qquad h^0(\tilde{Q}, R - \tilde{E}_L) = h^0(\tilde{Q}, \Phi) = 2. \tag{6.6}$$

Moreover, $R - E_C = 3\Phi - E_\Gamma$, and any section of $3\Phi = \varpi^*\mathcal{O}_{\mathbb{P}^1}(3)$ is the pullback of a section of $\mathcal{O}_{\mathbb{P}^1}(3)$, which vanishes on Γ' only if it is zero, as $\varpi(\Gamma') = \mathbb{P}^1$. Thus $h^0(\tilde{Q}, R - E_C) = 0$. Hence neither $\tilde{E}_L$ nor E_C is a fixed divisor of $|R|$.

Let s_L be a section of $\mathcal{O}_{\tilde{Q}}(\tilde{E}_L)$ with divisor $\tilde{E}_L$. Since $R+\tilde{E}_L = 5h-2E_\Gamma-3E_C$, multiplication by s_L and (6.5) for $m=1$ and $j=3$ embed $H^0(\tilde{Q},R)$ into $H^0(Y,\pi^*H-3E)\subset H^0(X,H)$. Suppose that a prime divisor $T\neq \tilde{E}_L, E_C$ on $\tilde{Q}$ is a fixed divisor of $|R|$. Then T is not exceptional over X by **Step 3**, and the image T_X of T in X is a prime divisor contained in the divisor of any element of $s_L\cdot H^0(\tilde{Q},R)$, a subspace of $H^0(X,H)$ of dimension 3. Since $\operatorname{Pic} X=\mathbb{Z}H$, we have $T_X\in|cH|$ for some $c\geq 1$. Thus $s_L\cdot H^0(\tilde{Q},R)\subset s_T\cdot H^0(X,(1-c)H)$ for a section s_T with divisor T_X, and $h^0(X,(1-c)H)\leq 1$, a contradiction. Hence $|R|$ has no fixed divisor.

Let $b\in\mathbb{P}^1$ be general. Then Q_b is a smooth quadric surface, and $\Gamma'\cap Q_b$ consists of two distinct points p_1,p_2, at which Γ' intersects Q_b transversally, as $\varpi|_{\Gamma'}$ is unramified over b. Let $\tilde{Q}_b$ be the fiber of $\tilde{Q}$ over b. Then $\tilde{Q}_b=\operatorname{Bl}_{p_1,p_2}Q_b$. Let e_1,e_2 be the exceptional curves, let h_b be the pullback of $\mathcal{O}_{Q_b}(h)$, and let

$$\theta := h_b-e_1-e_2.$$

Then $R|_{\tilde{Q}_b}=\theta$ and $E_C|_{\tilde{Q}_b}=h_b$. Since the sequence $0\to\mathcal{K}\to\mathcal{V}\to\mathcal{U}\to 0$ of vector bundles remains exact on fibers, $\mathcal{K}(2)\otimes k(b)$ is the space of sections of $\mathcal{O}_{Q_b}(h)$ vanishing at p_1 and p_2. Since $\mathcal{K}(2)$ is globally generated by (6.3), the restrictions to $\tilde{Q}_b$ of the members of $|R|$ not containing $\tilde{Q}_b$ are the members $A-e_1-e_2$ of $|\theta|$, where A runs through the pullbacks of the hyperplane sections of Q_b through p_1 and p_2. Suppose that the line ℓ_{12} through p_1 and p_2 is contained in Q_b for a general b. Then any member of $|R|$ contains the strict transform of ℓ_{12} in $\tilde{Q}_b$ for a general b, and the closure of the union of the strict transforms of ℓ_{12} over a general b is a surface contained in any member of $|R|$, a contradiction. Thus $\ell_{12}\not\subset Q_b$ for a general b, that is, p_1 and p_2 are contained in different rulings of Q_b.

Step 5. In this step we prove that $H^0(Y,m\pi^*H-jE)=0$ for $m\geq 1$ and $j>3m$.

Let $b\in\mathbb{P}^1$ be general. Since $\ell_{12}\not\subset Q_b$, the scheme $\ell_{12}\cap Q_b$ is the reduced scheme $\{p_1,p_2\}$. Thus the planes containing ℓ_{12} cut on Q_b a pencil of conics with base scheme $\{p_1,p_2\}$, and two members of the pencil intersect transversally at p_1 and p_2, as $h_b^2=2$. Hence the blow-up of p_1 and p_2 resolves the pencil, and θ is base point free, in particular nef. Since $h_b^2=2$, $(h_b\cdot e_i)=0$, and $e_i^2=-1$, the restriction of $5mh-2mE_\Gamma-jE_C$ to $\tilde{Q}_b$ satisfies

$$\big(((5m-j)h_b-2m(e_1+e_2))\cdot\theta\big)=2(5m-j)-4m=2(3m-j).$$

Suppose that $j>3m$ and that $\tilde{Q}$ carries a divisor $D_0\geq 0$ with $D_0\sim 5mh-2mE_\Gamma-jE_C$. For a general b, $\tilde{Q}_b\not\subset\operatorname{Supp}D_0$, so $D_0|_{\tilde{Q}_b}\geq 0$ and $(D_0|_{\tilde{Q}_b}\cdot\theta)\geq 0$, as θ is nef, a contradiction with $2(3m-j)<0$. By (6.5), $H^0(Y,m\pi^*H-jE)=0$.

Step 6. In this step we bound $S(W;v)$.

Let $r_{m,j}:=\dim W_{m,j}$ and $N_m=h^0(X,mH)$. By Lemma 2.12(2), $N_m=3m^3+O(m^2)$. The kernel of $H^0(Y,m\pi^*H-jE)\to W_{m,j}$ is $H^0(Y,m\pi^*H-(j+1)E)$, so **Step 5** implies $\sum_{0\leq j\leq 3m}r_{m,j}=N_m$. By Proposition 4.5 with $d=18$ and $k=1$, $W_{m,j}=H^0(E,\mathcal{O}_E(j))$ for $0\leq j\leq m-2$ and $r_{m,j}=\frac{j^2}{2}-\frac{1}{4}(j-m)^2+O(m)$ uniformly for $m\leq j<2m$. Thus

$$\sum_{0\leq j<2m}r_{m,j}=\sum_{0\leq j<2m}\frac{j^2}{2}-\sum_{m\leq j<2m}\frac{(j-m)^2}{4}+O(m^2)=\frac{5}{4}m^3+O(m^2),$$

and

$$\sum_{2m\leq j\leq 3m}r_{m,j}=\frac{7}{4}m^3+O(m^2). \tag{6.7}$$

Let v be a prime divisor over E, let $a:=A_E(v)$, and for a linear subspace $U\subset H^0(E,\mathcal{O}_E(j))$ let $w_v(U)$ be the sum of the jumping numbers of the filtration induced by v on U. Then $S(W;v)=\lim_{m\to+\infty}\frac{1}{mN_m}\sum_j w_v(W_{m,j})$ [AZ22, Lemma 2.9]. Let $\mathcal{F}_v^t\subset H^0(E,\mathcal{O}_E(j))$ be the

subspace of sections s with $v(s) \geq t$. Since $w_v(U) = \int_0^\infty \dim(U \cap \mathcal{F}_v^t)dt$, we have $w_v(W_{m,j}) \leq w_v(H^0(E, \mathcal{O}_E(j)))$. By Lemma 3.2(1) with $\rho = 1$,

$$w_v(H^0(E, \mathcal{O}_E(j))) \leq j\binom{j+2}{2} S_{\mathcal{O}(1)}(v) + o(j^3) \leq \frac{a}{3} j\binom{j+2}{2} + o(j^3),$$

where $S_{\mathcal{O}(1)}(v) := S(V; v)$ for the complete series V of $\mathcal{O}_{\mathbb{P}^2}(1)$. Since $\int_0^2 \frac{t^3}{6} dt = \frac{2}{3}$, we obtain

$$\sum_{0 \leq j < 2m} w_v(W_{m,j}) \leq \frac{2a}{3} m^4 + o(m^4). \tag{6.8}$$

For any real numbers $r \geq 0$ and $j \geq 2m/3$,

$$jr - \frac{2\sqrt{2}}{3} r^{3/2} \leq \frac{2m}{3} r + \frac{1}{6}\Big(j - \frac{2m}{3}\Big)^3. \tag{6.9}$$

Inequality (6.9) holds because the function $r \to (j - \frac{2m}{3})r - \frac{2\sqrt{2}}{3} r^{3/2}$ on $[0, +\infty)$ attains the maximum $\frac{1}{6}(j - \frac{2m}{3})^3$ at $r = \frac{1}{2}(j - \frac{2m}{3})^2$. For $2m \leq j \leq 3m$, Lemma 3.2(2) applied to $U = W_{m,j}$ and (6.9) for $r = r_{m,j}$ imply

$$w_v(W_{m,j}) \leq a\Big(\frac{2m}{3} r_{m,j} + \frac{1}{6}\Big(j - \frac{2m}{3}\Big)^3\Big) + o(m^3),$$

where the error term is $o(m^3)$ uniformly in $2m \leq j \leq 3m$, since the error term of Lemma 3.2(2) depends on v but not on r or U, and $j \leq 3m$. Summing over $2m \leq j \leq 3m$, by (6.7) we obtain

$$\begin{aligned} \sum_{2m \leq j \leq 3m} w_v(W_{m,j}) &\leq a\Big(\frac{2}{3} \cdot \frac{7}{4} + \frac{1}{6} \int_2^3 \Big(t - \frac{2}{3}\Big)^3 dt\Big) m^4 + o(m^4) \\ &= \Big(\frac{7}{6} + \frac{715}{648}\Big) a m^4 + o(m^4) = \frac{1471}{648} a m^4 + o(m^4). \end{aligned}$$

Here the Riemann sums differ from the integrals by $O(m^3)$. Together with (6.8) and $mN_m = 3m^4 + O(m^3)$, we obtain

$$S(W; v) \leq \frac{1}{3}\Big(\frac{2}{3} + \frac{1471}{648}\Big) a = \frac{1903}{1944} A_E(v).$$

Step 7. In this step we prove that $\delta_x(X) \geq 1944/1903$.

By [AZ22, Lemma 2.9], $S_X(E) = \lim_{m \to +\infty} \frac{1}{mN_m} \sum_j j r_{m,j}$. By **Step 6**, the grades $j < 2m$ have total dimension $\frac{5}{4} m^3 + O(m^2)$, and the grades $2m \leq j \leq 3m$ have total dimension $\frac{7}{4} m^3 + O(m^2)$. Thus

$$S_X(E) \leq \frac{1}{3}\Big(2 \cdot \frac{5}{4} + 3 \cdot \frac{7}{4}\Big) = \frac{31}{12},$$

and $A_X(E)/S_X(E) \geq 3 \cdot \frac{12}{31} = \frac{36}{31}$. Since Y is smooth, (Y, E) is plt, $-E$ is ample over X, the different on E is zero, and $A_X(E) = 3$. By Theorem 2.9 applied to $Z = \{x\}$ and [AZ22, Lemma 2.9],

$$\delta_x(X) \geq \min\Big\{\frac{A_X(E)}{S_X(E)}, \inf_v \frac{A_E(v)}{S(W; v)}\Big\} \geq \min\Big\{\frac{36}{31}, \frac{1944}{1903}\Big\} = \frac{1944}{1903},$$

where v runs through the prime divisors over E. □

6.2. **Points on two lines.** For a point x on two lines L and M, we use the blow-up along L, with exceptional divisor $E \cong \mathbb{F}_1$, and the refinement by the fiber f_x (Definition 2.8). The second line M gives the point p of $Z_U \cap f_x$, and whether p is contained in the negative section s decides how hard the estimate is: Proposition 6.2 treats the case where $p \notin s$ for L or for M and gives $\delta_x(X) \geq 144/143$, and the case where $p \in s$ for both L and M is treated in Theorem 6.5.

Proposition 6.2. *Notation and conditions as in Set-up 5.1. Assume that $x \in L$ is a closed point such that $\Sigma(X,x)$ consists of two reduced points $[L]$ and $[M]$. Then L is non-special, and $Z_U \cap f_x$ is one reduced point p, which represents M. Assume moreover that $p \notin s$, and let $q := s \cap f_x$. Then the following hold.*

(1) $S(W; f_x) \le \frac{143}{144}$, $S(W^{f_x}; q) \le \frac{143}{144}$, and $S(W^{f_x}; a) \le \frac{7}{8}$ for $a \in f_x \backslash \{q\}$.
(2) Every prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies

$$A_X(F) \ge \frac{144}{143} S_X(F) + \frac{84}{143} \operatorname{mult}_F E.$$

In particular $\delta_x(X) \ge 144/143$.

Proof of Proposition 6.2. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that L is non-special, describe $Z_U \cap f_x$, and describe $\operatorname{Supp} Z_U \cap s$.

Since $\Sigma(X,x)$ is reduced at $[L]$, the line L is non-special by Proposition 4.1. Thus $E \cong \mathbb{F}_1$, and the notation of Set-up 5.1 for a non-special line is available. By Proposition 5.3,

$$Z_U \cap f_x \cong \Sigma(X,x) \backslash \{[L]\} = \{[M]\},$$

so $Z_U \cap f_x$ is one reduced point p, which represents M. Let Z_p be the component of Z_U supported at p, and let n be the length of Z_p. By Proposition 5.3, Z_U is a curvilinear scheme of length 4, n is the width of M, $\widehat{\mathcal{O}}_{Z_U,p} \cong \mathbb{C}[\epsilon]/(\epsilon^n)$, and the local ring of $Z_U \cap f_x$ at p is $\mathbb{C}[\epsilon]/(\epsilon^n, \psi(\epsilon))$, where $\psi(\epsilon)$ is the image of a local parameter of L at x. Since $Z_U \cap f_x$ is reduced at p, either $n = 1$ or ψ has order one. In both cases the image of a local parameter of L at x generates the maximal ideal of $\widehat{\mathcal{O}}_{Z_U,p}$, so the composite of $Z_p \subset E$ with $E \to L$ is a closed embedding. Moreover $1 \le n \le 4$.

By Construction-Proposition 5.4(2)(a), $D|_E \sim 5s + 7f$, and since $Z_U \cap f_x$ is the reduced point p, Construction-Proposition 5.4(4) implies that f_x is not a component of $D|_E$ and that $(D|_E)|_{f_x}$ has degree 5 and contains $3p$.

Let $p_T : T \to E$ be the four blow-ups resolving Z_U (Proposition 5.3). By Construction-Proposition 5.4(1)(c), at most one of the four centers of p_T is a point of s or of a strict transform of s. Since Z_U is curvilinear, the centers of p_T over a point $y \in \operatorname{Supp} Z_U$ are y and the points infinitely near y determined by the tangent directions of Z_U at y, and the subscheme of length two of Z_U at y is contained in s if and only if $y \in s$ and the second center over y is a point of the strict transform of s. Thus $\operatorname{Supp} Z_U \cap s$ has at most one point, and if $y \in \operatorname{Supp} Z_U \cap s$ and Z_U has length at least two at y, then the subscheme of length two of Z_U at y is not contained in s.

Step 2. In this step we fix coordinates on E and a one-parameter group of automorphisms of E.

Let z be an affine coordinate on $L \cong \mathbb{P}^1$ such that $x = \{z = 0\}$ and the point $z = +\infty$ is not in the image of $\operatorname{Supp} Z_U$, and let f_∞ be the fiber of $E \to L$ over $z = +\infty$. Over $L \backslash \{\infty\}$, the open set $E \backslash s$ is a line bundle, and we let u be a fiber coordinate of it such that, for any integers $0 \le \alpha \le \beta$, the sections of $\mathcal{O}_E(\alpha s + \beta f)$ are the polynomials

$$\sigma = \sum c_{d\ell} z^d u^\ell, \qquad 0 \le \ell \le \alpha, \quad 0 \le d \le \beta - \ell \tag{6.10}$$

[Har77, Exercise III.8.4(a), Proposition V.2.8 with Notation V.2.8.1, Proposition V.2.9, and Exercise II.5.1(d)]. Translating u by a section of the line bundle $E \backslash s$ preserves the form (6.10), so we may assume that $p = (z,u) = (0,0)$. Then $\operatorname{ord}_s(z^d u^\ell) = \alpha - \ell$, and $\operatorname{ord}_{f_x} \sigma$ is the least d with $c_{d\ell} \ne 0$ for some ℓ. Let

$$\zeta := \frac{1}{z}, \qquad u_\infty := \frac{u}{z}, \qquad v_\infty := \frac{1}{u_\infty}.$$

In the coordinates (ζ, u_∞) on $E \backslash (s \cup f_x)$, the monomial $z^d u^\ell$ in (6.10) is the function $\zeta^{\beta-\ell-d} u_\infty^\ell$, and in the coordinates (ζ, v_∞) near $s \cap f_\infty$ it is the function $\zeta^{\beta-\ell-d} v_\infty^{\alpha-\ell}$. Let $r \in f_\infty \backslash s$ be the point $\zeta = u_\infty = 0$, and let $t := s \cap f_\infty$, the point $\zeta = v_\infty = 0$.

For $\tau \in \mathbb{C}^*$ let $\varphi_\tau : E \to E$ be the automorphism given by $(z,u) \to (\tau^{-1}z, \tau u)$. In the other two charts,

$$\varphi_\tau(\zeta, u_\infty) = (\tau\zeta, \tau^2 u_\infty), \qquad \varphi_\tau(\zeta, v_\infty) = (\tau\zeta, \tau^{-2} v_\infty).$$

Thus φ_τ preserves s, f_x, and f_∞, so $\varphi_\tau^* \mathcal{O}_E(\alpha s + \beta f) \cong \mathcal{O}_E(\alpha s + \beta f)$. Similarly, for $c_1, c_2 \in \mathbb{C}^*$, the map $(z,u) \to (c_1 z, c_2 u)$ is an automorphism of E which fixes p, r, and t, preserves f_x, and multiplies the monomial $z^d u^\ell$ in (6.10) by $c_1^d c_2^\ell$, after linearizing $\mathcal{O}_E(\alpha s + \beta f)$.

Step 3. In this step we fix a subscheme $Z' \subset Z_U$ and a flat family of subschemes of E which degenerates Z' to a subscheme with monomial ideals.

By **Step 1**, $\mathcal{O}_{Z_p} \cong \mathbb{C}[z]/(z^n)$. Thus $u \equiv P(z)$ modulo the ideal of Z_p for a unique polynomial P with $\deg P < n$, and $P(0) = 0$ because $u(p) = 0$. Since $(u - P(z), z^n)$ is contained in the ideal of Z_p and both ideals have colength n, the ideal of Z_p is $(u - P(z), z^n)$. For $1 \le n' \le n$, the subscheme of length n' of Z_p has ideal $(u - P(z), z^{n'})$, where P is replaced by the sum of the terms of P of degree $< n'$. Let Z^{res} be the union of the components of Z_U other than Z_p. Then Z^{res} has length $4 - n$ and $\operatorname{Supp} Z^{\mathrm{res}} \cap (f_x \cup f_\infty) = \emptyset$.

We use the following families of ideals over $\mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[\tau]$. In each case the ideal at $\tau \neq 0$ is the ideal of the image under φ_τ of the subscheme at $\tau = 1$, and $\zeta_0, \zeta_1, \zeta_2 \in \mathbb{C}^*$ and $u_0, u_1, u_2, \gamma_0, \gamma_1 \in \mathbb{C}$ are constants.

(i) The subscheme of length n' of Z_p, with $P(z) = \sum_{i<n'} b_i z^i$:

$$\Big(u - \sum_{i<n'} \tau^{i+1} b_i z^i,\ z^{n'}\Big) \ \text{ with limit } \ (u, z^{n'}) \text{ at } p.$$

(ii) A reduced point (ζ_0, u_0) off s: $(u_\infty - \tau^2 u_0,\ \zeta - \tau\zeta_0)$, with limit (u_∞, ζ) at r.

(iii) A reduced point $(\zeta_0, 0)$ of s: $(v_\infty,\ \zeta - \tau\zeta_0)$, with limit (ζ, v_∞) at t.

(iv) A subscheme of length two off s which is not contained in a fiber, that is, two points with $\zeta_1 \neq \zeta_2$, or one point $\zeta_1 = \zeta_2$ with a tangent direction different from the fiber: $(u_\infty - \gamma_0 - \gamma_1\zeta,\ (\zeta - \zeta_1)(\zeta - \zeta_2))$ at $\tau = 1$, and

$$\big(u_\infty - \tau^2\gamma_0 - \tau\gamma_1\zeta,\ (\zeta - \tau\zeta_1)(\zeta - \tau\zeta_2)\big) \ \text{ with limit } \ (u_\infty, \zeta^2) \text{ at } r.$$

(v) A subscheme of length two off s contained in the fiber $\zeta = \zeta_0$, with $u_1 = u_2$ for a double point:

$$\big(\zeta - \tau\zeta_0,\ (u_\infty - \tau^2 u_1)(u_\infty - \tau^2 u_2)\big) \ \text{ with limit } \ (\zeta, u_\infty^2) \text{ at } r.$$

(vi) A subscheme of length two supported at a point $(\zeta_0, 0)$ of s and not contained in s: $(\zeta - \zeta_0 - \gamma_0 v_\infty,\ v_\infty^2)$ at $\tau = 1$, and

$$\big(\zeta - \tau\zeta_0 - \tau^3\gamma_0 v_\infty,\ v_\infty^2\big) \ \text{ with limit } \ (\zeta, v_\infty^2) \text{ at } t.$$

Any subscheme of length two of $E \backslash s$ supported in the chart (ζ, u_∞) is of type (iv) or (v), because two points with distinct values of ζ are cut out by the line through them and a double point is determined by the point of support and the tangent direction.

We now fix $Z' \subset Z_U$, the integer n', and a symbol ω, according to the following cases, which cover all possibilities by **Step 1**.

Here n' is the length of the part of Z' at p, and ω is one of the symbols $\emptyset$, (r), (h), (v), (r+t), and (t_v): the symbols (r), (h), (v), and (t_v) stand for the ideals (u_∞, ζ), (u_∞, ζ^2), (ζ, u_∞^2) at r and (ζ, v_∞^2) at t, and (r+t) for the union of the reduced points r and t. The symbol ω names the limit at $\tau = 0$, in the families (i)–(vi), of the part of Z' away from f_x. The cases are as follows.

(a) $n \ge 3$. Let Z' be the subscheme of length three of Z_p. Then $n' = 3$ and $\omega = \emptyset$.

(b) $n = 2$ and $\operatorname{Supp} Z^{\mathrm{res}} \not\subset s$. Let Z' be the union of Z_p and one reduced point of $\operatorname{Supp} Z^{\mathrm{res}} \backslash s$. Then $n' = 2$ and ω is (r).

(c) $n = 2$ and $\operatorname{Supp} Z^{\mathrm{res}} \subset s$. By **Step 1**, Z^{res} is supported at one point of s, has length two, and is not contained in s. Let $Z' := Z_U$. Then $n' = 2$ and ω is (t_v).

(d) $n = 1$ and the part of Z^{res} supported off s has length at least two. Let Z' be the union of p and a subscheme of length two of that part, which is of type (iv) or (v). Then $n' = 1$, and ω is (h) or (v), respectively.

(e) $n = 1$ and the part of Z^{res} supported off s has length one. By **Step 1** the rest of Z^{res} is supported at one point of s. Let Z' be the union of p, the point of $\operatorname{Supp} Z^{\mathrm{res}} \backslash s$, and the reduced point of $\operatorname{Supp} Z^{\mathrm{res}} \cap s$. Then $n' = 1$ and ω is (r+t).

(f) $n = 1$ and $\operatorname{Supp} Z^{\mathrm{res}} \subset s$. By **Step 1**, Z^{res} is supported at one point of s, and the subscheme of length two of Z^{res} is not contained in s. Let Z' be the union of p and the subscheme of length two of Z^{res}. Then $n' = 1$ and ω is (t_v).

Thus (n', ω) is one of seven pairs.

For $\tau \in \mathbb{A}^1$ let Z'_τ be the subscheme of E whose components are given by the ideals of (i)–(vi) at τ. Thus $Z'_1 = Z'$ and $Z'_\tau = \varphi_\tau(Z')$ for $\tau \neq 0$. Each family in (i)–(vi) has the form $(\xi - P_\tau(\eta), Q_\tau(\eta))$ in coordinates (ξ, η) on $\mathbb{A}^2$, with Q_τ monic in η and $\deg_\eta P_\tau < \deg_\eta Q_\tau$. By Lemma 3.5, for any integer $\nu \geq 1$ the quotient of $\mathbb{C}[\tau][\xi, \eta]$ by the ν-th power of each family is a free $\mathbb{C}[\tau]$-module of finite rank, so the corresponding subscheme of $\mathbb{A}^2 \times \mathbb{A}^1$ is finite over $\mathbb{A}^1$ and closed in $E \times \mathbb{A}^1$. For $\tau \neq 0$ the components of Z'_τ are pairwise disjoint because φ_τ is an automorphism, and for $\tau = 0$ they are supported at the distinct points p, r, and t. Since the ν-th power of the intersection of pairwise comaximal ideals is the intersection of their ν-th powers, the subscheme $\mathcal{Z}'_\nu \subset E \times \mathbb{A}^1$ defined by the ν-th power of the ideal of $\bigcup_\tau Z'_\tau \times \{\tau\}$ is finite and flat over $\mathbb{A}^1$, with fiber over τ defined by $\mathcal{I}^\nu_{Z'_\tau}$.

Step 4. In this step we bound the dimensions of the subspaces of sections with given order along f_x which vanish on $\mathcal{I}^\nu_{Z'}$.

Let $0 \leq \alpha \leq \beta$, $\nu \geq 0$, and $k \geq 0$ be integers, and for $\tau \in \mathbb{A}^1$ let

$$V_\tau(k) := \{\sigma \in H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathcal{I}^\nu_{Z'_\tau}) : \operatorname{ord}_{f_x} \sigma \geq k\}.$$

We prove that $\dim V_1(k) \leq \dim V_0(k)$, and that $V_0(k)$ is spanned by the monomials $z^d u^\ell$ of (6.10) with $d \geq k$ which satisfy the following conditions, one for each component of Z'_0:

$$\begin{aligned}
&(u, z^{n'})^\nu \text{ at } p: && d + n'\ell \geq n'\nu, && (\zeta, v_\infty)^\nu \text{ at } t: && \alpha + \beta - d - 2\ell \geq \nu,\\
&(u_\infty, \zeta)^\nu \text{ at } r: && \beta - d \geq \nu, && (\zeta, v_\infty^2)^\nu \text{ at } t: && \alpha + 2\beta - 2d - 3\ell \geq 2\nu,\\
&(u_\infty, \zeta^2)^\nu \text{ at } r: && \beta - d + \ell \geq 2\nu, && (\zeta, u_\infty^2)^\nu \text{ at } r: && 2\beta - 2d - \ell \geq 2\nu.
\end{aligned} \tag{6.11}$$

The space $V_\tau(k)$ is the kernel of the map

$$H^0(E, \mathcal{O}_E(\alpha s + \beta f - k f_x)) \to H^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathcal{O}_E/\mathcal{I}^\nu_{Z'_\tau}\big)$$

given by multiplication by z^k followed by restriction. By **Step 3**, as τ varies, the maps above form a morphism from a trivial vector bundle on $\mathbb{A}^1$ to the locally free sheaf obtained by pushing forward $\mathcal{O}_{\mathcal{Z}'_\nu} \otimes \mathcal{O}_E(\alpha s + \beta f)$. Hence $\tau \to \dim V_\tau(k)$ is upper semicontinuous [Har77, Exercise II.5.8(a)]. For $\tau \neq 0$, pullback by φ_τ maps $V_\tau(k)$ isomorphically onto $V_1(k)$, since $\varphi_\tau(Z') = Z'_\tau$ and $\varphi_\tau(f_x) = f_x$. Thus $\dim V_1(k) = \dim V_\tau(k) \leq \dim V_0(k)$.

The ideals of the components of Z'_0 are monomial in the coordinates (u, z), (u_∞, ζ), and (ζ, v_∞), so $V_0(k)$ is stable under the automorphisms $(z, u) \to (c_1 z, c_2 u)$ of **Step 2**. Since the monomials in (6.10) are eigenvectors of the automorphisms $(z, u) \to (c_1 z, c_2 u)$ with pairwise distinct characters, $V_0(k)$ is spanned by monomials. A monomial $\xi^i \eta^{i'}$ belongs to $(\xi, \eta^c)^\nu$ if and only if $ci + i' \geq c\nu$. Applying the criterion to the local functions $z^d u^\ell$, $\zeta^{\beta-\ell-d} u_\infty^\ell$, and $\zeta^{\beta-\ell-d} v_\infty^{\alpha-\ell}$ of **Step 2**, we obtain (6.11).

Step 5. In this step we prove that $S(W; f_x) \leq \frac{143}{144}$.

Let U_{5m-2j} be as in Construction-Proposition 5.4, and let d_E be the restriction to E of a section defining D. By Construction-Proposition 5.4(3),

$$W_{m,j} \subset H^0\big(E, \mathcal{O}_E(js + (m + j)f) \otimes \mathcal{I}^{j-m}_{Z_U}\big) \qquad \text{for } m \leq j \leq 2m, \tag{6.12}$$

$$W_{m,j} = d_E^{j-2m} \cdot U_{5m-2j} \qquad \text{for } 2m \le j \le \lfloor 5m/2 \rfloor, \tag{6.13}$$

and $U_{5m-2j} \subset H^0(E, \mathcal{O}_E((5m-2j)(2s+3f)) \otimes \mathcal{I}_{Z_U}^{5m-2j})$ for $2m \le j \le \lfloor 5m/2 \rfloor$. By Lemma 5.2, $W_{m,j} = 0$ for $j > 5m/2$, $N_m = 3m^3 + O(m^2)$, $A_X(E) = 2$, $S_X(E) = \frac{101}{72}$, and

$$J := \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{2m<j\le 5m/2} (j-2m) \dim W_{m,j} = \frac{1}{72}.$$

Here $\dim W_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$, because the kernel of the restriction map $H^0(Y, mH - jE) \to H^0(E, \mathcal{O}_E(js + (m+j)f))$ is $H^0(Y, mH - (j+1)E)$, so J is the limit in the last formula of Lemma 5.2.

For $k \ge 0$ let $W_{m,j}^{\ge k} := \{\sigma \in W_{m,j} : \operatorname{ord}_{f_x} \sigma \ge k\}$. Then

$$S_m(W; f_x) = \frac{1}{mN_m} \sum_{j\ge 0} \sum_{k\ge 1} \dim W_{m,j}^{\ge k},$$

and $S(W; f_x) = \lim_{m\to+\infty} S_m(W; f_x)$ [AZ22, Lemma 2.9]. Let (n', ω) be the pair of **Step 3**.

Let $m \le j \le 2m$. Since $\mathcal{I}_{Z_U} \subset \mathcal{I}_{Z'}$, (6.12) implies $W_{m,j}^{\ge k} \subset V_1(k)$ for $(\alpha, \beta, \nu) = (j, m+j, j-m)$. By **Step 4**,

$$\sum_{k\ge 1} \dim W_{m,j}^{\ge k} \le \sum_{k\ge 1} \dim V_0(k) = \sum d,$$

where the last sum runs over the monomials $z^d u^\ell$ of (6.10) which satisfy the conditions of (6.11) for the components of Z_0'. Let $w := (j-m)/m$, $y := \ell/m$, and $\delta := d/m$. For $w \in [0,1]$ let $\mathcal{P}(w) \subset \mathbb{R}^2$ be the polygon of the points (δ, y) with

$$0 \le y \le 1 + w, \qquad 0 \le \delta \le 2 + w - y, \qquad \delta + n'y \ge n'w,$$

which satisfy the conditions of (6.11) for the components of Z_0' at r and t with $(\alpha, \beta, \nu, \ell, d)$ replaced by $(1+w, 2+w, w, y, \delta)$. For example, for (t_v) the condition at t is $2\delta \le 5 + w - 3y$. Then the monomials in the last sum correspond to the points of $\mathcal{P}(w) \cap \frac{1}{m}\mathbb{Z}^2$, and we let

$$\mu(w) := \int_{\mathcal{P}(w)} \delta dy d\delta.$$

Since the lines bounding $\mathcal{P}(w)$ have fixed rational slopes and depend linearly on w, the last sum is $m^3\mu(w) + O(m^2)$, uniformly in j.

Let $0 \le j < m$. Counting all monomials in (6.10) with $(\alpha, \beta) = (j, m+j)$, we obtain

$$\sum_{k\ge 1} \dim W_{m,j}^{\ge k} \le m^3 \int_0^{j/m} \frac{1}{2}\Big(1 + \frac{j}{m} - y\Big)^2 dy + O(m^2).$$

Let $2m < j \le 5m/2$. For $k \ge 0$ let $U_{5m-2j}^{\ge k} := \{\sigma \in U_{5m-2j} : \operatorname{ord}_{f_x} \sigma \ge k\}$. Since f_x is not a component of $D|_E$ by Construction-Proposition 5.4(4), (6.13) implies $W_{m,j}^{\ge k} = d_E^{j-2m} \cdot U_{5m-2j}^{\ge k}$. Since $U_{5m-2j} \subset H^0(E, \mathcal{O}_E((5m-2j)(2s+3f)) \otimes \mathcal{I}_{Z_U}^{5m-2j})$ and $\mathcal{I}_{Z_U} \subset \mathcal{I}_{Z'}$, we have $U_{5m-2j}^{\ge k} \subset V_1(k)$ for

$$(\alpha, \beta, \nu) = \big(2(5m-2j), 3(5m-2j), 5m-2j\big).$$

By **Step 4** and the count for $m \le j \le 2m$ at $w = 1$, with m replaced by $5m - 2j$, we obtain $\sum_{k\ge 1} \dim W_{m,j}^{\ge k} \le (5m-2j)^3 \mu(1) + O(m^2)$.

Summing over j and using

$$\int_0^1 \int_0^\theta \frac{1}{2}(1 + \theta - y)^2 dy d\theta = \frac{11}{24}, \qquad \int_2^{5/2} (5 - 2\theta)^3 d\theta = \frac{1}{8},$$

we obtain

$$\sum_{j\ge 0} \sum_{k\ge 1} \dim W_{m,j}^{\ge k} \le m^4 \Big(\frac{11}{24} + \int_0^1 \mu(w) dw + \frac{1}{8}\mu(1)\Big) + O(m^3).$$

Since $mN_m = 3m^4 + O(m^3)$, we have $S_m(W; f_x) \le \Psi + O(1/m)$, where

$$\Psi := \frac{1}{3}\Big(\frac{11}{24} + \int_0^1 \mu(w)dw + \frac{1}{8}\mu(1)\Big).$$

The function μ is piecewise polynomial in w. A direct calculation of the integrals of δ over the polygons $\mathcal{P}(w)$ and of μ over $[0,1]$ shows that $\Psi \le \frac{143}{144}$ for each of the seven pairs (n', ω) of **Step 3**. The values are listed in Table 76 (Remark A.11), and the polygons $\mathcal{P}(w)$ for (t_v) are described in Remark A.12 and Table 77. Thus $S(W; f_x) \le \frac{143}{144}$.

Step 6. In this step we prove the bounds for $S(W^{f_x}; q)$ and $S(W^{f_x}; a)$.

Since $((js + (m+j)f) \cdot f) = j$, the refinement W^{f_x} of W by f_x (Definition 2.8) is given by

$$W^{f_x}_{m,j,k} := \mathrm{Im}\left(W^{\ge k}_{m,j} \to H^0(f_x, \mathcal{O}_{f_x}(j))\right), \qquad \sigma \to (z^{-k}\sigma)|_{f_x}.$$

For a point $a \in f_x$ and a finite-dimensional space V of sections of a line bundle on f_x, let $\mathcal{T}_a(V) := \sum_{i\ge 1} \dim\{\gamma \in V : \mathrm{ord}_a \gamma \ge i\}$, the sum of the values of ord_a on a basis of V adapted to the filtration by ord_a. Then

$$S_m(W^{f_x}; a) = \frac{1}{mN_m} \sum_{j,k\ge 0} \mathcal{T}_a(W^{f_x}_{m,j,k}),$$

$S(W^{f_x}; a) = \lim_{m\to+\infty} S_m(W^{f_x}; a)$ [AZ22, Lemma 2.9], and $\mathcal{T}_a(V) \le \mathcal{T}_a(V')$ if $V \subset V'$. The function u is a coordinate on $f_x \backslash \{q\}$. For $\sigma = \sum c_{d\ell} z^d u^\ell \in W^{\ge k}_{m,j}$, we have $(z^{-k}\sigma)|_{f_x} = \sum_\ell c_{k\ell} u^\ell$, and $\mathrm{ord}_q(u^\ell) = j - \ell$.

We first treat the grades $0 \le j \le 2m$. Since $p \in \mathrm{Supp}\, Z_U$, (6.12) implies $W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m+j)f) \otimes \mathfrak{m}_p^{\max\{0, j-m\}})$. Since $\mathfrak{m}_p = (z, u)$ and $(z,u)^\nu$ is a monomial ideal primary to $\mathfrak{m}_p$, a polynomial σ belongs to $\mathfrak{m}_p^\nu$ if and only if $c_{d\ell} = 0$ whenever $d + \ell < \nu$. Together with (6.10), we obtain

$$W^{f_x}_{m,j,k} \subset \mathrm{Span}\left\{u^\ell : \max\{0, j-m-k\} \le \ell \le \min\{j, m+j-k\}\right\}. \tag{6.14}$$

Hence $\mathcal{T}_q(W^{f_x}_{m,j,k}) \le \sum_\ell (j - \ell)$, where ℓ runs over the range in (6.14). With $y = \ell/m$, the order k/m runs over $[\max\{0, w-y\}, 2 + w - y]$ for $w = (j-m)/m \ge 0$, and over $[0, 1 + j/m - y]$ for $j < m$. Summing over k and ℓ, we obtain

$$\sum_{k\ge 0} \mathcal{T}_q(W^{f_x}_{m,j,k}) \le m^3\Big(\frac{1}{3}\theta^3 + \frac{1}{2}\theta^2\Big) + O(m^2) \qquad \text{for } \theta := j/m \in [0,1],$$

$$\sum_{k\ge 0} \mathcal{T}_q(W^{f_x}_{m,j,k}) \le m^3\Big(w^2 + 2w + \frac{5}{6}\Big) + O(m^2) \qquad \text{for } w := (j-m)/m \in [0,1].$$

We now treat the grades $2m < j \le 5m/2$. For $k \ge 0$ let

$$U^{f_x}_{5m-2j,k} := \mathrm{Im}\left(U^{\ge k}_{5m-2j} \to H^0(f_x, \mathcal{O}_{f_x}(2(5m-2j)))\right), \qquad \sigma \to (z^{-k}\sigma)|_{f_x}.$$

Since f_x is not a component of $D|_E$, (6.13) implies

$$W^{f_x}_{m,j,k} = (d_E|_{f_x})^{j-2m} \cdot U^{f_x}_{5m-2j,k}.$$

By Construction-Proposition 5.4(4), $(D|_E)|_{f_x}$ has degree 5 and contains $3p$, and $p \ne q$, so $\mathrm{ord}_q(d_E|_{f_x}) \le 2$. Thus

$$\mathcal{T}_q(W^{f_x}_{m,j,k}) \le 2(j-2m)\dim W^{f_x}_{m,j,k} + \mathcal{T}_q(U^{f_x}_{5m-2j,k}).$$

Since $U_{5m-2j} \subset H^0(E, \mathcal{O}_E((5m-2j)(2s+3f)) \otimes \mathfrak{m}_p^{5m-2j})$, the description of $\mathfrak{m}_p^\nu$ above shows that (6.14) holds with $W^{f_x}_{m,j,k}$ replaced by $U^{f_x}_{5m-2j,k}$ and (m, j) replaced by $(5m-2j, 2(5m-2j))$. Thus

the second bound above at $w=1$ implies $\sum_k \mathcal{T}_q(U^{f_x}_{5m-2j,k}) \le \frac{23}{6}(5m-2j)^3 + O(m^2)$. Moreover $\sum_k \dim W^{f_x}_{m,j,k} = \dim W_{m,j}$. The integrals of the three bounds over j/m are

$$\int_0^1 \Big(\frac{1}{3}\theta^3 + \frac{1}{2}\theta^2\Big)d\theta = \frac{1}{4}, \quad \int_0^1 \Big(w^2 + 2w + \frac{5}{6}\Big)dw = \frac{13}{6}, \quad \frac{23}{6}\int_2^{5/2}(5-2\theta)^3 d\theta = \frac{23}{48}$$

(Remark A.13 and Table 78). Summing over j and using the definition of J, we obtain

$$S(W^{f_x};q) \le \frac{1}{3}\Big(\frac{1}{4} + \frac{13}{6} + \frac{23}{48}\Big) + 2J = \frac{139}{144} + \frac{1}{36} = \frac{143}{144}.$$

Let $a \in f_x \backslash \{q\}$ and $u_a := u(a)$. Dropping the lower bound in (6.14), we have $W^{f_x}_{m,j,k} \subset \mathrm{Span}\{1, u, \dots, u^{e_k}\}$ with $e_k := \min\{j, m+j-k\}$ for $0 \le j \le 2m$. Using the basis $(u-u_a)^\ell$, we have $\mathcal{T}_a(\mathrm{Span}\{1,\dots,u^{e_k}\}) = \frac{1}{2}e_k(e_k+1)$. Summing over k, with $\theta = j/m \in [0,2]$, we obtain

$$\sum_{k\ge 0} \mathcal{T}_a(W^{f_x}_{m,j,k}) \le m^3\Big(\frac{\theta^2}{2} + \frac{\theta^3}{6}\Big) + O(m^2).$$

For $2m < j \le 5m/2$, the bound on $\sum_{k\ge0}\mathcal{T}_a(W^{f_x}_{m,j,k})$ at $\theta = 2$, applied to $U^{f_x}_{5m-2j,k}$ in place of $W^{f_x}_{m,j,k}$ and to $5m-2j$ in place of m, implies $\sum_k \mathcal{T}_a(U^{f_x}_{5m-2j,k}) \le \frac{10}{3}(5m-2j)^3 + O(m^2)$, and $\mathrm{ord}_a(d_E|_{f_x}) \le 5$. Since

$$\int_0^2 \Big(\frac{\theta^2}{2} + \frac{\theta^3}{6}\Big)d\theta = 2, \qquad \frac{10}{3}\int_2^{5/2}(5-2\theta)^3 d\theta = \frac{5}{12}$$

(Remark A.13 and Table 78), we obtain

$$S(W^{f_x};a) \le \frac{1}{3}\Big(2 + \frac{5}{12}\Big) + 5J = \frac{29}{36} + \frac{5}{72} = \frac{7}{8}.$$

Together with **Step 5**, this proves (1).

Step 7. In this step we prove (2).

Since E and f_x are smooth, the pair (E, f_x) is plt, the different of (E, f_x) on f_x is zero by the adjunction formula [Har77, Proposition II.8.20], $A_E(f_x) = 1$, and $A_{f_x}(a) = 1$ for any closed point $a \in f_x$. Let $a \in f_x$ be a closed point. By Theorem 2.9 applied to $(E, 0)$, the series W, $Z = \{a\}$, and the prime divisor f_x on E,

$$\delta_a(E;W) \ge \min\Big\{\frac{1}{S(W;f_x)}, \frac{1}{S(W^{f_x};a)}\Big\} \ge \frac{144}{143}$$

by (1). Thus $\frac{144}{143}S(W;v) \le A_E(v)$ for any prime divisor v over E with $\mathrm{center}_E v \cap f_x \ne \emptyset$.

The divisor E is of plt type over X with plt extraction $\pi : Y \to X$, since Y and E are smooth and $-E$ is π-ample, and the different of (Y, E) on E is zero [Har77, Proposition II.8.20]. By Lemma 3.1, applied with $G := f_x$ and with $\lambda = 144/143$,

$$A_X(F) \ge \frac{144}{143}S_X(F) + \Big(A_X(E) - \frac{144}{143}S_X(E)\Big)\,\mathrm{mult}_F E$$

for any prime divisor F over X with $\mathrm{center}_Y F \cap f_x \ne \emptyset$. Since $A_X(E) = 2$ and $S_X(E) = \frac{101}{72}$ by Lemma 5.2,

$$A_X(E) - \frac{144}{143}S_X(E) = 2 - \frac{144}{143}\cdot\frac{101}{72} = \frac{84}{143}.$$

If $x \in \mathrm{center}_X F$, then $\pi(\mathrm{center}_Y F) = \mathrm{center}_X F$ implies $\mathrm{center}_Y F \cap \pi^{-1}(x) \ne \emptyset$, and $\pi^{-1}(x) = f_x$. This proves the inequality in (2). Since $\mathrm{mult}_F E \ge 0$ and $S_X(F) > 0$, we have $A_X(F)/S_X(F) \ge \frac{144}{143}$ for any prime divisor F over X with $x \in \mathrm{center}_X F$, that is, $\delta_x(X) \ge \frac{144}{143}$. □

To treat the case where $p \in s$ for both lines through x, we compare the positions of the points of $\Sigma(X,x)$ on the exceptional divisors of both lines. By Lemma 6.3(2), in this case Z_U has length at least two at $s \cap f_x$, and Construction 6.4 fixes coordinates for this case.

Lemma 6.3. *Let X be a smooth prime Fano threefold of genus 10 and $x \in X$ a closed point. Assume that $\Sigma(X,x)$ consists of two reduced points $[L_1]$ and $[L_2]$. Then the following hold.*

(1) L_1 and L_2 are non-special.
(2) Use the notation of Set-up 5.1 for L_i with an index i, and for $\{i,j\} = \{1,2\}$ let $p_{ij} \in f_{i,x}$ be the point representing L_j (Proposition 5.3) and n_{ij} the length of Z_{U_i} at p_{ij}. Then $n_{ij} > 1$ if and only if $p_{ji} \in s_j$.

Proof. By Proposition 4.1, a point of $\Sigma(X,x)$ with local ring $\mathbb{C}$ corresponds to a non-special line. Thus (1) holds.

For (2), let Inc_{L_i} be the incidence scheme of Proposition 5.3 for L_i. By Proposition 5.3, the isomorphism $\mathrm{Inc}_{L_i} \cong Z_{U_i}$ sends (L_j,x) to p_{ij}, and the completed local ring of Inc_{L_i} at (L_j,x) is $\mathbb{C}[t]/(t^{n_{ij}})$. Thus $n_{ij} > 1$ if and only if the Zariski tangent space $T_{(L_j,x)}\mathrm{Inc}_{L_i}$ is non-zero.

Let $\mathcal{L} \to \Sigma(X)$ be the universal line and $g : \mathcal{L} \to X$ the evaluation map. Then

$$T_{(L_j,x)}\mathrm{Inc}_{L_i} = \{\xi \in T_{(L_j,x)}\mathcal{L} \mid dg(\xi) \in T_x L_i\}.$$

Since L_j is non-special, $N_{L_j/X} \cong \mathcal{O}_{L_j} \oplus \mathcal{O}_{L_j}(-1)$ [KPS18, Lemma 2.1.4]. By [KPS18, Corollary 2.1.6(ii)] and [KPS18, the paragraph following Lemma 2.1.4], $\Sigma(X)$ is smooth of dimension one at $[L_j]$ and $T_{[L_j]}\Sigma(X) = H^0(L_j, N_{L_j/X}) = \mathbb{C}\sigma_j$, where σ_j is a nowhere vanishing section of the summand $\mathcal{O}_{L_j}$ of $N_{L_j/X}$. By [KPS18, (2.2.2)], if $\xi \in T_{(L_j,x)}\mathcal{L}$ maps to $c\sigma_j$ with $c \in \mathbb{C}$, then the image of $dg(\xi)$ in $N_{L_j/X,x} = T_xX/T_xL_j$ is $c\sigma_j(x)$. Since L_i and L_j are distinct lines in the anti-canonical embedding, $T_xL_i \neq T_xL_j$, and the image $\overline{T_xL_i}$ of T_xL_i in $N_{L_j/X,x} = T_xX/T_xL_j$ is a line.

Let $\xi \in T_{(L_j,x)}\mathrm{Inc}_{L_i}$. If ξ maps to 0 in $T_{[L_j]}\Sigma(X)$, then ξ is tangent to the fiber L_j of $\mathcal{L}$, so $dg(\xi) \in T_xL_j \cap T_xL_i = 0$ and $\xi = 0$. Thus $T_{(L_j,x)}\mathrm{Inc}_{L_i} \neq 0$ implies $\sigma_j(x) \in \overline{T_xL_i}$. Conversely, assume that $\sigma_j(x) \in \overline{T_xL_i}$. Let $\xi_0 \in T_{(L_j,x)}\mathcal{L}$ map to σ_j and $\eta \in T_xL_i$ with $dg(\xi_0) - \eta \in T_xL_j$. There exists a vector ξ_1 tangent to the fiber L_j with $dg(\xi_1) = dg(\xi_0) - \eta$, and $\xi_0 - \xi_1$ is a non-zero element of $T_{(L_j,x)}\mathrm{Inc}_{L_i}$. Hence $n_{ij} > 1$ if and only if $\overline{T_xL_i} = \mathbb{C}\sigma_j(x)$.

Now $E_j = \mathbb{P}(N_{L_j/X})$. Let $\tilde{L}_i \subset Y_j$ be the strict transform of L_i. The point of $f_{j,x} = \mathbb{P}(N_{L_j/X,x})$ given by $\overline{T_xL_i}$ is $\tilde{L}_i \cap E_j$. Since $L_i \cap L_j = \{x\}$ is a reduced point, $(E_j \cdot \tilde{L}_i) = 1$ and

$$((H - 2E_j) \cdot \tilde{L}_i) = 1 - 2 = -1,$$

so any member of $|H - 2E_j|$ contains $\tilde{L}_i$. Thus $\tilde{L}_i \cap E_j$ is a point of $\mathrm{Supp}\, Z_{U_j} \cap f_{j,x}$. By Proposition 5.3, $Z_{U_j} \cap f_{j,x} \cong \Sigma(X,x)\backslash\{[L_j]\}$ is one point, namely p_{ji}. Hence $\tilde{L}_i \cap E_j = p_{ji}$. The summand $\mathcal{O}_{L_j} \subset N_{L_j/X}$ defines a section s'_j of $E_j \to L_j$ with normal bundle $\mathcal{H}om(\mathcal{O}_{L_j}, N_{L_j/X}/\mathcal{O}_{L_j}) \cong \mathcal{O}_{L_j}(-1)$. Thus $(s'_j)^2 = -1$ and $s'_j = s_j$, the unique curve on $E_j \cong \mathbb{F}_1$ with negative self-intersection. Hence $\overline{T_xL_i} = \mathbb{C}\sigma_j(x)$ if and only if $p_{ji} \in s_j$, which proves (2). □

When, for both lines through a point x of Theorem 6.5, the point of $Z_U \cap f_x$ is $s \cap f_x$, after rescaling the fiber coordinate u by a non-zero constant (see (6.16)), the ideal of Z_U near $s \cap f_x$ is contained in the ideal $\mathfrak{a}$ of Construction 6.4(3), which defines a length two subscheme tangent to neither s nor f_x. Using only that $I_{Z_U} \subset \mathfrak{a}$ to bound the filtration by the order of vanishing at $s \cap f_x$, without Lemma 3.3, one gets the bound 53/48 for $S(W^{f_x}; s \cap f_x)$, which is larger than one, whereas the bound needed for $\delta_x(X) \geq 216/215$ in Theorem 6.5 is smaller than 215/216. So for the orders of vanishing at $s \cap f_x$ up to a cutoff given by a piecewise linear function of (m,j), we use $\dim W_{m,j}$ instead (Lemma 3.3), and obtain the bound $356827303/358612992 < 215/216$. The following construction fixes the coordinates on E and the data used to bound $S(W; f_x)$ and $S(W^{f_x}; s \cap f_x)$.

Construction 6.4. Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point. Assume that $Z_U \cap f_x$ is the reduced point $q := s \cap f_x$, and that the length n of Z_U at q satisfies $n \geq 2$.

(1) Since $E \cong \mathbb{F}_1$, the open set $E\backslash s$ is the total space of $\mathcal{O}_L(1)$. Let z be an affine coordinate on $L \cong \mathbb{P}^1$, viewed also as a rational function on E, such that $f_x = (z = 0)$ and $Z_U \cap (z = +\infty) = \emptyset$. Let u be a fiber coordinate of $\mathcal{O}_L(1)$ over $(z \neq +\infty)$, linear on the fibers, let $v := 1/u$, so that (z, v) are coordinates near q and $s = (v = 0)$, and let $\zeta := 1/z$ and $u_\infty := u/z$, which are coordinates on $(E\backslash s) \cap (z \neq 0)$. Let r be the point $\zeta = u_\infty = 0$. For integers $\beta \geq \alpha \geq 0$, the sections

$$z^i v^\ell, \qquad 0 \leq \ell \leq \alpha, \quad 0 \leq i \leq \beta - \alpha + \ell,$$

form a basis of $H^0(E, \mathcal{O}_E(\alpha s + \beta f))$, since E is a toric surface with torus coordinates (z, u) and the exponents $(i, \alpha - \ell)$ of $z^i u^{\alpha-\ell} = z^i v^\ell u^\alpha$ are the lattice points of the polygon of $\alpha s + \beta f$. Near r the section $z^i v^\ell$ is $\zeta^{\beta-\alpha+\ell-i} u_\infty^{\alpha-\ell}$ times a local generator.

(2) For $\tau \in \mathbb{C}^*$ let $\varphi_\tau : E \to E$ be the automorphism $z \to \tau^{-1} z$, $u \to \tau u$. In the other coordinates it is $(z, v) \to (\tau^{-1} z, \tau^{-1} v)$ and $(\zeta, u_\infty) \to (\tau\zeta, \tau^2 u_\infty)$. We have $\varphi_\tau(s) = s$, $\varphi_\tau(f_x) = f_x$, $\varphi_\tau(q) = q$, $\varphi_\tau(r) = r$, and $\varphi_\tau^* \mathcal{O}_E(\alpha s + \beta f) \cong \mathcal{O}_E(\alpha s + \beta f)$.

(3) For an integer $n' \geq 2$ let $\mathfrak{b}_{n'} := (v - z, z^{n'})$, an ideal supported at q, and let $\mathfrak{a} := \mathfrak{b}_2 = (v - z, z^2)$. Let

$$\mathfrak{r} := (\zeta, u_\infty), \qquad \mathfrak{r}' := (u_\infty, \zeta^2), \qquad \mathfrak{r}'' := (\zeta, u_\infty^2)$$

be ideals supported at r, and let

$$\mathfrak{c}_4 := \mathfrak{b}_4, \qquad \mathfrak{c}_3 := \mathfrak{b}_3 \cap \mathfrak{r}, \qquad \mathfrak{c}_2' := \mathfrak{b}_2 \cap \mathfrak{r}', \qquad \mathfrak{c}_2'' := \mathfrak{b}_2 \cap \mathfrak{r}''.$$

(4) For real numbers $0 \leq w \leq 1$ and $\delta \geq 0$ let $\ell_+(w, \delta) := \min(1 + w, \delta)$ and

$$\begin{aligned}
\ell_-^{\mathfrak{c}_4}(w, \delta) &:= \max\left(0, \tfrac{\delta-1}{2}\right) + \max\left(0, \tfrac{4w-\delta}{3}\right), \\
\ell_-^{\mathfrak{c}_3}(w, \delta) &:= \max\left(0, \tfrac{\delta-1}{2}, \delta - 2\right) + \max\left(0, \tfrac{3w-\delta}{2}\right), \\
\ell_-^{\mathfrak{c}_2'}(w, \delta) = \ell_-^{\mathfrak{a}}(w, \delta) &:= \max\left(0, \tfrac{\delta-1}{2}\right) + \max(0, 2w - \delta), \\
\ell_-^{\mathfrak{c}_2''}(w, \delta) &:= \max\left(0, \tfrac{\delta-1}{2}, \tfrac{2\delta-3+w}{3}\right) + \max(0, 2w - \delta).
\end{aligned}$$

For $\mathfrak{c} \in \{\mathfrak{c}_4, \mathfrak{c}_3, \mathfrak{c}_2', \mathfrak{c}_2'', \mathfrak{a}\}$ let

$$\Omega_{\mathfrak{c}}(w) := \{(\delta, y) \in \mathbb{R}^2 \mid 0 \leq \delta \leq 3 + 2w,\ \ell_-^{\mathfrak{c}}(w, \delta) \leq y \leq \ell_+(w, \delta)\},$$

except that $\Omega_{\mathfrak{c}_2'}(w)$ is cut out by the additional inequality $\delta \leq 3$.

(5) Let $\rho(w) := (3 + 4w - 3w^2)/2$, and let $\kappa : [0, 1] \to \mathbb{R}$ be the piecewise affine interpolation of $\frac{5}{4}w^2$ at $w = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$, that is,

$$\kappa(w) = \frac{5w}{16}, \quad \frac{15w}{16} - \frac{5}{32}, \quad \frac{25w}{16} - \frac{15}{32}, \quad \frac{35w}{16} - \frac{15}{16}$$

on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$, $[\frac{3}{4}, 1]$, respectively.

In (5), $m^2\rho(w) + O(m)$ is $\dim W_{m,j}$ for $j = (1 + w)m$ (Lemma 5.2), and $\lfloor m\kappa(w) \rfloor$ is the integer c to which Lemma 3.3 is applied in **Step 5** of the proof of Theorem 6.5. Lemma 3.3 holds for every integer $c \geq 0$, so κ is taken piecewise affine only to make each term of the bound the integral of a piecewise affine function over a polygon (Appendix A).

Theorem 6.5. *Let X be a smooth prime Fano threefold of genus 10 and $x \in X$ a closed point such that $\Sigma(X, x)$ consists of two reduced points. Then the following hold.*

(1) $\delta_x(X) \geq 216/215$. More precisely, one of the lines L through x can be taken so that, with the notation of Set-up 5.1 for L, every prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies

$$A_X(F) \geq \frac{216}{215} S_X(F) + \frac{127}{215} \operatorname{mult}_F E.$$

(2) Assume that for both lines through x the point of $Z_U \cap f_x$ is $s \cap f_x$. Then either line may be taken in (1), and, with $q := s \cap f_x$,

$$S(W; f_x) \le \frac{533}{576}, \qquad S(W^{f_x}; q) \le \frac{356827303}{358612992} < \frac{215}{216}, \qquad S(W^{f_x}; a) \le \frac{7}{8} \quad \text{for } a \in f_x \backslash \{q\}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that three bounds on W imply the inequality of (1) for a line through x.

Let L be a line through x, which is non-special by Lemma 6.3(1), use the notation of Set-up 5.1 for L, let $q := s \cap f_x$, and assume that there exists a real number $\sigma < 215/216$ with

$$S(W; f_x) \le \sigma, \qquad S(W^{f_x}; q) \le \sigma, \qquad S(W^{f_x}; a) \le \sigma \quad \text{for } a \in f_x \backslash \{q\}. \tag{6.15}$$

Then the inequality of (1) holds for L.

The surface $E \cong \mathbb{F}_1$ and the curve f_x are smooth, so f_x is a plt type divisor over E with zero different, and the plt extraction of f_x is the identity of E. Let $a \in f_x$. By Theorem 2.9 applied to the klt pair $(E, 0)$, the series W, $Z = \{a\}$, and the plt type divisor f_x, since $\{a\}$ is the only irreducible closed subset of f_x with image $\{a\}$, and since ord_a is the only divisorial valuation of f_x centered at a up to scaling, with $A_{f_x}(\mathrm{ord}_a) = 1$,

$$\delta_a(E; W) \ge \min\left\{ \frac{A_E(f_x)}{S(W; f_x)}, \frac{1}{S(W^{f_x}; a)} \right\} \ge \frac{1}{\sigma} > \frac{216}{215}.$$

Let F' be a prime divisor over E with $\mathrm{center}_E F' \cap f_x \ne \emptyset$, and let $a \in \mathrm{center}_E F' \cap f_x$. Then $\frac{216}{215} S(W; F') \le A_E(F')$ by the definition of $\delta_a(E; W)$. Thus the assumption of Lemma 3.1 holds with $G = f_x$ and $\lambda = 216/215$. Since $A_X(E) = 2$ and $S_X(E) = 101/72$ (Lemma 5.2), Lemma 3.1 implies

$$A_X(F) \ge \frac{216}{215} S_X(F) + \left(2 - \frac{216}{215} \cdot \frac{101}{72}\right) \mathrm{mult}_F E = \frac{216}{215} S_X(F) + \frac{127}{215} \mathrm{mult}_F E$$

for any prime divisor F over X with $\mathrm{center}_Y F \cap f_x \ne \emptyset$. If $x \in \mathrm{center}_X F$, then $\mathrm{center}_Y F \cap f_x \ne \emptyset$, since $\pi(\mathrm{center}_Y F) = \mathrm{center}_X F$ and $\pi^{-1}(x) = f_x$. This proves the inequality of (1) for L. Since $\mathrm{mult}_F E \ge 0$, it implies $A_X(F) \ge \frac{216}{215} S_X(F)$ for any F with $x \in \mathrm{center}_X F$, that is, $\delta_x(X) \ge 216/215$.

Step 2. In this step we reduce the theorem to the three bounds in (2), and we describe Z_U near q.

Let L_1 and L_2 be the lines with $\Sigma(X, x) = \{[L_1], [L_2]\}$. By Lemma 6.3, L_1 and L_2 are non-special, so the notation of Set-up 5.1 with $E \cong \mathbb{F}_1$ is available for both lines. By Proposition 6.2, for $i = 1, 2$ the scheme $Z_{U_i} \cap f_{i,x}$ is one reduced point $p_i := p_{ij}$ ($\{i, j\} = \{1, 2\}$) of Lemma 6.3, which represents the other line L_j.

If $p_i \notin s_i$ for some i, then Proposition 6.2(1) implies (6.15) for L_i with $\sigma = 143/144$, since $7/8 < 143/144 < 215/216$. Thus from now on we assume $p_i = s_i \cap f_{i,x}$ for $i = 1, 2$. By symmetry it suffices to prove the three bounds of (2) for $L := L_1$, since then (6.15) holds for either line with $\sigma = 356827303/358612992$, as $7/8 < 533/576 < 356827303/358612992$.

We use the notation of Set-up 5.1 for L, and let $n := n_{12}$ be the length of Z_U at $q = p_1$. Since $p_2 = p_{21} \in s_2$, Lemma 6.3(2) with $(i, j) = (1, 2)$ implies $n > 1$. By Proposition 5.3, n is the width of L_2, and the widths of the lines intersecting L sum to 4. Thus $n \in \{2, 3, 4\}$, the length of Z_U is 4, and the conditions of Construction 6.4 hold for L and x. We use the notation of Construction 6.4. Write $Z_U = Z_q \sqcup Z^\circ$, where Z_q is the component supported at q, so that Z° has length $4 - n$.

By Proposition 5.3, $\widehat{\mathcal{O}}_{Z_U, q} \cong \mathbb{C}[t]/(t^n)$ and the local ring of $Z_U \cap f_x$ at q is $\mathbb{C}[t]/(t^n, \psi(t))$. Since $Z_U \cap f_x$ is reduced at q, ψ has order one. Thus z restricts to a generator of the maximal ideal of $\mathcal{O}_{Z_q}$, $\mathcal{O}_{Z_q} = \mathbb{C}[z]/(z^n)$, and $I_{Z_q} = (v - g(z), z^n)$ for a polynomial g with $g(0) = 0$ and $\deg g \le n - 1$.

Let $p_T : T \to E$ be the four blow-ups at simple free points resolving the base points of U (Proposition 5.3), let $e_1, \dots, e_4$ be the total transforms of the exceptional curves, let $\Lambda_T := p_T^*(2s+3f) - \sum_i e_i$, let s_T be the strict transform of s, and let k_s be the number of centers of p_T on the strict transforms of s. Since Λ_T is free, it is nef, and

$$0 \le (\Lambda_T \cdot s_T) = ((2s+3f)\cdot s) - k_s = 1 - k_s.$$

Thus $k_s \le 1$ (see also Construction-Proposition 5.4(1)). The first center of p_T over q is $q \in s$. So $k_s \ge 1$, hence $k_s = 1$, and no other center of p_T is on a strict transform of s. In the chart of the blow-up of q with coordinates (a_1, z), $v - g(z) = a_1 z$, the total transform of $I_{Z_q} = (v - g(z), z^n)$ is $z \cdot (a_1, z^{n-1})$. Since $n \ge 2$, the second center over q is the point $a_1 = z = 0$, that is, the point of the exceptional curve given by the tangent direction $v = g'(0)z$ of Z_q. In the chart (a_1, z) the curve s_T is $a_1 z = -g(z)$ with the exceptional curve $(z = 0)$ removed, that is, $a_1 = -g(z)/z$. Thus if $g'(0) = 0$, then the second center over q is on s_T, so $k_s \ge 2$, a contradiction. Thus $g'(0) \ne 0$, and no point of $\operatorname{Supp} Z^\circ$ is on s, since every point of $\operatorname{Supp} Z^\circ$ is a center of p_T. Since replacing u with a non-zero constant multiple keeps the conditions of Construction 6.4(1), possibly replacing u with $g'(0)u$ we may assume that $g'(0) = 1$. Then

$$I_{Z_q} = (v - z - \gamma_2 z^2 - \dots - \gamma_{n-1} z^{n-1}, z^n), \qquad \gamma_2, \dots, \gamma_{n-1} \in \mathbb{C}. \tag{6.16}$$

In particular $I_{Z_U} \subset \mathfrak{a}$. Moreover $\operatorname{Supp} Z^\circ$ is disjoint from s, from f_x, and from $(z = +\infty)$, so Z° is contained in the chart with coordinates (ζ, u_∞) and $\zeta \ne 0$ on $\operatorname{Supp} Z^\circ$.

Let d_E be the restriction to E of a section defining D. Since $Z_U \cap f_x$ is one reduced point, Construction-Proposition 5.4(4) implies that $f_x \not\subset \operatorname{Supp}(D|_E)$ and that $(D|_E)|_{f_x}$ has degree 5. Thus

$$\operatorname{ord}_{f_x} d_E = 0, \qquad \operatorname{ord}_a \left(d_E|_{f_x}\right) \le 5 \quad \text{for } a \in f_x. \tag{6.17}$$

By Construction-Proposition 5.4(3) and Lemma 5.2,

$$\begin{aligned} &W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m+j)f)) && 0 \le j \le m, \\ &W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m+j)f) \otimes I_{Z_U}^{j-m}) && m \le j \le 2m, \\ &W_{m,j} = d_E^{j-2m} \cdot U_{5m-2j} && 2m \le j \le \lfloor 5m/2 \rfloor, \end{aligned} \tag{6.18}$$

where $U_{5m-2j} \subset H^0(E, \mathcal{O}_E((5m-2j)(2s+3f)) \otimes I_{Z_U}^{5m-2j})$, and $W_{m,j} = 0$ for $j > 5m/2$. Since $\dim W_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$, Lemma 5.2 also implies $N_m = 3m^3 + O(m^2)$,

$$\begin{aligned} &\dim W_{m,j} = m^2 \rho(w) + O(m) && m < j < 2m,\ w := \tfrac{j}{m} - 1, \\ &\dim W_{m,j} = 2(5m-2j)^2 + O(m) && 2m < j < \tfrac{5m}{2}, \end{aligned}$$

uniformly in j, and

$$J := \lim_{m \to +\infty} \frac{1}{m N_m} \sum_{j > 2m} (j - 2m) \dim W_{m,j} = \frac{1}{72}. \tag{6.19}$$

Step 3. In this step we bound the filtration by the order along f_x on the spaces of (6.18) by the filtration by the order along f_x on $H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathfrak{c}^\nu)$ for $\mathfrak{c}$ one of the ideals $\mathfrak{c}_4, \mathfrak{c}_3, \mathfrak{c}_2', \mathfrak{c}_2''$ of Construction 6.4(3).

For a space V of sections of a line bundle on E and $k \ge 0$, let $V^{\ge k}$ be the subspace of elements of V whose order along f_x is at least k. Let $\mathfrak{c} := \mathfrak{c}_n$ if $n \in \{3, 4\}$. If $n = 2$, then Z° has length two. Let $\mathfrak{c} := \mathfrak{c}_2'$ if $\zeta|_{Z^\circ}$ generates the $\mathbb{C}$-algebra $H^0(Z^\circ, \mathcal{O}_{Z^\circ})$, and $\mathfrak{c} := \mathfrak{c}_2''$ otherwise. We show that for any integers $\beta \ge \alpha \ge 0$, $\nu \ge 0$, and $k \ge 0$,

$$\dim H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes I_{Z_U}^\nu)^{\ge k} \le \dim H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathfrak{c}^\nu)^{\ge k}. \tag{6.20}$$

For $\tau \in \mathbb{C}^*$ let $Z_\tau := \varphi_\tau(Z_U)$. A regular function χ vanishes on $\varphi_\tau(Z_U)$ if and only if $\chi \circ \varphi_\tau$ vanishes on Z_U. Thus by (6.16), near q,

$$I_{Z_\tau} = \Big(v - z - \sum_{i=2}^{n-1} \gamma_i \tau^{i-1} z^i,\ z^n\Big),$$

since $\big(v - z - \sum_i \gamma_i \tau^{i-1} z^i\big) \circ \varphi_\tau = \tau^{-1}\big(v - z - \sum_i \gamma_i z^i\big)$. That is, γ_i is multiplied by τ^{i-1}. Near $\varphi_\tau(Z^\circ)$ the ideal of Z_τ is as follows.

- If $n = 3$, then Z° is a reduced point (ζ_0, u_0), and $\varphi_\tau(Z^\circ)$ is the point $(\tau\zeta_0, \tau^2 u_0)$.
- If $n = 2$ and $\mathfrak{c} = \mathfrak{c}_2'$, then $\mathcal{O}_{Z^\circ} = \mathbb{C}[\zeta]/(Q(\zeta))$ for a monic quadratic polynomial Q, and $u_\infty = \mu_0 + \mu_1\zeta$ on Z° for some $\mu_0, \mu_1 \in \mathbb{C}$. Then $I_{\varphi_\tau(Z^\circ)} = (u_\infty - \tau^2\mu_0 - \tau\mu_1\zeta,\ \tau^2 Q(\tau^{-1}\zeta))$.
- If $n = 2$ and $\mathfrak{c} = \mathfrak{c}_2''$, then the image of $\mathbb{C}[\zeta]$ in the two-dimensional algebra $H^0(Z^\circ, \mathcal{O}_{Z^\circ})$ is $\mathbb{C}$, so $Z^\circ \subset (\zeta = \zeta_0)$ for some ζ_0 and $I_{Z^\circ} = (\zeta - \zeta_0, Q(u_\infty))$ for a monic quadratic polynomial Q. Then $I_{\varphi_\tau(Z^\circ)} = (\zeta - \tau\zeta_0,\ \tau^4 Q(\tau^{-2}u_\infty))$.

In each case the generators have coefficients polynomial in τ, and at $\tau = 0$ they generate $\mathfrak{b}_n$ near q and, if $n \le 3$, the factor $\mathfrak{r}$, $\mathfrak{r}'$, or $\mathfrak{r}''$ of $\mathfrak{c}$ at r. Thus we obtain a family $(Z_\tau)_{\tau\in\mathbb{A}^1}$ with $Z_1 = Z_U$ and $I_{Z_0} = \mathfrak{c}$. Each local piece of the family has, in local coordinates (ξ, η), the ideal $(\xi - P_\tau(\eta), Q_\tau(\eta))$ with Q_τ monic in η and $\deg_\eta P_\tau < \deg_\eta Q_\tau$. So by Lemma 3.5, with (ξ, η, τ) in place of (a, b, λ), for $\nu \ge 1$ the quotient by the ν-th power of the ideal $(\xi - P_\tau(\eta), Q_\tau(\eta))$ is a free $\mathbb{C}[\tau]$-module of finite rank. The supports of the piece at q and of the piece in the chart (ζ, u_∞) are disjoint for any $\tau \in \mathbb{A}^1$, since ζ is finite on the second one. Hence the subschemes defined by $I^\nu_{Z_\tau}$ form a finite flat family over $\mathbb{A}^1$, with fiber at 0 defined by $\mathfrak{c}^\nu$.

The space on the left of (6.20), with I_{Z_U} replaced by I_{Z_τ}, is the kernel of the map

$$H^0(E, \mathcal{O}_E(\alpha s + (\beta - k)f)) \to H^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathcal{O}_E/I^\nu_{Z_\tau}\big)$$

given by multiplication by z^k. By the flatness, the spaces $H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathcal{O}_E/I^\nu_{Z_\tau})$ form a vector bundle over $\mathbb{A}^1$, so the dimension of the kernel is upper semicontinuous in τ. For $\tau \ne 0$, φ_τ is an automorphism fixing f_x with $\varphi_\tau^*\mathcal{O}_E(\alpha s + \beta f) \cong \mathcal{O}_E(\alpha s + \beta f)$, so the dimension of the kernel at τ equals the dimension of the kernel at $\tau = 1$. This proves (6.20).

Step 4. In this step we prove $S(W; f_x) \le 533/576$.

Let $\beta \ge \alpha \ge 0$ and $\nu \ge 0$ be integers, and let $C_{\alpha,\beta,\nu} := H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes \mathfrak{c}^\nu)$. Since $\mathfrak{c}$ is invariant under φ_τ for any $\tau \in \mathbb{C}^*$ and $z^i v^\ell \circ \varphi_\tau = \tau^{-i-\ell} z^i v^\ell$, the space $C_{\alpha,\beta,\nu}$ is the direct sum over $d \ge 0$ of the spaces $C_{\alpha,\beta,\nu,d}$ of elements of $C_{\alpha,\beta,\nu}$ of the form $z^d P(\theta)$, where $\theta := v/z$ and $P(\theta) = \sum_\ell p_\ell \theta^\ell$. By Construction 6.4(1), $p_\ell = 0$ unless

$$\max\big(0, \lceil \tfrac{d-\beta+\alpha}{2} \rceil\big) \le \ell \le \min(\alpha, d),$$

since $z^d\theta^\ell = z^i v^\ell$ with $i = d - \ell$ and $0 \le i \le \beta - \alpha + \ell$. We bound $C_{\alpha,\beta,\nu,d}$ by the conditions at q and at r.

Write $\mathfrak{c} = \mathfrak{b}_n \cap \mathfrak{r}_0$ with $\mathfrak{r}_0 \in \{(1), \mathfrak{r}, \mathfrak{r}', \mathfrak{r}''\}$, where (1) is the unit ideal. Since $\mathfrak{b}_n$ and $\mathfrak{r}_0$ have disjoint supports, $\mathfrak{c}^\nu = \mathfrak{b}_n^\nu \cap \mathfrak{r}_0^\nu$. At q, in the coordinates $\xi := v - z$ and $\eta := z$, the element $z^d P(\theta)$ is a form of degree d, the ideal $\mathfrak{b}_n^\nu = (\xi, \eta^n)^\nu$ is monomial, and $\xi^c\eta^{d-c} \in (\xi, \eta^n)^\nu$ if and only if $(n-1)c \ge n\nu - d$. Since $\xi = z(\theta - 1)$, we obtain that $z^d P(\theta) \in \mathfrak{b}_n^\nu$ if and only if $(\theta - 1)^{\epsilon_n(\nu,d)}$ divides P, where

$$\epsilon_n(\nu, d) := \max\big(0, \lceil \tfrac{n\nu - d}{n-1} \rceil\big).$$

At r, the section $z^{d-\ell}v^\ell$ is $\zeta^A u_\infty^B$ times a local generator, with $A = \beta - \alpha + 2\ell - d$ and $B = \alpha - \ell$ (Construction 6.4(1)). Since $\mathfrak{r}$, $\mathfrak{r}'$, $\mathfrak{r}''$ are monomial, $z^d P(\theta) \in \mathfrak{r}_0^\nu$ if and only if each monomial of $z^d P(\theta)$ belongs to $\mathfrak{r}_0^\nu$, and:

- $\zeta^A u_\infty^B \in \mathfrak{r}^\nu$ implies $A + B \ge \nu$, that is, $\ell \ge d - \beta + \nu$;
- $\zeta^A u_\infty^B \in (\mathfrak{r}')^\nu$ implies $A + 2B \ge 2\nu$, since any generator $u_\infty^c \zeta^{2(\nu-c)}$ satisfies equality; that is, $d \le \alpha + \beta - 2\nu$;

- $\zeta^A u_\infty^B \in (\mathfrak{r}'')^\nu$ implies $2A+B \geq 2\nu$, since any generator $\zeta^c u_\infty^{2(\nu-c)}$ satisfies equality; that is, $\ell \geq (2d-2\beta+\alpha+2\nu)/3$.

Let $\ell_0(d)$ be the maximum of 0, $\lceil (d-\beta+\alpha)/2 \rceil$, and the lower bound on ℓ above for $\mathfrak{r}_0$ (rounded up), and let $\ell_1(d) := \min(\alpha, d)$. Since θ and $\theta-1$ are coprime, any element of $C_{\alpha,\beta,\nu,d}$ is $z^d \theta^{\ell_0(d)}(\theta-1)^{\epsilon_n(\nu,d)} R(\theta)$ with $\deg R \leq \ell_1(d) - \ell_0(d) - \epsilon_n(\nu,d)$, and $C_{\alpha,\beta,\nu,d} = 0$ if $\mathfrak{r}_0 = \mathfrak{r}'$ and $d > \alpha+\beta-2\nu$. The order of $z^d P(\theta)$ along f_x is $d - \deg P$, and the order along f_x of a sum of elements of distinct $C_{\alpha,\beta,\nu,d}$ is the minimum of the orders of the summands. Taking $R = \theta^e$ for $e \geq 0$, we obtain

$$\sum_{k\geq 1} \dim C_{\alpha,\beta,\nu}^{\geq k} \leq \sum_d \sum_{\ell=\ell_0(d)+\epsilon_n(\nu,d)}^{\ell_1(d)} (d-\ell), \tag{6.21}$$

where for $\mathfrak{r}_0 = \mathfrak{r}'$ the sum runs over $d \leq \alpha+\beta-2\nu$.

We apply (6.20) and (6.21) to (6.18). For $m \leq j \leq 2m$ take $(\alpha,\beta,\nu) = (j, m+j, j-m)$, and write $w := j/m - 1$, $d = m\delta$, $\ell = my$. Then the conditions on (d,ℓ) become $(\delta, y) \in \Omega_{\mathfrak{c}}(w)$ up to $O(1)$, so (6.21) is at most $m^3 \mu_{\mathfrak{c}}(w) + O(m^2)$, uniformly in j, where

$$\mu_{\mathfrak{c}}(w) := \int_{\Omega_{\mathfrak{c}}(w)} (\delta - y)\, dy d\delta,$$

that is, the integral over δ of $\delta(\ell_+ - \ell_-^{\mathfrak{c}}) - \frac{1}{2}(\ell_+^2 - (\ell_-^{\mathfrak{c}})^2)$ where $\ell_-^{\mathfrak{c}} \leq \ell_+$. For $2m \leq j \leq \lfloor 5m/2 \rfloor$, (6.17) implies that the filtration of $W_{m,j} = d_E^{j-2m} \cdot U_{5m-2j}$ by the order along f_x is the filtration of U_{5m-2j}. We take $(\alpha,\beta,\nu) = (2(5m-2j), 3(5m-2j), 5m-2j)$, which is the case $w=1$ scaled by $5m-2j$, so (6.21) is at most $(5m-2j)^3 \mu_{\mathfrak{c}}(1) + O((5m-2j)^2)$. For $0 \leq j \leq m$ the complete space has orders i for $0 \leq \ell \leq j$, $0 \leq i \leq m+\ell$, with sum $m^3((1+t)^3-1)/6 + O(m^2)$, $t = j/m$. Summing over j, we use $\int_0^1 ((1+t)^3-1)/6\, dt = 11/24$, $\int_2^{5/2} (5-2t)^3 dt = 1/8$, and $mN_m = 3m^4 + O(m^3)$. The lattice point counts differ from the integrals by $O(m^2)$ for each of the $O(m)$ values of j, so their contribution to the normalized sums is $O(1/m)$. Since $S(W; f_x) = \lim_m S_m(W; f_x)$ [AZ22, Lemma 2.9], we obtain

$$S(W; f_x) \leq \frac{1}{3}\Big(\frac{11}{24} + \int_0^1 \mu_{\mathfrak{c}}(w) dw + \frac{1}{8}\mu_{\mathfrak{c}}(1)\Big). \tag{6.22}$$

On the polygons in Appendix A the functions $\ell_\pm$ are affine, and exact integration of the polynomial integrands over the polygons with rational vertices yields the values of the right side of (6.22) for the four ideals in Table 79 of Appendix A. Their maximum is $533/576$, so $S(W; f_x) \leq 533/576$.

Step 5. In this step we prove the bounds for $S(W^{f_x}; a)$ and $S(W^{f_x}; q)$.

Recall that the refinement W^{f_x} of W by f_x (Definition 2.8) has $W^{f_x}_{m,j,k}$ equal to the image of $W_{m,j}^{\geq k}$ in $H^0(f_x, \mathcal{O}_E(js + (m+j)f - kf_x)|_{f_x})$. In the coordinate u, an element of $W^{f_x}_{m,j,k}$ is a polynomial in u, and the image of $z^i u^{\alpha-\ell}$ with $i = k$ is $u^{\alpha-\ell}$. Thus for $j \leq 2m$, by Construction 6.4(1) with $(\alpha,\beta) = (j, m+j)$, the space $W^{f_x}_{m,j,k}$ is contained in the polynomials of degree at most $\min(\alpha, \beta-k)$, and for $2m < j \leq 5m/2$, $W^{f_x}_{m,j,k}$ is the product of $(d_E|_{f_x})^{j-2m}$ and a subspace of the polynomials of degree at most $\min(2(5m-2j), 3(5m-2j)-k)$.

Let $a \in f_x \backslash \{q\}$, so that $u(a)$ is finite. For a subspace of the polynomials of degree at most d_0, the sum of the orders at a over a basis adapted to the order at a is at most $0+1+\cdots+d_0 = d_0(d_0+1)/2$. By (6.17), the factor $(d_E|_{f_x})^{j-2m}$ adds at most $5(j-2m)\dim W_{m,j}$ for each $j > 2m$. For integers $\beta \geq \alpha \geq 0$, $\sum_{0\leq k\leq\beta} \min(\alpha, \beta-k)^2/2$ is $\alpha^2(\beta-\alpha)/2 + \alpha^3/6$ up to lower order terms. A direct calculation then shows that the contributions of $0 \leq j \leq m$, $m \leq j \leq 2m$, and $2m \leq j \leq 5m/2$ are m^4 times $5/24$, $43/24$, and $\frac{1}{8}\cdot\frac{10}{3} = \frac{5}{12}$, up to $O(m^3)$. Thus by (6.19) and [AZ22, Lemma 2.9],

$$S(W^{f_x}; a) \leq \frac{1}{3}\Big(\frac{5}{24} + \frac{43}{24} + \frac{5}{12}\Big) + 5J = \frac{29}{36} + \frac{5}{72} = \frac{7}{8}.$$

Now consider q. On f_x the function v is a coordinate at q, so $\operatorname{ord}_q v^\ell = \ell$. For $m \le j \le 5m/2$ let

$$\overline{W}^{f_x}_{m,j,k} := \begin{cases} \text{image of } H^0\big(E, \mathcal{O}_E(js + (m+j)f) \otimes \mathfrak{a}^{j-m}\big)^{\ge k} & \text{if } m \le j \le 2m, \\ \text{image of } H^0\big(E, \mathcal{O}_E((5m-2j)(2s+3f)) \otimes \mathfrak{a}^{5m-2j}\big)^{\ge k} & \text{if } 2m < j \le 5m/2, \end{cases}$$

where the image is taken in $H^0(f_x, \mathcal{M}(-kf_x)|_{f_x})$ for the respective line bundle $\mathcal{M}$ on E, by the restriction map of Definition 2.8. Since $I_{Z_U} \subset \mathfrak{a}$ by (6.16), (6.18) implies $W^{f_x}_{m,j,k} \subset \overline{W}^{f_x}_{m,j,k}$ for $m \le j \le 2m$, and for $j > 2m$, $W^{f_x}_{m,j,k}$ is the product of $(d_E|_{f_x})^{j-2m}$ and a subspace of $\overline{W}^{f_x}_{m,j,k}$ of dimension $\dim W^{f_x}_{m,j,k}$. Apply the analysis of **Step 4** to $\mathfrak{a} = \mathfrak{b}_2$ with $\mathfrak{r}_0 = (1)$. Then $\mathfrak{a}$ gives no condition at r, and $\overline{W}^{f_x}_{m,j,k}$ is spanned by the monomials v^ℓ with $\ell_0(k+\ell) + \epsilon_2(\nu, k+\ell) \le \ell \le \ell_1(k+\ell)$, since dividing $z^d\theta^{\ell'}$ with $\ell' = \deg P$ by $z^{d-\ell'}$ and restricting to $z = 0$ leaves $v^{\ell'}$.

Fix j with $m \le j \le 2m$. For any integer $i \ge 0$,

$$\sum_k \dim \big(W^{f_x}_{m,j,k}\big)^{\operatorname{ord}_q \ge i} \le \min\Big\{ \dim W_{m,j},\ \sum_k \dim \big(\overline{W}^{f_x}_{m,j,k}\big)^{\operatorname{ord}_q \ge i} \Big\}, \tag{6.23}$$

since $\sum_k \dim W^{f_x}_{m,j,k} = \dim W_{m,j}$ and $W^{f_x}_{m,j,k} \subset \overline{W}^{f_x}_{m,j,k}$. By Lemma 3.3, applied, in the notation of that lemma, with $\bigoplus_k W^{f_x}_{m,j,k}$ in place of U' and $\bigoplus_k \overline{W}^{f_x}_{m,j,k}$ in place of U, with the filtration by $\operatorname{ord}_q \ge i$ (so that i is the index h of Lemma 3.3), and with the integer $\lfloor m\kappa(w) \rfloor$, $w = j/m - 1$, in place of c, and since each $\overline{W}^{f_x}_{m,j,k}$ is spanned by monomials, the sum over $k \ge 0$ of the orders at q over a basis of $W^{f_x}_{m,j,k}$ adapted to the order at q is at most

$$\sum_{(d,\ell)} \ell - \Big(\sum_{(d,\ell)} \min\big(\ell, \lfloor m\kappa(w) \rfloor\big) - \lfloor m\kappa(w) \rfloor \dim W_{m,j} \Big),$$

where (d, ℓ) runs over the pairs in (6.21) for $(\alpha, \beta, \nu) = (j, m+j, j-m)$ and $\mathfrak{a}$. By **Step 2** and the approximation of lattice point counts by integrals, this is

$$m^3 \Big(\int_{\Omega_{\mathfrak{a}}(w)} y dy d\delta - \Big(\int_{\Omega_{\mathfrak{a}}(w)} \min(\kappa(w), y) dy d\delta - \kappa(w)\rho(w) \Big) \Big) + O(m^2).$$

For $2m < j \le 5m/2$ we apply (6.23) and Lemma 3.3 to the subspaces of $\overline{W}^{f_x}_{m,j,k}$ whose products with $(d_E|_{f_x})^{j-2m}$ are $W^{f_x}_{m,j,k}$, $k \ge 0$, and to $\overline{W}^{f_x}_{m,j,k}$, with $(\alpha, \beta, \nu) = (2(5m-2j), 3(5m-2j), 5m-2j)$, the cutoff $\lfloor 5(5m-2j)/4 \rfloor$, and $\dim U_{5m-2j} = \dim W_{m,j} = 2(5m-2j)^2 + O(m)$. The factor $(d_E|_{f_x})^{j-2m}$ adds at most $5(j-2m)\dim W_{m,j}$ by (6.17). For $0 \le j \le m$ we bound $W^{f_x}_{m,j,k}$ by the image of $H^0(E, \mathcal{O}_E(js + (m+j)f))^{\ge k}$, which is spanned by monomials v^ℓ, with sum of orders $\sum_{\ell=0}^{j} \ell(m+\ell+1) = m^3(t^2/2 + t^3/3) + O(m^2)$, $t = j/m$, and $\int_0^1 (t^2/2 + t^3/3) dt = 1/4$.

Exact integration over the polygons of Appendix A implies

$$\int_0^1 \int_{\Omega_{\mathfrak{a}}(w)} y dy d\delta dw = \frac{223}{96}, \qquad \frac{1}{8} \int_{\Omega_{\mathfrak{a}}(1)} y dy d\delta = \frac{17}{32},$$

and the values of

$$\int \Big(\int_{\Omega_{\mathfrak{a}}(w)} \min(\kappa(w), y) dy d\delta - \kappa(w)\rho(w) \Big) dw$$

over the four intervals $[0, \frac14]$, $[\frac14, \frac12]$, $[\frac12, \frac34]$, and $[\frac34, 1]$ of Construction 6.4(5) are 3305/25165824, 148765/25165824, 1104181/25165824, and 74328067/478150656, with sum 24549209/119537664 (Tables 79 and 80 of Appendix A). In the tail, $\int_{\Omega_{\mathfrak{a}}(1)} \min(5/4, y) dy d\delta = 445/128$, the sum of the contributions 25/512, 387/512, 63/32, and 45/64 of $\delta \in [1, \frac54]$, $[\frac54, 2]$, $[2, \frac72]$, and $[\frac72, 5]$, so the tail term is

$$\frac{1}{8}\Big(\frac{445}{128} - \frac{5}{4} \cdot 2 \Big) = \frac{125}{1024}.$$

As in **Step 4**, the lattice point counts, the integer parts of the cutoffs, and the $O(m)$ terms of $\dim W_{m,j}$ change the normalized sums by $O(1/m)$. Hence, by (6.19) and [AZ22, Lemma 2.9],

$$S(W^{f_x};q) \le \frac{1}{3}\Big(\frac{1}{4}+\frac{223}{96}+\frac{17}{32}-\frac{24549209}{119537664}-\frac{125}{1024}\Big)+5J=\frac{356827303}{358612992}, \tag{6.24}$$

where $5J = 5/72$. Since

$$215\cdot 358612992 - 216\cdot 356827303 = 27095832 > 0,$$

we have $356827303/358612992 < 215/216$. This proves the three bounds of (2).

By **Step 2**, (6.15) holds for one of the lines through x, and, under the assumption of (2), for either line by **Steps 4 and 5**. Thus **Step 1** proves (1) and (2). □

6.3. **Points on three lines.** In this subsection, we study the points p of a smooth prime Fano threefold X of genus ten such that $\Sigma(X,p)$ consists of three distinct reduced points. At such a point, Theorem 6.18 below gives only $S(W;w) \le A_E(w)$ for prime divisors w over the exceptional plane E of the blow-up of p, with no margin, so the refined series W has to be computed exactly. A one-parameter subgroup that fixes p and preserves $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ degenerates X to an integral linear section of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ whose section ring is determined exactly by the Koszul complex and the Weyl character formula (Proposition 6.9).

By [Kap11, Section 2, Lemma 2.1], the alternating matrix of Set-up 2.14 realizes $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ as the five-dimensional section of $\mathrm{Gr}(2,7)$ by a non-degenerate four-form. We normalize the point and the linear forms in the coordinates of Set-up 2.14 as follows.

Set-up 6.6. Notation as in Set-up 2.14. Assign to the coordinates the weights 0 for h, 1 for i,a,e,n, 2 for f,g,k,c, 3 for l,b,d,M, and 4 for J. Let X be a smooth codimension two linear section of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ containing $p := [h]$, whose two defining linear forms have lowest-weight parts $n+i$ and $ug+vk+\lambda f+\mu c$ with $u-v=1$, where all remaining terms of the first form have weight at least 2 and those of the second form have weight at least 3. Let $\pi : Y := \mathrm{Bl}_p X \to X$ with exceptional plane E with coordinates $(i:a:e)$, and for integers $m, r \ge 0$ let

$$W_{m,r} := \mathrm{Im}\big(H^0(Y, m\pi^*H - rE) \to H^0(E, \mathcal{O}_E(r))\big).$$

Let

$$f_2 := ie - a^2, \quad c_2 := e^2 - ai, \quad Q_2 := ae - i^2,$$
$$G_2 := vQ_2 - \lambda f_2 - \mu c_2, \quad J_4 := f_2c_2 - G_2^2 - G_2Q_2,$$

and let $B_4 := \mathrm{div}(J_4)$ be the divisor of J_4 on E, a plane quartic counted with multiplicities. We write (C) for the condition that

$$(u-\mu\zeta, v\zeta+\lambda) \ne (0,0) \quad \text{for every } \zeta \in \mathbb{C} \text{ with } \zeta^3 = 1.$$

The weights in Set-up 6.6 are the weights of the $\mathbb{G}_m$-action of Lemma 6.8, namely the action of a one-parameter subgroup of G_2 fixing p, with the weights shifted so that h has weight 0 (Proposition 6.9). The relation $u-v=1$ is the normalization $d\ell_2(p) = dg$ for the second defining linear form ℓ_2, since $dk(p) = -dg(p)$ on the chart $h=1$ (Lemma 2.15(1)).

By Lemma 6.7(2), any smooth X as in Set-up 6.6 satisfies (C): if (C) fails for a cube root of unity, then X contains an explicit line L through p and is singular at a point of L.

Lemma 6.7. *The following hold.*

(1) Let X be a smooth prime Fano threefold of genus 10 and $p \in X$ a closed point such that $\Sigma(X,p)$ consists of three distinct reduced points. Then X and p are isomorphic to a codimension two linear section and the point $[h]$ as in Set-up 6.6.

(2) In Set-up 6.6, condition (C) holds, $H^3 = 18$, and $\Sigma(X,p)$ consists of three distinct reduced points.

Proof. The proof is long, so we divide it into several steps. For a closed subscheme $V \subset \mathbb{P}^{13}$ and a point $x \in V$, we write $\mathbb{T}_x V \subset \mathbb{P}^{13}$ for the embedded tangent space of V at x and $\mathbb{P}(T_x V)$ for the projective space of tangent directions of V at x. We use the Pfaffians $\mathrm{Pf}_{j_1j_2j_3j_4}$ of Set-up 2.14.

Step 1. In this step, we describe the lines through p on a linear section. Let $\Gamma_p \subset \mathbb{P}(T_p \mathrm{G}_2 \mathrm{Gr}(2,7))$ be the twisted cubic of Lemma 2.15(3). Under the isomorphism $\mathbb{P}^1 \to \Gamma_p$ of Lemma 2.15(3), the coordinates i, a, e, n pull back to the linearly independent cubic forms $\sigma^3, \sigma^2\tau, \sigma\tau^2, -\tau^3$, so Γ_p spans the hyperplane $\{g = 0\} \cong \mathbb{P}^3$ of $\mathbb{P}(T_p \mathrm{G}_2 \mathrm{Gr}(2,7))$. Since the binary cubic forms vanishing at three distinct points of $\mathbb{P}^1$ form a one-dimensional space, any three distinct points of Γ_p span a plane.

Let $\Lambda \subset \mathbb{P}^{13}$ be a codimension two linear subspace containing p such that $X' := \mathrm{G}_2 \mathrm{Gr}(2,7) \cap \Lambda$ is a smooth threefold. We show that

$$\mathbb{T}_p X' = \mathbb{T}_p \mathrm{G}_2 \mathrm{Gr}(2,7) \cap \Lambda \quad \text{and} \quad \Sigma(X', p) = \Gamma_p \cap \mathbb{P}(T_p X') \tag{6.25}$$

as subschemes of $\mathbb{P}(T_p \mathrm{G}_2 \mathrm{Gr}(2,7))$. The Zariski tangent space of an intersection is the intersection of the Zariski tangent spaces, which implies the first equality. The variety $\mathrm{G}_2 \mathrm{Gr}(2,7)$ is cut out by the quadrics $\mathrm{Pf}_{j_1j_2j_3j_4}$, and $X' \subset \Lambda \cong \mathbb{P}^{11}$ is cut out by their restrictions. By Lemma 2.15(3), $\mathrm{G}_2 \mathrm{Gr}(2,7) \cap \mathbb{T}_p \mathrm{G}_2 \mathrm{Gr}(2,7)$ is the cone with vertex p over Γ_p. Since $\mathbb{T}_p X'$ is a linear subspace of $\mathbb{T}_p \mathrm{G}_2 \mathrm{Gr}(2,7)$ through p and $\mathbb{T}_p X' \subset \Lambda$, we obtain

$$X' \cap \mathbb{T}_p X' = \big(\mathrm{G}_2 \mathrm{Gr}(2,7) \cap \mathbb{T}_p \mathrm{G}_2 \mathrm{Gr}(2,7)\big) \cap \mathbb{T}_p X',$$

which is the cone with vertex p over $\Gamma_p \cap \mathbb{P}(T_p X')$. By [KP23, Lemma 5.6], if $\Sigma(X', p) \neq \emptyset$, then $X' \cap \mathbb{T}_p X'$ is the cone with vertex p over $\Sigma(X', p) \subset \mathbb{P}(T_p X')$, and if $\Sigma(X', p) = \emptyset$, then $X' \cap \mathbb{T}_p X'$ is the first-order neighborhood of p in X'.

Suppose first that $\Sigma(X', p) \neq \emptyset$. A closed subscheme of $\mathbb{P}(T_p X')$ is determined by the cone over it, since the homogeneous ideal of the cone is generated by the homogeneous ideal of the subscheme in the coordinates $(i : a : e : n : g)$. Thus $\Sigma(X', p) = \Gamma_p \cap \mathbb{P}(T_p X')$.

Suppose next that $\Sigma(X', p) = \emptyset$. Then $X' \cap \mathbb{T}_p X'$ is the first-order neighborhood of p in X', which contains no line. The cone over $\Gamma_p \cap \mathbb{P}(T_p X')$ contains the line through p in each direction of $\Gamma_p \cap \mathbb{P}(T_p X')$, so $\Gamma_p \cap \mathbb{P}(T_p X') = \emptyset = \Sigma(X', p)$. Thus the second equality of (6.25) holds in both cases.

Step 2. In this step, we prove (1). By Theorem 2.13, we may assume that $X = \mathrm{G}_2 \mathrm{Gr}(2,7) \cap \Lambda$ for a codimension two linear subspace $\Lambda \subset \mathbb{P}^{13}$, with H the restriction of the hyperplane class. Let $\epsilon_1, \dots, \epsilon_7$ be the standard basis of $\mathbb{C}^7$, and let

$$\omega' := -\epsilon_1\epsilon_2\epsilon_3\epsilon_7 + \epsilon_4\epsilon_5\epsilon_6\epsilon_7 + \epsilon_2\epsilon_3\epsilon_5\epsilon_6 + \epsilon_1\epsilon_3\epsilon_4\epsilon_6 + \epsilon_1\epsilon_2\epsilon_4\epsilon_5 \in \Lambda^4\mathbb{C}^7$$

(products are exterior products). The element $\mathrm{diag}(1, -1, -1, -1, 1, 1, -1) \in \mathrm{SL}_7$ sends the non-degenerate four-form of [Kap11, Section 2] to ω', changing only the sign of the monomial $\epsilon_1\epsilon_2\epsilon_3\epsilon_7$, so ω' is non-degenerate. Identifying the matrix A with $\sum_{j<k} A_{jk}\epsilon_j \wedge \epsilon_k$, a direct calculation shows $\mathcal{W} = \{\xi \in \Lambda^2\mathbb{C}^7 : \xi \wedge \omega' = 0\}$. Since $\mathrm{G}_2 \mathrm{Gr}(2,7)$ is the locus in $\mathbb{P}(\mathcal{W})$ where A has rank at most two, it consists of the points $[v_1 \wedge v_2]$ of $\mathrm{Gr}(2,7)$ with $v_1 \wedge v_2 \wedge \omega' = 0$ (compare [Kap11, Lemma 2.1]). Hence the stabilizer $G_{\omega'}$ of ω' in SL_7, acting by $A \to \alpha A \alpha^t$, preserves $\mathcal{W}$ and $\mathrm{G}_2 \mathrm{Gr}(2,7)$, so $G_{\omega'}$ is contained in the group $G_{\mathcal{W}}$ of Set-up 2.14(5). Under the SL_7-equivariant isomorphism $\Lambda^4\mathbb{C}^7 \cong \Lambda^3(\mathbb{C}^7)^\vee$, ω' corresponds to a general alternating trilinear form, so the Lie algebra of $G_{\omega'}$ is $\mathfrak{g}_2$ [FH91, Proposition 22.12], the identity component of $G_{\omega'}$ is G_2, and $\mathrm{G}_2 \subset G^\circ_{\mathcal{W}}$. By [Lat23, Theorem 2.1], $\mathrm{G}_2 \mathrm{Gr}(2,7)$ is a homogeneous space of G_2, so $G^\circ_{\mathcal{W}}$, which preserves $\mathrm{G}_2 \mathrm{Gr}(2,7)$ by Lemma 2.15(4), acts transitively on it. An element of $G^\circ_{\mathcal{W}}$ maps X isomorphically onto another smooth codimension two linear section of $\mathrm{G}_2 \mathrm{Gr}(2,7)$. Possibly replacing X with $\alpha(X)$ for an element $\alpha \in G^\circ_{\mathcal{W}}$ with $\alpha(p) = [h]$, we may assume that $p = [h]$.

A line on X through p is a line on $\mathrm{G}_2 \mathrm{Gr}(2,7)$ through p whose tangent direction at p belongs to $\mathbb{P}(T_p X)$, and a line through p is determined by the tangent direction at p. Since $\Sigma(X, p)$ consists of three distinct points, $\Gamma_p \cap \mathbb{P}(T_p X)$ contains three distinct points y_1, y_2, y_3 of Γ_p. By

Step 1, y_1, y_2, y_3 span a plane contained in $\{g = 0\}$. Since $\dim \mathbb{P}(T_pX) = 2$, $\mathbb{P}(T_pX)$ is the plane spanned by y_1, y_2, y_3. By Lemma 2.15(4), the stabilizer of p in $G^\circ_{\mathcal{W}}$ acts triply transitively on Γ_p. It preserves $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ and the hyperplane $\{g = 0\}$ spanned by Γ_p. The points

$$z_\zeta := (1 : \zeta : \zeta^2 : -1 : 0), \qquad \zeta^3 = 1,$$

are the points of Γ_p with $\sigma^3 = \tau^3$, and they span the plane $\{g = n + i = 0\}$. Possibly replacing X with $\alpha(X)$ for an element α of the stabilizer of p in $G^\circ_{\mathcal{W}}$ that maps $\{y_1, y_2, y_3\}$ onto $\{z_\zeta\}_{\zeta^3=1}$, we may assume that $\mathbb{P}(T_pX) = \{g = n + i = 0\}$.

Let ℓ_1 and ℓ_2 be linear forms defining Λ. Since $p \in \Lambda$, the coefficients of h in ℓ_1 and ℓ_2 are zero. For a linear form $\ell = \sum_\xi \beta_\xi \xi$ without h, ξ running over the coordinates other than h, regarded as a regular function on $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{h \neq 0\}$, Lemma 2.15(1) implies

$$d\ell(p) = \beta_i di + \beta_a da + \beta_e de + \beta_n dn + (\beta_g - \beta_k) dg.$$

By (6.25), the span of $d\ell_1(p)$ and $d\ell_2(p)$ is the span of $di + dn$ and dg. Possibly replacing ℓ_1 and ℓ_2 with linear combinations of ℓ_1 and ℓ_2, we may assume that $d\ell_1(p) = di + dn$ and $d\ell_2(p) = dg$. Thus the weight-one part of ℓ_1 is $n + i$, the form ℓ_2 has no weight-one part, and the coefficients u of g and v of k in ℓ_2 satisfy $u - v = 1$. Writing λ and μ for the coefficients of f and c in ℓ_2, the lowest-weight parts of ℓ_1 and ℓ_2 are $n + i$ and $ug + vk + \lambda f + \mu c$, and the remaining terms have weight at least 2 and at least 3, respectively. This proves (1).

Step 3. In this step, we prove that $H^3 = 18$ and that $\Sigma(X, p)$ consists of three distinct reduced points in Set-up 6.6. By [Lat23, p. 568], $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is a Fano fivefold of index 3 and degree 18. Thus adjunction implies that $-K_X$ is the restriction of the hyperplane class, and $H^3 = 18$.

Let ℓ_1 and ℓ_2 be the defining linear forms of X in Set-up 6.6, with lowest-weight parts $n+i$ and $ug+vk+\lambda f+\mu c$. By Lemma 2.15(1), f, c, $g+k$, and the coordinates of weight at least 3 vanish to order at least 2 at p. Hence $d\ell_1(p) = di+dn+\gamma dg$ for some $\gamma \in \mathbb{C}$ and $d\ell_2(p) = (u-v)dg = dg$. Thus $\mathbb{P}(T_pX) = \{g = n + i = 0\}$, and (6.25) implies $\Sigma(X, p) = \Gamma_p \cap \{n + i = 0\}$. The pullback of $n + i$ under the isomorphism $\mathbb{P}^1 \to \Gamma_p$ of Lemma 2.15(3) is $\sigma^3 - \tau^3$, which has three distinct simple zeros $[1 : \zeta]$ with $\zeta^3 = 1$. Thus $\Sigma(X, p)$ consists of three distinct reduced points.

Step 4. In this step, we start the proof of (C). We use the following consequence of the Jacobian criterion [Har77, Theorem I.5.1 and Exercise I.5.10]: if $x \in X$ and the differential of ℓ_2 at x, regarded as a regular function on an affine chart of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, is zero, then the Zariski tangent space T_xX is the kernel of $d\ell_1(x)$ in the five-dimensional space $T_x\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$, so $\dim T_xX \geq 4$, and X is singular at x.

Suppose that (C) fails for some $\zeta \in \mathbb{C}$ with $\zeta^3 = 1$, that is, $u = \mu\zeta$ and $\lambda = -v\zeta$. Since $u - v = 1$, we have $\mu = u\zeta^2 = (v + 1)\zeta^2$. Write

$$\ell_1 = n + i + \ell_1', \qquad \ell_2 = ug + vk + \lambda f + \mu c + \beta_b b + \beta_d d + \beta_l l + \beta_M M + \beta_J J,$$

where ℓ_1' is a linear combination of the coordinates of weight at least 2 and $\beta_b, \beta_d, \beta_l, \beta_M, \beta_J \in \mathbb{C}$. Let

$$\rho := -\zeta\beta_b + \zeta^2\beta_d - \beta_l - \beta_M.$$

For $t \in \mathbb{C}$, let $x_t \in \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{h \neq 0\}$ be the point with

$$(i, a, e, n, g) = (t, \zeta t, \zeta^2 t, -t, 0).$$

Since $\zeta^3 = 1$, (2.1) implies that f, c, $ae + in$, k, b, d, l, M, and J vanish at x_t. Hence $\ell_1(x_t) = \ell_2(x_t) = 0$ and $x_t \in X$. The points x_t with $t \in \mathbb{C}$ form the complement of the point

$$q_\zeta : \quad h = 0,\ i = 1,\ a = \zeta,\ e = \zeta^2,\ n = -1,\ \text{all other coordinates zero}$$

in the line $\overline{pq_\zeta} \subset \mathbb{P}^{13}$. Since X is closed, $\overline{pq_\zeta} \subset X$ and $q_\zeta \in X$.

We compute the differentials at x_t in the basis di, da, de, dn, dg of the cotangent space of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ at x_t. By (2.1),

$$df = t(\zeta^2 di - 2\zeta da + de), \qquad dc = t(-da + 2\zeta^2 de + \zeta dn),$$

$$d(ae+in) = t(-di+\zeta^2 da+\zeta de+dn) = -\zeta df+\zeta^2 dc.$$

Let $\vartheta_t := dg+\zeta^2 dc$. The coefficient of dg in ϑ_t is 1, so $\vartheta_t \neq 0$. Since $dk = -dg - d(ae+in)$,

$$d(ug+vk+\lambda f+\mu c) = (u-v)dg+(\lambda+v\zeta)df+(\mu-v\zeta^2)dc = dg+\zeta^2 dc = \vartheta_t.$$

Since f, c, g, and k vanish at x_t, (2.1) implies

$$db = -adg-idc = -\zeta t\vartheta_t, \qquad dd = edg+adc = \zeta^2 t\vartheta_t,$$
$$dl = -adf+idk = -t\vartheta_t, \qquad dM = -edc+ndg = -t\vartheta_t,$$

and $dJ = 0$. For instance, $dl = -\zeta t df - tdg - td(ae+in) = -t(dg+\zeta^2 dc)$. Therefore

$$d\ell_2(x_t) = (1+\rho t)\vartheta_t. \tag{6.26}$$

If $\rho \neq 0$, then $d\ell_2(x_{-1/\rho}) = 0$ by (6.26), and X is singular at $x_{-1/\rho} \in X$. This is a contradiction.

Step 5. In this step, we finish the proof of (C). By **Step 4**, it suffices to treat the case $\rho = 0$. Recall that $q_\zeta \in X$ is the point of $\overline{pq_\zeta}$ with $h = 0$. In the coordinates (a, f, g, h, l) of $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{i \neq 0\}$ of Lemma 2.15(2), the point q_ζ is $(\zeta, 0, 0, 0, 0)$. By (2.2), at q_ζ,

$$dk = \zeta df, \quad dc = -\zeta dg, \quad db = \zeta dl,$$
$$dd = -\zeta db = -\zeta^2 dl, \quad dM = \zeta^2 db = dl, \quad dJ = 0.$$

Thus, regarding ℓ_2 as a regular function on $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{i \neq 0\}$,

$$\begin{aligned} d\ell_2(q_\zeta) &= (u-\mu\zeta)dg+(v\zeta+\lambda)df+(\zeta\beta_b-\zeta^2\beta_d+\beta_l+\beta_M)dl \\ &= -\rho dl = 0. \end{aligned}$$

Hence X is singular at q_ζ by **Step 4**. This is a contradiction.

By **Steps 4 and 5**, (C) holds for any $\zeta \in \mathbb{C}$ with $\zeta^3 = 1$. Together with **Step 3**, this proves (2). □

We now compute the refined series $W_{m,r}$ of Set-up 6.6 by degenerating X to the linear section of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ cut out by the lowest-weight parts of the defining linear forms of X.

Lemma 6.8. *Notation and conditions as in Set-up 6.6. Let*

$$Z := \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{n+i = 0,\ ug+vk+\lambda f+\mu c = 0\},$$

and let $\mathbb{G}_m$ act on $\mathbb{P}^{13}$ by multiplying each coordinate of weight w by t^w. Then the following hold.

(1) Z is a $\mathbb{G}_m$-invariant integral Cohen–Macaulay threefold, and $Z \cap \{h \neq 0\}$ is the affine space with coordinates (i, a, e), on which

$$\begin{gathered} h = 1, \quad n = -i, \quad f = f_2, \quad c = c_2, \quad g = G_2, \quad k = -G_2 - Q_2, \\ b = -aG_2 - ic_2, \quad d = eG_2 + ac_2, \quad l = -af_2 - i(G_2+Q_2), \\ M = -ec_2 - iG_2, \quad J = J_4. \end{gathered} \tag{6.27}$$

(2) The part of $Z \cap \{h = 0\}$ in the chart $\{i \neq 0\}$ is contained in the union of the loci

$$\{a = \zeta i,\ (u-\mu\zeta)g+(v\zeta+\lambda)f = 0\}, \qquad \zeta^3 = 1,$$

and $Z \cap \{h = i = 0\}$ is contained in the cone

$$\{a = e = f = g = k = c = n = 0,\ ld+b^2 = lM+bd = bM-d^2 = 0\}.$$

(3) For any integer $m \geq 0$, the map sending $s \in H^0(Z, \mathcal{O}_Z(m))$ to the polynomial s/h^m on $Z \cap \{h \neq 0\}$ is a $\mathbb{G}_m$-equivariant isomorphism

$$H^0(Z, \mathcal{O}_Z(m)) \to \bigoplus_{r\geq 0} W_{m,r} \subset \bigoplus_{r\geq 0} H^0(E, \mathcal{O}_E(r)) = \mathbb{C}[i, a, e],$$

under which $W_{m,r}$ is the part of weight r.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that Z is $\mathbb{G}_m$-invariant and compute the chart $Z \cap \{h \neq 0\}$.

By Set-up 2.14(3), $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is cut out by the 35 Pfaffians $\mathrm{Pf}_{j_1j_2j_3j_4}$ of Set-up 2.14(4). Let

$$(\epsilon_1, \dots, \epsilon_7) := (-1, 1, 0, 1, -1, 0, 0).$$

Going through the 21 entries above the diagonal, the weight of A_{jk} in Set-up 6.6 equals $\epsilon_j + \epsilon_k + 2$. For example, $A_{15} = h$ has weight 0, $A_{24} = J$ has weight 4, and $A_{36} = -g - k$ has weight 2. Thus $\mathrm{Pf}_{j_1j_2j_3j_4}$ is homogeneous of weight $\epsilon_{j_1} + \epsilon_{j_2} + \epsilon_{j_3} + \epsilon_{j_4} + 4$, and $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is $\mathbb{G}_m$-invariant. Since $n + i$ and $ug + vk + \lambda f + \mu c$ are homogeneous of weights 1 and 2, Z is $\mathbb{G}_m$-invariant.

By Lemma 2.15(1), $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{h \neq 0\}$ is the affine space with coordinates (i, a, e, n, g), on which the other coordinates are given by (2.1). Giving i, a, e, n weight 1 and g weight 2, any coordinate of weight w restricts to a polynomial in (i, a, e, n, g) of weight w. Since $u - v = 1$, (2.1) implies

$$ug + vk + \lambda f + \mu c = g - \widehat{G}, \qquad \widehat{G} := v(ae + in) - \lambda(ie - a^2) - \mu(e^2 + an). \tag{6.28}$$

At $n = -i$ we have $\widehat{G} = vQ_2 - \lambda f_2 - \mu c_2 = G_2$. Thus $Z \cap \{h \neq 0\}$ is the affine space with coordinates (i, a, e), cut out by $n = -i$ and $g = G_2$, and substituting $n = -i$ and $g = G_2$ into (2.1) implies (6.27).

Step 2. In this step we prove (2) and the remaining part of (1).

By Lemma 2.15(2), on $\{i = 1\}$ the coordinates e, k, b, c, n, d, M, J are the polynomials in the free coordinates (a, f, g, h, l) given by (2.2). At $h = 0$ we obtain $e = a^2$, $k = af$, $c = -ag$, and $n = -a^3$. Thus on $Z \cap \{h = 0,\ i = 1\}$ the equation $n + i = 0$ becomes $a^3 = 1$, so $a = \zeta$ with $\zeta^3 = 1$, and the second equation becomes

$$(u - \mu\zeta)g + (v\zeta + \lambda)f = 0.$$

This proves the first part of (2). By Lemma 6.7(2), condition (C) holds, so for any ζ with $\zeta^3 = 1$ the last equation is a non-zero linear equation in the free coordinates (f, g, l). Thus $\dim\big(Z \cap \{h = 0,\ i \neq 0\}\big) \leq 2$.

On $Z \cap \{h = i = 0\}$ we work on the reduced support. There

$$\mathrm{Pf}_{1567} = a^2 - ie + fh, \qquad \mathrm{Pf}_{1357} = e^2 - hc + an, \qquad \mathrm{Pf}_{1267} = al - bi - f^2$$

imply $a = e = f = 0$, and $n + i = 0$ implies $n = 0$. Then

$$\mathrm{Pf}_{1245} = Jh - cf - gk, \qquad \mathrm{Pf}_{1346} = Mi - be + g(g + k), \qquad \mathrm{Pf}_{2356} = -ad + k(g + k) + ln$$

imply $gk = g(g + k) = k(g + k) = 0$, hence $g = k = 0$, and $\mathrm{Pf}_{3457} = Me + c^2 - dn$ implies $c = 0$. Finally

$$\mathrm{Pf}_{2467} = Jf - ld - b^2, \qquad \mathrm{Pf}_{2346} = J(g + k) + lM + bd, \qquad \mathrm{Pf}_{2347} = -Jc + bM - d^2$$

imply $ld + b^2 = lM + bd = bM - d^2 = 0$. This proves the second part of (2). The cone in (2) is the cone with vertex $[J]$ over a twisted cubic in the coordinates $(l : b : d : M)$, so $\dim\big(Z \cap \{h = i = 0\}\big) \leq 2$. Thus $\dim\big(Z \cap \{h = 0\}\big) \leq 2$.

The linear forms $n + i$ and $ug + vk + \lambda f + \mu c$ are linearly independent. By Lemma 2.16, Z is a Gorenstein scheme of pure dimension three, in particular Cohen–Macaulay. Any irreducible component of Z is three-dimensional, so it is not contained in $Z \cap \{h = 0\}$, and it contains a non-empty open subset of the irreducible affine space $Z \cap \{h \neq 0\}$. Thus Z is irreducible and generically reduced. Since a generically reduced Cohen–Macaulay scheme is reduced [Mat89, Section 23, p. 183], Z is integral. This proves (1).

Step 3. In this step we compare the local equations of X at p with those of Z.

Write the defining linear forms ℓ_1, ℓ_2 of X as

$$\ell_1 = n + i + \sum_{w \geq 2} \ell_{1,w}, \qquad \ell_2 = ug + vk + \lambda f + \mu c + \sum_{w \geq 3} \ell_{2,w},$$

where $\ell_{j,w}$ is a linear form in the coordinates of weight w. By **Step 1**, on $\{h = 1\}$ the restriction of $\ell_{j,w}$ is a polynomial in (i, a, e, n, g) of weight w. By (2.1), the coordinates f, c, b, d, J, l, M restrict to polynomials without constant and linear terms, while $k + g$ restricts to a quadratic polynomial. By (6.28), $d\ell_1(p) = di + dn + \gamma dg$ for some $\gamma \in \mathbb{C}$ and $d\ell_2(p) = dg$. Thus i, a, e are regular parameters of X at p, and by the formal implicit function theorem $\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[i, a, e]]$, where $n, g \in \mathbb{C}[[i, a, e]]$ are determined by $\ell_1 = \ell_2 = 0$. Let $\mathfrak{m} := (i, a, e) \subset \mathbb{C}[[i, a, e]]$.

Since $n, g \in \mathfrak{m}$, a monomial $i^{\alpha_1} a^{\alpha_2} e^{\alpha_3} n^{\beta_1} g^{\beta_2}$ of weight $w = \alpha_1 + \alpha_2 + \alpha_3 + \beta_1 + 2\beta_2$ belongs to $\mathfrak{m}^{w-\beta_2}$. By (6.28), $\ell_2 = 0$ reads

$$g = \widehat{G} - \sum_{w \geq 3} \ell_{2,w}.$$

Since $\widehat{G}$ is quadratic in (i, a, e, n), and $w - \beta_2 \geq w/2 > 1$ for any monomial of weight $w \geq 3$, we obtain $g \in \mathfrak{m}^2$. Then any monomial of weight w belongs to $\mathfrak{m}^w$, so $\ell_1 = 0$ implies $n + i \in \mathfrak{m}^2$, and $\ell_2 = 0$ implies $g - \widehat{G} \in \mathfrak{m}^3$. Since $\widehat{G}$ is linear in n with coefficient $vi - \mu a$, we have

$$\widehat{G} - G_2 = (vi - \mu a)(n + i) \in \mathfrak{m}^3.$$

Thus

$$n = -i + n', \qquad g = G_2 + g', \qquad n' \in \mathfrak{m}^2, \quad g' \in \mathfrak{m}^3.$$

Let P be a polynomial in (i, a, e, n, g) of weight w. Substituting $n = -i + n'$ and $g = G_2 + g'$ into a monomial of P, any term involving n' or g' belongs to $\mathfrak{m}^{w+1}$. Therefore

$$P|_X - P(i, a, e, -i, G_2) \in \mathfrak{m}^{w+1}. \tag{6.29}$$

Step 4. In this step we prove (3).

Let $V \in \{X, Z\}$. By Lemma 2.16, V has pure dimension three, and $\mathrm{G}_2 \operatorname{Gr}(2, 7)$ is a smooth fivefold, so the defining linear forms of V form a regular sequence and the Koszul complex

$$0 \to \mathcal{O}(m-2) \to \mathcal{O}(m-1)^{\oplus 2} \to \mathcal{O}(m) \to \mathcal{O}_V(m) \to 0$$

on $\mathrm{G}_2 \operatorname{Gr}(2, 7)$ is exact. The canonical divisor of $\mathrm{G}_2 \operatorname{Gr}(2, 7)$ is -3 times the hyperplane class [Lat23, p. 568], so Kodaira vanishing [Laz04a, Theorem 4.2.1] implies $H^j(\mathrm{G}_2 \operatorname{Gr}(2, 7), \mathcal{O}(m')) = 0$ for $j > 0$ and $m' \geq -2$. Thus the restriction map from $H^0(\mathrm{G}_2 \operatorname{Gr}(2, 7), \mathcal{O}(m))$ to $H^0(V, \mathcal{O}_V(m))$ is surjective, and

$$h^0(V, \mathcal{O}_V(m)) = h^0(\mathcal{O}(m)) - 2h^0(\mathcal{O}(m-1)) + h^0(\mathcal{O}(m-2)).$$

Since H is the hyperplane class of X (Theorem 2.13), we obtain

$$h^0(X, mH) = h^0(Z, \mathcal{O}_Z(m)). \tag{6.30}$$

Since Z is integral and $Z \cap \{h \neq 0\}$ is dense in Z, the map $s \to s/h^m$ is injective on $H^0(Z, \mathcal{O}_Z(m))$. By (1), i, a, e have weight 1 on $Z \cap \{h \neq 0\}$ and h has weight 0, so the map is $\mathbb{G}_m$-equivariant and sends the weight-r part $H^0(Z, \mathcal{O}_Z(m))_r$ into $\mathbb{C}[i, a, e]_r = H^0(E, \mathcal{O}_E(r))$.

Let $s \in H^0(Z, \mathcal{O}_Z(m))_r$. The restriction map above is $\mathbb{G}_m$-equivariant and surjective, so s is the restriction of a section $\tilde{s}$ of $\mathcal{O}(m)$ on $\mathrm{G}_2 \operatorname{Gr}(2, 7)$ of weight r. By **Step 1**, $P := \tilde{s}/h^m$ is a polynomial in (i, a, e, n, g) of weight r, and $P(i, a, e, -i, G_2) = s/h^m$ by (1). By (6.29),

$$\tilde{s}|_X/h^m - s/h^m \in \mathfrak{m}^{r+1}.$$

Since E has coordinates $(i : a : e)$, the section $\pi^*(\tilde{s}|_X)$ belongs to $H^0(Y, m\pi^*H - rE)$, and the image of $\pi^*(\tilde{s}|_X)$ in $H^0(E, \mathcal{O}_E(r)) = \mathbb{C}[i, a, e]_r$ is the degree-r part of $\tilde{s}|_X/h^m$, which is s/h^m. Thus the map sends $H^0(Z, \mathcal{O}_Z(m))_r$ injectively into $W_{m,r}$.

The kernel of $H^0(Y, m\pi^*H - rE) \to H^0(E, \mathcal{O}_E(r))$ is $H^0(Y, m\pi^*H - (r+1)E)$, and $H^0(Y, m\pi^*H - rE) = 0$ for $r \gg 0$ because a non-zero section of mH vanishes to finite order at p. Thus, by (6.30),

$$\sum_{r \geq 0} \dim W_{m,r} = h^0(X, mH) = h^0(Z, \mathcal{O}_Z(m)) = \sum_{r \geq 0} \dim H^0(Z, \mathcal{O}_Z(m))_r.$$

Since $\dim H^0(Z,\mathcal{O}_Z(m))_r \le \dim W_{m,r}$ for any r, equality holds for any r, and the map sends $H^0(Z,\mathcal{O}_Z(m))_r$ onto $W_{m,r}$. This proves (3). □

Proposition 6.9. *Notation and conditions as in Set-up 6.6. For an integer N let $\gamma_N := (\lfloor N/2\rfloor+1)(N+1-\lfloor N/2\rfloor)$ if $N\ge 0$ and $\gamma_N := 0$ if $N<0$. Then the following hold.*

(1) The bigraded ring $\bigoplus_{m,r\ge 0} W_{m,r}$, the associated graded ring of $\bigoplus_m H^0(X,mH)$ with respect to the order of vanishing along E, is isomorphic to the bigraded quotient of the homogeneous coordinate ring of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ by $(n+i, ug+vk+\lambda f+\mu c)$. More precisely, for a variable τ, sending a section s of degree m to $(s/h^m)\tau^m$ defines an isomorphism from the quotient ring onto the subring $\bigoplus_{m,r} W_{m,r}\tau^m$ of $\mathbb{C}[i,a,e][\tau]$, under which i,a,e are sent to $\tau i,\tau a,\tau e$, and each of the coordinates h,n,f,c,g,k,b,d,l,M,J is sent to τ times the polynomial in i,a,e given for it in (6.27).

(2) For any integer $m\ge 0$,

$$\dim W_{m,r} = \binom{r+2}{2} - 3\gamma_{r-m-1} \quad \text{for } 0\le r\le 3m,$$

$$W_{m,r} = J_4^{r-3m} W_{4m-r,3(4m-r)} \quad \text{and} \quad \dim W_{m,r} = \frac{3q^2+3q+2}{2}$$

for $3m\le r\le 4m$, where $q := 4m-r$, and $W_{m,r}=0$ for $r>4m$. In particular, $\dim W_{m,r} = m^2\varphi(r/m)+O(m)$ uniformly for $0\le r\le 4m$, where $\varphi:[0,4]\to\mathbb{R}$ equals $\frac{t^2}{2}$, $\frac{t^2}{2}-\frac{3}{4}(t-1)^2$, and $\frac{3}{2}(4-t)^2$ on $[0,1]$, $[1,3]$, and $[3,4]$, respectively.

(3) The pair $(E,\frac{1}{2}B_4)$ is log canonical.

(4) We have

$$\mathrm{vol}(\pi^*H - tE) = \begin{cases} 18-t^3 & 0\le t\le 1,\\ 18-t^3+\frac{3}{2}(t-1)^3 & 1\le t\le 3,\\ 3(4-t)^3 & 3\le t\le 4,\\ 0 & t\ge 4,\end{cases}$$

and $S_X(E) = 9/4$.

Proof. The proof is long, so we divide it into several steps. Let Z be the threefold of Lemma 6.8, and let $R := \bigoplus_{m\ge 0} H^0(\mathrm{G}_2\,\mathrm{Gr}(2,7),\mathcal{O}(m))$.

Step 1. In this step we prove (1).

By the Borel–Weil theorem [FH91, §23.3, Claim 23.57], $H^0(\mathrm{G}_2\,\mathrm{Gr}(2,7),\mathcal{O}(m))$ is an irreducible representation of G_2 with highest weight $m\theta$, where θ is the highest root. The multiplication map $\mathrm{Sym}^m H^0(\mathcal{O}(1)) \to H^0(\mathcal{O}(m))$ is G_2-equivariant and non-zero, since $h^m\ne 0$, so it is surjective. Thus R is the homogeneous coordinate ring of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$. The Koszul resolution of $\mathcal{O}_Z$ by $\mathcal{O}(-1)^{\oplus 2}$ and $\mathcal{O}(-2)$, together with $H^j(\mathrm{G}_2\,\mathrm{Gr}(2,7),\mathcal{O}(m'))=0$ for $j>0$ and $m'\ge -2$ ([Lat23, p. 568] and [Laz04a, Theorem 4.2.1]), implies the exact sequence

$$0\to R_{m-2}\to R_{m-1}^{\oplus 2}\to R_m\to H^0(Z,\mathcal{O}_Z(m))\to 0,$$

where $R_{m-1}^{\oplus 2}\to R_m$ is $(s_1,s_2)\to (n+i)s_1+(ug+vk+\lambda f+\mu c)s_2$. Thus $\bigoplus_m H^0(Z,\mathcal{O}_Z(m)) = R/(n+i, ug+vk+\lambda f+\mu c)$, bigraded by degree and weight.

Order of vanishing at p is a valuation, so the product of the leading forms of two sections is the leading form of their product, and the associated graded ring $\bigoplus_{m,r} W_{m,r}$ is the subring $\bigoplus_{m,r} W_{m,r}\tau^m$ of $\mathbb{C}[i,a,e][\tau]$. The map $s\to (s/h^m)\tau^m$ from $\bigoplus_m H^0(Z,\mathcal{O}_Z(m))$ to $\mathbb{C}[i,a,e][\tau]$ is a ring homomorphism, and by Lemma 6.8(3) it is injective with image $\bigoplus_{m,r} W_{m,r}\tau^m$, sending the part of weight r in degree m onto $W_{m,r}\tau^m$. By (6.27), it sends i,a,e to $\tau i,\tau a,\tau e$, and each of h,n,f,c,g,k,b,d,l,M,J to τ times the polynomial given for it in (6.27). This proves (1).

Step 2. In this step we compute the weight character

$$\Psi_m(y) := \sum_{r\geq 0} \dim H^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(m))_r y^r,$$

and we let $\Psi_m := 0$ for $m < 0$. Let

$$T_m(y) := 1 - (3m+4)y^{m+1} + (3m+5)y^{m+2} - (3m+5)y^{3m+4} + (3m+4)y^{3m+5} - y^{4m+6}.$$

We show that

$$\Psi_m(y) = \frac{T_m(y)}{(1-y)^4(1-y^2)} \qquad (m \geq 0). \tag{6.31}$$

Let λ_0 be the diagonal one-parameter subgroup of GL_7 with weights $(\epsilon_1, \dots, \epsilon_7) = (-1,1,0,1,-1,0,0)$ on the basis vectors $x_1, \dots, x_7$ of $\mathbb{C}^7$, and let

$$\omega' := -x_1x_2x_3x_7 + x_4x_5x_6x_7 + x_2x_3x_5x_6 + x_1x_3x_4x_6 + x_1x_2x_4x_5 \in \Lambda^4\mathbb{C}^7$$

(products are exterior products). The element $\operatorname{diag}(1,-1,-1,-1,1,1,-1) \in \mathrm{SL}_7$ sends the four-form of [Kap11, Section 2] to ω', changing only the sign of the monomial $x_1x_2x_3x_7$, so ω' is non-degenerate. Identifying the matrix A with $\sum_{j<k} A_{jk}x_j \wedge x_k \in \Lambda^2\mathbb{C}^7$, a direct calculation shows

$$\mathcal{W} = \{\xi \in \Lambda^2\mathbb{C}^7 : \xi \wedge \omega' = 0\}.$$

Each monomial of ω' has weight sum 0, so λ_0 preserves ω'. The Lie algebra of the stabilizer of a non-degenerate four-form on $\mathbb{C}^7$ is $\mathfrak{g}_2$ [FH91, Proposition 22.12], applied to the dual three-form, so the identity component of the stabilizer of ω' is G_2, and λ_0, whose image is connected, is a one-parameter subgroup of G_2. The stabilizer of ω' preserves $\mathcal{W}$, so G_2 acts on $\mathbb{P}^{13} = \mathbb{P}(\mathcal{W})$, and $\mathcal{W}$ is the adjoint representation of G_2 [Kap11, Sections 1 and 2]. On $\mathbb{P}^{13}$, $\lambda_0(t)$ multiplies the entry A_{jk} by $t^{\epsilon_j+\epsilon_k}$, which is the action of Lemma 6.8 composed with the scalar t^{-2}. Thus $\Psi_m(y) = y^{2m}\chi_m(y)$, where $\chi_m(y)$ is the character of $H^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(m))$ evaluated on λ_0. The coordinates are dual to $\mathfrak{g}_2$, so their λ_0-weights are the negatives of those of $\mathfrak{g}_2$. Since -1 belongs to the Weyl group of G_2, $\chi_m(y) = \chi_m(y^{-1})$, so the sign of the weights does not matter. The weights of λ_0 on the adjoint representation are $-2,-1,0,1,2$ with multiplicities $1,4,4,4,1$. Choose simple roots α (short) and β (long) such that $\langle\alpha,\lambda_0\rangle, \langle\beta,\lambda_0\rangle \geq 0$. Then $\theta = 3\alpha + 2\beta$ and $\langle\theta,\lambda_0\rangle = 2$, so $\langle\alpha,\lambda_0\rangle = 0$ and $\langle\beta,\lambda_0\rangle = 1$.

We apply the Weyl character formula [FH91, Theorem 24.2] with highest weight $m\theta = 3m\alpha + 2m\beta$, where half the sum of the positive roots is $5\alpha + 3\beta$. In the coordinates $x\alpha + z\beta$, the simple reflections are $s_\alpha(x,z) = (-x+3z, z)$ and $s_\beta(x,z) = (x, x-z)$. Since $\langle\alpha,\lambda_0\rangle = 0$, the numerator and the denominator vanish after the substitution $e^\alpha = 1$, $e^\beta = y$. We pair the twelve terms of the numerator and of the denominator under s_α, differentiate with respect to e^α, and then substitute $e^\alpha = 1$, $e^\beta = y$. For w in the Weyl group, write $w(m\theta + 5\alpha + 3\beta) = x\alpha + z\beta$. The derivative of $e^{x\alpha+z\beta} - e^{(3z-x)\alpha+z\beta}$ at $e^\alpha = 1$ is $(2x-3z)y^z$, so the pair $\{w, s_\alpha w\}$ contributes $\operatorname{sgn}(w)(2x-3z)y^z$ to the derivative of the numerator. The six pairs and their contributions are listed in Table 2, where the first entry of each pair is the (x,z) with $2x-3z > 0$. The sum of the contributions is $-y^{-2m-3}T_m(y)$. The denominator is the numerator for $m = 0$, so the derivative of the denominator becomes

$$y^3 - 4y^2 + 5y - 5y^{-1} + 4y^{-2} - y^{-3} = -y^{-3}(1-y)^4(1-y^2).$$

Since $\Psi_m(y) = y^{2m}\chi_m(y)$, this implies (6.31).

Step 3. In this step we prove (2).

The linear forms $n+i$ and $ug + vk + \lambda f + \mu c$ have weights 1 and 2, so the exact sequence of **Step 1** is $\mathbb{G}_m$-equivariant after shifting the weights on the first and second summands of $R_{m-1}^{\oplus 2}$ by 1 and 2 respectively, and the weights on R_{m-2} by 3. By Lemma 6.8(3),

$$\sum_{r\geq 0} \dim W_{m,r} y^r = \Psi_m(y) - (y+y^2)\Psi_{m-1}(y) + y^3\Psi_{m-2}(y). \tag{6.32}$$

(x,z)	$(3z-x,z)$	$\operatorname{sgn}(w)(2x-3z)y^z$
$(3m+5,2m+3)$	$(3m+4,2m+3)$	y^{2m+3}
$(3m+5,m+2)$	$(1,m+2)$	$-(3m+4)y^{m+2}$
$(3m+4,m+1)$	$(-1,m+1)$	$(3m+5)y^{m+1}$
$(1,-m-1)$	$(-3m-4,-m-1)$	$-(3m+5)y^{-m-1}$
$(-1,-m-2)$	$(-3m-5,-m-2)$	$(3m+4)y^{-m-2}$
$(-3m-4,-2m-3)$	$(-3m-5,-2m-3)$	$-y^{-2m-3}$

TABLE 2. The pairs $\{w, s_\alpha w\}$ in the Weyl character formula for $m\theta$ and their contributions to the derivative of the numerator.

For $m=0$ and $m=1$, (6.32) and (6.31) imply 1 and $1+3y+3y^2+4y^3+y^4$, which is (2) for $m\le 1$. Let $m\ge 2$. A direct calculation shows

$$T_m-(y+y^2)T_{m-1}+y^3T_{m-2}\equiv(1-y)(1-y^2)-3y^{m+1}(1-y)^2 \pmod{y^{3m+1}}. \tag{6.33}$$

Dividing (6.33) by $(1-y)^4(1-y^2)$, we see from (6.32) and (6.31) that, modulo y^{3m+1},

$$\sum_{r\ge 0}\dim W_{m,r}y^r\equiv\frac{1}{(1-y)^3}-\frac{3y^{m+1}}{(1-y)^2(1-y^2)}.$$

The coefficient of y^N in $1/((1-y)^2(1-y^2))$ is $\sum_{j=0}^{\lfloor N/2\rfloor}(N-2j+1)=\gamma_N$. This proves the first formula in (2).

By **Step 1**, R is generated in degree one. Any coordinate has weight at most 3, except J, which has weight 4. Thus a monomial of degree m in the coordinates has weight at most $4m$, which proves $W_{m,r}=0$ for $r>4m$. If such a monomial has weight $r\ge 3m$, then it contains J at least $r-3m$ times, and after removing J^{r-3m} we obtain a monomial of degree $4m-r$ and weight $r-4(r-3m)=3(4m-r)$. By (1), this proves $W_{m,r}=J_4^{r-3m}W_{4m-r,3(4m-r)}$ for $3m\le r\le 4m$. Since $\dim W_{1,4}=1$ and J is the only coordinate of weight 4, we have $J_4\ne 0$, so multiplication by J_4^{r-3m} in $\mathbb{C}[i,a,e]$ is injective. With $q=4m-r$, the first formula of (2) implies

$$\dim W_{m,r}=\dim W_{q,3q}=\binom{3q+2}{2}-3\gamma_{2q-1}=\frac{3q^2+3q+2}{2}.$$

Since $\gamma_N=N^2/4+O(N)$, the formulas $\dim W_{m,r}=\binom{r+2}{2}-3\gamma_{r-m-1}$ for $r\le 3m$ and $\dim W_{m,r}=\frac{3q^2+3q+2}{2}$ for $r\ge 3m$ imply $\dim W_{m,r}=m^2\varphi(r/m)+O(m)$ uniformly for $0\le r\le 4m$. This proves (2).

Step 4. In this step we prove (4).

The kernel of $H^0(Y,m\pi^*H-rE)\to W_{m,r}$ is $H^0(Y,m\pi^*H-(r+1)E)$, and $W_{m,r}=0$ for $r>4m$, so

$$h^0(Y,m\pi^*H-rE)=\sum_{s\ge r}\dim W_{m,s}.$$

For a rational number $t\ge 0$ and m with $mt\in\mathbb{Z}$, (2) and the Riemann sums of φ imply

$$h^0(Y,m\pi^*H-mtE)=m^3\int_t^4\varphi(s)ds+O(m^2),$$

where the integral is 0 for $t\ge 4$. Thus

$$\operatorname{vol}(\pi^*H-tE)=6\int_t^4\varphi(s)ds$$

for rational $t\ge 0$, and for real $t\ge 0$ by the continuity of the volume [Laz04a, Theorem 2.2.44 and Corollary 2.2.45]. We have $\int_0^1\varphi=\frac{1}{6}$, $\int_1^3\varphi=\frac{7}{3}$, and $\int_3^4\varphi=\frac{1}{2}$, so $\int_0^4\varphi=3$. Integrating the

three quadratic polynomials implies the formula for $\mathrm{vol}(\pi^*H - tE)$ in (4). By Lemma 6.7(2), $H^3 = 18$, and

$$\int_0^4 \mathrm{vol}(\pi^*H - tE)dt = \left(18 - \frac{1}{4}\right) + 22 + \frac{3}{4} = \frac{81}{2}.$$

Thus $S_X(E) = \frac{1}{18} \cdot \frac{81}{2} = \frac{9}{4}$.

Step 5. In this step we prove (3).

The polynomials $f_2 = ie - a^2$, $-Q_2 = i^2 - ae$, and $-c_2 = ai - e^2$ are the 2×2 minors of the matrix

$$\begin{pmatrix} i & a & e \\ a & e & i \end{pmatrix}.$$

The rank of the matrix is at most one exactly when (a, e, i) is proportional to (i, a, e), that is, at the three points $(i : a : e) = (1 : \zeta : \zeta^2)$ with $\zeta^3 = 1$. The three points are not collinear, since the Vandermonde determinant of $1, \zeta, \zeta^2$ is non-zero. The conics f_2, c_2, Q_2 are linearly independent and vanish at the three points, so they span the conics through the three points. By [Har77, Example V.4.2.3], the rational map $(f_2 : c_2 : Q_2)$ from E to $\mathbb{P}^2$ is resolved by the blow-up $\rho_0 : S_0 \to E$ of the three points, with exceptional curves e^0_ζ, and the induced morphism $\kappa : S_0 \to \mathbb{P}^2$ is the blow-up of three non-collinear points of $\mathbb{P}^2$.

Let $(z_1 : z_2 : z_3)$ be the coordinates of $\mathbb{P}^2$, and let

$$G' := vz_3 - \lambda z_1 - \mu z_2, \qquad \Phi_0 := z_1z_2 - G'^2 - G'z_3 = z_1z_2 + \frac{z_3^2}{4} - \left(G' + \frac{z_3}{2}\right)^2.$$

Then $G_2 = G'(f_2, c_2, Q_2)$ and $J_4 = \Phi_0(f_2, c_2, Q_2)$. The quadratic form $z_1z_2 + z_3^2/4$ has rank 3, so Φ_0 has rank at least 2, and $D_2 := \{\Phi_0 = 0\}$ is a conic. Since $\kappa^*\mathcal{O}_{\mathbb{P}^2}(1) = \rho_0^*\mathcal{O}_E(2) - \sum_\zeta e^0_\zeta$, we obtain

$$\rho_0^*B_4 = \kappa^*D_2 + 2\sum_\zeta e^0_\zeta, \qquad K_{S_0} = \rho_0^*K_E + \sum_\zeta e^0_\zeta.$$

Thus $K_{S_0} + \frac{1}{2}\kappa^*D_2 = \rho_0^*(K_E + \frac{1}{2}B_4)$, and $(E, \frac{1}{2}B_4)$ is log canonical if and only if $(S_0, \frac{1}{2}\kappa^*D_2)$ is log canonical [KM98, Lemma 2.30].

Since Φ_0 has rank at least 2, the conic D_2 is smooth or a union of two distinct lines, and κ is the blow-up of three distinct points. If D_2 is smooth, then κ^*D_2 is the strict transform of D_2 plus the exceptional curves over the blown-up points on D_2, each with coefficient 1 and meeting the strict transform transversally. If $D_2 = \Delta' + \Delta''$ with $\Delta' \cap \Delta'' = \{o\}$, then the exceptional curve over o, if o is blown up, has coefficient 2 and intersects the strict transforms of Δ' and Δ'' at two distinct points, and the exceptional curves over the other blown-up points on $\Delta' \cup \Delta''$ have coefficient 1. In both cases κ^*D_2 has simple normal crossing support and coefficients at most 2. Thus $\frac{1}{2}\kappa^*D_2$ has simple normal crossing support and coefficients at most 1, and $(S_0, \frac{1}{2}\kappa^*D_2)$ is log canonical [KM98, Corollary 2.31(3)]. This proves (3). □

We first describe the cubic part $W_{1,3}$ of the refined series and the diagonal series $W_{m,3m}$, $m \geq 0$.

Construction-Proposition 6.10. Notation and conditions as in Set-up 6.6. For $\zeta \in \mathbb{C}$ with $\zeta^3 = 1$, let $p_\zeta := (1 : \zeta : \zeta^2) \in E$, and let t_ζ be the linear part at p_ζ, in the chart $i = 1$, of $-(v\zeta + \lambda)f_2 + \zeta^2(u - \mu\zeta)c_2$. Then the following hold.

(1) The linear system $W_{1,3}$ of plane cubics has no fixed curve. The base points of $W_{1,3}$ are exactly the three points p_ζ, each of multiplicity one, and over each p_ζ one simple infinitely near base point, in the direction $t_\zeta = 0$.

(2) Let $\rho : \widetilde{E} \to E$ be the blow-up of the six base points of $W_{1,3}$, let e_ζ be the total transform of the exceptional curve over p_ζ, e'_ζ the exceptional curve over the infinitely near point,

$R_\zeta := e_\zeta - e'_\zeta$, and $\eta := \rho^*\mathcal{O}_E(1)$. Then

$$-K_{\widetilde{E}} = 3\eta - \sum_\zeta (e_\zeta + e'_\zeta)$$

is base point free with $(-K_{\widetilde{E}})^2 = 3$, and for any integer $m \geq 0$ the space $\rho^*W_{m,3m}$ consists of the products of all sections of $\mathcal{O}_{\widetilde{E}}(-mK_{\widetilde{E}})$ with a fixed section whose divisor is $m\sum_\zeta (R_\zeta + 2e'_\zeta)$. The morphism defined by $|-K_{\widetilde{E}}|$ is birational onto a normal cubic surface in $\mathbb{P}^3$, contracts finitely many curves, and is an isomorphism over a neighborhood of the image of $\rho^{-1}(E\backslash B_4)$.

(3) For each ζ, let (t, y) be regular local parameters of E at p_ζ whose linear parts are t_ζ and a form independent of t_ζ, and let w_ζ be the monomial valuation with weights $(2, 1)$ on (t, y). Then $w_\zeta(s) \geq r - m$ for any integers $m, r \geq 0$ and any $s \in W_{m,r}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the diagonal algebra $T := \bigoplus_{m\geq 0} W_{m,3m}$ is generated by $W_{1,3}$ and is the homogeneous coordinate ring of an integral cubic surface.

By Proposition 6.9(1), $W_{m,r}$ is spanned by the products $\overline{x_1}\cdots\overline{x_m}$ over the monomials $x_1\cdots x_m$ of degree m and weight r in the Kapustka coordinates, where for a coordinate x we let $\overline{x} \in \mathbb{C}[i, a, e]$ be the polynomial such that the isomorphism of Proposition 6.9(1) sends x to $\overline{x}\tau$. Thus $\overline{h} = 1$, $\overline{n} = -i$, $\overline{f} = f_2$, $\overline{c} = c_2$, $\overline{g} = G_2$, $\overline{k} = -G_2 - Q_2$, $\overline{J} = J_4$, and we let

$$l_3 := \overline{l} = -af_2 - i(G_2 + Q_2), \qquad b_3 := \overline{b} = -aG_2 - ic_2,$$
$$d_3 := \overline{d} = eG_2 + ac_2, \qquad M_3 := \overline{M} = -ec_2 - iG_2.$$

Since l, b, d, M are the coordinates of weight three, $W_{1,3}$ is spanned by l_3, b_3, d_3, M_3, and $\dim W_{1,3} = \binom{5}{2} - 3\gamma_1 = 4$ by Proposition 6.9(2).

We regard l, b, d, M as homogeneous coordinates on $\mathbb{P}^3$, and let

$$\Phi := ld + b^2, \quad \Psi := bM - d^2, \quad \Xi := lM + bd, \quad \Lambda := v\Xi - \lambda\Phi - \mu\Psi, \quad \Theta := \Phi\Psi - \Lambda(\Lambda + \Xi).$$

For a form P in l, b, d, M we write $\overline{P} := P(l_3, b_3, d_3, M_3) \in \mathbb{C}[i, a, e]$. A direct calculation with the formulas of Proposition 6.9(1) shows

$$\overline{\Phi} = J_4 f_2, \qquad \overline{\Psi} = J_4 c_2, \qquad \overline{\Xi} = J_4 Q_2, \qquad \overline{\Lambda} = J_4 G_2, \qquad \overline{\Theta} = J_4^3, \tag{6.34}$$

and

$$l_3G_2 - b_3f_2 = J_4 i, \qquad d_3f_2 + b_3G_2 = J_4 a, \qquad -M_3f_2 - d_3G_2 = J_4 e. \tag{6.35}$$

In particular $J_4(-G_2 - Q_2) = -\overline{\Lambda} - \overline{\Xi}$, and multiplying (6.35) by J_4 and using (6.34) we obtain

$$\overline{\Lambda l - \Phi b} = J_4^2 i, \qquad \overline{\Phi d + \Lambda b} = J_4^2 a, \qquad \overline{-\Phi M - \Lambda d} = J_4^2 e. \tag{6.36}$$

Let $x_1\cdots x_m$ be a monomial of degree m and weight $3m$ in the Kapustka coordinates, and let θ_j be the number of factors of weight j, $0 \leq j \leq 4$. Then $\sum_j \theta_j = m$ and $\sum_j j\theta_j = 3m$, so

$$\theta_4 = 3\theta_0 + 2\theta_1 + \theta_2.$$

By (6.34) and (6.36), $J_4^3\overline{h}$, the product of J_4^2 with the image of any coordinate of weight one, and the product of J_4 with the image of any coordinate of weight two are polynomials in l_3, b_3, d_3, M_3. Grouping each factor of weight 0, 1, and 2 with 3, 2, and 1 factors J, respectively, we see that $\overline{x_1}\cdots\overline{x_m}$ is a polynomial in l_3, b_3, d_3, M_3, homogeneous of degree m in l_3, b_3, d_3, M_3 since $\overline{x_1}\cdots\overline{x_m}$ has degree $3m$ in (i, a, e). Thus $W_{m,3m}$ is spanned by the products of m elements of $W_{1,3}$, and $T = \mathbb{C}[l_3, b_3, d_3, M_3]$ with $T_m = W_{m,3m}$.

Hence the substitution defines a surjection $\mathbb{C}[l, b, d, M] \to T$ of graded rings, and T is a domain as a subring of $\mathbb{C}[i, a, e]$. By Proposition 6.9(2) with $r = 3m$, so that $q = m$,

$$\dim T_m = \dim W_{m,3m} = 1 + \frac{3}{2}m(m + 1).$$

Thus $\operatorname{Proj} T$ is an integral surface, and the kernel of the surjection is a homogeneous prime ideal of height one. Since $\mathbb{C}[l,b,d,M]$ is a unique factorization domain, the kernel is generated by an irreducible form of some degree s, and the Hilbert polynomial of $\mathbb{C}[l,b,d,M]/(F)$ for a form F of degree s has leading term $\frac{s}{2}m^2$. Hence $s=3$. Then $\operatorname{Proj} T \subset \mathbb{P}^3$ is an integral cubic surface, and it is the closure of the image of the rational map $E \dashrightarrow \mathbb{P}^3$ defined by $(l_3 : b_3 : d_3 : M_3)$.

Step 2. In this step we show that $W_{1,3}$ has no fixed curve.

Suppose that an irreducible form $\nu \in \mathbb{C}[i,a,e]$ divides l_3, b_3, d_3, and M_3. By (6.34), ν^2 divides $J_4 f_2$, $J_4 c_2$, and $J_4 Q_2$. If ν^2 does not divide J_4, then ν divides f_2, c_2, and Q_2, since ν is a prime element of the factorial ring $\mathbb{C}[i,a,e]$, and this is impossible because f_2 is irreducible and does not divide c_2. Hence ν^2 divides J_4.

The three points p_ζ are not collinear, and f_2, c_2, Q_2 are linearly independent conics vanishing at them. Thus $(f_2 : c_2 : Q_2)$ is a basis of the net of conics through the p_ζ, and the rational map $E \dashrightarrow \mathbb{P}^2$ defined by $(f_2 : c_2 : Q_2)$ is a quadratic transformation. It restricts to an isomorphism from the complement of the three lines

$$L_\zeta := (i + \zeta^2 a + \zeta e = 0), \qquad \zeta^3 = 1,$$

onto an open subset of $\mathbb{P}^2$ ([Har77, Example V.4.2.3]). Here L_ζ is the line through the points $p_{\zeta'}$ with $\zeta' \neq \zeta$. By the definition of J_4,

$$J_4 = \Phi_0(f_2, c_2, Q_2), \qquad \Phi_0(z_1,z_2,z_3) := z_1 z_2 + \frac{1}{4} z_3^2 - \Big(v z_3 - \lambda z_1 - \mu z_2 + \frac{1}{2} z_3\Big)^2.$$

The quadratic form $z_1 z_2 + \frac{1}{4} z_3^2$ has rank three, so Φ_0 has rank at least two and the conic $(\Phi_0 = 0)$ is reduced. Hence the divisor B_4 of J_4 is reduced outside $L_1 \cup L_\omega \cup L_{\omega^2}$, where ω is a primitive cube root of unity, and ν defines one of the lines L_ζ.

Let $\chi_\zeta := i^2 + \zeta^2 i a + \zeta a^2$ and $\alpha_\zeta := \zeta^2 \lambda - v - \zeta \mu$. Substituting $e = -\zeta^2 i - \zeta a$, a direct calculation shows that the restrictions to L_ζ, as forms in (i,a), are

$$f_2|_{L_\zeta} = -\zeta^2 \chi_\zeta, \quad c_2|_{L_\zeta} = \zeta \chi_\zeta, \quad Q_2|_{L_\zeta} = -\chi_\zeta, \quad G_2|_{L_\zeta} = \alpha_\zeta \chi_\zeta, \quad b_3|_{L_\zeta} = -(\alpha_\zeta a + \zeta i)\chi_\zeta.$$

Since $\chi_\zeta \neq 0$ and $\alpha_\zeta a + \zeta i \neq 0$, $b_3|_{L_\zeta} \neq 0$, so ν does not define L_ζ, a contradiction. Thus $W_{1,3}$ has no fixed curve.

Step 3. In this step we prove (1).

Since $f_2(p_\zeta) = c_2(p_\zeta) = Q_2(p_\zeta) = 0$, the forms G_2, J_4, l_3, b_3, d_3, and M_3 vanish at p_ζ. In the chart $i = 1$ with affine coordinates (a,e), the linear parts of f_2 and c_2 at p_ζ are

$$f_\zeta^{(1)} := (e - \zeta^2) - 2\zeta(a - \zeta), \qquad c_\zeta^{(1)} := 2\zeta^2(e - \zeta^2) - (a - \zeta),$$

which are linearly independent, and the linear part of Q_2 is $-\zeta f_\zeta^{(1)} + \zeta^2 c_\zeta^{(1)}$. Let $\lambda_\zeta := v\zeta + \lambda$ and $\mu_\zeta := u - \mu\zeta$, so that

$$t_\zeta = -\lambda_\zeta f_\zeta^{(1)} + \zeta^2 \mu_\zeta c_\zeta^{(1)}.$$

By Lemma 6.7(2), condition (C) holds, that is, $(\mu_\zeta, \lambda_\zeta) \neq (0,0)$, so $t_\zeta \neq 0$. Using $u - v = 1$, a direct calculation shows that the linear parts of l_3, b_3, d_3, and M_3 at p_ζ are

$$-t_\zeta, \qquad -\zeta t_\zeta, \qquad \zeta^2 t_\zeta, \qquad -t_\zeta.$$

Thus p_ζ is a base point of $W_{1,3}$ of multiplicity one. After the blow-up of p_ζ, the pullback of any member of $W_{1,3}$, with the exceptional curve removed once, passes through the point p'_ζ of the exceptional curve corresponding to the direction $t_\zeta = 0$. Hence p'_ζ is a base point of multiplicity at least one.

Let $\rho' : E' \to E$ be a composite of point blow-ups resolving the base points of $W_{1,3}$, let $\beta_1, \dots, \beta_\ell$ be the multiplicities of the base points, proper or infinitely near, and let $\epsilon_1, \dots, \epsilon_\ell$ be the total transforms of the exceptional curves. Since ρ' resolves the base points, the moving part of $\rho'^* W_{1,3}$ is base point free, and by **Step 2** its class is $3\rho'^* \mathcal{O}_E(1) - \sum_j \beta_j \epsilon_j$. By **Step 1**

it defines a generically finite morphism $E' \to \operatorname{Proj} T$. By [Har77, Proposition V.3.2] and the projection formula [Har77, Appendix A, §1, A4],

$$9 - \sum_j \beta_j^2 = \Big(3\rho'^*\mathcal{O}_E(1) - \sum_j \beta_j \epsilon_j\Big)^2 = \deg(E'/\operatorname{Proj} T) \cdot \deg \operatorname{Proj} T = 3\deg(E'/\operatorname{Proj} T).$$

The six base points p_ζ and p'_ζ contribute at least 6 to $\sum_j \beta_j^2$, and $\deg(E'/\operatorname{Proj} T) \geq 1$. Hence $\deg(E'/\operatorname{Proj} T) = 1$ and $\sum_j \beta_j^2 = 6$. Thus the base points of $W_{1,3}$ are exactly the p_ζ and the p'_ζ, all of multiplicity one, and $E' \to \operatorname{Proj} T$ is birational. This proves (1).

Step 4. In this step we prove (2) except for the isomorphism over a neighborhood of the image of $\rho^{-1}(E\backslash B_4)$.

By **Step 3** we may take $E' = \widetilde{E}$ and $\rho' = \rho$, and the moving part of $\rho^* W_{1,3}$ is $3\eta - \sum_\zeta (e_\zeta + e'_\zeta)$. By [Har77, Proposition V.3.3] applied to each blow-up,

$$K_{\widetilde{E}} = \rho^* K_E + \sum_\zeta (e_\zeta + e'_\zeta) = -3\eta + \sum_\zeta (e_\zeta + e'_\zeta).$$

Thus $-K_{\widetilde{E}}$ is base point free. By [Har77, Proposition V.3.2], $(\eta^2) = 1$, $(e_\zeta^2) = (e_\zeta'^2) = -1$, and the other products of η, e_ζ, e'_ζ are zero, so

$$(-K_{\widetilde{E}})^2 = 9 - 6 = 3.$$

In particular $-K_{\widetilde{E}}$ is nef and big, and $\widetilde{E}$ is a weak del Pezzo surface of degree three.

As divisors, the total transform e_ζ equals $R_\zeta + e'_\zeta$, so $e_\zeta + e'_\zeta = R_\zeta + 2e'_\zeta$. Let s_0 be a section of $\mathcal{O}_{\widetilde{E}}(\sum_\zeta (e_\zeta + e'_\zeta))$ with divisor $\sum_\zeta (R_\zeta + 2e'_\zeta)$. By (1), $\rho^* s \in s_0 \cdot H^0(\widetilde{E}, -K_{\widetilde{E}})$ for any $s \in W_{1,3}$. Since $W_{m,3m}$ is spanned by the products of m elements of $W_{1,3}$ (**Step 1**),

$$\rho^* W_{m,3m} \subset s_0^m \cdot H^0(\widetilde{E}, -mK_{\widetilde{E}}).$$

Since $-mK_{\widetilde{E}} - K_{\widetilde{E}} = (m+1)(-K_{\widetilde{E}})$ is nef and big, [KM98, Theorem 2.64] implies $H^i(\widetilde{E}, (m+1)K_{\widetilde{E}}) = 0$ for $i < 2$, and by Serre duality $H^j(\widetilde{E}, -mK_{\widetilde{E}}) = 0$ for $j > 0$. Since $\widetilde{E}$ is rational, the Riemann–Roch theorem [Har77, Theorem V.1.6] implies

$$h^0(\widetilde{E}, -mK_{\widetilde{E}}) = \frac{1}{2}(-mK_{\widetilde{E}}) \cdot (-(m+1)K_{\widetilde{E}}) + 1 = 1 + \frac{3}{2}m(m+1) = \dim W_{m,3m}.$$

Since ρ^* is injective, $\rho^* W_{m,3m} = s_0^m \cdot H^0(\widetilde{E}, -mK_{\widetilde{E}})$, which is the description of $\rho^* W_{m,3m}$ in (2).

Thus multiplication by s_0^{-m} in degree m is an isomorphism from T onto the anti-canonical ring $\bigoplus_m H^0(\widetilde{E}, -mK_{\widetilde{E}})$, and the morphism $\phi : \widetilde{E} \to \mathbb{P}^3$ defined by $|-K_{\widetilde{E}}|$ is the morphism $E' \to \operatorname{Proj} T$ of **Step 3**. Hence ϕ is birational onto $\operatorname{Proj} T$, and $\operatorname{Proj} T$ is the anti-canonical model of the weak del Pezzo surface $\widetilde{E}$. In particular $\operatorname{Proj} T$ is a normal cubic surface and ϕ contracts finitely many curves ([Dol12, Section 8.3]).

Step 5. In this step we show that ϕ is an isomorphism over a neighborhood of $\phi(\rho^{-1}(E\backslash B_4))$.

Since $J_4(p_\zeta) = 0$ (**Step 3**), $p_\zeta \in B_4$, so ρ is an isomorphism over $E\backslash B_4$, and no point of $E\backslash B_4$ is a base point of $W_{1,3}$. Let $\psi : \mathbb{P}^3 \dashrightarrow E$ be the rational map defined by the cubic forms

$$(\Lambda l - \Phi b : \Phi d + \Lambda b : -\Phi M - \Lambda d).$$

For a closed point $z' \in E\backslash B_4$ with representative $\hat{z}' \in \mathbb{C}^3\backslash\{0\}$, the point $\phi(\rho^{-1}(z'))$ has the representative $(l_3, b_3, d_3, M_3)(\hat{z}') \neq 0$, and by (6.36) the three cubic forms take at $(l_3, b_3, d_3, M_3)(\hat{z}')$ the values $J_4(\hat{z}')^2 \hat{z}' \neq 0$. Thus ψ is a morphism near $\phi(\rho^{-1}(z'))$, and $\psi(\phi(\rho^{-1}(z'))) = z'$.

Fix a closed point $z \in E\backslash B_4$ and let $\tilde{z} := \rho^{-1}(z)$. Let $U \subset \mathbb{P}^3$ be an open neighborhood of $\phi(\tilde{z})$ on which ψ is a morphism. Then $\psi \circ \phi$ and ρ coincide on the dense open subset $\phi^{-1}(U) \cap \rho^{-1}(E\backslash B_4)$ of $\phi^{-1}(U)$, hence on $\phi^{-1}(U)$. If $\phi^{-1}(\phi(\tilde{z}))$ contained a curve Γ, then $\rho(\Gamma) = \psi(\phi(\Gamma)) = \{z\}$, which contradicts $\rho^{-1}(z) = \{\tilde{z}\}$. Thus $\phi^{-1}(\phi(\tilde{z}))$ is finite. Since ϕ is birational and projective and $\operatorname{Proj} T$ is normal, the fibers of ϕ are connected by Zariski's Main

Theorem [Har77, Corollary III.11.4], so ϕ is an isomorphism over a neighborhood of $\phi(\tilde{z})$. Since $z \in E \backslash B_4$ is arbitrary, this proves (2).

Step 6. In this step we prove (3).

Fix ζ and (t, y). Since $i(p_\zeta) \neq 0$, for a section s of $\mathcal{O}_E(r)$ we evaluate w_ζ on s/i^r. A function vanishing at p_ζ has $w_\zeta \geq 1$. A function vanishing at p_ζ whose linear part is a multiple of t_ζ has $w_\zeta \geq 2$, since the expansion of the function in (t, y) is a multiple of t plus terms of order at least two. By **Step 3** we obtain the following bounds for the images of the Kapustka coordinates. The image $\overline{h} = 1$ has $w_\zeta(\overline{h}) = 0$. The images $i, a, e, -i$ of the coordinates of weight one have $w_\zeta \geq 0$. The images $f_2, c_2, G_2, -G_2 - Q_2$ of the coordinates of weight two vanish at p_ζ, so they have $w_\zeta \geq 1$. The images l_3, b_3, d_3, M_3 have $w_\zeta \geq 2$. Since $f_\zeta^{(1)}$ and $c_\zeta^{(1)}$ are linearly independent, one of them is not a multiple of t_ζ, so $w_\zeta(f_2) = 1$ or $w_\zeta(c_2) = 1$. By (6.34), $w_\zeta(J_4 f_2) = w_\zeta(\overline{\Phi}) \geq 4$ and $w_\zeta(J_4 c_2) = w_\zeta(\overline{\Psi}) \geq 4$, hence $w_\zeta(J_4) \geq 3$.

Thus $w_\zeta(\overline{x}) \geq \mathrm{wt}(x) - 1$ for any Kapustka coordinate x of weight $\mathrm{wt}(x) \geq 1$, while $\overline{h} = 1$. For a monomial $x_1 \cdots x_m$ of degree m and weight r,

$$w_\zeta(\overline{x_1} \cdots \overline{x_m}) \geq \sum_{j:\mathrm{wt}(x_j)\geq 1} (\mathrm{wt}(x_j) - 1) = r - \#\{j : \mathrm{wt}(x_j) \geq 1\} \geq r - m.$$

Since $W_{m,r}$ is spanned by the images of such monomials (Proposition 6.9(1)), $w_\zeta(s) \geq r - m$ for any $s \in W_{m,r}$. This proves (3). □

We next describe the fixed parts of the spaces $\rho^* W_{m,r}$ on $\widetilde{E}$, and we use them to bound $S(W; w)$ for the prime divisors w over E whose centers are not contained in the image of the curves on $\widetilde{E}$ with negative self-intersection.

Construction-Proposition 6.11. Notation and conditions as in Construction-Proposition 6.10. Then η and $2\eta - \sum_\zeta e_\zeta$ are base point free, and for all integers $m, r \geq 0$ the space $\rho^* W_{m,r}$ consists of the sections of a complete base point free linear system, multiplied by a fixed section with divisor $\Theta_{m,r}$, as follows.

(1) If $0 \leq r \leq m$, then $\Theta_{m,r} = 0$ and the linear system is $|r\eta|$.
(2) If $m < r \leq 3m$, then $\Theta_{m,r} = \sum_\zeta \left(\lceil \frac{r-m}{2} \rceil R_\zeta + (r-m) e'_\zeta \right)$ and the linear system is $\left| r\eta - \sum_\zeta \left(\lceil \frac{r-m}{2} \rceil e_\zeta + \lfloor \frac{r-m}{2} \rfloor e'_\zeta \right) \right|$.
(3) If $3m < r \leq 4m$, then $\Theta_{m,r} = (r-3m)\rho^* B_4 + (4m-r) \sum_\zeta (R_\zeta + 2e'_\zeta)$ and the linear system is $|-(4m-r)K_{\widetilde{E}}|$.
(4) If $r > 4m$, then $W_{m,r} = 0$.

For a prime divisor w over $\widetilde{E}$, the limit $S_{\mathrm{mov}}(w)$ of the sums of the jumping numbers of w on the complete linear systems above, divided by mN_m, exists, and

$$S(W; w) = S_{\mathrm{mov}}(w) + \frac{41}{72} \sum_\zeta \mathrm{mult}_w R_\zeta + \frac{41}{36} \sum_\zeta \mathrm{mult}_w e'_\zeta + \frac{1}{24} \mathrm{mult}_w \rho^* B_4.$$

Proof. We number the cube roots of unity as $\zeta_1, \zeta_2, \zeta_3$, and we write p_j, e_j, e'_j, R_j for p_{ζ_j}, e_{ζ_j}, e'_{ζ_j}, R_{ζ_j}. For a non-zero section σ of a line bundle on E or on $\widetilde{E}$ and a prime divisor w over $\widetilde{E}$, let $w(\sigma) := \mathrm{mult}_w \mathrm{div}(\sigma)$. Then $w(\rho^*\sigma) = w(\sigma)$. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that η and $2\eta - \sum_j e_j$ are base point free, and we prove (1) and (4).

Since $\eta = \rho^* \mathcal{O}_E(1)$, η is base point free. The points p_1, p_2, p_3 are not collinear, because

$$\det \begin{pmatrix} 1 & \zeta_1 & \zeta_1^2 \\ 1 & \zeta_2 & \zeta_2^2 \\ 1 & \zeta_3 & \zeta_3^2 \end{pmatrix} = (\zeta_2 - \zeta_1)(\zeta_3 - \zeta_1)(\zeta_3 - \zeta_2) \neq 0.$$

Let $\rho_0 : S_0 \to E$ be the blow-up of p_1, p_2, p_3 with exceptional curves e^0_ζ, write $e^0_j := e^0_{\zeta_j}$, and let $\rho_1 : \widetilde{E} \to S_0$ be the morphism with $\rho = \rho_0 \circ \rho_1$. Then $\rho_1^* e^0_j = e_j$, so

$$2\eta - \sum_j e_j = \rho_1^* D^0, \qquad D^0 := 2\rho_0^* \mathcal{O}_E(1) - e^0_1 - e^0_2 - e^0_3,$$

and it suffices to prove that $|D^0|$ is base point free. Let $\{j, j', j''\} = \{1, 2, 3\}$, and let $\ell^0_{jj'} \subset S_0$ be the strict transform of the line through p_j and $p_{j'}$. Then $\ell^0_{jj'} \sim \rho_0^* \mathcal{O}_E(1) - e^0_j - e^0_{j'}$. The linear system $|\rho_0^* \mathcal{O}_E(1) - e^0_{j''}|$ is the pullback of the base point free pencil of lines through $p_{j''}$ on the blow-up of $p_{j''}$ [Har77, Example V.2.11.5], so it is base point free. Since

$$\ell^0_{jj'} + |\rho_0^* \mathcal{O}_E(1) - e^0_{j''}| \subset |D^0|,$$

the base locus of $|D^0|$ is contained in $\ell^0_{jj'}$. The curves ℓ^0_{12}, ℓ^0_{13}, and ℓ^0_{23} are distinct and irreducible, and

$$(\ell^0_{jj'} \cdot \ell^0_{jj''}) = 1 - 1 = 0,$$

so they are pairwise disjoint. Thus $|D^0|$ is base point free, and so is $2\eta - \sum_j e_j$.

If $0 \le r \le m$, then $\gamma_{r-m-1} = 0$, so $\dim W_{m,r} = \binom{r+2}{2}$ by Proposition 6.9(2), that is, $W_{m,r} = H^0(E, \mathcal{O}_E(r))$. Since $\rho_* \mathcal{O}_{\widetilde{E}} = \mathcal{O}_E$, we have $\rho^* W_{m,r} = H^0(\widetilde{E}, r\eta)$. This proves (1). Statement (4) is the vanishing $W_{m,r} = 0$ for $r > 4m$ in Proposition 6.9(2).

Step 2. In this step, we fix integers $m < r \le 3m$ and show that the divisor of any non-zero element of $\rho^* W_{m,r}$ contains $\Theta_{m,r}$.

Let $s := r - m$, $s_+ := \lceil s/2 \rceil$, and $s_- := \lfloor s/2 \rfloor$, so that $\Theta_{m,r} = \sum_j (s_+ R_j + s e'_j)$. Fix j, let (t, y) be the regular local parameters of E at p_j of Construction-Proposition 6.10(3), and let w_{ζ_j} be the monomial valuation with weights $(2, 1)$ on (t, y), that is, the valuation w_ζ of Construction-Proposition 6.10(3) at $\zeta = \zeta_j$. By Construction-Proposition 6.10(1), the base point of $W_{1,3}$ infinitely near to p_j is the point of the exceptional curve of the blow-up of p_j in the direction $t = 0$. In the chart of that blow-up with coordinates (t_1, y) and $t = t_1 y$, the infinitely near base point is the origin, and e'_j is the exceptional curve of the blow-up of the origin. The substitution $t = t_1 y$ sends the monomial $t^\alpha y^\beta$ to $t_1^\alpha y^{\alpha+\beta}$, and distinct monomials go to distinct monomials. Hence, for a non-zero function ψ regular near p_j, the multiplicity $\operatorname{mult}_{e'_j} \rho^* \operatorname{div}(\psi)$ is the minimum of $2\alpha + \beta$ over the monomials $t^\alpha y^\beta$ with non-zero coefficient in the expansion of ψ in $\mathbb{C}[[t, y]]$, that is,

$$\operatorname{mult}_{e'_j} \rho^* \operatorname{div}(\psi) = w_{\zeta_j}(\psi).$$

Moreover, $\operatorname{mult}_{R_j} \rho^* \operatorname{div}(\psi) = \operatorname{mult}_{p_j} \psi$, because R_j is the strict transform of the exceptional curve of the blow-up of p_j.

Let $\sigma \in W_{m,r}$ be non-zero, let ψ be a local equation of $\operatorname{div}(\sigma)$ at p_j, and let $\nu := \operatorname{mult}_{p_j} \psi$. By Construction-Proposition 6.10(3), $w_{\zeta_j}(\psi) \ge s$. The expansion of ψ has a monomial $t^\alpha y^\beta$ with non-zero coefficient and $\alpha + \beta = \nu$, and $2\alpha + \beta \le 2\nu$. Thus

$$s \le w_{\zeta_j}(\psi) \le 2\nu,$$

so $\nu \ge s_+$, as ν is an integer. Hence $\operatorname{div}(\rho^* \sigma) \ge s_+ R_j + s e'_j$ near $\rho^{-1}(p_j)$. The sets $\rho^{-1}(p_1)$, $\rho^{-1}(p_2)$, and $\rho^{-1}(p_3)$ are pairwise disjoint, so

$$\operatorname{div}(\rho^* \sigma) \ge \Theta_{m,r}.$$

Since $R_j \sim e_j - e'_j$ and $s_+ + s_- = s$, we have $\Theta_{m,r} \sim \sum_j (s_+ e_j + s_- e'_j)$. Let

$$M_{m,r} := r\eta - \sum_j (s_+ e_j + s_- e'_j),$$

and let $\sigma_{m,r}$ be a section of $\mathcal{O}_{\widetilde{E}}(\Theta_{m,r})$ with divisor $\Theta_{m,r}$. Then division by $\sigma_{m,r}$ defines an injective linear map

$$\rho^*W_{m,r} \to H^0(\widetilde{E}, M_{m,r}).$$

Step 3. In this step, we show that $|M_{m,r}|$ is base point free and $h^0(\widetilde{E}, M_{m,r}) = \dim W_{m,r}$ for $m < r \leq 3m$, which proves (2).

If $s = 2s_-$, then $s_+ = s_-$ and

$$M_{m,r} = (r - 3s_-)\eta + s_-(-K_{\widetilde{E}}), \qquad r - 3s_- = \frac{3m - r}{2} \geq 0.$$

If $s = 2s_- + 1$, then $s_+ = s_- + 1$ and

$$M_{m,r} = (r - 3s_- - 2)\eta + s_-(-K_{\widetilde{E}}) + \Big(2\eta - \sum_j e_j\Big), \qquad r - 3s_- - 2 = \frac{3m - r - 1}{2} \geq 0,$$

where the last inequality holds because $r - 3m = s - 2m$ is odd, so $r \leq 3m - 1$. Here $-K_{\widetilde{E}} = 3\eta - \sum_j (e_j + e'_j)$ is base point free and $(K_{\widetilde{E}}^2) = 3$ by Construction-Proposition 6.10(2). By **Step 1**, $M_{m,r}$ is a sum of base point free divisors with non-negative integer coefficients, so $|M_{m,r}|$ is base point free. In particular $M_{m,r} - K_{\widetilde{E}}$ is nef and big, and $H^i(\widetilde{E}, M_{m,r}) = 0$ for $i > 0$ by the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64]. The surface $\widetilde{E}$ is rational, so the Riemann–Roch theorem [Har77, Theorem V.1.6] implies

$$h^0(\widetilde{E}, M_{m,r}) = 1 + \frac{1}{2}\big(M_{m,r} \cdot (M_{m,r} - K_{\widetilde{E}})\big).$$

The classes $\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3$ are pairwise orthogonal with $(\eta^2) = 1$ and $(e_j^2) = (e_j'^2) = -1$ [Har77, Proposition V.3.2]. Since

$$M_{m,r} - K_{\widetilde{E}} = (r + 3)\eta - \sum_j \big((s_+ + 1)e_j + (s_- + 1)e'_j\big),$$

we obtain

$$h^0(\widetilde{E}, M_{m,r}) = 1 + \frac{r(r+3)}{2} - \frac{3}{2}\big(s_+(s_+ + 1) + s_-(s_- + 1)\big) = \binom{r+2}{2} - \frac{3}{2}\big(s_+(s_+ + 1) + s_-(s_- + 1)\big).$$

Now $\frac{1}{2}\big(s_+(s_+ + 1) + s_-(s_- + 1)\big)$ equals $s_-(s_- + 1)$ if $s = 2s_-$ and $(s_- + 1)^2$ if $s = 2s_- + 1$, which is $\gamma_{s-1} = \gamma_{r-m-1}$ in both cases. Therefore $h^0(\widetilde{E}, M_{m,r}) = \dim W_{m,r}$ by Proposition 6.9(2), and the injective map of **Step 2** is an isomorphism. Thus

$$\rho^*W_{m,r} = \sigma_{m,r} \cdot H^0(\widetilde{E}, M_{m,r}),$$

which is (2).

Step 4. In this step, we prove (3).

Let $3m < r \leq 4m$ and $q := 4m - r$. By Proposition 6.9(2), $W_{m,r} = J_4^{r-3m} W_{q,3q}$. By Construction-Proposition 6.10(2), $\rho^*W_{q,3q} = \sigma' \cdot H^0(\widetilde{E}, -qK_{\widetilde{E}})$ for a section σ' with divisor $q\sum_j (R_j + 2e'_j)$. Since $\operatorname{div}(\rho^* J_4^{r-3m}) = (r - 3m)\rho^* B_4$, we obtain

$$\rho^*W_{m,r} = \sigma_{m,r} \cdot H^0(\widetilde{E}, -qK_{\widetilde{E}}), \qquad \sigma_{m,r} := \rho^* J_4^{r-3m} \cdot \sigma',$$

and $\operatorname{div}(\sigma_{m,r}) = (r - 3m)\rho^* B_4 + q\sum_j (R_j + 2e'_j)$. This is (3). We set $M_{m,r} := -qK_{\widetilde{E}}$ for $3m < r \leq 4m$, and $M_{m,r} := r\eta$ and $\sigma_{m,r} := 1$ for $0 \leq r \leq m$.

Step 5. In this step, we prove the formula for $S(W; w)$.

For a non-zero space V of sections of a line bundle on $\widetilde{E}$, let

$$\mathcal{T}_w(V) := \sum_{k \geq 1} \dim\{\sigma \in V \mid w(\sigma) \geq k\}$$

be the sum of the jumping numbers of w on V. The spaces $W_{m,r}$ are the graded pieces of the filtration of $H^0(X, mH)$ by the order of vanishing along E, so $N_m = \sum_r \dim W_{m,r}$. By [AZ22, Lemma 2.9], since W has bounded support and contains an ample series [AZ22, Examples 2.5 and 2.6], the limit $S(W; w) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_r \mathcal{T}_w(W_{m,r})$ exists.

If $V = \sigma_0 \cdot V'$ for a fixed section σ_0, then

$$\{\sigma \in V \mid w(\sigma) \geq k\} = \sigma_0 \cdot \{\tau \in V' \mid w(\tau) \geq k - w(\sigma_0)\},$$

so

$$\mathcal{T}_w(\sigma_0 \cdot V') = \mathcal{T}_w(V') + w(\sigma_0) \dim V'.$$

By **Steps 1–4**, $\rho^* W_{m,r} = \sigma_{m,r} \cdot H^0(\widetilde{E}, M_{m,r})$ with $\operatorname{div}(\sigma_{m,r}) = \Theta_{m,r}$ for $0 \leq r \leq 4m$. Since $w(\rho^*\sigma) = w(\sigma)$, we obtain

$$\sum_r \mathcal{T}_w(W_{m,r}) = \sum_r \mathcal{T}_w(H^0(\widetilde{E}, M_{m,r})) + c_m^{(1)} \sum_j \operatorname{mult}_w R_j + c_m^{(2)} \sum_j \operatorname{mult}_w e'_j + c_m^{(3)} \operatorname{mult}_w \rho^* B_4,$$

where

$$\begin{aligned} c_m^{(1)} &:= \sum_{m<r\leq 3m} \left\lceil \frac{r-m}{2} \right\rceil \dim W_{m,r} + \sum_{3m<r\leq 4m} (4m-r) \dim W_{m,r}, \\ c_m^{(2)} &:= \sum_{m<r\leq 3m} (r-m) \dim W_{m,r} + \sum_{3m<r\leq 4m} 2(4m-r) \dim W_{m,r}, \\ c_m^{(3)} &:= \sum_{3m<r\leq 4m} (r-3m) \dim W_{m,r}. \end{aligned}$$

With φ as in Proposition 6.9(2), $N_m = m^3 \int_0^4 \varphi(t)dt + O(m^2) = 3m^3 + O(m^2)$, and $c_m^{(i)}/(mN_m)$ is a Riemann sum which converges, with error $O(1/m)$, to the integral in the row of Table 3 with value $\frac{41}{72}$, $\frac{41}{36}$, or $\frac{1}{24}$ for $i = 1, 2, 3$. A direct calculation shows the values in Table 3. Thus

$$\lim_{m\to+\infty} \frac{c_m^{(1)}}{mN_m} = \frac{41}{72}, \qquad \lim_{m\to+\infty} \frac{c_m^{(2)}}{mN_m} = \frac{41}{36}, \qquad \lim_{m\to+\infty} \frac{c_m^{(3)}}{mN_m} = \frac{1}{24}.$$

Since $S(W; w)$ exists, the limit

$$S_{\mathrm{mov}}(w) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_r \mathcal{T}_w(H^0(\widetilde{E}, M_{m,r}))$$

exists, and the formula for $S(W; w)$ holds. □

TABLE 3. Integrals of $f\varphi$, with φ as in Proposition 6.9(2).

f on $[0,1]$	f on $[1,3]$	f on $[3,4]$	$\int_0^1 f\varphi dt$	$\int_1^3 f\varphi dt$	$\int_3^4 f\varphi dt$	c	$c\int_0^4 f\varphi dt$
1	1	1	$\frac{1}{6}$	$\frac{7}{3}$	$\frac{1}{2}$	1	3
0	$\frac{t-1}{2}$	$4-t$	0	$\frac{4}{3}$	$\frac{3}{8}$	$\frac{1}{3}$	$\frac{41}{72}$
0	$t-1$	$2(4-t)$	0	$\frac{8}{3}$	$\frac{3}{4}$	$\frac{1}{3}$	$\frac{41}{36}$
0	0	$t-3$	0	0	$\frac{1}{8}$	$\frac{1}{3}$	$\frac{1}{24}$
t	$\frac{3}{2}$	$\frac{3}{2}(4-t)$	$\frac{1}{8}$	$\frac{7}{2}$	$\frac{9}{16}$	$\frac{2}{9}$	$\frac{67}{72}$
t	$\frac{t+1}{2}$	$2(4-t)$	$\frac{1}{8}$	$\frac{11}{3}$	$\frac{3}{4}$	$\frac{1}{6}$	$\frac{109}{144}$

The next lemma bounds $S(W; w)$ away from the curves on $\widetilde{E}$ with negative self-intersection.

Lemma 6.12. *Notation and conditions as in Construction-Proposition 6.10, and let $C^- \subset \widetilde{E}$ be the union of the irreducible curves with negative self-intersection. Let ρ^*W be the series on $\widetilde{E}$ with pieces $\rho^*W_{m,r} := \{\rho^*s : s \in W_{m,r}\} \subset H^0(\widetilde{E}, r\eta)$, and for a smooth irreducible curve $C \subset \widetilde{E}$ let $(\rho^*W)^C$ be the refinement of ρ^*W by C as in Definition 2.8. Then C^- is a union of finitely many curves, and*

$$S(W;w) \le \frac{67}{72}A_E(w)$$

for every prime divisor w over E whose center is not contained in $\rho(C^-)$. More precisely, for every closed point $z \in \widetilde{E}\backslash C^-$, one of the three pencils of lines through the three points p_ζ has a smooth irreducible member whose strict transform $C \ni z$ is not contained in ρ^*B_4 and satisfies $S(\rho^*W;\mathrm{ord}_C) \le 67/72$ and $S((\rho^*W)^C;z) \le 133/144$.

Proof. We number the cube roots of unity as $\zeta_1, \zeta_2, \zeta_3$, and we write p_j, e_j, e'_j, R_j for p_{ζ_j}, e_{ζ_j}, e'_{ζ_j}, R_{ζ_j}. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that C^- is a finite union of smooth rational curves, and that C^- contains the exceptional curves R_j and e'_j of ρ.

By Construction-Proposition 6.10(2), $-K_{\widetilde{E}}$ is base point free and $(K^2_{\widetilde{E}}) = 3$, so $-K_{\widetilde{E}}$ is nef and big. Let $D \subset \widetilde{E}$ be an irreducible curve with $(D^2) < 0$. Since $2p_a(D) - 2 = (D \cdot (D + K_{\widetilde{E}}))$ [Har77, Exercise V.1.3(a)] and $(-K_{\widetilde{E}} \cdot D) \ge 0$, we have

$$-2 \le 2p_a(D) - 2 = (D^2) - (-K_{\widetilde{E}} \cdot D) \le (D^2) < 0.$$

Thus $p_a(D) = 0$, so D is a smooth rational curve, and

$$\big((D^2), (-K_{\widetilde{E}} \cdot D)\big) \in \{(-1,1), (-2,0)\}$$

(see also [Dol12, Section 8.2]). In the basis $\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3$ of $\mathrm{Pic}(\widetilde{E})$, the intersection form is diagonal with entries $1, -1, \dots, -1$ [Har77, Proposition V.3.2]. Since $(K^2_{\widetilde{E}}) = 3 > 0$, the intersection form is negative definite on the orthogonal complement $K^\perp_{\widetilde{E}}$ of $K_{\widetilde{E}}$ in $\mathrm{Pic}(\widetilde{E}) \otimes \mathbb{R}$. Write $D = \frac{1}{3}(-K_{\widetilde{E}} \cdot D)(-K_{\widetilde{E}}) + D_0$ with $D_0 \in K^\perp_{\widetilde{E}}$. Then $3D_0 \in \mathrm{Pic}(\widetilde{E})$ and

$$(D_0^2) = (D^2) - \frac{1}{3}(-K_{\widetilde{E}} \cdot D)^2 \in \Big\{-\frac{4}{3}, -2\Big\}.$$

A negative definite lattice has finitely many vectors of a given square, so there exist finitely many classes of irreducible curves with negative self-intersection. Two distinct irreducible curves $D \ne D'$ with $D \sim D'$ and $(D^2) < 0$ would satisfy $0 \le (D \cdot D') = (D^2) < 0$, which is impossible. Thus C^- is a union of finitely many curves. Since $(R_j^2) = ((e_j - e'_j)^2) = -2$ and $(e_j'^2) = -1$, the curves R_j and e'_j are contained in C^-. In particular ρ restricts to an isomorphism $\widetilde{E}\backslash C^- \to E\backslash\rho(C^-)$.

Step 2. In this step, we fix a closed point $z \in \widetilde{E}\backslash C^-$ and construct the curve C.

Let $x := \rho(z)$. By **Step 1**, $x \notin \{p_1, p_2, p_3\}$. For each j, let $\ell_j \subset E$ be the line through p_j and x, and let $\tilde{\ell}_j \subset \widetilde{E}$ be the strict transform of ℓ_j. Then $z \in \tilde{\ell}_j$. If ℓ_j passes through one of the six base points of $W_{1,3}$ other than p_j, that is, through some $p_{j'}$ with $j' \ne j$ or through the base point infinitely near to p_j, then

$$(\tilde{\ell}_j^2) \le 1 - 2 = -1$$

[Har77, Propositions V.3.2 and V.3.6], so $\tilde{\ell}_j \subset C^-$, which contradicts $z \notin C^-$. Hence $\rho^*\ell_j = \tilde{\ell}_j + e_j$, and

$$\tilde{\ell}_j \sim \eta - e_j, \qquad (\tilde{\ell}_j^2) = 0, \qquad (\eta \cdot \tilde{\ell}_j) = 1, \qquad (-K_{\widetilde{E}} \cdot \tilde{\ell}_j) = 2.$$

The curve $\tilde{\ell}_j \cong \ell_j \cong \mathbb{P}^1$ is smooth, and it is a member of the base point free pencil $|\eta - e_j|$, the pullback of the pencil of lines through p_j. In particular $\ell_j \ne \ell_{j'}$ for $j \ne j'$, since $p_{j'} \notin \ell_j$.

We claim that $\ell_j \not\subset B_4$ for some j. Since G_2 is a linear combination of f_2, c_2, and Q_2, the quartic $J_4 = f_2c_2 - G_2^2 - G_2Q_2$ is a quadratic form in f_2, c_2, and Q_2. At $p_j = (1 : \zeta_j : \zeta_j^2)$ we have

$$f_2 = \zeta_j^2 - \zeta_j^2 = 0, \qquad c_2 = \zeta_j^4 - \zeta_j = 0, \qquad Q_2 = \zeta_j^3 - 1 = 0,$$

so $\mathrm{mult}_{p_j} B_4 \geq 2$. Suppose that $\ell_1+\ell_2+\ell_3 \leq B_4$. Then $B_4-\ell_1-\ell_2-\ell_3$ is a line ℓ_0. Since $p_j \in \ell_{j'}$ only for $j' = j$, we have $\mathrm{mult}_{p_j}(\ell_1 + \ell_2 + \ell_3) = 1$, so $p_j \in \ell_0$ for $j = 1, 2, 3$. This contradicts the fact that p_1, p_2, p_3 are not collinear, which holds because the Vandermonde determinant of $\zeta_1, \zeta_2, \zeta_3$ is non-zero. Thus the claim holds.

Fix j with $\ell_j \not\subset B_4$, and let $C := \widetilde{\ell}_j$. The components of ρ^*B_4 are the strict transforms of the components of B_4 and exceptional curves of ρ, so $C \not\subset \rho^*B_4$. Since $(\rho^*B_4 \cdot C) = (4\eta \cdot (\eta - e_j)) = 4$, the local intersection number satisfies

$$(\rho^*B_4 \cdot C)_z \leq 4$$

[Har77, Proposition V.1.4]. By Construction-Proposition 6.11, any component of $\Theta_{m,r}$ is one of the curves $R_{j'}$ and $e'_{j'}$ or a component of ρ^*B_4. Since $(C^2) = 0$, the curve C is none of the curves $R_{j'}$ and $e'_{j'}$, and $C \not\subset \rho^*B_4$, so $C \not\subset \Theta_{m,r}$ for all m, r. Since $z \notin C^-$ and $R_{j'}, e'_{j'} \subset C^-$ (**Step 1**), any component of $\Theta_{m,r}$ through z is a component of ρ^*B_4.

Step 3. In this step, we introduce the moving classes P_t and the Zariski decompositions of $P_t - yC$.

By Construction-Proposition 6.11, for integers $m \geq 1$ and $0 \leq r \leq 4m$ we have $\rho^*W_{m,r} = \sigma_{m,r} \cdot H^0(\widetilde{E}, M_{m,r})$, where $\mathrm{div}(\sigma_{m,r}) = \Theta_{m,r}$, $|M_{m,r}|$ is base point free, and

$$M_{m,r} = \begin{cases} r\eta & 0 \leq r \leq m, \\ r\eta - \sum_{j'} \left(\lceil \frac{r-m}{2} \rceil e_{j'} + \lfloor \frac{r-m}{2} \rfloor e'_{j'}\right) & m < r \leq 3m, \\ -(4m-r)K_{\widetilde{E}} & 3m < r \leq 4m. \end{cases}$$

For $t \in [0, 4]$ let

$$P_t := \begin{cases} t\eta & 0 \leq t \leq 1, \\ t\eta - \frac{t-1}{2}\sum_{j'}(e_{j'} + e'_{j'}) = \frac{3-t}{2}\eta + \frac{t-1}{2}(-K_{\widetilde{E}}) & 1 \leq t \leq 3, \\ (4-t)(-K_{\widetilde{E}}) & 3 \leq t \leq 4. \end{cases}$$

Then P_t is nef, and with $s := r - m$, $mP_{r/m} = r\eta - \frac{s}{2}\sum_{j'}(e_{j'} + e'_{j'})$ for $m \leq r \leq 3m$. Thus

$$M_{m,r} - mP_{r/m} \in \left\{0, -\frac{1}{2}\sum_{j'} R_{j'}\right\}.$$

By **Step 2**, $(\eta \cdot C) = 1$, $(e_{j'} \cdot C) = \delta_{jj'}$, $(e'_{j'} \cdot C) = 0$, and $(-K_{\widetilde{E}} \cdot C) = 2$, so

$$\frac{1}{2}(P_t^2) = \varphi(t), \qquad d_0(t) := (P_t \cdot C) = \begin{cases} t & 0 \leq t \leq 1, \\ \frac{t+1}{2} & 1 \leq t \leq 3, \\ 2(4-t) & 3 \leq t \leq 4, \end{cases}$$

where φ is as in Proposition 6.9(2). In particular P_t is big for $0 < t < 4$.

For $0 < t < 4$, let $\tau_t := \sup\{y \geq 0 : P_t - yC \text{ is pseudo-effective}\}$. Since η and $-K_{\widetilde{E}}$ are nef, a pseudo-effective class $P_t - yC$ satisfies $(\eta \cdot (P_t - yC)) \geq 0$ and $(-K_{\widetilde{E}} \cdot (P_t - yC)) \geq 0$. Since $(\eta \cdot P_t) = t$ for $t \leq 1$, $(-K_{\widetilde{E}} \cdot P_t) = 3$ for $1 \leq t \leq 3$, and $(-K_{\widetilde{E}} \cdot P_t) = 3(4-t)$ for $t \geq 3$, we obtain

$$\tau_t \leq \bar{\tau}_t := \begin{cases} t & 0 < t \leq 1, \\ \frac{3}{2} & 1 \leq t \leq 3, \\ \frac{3}{2}(4-t) & 3 \leq t < 4. \end{cases}$$

For $0 \leq y < \tau_t$, the class $P_t - yC$ is big. Let

$$P_t - yC = P_{t,y} + N_{t,y}$$

be the Zariski decomposition [Laz04a, Theorem 2.3.19], and let $d(t,y) := (P_{t,y} \cdot C)$. The intersection matrix of the components of $N_{t,y}$ is negative definite, so any component of $N_{t,y}$ has negative self-intersection, and $\operatorname{Supp} N_{t,y} \subset C^-$. The curve C is the strict transform of a line through p_j with $(C^2) = 0$, so $C \not\subset C^-$. In particular $C \not\subset \operatorname{Supp} N_{t,y}$ and $z \notin \operatorname{Supp} N_{t,y}$. Since $(C^2) = 0$, $(N_{t,y} \cdot C) \geq 0$, and $P_{t,y}$ is nef,

$$0 \leq d(t,y) = d_0(t) - (N_{t,y} \cdot C) \leq d_0(t).$$

By [BFJ09, Theorem A and Section 3.4], the function $y \to \operatorname{vol}(P_t - yC)$ is C^1 on $[0, \tau_t)$ with derivative $-2d(t,y)$. Since $\operatorname{vol}(P_t) = (P_t^2)$ and the volume is non-negative, for $0 < y_1 < \tau_t$ we have $2\int_0^{y_1} d(t,y)dy = (P_t^2) - \operatorname{vol}(P_t - y_1 C) \leq (P_t^2)$. Letting $y_1 \to \tau_t$, we obtain

$$\int_0^{\tau_t} d(t,y)dy \leq \frac{1}{2}(P_t^2) = \varphi(t).$$

Step 4. In this step, we prove $S(\rho^*W; \operatorname{ord}_C) \leq 67/72$.

For a non-zero space V of sections of a line bundle on $\widetilde{E}$ and a prime divisor w over $\widetilde{E}$, let

$$\mathcal{T}_w(V) := \sum_{k \geq 1} \dim\{\sigma \in V \mid \operatorname{mult}_w \operatorname{div}(\sigma) \geq k\}$$

be the sum of the jumping numbers of w on V. Since $C \not\subset \Theta_{m,r}$, we have $\mathcal{T}_C(\rho^*W_{m,r}) = \mathcal{T}_C(H^0(\widetilde{E}, M_{m,r}))$, and

$$\mathcal{T}_C(H^0(\widetilde{E}, M_{m,r})) = \sum_{k \geq 1} h^0(\widetilde{E}, M_{m,r} - kC).$$

The series ρ^*W has bounded support, since $W_{m,r} = 0$ for $r > 4m$. It contains an ample series by Kodaira's lemma [KM98, Lemma 2.60], since $\rho^*W_{2k,k} = H^0(\widetilde{E}, k\eta)$ for $k \geq 1$ by Construction-Proposition 6.11(1) and η is big. By [AZ22, Lemma 2.9] and the formula for $\mathcal{T}_C(\rho^*W_{m,r})$ above,

$$S(\rho^*W; \operatorname{ord}_C) = \lim_{m \to +\infty} \frac{1}{mN_m} \sum_{r=0}^{4m} \sum_{k \geq 1} h^0(\widetilde{E}, M_{m,r} - kC).$$

For $y \geq 0$, let $(\rho^*W)^{(y)}$ be the series with pieces $(\rho^*W)^{(y)}_{m,r} := \{\sigma \in \rho^*W_{m,r} \mid \operatorname{ord}_C(\sigma) \geq my\}$. By [AZ22, Corollary 2.13],

$$S(\rho^*W; \operatorname{ord}_C) = \frac{1}{\operatorname{vol}(\rho^*W)} \int_0^\infty \operatorname{vol}\left((\rho^*W)^{(y)}\right) dy, \qquad \operatorname{vol}(\rho^*W) = \lim_{m \to +\infty} \frac{6N_m}{m^3} = 18.$$

By [AZ22, Lemma 2.10], applied with $n = 2$ and one extra grading,

$$\operatorname{vol}\left((\rho^*W)^{(y)}\right) = 3\int_0^4 \theta_y(t)dt,$$

where $\theta_y(t)$ is the volume function of the slice of $(\rho^*W)^{(y)}$ at t, the limit of $\frac{2}{m^2} h^0(\widetilde{E}, M_{m,r} - \lceil my \rceil C)$ as $r/m \to t$. We show that $\theta_y(t) = \operatorname{vol}(P_t - yC)$ for $0 < t < 4$. By **Step 3**,

$$mP_{r/m} - \frac{1}{2}\sum_{j'} R_{j'} \leq M_{m,r} \leq mP_{r/m},$$

where $\leq$ means that the difference is effective, so

$$\operatorname{vol}\left(P_{r/m} - \frac{k}{m}C - \frac{1}{2m}\sum_{j'} R_{j'}\right) \leq \frac{1}{m^2}\operatorname{vol}(M_{m,r} - kC) \leq \operatorname{vol}\left(P_{r/m} - \frac{k}{m}C\right).$$

Since the volume is continuous on $N^1(\widetilde{E})_{\mathbb{R}}$ [Laz04a, Corollary 2.2.45], both bounds converge to $\operatorname{vol}(P_t - yC)$ as $r/m \to t$ and $k/m \to y$, and $\frac{2}{m^2} h^0(\widetilde{E}, M_{m,r} - kC) = \frac{1}{m^2}\operatorname{vol}(M_{m,r} - kC) + O(1/m)$ uniformly in r and k, by the Riemann–Roch theorem applied to the round-down of the positive part of the Zariski decomposition [Laz04a, Theorem 2.3.19 and Proposition 2.3.21], whose first

cohomology is $O(m)$ by [KM98, Theorem 2.64] and Serre duality, since the negative parts are supported on the finitely many curves of C^-. Thus $\theta_y(t) = \mathrm{vol}(P_t - yC)$ for $0 < t < 4$, and $\mathrm{vol}(P_t - yC) = 0$ for $y \geq \tau_t$, so

$$S(\rho^*W; \mathrm{ord}_C) = \frac{3}{18}\int_0^4 \int_0^{\tau_t} \mathrm{vol}(P_t - yC)dydt = \frac{1}{6}\int_0^4 \int_0^{\tau_t} \mathrm{vol}(P_t - yC)dydt.$$

Fix $0 < t < 4$ and an ample divisor A' on $\widetilde{E}$, and let $L_\epsilon := P_t + \epsilon A'$ for rational $\epsilon > 0$. Then L_ϵ is ample, and [AZ22, Lemma 4.1] implies

$$\int_0^\infty \mathrm{vol}(L_\epsilon - yC)dy \leq \frac{2}{3}(L_\epsilon^2)\tau(L_\epsilon),$$

where $\tau(L_\epsilon)$ is the pseudo-effective threshold of C with respect to L_ϵ. The argument of **Step 3** with L_ϵ in place of P_t implies $\tau(L_\epsilon) \leq \bar{\tau}_t + c\epsilon$ with $c := \max\{(\eta \cdot A'), \frac{1}{2}(-K_{\widetilde{E}} \cdot A')\}$. Since A' is ample, $\mathrm{vol}(L_\epsilon - yC) \geq \mathrm{vol}(P_t - yC)$. Letting $\epsilon \to 0$, we obtain

$$\int_0^{\tau_t} \mathrm{vol}(P_t - yC)dy \leq \frac{2}{3}(P_t^2)\bar{\tau}_t = \frac{4}{3}\varphi(t)\bar{\tau}_t.$$

Therefore, by Table 3,

$$S(\rho^*W; \mathrm{ord}_C) \leq \frac{2}{9}\int_0^4 \varphi(t)\bar{\tau}_t dt = \frac{2}{9}\Big(\int_0^1 t\varphi dt + \frac{3}{2}\int_1^3 \varphi dt + \frac{3}{2}\int_3^4 (4-t)\varphi dt\Big) = \frac{67}{72}.$$

Step 5. In this step, we prove $S((\rho^*W)^C; z) \leq 133/144$.

By Definition 2.8, the pieces of $(\rho^*W)^C$ are

$$(\rho^*W)^C_{m,r,k} = \mathrm{Im}\left(\{\sigma \in \rho^*W_{m,r} \mid \mathrm{ord}_C(\sigma) \geq k\} \to H^0(C, (r\eta - kC)|_C)\right),$$

where a section σ is sent to the restriction of σ/σ_C^k to C, and σ_C is a section with divisor C. For fixed (m, r), $\sum_k \dim(\rho^*W)^C_{m,r,k} = \dim W_{m,r}$, so

$$S((\rho^*W)^C; z) = \lim_{m\to+\infty} \frac{1}{mN_m}\sum_{r,k} \mathcal{T}_z\big((\rho^*W)^C_{m,r,k}\big),$$

where $\mathcal{T}_z$ is the sum of the jumping numbers of ord_z on a space of sections on C. Since $C \not\subset \Theta_{m,r}$,

$$(\rho^*W)^C_{m,r,k} = (\sigma_{m,r}|_C) \cdot V_{m,r,k},$$

$$V_{m,r,k} := \mathrm{Im}\left(H^0(\widetilde{E}, M_{m,r} - kC) \to H^0(C, (M_{m,r} - kC)|_C)\right).$$

By **Step 2**, $\mathrm{ord}_z(\sigma_{m,r}|_C) = \max\{r - 3m, 0\} \cdot (\rho^*B_4 \cdot C)_z$. Hence

$$\sum_{r,k} \mathcal{T}_z((\rho^*W)^C_{m,r,k}) = \sum_{r,k} \mathcal{T}_z(V_{m,r,k}) + (\rho^*B_4 \cdot C)_z \sum_{3m<r\leq 4m} (r - 3m)\dim W_{m,r}.$$

By Proposition 6.9(2), $\dim W_{m,r} = m^2\varphi(r/m) + O(m)$ uniformly in r, so the normalized last sum converges to $\frac{1}{24}(\rho^*B_4 \cdot C)_z$ by Table 3. By **Step 2**, $\frac{1}{24}(\rho^*B_4 \cdot C)_z \leq \frac{4}{24} = \frac{1}{6}$.

For $0 < t < 4$, consider the complete series of P_t and the flag $\widetilde{E} \supset C \supset \{z\}$. By [Fuj26b, Corollary 9.1], the expected vanishing order of ord_z on the refinement by C of the complete series of P_t equals

$$\frac{2}{(P_t^2)}\int_0^{\tau_t} d(t,y)\Big(\frac{d(t,y)}{2} + \mathrm{mult}_z(N_{t,y}|_C)\Big)dy = \frac{1}{(P_t^2)}\int_0^{\tau_t} d(t,y)^2 dy,$$

because $z \notin \mathrm{Supp}\, N_{t,y}$ by **Step 3**. By [AZ22, Corollary 2.13 and Lemma 2.10], applied with $n = 1$ and two extra gradings to $(\rho^*W)^C$, a series on the curve C graded by (m, r, k), we have $S((\rho^*W)^C; z) = \frac{1}{3}\int_0^4 \varphi(t)S'_t dt$, where $\mathrm{vol}((\rho^*W)^C) = \mathrm{vol}(\rho^*W) = 18$, the slice of $(\rho^*W)^C$ at t has volume $(P_t^2) = 2\varphi(t)$, and S'_t is the expected vanishing order of ord_z on that slice. For $0 < t < 4$ the slice is the refinement by C of the complete series of P_t multiplied by the fixed factor coming from $\sigma_{m,r}|_C$, so S'_t is the value computed above plus $(t-3)_+(\rho^*B_4 \cdot C)_z$.

Since $\frac{1}{3}\int_3^4(t-3)\varphi(t)dt=\frac{1}{24}$, the second summand contributes the term $\frac{1}{24}(\rho^*B_4\cdot C)_z$ of the previous paragraph, and the first summand contributes the limit of $\frac{1}{mN_m}\sum_{r,k}\mathcal{T}_z(V_{m,r,k})$. Here the comparison of $M_{m,r}$ with $mP_{r/m}$ in **Step 4** shows that the slice at t of the series formed by the spaces $V_{m,r,k}$ is the refinement by C of the complete series of P_t. Thus

$$\lim_{m\to+\infty}\frac{1}{mN_m}\sum_{r,k}\mathcal{T}_z(V_{m,r,k})=\frac{1}{3}\int_0^4\varphi(t)\frac{1}{(P_t^2)}\int_0^{\tau_t}d(t,y)^2dydt=\frac{1}{6}\int_0^4\int_0^{\tau_t}d(t,y)^2dydt.$$

By **Step 3**, $0\le d(t,y)\le d_0(t)$ and $\int_0^{\tau_t}d(t,y)dy\le\varphi(t)$. Therefore, by Table 3,

$$\frac{1}{6}\int_0^4\int_0^{\tau_t}d(t,y)^2dydt\le\frac{1}{6}\int_0^4d_0(t)\varphi(t)dt=\frac{109}{144}.$$

Thus

$$S((\rho^*W)^C;z)\le\frac{109}{144}+\frac{1}{6}=\frac{133}{144}.$$

Step 6. In this step, we conclude the proof.

The pair $(\widetilde{E},C)$ is plt, since C is a smooth curve on the smooth surface $\widetilde{E}$, and the different of $(\widetilde{E},C)$ on C is zero by adjunction. The only prime divisor over C whose center contains z is z itself, and $A_C(z)=1$. Theorem 2.9, applied to $\widetilde{E}$, the series ρ^*W, $Z=\{z\}$, and the prime divisor C, together with **Steps 4 and 5**, implies

$$\delta_z(\widetilde{E};\rho^*W)\ge\min\left\{\frac{A_{\widetilde{E}}(C)}{S(\rho^*W;\mathrm{ord}_C)},\frac{1}{S((\rho^*W)^C;z)}\right\}\ge\min\left\{\frac{72}{67},\frac{144}{133}\right\}=\frac{72}{67}.$$

By [AZ22, Lemma 2.9], $\delta_z(\widetilde{E};\rho^*W)$ is the infimum of $A_{\widetilde{E}}(w)/S(\rho^*W;w)$ over the prime divisors w over $\widetilde{E}$ with $z\in\mathrm{center}_{\widetilde{E}}w$. Thus

$$S(\rho^*W;w)\le\frac{67}{72}A_{\widetilde{E}}(w)$$

for any prime divisor w over $\widetilde{E}$ with $z\in\mathrm{center}_{\widetilde{E}}w$.

Now let w be a prime divisor over E with $\mathrm{center}_E w\not\subset\rho(C^-)$. Choose a closed point $x\in\mathrm{center}_E w\backslash\rho(C^-)$. By **Step 1**, $\rho^{-1}(x)$ is a single point $z\in\widetilde{E}\backslash C^-$, and $z\in\mathrm{center}_{\widetilde{E}}w$ because $\rho(\mathrm{center}_{\widetilde{E}}w)=\mathrm{center}_E w$. The jumping numbers of w on $W_{m,r}$ and on $\rho^*W_{m,r}$ coincide, so $S(W;w)=S(\rho^*W;w)$. The blow-ups of p_j and of the base point infinitely near to p_j imply

$$K_{\widetilde{E}}=\rho^*K_E+\sum_j(R_j+2e'_j)$$

[Har77, Proposition V.3.3], so

$$A_E(w)=A_{\widetilde{E}}(w)+\sum_j\left(\mathrm{mult}_w R_j+2\,\mathrm{mult}_w e'_j\right)\ge A_{\widetilde{E}}(w).$$

Therefore

$$S(W;w)=S(\rho^*W;w)\le\frac{67}{72}A_{\widetilde{E}}(w)\le\frac{67}{72}A_E(w).$$

□

In the rest of this subsection, we prove Theorem 6.18. Lemma 6.12 treats the prime divisors over E whose centers are not contained in $\rho(C^-)$. We first list the curves of negative self-intersection on $\widetilde{E}$, which contain the centers on $\widetilde{E}$ of the remaining prime divisors.

Set-up 6.13. Notation and conditions as in Construction-Proposition 6.10. Write $\zeta_1=1,\zeta_2,\zeta_3$ for the cube roots of unity and $e_j:=e_{\zeta_j}$, $e'_j:=e'_{\zeta_j}$, $R_j:=R_{\zeta_j}$ for $j=1,2,3$. In $\mathrm{Pic}\,\widetilde{E}$ let

$$L_{jj'}=L_{j'j}:=\eta-e_j-e_{j'}\ (j\ne j'),\quad T_j:=\eta-e_j-e'_j,\quad \mathcal{C}_j:=2\eta-\sum_{j'}e_{j'}-\sum_{j'\ne j}e'_{j'},$$

$$D_{jj'} := \eta - e_j - e'_j - e_{j'} \ (j \neq j'), \quad \mathcal{C}_0 := 2\eta - \sum_{j'}(e_{j'} + e'_{j'}).$$

For $j \neq j'$ we write $j \to j'$ if the infinitely near base point over p_{ζ_j} is in the direction of the line through p_{ζ_j} and $p_{\zeta_{j'}}$.

Lemma 6.14. *Notation and conditions as in Set-up 6.13. Then every irreducible curve $C \subset \widetilde{E}$ with $C^2 < 0$ is smooth and rational, with $(C^2, -K_{\widetilde{E}} \cdot C) = (-1, 1)$ or $(-2, 0)$, and it is one of R_j, e'_j, or a curve of one of the classes $L_{jj'}$, T_j, $\mathcal{C}_j$, $D_{jj'}$, $\mathcal{C}_0$, the first three of self-intersection -1 and the last two of self-intersection -2. The class $D_{jj'}$ contains an irreducible curve if and only if $j \to j'$. Up to relabeling, exactly one of the following six configurations occurs.*

(1) No arrow and no irreducible curve in $\mathcal{C}_0$ (generic).
(2) No arrow and an irreducible curve in $\mathcal{C}_0$ (conic).
(3) The arrow $1 \to 2$ (edge).
(4) The arrows $2 \to 1$ and $3 \to 1$ (star).
(5) The arrows $1 \to 2$ and $2 \to 3$ (path).
(6) The arrows $1 \to 2$, $2 \to 3$, and $3 \to 1$ (cycle).

A class among e'_j, $L_{jj'}$, T_j, $\mathcal{C}_j$ contains an irreducible curve if and only if it has non-negative intersection with every irreducible curve of self-intersection -2 in the configuration. Two distinct irreducible curves of negative self-intersection have intersection number 0 or 1.

Proof. By Construction-Proposition 6.10(2), $-K_{\widetilde{E}} = 3\eta - \sum_j (e_j + e'_j)$ is base point free, hence nef, and $(-K_{\widetilde{E}})^2 = 3$. In the basis $(\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3)$ of $\operatorname{Pic} \widetilde{E}$ we have $\eta^2 = 1$ and $e_j^2 = e_j'^2 = -1$, and the product of two distinct elements of the basis is 0 [Har77, Proposition V.3.2]. In particular $R_j^2 = -2$ and $(R_j \cdot e'_j) = 1$.

Let $C \subset \widetilde{E}$ be an irreducible curve with $C^2 < 0$. Since $-K_{\widetilde{E}}$ is nef, $(-K_{\widetilde{E}} \cdot C) \geq 0$, and the adjunction formula [Har77, Exercise V.1.3(a)] implies

$$2p_a(C) - 2 = C^2 + (K_{\widetilde{E}} \cdot C) \leq -1.$$

Thus $p_a(C) = 0$, $C^2 + (K_{\widetilde{E}} \cdot C) = -2$, and $(C^2, -K_{\widetilde{E}} \cdot C)$ is $(-1, 1)$ or $(-2, 0)$. Since $p_a(C) = 0$, $C \cong \mathbb{P}^1$ [Har77, Exercise IV.1.8(b)].

The exceptional curves of ρ are R_j and e'_j. Assume that C is not ρ-exceptional and write $C \equiv d\eta - \sum_j (m_j e_j + n_j e'_j)$. Then $d = (C \cdot \eta) \geq 1$, $n_j = (C \cdot e'_j) \geq 0$, and $m_j - n_j = (C \cdot R_j) \geq 0$. If $(C^2, -K_{\widetilde{E}} \cdot C) = (-1, 1)$, then

$$\sum_j (m_j + n_j) = 3d - 1, \qquad \sum_j (m_j^2 + n_j^2) = d^2 + 1,$$

and the Cauchy–Schwarz inequality $(3d-1)^2 \leq 6(d^2+1)$ implies $3d^2 - 6d - 5 \leq 0$, hence $d \leq 2$. If $(C^2, -K_{\widetilde{E}} \cdot C) = (-2, 0)$, then $\sum_j (m_j + n_j) = 3d$ and $\sum_j (m_j^2 + n_j^2) = d^2 + 2$, and $9d^2 \leq 6(d^2+2)$ implies $d \leq 2$. For $d \in \{1, 2\}$ we have $3d - 1 = d^2 + 1$ and $3d = d^2 + 2$, so $\sum_j (m_j^2 - m_j + n_j^2 - n_j) = 0$ and all m_j, n_j belong to $\{0, 1\}$. Using $m_j \geq n_j$, the solutions are the classes $L_{jj'}$ and T_j ($d = 1$, two non-zero entries), $D_{jj'}$ and $\eta - e_1 - e_2 - e_3$ ($d = 1$, three non-zero entries), $\mathcal{C}_j$ ($d = 2$, five non-zero entries), and $\mathcal{C}_0$ ($d = 2$, six non-zero entries). The class $\eta - e_1 - e_2 - e_3$ contains no irreducible curve, since the points $p_\zeta = (1 : \zeta : \zeta^2)$ are not collinear by the Vandermonde determinant. A direct calculation shows $L_{jj'}^2 = T_j^2 = \mathcal{C}_j^2 = -1$ and $D_{jj'}^2 = \mathcal{C}_0^2 = -2$.

Let $\ell \subset E$ be the line through p_{ζ_j} and $p_{\zeta_{j'}}$ and $\widetilde{\ell}$ the strict transform of ℓ. An irreducible curve of class $D_{jj'}$ has degree $(D_{jj'} \cdot \eta) = 1$ over E and passes through the points over p_{ζ_j} and $p_{\zeta_{j'}}$, so it is $\widetilde{\ell}$. For $k \in \{j, j'\}$ let $\epsilon_k := 1$ if the infinitely near base point over p_{ζ_k} is in the direction of ℓ, and $\epsilon_k := 0$ otherwise. By [Har77, Proposition V.3.6], $\widetilde{\ell} \equiv \eta - e_j - e_{j'} - \epsilon_j e'_j - \epsilon_{j'} e'_{j'}$. If

$\epsilon_j = \epsilon_{j'} = 1$, then $(-K_{\widetilde{E}} \cdot \widetilde{\ell}) = -1$, which is impossible. Thus $\widetilde{\ell} \in D_{jj'}$ if and only if $\epsilon_j = 1$, and the arrows $j \to j'$ and $j' \to j$ do not both occur. Since the three points p_ζ are not collinear, the infinitely near base point over p_{ζ_j} is in the direction of at most one of the lines through p_{ζ_j} and p_{ζ_k} with $k \neq j$, so at most one arrow starts at j. Two arrows have distinct starting points, so they form a star if their end points coincide and a path otherwise. Three arrows form a cycle. If $j \to j'$, then $(\mathcal{C}_0 \cdot D_{jj'}) = -1$, so every divisor $D' \geq 0$ of class $\mathcal{C}_0$ contains the curve in $D_{jj'}$ and $\mathcal{C}_0$ contains no irreducible curve. Hence exactly one of the six configurations occurs.

Let D be one of the classes e'_j, $L_{jj'}$, T_j, $\mathcal{C}_j$. Then $D^2 = -1$ and $(-K_{\widetilde{E}} \cdot D) = 1$, so $\chi(\mathcal{O}_{\widetilde{E}}(D)) = 1$ by the Riemann–Roch theorem [Har77, Theorem V.1.6]. Since $(-K_{\widetilde{E}} \cdot (K_{\widetilde{E}} - D)) = -4 < 0$ and $-K_{\widetilde{E}}$ is nef, $h^0(K_{\widetilde{E}} - D) = 0$, and Serre duality implies $h^0(D) \geq 1$. If $(D \cdot V) < 0$ for an irreducible curve V with $V^2 = -2$, then every divisor $D' \geq 0$ of class D contains V, and since $(-K_{\widetilde{E}} \cdot V) = 0 \neq (-K_{\widetilde{E}} \cdot D)$ it is reducible. Conversely, assume that $(D \cdot V) \geq 0$ for every irreducible curve V with $V^2 = -2$, and let $D' \geq 0$ be a divisor of class D. Since $-K_{\widetilde{E}}$ is nef and $(-K_{\widetilde{E}} \cdot D') = 1$, we can write $D' = C' + V'$ with C' irreducible, $(-K_{\widetilde{E}} \cdot C') = 1$, and $V' \geq 0$ with $(-K_{\widetilde{E}} \cdot V') = 0$. By the Hodge index theorem [Har77, Theorem V.1.9 and Remark V.1.9.1], applied with the class $-K_{\widetilde{E}}$ of positive square, the intersection form is negative definite on the classes orthogonal to $-K_{\widetilde{E}}$. Hence every component of V' has negative self-intersection, and it is a curve with self-intersection -2 by the first part of the proof. The Hodge index theorem [Har77, Theorem V.1.9 and Remark V.1.9.1] implies $3C'^2 \leq (-K_{\widetilde{E}} \cdot C')^2 = 1$, while $C'^2 = 2p_a(C') - 1 \geq -1$ by adjunction, so $C'^2 = -1$. From $-1 = D'^2 = C'^2 + 2(C' \cdot V') + V'^2$ we get $V'^2 = -2(C' \cdot V')$, hence

$$0 \leq (D \cdot V') = (C' \cdot V') + V'^2 = -(C' \cdot V') \leq 0.$$

Thus $V'^2 = 0$, so $V' = 0$ by negative definiteness, and $D' = C'$ is irreducible.

Finally, let C, C' be distinct irreducible curves with negative self-intersection. Then $(C \cdot C') \geq 0$, and the Hodge index theorem implies $3(C + C')^2 \leq (-K_{\widetilde{E}} \cdot (C + C'))^2$. If $C^2 = C'^2 = -1$, then $3(2(C \cdot C') - 2) \leq 4$. If $C^2 = -1$ and $C'^2 = -2$, then $3(2(C \cdot C') - 3) \leq 1$. If $C^2 = C'^2 = -2$, then $C + C'$ is orthogonal to $-K_{\widetilde{E}}$ and non-zero, so $2(C \cdot C') - 4 < 0$. In all three cases $(C \cdot C') \leq 1$. □

The pullback $\rho^* B_4$ has the following coefficients along the exceptional curves of ρ, and the following intersection numbers with them hold for its strict transform.

Lemma 6.15. *Notation and conditions as in Set-up 6.13, and let $\widetilde{B}_4$ be the strict transform of B_4 on $\widetilde{E}$. Then there exist $b_1, b_2, b_3 \in \{2, 3\}$ such that*

$$\rho^* B_4 = \widetilde{B}_4 + \sum_j \left(b_j R_j + 3e'_j\right).$$

Moreover, every component of $\widetilde{B}_4$ has coefficient at most 2, and

$$(\widetilde{B}_4 \cdot R_j) = 2b_j - 3, \qquad (\widetilde{B}_4 \cdot e'_j) = 3 - b_j \qquad \text{for } j = 1, 2, 3.$$

Proof. Let $\rho_0 : S_0 \to E$ be the blow-up of the three points p_ζ, with exceptional curves e^0_ζ, and let $x_\zeta \in e^0_\zeta$ be the infinitely near base point of $W_{1,3}$ over p_ζ. Then $\rho = \rho_0 \circ \rho_1$, where ρ_1 is the blow-up of the three points x_ζ, $e_\zeta = \rho_1^* e^0_\zeta$, and R_ζ is the strict transform of e^0_ζ. By [Har77, Proposition V.3.6], the coefficient of e'_ζ in $\rho^* B_4$ is $\operatorname{mult}_{x_\zeta} \rho_0^* B_4$, and the coefficient of R_ζ in $\rho^* B_4$ is the coefficient of e^0_ζ in $\rho_0^* B_4$.

The quadrics f_2, c_2, Q_2 vanish at the three non-collinear points p_ζ and are linearly independent, so they span the plane conics through the points p_ζ. By [Har77, Example V.4.2.3], $(z_1 : z_2 : z_3) = (f_2 : c_2 : Q_2)$ defines a morphism $\kappa : S_0 \to \mathbb{P}^2$, which is the blow-up of three points of $\mathbb{P}^2$, the vertices, and contracts the strict transforms of the three lines through two of the points p_ζ. Let

$$G' := v z_3 - \lambda z_1 - \mu z_2, \qquad \Phi_0 := z_1 z_2 - G'^2 - G' z_3,$$

and $D_2 := (\Phi_0 = 0) \subset \mathbb{P}^2$. Since $J_4 = \Phi_0(f_2, c_2, Q_2)$ and $\kappa^*\mathcal{O}_{\mathbb{P}^2}(1) = \rho_0^*\mathcal{O}_E(2) - \sum_\zeta e_\zeta^0$, we have

$$\rho_0^* B_4 = \kappa^* D_2 + 2\sum_\zeta e_\zeta^0. \tag{6.37}$$

Write $\Phi_0 = \Psi_0 - \Xi_0^2$ with $\Psi_0 := z_1z_2 + \frac{1}{4}z_3^2$ and $\Xi_0 := G' + \frac{1}{2}z_3$. The quadratic form Ψ_0 is non-degenerate and Ξ_0 is linear, so Φ_0 has rank at least 2 and the conic D_2 is reduced.

Fix ζ. In the chart $i = 1$ of E with coordinates (a, e), the differentials at p_ζ satisfy

$$dQ_2 = \zeta^2 da + \zeta de = -\zeta df_2 + \zeta^2 dc_2.$$

Since κ sends a tangent direction at p_ζ to $(df_2 : dc_2 : dQ_2)$ and $(\kappa^*\mathcal{O}_{\mathbb{P}^2}(1)\cdot e_\zeta^0) = 1$, the morphism κ maps e_ζ^0 isomorphically onto the line $\Delta_\zeta := (z_3 = -\zeta z_1 + \zeta^2 z_2)$, on which $(z_1 : z_2)$ are coordinates. By Construction-Proposition 6.10(1), x_ζ is the direction $t_\zeta = 0$. Since t_ζ is the linear part of $-(v\zeta + \lambda)f_2 + \zeta^2(u - \mu\zeta)c_2$ (Construction-Proposition 6.10), the point $y_\zeta := \kappa(x_\zeta)$ of Δ_ζ is a zero of $T_\zeta^0 := -(v\zeta + \lambda)z_1 + \zeta^2(u - \mu\zeta)z_2$. Let $U_\zeta^0 := \zeta z_1 + \zeta^2 z_2$. Since $u - v = 1$, on Δ_ζ we have $G' = T_\zeta^0 - \zeta^2 z_2$ and $G' + z_3 = T_\zeta^0 - \zeta z_1$, hence

$$\Phi_0|_{\Delta_\zeta} = T_\zeta^0(U_\zeta^0 - T_\zeta^0), \qquad \Psi_0|_{\Delta_\zeta} = \tfrac{1}{4}(U_\zeta^0)^2, \qquad \Xi_0|_{\Delta_\zeta} = T_\zeta^0 - \tfrac{1}{2}U_\zeta^0.$$

By condition (C) of Set-up 6.6, T_ζ^0 is a non-zero linear form on Δ_ζ, and y_ζ is the only zero of T_ζ^0 on Δ_ζ.

We show that D_2 is smooth at y_ζ, and transversal to Δ_ζ at y_ζ if $U_\zeta^0(y_\zeta) \neq 0$. If $U_\zeta^0(y_\zeta) \neq 0$, then $\Phi_0|_{\Delta_\zeta}$ has a simple zero at y_ζ, so $(D_2 \cdot \Delta_\zeta)_{y_\zeta} = 1$. If $U_\zeta^0(y_\zeta) = 0$, then $\Xi_0(y_\zeta) = 0$, hence $d\Phi_0(y_\zeta) = d\Psi_0(y_\zeta) \neq 0$ since Ψ_0 is non-degenerate.

Since $\Psi_0|_{\Delta_\zeta} = \frac{1}{4}(U_\zeta^0)^2$, the line Δ_ζ is tangent to the smooth conic $(\Psi_0 = 0)$ at the point $(\zeta : -1 : -2\zeta^2)$. The points $(\zeta : -1 : -2\zeta^2)$ for the three values of ζ are distinct, so the lines Δ_ζ are three distinct tangent lines of $(\Psi_0 = 0)$, and $\Delta_\zeta \cap \Delta_{\zeta'} \notin (\Psi_0 = 0)$ for $\zeta \neq \zeta'$. The strict transform of the line through p_ζ and $p_{\zeta'}$ intersects e_ζ^0 and $e_{\zeta'}^0$, so the vertex to which it is contracted is $\Delta_\zeta \cap \Delta_{\zeta'}$. Therefore $\Psi_0 \neq 0$, hence $U_\zeta^0 \neq 0$, at any vertex on Δ_ζ.

Next we show $\mathrm{mult}_{x_\zeta} \kappa^* D_2 = 1$. If y_ζ is not a vertex, then κ is an isomorphism near x_ζ and $\mathrm{mult}_{x_\zeta} \kappa^* D_2 = \mathrm{mult}_{y_\zeta} D_2 = 1$. If y_ζ is a vertex, let Γ be the exceptional curve of κ over y_ζ. Then $\kappa^* D_2 = \widetilde{D}_2 + \Gamma$ with $\widetilde{D}_2$ the strict transform. The point $x_\zeta = e_\zeta^0 \cap \Gamma$ corresponds to the tangent direction of Δ_ζ at y_ζ, and $\widetilde{D}_2 \cap \Gamma$ to that of D_2. Since $U_\zeta^0(y_\zeta) \neq 0$, the curves D_2 and Δ_ζ have different tangent directions at y_ζ, so $x_\zeta \notin \widetilde{D}_2$ and $\mathrm{mult}_{x_\zeta} \kappa^* D_2 = 1$.

The curves $e_{\zeta'}^0$ with $\zeta' \neq \zeta$ do not contain x_ζ. Hence (6.37) implies $\mathrm{mult}_{x_\zeta} \rho_0^* B_4 = 1 + 2 = 3$. The coefficient of e_ζ^0 in $\kappa^* D_2$ is the coefficient of Δ_ζ in the reduced conic D_2, which is 0 or 1, so the coefficient of e_ζ^0 in $\rho_0^* B_4$ is $b_\zeta \in \{2, 3\}$. Writing $b_j := b_{\zeta_j}$, we obtain the decomposition of $\rho^* B_4$ in the lemma.

By the projection formula, $(\rho^* B_4 \cdot R_j) = (\rho^* B_4 \cdot e_j') = 0$. Since $R_j^2 = -2$, $e_j'^2 = -1$, $(R_j \cdot e_j') = 1$, and R_j, e_j' are disjoint from R_k, e_k' for $k \neq j$, we obtain $(\widetilde{B}_4 \cdot R_j) = 2b_j - 3$ and $(\widetilde{B}_4 \cdot e_j') = 3 - b_j$. Finally, the pair $(E, \frac{1}{2}B_4)$ is log canonical by Proposition 6.9, so every component of B_4, and hence of $\widetilde{B}_4$, has coefficient at most 2. □

Up to rounding, the moving classes of the linear systems in Construction-Proposition 6.11 are the classes $mP_{r/m}$ of the following definition, which also defines the moments used in the local estimates.

Definition 6.16. Notation as in Set-up 6.13. For $0 \leq t \leq 4$ let

$$P_t := \begin{cases} t\eta & 0 \leq t \leq 1, \\ t\eta - \frac{t-1}{2}\sum_j (e_j + e_j') & 1 \leq t \leq 3, \\ -(4-t)K_{\widetilde{E}} & 3 \leq t \leq 4. \end{cases}$$

For an irreducible curve $C \subset \widetilde{E}$ with $C^2 < 0$, a real number $t \in [0,4]$, and $y \geq 0$ such that $P_t - yC$ is big, let $P_t - yC = P_{t,y} + N_{t,y}$ be the Zariski decomposition [Laz04a, Theorem 2.3.19]. Define

$$S_C := \frac{1}{6}\iint P_{t,y}^2 dydt, \qquad \theta_C := \frac{1}{6}\iint (P_{t,y}\cdot C)^2 dydt,$$
$$\kappa_{C,D} := \frac{1}{3}\iint (P_{t,y}\cdot C)\operatorname{mult}_D N_{t,y} dydt$$

for every curve $D \neq C$, where t runs over $[0,4]$ and y over the interval on which $P_t - yC$ is big.

For an irreducible curve C with $C^2 < 0$, the next lemma uses the flag given by C. In its proof only the coefficient along C is removed from the boundary, so the remaining boundary is effective and inversion of adjunction can be used.

Lemma 6.17. *Notation and conditions as in Construction-Proposition 6.11 and Definition 6.16. Let $C \subset \widetilde{E}$ be an irreducible curve with $C^2 < 0$ and $z \in C$ a closed point. Let $a_C := \frac{31}{72}$ if $C = R_j$ for some j, $a_C := \frac{31}{36}$ if $C = e'_j$ for some j, and $a_C := 0$ otherwise. For an irreducible curve $D \neq C$, let $(C\cdot D)_z$ be the local intersection number of C and D at z. Assume that*

$$S_C + \frac{1}{24}\operatorname{mult}_C \rho^* B_4 - a_C < 1 \tag{6.38}$$

and

$$\theta_C + \sum_{D\neq C} \kappa_{C,D}(C\cdot D)_z + \frac{1}{24}\big(C\cdot(\rho^*B_4 - (\operatorname{mult}_C \rho^* B_4)C)\big)_z < 1, \tag{6.39}$$

where D runs over the irreducible curves on $\widetilde{E}$ different from C. Then

$$S_{\mathrm{mov}}(w) + \frac{1}{24}\operatorname{mult}_w \rho^* B_4 \leq A_{\widetilde{E}}(w) + a_C \operatorname{mult}_w C$$

for any prime divisor w over $\widetilde{E}$ whose center on $\widetilde{E}$ contains z.

The numbers a_C are $\frac{31}{72} = 1 - \frac{41}{72}$ and $\frac{31}{36} = 2 - \frac{41}{36}$, where $\frac{41}{72}$ and $\frac{41}{36}$ are the coefficients of $\operatorname{mult}_w R_\zeta$ and $\operatorname{mult}_w e'_\zeta$ in the formula for $S(W;w)$ in Construction-Proposition 6.11, and 1 and 2 are their coefficients in $A_E(w) - A_{\widetilde{E}}(w)$, see (6.41) below.

Proof of Lemma 6.17. The support of $N_{t,y}$ has negative definite intersection matrix, so every component of $N_{t,y}$ has negative self-intersection. By Lemma 6.14 there exist only finitely many such curves, and the sum in (6.39) is finite.

For integers $m \geq 1$ and $0 \leq r \leq 4m$, let $M_{m,r}$ be the class of the complete linear system in Construction-Proposition 6.11, and let $t := r/m$. Comparing with Definition 6.16,

$$M_{m,r} = mP_t - \frac{\epsilon_{m,r}}{2}\sum_j R_j, \tag{6.40}$$

where $\epsilon_{m,r} = 1$ if $m < r < 3m$ and $r - m$ is odd, and $\epsilon_{m,r} = 0$ otherwise. Let w be a prime divisor over $\widetilde{E}$ whose center contains z. By [AZ22, Lemma 3.1], there exists a basis $\mathcal{B}_{m,r}$ of $H^0(\widetilde{E}, M_{m,r})$ compatible with the filtrations by mult_w and by mult_C. Let

$$\Delta_m := \frac{1}{mN_m}\sum_{r=0}^{4m}\sum_{s\in\mathcal{B}_{m,r}} \operatorname{div}(s).$$

Since $\mathcal{B}_{m,r}$ is compatible with mult_w, $\operatorname{mult}_w \Delta_m$ is the sum of the jumping numbers of w on the linear systems $|M_{m,r}|$, divided by mN_m. Thus $\operatorname{mult}_w \Delta_m \to S_{\mathrm{mov}}(w)$ as $m \to +\infty$, by Construction-Proposition 6.11.

We first show that $\operatorname{mult}_C \Delta_m \to S_C$. Since $\mathcal{B}_{m,r}$ is compatible with mult_C,

$$\operatorname{mult}_C \Delta_m = \frac{1}{mN_m} \sum_{r=0}^{4m} \sum_{k\geq 1} h^0(M_{m,r} - kC).$$

With $y := k/m$, (6.40) shows $M_{m,r} - kC = m(P_t - yC) - \frac{\epsilon_{m,r}}{2} \sum_j R_j$. By the Riemann–Roch theorem on the positive parts of the Zariski decompositions [Laz04a, Theorem 2.3.19 and Proposition 2.3.21] and the continuity of the volume [Laz04a, Theorem 2.2.44], $h^0(M_{m,r} - kC) = \frac{m^2}{2} P_{t,y}^2 + o(m^2)$ uniformly in (r,k). Since $H^3 = 18$, we have $N_m = 3m^3 + O(m^2)$, and the Riemann sum over (r,k) converges to $\frac{1}{6} \iint P_{t,y}^2 dy dt = S_C$.

Next we bound the restriction to C. Fix r and k, and let $\mathcal{B}_{m,r,k}$ be the set of $s \in \mathcal{B}_{m,r}$ with $\operatorname{mult}_C \operatorname{div}(s) = k$. For $s \in \mathcal{B}_{m,r,k}$ let s' be the section of $M_{m,r} - kC$ with $\operatorname{div}(s) = kC + \operatorname{div}(s')$. Since $\mathcal{B}_{m,r}$ is compatible with mult_C, the restrictions $s'|_C$ with $s \in \mathcal{B}_{m,r,k}$ are linearly independent. The integral divisor $2\operatorname{div}(s') + \epsilon_{m,r} \sum_j R_j \geq 0$ has class $2m(P_t - yC)$, so it is at least $2mN_{t,y}$ [Laz04a, Proposition 2.3.21]. Hence $\operatorname{div}(s') \geq N'$, where

$$N' := \sum_{D \neq C} \left\lceil \max\left(0, m \operatorname{mult}_D N_{t,y} - \frac{1}{2} \sum_j \operatorname{mult}_D R_j\right) \right\rceil D.$$

Since $C \not\subset \operatorname{div}(s')$, we get $\operatorname{div}(s'|_C) \geq N'|_C$. Let $d' := ((M_{m,r} - kC - N') \cdot C)$. As $C \cong \mathbb{P}^1$ by Lemma 6.14, the sections $s'|_C$ divided by a section with divisor $N'|_C$ are linearly independent elements of $H^0(\mathbb{P}^1, \mathcal{O}(d'))$. There are at most $d'+1$ of them, and the sum of their orders at z is at most $0 + 1 + \cdots + d'$. Thus

$$\sum_{s \in \mathcal{B}_{m,r,k}} \operatorname{mult}_z \left(\operatorname{div}(s')|_C\right) \leq (d'+1) \operatorname{mult}_z(N'|_C) + \frac{d'(d'+1)}{2},$$

and the left-hand side is 0 if $d' < 0$. Each coefficient of N' differs from $m \operatorname{mult}_D N_{t,y}$ by at most 1, and $C^2 < 0$, so

$$d' \leq m(P_{t,y} \cdot C) + c, \qquad \operatorname{mult}_z(N'|_C) \leq m \sum_{D \neq C} \operatorname{mult}_D N_{t,y} (C \cdot D)_z + c$$

for a constant c independent of m, r, and k. If $h^0(M_{m,r} - kC) \neq 0$, then $P_t - yC$ is pseudo-effective by (6.40). For $0 < t < 4$ the class P_t is nef with $P_t^2 > 0$, hence big, so $P_t - y'C$ is big for $0 \leq y' < y$ whenever $P_t - yC$ is pseudo-effective. Thus for each r with $0 < r < 4m$ at most one value of k has $P_t - yC$ pseudo-effective but not big, and for $r \in \{0, 4m\}$ we have $M_{m,r} = 0$ and $h^0(M_{m,r} - kC) = 0$ for $k \geq 1$. The pairs (r,k) with $P_t - yC$ not big and $h^0(M_{m,r} - kC) \neq 0$ contribute $O(m^3)$ in total. Summing over (r,k), dividing by mN_m, and passing to the Riemann integral, we obtain

$$\limsup_{m \to +\infty} \operatorname{mult}_z \left((\Delta_m - (\operatorname{mult}_C \Delta_m)C)|_C\right) \leq \theta_C + \sum_{D \neq C} \kappa_{C,D} (C \cdot D)_z.$$

Now let $\alpha_m := \operatorname{mult}_C \Delta_m + \frac{1}{24} \operatorname{mult}_C \rho^* B_4$ and

$$\Lambda_m := \Delta_m + \frac{1}{24} \rho^* B_4 - \alpha_m C.$$

Then $\Lambda_m \geq 0$ is a $\mathbb{Q}$-divisor not containing C. By the limits of $\operatorname{mult}_C \Delta_m$ and of $\operatorname{mult}_z((\Delta_m - (\operatorname{mult}_C \Delta_m)C)|_C)$ above, (6.38), and (6.39), for $m \gg 0$ we have $\alpha_m - a_C < 1$ and $\operatorname{mult}_z(\Lambda_m|_C) < 1$. Thus $(C, \Lambda_m|_C)$ is klt near z, and by inversion of adjunction [KM98, Theorem 5.50] the pair $(\widetilde{E}, C + \Lambda_m)$ is plt near z. Since the center of w contains z, $A_{\widetilde{E}}(w) \geq \operatorname{mult}_w C + \operatorname{mult}_w \Lambda_m$. Therefore

$$\begin{aligned} \operatorname{mult}_w \Delta_m + \frac{1}{24} \operatorname{mult}_w \rho^* B_4 &= \alpha_m \operatorname{mult}_w C + \operatorname{mult}_w \Lambda_m \\ &\leq A_{\widetilde{E}}(w) + (\alpha_m - 1) \operatorname{mult}_w C \leq A_{\widetilde{E}}(w) + a_C \operatorname{mult}_w C. \end{aligned}$$

Letting $m \to +\infty$ proves the lemma. □

The theorem combines the surface inequality on E with the margin $A_X(E) - S_X(E)$ kept by Lemma 3.1.

Theorem 6.18. *Let X be a smooth prime Fano threefold of genus 10 and $p \in X$ a closed point such that $\Sigma(X,p)$ consists of three distinct reduced points. Let $\pi : Y := \mathrm{Bl}_p X \to X$ with exceptional plane E, and let W be the refinement of $(H^0(X,mH))_m$ by E. Then $S_X(E) = 9/4$, and $S(W;w) \le A_E(w)$ for every prime divisor w over E. Hence every prime divisor F over X with $p \in \mathrm{center}_X F$ satisfies*

$$A_X(F) \ge S_X(F) + \frac{3}{4}\,\mathrm{mult}_F E.$$

In particular $\delta_p(X) \ge 1$, and $A_X(F) > S_X(F)$ if $\mathrm{center}_X F = \{p\}$.

Proof of Theorem 6.18. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the surface inequality to an inequality on $\widetilde{E}$.

By Lemma 6.7(1), X and p are isomorphic to a codimension two linear section and the point $[h]$ as in Set-up 6.6. All quantities in the theorem are invariant under isomorphism. Possibly replacing X and p with the linear section and $[h]$, we may assume that X, p, π, E, and W are as in Set-up 6.6. By Lemma 6.7(2), condition (C) holds and $H^3 = 18$. By Proposition 6.9, $S_X(E) = 9/4$.

Since Y and $E \cong \mathbb{P}^2$ are smooth, the different Δ_E is 0 [Har77, Proposition II.8.20], so $A_{E,\Delta_E} = A_E$. By [Har77, Proposition V.3.3], applied to the blow-up of the points p_ζ and then to the blow-up of the three infinitely near base points,

$$K_{\widetilde{E}} = \rho^* K_E + \sum_\zeta (e_\zeta + e'_\zeta) = \rho^* K_E + \sum_\zeta (R_\zeta + 2e'_\zeta).$$

Hence

$$A_E(w) = A_{\widetilde{E}}(w) + \sum_\zeta \mathrm{mult}_w R_\zeta + 2\sum_\zeta \mathrm{mult}_w e'_\zeta$$

for any prime divisor w over $\widetilde{E}$. By Construction-Proposition 6.11,

$$S(W;w) = S_{\mathrm{mov}}(w) + \frac{41}{72}\sum_\zeta \mathrm{mult}_w R_\zeta + \frac{41}{36}\sum_\zeta \mathrm{mult}_w e'_\zeta + \frac{1}{24}\,\mathrm{mult}_w \rho^* B_4,$$

so $S(W;w) \le A_E(w)$ is equivalent to

$$S_{\mathrm{mov}}(w) + \frac{1}{24}\,\mathrm{mult}_w \rho^* B_4 \le A_{\widetilde{E}}(w) + \frac{31}{72}\sum_\zeta \mathrm{mult}_w R_\zeta + \frac{31}{36}\sum_\zeta \mathrm{mult}_w e'_\zeta, \tag{6.41}$$

where $\frac{31}{72} = 1 - \frac{41}{72}$ and $\frac{31}{36} = 2 - \frac{41}{36}$.

Step 2. In this step we collect the bounds on S_C, θ_C, and $\kappa_{C,D}$ and verify (6.38).

By Lemma 6.14, the curves with negative self-intersection on $\widetilde{E}$ form one of six configurations. For each configuration and each such curve C, the tables in Appendix A, Subsection A.1, list the Zariski decomposition of $P_t - yC$ on finitely many polygons in the (t,y)-plane and the values of S_C, θ_C, and $\kappa_{C,D}$. Remark A.2 explains how the tables are checked and integrated.

The tables list one curve C in each orbit of the group of permutations of $\{1,2,3\}$ preserving the arrows. Such a permutation τ permutes the classes e_j and e'_j, fixes η and hence every P_t, and permutes the curves of negative self-intersection by Lemma 6.14. Hence it maps the Zariski decomposition of $P_t - yC$ to that of $P_t - y\tau(C)$, so $S_{\tau(C)} = S_C$, $\theta_{\tau(C)} = \theta_C$, and $\kappa_{\tau(C),\tau(D)} = \kappa_{C,D}$. Reading the tables, we obtain the following bounds in all six configurations.

(1) $S_{e'_j} \le \frac{17}{18}$, $S_{R_j} \le \frac{65}{72}$, and $S_C \le \frac{17}{24}$ for any other C.
(2) $\theta_{e'_j} \le \frac{5}{12}$, and $\kappa_{e'_j,D} \le \frac{121}{288}$ for $D \ne R_j$. Moreover $\theta_{e'_j} + \kappa_{e'_j,R_j} \le \frac{59}{72}$, except for $j = 1$ in the star configuration.

(3) $\theta_{R_j} \le \frac{19}{36}$, and $\kappa_{R_j,D} \le \frac{7}{24}$ for $D \ne e'_j$. In the star configuration, $\theta_{R_1} = \frac{5}{12}$ and $\kappa_{R_1,e'_1} = \frac{7}{18}$.
(4) Let C be not ρ-exceptional, and let Σ_C be the sum of $\kappa_{C,D}$ over the curves D that are not ρ-exceptional and satisfy $(C \cdot D) > 0$. If C has class $L_{jj'}$, then $\theta_C \le \frac{5}{12}$ and $\Sigma_C \le \frac{1}{18}$. If C has class T_j, then $\theta_C \le \frac{11}{18}$ and $\Sigma_C \le \frac{43}{216}$. In the edge configuration with $1 \to 2$, the non-zero terms of Σ_{T_3} are $\kappa_{T_3,T_2} = \frac{1}{72}$ and $\kappa_{T_3,D_{12}} = \frac{5}{27}$. If C has class $\mathcal{C}_j$, then $\theta_C \le \frac{47}{72}$ and $\Sigma_C \le \frac{1}{216}$. If C has class $D_{jj'}$, then $\theta_C \le \frac{4}{9}$ and $\Sigma_C \le \frac{5}{72}$. If C has class $\mathcal{C}_0$, then $\theta_C \le \frac{2}{3}$ and $\Sigma_C = 0$.

Moreover $\kappa_{C,D} \ge 0$, since $P_{t,y}$ is nef and $N_{t,y} \ge 0$. By Lemma 6.15, $\operatorname{mult}_{e'_j} \rho^* B_4 = 3$, $\operatorname{mult}_{R_j} \rho^* B_4 = b_j \le 3$, and $\operatorname{mult}_C \rho^* B_4 \le 2$ for any other C. Thus the left-hand side of (6.38) is at most

$$\frac{17}{18} + \frac{3}{24} - \frac{31}{36} = \frac{5}{24}, \qquad \frac{65}{72} + \frac{3}{24} - \frac{31}{72} = \frac{43}{72}, \qquad \frac{17}{24} + \frac{2}{24} = \frac{19}{24}$$

for $C = e'_j$, $C = R_j$, and the other curves respectively. Hence (6.38) holds for any irreducible curve C with $C^2 < 0$.

Step 3. In this step we verify (6.39) at the points of $R_j \cup e'_j$. The bounds obtained in this step and in **Step 4** are collected in Table 4.

TABLE 4. Upper bounds for the terms θ_C, $\sum_D \kappa_{C,D}(C \cdot D)_z$, and $\frac{1}{24}(C \cdot (\rho^* B_4 - (\operatorname{mult}_C \rho^* B_4) C))_z$ of (6.39) at a point z of a curve of negative self-intersection, where the third row excludes $j = 1$ in the star configuration and in the last five rows z is not on any R_j or e'_j.

Point z	Curve C	θ_C	κ-term	B_4-term	Total
$e'_j \backslash R_j$	e'_j	5/12	121/288	1/24	253/288
$R_j \backslash e'_j$	R_j	19/36	7/24	3/24	17/18
$R_j \cap e'_j$	e'_j	59/72		3/24	17/18
$R_1 \cap e'_1$, star	R_1	5/12	7/18	4/24	35/36
on $L_{jj'}$	$L_{jj'}$	5/12	1/18	2/24	5/9
on T_j	T_j	11/18	43/216	3/24	101/108
on $\mathcal{C}_j$	$\mathcal{C}_j$	47/72	1/216	2/24	20/27
on $D_{jj'}$	$D_{jj'}$	4/9	5/72	3/24	23/36
on $\mathcal{C}_0$	$\mathcal{C}_0$	2/3	0	3/24	19/24

For a curve $C = d\eta - \sum_k (m_k e_k + n_k e'_k)$ in the list of Lemma 6.14 that is not ρ-exceptional, we have $(C \cdot e'_j) = n_j$ and $(C \cdot R_j) = m_j - n_j$ with $m_j, n_j \in \{0, 1\}$. Thus a curve with $(C \cdot e'_j) > 0$ satisfies $(C \cdot R_j) = 0$, and a curve with $(C \cdot R_j) > 0$ satisfies $(C \cdot e'_j) = 0$. The curves $C \ne R_j$ with $C^2 < 0$ and $(C \cdot e'_j) > 0$ have classes among T_j, D_{jk}, $\mathcal{C}_k$ $(k \ne j)$, and $\mathcal{C}_0$. A direct calculation shows

$$(T_j \cdot \mathcal{C}_k) = (T_j \cdot \mathcal{C}_0) = (\mathcal{C}_k \cdot \mathcal{C}_{k'}) = 0,$$
$$(T_j \cdot D_{jl}) = (\mathcal{C}_k \cdot D_{jl}) = (\mathcal{C}_k \cdot \mathcal{C}_0) = (D_{jl} \cdot \mathcal{C}_0) = -1$$

for distinct k, k' different from j and any $l \ne j$. Two distinct irreducible curves have non-negative intersection, and at most one arrow starts at j. Hence the curves $C \ne R_j$ with $C^2 < 0$ and $(C \cdot e'_j) > 0$ are pairwise disjoint and disjoint from R_j. Similarly, the curves $C \ne e'_j$ with $C^2 < 0$ and $(C \cdot R_j) > 0$ have classes among L_{jk}, $\mathcal{C}_j$, and D_{kj}, and

$$(L_{jk} \cdot L_{jk'}) = (L_{jk} \cdot \mathcal{C}_j) = (L_{jk} \cdot D_{k'j}) = (D_{kj} \cdot D_{k'j}) = 0,$$
$$(L_{jk} \cdot D_{kj}) = (\mathcal{C}_j \cdot D_{kj}) = -1$$

for distinct k, k' different from j. Hence the curves $C \neq e'_j$ with $C^2 < 0$ and $(C \cdot R_j) > 0$ are pairwise disjoint and disjoint from e'_j. By Lemma 6.14, $(C \cdot D)_z \leq (C \cdot D) \leq 1$ for distinct curves C, D of negative self-intersection, and $\kappa_{C,D} \neq 0$ only if D has negative self-intersection.

Let $z \in e'_j \backslash R_j$, and let $C := e'_j$. At most one curve $D \neq R_j$ with $D^2 < 0$ and $(D \cdot e'_j) > 0$ passes through z, so the second term of (6.39) is at most $\frac{121}{288}$. The curves R_k and e'_k with $k \neq j$ are disjoint from e'_j, and $z \notin R_j$, so near z we have $\rho^* B_4 - 3e'_j = \widetilde{B}_4$. By Lemma 6.15, the third term is $\frac{1}{24}(e'_j \cdot \widetilde{B}_4)_z \leq \frac{1}{24}(3 - b_j) \leq \frac{1}{24}$.

Let $z \in R_j \backslash e'_j$, and let $C := R_j$. At most one curve $D \neq e'_j$ with $D^2 < 0$ and $(D \cdot R_j) > 0$ passes through z, so the second term is at most $\frac{7}{24}$. Near z we have $\rho^* B_4 - b_j R_j = \widetilde{B}_4$, so by Lemma 6.15 the third term is $\frac{1}{24}(R_j \cdot \widetilde{B}_4)_z \leq \frac{1}{24}(2b_j - 3) \leq \frac{3}{24}$.

Let $z = R_j \cap e'_j$, where $j \neq 1$ if the configuration is the star, and let $C := e'_j$. By the disjointness above, R_j and e'_j are the only curves of negative self-intersection through z, so the first two terms are at most $\theta_{e'_j} + \kappa_{e'_j, R_j} \leq \frac{59}{72}$. We have $\rho^* B_4 - 3e'_j \geq 0$, and e'_j is not a component of $\rho^* B_4 - 3e'_j$, so by [Har77, Proposition V.1.4] the third term is at most

$$\frac{1}{24}\big(e'_j \cdot (\rho^* B_4 - 3e'_j)\big) = \frac{1}{24}\big(0 + 3\big) = \frac{3}{24}.$$

Let $z = R_1 \cap e'_1$ in the star configuration with arrows $2 \to 1$ and $3 \to 1$. Here $\theta_{e'_1} + \kappa_{e'_1, R_1} = \frac{25}{72} + \frac{5}{9} = \frac{65}{72}$ by the tables, and we use $C := R_1$ instead. The only curve $D \neq R_1$ of negative self-intersection through z is e'_1, so the first two terms are $\frac{5}{12} + \frac{7}{18}$. Near z we have $\rho^* B_4 - b_1 R_1 = \widetilde{B}_4 + 3e'_1$. If $b_1 = 2$, then $(R_1 \cdot \widetilde{B}_4)_z \leq 2b_1 - 3 = 1$. If $b_1 = 3$, then $(\widetilde{B}_4 \cdot e'_1) = 0$, so $\widetilde{B}_4$ is disjoint from e'_1 and $(R_1 \cdot \widetilde{B}_4)_z = 0$. In both cases the third term is at most $\frac{1}{24}(1 + 3) = \frac{4}{24}$.

In each of the four cases above, the left-hand side of (6.39) is at most the total in the corresponding row of Table 4, which is less than 1. Thus for any point $z \in R_j \cup e'_j$ there exists an irreducible curve $C \ni z$ with $C^2 < 0$ satisfying (6.39) at z.

Step 4. In this step we verify (6.39) at the other points of C^-.

Let $z \in C^-$ with $z \notin \bigcup_j (R_j \cup e'_j)$, and let $C \ni z$ be an irreducible curve with $C^2 < 0$, so C is not ρ-exceptional. Let $b_C := \operatorname{mult}_C \rho^* B_4$, so $b_C \leq 2$ by Lemma 6.15. Near z we have $\rho^* B_4 - b_C C = \widetilde{B}_4 - b_C C$, which is ≥ 0 and does not contain C. Let $B_0 := 4\eta - 2\sum_k e_k - \sum_k e'_k$. Since $\rho^* B_4 \equiv 4\eta$ and $2R_k + 3e'_k = 2e_k + e'_k$, Lemma 6.15 implies $\widetilde{B}_4 \equiv B_0 - \sum_k (b_k - 2) R_k$. As $b_k \geq 2$ and $(C \cdot R_k) \geq 0$,

$$\big(C \cdot (\widetilde{B}_4 - b_C C)\big)_z \leq \big(C \cdot (\widetilde{B}_4 - b_C C)\big) \leq (C \cdot B_0) - b_C C^2.$$

A direct calculation shows

$$(L_{jj'} \cdot B_0) = (\mathcal{C}_j \cdot B_0) = 0, \qquad (T_j \cdot B_0) = 1, \qquad (D_{jj'} \cdot B_0) = (\mathcal{C}_0 \cdot B_0) = -1.$$

With $b_C \leq 2$ and $C^2 \in \{-1, -2\}$, the third term of (6.39) is at most $\frac{2}{24}$ for $L_{jj'}$ and $\mathcal{C}_j$, and at most $\frac{3}{24}$ for T_j, $D_{jj'}$, and $\mathcal{C}_0$. Only curves D through z contribute to the second term, and such a curve D is not ρ-exceptional and satisfies $0 < (C \cdot D)_z \leq (C \cdot D) \leq 1$. Thus the second term is at most Σ_C of **Step 2**(4). Therefore the left-hand side of (6.39) is at most the total in the corresponding row of Table 4, which is less than 1, and (6.39) holds at z.

Step 5. In this step we prove $S(W; w) \leq A_E(w)$ for every prime divisor w over E.

Let w be a prime divisor over E, and let Z_w be the center of w on $\widetilde{E}$. The morphism ρ is an isomorphism over $E \backslash \{p_\zeta\}_\zeta$, and the exceptional curves R_j, e'_j are contained in C^-. Thus the center $\rho(Z_w)$ of w on E is contained in $\rho(C^-)$ if and only if $Z_w \subset C^-$. If $Z_w \not\subset C^-$, then Lemma 6.12 implies $S(W; w) \leq \frac{67}{72} A_E(w) \leq A_E(w)$. Assume that $Z_w \subset C^-$, and fix a closed point $z \in Z_w$. By **Steps 3 and 4**, there exists an irreducible curve $C \ni z$ with $C^2 < 0$ satisfying

(6.39) at z, and C satisfies (6.38) by **Step 2**. Then Lemma 6.17 implies

$$S_{\mathrm{mov}}(w) + \frac{1}{24}\operatorname{mult}_w \rho^* B_4 \le A_{\widetilde{E}}(w) + a_C \operatorname{mult}_w C.$$

The term $a_C \operatorname{mult}_w C$ is 0 or one of the terms on the right-hand side of (6.41), and all terms on that side are non-negative. Hence (6.41) holds for w, and $S(W;w) \le A_E(w)$ by **Step 1**.

Step 6. In this step we return to X.

Since X is smooth, π is the plt extraction of E, $A_X(E) = 3$ [KM98, Lemma 2.29], and $\Delta_E = 0$ by **Step 1**. In Lemma 3.1 we take $\Delta = 0$, the complete series of $H = -K_X$, the closed subset $G := E$, and $\lambda := 1$. By **Steps 1–5**, $S(W;w) \le A_{E,\Delta_E}(w)$ for any prime divisor w over E, which is the assumption of Lemma 3.1. Since $\pi^{-1}(p) = E$, a prime divisor F over X satisfies $p \in \operatorname{center}_X F$ if and only if $\operatorname{center}_Y F \cap E \ne \emptyset$. Hence every such F satisfies

$$A_X(F) \ge S_X(F) + \big(A_X(E) - S_X(E)\big)\operatorname{mult}_F E = S_X(F) + \frac{3}{4}\operatorname{mult}_F E,$$

since $A_X(E) - S_X(E) = 3 - \frac{9}{4} = \frac{3}{4}$. As $\operatorname{mult}_F E \ge 0$, we get $A_X(F) \ge S_X(F)$, so $\delta_p(X) \ge 1$. If $\operatorname{center}_X F = \{p\}$, then $\operatorname{center}_Y F \subset E$, so $\operatorname{mult}_F E > 0$ and $A_X(F) > S_X(F)$. □

Prime divisors over X whose centers are curves through p are treated in Section 4 and Corollary 5.10.

7. Points on a doubled special line in genus ten

In this section, we prove Theorem 7.2, which concerns the closed points x of a smooth prime Fano threefold X of genus 10 such that $\Sigma(X,x)$ has length two and is supported at one line. As for the special lines in Section 5, we work with the prime divisor P, and the problem reduces to estimates for linear series on a surface.

Construction-Proposition 7.1. Notation and conditions as in Set-up 5.1, with L special, so that $E \cong \mathbb{F}_3$ with negative section s. Let $\hat\pi : \hat X \to X$ be the blow-up of the unique non-reduced conic supported on L. Then $\hat X$ is normal, the reduced exceptional divisor of $\hat\pi$ is the prime divisor P of Set-up 5.1, viewed over X, and the exceptional Cartier divisor of $\hat\pi$ is $2P$. The pair $(\hat X, P)$ is plt and $-P$ is $\hat\pi$-ample, so that $\hat\pi$ is the plt extraction of P. Along P, the threefold $\hat X$ is smooth outside a smooth curve $B \subset P$ along which it has transverse A_1 singularities. The curve B maps isomorphically onto L, and $\operatorname{Diff}_P(0) = \frac{1}{2}B$. Moreover, the morphism $Y_1 := \operatorname{Bl}_s Y \to X$ factors through a resolution $\mu : Y_1 \to \hat X$, which maps the exceptional divisor of $Y_1 \to Y$ isomorphically onto P and contracts the strict transform E_1 of E onto B. Writing P also for the exceptional divisor of $Y_1 \to Y$, we have $\mu^*(2P) = 2P + E_1$, and $E_1|_P = B$ under the isomorphism $\mu|_P$.

Proof. Since $E \cong \mathbb{F}_3$ and $\deg N_{L/X} = -1$, we have $N_{L/X} \cong \mathcal{O}_L(1) \oplus \mathcal{O}_L(-2)$. By [KPS18, Lemma 2.1.1, Remarks 2.1.2 and 2.1.7], there is a unique non-reduced conic C supported on L. It is the subscheme with $I_L^2 \subset I_C$ and $I_L/I_C \cong \mathcal{O}_L(-1)$. Since I_C/I_L^2 is a subbundle of I_L/I_L^2 with locally free quotient I_L/I_C, any point of L has an analytic neighborhood with coordinates (x,y,w) such that

$$I_L = (x,y) \qquad \text{and} \qquad I_C = (y, x^2).$$

Here x is a local section of I_L whose class generates I_L/I_C, and y is a local section of I_C whose class generates I_C/I_L^2.

Over such a neighborhood, $\hat X$ is covered by two charts. On the first chart, $\hat X$ has coordinates (x,t,w) with $y = x^2 t$, and the pullback of I_C is (x^2). The second chart is the hypersurface $x^2 = yt'$ in the space with coordinates (x,y,t',w), and the pullback of I_C is (y). On the overlap $tt' = 1$. The morphism

$$\mathbb{A}^3 \to \{x^2 = yt'\}, \qquad (\alpha,\beta,w) \to (x,y,t',w) = (\alpha\beta, \alpha^2, \beta^2, w),$$

is the quotient by the involution $(\alpha,\beta,w)\to(-\alpha,-\beta,w)$, and the pullback of I_C becomes (α^2). Thus $\hat{X}$ is normal, the reduced preimage of L is the prime divisor P given by $(x=0)$ on the first chart and by the image of $(\alpha=0)$ on the second chart, and the exceptional Cartier divisor of $\hat{\pi}$ is $2P$. Since $\mathcal{O}_{\hat{X}}(-2P)=I_C\cdot\mathcal{O}_{\hat{X}}$ is $\hat{\pi}$-ample, $-P$ is $\hat{\pi}$-ample. The surface P is a $\mathbb{P}^1$-bundle over L with fiber coordinate $t=1/t'$. The first chart is smooth, and the singular locus of the second chart is the curve $B:=(x=y=t'=0)$, along which $\hat{X}$ is locally the product of an A_1 surface singularity with a line. The curve B is the section $(t'=0)$ of $P\to L$, so $B\cong L$.

The quotient map above is étale in codimension one, so $K_{\mathbb{A}^3}+(\alpha=0)$ is the pullback of $K_{\hat{X}}+P$. The induced map $(\alpha=0)\to P$ is $(\beta,w)\to(t',w)=(\beta^2,w)$, a double cover branched along B. Thus the Hurwitz formula implies $\operatorname{Diff}_P(0)=\frac{1}{2}B$ on the second chart. On the first chart $\hat{X}$ and P are smooth, so $\operatorname{Diff}_P(0)=0$ there. Hence $\operatorname{Diff}_P(0)=\frac{1}{2}B$. Since $(P,\frac{1}{2}B)$ is klt, $K_{\hat{X}}+P$ is $\mathbb{Q}$-Cartier, and $\hat{X}\backslash P\cong X\backslash L$ is smooth, the pair $(\hat{X},P)$ is plt by inversion of adjunction [Kol13, Theorem 4.9(1)].

It remains to compare $\hat{X}$ with $Y_1=\operatorname{Bl}_s Y$. Let P also denote the exceptional divisor of $Y_1\to Y$, and let E_1 be the strict transform of E. The negative section s corresponds to the subbundle $\mathcal{O}_L(1)\subset N_{L/X}$, which is the annihilator of I_C/I_L^2. Hence s is contained in the chart of Y with coordinates (x,z,w) and $y=xz$, where $E=(x=0)$ and $s=(x=z=0)$. Over the chart with $y=xz$, Y_1 is covered by the chart with coordinates (x,t,w) and $z=xt$, where $y=x^2t$ and $P=(x=0)$, and the chart with coordinates (z,t',w) and $x=zt'$, where $y=z^2t'$, $P=(z=0)$, and $E_1=(t'=0)$. Thus $I_C\cdot\mathcal{O}_{Y_1}$ is (x^2) on the first chart and (z^2t') on the second, and $I_C\cdot\mathcal{O}_Y=(y)$ on the chart of Y with $x=yz''$. Therefore $I_C\cdot\mathcal{O}_{Y_1}$ is invertible, and $Y_1\to X$ factors through a morphism $\mu:Y_1\to\hat{X}$ [Har77, Proposition II.7.14]. In the coordinates above, μ is the identity $(x,t,w)\to(x,t,w)$ on the first charts and

$$(z,t',w)\to(x,y,t',w)=(zt',z^2t',t',w)$$

on the second charts. Thus the exceptional divisor of $Y_1\to Y$ maps isomorphically onto $P\subset\hat{X}$, and E_1 is contracted onto B. In particular the exceptional divisor of $Y_1\to Y$ and $P\subset\hat{X}$ coincide as prime divisors over X, and both are the prime divisor P of Set-up 5.1. Finally, $\mu^*(2P)$ is the divisor of $I_C\cdot\mathcal{O}_{Y_1}$, which is (x^2) on the first chart and (z^2t') on the second, so $\mu^*(2P)=2P+E_1$. On the second chart, $P=(z=0)$ and $E_1=(t'=0)$ meet transversally along $E_1\cap P=(z=t'=0)$, which μ maps onto $B=(x=y=t'=0)$. Hence $E_1|_P=B$ under the isomorphism $\mu|_P$. □

The prime divisors over X centered at a point of L are controlled by surface series on the exceptional surface P of $\hat{\pi}$, which are treated in Proposition 7.11 and Theorem 7.2, and the margin of P is computed in Proposition 5.7. The main result of this section is the following.

Theorem 7.2. *Let X be a smooth prime Fano threefold of genus 10 and $x\in X$ a closed point. Assume that $\Sigma(X,x)$ is of length two and supported at one line L. Then L is special, and, with the notation of Construction-Proposition 7.1, any prime divisor F over X with $\operatorname{center}_X F=\{x\}$ satisfies*

$$A_X(F)-S_X(F)\geq\frac{27}{95}\operatorname{mult}_F P>0,$$

where $\operatorname{mult}_F P$ is taken for the $\mathbb{Q}$-Cartier divisor P on $\hat{X}$. If the normalization of the image of E in Q under the line link $Y\dashrightarrow Y^+\to Q$ of Set-up 5.1 is $\mathbb{F}_2$, then more strongly $A_X(F)\geq\frac{108}{95}S_X(F)+\frac{27}{95}\operatorname{mult}_F P$.

The proof of Theorem 7.2 is given at the end of Subsection 7.3.

7.1. **Surface series.** In this subsection, we prove estimates for linear series on $\mathbb{P}^1\times\mathbb{P}^1$ and on $\mathbb{F}_2$ with boundary $\frac{1}{2}B$ and base conditions along a curvilinear subscheme. The statements concern surfaces only, and they correspond to the possible positions of the base conditions on $\mathbb{F}_0$ and on $\mathbb{F}_2$.

Definition 7.3. Let S be a smooth projective surface and $B \subset S$ a reduced curve, and for integers $m \geq 1$ and $0 \leq k \leq k_m$ let $R_{m,k}$ be a subspace of the space of sections of a line bundle $\mathcal{L}_{m,k}$ on S. We set $n_m := \sum_k \dim R_{m,k}$. For a prime divisor v over S, let $S(R;v)$ be the lim sup as $m \to +\infty$ of the sum of the jumping numbers (Definition 2.8) of v on all $R_{m,k}$, divided by mn_m, and let $A_{S,B/2}(v) := A_S(v) - \frac{1}{2}\operatorname{mult}_v B$. For a closed point $q \in S$, the *local threshold* of R at q is $\delta_q(S, B/2; R) := \inf\{A_{S,B/2}(v)/S(R;v) : q \in \operatorname{center}_S v\}$. Let $h : S' \to S$ be either the identity or a projective birational morphism from a normal surface S' whose exceptional locus is one curve lying over a closed point of S, for instance the blow-up of the point or a weighted blow-up, and let $C \subset S'$ be a normal curve, equal to the exceptional curve of h in the second case. When C is not Cartier, we assume that (S', C) is plt along C, as for the exceptional curve of a weighted blow-up of a smooth point. In the second case each $R_{m,k}$ is replaced by the pullback of $R_{m,k}$ to S'. For $j \geq 0$ let

$$R^C_{m,k,j} := \operatorname{Im}\big(\{r \in R_{m,k} : \operatorname{ord}_C(r) \geq j\} \to H^0(C, (h^*\mathcal{L}_{m,k} - jC)|_C)\big),$$

where $(h^*\mathcal{L}_{m,k} - jC)|_C$ is the $\mathbb{Q}$-divisor class of [AZ22, Lemma 2.1] when C is not Cartier. For a closed point $q \in C$, the *secondary expected vanishing order* $S(R^C; q)$ is the lim sup as $m \to +\infty$ of the sum, over all k and j, of the jumping numbers of ord_q on $R^C_{m,k,j}$, divided by mn_m, where the order at q of a section is the multiplicity at q of the associated $\mathbb{Q}$-divisor on C. We write $S_m(R;v)$ and $S_m(R^C;q)$ for the quotients before the lim sup is taken.

In Subsections 7.2 and 7.3, S is the surface P of Construction-Proposition 7.1, the boundary $\frac{1}{2}B$ is the different $\operatorname{Diff}_P(0)$, and the base conditions of R come from the lines of X intersecting L. The refinements of the anti-canonical series of X by P are compared with such series R in Proposition 7.11 and Theorem 7.2.

Lemma 7.4. *Notation as in Definition 7.3, with $h : S' \to S$ and $C \subset S'$ as there. Let B'_C be the strict transform of B on S' with the component C removed if C is a component of B, let $q \in C$ be a closed point such that $(S', \frac{1}{2}B'_C + C)$ is plt near q, and let Δ_C be the different of $(S', \frac{1}{2}B'_C + C)$ on C. Let $\lambda > 0$ be a real number such that*

$$\lambda S(R; \operatorname{ord}_C) \leq A_{S,B/2}(\operatorname{ord}_C) \qquad \text{and} \qquad \lambda S(R^C; q) \leq 1 - \operatorname{mult}_q \Delta_C.$$

Then $A_{S,B/2}(v) \geq \lambda S(R;v)$ for any prime divisor v over S with $q \in \operatorname{center}_{S'} v$. In particular, if h is the identity, then $\delta_q(S, B/2; R) \geq \lambda$.

Proof. Fix a real number $\lambda' \in (0, \lambda)$ and a prime divisor v over S with $q \in \operatorname{center}_{S'} v$. For each m and k, let $(r_{k,i})_i$ be a basis of $R_{m,k}$ compatible with both v and ord_C [AZ22, Lemma 3.1], and let $D_m := \frac{1}{mn_m}\sum_{k,i} \operatorname{div}(r_{k,i})$. Then $\operatorname{mult}_v D_m = S_m(R;v)$, and

$$h^*D_m = S_m(R; \operatorname{ord}_C)C + \Gamma_m,$$

where $\Gamma_m \geq 0$ is a $\mathbb{Q}$-divisor whose support does not contain C. Since $(r_{k,i})_i$ is compatible with ord_C, the elements $r_{k,i}$ with $\operatorname{ord}_C(r_{k,i}) = j$ map to a basis of $R^C_{m,k,j}$, and the image of such an $r_{k,i}$ has divisor $(h^*\operatorname{div}(r_{k,i}) - jC)|_C$. For any basis of a space of sections, the sum of the orders at q is at most the sum of the jumping numbers of ord_q. Hence

$$\operatorname{mult}_q(\Gamma_m|_C) \leq S_m(R^C; q).$$

Since $(S', \frac{1}{2}B'_C + C)$ is plt near q, the pair (C, Δ_C) is klt near q [Kol13, Theorem 4.9], that is, $\operatorname{mult}_q \Delta_C < 1$, so $\lambda' S(R^C; q) < 1 - \operatorname{mult}_q \Delta_C$ as $\lambda' < \lambda$. Thus there exists an integer m_0, independent of v, such that $\lambda' \operatorname{mult}_q(\Gamma_m|_C) < 1 - \operatorname{mult}_q \Delta_C$ for any $m \geq m_0$.

Let $m \geq m_0$. The different of $(S', \frac{1}{2}B'_C + C + \lambda'\Gamma_m)$ on C is $\Delta_C + \lambda'\Gamma_m|_C$, whose coefficients are smaller than 1 on a neighborhood of q in C. As $(S', \frac{1}{2}B'_C + C)$ is plt near q, the surface S' has quotient singularities near q, so any Weil divisor on S' is $\mathbb{Q}$-Cartier near q. By inversion of

adjunction [Kol13, Theorem 4.9(1)], the pair $(S', \frac{1}{2}B'_C + C + \lambda'\Gamma_m)$ is plt near q. On the other hand,

$$h^*\big(K_S + \tfrac{1}{2}B\big) = K_{S'} + \tfrac{1}{2}B'_C + \big(1 - A_{S,B/2}(\mathrm{ord}_C)\big)C,$$

so with $c_m := 1 - A_{S,B/2}(\mathrm{ord}_C) + \lambda' S_m(R; \mathrm{ord}_C)$ we have

$$h^*\big(K_S + \tfrac{1}{2}B + \lambda' D_m\big) = K_{S'} + \tfrac{1}{2}B'_C + c_m C + \lambda'\Gamma_m.$$

Since $q \in \mathrm{center}_{S'} v$ and $(S', \frac{1}{2}B'_C + C + \lambda'\Gamma_m)$ is plt near q, we obtain

$$\begin{aligned} A_{S,B/2}(v) - \lambda' S_m(R; v) &= a\big(v, S', \tfrac{1}{2}B'_C + C + \lambda'\Gamma_m\big) - (c_m - 1)\,\mathrm{mult}_v C \\ &\geq -(c_m - 1)_+\,\mathrm{mult}_v C. \end{aligned}$$

By assumption $\lambda' S(R; \mathrm{ord}_C) \leq \lambda S(R; \mathrm{ord}_C) \leq A_{S,B/2}(\mathrm{ord}_C)$, so $\limsup_m c_m \leq 1$ and $(c_m - 1)_+ \to 0$. Taking the lim sup as $m \to +\infty$ implies $A_{S,B/2}(v) \geq \lambda' S(R; v)$. Letting $\lambda' \to \lambda$, we obtain $A_{S,B/2}(v) \geq \lambda S(R; v)$. If h is the identity, then $\mathrm{center}_{S'} v = \mathrm{center}_S v$, so $\delta_q(S, B/2; R) \geq \lambda$. □

Lemma 7.5. *Let $S := \mathbb{P}^1 \times \mathbb{P}^1$ with ruling classes σ and f, and let $B \subset S$ be a smooth curve of class $\sigma + 2f$. The ruling with fibers of class σ restricts to a double cover $B \to \mathbb{P}^1$, with involution ι. Let $Z \subset B$ be a divisor of degree three, viewed as a curvilinear subscheme of S, such that both ruling projections restrict to closed immersions on Z. For integers $m \geq 1$ and $0 \leq k \leq 2m$ let $R_{m,k} := H^0(S, \mathcal{O}_S(k\sigma + mf) \otimes I_Z^{\max(k-m,0)})$, viewed as a space of sections of $\mathcal{O}_S(k\sigma + mf)$, and let n_m, $S(R; v)$, $A_{S,B/2}(v)$, and $\delta_q(S, B/2; R)$ be as in Definition 7.3. Then $n_m = \frac{3}{2}m^3 + O(m^2)$, and $S(R; v) \leq A_{S,B/2}(v)$ for any prime divisor v over S whose center is not contained in* $\mathrm{Supp}\, Z$. *More precisely, $S(R; \mathrm{ord}_B) = 5/24$, and for a closed point $q \in S$ the local threshold $\delta_q(S, B/2; R)$ is at least $9/8$ if $q \notin B$, at least $72/71$ if $q \in B \setminus (\mathrm{Supp}\, Z \cup \mathrm{Supp}\, \iota Z)$, and at least 1 if $q \in B \setminus \mathrm{Supp}\, Z$.*

Proof. The proof is long, so we divide it into several steps. Let $p_1, p_2 : S \to \mathbb{P}^1$ be the projections whose fibers have classes σ and f, respectively. Then $(B \cdot \sigma) = 2$ and $(B \cdot f) = 1$, so $p_1|_B$ is the double cover with involution ι and $p_2|_B$ is an isomorphism.

Step 1. In this step, we resolve Z, describe $R_{m,k}$ on the resolution, and compute n_m.

Let $b : \tilde{S} \to S$ be the composition of three point blow-ups defined as follows: over a point $z \in \mathrm{Supp}\, Z$ of multiplicity r in Z, we blow up z and then, $r - 1$ times, the point of the last exceptional curve on the strict transform of B. Let e_1, e_2, e_3 be the total transforms of the exceptional curves, numbered so that those over one point of $\mathrm{Supp}\, Z$ are consecutive, and let $e := e_1 + e_2 + e_3$. We write σ and f also for $b^*\sigma$ and b^*f. Then $(e_i \cdot e_j) = -\delta_{ij}$, $(\sigma \cdot e_i) = (f \cdot e_i) = 0$, and the strict transform $\tilde{B}$ of B satisfies $\tilde{B} \sim \sigma + 2f - e$. Under the isomorphism $\tilde{B} \cong B$ we have $e|_{\tilde{B}} = Z$.

Let $z \in \mathrm{Supp}\, Z$ have multiplicity r, and let (y, t) be local parameters at z with $B = (y = 0)$, so that $I_Z = (y, t^r)$ near z. For $1 \leq j \leq r$, the order of vanishing along the j-th exceptional curve over z is the monomial valuation μ_j with $\mu_j(y) = j$ and $\mu_j(t) = 1$, and the coefficient of the j-th exceptional curve over z in e is j. Hence, for an integer $c \geq 0$, the ideal $b_*\mathcal{O}_{\tilde{S}}(-ce)$ is generated near z by the monomials $y^\alpha t^\beta$ with $j\alpha + \beta \geq jc$ for $1 \leq j \leq r$, that is, with $r\alpha + \beta \geq rc$, and the monomials with $r\alpha + \beta \geq rc$ generate $(y, t^r)^c$. Thus $b_*\mathcal{O}_{\tilde{S}}(-ce) = I_Z^c$, and

$$R_{m,k} = H^0\big(\tilde{S}, \mathcal{O}_{\tilde{S}}(k\sigma + mf - (k - m)_+ e)\big). \tag{7.1}$$

A computation in the charts of the blow-ups also shows $I_Z \cdot \mathcal{O}_{\tilde{S}} = \mathcal{O}_{\tilde{S}}(-e)$.

We show that $M_1 := 2\sigma + f - e$ and $M_2 := \sigma + 2f - e$ are base point free. Since $p_1^*\mathcal{O}(1) = \mathcal{O}_S(\sigma)$ and $(f \cdot \sigma) = 1$, we have $p_{1*}\mathcal{O}_S(2\sigma + f) \cong \mathcal{O}(2)^{\oplus 2}$. As $p_1|_Z$ is a closed immersion, the evaluation map $p_{1*}\mathcal{O}_S(2\sigma + f) \to p_{1*}(\mathcal{O}_Z \otimes \mathcal{O}_S(2\sigma + f))$ is surjective, and $p_{1*}(\mathcal{O}_Z \otimes \mathcal{O}_S(2\sigma + f))$ has length three. Hence the kernel $\mathcal{K} = p_{1*}(I_Z \otimes \mathcal{O}_S(2\sigma + f))$ is a vector bundle of rank two and degree one. Write $\mathcal{K} \cong \mathcal{O}(i) \oplus \mathcal{O}(1 - i)$ with $i \geq 1$. The inclusion $\mathcal{K} \subset \mathcal{O}(2)^{\oplus 2}$ implies $i \leq 2$. If $i = 2$, then the inclusion of the summand $\mathcal{O}(2)$ into $\mathcal{O}(2)^{\oplus 2}$ is given by a constant vector, so there exists

a non-zero $l \in H^0(S, \mathcal{O}_S(f))$ such that $p_1^*\phi \cdot l$ vanishes on Z for any $\phi \in H^0(\mathbb{P}^1, \mathcal{O}(2))$. Thus l vanishes on Z, and Z is contained in a fiber of p_2, which contradicts the assumption on $p_2|_Z$. Therefore $\mathcal{K} \cong \mathcal{O}(1) \oplus \mathcal{O}$ is globally generated. Near a point of $p_1(Z)$, let t be a coordinate on the base and y an affine coordinate on the fibers of p_1. Since $p_1|_Z$ is a closed immersion, $I_Z = (y - g_1(t), g_2(t))$ for functions g_1 and g_2, and both generators are local sections of $\mathcal{K}$. Hence $I_Z \otimes \mathcal{O}_S(2\sigma + f)$ is generated by $H^0(S, I_Z \otimes \mathcal{O}_S(2\sigma + f))$, and since $I_Z \cdot \mathcal{O}_{\tilde{S}} = \mathcal{O}_{\tilde{S}}(-e)$, the divisor M_1 is base point free. Interchanging p_1 and p_2 shows that M_2 is base point free.

Thus $-K_{\tilde{S}} = 2\sigma + 2f - e = M_1 + f$ is nef and big, with $(-K_{\tilde{S}})^2 = 5$. Let $D = a_1\sigma + a_2 f - ce$ with real numbers $c \geq 0$ and

$$a_1 \geq c, \qquad a_2 \geq c, \qquad a_1 + a_2 \geq 3c. \tag{7.2}$$

Then there exists $\theta \in [0, 1]$ with $a_1 \geq c(1 + \theta)$ and $a_2 \geq c(2 - \theta)$, and

$$D = c\big(\theta M_1 + (1 - \theta)M_2\big) + \big(a_1 - c(1 + \theta)\big)\sigma + \big(a_2 - c(2 - \theta)\big)f$$

is nef. If moreover a_1, a_2, and c are integers, then $D - K_{\tilde{S}}$ is nef and big, so $H^i(\tilde{S}, \mathcal{O}_{\tilde{S}}(D)) = 0$ for $i > 0$ by Kawamata–Viehweg vanishing [KM98, Theorem 2.64] combined with Serre duality, and by Riemann–Roch [Har77, Theorem V.1.6],

$$h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(D)) = \tfrac{1}{2}\big(D \cdot (D - K_{\tilde{S}})\big) + 1. \tag{7.3}$$

For $k \leq m$, (7.1) implies $\dim R_{m,k} = (k + 1)(m + 1)$. For $m \leq k \leq 2m$, the divisor $k\sigma + mf - (k - m)e$ satisfies (7.2), and (7.3) implies $\dim R_{m,k} = km - \frac{3}{2}(k - m)^2 + O(m)$. Thus

$$n_m = m^3\Big(\int_0^1 a\,da + \int_1^2 \big(a - \tfrac{3}{2}(a - 1)^2\big)da\Big) + O(m^2) = \frac{3}{2}m^3 + O(m^2).$$

Step 2. In this step, we compute the refinement of R by ord_B, the value $S(R; \mathrm{ord}_B)$, and an upper bound for $S(R^B; q)$ at points $q \in B \setminus \operatorname{Supp} Z$.

Let s_B be a section defining B, and let $0 \leq k \leq 2m$ and $\ell \geq 0$. Near a point of $\operatorname{Supp} Z$, with (y, t) and r as in **Step 1**, a monomial computation shows $\big((y, t^r)^c : y^\ell\big) = (y, t^r)^{(c-\ell)_+}$ for any integer $c \geq 0$. Hence

$$\{\phi \in R_{m,k} : \mathrm{ord}_B(\phi) \geq \ell\} = s_B^\ell \cdot H^0\big(S, \mathcal{O}_S((k - \ell)\sigma + (m - 2\ell)f) \otimes I_Z^{(k-m-\ell)_+}\big).$$

For real numbers a and u let

$$D_{a,u} := (a - u)\sigma + (1 - 2u)f - (a - 1 - u)_+ e.$$

With $a = k/m$ and $u = \ell/m$, the $\mathbb{Q}$-divisor $mD_{a,u}$ is integral, and since $b_*\mathcal{O}_{\tilde{S}}(-ce) = I_Z^c$ for $c = (k - m - \ell)_+$,

$$\dim\{\phi \in R_{m,k} : \mathrm{ord}_B(\phi) \geq \ell\} = h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{a,u})).$$

Thus $\{\phi \in R_{m,k} : \mathrm{ord}_B(\phi) \geq \ell\} \neq 0$ only if (a, u) is contained in $\Theta := \{(a, u) : 0 \leq a \leq 2,\ 0 \leq u \leq \min\{a, \frac{1}{2}\}\}$. We divide Θ into the three chambers of Table 5. In the first chamber $D_{a,u} = (a - u)\sigma + (1 - 2u)f$ is nef, and in the second chamber $D_{a,u}$ satisfies (7.2) with $c = a - 1 - u$, since $a_2 \geq c$ is equivalent to $a + u \leq 2$ and the other two inequalities hold on Θ.

Consider the third chamber $a + u \geq 2$, so that $a \geq \frac{3}{2}$ and $u \leq \frac{1}{2} \leq a - 1$. Set

$$D_{a,u}^+ := (1 - 2u)M_1 + (4 - 2a)\sigma, \qquad D_{a,u}^- := (a + u - 2)(3\sigma - e),$$

so that $D_{a,u} = D_{a,u}^+ + D_{a,u}^-$. Let $z \in \operatorname{Supp} Z$ have multiplicity r, let $e_{i+1}, \ldots, e_{i+r}$ be the classes over z, and let c_0 be the strict transform of the fiber of p_1 through z. If $r \geq 2$, the fiber is transverse to B at z, since otherwise the intersection of the fiber with Z would have length at least two, which contradicts the assumption on $p_1|_Z$. If $r = 1$, only z is blown up over z, and the fiber may be tangent to B at z, that is, z may be a branch point of $p_1|_B$. In both cases $c_0 \sim \sigma - e_{i+1}$, and with $d_j := e_{i+j} - e_{i+j+1}$ for $1 \leq j \leq r - 1$, the curves $c_0, d_1, \ldots, d_{r-1}$ form a chain with

$$c_0^2 = -1, \qquad d_j^2 = -2, \qquad (c_0 \cdot d_1) = (d_j \cdot d_{j+1}) = 1,$$

and

$$r\sigma - \sum_{j=1}^{r} e_{i+j} = rc_0 + (r-1)d_1 + \cdots + d_{r-1}.$$

The chains over distinct points of $\operatorname{Supp} Z$ are disjoint, because $p_1|_Z$ is a closed immersion. Thus $D^-_{a,u} \geq 0$, and the intersection matrix of the components of $\operatorname{Supp} D^-_{a,u}$ is negative definite, since contracting c_0, then the image of d_1, and so on contracts each chain. Moreover

$$(M_1 \cdot c_0) = (M_1 \cdot d_j) = (\sigma \cdot c_0) = (\sigma \cdot d_j) = 0,$$

and $D^+_{a,u}$ is nef because M_1 is base point free. Thus $D_{a,u} = D^+_{a,u} + D^-_{a,u}$ is the Zariski decomposition [Laz04a, Theorem 2.3.19]. In the first two chambers we set $D^+_{a,u} := D_{a,u}$ and $D^-_{a,u} := 0$.

For $(a,u) = (k/m, \ell/m)$, both $mD^+_{a,u}$ and $mD^-_{a,u}$ are integral, so $h^0(\tilde{S}, mD_{a,u}) = h^0(\tilde{S}, \lfloor mD^+_{a,u} \rfloor) = h^0(\tilde{S}, mD^+_{a,u})$ [Laz04a, Proposition 2.3.21]. By (7.3),

$$\dim\{\phi \in R_{m,k} : \operatorname{ord}_B(\phi) \geq \ell\} = \frac{m^2}{2}\operatorname{vol}(D_{a,u}) + O(m), \qquad \operatorname{vol}(D_{a,u}) = (D^+_{a,u})^2,$$

uniformly on Θ. The sums over k and ℓ below are Riemann sums of continuous piecewise polynomial functions on Θ, and replacing them by integrals changes the quotients by mn_m by $O(1/m)$. Since the sum of the jumping numbers of ord_B on $R_{m,k}$ is $\sum_{\ell \geq 1} \dim\{\phi \in R_{m,k} : \operatorname{ord}_B(\phi) \geq \ell\}$, **Step 1** implies

$$S(R; \operatorname{ord}_B) = \frac{2}{3}\iint_\Theta \tfrac{1}{2}\operatorname{vol}(D_{a,u})\,du\,da = \frac{2}{3}\cdot\frac{5}{16} = \frac{5}{24}, \tag{7.4}$$

where the integral is a direct calculation from the formulas for $D^+_{a,u}$ in Table 5, using $M_1^2 = 1$, $(M_1 \cdot \sigma) = 1$, and $\sigma^2 = 0$.

We now describe $R^B_{m,k,\ell}$. A non-zero element of $R^B_{m,k,\ell}$ is the restriction $\phi'|_B$ of a section ϕ' of $\mathcal{O}_S((k-\ell)\sigma + (m-2\ell)f) \otimes I_Z^{(k-m-\ell)_+}$ with $B \not\subset \operatorname{div}(\phi')$. Since $I_Z \cdot \mathcal{O}_{\tilde{S}} = \mathcal{O}_{\tilde{S}}(-e)$, we can write $b^* \operatorname{div}(\phi') = D' + (k-m-\ell)_+ e$ with $D' \in |mD_{a,u}|$. Since $e|_{\tilde{B}} = Z$, we obtain

$$\operatorname{div}(\phi'|_B) = D'|_{\tilde{B}} + (k-m-\ell)_+ Z.$$

By [Laz04a, Proposition 2.3.21], $D' \geq mD^-_{a,u}$. The curve c_0 is the strict transform of the fiber of p_1 through z, which intersects B in the divisor $z + \iota(z)$. If the fiber is transverse to B at z, which holds if $r \geq 2$, then $z \neq \iota(z)$ and the intersection at z is removed by the first blow-up over z. Otherwise $r = 1$, the point $z = \iota(z)$ is a branch point, and one blow-up lowers the intersection multiplicity at z from 2 to 1. In both cases $c_0|_{\tilde{B}} = \iota(z)$, while $(d_j \cdot \tilde{B}) = 0$. Thus $(3\sigma - e)|_{\tilde{B}} = \iota Z$, and any non-zero element of $R^B_{m,k,\ell}$ vanishes on the divisor

$$F_{k,\ell} := (k-m-\ell)_+ Z + (k+\ell-2m)_+ \iota Z.$$

The degree of $(k-\ell)\sigma + (m-2\ell)f$ on B is $2k+m-4\ell$. Hence $R^B_{m,k,\ell}$ is isomorphic, by division by a section defining $F_{k,\ell}$, to a subspace of $H^0(\mathbb{P}^1, \mathcal{O}(mn(a,u)))$, where

$$n(a,u) := 2a + 1 - 4u - 3(a-1-u)_+ - 3(a+u-2)_+$$

is given in Table 5.

TABLE 5. The Zariski decomposition of $D_{a,u}$ and the degree $n(a,u)$ of the moving part on B in the proof of Lemma 7.5.

chamber of Θ	$D^+_{a,u}$	$D^-_{a,u}$	$n(a,u)$
$u \geq a-1$	$(a-u)\sigma + (1-2u)f$	0	$2a+1-4u$
$u \leq a-1$, $a+u \leq 2$	$D_{a,u}$	0	$4-a-u$
$a+u \geq 2$	$(1-2u)M_1 + (4-2a)\sigma$	$(a+u-2)(3\sigma - e)$	$10-4a-4u$

For a subspace of $H^0(\mathbb{P}^1, \mathcal{O}(d))$, the sum of the jumping numbers of the order at a point is at most $0 + 1 + \cdots + d = \frac{d(d+1)}{2}$. Let $q \in B \setminus \operatorname{Supp} Z$, so that $\operatorname{mult}_q F_{k,\ell} = (k + \ell - 2m)_+ \operatorname{mult}_q \iota Z$. Then the sum of the jumping numbers of ord_q on $R^B_{m,k,\ell}$ is at most

$$\big(mn(a,u)+1\big)(k+\ell-2m)_+ \operatorname{mult}_q \iota Z + \tfrac{1}{2} mn(a,u)\big(mn(a,u)+1\big).$$

A direct calculation shows

$$\iint_\Theta \tfrac{1}{2} n(a,u)^2 du da = \frac{71}{48} \qquad \text{and} \qquad \iint_\Theta (a+u-2)_+ n(a,u) du da = \frac{1}{48}. \tag{7.5}$$

Summing over k and ℓ and dividing by $mn_m = \frac{3}{2}m^4 + O(m^3)$, we obtain

$$S(R^B; q) \le \frac{2}{3}\Big(\frac{71}{48} + \frac{\operatorname{mult}_q \iota Z}{48}\Big) = \frac{71}{72} + \frac{\operatorname{mult}_q \iota Z}{72} \qquad \text{for } q \in B \setminus \operatorname{Supp} Z. \tag{7.6}$$

Step 3. In this step, we treat the points $q \in B \setminus \operatorname{Supp} Z$ with $\operatorname{mult}_q \iota Z \le 1$.

In Lemma 7.4, let h be the identity and $C = B$. Then $B'_C = 0$, the pair (S, B) is plt since B is smooth, and $\Delta_C = 0$. Moreover $A_{S,B/2}(\operatorname{ord}_B) = \frac{1}{2}$. If $q \notin \operatorname{Supp} \iota Z$, let $\lambda := \frac{72}{71}$. Then (7.4) and (7.6) imply

$$\lambda S(R; \operatorname{ord}_B) = \frac{72}{71} \cdot \frac{5}{24} \le \frac{1}{2} \qquad \text{and} \qquad \lambda S(R^B; q) \le \frac{72}{71} \cdot \frac{71}{72} = 1.$$

Thus $\delta_q(S, B/2; R) \ge \frac{72}{71}$ by Lemma 7.4. If $\operatorname{mult}_q \iota Z = 1$, then (7.6) implies $S(R^B; q) \le 1$, and since $S(R; \operatorname{ord}_B) \le \frac{1}{2}$, Lemma 7.4 with $\lambda = 1$ implies $\delta_q(S, B/2; R) \ge 1$.

Step 4. In this step, we treat the points $q \in B \setminus \operatorname{Supp} Z$ with $r := \operatorname{mult}_q \iota Z \ge 2$.

Let $q' := \iota(q)$, so that $q' \in \operatorname{Supp} Z$ has multiplicity r in Z. Since $r \ge 2$, the point q' is not a branch point of $p_1|_B$, so $q \ne q'$. Let C be the fiber of p_1 through q and q'. Since $(B \cdot \sigma) = 2$, the curve C is transverse to B at q and at q'.

First, $R_{m,k} \subset H^0(S, \mathcal{O}_S(k\sigma + mf))$, and the sum of the jumping numbers of ord_C on $H^0(S, \mathcal{O}_S(k\sigma + mf))$ is $(m+1)\frac{k(k+1)}{2}$. Since passing to a subspace does not increase the sum of the jumping numbers,

$$S(R; \operatorname{ord}_C) \le \lim_{m \to +\infty} \frac{1}{mn_m} \sum_{k=0}^{2m} (m+1)\frac{k(k+1)}{2} = \frac{2}{3} \cdot \frac{8}{6} = \frac{8}{9}.$$

Next, the restriction of $k\sigma + mf$ to C has degree m, and $R^C_{m,k,\ell} = 0$ for $\ell > k$. Let $k \ge m$. Near q', let t be the pullback of a coordinate on the base of p_1 and y a local parameter with $B = (y = 0)$, so that $C = (t = 0)$, $y|_C$ is a local parameter of C at q', and $I_Z = (y, t^r)$. The ideal $(y, t^r)^{k-m}$ is generated by the monomials $y^{k-m-i}t^{ri}$, so the colon ideal $\big((y, t^r)^{k-m} : t^\ell\big)$ consists of the functions whose monomials $y^\alpha t^\beta$ satisfy $r\alpha + \beta + \ell \ge r(k-m)$. Restricting to $C = (t = 0)$, we find that any element of $R^C_{m,k,\ell}$ vanishes at q' to order at least

$$\nu := \max\big(k - m - \lfloor \ell/r \rfloor, 0\big) \ge \max\big(k - m - \tfrac{\ell}{2}, 0\big).$$

Since $q \ne q'$, the sum of the jumping numbers of ord_q on $R^C_{m,k,\ell}$ is at most $\frac{1}{2}(m-\nu)(m-\nu+1)$. For $a = k/m \in [1, 2]$ and $u = \ell/m \in [0, a]$, let

$$d(a,u) := 1 - \max\big(a - 1 - \tfrac{u}{2}, 0\big),$$

so that $m - \nu \le m d(a,u)$. Since $2(a-1) \le a$,

$$\int_0^a d(a,u)^2 du = a - 2(a-1)^2 + \tfrac{2}{3}(a-1)^3 \qquad \text{and} \qquad \int_1^2 \int_0^a d(a,u)^2 du da = 1.$$

For $k \leq m$, the sum over ℓ is at most $(k+1)\frac{m(m+1)}{2}$, and $\int_0^1 ada = \frac{1}{2}$. Summing over k and ℓ and dividing by mn_m, we obtain

$$S(R^C; q) \leq \frac{2}{3} \cdot \frac{1}{2}\Big(\frac{1}{2} + 1\Big) = \frac{1}{2}.$$

Near q, the pair $(S, \frac{1}{2}B + C)$ is plt, and since B is transverse to C at q, the different Δ_C satisfies $\operatorname{mult}_q \Delta_C = \frac{1}{2}$. As $A_{S,B/2}(\operatorname{ord}_C) = 1$, we have

$$S(R; \operatorname{ord}_C) \leq \frac{8}{9} \leq 1 \qquad \text{and} \qquad S(R^C; q) \leq \frac{1}{2} = 1 - \operatorname{mult}_q \Delta_C.$$

Thus $\delta_q(S, B/2; R) \geq 1$ by Lemma 7.4 with $\lambda = 1$.

Step 5. In this step, we treat the points $q \notin B$ and finish the proof.

Let $q \in S \backslash B$, and let C be the fiber of p_2 through q, of class f. The sum of the jumping numbers of ord_C on $H^0(S, \mathcal{O}_S(k\sigma + mf))$ is $(k+1)\frac{m(m+1)}{2}$, so as in **Step 4**,

$$S(R; \operatorname{ord}_C) \leq \lim_{m \to +\infty} \frac{1}{mn_m} \sum_{k=0}^{2m} (k+1)\frac{m(m+1)}{2} = \frac{2}{3}.$$

The restriction of $k\sigma + mf$ to C has degree k, and $R^C_{m,k,\ell} = 0$ for $\ell > m$. Thus the sum over ℓ of the sums of the jumping numbers of ord_q on $R^C_{m,k,\ell}$ is at most $(m+1)\frac{k(k+1)}{2}$, and

$$S(R^C; q) \leq \lim_{m \to +\infty} \frac{1}{mn_m} \sum_{k=0}^{2m} (m+1)\frac{k(k+1)}{2} = \frac{8}{9}.$$

Near q, the curve B is absent, so $(S, \frac{1}{2}B + C)$ is plt near q and $\Delta_C = 0$ near q. With $\lambda := \frac{9}{8}$,

$$\lambda S(R; \operatorname{ord}_C) \leq \frac{9}{8} \cdot \frac{2}{3} < 1 = A_{S,B/2}(\operatorname{ord}_C) \qquad \text{and} \qquad \lambda S(R^C; q) \leq \frac{9}{8} \cdot \frac{8}{9} = 1.$$

Thus $\delta_q(S, B/2; R) \geq \frac{9}{8}$ by Lemma 7.4.

Finally, let v be a prime divisor over S whose center is not contained in $\operatorname{Supp} Z$. Then $\operatorname{center}_S v$ contains a closed point $q \notin \operatorname{Supp} Z$, and **Steps 3, 4, and 5** imply $\delta_q(S, B/2; R) \geq 1$. Hence $S(R; v) \leq A_{S,B/2}(v)$. Together with **Step 1** and (7.4), this proves the lemma. □

We now turn to the surface series on $\mathbb{F}_2$. The following lemma reduces the invariants of Definition 7.3 to integrals of piecewise affine functions over polygons.

Lemma 7.6. *Notation as in Definition 7.3, with $h : S' \to S$ and $C \subset S'$ as there. Let $\gamma > 0$ and $c > 0$ be real numbers, let $\Omega \subset \mathbb{R}^2$ be the union of compact convex polygons $\Omega_1, \dots, \Omega_N$ with pairwise disjoint interiors, let $\varphi, \mu_1, \dots, \mu_r$ be non-negative functions on Ω which are affine on the interior of each Ω_l, and let $q_1, \dots, q_r \in C$ be pairwise distinct closed points. Assume that the following hold.*

(1) S' is smooth, $C \cong \mathbb{P}^1$, $n_m = \gamma m^3 + O(m^2)$, and $\int_\Omega \varphi dadt = \gamma$.

(2) If $R^C_{m,k,j} \neq 0$, then $(k/m, j/m) \in \Omega$ and $\deg((h^*\mathcal{L}_{m,k} - jC)|_C) \leq cm$.

(3) If $(a,t) := (k/m, j/m)$ is contained in the interior of one of $\Omega_1, \dots, \Omega_N$, then

$$\deg((h^*\mathcal{L}_{m,k} - jC)|_C) \leq m\Big(\varphi(a,t) + \sum_{i=1}^r \mu_i(a,t)\Big) + c,$$

and the divisor of any non-zero element of $R^C_{m,k,j}$ is at least $\sum_{i=1}^r m\mu_i(a,t)q_i$.

Then

$$S(R; \operatorname{ord}_C) = \frac{1}{\gamma}\int_\Omega t\varphi dadt \qquad \text{and} \qquad S(R^C; q_i) \leq \frac{1}{\gamma}\int_\Omega \Big(\frac{\varphi^2}{2} + \varphi\mu_i\Big)dadt \quad \text{for } 1 \leq i \leq r,$$

and $S(R^C; q) \leq \frac{1}{\gamma}\int_\Omega \frac{\varphi^2}{2} dadt$ for any closed point $q \in C \backslash \{q_1, \dots, q_r\}$.

Proof. Fix m, and for $0 \le k \le k_m$ and $j \ge 0$ let $\rho_{k,j} := \dim R^C_{m,k,j}$. Since C is Cartier, the kernel of the map $\{s \in R_{m,k} : \mathrm{ord}_C(s) \ge j\} \to R^C_{m,k,j}$ is $\{s \in R_{m,k} : \mathrm{ord}_C(s) \ge j+1\}$. Thus $\rho_{k,j}$ is the multiplicity of j as a jumping number of ord_C on $R_{m,k}$. Hence $\sum_{k,j} \rho_{k,j} = n_m$, and the sum of the jumping numbers of ord_C on all $R_{m,k}$ is $\sum_{k,j} j\rho_{k,j}$.

Let Λ°_m be the set of pairs (k,j) such that $(k/m, j/m)$ is contained in the interior of one of $\Omega_1, \dots, \Omega_N$, and let Λ^∂_m be the set of the other pairs (k,j) with $(k/m, j/m) \in \Omega$. Since $\Omega_1, \dots, \Omega_N$ are compact convex polygons, $|\Lambda^\partial_m| = O(m)$, there exists $t_0 > 0$ with $t \le t_0$ on Ω, and

$$\sum_{(k,j)\in\Lambda^\circ_m} F(k/m, j/m) = m^2 \int_\Omega F da dt + O(m) \tag{7.7}$$

for any bounded function F on Ω which is a polynomial of degree at most two on the interior of each Ω_l.

Let $(k,j) \in \Lambda^\circ_m$, $(a,t) := (k/m, j/m)$, and $\delta := \deg((h^*\mathcal{L}_{m,k} - jC)|_C)$. By (3), $R^C_{m,k,j}$ is contained in the space of sections of a line bundle of degree δ on $C \cong \mathbb{P}^1$ vanishing to order at least $m\mu_i(a,t)$ at q_i for $1 \le i \le r$. Thus, if $R^C_{m,k,j} \ne 0$, then $\rho_{k,j} \le \delta - \sum_{i=1}^r m\mu_i(a,t) + 1$, and in any case

$$\rho_{k,j} \le m\varphi(a,t) + c + 1. \tag{7.8}$$

Let $q \in C$ be a closed point, and let $\mu_q := \mu_i$ if $q = q_i$ and $\mu_q := 0$ if $q \notin \{q_1, \dots, q_r\}$. On $C \cong \mathbb{P}^1$, the sections vanishing to order at least $p+1$ at q form a subspace of codimension at most one in the sections vanishing to order at least p. Thus the jumping numbers of ord_q on $R^C_{m,k,j}$ are $\rho_{k,j}$ pairwise distinct integers, and by (3) each of them is at most $\delta - \sum_{q_i \ne q} m\mu_i(a,t) \le m(\varphi + \mu_q)(a,t) + c$. By (7.8), their sum is at most

$$\sum_{p=0}^{\rho_{k,j}-1} \big(m(\varphi+\mu_q)(a,t) + c - p\big) \le m^2\Big(\frac{\varphi^2}{2} + \varphi\mu_q\Big)(a,t) + c'm,$$

where c' depends only on c and the bounds of $\varphi, \mu_1, \dots, \mu_r$. For $(k,j) \in \Lambda^\partial_m$, (2) implies $\rho_{k,j} \le cm + 1$, and all jumping numbers of ord_C and of ord_q involved are at most $\max\{c, t_0\}m$. By (2), $\rho_{k,j} = 0$ if $(k,j) \notin \Lambda^\circ_m \cup \Lambda^\partial_m$.

By (7.8), the numbers $\epsilon_{k,j} := m\varphi(k/m, j/m) + c + 1 - \rho_{k,j}$, $(k,j) \in \Lambda^\circ_m$, are non-negative, and by (7.7) and (1),

$$\sum_{(k,j)\in\Lambda^\circ_m} \epsilon_{k,j} = \gamma m^3 - n_m + \sum_{(k,j)\in\Lambda^\partial_m} \rho_{k,j} + O(m^2) = O(m^2).$$

Since $j \le t_0 m$, we obtain

$$\sum_{k,j} j\rho_{k,j} = \sum_{(k,j)\in\Lambda^\circ_m} j\big(m\varphi(k/m, j/m) + c + 1\big) - \sum_{(k,j)\in\Lambda^\circ_m} j\epsilon_{k,j} + O(m^3) = m^4 \int_\Omega t\varphi da dt + O(m^3).$$

Dividing by $mn_m = \gamma m^4 + O(m^3)$ and letting $m \to +\infty$ implies the formula for $S(R; \mathrm{ord}_C)$. Summing the bound on the jumping numbers of ord_q over Λ°_m by (7.7), and over Λ^∂_m by $\rho_{k,j} \le cm + 1$ and $|\Lambda^\partial_m| = O(m)$, the sum over all k and j of the jumping numbers of ord_q on $R^C_{m,k,j}$ is at most $m^4 \int_\Omega (\frac{\varphi^2}{2} + \varphi\mu_q) da dt + O(m^3)$, which implies the bounds for $S(R^C; q)$. □

The surface series of Lemma 7.8 are computed on the following blow-ups of $\mathbb{F}_2$.

Construction-Proposition 7.7. Let S, σ, f, B, q_0, Z_B, and q_3 be as follows.

(i) $S := \mathbb{F}_2$, with negative section σ and fiber class f.
(ii) $B \subset S$ is a smooth curve of class $\sigma + 3f$, and $q_0 := B \cap \sigma$.
(iii) $Z_B \subset B \backslash \sigma$ is a divisor of degree two on which the ruling restricts to a closed immersion.
(iv) $q_3 \in \sigma \backslash \{q_0\}$ is a point whose fiber does not meet Z_B.

Let $\tilde{S} \to S$ be the blow-up of $\operatorname{Supp} Z_B$ and q_3 if Z_B is reduced, and otherwise the composition of the blow-up of $\operatorname{Supp} Z_B$, the blow-up of the point of the exceptional curve given by the tangent direction of B, and the blow-up of q_3. Let e_1, e_2 be the total exceptional classes over Z_B, e_3 the exceptional class over q_3, $\tilde{\sigma} := \sigma - e_3$, and $c_i := f - e_i$ for $i = 1, 2, 3$. Then the following hold.

(1) If Z_B is reduced, the irreducible curves of negative self-intersection on $\tilde{S}$ are the curves of classes $\tilde{\sigma}, c_1, c_2, c_3, e_1, e_2, e_3$, and their classes generate the cone of curves of $\tilde{S}$. We have $\tilde{\sigma}^2 = -3$, $c_i^2 = e_i^2 = -1$, $(\tilde{\sigma} \cdot c_1) = (\tilde{\sigma} \cdot c_2) = (\tilde{\sigma} \cdot e_3) = (c_i \cdot e_i) = 1$, and all other intersection numbers of two distinct curves among them vanish.

(2) If Z_B is not reduced, the irreducible curves of negative self-intersection on $\tilde{S}$ are the curves of classes $\tilde{\sigma}$, $C_1 := f - e_1$, $D_1 := e_1 - e_2$, c_3, e_2, e_3, and their classes generate the cone of curves of $\tilde{S}$. We have $\tilde{\sigma}^2 = -3$, $D_1^2 = -2$, $C_1^2 = c_3^2 = e_2^2 = e_3^2 = -1$, $(\tilde{\sigma} \cdot C_1) = (\tilde{\sigma} \cdot e_3) = (C_1 \cdot D_1) = (D_1 \cdot e_2) = (c_3 \cdot e_3) = 1$, and all other intersection numbers of two distinct curves among them vanish.

(3) Let $\tilde{S}' \to \tilde{S}$ be the blow-up of the preimage of q_0, with exceptional class e_4, and let $\tilde{\sigma}' := \sigma - e_3 - e_4$. Then the statements (1) and (2) hold on $\tilde{S}'$ after replacing $\tilde{\sigma}$ by $\tilde{\sigma}'$, of self-intersection -4, and adding $c_4 := f - e_4$ and e_4. The curve $\tilde{\sigma}'$ satisfies $(\tilde{\sigma}' \cdot e_4) = 1$ and is disjoint from c_3 and c_4, and c_4 intersects e_4 and no other exceptional curve.

Proof. Recall that $(\sigma^2) = -2$, $(\sigma \cdot f) = 1$, and $(f^2) = 0$, and that any irreducible curve on S other than σ and the fibers has class $\alpha\sigma + \beta f$ with $\alpha \geq 1$ and $\beta \geq 2\alpha$ [Har77, Proposition V.2.3, Proposition V.2.9, and Proposition V.2.20]. By [Har77, Proposition V.3.2 and Proposition V.3.6], applied to each blow-up, the total exceptional classes satisfy $(e_i \cdot e_{i'}) = -1$ if $i = i'$ and 0 otherwise, $(e_i \cdot \sigma) = (e_i \cdot f) = 0$, and the strict transform of a curve through a center with multiplicity m' is obtained by subtracting m' times the total exceptional class of that center. All intersection numbers in (1), (2), and (3) follow from the intersection numbers of σ, f, and $e_1, \dots, e_4$ and the description of strict transforms in [Har77, Proposition V.3.6], for example $(\tilde{\sigma} \cdot c_3) = (\sigma \cdot f) + (e_3^2) = 0$ and $(D_1^2) = (e_1^2) + (e_2^2) = -2$.

Since $(B \cdot \sigma) = 1$, we have $B \cap \sigma = \{q_0\}$, so $q_0 \notin \operatorname{Supp} Z_B$. Since $(B \cdot f) = 1$, the fiber through q_0 intersects B only at q_0, and B is transverse to any fiber. Together with the assumptions on Z_B and q_3, the fibers through q_0, q_3, and the points of $\operatorname{Supp} Z_B$ are pairwise distinct, and each of them contains exactly one point of $\{q_0, q_3\} \cup \operatorname{Supp} Z_B$.

(1) Let q_1, q_2 be the points of $\operatorname{Supp} Z_B$. The curve σ contains q_3 and neither q_1 nor q_2, and the fiber through q_i contains no center other than q_i. Thus the strict transforms of σ and of the fibers through q_1, q_2, q_3 have classes $\tilde{\sigma}$, c_1, c_2, and c_3, and e_1, e_2, e_3 are the exceptional curves. Let $\Gamma \subset \tilde{S}$ be an irreducible curve different from the seven curves above. Then Γ is the strict transform of an irreducible curve $\Gamma_0 \subset S$. If Γ_0 is a fiber, then $[\Gamma] = f = c_3 + e_3$. Otherwise $[\Gamma_0] = \alpha\sigma + \beta f$ with $\alpha \geq 1$ and $\beta \geq 2\alpha$, and $[\Gamma] = \alpha\sigma + \beta f - m_1 e_1 - m_2 e_2 - m_3 e_3$, where m_i is the multiplicity of Γ_0 at q_i. Since $\Gamma \neq c_i$, we have $([\Gamma] \cdot c_i) = \alpha - m_i \geq 0$. Thus

$$[\Gamma] = \alpha\tilde{\sigma} + m_1 c_1 + m_2 c_2 + (\beta - m_1 - m_2)(c_3 + e_3) + (\alpha - m_3)e_3,$$

and all coefficients are non-negative. Hence the seven classes generate the cone of curves. Suppose that Γ has $(\Gamma^2) < 0$, and write $[\Gamma] = \sum_G x_G[G]$ with $x_G \geq 0$, where G runs through the seven curves. Since $(\Gamma \cdot G) \geq 0$ for any such G, we have $(\Gamma^2) = \sum_G x_G(\Gamma \cdot G) \geq 0$, a contradiction.

(2) Let q_1 be the point of $\operatorname{Supp} Z_B$. Since B is transverse to the fiber through q_1, the strict transform of the fiber through q_1 after the first blow-up does not contain the second center, so it has class C_1. The strict transform of the first exceptional curve has class D_1, and the second exceptional curve has class e_2. Let Γ_0 be as in (1), and let m_1 and m_2 be the multiplicities of Γ_0 at q_1 and at the second center. Then $([\Gamma] \cdot C_1) = \alpha - m_1 \geq 0$ and $([\Gamma] \cdot D_1) = m_1 - m_2 \geq 0$. Since $c_2 = C_1 + D_1$, we have

$$[\Gamma] = \alpha\tilde{\sigma} + m_1 C_1 + m_2(C_1 + D_1) + (\beta - m_1 - m_2)(c_3 + e_3) + (\alpha - m_3)e_3,$$

and the rest of the argument is the argument of (1) with the seven curves replaced by the six curves of (2).

(3) The preimage of q_0 is not a center of $\tilde{S} \to S$, and it is contained in the strict transforms of σ and of the fiber through q_0. Thus the strict transforms of σ and of the fiber through q_0 on $\tilde{S}'$ have classes $\tilde{\sigma}'$ and c_4, and $(\tilde{\sigma}'^2) = -4$. For Γ_0 as in (1), let m_4 be the multiplicity of Γ_0 at q_0, so that $([\Gamma] \cdot c_4) = \alpha - m_4 \geq 0$. Adding $(\alpha - m_4)e_4$ to the expressions of $[\Gamma]$ in (1) and (2), and replacing $\alpha\tilde{\sigma}$ by $\alpha\tilde{\sigma}'$, the argument of (1) and (2) proves the cone statement and the list of negative curves on $\tilde{S}'$. Finally $(\tilde{\sigma}' \cdot e_4) = 1$, $(\tilde{\sigma}' \cdot c_3) = (\tilde{\sigma}' \cdot c_4) = 1 - 1 = 0$, $(c_4 \cdot e_4) = 1$, and $(c_4 \cdot e_i) = 0$ for $i \leq 3$. □

Lemma 7.8. *Let S, σ, f, B, q_0, Z_B, and q_3 be as in Construction-Proposition 7.7, let $Z := Z_B \sqcup \{q_3\}$, and let T_Z be the union of the fibers through* Supp Z. *For $m \geq 1$ and $0 \leq k \leq 2m$ let $R_{m,k} := H^0(S, \mathcal{O}_S(k\sigma+(m+k)f))$ for $k \leq m$ and $R_{m,k} := s_\sigma^{k-m} H^0(S, \mathcal{O}_S(m\sigma+(m+k)f) \otimes I_Z^{k-m})$ for $k \geq m$, where s_σ defines σ, viewed as a space of sections of $\mathcal{O}_S(k\sigma + (m+k)f)$, and let n_m, $S(R; v)$, $A_{S,B/2}(v)$, $\delta_q(S, B/2; R)$, and $S(R^C; q)$ be as in Definition 7.3. Then $n_m = \frac{3}{2}m^3 + O(m^2)$ and*

$$A_{S,B/2}(v) \geq \frac{54}{53} S(R; v)$$

for every prime divisor v over S whose center is not contained in T_Z. More precisely, $\delta_q(S, B/2; R) \geq \frac{36}{31}$ for every closed point $q \in S \backslash (T_Z \cup \{q_0\})$, and $\delta_{q_0}(S, B/2; R) \geq \frac{54}{53}$. Let $h : \mathrm{Bl}_{q_0} S \to S$ be the blow-up of q_0, with exceptional curve E_0, and take $C = E_0$ in Definition 7.3. Then $S(R; \mathrm{ord}_{E_0}) = \frac{53}{36} < \frac{3}{2} = A_{S,B/2}(\mathrm{ord}_{E_0})$, and $S(R^{E_0}; q)$ is at most $\frac{31}{36}$ for the point $q \in E_0$ given by the tangent direction of σ, at most $\frac{11}{18}$ for the point given by the tangent direction of the fiber through q_0, and at most $\frac{11}{48}$ for the remaining closed points $q \in E_0$, among them the point given by the tangent direction of B.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe $R_{m,k}$ on $\tilde{S}$ and compute n_m.

Let $\pi : \tilde{S} \to S$, $\tilde{S}' \to \tilde{S}$, and the classes $e_1, \dots, e_4$, $\tilde{\sigma}$, $c_1, \dots, c_4$, C_1, D_1 be as in Construction-Proposition 7.7, and identify divisors on S and on $\tilde{S}$ with their pullbacks to $\tilde{S}$ and to $\tilde{S}'$. By Construction-Proposition 7.7, a divisor on $\tilde{S}$ (resp. $\tilde{S}'$) is nef if and only if it has non-negative intersection numbers with the curves of negative self-intersection listed there.

Let $p \geq 0$ be an integer. We have $I_Z^p = \pi_* \mathcal{O}_{\tilde{S}}(-p(e_1 + e_2 + e_3))$. At a reduced point q_i of Z, this is the equality $\mathfrak{m}_{q_i}^p = \pi_* \mathcal{O}_{\tilde{S}}(-pe_i)$. If Z_B is supported at one point q_1, let (y_1, y_2) be local coordinates at q_1 such that y_1 defines B. Then the ideal of Z_B at q_1 is (y_1, y_2^2), the first and the second exceptional curves define the monomial valuations with weights $(1,1)$ and $(2,1)$ on (y_1, y_2), and $-p(e_1 + e_2)$ is $-p$ times the strict transform of the first exceptional curve minus $2p$ times the second one. Since $i + i' \geq p$ whenever $2i + i' \geq 2p$, $\pi_* \mathcal{O}_{\tilde{S}}(-p(e_1 + e_2))$ is generated near q_1 by the monomials $y_1^i y_2^{i'}$ with $2i + i' \geq 2p$, which generate $(y_1, y_2^2)^p$.

For $0 \leq a \leq 2$ let $(a-1)_+ := \max\{a-1, 0\}$, and let $D_{a,0} := a\sigma + (a+1)f$ if $a \leq 1$ and $D_{a,0} := \sigma + (a+1)f - (a-1)(e_1 + e_2 + e_3)$ if $a \geq 1$. For $a := k/m$, the map $g \to s_\sigma^{m(a-1)_+} g$ induces an isomorphism

$$H^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{a,0})) \cong R_{m,k}. \tag{7.9}$$

For $a \leq 1$ we have $(D_{a,0} \cdot \tilde{\sigma}) = 1 - a$, $(D_{a,0} \cdot c_i) = a$, and $(D_{a,0} \cdot e_i) = 0$. For $a \geq 1$ we have $(D_{a,0} \cdot \tilde{\sigma}) = 0$, $(D_{a,0} \cdot c_i) = 2 - a$, $(D_{a,0} \cdot e_i) = a - 1$, and $(D_{a,0} \cdot D_1) = 0$ if Z_B is not reduced. Thus $D_{a,0}$ is nef. By [Har77, Corollary V.2.11 and Proposition V.3.3], $K_{\tilde{S}} = -2\sigma - 4f + e_1 + e_2 + e_3$, so

$$-\left(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma}\right) = \frac{3}{2}\sigma + 4f - e_1 - e_2 - \frac{1}{2}e_3.$$

The intersection numbers of $-(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma})$ with $\tilde{\sigma}, c_1, c_2, c_3, e_1, e_2, e_3$ are $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1, 1, 1, \frac{1}{2}$, those with C_1, D_1 are $\frac{1}{2}, 0$, and the self-intersection number is $\frac{21}{4}$. Thus $-(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma})$ is nef and big.

For a nef divisor L on $\tilde S$, the $\mathbb{Q}$-divisor $L-K_{\tilde S}-\frac12\tilde\sigma$ is nef and big. Since $\tilde S$ is smooth and $\tilde\sigma$ is a smooth curve, [KM98, Theorem 2.64], applied to $\mathcal{O}_{\tilde S}(L-K_{\tilde S})$ and $L-K_{\tilde S}\equiv(L-K_{\tilde S}-\frac12\tilde\sigma)+\frac12\tilde\sigma$, gives $H^i(\tilde S,\mathcal{O}_{\tilde S}(K_{\tilde S}-L))=0$ for $i<2$, hence $H^i(\tilde S,\mathcal{O}_{\tilde S}(L))=0$ for $i>0$ by Serre duality. Thus $h^0(\tilde S,\mathcal{O}_{\tilde S}(L))=\frac12(L\cdot(L-K_{\tilde S}))+1$ by the Riemann–Roch theorem [Har77, Theorem V.1.6]. Since $(D_{a,0}^2)=2a$ for $a\le1$ and $(D_{a,0}^2)=2a-3(a-1)^2$ for $a\ge1$, (7.9) implies

$$n_m=m^3\Big(\int_0^1 a\,da+\int_1^2\Big(a-\frac32(a-1)^2\Big)da\Big)+O(m^2)=\frac32m^3+O(m^2).$$

Step 2. In this step we show that $\delta_q(S,B/2;R)\ge\frac{36}{31}$ for any closed point $q\in S\backslash(T_Z\cup\{q_0\})$.

Let $C\subset S$ be the fiber through q. In this step we use Definition 7.3 with the identity of S in place of h and with the fiber C. Let $q_\sigma:=C\cap\sigma$ and $q_B:=C\cap B$. Since $q\notin T_Z$, C contains no point of $\operatorname{Supp}Z$. Let $\ell\ge0$ be an integer and $u:=\ell/m$. Under (7.9), the sections $r\in R_{m,k}$ with $\operatorname{ord}_C(r)\ge\ell$ correspond to $H^0(\tilde S,\mathcal{O}_{\tilde S}(mD_{a,u}))$, where $D_{a,u}:=D_{a,0}-uf$. Let $\Omega\subset\mathbb{R}^2$ be the union of the five triangles $\Omega_1,\dots,\Omega_5$ of Table 88 in Appendix A, that is,

$$\Omega=\{(a,u):0\le a\le2,\ 0\le u\le\min\{a+1,3-a\}\}.$$

The divisor $M:=\sigma+2f-e_1-e_2$ is nef, and $(D_{a,u}\cdot M)=a+1-u$ for $a\le1$ and $(D_{a,u}\cdot M)=3-a-u$ for $a\ge1$. Thus $D_{a,u}$ is not pseudo-effective for $(a,u)\in([0,2]\times[0,+\infty))\backslash\Omega$.

For $(a,u)\in\Omega$, let $D_{a,u}=D_{a,u}^+ + D_{a,u}^-$ be the Zariski decomposition [Laz04a, Theorem 2.3.19], and let w be the coefficient of $\tilde\sigma$ in $D_{a,u}^-$. Table 88 lists $D_{a,u}^-$, w, and $d:=(D_{a,u}^+\cdot f)$, and Remark A.27 lists the coefficients of $D_{a,u}^-$ at e_3, c_1, and c_2.

On each triangle, $D_{a,u}^-$ is the solution of the linear equations $((D_{a,u}-D_{a,u}^-)\cdot G)=0$, where G runs through the curves in the support listed in the table. A direct calculation shows that the coefficients of $D_{a,u}^-$ are non-negative on the triangle, the intersection matrix of the support is negative definite, and $D_{a,u}-D_{a,u}^-$ has non-negative intersection numbers with the curves of negative self-intersection on $\tilde S$, hence is nef. This proves that Table 88 lists the Zariski decompositions. If Z_B is not reduced, then $c_1=C_1$ and $c_2=C_1+D_1$ as classes, the positive parts have equal coefficients at e_1 and e_2, and hence $(D_{a,u}^+\cdot D_1)=0$. Thus Table 88 holds after replacing c_1+c_2 by $2C_1+D_1$, and the support matrices remain negative definite.

We verify the conditions of Lemma 7.6 for the identity of S in place of h, the curve C, $\gamma=\frac32$, $c=2$, the polygons $\Omega_1,\dots,\Omega_5$ with coordinates $(a,t)=(a,u)$, $\varphi:=d$, the point $q_1:=q_\sigma$, and $\mu_1:=(a-1)_+ + w$. Condition (1) holds by **Step 1** and the first integral in (7.10) below. Condition (2) holds since $\deg((k\sigma+(m+k)f-\ell C)|_C)=k\le2m$ and since $D_{a,u}$ is pseudo-effective if $R^C_{m,k,\ell}\ne0$. For (3), let (a,u) be contained in the interior of some Ω_l. Then $k=m(d+(a-1)_+ +w)$, since $d+w=(D_{a,u}\cdot f)$. Any member of $|mD_{a,u}|$ contains $\lceil mD_{a,u}^-\rceil$ [Laz04a, Proposition 2.3.21], and the only component of $D_{a,u}^-$ intersecting C is $\tilde\sigma$, which intersects C transversally at q_σ. The factor $s_\sigma^{m(a-1)_+}$ vanishes at q_σ to order $m(a-1)_+$ on C. Thus any non-zero element of $R^C_{m,k,\ell}$ has divisor at least $m((a-1)_+ +w)q_\sigma$. A direct calculation over the five triangles shows

$$\int_\Omega d=\frac32,\qquad \int_\Omega ud=\frac{11}{12},\qquad \int_\Omega\frac{d^2}{2}=\frac12,\qquad \int_\Omega d((a-1)_+ +w)=\frac{19}{24},\tag{7.10}$$

where all integrals are taken with respect to $dadu$. By Lemma 7.6,

$$S(R;\operatorname{ord}_C)=\frac23\cdot\frac{11}{12}=\frac{11}{18},\qquad S(R^C;q_\sigma)\le\frac23\Big(\frac12+\frac{19}{24}\Big)=\frac{31}{36},\qquad S(R^C;q')\le\frac23\cdot\frac12=\frac13$$

for any closed point $q'\in C\backslash\{q_\sigma\}$.

Let $\lambda:=\frac{36}{31}$. Since $(B\cdot f)=1$, B intersects C transversally at q_B, so $(S,\frac12B+C)$ is plt and the different of $(S,\frac12B+C)$ on C is $\frac12q_B$. We have $A_{S,B/2}(\operatorname{ord}_C)=1$ and $\lambda S(R;\operatorname{ord}_C)=\frac{22}{31}\le1$. If $q=q_\sigma$, then $q\notin B$, since $B\cap\sigma=\{q_0\}$ and $q\ne q_0$, and $\lambda S(R^C;q)\le\frac{36}{31}\cdot\frac{31}{36}=1$. If $q\ne q_\sigma$,

then $\lambda S(R^C;q) \le \frac{12}{31} \le \frac{1}{2}$. Hence the assumptions of Lemma 7.4 hold for the identity of S in place of h, C, q, and λ, and $\delta_q(S,B/2;R) \ge \frac{36}{31}$.

Step 3. In this step we compute the invariants of ord_{E_0} and show that $\delta_{q_0}(S,B/2;R) \ge \frac{54}{53}$.

Write $S' := \mathrm{Bl}_{q_0} S$. The morphism $\tilde{S}' \to S$ factors through S', the curve e_4 is the strict transform of E_0, and $\tilde{S}' \to S'$ is an isomorphism near E_0. Since σ is smooth at q_0, we have $\mathrm{ord}_{E_0}(s_\sigma^{m(a-1)_+}g) = m(a-1)_+ + \mathrm{ord}_{E_0}(g)$. Thus, under (7.9), the sections $r \in R_{m,k}$ with $\mathrm{ord}_{E_0}(r) \ge m(a-1)_+ + \ell$ correspond to $H^0(\tilde{S}', \mathcal{O}_{\tilde{S}'}(mD_{a,u}))$, where $u := \ell/m$ and, in this step, $D_{a,u} := D_{a,0} - ue_4$, and $R^{E_0}_{m,k,j} = 0$ for $j < m(a-1)_+$. Let Ω' be the union of the eleven polygons $\Omega'_1, \dots, \Omega'_{11}$ of Table 89 in Appendix A, that is,

$$\Omega' = \{(a,u) : 0 \le a \le 2,\ 0 \le u \le \min\{2a+1, 4-a\}\}.$$

The divisor $M' := \sigma + 3f - e_1 - e_2 - e_4$ on $\tilde{S}'$ is nef, and $(D_{a,u} \cdot M') = 2a+1-u$ for $a \le 1$ and $(D_{a,u} \cdot M') = 4-a-u$ for $a \ge 1$. Thus $D_{a,u}$ is not pseudo-effective for $(a,u) \in ([0,2]\times[0,+\infty))\backslash\Omega'$.

Let $\tilde{\sigma}'$ be the curve of Construction-Proposition 7.7(3). For $(a,u) \in \Omega'$, let $D_{a,u} = D^+_{a,u} + D^-_{a,u}$ be the Zariski decomposition on $\tilde{S}'$, and let w and z be the coefficients of $\tilde{\sigma}'$ and c_4 in $D^-_{a,u}$. Table 89 lists the support of $D^-_{a,u}$, w, z, and $d := (D^+_{a,u} \cdot e_4) = u - w - z$, and Remark A.27 lists the coefficients of $D^-_{a,u}$ at e_3, c_1, and c_2.

The table is verified by the criteria of **Step 2**: on each polygon, $D^-_{a,u}$ solves the linear equations $((D_{a,u} - D^-_{a,u}) \cdot G) = 0$ for the curves G in the support, a direct calculation shows that the coefficients of $D^-_{a,u}$ are non-negative there, the support matrix is negative definite, and $D_{a,u} - D^-_{a,u}$ is nef by Construction-Proposition 7.7(3). If Z_B is not reduced, $c_1 + c_2$ is again replaced by $2C_1 + D_1$.

Let q_σ and q_f be the points of E_0 given by the tangent directions of σ and of the fiber through q_0. They are the intersection points of e_4 with $\tilde{\sigma}'$ and with c_4, and by Construction-Proposition 7.7(3), no other curve in the support of $D^-_{a,u}$ intersects e_4. We verify the conditions of Lemma 7.6 for h, $C = E_0$, $\gamma = \frac{3}{2}$, $c = 4$, the coordinates $(a,t) := (a, (a-1)_+ + u)$, the polygons $\{(a,(a-1)_+ + u) : (a,u) \in \Omega'_l\}$, $\varphi := d$, the points $q_1 := q_\sigma$ and $q_2 := q_f$, and $\mu_1 := (a-1)_+ + w$ and $\mu_2 := z$. Since each Ω'_l is contained in $\{a \le 1\}$ or in $\{a \ge 1\}$, the map $(a,u) \to (a,(a-1)_+ + u)$ is affine on each Ω'_l and preserves $dadu$. Condition (1) holds by **Step 1** and the first integral in (7.11) below. Condition (2) holds since $R^{E_0}_{m,k,j} \ne 0$ implies $j \ge m(a-1)_+$ and that $D_{a,u}$ is pseudo-effective for $u = j/m - (a-1)_+$, and since then $\deg((h^*(k\sigma + (m+k)f) - jE_0)|_{E_0}) = j \le m((a-1)_+ + 3) \le 4m$. For (3), we have $j = m((a-1)_+ + u) = m(d + \mu_1 + \mu_2)$. Any member of $|mD_{a,u}|$ contains $\lceil mD^-_{a,u} \rceil$ [Laz04a, Proposition 2.3.21], and the strict transform of σ intersects E_0 transversally at q_σ. Thus any non-zero element of $R^{E_0}_{m,k,j}$ has divisor at least $m((a-1)_+ + w)q_\sigma + mzq_f$. A direct calculation over the eleven polygons shows

$$\begin{aligned} &\int_{\Omega'} d = \frac{3}{2}, \qquad \int_{\Omega'} ((a-1)_+ + u)d = \frac{53}{24}, \qquad \int_{\Omega'} \frac{d^2}{2} = \frac{11}{32}, \\ &\int_{\Omega'} \Big(\frac{d^2}{2} + d((a-1)_+ + w)\Big) = \frac{31}{24}, \qquad \int_{\Omega'} \Big(\frac{d^2}{2} + dz\Big) = \frac{11}{12}, \end{aligned} \tag{7.11}$$

with respect to $dadu$. By Lemma 7.6,

$$S(R;\mathrm{ord}_{E_0}) = \frac{53}{36}, \qquad S(R^{E_0};q_\sigma) \le \frac{31}{36}, \qquad S(R^{E_0};q_f) \le \frac{11}{18}, \qquad S(R^{E_0};q') \le \frac{11}{48}$$

for any closed point $q' \in E_0\backslash\{q_\sigma, q_f\}$.

Since B is smooth at q_0, $A_{S,B/2}(\mathrm{ord}_{E_0}) = 2 - \frac{1}{2} = \frac{3}{2} > \frac{53}{36}$. Since $(B \cdot \sigma) = (B \cdot f) = 1$, B is transverse to σ and to the fiber at q_0. Thus the strict transform B' of B intersects E_0 transversally at one point $q_B \notin \{q_\sigma, q_f\}$, the pair $(S', \frac{1}{2}B' + E_0)$ is plt, and the different on E_0 is

$\frac{1}{2}q_B$. Let $\lambda := \frac{54}{53}$. Then

$$\lambda S(R;\operatorname{ord}_{E_0}) = \frac{3}{2}, \qquad \lambda\cdot\frac{31}{36} = \frac{93}{106} \le 1, \qquad \lambda\cdot\frac{11}{18} = \frac{33}{53} \le 1, \qquad \lambda\cdot\frac{11}{48} = \frac{99}{424} \le \frac{1}{2}.$$

By Lemma 7.4, applied to h, E_0, and any closed point of E_0, we obtain $A_{S,B/2}(v) \ge \lambda S(R;v)$ for any prime divisor v over S such that $\operatorname{center}_{S'} v$ intersects E_0. If $q_0 \in \operatorname{center}_S v$, then $\operatorname{center}_{S'} v$ intersects $E_0 = h^{-1}(q_0)$, since $h(\operatorname{center}_{S'} v) = \operatorname{center}_S v$. Hence $\delta_{q_0}(S, B/2; R) \ge \frac{54}{53}$.

Step 4. In this step we conclude the proof.

Let v be a prime divisor over S such that $\operatorname{center}_S v$ is not contained in T_Z. Since T_Z is closed, $\operatorname{center}_S v$ contains a closed point $q \notin T_Z$. By **Steps 2 and 3**,

$$A_{S,B/2}(v) \ge \min\left\{\frac{36}{31}, \frac{54}{53}\right\} S(R;v) = \frac{54}{53} S(R;v).$$

The remaining assertions are proved in **Steps 1, 2, and 3**. □

The next lemma treats a curvilinear base scheme of length three at q_0.

Lemma 7.9. *Let $S := \mathbb{F}_2$ with negative section σ and fiber class f, let $B \subset S$ be a smooth curve of class $\sigma + 3f$, let $q_0 := B \cap \sigma$, and let $\Delta \subset S$ be a curvilinear subscheme of length three supported at q_0 whose tangent line differs from the tangent lines of σ and of the fiber through q_0. For $m \ge 1$ and $0 \le k \le 2m$ let $R_{m,k} := H^0(S, \mathcal{O}_S(k\sigma + (m+k)f))$ for $k \le m$ and $R_{m,k} := s_\sigma^{k-m} H^0(S, \mathcal{O}_S(m\sigma + (m+k)f) \otimes I_\Delta^{k-m})$ for $k \ge m$, where s_σ defines σ, viewed as a space of sections of $\mathcal{O}_S(k\sigma + (m+k)f)$, and let n_m, $S(R;v)$, $A_{S,B/2}(v)$, and $S(R^C;q)$ be as in Definition 7.3. Then $n_m = \frac{3}{2}m^3 + O(m^2)$, and for every fiber C not through q_0,*

$$S(R;\operatorname{ord}_C) = \frac{23}{36}, \qquad S(R^C; C\cap\sigma) \le \frac{95}{108}, \qquad S(R^C;q) \le \frac{17}{54} \quad \text{for } q \in C\backslash\sigma.$$

In particular $A_{S,B/2}(v) \ge \frac{108}{95} S(R;v)$ for every prime divisor v over S whose center intersects a fiber not through q_0.

Proof. Since Δ is curvilinear, there are local coordinates (y_1, y_2) at q_0 with $I_\Delta = (y_1, y_2^3)$. Let $\pi : \tilde{S} \to S$ be the composition of the blow-up of q_0, the blow-up of the intersection point of the first exceptional curve with the strict transform of $\{y_1 = 0\}$, and the blow-up of the intersection point of the second exceptional curve with the strict transform of $\{y_1 = 0\}$, with total exceptional classes e_1, e_2, e_3. The second center is the transversal intersection point of the first exceptional curve and the strict transform of $\{y_1 = 0\}$. After the second blow-up, the strict transforms of the first exceptional curve and of $\{y_1 = 0\}$ are disjoint, so the third center, which is contained in the strict transform of $\{y_1 = 0\}$, is not contained in the strict transform of the first exceptional curve. Let $p \ge 0$ be an integer. The three exceptional curves define the monomial valuations with weights $(1,1)$, $(2,1)$, and $(3,1)$ on (y_1, y_2), and $-p(e_1+e_2+e_3)$ is the sum of $-p$, $-2p$, and $-3p$ times the strict transforms of the three exceptional curves. Since $i + i' \ge p$ and $2i + i' \ge 2p$ whenever $3i + i' \ge 3p$, $\pi_*\mathcal{O}_{\tilde{S}}(-p(e_1+e_2+e_3))$ is generated near q_0 by the monomials $y_1^i y_2^{i'}$ with $3i + i' \ge 3p$. Thus $\pi_*\mathcal{O}_{\tilde{S}}(-p(e_1+e_2+e_3)) = I_\Delta^p$.

Since the tangent line of Δ differs from the tangent lines of σ and of the fiber through q_0, the strict transforms of σ and of the fiber through q_0 have classes $\tilde{\sigma} := \sigma - e_1$ and $C_1 := f - e_1$. Let $D_1 := e_1 - e_2$ and $D_2 := e_2 - e_3$. Then

$$(\tilde{\sigma}^2) = -3, \qquad (C_1^2) = -1, \qquad (D_1^2) = (D_2^2) = -2, \qquad (e_3^2) = -1,$$
$$(\tilde{\sigma}\cdot D_1) = (C_1 \cdot D_1) = (D_1 \cdot D_2) = (D_2 \cdot e_3) = 1,$$

and all other intersection numbers of two distinct curves among $\tilde{\sigma}, C_1, D_1, D_2, e_3$ vanish [Har77, Proposition V.3.2]. Let Γ be the strict transform of an irreducible curve $\Gamma_0 \subset S$ which is neither σ nor a fiber. Then $[\Gamma] = \alpha\sigma + \beta f - m_1 e_1 - m_2 e_2 - m_3 e_3$ with $\beta \ge 2\alpha$ and $\alpha \ge m_1 \ge m_2 \ge m_3 \ge 0$,

where the inequalities $m_1 \le \alpha$, $m_2 \le m_1$, and $m_3 \le m_2$ follow from $([\Gamma]\cdot C_1) \ge 0$, $([\Gamma]\cdot D_1) \ge 0$, and $([\Gamma]\cdot D_2) \ge 0$. Thus

$$[\Gamma] = \alpha\tilde\sigma + m_2(C_1+D_1) + m_3(C_1+D_1+D_2) \\ + (\beta - m_2 - m_3)(C_1+D_1+D_2+e_3) + (\alpha - m_1)(D_1+D_2+e_3)$$

with non-negative coefficients. Since a fiber not through q_0 has class $f = C_1+D_1+D_2+e_3$, and the strict transform of the fiber through q_0 and the exceptional curves are among C_1, D_1, D_2, e_3, the classes of $\tilde\sigma, C_1, D_1, D_2, e_3$ generate the cone of curves of $\tilde S$. In particular, a divisor on $\tilde S$ is nef if and only if it has non-negative intersection numbers with $\tilde\sigma, C_1, D_1, D_2, e_3$.

For $0 \le a \le 2$ let $D_{a,0} := a\sigma + (a+1)f$ if $a \le 1$ and $D_{a,0} := \sigma + (a+1)f - (a-1)(e_1+e_2+e_3)$ if $a \ge 1$. By the description of I^p_Δ above, the map $g \to s_\sigma^{m(a-1)_+}g$ induces an isomorphism $H^0(\tilde S, \mathcal{O}_{\tilde S}(mD_{a,0})) \cong R_{m,k}$ for $a = k/m$. For $a \le 1$ the intersection numbers of $D_{a,0}$ with $\tilde\sigma, C_1, D_1, D_2, e_3$ are $1-a, a, 0, 0, 0$, and for $a \ge 1$ they are $0, 2-a, 0, 0, a-1$. Thus $D_{a,0}$ is nef, and $(D^2_{a,0})$ is $2a$ for $a \le 1$ and $2a-3(a-1)^2$ for $a \ge 1$. Let $\Theta := \frac12\tilde\sigma + \frac13 D_1 + \frac16 D_2$. The intersection numbers of $-(K_{\tilde S}+\Theta)$ with $\tilde\sigma, C_1, D_1, D_2, e_3$ are $\frac16, \frac23, 0, 0, \frac56$, and the self-intersection number is $\frac{65}{12}$. As Θ has simple normal crossing support and coefficients in $[0,1)$, and $mD_{a,0} - K_{\tilde S} - \Theta$ is nef and big, [KM98, Theorem 2.64] gives $H^i(\tilde S, \mathcal{O}_{\tilde S}(K_{\tilde S} - mD_{a,0})) = 0$ for $i < 2$, hence $H^i(\tilde S, \mathcal{O}_{\tilde S}(mD_{a,0})) = 0$ for $i > 0$ by Serre duality, and the Riemann–Roch theorem [Har77, Theorem V.1.6] gives $h^0(\tilde S, \mathcal{O}_{\tilde S}(mD_{a,0})) = \frac12 m^2(D^2_{a,0}) + O(m)$. Hence $n_m = \frac32 m^3 + O(m^2)$.

Let C be a fiber not through q_0, let $q_\sigma := C \cap \sigma$, and let $u := \ell/m$ for an integer $\ell \ge 0$. Since C does not contain q_0, the sections $r \in R_{m,k}$ with $\mathrm{ord}_C(r) \ge \ell$ correspond to $H^0(\tilde S, \mathcal{O}_{\tilde S}(mD_{a,u}))$, where $D_{a,u} := D_{a,0} - uf$. The divisors $\sigma+2f$ and $\sigma+3f-e_1-e_2-e_3$ are nef, and $(D_{a,u}\cdot(\sigma+2f)) = a+1-u$ and, for $a \ge 1$, $(D_{a,u}\cdot(\sigma+3f-e_1-e_2-e_3)) = 5-2a-u$. Thus $D_{a,u}$ is not pseudo-effective for $(a,u) \in ([0,2]\times[0,+\infty))\backslash\Omega$, where

$$\Omega = \{(a,u) : 0 \le a \le 2,\ 0 \le u \le \min\{a+1, 5-2a\}\},$$

the union of the five triangles of Table 90 in Appendix A. The Zariski decomposition $D_{a,u} = D^+_{a,u} + D^-_{a,u}$ on each triangle is listed in Table 90. On each triangle, $D^-_{a,u}$ is the solution of the linear equations $((D_{a,u} - D^-_{a,u})\cdot G) = 0$ for the curves G in the support. A direct calculation shows that the coefficients of $D^-_{a,u}$ are non-negative there, the support matrices are negative definite, and $D^+_{a,u}$ has non-negative intersection numbers with $\tilde\sigma, C_1, D_1, D_2, e_3$. For example, on the third triangle, $D^-_{a,u} = \frac{3u}{7}\tilde\sigma + \frac{2u}{7}D_1 + \frac{u}{7}D_2$. Let w be the coefficient of $\tilde\sigma$ in $D^-_{a,u}$, and let $d := (D^+_{a,u}\cdot f)$. Since C_1, D_1, D_2, and e_3 are disjoint from C, and $\tilde\sigma$ intersects C transversally at q_σ, any non-zero element of $R^C_{m,k,\ell}$ has divisor at least $m((a-1)_+ + w)q_\sigma$ [Laz04a, Proposition 2.3.21], and $k = m(d + (a-1)_+ + w)$. Moreover $\deg((k\sigma + (m+k)f - \ell C)|_C) = k \le 2m$, and $R^C_{m,k,\ell} \ne 0$ implies $(a,u) \in \Omega$. A direct calculation over the five triangles shows

$$\int_\Omega d = \frac32, \qquad \int_\Omega ud = \frac{23}{24}, \qquad \int_\Omega \frac{d^2}{2} = \frac{17}{36}, \qquad \int_\Omega d((a-1)_+ + w) = \frac{61}{72},$$

with respect to $dadu$. By Lemma 7.6, applied to $h = \mathrm{id}_S$, C, $\gamma = \frac32$, $c = 2$, the five triangles, $\varphi := d$, $q_1 := q_\sigma$, and $\mu_1 := (a-1)_+ + w$,

$$S(R;\mathrm{ord}_C) = \frac23\cdot\frac{23}{24} = \frac{23}{36}, \qquad S(R^C;q_\sigma) \le \frac23\Big(\frac{17}{36} + \frac{61}{72}\Big) = \frac{95}{108}, \qquad S(R^C;q) \le \frac23\cdot\frac{17}{36} = \frac{17}{54}$$

for $q \in C\backslash\sigma$.

Let $\lambda := \frac{108}{95}$. Since $(B\cdot f) = 1$, B intersects C transversally at one point q_B, and $q_B \ne q_\sigma$ since $B \cap \sigma = \{q_0\}$ and $q_0 \notin C$. Thus $(S, \frac12 B + C)$ is plt, and the different on C is $\frac12 q_B$. We have $A_{S,B/2}(\mathrm{ord}_C) = 1$ and

$$\lambda S(R;\mathrm{ord}_C) = \frac{69}{95} \le 1, \qquad \lambda S(R^C;q_\sigma) \le 1, \qquad \lambda S(R^C;q) \le \frac{34}{95} \le \frac12 \quad \text{for } q \in C\backslash\sigma.$$

Let v be a prime divisor over S whose center intersects a fiber C not through q_0, and let $q \in \operatorname{center}_S v \cap C$ be a closed point. By Lemma 7.4, applied to $h = \mathrm{id}_S$, C, q, and λ, we obtain $A_{S,B/2}(v) \geq \frac{108}{95} S(R; v)$. □

The next lemma is the surface estimate on $\mathbb{F}_2$ for a subscheme of length three of B with a part of length two at the point $B \cap \sigma$.

Lemma 7.10. *Let $S := \mathbb{F}_2$ with negative section σ and fiber class f, let $B \subset S$ be a smooth curve of class $\sigma + 3f$, let $q_0 := B \cap \sigma$, let Z_0 be a curvilinear subscheme of length two supported at q_0 whose tangent line differs from the tangent lines of σ and of the fiber through q_0, let $q_3 \in B$ be a point on another fiber, and let $Z := Z_0 \sqcup \{q_3\}$ and T_Z the union of the fibers through* $\operatorname{Supp} Z$. *For $m \geq 1$ and $0 \leq k \leq 2m$ let $R_{m,k} := H^0(S, \mathcal{O}_S(k\sigma + (m+k)f))$ for $k \leq m$ and $R_{m,k} := s_\sigma^{k-m} H^0(S, \mathcal{O}_S(m\sigma + (m+k)f) \otimes I_Z^{k-m})$ for $k \geq m$, where s_σ defines σ, viewed as a space of sections of $\mathcal{O}_S(k\sigma + (m+k)f)$, and let n_m, $S(R; v)$, $A_{S,B/2}(v)$, and $S(R^C; q)$ be as in Definition 7.3. Then $n_m = \frac{3}{2}m^3 + O(m^2)$, and*

$$A_{S,B/2}(v) \geq \frac{8}{7} S(R; v)$$

for every prime divisor v over S whose center is not contained in T_Z. More precisely, for every fiber C disjoint from Z,

$$S(R; \operatorname{ord}_C) = \frac{17}{27}, \qquad S(R^C; C \cap \sigma) \leq \frac{7}{8}, \qquad S(R^C; q) \leq \frac{23}{72} \quad \text{for } q \in C \backslash \sigma.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we resolve Z and describe the cone of curves of the resolution.

Let $\tilde{S} \to S$ be the composition of the blow-up of q_0, the blow-up of the point of the exceptional curve given by the tangent line of Z_0, and the blow-up of q_3. Let e_1, e_2, e_3 be the total transforms on $\tilde{S}$ of the three exceptional curves, and write σ and f also for the pullbacks of σ and f. Then $(e_i \cdot e_l) = -\delta_{il}$ and $(e_i \cdot \sigma) = (e_i \cdot f) = 0$. In local coordinates (y_1, y_2) at q_0 with $I_{Z_0} = (y_1, y_2^2)$, the exceptional curves over q_0 define the monomial valuations of weights $(1,1)$ and $(2,1)$ on (y_1, y_2), with coefficients 1 and 2 in $e_1 + e_2$, respectively. Hence, for $p \geq 0$, the pushforward of $\mathcal{O}_{\tilde{S}}(-p(e_1 + e_2))$ near q_0 is spanned by the monomials $y_1^i y_2^{i'}$ with $i + i' \geq p$ and $2i + i' \geq 2p$, that is, with $2i + i' \geq 2p$, and the ideal spanned by the monomials $y_1^i y_2^{i'}$ with $2i + i' \geq 2p$ is $(y_1, y_2^2)^p$. Thus the pushforward of $\mathcal{O}_{\tilde{S}}(-p(e_1 + e_2 + e_3))$ is I_Z^p.

Since $(B \cdot \sigma) = 1$, we have $B \cap \sigma = \{q_0\}$, so $q_3 \notin \sigma$. Let $\tilde{\sigma}$, C_1, and c_3 be the strict transforms of σ and of the fibers through q_0 and q_3, and let D_1 be the strict transform of the first exceptional curve. By the assumption on the tangent line of Z_0, neither $\tilde{\sigma}$ nor C_1 passes through the second center. Hence

$$\tilde{\sigma} \equiv \sigma - e_1, \qquad C_1 \equiv f - e_1, \qquad c_3 \equiv f - e_3, \qquad D_1 \equiv e_1 - e_2,$$

the self-intersection numbers of $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ are $-3, -1, -1, -2, -1, -1$, and the non-zero intersection numbers of two distinct curves among them are

$$(\tilde{\sigma} \cdot c_3) = (\tilde{\sigma} \cdot D_1) = (C_1 \cdot D_1) = (c_3 \cdot e_3) = (D_1 \cdot e_2) = 1.$$

In particular $(\tilde{\sigma} \cdot C_1) = 0$.

We show that $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ generate the cone of curves of $\tilde{S}$. The irreducible curves contained in fibers of $\tilde{S} \to \mathbb{P}^1$ are C_1, D_1, e_2, c_3, e_3 and the fibers of class $f \equiv c_3 + e_3$. Let C' be an irreducible curve on $\tilde{S}$ not contained in a fiber and different from $\tilde{\sigma}$, and write $C' \equiv \alpha\sigma + \beta f - m_1 e_1 - m_2 e_2 - m_3 e_3$. Then

$$\alpha = (C' \cdot f) \geq 1, \quad \beta - 2\alpha - m_1 = (C' \cdot \tilde{\sigma}) \geq 0, \quad \alpha - m_1 = (C' \cdot C_1) \geq 0, \quad \alpha - m_3 = (C' \cdot c_3) \geq 0,$$

$$m_1 - m_2 = (C' \cdot D_1) \geq 0, \qquad m_2 = (C' \cdot e_2) \geq 0, \qquad m_3 = (C' \cdot e_3) \geq 0.$$

Thus

$$C' \equiv \alpha\tilde{\sigma} + m_2(C_1 + D_1) + m_3c_3 + (\beta - m_2 - m_3)(c_3 + e_3) + (\alpha - m_1)(D_1 + e_2)$$

with non-negative coefficients, as $\beta - m_2 - m_3 \geq (2\alpha + m_1) - m_1 - \alpha = \alpha \geq 0$, using $\beta \geq 2\alpha + m_1$, $m_2 \leq m_1$, and $m_3 \leq \alpha$. Therefore an $\mathbb{R}$-divisor D on $\tilde{S}$ is nef if and only if $(D \cdot \gamma) \geq 0$ for $\gamma \in \{\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3\}$.

Let $\Theta_0 := \frac{1}{2}\tilde{\sigma} + \frac{1}{4}D_1$. Its support $\tilde{\sigma} \cup D_1$ is a simple normal crossing divisor, as $\tilde{\sigma}$ and D_1 are smooth and intersect transversally in one point. Since $-K_{\tilde{S}} \equiv 2\sigma + 4f - e_1 - e_2 - e_3$, the intersection numbers of $-(K_{\tilde{S}} + \Theta_0)$ with $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ are $\frac{1}{4}, \frac{3}{4}, \frac{1}{2}, 0, \frac{3}{4}, 1$, and

$$\big((K_{\tilde{S}} + \Theta_0)^2\big) = \frac{43}{8}.$$

Thus $-(K_{\tilde{S}} + \Theta_0)$ is nef and big. Let D be a nef divisor on $\tilde{S}$. Then $D - K_{\tilde{S}} - \Theta_0$ is nef and big, and the coefficients of Θ_0 are less than 1. Hence $H^i(\tilde{S}, \mathcal{O}_{\tilde{S}}(D)) = 0$ for $i > 0$ by Kawamata–Viehweg vanishing [KM98, Theorem 2.64], applied to $D - K_{\tilde{S}} = (D - K_{\tilde{S}} - \Theta_0) + \Theta_0$ and combined with Serre duality, so that

$$h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(D)) = \frac{1}{2}(D^2) - \frac{1}{2}(D \cdot K_{\tilde{S}}) + 1$$

by the Riemann–Roch theorem [Har77, Theorem V.1.6].

Step 2. In this step we compute n_m.

For $0 \leq a \leq 2$ let $(a-1)_+ := \max(a-1, 0)$ and

$$D_{a,0} := a\sigma + (a+1)f \quad (a \leq 1), \qquad D_{a,0} := \sigma + (a+1)f - (a-1)(e_1 + e_2 + e_3) \quad (a \geq 1).$$

By **Step 1**, multiplication by $s_\sigma^{(k-m)_+}$ is an isomorphism from $H^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{k/m,0}))$ onto $R_{m,k}$. For $a \leq 1$ the $\mathbb{R}$-divisor $D_{a,0}$ is nef, as $(D_{a,0} \cdot \sigma) = 1 - a$ and $(D_{a,0} \cdot f) = a$ on S. For $a \geq 1$ the intersection numbers of $D_{a,0}$ with $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ are $0, 2-a, 2-a, 0, a-1, a-1$, so $D_{a,0}$ is nef by **Step 1**. Moreover

$$(D_{a,0}^2) = 2a \quad (a \leq 1), \qquad (D_{a,0}^2) = 2a - 3(a-1)^2 \quad (a \geq 1).$$

By **Step 1**, $h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{a,0})) = \frac{m^2}{2}(D_{a,0}^2) + O(m)$ uniformly in a, and the Riemann sum over $a = k/m$ implies

$$n_m = \sum_{k=0}^{2m} h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{k/m,0})) = m^3\Big(\int_0^1 a\,da + \int_1^2 \big(a - \tfrac{3}{2}(a-1)^2\big)da\Big) + O(m^2) = \frac{3}{2}m^3 + O(m^2).$$

Step 3. In this step we compute the Zariski decompositions of the $\mathbb{R}$-divisors $D_{a,0} - uf$.

For $u \geq 0$ let $D_{a,u} := D_{a,0} - uf$. Let Ω be the union of the seven triangles $\Omega_1, \dots, \Omega_7$ with rational vertices listed in Table 91, that is,

$$\Omega = \{(a, u) : 0 \leq a \leq 2,\ 0 \leq u \leq \min(a+1, 2, 5-2a)\}.$$

For $(a, u) \in \Omega$ the computation below writes $D_{a,u}$ as the sum of a nef $\mathbb{R}$-divisor and an $\mathbb{R}$-divisor ≥ 0, so $D_{a,u}$ is pseudo-effective, and we let $D_{a,u} = D_{a,u}^+ + D_{a,u}^-$ be its Zariski decomposition [Laz04a, Theorem 2.3.19], let $w(a, u)$ be the coefficient of $\tilde{\sigma}$ in $D_{a,u}^-$, and let $d(a, u) := (D_{a,u}^+ \cdot f)$.

The components of $D_{a,u}^-$ are negative curves, hence are among $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ by **Step 1**. On each Ω_i the support of $D_{a,u}^-$ is a fixed set I_i of curves among $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$. On Ω_i the vector of coefficients of $D_{a,u}^-$ is $M_{I_i}^{-1}\mathbf{d}_{I_i}$, where M_{I_i} is the intersection matrix of the curves in I_i, which is negative definite, and $\mathbf{d}_{I_i}$ is the vector of the intersection numbers of $D_{a,u}$ with the curves in I_i. To verify the Zariski decomposition on Ω_i it suffices to check that the coefficients of $D_{a,u}^-$ are non-negative and that $D_{a,u} - D_{a,u}^-$ has non-negative intersection numbers with $\tilde{\sigma}, C_1, c_3, D_1, e_2, e_3$ (**Step 1**).

For example, let Ω_3 be the triangle with vertices $(1,0)$, $(2,0)$, and $(4/3,5/3)$, and $I_3=\{\tilde\sigma, D_1\}$. Then $(D_{a,u}\cdot\tilde\sigma)=-u$ and $(D_{a,u}\cdot D_1)=0$, so

$$D^-_{a,u}=\frac{2u}{5}\tilde\sigma+\frac{u}{5}D_1,\qquad M_{I_3}=\begin{pmatrix}-3 & 1\\ 1 & -2\end{pmatrix}.$$

The intersection numbers of $D^+_{a,u}$ with $\tilde\sigma, C_1, c_3, D_1, e_2, e_3$ are

$$0,\qquad 2-a-\frac{u}{5},\qquad 2-a-\frac{2u}{5},\qquad 0,\qquad a-1-\frac{u}{5},\qquad a-1,$$

which are non-negative exactly on Ω_3. Thus on Ω_3 we have $w=\frac{2u}{5}$ and $d=1-\frac{2u}{5}$. The remaining chambers are listed in Table 91.

On each Ω_i the functions d and w are affine in (a,u). Direct integration over the seven triangles with respect to $dadu$ implies

$$\iint_\Omega d=\frac{3}{2},\qquad \iint_\Omega ud=\frac{17}{18},\qquad \iint_\Omega \frac{d^2}{2}=\frac{23}{48},\qquad \iint_\Omega d\big((a-1)_+ +w\big)=\frac{5}{6}. \tag{7.12}$$

Step 4. In this step we compute $S(R;\mathrm{ord}_C)$ and bound the secondary expected vanishing orders along C.

Let C be a fiber disjoint from Z, and let s_C define C. Since $C\neq\sigma$ and $C\cap Z=\emptyset$, for $\ell\geq 0$ we have

$$\{\xi\in R_{m,k}:\mathrm{ord}_C(\xi)\geq\ell\}=s_\sigma^{(k-m)_+}s_C^\ell H^0(\tilde S,\mathcal{O}_{\tilde S}(mD_{k/m,\ell/m})).$$

We check the conditions of Lemma 7.6 for h the identity, $\gamma=\frac{3}{2}$, $c=2$, the triangles $\Omega_1,\dots,\Omega_7$ in the coordinates (a,u) in place of (a,t), $\varphi=d$, $r=1$, $q_1=C\cap\sigma$, and $\mu_1=(a-1)_+ +w$. The functions d and μ_1 are non-negative and affine on each Ω_i by **Step 3**. Condition (1) of Lemma 7.6 holds by **Step 2** and (7.12).

We check condition (2). Let $(a,u):=(k/m,\ell/m)$ and assume that $R^C_{m,k,\ell}\neq 0$. Then $|mD_{a,u}|\neq\emptyset$ on $\tilde S$, and $((k\sigma+(m+k-\ell)f)\cdot f)=k\leq 2m$. The divisors

$$N_1:=\sigma+2f,\qquad N_2:=\sigma+3f-e_1-e_2-e_3,\qquad N_3:=\sigma+2f-e_3$$

have intersection numbers $0,1,1,0,0,0$, resp. $0,0,0,0,1,1$, resp. $0,1,0,0,0,1$ with $\tilde\sigma, C_1, c_3, D_1, e_2, e_3$, so they are nef by **Step 1**. For $a\leq 1$ we have

$$(D_{a,u}\cdot N_1)=(D_{a,u}\cdot N_3)=a+1-u,\qquad (D_{a,u}\cdot N_2)=2a+1-u,$$

and for $a\geq 1$ we have

$$(D_{a,u}\cdot N_1)=a+1-u,\qquad (D_{a,u}\cdot N_2)=5-2a-u,\qquad (D_{a,u}\cdot N_3)=2-u.$$

As $|mD_{a,u}|\neq\emptyset$, we have $(D_{a,u}\cdot N_i)\geq 0$ for $i=1,2,3$, so $u\leq\min(a+1,2,5-2a)$ and $(a,u)\in\Omega$.

We check condition (3). Let $(a,u)=(k/m,\ell/m)$ be contained in the interior of some Ω_i. Since $(D_{a,u}\cdot f)=a-(a-1)_+$, the curves in $\mathrm{Supp}\,D^-_{a,u}$ other than $\tilde\sigma$ are contained in fibers, and $(\tilde\sigma\cdot f)=1$, we have

$$a=(a-1)_+ +w+d,\qquad \text{so}\qquad \deg\big((k\sigma+(m+k-\ell)f)|_C\big)=k=m\big(\varphi(a,u)+\mu_1(a,u)\big).$$

Every divisor $G\geq 0$ with $G\sim mD_{a,u}$ satisfies $G\geq mD^-_{a,u}$ [Laz04a, Proposition 2.3.21]. The curves in $\mathrm{Supp}\,D^-_{a,u}$ other than $\tilde\sigma$ are contained in the fibers through q_0 and q_3, so they are disjoint from C, and $(\tilde\sigma\cdot C)=1$ with $\tilde\sigma\cap C=\{C\cap\sigma\}$. Hence every non-zero element of $R^C_{m,k,\ell}$ vanishes at $C\cap\sigma$ to order at least $(k-m)_+ +mw=m\mu_1(a,u)$.

Lemma 7.6 and (7.12) imply

$$S(R;\mathrm{ord}_C)=\frac{2}{3}\iint_\Omega ud=\frac{2}{3}\cdot\frac{17}{18}=\frac{17}{27},\qquad S(R^C;C\cap\sigma)\leq\frac{2}{3}\iint_\Omega\Big(\frac{d^2}{2}+d\big((a-1)_+ +w\big)\Big)=\frac{2}{3}\Big(\frac{23}{48}+\frac{5}{6}\Big)=\frac{7}{8},$$

and

$$S(R^C;q)\leq\frac{2}{3}\iint_\Omega\frac{d^2}{2}=\frac{23}{72}\quad\text{for } q\in C\backslash\sigma.$$

Step 5. In this step we conclude the proof.

Let v be a prime divisor over S whose center is not contained in T_Z, and let $q \in \operatorname{center}_S v \backslash T_Z$ be a closed point. Let C be the fiber through q, so $C \cap Z = \emptyset$. Since $(B \cdot C) = 1$, the curve B intersects C transversally in one point q_B, and $q_B \notin \sigma$ because $B \cap \sigma = \{q_0\} \subset T_Z$. Hence $(S, \frac{1}{2}B + C)$ is plt near q, and the different of $(S, \frac{1}{2}B + C)$ on C is $\frac{1}{2}q_B$. By **Step 4**,

$$\frac{8}{7}S(R; \operatorname{ord}_C) = \frac{136}{189} < 1 = A_{S,B/2}(\operatorname{ord}_C), \qquad \frac{8}{7}S(R^C; C \cap \sigma) \le 1, \qquad \frac{8}{7}S(R^C; q_B) \le \frac{23}{63} < \frac{1}{2},$$

and $\frac{8}{7}S(R^C; q') \le \frac{23}{63} < 1$ for any $q' \in C \backslash (\sigma \cup \{q_B\})$. By Lemma 7.4, applied with h the identity and $\lambda = \frac{8}{7}$, we obtain $A_{S,B/2}(v) \ge \frac{8}{7}S(R; v)$. □

7.2. **The $\mathbb{F}_0$ case.** In this subsection, we prove Proposition 7.11. The assumption that L is the only line of X through x is used for Lemma 7.5 along the fiber of $P \to L$ over x.

Proposition 7.11. *Notation as in Construction-Proposition 7.1. Let $x \in L$ be a closed point. Assume that the normalization of the image of E in Q is $\mathbb{F}_0$ and that L is the only line of X through x. Then every prime divisor F over X with $\operatorname{center}_X F = \{x\}$ satisfies*

$$A_X(F) - S_X(F) \ge \frac{11}{18} \operatorname{mult}_F P > 0,$$

where $\operatorname{mult}_F P$ is taken for the $\mathbb{Q}$-Cartier divisor P on $\hat{X}$.

Proof. We use the notation of Set-up 5.1 and Construction-Proposition 5.6. Recall that $\Lambda \subset Q$ is the singular line of the image E_Q of E in Q, that $Q' = \operatorname{Bl}_\Lambda Q$ has exceptional divisor $P_{Q'} \cong \mathbb{F}_1$ with negative section g and ruling $\varrho : P_{Q'} \to \Lambda$ of fiber class η, and that $l = h - P_{Q'}$. The strict transform $\bar{E}_Q \subset Q'$ of E_Q is the normalization of E_Q and the preimage of the conic $\kappa \subset \mathbb{P}^2$, with classes $\bar{\sigma}$ and $\bar{f}$. Moreover Γ' is the strict transform of Γ in Q', and $Q'' = \operatorname{Bl}_{\Gamma'} Q'$ has exceptional divisor D'', with $\bar{E}_{Q''}$ and P the strict transforms of $\bar{E}_Q$ and $P_{Q'}$, and $H_{Q''} = 5h - 2D'' - P$. Let $Y_1 := \operatorname{Bl}_s Y$, and let E_1 be the strict transform of E in Y_1. Let $\Lambda^+ \subset Y^+$ be the strict transform of Λ, which is the flopped curve of s. Let W be the refinement of the complete series of H by P with respect to the plt extraction $\hat{\pi} : \hat{X} \to X$ (Definition 2.8), so that $W_{m,j}$ is the image of $\{\phi \in H^0(X, mH) : \operatorname{ord}_P(\phi) \ge j\}$ in $H^0(P, (m\hat{\pi}^*H - jP)|_P)$.

By Construction-Proposition 7.1, there is a morphism $\mu : Y_1 \to \hat{X}$ over X which maps the exceptional divisor of $Y_1 \to Y$ isomorphically onto P and contracts E_1 onto B. We write P also for the exceptional divisor of $Y_1 \to Y$. Then $\mu^*(2P) = 2P + E_1$, that is, $\mu^*P = P + \frac{1}{2}E_1$, and $E_1|_P = B$ under the isomorphism $\mu|_P$.

Next we introduce curves on $P_{Q'}$. Let $B' := \bar{E}_Q \cap P_{Q'}$ and $\Xi := \Gamma' \cap P_{Q'}$. By Construction-Proposition 5.6(2), B' is a curve of class $\bar{\sigma}$ on $\bar{E}_Q \cong \mathbb{F}_0$, hence a smooth rational curve. Since $\bar{E}_Q$ is the preimage of the conic κ, we have $\bar{E}_Q \sim 2l = 2h - 2P_{Q'}$, so $B' \sim 2g + 2\eta$ on $P_{Q'}$. As B' is irreducible, $B' \ne g$, and $(B' \cdot g) = 0$, the curves B' and g are disjoint. Since $(B' \cdot \eta) = 2$, the restriction to B' of ϱ is a double cover $B' \to \Lambda$. Let ι be the involution of B' over Λ. The morphism $P_{Q'} \to \mathbb{P}^2$ is an isomorphism near B' and maps B' onto κ. Since $\Gamma' \subset \bar{E}_Q$, we have $\Xi \subset B'$, and Ξ is a divisor on B' of degree $(\Gamma' \cdot \bar{\sigma}) = ((2\bar{\sigma} + 3\bar{f}) \cdot \bar{\sigma}) = 3$ on which ϱ restricts to a closed immersion (Construction-Proposition 5.6(3)). Let $\Xi_\Lambda := \varrho(\Xi)$.

Claim 7.12. *(1) The surface P is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$. Let f be the class of the fibers of $\hat{\pi}|_P : P \to L$ and σ the class of the fibers of the other ruling. Then $B \in |\sigma + 2f|$.*

(2) There exists a divisor $\Delta \subset B$ of degree three on which both ruling projections of P restrict to closed immersions, such that, for all integers $m \ge 1$ and $k \ge 0$, the following hold.

(i) $W_{m,2k}$ is a subspace of

$$H^0\big(P, \mathcal{O}_P(k\sigma + mf) \otimes I_\Delta^{\max(k-m,0)}\big).$$

(ii) There is an injective linear map

$$\theta_{m,k} : W_{m,2k+1} \to H^0\big(P, \mathcal{O}_P(k\sigma + (m-1)f) \otimes I_\Delta^{\max(k-m,0)}\big)$$

such that the $\mathbb{Q}$-divisor on P of any non-zero $\omega \in W_{m,2k+1}$ is $\frac{1}{2}B + \operatorname{div}(\theta_{m,k}(\omega))$.

(iii) $W_{m,j} = 0$ for $j > 4m$.

(3) The birational map $Q'' \dashrightarrow Y_1$ restricts to an isomorphism from the strict transform of B' in Q'' onto $E_1 \cap P$. Let $\beta_B : B' \to B$ be the composite of the isomorphism of B' with the strict transform of B' in Q'', the isomorphism onto $E_1 \cap P$, and $\mu|_{E_1 \cap P}$. Then $\beta_B(\iota(\Xi)) = \Delta$.

Proof. We first compute the conormal bundle of Λ^+ in Y^+. Since $P_{Q'}$ is the exceptional divisor of $\operatorname{Bl}_\Lambda Q$ and $-P_{Q'}|_{P_{Q'}} = g$, we have $\varrho_* \mathcal{O}_{P_{Q'}}(g) = N^*_{\Lambda/Q}$ [Har77, Theorem II.8.24], and, as $h|_{P_{Q'}} = \eta$,

$$\varrho_* \mathcal{O}_{P_{Q'}}(g + 3\eta) \cong \mathcal{O}_\Lambda(3) \oplus \mathcal{O}_\Lambda(2)$$

by [Har77, Exercise III.8.4(a) and Proposition V.2.8], where the summand $\mathcal{O}_\Lambda(3)$ consists of the sections vanishing on g. Let $\mathcal{K}_0 := \varrho_*(I_\Xi \otimes \mathcal{O}_{P_{Q'}}(g + 3\eta))$. As $\Xi \to \Lambda$ is a closed immersion, $\mathcal{K}_0$ is the kernel of the surjection $\varrho_* \mathcal{O}_{P_{Q'}}(g + 3\eta) \to \mathcal{O}_\Xi$, so $\mathcal{K}_0$ is a vector bundle of rank two and degree two. Let $p \in \Gamma \cap \Lambda$, and let r be the length of $\Gamma \cap \Lambda$ at p. There are local analytic coordinates (t_1, t_2, t_3) of Q at p with $\Lambda = \{t_1 = t_2 = 0\}$ and $\Gamma = \{t_1 = 0,\ t_2 = t_3^r\}$. To see this, let (z_1, z_2, z_3) be local analytic coordinates of Q at p with $\Lambda = \{z_1 = z_2 = 0\}$. If $r = 1$, then Γ is transverse to Λ, so after a linear change of (z_1, z_2) we have $\Gamma = \{z_1 = \gamma_1(z_2),\ z_3 = \gamma_3(z_2)\}$ with $\gamma_1(0) = \gamma_3(0) = 0$, and $t_1 := z_1 - \gamma_1(z_2)$, $t_2 := z_2$, $t_3 := z_2 + z_3 - \gamma_3(z_2)$ are such coordinates. If $r \geq 2$, then Γ is tangent to Λ, so $\Gamma = \{z_1 = \gamma_1(z_3),\ z_2 = \gamma_2(z_3)\}$ with $r = \min(\operatorname{ord}\gamma_1, \operatorname{ord}\gamma_2)$. After exchanging z_1 and z_2 we may assume that $\operatorname{ord}\gamma_2 = r \leq \operatorname{ord}\gamma_1$, and $t_1 := z_1 - (\gamma_1/\gamma_2)(z_3)z_2$, $t_2 := z_2$, $t_3 := z_3(\gamma_2(z_3)/z_3^r)^{1/r}$ are such coordinates. Near p, the scheme Ξ is the point $[0 : 1]$ of the fibers of ϱ (in the basis $\partial_{t_1}, \partial_{t_2}$ of $N_{\Lambda/Q}$) over $\{t_3^r = 0\}$, so the subsheaf of $N^*_{\Lambda/Q}$ of forms vanishing on Ξ is generated by dt_1 and $t_3^r dt_2$. In the chart $t_1 = (t_2 - t_3^r)v^+$ of Y^+ we have $\Lambda^+ = \{t_2 = v^+ = 0\}$, so $N^*_{\Lambda^+/Y^+}$ is generated near p by dt_2 and $dv^+|_{\Lambda^+} = -t_3^{-r} dt_1$. Here $\mathcal{K}_0 \otimes \mathcal{O}_\Lambda(-3) = \varrho_*(I_\Xi \otimes \mathcal{O}_{P_{Q'}}(g))$ is the subsheaf of $N^*_{\Lambda/Q}$ of forms vanishing on Ξ, and $N^*_{\Lambda^+/Y^+}$ agrees with $N^*_{\Lambda/Q}$ over $\Lambda \backslash \Xi_\Lambda$. Comparing the generators $dt_1, t_3^r dt_2$ and $t_3^{-r} dt_1, dt_2$ at each point of Ξ_Λ, we obtain

$$N^*_{\Lambda^+/Y^+} \cong \mathcal{K}_0 \otimes \mathcal{O}_\Lambda(-3) \otimes \mathcal{O}_\Lambda(\Xi_\Lambda),$$

where Ξ_Λ is viewed as a divisor $\Xi_\Lambda \geq 0$ of degree three on Λ (Construction-Proposition 5.6(3)). As $\deg \Xi_\Lambda = 3$, we obtain

$$\mathcal{K}_0 \cong N^*_{\Lambda^+/Y^+}.$$

We show that $\mathcal{K}_0 \cong \mathcal{O}_\Lambda(1)^{\oplus 2}$. Write $\mathcal{K}_0 \cong \mathcal{O}_\Lambda(a) \oplus \mathcal{O}_\Lambda(2-a)$ with $a \geq 1$ [Har77, Exercise V.2.6]. Since $\mathcal{K}_0 \subset \mathcal{O}_\Lambda(3) \oplus \mathcal{O}_\Lambda(2)$, we have $a \leq 3$. If $a = 3$, then the summand $\mathcal{O}_\Lambda(3)$ of $\mathcal{K}_0$ is the summand $\mathcal{O}_\Lambda(3)$ of $\varrho_* \mathcal{O}_{P_{Q'}}(g + 3\eta)$, so the sections vanishing on g vanish on Ξ, and $\Xi \subset g$. This is impossible because $\Xi \subset B'$ and $B' \cap g = \emptyset$. If $a = 2$, then $H^0(P_{Q'}, I_\Xi \otimes \mathcal{O}_{P_{Q'}}(g + \eta)) = H^0(\Lambda, \mathcal{K}_0(-2)) \neq 0$. Since $g + \eta = l|_{P_{Q'}}$, the image of Ξ in $\mathbb{P}^2$ is then contained in a line. But the image of Ξ is a subscheme of length three of the smooth conic κ, which is not contained in a line. Thus $a = 1$, and $N_{\Lambda^+/Y^+} \cong \mathcal{O}_\Lambda(-1)^{\oplus 2}$.

By Theorem 2.22 the flopping curves are pairwise disjoint, so the flop $Y \dashrightarrow Y^+$ near Λ^+ is the flop of a curve with normal bundle $\mathcal{O}(-1)^{\oplus 2}$, that is, the Atiyah flop [Rei83, Definition (5.3) and the sentence after it, and (5.5)]. Hence $Y_1 = \operatorname{Bl}_s Y \cong \operatorname{Bl}_{\Lambda^+} Y^+$ over $Y \dashrightarrow Y^+$, and $P \cong \mathbb{P}^1 \times \mathbb{P}^1$, whose rulings are $P \to s \cong L$ and $P \to \Lambda^+ \cong \Lambda$. As $(P \cdot f) = (P \cdot \sigma) = -1$, we have $P|_P \sim -\sigma - f$.

Next we compare P with the strict transform of $P_{Q'}$ in Q'', which is the blow-up of $P_{Q'}$ along Ξ. Let $\tilde{P} \to P_{Q'}$ be the composite of the blow-ups of the points of Ξ and of the infinitely near points of Ξ on the strict transforms of B', three blow-ups in total, so that $\tilde{P}$ is the minimal resolution of the strict transform of $P_{Q'}$ in Q''. Let e_1, e_2, e_3 be the total transforms on $\tilde{P}$ of the exceptional curves, $e := e_1 + e_2 + e_3$, and $\tilde{B}'$ the strict transform of B', so $\tilde{B}' \equiv 2g + 2\eta - e$. Since Ξ is a curvilinear subscheme of the smooth curve B', the pullback of $I_\Xi = I_{\Gamma'} \mathcal{O}_{P_{Q'}}$ to $\tilde{P}$ is

$\mathcal{O}_{\tilde{P}}(-e)$. By Construction-Proposition 5.6(1) and (4), the pullbacks to $\tilde{P}$ of the restrictions of divisors on Q'' to the strict transform of $P_{Q'}$ in Q'' satisfy

$$P|_{\tilde{P}} = -g, \quad D''|_{\tilde{P}} = e, \quad -K_{Q''}|_{\tilde{P}} = g + 3\eta - e, \quad l|_{\tilde{P}} = g + \eta, \quad \bar{E}_{Q''}|_{\tilde{P}} = 2g + 2\eta - e. \tag{7.13}$$

Over $Q\backslash\Xi_\Lambda$, both Q'' and Y_1 are the blow-up of Q along $\Gamma \cup \Lambda$, which is a disjoint union there. Let $p \in \Xi_\Lambda$ and let (t_1, t_2, t_3) be the local analytic coordinates of Q at p with $\Lambda = \{t_1 = t_2 = 0\}$ and $\Gamma = \{t_1 = 0,\ t_2 = t_3^r\}$. On the chart $t_1 = t_2 v$ of Q', $P_{Q'} = \{t_2 = 0\}$ has coordinates (v, t_3), $\Gamma' = \{v = 0,\ t_2 = t_3^r\}$, and $I_\Xi = (v, t_3^r)$. On the chart $v = (t_2 - t_3^r)w$ of Q'', with coordinates (w, t_2, t_3), the strict transform of $P_{Q'}$ is $\{t_2 = 0\}$, mapped to $P_{Q'}$ by $(w, t_3) \to (v, t_3) = (-t_3^r w, t_3)$. On the chart $t_1 = (t_2 - t_3^r)v^+$ of Y^+ and the chart $v^+ = t_2 z$ of $Y_1 = \mathrm{Bl}_{\Lambda^+} Y^+$, with coordinates (z, t_2, t_3), we have $P = \{t_2 = 0\}$. Both charts map to Q by $t_1 = t_2(t_2 - t_3^r)w$, resp. $t_1 = t_2(t_2 - t_3^r)z$, so they are isomorphic over Q via $z = w$. Near p, the complement in P of the chart with coordinates (z, t_2, t_3) is the section $\{z' = 0\}$ of $P \to \Lambda^+$ in the chart $t_2 = v^+ z'$ of Y_1, and its points other than the one over p map to $Q\backslash\Xi_\Lambda$. Hence $P_{Q'} \dashrightarrow P$ is $(v, t_3) \to (z, t_3) = (-v/t_3^r, t_3)$ near p, which is the composite of r elementary transformations of ϱ at the points of Ξ over p [Har77, Example V.5.7.1]: it blows up Ξ and its infinitely near points and contracts the strict transforms of the fiber $\{t_3 = 0\}$ and of the first $r - 1$ exceptional curves. So $Q'' \dashrightarrow Y_1$ induces a birational morphism $b : \tilde{P} \to P$ over Λ, and $Q'' \dashrightarrow Y_1$ is an isomorphism from a neighborhood in Q'' of the image of $\tilde{P}\backslash \mathrm{Exc}(b)$ onto a neighborhood of $P\backslash b(\mathrm{Exc}(b))$ in Y_1. The curves contracted by b are the strict transforms of the fibers of ϱ through $\operatorname{Supp}\Xi$ and, over each point of Ξ_Λ at which Ξ has length r, the strict transforms of the first $r - 1$ exceptional curves, which are the exceptional curves of the minimal resolution of the strict transform of $P_{Q'}$ in Q''. The sum of the total transforms of the exceptional curves of b has class $e_b := 3\eta - e$, and we let $\mathcal{E}_b \geq 0$ be the divisor of class e_b supported on the exceptional curves of b, which is unique because the intersection matrix of the exceptional curves of b is negative definite. Since b is a morphism over Λ, we have $b^*\sigma = \eta$. The class b^*f is orthogonal to the exceptional curves of b and satisfies $((b^*f)^2) = 0$ and $(b^*f \cdot \eta) = 1$, and the only such class is

$$b^*f = g + 2\eta - e.$$

Together with (7.13), we obtain

$$\begin{aligned} -K_{Q''}|_{\tilde{P}} &= b^*(\sigma + f), \qquad l|_{\tilde{P}} = b^*(2\sigma + f) - e_b, \\ \tilde{B}' &\equiv b^*(\sigma + 2f) - e_b, \qquad (-P - \bar{E}_{Q''})|_{\tilde{P}} = -b^*f. \end{aligned} \tag{7.14}$$

We prove (1) and (3). The divisors $\bar{E}_{Q''}$ and E_1 are the centers of E on Q'' and Y_1, and $Q'' \dashrightarrow Y_1$ is an isomorphism near $P\backslash b(\mathrm{Exc}(b))$, so $b(\tilde{B}') = E_1 \cap P$, and $\mu(b(\tilde{B}')) = B$. Since $(e_b \cdot \tilde{B}') = 3$ and $\tilde{B}' \equiv b^*(\sigma + 2f) - e_b$, we have $B \sim \sigma + 2f$, and $b|_{\tilde{B}'} : \tilde{B}' \to B$ is a birational morphism of smooth curves, hence an isomorphism, which is β_B. Let $\Delta := b(\mathcal{E}_b|_{\tilde{B}'})$, a divisor of degree three on B. Let $\mathcal{E} \geq 0$ be the sum of the total transforms on $\tilde{P}$ of the exceptional curves of $\tilde{P} \to P_{Q'}$, a divisor of class e. Then the pullback of Ξ_Λ to $\tilde{P}$ is $\mathcal{E}_b + \mathcal{E}$, and $\mathcal{E}|_{\tilde{B}'}$ corresponds to Ξ under $\tilde{B}' \cong B'$. As $\varrho|_\Xi$ is a closed immersion, the pullback of Ξ_Λ to B' is $\Xi + \iota(\Xi)$, so $\mathcal{E}_b|_{\tilde{B}'}$ corresponds to $\iota(\Xi)$, that is, $\beta_B(\iota(\Xi)) = \Delta$. This proves (1) and (3). Since $\varrho \circ \iota = \varrho$ on B', the divisor Δ maps isomorphically onto Ξ_Λ under the ruling $P \to \Lambda^+ \cong \Lambda$, and Δ maps isomorphically into L because $B \to L$ is an isomorphism. Locally at each point q of $\operatorname{Supp}\Delta$, the morphism b is the composite of the blow-ups of the points of Δ and of the infinitely near points of Δ on the strict transforms of B. Hence, for $n \geq 0$,

$$b_*\mathcal{O}_{\tilde{P}}(-n\mathcal{E}_b) = I_\Delta^n,$$

since in local coordinates (u, t) of P at q with $B = \{u = 0\}$ and $I_\Delta = (u, t^r)$ both sides are spanned by the monomials $u^i t^{i'}$ with $ri + i' \geq rn$, where r is the length of Δ at q.

We prove (2). Let $\phi \in H^0(X, mH)$ with $\mathrm{ord}_P(\phi) \geq j$, where $j \in \{2k, 2k + 1\}$, and assume that the image of ϕ in $W_{m,j}$ is non-zero, that is, $\mathrm{ord}_P(\phi) = j$. Let $\mathrm{div}_{\hat{X}}(\phi) \in |m\hat{\pi}^*H|$ and

$\operatorname{div}_{Y_1}(\phi) = \mu^* \operatorname{div}_{\hat{X}}(\phi) \in |mH|$ be the divisors of ϕ as a section of $\mathcal{O}_{\hat{X}}(m\hat{\pi}^*H)$ and of $\mathcal{O}_{Y_1}(mH)$. We have $\mu^*(\operatorname{div}_{\hat{X}}(\phi) - jP) = \operatorname{div}_{Y_1}(\phi) - jP - \frac{j}{2}E_1 \geq 0$, so $\operatorname{ord}_E(\phi) \geq \lceil j/2 \rceil$, and

$$\Theta_\phi := \operatorname{div}_{Y_1}(\phi) - jP - \lceil j/2 \rceil E_1$$

satisfies $\Theta_\phi \geq 0$ and does not contain P. The $\mathbb{Q}$-divisor on P of the image of ϕ in $W_{m,j}$ is

$$\big(\mu^*(\operatorname{div}_{\hat{X}}(\phi) - jP)\big)|_P = \Theta_\phi|_P + \big(\lceil j/2 \rceil - \tfrac{j}{2}\big)B.$$

Since $H|_P = f$, $P|_P \sim -\sigma - f$, and $E_1|_P = B \sim \sigma + 2f$, the divisor $\Theta_\phi|_P$ belongs to $|k\sigma + mf|$ for $j = 2k$ and to $|k\sigma + (m-1)f|$ for $j = 2k+1$. The section of $\mathcal{O}_P(\Theta_\phi|_P)$ defining $\Theta_\phi|_P$ is the restriction to P of ϕ divided by the equations of jP and $\lceil j/2 \rceil E_1$, which is linear in ϕ. Sending ϕ to this section defines the inclusion in (i) and the map $\theta_{m,k}$ in (ii), without the base conditions. In particular (i) and (ii) hold for $k \leq m$.

Assume that $k \geq m$. By Construction-Proposition 5.6(4), $\phi \in H^0(Q'', mH_{Q''} - jP)$, and the coefficient of $\bar{E}_{Q''}$ in the divisor $\operatorname{div}_{Q''}(\phi) \in |mH_{Q''}|$ of ϕ as a section of $\mathcal{O}_{Q''}(mH_{Q''})$ is $\operatorname{ord}_E(\phi) \geq \lceil j/2 \rceil$. Hence

$$\Theta''_\phi := \operatorname{div}_{Q''}(\phi) - jP - \lceil j/2 \rceil \bar{E}_{Q''}$$

satisfies $\Theta''_\phi \geq 0$ and does not contain P, and Θ_ϕ and Θ''_ϕ are strict transforms of each other. By Construction-Proposition 5.6(4),

$$mH_{Q''} - 2kP - k\bar{E}_{Q''} \sim (2m-k)(-K_{Q''}) + (k-m)l,$$

so by (7.14) the pullback D_ϕ to $\tilde{P}$ of the restriction of Θ''_ϕ to the strict transform of $P_{Q'}$ in Q'' belongs to $|b^*(k\sigma + mf) - (k-m)e_b|$ for $j = 2k$ and to $|b^*(k\sigma + (m-1)f) - (k-m)e_b|$ for $j = 2k+1$. As $Q'' \dashrightarrow Y_1$ is an isomorphism near $P \backslash b(\operatorname{Exc}(b))$, the divisors b_*D_ϕ and $\Theta_\phi|_P$ coincide outside the finite set $b(\operatorname{Exc}(b)) = \operatorname{Supp}\Delta$, hence $b_*D_\phi = \Theta_\phi|_P$. The divisor $b^*b_*D_\phi - D_\phi$ is supported on the exceptional curves of b and has class $(k-m)e_b$, so $b^*b_*D_\phi = D_\phi + (k-m)\mathcal{E}_b$. Thus the section defining $\Theta_\phi|_P$ belongs to

$$H^0\big(\tilde{P}, b^*\mathcal{O}_P(\Theta_\phi|_P) \otimes \mathcal{O}_{\tilde{P}}(-(k-m)\mathcal{E}_b)\big) = H^0\big(P, \mathcal{O}_P(\Theta_\phi|_P) \otimes I_\Delta^{k-m}\big)$$

by the projection formula. This proves (i) and (ii).

Finally, the strict transforms of the fibers of $Q' \to \mathbb{P}^2$ cover Q''. The strict transform Φ'' of a general fiber Φ satisfies $(h \cdot \Phi'') = (P \cdot \Phi'') = 1$ and $(D'' \cdot \Phi'') = 0$, as Φ is disjoint from Γ'. Hence $((mH_{Q''} - jP) \cdot \Phi'') = 5m - m - j = 4m - j$. Any divisor $G \geq 0$ on Q'' satisfies $(G \cdot \Phi'') \geq 0$, so $H^0(Q'', mH_{Q''} - jP) = 0$ for $j > 4m$. This proves (iii). □

Proof of Proposition 7.11 continued. We show that the rulings of $\bar{E}_{Q''} \cong \bar{E}_Q$ and of $E_1 \cong E$ correspond. The prime divisors on Q'' and on Y_1 exceptional over X are the centers of E and P (Construction-Proposition 5.6(4)), so the birational map $Y_1 \dashrightarrow Q''$ contracts no divisor and induces an isomorphism $H^0(Y_1, -K_{Y_1}) \cong H^0(Q'', -K_{Q''})$. By Construction-Proposition 5.6(2)–(4), $|-K_{Q''}|$ is base point free and

$$-K_{Q''}|_{\bar{E}_{Q''}} \sim 3(\bar{\sigma} + 2\bar{f}) - \bar{\sigma} - (2\bar{\sigma} + 3\bar{f}) = 3\bar{f}.$$

Let f_E be the class of the fibers of $E_1 \cong E \to L$. Since $(E|_E \cdot f_E) = -1$ and $((E|_E)^2) = (E^3) = -\deg N_{L/X} = 1$ [IP99, Lemma 2.2.14], we have $E|_E \sim -s - 2f_E$. As $-K_{Y_1} = H - E_1 - 2P$, the pullback of E is $E_1 + P$, $H|_{E_1} = f_E$, and $P|_{E_1} = s$, we obtain $E_1|_{E_1} \sim -2s - 2f_E$ and $-K_{Y_1}|_{E_1} \sim 3f_E$. The restrictions of a general member of $|-K_{Q''}|$ to $\bar{E}_{Q''}$ and of the corresponding member of $|-K_{Y_1}|$ to E_1 are sums of three fibers, and they coincide on the open subset of E over which $Y_1 \dashrightarrow Q''$ is an isomorphism. Hence the birational map $\psi : \bar{E}_{Q''} \dashrightarrow E_1$ induced by $Q'' \dashrightarrow Y_1$ maps general fibers of $\bar{E}_{Q''} \to \kappa$ to fibers of $E_1 \to L$, and there is an isomorphism $\epsilon : \kappa \to L$ such that ψ is a birational map over ϵ. The strict transform of B' in Q'' is a section of $\bar{E}_{Q''} \to \kappa$, as $(\bar{\sigma} \cdot \bar{f}) = 1$, and ψ is defined at the generic point of the strict transform of B' and maps it onto $E_1 \cap P$, which μ maps isomorphically onto B. For $z \in B'$ let $\bar{z} \in \kappa$ be the image

of z. By Claim 7.12(3), the morphisms $z \to \hat{\pi}(\beta_B(z))$ and $z \to \epsilon(\bar{z})$ from B' to L coincide at general points of B', hence

$$\hat{\pi}(\beta_B(z)) = \epsilon(\bar{z}) \qquad \text{for any } z \in B'. \tag{7.15}$$

Claim 7.13. *For any closed point $q \in \operatorname{Supp}\Delta$, there exists a line $L' \neq L$ of X through $\hat{\pi}(q)$.*

Proof. By Claim 7.12(3), $q = \beta_B(\zeta)$ for a point $\zeta \in \iota(\operatorname{Supp}\Xi)$. Let $\zeta' := \iota(\zeta) \in \operatorname{Supp}\Xi$. Let $\bar{f}_\zeta \subset \bar{E}_Q$ be the fiber of $\bar{E}_Q \to \kappa$ through ζ, and let $R \subset E_Q$ be the image of $\bar{f}_\zeta$ in Q, a line intersecting Λ in $\varrho(\zeta)$. Since $(B' \cdot \bar{f}_\zeta) = (\bar{\sigma} \cdot \bar{f}) = 1$, we have $\bar{f}_\zeta \cap B' = \{\zeta\}$, and $(\Gamma' \cdot \bar{f}_\zeta) = ((2\bar{\sigma} + 3\bar{f}) \cdot \bar{f}) = 2$ on $\bar{E}_Q$.

Suppose that $\zeta \neq \zeta'$. As $\varrho(\zeta) = \varrho(\zeta')$ and $\varrho|_\Xi$ is a closed immersion, $\zeta \notin \operatorname{Supp}\Xi$, so $\zeta \notin \Gamma'$. Hence $\Gamma' \cap \bar{f}_\zeta$ has length two and is disjoint from B', and $\bar{E}_Q \to E_Q$ is an isomorphism near $\Gamma' \cap \bar{f}_\zeta$. Moreover $\varrho(\zeta) = \varrho(\zeta') \in \Gamma \cap \Lambda$ is a further point of $R \cap \Gamma$. Thus $R \cap \Gamma$ has length at least three.

Suppose that $\zeta = \zeta'$, so that ζ is a ramification point of $B' \to \Lambda$ and $\zeta \in \Gamma'$. There are local analytic coordinates (u, t) of $\bar{E}_Q$ at ζ and (z_1, z_2, z_3) of Q at $\varrho(\zeta)$ in which $\bar{E}_Q \to Q$ is $(u, t) \to (ut, u, t^2)$, with $B' = \{u = 0\}$ and $\bar{f}_\zeta = \{t = 0\}$, so that $R = \{z_1 = z_3 = 0\}$ near $\varrho(\zeta)$. If Γ' were tangent to B' at ζ, then $\Gamma' = \{u = \vartheta_0(t)\}$ with $\operatorname{ord}_t \vartheta_0 \geq 2$, and the parametrization $t \to (t\vartheta_0(t), \vartheta_0(t), t^2)$ of Γ would have zero differential at $t = 0$, which contradicts the smoothness of $\Gamma \cong \Gamma'$. Hence $\Gamma' = \{t = \vartheta(u)\}$ with $\vartheta(0) = 0$. Let $\nu := \operatorname{ord}_u \vartheta$, which is finite and positive because $\Gamma' \neq \bar{f}_\zeta$. The intersection multiplicity of Γ' and $\bar{f}_\zeta$ at ζ is ν, so $\Gamma' \cap \bar{f}_\zeta$ has length $2 - \nu$ outside B'. In the parametrization $u \to (u\vartheta(u), u, \vartheta(u)^2)$ of Γ, the ideal of R restricts to $(u\vartheta(u), \vartheta(u)^2)$, of colength $\min(\nu + 1, 2\nu) = \nu + 1$. Thus $R \cap \Gamma$ has length at least $(2 - \nu) + (\nu + 1) = 3$.

In both cases $R \not\subset \Gamma$, and the strict transform R^+ of R in Y^+ satisfies $(D' \cdot R^+) = \operatorname{length}(R \cap \Gamma) \geq 3$, since $\mathcal{O}_{Y^+}(-D')|_{R^+} = I_\Gamma \mathcal{O}_{R^+}$. The divisor $-K_{Y^+} = 3h - D'$ is nef [KPZ14, Lemma 2.7], and $(h \cdot R^+) = 1$. Hence $(-K_{Y^+} \cdot R^+) = 3 - \operatorname{length}(R \cap \Gamma) = 0$. As $R \neq \Lambda$, by [KPZ14, Proposition 2.12] and Theorem 2.22, R^+ is the flopped curve of the strict transform $L'_Y \subset Y$ of a line $L' \neq L$ of X intersecting L. Let $x' := L \cap L'$, and let p' be the point $L'_Y \cap E$, which belongs to the fiber of E over x'.

Let $\tilde{E}^+$ be the normalization of the strict transform E^+ of E in Y^+, and let $\chi : \tilde{E}^+ \dashrightarrow E$ be the birational map induced by the flop $Y^+ \dashrightarrow Y$. The morphism $Y^+ \to Q$ induces a morphism $\tilde{E}^+ \to \bar{E}_Q$, and χ is the composite of the morphism $\tilde{E}^+ \to \bar{E}_Q$, the isomorphism $\bar{E}_Q \cong \bar{E}_{Q''}$, ψ, and $E_1 \cong E$. Thus χ is a birational map over ϵ, where $\tilde{E}^+$ is viewed over κ. Let $\tilde{R}^+ \subset \tilde{E}^+$ be a curve mapping onto R^+. Then $\tilde{R}^+$ maps onto $\bar{f}_\zeta$, the fiber over $\bar{\zeta}$. Since $Y^+ \dashrightarrow Y$ is an isomorphism outside the flopped and flopping curves, the total transform of $\tilde{R}^+$ under χ is contained in $L'_Y \cap E = \{p'\}$. As χ is a birational map over ϵ, the point p' belongs to the fiber over $\epsilon(\bar{\zeta})$. Hence $x' = \epsilon(\bar{\zeta}) = \hat{\pi}(\beta_B(\zeta)) = \hat{\pi}(q)$ by (7.15), and L' passes through $\hat{\pi}(q)$. □

Proof of Proposition 7.11 continued. Let P_x be the fiber of $\hat{\pi}|_P : P \to L$ over x. Since L is the only line of X through x, Claim 7.13 implies $P_x \cap \operatorname{Supp}\Delta = \emptyset$.

For $m \geq 1$ and $0 \leq k \leq 2m$ let $R_{m,k} := H^0(P, \mathcal{O}_P(k\sigma + mf) \otimes I_\Delta^{\max(k-m,0)})$, and let n_m, $S(R; v)$, and $A_{P,B/2}(v)$ be as in Definition 7.3. By Claim 7.12 and Lemma 7.5 for $P \cong \mathbb{P}^1 \times \mathbb{P}^1$, the curve B, and $Z := \Delta$, we have $n_m = \frac{3}{2}m^3 + O(m^2)$, and $S(R; v) \leq A_{P,B/2}(v)$ for any prime divisor v over P whose center is not contained in $\operatorname{Supp}\Delta$.

We compare W with R. Let v be a prime divisor over P. For a subspace $\mathcal{W}' \subset \mathcal{W}$ of a space $\mathcal{W}$ of sections, let $\mathcal{F}^t\mathcal{W} := \{\omega \in \mathcal{W} : v(\omega) \geq t\}$ and let $J(\mathcal{W})$ be the sum of the jumping numbers of v on $\mathcal{W}$. Since $\dim(\mathcal{W}' \cap \mathcal{F}^t\mathcal{W}) \leq \dim \mathcal{F}^t\mathcal{W}$ for any t, the i-th largest jumping number of v on $\mathcal{W}'$ is at most the i-th largest jumping number of v on $\mathcal{W}$. As jumping numbers are non-negative, $J(\mathcal{W}') \leq J(\mathcal{W})$. By Claim 7.12(2)(i), $J(W_{m,2k}) \leq J(R_{m,k})$. Let $s_0 \in H^0(P, \mathcal{O}_P(f))$ define a fiber f_0 of $\hat{\pi}|_P$. By Claim 7.12(2)(ii), $\omega \to s_0\theta_{m,k}(\omega)$ is an injective linear map from $W_{m,2k+1}$ to

$R_{m,k}$ with $v(\omega) = v(s_0\theta_{m,k}(\omega)) + \frac{1}{2}\operatorname{mult}_v B - \operatorname{mult}_v f_0$, so

$$J(W_{m,2k+1}) \le J(R_{m,k}) + \frac{1}{2}\operatorname{mult}_v B \cdot \dim W_{m,2k+1}.$$

By Claim 7.12(2)(iii), $W_{m,j} = 0$ for $j > 4m$, so $k \le 2m$ in both cases. Since $\sum_j \dim W_{m,j} = N_m = h^0(X, mH) = 3m^3 + O(m^2) = 2n_m + O(m^2)$, we obtain

$$S(W; v) = \lim_{m\to+\infty} \frac{\sum_j J(W_{m,j})}{mN_m} \le \limsup_{m\to+\infty} \frac{2\sum_{k=0}^{2m} J(R_{m,k}) + \frac{1}{2}\operatorname{mult}_v B \cdot N_m}{m(2n_m + O(m^2))} = S(R; v).$$

Let v be a prime divisor over P with $\operatorname{center}_P v \cap P_x \ne \emptyset$. Then $\operatorname{center}_P v \not\subset \operatorname{Supp}\Delta$, so

$$S(W; v) \le S(R; v) \le A_{P,B/2}(v),$$

and $A_{P,B/2}(v)$ is the log discrepancy of v with respect to $(P, \frac{1}{2}B)$, where $\frac{1}{2}B$ is the different of $(\hat{X}, P)$ on P (Construction-Proposition 7.1). Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$. Then $\operatorname{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(x) = P_x$. By Lemma 3.1, applied to the plt extraction $\hat{\pi} : \hat{X} \to X$ of P, the closed subset P_x, and $\lambda = 1$, together with $A_X(P) = 3$ and $S_X(P) = \frac{43}{18}$ (Proposition 5.7), we obtain

$$A_X(F) - S_X(F) \ge \left(3 - \frac{43}{18}\right)\operatorname{mult}_F P = \frac{11}{18}\operatorname{mult}_F P.$$

Finally $\operatorname{mult}_F P > 0$, since $\operatorname{center}_{\hat{X}} F \subset P$. □

7.3. **The $\mathbb{F}_2$ case.** In this subsection, we prove Theorem 7.2. Proposition 7.11 treats the case where the normalization of the image of E in Q is $\mathbb{F}_0$, and here we reduce the remaining $\mathbb{F}_2$ case to Lemmas 7.8, 7.16, 7.9, and 7.10, after excluding one point of L by Lemma 7.15.

We first describe the powers of a curvilinear ideal on a smooth surface.

Lemma 7.14. *Let T be a smooth surface, $q \in T$ a closed point, (u, t) local coordinates of T at q, and $n \ge 1$ an integer. Let $\Delta \subset T$ be the subscheme supported at q with ideal (u, t^n), let $\varphi : T' \to T$ be the composition of n blow-ups, the first of which is the blow-up of q and the i-th of which is the blow-up of the intersection point of the exceptional curve of the $(i-1)$-th blow-up with the strict transform of $(u = 0)$, and let J' be the sum of the total transforms on T' of the n exceptional curves. Then $I_\Delta \cdot \mathcal{O}_{T'} = \mathcal{O}_{T'}(-J')$ and $\varphi_*\mathcal{O}_{T'}(-\nu J') = I_\Delta^\nu$ for any integer $\nu \ge 0$.*

Proof. In the chart $u = u_1 t$ of the blow-up of q we have $(u, t^n) = t \cdot (u_1, t^{n-1})$, and in the other chart the pullback of (u, t^n) is generated by u. By induction on n, $I_\Delta \cdot \mathcal{O}_{T'} = \mathcal{O}_{T'}(-J')$. Let E_i be the strict transform on T' of the i-th exceptional curve. Then $J' = \sum_{i=1}^n iE_i$, and ord_{E_i} is the monomial valuation with $\operatorname{ord}_{E_i}(u) = i$ and $\operatorname{ord}_{E_i}(t) = 1$. Hence $\varphi_*\mathcal{O}_{T'}(-\nu J')$ is the monomial ideal spanned by the monomials $u^\alpha t^b$ with $i\alpha + b \ge \nu i$ for $1 \le i \le n$, that is, with $\alpha + b/n \ge \nu$. The monomials $u^\alpha t^b$ with $\alpha + b/n \ge \nu$ also span $(u, t^n)^\nu$. □

In **Step 5** of the proof of Theorem 7.2, the following lemma shows that x differs from the point p_0 of L defined in it, so that the fiber P_x does not contain the point $q_0 = B \cap \sigma$ of **Step 2**.

Lemma 7.15. *Notation as in Construction-Proposition 7.1. Assume that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, and let $p_0 \in L$ be the zero of the image of a non-zero section of $N_{s/Y}$ in $H^0(N_{s/Y}/N_{s/E}) \cong H^0(\mathcal{O}_{\mathbb{P}^1}(1))$, where s is identified with L through π. Then the local ring of $\Sigma(X, p_0)$ at $[L]$ is not isomorphic to $\mathbb{C}[t]/(t^2)$.*

Proof. Since L is special, $N_{L/X} = M \oplus N_-$ with $M \cong \mathcal{O}_{\mathbb{P}^1}(1)$ and $N_- \cong \mathcal{O}_{\mathbb{P}^1}(-2)$, the divisor E is the projectivization of $N_{L/X}$, and s corresponds to the subbundle M. Thus $N_{s/E} \cong \operatorname{Hom}(M, N_-) \cong \mathcal{O}_{\mathbb{P}^1}(-3)$, $N_{E/Y}|_s \cong M$, and there exists an exact sequence

$$0 \to N_{s/E} \to N_{s/Y} \to N_{E/Y}|_s \to 0. \tag{7.16}$$

Let $\delta : H^0(L, M) \to H^1(L, N_{s/E})$ be the connecting map of (7.16). Since $h^0(N_{s/Y}) = 1$ and $h^0(N_{s/E}) = 0$, we have $\operatorname{Ker}\delta = \mathbb{C}\ell_0$, where $\ell_0 \neq 0$ is the image of a non-zero section of $N_{s/Y}$, and p_0 is the zero of ℓ_0.

Let $x' \in L$ be a closed point, and cover L by two affine open subsets L_1, L_2 with $x' \in L_1 \backslash L_2$. For $i = 1, 2$ choose functions λ_i, u_i, v_i on the formal completion of X along L_i such that λ_i restricts to a coordinate of L_i, u_i and v_i generate the ideal of L_i, and the classes of u_i and v_i in $I_L/I_L^2 = N_{L/X}^\vee$ generate $M^\vee$ and $N_-^\vee$ over L_i. On the formal completion along $L_1 \cap L_2$ we have

$$\begin{aligned} u_2 &= \tau_M u_1 + \xi u_1^2 + (\text{element of } (u_1v_1, v_1^2) + I_L^3), \\ v_2 &= \tau_N v_1 + a u_1^2 + (\text{element of } (u_1v_1, v_1^2) + I_L^3), \end{aligned} \tag{7.17}$$

and $\lambda_2 = \mu(\lambda_1) + \zeta u_1 + (\text{element of } (v_1) + I_L^2)$, where $\xi, a, \zeta, \tau_M, \tau_N$ are functions of λ_1 on $L_1 \cap L_2$, the invertible functions τ_M, τ_N are the transition functions of $M^\vee$ and $N_-^\vee$, and a is a section over $L_1 \cap L_2$ of

$$\operatorname{Hom}(N_-^\vee, \operatorname{Sym}^2 M^\vee) \cong \operatorname{Hom}(M \otimes M, N_-) \cong \mathcal{O}_{\mathbb{P}^1}(-4).$$

For $\ell \in H^0(L, M)$ we write $a\ell \in \operatorname{Hom}(M, N_-) = N_{s/E}$ and $a\ell^2 \in N_-$ for the contractions over $L_1 \cap L_2$.

We first compute δ. Over $\pi^{-1}(L_i)$ near s, the threefold Y has coordinates λ_i, u_i, w_i with $v_i = u_i w_i$, $E = (u_i = 0)$, and $s = (u_i = w_i = 0)$. Since $I_L^2 \mathcal{O}_Y \subset (u_1^2)$ and $(u_1v_1, v_1^2) + I_L^3 \subset (u_1^2 w_1, u_1^3)$, (7.17) implies

$$w_2 = \frac{v_2}{u_2} = \frac{\tau_N w_1 + a u_1 + (\text{element of } (u_1 w_1, u_1^2))}{\tau_M + (\text{element of } (u_1))} \equiv \frac{\tau_N}{\tau_M} w_1 + \frac{a}{\tau_M} u_1 \mod I_s^2.$$

Thus in $N_{s/Y}^\vee = I_s/I_s^2$ we have $[u_2] = \tau_M[u_1]$ and $[w_2] = \frac{\tau_N}{\tau_M}[w_1] + \frac{a}{\tau_M}[u_1]$. The classes $[u_i]$ generate $N_{E/Y}^\vee|_s$, and the classes $[w_i]$ map to generators of $N_{s/E}^\vee$. Hence the extension class of (7.16) is represented by the 1-cocycle a, and $\delta(\ell)$ is, up to sign, the class of the 1-cocycle $a\ell$ with values in $N_{s/E}$.

Next let $\ell \in H^0(L, M(-x'))$ be non-zero, and let ℓ_i be the function $u_i(\ell)$ on L_i. For functions r_i, b_i on L_i, the ideal

$$\left(u_i - t\ell_i - t^2 r_i,\ v_i - t^2 b_i\right) \subset \widehat{\mathcal{O}}_{X,L_i}[t]/(t^3)$$

contains I_L^3, so it defines a closed subscheme of $X \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$ over L_i. The subscheme is flat over $\mathbb{C}[t]/(t^3)$ with fiber L_i over $t = 0$, and for $i = 1$ it contains $\{x'\} \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$ if and only if $r_1(x') = b_1(x') = 0$. On the subscheme for $i = 1$, (7.17) implies

$$v_2 \equiv t^2(\tau_N b_1 + a\ell_1^2) \mod t^3, \qquad u_2 \equiv t\tau_M \ell_1 + t^2(\tau_M r_1 + \xi \ell_1^2) \mod t^3,$$
$$\lambda_2 \equiv \mu(\lambda_1) + t\zeta\ell_1 \mod t^2.$$

Since $\ell_2 \circ \mu = \tau_M \ell_1$, the subschemes for $i = 1$ and $i = 2$ coincide over $L_1 \cap L_2$ if and only if

$$b_2 \circ \mu - \tau_N b_1 = a\ell_1^2 \qquad \text{and} \qquad r_2 \circ \mu - \tau_M r_1 = \xi\ell_1^2 - \zeta\ell_1(\ell_2' \circ \mu),$$

where ℓ_2' denotes the derivative of ℓ_2 with respect to λ_2. The right-hand side of the second equation is a 1-cochain with values in M, and the right-hand side of the first is the 1-cocycle $a\ell^2$ with values in N_-. Since $x' \notin L_2$, the sheaves $M(-x')$ and M, and $N_-(-x')$ and N_-, coincide on $L_1 \cap L_2$. Since $H^1(L, M(-x')) = 0$, the second equation has a solution with $r_1(x') = 0$. The first equation has a solution with $b_1(x') = 0$ if and only if the class of $a\ell^2$ in $H^1(L, N_-(-x'))$ vanishes. Since ℓ is a nowhere vanishing section of $M(-x') \cong \mathcal{O}_L$, multiplication by ℓ is an isomorphism $N_{s/E} = \operatorname{Hom}(M, N_-) \to N_-(-x')$ which maps $a\ell$ to $a\ell^2$. Hence the first equation has a solution with $b_1(x') = 0$ if and only if $\delta(\ell) = 0$.

Now let $x' = p_0$ and $\ell = \ell_0$. Then $\delta(\ell_0) = 0$, and the subschemes above glue to a closed subscheme of $X \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$, flat over $\mathbb{C}[t]/(t^3)$, containing $\{p_0\} \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$, with fiber L over $t = 0$. By flatness the glued subscheme has the Hilbert polynomial of L, so it defines a morphism $\operatorname{Spec}\mathbb{C}[t]/(t^3) \to \Sigma(X, p_0)$ mapping the closed point to $[L]$, whose tangent vector is

$\ell_0 \neq 0$ in $H^0(L, N_{L/X}(-p_0))$, the tangent space of $\Sigma(X, p_0)$ at $[L]$ (Proposition 4.1). If the local ring of $\Sigma(X, p_0)$ at $[L]$ were $\mathbb{C}[t]/(t^2)$, then any local homomorphism $\mathbb{C}[t]/(t^2) \to \mathbb{C}[t]/(t^3)$ would map t to $\alpha t + \beta t^2$ with $\alpha^2 = 0$, and the induced map of tangent spaces would vanish. This is a contradiction. □

The next lemma treats the fiber through the point q_3 of Construction-Proposition 7.7 on the negative section.

Lemma 7.16. *Notation as in Construction-Proposition 7.7. Let $Z := Z_B \sqcup \{q_3\}$, let C be the fiber through q_3, and for $m \geq 1$ and $0 \leq k \leq 2m$ let $R_{m,k} := H^0(S, \mathcal{O}_S(k\sigma + (m+k)f))$ for $k \leq m$ and $R_{m,k} := s_\sigma^{k-m} H^0(S, \mathcal{O}_S(m\sigma + (m+k)f) \otimes I_Z^{k-m})$ for $k \geq m$, where s_σ defines σ. Let n_m, $S(R; v)$, $A_{S,B/2}(v)$, and $S(R^C; q)$ be as in Definition 7.3, with h the identity. Then*

$$S(R; \mathrm{ord}_C) = \frac{2}{3}, \qquad S(R^C; q_3) \leq \frac{31}{36}, \qquad S(R^C; q) \leq \frac{1}{3} \quad \text{for } q \in C \backslash \{q_3\}.$$

In particular $A_{S,B/2}(v) \geq \frac{36}{31} S(R; v)$ for any prime divisor v over S whose center intersects C.

Proof. We use the surface $\tilde{S}$ and the classes e_1, e_2, e_3, $\tilde{\sigma} = \sigma - e_3$, and $c_i = f - e_i$ of Construction-Proposition 7.7, and we write σ and f also for their pullbacks to $\tilde{S}$. The curve c_3 is the strict transform of C. By Lemma 7.14, applied at the points of $\mathrm{Supp}\, Z_B$ and at q_3, the powers $I_{Z_B}^p$ and $I_{q_3}^p$, $p \geq 0$, are the push-forwards of $\mathcal{O}_{\tilde{S}}(-p(e_1 + e_2))$ and of $\mathcal{O}_{\tilde{S}}(-pe_3)$.

Let $0 \leq k \leq 2m$ and $\ell \geq 0$ be integers, and let $a := k/m$, $u := \ell/m$, $\gamma := \max(a - 1, 0)$, and $h_3 := \max(a - 1 - u, 0)$. Since $C \neq \sigma$, multiplication by s_σ^{k-m} does not change the order along C. The fiber C is disjoint from Z_B, and a local equation of C at q_3 is part of a system of local coordinates, so $(I_{q_3}^{k-m} : s_C^\ell) = I_{q_3}^{\max(k-m-\ell, 0)}$, where s_C defines C. Hence $\{r \in R_{m,k} : \mathrm{ord}_C(r) \geq \ell\}$ is isomorphic to $H^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(mD_{a,u}))$, where

$$D_{a,u} := \begin{cases} a\sigma + (a + 1 - u)f & (a \leq 1), \\ \sigma + (a + 1 - u)f - \gamma(e_1 + e_2) - h_3 e_3 & (a \geq 1). \end{cases}$$

Let $D_{a,u} = D_{a,u}^+ + D_{a,u}^-$ be the Zariski decomposition of the $\mathbb{R}$-divisor $D_{a,u}$ [Laz04a, Theorem 2.3.19 and Proposition 2.3.21]. Table 92 in Appendix A lists the four chambers of the region where $D_{a,u}$ is big, the support of $D_{a,u}^-$, the coefficient w of $D_{a,u}^-$ along $\tilde{\sigma}$, the intersection number $d = d(a, u) := (D_{a,u}^+ \cdot c_3)$, and h_3. When $w > 0$ we have $D_{a,u}^- = w(\tilde{\sigma} + e_3)$.

To verify the table, note that the intersection matrix of $\tilde{\sigma}$ and e_3 is $\left(\begin{smallmatrix} -3 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, which is negative definite. The four chambers Ω_1, Ω_2, Ω_3, and Ω_4 are the triangles with vertices $(0,0), (1,0), (0,1)$, with vertices $(1,0), (1,2), (0,1)$, with vertices $(1,0), (2,0), (2,1)$, and with vertices $(1,0), (2,1), (1,2)$, respectively. On Ω_1 we have $w = h_3 = 0$ and $D_{a,u}^- = 0$, and on Ω_2 and Ω_4 the coefficient w is determined by $(D_{a,u}^+ \cdot \tilde{\sigma}) = 0$. On all four chambers

$$D_{a,u}^+ = (\min(a, 1) - w)\sigma + (a + 1 - u)f - \gamma(e_1 + e_2) - h_3 e_3,$$

so that

$$(D_{a,u}^+ \cdot \tilde{\sigma}) = -2(\min(a, 1) - w) + (a + 1 - u) - h_3, \qquad (D_{a,u}^+ \cdot c_i) = \min(a, 1) - w - \gamma, \qquad (D_{a,u}^+ \cdot e_i) = \gamma \quad (i =$$
$$(D_{a,u}^+ \cdot c_3) = \min(a, 1) - w - h_3 = d(a, u), \qquad (D_{a,u}^+ \cdot e_3) = h_3.$$

If Z_B is non-reduced, then $(D_{a,u}^+ \cdot C_1) = \min(a, 1) - w - \gamma$, $(D_{a,u}^+ \cdot e_2) = \gamma$, and $(D_{a,u}^+ \cdot D_1) = 0$ for the curves C_1 and D_1 of Construction-Proposition 7.7(2). Substituting the values of w and h_3 from the table, we obtain the following intersection numbers of $D_{a,u}^+$, where the columns c_i

and e_i stand for $i = 1, 2$:

	$\tilde{\sigma}$	c_i	e_i	c_3	e_3
Ω_1	$1-a-u$	a	0	a	0
Ω_2	0	$\frac{a-u+1}{2}$	0	$\frac{a-u+1}{2}$	0
Ω_3	0	$2-a$	$a-1$	$2-a+u$	$a-1-u$
Ω_4	0	$\frac{3-a-u}{2}$	$a-1$	$\frac{1+a-u}{2}$	0

All of them are non-negative on the respective chambers, so $D^+_{a,u}$ is nef by the description of the cone of curves in Construction-Proposition 7.7(1) and (2), and $(D^+_{a,u} \cdot \tilde{\sigma}) = (D^+_{a,u} \cdot e_3) = 0$ whenever $w > 0$. Outside the four chambers $D_{a,u}$ is not big, since $((\sigma+2f)\cdot D_{a,u}) = a+1-u < 0$ for $a \le 1$ and $u > a+1$, and $((\sigma+2f-e_1-e_2)\cdot D_{a,u}) = 3-a-u < 0$ for $a \ge 1$ and $u > 3-a$, where $\sigma+2f$ and $\sigma+2f-e_1-e_2$ are nef by the description of the cone of curves in Construction-Proposition 7.7(1) and (2).

On each chamber $D^+_{a,u}$ is affine in (a,u), and $\frac{\partial}{\partial u}\operatorname{vol}(D_{a,u}) = -2d(a,u)$. Riemann–Roch on $\tilde{S}$ and the Zariski decomposition determine the dimensions of the spaces above up to $O(m)$, uniformly on the four chambers, and the Riemann sums of the normalized sums converge to the corresponding integrals over the four chambers with error $O(1/m)$. Integrating the affine functions of Table 92 over the four chambers, the integrals of d, ud, $d^2/2$, and $d(\gamma+w+h_3)$ are $\frac{3}{2}$, 1, $\frac{1}{2}$, and $\frac{19}{24}$ (Remark A.36). The first integral is $\lim n_m/m^3$. Since $\int_0^\infty \operatorname{vol}(D_{a,u})du/2 = \int_0^\infty ud(a,u)du$, the second integral implies $S(R;\operatorname{ord}_C) = \frac{2}{3}\cdot 1 = \frac{2}{3}$.

For the secondary orders, note that c_3 intersects e_3 in one point, which maps to q_3, and that $(c_3\cdot\tilde{\sigma}) = (c_3\cdot e_1) = (c_3\cdot e_2) = 0$. The restriction to C of an element of $R_{m,k}$ with order ℓ along C is a section of $\mathcal{O}_C(k)$ which vanishes at q_3 to order at least $m(\gamma+w+h_3)-O(1)$: the factor s_σ^{k-m} contributes $m\gamma$, the negative part contributes mw through e_3, and the condition $I_{q_3}^{mh_3}$ contributes mh_3. The remaining factor is a section of a line bundle of degree $md+O(1)$ on $C \cong \mathbb{P}^1$. Hence the sum of the jumping numbers of ord_{q_3} on $R^C_{m,k,\ell}$ is at most $m^2\big(d(\gamma+w+h_3)+d^2/2\big)+O(m)$, and at any other point $q \in C$ it is at most $m^2d^2/2+O(m)$. Summing over k and ℓ and dividing by $mn_m = \frac{3}{2}m^4+O(m^3)$,

$$S(R^C;q_3) \le \frac{2}{3}\Big(\frac{1}{2}+\frac{19}{24}\Big) = \frac{31}{36}, \qquad S(R^C;q) \le \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3} \quad (q \ne q_3).$$

Finally let $\lambda := 36/31$. Since C is not a component of B, $A_{S,B/2}(\operatorname{ord}_C) = 1$. The curve B does not contain q_3, since $B\cap\sigma = \{q_0\}$, and B intersects C transversally in one point q_B, since $(B\cdot f) = 1$. Thus $(S, \frac{1}{2}B+C)$ is plt near any point of C, the different Δ_C of $(S,\frac{1}{2}B+C)$ on C is $\frac{1}{2}q_B$, and

$$\lambda S(R;\operatorname{ord}_C) \le \frac{24}{31} \le 1, \qquad \lambda S(R^C;q_3) \le 1,$$
$$\lambda S(R^C;q_B) \le \frac{12}{31} \le \frac{1}{2}, \qquad \lambda S(R^C;q) \le \frac{12}{31} \le 1 \quad (q \ne q_3, q_B).$$

Lemma 7.4, applied with h the identity at any point of C, implies $A_{S,B/2}(v) \ge \lambda S(R;v)$ for any prime divisor v over S whose center contains a point of C. □

Proof of Theorem 7.2. The proof is long, so we divide it into several steps.

Step 1. In this step we show that L is special and reduce to the case when the normalization of the image of E in Q is $\mathbb{F}_2$.

The local ring of $\Sigma(X,x)$ at $[L]$ has length two, so by Proposition 4.1 it is isomorphic to $\mathbb{C}[t]/(t^2)$ and L is special. We use the notation of Set-up 5.1 and Construction-Propositions 5.6 and 7.1. Since $\Sigma(X,x)$ is supported at $[L]$, the line L is the only line of X through x.

By Proposition 5.7,

$$A_X(P) = 3, \qquad S_X(P) = \frac{43}{18}. \tag{7.18}$$

If the normalization $\bar{E}_Q$ of E_Q is $\mathbb{F}_0$, then Proposition 7.11 implies

$$A_X(F) - S_X(F) \geq \frac{11}{18}\operatorname{mult}_F P \geq \frac{27}{95}\operatorname{mult}_F P.$$

Thus we may assume that $\bar{E}_Q \cong \mathbb{F}_2$, so that $h|_{\bar{E}_Q} = \bar{\sigma} + 3\bar{f}$, $P_{Q'}|_{\bar{E}_Q} = \bar{\sigma} + \bar{f}$, and $\Gamma' \sim 2\bar{\sigma} + 5\bar{f}$ (Construction-Proposition 5.6(2)(3)).

We also compare the orders along P and along E. Let $Y_1 := \operatorname{Bl}_s Y$, with exceptional divisor P, and let E_1 be the strict transform of E. At a point of L choose local coordinates (z_1, z_2, z_3) of X such that $L = (z_1 = z_2 = 0)$ and the non-reduced conic on L has ideal (z_2, z_1^2) [KPS18, Lemma 2.1.1, Remarks 2.1.2 and 2.1.7]. By Construction-Proposition 7.1, $Y_1 \to X$ factors through a morphism $Y_1 \to \hat{X}$ which contracts E_1 onto $B = E_1 \cap P$ and maps $P \subset Y_1$ isomorphically onto $P \subset \hat{X}$, and we identify $P \subset Y_1$ with $P \subset \hat{X}$ through $Y_1 \to \hat{X}$. Moreover ord_P and ord_E are the monomial valuations with weights $(1,2)$ and $(1,1)$ on (z_1, z_2) over the generic point of L. Taking a monomial $z_1^{\alpha_1} z_2^{\alpha_2}$ of a function ϕ with $\operatorname{ord}_E(\phi) = \alpha_1 + \alpha_2$, we obtain $\operatorname{ord}_P(\phi) \leq \alpha_1 + 2\alpha_2 \leq 2\operatorname{ord}_E(\phi)$. Thus

$$\operatorname{ord}_P(\phi) \geq j \implies \operatorname{ord}_E(\phi) \geq \lceil j/2 \rceil \tag{7.19}$$

for any $\phi \in \mathcal{O}_{X,\eta_L}$, where η_L is the generic point of L, in particular for any $\phi \in H^0(X, mH)$, and for any integer $j \geq 0$.

Step 2. In this step we show that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and that the flop $Y_1 \dashrightarrow Y_1^+$ of σ maps $P \cong \mathbb{F}_2$ isomorphically onto the exceptional divisor P^+ of $Y_1^+ = \operatorname{Bl}_{\Lambda^+} Y^+ \to Y^+$.

Let $\varrho : P_{Q'} \to \Lambda$ be the ruling and let $\Xi := \Gamma' \cap P_{Q'}$. By Construction-Proposition 5.6(3), Ξ has length three and $\varrho|_\Xi$ is a closed immersion onto $\Gamma \cap \Lambda$. The fibers of $\bar{E}_Q \to \kappa$ are fibers of $Q' \to \mathbb{P}^2$, and the only fiber of $Q' \to \mathbb{P}^2$ contained in $P_{Q'}$ is g. Hence the fiber component of $P_{Q'}|_{\bar{E}_Q} = \bar{\sigma} + \bar{f}$ is g, and

$$P_{Q'} \cap \bar{E}_Q = g \cup B',$$

where $B' \subset P_{Q'}$ is the curve $\bar{\sigma}$. Since $\bar{E}_Q \sim 2l$ and $l|_{P_{Q'}} = g + \eta$, we have $B' \sim g + 2\eta$ on $P_{Q'}$, so $(g \cdot B') = 1$, and g and B' intersect transversally in one point c. Since $\Xi \subset \bar{E}_Q \cap P_{Q'}$ and intersections of curves in $\bar{E}_Q$ can be computed on $\bar{E}_Q$,

$$\Xi \subset g \cup B', \qquad \operatorname{length}(\Xi \cap g) = (\Gamma' \cdot \bar{f}) = 2, \qquad \operatorname{length}(\Xi \cap B') = (\Gamma' \cdot \bar{\sigma}) = 1. \tag{7.20}$$

Let $\Lambda^+ \subset Y^+$ be the strict transform of Λ, the flopped curve of s. We compute the conormal sheaf of Λ^+ near a point $o \in \Gamma \cap \Lambda$. Choose local coordinates (t, x_1, x_2) of Q at o with $\Lambda = (x_1 = x_2 = 0)$, and let $y := x_2/x_1$, a fiber coordinate of ϱ, normalized so that $g = (y = 0)$ near $\varrho^{-1}(o)$. Since $\varrho|_\Xi$ is a closed immersion, the part of Ξ over o has ideal $(t^n, y - \rho(t))$ for an integer $n \geq 1$ and a polynomial ρ of degree less than n, and the tangent line of Γ' at the point of Ξ over o is not the tangent line of the fiber of ϱ.

Hence, in the chart of Q' with coordinates (t, x_1, y), Γ' is defined by $y = \psi(t, x_1)$ and $F(t, x_1) = 0$, so that $I_\Gamma = (F, x_2 - \psi x_1)$, $F(t, 0) = t^n \epsilon(t)$ with $\epsilon(0) \neq 0$, and $\psi(t, 0) \equiv \rho(t)$ modulo t^n. In the chart of $\operatorname{Bl}_\Gamma Q$ with coordinates (t, x_1, ζ), $x_2 - \psi x_1 = \zeta F$, the curve Λ^+ is $(x_1 = \zeta = 0)$, and modulo $I_{\Lambda^+}^2$,

$$\zeta \equiv \frac{x_2 - \rho(t)x_1}{t^n \epsilon(t)} + (\text{multiple of } x_1).$$

Thus $N^\vee_{\Lambda^+/Y^+}$ is generated near o by the classes of x_1 and $(x_2 - \rho(t)x_1)/t^n$. Since $N^\vee_{\Lambda/Q} = \varrho_* \mathcal{O}_{P_{Q'}}(g)$, the conormal form $\alpha x_1 + \beta x_2$ corresponds to the section $\alpha + \beta y$ of $\mathcal{O}_{P_{Q'}}(g)$, and $\Gamma \cap \Lambda$ is a divisor of degree three on Λ, we obtain

$$N^\vee_{\Lambda^+/Y^+} \cong \mathcal{K}_0 := \varrho_*\big(I_\Xi \otimes \mathcal{O}_{P_{Q'}}(g + 3\eta)\big) \subset \varrho_* \mathcal{O}_{P_{Q'}}(g + 3\eta) \cong \mathcal{O}_{\mathbb{P}^1}(3) \oplus \mathcal{O}_{\mathbb{P}^1}(2), \tag{7.21}$$

where the summand $\mathcal{O}_{\mathbb{P}^1}(3)$ is $s_g \cdot \varrho^* \mathcal{O}_\Lambda(3)$ for the section s_g defining g. The sheaf $\mathcal{K}_0$ has rank two and degree $5 - 3 = 2$. By (7.20), the ideal $(I_\Xi : s_g)$ has colength one, say at a point z_0. If s' is a section of $\mathcal{O}_{P_{Q'}}(\eta)$ vanishing on $\varrho^{-1}(\varrho(z_0))$, then $s_g s' \in H^0(I_\Xi(g + \eta)) = H^0(\mathcal{K}_0(-2))$ is

non-zero. By [Gro57, Théorème 2.1], $\mathcal{K}_0 \cong \mathcal{O}(a_1) \oplus \mathcal{O}(2-a_1)$ with $a_1 \geq 2$, and $a_1 \leq 3$ by (7.21). If $a_1 = 3$, then the summand $\mathcal{O}(3)$ of $\mathcal{K}_0$ is the summand $s_g \cdot \varrho^* \mathcal{O}_\Lambda(3)$, so $s_g \in I_\Xi$ and $\Xi \subset g$, contradicting (7.20). Thus $\mathcal{K}_0 \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}$ and

$$N_{\Lambda^+/Y^+} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2).$$

The image of s in the common contraction $\bar{Y}$ of the flop $Y \dashrightarrow Y^+$ is a point, which is also the image of Λ^+, so $w(s) = w(\Lambda^+)$ by Definition-Proposition 2.21. By Theorem 2.22, $N_{s/Y}$ is $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, so s satisfies the hypothesis of Definition-Proposition 2.21, and so does Λ^+ by the computation above. For such a curve the width is one if and only if the normal bundle is $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ [Rei83, Definition (5.3) and the sentence after it]. Hence $w(\Lambda^+) \geq 2$, so $w(s) \geq 2$ and

$$N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2).$$

In particular $P = \mathbb{P}(N_{s/Y}) \cong \mathbb{F}_2$. Let σ be the negative section and f the class of a fiber of $P \to s$.

By [Rei83, Definition (5.3), Theorem (5.4), (5.7), Pagoda (5.8)], near s the flop $Y \dashrightarrow Y^+$ is the composite of the inverse of the blow-up $Y_1 \to Y$, the flop $Y_1 \dashrightarrow Y_1^+$ of σ, and the contraction $Y_1^+ \to Y^+$, where $Y_1^+ := \mathrm{Bl}_{\Lambda^+} Y^+$ and the image P^+ of P is the exceptional divisor of $Y_1^+ \to Y^+$. The other flopping curves of $Y \dashrightarrow Y^+$ are disjoint from s (Theorem 2.22). Since $(\sigma^2)_P = -2$, [Rei83, (6.1) and Theorem (6.3)] implies that $Y_1 \dashrightarrow Y_1^+$ maps P isomorphically onto P^+, and that for any divisor D on Y_1 with strict transform D^+ on Y_1^+,

$$D^+|_{P^+} = D|_P + (D \cdot \sigma)\sigma \tag{7.22}$$

under the isomorphism $P \cong P^+$. Moreover $Y_1 \dashrightarrow Y_1^+$ is an isomorphism near $P \backslash \sigma$. The isomorphism $P \cong P^+$ maps fibers to fibers, and we identify $L \cong s$ with $\Lambda \cong \Lambda^+$ accordingly.

On Y_1 we have $(H \cdot s) = (H \cdot L) = 1$, and $(E \cdot s) = \deg N_{E/Y}|_s = \deg N_{s/Y} - \deg N_{s/E} = 1$ by (7.16). Since $-P|_P$ is the tautological class of $\mathbb{P}(N_{s/Y})$ and E pulls back to $E_1 + P$,

$$H|_P = f, \qquad P|_P = -\sigma - 2f, \qquad E_1|_P = B \sim \sigma + 3f. \tag{7.23}$$

The negative section σ corresponds to the subbundle $\mathcal{O}_{\mathbb{P}^1}$ of $N_{s/Y}$, and $B = E_1 \cap P$ corresponds to the subbundle $N_{s/E}$. Hence the unique point $q_0 := B \cap \sigma$ is the point over which the image of $\mathcal{O}_{\mathbb{P}^1}$ in $N_{s/Y}/N_{s/E}$ vanishes, that is, q_0 is contained in the fiber of P over the point p_0 of Lemma 7.15.

Step 3. In this step we describe the indeterminacy scheme $\Delta \subset P^+$ of the inverse map $P^+ \dashrightarrow P_{Q'}$ in three configurations.

By (7.21) and the local generators above, the birational map $P_{Q'} \dashrightarrow P^+$ induced by the identification of both surfaces with the divisor P is, near $\varrho^{-1}(o)$,

$$(t, y) \to \big(t, y^+\big), \qquad y^+ := \frac{y - \rho(t)}{t^n}.$$

Let $u := 1/y^+$. Then $y = \rho(t) + t^n/u$, so the inverse map is undefined exactly on the subscheme $\Delta_o \subset P^+$ with ideal $(u, u\rho(t) + t^n) = (u, t^n)$, and near $(t, u) = (0, 0)$ the strict transform of a curve $(y = \gamma(t))$ of $P_{Q'}$ is the closure of the curve

$$u\big(\gamma(t) - \rho(t)\big) = t^n, \qquad t \neq 0. \tag{7.24}$$

Let $\Delta \subset P^+$ be the union of the subschemes Δ_o, $o \in \Gamma \cap \Lambda$. Let $\tilde{P}$ be the minimal resolution of $P_{Q'} \dashrightarrow P^+$, with morphisms $\tilde{P} \to P_{Q'}$ and $b : \tilde{P} \to P^+$. Near the part of Ξ over o, the morphism $\tilde{P} \to P_{Q'}$ is the composite of blow-ups of Lemma 7.14 for Ξ along $(y = \rho(t))$, and near Δ_o the morphism b is the composite of blow-ups of Lemma 7.14 for Δ_o along $(u = 0)$. Let e and e_b be the sums of the total transforms of the exceptional curves of $\tilde{P} \to P_{Q'}$ and of b, and write σ and f also for the negative section and the fiber class of $P^+ \cong P$. We show that

$$b^* f = \eta, \qquad b^* \sigma = g + \eta - e, \qquad e_b = 3\eta - e. \tag{7.25}$$

Over o, let $e_{o,1},\dots,e_{o,n}$ be the total transforms on $\tilde P$ of the n exceptional curves of $\tilde P\to P_{Q'}$, let $E_{o,k}:=e_{o,k}-e_{o,k+1}$ for $1\le k<n$ and $E_{o,n}:=e_{o,n}$, and let $\varrho^{-1}(o)'$ be the strict transform of the fiber $\varrho^{-1}(o)$. Then $e=\sum_o\sum_k e_{o,k}$, and $(e^2)=-3$ since the integers n sum to length $\Xi=3$.

(i) Since b is a morphism over Λ, $b^*f=\eta$.

(ii) Over o, the curves contracted by b are $\varrho^{-1}(o)'$, of class $\eta-e_{o,1}$, and $E_{o,1},\dots,E_{o,n-1}$. The class $g+\eta-e$ is orthogonal to all of them, and

$$((g+\eta-e)\cdot\eta)=1,\qquad((g+\eta-e)^2)=-1+2-3=-2.$$

The subgroup $b^*\operatorname{Pic}(P^+)$ is the orthogonal complement of the curves contracted by b, and it is spanned by $b^*\sigma$ and η. Since $(b^*\sigma\cdot\eta)=1$ and $(\eta^2)=0$, we obtain $g+\eta-e=b^*\sigma+c\eta$ for some c, and $((b^*\sigma+c\eta)^2)=-2+2c$ gives $c=0$. Thus $b^*\sigma=g+\eta-e$.

(iii) Over o, the curves contracted by b form the chain $\varrho^{-1}(o)'-E_{o,1}-\dots-E_{o,n-1}$, and the exceptional curve of the last blow-up of b is $\varrho^{-1}(o)'$, with $((\varrho^{-1}(o)')^2)=-1$. Hence the part of e_b over o is $n\varrho^{-1}(o)'+\sum_{k<n}(n-k)E_{o,k}$. The part of e over o is $\sum_k e_{o,k}=\sum_k kE_{o,k}$. Hence the part of $e+e_b$ over o is $n\big(\varrho^{-1}(o)'+\sum_k E_{o,k}\big)$, which is n times the pullback of $\varrho^{-1}(o)$. Summing over o, we obtain $e+e_b\sim 3\eta$.

Near the point c we choose t and x_1 such that $B'=(y=t)$. Such coordinates exist since B' is a section of ϱ through c transversal to g. A curvilinear subscheme of length n at c contained in $g\cup B'=(y(y-t)=0)$ and mapped isomorphically to Λ is either (t,y), or $(t^2,y-dt)$ with $d\in\mathbb{C}$, or $(t^3,y-dt^2)$ or $(t^3,y-t-dt^2)$ with $d\in\mathbb{C}$. Comparing with (7.20), which allots length two to g and length one to B', the scheme Ξ is one of the following.

(I) $c\notin\Xi$, and $\Xi=\Xi_g\sqcup\{z_3\}$ with $\Xi_g\subset g\backslash\{c\}$ of length two and $z_3\in B'\backslash\{c\}$.

(II) $\Xi=(t^3,y-dt^2)$ with $d\ne0$.

(III) $\Xi=\Xi_c\sqcup\{z\}$, where $\Xi_c=(t^2,y-dt)$ with $d\notin\{0,1\}$ and $z\in g\backslash\{c\}$.

If Ξ contains c as a reduced point, then the rest of Ξ has length one on $g\backslash\{c\}$ and length zero on $B'\backslash\{c\}$, so Ξ has length two. The scheme $(t^2,y-t)$ has length two on B', and (t^3,y) has length three on g. The scheme (t^2,y) has length two on g and length one on B', so Ξ would have length two. The scheme $(t^3,y-t-dt^2)$ has length at least two on B'. The remaining possibilities are (II) and (III), and (I) when $c\notin\Xi$.

By **Step 2** and the reflection (7.22) applied to $D=E_1$, for which $(E_1\cdot\sigma)=(B\cdot\sigma)=1$, the strict transform E_1^+ of E on Y_1^+ satisfies $E_1^+|_{P^+}=B+\sigma$, and $E_1^+|_{P^+}$ coincides with B off σ. Since $E_1^+\cap P^+$ is the image of $\bar E_Q\cap P_{Q'}=g\cup B'$ and the image of g is σ, we obtain $B=b(B')$. Now (7.24) with $\gamma=0$ for g and $\gamma=t$ or a section γ with $\gamma(0)\ne0$ for B' describes Δ in the three configurations.

(I) Over a point of Ξ_g we have $\rho=0$; the curve σ avoids Δ_o, and B contains Δ_o. Over $\varrho(z_3)$ we have $n=1$ and $\rho=\gamma(0)\ne0$ for the equation $y=\gamma(t)$ of B'; the curve σ is $u=-t/\gamma(0)$ and contains Δ_o, and B avoids Δ_o. Hence $\Delta=Z_B\sqcup\{q_3\}$, where $Z_B\subset B\backslash\sigma$ has length two, $q_3\in\sigma\backslash\{q_0\}$, and the fibers through Z_B and through q_3 are distinct; the ruling restricts to a closed immersion on Z_B, since $\varrho|_\Xi$ is a closed immersion.

(II) Δ has ideal (u,t^3) at q_0, the curve σ is $u=-t/d$, and B is $u=t^2/(1-dt)$. Hence Δ is curvilinear of length three at q_0 with tangent line $(u=0)$, different from the tangent lines of σ and of the fiber through q_0.

(III) Over $\varrho(c)$, Δ_o has ideal (u,t^2), the curve σ is $u=-t/d$, and B is $u=t/(1-d)$. Hence $\Delta=\Delta_0\sqcup\{q_3\}$, where Δ_0 is curvilinear of length two at q_0 with tangent line $(u=0)$, different from the tangent lines of σ and of the fiber $(t=0)$, and $q_3\in B$ is the point over $\varrho(z)$, on another fiber.

Step 4. In this step we show that each $W_{m,j}$ embeds into the space $R_{m,\lfloor j/2\rfloor}$ of the surface series (7.26), which is the series of Lemmas 7.8, 7.16, 7.9, and 7.10.

For $m \geq 1$ and $j \geq 0$ let $W_{m,j}$ be the refinement of $H^0(X, mH)$ by P, a subspace of $H^0(P, (mH - jP)|_P)$ on $\hat{X}$, and for $0 \leq k \leq 2m$ let

$$R_{m,k} := \begin{cases} H^0(P, \mathcal{O}_P(k\sigma + (m+k)f)) & (k \leq m), \\ s_\sigma^{k-m} H^0\big(P, \mathcal{O}_P(m\sigma + (m+k)f) \otimes I_\Delta^{k-m}\big) & (k \geq m), \end{cases} \tag{7.26}$$

where s_σ defines σ and Δ is transported to P through $P \cong P^+$.

Let $\phi \in H^0(X, mH)$ with $\operatorname{ord}_P(\phi) \geq j$. If $j = 2k$, then $\operatorname{ord}_E(\phi) \geq k$ by (7.19), so $\phi \in H^0(Y_1, mH - 2kP - kE_1)$, the pullback of $mH - 2kP$ from $\hat{X}$. By (7.23) the restriction ψ of ϕ to P is a section of $\mathcal{O}_P(k\sigma + (m+k)f)$. If $j = 2k+1$, then $\operatorname{ord}_E(\phi) \geq k+1$, the pullback of $mH - (2k+1)P$ from $\hat{X}$ is $mH - (2k+1)P - (k+1)E_1 + \frac{1}{2}E_1$, and the restriction ψ of ϕ as a section of $mH - (2k+1)P - (k+1)E_1$ is a section of $\mathcal{O}_P(k\sigma + (m+k-1)f)$, while the $\mathbb{Q}$-divisor of the element of $W_{m,2k+1}$ is $\operatorname{div}\psi + \frac{1}{2}B$. If $\operatorname{ord}_P(\phi) > j$, then $\psi = 0$.

Let $k \geq m$ and $j = 2k$. By Construction-Proposition 5.6(4) and (7.19), ϕ is a section on Q'' of

$$mH_{Q''} - 2kP - k\bar{E}_{Q''} = (5m - 2k)h - mP - (2m - k)D'' = (2m - k)A + (k - m)l,$$

where $A := 3h - P - D'' = -K_{Q''}$ and $l = h - P$. By [Har77, Corollary II.7.15], the strict transform $P \subset Q''$ of $P_{Q'}$ is the blow-up of $P_{Q'}$ along $\Xi = \Gamma' \cap P_{Q'}$. Since $I_\Xi \mathcal{O}_{\tilde{P}} = \mathcal{O}_{\tilde{P}}(-e)$ by Lemma 7.14, the morphism $\tilde{P} \to P_{Q'}$ factors through $P \subset Q''$, and restrictions to $\tilde{P}$ below are pullbacks under $\tilde{P} \to P \subset Q''$. On $\tilde{P}$ we have $h|_{\tilde{P}} = \eta$, $P|_{\tilde{P}} = -g$, and $D''|_{\tilde{P}} = e$, so by (7.25)

$$A|_{\tilde{P}} = g + 3\eta - e = b^*(\sigma + 2f), \qquad l|_{\tilde{P}} = g + \eta = b^*(\sigma + 3f) - e_b.$$

Hence the restriction $\tilde{\psi}$ of ϕ to $\tilde{P}$ is a section of

$$b^*\mathcal{O}_{P^+}\big(m\sigma + (m+k)f\big) \otimes \mathcal{O}_{\tilde{P}}(-(k-m)e_b),$$

and by Lemma 7.14 and the projection formula, $\psi^+ := b_*\tilde{\psi}$ is a section of

$$\mathcal{O}_{P^+}\big(m\sigma + (m+k)f\big) \otimes I_\Delta^{k-m}.$$

The threefold Q'' is the blow-up of Q along Λ followed by the blow-up along the strict transform of Γ, and Y_1^+ is the blow-up of Q along Γ followed by the blow-up along the strict transform of Λ. The threefolds Q'' and Y_1^+ are isomorphic over $Q \backslash (\Gamma \cap \Lambda)$, and over a point of $\Gamma \cap \Lambda$ the birational map between them is an isomorphism near the generic point of the image in Q'' of any curve of $\tilde{P}$ not contracted by b. Thus ψ^+ is the restriction of ϕ to P^+, as a section of the strict transform D^+ on Y_1^+ of $D := mH - 2kP - kE_1$. Since $Y_1 \dashrightarrow Y_1^+$ is an isomorphism near $P \backslash \sigma$, the divisors of ψ and ψ^+ coincide on $P \backslash \sigma$. By (7.22), with $(D \cdot \sigma) = ((k\sigma + (m+k)f) \cdot \sigma) = m - k$, their classes differ by $(k-m)\sigma$. Since $(\sigma^2) \neq 0$,

$$\operatorname{div}\psi = \operatorname{div}\psi^+ + (k-m)\sigma,$$

so

$$\psi \in s_\sigma^{k-m} H^0\big(P, \mathcal{O}_P(m\sigma + (m+k)f) \otimes I_\Delta^{k-m}\big) = R_{m,k}.$$

Let $k \geq m$ and $j = 2k+1$. By Construction-Proposition 5.6(4) and (7.19), ϕ is a section on Q'' of

$$mH_{Q''} - (2k+1)P - (k+1)\bar{E}_{Q''} = (2m - k)A + (k - m)l - (P + \bar{E}_{Q''}).$$

Since $(P + \bar{E}_{Q''})|_{\tilde{P}} = g + 2\eta - e = b^*(\sigma + f)$ by (7.25),

$$\big(mH_{Q''} - (2k+1)P - (k+1)\bar{E}_{Q''}\big)|_{\tilde{P}} = b^*\big((m-1)\sigma + (m+k-1)f\big) - (k-m)e_b.$$

Hence the restriction $\tilde{\psi}$ of ϕ to $\tilde{P}$ is a section of

$$b^*\mathcal{O}_{P^+}\big((m-1)\sigma + (m+k-1)f\big) \otimes \mathcal{O}_{\tilde{P}}(-(k-m)e_b),$$

and by Lemma 7.14 and the projection formula, $\psi^+ := b_*\tilde{\psi}$ is a section of

$$\mathcal{O}_{P^+}\big((m-1)\sigma + (m+k-1)f\big) \otimes I_\Delta^{k-m}.$$

The description of the birational map between Q'' and Y_1^+ in the case $j = 2k$ does not involve the divisor, so ψ^+ is the restriction of ϕ to P^+, as a section of the strict transform D^+ on Y_1^+ of $D := mH - (2k+1)P - (k+1)E_1$, and the divisors of ψ and ψ^+ coincide on $P\backslash\sigma$. Here $(D \cdot \sigma) = ((k\sigma + (m+k-1)f) \cdot \sigma) = m - k - 1$, so by (7.22) and $(\sigma^2) \neq 0$

$$\operatorname{div} \psi = \operatorname{div} \psi^+ + (k+1-m)\sigma,$$

and

$$\psi \in s_\sigma^{k+1-m} H^0\big(P, \mathcal{O}_P((m-1)\sigma + (m+k-1)f) \otimes I_\Delta^{k-m}\big) \\ \subset s_\sigma^{k-m} H^0\big(P, \mathcal{O}_P(m\sigma + (m+k-1)f) \otimes I_\Delta^{k-m}\big).$$

Moreover $W_{m,j} = 0$ for $j > 4m$. Let $\Phi'' \subset Q''$ be the strict transform of a general fiber of $Q' \to \mathbb{P}^2$. Then $(h \cdot \Phi'') = (P \cdot \Phi'') = 1$ and $(D'' \cdot \Phi'') = 0$, so $((mH_{Q''} - jP) \cdot \Phi'') = 4m - j$, and the curves Φ'' cover Q''.

Fix a fiber f_0 of P, with defining section s_{f_0}. Multiplication by s_{f_0} maps the sections ψ for $j = 2k+1$ into $R_{m,k}$, and for $j = 2k$ the sections ψ are elements of $R_{m,k}$. Let v be a prime divisor over P. The injective maps $W_{m,j} \to R_{m,\lfloor j/2 \rfloor}$ change the order along v by at most $c_v := \operatorname{mult}_v f_0 + \frac{1}{2}\operatorname{mult}_v B$. For a subspace $\mathcal{W}'$ of a finite-dimensional space $\mathcal{W}$ of sections and $t \geq 0$, let $\mathcal{F}^t\mathcal{W} \subset \mathcal{W}$ be the subspace of elements of order at least t along v. Since $\dim(\mathcal{W}' \cap \mathcal{F}^t\mathcal{W}) \leq \dim \mathcal{F}^t\mathcal{W}$ for any t, the i-th largest jumping number of v on $\mathcal{W}'$ is at most that on $\mathcal{W}$. As jumping numbers are non-negative, the sum does not increase. Hence the sum of the jumping numbers of v on $W_{m,j}$ is at most the sum on $R_{m,\lfloor j/2 \rfloor}$ plus $c_v \dim W_{m,j}$. We sum over $0 \leq j \leq 4m$. Since each $R_{m,k}$ receives $W_{m,2k}$ and $W_{m,2k+1}$, the sum of the jumping numbers of v on all $W_{m,j}$ is at most twice the sum on all $R_{m,k}$ plus $c_v N_m$, and $N_m = 3m^3 + O(m^2) = 2n_m + O(m^2)$ (Lemma 2.12(2), and $n_m = \frac{3}{2}m^3 + O(m^2)$ by Lemmas 7.8, 7.9, and 7.10). After taking limits [AZ22, Lemma 2.9], we obtain

$$S(W; v) \leq S(R; v) \qquad \text{for any prime divisor } v \text{ over } P. \tag{7.27}$$

Step 5. In this step we show that the fiber P_x of P over x does not contain q_0 or a point of $\Delta \cap (B\backslash\sigma)$.

By **Step 1** the local ring of $\Sigma(X, x)$ at $[L]$ is $\mathbb{C}[t]/(t^2)$, and by **Step 2** $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Hence $x \neq p_0$ by Lemma 7.15, and $q_0 \notin P_x$ by **Step 2**.

Let $q \in \Delta \cap (B\backslash\sigma)$. By **Step 3**, q is the image of a point $z \in \Xi \cap (g\backslash\{c\})$, in configuration (I) or (III). Let $o \in \Gamma \cap \Lambda$ be the image of z, and let $z' \in B'$ be the other point of $\bar{E}_Q$ over o. Then $z' \neq c$ and $\varrho(z') = \varrho(z)$. Let $\Phi_{z'} \neq g$ be the fiber of $\bar{E}_Q \to \kappa$ through z', let $\bar{z}' \in \kappa$ be its image, and let $\Phi_{z',Q}$ be the image of $\Phi_{z'}$, a line in Q different from Λ through o. Since $\Gamma' \to \Gamma$ is an isomorphism, $z' \notin \Gamma'$, so $\Gamma' \cap \Phi_{z'}$ is disjoint from $g \cup B'$, and $\operatorname{length}(\Gamma \cap \Phi_{z',Q}) \geq (\Gamma' \cdot \bar{f}) + 1 = 3$. On Y^+ the strict transform $\Phi_{z'}^+$ of $\Phi_{z',Q}$ satisfies $(-K_{Y^+} \cdot \Phi_{z'}^+) = 3 - \operatorname{length}(\Gamma \cap \Phi_{z',Q}) \leq 0$, and $-K_{Y^+}$ is nef, so $\Phi_{z'}^+$ is a flopped curve of the link different from Λ^+. By Theorem 2.22, $\Phi_{z'}^+$ is the flop of the strict transform $L'_Y \subset Y$ of a line $L' \neq L$ intersecting L. We show that $L \cap L'$ is the point $\hat{\pi}(q)$ of L under q.

Let f_E be the class of a fiber of $E_1 \cong E \to L$. Since $-K_{Y_1} = H - E_1 - 2P$, $H|_{E_1} = f_E$, $P|_{E_1} = s$, and $E_1|_{E_1} = E|_E - s$ with $E|_E \sim -s - 2f_E$, we have $-K_{Y_1}|_{E_1} \sim 3f_E$. Since $\bar{E}_{Q''} \cong \bar{E}_Q$ and $D''|_{\bar{E}_{Q''}} = \Gamma'$, Construction-Proposition 5.6(2)(3) gives

$$A|_{\bar{E}_{Q''}} \sim 3(\bar{\sigma} + 3\bar{f}) - (\bar{\sigma} + \bar{f}) - (2\bar{\sigma} + 5\bar{f}) = 3\bar{f}.$$

The prime divisors on Q'' exceptional over X are $\bar{E}_{Q''}$ and P (Construction-Proposition 5.6(4)(c)), and those on Y_1 are E_1 and P. Hence the birational map $Q'' \dashrightarrow Y_1$ over X is an isomorphism in codimension one, A corresponds to $-K_{Y_1}$, and $H^0(Q'', A) \cong H^0(Y_1, -K_{Y_1})$. Let $\chi : \bar{E}_{Q''} \dashrightarrow E_1$ be the birational map induced by $Q'' \dashrightarrow Y_1$. Since $|A|$ is base point free (Construction-Proposition 5.6(4)(a)), a general member of $|A|$ restricts to $\bar{E}_{Q''}$ as a sum of three distinct fibers of $\bar{E}_{Q''} \to \kappa$. The corresponding member of $|-K_{Y_1}|$ restricts to E_1 as a member of $|3f_E|$, that is,

a sum of fibers of $E_1 \to L$. The restrictions to $\bar{E}_{Q''}$ and to E_1 correspond under χ on the dense open subset where $Q'' \dashrightarrow Y_1$ is an isomorphism. Thus χ maps general fibers of $\bar{E}_{Q''} \to \kappa$ onto fibers of $E_1 \to L$, and it induces a birational map $\kappa \dashrightarrow L$, which is an isomorphism $\epsilon : \kappa \to L$. At any point where χ is defined, the composite of χ and $E_1 \to L$ equals the composite of $\bar{E}_{Q''} \to \kappa$ and ϵ.

We regard B' and $\Phi_{z'}$ also as curves in $\bar{E}_{Q''} \cong \bar{E}_Q$. The threefolds Q'' and Y_1^+ are isomorphic over $Q \backslash (\Gamma \cap \Lambda)$ (**Step 4**), $Y_1^+ \dashrightarrow Y_1$ is an isomorphism near $P^+ \backslash \sigma$ (**Step 2**), and $b(B') = B \neq \sigma$ (**Step 3**). Hence $Q'' \dashrightarrow Y_1$ is an isomorphism near a general point of B', and there χ coincides with the map $P_{Q'} \dashrightarrow P^+ \cong P$, whose restriction to B' is a morphism $b_{B'} : B' \to B$. Since $B \subset E_1 \cap P$, we have $(P \to L)|_B = (E_1 \to L)|_B$. Thus for a general point $z_1 \in B'$ with image $\bar{z}_1 \in \kappa$ we have $\hat{\pi}(b_{B'}(z_1)) = \epsilon(\bar{z}_1)$. Both sides are morphisms $B' \to L$, so this holds for all $z_1 \in B'$, in particular for $z_1 = z'$.

Next we locate $\chi(\Phi_{z'})$. Over $Q \backslash \Lambda$, both Q'' and Y^+ are the blow-up of Q along Γ, and $\Phi_{z',Q} \cap \Lambda = \{o\}$, so $Q'' \dashrightarrow Y^+$ is an isomorphism near a general point of $\Phi_{z'}$. The flop $Y \dashrightarrow Y^+$ is a birational map over the common contraction $\bar{Y}$, and the flopping curves are disjoint (Theorem 2.22), so the fiber of $Y \to \bar{Y}$ over the image of $\Phi_{z'}^+$ is L'_Y. Moreover $Y_1 \to Y$ is an isomorphism near L'_Y, since $L'_Y \cap s = \emptyset$. Hence the closure of the graph of $Q'' \dashrightarrow Y_1$ maps a general point $\zeta \in \Phi_{z'}$ into the strict transform L'_{Y_1} of L'_Y in Y_1. The map χ is defined at a general point $\zeta \in \Phi_{z'}$, since $\bar{E}_{Q''}$ is a smooth surface, and then $\chi(\zeta) \in L'_{Y_1}$. The curve L'_{Y_1} intersects E_1 only in the fiber over $L \cap L'$, so $\chi(\zeta)$ maps to $L \cap L'$ under $E_1 \to L$. Since $\zeta \in \Phi_{z'}$ maps to $\bar{z}' \in \kappa$, the point $\chi(\zeta)$ also maps to $\epsilon(\bar{z}')$. Hence $L \cap L' = \epsilon(\bar{z}') = \hat{\pi}(b_{B'}(z'))$.

Finally $b_{B'}(z')$ is the point of B over o, since $P_{Q'} \dashrightarrow P^+$ is a map over Λ. The point $q \in B$ is also contained in the fiber over o (**Step 3**), and B is a section of $P \to s$ since $(B \cdot f) = 1$, so $q = b_{B'}(z')$. Thus $L \cap L' = \hat{\pi}(q)$. Since L is the only line through x, $\hat{\pi}(q) \neq x$, so $q \notin P_x$.

Step 6. In this step we show that $A_{P,B/2}(v) \geq \frac{108}{95} S(W; v)$ for any prime divisor v over P whose center intersects P_x.

By Construction-Proposition 7.1, $\mathrm{Diff}_P(0) = \frac{1}{2}B$, and B is a smooth curve of class $\sigma + 3f$ with $B \cap \sigma = \{q_0\}$ by (7.23). We apply the surface lemmas with $S := P$ and the series (7.26), with the scheme Δ of **Step 3**.

In configuration (I), $\Delta = Z_B \sqcup \{q_3\}$ satisfies the hypotheses of Lemma 7.8 and of Construction-Proposition 7.7. By **Step 5**, P_x does not contain q_0 or a point of Z_B. If P_x is the fiber through q_3, Lemma 7.16 implies $A_{P,B/2}(v) \geq \frac{36}{31} S(R; v)$. Otherwise P_x is disjoint from the union of the fibers through $\mathrm{Supp}\, \Delta$ and from q_0, and Lemma 7.8 implies that the local threshold is at least $\frac{36}{31}$ at any point of P_x, so again $A_{P,B/2}(v) \geq \frac{36}{31} S(R; v)$.

In configuration (II), Δ satisfies the hypotheses of Lemma 7.9, and P_x does not contain q_0 by **Step 5**. Hence $A_{P,B/2}(v) \geq \frac{108}{95} S(R; v)$.

In configuration (III), $\Delta = \Delta_0 \sqcup \{q_3\}$ satisfies the hypotheses of Lemma 7.10, and P_x is disjoint from Δ by **Step 5**. Hence $A_{P,B/2}(v) \geq \frac{8}{7} S(R; v)$.

Since $\frac{36}{31} > \frac{8}{7} > \frac{108}{95}$, (7.27) implies $A_{P,B/2}(v) \geq \frac{108}{95} S(W; v)$ in all three configurations.

Step 7. In this step we conclude the proof.

By Construction-Proposition 7.1, $\hat{\pi} : \hat{X} \to X$ is the plt extraction of P with different $\frac{1}{2}B$. Let F be a prime divisor over X with $\mathrm{center}_X F = \{x\}$. Then $\mathrm{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(x) = P_x$. By **Step 6**, Lemma 3.1 with $\lambda = \frac{108}{95}$ and the closed subset P_x, together with (7.18), implies

$$A_X(F) \geq \frac{108}{95} S_X(F) + \left(3 - \frac{108}{95} \cdot \frac{43}{18}\right) \mathrm{mult}_F P = \frac{108}{95} S_X(F) + \frac{27}{95} \mathrm{mult}_F P.$$

Since $S_X(F) \geq 0$, we obtain $A_X(F) - S_X(F) \geq \frac{27}{95} \mathrm{mult}_F P$. Finally $\mathrm{mult}_F P > 0$, since $\mathrm{center}_{\hat{X}} F \subset P$ and $2P$ is Cartier. □

8. Mixed points in genus ten

In this section, we prove strict local inequalities between A_X and S_X at the points of incidence type $2+1$ (Definition 2.11) on a smooth prime Fano threefold X of genus ten. At such a point x, the scheme $\Sigma(X,x)$ consists of a reduced point $[L]$ and a curvilinear double point at a line M, where L is non-special and M is special (Proposition 4.1). We divide the points x according to the width $n \in \{2,3,4\}$ of the strict transform of M in $\mathrm{Bl}_L X$ (Definition 2.21, Proposition 5.3). The widths two and three are treated in Propositions 8.8 and 8.9, and the width four is Theorem 8.12.

8.1. **Configurations of the base scheme.** In this subsection, we describe the Λ_T-trivial curves on the surface T of Construction-Proposition 5.4, determine the configuration of width four, and show that for widths three and four the base scheme Z_U is disjoint from s.

Lemma 8.1. *Notation and conditions as in Construction-Proposition 5.4. Then $\Lambda_T = -K_T$, Λ_T is nef and big, and $D_T^2 = 9$. Write $D_T = C + \sum_i \nu_i R_i$, where C is the component of D_T of positive Λ_T-degree and the R_i are distinct irreducible curves with $(\Lambda_T \cdot R_i) = 0$ and $\nu_i \geq 0$ integers. Then C is a smooth curve isomorphic to Γ with $(\Lambda_T \cdot C) = 7$ and $C^2 = 9$. Any irreducible curve $R \subset T$ with $(\Lambda_T \cdot R) = 0$ is a smooth rational curve with $R^2 = -2$, and either R is the strict transform of the exceptional curve of the blow-up at a center c of p_T and $R \sim e_i - e_{i'}$, where e_i and $e_{i'}$ are the total exceptional classes over c and over the center lying on the exceptional curve of the blow-up at c, or R is the strict transform of a curve of class s, f, or $s+f$ passing through one, two, or three of the centers, respectively. For any connected configuration of such curves, with Cartan matrix A (the negative of their intersection matrix) and vector $\mathbf{c}$ of intersection numbers with C, we have $\mathbf{c}^t \mathsf{A}^{-1} \mathbf{c} \leq 13/4$.*

Proof. By Proposition 5.3, at a point $y \in \operatorname{Supp} Z_U$ corresponding to a line of width n, the scheme Z_U is curvilinear of length n. Thus, if Δ_y is a smooth curve germ at y containing Z_U near y, the n centers of p_T over y are $y_1 := y$ and, for $1 \leq j < n$, the point y_{j+1} where the strict transform of Δ_y intersects the exceptional curve of the blow-up at y_j. Any of them but the first is a point of the exceptional curve of the blow-up at the previous one, and no center is a point of two exceptional curves.

Since $K_E = -2s - 3f$ and $K_T = p_T^* K_E + \sum_i e_i$ [Har77, Corollary V.2.11 and Proposition V.3.3], we have $\Lambda_T = -K_T$. By Construction-Proposition 5.4(1), Λ_T is nef and $\Lambda_T^2 = 4 > 0$, so Λ_T is big. Since $D|_E \sim 5s + 7f$ (Construction-Proposition 5.4(2)) and $(e_i \cdot e_j) = -\delta_{ij}$,

$$D_T^2 = (5s+7f)^2 + 9\sum_i e_i^2 = 45 - 36 = 9.$$

Let $\phi_T : T \to E_Q$ be the morphism defined by $|\Lambda_T|$, where $E_Q \subset Q$ is the image of E. By Construction-Proposition 5.4(2), exactly one irreducible component C of D_T has positive Λ_T-degree, the coefficient of C in D_T is one, and $\phi_T|_C : C \to \Gamma$ is birational. Since Λ_T is nef, any other irreducible component R of D_T satisfies $(\Lambda_T \cdot R) = 0$, and since D_T has integer coefficients, $D_T = C + \sum_i \nu_i R_i$ as in the statement. The curve Γ is smooth of degree seven and genus two (Theorem 2.22, [KPZ14, Theorem 2.6]). Since $\phi_T|_C$ is projective and birational onto the curve Γ, the fibers of $\phi_T|_C$ are finite, so $\phi_T|_C$ is finite [Har77, Exercise III.11.2]. Hence $(\phi_T|_C)_* \mathcal{O}_C$ is a coherent $\mathcal{O}_\Gamma$-algebra contained in the function field of Γ, and it equals $\mathcal{O}_\Gamma$ since Γ is normal. Thus $\phi_T|_C$ is an isomorphism. Hence C is smooth of genus two, and

$$(\Lambda_T \cdot C) = (\Lambda_T \cdot D_T) = 7.$$

By the adjunction formula [Har77, Proposition V.1.5],

$$2 = 2p_a(C) - 2 = C^2 + (K_T \cdot C) = C^2 - 7,$$

so $C^2 = 9$.

Let $R \subset T$ be an irreducible curve with $(\Lambda_T \cdot R) = 0$. The intersection form on $N^1(T)_{\mathbb{R}}$ has exactly one positive eigenvalue [Har77, Theorem V.1.9 and Remark V.1.9.1] and $\Lambda_T^2 > 0$, so

the intersection form is negative definite on $\Lambda_T^{\perp}$. Hence $R^2 < 0$. By [Har77, Exercise V.1.3(a)], $R^2 = R^2 + (K_T \cdot R) = 2p_a(R) - 2$. As $p_a(R) \geq 0$ and $R^2 < 0$, we have $p_a(R) = 0$ and $R^2 = -2$, so R is a smooth rational curve [Har77, Exercise IV.1.8(b)].

Assume that R is contracted by p_T. Then R is the strict transform of the exceptional curve of the blow-up at a center, say the center c with total exceptional class e_i. By the first paragraph, $R \sim e_i - e_{i'}$ if the center with total exceptional class $e_{i'}$ follows c in the chain of c, and $R \sim e_i$ otherwise [Har77, Propositions V.3.2 and V.3.6]. Since $(\Lambda_T \cdot e_i) = 1$ and $(\Lambda_T \cdot (e_i - e_{i'})) = 0$, we have $R \sim e_i - e_{i'}$ as in the statement.

Assume that R is not contracted by p_T. Then $p_T(R)$ is an integral curve on E of class $as + bf$ with integers $a, b \geq 0$, and $R \sim p_T^*(as + bf) - \sum_i m_i e_i$, where $m_i \geq 0$ is the multiplicity at the i-th center of the strict transform of $p_T(R)$ [Har77, Proposition V.3.6]. Hence

$$0 = (\Lambda_T \cdot R) = a + 2b - \sum_i m_i, \qquad -2 = R^2 = -a^2 + 2ab - \sum_i m_i^2.$$

By the Cauchy–Schwarz inequality $(\sum_i m_i)^2 \leq 4 \sum_i m_i^2$, we have $(a + 2b)^2 \leq 4(-a^2 + 2ab + 2)$, that is,

$$(2b - a)^2 + 4a^2 = 5a^2 - 4ab + 4b^2 \leq 8.$$

Thus $a \leq 1$. If $a = 0$, then $b = 1$, since R is a curve. If $a = 1$, then $(2b - 1)^2 \leq 4$, so $b \in \{0, 1\}$. For $(a, b) = (1, 0)$, $(0, 1)$, and $(1, 1)$ we obtain $\sum_i m_i = \sum_i m_i^2 = 1$, 2, and 3, respectively, so $m_i \in \{0, 1\}$ for any i. Since s is the only integral curve of class s, this proves the description of R.

Finally, let $R_1, \dots, R_k$ be distinct irreducible curves with $(\Lambda_T \cdot R_i) = 0$ forming a connected configuration, let $\mathsf{A} := (-(R_i \cdot R_j))_{i,j}$, and let $\mathbf{c} := ((C \cdot R_i))_i$. We show that the classes of $R_1, \dots, R_k$ are linearly independent. Let a_i be real numbers with $\sum_i a_i R_i \equiv 0$. Then $P_+ := \sum_{a_i > 0} a_i R_i$ and $P_- := \sum_{a_i < 0} (-a_i) R_i$ satisfy $P_+ \equiv P_-$ and $P_+^2 = (P_+ \cdot P_-) \geq 0$, so $P_+ \equiv 0$ by the negative definiteness on $\Lambda_T^{\perp}$. As $P_+ \geq 0$, $P_+ = 0$, and likewise $P_- = 0$. Thus A is positive definite. Let

$$\alpha := C - \frac{7}{4}\Lambda_T, \qquad \alpha_0 := -\sum_i (\mathsf{A}^{-1}\mathbf{c})_i R_i.$$

Then $(\alpha \cdot \Lambda_T) = 0$ and

$$\alpha^2 = C^2 - \frac{7}{2}(\Lambda_T \cdot C) + \frac{49}{16}\Lambda_T^2 = 9 - \frac{49}{2} + \frac{49}{4} = -\frac{13}{4}.$$

Moreover, for any j,

$$(\alpha_0 \cdot R_j) = (\mathsf{A}\mathsf{A}^{-1}\mathbf{c})_j = (C \cdot R_j) = (\alpha \cdot R_j),$$

so $\alpha - \alpha_0 \in \Lambda_T^{\perp}$ and $((\alpha - \alpha_0) \cdot \alpha_0) = 0$. Hence

$$-\frac{13}{4} = \alpha^2 = \alpha_0^2 + (\alpha - \alpha_0)^2 \leq \alpha_0^2 = -(\mathsf{A}^{-1}\mathbf{c})^t \mathsf{A} (\mathsf{A}^{-1}\mathbf{c}) = -\mathbf{c}^t \mathsf{A}^{-1} \mathbf{c},$$

that is, $\mathbf{c}^t \mathsf{A}^{-1} \mathbf{c} \leq 13/4$. □

Lemma 8.2. *Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point such that $\Sigma(X, x)$ consists of $[L]$ and a curvilinear double point at a line M of width four, and let $p_T : T \to E$ and $e_1, \dots, e_4$ be as in Construction-Proposition 5.4. Assume that the point of Z_U representing M does not belong to s. Then the following hold.*

(1) There exist coordinates (z, u) on an open subset of E containing $f_x \backslash s$, in which $f_x \backslash s = (z = 0)$ and Z_U has ideal $(z - u^2, u^4)$.

(2) The image E_Q of E in Q is a quartic del Pezzo surface with exactly one singular point o, of type D_4, whose minimal resolution is T, and $o \in \Gamma$.

(3) $D|_E = 2f_x + G_5$, where G_5 is an integral curve of class $5s + 5f$ disjoint from s with multiplicity two at each of the four (proper or infinitely near) centers of Z_U, whose strict transform on T is a smooth curve of genus two, isomorphic to Γ, of class $p_T^*(5s + 5f) - 2\sum_i e_i$, disjoint from the curves contracted by p_T of classes $e_i - e_{i+1}$ for $i = 1, 2, 3$, and

intersecting the strict transform $\tilde{f}_x$ of f_x, of class $p_T^*f - e_1 - e_2$, transversally in one point.

(4) The surface E_Q is the zero scheme of a section of $\mathcal{O}_Q(2)$, which belongs to $H^0(Q, I_\Gamma(2))$ since $\Gamma \subset E_Q$. The map $N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)$ induced by this section has image $\mathcal{O}_\Gamma(2)(-2o)$, and there exists an exact sequence of line bundles

$$0 \to K_\Gamma \otimes \mathcal{O}_\Gamma(1)(2o) \to N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)(-2o) \to 0$$

of degrees 11 *and* 12.

Proof. The proof is long, so we divide it into several steps. Let p_0 be the point of Z_U representing M, and let $\phi_T : T \to E_Q$ be the morphism defined by $|\Lambda_T|$ (Construction-Proposition 5.4(1)).

Step 1. In this step, we prove (1).

By Proposition 5.3, the widths of the lines intersecting L sum to four. Since M has width four, Z_U is supported at p_0, $\widehat{\mathcal{O}}_{Z_U,p_0} \cong \mathbb{C}[t]/(t^4)$, and $Z_U \cap f_x \cong \Sigma(X,x)\backslash\{[L]\}$ has length two.

The surface $E \cong \mathbb{F}_1$ is the blow-up of $\mathbb{P}^2$ at a point, with exceptional curve s, and the fibers of $E \to L$ are the strict transforms of the lines through the blown-up point [Har77, Example V.2.11.5]. Choose homogeneous coordinates $[X_0 : X_1 : X_2]$ on $\mathbb{P}^2$ such that the blown-up point is $[0:0:1]$ and f_x is the strict transform of $(X_1 = 0)$, and let $z := X_1/X_0$ and $u := X_2/X_0$. Then (z,u) are coordinates on the complement E° in E of s and of the strict transform of $(X_0 = 0)$. The open subset E° contains $f_x\backslash s$, because the lines $(X_1 = 0)$ and $(X_0 = 0)$ meet only at $[0:0:1]$, with distinct tangent directions, so that f_x and the strict transform of $(X_0 = 0)$ are disjoint. The fibers of $E \to L$ intersect E° in the curves $(z = \text{const})$, and $f_x\backslash s = (z = 0)$. Since $s + f$ is the class of the pullback of a line and $2s + 3f = 3(s + f) - s$, the space $H^0(E, \mathcal{O}_E(2s + 3f))$ is the space of plane cubics through $[0:0:1]$, which restricts to the span of the monomials $z^i u^\ell$ with $0 \le \ell \le 2$ and $i + \ell \le 3$. For complex numbers $a_1, \dots, a_4$ with $a_1 a_2 \neq 0$, the substitution $(z,u) \to (a_1 z, a_2 u + a_3 z + a_4)$ is induced by the automorphism $[X_0 : X_1 : X_2] \to [X_0 : a_1 X_1 : a_2 X_2 + a_3 X_1 + a_4 X_0]$ of $\mathbb{P}^2$, which fixes $[0:0:1]$ and preserves the lines $(X_0 = 0)$ and $(X_1 = 0)$, hence extends to an automorphism of E preserving s and f_x.

Since $p_0 \in f_x\backslash s$, after such a substitution we may assume that $p_0 = (0,0)$. The maximal ideal of $\mathcal{O}_{Z_U,p_0} \cong \mathbb{C}[t]/(t^4)$ is generated by the image of z or by the image of u. If it is generated by the image of z, then the ideal of Z_U contains $u - r(z)$ for a polynomial r with $r(0) = 0$, and $Z_U \cap f_x$ has length one, a contradiction. Thus the ideal of Z_U is $(z - b_1 u - b_2 u^2 - b_3 u^3, u^4)$ for some $b_1, b_2, b_3 \in \mathbb{C}$. Since $Z_U \cap f_x$ has ideal $(b_1 u + b_2 u^2 + b_3 u^3, u^4)$ in $\mathbb{C}[u]$ and length two, $b_1 = 0$ and $b_2 \neq 0$. Replacing z with z/b_2, we may assume that $b_2 = 1$. Let $u' := u + \frac{b_3}{2} z$. A direct calculation shows that $z - u'^2$ and u'^4 belong to $(z - u^2 - b_3 u^3, u^4)$. Since both $(z - u'^2, u'^4)$ and $(z - u^2 - b_3 u^3, u^4)$ have colength four, they coincide. Replacing u with u', we obtain (1).

Step 2. In this step, we determine the curves $R \subset T$ with $(\Lambda_T \cdot R) = 0$.

By (1), Z_U is the subscheme of length four at p_0 of the smooth curve $(z = u^2)$. Thus the four centers of p_T are $c_1 := p_0$ and, for $i = 1, 2, 3$, the point c_{i+1} where the strict transform of $(z = u^2)$ intersects the exceptional curve of the blow-up at c_i, and we index the exceptional classes so that e_i is the total exceptional class over c_i. Since $((z = u^2) \cdot f_x)_{p_0} = 2$, the strict transforms of f_x pass through c_1 and c_2 and not through c_3 [Har77, Proposition V.3.6 and Exercise V.3.2], so $\tilde{f}_x \sim p_T^* f - e_1 - e_2$. Since $p_0 \notin s$, no center is a point of the strict transform of s. An irreducible member G of $|s + f|$ satisfies $(G \cdot f_x) = 1$, so G is not tangent to f_x at p_0, and the strict transform of G does not pass through c_2. Hence G passes through at most one center. By Lemma 8.1, the curves R with $(\Lambda_T \cdot R) = 0$ are $\tilde{f}_x$ and the strict transforms R_1, R_2, R_3 of the exceptional curves over c_1, c_2, c_3, with $R_i \sim e_i - e_{i+1}$. A direct calculation shows that

$$(R_1 \cdot R_2) = (R_2 \cdot R_3) = (R_2 \cdot \tilde{f}_x) = 1, \qquad (R_1 \cdot R_3) = (R_1 \cdot \tilde{f}_x) = (R_3 \cdot \tilde{f}_x) = 0,$$

so R_1, R_2, R_3, and $\tilde{f}_x$ form a configuration of type D_4 with central curve R_2.

Step 3. In this step, we prove (2) except for $o \in \Gamma$.

By Lemma 5.2, $h^0(Y, H-2E) = h^0(Y, h) = h^0(Q, \mathcal{O}_Q(1)) = 5$ and $H^0(Y, H-3E) = 0$. Thus $\dim U = 5$. By Construction-Proposition 5.4(3) for $r = 1$, $U \subset H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U})$. In the coordinates of **Step 1**, the restriction map from the span of $z^i u^\ell$ $(0 \le \ell \le 2,\ i + \ell \le 3)$ to $\mathbb{C}[z,u]/(z-u^2, u^4)$ maps $1, u, u^2, zu$ to the basis $1, u, u^2, u^3$, so it is surjective with kernel of dimension $9 - 4 = 5$. Hence $U = H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U})$, and a direct calculation shows that

$$z - u^2, \quad z^2, \quad z^3, \quad z^2u, \quad zu^2$$

is a basis of U. Since U is the restriction of $H^0(Y, h) \cong H^0(Q, \mathcal{O}_Q(1))$, the rational map $E \dashrightarrow Q \subset \mathbb{P}^4$ is defined by U. Let $[\xi_0 : \cdots : \xi_4]$ be the homogeneous coordinates on $\mathbb{P}^4$ corresponding to $z - u^2, z^2, z^3, z^2u, zu^2$. Then E_Q is contained in $V(q_1, q_2)$, where

$$q_1 := \xi_0\xi_2 - \xi_1(\xi_1 - \xi_4), \qquad q_2 := \xi_2\xi_4 - \xi_3^2. \tag{8.1}$$

We describe $V(q_1, q_2)$ on affine charts.

- On $(\xi_0 = 1)$, we have $\xi_2 = \xi_1(\xi_1 - \xi_4)$ and $\xi_3^2 = \xi_1\xi_4(\xi_1 - \xi_4)$. Since the cubic $\xi_1\xi_4(\xi_1 - \xi_4)$ has three distinct linear factors, the surface $(\xi_3^2 = \xi_1\xi_4(\xi_1 - \xi_4)) \subset \mathbb{A}^3$ is smooth outside the origin and has a singularity of type D_4 at the origin [KM98, Theorem 4.22].
- On $(\xi_2 = 1)$, we have $\xi_4 = \xi_3^2$ and $\xi_0 = \xi_1(\xi_1 - \xi_3^2)$, an affine plane.
- On $(\xi_1 = 1)$, we have $\xi_4 = 1 - \xi_0\xi_2$ and $\xi_2 - \xi_0\xi_2^2 - \xi_3^2 = 0$, whose partial derivative in ξ_2 equals 1 at the points with $\xi_2 = 0$.
- On $(\xi_4 = 1)$, we have $\xi_2 = \xi_3^2$ and $\xi_0\xi_3^2 = \xi_1(\xi_1 - 1)$. At the points with $\xi_3 = 0$, we have $\xi_1 \in \{0, 1\}$, and the partial derivative in ξ_1 equals $2\xi_1 - 1 = \pm 1$.

Let $y \in V(q_1, q_2)$. If $\xi_2(y) \ne 0$, then y belongs to the second chart. Otherwise $\xi_3(y) = 0$ by $q_2 = 0$, and y belongs to the first chart, or to the third chart with $\xi_2(y) = 0$, or to the fourth chart with $\xi_3(y) = 0$, according as $\xi_0(y) \ne 0$, or $\xi_0(y) = 0$ and $\xi_1(y) \ne 0$, or $\xi_0(y) = \xi_1(y) = 0$. Thus $V(q_1, q_2)$ is a complete intersection surface of degree four, smooth outside $o := [1 : 0 : 0 : 0 : 0]$, with a singularity of type D_4 at o. In particular $V(q_1, q_2)$ is reduced, being Cohen–Macaulay [Har77, Proposition II.8.23(a)] and generically reduced. Since E_Q is an integral quartic surface (Construction-Proposition 5.4(1)) contained in $V(q_1, q_2)$, we have $E_Q = V(q_1, q_2)$. Since E_Q is a complete intersection in $\mathbb{P}^4$ whose only singular point is o, it is normal [Har77, Proposition II.8.23(b)].

By Lemma 8.1, $\Lambda_T = -K_T$ is nef and big with $K_T^2 = 4$, so T is a weak del Pezzo surface of degree four and ϕ_T is the anti-canonical morphism of T. Since the degree is at least three, ϕ_T contracts exactly the (-2)-curves of T [Dol12, Section 8.3], that is, the curves R_1, R_2, R_3, and $\tilde{f}_x$ of **Step 2**. As ϕ_T is birational and E_Q is normal, ϕ_T is an isomorphism over the complement of $\phi_T(R_1 \cup R_2 \cup R_3 \cup \tilde{f}_x)$, which is one point because the configuration is connected. As T is smooth and E_Q is singular at o, we have $\phi_T(R_1 \cup R_2 \cup R_3 \cup \tilde{f}_x) = \{o\}$. Since ϕ_T contracts no (-1)-curve, ϕ_T is the minimal resolution of E_Q. By [Dol12, Section 8.6], E_Q is a del Pezzo surface of degree four.

Step 4. In this step, we prove (3) and $o \in \Gamma$.

By Lemma 8.1 and **Step 2**,

$$D_T = C + \mu\tilde{f}_x + \nu_1 R_1 + \nu_2 R_2 + \nu_3 R_3$$

for integers $\mu, \nu_1, \nu_2, \nu_3 \ge 0$. Here C is the unique irreducible component of D_T of positive Λ_T-degree, it has coefficient one, and ϕ_T maps it birationally onto Γ (Construction-Proposition 5.4(2)). By Lemma 8.1, C is smooth of genus two, $\phi_T|_C : C \to \Gamma$ is an isomorphism, $(\Lambda_T \cdot C) = 7$, and $C^2 = 9$. Since $p_{T*}D_T = D|_E$ and p_T contracts R_1, R_2, and R_3, we have $D|_E = \mu f_x + G_5$, where $G_5 := p_T(C)$ is an integral curve of class $5s + (7-\mu)f$. Since C is not rational, $G_5 \ne s$, so

$$0 \le (G_5 \cdot s) = 2 - \mu.$$

Thus $\mu \in \{0, 1, 2\}$. Let m_i be the multiplicity at c_i of the strict transform of G_5, so that $C \sim p_T^*(5s + (7-\mu)f) - \sum_i m_i e_i$ [Har77, Proposition V.3.6]. Since C is different from R_1, R_2,

R_3, and $\tilde{f}_x$,

$$(C \cdot R_i) = m_i - m_{i+1} \geq 0 \quad (i = 1, 2, 3), \qquad (C \cdot \tilde{f}_x) = 5 - m_1 - m_2 \geq 0. \tag{8.2}$$

Moreover

$$7 = (\Lambda_T \cdot C) = 19 - 2\mu - \sum_i m_i, \qquad 9 = C^2 = 45 - 10\mu - \sum_i m_i^2,$$

that is,

$$\sum_i m_i = 12 - 2\mu, \qquad \sum_i m_i^2 = 36 - 10\mu. \tag{8.3}$$

If $\mu = 0$, then $(\sum_i m_i)^2 = 144 = 4\sum_i m_i^2$, so equality holds in the Cauchy–Schwarz inequality and $m_i = 3$ for any i. Then $(C \cdot \tilde{f}_x) = -1$, which contradicts (8.2). If $\mu = 1$, then

$$\sum_i \left(m_i - \frac{5}{2}\right)^2 = 26 - 5 \cdot 10 + 25 = 1,$$

so $m_i \in \{2, 3\}$ for any i. By (8.3) and (8.2), $(m_1, m_2, m_3, m_4) = (3, 3, 2, 2)$, and $(C \cdot \tilde{f}_x) = -1$, a contradiction again. Thus $\mu = 2$, $(\sum_i m_i)^2 = 64 = 4\sum_i m_i^2$, and $m_i = 2$ for any i. Hence $D|_E = 2f_x + G_5$ with $G_5 \sim 5s + 5f$, $(G_5 \cdot s) = 0$, so $G_5 \cap s = \emptyset$, and $C \sim p_T^*(5s + 5f) - 2\sum_i e_i$. By (8.2), $(C \cdot R_i) = 0$ for $i = 1, 2, 3$, so $C \cap R_i = \emptyset$, and $(C \cdot \tilde{f}_x) = 1$, so C intersects $\tilde{f}_x$ transversally in one point. This proves (3). Since $\phi_T(\tilde{f}_x) = o$ by **Step 3** and $\phi_T(C) = \Gamma$, we have $o \in \Gamma$, which completes the proof of (2).

Step 5. In this step, we prove (4).

Let $Q = V(q_Q)$ with q_Q a quadratic form. Since $E_Q = V(q_1, q_2)$ is a complete intersection, the homogeneous ideal of E_Q is (q_1, q_2), so $q_Q = \lambda_1 q_1 + \lambda_2 q_2$ for some $\lambda_1, \lambda_2 \in \mathbb{C}$. The differential of q_2 vanishes at o and the differential of q_1 at o is $d\xi_2$. Since Q is smooth (Theorem 2.22), $\lambda_1 \neq 0$, and we let $\lambda := \lambda_2/\lambda_1$. Thus $E_Q = Q \cap V(q_2)$, and E_Q is the zero scheme of the section $q_2|_Q$ of $\mathcal{O}_Q(2)$. On $(\xi_0 = 1)$ near o, the threefold Q is given by

$$\xi_2(1 + \lambda\xi_4) = \xi_1(\xi_1 - \xi_4) + \lambda\xi_3^2,$$

so (ξ_1, ξ_3, ξ_4) are local coordinates of Q at o, and

$$q_2|_Q = \frac{F_0}{1 + \lambda\xi_4}, \qquad F_0 := \xi_1\xi_4(\xi_1 - \xi_4) - \xi_3^2.$$

We compute the orders along Γ at o. Let c_0 be the point of $C \cap \tilde{f}_x$. Since $(\tilde{f}_x \cdot R_1) = (\tilde{f}_x \cdot R_3) = (\tilde{f}_x \cdot e_4) = 0$, the curve $\tilde{f}_x$ intersects the exceptional locus of p_T only along R_2, and $C \cap R_2 = \emptyset$ by **Step 4**. Thus p_T is an isomorphism near c_0, and $p_T(c_0) \in f_x \backslash (\{p_0\} \cup s)$ as $G_5 \cap s = \emptyset$. In the coordinates of **Step 1**, $p_T(c_0) = (0, u_0)$ with $u_0 \neq 0$. Since C and $\tilde{f}_x$ intersect transversally at c_0, the restriction of z to G_5 is a local parameter at $(0, u_0)$. As $\phi_T|_C$ is an isomorphism and $\phi_T(c_0) = o$, the restrictions to Γ of

$$\frac{\xi_1}{\xi_0} = \frac{z^2}{z - u^2}, \qquad \frac{\xi_3}{\xi_0} = \frac{z^2 u}{z - u^2}, \qquad \frac{\xi_4}{\xi_0} = \frac{zu^2}{z - u^2}$$

have orders 2, 2, and 1 at o, because u and $z - u^2$ are units near $(0, u_0)$. The partial derivatives of F_0 are

$$\frac{\partial F_0}{\partial \xi_1} = 2\xi_1\xi_4 - \xi_4^2, \qquad \frac{\partial F_0}{\partial \xi_3} = -2\xi_3, \qquad \frac{\partial F_0}{\partial \xi_4} = \xi_1^2 - 2\xi_1\xi_4,$$

and their restrictions to Γ have orders 2, 2, and 3 at o.

Since $\Gamma \subset E_Q$, the section $q_2|_Q$ belongs to $H^0(Q, I_\Gamma(2))$. For the map $N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)$ induced by $q_2|_Q \in H^0(Q, I_\Gamma(2))$, the composite $T_Q|_\Gamma \to N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)$ is the differential of $q_2|_Q$. Since $T_Q|_\Gamma \to N_{\Gamma/Q}$ is surjective and $F_0|_\Gamma = 0$, the image of $N_{\Gamma/Q}$ near o is generated by the restrictions to Γ of the partial derivatives of F_0, that is, it is $\mathcal{O}_\Gamma(2)(-2o)$ near o. At a point of $\Gamma \backslash \{o\}$, the

surface E_Q is smooth by **Step 3**, so the map $N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)$ is surjective there. Hence the image of $N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)$ is $\mathcal{O}_\Gamma(2)(-2o)$.

The kernel of $N_{\Gamma/Q} \to \mathcal{O}_\Gamma(2)(-2o)$ is a line bundle. Since $K_Q = \mathcal{O}_Q(-3)$ [Har77, Example II.8.20.3], the adjunction formula [Har77, Proposition II.8.20] implies $\det N_{\Gamma/Q} \cong K_\Gamma \otimes \mathcal{O}_\Gamma(3)$. Thus the kernel is

$$K_\Gamma \otimes \mathcal{O}_\Gamma(3) \otimes \big(\mathcal{O}_\Gamma(2)(-2o)\big)^{-1} \cong K_\Gamma \otimes \mathcal{O}_\Gamma(1)(2o).$$

Since $\deg K_\Gamma = 2$ and $\deg \mathcal{O}_\Gamma(1) = 7$, the degrees of the kernel and of the quotient are $2+7+2 = 11$ and $14 - 2 = 12$. This proves (4). □

The configurations of width three or four in which Z_U intersects s are not covered by the estimates of this section. The next lemma shows that they do not occur. The curve $C \cong \Gamma$ intersects any connected configuration of Λ_T-trivial curves at most once, and the equality $C^2 = D_T^2$ then contradicts the inverse Cartan matrices of A_4 and D_5.

Lemma 8.3. *Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point such that $\Sigma(X, x)$ consists of $[L]$ and a curvilinear double point at a line M, let $n \in \{2, 3, 4\}$ be the width of M, and let $p_0 \in f_x$ be the corresponding point of Z_U. If $n \geq 3$, then $Z_U \cap s = \emptyset$. In particular, for $n = 4$ the configuration is that of Lemma 8.2.*

Proof. Assume that $n \geq 3$, and let $p_T : T \to E$, $e_1, \dots, e_4$, Λ_T, and D_T be as in Construction-Proposition 5.4. Let $\phi_T : T \to E_Q$ be the morphism defined by $|\Lambda_T|$, and let C be the component of D_T of positive Λ_T-degree (Lemma 8.1). We show that none of the following configurations occurs:

(a) $n = 3$, $p_0 \notin s$, and the remaining point of Z_U belongs to s;
(b) $n = 4$ and $p_0 \in s$;
(c) $n = 3$ and $p_0 \in s$.

Step 1. In this step, we list the Λ_T-trivial curves in the configurations (a), (b), and (c).

By Proposition 5.3, $Z_U \cap f_x \cong \Sigma(X, x) \backslash \{[L]\}$ is a curvilinear scheme of length two supported at p_0, $\widehat{\mathcal{O}}_{Z_U, p_0} \cong \mathbb{C}[t]/(t^n)$, the image $\psi(t)$ of a local parameter of L at x has order exactly two, and the widths of the lines intersecting L sum to four. Thus $Z_U = \{p_0\}$ if $n = 4$, and $Z_U = \{p_0, p_1\}$ with $p_1 \notin f_x$ a reduced point if $n = 3$. Let Δ be a smooth curve germ at p_0 containing Z_U near p_0. Then $(\Delta \cdot f_x)_{p_0} = \operatorname{ord} \psi = 2$. We index the centers of p_T so that $c_1 = p_0$, for $1 \leq i < n$ the center c_{i+1} is the point where the strict transform of Δ intersects the exceptional curve of the blow-up at c_i, $c_4 = p_1$ if $n = 3$, and e_i is the total exceptional class over c_i. Since $(\Delta \cdot f_x)_{p_0} = 2 < n$, the strict transforms of f_x pass through c_1 and c_2 and not through c_3 [Har77, Proposition V.3.6 and Exercise V.3.2], so the strict transform $\tilde{f}_x$ of f_x on T satisfies $\tilde{f}_x \sim p_T^* f - e_1 - e_2$. Let $\tilde{s}$ be the strict transform of s on T. If $p_0 \in s$, then c_2 is not a point of the strict transform of s, because $(s \cdot f_x)_{p_0} = 1$. By Construction-Proposition 5.4(1), at most one center is a point of the strict transform of s.

An irreducible member G of $|s + f|$ satisfies $(G \cdot s) = 0$ and $(G \cdot f_x) = 1$. Thus $G \cap s = \emptyset$, G is not tangent to f_x at p_0, and the strict transform of G does not pass through c_2. Hence G passes through at most two centers, namely c_1 and, if $n = 3$, c_4. Any fiber of $E \to L$ other than f_x contains at most one center. By Lemma 8.1, the Λ_T-trivial curves are the strict transforms of the exceptional curves over $c_1, \dots, c_{n-1}$, of classes $e_i - e_{i+1}$, the curve $\tilde{f}_x$, and the curve $\tilde{s}$ if exactly one center is a point of the strict transform of s. We list them as follows.

(a) $\tilde{s} \sim p_T^* s - e_4$, and $R_1 \sim e_1 - e_2$, $R_2 \sim e_2 - e_3$, $R_3 := \tilde{f}_x$, $R_4 := \tilde{s}$.
(b) $\tilde{s} \sim p_T^* s - e_1$, and $R_1 := \tilde{s}$, $R_2 \sim e_1 - e_2$, $R_3 \sim e_2 - e_3$, $R_4 \sim e_3 - e_4$, $R_5 := \tilde{f}_x$.
(c) Since c_1 is a point of s, we have $p_1 \notin s$, $\tilde{s} \sim p_T^* s - e_1$, and $R_1 := \tilde{f}_x$, $R_2 \sim e_2 - e_3$, $R_3 \sim e_1 - e_2$, $R_4 := \tilde{s}$.

A direct calculation with $(e_i \cdot e_j) = -\delta_{ij}$, $s^2 = -1$, $(s \cdot f) = 1$, and $f^2 = 0$ shows that $R_i^2 = -2$ for any i, and that $(R_i \cdot R_j) = 1$ if $\{i, j\}$ is one of

$$\{1,2\},\ \{2,3\},\ \{3,4\} \text{ in (a) and (c)}, \qquad \{1,2\},\ \{2,3\},\ \{3,4\},\ \{3,5\} \text{ in (b)},$$

and $(R_i \cdot R_j) = 0$ for any other $i \neq j$. Thus the curves R_i form a connected configuration of type A_4 in (a) and (c), and of type D_5 in (b). Let $\mathsf{A} := (-(R_i \cdot R_j))_{i,j}$ be the Cartan matrix, $\mathbf{c} := ((C \cdot R_i))_i$, and $\mathbf{d} := ((D_T \cdot R_i))_i$. Since $D_T = p_T^*(D|_E) - 3\sum_i e_i$ with $D|_E \sim 5s + 7f$ (Construction-Proposition 5.4(2)), we have $(D_T \cdot (e_i - e_{i+1})) = 0$, $(D_T \cdot \tilde{f}_x) = ((5s+7f)\cdot f) - 6 = -1$, and $(D_T \cdot \tilde{s}) = ((5s + 7f) \cdot s) - 3 = -1$. Hence

$$\mathbf{d} = \begin{cases} (0,0,-1,-1)^t & \text{in (a)}, \\ (-1,0,0,0,-1)^t & \text{in (b)}, \\ (-1,0,0,-1)^t & \text{in (c)}. \end{cases} \tag{8.4}$$

Step 2. In this step, we prove that the entries of $\mathbf{c}$ are non-negative integers and

$$\sum_i (C \cdot R_i) \leq 1. \tag{8.5}$$

Since C is not contracted by ϕ_T, C is different from any R_i, so $(C \cdot R_i) \geq 0$. Since $(\Lambda_T \cdot R_i) = 0$, ϕ_T contracts any R_i, and since the configuration is connected, ϕ_T maps $\bigcup_i R_i$ to one point o'. We may assume that $C \cap \bigcup_i R_i \neq \emptyset$. Then $o' \in \phi_T(C) = \Gamma$. Since $\phi_T|_C : C \to \Gamma$ is an isomorphism (Lemma 8.1), C contains exactly one point c' over o', and $C \cap \bigcup_i R_i = \{c'\}$. Since Γ is a smooth curve in the smooth threefold Q, there exists a regular function φ on an open neighborhood V of o' in E_Q with $\varphi(o') = 0$ and $\mathrm{ord}_{o'}(\varphi|_\Gamma) = 1$. The function $\phi_T^*\varphi$ is regular on $\phi_T^{-1}(V) \supset \bigcup_i R_i$ and vanishes along any R_i, so $\mathrm{div}(\phi_T^*\varphi) - \sum_i R_i \geq 0$ on $\phi_T^{-1}(V)$. The divisor $\mathrm{div}(\phi_T^*\varphi) - \sum_i R_i \geq 0$ on $\phi_T^{-1}(V)$ does not contain C, since $\mathrm{ord}_{c'}((\phi_T^*\varphi)|_C) = \mathrm{ord}_{o'}(\varphi|_\Gamma) = 1$. By [Har77, Proposition V.1.4],

$$1 = \mathrm{ord}_{c'}((\phi_T^*\varphi)|_C) \geq \sum_i (R_i \cdot C)_{c'} = \sum_i (C \cdot R_i).$$

Step 3. In this step, we conclude the proof.

By Lemma 8.1 and **Step 1**, $D_T = C + \sum_i \nu_i R_i$ with $\boldsymbol{\nu} := (\nu_i)_i \in \mathbb{Z}^k_{\geq 0}$, where $k = 4$ or 5. Then $\mathbf{d} = \mathbf{c} - \mathsf{A}\boldsymbol{\nu}$. Since $C^2 = D_T^2 = 9$ (Lemma 8.1),

$$0 = C^2 - D_T^2 = -2\mathbf{d}^t\boldsymbol{\nu} - \boldsymbol{\nu}^t\mathsf{A}\boldsymbol{\nu},$$

and therefore

$$\mathbf{c}^t\mathsf{A}^{-1}\mathbf{c} = (\mathbf{d} + \mathsf{A}\boldsymbol{\nu})^t\mathsf{A}^{-1}(\mathbf{d} + \mathsf{A}\boldsymbol{\nu}) = \mathbf{d}^t\mathsf{A}^{-1}\mathbf{d} + 2\mathbf{d}^t\boldsymbol{\nu} + \boldsymbol{\nu}^t\mathsf{A}\boldsymbol{\nu} = \mathbf{d}^t\mathsf{A}^{-1}\mathbf{d}. \tag{8.6}$$

By **Step 2**, $\mathbf{c}$ is zero or a standard basis vector, so the left-hand side of (8.6) is zero or a diagonal entry of A^{-1}. In the orderings of **Step 1**, the inverse Cartan matrices are [Bou02, Ch. VI, Plates I and IV, item (VI)]

$$\mathsf{A}_{A_4}^{-1} = \frac{1}{5}\begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \qquad \mathsf{A}_{D_5}^{-1} = \frac{1}{4}\begin{pmatrix} 4 & 4 & 4 & 2 & 2 \\ 4 & 8 & 8 & 4 & 4 \\ 4 & 8 & 12 & 6 & 6 \\ 2 & 4 & 6 & 5 & 3 \\ 2 & 4 & 6 & 3 & 5 \end{pmatrix}.$$

The largest diagonal entry is $\frac{6}{5}$ for A_4 and 3 for D_5. By (8.4), the right-hand side of (8.6) is

$$\begin{cases} \frac{1}{5}(6 + 2\cdot 3 + 4) = \frac{16}{5} > \frac{6}{5} & \text{in (a)}, \\ 1 + 2\cdot\frac{1}{2} + \frac{5}{4} = \frac{13}{4} > 3 & \text{in (b)}, \\ \frac{1}{5}(4 + 2\cdot 1 + 4) = 2 > \frac{6}{5} & \text{in (c)}. \end{cases}$$

This contradicts (8.6), so none of (a), (b), and (c) occurs.

If $n = 4$, then $Z_U = \{p_0\}$ and $p_0 \notin s$ since (b) does not occur. If $n = 3$, then $p_0 \notin s$ since (c) does not occur, and $p_1 \notin s$ since (a) does not occur. Thus $Z_U \cap s = \emptyset$. For $n = 4$, the point p_0 of Z_U representing M does not belong to s, so the hypotheses of Lemma 8.2 hold. □

8.2. **Widths two and three.** In this subsection, we prove Propositions 8.8 and 8.9. We work in the following coordinates on E.

Set-up 8.4. Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ and $p_0 \in f_x$ be closed points, and let $q := s \cap f_x$. We view E as the blow-up of $\mathbb{P}^2$ at $[0:0:1]$ with exceptional curve s, such that $E \to L$ is induced by $[X_0 : X_1 : X_2] \to [X_0 : X_1]$ [Har77, Example V.2.11.5]. The stabilizer of $[0:0:1]$ in PGL_3 acts on the lines through $[0:0:1]$ as PGL_2, and it contains the maps $X_2 \to X_2 - cX_0$ for $c \in \mathbb{C}$. Thus we may assume that $f_x = \{X_1 = 0\}$, that the fiber $f_\infty := \{X_0 = 0\}$ is disjoint from Z_U, and that $p_0 = [1:0:0]$ if $p_0 \neq q$. Let

$$z := \frac{X_1}{X_0}, \qquad u := \frac{X_2}{X_0}, \qquad v := \frac{1}{u}, \qquad \zeta := \frac{1}{z}, \qquad u_\infty := \frac{u}{z}, \qquad v_\infty := \frac{z}{u}.$$

Then (z,u), (z,v), (ζ,u_∞), and (ζ,v_∞) are coordinates on the four affine charts of E that are invariant under the torus $\mathbb{G}_m^2$ acting by $(z,u) \to (\lambda_1 z, \lambda_2 u)$. Moreover $f_x = \{z = 0\}$ and $f_\infty = \{\zeta = 0\}$, and s is given by $v = 0$ and by $v_\infty = 0$ in the charts (z,v) and (ζ, v_∞). Let r and $t = s \cap f_\infty$ be the origins of the charts (ζ, u_∞) and (ζ, v_∞).

For integers $0 \le \alpha \le \beta$, the divisor $\alpha s + \beta f$ is β times the pullback of a line minus $(\beta - \alpha)s$, so the monomials $z^\ell u^d$ with $0 \le d \le \alpha$ and $0 \le \ell \le \beta - d$ form a basis of $H^0(E, \mathcal{O}_E(\alpha s + \beta f))$, and $\mathbb{G}_m^2$ acts on $z^\ell u^d$ by a character. Near q, r, and t, the section $z^\ell u^d$ is a local generator of $\mathcal{O}_E(\alpha s + \beta f)$ times

$$z^\ell v^{\alpha - d}, \qquad \zeta^{\beta - \ell - d} u_\infty^d, \qquad \zeta^{\beta - \ell - d} v_\infty^{\alpha - d},$$

respectively. In particular $z^\ell u^d$ vanishes to order ℓ along f_x. If $\sigma = \sum c_{\ell d} z^\ell u^d$ vanishes to order at least ℓ_0 along f_x, then the restriction of $z^{-\ell_0}\sigma$ to f_x is the polynomial $P = \sum_d c_{\ell_0 d} u^d$, viewed as a section of $\mathcal{O}_{f_x}(\alpha)$, and the order of P at q is $\alpha - \deg P$.

For a subspace $V \subset H^0(E, \mathcal{O}_E(\alpha s + \beta f))$ and an integer ℓ, let $\mathcal{F}^\ell V \subset V$ be the subspace of sections vanishing to order at least ℓ along f_x, so that $\mathcal{F}^\ell V = V$ for $\ell \le 0$. For integers $0 \le \alpha \le \beta$ and $\gamma \ge 0$, let

$$V_{\alpha,\beta,\gamma} := H^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes I_{Z_U}^\gamma\big).$$

By Construction-Proposition 5.4(3), $W_{m,j} \subset V_{j,m+j,0}$ for $0 \le j \le m$, $W_{m,j} \subset V_{j,m+j,j-m}$ for $m \le j \le 2m$, and $W_{m,j} = d_E^{j-2m} U_{5m-2j}$ for $2m < j \le 5m/2$, where $U_k \subset V_{2k,3k,k}$ for any integer $k \ge 0$.

The following lemma bounds $S(W^{f_x}; a)$ for $a \in f_x$. It uses only the subscheme $Z_U \cap f_x$, and not the rest of Z_U.

Lemma 8.5. *Notation as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point, and let $q := s \cap f_x$. Assume that $Z_U \cap f_x$ is a curvilinear scheme of length two supported at one point p_0. Then the following hold.*

(1) If $p_0 \neq q$, then $S(W^{f_x}; q) \le \frac{23}{24}$ and $S(W^{f_x}; a) \le \frac{7}{8}$ for any $a \in f_x \backslash \{q\}$.
(2) If $p_0 = q$, then $S(W^{f_x}; q) \le \frac{3323}{3360}$ and $S(W^{f_x}; a) \le \frac{7}{8}$ for any $a \in f_x \backslash \{q\}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we bound $S(W^{f_x}; a)$ by sums over the rows of explicit monomial spaces.

We use the coordinates of Set-up 8.4 for x and p_0. For a closed point $a \in f_x$ and a subspace $V \subset H^0(f_x, \mathcal{O}_{f_x}(n))$, let $\mathcal{F}_a^h V \subset V$ be the subspace of sections vanishing to order at least h at a, and let

$$\tau_a(V) := \sum_{h \ge 1} \dim \mathcal{F}_a^h V.$$

For a subspace $V \subset H^0(E, \mathcal{O}_E(\alpha s + \beta f))$ and an integer ℓ, let $V_\ell^{f_x} \subset H^0(f_x, \mathcal{O}_{f_x}(\alpha))$ be the image of $\mathcal{F}^\ell V$ under $\sigma \to (z^{-\ell}\sigma)|_{f_x}$ if $\ell \geq 0$, and $V_\ell^{f_x} := 0$ if $\ell < 0$. Then $W_{m,j,\ell}^{f_x} = (W_{m,j})_\ell^{f_x}$, and by [AZ22, Lemma 2.9],

$$S(W^{f_x}; a) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j\geq 0} \sum_{\ell \geq 0} \tau_a\big(W_{m,j,\ell}^{f_x}\big). \tag{8.7}$$

Let μ be the coefficient of f_x in $D|_E$, and let $\bar{d} \in H^0(f_x, \mathcal{O}_{f_x}(5))$ be the restriction of $z^{-\mu} d_E$ to f_x. The divisor of $\bar{d}$ is $(D|_E - \mu f_x)|_{f_x}$, and

$$\big((D|_E - \mu f_x) \cdot f\big) = \big((5s + (7-\mu)f) \cdot f\big) = 5$$

by Construction-Proposition 5.4(2). Let $2m < j \leq 5m/2$, and let $k := 5m - 2j$. By Set-up 8.4, $W_{m,j} = d_E^{j-2m} U_k$. Since d_E vanishes to order exactly μ along f_x,

$$W_{m,j,\ell}^{f_x} = \bar{d}^{j-2m} \cdot (U_k)_{\ell-\mu(j-2m)}^{f_x} \qquad \text{for any } \ell \geq 0.$$

Multiplication by $\bar{d}^{j-2m}$ raises the order at a of any section by $(j-2m)\operatorname{mult}_a\big((D|_E - \mu f_x)|_{f_x}\big) \leq 5(j-2m)$. Thus

$$\sum_{\ell\geq 0} \tau_a\big(W_{m,j,\ell}^{f_x}\big) \leq 5(j-2m) \dim W_{m,j} + \sum_{\ell \geq 0} \tau_a\big((U_k)_\ell^{f_x}\big).$$

The kernel of $H^0(Y, mH - jE) \to H^0(E, \mathcal{O}_E(js + (m+j)f))$ is $H^0(Y, mH - (j+1)E)$, so $\dim W_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$. Hence Lemma 5.2 implies

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j>2m} 5(j-2m) \dim W_{m,j} = \frac{5}{72}.$$

Let I be the ideal of $Z_U \cap f_x$ at p_0. For integers $0 \leq \alpha \leq \beta$ and $\gamma \geq 0$, let $V'_{\alpha,\beta,\gamma} \subset H^0(E, \mathcal{O}_E(\alpha s + \beta f))$ be the subspace of sections whose germ at p_0 belongs to I^γ, and let $\mathcal{R}_\ell(\alpha,\beta,\gamma) := (V'_{\alpha,\beta,\gamma})_\ell^{f_x}$. Here $V'_{\alpha,\beta,\gamma}$ is defined only by the condition at p_0 given by $Z_U \cap f_x$, whereas $V_{\alpha,\beta,\gamma}$ in Set-up 8.4 is defined by the powers of the ideal of Z_U. Since $Z_U \cap f_x \subset Z_U$, the ideal of Z_U at p_0 is contained in I, so $V_{\alpha,\beta,\gamma} \subset V'_{\alpha,\beta,\gamma}$. Thus Set-up 8.4 implies

$$W_{m,j,\ell}^{f_x} \subset \mathcal{R}_\ell\big(j, m+j, \max(0, j-m)\big)\ (0 \leq j \leq 2m), \qquad (U_k)_\ell^{f_x} \subset \mathcal{R}_\ell(2k, 3k, k). \tag{8.8}$$

If $V_1 \subset V_2$, then $\mathcal{F}_a^h V_1 \subset \mathcal{F}_a^h V_2$, so $\tau_a(V_1) \leq \tau_a(V_2)$.

All sums below are sums of polynomials over the lattice points of polygons whose vertices depend affinely on j/m, over $O(m)$ values of j. Replacing them by the corresponding integrals changes them by $O(m^3)$, while $mN_m = 3m^4 + O(m^3)$ by Lemma 5.2. Below we replace sums over lattice points by integrals without further mention.

Let $a \in f_x$, and let ϑ_a be a function on $\{(\alpha,\beta,\gamma) \in \mathbb{R}^3 : 0 \leq \alpha \leq \beta,\ \gamma \geq 0\}$, homogeneous of degree three and piecewise polynomial, such that

$$\sum_\ell \tau_a\big(\mathcal{R}_\ell(\alpha,\beta,\gamma)\big) \leq \vartheta_a(\alpha,\beta,\gamma) + O(\beta^2)$$

for integers $0 \leq \alpha \leq \beta$ and $\gamma \geq 0$. We divide j into the ranges $0 \leq j \leq m$, $m \leq j \leq 2m$, and $2m < j \leq 5m/2$. In the first two ranges, let $\rho := j/m$ and $w := (j-m)/m$, so that $(j, m+j, 0) = m(\rho, 1+\rho, 0)$ and $(j, m+j, j-m) = m(1+w, 2+w, w)$. In the third range, $(2k, 3k, k) = k(2,3,1)$ with $k = 5m - 2j$, and

$$\sum_{2m<j\leq 5m/2} k^3 = m^4 \int_2^{5/2} (5-2\rho)^3 d\rho + O(m^3) = \frac{m^4}{8} + O(m^3).$$

By (8.7), (8.8), the bound for $2m < j \le 5m/2$ above, and $mN_m = 3m^4 + O(m^3)$, we obtain

$$S(W^{f_x};a) \le \frac{1}{3}\Big(\int_0^1 \vartheta_a(\rho,1+\rho,0)d\rho + \int_0^1 \vartheta_a(1+w,2+w,w)dw + \frac{1}{8}\vartheta_a(2,3,1)\Big) + \frac{5}{72}. \tag{8.9}$$

Step 2. In this step we prove $S(W^{f_x};a) \le 7/8$ for $a \in f_x\backslash\{q\}$, in both cases (1) and (2).

Let $a \in f_x\backslash\{q\}$. Then a is the point $u = c$ of f_x for some $c \in \mathbb{C}$. By Set-up 8.4, $\mathcal{R}_\ell(\alpha,\beta,\gamma) \subset \mathcal{R}_\ell(\alpha,\beta,0)$, and $\mathcal{R}_\ell(\alpha,\beta,0)$ is the space of polynomials in u of degree at most $n_\ell := \min(\alpha, \beta-\ell)$ for $0 \le \ell \le \beta$. Since $\mathcal{F}_a^h\mathcal{R}_\ell(\alpha,\beta,0)$ is $(u-c)^h$ times the space of polynomials of degree at most $n_\ell - h$,

$$\tau_a\big(\mathcal{R}_\ell(\alpha,\beta,0)\big) = \sum_{h=1}^{n_\ell}(n_\ell+1-h) = \frac{n_\ell(n_\ell+1)}{2}.$$

Summing over ℓ, we obtain $\sum_\ell \tau_a(\mathcal{R}_\ell(\alpha,\beta,\gamma)) \le \theta_1(\alpha,\beta)+O(\beta^2)$, where $\theta_1(\alpha,\beta) := \frac{\alpha^2(\beta-\alpha)}{2} + \frac{\alpha^3}{6}$. By (8.9) with $\vartheta_a(\alpha,\beta,\gamma) = \theta_1(\alpha,\beta)$,

$$\begin{aligned} S(W^{f_x};a) &\le \frac{1}{3}\Big(\int_0^1 \theta_1(\rho,1+\rho)d\rho + \int_0^1 \theta_1(1+w,2+w)dw + \frac{\theta_1(2,3)}{8}\Big) + \frac{5}{72} \\ &= \frac{1}{3}\Big(\frac{5}{24} + \frac{43}{24} + \frac{5}{12}\Big) + \frac{5}{72} = \frac{7}{8}. \end{aligned}$$

Step 3. In this step we prove $S(W^{f_x};q) \le 23/24$ in case (1).

Assume that $p_0 \ne q$. Then p_0 is the origin of the chart (z,u), and $I = (z,u^2)$. The ideal I^γ is generated by the monomials $z^\ell u^d$ with $2\ell + d \ge 2\gamma$. Thus a section $\sum c_{\ell d} z^\ell u^d$ belongs to $V'_{\alpha,\beta,\gamma}$ if and only if $2\ell + d \ge 2\gamma$ whenever $c_{\ell d} \ne 0$, and

$$\mathcal{R}_\ell(\alpha,\beta,\gamma) = \mathrm{Span}\{u^d : \max(0,2(\gamma-\ell)) \le d \le \min(\alpha,\beta-\ell)\}.$$

Since u^d vanishes to order $\alpha - d$ at q, the space $\mathcal{F}_q^h\mathcal{R}_\ell(\alpha,\beta,\gamma)$ is spanned by the u^d in $\mathcal{R}_\ell(\alpha,\beta,\gamma)$ with $\alpha - d \ge h$. Hence $\tau_q(\mathcal{R}_\ell(\alpha,\beta,\gamma))$ is the sum of $\alpha - d$ over the d with $u^d \in \mathcal{R}_\ell(\alpha,\beta,\gamma)$, and $\sum_\ell \tau_q(\mathcal{R}_\ell(\alpha,\beta,\gamma)) = \theta_2(\alpha,\beta,\gamma) + O(\beta^2)$, where

$$\theta_2(\alpha,\beta,\gamma) := \int_0^\beta \int_{\max(0,2(\gamma-y))}^{\min(\alpha,\beta-y)} (\alpha-\delta)d\delta dy.$$

A direct calculation shows that $\theta_2(\rho,1+\rho,0) = \frac{\rho^2}{2} + \frac{\rho^3}{3}$,

$$\theta_2(1+w,2+w,w) = \frac{3w+w^3}{6} + \frac{(1-w)(1+w)^2}{2} + \frac{(1+w)^3}{3},$$

and $\theta_2(2,3,1) = \frac{10}{3}$. Thus, by (8.9) with $\vartheta_q = \theta_2$,

$$S(W^{f_x};q) \le \frac{1}{3}\Big(\frac{1}{4} + 2 + \frac{10}{3\cdot 8}\Big) + \frac{5}{72} = \frac{8}{9} + \frac{5}{72} = \frac{23}{24}.$$

Step 4. In this step we prove $S(W^{f_x};q) \le 3323/3360$ in case (2).

Assume that $p_0 = q$. Then p_0 is the origin of the chart (z,v), and $I = (z,v^2)$. By Set-up 8.4, near q the section $z^\ell u^{\alpha-d}$ is a local generator times $z^\ell v^d$. As in **Step 3**, a section belongs to $V'_{\alpha,\beta,\gamma}$ if and only if all the monomials $z^\ell u^{\alpha-d}$ occurring in it satisfy $2\ell + d \ge 2\gamma$, and

$$\mathcal{R}_\ell(\alpha,\beta,\gamma) = \mathrm{Span}\{u^{\alpha-d} : \max(0,\alpha-\beta+\ell,2(\gamma-\ell)) \le d \le \alpha\},$$

where $u^{\alpha-d}$ vanishes to order d at q. Let $\mathcal{M}(\alpha,\beta,\gamma)$ be the set of pairs (ℓ,d) with $u^{\alpha-d} \in \mathcal{R}_\ell(\alpha,\beta,\gamma)$. Then

$$\sum_\ell \tau_q\big(\mathcal{R}_\ell(\alpha,\beta,\gamma)\big) = \sum_{(\ell,d)\in\mathcal{M}(\alpha,\beta,\gamma)} d = \theta_3(\alpha,\beta,\gamma)+O(\beta^2), \qquad \theta_3(\alpha,\beta,\gamma) := \int_0^\beta \int_{\max(0,\alpha-\beta+y,2(\gamma-y))}^{\alpha} \delta d\delta dy,$$

where the inner integral is zero if its lower limit exceeds α. A direct calculation shows that $\theta_3(\rho, 1+\rho, 0) = \frac{\rho^2}{2} + \frac{\rho^3}{3}$,

$$\theta_3(1+w, 2+w, w) = \frac{(1+w)^2}{2} - \frac{2w^3}{3} + \frac{(1+w)^3}{3},$$

and $\theta_3(2,3,1) = 4$. Thus (8.9) with $\vartheta_q = \theta_3$ only gives

$$S(W^{f_x}; q) \le \frac{1}{3}\Big(\frac{1}{4} + \frac{9}{4} + \frac{4}{8}\Big) + \frac{5}{72} = 1 + \frac{5}{72}.$$

Lemma 3.3 improves the bound $1 + \frac{5}{72}$ as follows.

Let $m \le j \le 2m$, with $w = (j-m)/m$ as in **Step 1**, and let $c_j := \lfloor (j-m)^2/m \rfloor$. Let $\mathcal{R}_\ell := \mathcal{R}_\ell(j, m+j, j-m)$. Since $\sum_{h>c_j} \dim \mathcal{F}_q^h \mathcal{R}_\ell$ is the sum of $\max(0, d-c_j)$ over the d with $u^{j-d} \in \mathcal{R}_\ell$, Lemma 3.3, applied to $W^{f_x}_{m,j,\ell} \subset \mathcal{R}_\ell$ with the filtration $\mathcal{F}_q^h$ and $c = c_j$ and summed over ℓ, implies

$$\sum_\ell \tau_q\big(W^{f_x}_{m,j,\ell}\big) \le \sum_{\mathcal{M}(j,m+j,j-m)} d - \Big(\sum_{\mathcal{M}(j,m+j,j-m)} \min(d, c_j) - c_j \dim W_{m,j} \Big).$$

For fixed d, the ℓ with $(\ell, d) \in \mathcal{M}(j, m+j, j-m)$ form an interval of length $m h_w(d/m) + O(1)$, where

$$h_w(\delta) := 1 + \delta - \max\Big(0, w - \frac{\delta}{2}\Big), \qquad 0 \le \delta \le 1 + w.$$

Thus

$$|\mathcal{M}(j, m+j, j-m)| = m^2 \int_0^{1+w} h_w(\delta) d\delta + O(m) = m^2\Big(\frac{3}{2} + 2w - \frac{w^2}{2}\Big) + O(m),$$

while $\dim W_{m,j} = m^2\big(\frac{3}{2} + 2w - \frac{3w^2}{2}\big) + O(m)$ by Lemma 5.2. Since $w^2 \le 2w$, we have $h_w(\delta) = 1 - w + \frac{3}{2}\delta$ for $0 \le \delta \le w^2$, and the term in the large brackets is

$$\begin{aligned} &c_j\big(|\mathcal{M}(j, m+j, j-m)| - \dim W_{m,j}\big) - \sum_{\mathcal{M}(j,m+j,j-m), d<c_j} (c_j - d) \\ &\quad = m^3\Big(w^4 - \int_0^{w^2} (w^2 - \delta)\Big(1 - w + \frac{3}{2}\delta\Big) d\delta\Big) + O(m^2), \end{aligned}$$

which is $m^3\big(\frac{w^4}{2} + \frac{w^5}{2} - \frac{w^6}{4}\big) + O(m^2)$. Summing over $m \le j \le 2m$, the total saving is

$$m^4 \int_0^1 \Big(\frac{w^4}{2} + \frac{w^5}{2} - \frac{w^6}{4}\Big) dw + O(m^3) = \frac{31}{210} m^4 + O(m^3).$$

Let $2m < j \le 5m/2$, and let $k := 5m - 2j$. For fixed $d \le 2k$, the ℓ with $(\ell, d) \in \mathcal{M}(2k, 3k, k)$ form the interval $\max(0, k - d/2) \le \ell \le k + d$, of length $\frac{3}{2}d + O(1)$. Thus $|\mathcal{M}(2k, 3k, k)| = 3k^2 + O(k)$, while $\dim U_k = \dim W_{m,j} = 2k^2 + O(k)$ by Lemma 5.2. By Lemma 3.3, applied to $(U_k)^{f_x}_\ell \subset \mathcal{R}_\ell(2k, 3k, k)$ with the filtration $\mathcal{F}_q^h$ and $c = k$ and summed over ℓ as for $m \le j \le 2m$,

$$\begin{aligned} \sum_\ell \tau_q\big((U_k)^{f_x}_\ell\big) &\le \sum_{\mathcal{M}(2k,3k,k)} d - k^3\Big(1 - \int_0^1 (1-\delta)\frac{3\delta}{2} d\delta\Big) + O(k^2) \\ &= \sum_{\mathcal{M}(2k,3k,k)} d - \frac{3}{4}k^3 + O(k^2). \end{aligned}$$

Summing over j, the saving is $\frac{3}{4} \cdot \frac{1}{8} m^4 + O(m^3) = \frac{3}{32} m^4 + O(m^3)$. For $0 \le j \le m$ we keep the bound $\tau_q(W^{f_x}_{m,j,\ell}) \le \tau_q(\mathcal{R}_\ell(j, m+j, 0))$. Subtracting the savings $\frac{31}{210} m^4$ and $\frac{3}{32} m^4$ from the sums in the derivation of (8.9) with $\vartheta_q = \theta_3$, we obtain

$$S(W^{f_x}; q) \le \frac{1}{3}\Big(3 - \frac{31}{210} - \frac{3}{32}\Big) + \frac{5}{72} = 1 - \frac{811}{10080} + \frac{700}{10080} = \frac{3323}{3360}.$$

Together with **Step 2**, this proves (2). □

The next lemma reduces the bound of $S(W;f_x)$ to a count of the dimensions of the spaces $\mathcal{F}^\ell V_{\alpha,\beta,\gamma}$, which in turn is bounded by a degeneration of the components of Z_U off f_x.

Lemma 8.6. *Notation as in Set-up 8.4. Assume that $Z_U \cap f_x$ is supported at p_0. Let μ be the coefficient of f_x in $D|_E$, let Z_0 be the connected component of Z_U supported at p_0, and let Z_1 be the union of the other connected components of Z_U.*

(1) Let $(g_\lambda)_{\lambda\in\mathbb{C}^*}$ be a one-parameter subgroup of $\mathbb{G}_m^2$ with $g_\lambda(Z_0) = Z_0$, and let $Z_{1,\lambda} \subset E$, $\lambda \in \mathbb{A}^1$, be closed subschemes such that $Z_{1,\lambda} = g_\lambda(Z_1)$ for $\lambda \in \mathbb{C}^*$, $Z_{1,\lambda} \cap f_x = \emptyset$ for $\lambda \in \mathbb{A}^1$, and the subschemes $V(I^\gamma_{Z_{1,\lambda}})$ form a flat family over $\mathbb{A}^1$ for any $\gamma \geq 0$. Then for any integers $0 \leq \alpha \leq \beta$ and $\gamma, \ell \geq 0$,

$$\dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma} \leq h^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes (I^\gamma_{Z_0} \cap I^\ell_{f_x}) \cdot I^\gamma_{Z_{1,0}}\big).$$

(2) Let ϑ be a function on $\{(\alpha,\beta,\gamma) \in \mathbb{R}^3 : 0 \leq \alpha \leq \beta,\ \gamma \geq 0\}$, homogeneous of degree three and piecewise polynomial, such that $\sum_{\ell\geq 1} \dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma} \leq \vartheta(\alpha,\beta,\gamma) + O(\beta^2)$ for integers $0 \leq \alpha \leq \beta$ and $\gamma \geq 0$. Then

$$S(W;f_x) \leq \frac{\mu}{72} + \frac{1}{3}\Big(\frac{11}{24} + \int_0^1 \vartheta(1+w, 2+w, w)dw + \frac{\vartheta(2,3,1)}{8}\Big).$$

Proof. (1) For $\lambda \in \mathbb{A}^1$, let $J_\lambda := (I^\gamma_{Z_0} \cap I^\ell_{f_x}) \cdot I^\gamma_{Z_{1,\lambda}}$. Since $Z_U = Z_0 \sqcup Z_1$ and $Z_1 = Z_{1,1}$ is disjoint from f_x, we have $I^\gamma_{Z_U} \cap I^\ell_{f_x} = J_1$, so $H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes J_1) = \mathcal{F}^\ell V_{\alpha,\beta,\gamma}$. The subscheme $V(I^\gamma_{Z_0} \cap I^\ell_{f_x})$ is supported on f_x and does not depend on λ, and $Z_{1,\lambda} \cap f_x = \emptyset$. Thus $V(J_\lambda)$ is the disjoint union of $V(I^\gamma_{Z_0} \cap I^\ell_{f_x})$ and $V(I^\gamma_{Z_{1,\lambda}})$, and the subschemes $V(J_\lambda)$ form a flat family over $\mathbb{A}^1$. Since g_λ preserves Z_0, f_x, and s, and maps fibers to fibers, g_λ maps $V(J_1)$ onto $V(J_\lambda)$ and $g_\lambda^*\mathcal{O}_E(\alpha s + \beta f) \cong \mathcal{O}_E(\alpha s + \beta f)$ for $\lambda \in \mathbb{C}^*$. Hence $h^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes J_\lambda) = \dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma}$ for $\lambda \in \mathbb{C}^*$, and upper semicontinuity [Har77, Theorem III.12.8] gives the inequality in (1).

(2) By [AZ22, Lemma 2.9],

$$S(W;f_x) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j\geq 0} \sum_{\ell\geq 1} \dim \mathcal{F}^\ell W_{m,j}.$$

We use the three ranges of j in Set-up 8.4. Let $2m < j \leq 5m/2$, and let $k := 5m - 2j$. Since $W_{m,j} = d_E^{j-2m} U_k$ and d_E vanishes to order exactly μ along f_x,

$$\sum_{\ell\geq 1} \dim \mathcal{F}^\ell W_{m,j} = \mu(j-2m) \dim W_{m,j} + \sum_{\ell\geq 1} \dim \mathcal{F}^\ell U_k.$$

As $\dim W_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$, Lemma 5.2 implies that the first terms contribute $\frac{\mu}{72}$ to $S(W;f_x)$. Since $U_k \subset V_{2k,3k,k}$, the second term is at most $k^3\vartheta(2,3,1) + O(m^2)$, and

$$\sum_{2m<j\leq 5m/2} k^3 = m^4 \int_2^{5/2} (5-2\rho)^3 d\rho + O(m^3) = \frac{m^4}{8} + O(m^3).$$

For $m \leq j \leq 2m$, let $w := (j-m)/m$. Then $W_{m,j} \subset V_{j,m+j,j-m}$ and $(j, m+j, j-m) = m(1+w, 2+w, w)$. Since ϑ is piecewise polynomial, the sum of $\sum_{\ell\geq 1} \dim \mathcal{F}^\ell W_{m,j}$ over these j is at most $m^4 \int_0^1 \vartheta(1+w,2+w,w)dw + O(m^3)$. Let $0 \leq j \leq m$. By Set-up 8.4, $H^0(E, \mathcal{O}_E(js + (m+j)f))$ has the basis of the monomials $z^\ell u^d$ with $0 \leq d \leq j$ and $0 \leq \ell \leq m + j - d$, and $z^\ell u^d$ vanishes to order ℓ along f_x. Summing ℓ over the monomials in the basis, with $\rho := j/m$, shows that

$$\sum_{j=0}^{m} \sum_{\ell\geq 1} \dim \mathcal{F}^\ell W_{m,j} \leq m^4 \int_0^1 \int_0^\rho \frac{(1+\rho-\delta)^2}{2} d\delta d\rho + O(m^3) = \frac{11}{24} m^4 + O(m^3).$$

Since $mN_m = 3m^4 + O(m^3)$ by Lemma 5.2, we obtain the inequality in (2). □

The following lemma deduces the bound of $\delta_x(X)$ in Propositions 8.8 and 8.9 from bounds of $S(W;f_x)$ and $S(W^{f_x};a)$.

Lemma 8.7. *Notation as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point such that $S(W;f_x) \le \frac{3323}{3360}$ and $S(W^{f_x};a) \le \frac{3323}{3360}$ for any $a \in f_x$. Then any prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies*

$$A_X(F) \ge \frac{3360}{3323} S_X(F) + \frac{5798}{9969} \operatorname{mult}_F E.$$

In particular $\delta_x(X) \ge 3360/3323$.

Proof. Since Y and E are smooth, the different of (Y,E) on E is zero. The curve $f_x \cong \mathbb{P}^1$ is a smooth prime divisor on the smooth surface E, so (E,f_x) is plt [KM98, Corollary 2.31(3)] and the different on f_x is zero. Let $a \in f_x$. Then $A_E(f_x) = 1$, the only prime divisor over f_x whose center contains a is a, and $A_{f_x}(a) = 1$. By Theorem 2.9 applied to the klt pair $(E,0)$, the series W, $Z = \{a\}$, and the plt type divisor f_x,

$$\delta_a(E;W) \ge \min\left\{ \frac{A_E(f_x)}{S(W;f_x)}, \frac{1}{S(W^{f_x};a)} \right\} \ge \frac{3360}{3323}.$$

Thus $\frac{3360}{3323} S(W;v) \le A_E(v)$ for any prime divisor v over E with $\operatorname{center}_E v \cap f_x \ne \emptyset$.

Let F be a prime divisor over X with $x \in \operatorname{center}_X F$. Since $\pi(\operatorname{center}_Y F) = \operatorname{center}_X F$ and $\pi^{-1}(x) = f_x$, we have $\operatorname{center}_Y F \cap f_x \ne \emptyset$. By Lemma 3.1 with $G = f_x$ and $\lambda = \frac{3360}{3323}$, and since $A_X(E) = 2$ and $S_X(E) = \frac{101}{72}$ (Lemma 5.2),

$$A_X(F) \ge \frac{3360}{3323} S_X(F) + \left(2 - \frac{3360}{3323} \cdot \frac{101}{72}\right) \operatorname{mult}_F E = \frac{3360}{3323} S_X(F) + \frac{5798}{9969} \operatorname{mult}_F E.$$

Since $\operatorname{mult}_F E \ge 0$, we have $A_X(F) \ge \frac{3360}{3323} S_X(F)$. Thus $\delta_x(X) \ge 3360/3323$. □

Proposition 8.8. *Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point such that $\Sigma(X,x)$ consists of $[L]$ and a curvilinear double point at a line M of width two. Then the coefficient μ of f_x in $D|_E$ satisfies $1 \le \mu \le 2$, with $\mu = 1$ if the point of $Z_U \cap f_x$ belongs to s, and*

$$S(W;f_x) \le \frac{35}{36}, \qquad S(W^{f_x};a) \le \frac{3323}{3360} \quad \text{for every } a \in f_x.$$

Any prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies

$$A_X(F) \ge \frac{3360}{3323} S_X(F) + \frac{5798}{9969} \operatorname{mult}_F E.$$

In particular $\delta_x(X) \ge 3360/3323$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe Z_U near f_x and the strict transform of f_x on T.

By Proposition 5.3, $Z_U \cap f_x \cong \Sigma(X,x)\backslash\{[L]\}$ is a curvilinear scheme of length two supported at the point $p_0 \in f_x$ which represents M, and the completed local ring of Z_U at p_0 is $\mathbb{C}[\epsilon]/(\epsilon^2)$. Thus the connected component Z_0 of Z_U supported at p_0 is $Z_U \cap f_x$. Since the widths of the lines intersecting L sum to 4 (Proposition 5.3), the union Z_1 of the other connected components of Z_U has length two, and $Z_1 \cap f_x = \emptyset$. We use the coordinates of Set-up 8.4 for x and p_0, and write $q := s \cap f_x$. Then the ideal of Z_0 is (z,u^2) if $p_0 \ne q$, and (z,v^2) if $p_0 = q$.

In the chart $z = uz_1$ (respectively $z = vz_1$) of the blow-up of p_0, the ideal (z,u^2) (respectively (z,v^2)) becomes $u(z_1,u)$ (respectively $v(z_1,v)$). Thus p_T blows up p_0, then the point of the exceptional curve in the direction of f_x, and then two points over $\operatorname{Supp} Z_1$. Let e_1, e_2 be the total exceptional classes over p_0 and e_3, e_4 those over $\operatorname{Supp} Z_1$. The strict transform $\tilde{f}_x$ of f_x on T is $p_T^* f - e_1 - e_2$ [Har77, Proposition V.3.6], and by Construction-Proposition 5.4(2),

$$(\Lambda_T \cdot \tilde{f}_x) = 2 - 2 = 0, \qquad (\tilde{f}_x^2) = -2, \qquad (D_T \cdot \tilde{f}_x) = (D|_E \cdot f) - 6 = -1.$$

Step 2. In this step we prove $1 \le \mu \le 2$, and $\mu = 1$ if $p_0 \in s$.

Since $D_T \ge 0$ and $(D_T \cdot \tilde{f}_x) < 0$, the curve $\tilde{f}_x$ is a component of D_T, and the coefficient of $\tilde{f}_x$ in D_T is μ. Thus $\mu \ge 1$.

Let $\mathcal{C}_x$ be the connected component containing $\tilde{f}_x$ of the union of the irreducible curves $R \subset T$ with $(\Lambda_T \cdot R) = 0$. We determine $\mathcal{C}_x$ with the list of Lemma 8.1. First, $((e_1 - e_2) \cdot \tilde{f}_x) = 0$, and, if the fourth center is infinitely near the third, the strict transform of the exceptional curve over the third center, which has class $e_3 - e_4$, is disjoint from $\tilde{f}_x$, as are the strict transforms of fibers through two of the centers over $\operatorname{Supp} Z_1$, since $Z_1 \cap f_x = \emptyset$. Second, let $G \subset E$ be an irreducible curve of class $s + f$. Since $(G \cdot f) = 1$, the strict transform of G does not pass through the center over p_0 in the direction of f_x. If the strict transform of G passes through three centers, then the three centers are p_0 and the centers with classes e_3 and e_4, and the intersection number of the strict transform of G with $\tilde{f}_x$ is $1 - 1 = 0$. Third, let $\tilde{s}$ be the strict transform of s. By Construction-Proposition 5.4(1), $\tilde{s}$ passes through at most one center. If $p_0 \in s$, then $\tilde{s} = p_T^* s - e_1$, since the direction of f_x is transverse to s, and $(\tilde{s} \cdot \tilde{f}_x) = 1 - 1 = 0$. If $p_0 \notin s$ and $\tilde{s}$ passes through a center over $\operatorname{Supp} Z_1$, say the one with class e_3, then $\tilde{s} = p_T^* s - e_3$ and $(\tilde{s} \cdot \tilde{f}_x) = 1$.

Hence exactly one of the following holds.

(i) $p_0 \in s$, or $p_0 \notin s$ and $\tilde{s}$ passes through no center over $\operatorname{Supp} Z_1$. Then $\mathcal{C}_x = \tilde{f}_x$ is of type A_1.

(ii) $p_0 \notin s$, and Z_1 is reduced with exactly one point on s. Then $\mathcal{C}_x$ is the chain $(R_1, R_2) = (\tilde{f}_x, \tilde{s})$ of type A_2.

(iii) $p_0 \notin s$, and Z_1 is supported at one point of s. Then $\mathcal{C}_x$ is the chain $(R_1, R_2, R_3) = (\tilde{f}_x, \tilde{s}, R_3)$ of type A_3, where $R_3 \sim e_3 - e_4$ is the strict transform of the exceptional curve over the third center.

In cases (ii) and (iii), an irreducible curve of class $s + f$ is disjoint from s since $((s + f) \cdot s) = 0$, so no strict transform of such a curve passes through three centers, and the strict transform $p_T^* f - e_3 - e_4$ of a fiber through the third and fourth centers satisfies $((p_T^* f - e_3 - e_4) \cdot \tilde{s}) = 0$. Moreover, in case (iii), $(R_3 \cdot \tilde{s}) = 1$ and $((p_T^* f - e_3 - e_4) \cdot R_3) = 0$. In cases (ii) and (iii),

$$(D_T \cdot \tilde{s}) = ((5s + 7f) \cdot s) - 3 = -1,$$

and in case (iii), $(D_T \cdot R_3) = 0$.

Let $R_1 = \tilde{f}_x, \dots, R_k$ be the components of $\mathcal{C}_x$ listed in (i)–(iii), let A be the Cartan matrix of type A_k, and let $\mathbf{c}$, $\mathbf{d}$, and $\boldsymbol{\nu}$ be the vectors with entries $(C \cdot R_i)$, $(D_T \cdot R_i)$, and the coefficient of R_i in D_T. By Lemma 8.1, $D_T = C + \sum_{i=1}^{k} \nu_i R_i + D'$, where D' is the sum, with their coefficients, of the other Λ_T-trivial components of D_T. The other Λ_T-trivial components of D_T are disjoint from $\mathcal{C}_x$, since $\mathcal{C}_x$ is a connected component of the union of the Λ_T-trivial curves. Thus $\mathbf{d} = \mathbf{c} - \mathsf{A}\boldsymbol{\nu}$. Since A is positive definite, the Cauchy–Schwarz inequality for the form A^{-1} and Lemma 8.1 imply

$$\begin{aligned} \mu = \left(\mathsf{A}^{-1}(\mathbf{c} - \mathbf{d})\right)_1 &\le \left(\mathsf{A}^{-1}(-\mathbf{d})\right)_1 + \sqrt{(\mathsf{A}^{-1})_{11}\, \mathbf{c}^t \mathsf{A}^{-1} \mathbf{c}} \\ &\le \left(\mathsf{A}^{-1}(-\mathbf{d})\right)_1 + \sqrt{\tfrac{13}{4}(\mathsf{A}^{-1})_{11}}. \end{aligned} \tag{8.10}$$

In the cases (i), (ii), and (iii), we have $-\mathbf{d} = (1)$, $(1, 1)$, and $(1, 1, 0)$, and $(\mathsf{A}^{-1})_{11} = \frac{1}{2}$, $\frac{2}{3}$, and $\frac{3}{4}$, respectively [Bou02, Ch. VI, Plate I, item (VI)]. Thus the right-hand side of (8.10) is

$$\frac{1}{2} + \sqrt{\frac{13}{8}} < 2, \qquad 1 + \sqrt{\frac{13}{6}} < 3, \qquad \frac{5}{4} + \sqrt{\frac{39}{16}} < 3.$$

Since μ is an integer, $\mu = 1$ in case (i), in particular if $p_0 \in s$, and $\mu \le 2$ in any case.

Step 3. In this step we prove $S(W; f_x) \le 35/36$.

By **Step 1**, $Z_U \cap f_x$ is supported at p_0, Z_0 is the connected component of Z_U supported at p_0, and Z_1 is the union of the other ones. We bound $\sum_{\ell \ge 1} \dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma}$ by a degeneration of Z_1.

For $\lambda \in \mathbb{C}^*$, let g_λ be the automorphism $(z,u) \to (\lambda^{-1}z, \lambda u)$ of E. It preserves f_x, s, Z_0, and each line $\mathbb{C}z^\ell u^d$, and it acts by $(\zeta, u_\infty) \to (\lambda\zeta, \lambda^2 u_\infty)$ and $(\zeta, v_\infty) \to (\lambda\zeta, \lambda^{-2}v_\infty)$. Let $Z_{1,\lambda} := g_\lambda(Z_1)$. Since Z_1 is disjoint from f_x and f_∞ (Set-up 8.4), it is contained in the union of the charts (ζ, u_∞) and (ζ, v_∞), and $\zeta \neq 0$ on Z_1.

By Construction-Proposition 5.4(1), $\tilde{s}$ passes through at most one center, and through the center p_0 if $p_0 \in s$. Thus exactly one of the following holds, and only (h) and (v) occur if $p_0 \in s$. The labels name the limit $Z_{1,0}$ of $Z_{1,\lambda}$ for $\lambda \to 0$ found below: (h) is a double point at r along the horizontal direction, (v) is a double point at r along the fiber, (r+t) consists of the simple points r and t, and (t_v) is a double point at t along the fiber.

- (h) $Z_1 \cap s = \emptyset$ and Z_1 is not contained in a fiber. Then ζ restricts to a closed embedding of Z_1 into $\mathbb{A}^1$, so the ideal of Z_1 is $(u_\infty - P(\zeta), Q(\zeta))$ with Q monic of degree two and $\deg P \leq 1$, and the ideal of $Z_{1,\lambda}$ is $(u_\infty - \lambda^2 P(\zeta/\lambda), \lambda^2 Q(\zeta/\lambda))$.
- (v) $Z_1 \cap s = \emptyset$ and $Z_1 \subset \{\zeta = \zeta_1\}$. Then the ideal of Z_1 is $(\zeta - \zeta_1, Q(u_\infty))$ with Q monic of degree two, and the ideal of $Z_{1,\lambda}$ is $(\zeta - \lambda\zeta_1, \lambda^4 Q(u_\infty/\lambda^2))$.
- (r+t) Z_1 consists of the reduced points $(\zeta, u_\infty) = (\zeta_1, u_1)$ and $(\zeta, v_\infty) = (\zeta_2, 0)$. The ideals of the points of $Z_{1,\lambda}$ are $(\zeta - \lambda\zeta_1, u_\infty - \lambda^2 u_1)$ and $(\zeta - \lambda\zeta_2, v_\infty)$.
- (t_v) Z_1 is supported at the point $(\zeta, v_\infty) = (\zeta_1, 0)$ of s. Since the second center over $(\zeta_1, 0)$ is not on $\tilde{s}$, the ideal of Z_1 is $(\zeta - \zeta_1 - \kappa v_\infty, v_\infty^2)$ for some $\kappa \in \mathbb{C}$, and the ideal of $Z_{1,\lambda}$ is $(\zeta - \lambda\zeta_1 - \kappa\lambda^3 v_\infty, v_\infty^2)$.

In each case, the ideal of $Z_{1,\lambda}$, or the ideal of each point of $Z_{1,\lambda}$ in case (r+t), is of the form of Lemma 3.5 with coefficients in $\mathbb{C}[\lambda]$. Thus $Z_{1,\lambda}$ extends to $\lambda \in \mathbb{A}^1$, the subschemes $V(I^\gamma_{Z_{1,\lambda}})$ form a flat family over $\mathbb{A}^1$ for any $\gamma \geq 0$, and $Z_{1,0}$ is $V(u_\infty, \zeta^2)$ at r, $V(\zeta, u_\infty^2)$ at r, $\{r,t\}$, and $V(\zeta, v_\infty^2)$ at t in the cases (h), (v), (r+t), and (t_v), respectively. Moreover $Z_{1,\lambda} \cap f_x = \emptyset$ for any $\lambda \in \mathbb{A}^1$. Thus Lemma 8.6(1) implies

$$\dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma} \leq h^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes J_0\big), \qquad J_0 := (I^\gamma_{Z_0} \cap I^\ell_{f_x}) \cdot I^\gamma_{Z_{1,0}}.$$

The ideals I_{Z_0}, I_{f_x}, and $I_{Z_{1,0}}$ are generated by monomials in the coordinates of Set-up 8.4, so the right-hand side is the number of monomials of $H^0(E, \mathcal{O}_E(\alpha s + \beta f))$ in J_0. Hence $\sum_{\ell \geq 1} \dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma}$ is at most the sum of ℓ over the monomials $z^\ell u^d$ (if $p_0 \notin s$), respectively $z^\ell u^{\alpha - d}$ (if $p_0 \in s$), whose local forms in Set-up 8.4 belong to $I^\gamma_{Z_0}$ and $I^\gamma_{Z_{1,0}}$. In terms of ℓ and d, the conditions are $0 \leq d \leq \alpha$, $\ell \geq \max(0, \gamma - d/2)$, $\ell \leq \beta - d$ if $p_0 \notin s$, $\ell \leq \beta - \alpha + d$ if $p_0 \in s$, and

- (h) $\ell \leq \beta + d - 2\gamma$ if $p_0 \notin s$, and $\ell \leq \alpha + \beta - d - 2\gamma$ if $p_0 \in s$;
- (v) $\ell \leq \beta - \gamma - d/2$ if $p_0 \notin s$, and $\ell \leq \beta - \gamma - (\alpha - d)/2$ if $p_0 \in s$;
- (r+t) $\ell \leq \beta - \gamma$ and $\ell \leq \alpha + \beta - \gamma - 2d$;
- (t_v) $\ell \leq \beta - \gamma - d + (\alpha - d)/2$.

For real numbers $0 \leq \alpha \leq \beta$, $\gamma \geq 0$, and $0 \leq \delta \leq \alpha$, let $\ell_-(\delta) := \max(0, \gamma - \delta/2)$ and let $\ell_+(\delta)$ be the minimum of the upper bounds above with $d = \delta$. Let

$$\Psi(\alpha,\beta,\gamma) := \frac{1}{2} \int_{\{0 \leq \delta \leq \alpha:\ \ell_-(\delta) \leq \ell_+(\delta)\}} \big(\ell_+(\delta)^2 - \ell_-(\delta)^2\big) d\delta.$$

Then the sum above is $\Psi(\alpha,\beta,\gamma) + O(\beta^2)$, and Ψ is homogeneous of degree three and piecewise polynomial. By Lemma 8.6(2) with $\vartheta = \Psi$,

$$S(W; f_x) \leq \frac{\mu}{72} + \frac{1}{3}\Big(\frac{11}{24} + \int_0^1 \Psi(1+w, 2+w, w) dw + \frac{\Psi(2,3,1)}{8}\Big). \tag{8.11}$$

Write $\ell_\pm(w,\delta)$ for the functions $\ell_\pm$ at $(\alpha,\beta,\gamma) = (1+w, 2+w, w)$. The functions $\ell_\pm$ are piecewise affine, and a direct calculation of the integrals of $\frac{1}{2}(\ell_+^2 - \ell_-^2)$ over the polygons on which $\ell_\pm$ are affine gives the integrals in (8.11). For example, in case (t_v), where $p_0 \notin s$, we have $\ell_-(w,\delta) = \max(0, w - \delta/2)$ and $\ell_+(w,\delta) = \min(2 + w - \delta, (5 + w - 3\delta)/2)$, the lines $\delta = 2w$ and $\delta = 1 - w$ cut the domain $0 \leq w \leq 1$, $0 \leq \delta \leq 1 + w$ into the triangles with vertices

$\{(0,0),(\frac{1}{3},\frac{2}{3}),(0,1)\}$, $\{(\frac{1}{3},\frac{2}{3}),(1,2),(0,1)\}$, $\{(0,0),(1,0),(\frac{1}{3},\frac{2}{3})\}$, and $\{(1,0),(1,2),(\frac{1}{3},\frac{2}{3})\}$ in the (w,δ)-plane, and the four integrals are $\frac{133}{648}$, $\frac{49}{324}$, $\frac{131}{162}$, and $\frac{71}{81}$, with sum $\frac{49}{24}$. The polygons of all six models, namely the cases (h), (v), (r+t), and (t_v) with $p_0 \notin s$ and the cases (h) and (v) with $p_0 \in s$, are listed in Table 82, and the terms of (8.11) in Table 83.

By **Step 2** and Table 83, (8.11) implies $S(W;f_x) \le \frac{2}{72} + \frac{17}{18} = \frac{35}{36}$.

Step 4. In this step we conclude the proof.

By **Step 1**, $Z_U \cap f_x$ is a curvilinear scheme of length two supported at p_0. Since $\frac{7}{8} < \frac{23}{24} < \frac{3323}{3360}$, Lemma 8.5 implies $S(W^{f_x};a) \le \frac{3323}{3360}$ for any $a \in f_x$. By **Step 3**, $S(W;f_x) \le \frac{35}{36} < \frac{3323}{3360}$. Thus Lemma 8.7 implies the remaining assertions. □

Proposition 8.9. *Notation and conditions as in Set-up 5.1, with L non-special. Let $x \in L$ be a closed point such that $\Sigma(X,x)$ consists of $[L]$ and a curvilinear double point at a line M of width three. Then the coefficient μ of f_x in $D|_E$ satisfies $1 \le \mu \le 2$,*

$$S(W;f_x) \le \frac{271}{288}, \qquad S(W^{f_x};a) \le \frac{3323}{3360} \quad \text{for every } a \in f_x,$$

and any prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies

$$A_X(F) \ge \frac{3360}{3323} S_X(F) + \frac{5798}{9969} \operatorname{mult}_F E.$$

In particular $\delta_x(X) \ge 3360/3323$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe Z_U near f_x.

Let $p_0 \in f_x$ be the point which represents M, and let Z_0 be the connected component of Z_U supported at p_0. By Proposition 5.3, $\widehat{\mathcal{O}}_{Z_0,p_0} \cong \mathbb{C}[\epsilon]/(\epsilon^3)$, the image $\psi(\epsilon)$ of z has order exactly two, and $Z_U \cap f_x = Z_0 \cap f_x$ is a curvilinear scheme of length two supported at p_0. Since the widths of the lines intersecting L sum to 4 (Proposition 5.3), $Z_U = Z_0 \sqcup \{p_1\}$ for a reduced point $p_1 \notin f_x$. By Lemma 8.3, $Z_U \cap s = \emptyset$, so $p_0, p_1 \notin s$. We use the coordinates of Set-up 8.4 for x and p_0. Then p_0 is the origin of the chart (z,u), and p_1 is a point (ζ_1,u_1) of the chart (ζ,u_∞) with $\zeta_1 \ne 0$.

Since $\mathbb{C}[[z,u]] \to \widehat{\mathcal{O}}_{Z_0,p_0}$ is surjective and ψ has order two, the image of u has order one, and we may take it as ϵ. Then $z \to \kappa\epsilon^2$ for some $\kappa \in \mathbb{C}^*$, so the ideal of Z_0 is $(z - \kappa u^2, u^3)$. Since $u \to \kappa^{1/2}u$ is the action of an element of $\mathbb{G}_m^2$, possibly replacing u with $\kappa^{1/2}u$, we may assume that the ideal of Z_0 is $(z-u^2,u^3)$. Thus p_T blows up p_0 and two points infinitely near p_0 in a chain. Write $z = z_1u$ on the blow-up of p_0 and $z_1 = z_2u$ on the next blow-up. The strict transforms of the curve $\{z = u^2\}$ are $\{z_1 = u\}$ and $\{z_2 = 1\}$, so the three centers over p_0 are the origin of (z,u), the origin of (z_1,u), and the point $(z_2,u) = (1,0)$. The strict transforms of $f_x = \{z = 0\}$ are $\{z_1 = 0\}$ and $\{z_2 = 0\}$, which pass through the first two centers and not through the third. Let e_1, e_2, e_3 be the total exceptional classes over p_0 and e_4 the one over p_1. Then the strict transform $\tilde{f}_x$ of f_x is $p_T^*f - e_1 - e_2$ [Har77, Proposition V.3.6], and

$$(\Lambda_T \cdot \tilde{f}_x) = ((2s+3f)\cdot f) - 2 = 0, \qquad (D_T \cdot \tilde{f}_x) = ((5s+7f)\cdot f) - 6 = -1.$$

Step 2. In this step we prove $1 \le \mu \le 2$.

Since $D_T \ge 0$ and $(D_T \cdot \tilde{f}_x) < 0$, the coefficient μ of $\tilde{f}_x$ in D_T is at least one. Let $\mathcal{C}_x$ be the connected component containing $\tilde{f}_x$ of the union of the irreducible curves $R \subset T$ with $(\Lambda_T \cdot R) = 0$. Let $R_1 := \tilde{f}_x$, and let $R_2 \sim e_2 - e_3$ and $R_3 \sim e_1 - e_2$ be the strict transforms of the exceptional curves over the second and the first center. We have

$$(R_2 \cdot R_1) = 1, \qquad (R_3 \cdot R_2) = 1, \qquad (R_3 \cdot R_1) = 0.$$

By Lemma 8.1, any other irreducible curve R with $(\Lambda_T \cdot R) = 0$ is the strict transform of a curve of class s, f, or $s+f$ through one, two, or three centers. No center is on s, the only fiber through two centers is f_x, and the strict transform of an irreducible curve G of class $s+f$ does

not pass through the second center over p_0 since $(G \cdot f) = 1$, so it passes through at most two centers. Thus $\mathcal{C}_x$ is the chain (R_1, R_2, R_3) of type A_3, and the vector of the intersection numbers $(D_T \cdot R_i)$ is $\mathbf{d} = (-1, 0, 0)$, since $((e_i - e_{i+1}) \cdot \sum_k e_k) = 0$. Let A be the Cartan matrix of type A_3, and let $\mathbf{c}$ and $\boldsymbol{\nu}$ be the vectors with entries $(C \cdot R_i)$ and the coefficient of R_i in D_T. By Lemma 8.1, $\mathbf{d} = \mathbf{c} - \mathsf{A}\boldsymbol{\nu}$ and $\mathbf{c}^t \mathsf{A}^{-1} \mathbf{c} \le \frac{13}{4}$, so by the Cauchy–Schwarz inequality for the form A^{-1} and [Bou02, Ch. VI, Plate I, item (VI)],

$$\mu = \big(\mathsf{A}^{-1}(\mathbf{c} - \mathbf{d})\big)_1 \le \big(\mathsf{A}^{-1}(-\mathbf{d})\big)_1 + \sqrt{\tfrac{13}{4}(\mathsf{A}^{-1})_{11}} = \frac{3}{4} + \sqrt{\frac{39}{16}} < 3.$$

Since μ is an integer, $1 \le \mu \le 2$.

Step 3. In this step we prove $S(W; f_x) \le 271/288$.

By **Step 1**, $Z_U \cap f_x$ is supported at p_0, Z_0 is the connected component of Z_U supported at p_0, and the union of the other ones is $\{p_1\}$. For $\lambda \in \mathbb{C}^*$, let g_λ be the automorphism $(z, u) \to (\lambda^{-2} z, \lambda^{-1} u)$ of E. It preserves f_x, s, and the ideal $(z - u^2, u^3)$ of Z_0, and it acts by $(\zeta, u_\infty) \to (\lambda^2 \zeta, \lambda u_\infty)$. Thus the ideal of $g_\lambda(p_1)$ is $(u_\infty - \lambda u_1, \zeta - \lambda^2 \zeta_1)$, which is of the form of Lemma 3.5, and the limit of $g_\lambda(p_1)$ for $\lambda \to 0$ is r. Let $g_0(p_1) := r$. By Lemma 3.5, the subschemes $V(\mathfrak{m}^\gamma_{g_\lambda(p_1)})$, $\lambda \in \mathbb{A}^1$, form a flat family over $\mathbb{A}^1$ for any $\gamma \ge 0$, and they are contained in the chart (ζ, u_∞), which is disjoint from f_x. Thus Lemma 8.6(1), with $Z_{1,\lambda} := g_\lambda(p_1)$, implies

$$\dim \mathcal{F}^\ell V_{\alpha,\beta,\gamma} \le h^0\big(E, \mathcal{O}_E(\alpha s + \beta f) \otimes (I^\gamma_{Z_0} \cap I^\ell_{f_x}) \cdot \mathfrak{m}^\gamma_r\big) = \dim \mathcal{F}^\ell V^0_{\alpha,\beta,\gamma},$$

where $V^0_{\alpha,\beta,\gamma} := H^0(E, \mathcal{O}_E(\alpha s + \beta f) \otimes I^\gamma_{Z_0} \otimes \mathfrak{m}^\gamma_r)$.

Since the one-parameter group $(g_\lambda)_{\lambda \in \mathbb{C}^*}$ preserves Z_0 and r, and acts on $z^\ell u^e$ with weight $-(2\ell + e)$, the space $V^0_{\alpha,\beta,\gamma}$ is the direct sum of the spaces V^0_d of the sections in $V^0_{\alpha,\beta,\gamma}$ that are linear combinations of the monomials $z^\ell u^{d-2\ell}$, and $\mathcal{F}^\ell V^0_{\alpha,\beta,\gamma} = \bigoplus_d \mathcal{F}^\ell V^0_d$. Fix d. By Set-up 8.4, $z^\ell u^{d-2\ell}$ is a section of $\mathcal{O}_E(\alpha s + \beta f)$ if and only if $\ell \ge \max(0, (d-\alpha)/2, d - \beta)$ and $\ell \le d/2$, and it belongs to $\mathfrak{m}^\gamma_r$ if and only if $(\beta - \ell - (d - 2\ell)) + (d - 2\ell) = \beta - \ell \ge \gamma$. Let L_0 and U_0 be the smallest and largest integers ℓ satisfying both conditions. A linear combination $\sigma = \sum_{\ell=L_0}^{U_0} c_\ell z^\ell u^{d-2\ell}$ equals $u^d P(z/u^2)$ with $P(T) := \sum_\ell c_\ell T^\ell$. Let $y := z - u^2$. Then

$$\sigma = u^d P\Big(1 + \frac{y}{u^2}\Big) = \sum_{k \ge 0} \frac{P^{(k)}(1)}{k!} y^k u^{d-2k},$$

and $(y, u^3)^\gamma$ is generated by the monomials $y^k u^e$ with $3k + e \ge 3\gamma$. Thus $\sigma \in I^\gamma_{Z_0}$ if and only if $P^{(k)}(1) = 0$ for $k < 3\gamma - d$, that is, $(T-1)^h$ divides P for $h := \max(0, 3\gamma - d)$. Writing $P = T^{L_0}(T-1)^h P_1$ with $\deg P_1 \le U_0 - L_0 - h$, the order of σ along f_x is L_0 plus the order of P_1 at $T = 0$. Hence

$$\sum_{\ell \ge 1} \dim \mathcal{F}^\ell V^0_d = \sum_{\ell=L_0}^{U_0 - h} \ell.$$

In this proof, for real numbers $0 \le \alpha \le \beta$, $\gamma \ge 0$, and $0 \le \delta \le 2\beta$, let

$$\ell_-(\delta) := \max\Big(0, \frac{\delta - \alpha}{2}, \delta - \beta\Big), \qquad \ell_+(\delta) := \min\Big(\frac{\delta}{2}, \beta - \gamma\Big) - \max(0, 3\gamma - \delta),$$

and let $\Psi(\alpha, \beta, \gamma)$ denote the integral of $\frac{1}{2}(\ell_+^2 - \ell_-^2)$ over $\{0 \le \delta \le 2\beta : \ \ell_-(\delta) \le \ell_+(\delta)\}$. Then $\sum_{\ell \ge 1} \dim \mathcal{F}^\ell V^0_{\alpha,\beta,\gamma} = \Psi(\alpha, \beta, \gamma) + O(\beta^2)$, and Ψ is homogeneous of degree three and piecewise polynomial. By Lemma 8.6(2) with $\vartheta = \Psi$,

$$S(W; f_x) \le \frac{\mu}{72} + \frac{1}{3}\Big(\frac{11}{24} + \int_0^1 \Psi(1 + w, 2 + w, w)\, dw + \frac{\Psi(2, 3, 1)}{8}\Big).$$

At $(\alpha,\beta,\gamma) = (1+w, 2+w, w)$ the functions $\ell_\pm$ are piecewise affine in (w,δ), and a direct calculation of the integrals over the polygons on which they are affine shows that

$$\int_0^1 \Psi(1+w,2+w,w)dw = \frac{95}{48}, \qquad \Psi(2,3,1) = \frac{29}{12}. \tag{8.12}$$

The polygons are listed in Table 85, and the sums of the integrals over them in Table 86. Thus, by **Step 2**,

$$S(W;f_x) \le \frac{2}{72} + \frac{1}{3}\Big(\frac{11}{24} + \frac{95}{48} + \frac{29}{96}\Big) = \frac{8}{288} + \frac{263}{288} = \frac{271}{288}.$$

Step 4. In this step we conclude the proof.

By **Step 1**, $Z_U \cap f_x$ is a curvilinear scheme of length two supported at $p_0 \ne s \cap f_x$. By Lemma 8.5(1), $S(W^{f_x};a) \le \frac{23}{24} < \frac{3323}{3360}$ for any $a \in f_x$. By **Step 3**, $S(W;f_x) \le \frac{271}{288} < \frac{3323}{3360}$. Thus Lemma 8.7 implies the remaining assertions. □

8.3. **Width four.** In this subsection, we construct the weighted extraction used at a point p of incidence type $2+1$ whose special line M has width four. Lemma 8.2 describes the line link of L in this case. For the estimate we use the weighted blow-up of p with weights $(2,3,2)$: in the coordinates of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ the width-four condition has a short normal form, and the associated graded ring of the extraction has twelve explicit generators. In this subsection, E denotes the exceptional divisor of the weighted extraction $\hat\pi : \hat X \to X$ of Construction-Proposition 8.10, not that of $\mathrm{Bl}_L X \to X$, and T denotes the variable of degree one introduced there. The exact description in Construction-Proposition 8.10 is the input of Theorem 8.12.

Construction-Proposition 8.10. Let X be a smooth prime Fano threefold of genus 10 and $p \in X$ a closed point of incidence type $2+1$, with non-special line L and special line M through p, such that the strict transform of M in $\mathrm{Bl}_L X$ has width four. Then there exist regular parameters (x,y,z) at p and a section $\tau \in H^0(X,H)$ with $\tau(p) \ne 0$ such that the monomial valuation with weights $(2,3,2)$ defines a plt extraction $\hat\pi : \hat X \to X$ with exceptional divisor E, $A_X(E) = 7$, and $S_X(E) = 67/12$. The coarse space of E is $\mathbb{P}(1,1,3)$ with coordinates $(x:z:w)$, $w = y^2$, and the different is $\Delta_E = \frac{1}{2}(w=0)$. With T the leading form of τ, $\varphi_1 := x^2z - y^2$, and $\varphi_2 := \varphi_1 - xz^2$, the associated graded ring of $\bigoplus_m H^0(X,mH)$ for the weighted valuation is

$$\mathbb{C}[T, Tx, Ty, Tz, Txz, Tyz, T\varphi_1, Txz^2, Tx\varphi_2, Tz\varphi_1, Ty\varphi_2, Txz\varphi_2] \subset \mathbb{C}[T,x,y,z].$$

Proof. In this proof we write m and j for the coordinates M and J of Set-up 2.14, since M denotes the special line, and we write r for the section degree. The proof is long, so we divide it into several steps.

Step 1. In this step we choose coordinates on $\mathbb{P}^{13}$ adapted to p, L, and M, and we write the width-four condition as conditions on the linear equations of X.

Notation as in Set-up 2.14. By Theorem 2.13, $X = \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \Lambda$ for a linear subspace $\Lambda \subset \mathbb{P}^{13}$ of codimension two. Let $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),p)$ be the scheme of lines in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ through p. Since $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is a homogeneous space of $G^\circ_{\mathcal{W}}$, Lemma 2.15(3) shows that $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),p)$ is a twisted cubic in $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)) \cong \mathbb{P}^4$. We call the linear span of $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),p)$ the *contact* $\mathbb{P}^3$. Since X is an intersection of quadrics (Lemma 2.12), $\Sigma(X,p) \subset \mathbb{P}(T_pX)$ by [KP23, Lemma 5.6]. Since Λ is a linear subspace containing p, a line through p contained in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is contained in X if and only if the line is contained in Λ. Thus $\Sigma(X,p)$ is the scheme-theoretic intersection of $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),p)$ with the plane $\mathbb{P}(T_pX)$. Since p has incidence type $2+1$, $\Sigma(X,p)$ is the union of the reduced point $[L]$ and a curvilinear double point at $[M]$, so $\Sigma(X,p)$ has length three.

We show that $\mathbb{P}(T_pX)$ is the linear span of $\Sigma(X,p)$ in the contact $\mathbb{P}^3$. First, we show that $\Sigma(X,p)$ is not contained in a line. Suppose that $\Sigma(X,p)$ is contained in a line. The line is contained in the contact $\mathbb{P}^3$. The twisted cubic is cut out by quadrics in the contact $\mathbb{P}^3$, and the restriction to the line of any such quadric vanishes on the length-three subscheme $\Sigma(X,p)$

of the line, hence vanishes on the line. Thus the line would be contained in the twisted cubic, which is impossible. If $\mathbb{P}(T_pX)$ is not contained in the contact $\mathbb{P}^3$, then $\mathbb{P}(T_pX)$ intersects the contact $\mathbb{P}^3$ in a line containing $\Sigma(X,p)$, a contradiction. Hence $\mathbb{P}(T_pX)$ is a plane in the contact $\mathbb{P}^3$ containing $\Sigma(X,p)$, and $\mathbb{P}(T_pX)$ is the linear span of $\Sigma(X,p)$.

Let $[h] \in \mathrm{G}_2\,\mathrm{Gr}(2,7)$ be the point where all coordinates except h vanish. By Lemma 2.15(1), on the chart $(h=1)$ the functions (i,a,e,n,g) are free coordinates on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, and the other coordinates are given by (2.1). By Lemma 2.15(3), $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),[h])$ is the twisted cubic $\{[1:\rho:\rho^2:-\rho^3:0] : \rho\in\mathbb{C}\} \cup \{[0:0:0:1:0]\}$ in the coordinates $(i:a:e:n:g)$ of $\mathbb{P}(T_{[h]}\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$, and the contact $\mathbb{P}^3$ at $[h]$ is $(g=0)$.

By Lemma 2.15(4), the group $G^\circ_{\mathcal{W}}$ acts on $\mathbb{P}^{13}$ preserving $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, and the stabilizer of $[h]$ in $G^\circ_{\mathcal{W}}$ acts on the twisted cubic $\Sigma(\mathrm{G}_2\,\mathrm{Gr}(2,7),[h]) \cong \mathbb{P}^1$ through PGL_2, which is transitive on ordered pairs of distinct points of $\mathbb{P}^1$. Since any element of $G^\circ_{\mathcal{W}}$ maps Λ to a linear subspace Λ' with $\mathrm{G}_2\,\mathrm{Gr}(2,7)\cap\Lambda' \cong X$, possibly replacing Λ with such a Λ', we may assume that $p=[h]$, that the double point of $\Sigma(X,p)$ is supported at $[1:0:0:0:0]$, and that $[L]=[0:0:0:1:0]$ in the coordinates $(i:a:e:n:g)$. Then M is the line where all coordinates except h,i vanish, and L is the line where all coordinates except h,n vanish. The tangent line of the twisted cubic at $[1:0:0:0:0]$ is spanned by $[1:0:0:0:0]$ and $[0:1:0:0:0]$, so $\mathbb{P}(T_pX)=(e=g=0)$. In particular, the restrictions of (i,a,n) to X are regular parameters at p.

We write the linear equations of X. Each linear form vanishing on Λ has no h-term, since $p\in\Lambda$. By (2.1), at p we have $dk=-dg$, and $df, dc, db, dd, dj, dl, dm$ vanish on $T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$. Since X is smooth of dimension three at p, the differentials at p of the linear forms vanishing on Λ span the annihilator $\langle de, dg\rangle$ of T_pX in the cotangent space of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ at p. Thus the linear forms vanishing on Λ have no i,a,n-terms, and after row operations $\Lambda=(F_1=F_2=0)$ with

$$\begin{aligned} F_1 &= e+\alpha_1 f+\alpha_2(g+k)+\alpha_3 c+\alpha_4 l+\alpha_5 b+\alpha_6 d+\alpha_7 m+\alpha_8 j,\\ F_2 &= \beta_1 g+\beta_2 k+\beta_3 f+\beta_4 c+\beta_5 l+\beta_6 b+\beta_7 d+\beta_8 m+\beta_9 j, \qquad \beta_1-\beta_2=1, \end{aligned} \tag{8.13}$$

where $\alpha_1,\dots,\alpha_8,\beta_1,\dots,\beta_9\in\mathbb{C}$.

We compute the scheme Inc_L of Proposition 5.3 near the pair (M,p). A point of L near p is $(i,a,e,n,g)=(0,0,0,\eta,0)$ with $\eta\in\mathbb{C}$, and a line near M through $(0,0,0,\eta,0)$ has the form $i=\kappa$, $a=\rho\kappa$, $e=\sigma_1\kappa$, $n=\eta+\sigma_2\kappa$, $g=\sigma_3\kappa$ in the chart $(h=1)$, with parameter κ. By Lemma 2.15(1), on the chart $(h=1)$, $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is the graph of the polynomial map given by (2.1), so the line is contained in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ if and only if the coordinates f,k,c,b,d,j,l,m given by (2.1) are affine linear in κ. The coefficients of κ^2 in f, k, and l are $\sigma_1-\rho^2$, $-\rho\sigma_1-\sigma_2$, and $-(\sigma_3+\eta)$. Thus $\sigma_1=\rho^2$, $\sigma_2=-\rho^3$, $\sigma_3=-\eta$, and then

$$f=k=l=b=d=j=0, \qquad c=\rho\eta\kappa, \qquad m=-\eta^2\kappa.$$

Conversely, the coordinates above satisfy (2.1), so the line is contained in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ if and only if $\sigma_1=\rho^2$, $\sigma_2=-\rho^3$, and $\sigma_3=-\eta$. Hence near (M,p) the scheme of pairs (M',p'), with M' a line in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ and $p'\in M'\cap L$, is smooth with coordinates (ρ,η). Substituting into (8.13), the line M' is contained in X if and only if

$$\rho^2-\alpha_2\eta+\alpha_3\rho\eta-\alpha_7\eta^2=0, \qquad \eta(-\beta_1+\beta_4\rho-\beta_8\eta)=0. \tag{8.14}$$

Thus (8.14) defines Inc_L near (M,p). By Proposition 5.3, the completed local ring of Inc_L at (M,p) is $\mathbb{C}[t]/(t^4)$, since the strict transform of M in $\mathrm{Bl}_L X$ has width four.

If $\beta_1\neq 0$, then the second equation of (8.14) implies $\eta=0$, and the local ring is $\mathbb{C}[[\rho]]/(\rho^2)$, of length two. Thus $\beta_1=0$. If $\alpha_2=0$, then both equations of (8.14) belong to $(\rho,\eta)^2$, so the local ring has embedding dimension two and is not curvilinear. Thus $\alpha_2\neq 0$, and by the implicit function theorem the first equation is equivalent to $\eta=\eta(\rho)$ for a power series $\eta(\rho)=\rho^2/\alpha_2+O(\rho^3)$. The local ring is then $\mathbb{C}[[\rho]]/(\eta(\rho)(\beta_4\rho-\beta_8\eta(\rho)))$. The order of $\eta(\rho)(\beta_4\rho-\beta_8\eta(\rho))$ is 3 if $\beta_4\neq 0$, and 4 if $\beta_4=0\neq\beta_8$, and the series is zero if $\beta_4=\beta_8=0$. Hence

$$\beta_1=\beta_4=0, \qquad \beta_2=-1, \qquad \alpha_2\beta_8\neq 0. \tag{8.15}$$

We show that $\beta_3 \neq 0$. By Lemma 2.15(2), on the chart $(i = 1)$ the functions (a, f, g, h, l) are free coordinates on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, and the other coordinates are given by (2.2). On the chart $(i = 1)$, M is the line $(a = f = g = l = 0)$ with coordinate h. At the point $p_h \in M$ with coordinate h, the differentials of b, d, j, m vanish and $dk = h\,dl$. By (8.13) and (8.15),

$$dF_2|_{T_{p_h}\,\mathrm{G}_2\,\mathrm{Gr}(2,7)} = \beta_3 df + (\beta_5 - h)dl.$$

If $\beta_3 = 0$, then dF_2 vanishes on $T_{p_h}\,\mathrm{G}_2\,\mathrm{Gr}(2,7)$ at the point $p_h \in M$ with $h = \beta_5$. Since X is the scheme-theoretic intersection of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ with $(F_1 = F_2 = 0)$, this implies $\dim T_{p_h} X \geq 4$, contradicting the smoothness of X. Thus $\beta_3 \neq 0$.

Step 2. In this step we define (x, y, z), τ, and the extraction $\hat{\pi}$, and we compute $A_X(E)$, the coarse space of E, and Δ_E.

Let $\nu_0 := \beta_8/\beta_3 \neq 0$, and let $\tau \in H^0(X, H)$ be the restriction of the coordinate h, so that $\tau(p) \neq 0$. Let

$$x := \frac{i}{\nu_0 h}\Big|_X, \qquad y := \frac{a}{\alpha_2 \nu_0 h}\Big|_X, \qquad z := \frac{n}{\alpha_2 h}\Big|_X.$$

By **Step 1**, (x, y, z) are regular parameters at p, and $\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x, y, z]]$. On the chart $(h = 1)$,

$$i = \nu_0 x, \qquad a = \alpha_2 \nu_0 y, \qquad n = \alpha_2 z. \tag{8.16}$$

Let $\hat{\pi} : \hat{X} \to X$ be the weighted blow-up of p with weights $(2, 3, 2)$ on (x, y, z), and let E be the exceptional divisor. Since $\gcd(2, 3, 2) = 1$, ord_E is the monomial valuation with weights $(2, 3, 2)$, and $A_X(E) = 2 + 3 + 2 = 7$ [JM12, Proposition 5.1].

The divisor E is $\operatorname{Proj} \mathbb{C}[x, y, z]$ with weights $(2, 3, 2)$. The degree-two Veronese subring of $\mathbb{C}[x, y, z]$ is $\mathbb{C}[x, z, w]$ with $w = y^2$, of degrees $(2, 2, 6)$. Thus the coarse space of E is $\mathbb{P}(1, 1, 3)$ with coordinates $(x : z : w)$. The variety $\hat{X}$ is covered near E by three charts. The charts over $x \neq 0$ and $z \neq 0$ are $\mathbb{A}^3/\mu_2$ with weights $(1, 1, 0)$, where E is the image of the first coordinate hyperplane, and the chart over $y \neq 0$ is $\mathbb{A}^3/\mu_3$ with weights $(1, 1, 1)$. Thus $\hat{X}$ has a transversal A_1 singularity along the curve $C_w = (w = 0) \subset E$, the only other singular point of $\hat{X}$ is the point $o_w = [0 : 0 : 1]$ of E, and $\hat{X}$ is $\mathbb{Q}$-factorial. By [Kol+92, Proposition 16.6],

$$\Delta_E = \mathrm{Diff}_E(0) = \tfrac{1}{2} C_w = \tfrac{1}{2}(w = 0).$$

The surface E has quotient singularities, and C_w is a smooth rational curve not containing the unique singular point o_w of E. Thus (E, Δ_E) is klt. By [Kol13, Theorem 4.9(1)], $(\hat{X}, E)$ is plt near E. Since $\hat{\pi}$ is an isomorphism over $X \backslash \{p\}$, $(\hat{X}, E)$ is plt, and $\hat{\pi}$ is a plt extraction.

Step 3. In this step we compute the leading forms of the coordinates of $\mathbb{P}^{13}$ on X at p.

For a non-zero $\varphi \in \mathbb{C}[[x, y, z]]$, let $\overline{\varphi}$ be the leading form of φ, that is, the non-zero weighted-homogeneous component of φ of lowest weight, where x, y, z have weights $2, 3, 2$. The weight of $\overline{\varphi}$ is $\mathrm{ord}_E(\varphi)$. By (8.16), $\mathrm{ord}_E(i) = \mathrm{ord}_E(n) = 2$ and $\mathrm{ord}_E(a) = 3$. By the implicit function theorem, on X near p the coordinates e and g are power series in (i, a, n). By (8.13), (8.15), and the relation $g + k = -ae - in$ of (2.1), the equations $F_1 = F_2 = 0$ read

$$\begin{aligned} e &= \alpha_2(ae + in) - \alpha_1 f - \alpha_3 c - \alpha_4 l - \alpha_5 b - \alpha_6 d - \alpha_7 m - \alpha_8 j, \\ k &= \beta_3 f + \beta_5 l + \beta_6 b + \beta_7 d + \beta_8 m + \beta_9 j, \qquad g = -k - ae - in. \end{aligned} \tag{8.17}$$

We show that

$$\begin{gathered} \overline{e} = \alpha_2 in, \quad \overline{g} = -in, \quad \overline{c} = an, \quad \overline{f} = \alpha_2 i^2 n - a^2, \quad \overline{m} = -in^2, \quad \overline{k} = \beta_3 \overline{f} - \beta_8 in^2, \\ \overline{l} = i\overline{k}, \quad \overline{d} = -n\overline{f}, \quad \overline{b} = a\overline{k}, \quad \overline{j} = -in\overline{k}. \end{gathered} \tag{8.18}$$

Since de and dg vanish on $T_p X$, we have $\mathrm{ord}_E(e) \geq 4$ and $\mathrm{ord}_E(g) \geq 4$. By (2.1), $\mathrm{ord}_E(f) \geq 6$, $\mathrm{ord}_E(c) \geq 5$, $\mathrm{ord}_E(k) \geq 4$, $\mathrm{ord}_E(l) \geq 6$, $\mathrm{ord}_E(b) \geq 7$, $\mathrm{ord}_E(d) \geq 8$, $\mathrm{ord}_E(j) \geq 8$, and $\mathrm{ord}_E(m) \geq 6$. By (8.17), $\mathrm{ord}_E(e - \alpha_2 in) \geq 5$ and $\mathrm{ord}_E(k) \geq 6$, so $\mathrm{ord}_E(g + in) \geq 6$. By (2.1) again,

$$\mathrm{ord}_E(f - \alpha_2 i^2 n + a^2) \geq 7, \quad \mathrm{ord}_E(c - an) \geq 8, \quad \mathrm{ord}_E(m + in^2) \geq 8, \quad \mathrm{ord}_E(l) \geq 8.$$

By (8.17), $\mathrm{ord}_E(k-\beta_3(\alpha_2 i^2 n-a^2)+\beta_8 i n^2)\geq 7$. The polynomial $\beta_3(\alpha_2 i^2 n-a^2)-\beta_8 i n^2$ is non-zero since $\beta_3\neq 0$, so $\overline{k}$ is as in (8.18). Finally, by (2.1),

$$\mathrm{ord}_E(l-i\overline{k})\geq 9,\quad \mathrm{ord}_E(d+n\overline{f})\geq 9,\quad \mathrm{ord}_E(b-a\overline{k})\geq 10,\quad \mathrm{ord}_E(j+in\overline{k})\geq 11,$$

using $l=-af+ik$, $j=fc+gk$, and the identities $d=-nf-ek$ and $b=-ef+ak$, which follow from (2.1). This proves (8.18).

By (8.16) and $\nu_0=\beta_8/\beta_3$,

$$\alpha_2 i^2 n-a^2=\alpha_2^2\nu_0^2\varphi_1,\qquad \beta_3(\alpha_2 i^2 n-a^2)-\beta_8 i n^2=\alpha_2^2\nu_0^2\beta_3\varphi_2.$$

Hence, by (8.18), the leading forms of the restrictions to X of

$$h/h,\ i/h,\ a/h,\ n/h,\ g/h,\ c/h,\ f/h,\ m/h,\ l/h,\ d/h,\ b/h,\ j/h$$

are non-zero constant multiples of

$$1,\ x,\ y,\ z,\ xz,\ yz,\ \varphi_1,\ xz^2,\ x\varphi_2,\ z\varphi_1,\ y\varphi_2,\ xz\varphi_2, \tag{8.19}$$

of weights $0,2,3,2,4,5,6,6,8,8,9,10$, respectively.

Step 4. In this step we prove the description of the associated graded ring.

For $\vartheta\in H^0(X,rH)\backslash\{0\}$, $\mathrm{ord}_E(\vartheta)=\mathrm{ord}_E(\vartheta/\tau^r)$ since $\tau(p)\neq 0$. Let $\mathrm{gr}_r\subset T^r\mathbb{C}[x,y,z]$ be the span of the elements $T^r\overline{\vartheta/\tau^r}$, $\vartheta\in H^0(X,rH)\backslash\{0\}$. The weight-$\lambda$ part of gr_r is the image of the injective map from the λ-th graded piece of the filtration of $H^0(X,rH)$ by ord_E, given by $\vartheta\to T^r\overline{\vartheta/\tau^r}$. Thus $\bigoplus_r\mathrm{gr}_r$ is the associated graded ring of $\bigoplus_r H^0(X,rH)$ for ord_E, the leading form of τ is T, and

$$\dim\mathrm{gr}_r=N_r=3r^3+\tfrac{9}{2}r^2+\tfrac{7}{2}r+1$$

by Lemma 2.12. By **Step 3**, gr_1 contains T times each element of (8.19). Let $\mathrm{gr}'_r\subset\mathrm{gr}_r$ be the span of the products of r elements of the form $T\psi$ with ψ in (8.19). It suffices to show $\dim\mathrm{gr}'_r\geq N_r$ for any $r\geq 0$.

For a non-zero $\varphi\in\mathbb{C}[x,y,z]$, let $\epsilon(\varphi)\in\mathbb{Z}^3_{\geq 0}$ be the exponent of the lexicographically smallest monomial of φ, where (q_1,q_2,q_3) is the exponent of $x^{q_1}y^{q_2}z^{q_3}$, and let $\epsilon(T^r\varphi):=\epsilon(\varphi)$. Then $\epsilon(\varphi\psi)=\epsilon(\varphi)+\epsilon(\psi)$, and for any subspace $V\subset\mathbb{C}[x,y,z]$ with $\dim V<+\infty$, $\#\epsilon(V\backslash\{0\})=\dim V$. The lexicographically smallest monomial of both φ_1 and φ_2 is y^2. Thus the exponents ϵ of the elements of (8.19) are

$$\begin{aligned}&(0,0,0),\ (1,0,0),\ (0,1,0),\ (0,0,1),\ (1,0,1),\ (0,1,1),\\ &(0,2,0),\ (1,0,2),\ (1,2,0),\ (0,2,1),\ (0,3,0),\ (1,2,1).\end{aligned} \tag{8.20}$$

Let $\Xi_r\subset\mathbb{Z}^3_{\geq 0}$ be the set of sums of r elements of (8.20), with repetitions allowed. Then $\Xi_r\subset\epsilon(\mathrm{gr}'_r\backslash\{0\})$, so $\dim\mathrm{gr}'_r\geq\#\Xi_r$. For $0\leq q_1\leq r$ and $0\leq q_2\leq 3r-q_1$, let

$$d_{\max}(q_1,q_2):=\min\{r+q_1,\ 2r-\lceil q_2/2\rceil,\ 3r-q_2\}.$$

We show that Ξ_r contains the set

$$\begin{aligned}\Xi'_r:=\{(q_1,q_2,q_3)\in\mathbb{Z}^3_{\geq 0}:\ &q_1\leq r,\ q_2\leq 3r-q_1,\ q_3\leq d_{\max}(q_1,q_2),\\ &q_2\in 2\mathbb{Z}\text{ if }q_1=r\}.\end{aligned} \tag{8.21}$$

Write a sum of r elements of (8.20) with first coordinate q_1 as a sum of q_1 elements with first coordinate 1 and $r-q_1$ elements with first coordinate 0. Among the five elements of (8.20) with first coordinate 1, the second coordinate is 0 with third coordinate in $\{0,1,2\}$, or 2 with third coordinate in $\{0,1\}$. Thus the sums of q_1 of them with exactly ℓ summands of second coordinate 2 are the triples $(q_1,2\ell,q'_3)$ with $0\leq q'_3\leq 2q_1-\ell$. Among the seven elements with first coordinate 0, the second coordinate is any integer in $[0,3]$, and the third coordinate is 0 or 1 when the second coordinate is at most 2, and 0 otherwise. Thus the sums of $r-q_1$ of them are the triples $(0,q''_2,q''_3)$ with $0\leq q''_2\leq 3(r-q_1)$ and $0\leq q''_3\leq\min\{r-q_1,3(r-q_1)-q''_2\}$. Hence Ξ_r contains

$(q_1, 2\ell + q_2'', q_3)$ for any $0 \le q_3 \le 2q_1 - \ell + \min\{r - q_1, 3(r - q_1) - q_2''\}$. Let $(q_1, q_2, q_3) \in \Xi_r'$. If $q_1 = r$, take $\ell = q_2/2$ and $q_2'' = 0$. If $q_1 < r$, take

$$\ell = \begin{cases} 0 & \text{if } q_2 \le 2(r - q_1), \\ \lceil q_2/2 \rceil - (r - q_1) & \text{if } 2(r - q_1) \le q_2 \le 2r, \\ q_1 & \text{if } 2r \le q_2 \le 3r - q_1, \end{cases}$$

and $q_2'' = q_2 - 2\ell$. In each case $0 \le \ell \le q_1$, $0 \le q_2'' \le 3(r - q_1)$, and $2q_1 - \ell + \min\{r - q_1, 3(r - q_1) - q_2''\} = d_{\max}(q_1, q_2)$. Thus $(q_1, q_2, q_3) \in \Xi_r$, which proves $\Xi_r' \subset \Xi_r$.

We count Ξ_r'. For $0 \le q_1 < r$, a direct summation over q_2 shows

$$\sum_{q_2=0}^{3r-q_1} \big(d_{\max}(q_1, q_2) + 1\big) = 3r^2 + 3r + 1 + r(r - q_1) + \frac{(r - q_1) - 3(r - q_1)^2}{2},$$

and the sum of the right side over $0 \le q_1 \le r - 1$ is $3r^3 + 3r^2 + r$. For $q_1 = r$, the number of points is $\sum_{\ell=0}^{r}(2r - \ell + 1) = (r + 1)(3r + 2)/2$. Hence

$$\#\Xi_r' = 3r^3 + 3r^2 + r + \frac{(r + 1)(3r + 2)}{2} = 3r^3 + \tfrac{9}{2}r^2 + \tfrac{7}{2}r + 1 = N_r.$$

Therefore

$$N_r = \#\Xi_r' \le \#\Xi_r \le \dim \mathrm{gr}_r' \le \dim \mathrm{gr}_r = N_r.$$

Thus $\mathrm{gr}_r' = \mathrm{gr}_r$ for any $r \ge 0$, which is the description of the associated graded ring in the statement. Moreover,

$$\Xi_r = \Xi_r' = \epsilon(\mathrm{gr}_r \setminus \{0\}). \tag{8.22}$$

Step 5. In this step we compute $S_X(E)$.

By Definition 2.8, the refinement $W_{r,\lambda}$ of $(H^0(X, rH))_r$ by E is identified with the weight-λ part of gr_r, so that $\mathrm{gr}_r = \bigoplus_\lambda W_{r,\lambda}$ and $\dim W_{r,\lambda}$ is the multiplicity of λ as a jumping number of the filtration of $H^0(X, rH)$ by ord_E. Any $q \in \epsilon(W_{r,\lambda} \setminus \{0\})$ satisfies $2q_1 + 3q_2 + 2q_3 = \lambda$. Thus the sets $\epsilon(W_{r,\lambda} \setminus \{0\})$ are pairwise disjoint, and since the sum of their cardinalities is $\dim \mathrm{gr}_r = \#\epsilon(\mathrm{gr}_r \setminus \{0\})$, the set $\epsilon(\mathrm{gr}_r \setminus \{0\})$ is the union of the sets $\epsilon(W_{r,\lambda} \setminus \{0\})$. By (8.22), the sum of the jumping numbers of the filtration of $H^0(X, rH)$ by ord_E is $\sum_{q \in \Xi_r}(2q_1 + 3q_2 + 2q_3)$. Hence, by [AZ22, Section 2.2],

$$S_r(E) = \frac{1}{rN_r} \sum_{q \in \Xi_r} (2q_1 + 3q_2 + 2q_3).$$

Let $\Xi \subset \mathbb{Z}_{\ge 0} \times \mathbb{Z}^3$ be the semigroup generated by $(1, q)$ for q in (8.20). Then $\Xi_r = \{q : (r, q) \in \Xi\}$, $\Xi \cap (\{0\} \times \mathbb{Z}^3) = \{0\}$, and Ξ generates $\mathbb{Z}^4$, since $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, and $(1, 0, 0, 1)$ belong to Ξ. By (8.21), the polytope of Lemma 3.4 for Ξ is

$$\{(t_1, t_2, t_3) : 0 \le t_1 \le 1,\ 0 \le t_2 \le 3 - t_1,\ 0 \le t_3 \le \min\{1 + t_1,\ 2 - t_2/2,\ 3 - t_2\}\}.$$

The minimum is $1 + t_1$ if $t_2 \le 2 - 2t_1$, $2 - t_2/2$ if $2 - 2t_1 \le t_2 \le 2$, and $3 - t_2$ if $2 \le t_2 \le 3 - t_1$. Hence the volume of the polytope is

$$\int_0^1 \Big(2(1 - t_1^2) + t_1(2 + t_1) + \tfrac{1}{2}(1 - t_1^2)\Big) dt_1 = 3,$$

which is the leading coefficient of N_r, and

$$\begin{aligned}\int(2t_1+3t_2+2t_3)dt &= \int_0^1\Big(\int_0^{2-2t_1}\int_0^{1+t_1}+\int_{2-2t_1}^{2}\int_0^{2-t_2/2}\\ &\qquad+\int_2^{3-t_1}\int_0^{3-t_2}\Big)(2t_1+3t_2+2t_3)dt_3dt_2dt_1\\ &=\int_0^1\Big(8(1-t_1^2)+\tfrac{2}{3}t_1(21+9t_1-2t_1^2)\\ &\qquad+\tfrac{1}{6}(1-t_1)(23+29t_1+2t_1^2)\Big)dt_1=\frac{67}{4}.\end{aligned}$$

By Lemma 3.4 applied to the linear function $(r,q)\to 2q_1+3q_2+2q_3$, $\sum_{q\in\Xi_r}(2q_1+3q_2+2q_3)=\frac{67}{4}r^4+O(r^3)$, and $N_r=3r^3+O(r^2)$. Thus $S_r(E)=\frac{67}{12}+O(1/r)$, and

$$S_X(E)=\lim_{r\to+\infty}S_r(E)=\frac{67}{12}$$

by [AZ22, Lemma 2.9]. □

We fix notation for the curves and points on E and for a degeneration of the series W.

Set-up 8.11. Notation as in Construction-Proposition 8.10, and let W be the refinement of $(H^0(X,mH))_m$ by E (Definition 2.8).

(1) On E with coordinates $(x:z:w)$, let $C_x=(x=0)$, $C_z=(z=0)$, $C_w=(w=0)$, and $C_\theta=(w-\theta x^2z=0)$ for $\theta\in\mathbb{C}^*$. Thus $\Delta_E=\frac{1}{2}C_w$.
(2) Let $o_x=[1:0:0]$, $o_z=[0:1:0]$, and $o_w=[0:0:1]$.
(3) Let $Q:=w/(x^2z)$, a rational function on E. Thus C_θ is the closure of $(Q=\theta)$.
(4) Let $\mathbb{C}[x,y,z]_j$ be the space of polynomials of weighted degree j for the weights $(2,3,2)$. We regard $W_{m,j}$ as the space of $f\in\mathbb{C}[x,y,z]_j$ such that T^mf belongs to the ring of Construction-Proposition 8.10.
(5) For $f\in\mathbb{C}[x,y,z]_j$, $\operatorname{div}_E(f)\ge 0$ denotes the $\mathbb{Q}$-divisor of f on E. Thus $\operatorname{div}_E(fg)=\operatorname{div}_E(f)+\operatorname{div}_E(g)$, $\operatorname{div}_E(x)=C_x$, $\operatorname{div}_E(z)=C_z$, $\operatorname{div}_E(y)=\frac{1}{2}C_w$, and $\operatorname{div}_E(y^2-\theta x^2z)=C_\theta$.
(6) For a prime divisor v over E, let $v(f):=\operatorname{mult}_v\operatorname{div}_E(f)$.
(7) Let $\mathbb{G}_m^2$ act on $\mathbb{C}[x,y,z]$ with characters $\chi(x)=(1,0)$, $\chi(y)=(1,1)$, and $\chi(z)=(0,2)$.
(8) Let ω be the weight with $\omega(x)=0$, $\omega(y)=1$, and $\omega(z)=2$. A monomial of character (u_1,u_2) has weighted degree $2u_1+u_2$ and ω-weight u_2.
(9) For a non-zero polynomial f, let $\omega_0(f)$ be the least ω-weight of a monomial of f, and let $\operatorname{in}_\omega f$ be the sum of the terms of f of ω-weight $\omega_0(f)$.
(10) Let $W'_{m,j}\subset\mathbb{C}[x,y,z]_j$ be the span of $\{\operatorname{in}_\omega f : f\in W_{m,j}\backslash\{0\}\}$, and let $W'=\bigoplus_{m,j}W'_{m,j}$.
(11) Let $\lambda:\mathbb{G}_m\to\operatorname{Aut}(E)$ be the one-parameter subgroup whose action on $\mathbb{C}[x,y,z]$ is $f(x,y,z)\to f(x,ty,t^2z)$.
(12) Let $h:E'\to E$ be the weighted blow-up of o_z with weights $(1,2)$ for the local coordinates (x,w) of the chart $(z=1)$, and let Θ be the exceptional curve of h.
(13) The restriction of h^*Q to Θ is a coordinate on $\Theta\cong\mathbb{P}^1$, also denoted by Q.
(14) Let $\gamma_-(u_1):=\max\{0,(u_1-1)/2\}$, $\gamma_+(u_1,u_2):=\min\{u_1/2,u_2/2,1+u_1-u_2/2,3-u_2/2\}$, and

$$\mathcal{P}:=\{(u_1,u_2,\gamma)\in\mathbb{R}^3:\gamma_-(u_1)\le\gamma\le\gamma_+(u_1,u_2),\ u_1\le 3,\ u_2\le 4\}.$$

In Theorem 8.12 the different $\Delta_E=\frac{1}{2}C_w$ is non-zero. We degenerate W by the one-parameter subgroup λ, which preserves C_w, and use Lemma 3.6, which allows a boundary invariant under the one-parameter subgroups.

Theorem 8.12. *Notation as in Construction-Proposition 8.10, and let W be the refinement of $(H^0(X,mH))_m$ by E. Then*

$$S(W;v) \le \frac{71}{72} A_{E,\Delta_E}(v)$$

for any prime divisor v over E. Hence any prime divisor F over X with $\operatorname{center}_X F = \{p\}$ satisfies

$$S_X(F) \le \frac{71}{72} A_X(F) - \frac{95}{72} \operatorname{mult}_F E,$$

and in particular $\beta_X(F) > 0$.

Proof. The proof is long, so we divide it into several steps. We use the notation of Set-up 8.11.

Step 1. In this step we define sums of jumps, construct a subalgebra $R_\bullet \subset W'$ and a space $\widetilde{W}_m$ of polynomials containing R_m, and show that the values of s_m and t_m (defined below) on W' and on $\widetilde{W}$ differ by $O(1/m)$.

For a filtration $\mathcal{F}$ of a vector space V with $\dim V < +\infty$, indexed by $\mathbb{R}$, let

$$\sigma(V;\mathcal{F}) := \sum_{t\in\mathbb{R}} t \dim\left(\mathcal{F}^t V/\mathcal{F}^{>t}V\right)$$

be the sum of the jumps of $\mathcal{F}$. For a subspace $V \subset \mathbb{C}[x,y,z]_j$ and a prime divisor v over E, we write $\sigma(V;v)$ for σ of the filtration $\mathcal{F}_v^t V = \{f \in V : v(f) \ge t\}$. Then

$$S_m(W;v) = \frac{1}{mN_m} \sum_j \sigma(W_{m,j};v)$$

by Definition 2.8, and we let $S_m(W';v) := \frac{1}{mN_m}\sum_j \sigma(W'_{m,j};v)$. Let C be one of C_x, C_z, C_w, C_θ ($\theta \in \mathbb{C}^*$), or Θ. For $f \in \mathbb{C}[x,y,z]_j$ with $\operatorname{ord}_C(f) = k$, let $r_C(f) \ge 0$ be the $\mathbb{Q}$-divisor

$$r_C(f) := \left(\operatorname{div}_E(f) - kC\right)|_C \quad (C \subset E), \qquad r_\Theta(f) := \left(h^* \operatorname{div}_E(f) - k\Theta\right)|_\Theta.$$

Let $V \subset \mathbb{C}[x,y,z]_j$ be a subspace, $k \ge 0$, and $q \in C$ a closed point. The multiplicity $\operatorname{mult}_q r_C(f)$ depends only on the class of f in $\operatorname{gr}^k_C V := \mathcal{F}^k_{\operatorname{ord}_C} V/\mathcal{F}^{>k}_{\operatorname{ord}_C} V$, and it defines a filtration of $\operatorname{gr}^k_C V$. Let $\sigma(V;C,q)$ be the sum, over $k \ge 0$, of the sums of the jumps of the filtrations of $\operatorname{gr}^k_C V$. For $V_\bullet \in \{W', \widetilde{W}\}$, with $\widetilde{W}$ defined below, let

$$s_m(V;C) := \frac{1}{mN_m}\sum_j \sigma(V_{m,j};\operatorname{ord}_C),$$

$$t_m(V;C,q) := \frac{1}{mN_m}\sum_j \sigma(V_{m,j};C,q).$$

Thus $s_m(W';C) = S_m(W';C)$ for $C \subset E$.

Since ω is additive on products of monomials, $\operatorname{in}_\omega(fg) = \operatorname{in}_\omega(f)\operatorname{in}_\omega(g)$. Thus W' is a graded algebra, and $W'_m := \bigoplus_j W'_{m,j}$. The space $W'_{m,j}$ is isomorphic to the graded space of the filtration of $W_{m,j}$ by ω_0, so $\dim W'_{m,j} = \dim W_{m,j}$. Since the $W_{m,j}$ are the graded pieces of the filtration of $H^0(X,mH)$ by ord_E,

$$\sum_j \dim W'_{m,j} = N_m = 3m^3 + \frac{9}{2}m^2 + \frac{7}{2}m + 1$$

by Lemma 2.12(2). A polynomial $\operatorname{in}_\omega f$ with $f \in W_{m,j}$ has weighted degree j and a single ω-weight, hence a single character. Thus $W'_{m,j} = \bigoplus_{2u_1+u_2=j} W'_{m,(u_1,u_2)}$, where $W'_{m,(u_1,u_2)}$ is the subspace of polynomials of character (u_1,u_2).

Since $\operatorname{in}_\omega(\varphi_2) = \varphi_1$, the lowest ω-parts of the twelve generators of the ring of Construction-Proposition 8.10, divided by T, are

$$1,\ x,\ y,\ z,\ xz,\ yz,\ \varphi_1,\ xz^2,\ x\varphi_1,\ z\varphi_1,\ y\varphi_1,\ xz\varphi_1 \in W'_1. \tag{8.23}$$

Let $R_\bullet = \bigoplus_m R_m \subset W'$ be the graded subalgebra generated by the polynomials (8.23). Then

$$\varphi_1^3 = (x\varphi_1)(xz\varphi_1) - (y\varphi_1)^2 \in R_2.$$

For integers u_1, u_2, let $\epsilon \in \{0,1\}$ with $\epsilon \equiv u_2 \pmod 2$, and let $\mu_{u_1,u_2} := x^{u_1-\epsilon}y^\epsilon z^{(u_2-\epsilon)/2}$. Since

$$\mu_{u_1,u_2}Q^i = x^{u_1-\epsilon-2i}y^{\epsilon+2i}z^{(u_2-\epsilon)/2-i},$$

any polynomial of character (u_1,u_2) is $\mu_{u_1,u_2}P(Q)$ for a polynomial P. Since $\varphi_1 = x^2z(1-Q)$, the largest integer γ with $\varphi_1^\gamma \mid \mu_{u_1,u_2}P(Q)$ is $\operatorname{ord}_{Q=1}P$. For such a non-zero $f = \mu_{u_1,u_2}P(Q)$, let $\ell(f) := (u_1,u_2,\operatorname{ord}_{Q=1}P)$. Then $\ell(fg) = \ell(f)+\ell(g)$, and the dimension of a space of polynomials of character (u_1,u_2) equals the number of values of $\operatorname{ord}_{Q=1}$ on the non-zero elements of the space. The values of ℓ on the polynomials (8.23), in the order listed there, are

$$\begin{aligned}&(0,0,0),\ (1,0,0),\ (1,1,0),\ (0,2,0),\ (1,2,0),\ (1,3,0),\\ &(2,2,1),\ (1,4,0),\ (3,2,1),\ (2,4,1),\ (3,3,1),\ (3,4,1),\end{aligned} \tag{8.24}$$

and $\ell(\varphi_1^3) = (6,6,3)$. Let $\Gamma \subset \mathbb{Z}_{\ge0}\times\mathbb{Z}^3$ be the semigroup generated by the vectors $(1,\ell(g))$ for g in (8.23) and by $(2,6,6,3)$. Then $\Gamma\cap(\{0\}\times\mathbb{Z}^3) = \{0\}$, and Γ generates $\mathbb{Z}^4$ because it contains $(1,0,0,0)$, $(1,1,0,0)$, $(1,1,1,0)$, and $(1,2,2,1)$. The closure of $\bigcup_m \frac{1}{m}\Gamma_m$ is the convex hull of the points (8.24) and $(3,3,\frac{3}{2})$ (Lemma 3.4). A direct calculation shows that $\mathcal{P}$ is the convex hull of

$$(0,0,0),\ (1,0,0),\ (0,2,0),\ (1,4,0),\ (2,4,1),\ (3,2,1),\ (3,4,1),\ (3,3,\tfrac{3}{2}),$$

which occur among the points (8.24) and $(3,3,\frac{3}{2})$, and that the points (8.24) satisfy the inequalities defining $\mathcal{P}$. Thus the closure of $\bigcup_m \frac{1}{m}\Gamma_m$ is $\mathcal{P}$. Decomposing $\mathcal{P}$ into the three regions

$$\begin{aligned}&0\le\gamma\le1,\quad 2\gamma\le u_1\le1+\gamma,\quad 2\gamma\le u_2\le2+2u_1-2\gamma,\\ &0\le\gamma\le1,\quad 1+\gamma\le u_1\le1+2\gamma,\quad 2\gamma\le u_2\le4,\\ &1\le\gamma\le\tfrac{3}{2},\quad 2\gamma\le u_1\le3,\quad 2\gamma\le u_2\le6-2\gamma,\end{aligned}$$

a direct calculation yields (Appendix A)

$$\int_{\mathcal{P}}1 = 3,\quad \int_{\mathcal{P}}u_1 = \frac{115}{24},\quad \int_{\mathcal{P}}u_2 = \frac{43}{6},\quad \int_{\mathcal{P}}\gamma = \frac{13}{8},\quad \int_{\mathcal{P}}\gamma_-(u_1) = 1. \tag{8.25}$$

By Lemma 3.4, $\#\Gamma_m = \operatorname{vol}(\mathcal{P})m^3 + O(m^2) = 3m^3 + O(m^2)$. Any element of Γ_m is $\ell(f)$ for a product $f \in R_m$ of polynomials (8.23) and of φ_1^3. Since $R_\bullet$ is generated by polynomials with a single character, R_m is the direct sum of the subspaces $R_{m,(u_1,u_2)}$ of polynomials of character (u_1,u_2), and so

$$\dim R_m \ge \#\Gamma_m = 3m^3 + O(m^2).$$

We now define $\widetilde{W}$. For $m \ge 1$ and integers $0 \le u_1 \le 3m$ and $0 \le u_2 \le 4m$, let

$$b_m^+(u_1,u_2) := \min\{u_1, u_2, 2m+2u_1-u_2, 6m-u_2\},$$

$$\gamma_m^+(u_1,u_2) := \left\lfloor \frac{b_m^+(u_1,u_2)-\epsilon}{2} \right\rfloor, \qquad \gamma_m^-(u_1) := \max\left\{0, \left\lceil \frac{u_1-m}{2}\right\rceil\right\}.$$

Let $\widetilde{W}_{m,(u_1,u_2)}$ be the space of polynomials f of character (u_1,u_2) such that $\varphi_1^{\gamma_m^-(u_1)} \mid f$ and any monomial $x^{u_1-b}y^bz^{(u_2-b)/2}$ of f satisfies $b \le b_m^+(u_1,u_2)$. Let $\widetilde{W}_{m,(u_1,u_2)} := 0$ for the other characters, $\widetilde{W}_{m,j} := \bigoplus_{2u_1+u_2=j}\widetilde{W}_{m,(u_1,u_2)}$, and $\widetilde{W}_m := \bigoplus_j \widetilde{W}_{m,j}$. Writing $\mu = \mu_{u_1,u_2}$ and $\gamma_m^\pm = \gamma_m^\pm(u_1,u_2)$, we obtain

$$\widetilde{W}_{m,(u_1,u_2)} = \mu(Q-1)^{\gamma_m^-}\cdot\{P\in\mathbb{C}[Q] : \deg P \le \gamma_m^+ - \gamma_m^-\}, \tag{8.26}$$

so $\dim \widetilde{W}_{m,(u_1,u_2)} = \max\{0, \gamma_m^+-\gamma_m^-+1\}$. A direct check shows that the polynomials (8.23) belong to $\widetilde{W}_1$. The inequalities $b\le u_1$, $b\le u_2$, $b\le 2m+2u_1-u_2$, $b\le 6m-u_2$, $u_1\le 3m$, and $u_2\le 4m$ on a monomial are linear and homogeneous in (m,u_1,u_2,b), and $\gamma_{m+m'}^-(u_1+u_1') \le \gamma_m^-(u_1)+\gamma_{m'}^-(u_1')$. Thus $\widetilde{W}_m\cdot\widetilde{W}_{m'} \subset \widetilde{W}_{m+m'}$, and so $R_m \subset \widetilde{W}_m$.

For $0 \le u_1 \le 3m$ and $0 \le u_2 \le 4m$, we have $b_m^+(u_1,u_2) = 2m\gamma_+(u_1/m, u_2/m)$, so $\gamma_m^+(u_1,u_2) = m\gamma_+(u_1/m,u_2/m) + O(1)$ and $\gamma_m^-(u_1) = m\gamma_-(u_1/m) + O(1)$. Thus $\dim \widetilde{W}_{m,(u_1,u_2)}$ differs from $m\max\{0,\gamma_+ - \gamma_-\}(u_1/m, u_2/m)$, the length of the corresponding fiber of $m\mathcal{P}$, by $O(1)$. The number of characters is $O(m^2)$, and the function $\max\{0,\gamma_+ - \gamma_-\}$ is Lipschitz on $[0,3]\times[0,4]$, so $\dim \widetilde{W}_m = \mathrm{vol}(\mathcal{P})m^3 + O(m^2) = 3m^3 + O(m^2)$. Since $R_m \subset W'_m \cap \widetilde{W}_m$, we conclude that

$$\dim W'_m/R_m = O(m^2), \qquad \dim \widetilde{W}_m/R_m = O(m^2). \tag{8.27}$$

Claim 8.13. *There exists $\kappa_1 > 0$ such that*

$$|s_m(W';C) - s_m(\widetilde{W};C)| \le \frac{\kappa_1}{m} \quad \text{and} \quad |t_m(W';C,q) - t_m(\widetilde{W};C,q)| \le \frac{\kappa_1}{m}$$

for any $m \ge 1$, any $C \in \{C_x, C_z, C_w, \Theta\} \cup \{C_\theta : \theta \in \mathbb{C}^*\}$, and any closed point $q \in C$.

Proof. Let $V \subset V'$ be vector spaces with $\dim V' < +\infty$, and let $\mathcal{F}$ be a filtration of V'. Since $\mathcal{F}^t V = \mathcal{F}^t V' \cap V$, the jumps of the induced filtration of V, counted with multiplicity, are among the jumps of $\mathcal{F}$, and $\sigma(V';\mathcal{F}) - \sigma(V;\mathcal{F})$ is a sum of $\dim V' - \dim V$ jumps of $\mathcal{F}$. Moreover $\mathrm{gr}_C^k V \subset \mathrm{gr}_C^k V'$ for subspaces $V \subset V' \subset \mathbb{C}[x,y,z]_j$, and $\sum_k \dim(\mathrm{gr}_C^k V'/\mathrm{gr}_C^k V) = \dim V'/V$.

The weighted degrees of the polynomials (8.23) are at most 10, and $2u_1 + u_2 \le 10m$ for $u_1 \le 3m$ and $u_2 \le 4m$. Thus $W'_{m,j} = \widetilde{W}_{m,j} = 0$ for $j > 10m$. For $f \in \mathbb{C}[x,y,z]_j$, $\mathrm{div}_E(f) \sim_{\mathbb{Q}} \frac{j}{2}C_x$. Since $(C_x \cdot C_x) = (C_x \cdot C_z) = \frac{1}{3}$, $(C_x \cdot C_w) = (C_x \cdot C_\theta) = 1$, and $C_z \sim C_x$, $C_w \sim C_\theta \sim 3C_x$, we obtain $\mathrm{ord}_C(f) \le \frac{j}{2}$ and $\deg r_C(f) \le \frac{j}{2}$ for $C \subset E$. On Θ, $\mathrm{ord}_\Theta(f) \le \frac{j}{2}$ and $\deg r_\Theta(f) = -\mathrm{ord}_\Theta(f)(\Theta^2) \le \frac{j}{4}$, as $(\Theta^2) = -\frac{1}{2}$. Hence all jumps of the filtrations defining s_m and t_m are at most $5m$ for $j \le 10m$.

Applying the first paragraph to $R_m \subset W'_m$ and to $R_m \subset \widetilde{W}_m$, (8.27) implies

$$|\sigma(W'_m;\mathrm{ord}_C) - \sigma(\widetilde{W}_m;\mathrm{ord}_C)| \le 5m \cdot O(m^2),$$
$$|\sigma(W'_m;C,q) - \sigma(\widetilde{W}_m;C,q)| \le 5m \cdot O(m^2),$$

where $\sigma(V_m;\cdot) := \sum_j \sigma(V_{m,j};\cdot)$. The constants do not depend on C or q. Dividing by $mN_m \ge 3m^4$ proves the claim. □

Proof of Theorem 8.12 continued.

Step 2. In this step we compute $s_m(\widetilde{W};C)$ and $t_m(\widetilde{W};C,q)$.

Fix (u_1,u_2) with $\widetilde{W}_{m,(u_1,u_2)} \ne 0$, and write $\mu = \mu_{u_1,u_2}$ and $\gamma^\pm = \gamma_m^\pm(u_1,u_2)$. By (8.26), each of the following families, indexed by the integers $\gamma^- \le \gamma \le \gamma^+$, is a basis of $\widetilde{W}_{m,(u_1,u_2)}$:

$$\text{(B1) } \mu(Q-1)^{\gamma^-}Q^{\gamma-\gamma^-}, \qquad \text{(B2) } \mu(Q-1)^{\gamma}, \qquad \text{(B3) } \mu(Q-1)^{\gamma^-}(Q-\theta)^{\gamma-\gamma^-},$$

where $\theta \ne 0,1$ in (B3). The element of (B1) indexed by γ has highest y-exponent $\epsilon + 2\gamma$ and lowest y-exponent $\epsilon + 2(\gamma - \gamma^-)$. Since ord_{C_x}, ord_{C_z}, and $2\,\mathrm{ord}_{C_w}$ are the least x-, z-, and y-exponents, (B1) is adapted to ord_{C_x}, ord_{C_z}, and ord_{C_w}, with

$$\mathrm{ord}_{C_x} = u_1 - \epsilon - 2\gamma, \qquad \mathrm{ord}_{C_z} = \frac{u_2-\epsilon}{2} - \gamma, \qquad \mathrm{ord}_{C_w} = \frac{\epsilon}{2} + \gamma - \gamma^-.$$

As μ does not vanish identically on C_θ, $\mathrm{ord}_{C_\theta}(\mu P(Q)) = \mathrm{ord}_{Q=\theta} P$. Thus (B2) is adapted to ord_{C_1} with values γ, and (B3) is adapted to ord_{C_θ} with values $\gamma - \gamma^-$ for $\theta \ne 1$. Since $\mathrm{ord}_\Theta(x) = \mathrm{ord}_\Theta(y) = 1$ and $\mathrm{ord}_\Theta(z) = 0$, $\mathrm{ord}_\Theta = u_1$ on $\widetilde{W}_{m,(u_1,u_2)}\backslash\{0\}$.

We now describe r_C. On $C_x \cong \mathbb{P}(1,3)_{z:w}$, r_{C_x} of the element of (B1) indexed by γ is the divisor of $y^{\epsilon+2\gamma}z^{(u_2-\epsilon)/2-\gamma}$, with multiplicity $\frac{\epsilon}{2}+\gamma$ at o_z and $\frac{1}{3}(\frac{u_2-\epsilon}{2} - \gamma)$ at o_w. On $C_z \cong \mathbb{P}(1,3)_{x:w}$, it is the divisor of $x^{u_1-\epsilon-2\gamma}y^{\epsilon+2\gamma}$, with multiplicity $\frac{\epsilon}{2}+\gamma$ at o_x and $\frac{1}{3}(u_1 - \epsilon - 2\gamma)$ at o_w. On $C_w \cong \mathbb{P}^1_{x:z}$, it is the divisor of $x^{u_1-\epsilon-2(\gamma-\gamma^-)}z^{(u_2-\epsilon)/2-(\gamma-\gamma^-)}$, with multiplicity $u_1 - \epsilon - 2(\gamma-\gamma^-)$ at o_z and $\frac{u_2-\epsilon}{2} - (\gamma - \gamma^-)$ at o_x. Parametrize C_θ by $x = \xi$, $z = \zeta^2$, $y = \theta^{1/2}\xi\zeta$. A polynomial of character (u'_1,u'_2) restricts to a constant multiple of $\xi^{u'_1}\zeta^{u'_2}$, and $w - \theta x^2 z$ has character $(2,2)$.

Since x is a local coordinate of C_θ at o_z and z is a local coordinate of C_θ at o_x, $r_{C_\theta}(f)$ for f of character (u_1,u_2) with $\operatorname{ord}_{C_\theta}(f)=k$ has multiplicity u_1-2k at o_z and $\frac{u_2}{2}-k$ at o_x, where $k=\gamma$ for (B2) and $k=\gamma-\gamma^-$ for (B3). On Θ, in the chart of E' with coordinates (x,Q) and $w=x^2Q$, $r_\Theta(\mu P(Q))$ is the divisor of $Q^{\epsilon/2}P(Q)$ on $\Theta\cong\mathbb{P}^1$, whose point $Q=+\infty$ is a singular point of E' of type $\frac12(1,1)$, with multiplicity $\frac{\epsilon}{2}+\operatorname{ord}_{Q=0}P$ at $Q=0$, $\frac{u_1-\epsilon}{2}-\deg P$ at $Q=+\infty$, and $\operatorname{ord}_{Q=\theta}P$ at $Q=\theta\in\mathbb{C}^*$.

For a curve $C\subset E$ and fixed (j,k), distinct characters yield distinct monomials, so the multiplicities above at the torus-fixed points o_x, o_z, o_w of C are the jumps of the filtrations of $\operatorname{gr}^k_C\widetilde{W}_{m,j}$. On Θ, fixed (j,k) determine the character, since $k=u_1$, and (B1), (B2), (B3) are adapted to the orders at $Q=0$ and $Q=+\infty$, at $Q=1$, and at $Q=\theta\notin\{0,1,\infty\}$, respectively. Each normalized sum $\frac{1}{mN_m}\sum$ of a linear function of $(u_1,u_2,\gamma,\gamma^-_m)$ over the bases equals $\frac13\int_{\mathcal{P}}$ of the corresponding function of $(u_1,u_2,\gamma,\gamma_-)$ up to $O(1/m)$: the range of γ in each character differs from $m[\gamma_-,\gamma_+]$ by $O(1)$, the number of characters is $O(m^2)$, the values are $O(m)$, and $N_m=3m^3+O(m^2)$. In the following two displays, $s(C)$ and $t(C,q)$ stand for $s_m(\widetilde{W};C)$ and $t_m(\widetilde{W};C,q)$, all integrals are over $\mathcal{P}$, $\theta\neq1$, and the equalities hold up to $O(1/m)$ with constants independent of θ. By (8.25), we obtain the values of s

$$\begin{aligned} s(C_x)&=\tfrac13\textstyle\int(u_1-2\gamma)=\tfrac{37}{72}, & s(C_z)&=\tfrac13\int(\tfrac{u_2}{2}-\gamma)=\tfrac{47}{72},\\ s(C_w)&=\tfrac13\textstyle\int(\gamma-\gamma_-)=\tfrac{5}{24}, & s(C_1)&=\tfrac13\int\gamma=\tfrac{13}{24},\\ s(C_\theta)&=\tfrac13\textstyle\int(\gamma-\gamma_-)=\tfrac{5}{24}, & s(\Theta)&=\tfrac13\int u_1=\tfrac{115}{72}, \end{aligned}\tag{8.28}$$

and the values of t

$$\begin{aligned} t(C_x,o_z)&=\tfrac13\textstyle\int\gamma=\tfrac{13}{24}, & t(C_x,o_w)&=\tfrac19\int(\tfrac{u_2}{2}-\gamma)=\tfrac{47}{216},\\ t(C_z,o_x)&=\tfrac13\textstyle\int\gamma=\tfrac{13}{24}, & t(C_z,o_w)&=\tfrac19\int(u_1-2\gamma)=\tfrac{37}{216},\\ t(C_w,o_z)&=\tfrac13\textstyle\int(u_1-2\gamma+2\gamma_-)=\tfrac{85}{72}, & t(C_w,o_x)&=\tfrac13\int(\tfrac{u_2}{2}-\gamma+\gamma_-)=\tfrac{71}{72},\\ t(C_1,o_z)&=\tfrac13\textstyle\int(u_1-2\gamma)=\tfrac{37}{72}, & t(C_1,o_x)&=\tfrac13\int(\tfrac{u_2}{2}-\gamma)=\tfrac{47}{72},\\ t(C_\theta,o_z)&=\tfrac13\textstyle\int(u_1-2\gamma+2\gamma_-)=\tfrac{85}{72}, & t(C_\theta,o_x)&=\tfrac13\int(\tfrac{u_2}{2}-\gamma+\gamma_-)=\tfrac{71}{72},\\ t(\Theta,\{Q=0\})&=\tfrac13\textstyle\int(\gamma-\gamma_-)=\tfrac{5}{24}, & t(\Theta,\{Q=+\infty\})&=\tfrac13\int(\tfrac{u_1}{2}-\gamma)=\tfrac{37}{144},\\ t(\Theta,\{Q=1\})&=\tfrac13\textstyle\int\gamma=\tfrac{13}{24}, & t(\Theta,\{Q=\theta\})&=\tfrac13\int(\gamma-\gamma_-)=\tfrac{5}{24}. \end{aligned}\tag{8.29}$$

It remains to bound $t_m(\widetilde{W};C,q)$ at the other points q of the curves $C\subset E$.

Claim 8.14. *Let $e_1<\cdots<e_{r_0}$ be integers, and let $V\subset\mathbb{C}(\eta)$ be the span of $\eta^{e_1},\dots,\eta^{e_{r_0}}$. Then for any $\eta_0\in\mathbb{C}^*$, the jumps of the order of vanishing at $\eta=\eta_0$ on V are $0,1,\dots,r_0-1$. In particular, their sum is at most $\sum_i(e_i-e_1)$ and at most $\sum_i(e_{r_0}-e_i)$.*

Proof. Let $g=\sum_i a_i\eta^{e_i}\in V$ vanish to order at least r_0 at η_0. Since $\eta\frac{d}{d\eta}$ lowers the order of vanishing at $\eta_0\neq0$ by at most one, $\sum_i a_ie_i^l\eta_0^{e_i}=(\eta\frac{d}{d\eta})^lg(\eta_0)=0$ for $0\le l<r_0$. The Vandermonde matrix $(e_i^l)_{i,l}$ is invertible, so $a_i\eta_0^{e_i}=0$ for any i, and $g=0$. Thus the jumps are r_0 distinct integers in $[0,r_0-1]$. The inequalities follow from $e_i-e_1\ge i-1$ and $e_{r_0}-e_i\ge r_0-i$. □

Proof of Theorem 8.12 continued. Let C be one of C_x, C_z, C_w, C_θ, let q_0,q_∞ be the torus-fixed points of C, and take the affine coordinate $\eta=w/z^3$, w/x^3, x/z, or z/x on C, respectively, so that $\eta(q_0)=0$ and $\eta(q_\infty)=+\infty$. For fixed (j,k), the $\mathbb{Q}$-divisors r_C computed above are $D_{j,k}+\operatorname{div}(\eta^{e_i})$ for a $\mathbb{Q}$-divisor $D_{j,k}$ supported on $\{q_0,q_\infty\}$ and, after moving a common factor into $D_{j,k}$, distinct integers e_i: for instance, on C_x we have $y^bz^{c'}=z^{(3b+2c')/2}\eta^{b_0/2}\cdot\eta^{(b-b_0)/2}$, where b_0 is the least value of b for the fixed (j,k). Since b has fixed parity, the exponents $e_i=(b-b_0)/2$ are integers, and $z^{(3b+2c')/2}\eta^{b_0/2}$ contributes to $D_{j,k}$. The multiplicities at q_0 and at q_∞ are non-negative, so their sums are at least $\sum_i(e_i-e_1)$ and $\sum_i(e_{r_0}-e_i)$, respectively. By Claim 8.14,

for any closed point $q \in C \backslash \{q_0, q_\infty\}$,

$$t_m(\widetilde{W}; C, q) \le \min\{t_m(\widetilde{W}; C, q_0), t_m(\widetilde{W}; C, q_\infty)\}. \tag{8.30}$$

Step 3. In this step we prove a finite-level criterion for log canonicity along the curves above. An *m-basis type divisor* of W' is $\frac{1}{mN_m}\sum_j \sum_i \operatorname{div}_E(f_{j,i})$, where $(f_{j,i})_i$ is a basis of $W'_{m,j}$ for each j.

Claim 8.15. *Let $c > 0$ and $m \ge 1$. In (1) and (2), let D be an m-basis type divisor of W' defined by bases $(f_{j,i})_i$ of the $W'_{m,j}$ compatible with the filtration by ord_C.*

(1) Let C be one of C_x, C_z, C_w, C_θ, let $q \in C$ be a closed point, and let d_q be the coefficient of q in $\operatorname{Diff}_C(\Delta_E - (\operatorname{mult}_C \Delta_E)C)$. Assume that $\operatorname{mult}_C \Delta_E + cs_m(W'; C) \le 1$ and $d_q + ct_m(W'; C, q) \le 1$. Then $(E, \Delta_E + cD)$ is log canonical near q.

(2) Let $C = \Theta$. For $q \in \Theta$ let $d_q = \frac{1}{2}$ if $Q(q) \in \{0, \infty\}$, and $d_q = 0$ otherwise. Assume that $cs_m(W'; \Theta) \le 2$ and $d_q + ct_m(W'; \Theta, q) \le 1$ for any closed point $q \in \Theta$. Then $(E, \Delta_E + cD)$ is log canonical near o_z.

Proof. (1) Write $D = (\operatorname{mult}_C D)C + D_C$ with $D_C \ge 0$ not containing C. Since the bases are compatible with ord_C, $\operatorname{mult}_C D = s_m(W'; C)$. Moreover $D_C|_C = \frac{1}{mN_m}\sum_{j,i} r_C(f_{j,i})$, and for each (j,k) the classes of the $f_{j,i}$ with $\operatorname{ord}_C(f_{j,i}) = k$ form a basis of $\operatorname{gr}^k_C W'_{m,j}$. Thus $\operatorname{mult}_q(D_C|_C) \le t_m(W'; C, q)$, and the coefficient of q in $\operatorname{Diff}_C(\Delta_E - (\operatorname{mult}_C \Delta_E)C) + cD_C|_C$ is at most 1. Since C is a smooth rational curve, the pair $(C, \operatorname{Diff}_C(\Delta_E - (\operatorname{mult}_C \Delta_E)C) + cD_C|_C)$ is log canonical near q. By inversion of adjunction [Kaw07, Theorem], $(E, C + (\Delta_E - (\operatorname{mult}_C \Delta_E)C) + cD_C)$ is log canonical near q. Since $\operatorname{mult}_C \Delta_E + c \operatorname{mult}_C D \le 1$,

$$\Delta_E + cD \le C + (\Delta_E - (\operatorname{mult}_C \Delta_E)C) + cD_C,$$

so $(E, \Delta_E + cD)$ is log canonical near q.

(2) Let $\Delta_{E'}$ be the strict transform of Δ_E. In the chart of E' with coordinates (x, Q) and $w = x^2 Q$, E' is smooth, $\Theta = (x = 0)$, and $\Delta_{E'} = \frac{1}{2}(Q = 0)$. The other chart is the quotient of $\mathbb{A}^2$ by μ_2 acting with weights $(1, 1)$, and the fixed point maps to $\{Q = +\infty\}$. Thus $\operatorname{Diff}_\Theta(\Delta_{E'}) = \frac{1}{2}\{Q = 0\} + \frac{1}{2}\{Q = +\infty\}$ [Kol+92, Proposition 16.6 and Corollary 16.7], and $(E', \Theta + \Delta_{E'})$ is plt near Θ by [Kol13, Theorem 4.9(1)]. By [JM12, Proposition 5.1], $A_E(\Theta) = 1 + 2 = 3$, and $\operatorname{ord}_\Theta(w) = 2$, so $A_{E,\Delta_E}(\Theta) = 3 - \frac{1}{2} \cdot 2 = 2$ and

$$h^*(K_E + \Delta_E) = K_{E'} + \Delta_{E'} - \Theta.$$

Write $h^*D = (\operatorname{mult}_\Theta D)\Theta + D_\Theta$ with $D_\Theta \ge 0$ not containing Θ. Since the bases are compatible with $\operatorname{ord}_\Theta$, $\operatorname{mult}_\Theta D = s_m(W'; \Theta)$. Moreover $D_\Theta|_\Theta = \frac{1}{mN_m}\sum_{j,i} r_\Theta(f_{j,i})$, and for each (j,k) the classes of the $f_{j,i}$ with $\operatorname{ord}_\Theta(f_{j,i}) = k$ form a basis of $\operatorname{gr}^k_\Theta W'_{m,j}$. Thus $\operatorname{mult}_q(D_\Theta|_\Theta) \le t_m(W'; \Theta, q)$ for $q \in \Theta$. Hence $(\Theta, \operatorname{Diff}_\Theta(\Delta_{E'}) + cD_\Theta|_\Theta)$ is log canonical, and $(E', \Theta + \Delta_{E'} + cD_\Theta)$ is log canonical near Θ by [Kaw07, Theorem]. Since $c \operatorname{mult}_\Theta D - 1 \le 1$, the pair $(E', \Delta_{E'} + (c \operatorname{mult}_\Theta D - 1)\Theta + cD_\Theta)$ is sub-log canonical near Θ. It is the crepant pullback of $(E, \Delta_E + cD)$, so $(E, \Delta_E + cD)$ is log canonical near $o_z = h(\Theta)$ [KM98, Lemma 2.30]. □

Proof of Theorem 8.12 continued.

Step 4. In this step we show that for any $0 < c < \frac{72}{71}$ there exists m_c such that $(E, \Delta_E + cD)$ is log canonical for any $m \ge m_c$ and any m-basis type divisor D of W'.

By Claim 8.13, (8.28), (8.29), and (8.30), there exists $\kappa_2 > 0$, independent of θ and q, such that $s_m(W'; C)$ and $t_m(W'; C, q)$ exceed the bounds listed below by at most κ_2/m. Let v be a prime divisor over E, and let $q \in \operatorname{center}_E v$ be a closed point. We take C as follows.

(a) If $x(q)z(q)w(q) \ne 0$, let $C = C_\theta$ with $\theta = Q(q)$. Then $q \notin C_w$, so $\operatorname{mult}_C \Delta_E = d_q = 0$. The bounds are $s_m \le \frac{13}{24}$ and $t_m \le \frac{71}{72}$.

(b) If $x(q) = 0$ and $q \neq o_z$, let $C = C_x$. Then $\operatorname{mult}_C \Delta_E = 0$ and $\operatorname{Diff}_{C_x}(\Delta_E) = \frac{1}{2}o_z + \frac{2}{3}o_w$ [Kol+92, Proposition 16.6 and Corollary 16.7], as o_w is a quotient singularity of type $\frac{1}{3}(1,1)$. The bounds are $s_m \leq \frac{37}{72}$ and $t_m \leq \frac{47}{216}$ at any point of $C_x \backslash \{o_z\}$, and $d_q = \frac{2}{3}$ for $q = o_w$.

(c) If $z(q) = 0$ and $w(q) \neq 0$, let $C = C_z$. Then $q \neq o_x$, $\operatorname{mult}_C \Delta_E = 0$, and $\operatorname{Diff}_{C_z}(\Delta_E) = \frac{1}{2}o_x + \frac{2}{3}o_w$ [Kol+92, Proposition 16.6 and Corollary 16.7]. The bounds are $s_m \leq \frac{47}{72}$ and $t_m \leq \frac{37}{216}$ at any point of $C_z \backslash \{o_x\}$, and $d_q = \frac{2}{3}$ for $q = o_w$.

(d) If $w(q) = 0$ and $q \neq o_z$, let $C = C_w$. Then $\operatorname{mult}_C \Delta_E = \frac{1}{2}$ and $d_q = 0$, as E is smooth along C_w. The bounds are $s_m \leq \frac{5}{24}$ and $t_m \leq \frac{71}{72}$.

(e) If $q = o_z$, let $C = \Theta$. The bounds are $s_m \leq \frac{115}{72}$, and $t_m \leq \frac{5}{24}$, $\frac{37}{144}$, $\frac{13}{24}$, and $\frac{5}{24}$ at $Q = 0$, at $Q = +\infty$, at $Q = 1$, and at the other points, respectively.

The cases (a)–(e) cover all closed points of E. Substituting $c = \frac{72}{71}$ and the bounds in (a)–(e) into $\operatorname{mult}_C \Delta_E + cs_m$ (respectively cs_m in case (e)) and into $d_q + ct_m$, we obtain

$$\begin{aligned} &\text{(a) } \tfrac{39}{71},\ 1, \quad \text{(b) } \tfrac{37}{71},\ \tfrac{189}{213},\ \tfrac{47}{213}, \quad \text{(c) } \tfrac{47}{71},\ \tfrac{179}{213},\ \tfrac{37}{213}, \\ &\text{(d) } \tfrac{101}{142},\ 1, \quad \text{(e) } \tfrac{115}{71},\ \tfrac{101}{142},\ \tfrac{54}{71},\ \tfrac{39}{71},\ \tfrac{15}{71}, \end{aligned} \tag{8.31}$$

which are at most 1, and at most 2 for the first entry of (e). Thus for $c < \frac{72}{71}$ the hypotheses of Claim 8.15 hold for $m \geq m_c$, where m_c depends only on c and κ_2. Choose bases of the $W'_{m,j}$ compatible with both v and ord_C [AZ22, Lemma 3.1], and let D_v be the corresponding m-basis type divisor. Then $v(D_v) = S_m(W'; v)$, and $(E, \Delta_E + cD_v)$ is log canonical near q by Claim 8.15. Since $q \in \operatorname{center}_E v$,

$$cS_m(W'; v) = cv(D_v) \leq A_{E,\Delta_E}(v).$$

Any m-basis type divisor D of W' satisfies $v(D) \leq S_m(W'; v)$, so $cv(D) \leq A_{E,\Delta_E}(v)$ for any prime divisor v over E. Thus $(E, \Delta_E + cD)$ is log canonical.

Step 5. In this step we prove the bound $S(W; v) \leq \frac{71}{72} A_{E,\Delta_E}(v)$ by Lemma 3.6(1) with $\lambda_m = \lambda$ for any m.

The pair (E, Δ_E) is projective and klt, since o_w is the only singular point of E, $o_w \notin C_w$, and C_w is smooth. The series W has bounded support and contains an ample series, since $(H^0(X, mH))_m$ has both properties and they are preserved by the refinement by E [AZ22, Examples 2.5 and 2.6]. Since $C_w = (y = 0)$ is λ-invariant, $\lambda(t)_* \Delta_E = \Delta_E$. The action $f(x, y, z) \to f(x, ty, t^2 z)$ of $\mathbb{G}_m$ on the graded ring $\mathbb{C}[x, y, z]$ preserves each $\mathbb{C}[x, y, z]_j$, the space of sections containing $W_{m,j}$, and so linearizes the reflexive sheaves of the series W.

The series W is associated with the $\mathbb{Q}$-divisor classes $\hat{\pi}^* H|_E$ and $-E|_E$ (Definition 2.8), not with Weil divisors: for odd j the class $-jE|_E$ has fractional part $\frac{1}{2}C_w$, as $\operatorname{div}_E(y) = \frac{1}{2}C_w$. The proof of Lemma 3.6 works verbatim, with the following changes. The sheaf containing $W_{m,j}$ is the sheaf $\mathcal{O}_E(j)$ on $E = \operatorname{Proj} \mathbb{C}[x, y, z]$ for the weights $(2, 3, 2)$, and $\mathcal{O}_E(6)$ is invertible. The divisor of a section f is $\operatorname{div}_E(f)$ (Set-up 8.11(5)). Finally, $\operatorname{div}_E(f(x, ty, t^2 z)) = \lambda(t)^* \operatorname{div}_E(f)$, and $\lambda(t)$ preserves Δ_E.

We show that $\operatorname{in}_\lambda W_{m,j} = W'_{m,j}$. By [AZ22, Lemma 3.1], there exists a basis $(f_i)_i$ of $W_{m,j}$ compatible with the filtration by ω_0, so that the polynomials $\operatorname{in}_\omega f_i$ are linearly independent. Then $\lambda(t) \cdot W_{m,j}$ is spanned by the polynomials $t^{-\omega_0(f_i)} f_i(x, ty, t^2 z)$, which converge to $\operatorname{in}_\omega f_i$ as $t \to 0$. Since $\dim W'_{m,j} = \dim W_{m,j}$, the limit is $W'_{m,j}$. The λ-eigenvectors in $W'_{m,j}$ are the polynomials of a single ω-weight, so an eigen-basis divisor at level m is an m-basis type divisor of W'.

Let $c^{(1)} < c^{(2)} < \cdots$ be a sequence with limit $\frac{72}{71}$. By **Step 4** we may take $m_{c^{(1)}} < m_{c^{(2)}} < \cdots$, and we let $c_m := c^{(k)}$ for $m_{c^{(k)}} \leq m < m_{c^{(k+1)}}$ and $c_m := c^{(1)}$ for $m < m_{c^{(1)}}$. Then $\lim_m c_m = \frac{72}{71}$, and $(E, \Delta_E + c_m D)$ is log canonical for any $m \geq m_{c^{(1)}}$ and any eigen-basis divisor D at level m. By Lemma 3.6(1),

$$\frac{72}{71} S(W; v) \leq A_{E,\Delta_E}(v)$$

for any prime divisor v over E. This proves the first inequality of the theorem.

Step 6. In this step we prove the bound on X.

Let F be a prime divisor over X with $\operatorname{center}_X F = \{p\}$. Then $\operatorname{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(p) = E$. By **Step 5**, the hypothesis of Lemma 3.1 holds with the real number $\frac{72}{71}$ (the constant called λ there), the complete series of $H = -K_X$, and $G = E$. Since $A_X(E) = 7$ and $S_X(E) = \frac{67}{12}$ (Construction-Proposition 8.10),

$$A_X(F) \geq \frac{72}{71} S_X(F) + \left(7 - \frac{72}{71} \cdot \frac{67}{12}\right) \operatorname{mult}_F E = \frac{72}{71} S_X(F) + \frac{95}{71} \operatorname{mult}_F E.$$

Multiplying by $\frac{71}{72}$, we obtain

$$S_X(F) \leq \frac{71}{72} A_X(F) - \frac{95}{72} \operatorname{mult}_F E.$$

Since $\operatorname{mult}_F E \geq 0$ and $A_X(F) > 0$, $\beta_X(F) \geq \frac{1}{72} A_X(F) > 0$. □

9. Curvilinear triple points in genus ten

In this section, we prove lower bounds for $\delta_x(X)$ at the points x of incidence type 3 on a smooth prime Fano threefold X of genus 10 (Theorems 9.9 and 9.13). Incidence type 3 is the last of the incidence types of Proposition 4.1 in genus 10. Any such point admits the normal form of Lemma 9.1, without any generality assumption, so the estimates of this section hold for any member, and they split into two cases (A) and (B).

9.1. The extraction.

In this subsection, we bring a point of incidence type 3 into a normal form, construct a plt extraction at the point, and compute the refinement of the anti-canonical series by the extracted divisor exactly.

Lemma 9.1. *Notation as in Set-up 2.14. Let X be a smooth prime Fano threefold of genus 10 and $x \in X$ a closed point such that $\Sigma(X, x)$ is curvilinear of length three and supported at one line, that is, x has incidence type 3. Then there exist complex numbers $\lambda, \mu, \beta, \kappa, \rho_y, \sigma_y$ $(y \in I)$, where $I := \{c, l, b, d, M, J\}$, and an isomorphism from X onto $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{\ell_1 = \ell_2 = 0\}$, where*

$$\ell_1 = n + \lambda f + \beta(g + k) + \sum_{y \in I} \rho_y y, \qquad \ell_2 = g + \mu f + \kappa(g + k) + \sum_{y \in I} \sigma_y y,$$

which sends x to $[h]$ and the line on X through x to the line joining $[h]$ and $[i]$. If $\mu = 0$, then $\lambda \neq 0$. We call such an isomorphism a normal form *at x. A normal form is in case (A) if $\mu = 0$, and in case (B) if $\mu \neq 0$.*

Proof. By Theorem 2.13, X is isomorphic to $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \Lambda$ for a linear subspace $\Lambda \subset \mathbb{P}^{13}$ of codimension two. We fix such an isomorphism and identify X with $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \Lambda$ through it. Since G_2 acts transitively on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ and preserves the set of codimension two linear sections, possibly replacing Λ with its translate by an element of G_2 and composing the isomorphism with that element, we may assume that $x = p := [h]$. Let $\mathbb{T}_x X \subset \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7) \subset \mathbb{P}^{13}$ be the embedded tangent spaces. Since $X \subset \Lambda$, we have $\mathbb{T}_x X \subset \Lambda$. Since $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is cut out by quadrics, so is X, and by [KP23, Lemma 5.6], applied to X and to $\mathrm{G}_2\,\mathrm{Gr}(2,7)$,

$$X \cap \mathbb{T}_x X = \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{T}_x X = \big(\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)\big) \cap \mathbb{T}_x X$$

is the cone over $\Sigma(X, x)$, and also the cone over $\Gamma \cap \mathbb{P}(T_x X)$, where $\Gamma \subset \mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7))$ is the scheme of lines through p. Hence $\Sigma(X, x) = \Gamma \cap \mathbb{P}(T_x X)$. By Lemma 2.15, Γ is the twisted cubic defined by $g = 0$ and $a^2 = ie$, $e^2 = -an$, $ae = -in$ in the coordinates (i, a, e, n, g) of $\mathbb{P}(T_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7)) \cong \mathbb{P}^4$, and $\mathbb{P}(T_x X)$ is a plane.

Suppose that $\mathbb{P}(T_x X) \not\subset \{g = 0\}$. Then $\mathbb{P}(T_x X) \cap \{g = 0\}$ is a line L_0, and $\Sigma(X, x) = \Gamma \cap L_0$. Since Γ is cut out by quadrics and contains no line, and a quadric vanishing on a subscheme of length three of L_0 vanishes on L_0, the length of $\Gamma \cap L_0$ is at most two, a contradiction. Thus $\mathbb{P}(T_x X)$ is a plane in $\{g = 0\} \cong \mathbb{P}^3$ which intersects Γ in a curvilinear scheme of length three

supported at one point q, that is, $\mathbb{P}(T_xX)$ is the osculating plane of Γ at q. By Lemma 2.15(4), the stabilizer of p in $G^\circ_{\mathcal{W}}$ acts transitively on Γ, hence on the osculating planes of Γ. Possibly replacing Λ with its translate by an element of the stabilizer and composing the isomorphism with that element, we may assume that $q = [1:0:0:0:0]$, the direction of i. The map $[s:t] \to (i,a,e,n,g) = (s^3, s^2t, st^2, -t^3, 0)$ parametrizes Γ, so the osculating plane at q is $\{n = g = 0\}$, and $\mathbb{P}(T_xX)$ has coordinates (i,a,e). On the chart $h = 1$ the points with $a = e = n = g = 0$ satisfy $f = k = c = b = d = J = l = M = 0$ by (2.1), so the line joining $[h]$ and $[i]$ is contained in $\mathrm{G}_2\,\mathrm{Gr}(2,7)$. The direction at p of the line joining $[h]$ and $[i]$ is q, so the line joining $[h]$ and $[i]$ is the line on X through x once we know that it is contained in Λ, which we check below.

We now determine Λ. By Lemma 2.15(1),

$$\mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7) = \{f = c = g + k = b = d = l = M = J = 0\},$$

and $\mathbb{T}_xX = \mathbb{T}_p\,\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{n = g = 0\}$, so the linear forms vanishing on $\mathbb{T}_xX$ are the combinations of $n, f, g, g+k$ and the coordinates in I. Any linear form defining Λ vanishes on $\mathbb{T}_xX$, so it is such a combination, and in particular it vanishes on the line joining $[h]$ and $[i]$. At p, the differentials of f, $g+k$ and of the coordinates in I vanish, so the differential of such a form ℓ is $\alpha dn + \gamma dg$, where α and γ are the coefficients of n and g when ℓ is written in the basis $n, f, g, g+k, c, l, b, d, M, J$. Since X is smooth of dimension three at p and the chart $h = 1$ is smooth of dimension five, the differentials at p of two linear forms ℓ, ℓ' defining Λ are linearly independent. Thus the 2×2 matrix of their coefficients of n and g is invertible, and after a change of basis of $\mathrm{Span}(\ell, \ell')$ we obtain forms ℓ_1, ℓ_2 as in the lemma with $\Lambda = \{\ell_1 = \ell_2 = 0\}$. Therefore the isomorphism from X onto $\mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \Lambda = \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{\ell_1 = \ell_2 = 0\}$ sends x to $[h]$ and the line on X through x to the line joining $[h]$ and $[i]$.

Finally assume that $\mu = 0$ and $\lambda = 0$. On the chart $i = 1$ the line joining $[h]$ and $[i]$ is $\{a = f = g = l = 0\}$ with free coordinate h. By (2.2), at the point of the line with coordinate h the differentials of b, c, d, J, M vanish, and

$$dn = -hdg - h^2dl, \qquad d(g+k) = dg + hdl.$$

Hence

$$d\ell_1 = \lambda df + (\beta - h)dg + (\beta h + \rho_l - h^2)dl, \qquad d\ell_2 = \mu df + (1+\kappa)dg + (\kappa h + \sigma_l)dl.$$

For $\lambda = \mu = 0$ the determinant of the coefficients of dg and dl equals

$$(\beta - h)(\kappa h + \sigma_l) - (1+\kappa)(\beta h + \rho_l - h^2) = h^2 - (\beta + \sigma_l)h + \beta\sigma_l - (1+\kappa)\rho_l.$$

The polynomial in h on the right-hand side is monic of degree two, so it has a complex root h_0. At the point of the line with coordinate h_0, which is a point of X, the differentials $d\ell_1$ and $d\ell_2$ are linearly dependent, so X is singular there, a contradiction. Therefore $\lambda \neq 0$ when $\mu = 0$. □

The weights of the extraction below are the second degrees of the coordinates, which come from the grading of the contact structure, and a degeneration of X by a cocharacter of G_2 turns the refined series into a bigraded quotient of the section ring of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$.

Construction-Proposition 9.2. Let X be a smooth prime Fano threefold of genus 10 and $x \in X$ a closed point such that $\Sigma(X,x)$ is curvilinear of length three and supported at one line, that is, x has incidence type 3, and fix a normal form at x (Lemma 9.1). Then there exists a prime divisor E over X of plt type with $\mathrm{center}_X E = \{x\}$, plt extraction $\hat{\pi} : \hat{X} \to X$, and $\mathrm{Diff}_E(0) = 0$, as follows.

(A) In case (A), $E \cong \mathbb{P}(1,2,3)$, $A_X(E) = 6$, and $S_X(E) = 21/4$.
(B) In case (B), $E \cong \mathbb{P}(1,3,10)$, $A_X(E) = 7$, and $S_X(E) = 11/2$.

Let W be the refinement of $(H^0(X, mH))_m$ by E (Definition 2.8). Assign to the coordinates of Set-up 2.14 the second degrees $h:0$, $i:1$, $a:2$, $e:3$, $n:4$, $f:4$, $g:5$, $k:5$, $c:6$, $l:6$, $b:7$, $d:8$, $M:9$, $J:10$, and let R_{G_2} be the homogeneous coordinate ring of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, bigraded by

the usual degree and the second degree. In case (A) let $R := R_{G_2}/(n + \lambda f, g + \kappa(g + k))$ and $Q := \mathbb{C}[i, a, e]$, graded by $1, 2, 3$. In case (B) let $R := R_{G_2}/(n, f)$ and $Q := \mathbb{C}[i, a, e, g]/(ie - a^2)$, graded by $1, 2, 3, 5$. Then the substitution $h = 1$ maps R into Q, is injective on each part of fixed usual degree, and for all integers $m, j \geq 0$ maps the bidegree (m, j) part of R isomorphically onto $W_{m,j}$, where $W_{m,j} \subset H^0(E, \mathcal{O}_E(j))$ in case (A) and $W_{m,j} \subset H^0(E, \mathcal{O}_{\mathbb{P}(1,3,10)}(2j))$ in case (B). For any integer $m \geq 1$, the finite-level expected vanishing order $S_m(E) := \frac{1}{mN_m} \sum_{j \geq 0} j \dim W_{m,j}$ equals

$$5 + q_m, \qquad q_m := \frac{h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-1)) - h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-2))}{mN_m},$$

in case (A), and $5 + 2q_m$ in case (B). Finally $\delta_x(X) \geq \min\{8/7, \delta(E; W)\}$ in case (A) and $\delta_x(X) \geq \min\{14/11, \delta(E; W)\}$ in case (B), where $\delta(E; W) := \inf_v A_E(v)/S(W; v)$ and v runs through the prime divisors over E.

Proof. The proof is long, so we divide it into several steps. In this proof we write $\mathrm{wt}(z)$ for the second degree of a coordinate z, and we use the chart $h = 1$ of $\mathrm{G}_2\,\mathrm{Gr}(2, 7)$, which is the affine space $\mathbb{A}^5$ with coordinates i, a, e, n, g (Lemma 2.15(1)). We write $\mathbb{C}[i, a, e, n, g]$ for the coordinate ring of the chart, graded by wt, and we view any coordinate z as the regular function z/h on the chart.

Step 1. In this step we compute the leading forms of the defining equations of X on the chart $h = 1$.

The ideal of $\mathrm{G}_2\,\mathrm{Gr}(2, 7)$ is homogeneous for the second degree: the second degree of an entry $A_{j_1 j_2}$ is 5 plus the sum of the weights of $\mathbf{e}_{j_1}$ and $\mathbf{e}_{j_2}$ under the cocharacter γ of **Step 4**, so each Pfaffian $\mathrm{Pf}_{j_1 j_2 j_3 j_4}$ is homogeneous of second degree $\mathrm{wt}(A_{j_1 j_2}) + \mathrm{wt}(A_{j_3 j_4})$. Thus R_{G_2} is bigraded. By (2.1), the function z on the chart is wt-homogeneous of degree $\mathrm{wt}(z)$ for any coordinate z: for instance $f = ie - a^2$ has degree 4, $g + k = -ae - in$ has degree 5, and $J = fc + gk$ has degree 10. In particular the substitution $h = 1$ maps the bidegree (m, j) part of R_{G_2} into the degree j part of $\mathbb{C}[i, a, e, n, g]$. The coordinates of degree 4 are n, f, those of degree 5 are g, k, and any coordinate in $I = \{c, l, b, d, M, J\}$ has degree at least 6. In case (A) we have $\mu = 0$, and the lowest-degree parts of ℓ_1 and ℓ_2 are

$$\ell_1^{\mathrm{lead}} := n + \lambda f = n + \lambda(ie - a^2), \qquad \ell_2^{\mathrm{lead}} := g + \kappa(g + k) = g - \kappa(ae + in),$$

of degrees $d_1 = 4$ and $d_2 = 5$. In case (B) we have $\mu \neq 0$, and we replace ℓ_1 with $\ell_1 - (\lambda/\mu)\ell_2$, which does not change X. The lowest-degree parts of the new ℓ_1 and of ℓ_2 are then $\ell_1^{\mathrm{lead}} := n$ and $\ell_2^{\mathrm{lead}} := \mu f = \mu(ie - a^2)$, both of degree $d_1 = d_2 = 4$. In both cases $(\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}})$ is a regular sequence in $\mathbb{C}[i, a, e, n, g]$, and

$$\mathbb{C}[i, a, e, n, g]/(\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}}) \cong Q$$

as graded rings: in case (A) ℓ_1^{lead} and ℓ_2^{lead} eliminate n and then g, and in case (B) they eliminate n and leave the irreducible quadric $ie - a^2$. In particular Q is a domain of dimension three, and the ideal of R_{G_2} defining R is generated by the homogenizations of ℓ_1^{lead} and ℓ_2^{lead}.

Step 2. In this step we prove that the linear section

$$X_0 := \mathrm{G}_2\,\mathrm{Gr}(2, 7) \cap \{\ell_1^{\mathrm{lead}} = \ell_2^{\mathrm{lead}} = 0\}$$

is integral, where $\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}}$ are viewed as linear forms on $\mathbb{P}^{13}$.

By **Step 1**, $X_0 \cap \{h \neq 0\} \cong \operatorname{Spec} Q$ is integral of dimension three. We claim that $\dim(X_0 \cap \{h = 0\}) \leq 2$. On $\{h = 0,\ i \neq 0\}$ we use the chart $i = 1$ with $h = 0$, where $n = -a^3$, $k = af$, and the free coordinates are a, f, g, l. In case (A), $n + \lambda f = 0$ implies $f = a^3/\lambda$ (recall $\lambda \neq 0$), and $g + \kappa(g + k) = 0$ becomes $(1 + \kappa)g + \kappa a^4/\lambda = 0$. If $\kappa \neq -1$, the equation determines g and leaves the free coordinates a, l. If $\kappa = -1$, it implies $a = 0$ on the reduced support, hence $f = 0$, and leaves g, l. In case (B), $n = f = 0$ implies $a = 0$ on the reduced support and leaves g, l. Thus $\dim(X_0 \cap \{h = 0,\ i \neq 0\}) = 2$. On $\{h = i = 0\}$ the Pfaffians of Table 6 vanish, and reading

the table from top to bottom shows that on the reduced support of $X_0 \cap \{h = i = 0\}$ we have $a = e = f = n = g = k = c = 0$ and

$$b^2 + dl = 0, \qquad lM + bd = 0, \qquad bM - d^2 = 0.$$

The left-hand sides are, up to sign, the 2×2 minors of

$$\begin{pmatrix} l & b & -d \\ b & -d & M \end{pmatrix},$$

so (l, b, d, M) is contained in the affine cone over a twisted cubic, of dimension two, and J is free. Hence $\dim(X_0 \cap \{h = i = 0\}) = 2$, and the claim follows.

TABLE 6. The equations of X_0 on $\{h = i = 0\}$, used from top to bottom: the Pfaffians $\mathrm{Pf}_{j_1j_2j_3j_4}$ of Set-up 2.14 and the linear equation ℓ_1^{lead}. The row of Pf_{2356} is combined with the rows of Pf_{1245} and Pf_{1346}.

equation	restriction to $\{h = i = 0\}$	consequence on the reduced support
Pf_{1567}	a^2	$a = 0$
Pf_{1357}	$e^2 + an$	$e = 0$
Pf_{1267}	$al - f^2$	$f = 0$
ℓ_1^{lead}	$n + \lambda f$ or n	$n = 0$
Pf_{1245}	$-cf - gk$	$gk = 0$
Pf_{1346}	$-be + g(g + k)$	$g(g + k) = 0$
Pf_{2356}	$-ad + k(g + k) + ln$	$k(g + k) = 0$, so $g = k = 0$
Pf_{3457}	$Me + c^2 - dn$	$c = 0$
Pf_{2467}	$Jf - b^2 - dl$	$b^2 + dl = 0$
Pf_{2346}	$J(g + k) + Ml + bd$	$lM + bd = 0$
Pf_{2347}	$-Jc + Mb - d^2$	$bM - d^2 = 0$

By Lemma 2.16, X_0 is a Gorenstein, hence Cohen–Macaulay, scheme of pure dimension three. Thus any irreducible component of X_0 intersects the dense open subset $X_0 \cap \{h \neq 0\}$, which is irreducible, so X_0 is irreducible. Since X_0 is generically reduced and Cohen–Macaulay, it is reduced [Mat89, Section 23, p. 183]. Therefore X_0 is integral.

Step 3. In this step we construct E and prove the assertions on E, $A_X(E)$, the plt property, and $\mathrm{Diff}_E(0)$.

Let $\widehat{\mathcal{O}} := \mathbb{C}[[i, a, e, n, g]]$ be the completion of the local ring of the chart at $p = [h]$, filtered by $F^{\geq r}\widehat{\mathcal{O}}$, the closure of the span of the monomials of degree at least r. For a non-zero $\theta \in \widehat{\mathcal{O}}$ let $\mathrm{wt}(\theta)$ be the largest r with $\theta \in F^{\geq r}\widehat{\mathcal{O}}$, and let θ^{lead} be the homogeneous component of θ of degree $\mathrm{wt}(\theta)$. Let $\mathfrak{J} := (\ell_1, \ell_2) \subset \widehat{\mathcal{O}}$. We claim that $\theta^{\mathrm{lead}} \in (\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}})$ for any non-zero $\theta \in \mathfrak{J}$. Write $\theta = \alpha_1\ell_1 + \alpha_2\ell_2$ and let $r := \min\{\mathrm{wt}(\alpha_1) + d_1, \mathrm{wt}(\alpha_2) + d_2\}$, where $\mathrm{wt}(0) = +\infty$. Then $r \leq \mathrm{wt}(\theta)$. Let τ_ϵ be the homogeneous component of α_ϵ of degree $r - d_\epsilon$ for $\epsilon = 1, 2$. If $\tau_1\ell_1^{\mathrm{lead}} + \tau_2\ell_2^{\mathrm{lead}} \neq 0$, then it is θ^{lead} and the claim holds. Otherwise, since $(\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}})$ is a regular sequence in $\mathbb{C}[i, a, e, n, g]$, the syzygies of $(\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}})$ are generated by the Koszul syzygy [Eis95, Corollary 17.5], so $(\tau_1, \tau_2) = \tau(\ell_2^{\mathrm{lead}}, -\ell_1^{\mathrm{lead}})$ for a homogeneous polynomial τ of degree $r - d_1 - d_2$. Replacing (α_1, α_2) with $(\alpha_1 - \tau\ell_2, \alpha_2 + \tau\ell_1)$ does not change θ and strictly increases r. Since $r \leq \mathrm{wt}(\theta) < +\infty$, after finitely many replacements the first alternative occurs, which proves the claim. Hence the graded ring associated with the filtration of $\widehat{\mathcal{O}}/\mathfrak{J} = \widehat{\mathcal{O}}_{X,x}$ induced by $F^{\geq\bullet}$ is

$$\mathbb{C}[i, a, e, n, g]/(\ell_1^{\mathrm{lead}}, \ell_2^{\mathrm{lead}}) \cong Q.$$

Since Q is a domain (**Step 1**), the function $\nu(\theta) := \max\{r : \theta \in F^{\geq r}\widehat{\mathcal{O}} + \mathfrak{J}\}$ is a valuation on $\widehat{\mathcal{O}}_{X,x}$, hence on $\mathcal{O}_{X,x}$, and $\nu(i) = 1$. For any $\theta \in \mathcal{O}_{X,x}$ with $\nu(\theta) = j$ we write $\bar{\theta} \in Q_j$ for the class of θ in the degree j part of the graded ring.

Let $\hat\pi : \hat X \to X$ be the strict transform of X in the weighted blow-up of the chart $h = 1$ at p with weights $(1, 2, 3, 4, 5)$ for (i, a, e, n, g), glued with $X\backslash\{x\}$, and let E be the exceptional locus of $\hat\pi$. By the claim, $E = \operatorname{Proj} Q$, which is irreducible, and $\nu(\theta) = \operatorname{ord}_E(\theta)$ for any $\theta \in \mathcal{O}_{X,x}$. The divisor $-E$ is $\hat\pi$-ample, since $\mathcal{O}_{\hat X}(-E)$ is the restriction of the tautological sheaf of the weighted blow-up, and $\mathcal{O}_{\hat X}(-jE)|_E \cong \mathcal{O}_E(j)$ for the grading of Q. Since $E = \operatorname{Proj} Q$ is the exceptional locus of the weighted blow-up, for any $j \geq 0$ the restriction map from $\{\theta \in \mathcal{O}_{X,x} : \operatorname{ord}_E(\theta) \geq j\}$ to $H^0(E, \mathcal{O}_E(j)) = Q_j$ sends θ to $\bar\theta$ if $\operatorname{ord}_E(\theta) = j$ and to 0 if $\operatorname{ord}_E(\theta) > j$. Since ℓ_1 has linear part n (or $n - (\lambda/\mu)g$ in case (B)) and ℓ_2 has linear part g, the functions i, a, e form a regular system of parameters of $\mathcal{O}_{X,x}$, and $\omega := di \wedge da \wedge de$ generates ω_X near x.

Case (A). Here $Q = \mathbb{C}[i, a, e]$ is generated by the classes of i, a, e, so ord_E is the monomial valuation in the parameters i, a, e with weights $(1, 2, 3)$, and $\hat\pi$ is the weighted blow-up of x with weights $(1, 2, 3)$ in i, a, e. Hence $E \cong \mathbb{P}(1, 2, 3)$. On the chart $i = u$, $a = u^2v_1$, $e = u^3w_1$, the variety $\hat X$ is smooth with coordinates (u, v_1, w_1), $E = \{u = 0\}$, and $\hat\pi^*\omega = u^5 du \wedge dv_1 \wedge dw_1$, so $A_X(E) = 6$. The charts where a or e is non-zero are the quotients of $\mathbb{A}^3$ by μ_2 acting with weights $(1, 1, 1)$ and by μ_3 acting with weights $(1, 2, 1)$, with E the image of a coordinate hyperplane. Since the weights are pairwise coprime, $\hat X$ is smooth along E outside the coordinate points of E of indices 2 and 3, and the quotient maps are unramified in codimension one.

Case (B). The graded ring $Q = \mathbb{C}[i, a, e, g]/(ie - a^2)$ is isomorphic, via $i = U^2$, $a = UV$, $e = V^2$, $g = G$, to the subring $\bigoplus_j \mathbb{C}[U, V, G]_{2j}$ of $\mathbb{C}[U, V, G]$ graded by $(1, 3, 10)$. Hence $E \cong \mathbb{P}(1, 3, 10)$ and $\mathcal{O}_E(j) = \mathcal{O}_{\mathbb{P}(1,3,10)}(2j)$. If $i = e = g = 0$ at a point of E, then $a^2 = ie = 0$, which is impossible, so E is covered by the charts where i, e, or g is non-zero. On the charts $i \neq 0$, $e \neq 0$, and $g \neq 0$, the coordinate n is eliminated by ℓ_1.

On the chart $i = u$, $a = u^2v_1$, $e = u^3w_1$, $n = u^4n_1$, $g = u^5g_1$, the strict transform is defined by $\ell_1/u^4 = n_1 + u\psi_1$ and

$$\ell_2/u^4 = \mu(w_1 - v_1^2) + u\big(g_1 - \kappa(v_1w_1 + n_1)\big) + u^2\psi_2$$

for regular functions ψ_1, ψ_2, since the part of ℓ_2 of degree 5 is $g - \kappa(ae + in)$. Solving for n_1 and w_1 we obtain

$$w_1 = v_1^2 - \frac{u}{\mu}g_1 + \frac{\kappa u}{\mu}v_1^3 + u^2\psi_3$$

for a regular function ψ_3. Thus $\hat X$ is smooth on the chart $i \neq 0$ with coordinates (u, v_1, g_1), $E = \{u = 0\}$ is smooth, and

$$\hat\pi^*\omega = u^5 du \wedge dv_1 \wedge dw_1 = u^5\frac{\partial w_1}{\partial g_1} du \wedge dv_1 \wedge dg_1 = u^6\Big(-\frac{1}{\mu} + u\psi_4\Big) du \wedge dv_1 \wedge dg_1$$

for a regular function ψ_4. Hence $A_X(E) = 7$.

On the chart where $e \neq 0$, with $e = s^3$, $\hat X$ is the quotient of a smooth threefold with coordinates (s, a_1, g_1), on which $E = \{s = 0\}$ is smooth, by μ_3 acting with weights $(1, 1, 1)$, with only the origin fixed. On the chart where $g \neq 0$, with $g = s^5$, $\hat X$ is the quotient of a threefold that is smooth along $E = \{s = 0\}$ by μ_5 acting on (s, i_1, a_1, e_1) with weights $(1, 4, 3, 2)$, with only the origin fixed. The cover is smooth at the origin since $\partial(\ell_2/s^4)/\partial s = 1$ there, and E is smooth on the cover except for an A_1 point $\{i_1e_1 = a_1^2\}$ at the origin.

In both cases, near each point of E, $\hat X$ is the quotient of a variety Y' by a cyclic group, possibly trivial, acting freely in codimension one, and the preimage E' of E is a normal Cartier divisor of Y' with canonical singularities: in case (A), E' is a coordinate hyperplane of $\mathbb{A}^3$, and in case (B), E' is smooth or has an A_1 point. A smooth surface is canonical [KM98, Corollary 2.31(2)], and the A_1 surface singularity is canonical [KM98, Theorem 4.20]. Hence (Y', E') is plt near E' by inversion of adjunction [KM98, Theorem 5.50(1)], and so is $(\hat X, E)$ [KM98, Proposition 5.20(2)]. In particular $\hat X$ is normal near E. By construction $\hat\pi$ is an isomorphism over $X\backslash\{x\}$. Since $\hat X$ has only isolated singularities along E, namely the coordinate points of E of indices 2 and 3 in case (A), and the points of E where $i = a = g = 0$ or $i = a = e = 0$ in case (B), we have $\operatorname{Diff}_E(0) = 0$.

In both cases E is a prime divisor over X of plt type with $\operatorname{center}_X E = \{x\}$ and plt extraction $\hat{\pi}$, and $\hat{\pi}^* H|_E$ is trivial, so $W_{m,j} \subset H^0(E, \mathcal{O}_E(j))$.

Step 4. In this step we degenerate X to X_0 by a cocharacter of G_2.

Let $\gamma : \mathbb{G}_m \to \mathrm{SL}(\mathbb{C}^7)$ act on the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_7$ with weights $(-3, 2, 1, 3, -2, -1, 0)$. It fixes the four-form

$$\Omega := \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_7 - \mathbf{e}_4 \wedge \mathbf{e}_5 \wedge \mathbf{e}_6 \wedge \mathbf{e}_7 - \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_5 \wedge \mathbf{e}_6 - \mathbf{e}_1 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4 \wedge \mathbf{e}_6 - \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 \in \wedge^4 \mathbb{C}^7,$$

since in each of the five terms the four weights sum to 0. Identifying the matrix A of Set-up 2.14 with the two-form $\sum_{j_1<j_2} A_{j_1 j_2} \mathbf{e}_{j_1} \wedge \mathbf{e}_{j_2}$, a direct calculation shows that $\mathcal{W} = \{\eta \in \wedge^2 \mathbb{C}^7 : \eta \wedge \Omega = 0\}$. Up to a scalar, Ω differs from the non-degenerate four-form of [Kap11, Section 2] only in the sign of the first term, which is the sign compatible with the matrix A. Since $\mathrm{G}_2 \operatorname{Gr}(2,7)$ is the locus where A has rank at most two, it consists of the points $[v_1 \wedge v_2]$ of $\operatorname{Gr}(2,7)$ such that $v_1 \wedge v_2 \wedge \Omega = 0$, so γ preserves $\mathrm{G}_2 \operatorname{Gr}(2,7)$. Under the $\mathrm{SL}(\mathbb{C}^7)$-equivariant isomorphism $\wedge^4 \mathbb{C}^7 \cong \wedge^3 (\mathbb{C}^7)^\vee$ Ω corresponds to a general alternating trilinear form, so the Lie algebra of the stabilizer of Ω in $\mathrm{SL}(\mathbb{C}^7)$ is $\mathfrak{g}_2$ [FH91, Proposition 22.12]. Hence the identity component of the stabilizer is G_2, and γ, whose image is connected, is a cocharacter of G_2. We fix the sign convention once: γ acts on the coordinate functions of $\mathbb{P}^{13}$, and the entry $A_{j_1 j_2}$ has weight equal to the sum of the weights of $\mathbf{e}_{j_1}$ and $\mathbf{e}_{j_2}$. A direct check over the entries of the matrix of Set-up 2.14 shows that any coordinate z has weight $\mathrm{wt}(z) - 5$: for instance $h = A_{15}$ has weight $-3 - 2 = -5$, $i = A_{16}$ has weight -4, g and k occur in A_{14}, A_{25}, and A_{36}, all of weight 0, and $J = A_{24}$ has weight 5. For a section s of $\mathcal{O}(r)$ on $\mathrm{G}_2 \operatorname{Gr}(2,7)$ which is a weight vector, we define the *local weight* of s as the weight of s plus $5r$, that is, the weight of s/h^r. Thus the local weight of a coordinate z is $\mathrm{wt}(z)$, and the local weight of a bihomogeneous element of R_{G_2} of bidegree (r, j) is j. Since G_2 is connected and semisimple, it is its own derived group [Bor91, Proposition IV.14.2(3)], so it has no non-trivial character, and G_2 acts trivially on $\det H^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(r))$. Hence the sum of the weights on $H^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(r))$ is 0, and

$$\sum (\text{local weights on } H^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(r))) = 5r h^0(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(r)). \tag{9.1}$$

Write $\ell_1 = \sum_z \xi_{1,z} z$ and $\ell_2 = \sum_z \xi_{2,z} z$, where z runs through the basis $n, f, g, g + k, c, l, b, d, M, J$, whose members are homogeneous, with $\mathrm{wt}(g + k) = 5$, and ℓ_1, ℓ_2 are the forms of **Step 1**. For $t \in \mathbb{A}^1$ and $\epsilon = 1, 2$ let

$$\ell_{\epsilon,t} := \sum_z \xi_{\epsilon,z} t^{\mathrm{wt}(z) - d_\epsilon} z.$$

Then $\ell_{\epsilon,1} = \ell_\epsilon$, $\ell_{\epsilon,0} = \ell_\epsilon^{\mathrm{lead}}$, and $\ell_{1,t}, \ell_{2,t}$ are linearly independent for any t. Let $\mathcal{X} \subset \mathrm{G}_2 \operatorname{Gr}(2,7) \times \mathbb{A}^1$ be defined by $\ell_{1,t}$ and $\ell_{2,t}$, and let $\phi : \mathcal{X} \to \mathbb{A}^1$ be the projection. Then ϕ is the pullback of the universal linear section along the morphism $t \to \operatorname{Span}(\ell_{1,t}, \ell_{2,t})$, so ϕ is flat with Gorenstein fibers of pure dimension three by Lemma 2.16, $\phi^{-1}(1) = X$, and $\phi^{-1}(0) = X_0$. For $\zeta \in \mathbb{G}_m$ we have

$$\ell_{\epsilon,t}(\gamma(\zeta) y) = \sum_z \xi_{\epsilon,z} t^{\mathrm{wt}(z) - d_\epsilon} \zeta^{\mathrm{wt}(z) - 5} z(y) = \zeta^{d_\epsilon - 5} \ell_{\epsilon, t\zeta}(y),$$

so the action $(y, t) \to (\gamma(\zeta) y, \zeta^{-1} t)$ preserves $\mathcal{X}$, and $\phi^{-1}(t) \cong X$ for any $t \neq 0$.

Step 5. In this step we compute the sections of $\mathcal{O}(m)$ on the fibers of ϕ and lift a basis from X_0 to X.

Since $-K_{\mathrm{G}_2 \operatorname{Gr}(2,7)} = \mathcal{O}(3)$ [Lat23, p. 568], adjunction gives $H = \mathcal{O}(1)|_X$. By Kodaira vanishing [Laz04a, Theorem 4.2.1], $H^q(\mathrm{G}_2 \operatorname{Gr}(2,7), \mathcal{O}(r)) = 0$ for $q > 0$ and $r \geq -2$. Since $\ell_{1,t}, \ell_{2,t}$ define a subscheme of pure codimension two in the Cohen–Macaulay variety $\mathrm{G}_2 \operatorname{Gr}(2,7)$, the Koszul complex

$$0 \to \mathcal{O}(m-2) \to \mathcal{O}(m-1)^{\oplus 2} \to \mathcal{O}(m) \to \mathcal{O}_{\phi^{-1}(t)}(m) \to 0$$

is exact for any t and $m \geq 0$. By the vanishing above and since R_{G_2} is the section ring of $\mathcal{O}(1)$ [RR85, Theorem 3], we obtain

$$H^0(\phi^{-1}(t), \mathcal{O}(m)) = R_{G_2,m}/(\ell_{1,t}R_{G_2,m-1} + \ell_{2,t}R_{G_2,m-1}),$$

where $R_{G_2,r}$ is the degree r part of R_{G_2}. In particular $H^0(X_0, \mathcal{O}(m))$ is the degree m part R_m of R, and

$$N_m = h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m)) - 2h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-1)) + h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-2))$$

for any t. Let $\mathcal{O}_{\mathcal{X}}(m)$ be the restriction to $\mathcal{X}$ of the pullback of $\mathcal{O}(m)$ on $\mathrm{G}_2\,\mathrm{Gr}(2,7)$. Thus $\mathcal{F}_m := \phi_*\mathcal{O}_{\mathcal{X}}(m)$ is the cokernel of a morphism of free $\mathbb{C}[t]$-modules whose fibers have constant rank, so $\mathcal{F}_m$ is locally free of rank N_m and $\mathcal{F}_m \otimes \mathbb{C}(t) = H^0(\phi^{-1}(t), \mathcal{O}(m))$ for any t. Since X_0 is integral (**Step 2**) and $\{h \neq 0\} \cap X_0$ is dense, the substitution $h = 1$ is injective on R_m. By **Step 1** it maps R into Q, and it maps the bidegree (m, j) part $R_{m,j}$ of R into Q_j.

Grade $\mathbb{C}[i, a, e, n, g][t]$ by wt and $\deg t = -1$. Then $\ell_{\epsilon,t}/h$ is homogeneous of degree d_ϵ, and the $\mathbb{C}[t]$-module $H^0(\mathbb{A}^1, \mathcal{F}_m)$ is graded, with homogeneous elements of degree j of the form $\sum_{u\geq 0} t^u\sigma_u$, where $\sigma_u \in R_{G_2,m}$ has local weight $j + u$. The grading is induced by the $\mathbb{G}_m$-action of **Step 4**. Fix a basis $(\sigma_{\alpha,0})_{1\leq\alpha\leq N_m}$ of R_m of bihomogeneous elements, of bidegrees (m, j_α). Since the restriction $H^0(\mathbb{A}^1, \mathcal{F}_m) \to R_m$ to $t = 0$ is surjective and graded, each $\sigma_{\alpha,0}$ lifts to a homogeneous element $\sigma_\alpha = \sum_{u\geq 0} t^u\sigma_{\alpha,u}$ of degree j_α. The determinant of $(\sigma_\alpha)_\alpha$ is a homogeneous section of $\det \mathcal{F}_m$ which is non-zero at 0, so the zero set of the determinant is a $\mathbb{G}_m$-stable closed subset of $\mathbb{A}^1$ not containing 0, hence empty. Therefore the restrictions $s_\alpha := \sum_u \sigma_{\alpha,u}|_X$ form a basis of $H^0(X, mH)$.

Step 6. In this step we prove the description of $W_{m,j}$, the formula for $S_m(E)$, and the bound on $\delta_x(X)$.

Recall from **Step 3** that ord_E is the weighted order modulo $\mathfrak{J}$, and recall the class $\bar\theta \in Q_j$ of a function $\theta \in \mathcal{O}_{X,x}$ with $\mathrm{ord}_E(\theta) = j$. Since $h(x) \neq 0$, we view sections of $\mathcal{O}(mH)$ as functions near x by dividing by h^m. The element $\sigma_{\alpha,u}$ restricted to the chart is homogeneous of degree $j_\alpha + u$, so $\mathrm{ord}_E(s_\alpha) \geq j_\alpha$, and the class of s_α in the degree j_α part of Q is the image $\sigma_{\alpha,0}|_{h=1}$, which is non-zero by **Step 5**. Thus $\mathrm{ord}_E(s_\alpha) = j_\alpha$ and $\bar s_\alpha = \sigma_{\alpha,0}|_{h=1}$. For any j the elements $\sigma_{\alpha,0}|_{h=1}$ with $j_\alpha = j$ are linearly independent in Q_j. Let $s = \sum_\alpha \chi_\alpha s_\alpha$ be non-zero and $j_0 := \min\{j_\alpha : \chi_\alpha \neq 0\}$. Then the class of s in the degree j_0 part of Q is $\sum_{j_\alpha=j_0} \chi_\alpha\bar s_\alpha \neq 0$, so $\mathrm{ord}_E(s) = j_0$. Hence $\{s \in H^0(X, mH) : \mathrm{ord}_E(s) \geq j\}$ has the basis $(s_\alpha)_{j_\alpha\geq j}$, and by **Step 3** the image $W_{m,j}$ of $\{s \in H^0(X, mH) : \mathrm{ord}_E(s) \geq j\}$ in $H^0(E, \mathcal{O}_E(j)) = Q_j$ is spanned by the elements $\sigma_{\alpha,0}|_{h=1}$ with $j_\alpha = j$, that is, $W_{m,j}$ is the image of $R_{m,j}$. Since the substitution $h = 1$ is injective on R_m, it maps $R_{m,j}$ isomorphically onto $W_{m,j}$, and in case (B) we have $Q_j = H^0(E, \mathcal{O}_{\mathbb{P}(1,3,10)}(2j))$ by **Step 3**.

The jumping numbers of ord_E on $H^0(X, mH)$ are the numbers j_α, so

$$mN_mS_m(E) = \sum_{j\geq 0} j \dim W_{m,j} = \sum_\alpha j_\alpha$$

is the sum of the local weights on R_m. The exact Koszul sequence of **Step 5** at $t = 0$ is γ-equivariant, and multiplication by $\ell_\epsilon^{\mathrm{lead}}$ raises the local weight by d_ϵ. By (9.1) we obtain

$$\begin{aligned}\sum_\alpha j_\alpha &= 5mh^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m)) - \big(10(m-1) + d_1 + d_2\big)h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-1))\\ &\quad + \big(5(m-2) + d_1 + d_2\big)h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-2))\\ &= 5mN_m + (10 - d_1 - d_2)\big(h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-1)) - h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-2))\big),\end{aligned}$$

where the second equality follows from the formula for N_m in **Step 5**. Since $d_1 + d_2 = 9$ in case (A) and $d_1 + d_2 = 8$ in case (B), dividing by mN_m yields the formula for $S_m(E)$.

By [Lat23, p. 568], $\mathrm{G}_2\,\mathrm{Gr}(2,7)$ is a fivefold of degree 18, so $h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(r)) = \frac{18}{5!}r^5 + O(r^4)$, and hence $h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-1)) - h^0(\mathrm{G}_2\,\mathrm{Gr}(2,7), \mathcal{O}(m-2)) = \frac{3}{4}m^4 + O(m^3)$ and

$N_m = 3m^3 + O(m^2)$. Thus the quotient in the formula for $S_m(E)$ is $\frac{1}{4} + O(1/m)$. Since $S_X(E) = \lim_{m\to+\infty} S_m(E)$ [AZ22, Lemma 2.9], we obtain $S_X(E) = 5 + \frac{1}{4} = \frac{21}{4}$ in case (A) and $S_X(E) = 5 + \frac{1}{2} = \frac{11}{2}$ in case (B).

Finally, $\mathrm{Diff}_E(0) = 0$ and $\delta_{Z'}(E, 0; W) \geq \delta(E; W)$ for any irreducible closed subset $Z' \subset E$. Hence, by Theorem 2.9 with $Z = \{x\}$, $\Delta = 0$, the complete series of $H = -K_X$, and the plt extraction $\hat{\pi}$ of E, we obtain

$$\delta_x(X) \geq \min\left\{\frac{A_X(E)}{S_X(E)}, \delta(E; W)\right\}.$$

Here $A_X(E)/S_X(E) = 6 \cdot \frac{4}{21} = \frac{8}{7}$ in case (A) and $A_X(E)/S_X(E) = 7 \cdot \frac{2}{11} = \frac{14}{11}$ in case (B). □

9.2. **Case (A).** In this subsection, we bound $S(W; v)$ for all prime divisors v over the exceptional divisor E of case (A), and we lift the bounds on $S(W; v)$ to bounds on $S_X(F)$ for prime divisors F over X. The subsection ends with Theorem 9.9. We treat the points of E region by region, in the notation of Set-up 9.3. At q, a weighted blow-up of q and the polytope of a semigroup of leading exponents (Lemma 3.4) compute the values of S exactly (Lemma 9.5), which bound S at q (Proposition 9.7). At a point of $U_E \backslash \{q\}$, a curve through the point and through q, smooth at q, is the flag (Proposition 9.8). At a point of $C_i \backslash \{p_2, p_3\}$, the curve C_i is the flag. At p_3, a toric blow-up of p_3 and a semigroup of leading exponents, enlarged by the leading exponents of two infinite families of sections when $\kappa \neq -2$, give S for the exceptional curve. The last two regions are treated in Theorem 9.9. In all cases we lift the bound by adjunction to a curve on E or on a blow-up of E, Lemma 3.1, and inversion of adjunction [Kaw07, Theorem].

Set-up 9.3. Let X, x, E, W, q, p_2, p_3, C_i, U_E, f, $\nu_{1,3}$, and $\nu_{1,2}$ be as follows.

(1) X and x are as in Construction-Proposition 9.2 in case (A), with the parameters $\lambda \neq 0$ and κ of Lemma 9.1.
(2) $E \cong \mathbb{P}(1, 2, 3)$ and W are the divisor and the refinement of Construction-Proposition 9.2, and $(i : a : e)$ are the coordinates of E.
(3) $q := (1 : 0 : 0)$, $p_2 := (0 : 1 : 0)$, $p_3 := (0 : 0 : 1)$, $C_i := (i = 0) \subset E$, and $U_E := (E \backslash C_i) \cup \{p_2\}$.
(4) $f := ie - a^2 \in \mathbb{C}[i, a, e]$, so that a and $f = e - a^2$ are regular parameters at q on the chart $i = 1$ of E.
(5) $\nu_{1,3}$ and $\nu_{1,2}$ are the monomial valuations at q with weights $(1, 3)$ and $(1, 2)$ on (a, f).

Set-up 9.4. Notation and conditions as in Set-up 9.3. Let ψ, z, $\varphi : \tilde{E} \to E$, C', p_*, and p_∞ be as follows.

(1) $\psi := ae - \lambda if \in \mathbb{C}[i, a, e]$.
(2) $z := \lambda f - a^3$ on the chart $i = 1$, so that (a, z) are regular parameters at q.
(3) If $\kappa \neq 0$, then $\varphi : \tilde{E} \to E$ is the weighted blow-up of q with weights $(1, 3)$ on (a, z), and if $\kappa = 0$, then $\varphi : \tilde{E} \to E$ is the weighted blow-up of q with weights $(1, 2)$ on (a, f).
(4) C' is the exceptional curve of φ.
(5) $p_* \in C'$ is the point where $z/a^3 = 0$ if $\kappa \neq 0$, and the point where $f/a^2 = 0$ if $\kappa = 0$.
(6) $p_\infty \in C'$ is the intersection point of C' with the strict transform of $(a = 0)$.

The next lemma computes the values of S for the valuations $\nu_{1,3}$ and $\nu_{1,2}$ at q. The list of tetrahedra for the polytopes Δ_{gen} and Δ_0 of the proof of Lemma 9.5 is in Appendix A.

Lemma 9.5. *Notation and conditions as in Set-up 9.4. Assume that $\kappa \neq -1$. Then the following hold.*

(1) The substitution $h = 1$ of Construction-Proposition 9.2 maps n, g, k, c, l, b, d, M, J to

$$\begin{aligned} &n \to -\lambda f, \quad g \to \kappa\psi, \quad k \to -(\kappa+1)\psi, \quad c \to e^2 - \lambda a f, \quad l \to -af - (\kappa+1)i\psi, \\ &b \to -\kappa a\psi - ic, \quad d \to \kappa e\psi + ac, \quad M \to -ec - \kappa\lambda f\psi, \quad J \to fc - \kappa(\kappa+1)\psi^2, \end{aligned} \tag{9.2}$$

where the letters c,l,b,d,M,J on the right denote the images of the coordinates c,l,b,d,M,J in $\mathbb{C}[i,a,e]$, and from now on c,l,b,d,M,J denote the images in $\mathbb{C}[i,a,e]$. Moreover, for any integers $m \geq 1$ and $j \geq 0$, $W_{m,j}$ is spanned by the products of m elements of $\{1,i,a,e,f,\psi,c,l,b,d,M,J\}$ whose second degrees $0,1,2,3,4,5,6,6,7,8,9,10$ add up to j.

(2) *The pair $(\tilde{E},C')$ is plt, $-C'$ is φ-ample, and $\operatorname{ord}_{C'} = \nu_{1,3}$, $A_E(C') = 4$, $C' \cong \mathbb{P}(1,3)$, $\operatorname{Diff}_{C'}(0) = \frac{2}{3}p_\infty$, $(C'^2) = -\frac{1}{3}$ if $\kappa \neq 0$, and $\operatorname{ord}_{C'} = \nu_{1,2}$, $A_E(C') = 3$, $C' \cong \mathbb{P}(1,2)$, $\operatorname{Diff}_{C'}(0) = \frac{1}{2}p_\infty$, $(C'^2) = -\frac{1}{2}$ if $\kappa = 0$. In particular, the $\mathbb{Q}$-divisor of zeros B on C' of any non-zero element of $W^{C'}_{m,j,k}$ satisfies $B \geq 0$, and $\deg B = \frac{k}{3}$ if $\kappa \neq 0$ and $\deg B = \frac{k}{2}$ if $\kappa = 0$.*

(3) *If $\kappa \neq 0$, then $S(W;\nu_{1,3}) = \frac{13}{4}$ and $S(W^{C'};p_*) = \frac{55}{72}$.*

(4) *If $\kappa = 0$, then $S(W;\nu_{1,2}) = \frac{91}{36}$, $S(W^{C'};p_*) = \frac{29}{36}$, and $S(W;\nu_{1,3}) = \frac{10}{3}$.*

Proof. (1) The ring R of Construction-Proposition 9.2 satisfies $n = -\lambda f$ and $g = -\kappa(g+k)$. On the chart $h = 1$, (2.1) gives $k = -g - ae - in$, so $g + k \to -ae + \lambda i f = -\psi$. Thus $g \to \kappa\psi$ and $k \to -(\kappa+1)\psi$, and the remaining identities of (2.1) imply (9.2). The ring R is generated by the fourteen coordinates, which have usual degree one, and they map to $1, i, a, e, -\lambda f, f, \kappa\psi, -(\kappa+1)\psi, c, l, b, d, M, J$. Since κ and $\kappa+1$ are not both zero, Construction-Proposition 9.2 implies (1).

(2) In the coordinates (a,z), respectively (a,f), the morphism φ is a toric morphism. Since $z - \lambda f = -a^3$, the monomial valuation with weights $(1,3)$ on (a,z) is $\nu_{1,3}$. A toric calculation implies that $\operatorname{ord}_{C'}$ is the monomial valuation with the weights of φ, $A_E(C') = 1+3$, respectively $1+2$, by [CLS11, Theorem 8.2.3], $(C'^2) = -\frac{1}{3}$, respectively $-\frac{1}{2}$, by [CLS11, Proposition 6.4.4(c)], and $-C'$ is φ-ample since $(C'^2) < 0$. The surface $\tilde{E}$ has exactly one singular point on C', the point p_∞, a cyclic quotient singularity of index 3, respectively 2, and C' is a toric boundary curve, so $(\tilde{E},C')$ is plt and $\operatorname{Diff}_{C'}(0) = \frac{2}{3}p_\infty$, respectively $\frac{1}{2}p_\infty$ [Kol13, (3.36.1)]. An element of $W^{C'}_{m,j,k}$ is the restriction to C' of a section of $\varphi^*\mathcal{O}_E(j) \otimes \mathcal{O}_{\tilde{E}}(-kC')$, and the restriction of $\varphi^*\mathcal{O}_E(j)$ to C' is trivial, so the divisor of a non-zero element of $W^{C'}_{m,j,k}$ has degree $(-kC' \cdot C')$.

(3) Assume that $\kappa \neq 0$. On the chart of $\tilde{E}$ with coordinates (a,t), $t := z/a^3$, we have $C' = (a = 0)$, $p_* = (a = t = 0)$, and $\varphi^*(a^u z^v) = a^{u+3v}t^v$. Hence, if the $\nu_{1,3}$-leading form of an element $s \in W_{m,j}$ is $\sum_{u+3v=k} \gamma_{uv} a^u z^v$, then the image of s in $W^{C'}_{m,j,k}$ vanishes at p_* to order $\min\{v : \gamma_{uv} \neq 0\}$. On the chart $i = 1$ we have $f = (z + a^3)/\lambda$ and $e = f + a^2$, and a direct calculation shows the $\nu_{1,3}$-leading forms of Table 7. The coefficients in Table 7 are non-zero since $\lambda \neq 0$ and $\kappa \neq 0, -1$. Let

$$\Phi := 1 \cdot M \cdot M + 2(\kappa+1)\lambda M\psi f - \frac{\kappa+1}{\kappa}\lambda^2 J f^2 \in W_{3,18},$$

where $1 \in W_{1,0}$ is the generator of (1), so that each of the three terms is a product of three generators of (1). Substituting the leading forms $\lambda f \to a^3 + z$, $M \to (a^3 + \kappa\lambda f)z$, $\psi \to -z$, and $J \to -\kappa(\kappa+1)z^2$, the $\nu_{1,3}$-leading form of Φ is

$$\big[(a^3 + \kappa\lambda f)^2 - 2(\kappa+1)\lambda f(a^3 + \kappa\lambda f) + (\kappa+1)^2\lambda^2 f^2\big]z^2 = (a^3 - \lambda f)^2 z^2 = z^4.$$

Thus $\nu_{1,3}(\Phi) = 12$ and the image of Φ vanishes at p_* to order 4.

Let $\Gamma \subset \mathbb{Z}_{\geq 0} \times \mathbb{Z}^3$ be the semigroup generated by the twelve tuples $(1,j,u,v)$ of Table 7 for $\kappa \neq 0$ and by $(3,18,0,4)$. The tuples of 1, i, a, ψ are $(1,0,0,0)$, $(1,1,0,0)$, $(1,2,1,0)$, $(1,5,0,1)$, so Γ is a semigroup generating $\mathbb{Z}^4$ as a group. Let $\Gamma_m := \{(j,u,v) : (m,j,u,v) \in \Gamma\}$ (the notation of Lemma 3.4). By Lemma 3.4, the closure Δ_{gen} of $\bigcup_m \frac{1}{m}\Gamma_m$ is the convex hull of the normalized generators, and the vertices of Δ_{gen} in the coordinates (j,u,v) are

$$(0,0,0),\ (1,0,0),\ (4,3,0),\ (6,0,\tfrac{4}{3}),\ (6,0,1),\ (9,3,1),\ (10,0,2).$$

TABLE 7. Leading forms on the chart $i = 1$ of the generators of Lemma 9.5(1), where (u, v) is the exponent of the monomial $a^u z^v$, respectively $a^u f^v$, of smallest degree in z, respectively f.

generator	j	$\nu_{1,3}$-leading form, $\kappa \neq 0$	(u,v)	$\nu_{1,2}$-leading form, $\kappa = 0$	(u,v)
1	0	1	$(0,0)$	1	$(0,0)$
i	1	1	$(0,0)$	1	$(0,0)$
a	2	a	$(1,0)$	a	$(1,0)$
e	3	a^2	$(2,0)$	a^2+f	$(2,0)$
f	4	$(a^3+z)/\lambda$	$(3,0)$	f	$(0,1)$
ψ	5	$-z$	$(0,1)$	$-\lambda f$	$(0,1)$
c	6	$-az$	$(1,1)$	$-\lambda af$	$(1,1)$
l	6	$(\kappa+1)z$	$(0,1)$	λf	$(0,1)$
b	7	$(\kappa+1)az$	$(1,1)$	λaf	$(1,1)$
d	8	$-(\kappa+1)a^2z$	$(2,1)$	$-\lambda a^2 f$	$(2,1)$
M	9	$((\kappa+1)a^3+\kappa z)z$	$(3,1)$	$\lambda af(a^2+f)$	$(3,1)$
J	10	$-\kappa(\kappa+1)z^2$	$(0,2)$	$-\lambda af^2$	$(1,2)$

Coning the boundary triangles of Δ_{gen} to the origin decomposes Δ_{gen} into the tetrahedra listed in Appendix A, and summing over them we obtain

$$\mathrm{vol}(\Delta_{\mathrm{gen}}) = 3, \qquad \int_{\Delta_{\mathrm{gen}}} u\,dx = \frac{23}{8}, \qquad \int_{\Delta_{\mathrm{gen}}} v\,dx = \frac{55}{24}. \tag{9.3}$$

For any $(m, j, u, v) \in \Gamma$ let $s_{m,j,u,v} \in W_{m,j}$ be a product of generators and of Φ with exponent sum (m, j, u, v). The $\nu_{1,3}$-leading form of a product is the product of the leading forms, so $\nu_{1,3}(s_{m,j,u,v}) = u + 3v$ and the image of $s_{m,j,u,v}$ in $W^{C'}_{m,j,u+3v}$ vanishes at p_* to order v. Distinct (u, v) have distinct pairs $(u + 3v, v)$. Hence, for fixed m, the elements $s_{m,j,u,v}$ are linearly independent, they form a basis of their span $V_m \subset \bigoplus_j W_{m,j}$ compatible with $\nu_{1,3}$, and their images in $\bigoplus_{j,k} W^{C'}_{m,j,k}$ form a basis compatible with the order at p_*. By Lemma 3.4 and (9.3),

$$\dim V_m = 3m^3 + O(m^2),$$
$$\sum_{(j,u,v)\in\Gamma_m} (u+3v) = m^4\Big(\frac{23}{8} + 3\cdot\frac{55}{24}\Big) + O(m^3),$$
$$\sum_{(j,u,v)\in\Gamma_m} v = \frac{55}{24}m^4 + O(m^3).$$

By Lemma 2.12(2), $N_m = 3m^3 + O(m^2)$, so V_m has codimension $O(m^2)$ in $\bigoplus_j W_{m,j}$, and the image of V_m has codimension $O(m^2)$ in $\bigoplus_{j,k} W^{C'}_{m,j,k}$. On the chart $i = 1$ any element of $W_{m,j}$ is a polynomial in (a, f) of degree at most $2j \le 20m$, so the $\nu_{1,3}$-order of a non-zero element of $W_{m,j}$ is at most $60m$, and by (2) the order at p_* of an element of $W^{C'}_{m,j,k}$ is at most $\frac{k}{3} \le 20m$. The sum of the jumping numbers of a filtration is additive for a subspace and the quotient with the induced filtrations. Thus replacing V_m by $\bigoplus_j W_{m,j}$ changes both sums of orders by $O(m^3)$. Dividing by $mN_m = 3m^4 + O(m^3)$, we obtain

$$S_m(W;\nu_{1,3}) = \frac{1}{3}\Big(\frac{23}{8} + \frac{55}{8}\Big) + O\Big(\frac{1}{m}\Big) = \frac{13}{4} + O\Big(\frac{1}{m}\Big), \qquad S_m(W^{C'};p_*) = \frac{55}{72} + O\Big(\frac{1}{m}\Big),$$

and (3) follows by letting $m \to +\infty$ [AZ22, Lemma 2.9].

(4) Assume that $\kappa = 0$. On the chart of $\tilde{E}$ with coordinates (a, t), $t := f/a^2$, we have $\varphi^*(a^u f^v) = a^{u+2v} t^v$, and the order at p_* of an image in $W^{C'}$ is the smallest degree in f of the $\nu_{1,2}$-leading form. A direct calculation shows the $\nu_{1,2}$-leading forms of Table 7, whose coefficients are non-zero since $\lambda \neq 0$. Let Γ_0 be the semigroup generated by the twelve tuples $(1, j, u, v)$ of

the last column. It generates $\mathbb{Z}^4$ as a group, since the tuples of 1, i, a, ψ in the last column are $(1,0,0,0)$, $(1,1,0,0)$, $(1,2,1,0)$, $(1,5,0,1)$. By Lemma 3.4, the closure Δ_0 of $\bigcup_m \frac{1}{m}(\Gamma_0)_m$ is the convex hull of the normalized generators, and it has the vertices

$$(0,0,0),\ (1,0,0),\ (3,2,0),\ (4,0,1),\ (6,0,1),\ (9,3,1),\ (10,1,2),$$

and the list of tetrahedra in Appendix A implies

$$\operatorname{vol}(\Delta_0) = 3, \qquad \int_{\Delta_0} u dx = \frac{11}{4}, \qquad \int_{\Delta_0} v dx = \frac{29}{12}. \tag{9.4}$$

The argument of (3) with $u+2v$ in place of $u+3v$ implies

$$S(W;\nu_{1,2}) = \frac{1}{3}\Big(\frac{11}{4} + 2\cdot\frac{29}{12}\Big) = \frac{91}{36}, \qquad S(W^{C'};p_*) = \frac{1}{3}\cdot\frac{29}{12} = \frac{29}{36}.$$

For $\nu_{1,3}$, the leading forms of a, e, f are a, a^2, f, and the leading forms of ψ, c, l, b, d, M, J are $(a^3-\lambda f)$ times 1, a, -1, $-a$, a^2, $-a^3$, af, respectively. In particular, the leading forms of c, M, J are $a^4-\lambda af$, $\lambda a^3 f - a^6$, $a^4 f - \lambda a f^2$. The monomial of largest degree in f of the $\nu_{1,3}$-leading form of a product is the product of the corresponding monomials of the factors, and for the twelve generators the exponents (u,v) of the monomials of largest degree in f are exactly the exponents of the last column of Table 7. Thus the argument of (3), applied to Γ_0 and the order $\nu_{1,3} = u+3v$, implies

$$S(W;\nu_{1,3}) = \frac{1}{3}\Big(\frac{11}{4} + 3\cdot\frac{29}{12}\Big) = \frac{10}{3}. \qquad \square$$

The next lemma lifts a bound on a curve to E and to X.

Lemma 9.6. *Notation and conditions as in Set-up 9.4. Let C be either a prime divisor on E such that (E,C) is plt, in which case let $E_C := E$ and $\varphi_C := \mathrm{id}_E$, or $C = C'$, in which case let $E_C := \tilde{E}$ and $\varphi_C := \varphi$. Let $\operatorname{Diff}_C(0)$ be the different of the pair (E_C, C) on C, let $o \in \varphi_C(C)$ be a closed point, and let $\lambda' > 0$ be a real number. Assume that $\kappa \neq -1$, that $\lambda' S(W;C) \le A_E(C)$, and that $\lambda' S(W^C;p) \le 1 - \operatorname{mult}_p \operatorname{Diff}_C(0)$ for any closed point $p \in C$ with $\varphi_C(p) = o$. Then*

(1) $\lambda' S(W;v) \le A_E(v)$ for any prime divisor v over E with $o \in \operatorname{center}_E v$, and

(2) $\lambda' S_X(F) + \big(6 - \frac{21}{4}\lambda'\big)\operatorname{mult}_F E \le A_X(F)$ for any prime divisor F over X with $o \in \operatorname{center}_{\hat{X}} F$.

Proof. The divisor C is of plt type over E with plt extraction φ_C (Lemma 9.5(2) for $C = C'$). Let $G := \varphi_C^{-1}(o)$. If C is a curve on E, then $G = \{o\} \subset C$, and if $C = C'$, then $o = q$ and $G = C'$. Since C is a normal curve, the prime divisors over C whose center intersects G are the closed points $p \in G$, and $A_{C,\operatorname{Diff}_C(0)}(p) = 1 - \operatorname{mult}_p \operatorname{Diff}_C(0)$. Thus Lemma 3.1, with $(X,\Delta,V,E) := (E,0,W,C)$, $Y = E_C$, and λ' in place of λ, implies

$$A_E(v) \ge \lambda' S(W;v) + \big(A_E(C) - \lambda' S(W;C)\big)\operatorname{mult}_v C \ge \lambda' S(W;v)$$

for any prime divisor v over E with $\operatorname{center}_{E_C} v \cap G \neq \emptyset$. If $o \in \operatorname{center}_E v$, then $\operatorname{center}_{E_C} v$ intersects $\varphi_C^{-1}(o) = G$, which proves (1). Since $\operatorname{Diff}_E(0) = 0$, (1) implies $\lambda' S(W;v) \le A_{E,\operatorname{Diff}_E(0)}(v)$ for any prime divisor v over E with $o \in \operatorname{center}_E v$. Lemma 3.1, with $Y = \hat{X}$, $G = \{o\}$, and λ' in place of λ, applied to the complete series of H, implies

$$A_X(F) \ge \lambda' S_X(F) + \big(A_X(E) - \lambda' S_X(E)\big)\operatorname{mult}_F E$$

for any prime divisor F over X with $o \in \operatorname{center}_{\hat{X}} F$. Since $A_X(E) = 6$ and $S_X(E) = \frac{21}{4}$ (Construction-Proposition 9.2), (2) holds. $\square$

Proposition 9.7. *Notation and conditions as in Set-up 9.4. Then $\kappa \neq -1$, and $S(W;v) \le \frac{23}{24}A_E(v)$ for any prime divisor v over E with $q \in \operatorname{center}_E v$, and $S(W;v) \le \frac{11}{12}A_E(v)$ for such v if $\kappa = 0$. Hence any prime divisor F over X with $\operatorname{center}_{\hat{X}} F = \{q\}$ satisfies $S_X(F) \le \frac{23}{24}A_X(F) - \frac{1}{2}\operatorname{mult}_F E < A_X(F)$, and $S_X(F) \le \frac{11}{12}A_X(F) - \frac{1}{4}\operatorname{mult}_F E$ if $\kappa = 0$.*

Proof. We first show that $\kappa \neq -1$. On the chart $i = 1$ of $\mathrm{G}_2\,\mathrm{Gr}(2,7)$, which is an affine space with coordinates a, f, g, h, l, the line of X through x intersects the chart in the line $a = f = g = l = 0$. On $a = f = g = l = 0$ we have $b = c = d = e = J = M = 0$ and $g + k = af + hl = 0$, and the identities (2.2) imply that the differentials of b, c, d, J, M vanish along the line and that $dk = hdl$, where h is the coordinate on the line. Since $\mu = 0$ in case (A), we obtain

$$d\ell_2 = (1+\kappa)dg + (\kappa h + \sigma_l)dl$$

along the line. Suppose that $\kappa = -1$. Then $d\ell_2 = 0$ at the point x_0 of the line with $h = \sigma_l$, and the Zariski tangent space of X at x_0 is the kernel of $d\ell_1$ in the 5-dimensional tangent space of the chart, so $\dim T_{x_0}X \geq 4$. This contradicts the smoothness of the threefold X. Thus $\kappa \neq -1$.

Since $\kappa \neq -1$, we may use Lemma 9.5. Let $r := 3$ if $\kappa \neq 0$ and $r := 2$ if $\kappa = 0$, and let $p \in C'\backslash\{p_*\}$ be a closed point. For any m, j, k, choose a basis of $W^{C'}_{m,j,k}$ compatible with the filtrations by the orders at p and at p_* [AZ22, Lemma 3.1]. By Lemma 9.5(2), any element s of the basis satisfies $\mathrm{ord}_p(s) + \mathrm{ord}_{p_*}(s) \leq \frac{k}{r}$. Summing over the bases and dividing by mN_m, we obtain

$$S_m(W^{C'};p) + S_m(W^{C'};p_*) \leq \frac{1}{r}S_m(W;C'),$$

because $\sum_{j,k} k \dim W^{C'}_{m,j,k} = mN_mS_m(W;C')$. Letting $m \to +\infty$ [AZ22, Lemma 2.9], Lemma 9.5(3)(4) implies

$$S(W^{C'};p) \leq \frac{13}{12} - \frac{55}{72} = \frac{23}{72} \quad \text{if } \kappa \neq 0, \qquad S(W^{C'};p) \leq \frac{91}{72} - \frac{29}{36} = \frac{11}{24} \quad \text{if } \kappa = 0. \tag{9.5}$$

Assume that $\kappa \neq 0$. We use Lemma 9.6 with $C = C'$, $o = q$, and $\lambda' = \frac{24}{23}$. By Lemma 9.5(2)(3) and (9.5),

$$\frac{24}{23}S(W;C') = \frac{78}{23} < 4 = A_E(C'), \qquad \frac{24}{23}S(W^{C'};p_*) = \frac{55}{69} < 1,$$
$$\frac{24}{23}S(W^{C'};p) \leq \frac{1}{3} \quad \text{for } p \neq p_*.$$

Since $\mathrm{Diff}_{C'}(0) = \frac{2}{3}p_\infty$ and $p_* \neq p_\infty$, we have $1 - \mathrm{mult}_{p_*}\mathrm{Diff}_{C'}(0) = 1$ and $1 - \mathrm{mult}_p\mathrm{Diff}_{C'}(0) \geq \frac{1}{3}$ for any $p \in C'$. Thus Lemma 9.6(1) implies $S(W;v) \leq \frac{23}{24}A_E(v)$ for any prime divisor v over E with $q \in \mathrm{center}_E v$, and Lemma 9.6(2) implies

$$\frac{24}{23}S_X(F) + \frac{12}{23}\mathrm{mult}_F E \leq A_X(F), \quad \text{that is,} \quad S_X(F) \leq \frac{23}{24}A_X(F) - \frac{1}{2}\mathrm{mult}_F E,$$

for any prime divisor F over X with $q \in \mathrm{center}_{\hat{X}} F$.

Assume that $\kappa = 0$. Let $\lambda' \in \{\frac{24}{23}, \frac{12}{11}\}$. By Lemma 9.5(2)(4) and (9.5),

$$\lambda' S(W;C') \leq \frac{12}{11}\cdot\frac{91}{36} = \frac{91}{33} < 3 = A_E(C'),$$
$$\lambda' S(W^{C'};p_*) \leq \frac{12}{11}\cdot\frac{29}{36} = \frac{29}{33} < 1,$$
$$\lambda' S(W^{C'};p) \leq \frac{12}{11}\cdot\frac{11}{24} = \frac{1}{2} \quad \text{for } p \neq p_*.$$

Since $\mathrm{Diff}_{C'}(0) = \frac{1}{2}p_\infty$ and $p_* \neq p_\infty$, we have $1 - \mathrm{mult}_{p_*}\mathrm{Diff}_{C'}(0) = 1$ and $1 - \mathrm{mult}_p\mathrm{Diff}_{C'}(0) \geq \frac{1}{2}$ for any $p \in C'$. Thus the assumptions of Lemma 9.6 hold for $C = C'$, $o = q$, and both values of λ'. For $\lambda' = \frac{12}{11}$, Lemma 9.6(1) implies $S(W;v) \leq \frac{11}{12}A_E(v) \leq \frac{23}{24}A_E(v)$ for any prime divisor v over E with $q \in \mathrm{center}_E v$, and Lemma 9.6(2) implies

$$\frac{12}{11}S_X(F) + \frac{3}{11}\mathrm{mult}_F E \leq A_X(F), \quad \text{that is,} \quad S_X(F) \leq \frac{11}{12}A_X(F) - \frac{1}{4}\mathrm{mult}_F E,$$

for any prime divisor F over X with $q \in \mathrm{center}_{\hat{X}} F$. For $\lambda' = \frac{24}{23}$, Lemma 9.6(2) implies $S_X(F) \leq \frac{23}{24}A_X(F) - \frac{1}{2}\mathrm{mult}_F E$ for any prime divisor F over X with $q \in \mathrm{center}_{\hat{X}} F$.

Finally, let F be a prime divisor over X with $\operatorname{center}_{\hat{X}} F = \{q\}$. Since $q \in E$, $\operatorname{mult}_F E > 0$, and $A_X(F) > 0$. Thus $\frac{23}{24}A_X(F) - \frac{1}{2}\operatorname{mult}_F E < A_X(F)$. □

At a point of U_E other than q we take as a flag a curve through the point and through q, smooth at q. The inequality (9.7) of **Step 1** of the proof below and the value $S(W;\nu_{1,3})$ bound the order of the refined series at q from below, and the degree of the refined series bounds the sum of the orders at q and at the point.

Proposition 9.8. *Notation and conditions as in Set-up 9.4. Then $S(W;v) \le \frac{47}{48}A_E(v)$ for any prime divisor v over E whose center intersects U_E. Hence any prime divisor F over X with $\operatorname{center}_X F = \{x\}$ and $\operatorname{center}_{\hat{X}} F \cap U_E \ne \emptyset$ satisfies $S_X(F) \le \frac{47}{48}A_X(F) - \frac{5}{8}\operatorname{mult}_F E < A_X(F)$.*

Proof. The proof is long, so we divide it into several steps. By Proposition 9.7, $\kappa \ne -1$, and Lemma 9.5(3)(4) implies

$$S(W;\nu_{1,3}) \ge \frac{13}{4}. \tag{9.6}$$

Step 1. In this step we show that

$$I_q(D_1, D_2) \ge \frac{1}{3}\nu_{1,3}(D_1)\nu_{1,3}(D_2) \tag{9.7}$$

for any $\mathbb{Q}$-divisors $D_1, D_2 \ge 0$ on a neighborhood of q in E without common components, where I_q is the local intersection number. By linearity we may assume that $D_1 = (\eta_1 = 0)$ and $D_2 = (\eta_2 = 0)$ for $\eta_1, \eta_2 \in \mathbb{C}[[a, f]]$. Let $\rho : \mathbb{C}[[a, f]] \to \mathbb{C}[[y_1, y_2]]$ be the homomorphism with $a = y_1$ and $f = y_2^3$. Then $\mathbb{C}[[y_1, y_2]]$ is a free $\mathbb{C}[[a, f]]$-module with basis $1, y_2, y_2^2$, so

$$\dim_{\mathbb{C}} \mathbb{C}[[y_1, y_2]]/(\rho\eta_1, \rho\eta_2) = 3\dim_{\mathbb{C}} \mathbb{C}[[a, f]]/(\eta_1, \eta_2) = 3I_q(D_1, D_2).$$

Since $\rho(a^u f^v) = y_1^u y_2^{3v}$, the multiplicity of $\rho\eta_\ell$ at the origin is $\nu_{1,3}(\eta_\ell)$ for $\ell = 1, 2$. A common component of $(\rho\eta_1 = 0)$ and $(\rho\eta_2 = 0)$ would map onto a common component of D_1 and D_2, so $\rho\eta_1$ and $\rho\eta_2$ have no common factor. After replacing $\rho\eta_\ell$ by their truncations of sufficiently high degree, which changes neither the multiplicities nor the quotient (the quotient has finite length), [Ful08, Section 3.3, property (5)] implies

$$\dim_{\mathbb{C}} \mathbb{C}[[y_1, y_2]]/(\rho\eta_1, \rho\eta_2) \ge \nu_{1,3}(\eta_1)\nu_{1,3}(\eta_2).$$

Thus (9.7) holds.

Step 2. In this step we bound $S(W^C;p)$ for a curve C through q. Let C be a prime divisor on E such that (E, C) is plt and C is smooth at q. Let L be the class of $\mathcal{O}_E(1)$, so that $(L^2) = \frac{1}{6}$. Let w be the integer with $C \sim wL$, and let $t := \nu_{1,3}(C)$ and $s := S(W;C)$. We show that

$$S(W^C;p) \le \frac{7w}{8} - \frac{13t}{12} + \Big(\frac{t^2}{3} - \frac{w^2}{6}\Big)s \tag{9.8}$$

for any closed point $p \in C\backslash\{q\}$. For any m, let $s_m := S_m(W;C)$, and recall that $S_m(E) = \frac{1}{mN_m}\sum_j j \dim W_{m,j}$ (Definition 2.8). For any j choose a basis of $W_{m,j}$ compatible with the filtrations by the order along C and by $\nu_{1,3}$ [AZ22, Lemma 3.1], and let D_m be the m-basis type divisor of W associated with the bases of $W_{m,j}$, $j \ge 0$ (Definition 2.8). Then $D_m = s_m C + D'_m$, where $D'_m \ge 0$ is a $\mathbb{Q}$-divisor with $C \not\subset \operatorname{Supp} D'_m$, $\nu_{1,3}(D_m) = S_m(W;\nu_{1,3})$, and $D'_m|_C$ is an m-basis type divisor of W^C. Since C is smooth at q, $\operatorname{mult}_q(D'_m|_C) = I_q(D'_m, C)$, and (9.7) implies

$$S_m(W^C;q) \ge \operatorname{mult}_q(D'_m|_C) \ge \frac{t}{3}\nu_{1,3}(D'_m) = \frac{t}{3}\big(S_m(W;\nu_{1,3}) - ts_m\big).$$

On the other hand, for any m, j, k choose a basis of $W^C_{m,j,k}$ compatible with the filtrations by the orders at p and at q. The $\mathbb{Q}$-divisor of zeros B on C of any element of the basis satisfies $B \ge 0$

and $\deg B = ((jL - kC)\cdot C) = \frac{jw-kw^2}{6}$, so the sum of the orders at p and at q is at most $\frac{jw-kw^2}{6}$. Summing over the bases and dividing by mN_m, we obtain

$$S_m(W^C;p) + S_m(W^C;q) \le \frac{w}{6}\big(S_m(E) - ws_m\big).$$

Thus

$$S_m(W^C;p) \le \frac{w}{6}\big(S_m(E) - ws_m\big) - \frac{t}{3}\big(S_m(W;\nu_{1,3}) - ts_m\big).$$

As $m \to +\infty$, $S_m(E) \to \frac{21}{4}$, $s_m \to s$, $S_m(W;\nu_{1,3}) \to S(W;\nu_{1,3})$, and $S_m(W^C;p) \to S(W^C;p)$ [AZ22, Lemma 2.9]. Since $t > 0$, (9.6) implies (9.8).

Step 3. In this step we introduce four kinds of curves through q and bound $S(W^C;p)$ on them. For $\tau \in \mathbb{C}\setminus\{0\}$ let

$$C_\tau := (ae - \tau if = 0), \qquad C_a := (a = 0), \qquad C_e := (e = 0), \qquad C_f := (ie - a^2 = 0)$$

on E. On the chart $i = 1$ the equation of C_τ is $(a-\tau)e + \tau a^2 = 0$. Since $a = \tau$ is impossible on C_τ, the curve $C_\tau \cap (i \ne 0)$ is the graph $e = \tau a^2/(\tau - a)$, and it is smooth. The equation $ae - \tau if$ has no factor i, so C_τ is integral, and $C_\tau \cap C_i = \{p_2, p_3\}$. In the coordinates (a, f) the equation of C_τ is $a^3 + (a-\tau)f$, so C_τ is smooth at q and $\nu_{1,3}(C_\tau) = 3$. The curves C_a, C_e, C_f are integral, and their intersections with the chart $i = 1$ are the smooth curves $a = 0$, $f + a^2 = 0$, $f = 0$, so they are smooth at q with $\nu_{1,3}$ equal to 1, 2, 3. Moreover, $C_a \cap C_i = \{p_3\}$, $C_e \cap C_i = \{p_2\}$, and $C_f \cap C_i = \{p_3\}$.

We show that (E, C) is plt for any $C \in \{C_\tau, C_a, C_e, C_f\}$. The four curves are smooth on the smooth chart $i = 1$. On the index-two cover at p_2, with coordinates (i, e) of weights $\frac{1}{2}(1,1)$, the preimages of C_e and C_τ are $e = 0$ and $e + \tau i - \tau i^2 e = 0$. On the index-three cover at p_3, with coordinates (i, a) of weights $\frac{1}{3}(1,2)$, the preimages of C_a, C_f, C_τ are $a = 0$, $i - a^2 = 0$, $a - \tau i^2 + \tau i a^2 = 0$. Each of the five preimages is a coordinate axis for equivariant coordinates, namely (i, e), $(i, e + \tau i - \tau i^2 e)$, (i, a), $(i - a^2, a)$, $(i, a - \tau i^2 + \tau i a^2)$. Thus each pair (E, C) is plt, and $\mathrm{Diff}_{C_e}(0)$ has coefficient $\frac{1}{2}$ at p_2 [Kol13, (3.36.1)].

We have $C_\tau \sim 5L$, $C_a \sim 2L$, $C_e \sim 3L$, $C_f \sim 4L$. Since $q \in C$ for any $C \in \{C_\tau, C_a, C_e, C_f\}$, Proposition 9.7 implies $0 \le S(W;C) \le \frac{23}{24}$. For C_τ we also have

$$S(W;C_\tau) \ge \frac{1}{8}. \tag{9.9}$$

To prove (9.9), let $\vartheta_\tau \in W_{2,5}$ be the product of $1 \in W_{1,0}$ and $\psi \in W_{1,5}$ plus $\lambda - \tau$ times the product of $i \in W_{1,1}$ and $f \in W_{1,4}$, where $\psi = ae - \lambda if$ (Lemma 9.5(1)). Then $\vartheta_\tau = ae - \tau if$, whose divisor is C_τ. Multiplication by ϑ_τ^k maps $W_{m-2k,j}$ injectively into the sections of $W_{m,j+5k}$ vanishing to order at least k along C_τ. Thus the sum of the orders along C_τ on $\bigoplus_j W_{m,j}$ is at least $\sum_{k\ge1} N_{m-2k}$. Since $N_m = 3m^3 + O(m^2)$ by Lemma 2.12(2), dividing by mN_m and letting $m \to +\infty$ we obtain $S(W;C_\tau) \ge \int_0^{1/2}(1-2y)^3 dy = \frac{1}{8}$.

Substituting $(w,t) = (5,3), (2,1), (3,2), (4,3)$ for C_τ, C_a, C_e, C_f in (9.8), and using (9.9) and $0 \le S(W;C) \le \frac{23}{24}$, we obtain, for any closed point $p \ne q$ of the respective curve,

$$\begin{aligned}
S(W^{C_\tau};p) &\le \frac{9}{8} - \frac{7}{6}S(W;C_\tau) \le \frac{47}{48},\\
S(W^{C_a};p) &\le \frac{2}{3} - \frac{1}{3}S(W;C_a) \le \frac{2}{3},\\
S(W^{C_e};p) &\le \frac{11}{24} - \frac{1}{6}S(W;C_e) \le \frac{11}{24},\\
S(W^{C_f};p) &\le \frac{1}{4} + \frac{1}{3}S(W;C_f) \le \frac{1}{4} + \frac{1}{3}\cdot\frac{23}{24} = \frac{41}{72}.
\end{aligned} \tag{9.10}$$

See also Table 95 in Appendix A.

Step 4. In this step we prove the bound on E. Let $p \in U_E \backslash \{q\}$ be a closed point. We choose a curve $C \ni p$ as follows. If $p = p_2$, let $C := C_e$. Otherwise p is a point of the chart $i = 1$. If $a(p) = 0$, $e(p) = 0$, or $f(p) = 0$, let C be C_a, C_e, or C_f, respectively. Otherwise let $\tau := a(p)e(p)/f(p) \neq 0$ and $C := C_\tau$, so that $p \in C_\tau$. In all cases C is smooth at p: for $p \neq p_2$, since C is smooth on the chart $i = 1$, and for $p = p_2$, since the preimage of C_e on the index-two cover is smooth. We use Lemma 9.6 with C, $o = p$, and $\lambda' = \frac{48}{47}$. First,

$$\frac{48}{47} S(W; C) \leq \frac{48}{47} \cdot \frac{23}{24} = \frac{46}{47} < 1 = A_E(C).$$

If $p \neq p_2$, then E and C are smooth at p, so $\operatorname{mult}_p \operatorname{Diff}_C(0) = 0$, and $\frac{48}{47} S(W^C; p) \leq \frac{48}{47} \cdot \frac{47}{48} = 1$ by (9.10). If $p = p_2$, then $\operatorname{mult}_{p_2} \operatorname{Diff}_{C_e}(0) = \frac{1}{2}$, and (9.10) implies

$$\frac{48}{47} S(W^{C_e}; p_2) \leq \frac{48}{47} \cdot \frac{11}{24} = \frac{22}{47} < \frac{1}{2}.$$

Thus Lemma 9.6(1) implies

$$S(W; v) \leq \frac{47}{48} A_E(v) \tag{9.11}$$

for any prime divisor v over E with $p \in \operatorname{center}_E v$.

Now let v be a prime divisor over E whose center intersects U_E. If $q \in \operatorname{center}_E v$, then $S(W; v) \leq \frac{23}{24} A_E(v) \leq \frac{47}{48} A_E(v)$ by Proposition 9.7. Otherwise any closed point $p \in \operatorname{center}_E v \cap U_E$ satisfies $p \neq q$, and (9.11) holds. This proves the bound $S(W; v) \leq \frac{47}{48} A_E(v)$ of Proposition 9.8.

Step 5. In this step we lift the bound to X. Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$ and $\operatorname{center}_{\hat{X}} F \cap U_E \neq \emptyset$. Then $\operatorname{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(x) = E$. If $p_2 \in \operatorname{center}_{\hat{X}} F$, let $p := p_2$. Otherwise $\operatorname{center}_{\hat{X}} F \cap (E \backslash C_i)$ is a non-empty open subset of the irreducible closed set $\operatorname{center}_{\hat{X}} F$, and we let p be a closed point of $\operatorname{center}_{\hat{X}} F \cap (E \backslash C_i)$. In both cases $p \in U_E$. The center of any prime divisor v over E with $p \in \operatorname{center}_E v$ intersects U_E, so **Step 4** implies $\frac{48}{47} S(W; v) \leq A_E(v) = A_{E, \operatorname{Diff}_E(0)}(v)$. Thus Lemma 3.1, with $Y = \hat{X}$, $G = \{p\}$, and $\frac{48}{47}$ in place of λ, applied to the complete series of H, implies

$$A_X(F) \geq \frac{48}{47} S_X(F) + \left(6 - \frac{48}{47} \cdot \frac{21}{4}\right) \operatorname{mult}_F E = \frac{48}{47} S_X(F) + \frac{30}{47} \operatorname{mult}_F E.$$

Thus $S_X(F) \leq \frac{47}{48} A_X(F) - \frac{5}{8} \operatorname{mult}_F E$. Since $\operatorname{center}_{\hat{X}} F \subset E$, $\operatorname{mult}_F E > 0$, so $\frac{47}{48} A_X(F) - \frac{5}{8} \operatorname{mult}_F E < A_X(F)$. □

By Proposition 9.8, it remains to treat the prime divisors over E whose centers are points of C_i other than p_2. The next theorem treats them and completes case (A).

Theorem 9.9. *Notation and conditions as in Set-up 9.3. Then $S(W; v) \leq \frac{47}{48} A_E(v)$ for any prime divisor v over E. Hence $\delta_x(X) \geq 48/47$, and any prime divisor F over X with $\operatorname{center}_X F = \{x\}$ satisfies*

$$S_X(F) \leq \frac{47}{48} A_X(F) - \frac{5}{8} \operatorname{mult}_F E < A_X(F).$$

At p_3 the stronger bound $S(W; v) \leq \frac{23}{24} A_E(v)$ holds for any v with $p_3 \in \operatorname{center}_E v$, and the toric valuation ν_{p_3} at p_3 with weights $\nu_{p_3}(i) = 1/3$ and $\nu_{p_3}(a) = 2/3$ on the index-three cover of the chart $e \neq 0$ satisfies $S(W; \nu_{p_3}) = 35/72$.

Proof. **Step 1.** In this step we prove $S(W; v) \leq \frac{47}{48} A_E(v)$ when $\operatorname{center}_E v$ contains a point $p \in C_i \backslash \{p_2, p_3\}$.

The surface $E \cong \mathbb{P}(1, 2, 3)$ is smooth at p, and $C_i \cong \mathbb{P}^1$ is smooth at p. Near p_2 and p_3, the pair (E, C_i) is the quotient of $\mathbb{A}^2$ with a coordinate line by μ_2, respectively μ_3, acting freely outside the origin. Thus (E, C_i) is plt [KM98, Definition 2.34 and Proposition 5.20(2)], and C_i is a prime divisor of plt type over E, with the identity of E as plt extraction. Since p is a smooth point of E, the different $\operatorname{Diff}_{C_i}(0)$ has coefficient 0 at p, so $A_{C_i, \operatorname{Diff}_{C_i}(0)}(p) = 1$.

We bound $S(W^{C_i};p)$. Since $C_i \in |\mathcal{O}_E(1)|$ and $(\mathcal{O}_E(1)^2) = \frac{1}{6}$, a section $s \in W_{m,j}$ with $\operatorname{ord}_{C_i}(s) = \ell$ defines an element of W^{C_i} whose $\mathbb{Q}$-divisor D_s on C_i satisfies $D_s \geq 0$ and has degree

$$\big((j-\ell)\mathcal{O}_E(1)\cdot C_i\big) = \frac{j-\ell}{6}.$$

Thus any m-basis type divisor D of W^{C_i} is a $\mathbb{Q}$-divisor on C_i with $D \geq 0$ and $\deg D = \frac{1}{6}(S_m(E) - S_m(W;C_i))$ [AZ22, Section 3.1]. Since $S_m(W^{C_i};p)$ is the largest value of $\operatorname{mult}_p D$ for an m-basis type divisor D of W^{C_i} [AZ22, Section 2.6], and $\operatorname{mult}_p D \leq \deg D$, we obtain

$$S_m(W^{C_i};p) \leq \frac{1}{6}\big(S_m(E) - S_m(W;C_i)\big) \leq \frac{1}{6}S_m(E),$$

where $S_m(E)$ is as in Definition 2.8. Since $\lim_m S_m(E) = S_X(E) = \frac{21}{4}$ [AZ22, Lemma 2.9], we have

$$\frac{48}{47}S(W^{C_i};p) \leq \frac{48}{47}\cdot\frac{7}{8} = \frac{42}{47} < 1 = A_{C_i,\operatorname{Diff}_{C_i}(0)}(p).$$

By Lemma 3.1 applied to $(E,0)$, the series W, the divisor C_i, $G = \{p\}$, and $48/47$ in place of λ, we obtain

$$A_E(v) \geq \frac{48}{47}S(W;v) + \Big(1 - \frac{48}{47}S(W;C_i)\Big)\operatorname{mult}_v C_i$$

for any prime divisor v over E with $p \in \operatorname{center}_E v$. Since $p_2 \in C_i \cap U_E$, Proposition 9.8 implies $S(W;C_i) \leq \frac{47}{48}A_E(C_i) = \frac{47}{48}$. Thus $S(W;v) \leq \frac{47}{48}A_E(v)$.

Step 2. In this step we normalize λ, write down the images of the generators, and introduce the toric valuation $\nu := \nu_{p_3}$ and a monomial order adapted to it.

By Construction-Proposition 9.2, the substitution $h = 1$ maps the bidegree (m,j) part of R isomorphically onto $W_{m,j} \subset H^0(E,\mathcal{O}_E(j)) = \mathbb{C}[i,a,e]_j$, the space of weighted homogeneous polynomials of degree j in (i,a,e). Thus the product of $s \in W_{m,j}$ and $s' \in W_{m',j'}$ is the product of polynomials, and $ss' \in W_{m+m',j+j'}$. By Proposition 9.7, $\kappa \neq -1$, so the bigraded algebra $\bigoplus_{m,j} W_{m,j}$ is generated by the images of the twelve elements of Lemma 9.5(1).

Write $a = \lambda a'$ and $e = \lambda^2 e'$. In the coordinates $(i : a' : e')$, the images of f, ψ, c, l, b, d, M, and J are λ^2, λ^3, λ^4, λ^3, λ^4, λ^5, λ^6, and λ^6 times the images for the parameter value $\lambda = 1$. The coordinate change $(i : a : e) \to (i : a' : e')$ is an automorphism of E which fixes p_2 and p_3, preserves C_i, U_E, and ν, and maps W onto the series obtained for $\lambda = 1$, since the coordinate change multiplies each generator by a non-zero scalar. Possibly replacing $(i : a : e)$ with $(i : a' : e')$, we may assume that $\lambda = 1$. Then the images of the generators in $\mathbb{C}[i,a,e]$ are the images (9.2) of Lemma 9.5(1) for $\lambda = 1$:

$$\begin{aligned} &\psi = ae - if, && c = e^2 - af, && l = -af - (\kappa+1)i\psi,\\ &b = -\kappa a\psi - ic, && d = ac + \kappa e\psi, && M = -ec - \kappa f\psi,\\ &J = fc - \kappa(\kappa+1)\psi^2. \end{aligned} \tag{9.12}$$

The chart $e \neq 0$ of E is the quotient of $\mathbb{A}^2$ with coordinates (i,a) by μ_3 acting with weights $(1,2)$. The toric lattice of the chart $e \neq 0$ is $\mathbb{Z}^2 + \mathbb{Z}(\frac{1}{3},\frac{2}{3})$, and p_3 corresponds to the cone generated by $(1,0)$ and $(0,1)$, whose rays correspond to $C_i = (i = 0)$ and $C_a := (a = 0)$. Let $\varphi_3 : E_3 \to E$ be the toric morphism given by the star subdivision along the primitive vector $(\frac{1}{3},\frac{2}{3})$ of $\mathbb{Z}^2 + \mathbb{Z}(\frac{1}{3},\frac{2}{3})$ [CLS11, Section 11.1 and Lemma 11.1.3], let C_ν be the exceptional curve, and let $\nu = \nu_{p_3}$ be the valuation $\operatorname{ord}_{C_\nu}$. Then $\nu(i) = \frac{1}{3}$ and $\nu(a) = \frac{2}{3}$, and for a non-zero $s \in \mathbb{C}[i,a,e]_j$,

$$\nu(s) = \frac{1}{3}\min\{\alpha_i + 2\alpha_a : i^{\alpha_i}a^{\alpha_a}e^{\alpha_e} \text{ is a monomial of } s\},$$

since on the index-three cover the section s is given by $s(i,a,1)$. The canonical divisor of a toric variety is $-\sum_\rho D_\rho$ [CLS11, Theorem 8.2.3], so $A_E(C_\nu)$ is the value at $(\frac{1}{3},\frac{2}{3})$ of the linear

function which equals 1 at $(1,0)$ and $(0,1)$:

$$A_E(C_\nu) = \frac{1}{3} + \frac{2}{3} = 1.$$

In $\mathbb{Z}^2 + \mathbb{Z}(\frac{1}{3}, \frac{2}{3})$, the cones generated by $\{(1,0), (\frac{1}{3}, \frac{2}{3})\}$ and by $\{(\frac{1}{3}, \frac{2}{3}), (0,1)\}$ have multiplicities 2 and 1, and the cone generated by $\{(1,0), (0,1)\}$ has multiplicity 3. Thus $C_\nu \cong \mathbb{P}^1$ contains exactly two torus-fixed points: the index-two point $o_i := C_\nu \cap \tilde{C}_i$ and the smooth point $o_a := C_\nu \cap \tilde{C}_a$, where $\tilde{C}_i$ and $\tilde{C}_a$ are the strict transforms. The function $\zeta := a/i^2$ has $\nu(\zeta) = 0$, restricts to a coordinate on C_ν with $o_a = (\zeta = 0)$ and $o_i = (\zeta = +\infty)$, and is a local equation of $\tilde{C}_a$ at o_a. In particular, $\tilde{C}_a$ and C_ν meet transversally at o_a. We have $(1,0) - 3(\frac{1}{3}, \frac{2}{3}) + 2(0,1) = 0$, and the cone generated by $\{(1,0), (\frac{1}{3}, \frac{2}{3})\}$ has multiplicity 2. Thus

$$(C_\nu^2) = \frac{-3}{1 \cdot 2} = -\frac{3}{2}$$

[CLS11, Proposition 6.4.4(c)]. The curve C_ν is a toric boundary curve of E_3 passing through the cyclic quotient point o_i of index 2 and the smooth point o_a, so (E_3, C_ν) is plt and $\mathrm{Diff}_{C_\nu}(0) = \frac{1}{2} o_i$ [Kol13, (3.36.1)]. Since $(C_\nu^2) < 0$, $-C_\nu$ is φ_3-ample, so C_ν is a prime divisor of plt type over E with plt extraction φ_3.

Let $\prec$ be the total order on $\mathbb{Z}^3$ defined by $(\alpha_i, \alpha_a, \alpha_e) \prec (\alpha_i', \alpha_a', \alpha_e')$ if $(\alpha_i + 2\alpha_a, \alpha_i, \alpha_e)$ is lexicographically smaller than $(\alpha_i' + 2\alpha_a', \alpha_i', \alpha_e')$. The order $\prec$ is compatible with addition. For a non-zero $s \in \mathbb{C}[i, a, e]$, let $\mathrm{in}_\nu(s)$ be the sum of the terms of s whose monomials have ν-value $\nu(s)$, let $\mathrm{lm}(s)$ be the $\prec$-smallest monomial of s, and let $\epsilon(s) \in \mathbb{Z}^3_{\geq 0}$ be the exponent of $\mathrm{lm}(s)$. Then $\mathrm{lm}(s)$ is a monomial of $\mathrm{in}_\nu(s)$, $\nu(s) = \nu(\mathrm{lm}(s))$,

$$\mathrm{in}_\nu(ss') = \mathrm{in}_\nu(s)\,\mathrm{in}_\nu(s'), \quad \text{and} \quad \epsilon(ss') = \epsilon(s) + \epsilon(s').$$

For $s \in \mathbb{C}[i, a, e]_j$ with $\epsilon(s) = (\alpha_i, \alpha_a, \alpha_e)$ we have $\alpha_i + 2\alpha_a + 3\alpha_e = j$, so $(\alpha_i + 2\alpha_a, \alpha_i)$ determines $\epsilon(s)$. If $s_1, \dots, s_r \in W_{m,j}$ have pairwise distinct $\epsilon(s_k)$, then for any non-zero linear combination $s = \sum_k c_k s_k$, $\mathrm{lm}(s)$ is the $\prec$-smallest of the $\mathrm{lm}(s_k)$ with $c_k \neq 0$. Thus $s_1, \dots, s_r$ are linearly independent, $\nu(s) = \min\{\nu(s_k) : c_k \neq 0\}$, and $s_1, \dots, s_r$ is a basis of $\mathrm{Span}(s_1, \dots, s_r)$ compatible with the filtration by ν.

A direct calculation from (9.12) yields Table 8. The entry $\epsilon(d) = (0,1,2)$ uses $\kappa + 1 \neq 0$, and the entry $\epsilon(l)$ depends on whether $\kappa + 2 = 0$.

TABLE 8. The twelve generators of W for $\lambda = 1$ with their bidegrees, ν-lowest parts, and leading exponents.

s	bidegree	$\mathrm{in}_\nu(s)$	$\epsilon(s)$
1	$(1,0)$	1	$(0,0,0)$
i	$(1,1)$	i	$(1,0,0)$
a	$(1,2)$	a	$(0,1,0)$
e	$(1,3)$	e	$(0,0,1)$
f	$(1,4)$	ie	$(1,0,1)$
ψ	$(1,5)$	$e(a - i^2)$	$(0,1,1)$
c	$(1,6)$	e^2	$(0,0,2)$
l	$(1,6)$	$ie\big((\kappa+1)i^2 - (\kappa+2)a\big)$	$(1,1,1)$ if $\kappa \neq -2$; $(3,0,1)$ if $\kappa = -2$
b	$(1,7)$	$-ie^2$	$(1,0,2)$
d	$(1,8)$	$e^2\big((\kappa+1)a - \kappa i^2\big)$	$(0,1,2)$
M	$(1,9)$	$-e^3$	$(0,0,3)$
J	$(1,10)$	ie^3	$(1,0,3)$

Step 3. In this step we assume $\kappa \neq -2$ and construct two families of sections P_m and Q_r of W with

$$\frac{\epsilon(P_m)}{m} \to (2,0,2) \quad \text{and} \quad \frac{\epsilon(Q_r)}{2r+1} \to \Big(2,0,\frac{1}{2}\Big).$$

Let $P_1 := -b$ and $P_2 := lM - (\kappa+2)\psi J$, and for $m \geq 2$ let

$$P_{m+1} := (\kappa+2)dP_m + (\kappa+1)lJP_{m-1}.$$

By induction on m, P_m is a polynomial in the generators of usual degree m and second degree $8m-1$, so $P_m \in W_{m,8m-1}$. We show

$$\mathrm{in}_\nu(P_m) = i^{2m-1}e^{2m}. \tag{9.13}$$

For $m=1$ this is Table 8. For $m=2$, the terms lM and $(\kappa+2)\psi J$ both have ν-value 1, and the sum of the ν-value 1 parts is

$$\begin{aligned} &-ie\big((\kappa+1)i^2 - (\kappa+2)a\big)e^3 - (\kappa+2)e(a-i^2)\cdot ie^3 \\ &= \big(-(\kappa+1)+(\kappa+2)\big)i^3e^4 + \big((\kappa+2)-(\kappa+2)\big)iae^4 = i^3e^4. \end{aligned}$$

Assume (9.13) for $m-1$ and m, where $m \geq 2$. Then dP_m and lJP_{m-1} both have ν-value $\frac{2m+1}{3}$, and the sum of the ν-value $\frac{2m+1}{3}$ parts of $(\kappa+2)dP_m$ and $(\kappa+1)lJP_{m-1}$ is

$$\begin{aligned} &(\kappa+2)e^2\big((\kappa+1)a - \kappa i^2\big)i^{2m-1}e^{2m} + (\kappa+1)ie\big((\kappa+1)i^2-(\kappa+2)a\big)ie^3i^{2m-3}e^{2m-2} \\ &= \big((\kappa+1)^2 - \kappa(\kappa+2)\big)i^{2m+1}e^{2m+2} + \big((\kappa+2)(\kappa+1)-(\kappa+1)(\kappa+2)\big)ai^{2m-1}e^{2m+2} \\ &= i^{2m+1}e^{2m+2}. \end{aligned}$$

All the other monomials of P_{m+1} have larger ν-value. Thus (9.13) holds for $m+1$. In particular, $\epsilon(P_m) = (2m-1,0,2m)$.

For the second family, let

$$\theta_1 := -il, \quad \theta_2 := a\psi, \quad \theta_3 := \frac{\kappa\theta_1 - (\kappa+2)^2\theta_2}{\kappa+1},$$

which belong to $W_{2,7}$. Let $Q_0 := i$, let

$$Q_1 := \frac{i^2l + (\kappa+2)\big(a^2f - ia\psi\big)}{\kappa+1},$$

and for $r \geq 1$ let

$$Q_{r+1} := \theta_3 Q_r + \frac{\theta_1^2 Q_{r-1}}{\kappa+1}.$$

By induction on r, $Q_r \in W_{2r+1,7r+1}$. We show

$$\mathrm{in}_\nu(Q_r) = i^{4r+1}e^r. \tag{9.14}$$

For $r=0$ this is clear. For $r=1$, (9.12) implies $a^2f - ia\psi = -a^4 + i^2af = -a^4 + i^3ae - i^2a^3$, and

$$i^2l = -(\kappa+2)i^3ae + (\kappa+1)i^5e + i^2a^3 - (\kappa+1)i^4a^2.$$

The monomials i^3ae and i^5e have ν-value $\frac{5}{3}$, and the other monomials have ν-value $\frac{8}{3}$. The coefficient of i^3ae in $i^2l + (\kappa+2)(a^2f - ia\psi)$ is $-(\kappa+2)+(\kappa+2) = 0$, so $\mathrm{in}_\nu(Q_1) = i^5e$.

For the induction step, we compute in $\mathbb{C}[i^{\pm1}, a, e]$ with $\xi := i^4e$ and $\zeta = a/i^2$, so that $\nu(\xi) = \frac{4}{3}$ and $\nu(\zeta) = 0$. By Table 8,

$$\mathrm{in}_\nu(\theta_1) = \xi\big((\kappa+2)\zeta - (\kappa+1)\big), \qquad \mathrm{in}_\nu(\theta_2) = \xi(\zeta^2 - \zeta),$$

and

$$\frac{\kappa\,\mathrm{in}_\nu(\theta_1) - (\kappa+2)^2\,\mathrm{in}_\nu(\theta_2)}{\kappa+1} = \xi\Big(-\frac{(\kappa+2)^2}{\kappa+1}\zeta^2 + 2(\kappa+2)\zeta - \kappa\Big).$$

The coefficient $-(\kappa+2)^2/(\kappa+1)$ of $\xi\zeta^2$ on the right-hand side is non-zero. Thus the right-hand side equals $\mathrm{in}_\nu(\theta_3)$, and a direct expansion shows

$$\xi^2 - \mathrm{in}_\nu(\theta_3)\xi - \frac{\mathrm{in}_\nu(\theta_1)^2}{\kappa+1} = 0. \tag{9.15}$$

Assume (9.14) for $r-1$ and r, where $r \geq 1$. Then $\theta_3 Q_r$ and $\theta_1^2 Q_{r-1}$ both have ν-value $\frac{4r+5}{3}$, and by (9.15) the sum of their ν-value $\frac{4r+5}{3}$ parts in Q_{r+1} is

$$\mathrm{in}_\nu(\theta_3) i\xi^r + \frac{\mathrm{in}_\nu(\theta_1)^2}{\kappa+1} i\xi^{r-1} = i\xi^{r-1}\xi^2 = i^{4r+5} e^{r+1} \neq 0.$$

Thus (9.14) holds for $r+1$. In particular, $\epsilon(Q_r) = (4r+1, 0, r)$.

Step 4. In this step we prove $S(W;\nu) = \frac{35}{72}$.

For a subspace $V' \subset W_{m,j}$ with the filtration $\mathcal{F}^t V' := \{s \in V' : \nu(s) \geq t\}$, let

$$\mathrm{wt}_\nu(V') := \int_0^\infty \dim \mathcal{F}^t V' dt,$$

which is the sum of $\nu(s)$ over a basis of V' compatible with $\mathcal{F}$. Then $mN_m S_m(W;\nu) = \sum_j \mathrm{wt}_\nu(W_{m,j})$. Any non-zero $s \in W_{m,j}$ satisfies $\nu(s) \leq \frac{j}{3}$. Since $\bigoplus_{m,j} W_{m,j}$ is generated by the twelve elements in Table 8, whose second degrees are at most 10, we have $W_{m,j} = 0$ for $j > 10m$. Since $\mathcal{F}^t V' \subset \mathcal{F}^t W_{m,j}$ and $\dim \mathcal{F}^t W_{m,j} \leq \dim \mathcal{F}^t V' + \dim W_{m,j} - \dim V'$, we obtain

$$\mathrm{wt}_\nu(V') \leq \mathrm{wt}_\nu(W_{m,j}) \leq \mathrm{wt}_\nu(V') + \frac{10m}{3}\big(\dim W_{m,j} - \dim V'\big). \tag{9.16}$$

Assume first that $\kappa \neq -2$. Let $\Delta_\infty \subset \mathbb{R}^3$ be the convex polytope with vertices

$$(0,0,0),\ (1,0,0),\ (0,1,0),\ \Big(2,0,\frac{1}{2}\Big),\ (1,1,1),\ (2,0,2),\ (0,1,2),\ (0,0,3),\ (1,0,3).$$

In the coordinates $(\alpha_i, \alpha_a, \alpha_e)$, Δ_∞ is defined by the inequalities

$$\alpha_i, \alpha_a, \alpha_e \geq 0, \quad \alpha_a \leq 1, \quad \alpha_a + \alpha_e \leq 3, \quad \alpha_i + \alpha_a \leq 2,$$
$$\alpha_i + \alpha_a - 2\alpha_e \leq 1, \quad \alpha_i + 2\alpha_a + \alpha_e \leq 4, \quad 2\alpha_i + 3\alpha_a - 2\alpha_e \leq 3.$$

A direct calculation, using the triangulation of Δ_∞ into seven tetrahedra obtained by coning the boundary triangles to the origin and listed in Appendix A, shows

$$\mathrm{vol}(\Delta_\infty) = 3, \quad \int_{\Delta_\infty} \alpha_i = \frac{25}{12}, \quad \int_{\Delta_\infty} \alpha_a = \frac{55}{48}, \quad \int_{\Delta_\infty} \frac{\alpha_i + 2\alpha_a}{3} = \frac{35}{24}. \tag{9.17}$$

Here the volume of a tetrahedron with vertices $0, x_1, x_2, x_3$ is $|\det(x_1, x_2, x_3)|/6$, and the integral of a linear function over a tetrahedron is the volume times the value at the barycenter.

Fix an integer $n_0 \geq 1$. Let $\Gamma^{(n_0)} \subset \mathbb{Z}_{\geq 0} \times \mathbb{Z}^3$ be the semigroup generated by $(1, \epsilon(s))$ for the twelve generators s in Table 8, by $(m, \epsilon(P_m))$ for $1 \leq m \leq n_0$, and by $(2r+1, \epsilon(Q_r))$ for $0 \leq r \leq n_0$. All generators of $\Gamma^{(n_0)}$ have positive first coordinate, and $\Gamma^{(n_0)}$ generates $\mathbb{Z}^4$ as a group, since the elements $(1, \epsilon(s))$ for $s = 1, i, a, e$ do. Let $\Gamma_m^{(n_0)}$ be the degree m slice of $\Gamma^{(n_0)}$, and let Δ_{n_0} be the closure of $\bigcup_{m \geq 1} \frac{1}{m}\Gamma_m^{(n_0)}$. By Lemma 3.4 applied to $\Gamma^{(n_0)}$ and to the linear function $(\alpha_i + 2\alpha_a)/3$, Δ_{n_0} is the convex hull of the points x/m' for the generators (m', x) of $\Gamma^{(n_0)}$, and

$$\#\Gamma_m^{(n_0)} = \mathrm{vol}(\Delta_{n_0}) m^3 + O_{n_0}(m^2), \qquad \sum_{(\alpha_i,\alpha_a,\alpha_e) \in \Gamma_m^{(n_0)}} \frac{\alpha_i + 2\alpha_a}{3} = m^4 \int_{\Delta_{n_0}} \frac{\alpha_i + 2\alpha_a}{3} + O_{n_0}(m^3).$$

By Table 8, **Step 3**, and the inequalities defining Δ_∞, each point x/m' belongs to Δ_∞, so $\Delta_{n_0} \subset \Delta_\infty$.

For any $(m, \alpha_i, \alpha_a, \alpha_e) \in \Gamma^{(n_0)}$, choose a product $s_{(m,\alpha_i,\alpha_a,\alpha_e)}$ of the generators, of the $P_{m'}$, and of the $Q_{r'}$ with $\epsilon(s_{(m,\alpha_i,\alpha_a,\alpha_e)}) = (\alpha_i, \alpha_a, \alpha_e)$. By **Step 2**, $s_{(m,\alpha_i,\alpha_a,\alpha_e)} \in W_{m,j}$ with $j =$

$\alpha_i + 2\alpha_a + 3\alpha_e$. Let $V'_{m,j}$ be the span of the $s_{(m,\alpha_i,\alpha_a,\alpha_e)}$ with $\alpha_i + 2\alpha_a + 3\alpha_e = j$. By **Step 2**, the $s_{(m,\alpha_i,\alpha_a,\alpha_e)}$ with $\alpha_i + 2\alpha_a + 3\alpha_e = j$ form a basis of $V'_{m,j}$ compatible with the filtration by ν. Thus $\sum_j \dim V'_{m,j} = \#\Gamma_m^{(n_0)}$ and

$$\sum_j \mathrm{wt}_\nu(V'_{m,j}) = \sum_{(\alpha_i,\alpha_a,\alpha_e)\in\Gamma_m^{(n_0)}} \frac{\alpha_i + 2\alpha_a}{3}.$$

We have $\sum_j \dim W_{m,j} = N_m = 3m^3 + O(m^2)$ by Lemma 2.12(2). Summing (9.16) over j, dividing by mN_m, and letting m tend to infinity, we obtain

$$\frac{1}{3}\int_{\Delta_{n_0}} \frac{\alpha_i + 2\alpha_a}{3} \le S(W;\nu) \le \frac{1}{3}\int_{\Delta_{n_0}} \frac{\alpha_i + 2\alpha_a}{3} + \frac{10}{9}\big(3 - \mathrm{vol}(\Delta_{n_0})\big).$$

The polytope Δ_{n_0} contains the convex hull of the points $\epsilon(s)$ for the twelve generators s and of $\frac{\epsilon(P_{n_0})}{n_0}$ and $\frac{\epsilon(Q_{n_0})}{2n_0+1}$, and $\Delta_{n_0} \subset \Delta_\infty$. By **Step 3**, as n_0 tends to infinity, the convex hull of the points $\epsilon(s)$ for the twelve generators s and of $\frac{\epsilon(P_{n_0})}{n_0}$ and $\frac{\epsilon(Q_{n_0})}{2n_0+1}$ converges to the convex hull of the points $\epsilon(s)$ for the twelve generators s and of $(2,0,2)$ and $(2,0,\frac{1}{2})$, which is Δ_∞. Thus $\mathrm{vol}(\Delta_{n_0}) \to 3$ and $\int_{\Delta_{n_0}} \frac{\alpha_i+2\alpha_a}{3} \to \frac{35}{24}$ by (9.17). Letting n_0 tend to infinity, both bounds tend to $\frac{1}{3}\cdot\frac{35}{24} = \frac{35}{72}$. Hence $S(W;\nu) = \frac{35}{72}$.

Assume now that $\kappa = -2$. Then $\epsilon(l) = (3,0,1)$, and the convex hull $\tilde{\Delta}$ of the points $\epsilon(s)$ for the twelve generators s has vertices

$$(0,0,0),\ (1,0,0),\ (0,1,0),\ (3,0,1),\ (0,1,2),\ (0,0,3),\ (1,0,3),$$

and is defined by the inequalities

$$\alpha_i,\alpha_a,\alpha_e \ge 0,\quad \alpha_a + \alpha_e \le 3,\quad \alpha_i + \alpha_a - 2\alpha_e \le 1,\quad \alpha_i + 2\alpha_a + \alpha_e \le 4,\quad \alpha_i + 3\alpha_a \le 3.$$

A direct calculation, using the triangulation of $\tilde{\Delta}$ into four tetrahedra listed in Appendix A, shows

$$\mathrm{vol}(\tilde{\Delta}) = 3,\quad \int_{\tilde{\Delta}} \alpha_i = \frac{19}{8},\quad \int_{\tilde{\Delta}} \alpha_a = 1.$$

Let $\Gamma \subset \mathbb{Z}_{\ge 0}\times\mathbb{Z}^3$ be the semigroup generated by $(1,\epsilon(s))$ for the twelve generators s, and let Γ_m be the degree m slice. Then Γ generates $\mathbb{Z}^4$ as a group. By Lemma 3.4, the closure of $\bigcup_{m\ge 1}\frac{1}{m}\Gamma_m$ is $\tilde{\Delta}$, and

$$\#\Gamma_m = 3m^3 + O(m^2),\qquad \sum_{(\alpha_i,\alpha_a,\alpha_e)\in\Gamma_m} \frac{\alpha_i + 2\alpha_a}{3} = m^4\int_{\tilde{\Delta}} \frac{\alpha_i + 2\alpha_a}{3} + O(m^3).$$

For any $(m,\alpha_i,\alpha_a,\alpha_e) \in \Gamma$, choose a product of m generators with $\epsilon = (\alpha_i,\alpha_a,\alpha_e)$, and let $V'_{m,j}$ be the span of the products with $\alpha_i + 2\alpha_a + 3\alpha_e = j$. By **Step 2**, $\sum_j \dim V'_{m,j} = \#\Gamma_m$, so $\sum_j(\dim W_{m,j} - \dim V'_{m,j}) = O(m^2)$, and

$$\sum_j \mathrm{wt}_\nu(V'_{m,j}) = \sum_{(\alpha_i,\alpha_a,\alpha_e)\in\Gamma_m} \frac{\alpha_i + 2\alpha_a}{3}.$$

Summing (9.16) over j, dividing by mN_m, and letting m tend to infinity, we obtain

$$S(W;\nu) = \frac{1}{3}\int_{\tilde{\Delta}} \frac{\alpha_i + 2\alpha_a}{3} = \frac{1}{9}\Big(\frac{19}{8} + 2\Big) = \frac{35}{72}.$$

Step 5. In this step we prove $S(W;v) \le \frac{23}{24}A_E(v)$ for any prime divisor v over E with $p_3 \in \mathrm{center}_E\, v$.

We first bound $S(W^{C_\nu};p)$ for the closed points $p \in C_\nu$. By the formula for ν in **Step 2**, ν takes values in $\frac{1}{3}\mathbb{Z}$ on the non-zero sections of W. We index the pieces of W^{C_ν} by m, j, and the value $t \in \frac{1}{3}\mathbb{Z}$ of ν, and the image in W^{C_ν} of $s \in W_{m,j}$ with $\nu(s) = t$ is $(\varphi_3^*\,\mathrm{div}(s) - tC_\nu)|_{C_\nu}$.

Strictly speaking, we use Definition 2.8 for the series $W' := (W_{6m,6j})_{m,j}$, since $\mathcal{O}_E(6)$ is Cartier and ν takes integer values on W'. As in the paragraph after Theorem 2.9, we run the rest of this step for W', with Lemma 3.1 applied with $\frac{1}{6}\cdot\frac{24}{23}$ in place of λ, and the bounds for W follow.

Let $s \in W_{m,j}$ with $\nu(s) = t$. Then $\varphi_3^*\operatorname{div}(s) - tC_\nu \geq 0$, the curve C_ν is not contained in $\operatorname{Supp}(\varphi_3^*\operatorname{div}(s) - tC_\nu)$, and

$$\big((\varphi_3^*\operatorname{div}(s) - tC_\nu)\cdot C_\nu\big) = -t(C_\nu^2) = \frac{3}{2}t.$$

Thus any m-basis type divisor D of W^{C_ν} is a $\mathbb{Q}$-divisor on C_ν with $D \geq 0$ and $\deg D = \frac{3}{2}S_m(W;\nu)$ [AZ22, Section 3.1]. For $p \neq o_a$, choose a basis of each piece of W^{C_ν} compatible with the filtrations by ord_p and by ord_{o_a} [AZ22, Lemma 3.1]. The corresponding m-basis type divisor of W^{C_ν} has coefficients $S_m(W^{C_\nu};p)$ at p and $S_m(W^{C_\nu};o_a)$ at o_a, so

$$S_m(W^{C_\nu};p) + S_m(W^{C_\nu};o_a) \leq \frac{3}{2}S_m(W;\nu). \tag{9.18}$$

Next we prove $S(W;C_a) \geq \frac{1}{4}$. For $1 \leq \ell \leq m$, multiplication by a^ℓ maps $\bigoplus_j W_{m-\ell,j}$ injectively into $\bigoplus_j W_{m,j}$, and the image consists of sections s with $\operatorname{ord}_{C_a}(s) \geq \ell$. Thus

$$S_m(W;C_a) \geq \frac{1}{mN_m}\sum_{\ell=1}^{m} N_{m-\ell} = \frac{1}{mN_m}\sum_{k=0}^{m-1} N_k,$$

and $N_k = 3k^3 + O(k^2)$ (Lemma 2.12(2)) implies $S(W;C_a) \geq \lim_m \frac{1}{3m^4}\cdot\frac{3m^4}{4} = \frac{1}{4}$.

Now choose a basis of each $W_{m,j}$ compatible with the filtrations by ν and by ord_{C_a} [AZ22, Lemma 3.1], and let D_m be the corresponding m-basis type divisor of W. Then $\operatorname{mult}_{C_a} D_m = S_m(W;C_a)$, and

$$\varphi_3^*D_m = S_m(W;\nu)C_\nu + D'_m,$$

where $D'_m \geq 0$ is a $\mathbb{Q}$-divisor whose support does not contain C_ν, $D'_m \geq S_m(W;C_a)\tilde{C}_a$ since $\varphi_3^*C_a = \tilde{C}_a + \frac{2}{3}C_\nu$, and $D'_m|_{C_\nu}$ is an m-basis type divisor of W^{C_ν} [AZ22, Section 3.1]. Since $\tilde{C}_a$ and C_ν meet transversally at the smooth point o_a, the coefficient of o_a in $D'_m|_{C_\nu}$ is at least $S_m(W;C_a)$. Thus $S_m(W^{C_\nu};o_a) \geq S_m(W;C_a)$. Together with (9.18), **Step 4**, and [AZ22, Lemma 2.9], we obtain

$$S(W^{C_\nu};p) \leq \frac{3}{2}\cdot\frac{35}{72} - \frac{1}{4} = \frac{23}{48} \quad \text{for } p \neq o_a, \qquad S(W^{C_\nu};o_a) \leq \frac{3}{2}\cdot\frac{35}{72} = \frac{35}{48}.$$

Since $\operatorname{Diff}_{C_\nu}(0) = \frac{1}{2}o_i$, we obtain

$$\frac{24}{23}S(W^{C_\nu};o_i) \leq \frac{1}{2} = A_{C_\nu,\operatorname{Diff}_{C_\nu}(0)}(o_i), \qquad \frac{24}{23}S(W^{C_\nu};o_a) \leq \frac{35}{46} < 1,$$

and $\frac{24}{23}S(W^{C_\nu};p) \leq \frac{1}{2} < 1$ for any other closed point $p \in C_\nu$.

By Lemma 3.1 applied to $(E,0)$, the series W, the divisor C_ν with plt extraction φ_3, $G = C_\nu$, and $24/23$ in place of λ, we obtain

$$A_E(v) \geq \frac{24}{23}S(W;v) + \Big(A_E(C_\nu) - \frac{24}{23}S(W;\nu)\Big)\operatorname{mult}_v C_\nu = \frac{24}{23}S(W;v) + \frac{34}{69}\operatorname{mult}_v C_\nu$$

for any prime divisor v over E with $\operatorname{center}_{E_3} v \cap C_\nu \neq \emptyset$. If $p_3 \in \operatorname{center}_E v$, then $\operatorname{center}_{E_3} v$ intersects $\varphi_3^{-1}(p_3) = C_\nu$. Thus $S(W;v) \leq \frac{23}{24}A_E(v)$.

Step 6. In this step we conclude the proof.

Let v be a prime divisor over E. If $\operatorname{center}_E v \cap U_E \neq \emptyset$, then $S(W;v) \leq \frac{47}{48}A_E(v)$ by Proposition 9.8. Otherwise $\operatorname{center}_E v \subset C_i\backslash\{p_2\}$. Since $\operatorname{center}_E v$ is closed and irreducible and $p_2 \in C_i$, $\operatorname{center}_E v$ is a closed point of $C_i\backslash\{p_2\}$, and **Steps 1 and 5** imply $S(W;v) \leq \frac{47}{48}A_E(v)$,

with $S(W;v) \le \frac{23}{24}A_E(v)$ if $\operatorname{center}_E v = \{p_3\}$. Thus $S(W;v) \le \frac{47}{48}A_E(v)$ for any prime divisor v over E, and $\delta(E;W) \ge \frac{48}{47}$. By Construction-Proposition 9.2,

$$\delta_x(X) \ge \min\left\{\frac{8}{7}, \frac{48}{47}\right\} = \frac{48}{47}.$$

Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$. Then $\operatorname{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(x) = E$ and $\operatorname{mult}_F E > 0$. Since $\operatorname{Diff}_E(0) = 0$, Lemma 3.1 applied with $48/47$ in place of λ and $G = E$ implies

$$A_X(F) \ge \frac{48}{47}S_X(F) + \left(6 - \frac{48}{47}\cdot\frac{21}{4}\right)\operatorname{mult}_F E = \frac{48}{47}S_X(F) + \frac{30}{47}\operatorname{mult}_F E.$$

Thus

$$S_X(F) \le \frac{47}{48}A_X(F) - \frac{5}{8}\operatorname{mult}_F E < A_X(F).$$

Together with **Steps 4 and 5**, this proves the theorem. □

9.3. **Case (B).** In this subsection, we prove that $\delta_x(X) \ge 9/8$ for any point x with a normal form in case (B) (Theorem 9.13). We use the weighted blow-up with weights $(1,2,3)$ in the regular parameters (i,a,e), and we first determine the associated graded ring of the anti-canonical ring for the corresponding valuation.

We use the following count of lattice points to show that twelve explicit sections generate the associated graded ring in Construction-Proposition 9.12.

Set-up 9.10. Let $U \subset \mathbb{Z}^3$ be the set of the twelve vectors

$$(0,0,0),\ (1,0,0),\ (0,1,0),\ (0,0,1),\ (0,2,0),\ (0,1,1),$$
$$(1,2,0),\ (0,0,2),\ (0,3,0),\ (0,2,1),\ (0,4,0),\ (0,0,3),$$

let $\Gamma \subset \mathbb{Z}_{\ge 0} \times \mathbb{Z}^3$ be the semigroup generated by $\{1\} \times U$, let $\Gamma_m := \{u \in \mathbb{Z}^3 : (m,u) \in \Gamma\}$ for any integer $m \ge 0$, and let

$$\Delta := \{(w_1,w_2,w_3) \in \mathbb{R}^3_{\ge 0} : 3w_1 + w_3 \le 3,\ 6w_1 + 3w_2 + 4w_3 \le 12\}.$$

Lemma 9.11. *Notation as in Set-up 9.10. Then*

(1) $\#\Gamma_m = 3m^3 + \frac{9}{2}m^2 + \frac{7}{2}m + 1$ *for any integer* $m \ge 0$,
(2) the closure of $\bigcup_{m\ge 1}\frac{1}{m}\Gamma_m$ *is* Δ, *and*
(3) $\operatorname{vol}(\Delta) = 3$, $\int_\Delta w_1 dw = 1$, $\int_\Delta w_2 dw = \frac{7}{2}$, *and* $\int_\Delta w_3 dw = \frac{9}{4}$.

Proof. (1) Since Γ is generated by $\{1\} \times U$, Γ_m is the set of sums of m elements of U, and $\Gamma_0 = \{0\}$. Thus (1) holds for $m = 0$, and we may assume that $m \ge 1$. Exactly two elements of U have first coordinate 1, namely $(1,0,0)$ and $(1,2,0)$, and the other ten have first coordinate 0. Let

$$\begin{aligned} V :=& \{(b,c) \in \mathbb{Z}^2 : (0,b,c) \in U\} \\ =& \{(b,0) : 0 \le b \le 4\} \cup \{(b,1) : 0 \le b \le 2\} \cup \{(0,2),(0,3)\}, \end{aligned}$$

and for any integer $p \ge 0$ let $\Lambda_p \subset \mathbb{Z}^2$ be the set of sums of p elements of V, so that $\Lambda_0 = \{(0,0)\}$. A sum of m elements of U with exactly $m-p$ summands of first coordinate 1 has first coordinate $m-p$. Thus

$$\Gamma_m = \bigcup_{p=0}^{m}\{(m-p, b+2k, c) : (b,c) \in \Lambda_p,\ 0 \le k \le m-p\}. \tag{9.19}$$

We claim that for any integer $p \ge 1$,

$$\Lambda_p = \{(0,3p)\} \cup \{(b,3s+r) : 0 \le s \le p-1,\ 0 \le r \le 2,\ 0 \le b \le 4p - 4s - 2r\}.$$

Any $(b',c') \in V$ satisfies $b' \le 4 - 2c'$ if $c' \le 2$, and $(b',c') = (0,3)$ if $c' = 3$. Let $(b,c) \in \Lambda_p$ be a sum of p elements of V, exactly k' of which are equal to $(0,3)$. Then $c \le 3p$ and

$$b \le 4(p-k') - 2(c - 3k') = 4p - 2c + 2k'.$$

If $c = 3p$, then all p summands are equal to $(0,3)$, and $(b,c) = (0,3p)$. Otherwise write $c = 3s+r$ with integers $s \geq 0$ and $0 \leq r \leq 2$. Then $s \leq p-1$ and $k' \leq s$, so $b \leq 4p-2c+2s = 4p-4s-2r$. Conversely, let s, r, b be integers with $0 \leq s \leq p-1$, $0 \leq r \leq 2$, and $0 \leq b \leq 4p-4s-2r$. Take s summands $(0,3)$, one summand (b_1, r) with $0 \leq b_1 \leq 4-2r$ if $r \geq 1$, and the remaining $p-s-1$ summands (or $p-s$ summands if $r = 0$) of the form $(b', 0)$ with $0 \leq b' \leq 4$. The sums of the first coordinates of such summands run through all integers from 0 to $4p-4s-2r$. Thus $(b, 3s+r) \in \Lambda_p$, which proves the claim.

Fix an integer $1 \leq p \leq m$, and let c be an integer with $0 \leq c \leq 3p$. By (9.19) and the claim, the elements of Γ_m with first coordinate $m-p$ and third coordinate c are the vectors $(m-p, y, c)$ with

$$y \in \begin{cases} \{b+2k : 0 \leq b \leq 4p-4s-2r,\ 0 \leq k \leq m-p\} & \text{if } c = 3s+r,\ 0 \leq s \leq p-1,\ 0 \leq r \leq 2, \\ \{2k : 0 \leq k \leq m-p\} & \text{if } c = 3p. \end{cases}$$

If $c \leq 3p-2$, then $4p-4s-2r \geq 2$, and the possible y are all integers from 0 to $2(m-p)+4p-4s-2r$. If $c = 3p-1$, then $4p-4s-2r = 0$, and the possible y are the $m-p+1$ even integers from 0 to $2(m-p)$. Hence the number of elements of Γ_m with first coordinate $m-p$ is

$$\begin{aligned} &\sum_{s=0}^{p-1} \sum_{r=0}^{2} \big(2(m-p) + 4p - 4s - 2r + 1\big) - (m-p) + (m-p+1) \\ &= \sum_{s=0}^{p-1} \big(6(m-p) + 12p - 12s - 3\big) + 1 = (6m+3)p + 1, \end{aligned}$$

where $-(m-p)$ corrects the term $c = 3p-1$ and $m-p+1$ counts $c = 3p$. The elements of Γ_m with first coordinate m are the vectors $(m, 2k, 0)$ with $0 \leq k \leq m$, and there are $m+1$ of them. Therefore

$$\begin{aligned} \#\Gamma_m &= (m+1) + \sum_{p=1}^{m} \big((6m+3)p + 1\big) \\ &= 2m + 1 + \frac{(6m+3)m(m+1)}{2} = 3m^3 + \frac{9}{2}m^2 + \frac{7}{2}m + 1. \end{aligned}$$

(2) The semigroup Γ is generated by $\{1\} \times U$, $\Gamma \cap (\{0\} \times \mathbb{Z}^3) = \{0\}$, and Γ generates $\mathbb{Z}^4$ as a group since it contains $(1,0,0,0)$, $(1,1,0,0)$, $(1,0,1,0)$, and $(1,0,0,1)$. Thus, by Lemma 3.4, the closure of $\bigcup_{m \geq 1} \frac{1}{m}\Gamma_m$ is the convex hull of U. A direct calculation shows that the vertices of Δ are $(0,0,0)$, $(1,0,0)$, $(1,2,0)$, $(0,4,0)$, and $(0,0,3)$, which belong to U, and that any element of U satisfies the inequalities defining Δ. Hence the convex hull of U is Δ.

(3) We have

$$\Delta = \Big\{0 \leq w_1 \leq 1,\ 0 \leq w_3 \leq 3(1-w_1),\ 0 \leq w_2 \leq 4 - 2w_1 - \frac{4}{3}w_3\Big\}.$$

For $0 \leq w_1 \leq 1$, let Δ_{w_1} be the slice of Δ over w_1. A direct calculation shows that

$$\int_{\Delta_{w_1}} dw_2 dw_3 = 6(1-w_1), \qquad \int_{\Delta_{w_1}} w_3 dw_2 dw_3 = (6+3w_1)(1-w_1)^2,$$

$$\int_{\Delta_{w_1}} w_2 dw_2 dw_3 = 6(1-w_1) + 2(1-w_1)^3.$$

Integrating over $0 \leq w_1 \leq 1$, we obtain

$$\mathrm{vol}(\Delta) = 3, \quad \int_{\Delta} w_1 dw = \int_0^1 6w_1(1-w_1)dw_1 = 1, \quad \int_{\Delta} w_3 dw = \frac{9}{4}, \quad \int_{\Delta} w_2 dw = \frac{7}{2}.$$

□

Construction-Proposition 9.12. Notation and conditions as in Construction-Proposition 9.2 in case (B). By Lemma 9.1, $X = \mathrm{G}_2\,\mathrm{Gr}(2,7) \cap \{\ell_1 = \ell_2 = 0\}$ with

$$\ell_1 = n + \lambda f + \beta(g+k) + \sum_{y\in I} \rho_y y, \qquad \ell_2 = g + \mu f + \kappa(g+k) + \sum_{y\in I} \sigma_y y, \qquad I = \{c, l, b, d, M, J\},$$

for complex numbers $\lambda, \mu, \beta, \kappa, \rho_y, \sigma_y$ with $\mu \neq 0$, where $f = ie - a^2$ on the chart $h = 1$ by (2.1). Then on the chart $h = 1$ the coordinates (i, a, e) are regular parameters at x. Let

$$\eta := 2\lambda - \mu\beta + \lambda\kappa, \qquad \theta := \lambda\mu\beta - \lambda^2\kappa - \lambda^2 + \mu^2\rho_l - \lambda\mu\sigma_l,$$
$$T_0 := e^3 - \eta aef - \theta i f^2,$$

let v_0 be the monomial valuation with weights $(1, 2, 3)$ in (i, a, e), and let $\hat\pi_0 : \hat X_0 \to X$ be the weighted blow-up with weights $(1, 2, 3)$ in (i, a, e), with exceptional divisor E_0. Then $\hat\pi_0$ is a plt extraction, $E_0 \cong \mathbb{P}(1, 2, 3)$, $\mathrm{Diff}_{E_0}(0) = 0$, $A_X(E_0) = 6$, and $S_X(E_0) = 59/12$. For $s \in H^0(X, mH)$ write $s = \phi_s h^m$ near x with $\phi_s \in \mathbb{C}[[i, a, e]]$, let $(\phi_s)_j$ be the part of ϕ_s of weight j, and let t be a variable. Sending the class of s with $v_0(\phi_s) \geq j$ in the j-th graded piece to $(\phi_s)_j t^m$ defines an isomorphism from the associated graded ring of $\bigoplus_m H^0(X, mH)$ for v_0 onto

$$B := \mathbb{C}[t, ti, ta, te, tf, tae, tif, te^2, taf, tef, tf^2, tT_0] \subset \mathbb{C}[i, a, e, t],$$

bigraded by the degree in t and the weights $1, 2, 3$ of i, a, e. The refinement W_0 of $(H^0(X, mH))_m$ by E_0 satisfies $(W_0)_{m,j} \subset H^0(E_0, \mathcal{O}_{E_0}(j)) = \mathbb{C}[i, a, e]_j$ and $(W_0)_{m,j} t^m = B_{m,j}$ for all integers $m, j \geq 0$. Moreover, let

$$B_0 := \mathbb{C}[t, ti, ta, te, tf, tae, tif, te^2, taf, tef, tf^2, te^3]$$

be the algebra B with $\eta = \theta = 0$. For the weights $(-1, 0, 1)$ on (i, a, e), which preserve the section degree and the weighted degree, the highest-weight parts of the elements of $(W_0)_{m,j} t^m$ span exactly the bidegree (m, j) part of B_0, for all integers $m, j \geq 0$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that (i, a, e) are regular parameters at x, that $\hat\pi_0$ is a plt extraction with the invariants of the statement except $S_X(E_0)$, and we compute the leading parts of n and g.

By Lemma 9.1, $\mu \neq 0$ since the normal form is in case (B), and x is the origin of the chart $h = 1$ of $\mathrm{G}_2\,\mathrm{Gr}(2, 7)$, which is an affine space with coordinates i, a, e, n, g. On the chart, X is defined by $\ell_1 = \ell_2 = 0$, where the remaining coordinates are the polynomials in i, a, e, n, g of (2.1). Since $g + k = -ae - in$ and f, c, l, b, d, J, M have no terms of degree ≤ 1, we have $\ell_1 \equiv n$ and $\ell_2 \equiv g$ modulo terms of degree ≥ 2 in i, a, e, n, g. Thus (i, a, e) are regular parameters at x, and $n, g \in \widehat{\mathcal{O}}_{X,x} = \mathbb{C}[[i, a, e]]$.

We consider the valuation v_0 of the statement on $\mathbb{C}[[i, a, e]]$. For $\phi \in \mathbb{C}[[i, a, e]]$ and an integer d, let ϕ_d be the part of ϕ of weight d. Since (i, a, e) are regular parameters at the smooth point x, the morphism (i, a, e) from a neighborhood of x to $\mathbb{A}^3$ is étale at x. The morphism $\hat\pi_0$ is the Proj of the Rees algebra $\bigoplus_{j\geq 0}\{\phi : v_0(\phi) \geq j\}$, whose graded pieces are ideals generated by monomials in (i, a, e). Thus near x, $\hat\pi_0$ is the pullback along (i, a, e) of the toric weighted blow-up of $\mathbb{A}^3$ with weights $(1, 2, 3)$, which has three charts. The chart $i \neq 0$ is $\mathbb{A}^3$. The chart $a \neq 0$ is $\mathbb{A}^3/\mu_2$, where $i = sy_1$, $a = s^2$, $e = s^3 y_3$, and μ_2 acts on (s, y_1, y_3) with weights $(1, 1, 1)$. The chart $e \neq 0$ is the quotient $q_e : \mathbb{A}^3 \to \mathbb{A}^3/\mu_3$, where $i = sy_1$, $a = s^2 y_2$, $e = s^3$, and μ_3 acts on (s, y_1, y_2) with weights $(1, 2, 1)$, which is the type $\frac{1}{3}(1, 2, 2)$ after the change of generator $\zeta \to \zeta^2$. The images of $(s = 0)$ on the three charts glue to $E_0 = \mathrm{Proj}\,\mathbb{C}[i, a, e] = \mathbb{P}(1, 2, 3)$, and $\mathrm{ord}_{E_0} = v_0$ on $\mathcal{O}_{X,x}$ since i, a, e vanish to orders $1, 2, 3$ along $(s = 0)$.

Thus $-E_0$ is $\hat\pi_0$-ample, and $\hat X_0$ has exactly two singular points on E_0, which are cyclic quotient singularities of types $\frac{1}{2}(1, 1, 1)$ and $\frac{1}{3}(1, 2, 2)$. In particular $\hat X_0$ is $\mathbb{Q}$-factorial and smooth in codimension two along E_0, so E_0 is Cartier in codimension two and $\mathrm{Diff}_{E_0}(0) = 0$. Since

$E_0 \cong \mathbb{P}(1,2,3)$ has only quotient singularities, $(E_0, 0)$ is klt, and $(\hat{X}_0, E_0)$ is plt near E_0 by [KM98, Theorem 5.50(1)]. On the chart $e \neq 0$, we have

$$di \wedge da \wedge de = 3s^5 dy_1 \wedge dy_2 \wedge ds.$$

All weights of the action of μ_3 on (s, y_1, y_2) are non-zero modulo 3, so μ_3 acts freely outside the origin. Thus q_e is étale in codimension one and $q_e^* E_0 = (s = 0)$. Since $di \wedge da \wedge de$ generates ω_X near x, we obtain $K_{\hat{X}_0} = \hat{\pi}_0^* K_X + 5E_0$, that is, $A_X(E_0) = 6$.

Let $Q := ae + in$, so that $g + k = -Q$ and $k = -g - Q$. Then $\ell_1 = \ell_2 = 0$ reads

$$n = -\lambda f + \beta Q - \sum_{y \in I} \rho_y y, \qquad g = -\mu f + \kappa Q - \sum_{y \in I} \sigma_y y \tag{9.20}$$

in $\mathbb{C}[[i, a, e]]$. We show that $v_0(n), v_0(g) \geq 4$, $v_0(Q) \geq 5$, and

$$\begin{gathered} n_4 = -\lambda f, \quad g_4 = -\mu f, \quad Q_5 = ae - \lambda i f, \quad l_5 = \mu i f, \\ n_5 = \beta Q_5 - \mu \rho_l i f, \quad g_5 = \kappa Q_5 - \mu \sigma_l i f. \end{gathered} \tag{9.21}$$

Let $\nu := \min\{v_0(n), v_0(g)\}$, so $\nu \geq 1$. The formulas

$$\begin{gathered} c = e^2 + an, \quad l = -af - ig - iQ, \quad b = -ag - ic, \\ d = eg + ac, \quad J = fc - g^2 - gQ, \quad M = -ec + gn \end{gathered}$$

of (2.1) imply that $v_0(Q) \geq \min\{5, \nu+1\}$ and $v_0(y) \geq \min\{5, \nu+1\}$ for any $y \in I$. Since $v_0(f) = 4$, (9.20) implies that $\nu \geq \min\{4, \nu+1\}$, hence $\nu \geq 4$. Then the formulas for c, l, b, d, J, M from (2.1) imply that $v_0(Q) \geq 5$, $v_0(l) \geq 5$ with $l_5 = -ig_4$, and $v_0(y) \geq 6$ for any $y \in I \backslash \{l\}$. Taking the parts of weight 4 of (9.20), we obtain $n_4 = -\lambda f$ and $g_4 = -\mu f$. Thus $Q_5 = ae + in_4 = ae - \lambda i f$ and $l_5 = \mu i f$. Taking the parts of weight 5 of (9.20), we obtain $n_5 = \beta Q_5 - \rho_l l_5$ and $g_5 = \kappa Q_5 - \sigma_l l_5$, which proves (9.21).

Step 2. In this step, we show that the twelve generators of the algebra B of the statement are leading forms of sections of H, so that B is contained in the image of the map of the statement.

The coordinate h is a section of H which does not vanish at x. For an integer $m \geq 0$ and $s \in H^0(X, mH)$, write $s = \phi_s h^m$ near x with $\phi_s \in \mathcal{O}_{X,x} \subset \mathbb{C}[[i, a, e]]$, and let $v_0(s) := v_0(\phi_s)$. For $s \neq 0$, let $\mathrm{in}_{v_0}(s) := (\phi_s)_{v_0(s)} t^m$, so that $t = \mathrm{in}_{v_0}(h)$. For any integer $j \geq 0$, let $F_m^j := \{s \in H^0(X, mH) : v_0(s) \geq j\}$. Sending the class of $s \in F_m^j$ to $(\phi_s)_j t^m$ defines an injective linear map $F_m^j / F_m^{j+1} \to \mathbb{C}[i, a, e]_j t^m$, and the maps $F_m^j / F_m^{j+1} \to \mathbb{C}[i, a, e]_j t^m$ form an injective homomorphism of bigraded algebras

$$\bigoplus_{m,j} F_m^j / F_m^{j+1} \to \mathbb{C}[i, a, e, t],$$

which is the map of the statement. Let $\mathrm{gr} \subset \mathbb{C}[i, a, e, t]$ be the image of the homomorphism, and let $\mathrm{gr}_{m,j}$ be the image of F_m^j / F_m^{j+1}. Then $\mathrm{in}_{v_0}(s) \in \mathrm{gr}_{m, v_0(s)}$ for any non-zero $s \in H^0(X, mH)$.

For any coordinate y of Set-up 2.14, the section y of H satisfies $\phi_y = y$. By (9.21) and the formulas in **Step 1**, $v_0(c), v_0(b) \geq 6$, $v_0(d) \geq 7$, $v_0(J), v_0(M) \geq 8$, and

$$c_6 = e^2 - \lambda a f, \quad b_6 = \mu a f, \quad d_7 = -\mu e f, \quad J_8 = -\mu^2 f^2, \quad M_8 = \lambda \mu f^2.$$

In particular $(M + \frac{\lambda}{\mu} J)_8 = 0$. We have

$$M + \frac{\lambda}{\mu} J = -ec + g\Big(n - \frac{\lambda}{\mu} g\Big) + \frac{\lambda}{\mu} fc - \frac{\lambda}{\mu} gQ.$$

Since $(n - \frac{\lambda}{\mu}g)_4 = 0$ and $v_0(fc) \geq 10$, (9.21) implies that

$$\begin{aligned}\Big(M + \frac{\lambda}{\mu}J\Big)_9 &= -ec_6 + g_4\Big(n_5 - \frac{\lambda}{\mu}g_5\Big) - \frac{\lambda}{\mu}g_4Q_5\\ &= -e^3 + \lambda aef + (\lambda - \mu\beta + \lambda\kappa)fQ_5 + (\mu^2\rho_l - \lambda\mu\sigma_l)if^2\\ &= -e^3 + \eta aef + \theta if^2 = -T_0,\end{aligned}$$

where the last equality uses $fQ_5 = aef - \lambda if^2$. The coefficient of e^3 in T_0 is 1, so $T_0 \neq 0$. Thus the sections in Table 9 have the values of v_0 and the leading forms listed there.

TABLE 9. Twelve sections of H near a point with a normal form in case (B), with their values under v_0 and their leading forms.

section s of H	$v_0(s)$	$\mathrm{in}_{v_0}(s)$
h	0	t
i	1	ti
a	2	ta
e	3	te
f	4	tf
$-(g+k) + \frac{\lambda}{\mu}l$	5	tae
$\frac{1}{\mu}l$	5	tif
$c + \frac{\lambda}{\mu}b$	6	te^2
$\frac{1}{\mu}b$	6	taf
$-\frac{1}{\mu}d$	7	tef
$-\frac{1}{\mu^2}J$	8	tf^2
$-M - \frac{\lambda}{\mu}J$	9	tT_0

Since gr is a subalgebra of $\mathbb{C}[i,a,e,t]$, we obtain $B \subset \mathrm{gr}$.

Step 3. In this step, we show that $B = \mathrm{gr}$ in any bidegree, and that $(W_0)_{m,j}t^m = B_{m,j}$.

For complex numbers η', θ', let $B(\eta',\theta') \subset \mathbb{C}[i,a,e,t]$ be the subalgebra generated by the twelve generators of B with tT_0 replaced by $t(e^3 - \eta' aef - \theta' if^2)$, so that $B = B(\eta,\theta)$ and $B_0 = B(0,0)$. All twelve generators are homogeneous for the bigrading, so $B(\eta',\theta') = \bigoplus_{m,j} B(\eta',\theta')_{m,j}$. Let U and Γ_m be as in Set-up 9.10, and for integers $m, j \geq 0$ let $\Gamma_{m,j} := \{u \in \Gamma_m : u_1 + 2u_2 + 3u_3 = j\}$. We claim that

$$\dim B(\eta',\theta')_{m,j} \geq \#\Gamma_{m,j} \quad \text{for all integers } m,j \geq 0. \tag{9.22}$$

Assign to (i,a,e,t) the weights $(0,4,7,0)$, and for a non-zero $\phi \in \mathbb{C}[i,a,e,t]$ let $\mathrm{in}_{(0,4,7)}(\phi)$ be the part of ϕ of the highest weight. Then

$$\mathrm{in}_{(0,4,7)}(\phi\phi') = \mathrm{in}_{(0,4,7)}(\phi)\,\mathrm{in}_{(0,4,7)}(\phi').$$

Since $f = ie - a^2$, $aef = iae^2 - a^3e$, and $if^2 = i^3e^2 - 2i^2a^2e + ia^4$, the highest-weight parts of the twelve generators, in the order of the list defining B, are

$$t,\ ti,\ ta,\ te,\ -ta^2,\ tae,\ -tia^2,\ te^2,\ -ta^3,\ -ta^2e,\ ta^4,\ te^3;$$

for the last generator, the monomial te^3 has weight 21, and the monomials of $taef$ and tif^2 have weights at most 19. The exponent vectors of (i,a,e) in the twelve monomials above are the elements of U in the order of Set-up 9.10. For $u \in \Gamma_m$, write u as a sum of m elements of U, and let P_u be the product of the corresponding m generators. Then $\mathrm{in}_{(0,4,7)}(P_u) = \pm t^m i^{u_1} a^{u_2} e^{u_3}$. Any generator γ is homogeneous for the weights $(1,2,3)$ of (i,a,e), and γ has $(1,2,3)$-weight equal to the $(1,2,3)$-weight of the monomial $\mathrm{in}_{(0,4,7)}(\gamma)$. Thus $P_u \in B(\eta',\theta')_{m,u_1+2u_2+3u_3}$. The elements P_u with $u \in \Gamma_m$ are linearly independent. To see this, consider a linear combination $\sum_u c_u P_u$ with some $c_u \neq 0$, and let w be the largest $(0,4,7)$-weight of the monomials $i^{u_1}a^{u_2}e^{u_3}$

with $c_u \neq 0$. Then the part of weight w of $\sum_u c_u P_u$ is a non-trivial linear combination of distinct monomials, so $\sum_u c_u P_u \neq 0$. This proves (9.22).

Since $F_m^0 = H^0(X, mH)$ and $F_m^j = 0$ for $j \gg 0$, we have $\sum_j \dim \mathrm{gr}_{m,j} = N_m$. By Lemma 2.12(2) with $H^3 = 18$, we have $N_m = 3m^3 + \frac{9}{2}m^2 + \frac{7}{2}m + 1$, so $N_m = \#\Gamma_m$ by Lemma 9.11(1). By **Step 2** and (9.22) with $(\eta', \theta') = (\eta, \theta)$,

$$N_m = \sum_j \dim \mathrm{gr}_{m,j} \geq \sum_j \dim B_{m,j} \geq \sum_j \#\Gamma_{m,j} = \#\Gamma_m = N_m,$$

and the inequalities hold termwise. Hence

$$B_{m,j} = \mathrm{gr}_{m,j} \quad \text{and} \quad \dim \mathrm{gr}_{m,j} = \#\Gamma_{m,j} \quad \text{for all integers } m, j \geq 0. \tag{9.23}$$

In particular $\mathrm{gr} = B$, so the map of the statement is an isomorphism onto B.

By Definition 2.8 (see also [AZ22, Example 2.6]), $(W_0)_{m,j}$ is the image of the restriction map

$$H^0(\hat{X}_0, m\hat{\pi}_0^* H - jE_0) \to H^0(E_0, \mathcal{O}_{E_0}(j)),$$

where $\hat{\pi}_0^* H|_{E_0} \sim 0$ and $-E_0|_{E_0} = \mathcal{O}_{E_0}(1)$. Since $\mathrm{ord}_{E_0} = v_0$, we have $H^0(\hat{X}_0, m\hat{\pi}_0^* H - jE_0) = F_m^j$, and the kernel of the restriction map is F_m^{j+1}. Under the isomorphism $H^0(E_0, \mathcal{O}_{E_0}(j)) \cong \mathbb{C}[i, a, e]_j$ given by $E_0 = \mathrm{Proj}\, \mathbb{C}[i, a, e]$, the restriction of $s \in F_m^j$ is $(\phi_s)_j$. Thus $(W_0)_{m,j} t^m = \mathrm{gr}_{m,j} = B_{m,j}$ for all integers $m, j \geq 0$.

Step 4. In this step, we show that $S_X(E_0) = 59/12$.

For any integer $m \geq 1$, let $S_m(E_0) := \frac{1}{mN_m} \sum_{j \geq 0} j \dim (W_0)_{m,j}$. By **Step 3** and (9.23),

$$S_m(E_0) = \frac{1}{mN_m} \sum_{u \in \Gamma_m} (u_1 + 2u_2 + 3u_3).$$

The semigroup Γ of Set-up 9.10 is finitely generated, $\Gamma \cap (\{0\} \times \mathbb{Z}^3) = \{0\}$ since all generators have first coordinate 1, and Γ generates $\mathbb{Z}^4$ as a group since it contains $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, and $(1, 0, 0, 1)$. By Lemma 3.4 applied to the linear function $(m, u) \to u_1 + 2u_2 + 3u_3$, and by Lemma 9.11(2)(3),

$$\sum_{u \in \Gamma_m} (u_1 + 2u_2 + 3u_3) = m^4 \Big(1 + 2 \cdot \frac{7}{2} + 3 \cdot \frac{9}{4}\Big) + O(m^3) = \frac{59}{4} m^4 + O(m^3).$$

Since $mN_m = 3m^4 + O(m^3)$, we obtain $S_m(E_0) = \frac{59}{12} + O(1/m)$. Thus, by [AZ22, Lemma 2.9],

$$S_X(E_0) = \lim_{m \to +\infty} S_m(E_0) = \frac{59}{12}.$$

Step 5. In this step, we prove the statement on the weights $(-1, 0, 1)$.

Assign to (i, a, e, t) the weights $(-1, 0, 1, 0)$, and for a non-zero $\phi \in \mathbb{C}[i, a, e, t]$ let $\mathrm{in}_{(-1,0,1)}(\phi)$ be the part of ϕ of the highest weight. The weights $(-1, 0, 1, 0)$, the weights $(1, 2, 3, 0)$, and the degree in t are additive on monomials, so $\mathrm{in}_{(-1,0,1)}(\phi) \in \mathbb{C}[i, a, e]_j t^m$ for any non-zero $\phi \in \mathbb{C}[i, a, e]_j t^m$. Moreover

$$\mathrm{in}_{(-1,0,1)}(\phi\phi') = \mathrm{in}_{(-1,0,1)}(\phi)\, \mathrm{in}_{(-1,0,1)}(\phi').$$

The polynomial $f = ie - a^2$ is homogeneous of weight 0 for the weights $(-1, 0, 1)$. Thus the first eleven generators of B are homogeneous for the weights $(-1, 0, 1, 0)$, and they are also the first eleven generators of B_0. The three terms te^3, $-\eta tae f$, and $-\theta t i f^2$ of tT_0 are homogeneous of weights 3, 1, and -1, so $\mathrm{in}_{(-1,0,1)}(tT_0) = te^3$. Hence, for any product P of generators of B, $\mathrm{in}_{(-1,0,1)}(P)$ is the product of the corresponding generators of B_0, and $\mathrm{in}_{(-1,0,1)}(P)$ has the bidegree of P since T_0 and e^3 are homogeneous of weight 9 for the weights $(1, 2, 3)$. Let $V_{m,j}$ be the span of the elements $\mathrm{in}_{(-1,0,1)}(\phi)$ with $0 \neq \phi \in B_{m,j}$. Since $(B_0)_{m,j}$ is spanned by products of generators of B_0 of bidegree (m, j), we obtain $(B_0)_{m,j} \subset V_{m,j}$.

For an integer w, let $B_{m,j}^{\le w} \subset B_{m,j}$ be the subspace of the elements whose monomials have weight at most w. Then $V_{m,j}$ is the direct sum over w of the images of $B_{m,j}^{\le w}$ in the parts of weight w, and the image for w is isomorphic to $B_{m,j}^{\le w}/B_{m,j}^{\le w-1}$. Thus $\dim V_{m,j} = \dim B_{m,j}$. By (9.22) with $(\eta', \theta') = (0,0)$ and by (9.23),

$$\dim(B_0)_{m,j} \ge \#\Gamma_{m,j} = \dim B_{m,j} = \dim V_{m,j}.$$

Hence $(B_0)_{m,j} = V_{m,j}$. Since $(W_0)_{m,j}t^m = B_{m,j}$ by **Step 3**, the highest-weight parts of the elements of $(W_0)_{m,j}t^m$ span exactly the bidegree (m,j) part of B_0, for all integers $m, j \ge 0$. □

We use the weighted blow-up $\hat{\pi}_0 : \hat{X}_0 \to X$ with weights $(1,2,3)$ of Construction-Proposition 9.12. The exceptional divisor of $\hat{\pi}_0$ is $E_0 \cong \mathbb{P}(1,2,3)$, with $A_X(E_0) = 6$ and $S_X(E_0) = 59/12$, and E_0 is different from the exceptional divisor $E \cong \mathbb{P}(1,3,10)$ of Construction-Proposition 9.2(B).

Theorem 9.13. *Let X and x be as in Construction-Proposition 9.2 in case (B), and let E_0 and W_0 be as in Construction-Proposition 9.12. Then $S(W_0; v) \le \frac{8}{9}A_{E_0}(v)$ for every prime divisor v over E_0. Hence every prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies*

$$S_X(F) \le \frac{8}{9}A_X(F) - \frac{5}{12}\operatorname{mult}_F E_0.$$

In particular $\delta_x(X) \ge 9/8$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe the limits of the spaces $(W_0)_{m,j}$ under a one-parameter subgroup of $\operatorname{Aut}(E_0)$.

By Construction-Proposition 9.12, $E_0 = \operatorname{Proj}\mathbb{C}[i,a,e]$, where i, a, and e have weights 1, 2, and 3, $(W_0)_{m,j} \subset H^0(E_0, \mathcal{O}_{E_0}(j)) = \mathbb{C}[i,a,e]_j$, and $(W_0)_{m,j}t^m = B_{m,j}$ for all integers $m, j \ge 0$. Recall that $f = ie - a^2$ and $T_0 = e^3 - \eta aef - \theta if^2$. We write $(B_0)_{m,j} \subset \mathbb{C}[i,a,e]_j$ for the bidegree (m,j) part of B_0, again omitting t^m.

Let $\rho : \mathbb{G}_m \to \operatorname{Aut}(E_0)$ be the one-parameter subgroup acting on weighted homogeneous polynomials by $(\rho(\tau)\cdot P)(i,a,e) = P(\tau i, a, \tau^{-1}e)$. Then

$$\rho(\tau)\cdot i^{k_1}a^{k_2}e^{k_3} = \tau^{k_1-k_3}i^{k_1}a^{k_2}e^{k_3}. \tag{9.24}$$

If $\rho(\tau)\cdot P = \tau^{-\ell}P$ for a polynomial $P \ne 0$ and an integer ℓ, we say that P is a ρ-eigenvector of ρ-weight ℓ. Thus the ρ-weight of a monomial is the weight for the weights $(-1,0,1)$ on (i,a,e). For a non-zero polynomial $P = \sum_\ell P_\ell$ with P_ℓ of ρ-weight ℓ and with largest index ℓ_0 such that $P_{\ell_0} \ne 0$, we have $\tau^{\ell_0}\rho(\tau)\cdot P \to P_{\ell_0}$ as $\tau \to 0$. Hence the limit $\operatorname{in}_\rho(W_0)_{m,j}$ of $\rho(\tau)\cdot(W_0)_{m,j}$ as $\tau \to 0$ is spanned by the parts of highest ρ-weight of the non-zero elements of $(W_0)_{m,j}$, that is, by the highest-weight parts for the weights $(-1,0,1)$ on (i,a,e). The weights $(-1,0,1)$ and the direction of the limit are those of Construction-Proposition 9.12, and the last sentence of Construction-Proposition 9.12 implies

$$\operatorname{in}_\rho(W_0)_{m,j} = (B_0)_{m,j} \quad \text{for all integers } m, j \ge 0. \tag{9.25}$$

For instance, the monomials of e^3, aef, and if^2 have ρ-weights 3, 1, and -1, so the part of T_0 of highest ρ-weight is e^3. The limit as $\tau \to +\infty$ keeps the term $-\theta if^2$ instead, and does not lead to B_0 when $\theta \ne 0$.

The pair $(E_0, 0)$ is a projective klt pair, since E_0 has quotient singularities [KM98, Proposition 5.20(4)], and $\operatorname{Diff}_{E_0}(0) = 0$ by Construction-Proposition 9.12. The one-parameter subgroup ρ is a diagonal torus action on E_0, and the reflexive sheaves $\mathcal{O}_{E_0}(k)$ carry the natural ρ-linearizations given by (9.24). Thus the hypotheses of Lemma 3.6 hold for W_0, with $\Delta_{E_0} = 0$ and $\lambda_m = \rho$ for all m. In **Step 5** we use Lemma 3.6(1), and the condition of Lemma 3.6(1) on the eigen-basis divisors is proved in **Step 4**. The flatness of the family of basis divisors and the semicontinuity of log canonical thresholds enter only through Lemma 3.6.

Step 2. In this step we show that each ρ-eigenspace of $(B_0)_{m,j}$ is contained in an explicit space of binary forms.

The substitution $f \to ie - a^2$ induces an isomorphism $\mathbb{C}[i,e,f][a]/(a^2 - ie + f) \cong \mathbb{C}[i,a,e]$. Thus $\mathbb{C}[i,a,e]$ is a free $\mathbb{C}[i,e,f]$-module with basis $1, a$, and i, e, f are algebraically independent. Hence the polynomials

$$i^p a^\epsilon e^q f^r, \qquad p,q,r \geq 0, \quad \epsilon \in \{0,1\}, \tag{9.26}$$

form a basis of $\mathbb{C}[i,a,e]$. Let $w := p + r$ and $y := q + r$. Since f has weighted degree 4 and ρ-weight 0, the polynomial $i^p a^\epsilon e^q f^r$ is weighted homogeneous of degree j and a ρ-eigenvector of ρ-weight ℓ, where

$$j = w + 3y + 2\epsilon \quad \text{and} \quad \ell = y - w. \tag{9.27}$$

We claim that for any integer $m \geq 0$, any element of $\bigoplus_j (B_0)_{m,j}$ is a linear combination of polynomials (9.26) with

$$3p + q + 2\epsilon \leq 3m \quad \text{and} \quad 3p + 2q + 3r + 3\epsilon \leq 6m. \tag{9.28}$$

The space $\bigoplus_j (B_0)_{m,j}$ is spanned by the products of m of the twelve polynomials in the first row below. For a monomial $i^p a^k e^q f^r$ with $k \geq 0$, let $\phi_1 := 3p + q + 2k$ and $\phi_2 := 3p + 2q + 3r + 3k$. Then

polynomial	1	i	a	e	f	ae	if	e^2	af	ef	f^2	e^3
ϕ_1	0	3	2	1	0	3	3	2	2	1	0	3
ϕ_2	0	3	3	2	3	5	6	4	6	5	6	6

so $\phi_1 \leq 3$ and $\phi_2 \leq 6$ for each of them. Both ϕ_1 and ϕ_2 are additive under products. Replacing a factor a^2 by ie keeps ϕ_1 and lowers ϕ_2 by 1, and replacing a^2 by $-f$ lowers ϕ_1 by 4 and ϕ_2 by 3. Induction on k proves the claim. Since $a^2 = ie - f$ is weighted homogeneous of degree 4 and ρ-weight 0, the reduction preserves the weighted degree and the ρ-weight.

Each product of the twelve polynomials is a ρ-eigenvector. Fix integers $m, j \geq 0$ and ℓ. By the claim, the ρ-eigenspace of ρ-weight ℓ of $(B_0)_{m,j}$ is contained in the span of the polynomials (9.26) satisfying (9.27) and (9.28). By (9.27), $j + \ell = 4y + 2\epsilon$, so (j,ℓ) determines ϵ, then y, and then w. In terms of (w,y,ϵ,r), we have $p = w - r$ and $q = y - r$, and

$$3p + q + 2\epsilon = 3w + y - 4r + 2\epsilon, \qquad 3p + 2q + 3r + 3\epsilon = 3w + 2y - 2r + 3\epsilon.$$

Thus the conditions $p, q \geq 0$ and (9.28) are equivalent to $r_- \leq r \leq r_+$, where

$$r_+ := \min\{w, y\},$$
$$r_- := \max\left\{0, \left\lceil \frac{3w + y + 2\epsilon - 3m}{4} \right\rceil, \left\lceil \frac{3w + 2y + 3\epsilon - 6m}{2} \right\rceil\right\}.$$

Let $R := r_+ - r_-$ and $(\xi_1, \xi_2) := (ie, f)$. Since $i^{w-r} e^{y-r} f^r = i^{w-r_+} e^{y-r_+} f^{r_-} \xi_1^{r_+ - r} \xi_2^{r - r_-}$, we obtain the following: if $r_- > r_+$, then the ρ-eigenspace of ρ-weight ℓ of $(B_0)_{m,j}$ is zero, and otherwise it is contained in

$$\mathcal{V}_{w,y,\epsilon} := a^\epsilon i^{w-r_+} e^{y-r_+} f^{r_-} \cdot \mathbb{C}[\xi_1, \xi_2]_R, \tag{9.29}$$

and $\dim \mathcal{V}_{w,y,\epsilon} = R+1$ by the linear independence of (9.26). Here $r_\pm$ and R depend on (m,w,y,ϵ), and distinct pairs (j,ℓ) correspond to distinct triples (w,y,ϵ). Since the polynomials (9.26) are ρ-eigenvectors, their span under (9.28) is ρ-stable and contains $\mathrm{in}_\rho(W_0)_{m,j}$ by (9.25).

Step 3. In this step we bound the coefficients and two local orders of any eigen-basis divisor of W_0 at level m.

Let $C_i = (i = 0)$, $C_a = (a = 0)$, and $C_e = (e = 0)$ be the coordinate curves of E_0, and for $\zeta \in \mathbb{C}\backslash\{1\}$ let $C_\zeta = (f - \zeta ie = 0) = ((1-\zeta)ie - a^2 = 0)$, an irreducible curve of weighted degree 4. In particular $C_0 = (f = 0)$. Let $\chi \in \mathbb{C}[\xi_1, \xi_2]_R$ be non-zero and write χ, up to a non-zero constant factor, as

$$\xi_1^{o_\infty(\chi)} \prod_{\zeta \in \mathbb{C}} (\xi_2 - \zeta \xi_1)^{o_\zeta(\chi)},$$

where $o_\infty(\chi)$ is the multiplicity of the factor ξ_1 in χ, $o_\zeta(\chi)$ is the multiplicity of the factor $\xi_2-\zeta\xi_1$ for $\zeta\in\mathbb{C}$, and only finitely many $o_\zeta(\chi)$ are non-zero. The factor with $\zeta=1$ is $\xi_2-\xi_1=-a^2$, which is why $o_1(\chi)$ enters the coefficient of C_a below, while C_ζ is defined only for $\zeta\neq 1$. Since $\xi_1=ie$ and $\xi_2-\zeta\xi_1$ defines C_ζ for $\zeta\neq 1$, the polynomial $P=a^\epsilon i^{w-r_+}e^{y-r_+}f^{r_-}\chi(\xi_1,\xi_2)\in\mathcal{V}_{w,y,\epsilon}$ satisfies

$$\begin{aligned}\operatorname{div}(P) &= (w-r_++o_\infty(\chi))C_i+(y-r_++o_\infty(\chi))C_e+(\epsilon+2o_1(\chi))C_a\\ &\quad+(r_-+o_0(\chi))C_0+\sum_{\zeta\in\mathbb{C}\setminus\{0,1\}}o_\zeta(\chi)C_\zeta.\end{aligned}\tag{9.30}$$

Let $V'\subset\mathbb{C}[\xi_1,\xi_2]_R$ be a subspace with basis $\chi_1,\dots,\chi_{n'}$, and let $\zeta\in\mathbb{C}\cup\{\infty\}$. Then

$$\sum_{k=1}^{n'}o_\zeta(\chi_k)\le\frac{R(R+1)}{2}.\tag{9.31}$$

To prove (9.31), we order the basis so that $o_\zeta(\chi_1)\ge\cdots\ge o_\zeta(\chi_{n'})$. For each k, the forms $\chi_1,\dots,\chi_k$ span a k-dimensional subspace of the space $\mathcal{U}_k\subset\mathbb{C}[\xi_1,\xi_2]_R$ of forms divisible by the $o_\zeta(\chi_k)$-th power of the linear factor $\xi_2-\zeta\xi_1$, or ξ_1 when ζ is ∞, and $\dim\mathcal{U}_k=R+1-o_\zeta(\chi_k)$. Thus $o_\zeta(\chi_k)\le R+1-k$, and (9.31) follows by summing over $k\le n'\le R+1$.

We also use two local orders. Let $z_i=[1:0:0]$, $z_a=[0:1:0]$, and $z_e=[0:0:1]$. The point z_i is a smooth point of E_0 with local coordinates (a,e) on the chart $i=1$. Let $\pi_i:Y_i\to E_0$ be the weighted blow-up of z_i with weights $(1,2)$ in (a,e), with exceptional divisor Q_i. The chart $e\neq 0$ of E_0 is $\mathbb{A}^2/\mu_3$, where μ_3 acts on the coordinates (i,a) with weights $(1,2)$. Let $q_e:\mathbb{A}^2\to\mathbb{A}^2/\mu_3$ be the quotient morphism, and let $\pi_e:Y_e\to\mathbb{A}^2$ be the weighted blow-up of the origin with weights $(2,1)$ in (i,a), with exceptional divisor Q_e. Let $P\in\mathcal{V}_{w,y,\epsilon}$ be non-zero. On the chart $i=1$ we have $f=e-a^2$, so $P(1,a,e)$ is weighted homogeneous of degree $\epsilon+2y$ for the weights $(1,2)$ on (a,e). On the chart $e=1$ we have $f=i-a^2$, so $P(i,a,1)$ is weighted homogeneous of degree $2w+\epsilon$ for the weights $(2,1)$ on (i,a). Thus

$$\operatorname{mult}_{Q_i}\operatorname{div}(P)=\epsilon+2y\quad\text{and}\quad\operatorname{mult}_{Q_e}q_e^*\operatorname{div}(P)=2w+\epsilon.\tag{9.32}$$

Fix an integer $m\ge 1$, and let $D_m=\frac{1}{mN_m}\sum_j\sum_k\operatorname{div}(s_{j,k})$ be an eigen-basis divisor of W_0 at level m in the sense of Lemma 3.6, with $\lambda_m=\rho$. By (9.25), $(s_{j,k})_k$ is a basis of $(B_0)_{m,j}$ consisting of ρ-eigenvectors for each j. For each (j,ℓ), the elements $s_{j,k}$ of ρ-weight ℓ form a basis of a subspace of $\mathcal{V}_{w,y,\epsilon}$ by **Step 2**, since the ρ-eigenspace of ρ-weight ℓ of $(B_0)_{m,j}$ is contained in (9.29). By (9.30), (9.31), and (9.32), the contribution of the elements $s_{j,k}$ of ρ-weight ℓ to the coefficients and to the local orders $\operatorname{mult}_{Q_i}$ and $\operatorname{mult}_{Q_e}q_e^*$ of mN_mD_m is bounded by the second column of Table 10.

To sum over (w,y,ϵ), let

$$\bar r_+(w,y):=\min\{w,y\},\quad \bar r_-(w,y):=\max\Big\{0,\frac{3w+y-3}{4},\frac{3w+2y-6}{2}\Big\},\quad R_0:=\bar r_+-\bar r_-$$

for real numbers $w,y\ge 0$. The set $\{R_0\ge 0\}$ is a polygon, and on $\{R_0\ge 0\}$ we have $0\le w\le 2$ and $0\le y\le 3$. We have $r_\pm=m\bar r_\pm(w/m,y/m)+O(1)$, there exist $O(m^2)$ triples (w,y,ϵ) with $r_-\le r_+$, and $N_m=3m^3+O(m^2)$ since $H^3=18$. Thus, for each line of Table 10, the sum of the second column over all (w,y,ϵ) divided by mN_m equals $\frac{2}{3}$ times the integral in the third column, up to $O(1/m)$. The factor 2 counts $\epsilon\in\{0,1\}$, and $\frac{1}{3}=\lim_{m\to+\infty}m^3/N_m$. The integrals in the third column are computed cell by cell in Table 98 of Appendix A: the domain $\{R_0\ge 0\}$ is divided into six cells according to which term attains the maximum in $\bar r_-$ and the minimum in $\bar r_+$, and on each cell each integrand is a polynomial of degree at most two.

TABLE 10. Bounds for the eigen-basis divisors D_m, where the integrals are taken with respect to $dwdy$ and the last column is $\frac{2}{3}$ times the integral.

Quantity	Bound for $\mathcal{V}_{w,y,\epsilon}$	Integral over $\{R_0 \geq 0\}$	Limit
mult_{C_i}	$(w-r_+)(R+1)+\frac{R(R+1)}{2}$	$\int((w-\bar{r}_+)R_0+\frac{R_0^2}{2})=\frac{1}{2}$	$\frac{1}{3}$
mult_{C_a}	$\epsilon(R+1)+R(R+1)$	$\int R_0^2=\frac{45}{64}$	$\frac{15}{32}$
mult_{C_e}	$(y-r_+)(R+1)+\frac{R(R+1)}{2}$	$\int((y-\bar{r}_+)R_0+\frac{R_0^2}{2})=\frac{9}{8}$	$\frac{3}{4}$
mult_{C_0}	$r_-(R+1)+\frac{R(R+1)}{2}$	$\int\frac{\bar{r}_+^2-\bar{r}_-^2}{2}=\frac{7}{8}$	$\frac{7}{12}$
mult_{C_ζ}, $\zeta\neq 0,1$	$\frac{R(R+1)}{2}$	$\int\frac{R_0^2}{2}=\frac{45}{128}$	$\frac{15}{64}$
mult_{Q_i}	$(\epsilon+2y)(R+1)$	$\int 2yR_0=4$	$\frac{8}{3}$
$\mathrm{mult}_{Q_e}\, q_e^*$	$(2w+\epsilon)(R+1)$	$\int 2wR_0=\frac{11}{4}$	$\frac{11}{6}$

The entries of Table 10 do not depend on ζ or on the basis. Hence, for any eigen-basis divisor D_m of W_0 at level m,

$$\begin{aligned}
&\mathrm{mult}_{C_i} D_m \leq \tfrac{1}{3}+O(\tfrac{1}{m}),\quad \mathrm{mult}_{C_a} D_m \leq \tfrac{15}{32}+O(\tfrac{1}{m}),\quad \mathrm{mult}_{C_e} D_m \leq \tfrac{3}{4}+O(\tfrac{1}{m}),\\
&\mathrm{mult}_{C_0} D_m \leq \tfrac{7}{12}+O(\tfrac{1}{m}),\quad \mathrm{mult}_{C_\zeta} D_m \leq \tfrac{15}{64}+O(\tfrac{1}{m})\quad (\zeta\in\mathbb{C}\backslash\{0,1\}),\\
&\mathrm{mult}_{Q_i} D_m \leq \tfrac{8}{3}+O(\tfrac{1}{m}),\quad \mathrm{mult}_{Q_e} q_e^* D_m \leq \tfrac{11}{6}+O(\tfrac{1}{m}),
\end{aligned} \tag{9.33}$$

where the constants in $O(1/m)$ depend neither on D_m nor on ζ. By (9.30), $\mathrm{Supp}\, D_m$ is contained in the union of C_i, C_a, C_e, and finitely many curves C_ζ with $\zeta\in\mathbb{C}\backslash\{1\}$.

Step 4. In this step we show that, for any real number c with $0<c<9/8$, the pair (E_0, cD_m) is klt for any $m\gg 0$ and any eigen-basis divisor D_m of W_0 at level m.

Recall that D_m is an eigen-basis divisor at level m as in **Step 3**. By (9.33), there exists m_0 depending only on c such that for any $m\geq m_0$ and any D_m,

$$\begin{aligned}
&c\,\mathrm{mult}_C D_m<1 \text{ for any prime divisor } C\subset E_0,\\
&c\,\mathrm{mult}_{Q_i} D_m<3,\qquad c\,\mathrm{mult}_{Q_e} q_e^* D_m<3.
\end{aligned} \tag{9.34}$$

Here we use that the largest constant in the first two lines of (9.33) is $\frac{3}{4}$ and that $\frac{3}{4}c<1$, $\frac{8}{3}c<3$, and $\frac{11}{6}c<3$. The first and the third inequality hold for $c<\frac{4}{3}$, and only $\frac{8}{3}c<3$, at z_i, requires $c<\frac{9}{8}$. Fix $m\geq m_0$ and D_m.

Away from z_i, z_a, and z_e. The surface E_0 is smooth outside $\{z_a, z_e\}$. A direct calculation shows that any two distinct curves among C_i, C_a, C_e, and C_ζ ($\zeta\in\mathbb{C}\backslash\{1\}$) intersect only at points of $\{z_i, z_a, z_e\}$: if $i=0$ or $e=0$ on C_ζ, then $a^2=0$, if $a=0$ on C_ζ, then $ie=0$, and on $C_\zeta\cap C_{\zeta'}$ with $\zeta\neq\zeta'$ we have $(\zeta'-\zeta)ie=0$ and then $a=0$. The curve C_ζ is the graph $e=a^2/(1-\zeta)$ on the chart $i=1$, and $C_\zeta\cap(i=0)=\{z_e\}$, so C_ζ is smooth outside z_e. The coordinate curves are smooth outside $\{z_i, z_a, z_e\}$. Thus $\mathrm{Supp}\, D_m$ is a simple normal crossing divisor outside $\{z_i, z_a, z_e\}$, and (E_0, cD_m) is klt there by (9.34) and [KM98, Corollary 2.31(3)].

At z_a. The chart $a\neq 0$ is $\mathbb{A}^2/\mu_2$, where μ_2 acts on the coordinates (i,e) by -1. The quotient morphism $q_a:\mathbb{A}^2\to\mathbb{A}^2/\mu_2$ is étale in codimension one. On the chart $a=1$ we have $f-\zeta ie=(1-\zeta)ie-1$, so C_ζ does not contain z_a. Near the origin, $\mathrm{Supp}\, q_a^* D_m$ is contained in the union of the coordinate axes $(i=0)$ and $(e=0)$, with coefficients $c\,\mathrm{mult}_{C_i} D_m<1$ and $c\,\mathrm{mult}_{C_e} D_m<1$. Thus $(\mathbb{A}^2, q_a^*(cD_m))$ is klt near the origin by [KM98, Corollary 2.31(3)], and (E_0, cD_m) is klt at z_a by [KM98, Proposition 5.20(4)].

At z_i. On the chart $i=1$ with coordinates (a,e), the curves through z_i in $\mathrm{Supp}\, D_m$ are $C_a=(a=0)$, $C_e=(e=0)$, and the curves $C_\zeta=(e=a^2/(1-\zeta))$. The variety Y_i is covered near Q_i by two charts. On the first chart, $e=a^2\gamma$ with coordinates (a,γ) and $Q_i=(a=0)$. The strict transforms of C_e and C_ζ are $(\gamma=0)$ and $(\gamma=1/(1-\zeta))$, and the constants $1/(1-\zeta)\neq 0$ are distinct for distinct ζ. The second chart is $\mathbb{A}^2/\mu_2$ with coordinates (α,σ), where $(a,e)=(\alpha\sigma,\sigma^2)$,

μ_2 acts by -1, and $Q_i = (\sigma = 0)$. Near the origin of the second chart, the only components are Q_i and the strict transform $(\alpha = 0)$ of C_a, since the strict transform of C_ζ is $(\alpha^2 = 1 - \zeta)$. Since mult_{Q_i} is the monomial valuation with weights $(1, 2)$ in the coordinates (a, e) at the smooth point z_i, we have $A_{E_0}(Q_i) = 1 + 2 = 3$ [JM12, Proposition 5.1], that is, $K_{Y_i} = \pi_i^* K_{E_0} + 2Q_i$. Define Δ_{Y_i} by $K_{Y_i} + \Delta_{Y_i} = \pi_i^*(K_{E_0} + cD_m)$. Then

$$\Delta_{Y_i} = (\pi_i^{-1})_*(cD_m) + (c\,\mathrm{mult}_{Q_i} D_m - 2)Q_i,$$

and all coefficients of Δ_{Y_i} are less than one by (9.34). On the first chart $\operatorname{Supp} \Delta_{Y_i}$ is a simple normal crossing divisor. On the second chart the pullback of Δ_{Y_i} to the double cover, which is étale in codimension one, is supported on the coordinate axes near the origin. Thus (Y_i, Δ_{Y_i}) is sub-klt near Q_i by [KM98, Corollary 2.31(3)] and [KM98, Proposition 5.20(4)], and (E_0, cD_m) is klt at z_i by [KM98, Lemma 2.30]. Here [KM98, Lemma 2.30, Corollary 2.31(3), and Proposition 5.20(4)] apply to sub-pairs, whose boundaries may have negative coefficients, such as the coefficient of Q_i in Δ_{Y_i}.

At z_e. Recall that $q_e : \mathbb{A}^2 \to \mathbb{A}^2/\mu_3$ is the index-three cover of the chart $e \neq 0$, with coordinates (i, a), and that q_e is étale in codimension one. Thus $K_{\mathbb{A}^2} + q_e^*(cD_m) = q_e^*(K_{E_0} + cD_m)$. On the chart $e = 1$ we have $f - \zeta ie = (1 - \zeta)i - a^2$, so near the origin the components of $q_e^* D_m$ are $(i = 0)$, $(a = 0)$, and the curves $(i = a^2/(1 - \zeta))$. The weighted blow-up π_e separates them as π_i separates C_a, C_e, and the curves C_ζ at z_i, with the roles of (a, e) and (C_a, C_e) played by (a, i) and (C_a, C_i): on the first chart $i = a^2\gamma$ with $Q_e = (a = 0)$, and on the second chart, a quotient of $\mathbb{A}^2$ by μ_2, only Q_e and the strict transform of $(a = 0)$ pass through the origin. Since mult_{Q_e} is the monomial valuation with weights $(2, 1)$ in (i, a), we have $A_{\mathbb{A}^2}(Q_e) = 2 + 1 = 3$ [JM12, Proposition 5.1], that is, $K_{Y_e} = \pi_e^* K_{\mathbb{A}^2} + 2Q_e$, so the coefficient of Q_e in the crepant pullback is $c\,\mathrm{mult}_{Q_e} q_e^* D_m - 2 < 1$ by (9.34). As at z_i, the pair $(\mathbb{A}^2, q_e^*(cD_m))$ is klt near the origin, and (E_0, cD_m) is klt at z_e by [KM98, Proposition 5.20(4)].

Step 5. In this step we conclude the proof.

Fix a real number c with $0 < c < 9/8$. By **Step 4**, (E_0, cD) is klt, hence log canonical, for any $m \gg 0$ and any eigen-basis divisor D of W_0 at level m. We deduce the bound for W_0 from Lemma 3.6: by Lemma 3.6(1) with $\Delta_{E_0} = 0$, $\lambda_m = \rho$, and $c_m = c$ for all m, we have $cS(W_0; v) \leq A_{E_0}(v)$ for any prime divisor v over E_0. Letting c tend to $9/8$, we obtain $S(W_0; v) \leq \frac{8}{9} A_{E_0}(v)$ for any prime divisor v over E_0.

Let F be a prime divisor over X with $x \in \mathrm{center}_X F$. Then the center of F on $\hat{X}_0$ intersects $\hat{\pi}_0^{-1}(x) = E_0$, so Lemma 3.1 can be used with $G = E_0$ and $9/8$ in place of λ, also when $\mathrm{center}_X F$ has positive dimension. Since $A_X(E_0) = 6$ and $S_X(E_0) = 59/12$, Lemma 3.1 implies

$$A_X(F) \geq \frac{9}{8} S_X(F) + \left(6 - \frac{9}{8} \cdot \frac{59}{12}\right) \mathrm{mult}_F E_0 = \frac{9}{8} S_X(F) + \frac{15}{32} \mathrm{mult}_F E_0.$$

Multiplying by $\frac{8}{9}$, we obtain

$$S_X(F) \leq \frac{8}{9} A_X(F) - \frac{5}{12} \mathrm{mult}_F E_0.$$

Since $\mathrm{mult}_F E_0 \geq 0$, we have $A_X(F) \geq \frac{9}{8} S_X(F)$ for any prime divisor F over X with $x \in \mathrm{center}_X F$. Thus $\delta_x(X) \geq 9/8$. □

10. Line links in genus twelve

In this section, we determine the flopping curves of the line link in genus twelve, their widths, and the local structure of the Hilbert scheme of lines at any line. The main result is Proposition 10.8, and the notation of this section is used in the following sections. Theorem 19.1 and the criterion of Theorem 1.3 use two consequences of the widths computed here: whether a special line intersects no other line, and the singularity of the Hilbert scheme of lines at a special line.

Set-up 10.1. Let X be a smooth prime Fano threefold of genus 12 and $L \subset X$ a line. Let $\pi : Y := \mathrm{Bl}_L X \to X$ with exceptional divisor E, write H also for π^*H, and let $Y \dashrightarrow Y^+$ be the flop of the line link of Theorem 2.23, where $Y^+ = \mathrm{Bl}_\Gamma V_5$ is the blow-up of the rational normal quintic curve $\Gamma \subset V_5$, h is the pullback of the hyperplane class of V_5, and R is the exceptional divisor of $Y^+ \to V_5$, so that $h = H - 2E$ and $R = H - 3E$ under strict transform. If L is non-special, then $E \cong \mathbb{F}_1$, and if L is special, then $E \cong \mathbb{F}_3$. In both cases s denotes the negative section of E and f the class of a fiber of $E \to L$, and for $x \in L$ we write f_x for the fiber over x.

The description of E follows from Lemma 2.12(3), since $E \cong \mathbb{P}(N_{L/X})$ with $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ or $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. The line link is the diagram

$$\begin{array}{ccc} Y & \dashrightarrow & Y^+ = \mathrm{Bl}_\Gamma V_5 \\ {\scriptstyle\pi}\downarrow & & \downarrow \\ X & & V_5. \end{array}$$

The next construction recalls Reid's pagoda resolution of a flop [Rei83, (5.3), (5.4), (5.8)] (see also [BO95, Section 3]), and Lemmas 10.3 and 10.4 compare divisors on both sides of a flop through Reid's pagoda resolution.

Construction-Proposition 10.2. Let $Y \dashrightarrow Y^+$ be a flop between smooth threefolds and $C \subset Y$ a connected component of the flopping locus which satisfies the hypotheses of Definition-Proposition 2.21, of width w, with flopped curve C^+. Let $T_0 := Y$ and $C_0 := C$. For $1 \le i \le w$, let $T_i \to T_{i-1}$ be the blow-up of C_{i-1}, let P_i be the exceptional divisor of $T_i \to T_{i-1}$, written P_i also for the strict transforms of P_i, and, for $i < w$, let $C_i \subset P_i$ be the negative section of P_i, which is isomorphic to $\mathbb{F}_2$ on T_i by [Rei83, Theorem (5.4)]. Let $W := T_w$ and $p : W \to Y$ the composition. Then the following hold over a neighborhood of C.

(1) $P_i \cong \mathbb{F}_2$ on W for $i < w$, $P_w \cong \mathbb{F}_0$, and the flop extends to a morphism $q : W \to Y^+$, which is the composition of w blow-ups of smooth rational curves starting with the blow-up of C^+; the strict transforms on W of the exceptional divisors of the w blow-ups composing q are $P_1, \dots, P_w$, in the order of the blow-ups.

(2) With σ_i and ℓ_i the classes of the negative section and of a fiber of $P_i \cong \mathbb{F}_2$ for $i < w$, ℓ_w the class of a fiber of $P_w \to C_{w-1}$, and ℓ_w^+ the class of the other ruling of P_w, we have $P_j|_{P_i} \equiv 0$ for $|i-j| \ge 2$ and

$$\begin{aligned} P_i|_{P_i} &\equiv -2\sigma_i - 2\ell_i, \quad P_{i+1}|_{P_i} \equiv \sigma_i, \quad P_{i-1}|_{P_i} \equiv \sigma_i + 2\ell_i \quad (i < w), \\ P_w|_{P_w} &\equiv -\ell_w - \ell_w^+, \quad P_{w-1}|_{P_w} \equiv \ell_w + \ell_w^+, \end{aligned}$$

where the terms with index 0 or $w+1$ are omitted.

(3) For $1 \le i \le w$, the morphism $p|_{P_i} : P_i \to C$ is the composition of the ruling with fiber class ℓ_i and an isomorphism onto C, and the morphism $q|_{P_i} : P_i \to C^+$ is the composition of the ruling with fiber class ℓ_i if $i < w$, and ℓ_w^+ if $i = w$, and an isomorphism onto C^+.

Proof. By [Rei83, Theorem (5.4)], for $i < w$ the curve C_i has width $w - i$ in T_i. By [Rei83, (5.1), Definition (5.3) and the sentence after it, and (5.9)], $N_{C_i/T_i} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ for $i < w-1$ and $N_{C_{w-1}/T_{w-1}} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, hence $P_w = \mathbb{P}(N_{C_{w-1}/T_{w-1}}) \cong \mathbb{F}_0$ on T_w (see also [Rei83, (5.5) and Pagoda (5.8)]). Since C_i is a Cartier divisor on the smooth surface $P_i \subset T_i$, the blow-up of C_i maps the strict transform of P_i isomorphically onto P_i.

We prove (2). A sub-line bundle M of a rank-two vector bundle N on $\mathbb{P}^1$ defines a section of $\mathbb{P}(N) \to \mathbb{P}^1$ with self-intersection $\deg N - 2\deg M$. For $2 \le i \le w$, the curve $P_{i-1} \cap P_i$ is the section of $P_i = \mathbb{P}(N_{C_{i-1}/T_{i-1}})$ given by $N_{C_{i-1}/P_{i-1}} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$, with self-intersection $-2 + 4 = 2$, while C_i is the section given by the summand $\mathcal{O}_{\mathbb{P}^1}$ if $i < w$. Thus $P_{i-1} \cap P_i$ is disjoint from C_i, hence $P_{i-1} \cap P_j = \emptyset$ for $j > i$, and $P_{i-1}|_{P_i}$ does not change under the later blow-ups. Since a section of $\mathbb{F}_2$ with self-intersection 2 has class $\sigma + 2\ell$ and a section of $\mathbb{P}^1 \times \mathbb{P}^1 \to C_{w-1}$ with self-intersection 2 has class $\ell_w + \ell_w^+$, we get $P_{i-1}|_{P_i}$, and $P_{i+1}|_{P_i} \equiv C_i = \sigma_i$ for $i < w$. Let

$\mathcal{I}$ be the ideal sheaf of C_{i-1} in T_{i-1}. By [Har77, Theorem II.8.24(b)(c)], $P_i \cong \mathbb{P}(\mathcal{I}/\mathcal{I}^2)$ on T_i and $P_i|_{P_i} \cong \mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)}(-1)$. Since $(\mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)}(1) \cdot \ell_i) = 1$ and $(\mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)}(1))^2 = \deg(\mathcal{I}/\mathcal{I}^2)$ [Har77, Appendix A, §3], on T_i we have

$$(P_i|_{P_i} \cdot \ell_i)_{P_i} = -1, \qquad (P_i|_{P_i})^2_{P_i} = \deg(\mathcal{I}/\mathcal{I}^2) = -\deg N_{C_{i-1}/T_{i-1}} = 2,$$

so $P_i|_{P_i} \equiv -\sigma_i - 2\ell_i$ for $i < w$ and $P_w|_{P_w} \equiv -\ell_w - \ell_w^+$. The blow-up of $C_i \subset P_i$ subtracts σ_i from $P_i|_{P_i}$ for $i < w$, and the later blow-ups do not meet P_i. This proves (2).

By [Rei83, (5.8)] and [BO95, Section 3], near C the flop extends to a morphism $W \to Y^+$ which first contracts $P_w \cong \mathbb{P}^1 \times \mathbb{P}^1$ along the ruling with fiber class ℓ_w^+, and then contracts the images of $P_{w-1}, \dots, P_1$ one by one onto smooth rational curves, the last one onto C^+. Since C^+ also has width w, [Rei83, (5.8)] applied to the inverse flop $Y^+ \dashrightarrow Y$ shows that the center of any blow-up composing q is mapped isomorphically onto C^+. Since $\mathbb{F}_2$ has a unique ruling, the image of P_i for $i < w$ is contracted along the ruling with fiber class ℓ_i. Read from Y^+, the description proves (1) and the statement on q in (3). The statement on p in (3) holds since $P_i \to C_{i-1}$ is the ruling with fiber class ℓ_i and C_{i-1} is a section over C_{i-2} for $i \geq 2$. □

Lemma 10.3. *Notation as in Construction-Proposition 10.2. Then for any Cartier divisor D on Y with strict transform D^+ on Y^+, we have $(D^+ \cdot C^+) = -(D \cdot C)$ and*

$$p^*D - q^*D^+ = -(D \cdot C) \sum_{i=1}^{w} iP_i$$

over a neighborhood of C.

Proof. Let $d := (D \cdot C)$ and $d^+ := (D^+ \cdot C^+)$, and let $\sigma_i, \ell_i, \ell_w^+$ be the classes of Construction-Proposition 10.2(2). Since p and q are isomorphisms outside $P_1 \cup \cdots \cup P_w$, there exist integers $c_1, \dots, c_w$ with

$$p^*D - q^*D^+ = \sum_{i=1}^{w} c_i P_i$$

over a neighborhood of C. By Construction-Proposition 10.2(3), $p^*D|_{P_i} \equiv d\ell_i$ for $1 \leq i \leq w$, $q^*D^+|_{P_i} \equiv d^+\ell_i$ for $i < w$, and $q^*D^+|_{P_w} \equiv d^+\ell_w^+$. We restrict both sides to each P_i and use Construction-Proposition 10.2(2). If $w = 1$, restricting to P_1 yields $-c_1(\ell_1 + \ell_1^+) \equiv d\ell_1 - d^+\ell_1^+$, so $c_1 = -d$ and $d^+ = -d$. Assume that $w \geq 2$. Restricting to P_w yields

$$(c_{w-1} - c_w)(\ell_w + \ell_w^+) \equiv d\ell_w - d^+\ell_w^+,$$

so $d^+ = -d$ and $c_{w-1} - c_w = d$. Restricting to P_1 yields

$$(c_2 - 2c_1)\sigma_1 - 2c_1\ell_1 \equiv (d - d^+)\ell_1 = 2d\ell_1,$$

so $c_1 = -d$ and $c_2 = -2d$. For $2 \leq i \leq w-1$, the coefficient of σ_i in the restriction to P_i yields $c_{i-1} - 2c_i + c_{i+1} = 0$. By induction on i, we get $c_i = -id$ for $1 \leq i \leq w$, and $c_{w-1} - c_w = d$ holds. □

Lemma 10.4 expresses the change of the cube of a divisor under a flop as a sum over the flopping curves weighted by their widths. For width one it is the classical formula for the Atiyah flop.

Lemma 10.4. *Let $Y \dashrightarrow Y^+$ be a flop between smooth projective threefolds whose flopping locus is a disjoint union of smooth rational curves $C_1, \dots, C_r$, each satisfying the hypotheses of Definition-Proposition 2.21, of widths $w_1, \dots, w_r$. Then for any Cartier divisor D on Y with strict transform D^+ on Y^+,*

$$D^3 - (D^+)^3 = \sum_{j=1}^{r} w_j (D \cdot C_j)^3.$$

Proof. Since $C_1, \dots, C_r$ are pairwise disjoint and the towers of Construction-Proposition 10.2 consist of blow-ups of smooth curves, they define a smooth projective threefold W with birational morphisms $p : W \to Y$ and $q : W \to Y^+$. Over C_j, let $P_{j,1}, \dots, P_{j,w_j}$ be the divisors and $\ell_{j,i}$, ℓ^+_{j,w_j} the classes of Construction-Proposition 10.2, and let $d_j := (D \cdot C_j)$. By Lemma 10.3 and Construction-Proposition 10.2(3),

$$D_W := q^*D^+ - p^*D = \sum_{j=1}^{r} d_j \sum_{i=1}^{w_j} iP_{j,i},$$

$p^*D|_{P_{j,i}} \equiv d_j\ell_{j,i}$, $D_W|_{P_{j,i}} \equiv -2d_j\ell_{j,i}$ for $i < w_j$, and $D_W|_{P_{j,w_j}} \equiv -d_j(\ell_{j,w_j} + \ell^+_{j,w_j})$. Since $\ell^2_{j,i} = 0$, we get $(p^*D)^2 \cdot P_{j,i} = 0$ for all i, and $p^*D \cdot D_W \cdot P_{j,i} = D_W^2 \cdot P_{j,i} = 0$ for $i < w_j$, while

$$p^*D \cdot D_W \cdot P_{j,w_j} = -d_j^2, \qquad D_W^2 \cdot P_{j,w_j} = 2d_j^2.$$

Since $D_W = \sum_j d_j \sum_{i=1}^{w_j} iP_{j,i}$ and only the terms with $i = w_j$ contribute, we get $(p^*D)^2 \cdot D_W = 0$, $p^*D \cdot D_W^2 = -\sum_j w_j d_j^3$, and $D_W^3 = 2\sum_j w_j d_j^3$. Since p and q are birational, $(p^*D)^3 = D^3$ and $(q^*D^+)^3 = (D^+)^3$. Thus

$$(D^+)^3 = (p^*D + D_W)^3 = D^3 + 3(p^*D)^2 \cdot D_W + 3p^*D \cdot D_W^2 + D_W^3 = D^3 - 3\sum_{j=1}^{r} w_j d_j^3 + 2\sum_{j=1}^{r} w_j d_j^3,$$

which proves the lemma. □

The next three lemmas compute the Hilbert scheme $\Sigma(X)$ of lines near $[L]$ for a special line L, in formal coordinates. For an isomorphism $t : L \to \mathbb{P}^1$, let $U_0 := L \backslash t^{-1}(\infty)$, $U_1 := L \backslash t^{-1}(0)$, and $t' := 1/t$. For $j = 0, 1$, let $X_j^\circ \subset X$ be an affine open subset with $X_j^\circ \cap L = U_j$, and let $\widehat{\mathcal{O}}_j$ be the I_L-adic completion of $\mathcal{O}_X(X_j^\circ)$.

Lemma 10.5. *Notation as in Set-up 10.1. Assume that L is special, and let $t : L \to \mathbb{P}^1$ be an isomorphism. Then there exist elements $x, y \in \widehat{\mathcal{O}}_0$ and $x', y' \in \widehat{\mathcal{O}}_1$, lifts of t and t' again written t and t', complex numbers g_1, g_0, g_{-1}, and $\gamma_3 \in \mathbb{C}[t, t^{-1}]$ satisfying the following.*

(1) $\widehat{\mathcal{O}}_0 = \mathbb{C}[t][[x, y]]$ and $\widehat{\mathcal{O}}_1 = \mathbb{C}[t'][[x', y']]$, with $I_L = (x, y)$, resp. $I_L = (x', y')$, and over $U_0 \cap U_1$ we have

$$\begin{aligned} &t' = t^{-1} + O(I_L^3), \qquad x' = t^{-1}x + O(I_L^3), \\ &y' = t^2y + (g_1t + g_0 + g_{-1}t^{-1})x^2 + \gamma_3(t)x^3 + O(I_L^4) + O(x^2y, xy^2, y^3). \end{aligned} \tag{10.1}$$

(2) Let $z := y/x$ and $z' := y'/x'$. Then (t, x, z) and (t', x', z') are formal coordinates of Y along $s \cap \pi^{-1}(U_0)$ and $s \cap \pi^{-1}(U_1)$, in which $s = \{x = z = 0\}$ and $s = \{x' = z' = 0\}$, and

$$\begin{aligned} &t' = t^{-1} + O(x^3), \quad x' = t^{-1}x + O(x^3), \\ &z' = t^3z + (g_1t^2 + g_0t + g_{-1})x + t\gamma_3(t)x^2 + O((x, z)^3). \end{aligned} \tag{10.2}$$

(3) Identify $H^0(s, N_{E/Y}|_s) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1))$ with $\mathbb{C}^2$ by $\lambda + \mu t \to (\lambda, \mu)$, and $H^1(s, N_{s/E}) \cong H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-3))$ with $\mathbb{C}^2$ by sending the class of a cocycle $\eta(t)$ in the trivialization z' to the coefficients of t^{-1} and t^{-2} in $t^{-3}\eta(t)$. Then the connecting map $\rho : H^0(s, N_{E/Y}|_s) \to H^1(s, N_{s/E})$ of the exact sequence $0 \to N_{s/E} \to N_{s/Y} \to N_{E/Y}|_s \to 0$ is

$$\rho(\lambda, \mu) = (g_1\lambda + g_0\mu,\ g_0\lambda + g_{-1}\mu).$$

Proof. Since X_j° is affine and X is smooth along U_j, lifts of t (resp. t') and of a basis of I_L/I_L^2 over U_j yield the isomorphisms in (1). Write $N := N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, so $N^* \cong \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(2)$. Choosing the bases compatible with the splitting of N^*, we get $t' \equiv t^{-1}$ modulo I_L, as well as $x' \equiv t^{-1}x$ and $y' \equiv t^2y$ modulo I_L^2. We improve the transition first in degree one and then in degree two. A correction term of degree $k \in \{1, 2\}$ in the transition of t', x', or y' is a Čech 1-cocycle of the cover $\{U_0, U_1\}$ with values in $T_L \otimes \operatorname{Sym}^k N^*$, $\mathcal{O}_{\mathbb{P}^1}(1) \otimes \operatorname{Sym}^k N^*$, or

$\mathcal{O}_{\mathbb{P}^1}(-2) \otimes \operatorname{Sym}^k N^*$, respectively. If a component of the cocycle in a direct summand has zero class in H^1, it is the difference of regular sections over U_0 and U_1, and changing the corresponding coordinate over U_0 and over U_1 by the corresponding sections removes the component. The change modifies the transition only in higher degree, and in degree three only in the terms left arbitrary in (10.1). We have

$$T_L \otimes N^* \cong \mathcal{O}(1) \oplus \mathcal{O}(4), \qquad T_L \otimes \operatorname{Sym}^2 N^* \cong \mathcal{O} \oplus \mathcal{O}(3) \oplus \mathcal{O}(6),$$
$$\mathcal{O}(1) \otimes \operatorname{Sym}^2 N^* \cong \mathcal{O}(-1) \oplus \mathcal{O}(2) \oplus \mathcal{O}(5), \qquad \mathcal{O}(-2) \otimes \operatorname{Sym}^2 N^* \cong \mathcal{O}(-4) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(2),$$

where in the last one the summands correspond to x^2, xy, and y^2. The degree-one corrections of x' and y' vanish by the choice of bases compatible with the splitting of N^*. All line bundles above on $\mathbb{P}^1$ have vanishing H^1 except $\mathcal{O}(-4)$. Thus we may assume (10.1) with some $\gamma_2(t) \in \mathbb{C}[t, t^{-1}]$ in place of $g_1 t + g_0 + g_{-1}t^{-1}$. Replacing y by $y + A(t)x^2$ and y' by $y' + B(t')x'^2$ with $A \in \mathbb{C}[t]$ and $B \in \mathbb{C}[t']$ replaces $\gamma_2(t)$ by $\gamma_2(t) - t^2A(t) + t^{-2}B(t^{-1})$ and changes the transition only in $O(I_L^4) + O(x^2y, xy^2, y^3)$. Thus we may assume that $\gamma_2 = g_1 t + g_0 + g_{-1}t^{-1}$. This proves (1).

On the chart of $Y = \operatorname{Bl}_L X$ with coordinates (t, x, z) we have $y = xz$, $I_L\mathcal{O}_Y = (x)$, and $E = \{x = 0\}$. Since $I_L^3 = x^3(1, z)^3$ on the chart, (10.1) yields $x' = t^{-1}x(1 + O(x^2))$ and

$$z' = \frac{y'}{x'} = t\big(t^2 z + \gamma_2(t)x + \gamma_3(t)x^2 + O((x, z)^3)\big)(1 + O(x^2)),$$

which is (10.2). On E we get $z' = t^3 z$, so $\{z = 0\} \subset E$ is a section of $E \to L$ with normal bundle $\mathcal{O}_{\mathbb{P}^1}(-3)$ in E, that is, the negative section s. This proves (2).

By (10.2), the transition of $N_{s/Y}$ in the frames given by (x, z) and (x', z') is $x' = t^{-1}x$ and $z' = t^3 z + t\gamma_2(t)x$, where x is the coordinate of $N_{E/Y}|_s \cong \mathcal{O}_{\mathbb{P}^1}(1)$ and z the coordinate of $N_{s/E} \cong \mathcal{O}_{\mathbb{P}^1}(-3)$. The section $\lambda + \mu t$ of $N_{E/Y}|_s$ lifts to $(x, z) = (\lambda + \mu t, 0)$ over U_0 and to $(x', z') = (\lambda t' + \mu, 0)$ over U_1, and the difference of the lifts over $U_0 \cap U_1$ is the cocycle $\eta(t) = t\gamma_2(t)(\lambda + \mu t)$ with values in $N_{s/E}$. A cocycle η is a coboundary if and only if $t^{-3}\eta \in t^{-3}\mathbb{C}[t^{-1}] + \mathbb{C}[t]$, so the identification in (3) is an isomorphism. The coefficients of t^{-1} and t^{-2} in $t^{-2}\gamma_2(t)(\lambda + \mu t)$ are $g_1\lambda + g_0\mu$ and $g_0\lambda + g_{-1}\mu$, which proves (3). □

Lemma 10.6. *Notation as in Set-up 10.1. Assume that L is special. Then the completed local ring of $\Sigma(X)$ at $[L]$ is isomorphic to $\mathbb{C}[[a, b]]/(\Phi)$ for some $\Phi \in (a, b)^2$, and the quadratic part of Φ is non-zero. The rank of the quadratic part of Φ equals the rank of the connecting map $H^0(s, N_{E/Y}|_s) \to H^1(s, N_{s/E})$ of the exact sequence*

$$0 \to N_{s/E} \to N_{s/Y} \to N_{E/Y}|_s \to 0,$$

where $N_{s/E} \cong \mathcal{O}_{\mathbb{P}^1}(-3)$ and $N_{E/Y}|_s \cong \mathcal{O}_{\mathbb{P}^1}(1)$. In particular the rank is 2 if $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, and the rank is 1 if $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

Proof. Let $\widehat{\mathcal{O}}$ be the completed local ring of $\Sigma(X)$ at $[L]$. By [Kol96, Theorem I.2.8], the Zariski tangent space of $\Sigma(X)$ at $[L]$ is $H^0(L, N_{L/X})$, and $h^0(L, N_{L/X}) = 2$ by Lemma 2.12(3). Thus $\widehat{\mathcal{O}} \cong \mathbb{C}[[a, b]]/I$ for an ideal $I \subset (a, b)^2$. Since $\Sigma(X)$ is a Gorenstein curve [KPS18, Corollary 5.2.9], $\widehat{\mathcal{O}}$ is a one-dimensional Cohen–Macaulay ring, so the associated primes of I have height one. Since $\mathbb{C}[[a, b]]$ is a unique factorization domain, a height-one prime is principal, (π_k) say, and a (π_k)-primary ideal is (π_k^m) for some m. Thus $I = (\Phi)$ for some $\Phi \in (a, b)^2$.

By the universal property of $\Sigma(X)$, for $n \geq 1$, local homomorphisms $\widehat{\mathcal{O}} \to \mathbb{C}[\epsilon]/(\epsilon^n)$ correspond to closed subschemes $Z \subset X \times \operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^n)$ flat over $\mathbb{C}[\epsilon]/(\epsilon^n)$ with closed fiber L. Fix t and the coordinates of Lemma 10.5. Since Z is flat over $\mathbb{C}[\epsilon]/(\epsilon^n)$ and its closed fiber is $\{x = y = 0\}$ over U_0, the subscheme Z is the graph $x = x_\epsilon(t)$, $y = y_\epsilon(t)$ with $x_\epsilon, y_\epsilon \in \epsilon\mathbb{C}[\epsilon, t]/(\epsilon^n)$ over U_0, and similarly $x' = x'_\epsilon(t')$, $y' = y'_\epsilon(t')$ over U_1, subject to (10.1). Since $I_L\mathcal{O}_Z \subset (\epsilon)$, modulo ϵ^3 the conditions are $x'_\epsilon = t^{-1}x_\epsilon$ and $y'_\epsilon = t^2 y_\epsilon + \gamma_2(t)x_\epsilon^2$, where $\gamma_2 := g_1 t + g_0 + g_{-1}t^{-1}$.

For $n = 2$, write $x_\epsilon = \epsilon X_1$ and $y_\epsilon = \epsilon Y_1$. Then $X_1' = t^{-1}X_1$ and $Y_1' = t^2 Y_1$, so $X_1 = \lambda + \mu t$ and $Y_1 = 0$. We choose the generators a, b so that the tangent vector $(a, b) \to (\lambda\epsilon, \mu\epsilon)$ corresponds

to $X_1 = \lambda + \mu t$. For $n = 3$, write $x_\epsilon = \epsilon(\lambda + \mu t) + \epsilon^2 X_2$ and $y_\epsilon = \epsilon^2 Y_2$. The condition on X_2 is $X_2' = t^{-1} X_2$, which has the solutions $X_2 \in H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1))$. The condition on Y_2 is

$$t^{-2} Y_2'(t^{-1}) - Y_2(t) = t^{-2} \gamma_2(t)(\lambda + \mu t)^2$$

with $Y_2 \in \mathbb{C}[t]$ and $Y_2' \in \mathbb{C}[t']$. The left-hand side runs through the Laurent polynomials without the monomial t^{-1}. Thus the tangent vector (λ, μ) extends to a $\mathbb{C}[\epsilon]/(\epsilon^3)$-point if and only if

$$\mathfrak{q}(\lambda, \mu) := g_1 \lambda^2 + 2 g_0 \lambda \mu + g_{-1} \mu^2$$

vanishes. On the other hand, let Φ_2 be the quadratic part of Φ. A local homomorphism $\widehat{\mathcal{O}} \to \mathbb{C}[\epsilon]/(\epsilon^3)$ with $a \to \lambda\epsilon + \lambda_2 \epsilon^2$ and $b \to \mu\epsilon + \mu_2 \epsilon^2$ exists if and only if $\Phi(\lambda\epsilon + \lambda_2\epsilon^2, \mu\epsilon + \mu_2\epsilon^2) \equiv \epsilon^2 \Phi_2(\lambda, \mu)$ vanishes modulo ϵ^3. Thus the zero set of $\mathfrak{q}$ in $\mathbb{C}^2$ equals the zero set of Φ_2 in $\mathbb{C}^2$.

The symmetric matrix of $\mathfrak{q}$ is the matrix of the connecting map ρ in Lemma 10.5(3). By Theorem 2.23, s is a flopping curve, so $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ by [KPS18, Lemma 5.2.5] and the paragraph following it. Since $H^0(s, N_{s/E}) = 0$, we have $\operatorname{Ker} \rho = H^0(s, N_{s/Y})$, so $\operatorname{rank} \rho = 2$ in the first case and $\operatorname{rank} \rho = 1$ in the second case. In particular $\mathfrak{q} \neq 0$, so $\Phi_2 \neq 0$, since otherwise any tangent vector would extend. A non-zero binary quadratic form is a product of two linear forms, and the zero set of the form is the union of the kernels of the linear forms. Thus Φ_2 and $\mathfrak{q}$ are proportional, and $\operatorname{rank} \Phi_2 = \operatorname{rank} \mathfrak{q} = \operatorname{rank} \rho$. □

Lemma 10.7. *Notation as in Set-up 10.1. Assume that L is special and $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Then $w(s) \geq 2$, and the completed local ring of $\Sigma(X)$ at $[L]$ is isomorphic to $\mathbb{C}[[a, b]]/(\Phi)$ for some $\Phi \in (a, b)^2$ with quadratic part a^2. For any such Φ, the coefficient of b^3 in $\Phi(0, b)$ is zero if and only if $w(s) \geq 3$.*

Proof. By Theorem 2.23, s is a flopping curve and the flopping curves are pairwise disjoint, so s is contracted to an isolated singular point and satisfies the hypotheses of Definition-Proposition 2.21. By [Rei83, Definition (5.3), Theorem (5.4) and (5.7)], $w(s)$ is the largest n such that there exists a closed subscheme $s_n \subset Y$ with $s \subset s_n$ and $s_n \cong s \times \operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^n)$. Such s_2 is an embedded deformation of s over $\mathbb{C}[\epsilon]/(\epsilon^2)$ whose section of $N_{s/Y}$ vanishes nowhere. Conversely, a nowhere vanishing section of $N_{s/Y}$ defines a deformation of s over $\mathbb{C}[\epsilon]/(\epsilon^2)$, which is abstractly trivial since $H^1(s, T_s) = 0$, and whose composition with the projection to Y is a closed immersion. A deformation of s over $\mathbb{C}[\epsilon]/(\epsilon^3)$ whose restriction to $\mathbb{C}[\epsilon]/(\epsilon^2)$ is given by a nowhere vanishing section is likewise abstractly trivial, and the composition of the deformation with the projection to Y is a closed immersion. Since $H^0(s, N_{s/Y}) \cong \mathbb{C}$ and the summand $\mathcal{O}_{\mathbb{P}^1}$ has a nowhere vanishing section, we get $w(s) \geq 2$, and $w(s) \geq 3$ if and only if the deformation of s over $\mathbb{C}[\epsilon]/(\epsilon^2)$ given by a non-zero element of $H^0(s, N_{s/Y})$ extends to an embedded deformation over $\mathbb{C}[\epsilon]/(\epsilon^3)$.

By Lemma 10.6, the connecting map ρ has rank one, and $\operatorname{Ker} \rho \subset H^0(s, N_{E/Y}|_s)$ is spanned by a section of $N_{E/Y}|_s \cong \mathcal{O}_{\mathbb{P}^1}(1)$ vanishing at a single point $e \in s$. We choose $t : L \to \mathbb{P}^1$ with $t(\pi(e)) = 0$, and Lemma 10.5 yields coordinates (t, x, y) and (t', x', y'), complex numbers g_1, g_0, g_{-1}, and γ_3. Then $(0, 1) \in \operatorname{Ker} \rho$, so $g_0 = g_{-1} = 0$ and $g_1 \neq 0$ by Lemma 10.5(3). Let c be the coefficient of t^{-2} in γ_3.

We show that the deformation of s extends over $\mathbb{C}[\epsilon]/(\epsilon^3)$ if and only if $c = 0$. By (10.2), a section of $N_{s/Y}$ is $(x, z) = (X_1, Z_1)$ and $(x', z') = (X_1', Z_1')$ with $X_1' = t^{-1} X_1$ and $Z_1' = t^3 Z_1 + g_1 t^2 X_1$. Thus $X_1 = \lambda + \mu t$, and $t^{-3} Z_1' - Z_1 = g_1 \lambda t^{-1} + g_1 \mu$ forces $\lambda = 0$, so $H^0(s, N_{s/Y})$ is spanned by $(X_1, Z_1) = (t, -g_1)$. An extension over $\mathbb{C}[\epsilon]/(\epsilon^3)$ is $x = \epsilon t + \epsilon^2 X_2$, $z = -g_1 \epsilon + \epsilon^2 Z_2$ over U_0 and $x' = \epsilon + \epsilon^2 X_2'$, $z' = \epsilon^2 Z_2'$ over U_1. The terms $O(x^3)$ and $O((x, z)^3)$ in (10.2) vanish modulo ϵ^3, so the conditions are $X_2' = t^{-1} X_2$ and

$$Z_2' = t^3 Z_2 + g_1 t^2 X_2 + t^3 \gamma_3(t).$$

Thus $X_2 = \lambda_2 + \mu_2 t$, and the second condition reads $t^{-3} Z_2' - Z_2 = g_1 \lambda_2 t^{-1} + g_1 \mu_2 + \gamma_3(t)$. The left-hand side runs through the Laurent polynomials without the monomials t^{-1} and t^{-2}. Since $g_1 \neq 0$, the coefficient of t^{-1} can be made zero by the choice of λ_2, and the coefficient of t^{-2} on the right-hand side is c. Thus an extension exists if and only if $c = 0$.

By the universal property of $\Sigma(X)$ and flatness, a $\mathbb{C}[\epsilon]/(\epsilon^n)$-point of $\Sigma(X)$ at $[L]$ is given by $x = x_\epsilon(t)$, $y = y_\epsilon(t)$ with $x_\epsilon, y_\epsilon \in \epsilon\mathbb{C}[\epsilon, t]/(\epsilon^n)$ over U_0 and $x' = x'_\epsilon(t')$, $y' = y'_\epsilon(t')$ over U_1, subject to (10.1). Let v be the tangent vector of $\Sigma(X)$ at $[L]$ given by $x_\epsilon = \epsilon t$ and $y_\epsilon = 0$. We show that v extends to a $\mathbb{C}[\epsilon]/(\epsilon^3)$-point, and that v extends to a $\mathbb{C}[\epsilon]/(\epsilon^4)$-point if and only if $c = 0$. Modulo ϵ^3, an extension of v is $x_\epsilon = \epsilon t + \epsilon^2 X_2$ and $y_\epsilon = \epsilon^2 Y_2$ with $X'_2 = t^{-1}X_2$ and $t^{-2}Y'_2 - Y_2 = g_1 t$. Thus $X_2 = \lambda_2 + \mu_2 t$, $Y_2 = -g_1 t$, and $Y'_2 = 0$, so v extends to a $\mathbb{C}[\epsilon]/(\epsilon^3)$-point. Modulo ϵ^4, write $x_\epsilon = \epsilon t + \epsilon^2 X_2 + \epsilon^3 X_3$ and $y_\epsilon = \epsilon^2 Y_2 + \epsilon^3 Y_3$. Since $y_\epsilon \in (\epsilon^2)$, the terms $O(I_L^4) + O(x^2y, xy^2, y^3)$ in (10.1) vanish modulo ϵ^4. The correction $O(I_L^3)$ of t' is in (ϵ^3) and changes $y'_\epsilon(t')$ only modulo ϵ^5 and $x'_\epsilon(t')$ only modulo ϵ^4, and the correction $O(I_L^3)$ of x' only changes the condition on X_3, which is solvable since $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1)) = 0$. The coefficient of ϵ^3 in $y' = t^2 y + g_1 t x^2 + \gamma_3(t)x^3$ yields the condition

$$t^{-2}Y'_3 - Y_3 = 2g_1 X_2 + t\gamma_3(t).$$

The left-hand side runs through the Laurent polynomials without the monomial t^{-1}, and the coefficient of t^{-1} on the right-hand side is c, for any X_2. Thus v extends to a $\mathbb{C}[\epsilon]/(\epsilon^4)$-point if and only if $c = 0$.

By Lemma 10.6, the completed local ring of $\Sigma(X)$ at $[L]$ is $\mathbb{C}[[a, b]]/(\Phi)$ with $\Phi \in (a, b)^2$ whose quadratic part Φ_2 has rank one. Thus Φ_2 is a non-zero multiple of the square of a linear form, and after a linear change of a and b and rescaling Φ, we get $\Phi_2 = a^2$. Let Φ be any equation with $\Phi_2 = a^2$, and write $\Phi = a^2 + \sum_{i+j\geq 3} \phi_{ij} a^i b^j$. Since $\Phi(\lambda\epsilon + \lambda_2\epsilon^2, \mu\epsilon + \mu_2\epsilon^2) \equiv \epsilon^2\Phi_2(\lambda, \mu)$ modulo ϵ^3, a tangent vector $(a, b) \to (\lambda\epsilon, \mu\epsilon)$ extends to a $\mathbb{C}[\epsilon]/(\epsilon^3)$-point if and only if $\lambda = 0$. Thus, after rescaling b, the vector v is $(a, b) \to (0, \epsilon)$. A $\mathbb{C}[\epsilon]/(\epsilon^4)$-point extending v sends b to an element $\tilde{\epsilon}$ with $\tilde{\epsilon} \equiv \epsilon$ modulo ϵ^2 and a to an element of (ϵ^2). Composing with the automorphism of $\mathbb{C}[\epsilon]/(\epsilon^4)$ sending $\tilde{\epsilon}$ to ϵ, we may assume that $b \to \epsilon$ and $a \to \alpha$ with $\alpha \in (\epsilon^2)$. Since $a^i b^j \to \alpha^i \epsilon^j \in (\epsilon^{2i+j})$ and $2i + j \geq 4$ for $i \geq 1$ and $i + j \geq 3$, we get

$$\Phi(\alpha, \epsilon) \equiv \phi_{03}\epsilon^3 \pmod{\epsilon^4}.$$

Thus v extends to a $\mathbb{C}[\epsilon]/(\epsilon^4)$-point if and only if $\phi_{03} = 0$. By the first and third paragraphs, $w(s) \geq 3$ if and only if $c = 0$, and by the fourth and fifth paragraphs, $c = 0$ if and only if $\phi_{03} = 0$. □

Proposition 10.8 relates the lines intersecting a special line L to the singularity of $\Sigma(X)$ at $[L]$.

Proposition 10.8. *Notation as in Set-up 10.1. The flopping locus of $Y \dashrightarrow Y^+$ consists of the pairwise disjoint strict transforms of the lines $L' \neq L$ intersecting L and, if L is special, of s. Each flopping curve satisfies the hypotheses of Definition-Proposition 2.21, so it has a width, and the widths of all flopping curves sum to 3. If L is special, let w_L be the width of s and $w_L(L')$ the width of the strict transform of a line $L' \neq L$ intersecting L. Then*

$$w_L + \sum_{L'\neq L,\ L'\cap L\neq\emptyset} w_L(L') = 3,$$

so $w_L \in \{1, 2, 3\}$, and $w_L = 3$ if and only if L intersects no other line. The completed local ring of $\Sigma(X)$ at $[L]$ is $\mathbb{C}[[t]]$ if L is non-special, and, if L is special, it is $\mathbb{C}[[a, b]]/(ab)$ for $w_L = 1$, $\mathbb{C}[[a, b]]/(a^2 + b^3)$ for $w_L = 2$, and $\mathbb{C}[[a, b]]/(a^2 + \varphi(b))$ with $\varphi \in b^4\mathbb{C}[[b]]$ for $w_L = 3$, where $\varphi = 0$ is allowed. If $\varphi \neq 0$ has order k, the germ is of type A_{k-1}, and if $\varphi = 0$ the completed local ring $\mathbb{C}[[a, b]]/(a^2)$ is non-reduced. In particular $\Sigma(X)$ has only smooth points and singular points of types A_1 and A_2 if and only if every special line on X intersects another line.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe the flopping curves and show that their widths sum to 3.

By Theorem 2.23 (see [KPS18, Theorem 5.2.2, Lemma 5.2.5] and [Fuj26a, Theorem 4.3(2)(3)(i), Proposition 4.5]), the flopping locus consists of the strict transforms of the lines $L' \neq L$ intersecting L and, if L is special, of s. The flopping curves are pairwise disjoint, and there are at most three of them. By [KPS18, Lemma 5.2.5] and the paragraph following it, each flopping curve C_j is

a smooth rational curve with $N_{C_j/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{C_j/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Since C_j is a connected component of the flopping locus, it is contracted by the flopping contraction to an isolated singular point. Thus C_j satisfies the hypotheses of Definition-Proposition 2.21 and has a width $w(C_j) \geq 1$.

Let H^+ be the strict transform of H on Y^+. Then $H^+ = 3h - 2R$ by Theorem 2.23. On $Y^+ = \mathrm{Bl}_\Gamma V_5$, we have $h^3 = 5$ and, by [IP99, Lemma 2.2.14],

$$h^2 \cdot R = 0, \qquad h \cdot R^2 = -(h \cdot \Gamma) = -5,$$
$$R^3 = -\deg N_{\Gamma/V_5} = -\big((-K_{V_5} \cdot \Gamma) - 2\big) = -(10 - 2) = -8.$$

Thus

$$(H^+)^3 = (3h - 2R)^3 = 27 \cdot 5 - 54 \cdot 0 + 36 \cdot (-5) - 8 \cdot (-8) = 135 - 180 + 64 = 19.$$

Since π maps the strict transform of a line L' isomorphically onto L' and maps s isomorphically onto L, we have $(H \cdot C_j) = 1$ for all j. Since $H^3 = 22$, Lemma 10.4 applied to $D = H$ yields

$$\sum_j w(C_j) = H^3 - (H^+)^3 = 22 - 19 = 3.$$

This proves the first two sentences of the proposition. If L is special, the left-hand side is $w_L + \sum_{L'} w_L(L')$, which proves the identity for w_L. Since any $w_L(L') \geq 1$, we get $w_L \in \{1, 2, 3\}$, and $w_L = 3$ if and only if no line $L' \neq L$ intersects L.

Step 2. In this step we treat the non-special lines.

Assume that L is non-special. Then $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ by Lemma 2.12(3), so the Zariski tangent space $H^0(L, N_{L/X})$ of $\Sigma(X)$ at $[L]$ [Kol96, Theorem I.2.8] is one-dimensional. Since $\Sigma(X)$ is a curve (Lemma 2.12(4)), the local ring of $\Sigma(X)$ at $[L]$ is regular of dimension one, and so is the completion of the local ring [Har77, Theorem I.5.4A(c)(d)]. By the Cohen structure theorem, the completed local ring is $\mathbb{C}[[t]]$.

Step 3. In this step we bring the equation of $\Sigma(X)$ at a special line into a normal form.

Assume that L is special. By Lemma 10.6, the completed local ring of $\Sigma(X)$ at $[L]$ is $\mathbb{C}[[a, b]]/(\Phi)$ with $\Phi \in (a, b)^2$ and with non-zero quadratic part Φ_2. After a linear change of a and b and rescaling Φ, we may assume that $\Phi_2 = a^2 + \kappa b^2$ with $\kappa = 1$ if $\operatorname{rank} \Phi_2 = 2$ and $\kappa = 0$ if $\operatorname{rank} \Phi_2 = 1$. Since Φ_2 has no monomial ab, we have $\partial\Phi/\partial a = 2a + r(a, b)$ with $r \in (a, b)^2$ and $r(0, b) \in (b^2)$. By the formal implicit function theorem (Hensel's lemma in $\mathbb{C}[[b]][[a]]$), there exists $\alpha \in b^2\mathbb{C}[[b]]$ with $(\partial\Phi/\partial a)(\alpha(b), b) = 0$. Let $a_1 := a - \alpha(b)$ and $\varphi(b) := \Phi(\alpha(b), b)$. Taylor's formula in a_1 yields $\Phi = \varphi(b) + a_1^2 u$ with a unit u, $u(0, 0) = 1$. Let $a_2 := a_1\sqrt{u}$. Then

$$\Phi = a_2^2 + \varphi(b).$$

Since $\alpha \in (b^2)$, any monomial $a^i b^j$ with $i \geq 1$ and $i + j \geq 3$ maps into (b^4) under $a \to \alpha(b)$, and so

$$\varphi(b) \equiv \kappa b^2 + c' b^3 \pmod{b^4},$$

where c' is the coefficient of b^3 in $\Phi(0, b)$.

Step 4. In this step we determine the germ at a special line from w_L.

By [Rei83, (5.1), Definition (5.3) and the sentence after it, and (5.9)], $w_L = 1$ if and only if $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, and then $\operatorname{rank} \Phi_2 = 2$ by Lemma 10.6. Thus $\kappa = 1$ and $\varphi = b^2 v(b)$ with a unit v, $v(0) = 1$. Let $b_2 := b\sqrt{v}$. Then $\Phi = a_2^2 + b_2^2 = (a_2 + \sqrt{-1}b_2)(a_2 - \sqrt{-1}b_2)$, and the completed local ring is $\mathbb{C}[[a, b]]/(ab)$.

If $w_L \geq 2$, then $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, so $\operatorname{rank} \Phi_2 = 1$, $\kappa = 0$, and $\Phi_2 = a^2$. By Lemma 10.7, $c' = 0$ if and only if $w_L \geq 3$, that is, $w_L = 3$ by **Step 1**. If $w_L = 2$, then $\varphi = c' b^3 v(b)$ with $c' \neq 0$ and a unit v, $v(0) = 1$. Let $b_2 := b(c'v)^{1/3}$ for a cube root of the unit $c'v$. Then $\Phi = a_2^2 + b_2^3$, and the completed local ring is $\mathbb{C}[[a, b]]/(a^2 + b^3)$. If $w_L = 3$, then $\varphi \in b^4\mathbb{C}[[b]]$.

Step 5. In this step we prove the remaining assertions.

Let $w_L = 3$. If $\varphi \neq 0$ has order $k \geq 4$, write $\varphi = b^k v(b)$ with a unit v. Let $b_2 := bv^{1/k}$ for a k-th root of v. Then $\Phi = a_2^2 + b_2^k$, so the germ is of type A_{k-1}, that is, the completed local ring is isomorphic to $\mathbb{C}[[a,b]]/(a^2+b^k)$. If $\varphi = 0$, the completed local ring is $\mathbb{C}[[a,b]]/(a^2)$, in which a is a non-zero nilpotent element.

If every special line intersects another line, then $w_L \leq 2$ for any special line L by **Step 1**, so $\Sigma(X)$ has only smooth points and singular points of types A_1 and A_2 by **Steps 2 and 4**. Conversely, if a special line L intersects no other line, then $w_L = 3$, and the completed local ring $\widehat{\mathcal{O}}$ of $\Sigma(X)$ at $[L]$ is non-reduced or of type A_{k-1} with $k \geq 4$. A smooth germ and germs of types A_1 and A_2 are reduced. For $\widehat{\mathcal{O}} = \mathbb{C}[[a,b]]/(\Phi)$, the module $\widehat{\Omega}$ of completed Kähler differentials of $\widehat{\mathcal{O}}$ over $\mathbb{C}$ is $(\widehat{\mathcal{O}}da \oplus \widehat{\mathcal{O}}db)/\widehat{\mathcal{O}}d\Phi$, so its first Fitting ideal is $(\partial\Phi/\partial a, \partial\Phi/\partial b)\widehat{\mathcal{O}}$, which depends only on $\widehat{\mathcal{O}}$ by [Eis95, Corollary-Definition 20.4]. Thus the length of $\widehat{\mathcal{O}}/(\partial\Phi/\partial a, \partial\Phi/\partial b)\widehat{\mathcal{O}}$ depends only on $\widehat{\mathcal{O}}$. For $\Phi = a^2 + b^k$ the length equals $k-1$, which is 0, 1, and 2 for the smooth germ and the germs of types A_1 and A_2. Thus $[L]$ is neither a smooth point nor a singular point of type A_1 or A_2. □

The following Definition-Proposition is the genus twelve form of Proposition 5.3: it describes the lines intersecting L in terms of the base scheme of the restriction of $|h|$ to E. The case analysis of Section 13 is based on it.

Definition-Proposition 10.9. Notation as in Set-up 10.1, and assume that L is non-special. Let

$$U := \operatorname{Im}\left(H^0(Y, H-2E) \to H^0(E, \mathcal{O}_E(2s+3f))\right)$$

and let $Z_U \subset E$ be the base scheme of U. Let Inc_L be the incidence scheme of pairs (L', y) with $L' \neq L$ a line and $y \in L' \cap L$, that is, the preimage of $\Sigma(X)\backslash\{[L]\}$ in the fiber product over X of the universal line over $\Sigma(X)$ and L. For such a pair let C be the strict transform of L' in Y and $p := C \cap E$. Then the completed local ring of Inc_L at (L', y) is isomorphic to the completed local ring of Z_U at p, by an isomorphism commuting with the maps to L. In particular, for any $x \in L$,

$$\Sigma(X, x)\backslash\{[L]\} \cong Z_U \cap f_x,$$

and the isomorphism sends $[L']$ to the normal direction of L' at x relative to L. If n is the width of C, then the completed local rings of Inc_L at (L', y) and of Z_U at p are isomorphic to $\mathbb{C}[t]/(t^n)$ with $1 \leq n \leq 3$, and the widths of the strict transforms of all lines $L' \neq L$ intersecting L sum to 3 (Proposition 10.8). If ξ is a local coordinate of L at y and $\psi(t) \in (t) \subset \mathbb{C}[t]/(t^n)$ is the pullback of ξ, then the completed local ring of $\Sigma(X, y)$ at $[L']$ is $\mathbb{C}[t]/(t^n, \psi(t))$, of length $\min\{n, \operatorname{ord}_t \psi\}$, where $\operatorname{ord}_t 0 := +\infty$.

Proof. The proof is long, so we divide it into several steps. Let $L' \neq L$ be a line intersecting L at y, let C and p be as in the statement, and let $n := w(C)$ be the width of C (Definition-Proposition 2.21).

Step 1. In this step we write down coordinates on the formal neighborhood of C.

Since L and L' are distinct lines in the anti-canonical embedding of X, the scheme $L \cap L'$ is the reduced point y. Thus the inverse image of the ideal of L in $\mathcal{O}_{L'}$ is the ideal of y, so π maps C isomorphically onto L' and $(E \cdot C) = 1$. In particular, E intersects C transversally at the single point p. Since $(H \cdot C) = 1$, we have

$$(h \cdot C) = (H \cdot C) - 2(E \cdot C) = -1.$$

By Theorem 2.23, C is a flopping curve of $Y \dashrightarrow Y^+$, and the flopped curve $C^+ \subset Y^+$ of C is the strict transform of a bisecant line of Γ. Thus the blow-up $Y^+ \to V_5$ maps C^+ isomorphically onto a line in $V_5 \subset \mathbb{P}^6$, and $(h \cdot C^+) = 1$. Let $g : Y \to \bar{Y}$ be the contraction of the flopping curves, let $g^+ : Y^+ \to \bar{Y}$ be the contraction of the flopped curves, and let $\bar{p} := g(C)$. By Proposition 10.8, C satisfies the hypotheses of Definition-Proposition 2.21, so it has a width, and

$$N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2} \quad \text{or} \quad N_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$$

by [KPS18, Lemma 5.2.5 and the paragraph after it].

By Definition-Proposition 2.21 and [Rei83, Remark (5.13)(b)] (see also [DW16, Example 3.10]), there exists an isomorphism of $\widehat{\mathcal{O}}_{\bar{Y},\bar{p}}$ with

$$\mathcal{O}_n := \mathbb{C}[[u,v,z,t]]/(uv - z(z-t^n))$$

under which the formal completion $\widehat{Y}$ of Y along C and the formal completion $\widehat{Y}^+$ of Y^+ along C^+ are the formal completions along the exceptional curves of $\mathrm{Bl}_{(u,z)}\operatorname{Spec}\mathcal{O}_n$ and $\mathrm{Bl}_{(u,z-t^n)}\operatorname{Spec}\mathcal{O}_n$, respectively, after possibly exchanging u and v. Since $(z-t^n)/u = v/z$ in the fraction field of $\mathcal{O}_n$, exchanging u and v exchanges $\mathrm{Bl}_{(u,z)}\operatorname{Spec}\mathcal{O}_n$ and $\mathrm{Bl}_{(u,z-t^n)}\operatorname{Spec}\mathcal{O}_n$. Thus $\widehat{Y}$ is covered by two charts with coordinates (a,t,λ) and (b,t,μ), the formal completions of $\mathbb{A}^3$ along the lines $\{a=t=0\}$ and $\{b=t=0\}$, glued by

$$\mu = \lambda^{-1}, \qquad b = \lambda^2 a - \lambda t^n,$$

and the morphism $\widehat{Y} \to \operatorname{Spec}\mathcal{O}_n$ is given by

$$(u,v,z,t) = (a,\ \lambda^2 a - \lambda t^n,\ \lambda a,\ t) \quad \text{and} \quad (u,v,z,t) = (\mu^2 b + \mu t^n,\ b,\ \mu b + t^n,\ t)$$

on the first and on the second chart, respectively. A direct calculation shows that $uv = z(z-t^n)$ holds on both charts and that the expressions on the first and on the second chart coincide under the gluing. The curve C is $\{a=t=0\}$ on the first chart and $\{b=t=0\}$ on the second chart. Since $g_*\mathcal{O}_Y = \mathcal{O}_{\bar{Y}}$, the theorem on formal functions [Har77, Theorem III.11.1] implies

$$H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = \widehat{\mathcal{O}}_{\bar{Y},\bar{p}} = \mathcal{O}_n.$$

Step 2. In this step we show that the completed local ring of $\mathrm{Hilb}(Y)$ at $[C]$ is $\mathbb{C}[\tau]/(\tau^n)$, with universal family $\{a=b=0,\ t=\tau\}$.

Let A be a local Artin $\mathbb{C}$-algebra with maximal ideal $\mathfrak{m}_A$ and residue field $\mathbb{C}$, and let $C_A \subset Y \times \operatorname{Spec} A$ be a closed subscheme flat over A with closed fiber C. Since the underlying space of C_A is C, the subscheme C_A is contained in $\widehat{Y} \times \operatorname{Spec} A$. The functions λ and $\mu = \lambda^{-1}$ define a morphism $\zeta : \widehat{Y} \to \mathbb{P}^1$ whose restriction to C is an isomorphism. Let B be the coordinate ring of the affine scheme $C_A \cap \zeta^{-1}(\mathbb{A}^1_\lambda)$. The homomorphism $A[\lambda] \to B$ induced by ζ is an isomorphism modulo $\mathfrak{m}_A$. Since $\mathfrak{m}_A$ is nilpotent, it is surjective. Since B is flat over A, the kernel K of $A[\lambda] \to B$ satisfies $K = \mathfrak{m}_A K$, so $K = 0$. Let B' be the coordinate ring of $C_A \cap \zeta^{-1}(\mathbb{A}^1_\mu)$. Replacing λ by μ and B by B' in the argument above shows that $A[\mu] \to B'$ is an isomorphism. Thus ζ maps C_A isomorphically onto $\mathbb{P}^1_A$.

Hence the restriction of t to C_A is a global function on $\mathbb{P}^1_A$, that is, an element $\tau \in \mathfrak{m}_A$, and the restrictions of a and b to C_A are polynomials $a(\lambda) \in \mathfrak{m}_A[\lambda]$ and $b(\mu) \in \mathfrak{m}_A[\mu]$. The gluing relation implies

$$b(\lambda^{-1}) = \lambda^2 a(\lambda) - \lambda\tau^n \quad \text{in } A[\lambda,\lambda^{-1}].$$

The left-hand side has only non-positive powers of λ, and $\lambda^2 a(\lambda)$ has only powers at least 2. Comparing coefficients, we obtain $a=0$, $b=0$, and $\tau^n = 0$. Thus $C_A = \{a=b=0,\ t=\tau\}$ with $\tau^n = 0$. Conversely, for any $\tau \in \mathfrak{m}_A$ with $\tau^n = 0$, the subscheme $\{a=b=0,\ t=\tau\}$ is compatible with the gluing, since $\lambda^2 \cdot 0 - \lambda\tau^n = 0$, and it is isomorphic to $\mathbb{P}^1_A$, hence flat over A. Thus, functorially in A, the embedded deformations of C over A correspond to the homomorphisms $\mathbb{C}[\tau]/(\tau^n) \to A$. Since $\widehat{\mathcal{O}}_{\mathrm{Hilb}(Y),[C]}$ pro-represents the functor of embedded deformations of C, we have

$$\widehat{\mathcal{O}}_{\mathrm{Hilb}(Y),[C]} \cong \mathbb{C}[\tau]/(\tau^n),$$

and the universal family is $\{a=b=0,\ t=\tau\}$.

Step 3. In this step we show that, after completion along C, the base ideal $\mathfrak{b} \subset \mathcal{O}_Y$ of $|h|$ is (a,t^n) on the first chart and (b,t^n) on the second chart.

Let $\mathcal{L}$ be the formal completion of $\mathcal{O}_Y(h)$ along C and let $\mathcal{F} := H^0(\widehat{Y},\mathcal{L})$, an $\mathcal{O}_n$-module. By the theorem on formal functions, $\mathcal{F}$ is the completion at $\bar{p}$ of the stalk of $g_*\mathcal{O}_Y(h)$. Since Y and Y^+ are smooth and isomorphic outside the flopping curves and the flopped curves, and

sections of an invertible sheaf on a smooth variety extend across a closed subset of codimension two [Har77, Proposition II.6.3A], $g_*\mathcal{O}_Y(h) = g^+_*\mathcal{O}_{Y^+}(h)$. Thus $\mathcal{F} = H^0(\widehat{Y}^+, \mathcal{L}^+)$, where $\mathcal{L}^+$ is the formal completion of $\mathcal{O}_{Y^+}(h)$ along C^+.

We first determine $\mathcal{L}$. For $k \geq 1$, let $C_k \subset \widehat{Y}$ be the closed subscheme defined by the k-th power of the ideal of C. Since $N^\vee_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$ or $N^\vee_{C/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2)$, we have $H^1(C, \operatorname{Sym}^k N^\vee_{C/Y}) = 0$ for $k \geq 1$, so the restriction $\operatorname{Pic} C_{k+1} \to \operatorname{Pic} C_k$ is injective [Har77, Exercise III.4.6]. Moreover, $\operatorname{Pic} \widehat{Y} = \varprojlim_k \operatorname{Pic} C_k$ [Har77, Exercise II.9.6], where the Mittag-Leffler condition holds by part (d) of that exercise, since each C_k is projective. Thus the restriction $\operatorname{Pic} \widehat{Y} \to \operatorname{Pic} C \cong \mathbb{Z}$ is injective. Let $D \subset \widehat{Y}$ be the divisor $\{\mu = 0\}$ of the second chart. Then $(D \cdot C) = 1$, so $(-D \cdot C) = -1 = (h \cdot C)$, and

$$\mathcal{L} \cong \mathcal{O}_{\widehat{Y}}(-D).$$

Next we compute $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}(-D))$. By **Step 1**, it is the kernel of the restriction $\mathcal{O}_n \to H^0(D, \mathcal{O}_D) = \mathbb{C}[[b,t]]$, which is given by

$$(u, v, z, t) \to (0,\ b,\ t^n,\ t).$$

Let $J := (u, z - t^n) \subset \mathcal{O}_n$. Since $uv - z(z - t^n) \in J$, we have $\mathcal{O}_n/J \cong \mathbb{C}[[v,t]]$, and the restriction induces an isomorphism $\mathcal{O}_n/J \to \mathbb{C}[[b,t]]$. Thus $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}(-D)) = J$, and $\mathcal{F} \cong J$. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_n$. We have $\dim_{\mathbb{C}} J/\mathfrak{m}J = 2$: if $\dim_{\mathbb{C}} J/\mathfrak{m}J \leq 1$, then J is generated by u or by $z - t^n$ by Nakayama's lemma, while $u \notin (z - t^n)$, since $\mathcal{O}_n/(z - t^n) \cong \mathbb{C}[[u, v, t]]/(uv)$, and $z - t^n \notin (u)$, since $\mathcal{O}_n/(u) \cong \mathbb{C}[[v, z, t]]/(z(z - t^n))$.

Now we show that the image of $H^0(Y, h)$ generates the $\mathcal{O}_n$-module $\mathcal{F}$. The restriction

$$\mathcal{F} = H^0(\widehat{Y}^+, \mathcal{L}^+) \to H^0(C^+, \mathcal{O}_{C^+}(h)) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1))$$

vanishes on $\mathfrak{m}\mathcal{F}$, since $g^+(C^+) = \bar{p}$. Since h is the pullback of $\mathcal{O}_{V_5}(1)$ and C^+ maps isomorphically onto a line in $\mathbb{P}^6$, the restriction $H^0(Y^+, h) \to H^0(C^+, \mathcal{O}_{C^+}(h))$ is surjective. Thus the induced map $\mathcal{F}/\mathfrak{m}\mathcal{F} \to H^0(C^+, \mathcal{O}_{C^+}(h))$ is a surjection between vector spaces of dimension two, hence an isomorphism, and the image of $H^0(Y, h) = H^0(Y^+, h)$ spans $\mathcal{F}/\mathfrak{m}\mathcal{F}$. By Nakayama's lemma, the image of $H^0(Y, h)$ generates $\mathcal{F}$.

Thus $\mathfrak{b} \cdot \mathcal{O}_{\widehat{Y}}$ is the image of $\mathcal{F} \otimes \mathcal{L}^{-1} \to \mathcal{O}_{\widehat{Y}}$. Under $\mathcal{L} \cong \mathcal{O}_{\widehat{Y}}(-D)$, the subsheaf of $\mathcal{O}_{\widehat{Y}}(-D)$ generated by $\mathcal{F} = J$ is $J \cdot \mathcal{O}_{\widehat{Y}}$, so

$$\mathfrak{b} \cdot \mathcal{O}_{\widehat{Y}} = J \cdot \mathcal{O}_{\widehat{Y}}(D).$$

On the first chart, D is empty and

$$J \cdot \mathcal{O}_{\widehat{Y}} = (a,\ \lambda a - t^n) = (a, t^n).$$

On the second chart, μ is a local equation of D and

$$J \cdot \mathcal{O}_{\widehat{Y}} = (\mu(\mu b + t^n),\ \mu b) = \mu \cdot (b, t^n).$$

In particular, the base scheme of $|h|$ near C is the total space of the universal family of **Step 2**.

Step 4. In this step we prove that the completed local rings of Inc_L at (L', y) and of Z_U at p are isomorphic, compatibly with the maps to L.

Let $(c, \kappa) := (a, \lambda)$ if p belongs to the first chart, and $(c, \kappa) := (b, \mu)$ otherwise. Then near p, the curve C is $\{c = t = 0\}$, the ideal $\mathfrak{b} \cdot \mathcal{O}_{\widehat{Y}}$ is (c, t^n) by **Step 3**, and the universal family of **Step 2** is $\{c = 0,\ t = \tau\}$. Since E intersects C transversally at p, a local equation of E at p has non-zero derivative with respect to κ at p. Thus $E = \{\kappa = \varphi(c, t)\}$ near p for a power series φ, and c, t are regular parameters of E at p. Since $\mathcal{O}_E(h) \cong \mathcal{O}_E(2s + 3f)$ and U is the image of the restriction to E, the ideal of Z_U is $\mathfrak{b} \cdot \mathcal{O}_E$. Thus

$$\widehat{\mathcal{O}}_{Z_U, p} = \mathbb{C}[[c, t]]/(c, t^n) \cong \mathbb{C}[t]/(t^n).$$

The universal family over $\mathbb{C}[\tau]/(\tau^n)$ intersects $E \times \operatorname{Spec}\mathbb{C}[\tau]/(\tau^n)$ in the section

$$c = 0, \quad t = \tau, \quad \kappa = \varphi(0,\tau).$$

As a morphism $\operatorname{Spec}\mathbb{C}[\tau]/(\tau^n) \to E$, the section factors through Z_U and induces $\mathbb{C}[t]/(t^n) \to \mathbb{C}[\tau]/(\tau^n)$, $t \to \tau$, which is an isomorphism. Thus intersecting with E induces an isomorphism $\widehat{\mathcal{O}}_{Z_U,p} \cong \widehat{\mathcal{O}}_{\operatorname{Hilb}(Y),[C]}$.

Next we compare $\operatorname{Hilb}(Y)$ at $[C]$ with Inc_L at (L',y) on local Artin $\mathbb{C}$-algebras A with residue field $\mathbb{C}$. An A-point of Inc_L near (L',y) is a flat family $L'_A \subset X \times \operatorname{Spec} A$ of lines with closed fiber L', together with a section y_A of $L \times \operatorname{Spec} A$ contained in L'_A. Since $L \cap L'$ is the reduced point y, Nakayama's lemma implies that the inverse image of the ideal of $L \times \operatorname{Spec} A$ in $\mathcal{O}_{L'_A}$ is the ideal of y_A, which is invertible. By the universal property of blowing up [Har77, Proposition II.7.14], the embedding $L'_A \to X \times \operatorname{Spec} A$ lifts uniquely to $Y \times \operatorname{Spec} A$. The lift is a closed immersion, since the composite of the lift with π is a closed immersion, and the image of the lift is an embedded deformation C_A of C. Conversely, let C_A be an embedded deformation of C over A. By **Step 2**, $C_A \cong \mathbb{P}^1_A$, and $C_A \cap (E \times \operatorname{Spec} A)$ is the section $\{c = 0,\ t = \tau,\ \kappa = \varphi(0,\tau)\}$. Since $(H \cdot C) = 1$, we have $\pi^*H|_{C_A} \cong \mathcal{O}_{\mathbb{P}^1_A}(1)$. Since $\pi(C) = L'$ is a line, the restriction $H^0(X,H) \to H^0(C_A, \mathcal{O}_{\mathbb{P}^1_A}(1)) \cong A^{\oplus 2}$ is surjective modulo $\mathfrak{m}_A$, hence surjective by Nakayama's lemma. Thus π maps C_A isomorphically onto a flat family L'_A of lines, and L'_A contains the section

$$y_A := \pi\big(C_A \cap (E \times \operatorname{Spec} A)\big)$$

of $L \times \operatorname{Spec} A$. By the uniqueness of the lift, the constructions $(L'_A, y_A) \to C_A$ and $C_A \to (L'_A, y_A)$ are mutually inverse. Thus

$$\widehat{\mathcal{O}}_{\operatorname{Inc}_L,(L',y)} \cong \widehat{\mathcal{O}}_{\operatorname{Hilb}(Y),[C]} \cong \widehat{\mathcal{O}}_{Z_U,p},$$

and the map $\operatorname{Inc}_L \to L$, which sends (L'_A, y_A) to $y_A = \pi(C_A \cap (E \times \operatorname{Spec} A))$, corresponds to the composite $Z_U \subset E \to L$.

Step 5. In this step we prove the global statements.

Since L is non-special, by Theorem 2.23 the flopping curves of $Y \dashrightarrow Y^+$ are the strict transforms of the lines $L' \neq L$ intersecting L, and they are pairwise disjoint. By Proposition 10.8, their widths sum to 3. In particular, $1 \le n \le 3$, and $\widehat{\mathcal{O}}_{Z_U,p} \cong \mathbb{C}[t]/(t^n)$ by **Step 4**. Outside the flopping curves, $Y \dashrightarrow Y^+$ is an isomorphism, and h is base point free on Y^+, since h is the pullback of $\mathcal{O}_{V_5}(1)$. Thus, by **Step 3**, the support of Z_U consists of the points $C \cap E$, one for each flopping curve C, and Z_U has length 3. The closed points of Inc_L are the pairs $(L', L' \cap L)$, one for each line $L' \neq L$ intersecting L. Thus Inc_L and Z_U are finite schemes over $\mathbb{C}$ whose closed points correspond to each other, and by **Step 4** the corresponding completed local rings are isomorphic, compatibly with the maps to L. A finite scheme W over $\mathbb{C}$ is the disjoint union of the schemes $\operatorname{Spec}\mathcal{O}_{W,w}$ for the closed points $w \in W$ [AM69, Theorem 8.7], and the local rings $\mathcal{O}_{W,w}$ are Artinian, whose maximal ideals are nilpotent [AM69, p. 90], hence complete. Thus $\operatorname{Inc}_L \cong Z_U$ as schemes over L.

Let $x \in L$. The fiber of Inc_L over x is $\Sigma(X,x)\backslash\{[L]\}$, and the fiber of Z_U over x is $Z_U \cap f_x$. Thus

$$\Sigma(X,x)\backslash\{[L]\} \cong Z_U \cap f_x.$$

The points of $\Sigma(X,x)$ other than $[L]$ correspond to the points of the finite set $Z_U \cap f_x$, so $\Sigma(X,x)$ is finite, and $[L]$ is an isolated point of $\Sigma(X,x)$. Since f_x is the projectivization of the normal space of L at x, the point $C \cap E \in f_x$ is the normal direction of L' at x relative to L.

Finally, let ξ and ψ be as in the statement. Since ψ is the pullback of ξ under the isomorphism of **Step 4**, the completed local ring of $\Sigma(X,y)$ at $[L']$ is

$$\widehat{\mathcal{O}}_{\operatorname{Inc}_L,(L',y)}/(\xi) \cong \mathbb{C}[t]/(t^n, \psi(t)).$$

If $\psi \neq 0$ in $\mathbb{C}[t]/(t^n)$, then $r := \operatorname{ord}_t \psi < n$ and $\psi = t^r \cdot e$ for a unit e, so the length is $r = \min\{n, r\}$. If $\psi = 0$, the length is $n = \min\{n, \infty\}$. □

Set-up 10.10. Notation as in Set-up 10.1, with L special. Then $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Let $p_T : T := \mathrm{Bl}_s Y \to Y$ with exceptional divisor P, so that $P \cong \mathbb{F}_0$ or $P \cong \mathbb{F}_2$, respectively, and in the second case let $\sigma \subset P$ be the negative section. The curve s is a flopping curve of $Y \dashrightarrow Y^+$, and the flopped curve of s is the strict transform of a bisecant line $M \subset V_5$ of Γ, which we call the *inverse bisecant* of s. Let f_P be the class of a fiber of $P \to s$, and if $P \cong \mathbb{F}_0$, let u_P be the class of the other ruling. Let E' and R_T be the strict transforms on T of E and of R. Let $\phi_M : \mathrm{Bl}_M V_5 \to V_5$ be the blow-up of M with exceptional divisor E_M, write h also for the pullback to $\mathrm{Bl}_M V_5$ of the hyperplane class of V_5, and let $\psi_M : \mathrm{Bl}_M V_5 \to Q$ be the morphism onto a smooth quadric threefold $Q \subset \mathbb{P}^4$ defined by $|h - E_M|$. Let $S_M \subset V_5$ be the image under ϕ_M of the exceptional divisor of ψ_M, which is the unique member of $|h - 2E_M|$. Then S_M is the unique hyperplane section of V_5 singular along M.

Proof of the assertions in Set-up 10.10. By Proposition 10.8, s satisfies the hypotheses of Definition-Proposition 2.21, so it has a width, and $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ by [KPS18, Lemma 5.2.5 and the paragraph after it]. Since $P = \mathbb{P}(N_{s/Y})$, the case $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ corresponds to $P \cong \mathbb{F}_0$, and the case $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ to $P \cong \mathbb{F}_2$. By Theorem 2.23, s is a flopping curve of $Y \dashrightarrow Y^+$ whose flopped curve is the strict transform of a bisecant line of Γ. Since M is a line in V_5, [Fuj26a, Theorem 3.6(4)–(6)], which collects results of Iskovskikh [Isk77] and T. Fujita, shows that ψ_M is the contraction of the blow-up of a smooth quadric threefold along a twisted cubic curve, that the exceptional divisor J of ψ_M belongs to $|h - 2E_M|$, and that S_M is a hyperplane section of V_5 singular along M. Since J is ψ_M-exceptional, $\psi_{M*}\mathcal{O}(J) = \mathcal{O}_Q$ [Har77, Proposition II.6.3A], so $h^0(\mathcal{O}(J)) = 1$, and J is the unique member of $|h - 2E_M|$. If S is a hyperplane section of V_5 singular along M, then $\phi_M^* S - 2E_M$ is an effective member of $|h - 2E_M|$, so it equals J, and $S = \phi_{M*} J = S_M$. □

The inverse bisecant M is the counterpart on V_5 of the special line L. The scheme $M \cap \Gamma$ has length two. Among the prime divisors over X with center L, whether $M \cap \Gamma$ is reduced matters only when $P \cong \mathbb{F}_2$, for those F with $\mathrm{center}_T F = \sigma$. If $M \cap \Gamma$ consists of two distinct points, Proposition 11.12 proves $A_X(F) - S_X(F) \geq \frac{15}{44} \mathrm{mult}_F P > 0$ by an estimate along σ that uses both points of $M \cap \Gamma$. If $M \cap \Gamma$ is a single point of length two, the local estimates do not cover F, and we use a degeneration instead (Section 12). All other prime divisors over X with center L are covered by Propositions 11.2, 11.10, and 11.11.

11. Curves of degree one and two in genus twelve

In this section, we prove that for a smooth prime Fano threefold X of genus 12, any prime divisor F over X with $\mathrm{center}_X F$ a curve of degree one or two satisfies $A_X(F) > S_X(F)$, except possibly in one configuration of a special line $L = \mathrm{center}_X F$ (Corollary 12.8(3)), in which the center of F on $\mathrm{Bl}_L X$ is the negative section of $E \cong \mathbb{F}_3$. Section 12 treats that configuration by a degeneration to a member with a torus action. By Proposition 4.6, any integral curve $C \subset X$ with $H \cdot C \geq 3$ satisfies $A_X(F) > S_X(F)$ for any prime divisor F over X with $C \subset \mathrm{center}_X F$. Thus among the prime divisors over X whose center is a curve C, only those with $H \cdot C \in \{1, 2\}$ remain.

11.1. The four normal bundle types.

In this subsection, we compute the volume functions of the blow-ups of lines and conics on X and prove Proposition 11.2. We first collect the numerical data of the links of Theorems 2.23 and 2.24. They are used again in Propositions 11.4 and 11.6.

Lemma 11.1. *Let X be a smooth prime Fano threefold of genus 12 and $C \subset X$ a line or a smooth conic, and let $e := H \cdot C$. Let $\pi : Y := \mathrm{Bl}_C X \to X$ with exceptional divisor E, write H also for $\pi^* H$, and let h and R be the divisors on Y given by the link of Theorem 2.23 if $e = 1$ and of Theorem 2.24 if $e = 2$. Let $\tau_1 := 3$ and $\tau_2 := 5/2$. Then the following hold.*

(1) $H^3 = 22$, $H^2 \cdot E = 0$, $H \cdot E^2 = -e$, and $E^3 = 2 - e$ on Y. On the other side of the link, $h^3 = 5$ if $e = 1$ and $h^3 = 2$ if $e = 2$. The divisor $H - E = -K_Y$ is base point free.

(2) Let $u := t-1$. If $e=1$, then

$$\operatorname{vol}(H-tE) = \begin{cases} 22-3t^2-t^3 & \text{if } 0 \le t \le 1, \\ 18-9u-6u^2+2u^3 & \text{if } 1 \le t \le 2, \\ 5(3-t)^3 & \text{if } 2 \le t \le 3. \end{cases}$$

If $e=2$, then

$$\operatorname{vol}(H-tE) = \begin{cases} 22-6t^2 & \text{if } 0 \le t \le 1, \\ 16-12u-6u^2+4u^3 & \text{if } 1 \le t \le 2, \\ 2(5-2t)^3 & \text{if } 2 \le t \le 5/2. \end{cases}$$

Moreover $\operatorname{vol}(H-tE) = 0$ for $t \ge \tau_e$. In particular $S_X(E) = 17/11$ if $e=1$ and $S_X(E) = 117/88$ if $e=2$.

(3) Let $m \ge 1$ and $j \ge 0$ be integers. Then $H^0(Y, mH-jE) = 0$ if $j > \tau_e m$. If $2m \le j \le \tau_e m$, then multiplication by the $(j-2m)$-th power of a section defining R is an isomorphism $H^0(Y,qh) \to H^0(Y, mH-jE)$, where $q := 3m-j$ if $e=1$ and $q := 5m-2j$ if $e=2$.

(4) Let W be the refinement of $(H^0(X,mH))_m$ by E. Then

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j>2m} (j-2m) \dim W_{m,j} = J_e, \qquad J_1 := \frac{5}{88},\ J_2 := \frac{1}{88}.$$

Proof. Let $Y \dashrightarrow Y^+$ be the flop and $Y^+ = \operatorname{Bl}_\Gamma M \to M$ the blow-up of the link, where $M = V_5$ and Γ is a rational normal quintic if $e=1$, and $M = Q$ and Γ is a smooth rational sextic if $e=2$. We use h, R, and H also for the strict transforms on Y^+. Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one, $H^0(Y,D) = H^0(Y^+,D)$ for any divisor D on Y, compatibly with multiplication of sections, and in particular $\operatorname{vol}_Y(D) = \operatorname{vol}_{Y^+}(D)$. By Theorems 2.23 and 2.24,

$$H-tE = \begin{cases} (3-t)h + (t-2)R & \text{if } e=1, \\ (5-2t)h + (t-2)R & \text{if } e=2. \end{cases} \tag{11.1}$$

(1) The curve C is isomorphic to $\mathbb{P}^1$ (Lemma 2.12(3) for a line), so $\deg N_{C/X} = -K_X \cdot C - 2 = e-2$ by adjunction. By [IP99, Lemma 2.2.14],

$$H^2 \cdot E = 0, \qquad H \cdot E^2 = -(H \cdot C) = -e, \qquad E^3 = -\deg N_{C/X} = 2-e,$$

and $H^3 = 22$ by Definition 2.11. On Y^+, the curve Γ is smooth and rational of degree $d=5$ (resp. $d=6$) with respect to the hyperplane class h_M of M, and $-K_M = 2h_M$ (resp. $-K_M = 3h_M$). Thus $\deg N_{\Gamma/M} = -K_M \cdot \Gamma - 2$ is 8 (resp. 16), and [IP99, Lemma 2.2.14] implies

$$(h^3,\ h^2 \cdot R,\ h \cdot R^2,\ R^3) = \begin{cases} (5,0,-5,-8) & \text{if } e=1, \\ (2,0,-6,-16) & \text{if } e=2. \end{cases} \tag{11.2}$$

Since $K_Y = \pi^* K_X + E$, we have $-K_Y = H - E$. If $e=2$, $|-K_Y|$ is base point free by Theorem 2.24. If $e=1$, $|H|$ embeds X into $\mathbb{P}^{13}$ (Lemma 2.12(1)) and C is a line in $\mathbb{P}^{13}$, so the ideal sheaf of C in X is generated by linear forms $l_1, \dots, l_{12}$ on $\mathbb{P}^{13}$. Since $\pi^{-1}\mathcal{I}_C \cdot \mathcal{O}_Y = \mathcal{O}_Y(-E)$, the sections $\pi^* l_1, \dots, \pi^* l_{12}$ generate $\mathcal{O}_Y(H-E)$.

We prove (3) before (2), since the proof of (2) uses (3).

(3) Let $p := j-2m$ and let q be as in the statement. By (11.1), $mH - jE \sim qh + pR$ on Y. Let s_R be a section defining R, and let ℓ be a general fiber of $R \to \Gamma$ on Y^+. Then $(h \cdot \ell) = 0$ and $(R \cdot \ell) = -1$ [IP99, Lemma 2.2.14]. Let $D \in |qh+pR|$ on Y^+ and let $b := \operatorname{mult}_R D$. Since ℓ is general, $\ell \not\subset \operatorname{Supp}(D - bR)$, so

$$0 \le ((D-bR) \cdot \ell) = -p + b.$$

Thus any section of $\mathcal{O}_{Y^+}(qh+pR)$ is divisible by s_R^p, and multiplication by s_R^p is an isomorphism $H^0(Y^+, qh) \to H^0(Y^+, qh+pR)$. Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one

and $mH - jE \sim qh + pR$ on Y, multiplication by s_R^p is also an isomorphism $H^0(Y, qh) \to H^0(Y, mH - jE)$. If $j > \tau_e m$, then $j > 2m$ and $q < 0$, and $H^0(Y^+, qh) = H^0(M, \mathcal{O}_M(qh_M)) = 0$.

(2) For $0 \le t \le 1$, $H - tE = (1-t)H + t(H-E)$ is nef on Y by (1), so $\operatorname{vol}(H - tE) = (H - tE)^3 = 22 - 3et^2 - (2-e)t^3$ [Laz04a, Corollary 1.4.41]. For $1 \le t \le 2$,

$$H - tE = (2-t)(H-E) + (t-1)h.$$

The divisor h is nef on Y^+, being the pullback of h_M. Since $Y \dashrightarrow Y^+$ is a flop, it is an isomorphism outside the flopping curves and the flopped curves, the strict transform of $-K_Y$ is $-K_{Y^+}$, and $K_{Y^+} \cdot \gamma = 0$ for any flopped curve γ. By (1), $|-K_Y|$ is base point free. Since Y and Y^+ are isomorphic in codimension one, $H^0(Y, -K_Y) = H^0(Y^+, -K_{Y^+})$, so the base locus of $|-K_{Y^+}|$ is contained in the union of the flopped curves. Hence $-K_{Y^+} \cdot \gamma \ge 0$ for any curve $\gamma \subset Y^+$, that is, $H - E = -K_{Y^+}$ is nef on Y^+. Therefore $\operatorname{vol}(H - tE) = ((H - tE)^3)_{Y^+}$ for $1 \le t \le 2$. By (11.2), for real numbers a and b,

$$(ah + bR)^3 = \begin{cases} 5a^3 - 15ab^2 - 8b^3 & \text{if } e = 1, \\ 2a^3 - 18ab^2 - 16b^3 & \text{if } e = 2, \end{cases}$$

and substituting (11.1) implies the second line of both formulas. For rational $t \in [2, \tau_e]$ and m divisible enough, (3) implies $h^0(Y, m(H - tE)) = h^0(Y, m(3-t)h)$ if $e = 1$ and $h^0(Y, m(H-tE)) = h^0(Y, m(5-2t)h)$ if $e = 2$. Since h is nef on Y^+, $\operatorname{vol}(ch) = c^3h^3$ for $c \ge 0$, and (11.2) implies the third line of both formulas. For rational $t > \tau_e$, $\operatorname{vol}(H - tE) = 0$ by (3). The volume function is continuous [Laz04a, Theorem 2.2.44 and Corollary 2.2.45], so the formulas of (2) hold for all real t.

Since $S_X(E) = \frac{1}{22}\int_0^\infty \operatorname{vol}(H - tE)dt$ [AZ22, Section 2.2], a direct calculation shows

$$S_X(E) = \begin{cases} \frac{1}{22}\left(\frac{83}{4} + 12 + \frac{5}{4}\right) = \frac{17}{11} & \text{if } e = 1, \\ \frac{1}{22}\left(20 + 9 + \frac{1}{4}\right) = \frac{117}{88} & \text{if } e = 2. \end{cases}$$

(4) By Definition 2.8, $W_{m,j}$ is the image of $H^0(Y, mH - jE) \to H^0(E, (mH - jE)|_E)$, whose kernel is $H^0(Y, mH - (j+1)E)$. Let $a_j := h^0(Y, mH - jE)$. Then $\dim W_{m,j} = a_j - a_{j+1}$, and $a_j = 0$ for $j > \tau_e m$ by (3). Thus

$$\begin{aligned} \sum_{j>2m} (j - 2m) \dim W_{m,j} &= \sum_{k \ge 1} k(a_{2m+k} - a_{2m+k+1}) \\ &= \sum_{2m < j \le \tau_e m} a_j = \sum_{2m < j \le \tau_e m} h^0(Y, qh), \end{aligned}$$

with q as in (3). Since $h^0(Y, qh) = h^0(M, \mathcal{O}_M(qh_M))$ and $qh_M - K_M$ is ample for $q \ge 0$, Kodaira vanishing [Har77, Remark III.7.15] and Riemann–Roch [Har77, Appendix A, Exercise 6.7] show that $h^0(Y, qh)$ is a polynomial of degree three in $q \ge 0$ with leading coefficient $\frac{h^3}{6}$ by (1). If $e = 1$, then $q = 3m - j$ runs through $0, 1, \dots, m-1$, and the sum is $\frac{5}{24}m^4 + O(m^3)$. If $e = 2$, then $q = 5m - 2j$ runs through the integers $0 \le q < m$ with $q \equiv m \bmod 2$, and the sum is $\frac{1}{3} \cdot \frac{m^4}{8} + O(m^3) = \frac{1}{24}m^4 + O(m^3)$. By Lemma 2.12(2), $mN_m = \frac{11}{3}m^4 + O(m^3)$. Thus the limit is $\frac{5}{24} \cdot \frac{3}{11} = \frac{5}{88}$ if $e = 1$ and $\frac{1}{24} \cdot \frac{3}{11} = \frac{1}{88}$ if $e = 2$. □

The divisors left open by Proposition 11.2 are treated in Proposition 11.4 in case (1), in Propositions 11.10, 11.11, and 11.12 and Corollary 12.8 in case (2), and in Proposition 11.6 in case (4).

Proposition 11.2. *Let X be a smooth prime Fano threefold of genus 12 and $C \subset X$ an integral curve with $e := H \cdot C \in \{1, 2\}$. Let $\pi : Y := \operatorname{Bl}_C X \to X$ with exceptional divisor E. Then C is a smooth rational curve, and exactly one of the following holds:*

(1) $e = 1$, $N_{C/X} \cong \mathcal{O} \oplus \mathcal{O}(-1)$, and $E \cong \mathbb{F}_1$,
(2) $e = 1$, $N_{C/X} \cong \mathcal{O}(1) \oplus \mathcal{O}(-2)$, and $E \cong \mathbb{F}_3$,

(3) $e = 2$, $N_{C/X} \cong \mathcal{O} \oplus \mathcal{O}$, *and* $E \cong \mathbb{F}_0$,
(4) $e = 2$, $N_{C/X} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$, *and* $E \cong \mathbb{F}_2$.

In all cases $S_X(E) = 17/11$ *if* $e = 1$ *and* $S_X(E) = 117/88$ *if* $e = 2$. *For* $E \cong \mathbb{F}_n$ *with* $n > 0$ *let* s_E *be the negative section. Let* F *be a prime divisor over* X *with* $\operatorname{center}_X F = C$. *If* $F = E$, *or case (3) holds, or* $\operatorname{center}_Y F \neq s_E$ *in cases (1), (2), and (4), then*

$$A_X(F) - S_X(F) \geq \kappa_e \operatorname{mult}_F E > 0, \qquad \kappa_1 := \frac{5}{11},\ \kappa_2 := \frac{59}{88}.$$

Hence, if $A_X(F) \leq S_X(F)$, *then* $\operatorname{center}_Y F = s_E$ *and case (1), (2), or (4) holds.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that $C \cong \mathbb{P}^1$ and that exactly one of (1)–(4) holds, and we compute the restrictions to E of H, E, and the divisors h, R of Lemma 11.1.

By Lemma 2.12(1), $|H|$ embeds X into $\mathbb{P}^{13}$, and e is the degree of C in $\mathbb{P}^{13}$. If $e = 1$, then C is a line in $\mathbb{P}^{13}$. Let $e = 2$ and let $\Pi \subset \mathbb{P}^{13}$ be the linear span of C. If $\dim \Pi \geq 3$, there exists a hyperplane of $\mathbb{P}^{13}$ containing three general points of C but not C, so $H \cdot C \geq 3$, a contradiction. If $\dim \Pi = 1$, then $C = \Pi$ and $H \cdot C = 1$, a contradiction. Thus C is an integral plane conic, hence smooth. In both cases $C \cong \mathbb{P}^1$, and C is a line or a smooth conic in the sense of Definition 2.11.

By [KPS18, Lemma 2.1.4], there exists an integer $a \geq 0$ with $N_{C/X} \cong \mathcal{O}(a) \oplus \mathcal{O}(-1-a)$ if $e = 1$ and $N_{C/X} \cong \mathcal{O}(a) \oplus \mathcal{O}(-a)$ if $e = 2$. If $e = 1$, then $a \in \{0, 1\}$ by Lemma 2.12(3). Let $e = 2$. The Hilbert scheme of conics on X is isomorphic to $\mathbb{P}^2$ [KPS18, Proposition 2.3.6(v)], and the Zariski tangent space of the Hilbert scheme of conics at $[C]$ is $H^0(C, N_{C/X})$ [KPS18, Lemma 2.1.4 and the paragraph following it]. Thus $h^0(C, N_{C/X}) = 2$. Since $h^0(\mathbb{P}^1, \mathcal{O}(a) \oplus \mathcal{O}(-a)) = a+1 \geq 3$ for $a \geq 2$, we have $a \in \{0, 1\}$. Since $E \cong \mathbb{P}(N^\vee_{C/X})$ [Har77, Theorem II.8.24] and $\mathbb{P}(\mathcal{O}(-a_1) \oplus \mathcal{O}(-a_2)) \cong \mathbb{F}_{|a_1 - a_2|}$ for integers a_1 and a_2 [Har77, Proposition V.2.8], exactly one of (1)–(4) holds. Let n be the integer with $E \cong \mathbb{F}_n$.

Let f be the class of a fiber of $E \to C$, and let s be the class of the negative section s_E if $n > 0$ and of a fiber of the other ruling if $n = 0$. In this proof s denotes a class on E and s_E the curve. If $e = 1$, then s_E is the curve s of Set-up 10.1 with $L := C$. We have $\operatorname{Pic} E = \mathbb{Z}s \oplus \mathbb{Z}f$, $(s^2) = -n$, $(s \cdot f) = 1$, and $(f^2) = 0$ [Har77, Propositions V.2.3 and V.2.9]. Since $H|_E$ is the pullback of a divisor of degree e on C, $H|_E \sim ef$. Write $-E|_E \sim xs + cf$. Since $(E \cdot f) = -1$ [IP99, Lemma 2.2.14], we have $x = 1$. Since $((E|_E)^2)_E = E^3 = 2 - e$ by Lemma 11.1(1), we have $-n + 2c = 2 - e$. By Theorems 2.23 and 2.24, $h = H - 2E$, and $R = H - 3E$ if $e = 1$ and $R = 2H - 5E$ if $e = 2$. We obtain Table 11. In particular,

$$(R|_E \cdot f) = (R \cdot f) = \begin{cases} 3 & \text{if } e = 1, \\ 5 & \text{if } e = 2. \end{cases} \tag{11.3}$$

TABLE 11. Restrictions to $E \cong \mathbb{F}_n$ in the four cases of Proposition 11.2.

case	n	$-E\|_E$	$(H - tE)\|_E$	$h\|_E$	$(R \cdot f)$
(1)	1	$s + f$	$ts + (1+t)f$	$2s + 3f$	3
(2)	3	$s + 2f$	$ts + (1+2t)f$	$2s + 5f$	3
(3)	0	s	$ts + 2f$	$2s + 2f$	5
(4)	2	$s + f$	$ts + (2+t)f$	$2s + 4f$	5

Step 2. In this step, we prove the assertions on $S_X(E)$ and the case $F = E$.

By **Step 1**, C is a line or a smooth conic, so $S_X(E) = 17/11$ if $e = 1$ and $S_X(E) = 117/88$ if $e = 2$ by Lemma 11.1(2). Since $K_Y = \pi^* K_X + E$, $A_X(E) = 2$. Thus

$$A_X(E) - S_X(E) = \begin{cases} 2 - \frac{17}{11} = \frac{5}{11} = \kappa_1 & \text{if } e = 1, \\ 2 - \frac{117}{88} = \frac{59}{88} = \kappa_2 & \text{if } e = 2. \end{cases}$$

Since $\mathrm{mult}_E E = 1$, the inequality of the proposition holds for $F = E$.

Step 3. In this step, we bound $S(W; \mathrm{ord}_{C'})$ from above for a curve $C' \subset E$ dominating C, where W is the refinement of $(H^0(X, mH))_m$ by E (Definition 2.8).

Let $C' \subset E$ be a prime divisor with $\pi(C') = C$. If $n = 0$, let $\Xi := s$. If $n > 0$, assume that $C' \neq s_E$ and let $\Xi := s + nf$. Write $C' \sim \alpha s + \beta f$. Then $\alpha = (C' \cdot f) \geq 1$. If $n = 0$, then $\beta = (C' \cdot s_1) \geq 0$ for a fiber $s_1 \neq C'$ of the other ruling. If $n > 0$, then $\beta - n\alpha = (C' \cdot s_E) \geq 0$ [Har77, Proposition V.1.4]. In both cases $C' - \Xi$ is linearly equivalent to an effective divisor. Let $\mu_{C'} := \mathrm{mult}_{C'}(R|_E)$. Since $R \neq E$, $R|_E - \mu_{C'}C' \geq 0$, and since f is nef on E, (11.3) implies

$$\mu_{C'} \leq \mu_{C'}(C' \cdot f) \leq (R \cdot f) = \begin{cases} 3 & \text{if } e = 1, \\ 5 & \text{if } e = 2. \end{cases} \tag{11.4}$$

Define the $\mathbb{R}$-divisor $D_E(t)$ on E by

$$D_E(t) := \begin{cases} (H - tE)|_E & \text{if } 0 \leq t \leq 2, \\ (3-t)h|_E & \text{if } e = 1 \text{ and } 2 \leq t \leq 3, \\ (5-2t)h|_E & \text{if } e = 2 \text{ and } 2 \leq t \leq 5/2, \end{cases}$$

and let $I(t) := \int_0^\infty \mathrm{vol}_E(D_E(t) - v\Xi)dv$. We prove

$$S(W; \mathrm{ord}_{C'}) \leq \begin{cases} \frac{3}{22}\int_0^3 I(t)dt + \frac{5}{88}\mu_{C'} & \text{if } e = 1, \\ \frac{3}{22}\int_0^{5/2} I(t)dt + \frac{1}{88}\mu_{C'} & \text{if } e = 2. \end{cases} \tag{11.5}$$

For integers $m \geq 1$ and $j \geq 0$, the values of $\mathrm{ord}_{C'}$ on $W_{m,j}\backslash\{0\}$ are non-negative integers, so the sum of the jumping numbers of $\mathrm{ord}_{C'}$ on $W_{m,j}$ is $\sum_{k\geq 1} \dim\{w \in W_{m,j} : \mathrm{ord}_{C'}(w) \geq k\}$. Thus

$$S_m(W; \mathrm{ord}_{C'}) = \frac{1}{mN_m} \sum_{j \geq 0} \sum_{k \geq 1} \dim\{w \in W_{m,j} : \mathrm{ord}_{C'}(w) \geq k\},$$

and $S(W; \mathrm{ord}_{C'}) = \lim_{m\to+\infty} S_m(W; \mathrm{ord}_{C'})$ [AZ22, Lemma 2.9]. By Lemma 11.1(3), $W_{m,j} = 0$ for $j > 3m$ if $e = 1$ and for $j > 5m/2$ if $e = 2$.

Let $0 \leq j \leq 2m$. Then $W_{m,j} \subset H^0(E, mD_E(j/m))$. Since $kC' - k\Xi$ is linearly equivalent to an effective divisor,

$$\dim\{w \in W_{m,j} : \mathrm{ord}_{C'}(w) \geq k\} \leq h^0(E, mD_E(j/m) - kC') \leq h^0(E, mD_E(j/m) - k\Xi).$$

An upper bound suffices here, since the right-hand side of (11.5) is less than 1 by **Step 4**.

Let $2m < j \leq 3m$ if $e = 1$ and $2m < j \leq 5m/2$ if $e = 2$, let $p := j - 2m$, and let $q := 3m - j$ if $e = 1$ and $q := 5m - 2j$ if $e = 2$. Let $r \in H^0(E, R|_E)$ be the restriction of a section defining R, which is non-zero since $R \neq E$. By Lemma 11.1(3), $W_{m,j} = r^p \cdot U_q$, where $U_q \subset H^0(E, qh|_E) = H^0(E, mD_E(j/m))$ is the image of $H^0(Y, qh)$. Since $\mathrm{ord}_{C'}(r^p u) = p\mu_{C'} + \mathrm{ord}_{C'}(u)$ for $u \in U_q$,

$$\sum_{k\geq 1} \dim\{w \in W_{m,j} : \mathrm{ord}_{C'}(w) \geq k\} \leq (j - 2m)\mu_{C'} \dim W_{m,j} + \sum_{k\geq 1} h^0(E, mD_E(j/m) - k\Xi).$$

Combining the bounds for $j \leq 2m$ and for $j > 2m$, we obtain

$$S_m(W; \mathrm{ord}_{C'}) \leq \frac{1}{mN_m} \sum_{j} \sum_{k\geq 1} h^0(E, mD_E(j/m) - k\Xi) + \frac{\mu_{C'}}{mN_m} \sum_{j>2m} (j - 2m) \dim W_{m,j},$$

where j runs through the integers with $0 \le j \le 3m$ if $e = 1$ and $0 \le j \le 5m/2$ if $e = 2$. By Lemma 11.1(4), the second term tends to $\frac{5}{88}\mu_{C'}$ if $e = 1$ and to $\frac{1}{88}\mu_{C'}$ if $e = 2$. For the first term, for integers $a \ge 0$ and b,

$$h^0(\mathbb{F}_n, as + bf) = \sum_{i=0}^{a} \max\{b - ni + 1, 0\}$$

[Har77, Proposition V.2.8 and Exercise III.8.4(a)], and $h^0(\mathbb{F}_n, as + bf) = 0$ if $a < 0$. The right-hand side differs from $\int_0^a \max\{b - nx, 0\}dx = \frac{1}{2}\operatorname{vol}_E(as + bf)$ by $O(a + |b| + 1)$. For the classes $mD_E(j/m) - k\Xi$ we have $a, b = O(m)$, and the coefficient of s in $mD_E(j/m)$ is at most $2m$ by Table 11, so only $O(m^2)$ pairs (j, k) contribute, and the accumulated error is $O(m^3)$. The function $(t, v) \to \operatorname{vol}_E(D_E(t) - v\Xi)$ is continuous and piecewise polynomial on $[0, 3] \times [0, 2]$ if $e = 1$ and on $[0, 5/2] \times [0, 2]$ if $e = 2$, and it vanishes for $v > 2$. Thus the Riemann sum $\sum_{j,k} \frac{m^2}{2} \operatorname{vol}_E(D_E(j/m) - \frac{k}{m}\Xi)$ is $\frac{m^4}{2}\int_0^3 I(t)dt + O(m^3)$ if $e = 1$ and $\frac{m^4}{2}\int_0^{5/2} I(t)dt + O(m^3)$ if $e = 2$. Since $mN_m = \frac{22}{6}m^4 + O(m^3)$ by Lemma 2.12(2), the first term tends to $\frac{3}{22}\int_0^3 I(t)dt$ if $e = 1$ and to $\frac{3}{22}\int_0^{5/2} I(t)dt$ if $e = 2$. This proves (11.5).

Step 4. In this step, we compute the right-hand side of (11.5) in the four cases and prove $S(W; \operatorname{ord}_{C'}) < 1$.

For real numbers x and y, the volume of $xs+yf$ on $\mathbb{F}_n$ is computed by the Zariski decomposition [Laz04a, Theorem 2.3.19 and Corollary 2.3.22] if $x, y \in \mathbb{Q}$, and by continuity of the volume [Laz04a, Corollary 2.2.45] in general. If $n \ge 1$, then

$$\operatorname{vol}_E(xs + yf) = \begin{cases} 2xy - nx^2 & \text{if } x \ge 0 \text{ and } y \ge nx, \\ y^2/n & \text{if } 0 \le y \le nx, \end{cases}$$

where in the second range the negative part is $(x - \frac{y}{n})s$ and the positive part is $\frac{y}{n}(s + nf)$. If $n = 0$, then $\operatorname{vol}_E(xs + yf) = 2xy$ for $x, y \ge 0$. In all cases $\operatorname{vol}_E(xs + yf) = 0$ if $x < 0$ or $y < 0$. We use Table 11 for $D_E(t)$.

In case (1), $\Xi = s + f$. For $0 \le t \le 2$, $D_E(t) - v\Xi = (t - v)s + (1 + t - v)f$ is nef for $0 \le v \le t$. For $2 \le t \le 3$, $D_E(t) - v\Xi = (2(3 - t) - v)s + (3(3 - t) - v)f$ is nef for $0 \le v \le 2(3 - t)$.

In case (2), $\Xi = s + 3f$ and $D_E(t) - v\Xi = (t - v)s + (1 + 2t - 3v)f$ for $0 \le t \le 2$. If $0 \le t \le 1$, it is nef for $0 \le v \le t$. If $1 \le t \le 2$, the Zariski decomposition of $D_E(t) - v\Xi$ has negative part $\frac{t-1}{3}s$ for $0 \le v \le \frac{1+2t}{3}$. For $2 \le t \le 3$, $D_E(t) - v\Xi = (2(3 - t) - v)s + (5(3 - t) - 3v)f$, and the Zariski decomposition of $D_E(t) - v\Xi$ has negative part $\frac{3-t}{3}s$ for $0 \le v \le \frac{5(3-t)}{3}$.

In case (3), $\Xi = s$. For $0 \le t \le 2$, $D_E(t) - v\Xi = (t - v)s + 2f$ is nef for $0 \le v \le t$. For $2 \le t \le 5/2$, $D_E(t) - v\Xi = (2(5 - 2t) - v)s + 2(5 - 2t)f$ is nef for $0 \le v \le 2(5 - 2t)$.

In case (4), $\Xi = s+2f$. For $0 \le t \le 2$, $D_E(t)-v\Xi = (t-v)s+(2+t-2v)f$ is nef for $0 \le v \le t$, since $(2 + t - 2v) - 2(t - v) = 2 - t \ge 0$. For $2 \le t \le 5/2$, $D_E(t) - v\Xi = (2(5 - 2t) - v)(s + 2f)$ is nef for $0 \le v \le 2(5 - 2t)$.

In all four cases $\operatorname{vol}_E(D_E(t) - v\Xi) = 0$ beyond the given range of v. A direct calculation shows the values of $I(t)$ and of $\int_0^3 I(t)dt$ if $e = 1$ and $\int_0^{5/2} I(t)dt$ if $e = 2$ in Table 12. Together with (11.5) and (11.4), we obtain

$$S(W; \operatorname{ord}_{C'}) \le \begin{cases} \frac{3}{22}\int_0^3 I(t)dt + 3 \cdot \frac{5}{88} & \text{if } e = 1, \\ \frac{3}{22}\int_0^{5/2} I(t)dt + 5 \cdot \frac{1}{88} & \text{if } e = 2, \end{cases}$$

and the right-hand side is the last column of Table 12. In particular,

$$S(W; \operatorname{ord}_{C'}) < 1 \tag{11.6}$$

for any prime divisor $C' \subset E$ with $\pi(C') = C$, where $C' \ne s_E$ in cases (1), (2), and (4).

Step 5. In this step, we prove the inequality for $F \ne E$ and conclude the proof.

TABLE 12. The function $I(t)$ on the ranges $0 \le t \le 1$, $1 \le t \le 2$, and $2 \le t \le 3$ if $e = 1$ or $2 \le t \le 5/2$ if $e = 2$, the integral of $I(t)$ over $[0, 3]$ if $e = 1$ and over $[0, 5/2]$ if $e = 2$, and the upper bound for $S(W; \mathrm{ord}_{C'})$ from (11.5) and (11.4) in the four cases of Proposition 11.2.

case	Ξ	$0 \le t \le 1$	$1 \le t \le 2$	$t \ge 2$	$\int I(t)dt$	bound
(1)	$s + f$	$\frac{t^3}{3} + t^2$	$\frac{t^3}{3} + t^2$	$\frac{20}{3}(3-t)^3$	$\frac{17}{3}$	$\frac{83}{88}$
(2)	$s + 3f$	t^2	$\frac{(1+2t)^3}{27}$	$\frac{125}{27}(3-t)^3$	$\frac{433}{108}$	$\frac{71}{99}$
(3)	s	$2t^2$	$2t^2$	$8(5-2t)^3$	$\frac{19}{3}$	$\frac{81}{88}$
(4)	$s + 2f$	$2t^2 - \frac{t^3}{3}$	$2t^2 - \frac{t^3}{3}$	$\frac{16}{3}(5-2t)^3$	$\frac{14}{3}$	$\frac{61}{88}$

Let $F \ne E$ be a prime divisor over X with $\mathrm{center}_X F = C$, and assume that $\mathrm{center}_Y F \ne s_E$ in cases (1), (2), and (4). Since $\pi(\mathrm{center}_Y F) = C$, $\mathrm{center}_Y F \subset E$, and since Y is normal and $F \ne E$, $\mathrm{center}_Y F \ne E$. Thus $C' := \mathrm{center}_Y F$ is a prime divisor on E with $\pi(C') = C$, and C' satisfies the assumptions of **Step 3**. Recall that W is the refinement of $(H^0(X, mH))_m$ by E.

The divisor E is of plt type with plt extraction π, since $-E$ is π-ample and Y and E are smooth, and $\Delta_E = 0$. In Lemma 3.1, take the complete series of $-K_X$, $\lambda = 1$, and G the generic point of C'. A prime divisor v over E with $\mathrm{center}_E v \cap G \ne \emptyset$ has $\mathrm{center}_E v = C'$, so $v = \mathrm{ord}_{C'}$ and $A_{E,\Delta_E}(v) = 1$. By (11.6), $S(W; \mathrm{ord}_{C'}) < 1$, so the assumption of Lemma 3.1 holds. Since $\mathrm{center}_Y F = C'$, Lemma 3.1 and **Step 2** imply

$$A_X(F) \ge S_X(F) + (A_X(E) - S_X(E)) \,\mathrm{mult}_F E = S_X(F) + \kappa_e \,\mathrm{mult}_F E.$$

Since $\mathrm{center}_Y F \subset E$, we have $\mathrm{mult}_F E > 0$, so $\kappa_e \,\mathrm{mult}_F E > 0$. Together with **Step 2**, this proves the inequality of the proposition.

Finally, if $A_X(F) \le S_X(F)$, then $F \ne E$, case (3) does not hold, and $\mathrm{center}_Y F = s_E$, since otherwise $A_X(F) - S_X(F) \ge \kappa_e \,\mathrm{mult}_F E > 0$. □

11.2. **Non-special lines and conics.** In this subsection, we prove the inequality of Proposition 11.2 for the prime divisors F over X whose center on the blow-up of X along a non-special line (Proposition 11.4) or along a conic (Proposition 11.6) is the negative section of the exceptional divisor.

For a non-special line L, the exceptional divisor E of $\mathrm{Bl}_L X$ is isomorphic to $\mathbb{F}_1$, and the estimate along the negative section s of E depends on two properties of the line link of Theorem 2.23. First, the restriction to E of the projection to V_5 has three simple base points. We keep the vanishing conditions at the base points in the estimate, and a torus action on E degenerates them to a curvilinear subscheme of length three supported at a point of s and contained in the fiber through that point. Second, since Γ is a quintic curve, the coefficient of s in $R|_E$ is at most 2. For a conic of type (4) in Proposition 11.2, the restriction to $E \cong \mathbb{F}_2$ of the map to the quadric threefold has either four simple base points or one double base point, all away from the negative section, and we keep the vanishing conditions at two points over them (Lemma 11.5).

Construction-Proposition 11.3. Notation as in Set-up 10.1 and Definition-Proposition 10.9. Assume that L is non-special, so that $E \cong \mathbb{F}_1$. Let $I_{Z_U} \subset \mathcal{O}_E$ be the ideal sheaf of Z_U, let $R_0 := R|_E$, and let r_0 be the restriction to E of a section defining R. For integers $q \ge 0$ let

$$U_q := \mathrm{Im}\left(H^0(Y, qh) \to H^0(E, \mathcal{O}_E(q(2s + 3f)))\right),$$

and for integers $m, j \ge 0$ let

$$W_{m,j} := \mathrm{Im}\left(H^0(Y, mH - jE) \to H^0(E, \mathcal{O}_E(js + (m + j)f))\right).$$

Then the following hold.

(1) $R_0 \in |3s + 4f|$, and Z_U is a curvilinear subscheme of E of length three.

(2) The base points of U are resolved by a composition $p_E : T_E \to E$ of three blow-ups of points, possibly infinitely near, and the total transforms e_1, e_2, e_3 on T_E of the three exceptional curves satisfy the following.
  (a) $I_{Z_U} \cdot \mathcal{O}_{T_E} = \mathcal{O}_{T_E}(-e_1 - e_2 - e_3)$.
  (b) The divisor $\Lambda := p_E^*(2s + 3f) - e_1 - e_2 - e_3$ is base point free.
  (c) For any integer $w \geq 0$ and any divisor $D \geq 0$ on E, the local equations of D belong to $I_{Z_U}^w$ if and only if $p_E^* D - w(e_1 + e_2 + e_3) \geq 0$.

(3) $U_q \subset H^0(E, \mathcal{O}_E(q(2s + 3f)) \otimes I_{Z_U}^q)$ for any integer $q \geq 0$.

(4) Let $m \geq 1$ be an integer. Then the following hold.
  (a) $W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m + j)f) \otimes I_{Z_U}^{j-m})$ for $m \leq j \leq 2m$.
  (b) $W_{m,j} = r_0^{j-2m} \cdot U_{3m-j}$ for $2m \leq j \leq 3m$.
  (c) $W_{m,j} = 0$ for $j > 3m$.

(5) $p_E^* R_0 - 2(e_1 + e_2 + e_3) \geq 0$, that is, the local equations of R_0 belong to $I_{Z_U}^2$.

Proof. On $E \cong \mathbb{F}_1$ we have $s^2 = -1$, $(s \cdot f) = 1$, and $f^2 = 0$. Since $(H \cdot L) = 1$, $H|_E \sim f$. By Lemma 11.1(1), $(f \cdot E|_E) = (H \cdot E^2) = -1$ and $((E|_E)^2) = E^3 = 1$, so $E|_E \sim -s - f$. Hence

$$(H - E)|_E \sim s + 2f, \qquad h|_E \sim 2s + 3f, \qquad (mH - jE)|_E \sim js + (m + j)f. \tag{11.7}$$

For a non-zero section σ of a line bundle, $(\sigma = 0)$ denotes the divisor of zeros of σ.

Proof of (1). Since $R = H - 3E = 2h - (H - E)$, (11.7) implies $R_0 \sim 2(2s+3f) - (s+2f) = 3s+4f$. As R is a prime divisor different from E, $R_0 \geq 0$ is a Cartier divisor on E, so $R_0 \in |3s + 4f|$. By Definition-Proposition 10.9, the points of Z_U are the points $C \cap E$, where C is the strict transform of a line $L' \neq L$ intersecting L, the completed local ring of Z_U at $C \cap E$ is $\mathbb{C}[t]/(t^n)$, where n is the width of C, and the sum of the widths over all such C is 3. Thus Z_U is a curvilinear subscheme of E of length three.

Proof of (2). By (1), the base scheme Z_U of U is finite, so U has no fixed curve. Let E^+ be the strict transform of E on Y^+, and let $S_\Gamma \subset V_5$ be the unique hyperplane section containing Γ. By Theorem 2.23, E^+ is the strict transform of S_Γ, so $E^+ \to S_\Gamma$ is birational. Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one, $H^0(Y, h) = H^0(Y^+, h^+) = H^0(V_5, \mathcal{O}_{V_5}(1))$. Thus the rational map of E defined by U is the composition of $E \dashrightarrow E^+$ and $E^+ \to S_\Gamma \subset V_5 \subset \mathbb{P}^6$, and it is birational onto S_Γ. By Lemma 11.1(1), $\deg S_\Gamma = (h^+)^3 = 5$.

Let $p_E : T_E \to E$ be the composition of the blow-ups of the base points $p_1, p_2, \dots$ of U, proper or infinitely near, ordered so that p_k is a point of the blow-up of $p_1, \dots, p_{k-1}$. Let e_k be the total transform on T_E of the exceptional curve of the blow-up of p_k, and let $b_k \geq 1$ be the multiplicity at p_k of the strict transform of a general member of U. Let $\Lambda' := p_E^*(2s + 3f) - \sum_k b_k e_k$. Then the linear system $p_E^* U - \sum_k b_k e_k \subset |\Lambda'|$ is base point free, and the morphism it defines is birational onto S_Γ. By [Har77, Propositions V.3.2 and V.3.6] and the projection formula [Har77, Appendix A, §1, A4],

$$5 = \deg S_\Gamma = (\Lambda'^2) = ((2s + 3f)^2) - \sum_k b_k^2 = 8 - \sum_k b_k^2.$$

Thus U has exactly three base points p_1, p_2, p_3, and $b_1 = b_2 = b_3 = 1$. In particular $\Lambda' = \Lambda$, and Λ is base point free since its linear subsystem $p_E^* U - e_1 - e_2 - e_3$ is base point free.

Let Θ_k be the strict transform on T_E of the exceptional curve of the blow-up of p_k. We say that p_l is proximate to p_k if p_l is a point of the strict transform of the exceptional curve of p_k. Then $\Theta_k = e_k - \sum e_l$, where the sum runs over the p_l proximate to p_k, so

$$(\Lambda \cdot \Theta_k) = 1 - \#\{l : p_l \text{ is proximate to } p_k\}.$$

Since Λ is nef, $(\Lambda \cdot \Theta_k) \geq 0$, which is the proximity inequality [Cas00, Section 4.2]. Thus at most one base point is proximate to p_k. Suppose that p_l is proximate to p_k and to $p_{k'}$ with $k < k' < l$. Then the strict transform of the exceptional curve of p_k intersects the exceptional curve of $p_{k'}$, so $p_{k'}$ is proximate to p_k, and two base points are proximate to p_k, a contradiction. Hence any

infinitely near base point is a free point in the first neighborhood of exactly one base point, and the first neighborhood of any base point contains at most one base point. Therefore the base points form chains $p_{k_1}, \dots, p_{k_n}$, where $p := p_{k_1}$ is a point of E and $p_{k_{i+1}}$ is a free point in the first neighborhood of p_{k_i}, and different chains start at different points of E.

Fix such a chain. The points of the chain are the first n infinitely near points of a smooth curve germ at p, and the first n infinitely near points of a smooth curve germ only depend on the germ modulo the n-th power of the maximal ideal (after a change of formal coordinates the germ is $\{y_1 = g(y_2)\}$ with $g(0) = 0$, and in the chart $y_1 = y_2\tilde{y}_1$ of the blow-up of p its strict transform is $\{\tilde{y}_1 = g(y_2)/y_2\}$, so by induction the i-th point of the chain depends only on g modulo y_2^i). Hence there exist formal coordinates (y_1, y_2) of E at p and a polynomial χ with $\deg \chi < n$ such that, with $y_1' := y_1 - \chi(y_2)$, the point $p_{k_{i+1}}$ is the point of the strict transform of $\{y_1' = 0\}$ in the first neighborhood of p_{k_i}. Thus for $1 \le i \le n$ the divisor Θ_{k_i} defines the monomial valuation v_i in (y_1', y_2) with $v_i(y_1') = i$ and $v_i(y_2) = 1$. Since the chain consists of free points, $e_{k_i} = \Theta_{k_i} + e_{k_{i+1}}$, so $e_{k_i} = \Theta_{k_i} + \Theta_{k_{i+1}} + \dots + \Theta_{k_n}$, and the coefficient of Θ_{k_i} in $e_1 + e_2 + e_3$ is i. Let $w \ge 0$ be an integer. A monomial ${y_1'}^c y_2^d$ satisfies $ic + d \ge iw$ for $1 \le i \le n$ if and only if $nc + d \ge nw$, and the monomials with $nc + d \ge nw$ span the ideal $(y_1', y_2^n)^w$. Since $v_1, \dots, v_n$ are monomial valuations in (y_1', y_2), for any non-zero $g \in \widehat{\mathcal{O}}_{E,p}$,

$$g \in (y_1 - \chi(y_2), y_2^n)^w \quad \iff \quad p_E^*(g = 0) - w(e_1 + e_2 + e_3) \ge 0 \text{ over a neighborhood of } p. \tag{11.8}$$

Let $\mathcal{J} := p_{E*}\mathcal{O}_{T_E}(-e_1 - e_2 - e_3) \subset \mathcal{O}_E$. By (11.8) with $w = 1$, the completion of $\mathcal{J}$ at the first point p of a chain of length n is $(y_1 - \chi(y_2), y_2^n)$, and $\mathcal{J}$ is trivial outside the first points of the chains. Thus $\mathcal{O}_E/\mathcal{J}$ has length three. Since $b_1 = b_2 = b_3 = 1$, the pullback of any member of U is at least $e_1 + e_2 + e_3$, so $I_{Z_U} \subset \mathcal{J}$. By (1), $\mathcal{O}_E/I_{Z_U}$ has length three, so $I_{Z_U} = \mathcal{J}$. Since $\widehat{\mathcal{O}}_{E,p}$ is faithfully flat over $\mathcal{O}_{E,p}$, (11.8) implies (2)(c). Finally, I_{Z_U} is generated by the local equations of the members of U, and the pullback of any member of U to T_E is the sum of $e_1 + e_2 + e_3$ and a member of the base point free linear system $p_E^*U - e_1 - e_2 - e_3 \subset |\Lambda|$. Thus $I_{Z_U} \cdot \mathcal{O}_{T_E} = \mathcal{O}_{T_E}(-e_1 - e_2 - e_3)$. This proves (2).

Proof of (3) and (4). Let $Y' \to Y$ be the composition of the blow-ups of p_1, p_2, and p_3, viewed as points of Y and of the blow-ups of Y, so that the strict transform of E on Y' is T_E. Let $\alpha : Z \to Y$ and $\beta : Z \to Y^+$ be a common resolution such that α factors through Y' and the strict transform E_Z of E on Z is smooth, and let $\gamma : E_Z \to T_E$ be the induced birational morphism, so that $\alpha|_{E_Z} = p_E \circ \gamma$.

Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one between normal varieties, $H^0(Y, D) = H^0(Y^+, D^+)$ for any divisor D on Y with strict transform D^+ on Y^+. If D and D^+ are Cartier, pulling back induces isomorphisms $H^0(Y, D) \cong H^0(Z, \alpha^*D)$ and $H^0(Y^+, D^+) \cong H^0(Z, \beta^*D^+)$. Let $D_1^+ \in |h^+|$ be general, and let D_1 be the strict transform of D_1^+ on Y. Since h^+ is base point free, $\beta^*D_1^+$ is the strict transform of D_1^+ on Z, which is also the strict transform of D_1. Thus

$$\Phi := \alpha^*D_1 - \beta^*D_1^+ \ge 0$$

is an α-exceptional divisor, $E_Z \not\subset \operatorname{Supp}\Phi$, and $\alpha^*h \sim \beta^*h^+ + \Phi$. Let σ_Φ be a section defining Φ. Let $\phi : Y \to \bar{Y}$ and $\phi^+ : Y^+ \to \bar{Y}$ be the small contractions of the flop. By Lemma 11.1(1), $-K_Y = H - E$ is base point free, so ϕ is the Stein factorization of the morphism defined by $|-K_Y|$, and $-K_Y \sim \phi^*A$ for an ample Cartier divisor A on $\bar{Y}$. The divisors $-K_{Y^+}$ and $(\phi^+)^*A$ are linearly equivalent outside the exceptional locus of ϕ^+, which has codimension at least two, so $-K_{Y^+} \sim (\phi^+)^*A$. Since $\phi \circ \alpha = \phi^+ \circ \beta$, we obtain $\alpha^*(H - E) = \alpha^*(-K_Y) \sim \beta^*(-K_{Y^+})$.

Let $a, b \ge 0$ be integers, and let $D_{a,b} := a(H - E) + bh$. Then $D_{a,b}^+ = a(-K_{Y^+}) + bh^+$, and $\alpha^*D_{a,b} \sim \beta^*D_{a,b}^+ + b\Phi$. Multiplication by σ_Φ^b is an injective map $H^0(Z, \beta^*D_{a,b}^+) \to H^0(Z, \alpha^*D_{a,b})$ between two spaces, each of dimension $h^0(Y, D_{a,b})$, hence an isomorphism. Thus for any non-zero $\sigma \in H^0(Y, D_{a,b})$ there exists a non-zero $\sigma^+ \in H^0(Y^+, D_{a,b}^+)$ with

$$\alpha^*(\sigma = 0) = b\Phi + \beta^*(\sigma^+ = 0) \ge b\Phi. \tag{11.9}$$

We show that $\Phi|_{E_Z} = \gamma^*(e_1+e_2+e_3)$. Let $\sigma \in H^0(Y,h)$ with $\sigma|_E \neq 0$, and let $\Lambda_\sigma := p_E^*(\sigma|_E = 0) - e_1 - e_2 - e_3$, a member of $|\Lambda|$. Since $\alpha^*(\sigma = 0)$ and Φ do not contain E_Z, restricting (11.9) with $(a,b) = (0,1)$ to E_Z implies

$$\gamma^*\Lambda_\sigma + \gamma^*(e_1+e_2+e_3) = \Phi|_{E_Z} + \beta^*(\sigma^+ = 0)|_{E_Z}.$$

Let σ vary in the non-empty open subset of $H^0(Y,h)$ defined by $\sigma|_E \neq 0$. Since the linear system $p_E^*U - e_1 - e_2 - e_3$, whose members are the Λ_σ, is base point free (proof of (2)), the divisors $\gamma^*\Lambda_\sigma$ have no common component. Since h^+ is base point free, the divisors $\beta^*(\sigma^+ = 0)|_{E_Z}$ have no common component. Taking on both sides the largest divisor contained in all members, we obtain $\Phi|_{E_Z} = \gamma^*(e_1+e_2+e_3)$.

Let $a, b \geq 0$ be integers and let $\sigma \in H^0(Y, D_{a,b})$ with $\sigma|_E \neq 0$. Restricting (11.9) to E_Z implies

$$\gamma^* p_E^*(\sigma|_E = 0) \geq b\Phi|_{E_Z} = b\gamma^*(e_1+e_2+e_3).$$

Applying γ_*, we obtain $p_E^*(\sigma|_E = 0) - b(e_1+e_2+e_3) \geq 0$, so by (2), $\sigma|_E \in H^0(E, \mathcal{O}_E(D_{a,b}|_E) \otimes I_{Z_U}^b)$. For $(a,b) = (0,q)$, we obtain (3). For $m \leq j \leq 2m$ we have $mH - jE = (2m-j)(H-E) + (j-m)h = D_{2m-j,j-m}$, which implies the inclusion in (4).

Let $2m \leq j \leq 3m$ and $q := 3m - j$. By Lemma 11.1(3), multiplication by the $(j-2m)$-th power of a section defining R is an isomorphism $H^0(Y,qh) \to H^0(Y, mH - jE)$. Restricting to E, we obtain $W_{m,j} = r_0^{j-2m} \cdot U_q$. Finally, $H^0(Y, mH - jE) = 0$ for $j > 3m$ by Lemma 11.1(3), so $W_{m,j} = 0$. This proves (4).

Proof of (5). Let R^+ be the strict transform of R on Y^+. Since $R \sim 2h - (H-E)$, we have $R^+ \sim 2h^+ - (-K_{Y^+})$, and $\alpha^*(H-E) \sim \beta^*(-K_{Y^+})$ and $\alpha^*h \sim \beta^*h^+ + \Phi$ imply $\alpha^*R \sim \beta^*R^+ + 2\Phi$. The strict transforms of R and R^+ on Z coincide, and a prime divisor on Z is α-exceptional if and only if it is β-exceptional, so $\alpha^*R - \beta^*R^+ - 2\Phi = (\xi)$ for a rational function ξ, and (ξ) is α-exceptional. Then the divisor of ξ on Y is $\alpha_*(\xi) = 0$, so ξ is constant, as Y is normal and projective. Thus

$$\alpha^*R = \beta^*R^+ + 2\Phi.$$

Since $R \neq E$ and $R^+ \neq E^+$, neither side contains E_Z, and restricting to E_Z implies $\gamma^* p_E^* R_0 \geq 2\Phi|_{E_Z} = 2\gamma^*(e_1+e_2+e_3)$. Applying γ_*, we obtain $p_E^*R_0 - 2(e_1+e_2+e_3) \geq 0$, so the local equations of R_0 belong to $I_{Z_U}^2$ by (2)(c). This proves (5). □

Proposition 11.4. *Let X be a smooth prime Fano threefold of genus 12 and $L \subset X$ a non-special line, and let E be the exceptional divisor of $\mathrm{Bl}_L X \to X$. Then any prime divisor F over X with $\mathrm{center}_X F = L$ satisfies*

$$A_X(F) - S_X(F) \geq \frac{5}{11} \mathrm{mult}_F E > 0.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the proposition to the inequality $S(W; \mathrm{ord}_s) \leq 1$ for the refinement W by E.

We use the notation of Set-up 10.1 and Construction-Proposition 11.3. Since L is non-special, case (1) of Proposition 11.2 holds, and the negative section s_E in Proposition 11.2 is s. Thus, if $F = E$ or $\mathrm{center}_Y F \neq s$, the inequality follows from Proposition 11.2 with $\kappa_1 = 5/11$. Hence we may assume that $\mathrm{center}_Y F = s$. Then $\mathrm{mult}_F E > 0$, since $s \subset E$.

Let W be the refinement of the complete series $(H^0(X, mH))_m$ by E (Definition 2.8), so that $W_{m,j}$ is the space of Construction-Proposition 11.3. Since Y and E are smooth and $-E$ is π-ample, E is a prime divisor over X of plt type with plt extraction π and different $\Delta_E = 0$, so $A_{E,\Delta_E}(\mathrm{ord}_s) = 1$. Since E is a smooth surface, the only prime divisor over E whose center contains the generic point of s is s. By [KM98, Lemma 2.29], $A_X(E) = 2$, and $S_X(E) = 17/11$ by Lemma 11.1(2). Assume that

$$S(W; \mathrm{ord}_s) \leq 1. \tag{11.10}$$

Then Lemma 3.1, applied with $\lambda = 1$ and the generic point of s, implies

$$A_X(F) \geq S_X(F) + \left(2 - \frac{17}{11}\right) \operatorname{mult}_F E = S_X(F) + \frac{5}{11} \operatorname{mult}_F E.$$

Thus it suffices to prove (11.10).

For integers $m \geq 1$, $j \geq 0$, and $k \geq 0$, let $W_{m,j}^{\geq k} \subset W_{m,j}$ be the subspace of sections σ with $\operatorname{ord}_s \sigma \geq k$, let $U_q^{\geq k} \subset U_q$ be the subspace of sections σ with $\operatorname{ord}_s \sigma \geq k$, and let

$$B_m(s) := \frac{1}{mN_m} \sum_{j \geq 0} \sum_{k \geq 1} \dim W_{m,j}^{\geq k}.$$

The inner sum is the sum of the jumping numbers of ord_s on $W_{m,j}$, so $S(W; \operatorname{ord}_s) = \lim_m B_m(s)$ [AZ22, Lemma 2.9].

Step 2. In this step we fix coordinates on E in which I_{Z_U} is generated by $z - \varphi(u)$ and a monic cubic polynomial $\psi(u)$.

By [Har77, Example V.2.11.5], E is the blow-up of $\mathbb{P}^2$ at a point, s is the exceptional curve, and the fibers of $E \to L$ are the strict transforms of the lines through the point. Fix homogeneous coordinates x_0, x_1, x_2 of $\mathbb{P}^2$ such that the point is $[0 : 0 : 1]$. Let f_∞ and C_∞ be the strict transforms of the lines $\{x_0 = 0\}$ and $\{x_2 = 0\}$, and let $E^\circ := E \backslash (f_\infty \cup C_\infty)$. Then $E^\circ \cong \mathbb{A}^2$ with coordinates $z := x_1/x_0$ and $u := x_0/x_2$, and $s \cap E^\circ = \{u = 0\}$.

Since Z_U is finite by Construction-Proposition 11.3(1) and $((2s + 3f) \cdot s) = 1$, the restriction $U|_s$ is a non-zero space of sections of a line bundle of degree 1 on $s \cong \mathbb{P}^1$. Thus the base scheme $Z_U \cap s$ of $U|_s$ has length at most 1. Hence at most one point p of Z_U is contained in s, and for such p the local equation u of s generates the maximal ideal of $\mathcal{O}_{Z_U,p} \cong \mathbb{C}[t]/(t^{n_p})$, where n_p is the length of Z_U at p.

We may choose x_0, x_1 such that f_∞ contains no point of Z_U. For $(\nu, c_0, c_1) \in \mathbb{C}^* \times \mathbb{C}^2$, replacing x_2 with $\nu x_2 + c_0 x_0 + c_1 x_1$ does not change $[0 : 0 : 1]$, s, f_∞, and z, and replaces $1/u = x_2/x_0$ with $\nu/u + c_0 + c_1 z$. Let $p \in Z_U \backslash s$ and let y_p be the value of x_2/x_0 at p. After the replacement, the value of $1/u$ at p is $\nu y_p + c_0 + c_1 z(p)$, and if $n_p \geq 2$ and (ζ, η) is a non-zero tangent vector of Z_U at p in the coordinates $(z, x_2/x_0)$, the derivative of $1/u$ along (ζ, η) is $\nu\eta + c_1\zeta$. Each of the following conditions on (ν, c_0, c_1) excludes the zero set of a non-zero polynomial: $\nu y_p + c_0 + c_1 z(p) \neq 0$ for $p \in Z_U \backslash s$, $\nu(y_p - y_{p'}) + c_1(z(p) - z(p')) \neq 0$ for different $p, p' \in Z_U \backslash s$, and $\nu\eta + c_1\zeta \neq 0$ for $p \in Z_U \backslash s$ with $n_p \geq 2$. Possibly replacing x_2 with $\nu x_2 + c_0 x_0 + c_1 x_1$ for a general (ν, c_0, c_1), we may assume that $Z_U \subset E^\circ$, that the values $u(p)$ for $p \in Z_U$ are pairwise different, and that $u - u(p)$ generates the maximal ideal of $\mathcal{O}_{Z_U,p}$ for any $p \in Z_U$.

By the Chinese remainder theorem, $\mathbb{C}[u] \to \mathcal{O}_{Z_U}$ is surjective with kernel generated by

$$\psi(u) := \prod_{p \in Z_U} (u - u(p))^{n_p},$$

a monic polynomial of degree 3. Hence $z \equiv \varphi(u)$ in $\mathcal{O}_{Z_U}$ for a polynomial φ with $\deg \varphi \leq 2$. Since $\mathbb{C}[z, u]/(z - \varphi(u), \psi(u)) \cong \mathbb{C}[u]/(\psi(u))$ also has length three, we obtain

$$I_{Z_U}|_{E^\circ} = (z - \varphi(u), \psi(u)) \subset \mathbb{C}[z, u]. \tag{11.11}$$

Step 3. In this step we bound the dimensions of the spaces of sections vanishing along s and satisfying the conditions $I_{Z_U}^w$ by a count of lattice points.

Let a, b be integers with $0 \leq a \leq b$. Since $s + f$ is the pullback of the class of a line and $as + bf \sim b(s + f) - (b - a)s$, the space $H^0(E, \mathcal{O}_E(as + bf))$ is the space of forms of degree b in x_0, x_1, x_2 vanishing to order at least $b - a$ at $[0 : 0 : 1]$. As a section of $\mathcal{O}_E(as + bf)$, the form $x_0^{b-a} x_2^a$ has divisor of zeros $(b - a)f_\infty + aC_\infty$, which is disjoint from E°. Dividing by $x_0^{b-a} x_2^a$, the monomial $x_0^{b-a+\ell-c} x_1^c x_2^{a-\ell}$ corresponds to $z^c u^\ell$. Thus $H^0(E, \mathcal{O}_E(as + bf))$ is the span of the monomials $z^c u^\ell$ with $(c, \ell) \in \mathcal{Q}_{a,b}$, where

$$\mathcal{Q}_{a,b} := \{(c, \ell) \in \mathbb{Z}^2 : 0 \leq \ell \leq a,\ 0 \leq c \leq b - a + \ell\},$$

and $\operatorname{ord}_s \sigma$ is the minimal ℓ among the monomials $z^c u^\ell$ of σ. Let $w, k \ge 0$ be integers. We show that

$$\dim\{\sigma \in H^0(E, \mathcal{O}_E(as+bf)\otimes I_{Z_U}^w) : \operatorname{ord}_s \sigma \ge k\} \le \#\{(c,\ell) \in \mathcal{Q}_{a,b} : \ell \ge k,\ 3c+\ell \ge 3w\}. \tag{11.12}$$

For $\tau \in \mathbb{C}^*$, the automorphism $[x_0 : x_1 : x_2] \to [x_0 : \tau^3 x_1 : \tau^{-1} x_2]$ of $\mathbb{P}^2$ fixes $[0:0:1]$, so it lifts to an automorphism g_τ of E with $g_\tau(s) = s$, $g_\tau(E^\circ) = E^\circ$, and $g_\tau(z,u) = (\tau^3 z, \tau u)$. Let

$$\varphi_\tau := \tau^3 \varphi(u/\tau), \qquad \psi_\tau := \tau^3 \psi(u/\tau), \qquad \mathcal{I} := (z - \varphi_\tau, \psi_\tau) \subset \mathbb{C}[\tau][z,u].$$

Since $\deg \varphi \le 2$ and ψ is monic of degree 3, both φ_τ and ψ_τ are polynomials in τ and u, ψ_τ is monic of degree 3 in u, $\varphi_0 = 0$, and $\psi_0 = u^3$. For $\tau_0 \in \mathbb{C}$, let $\mathcal{I}_{\tau_0} \subset \mathbb{C}[z,u]$ be the specialization of $\mathcal{I}$ at $\tau = \tau_0$. By (11.11), $\mathcal{I}_{\tau_0}$ is the ideal of $g_{\tau_0}(Z_U)$ for $\tau_0 \ne 0$, $\mathcal{I}_1 = I_{Z_U}|_{E^\circ}$, and $\mathcal{I}_0 = (z, u^3)$. Let $z_\tau := z - \varphi_\tau$. Then $\mathbb{C}[\tau][z,u] = \mathbb{C}[\tau][z_\tau, u]$, and division by the monic polynomials ψ_τ^{w-c} in u shows that

$$\mathbb{C}[\tau][z,u]/\mathcal{I}^w \cong \bigoplus_{c=0}^{w-1} z_\tau^c \cdot \mathbb{C}[\tau][u]/(\psi_\tau^{w-c})$$

is a free $\mathbb{C}[\tau]$-module with basis $z_\tau^c u^d$, $0 \le c < w$, $0 \le d < 3(w-c)$. Hence the specialization of $\mathbb{C}[\tau][z,u]/\mathcal{I}^w$ at τ_0 is $\mathbb{C}[z,u]/\mathcal{I}_{\tau_0}^w$.

Let $H^0_{\ge k}$ be the span of the monomials $z^c u^\ell$ with $(c,\ell) \in \mathcal{Q}_{a,b}$ and $\ell \ge k$, that is, the space of $\sigma \in H^0(E, \mathcal{O}_E(as+bf))$ with $\operatorname{ord}_s \sigma \ge k$. Reduction modulo $\mathcal{I}^w$ defines a $\mathbb{C}[\tau]$-linear map from $H^0_{\ge k} \otimes \mathbb{C}[\tau]$ to the free $\mathbb{C}[\tau]$-module $\mathbb{C}[\tau][z,u]/\mathcal{I}^w$ of finite rank, that is, a matrix with entries in $\mathbb{C}[\tau]$. For $\tau_0 \ne 0$, the kernel of the specialization at τ_0 is the space of $\sigma \in H^0_{\ge k}$ whose local equations belong to the w-th power of the ideal of $g_{\tau_0}(Z_U)$. Since $g_{\tau_0}(s) = s$ and $g_{\tau_0}^* \mathcal{O}_E(as+bf) \cong \mathcal{O}_E(as+bf)$, the dimension of the kernel does not depend on $\tau_0 \ne 0$, and for $\tau_0 = 1$ it is the left side of (11.12). If the rank of the matrix at a general τ is ρ, then all minors of size $\rho+1$ vanish identically, so the rank at $\tau = 0$ is at most ρ. Thus the left side of (11.12) is at most the dimension of the kernel at $\tau = 0$. Since $(z, u^3)^w$ is the monomial ideal generated by $z^c u^{3(w-c)}$, $0 \le c \le w$, the kernel at $\tau = 0$ is spanned by the monomials $z^c u^\ell$ with $(c,\ell) \in \mathcal{Q}_{a,b}$, $\ell \ge k$, and $3c + \ell \ge 3w$. This proves (11.12).

Let $W' \subset H^0(E, \mathcal{O}_E(as+bf) \otimes I_{Z_U}^w)$ be a subspace, and let $W'^{\ge k} \subset W'$ be the subspace of sections σ with $\operatorname{ord}_s \sigma \ge k$. Let $\Omega(a,b,w) := \{(c,\ell) \in \mathcal{Q}_{a,b} : 3c + \ell \ge 3w\}$. Summing (11.12) over $k \ge 1$, we obtain

$$\sum_{k \ge 1} \dim W'^{\ge k} \le \sum_{(c,\ell) \in \Omega(a,b,w)} \ell.$$

Assume moreover that $w \le b - a$. In the row ℓ of $\mathcal{Q}_{a,b}$, the condition $3c + \ell \ge 3w$ means $c \ge \lceil w - \ell/3 \rceil$, and $\lceil w - \ell/3 \rceil \le w \le b - a + \ell$. Since $\lceil x \rceil \ge x$, we obtain

$$\sum_{(c,\ell) \in \Omega(a,b,w)} \ell \le \sum_{\ell=0}^{a} \ell \Big(b - a + \ell + 1 - \max\Big\{0, w - \frac{\ell}{3}\Big\}\Big). \tag{11.13}$$

Step 4. In this step we show that the coefficient $\mu_s := \operatorname{mult}_s R_0$ of s in R_0 satisfies $\mu_s \le 2$.

By Construction-Proposition 11.3(1), $R_0 \in |3s+4f|$. Since f is nef, $\mu_s = \mu_s(s \cdot f) \le (R_0 \cdot f) = 3$. Suppose that $\mu_s = 3$. Then $R_0 - 3s \in |4f|$, and any component C' of $R_0 - 3s$ satisfies $(C' \cdot f) = 0$, so C' is a fiber. Hence any component of R_0 is s or a fiber.

Let E^+ be the strict transform of E on Y^+. By Theorem 2.23, the strict transform of R on Y^+ is the exceptional divisor R of $Y^+ \to V_5$. Let f_R be a fiber of $R \to \Gamma$. Since $E = h - R$ on Y, $E^+ \sim h^+ - R$, so

$$(E^+ \cdot f_R) = (h^+ \cdot f_R) - (R \cdot f_R) = 0 + 1 = 1.$$

As $E^+ \ne R$, the restriction $E^+|_R \ge 0$ is a divisor on R with $(E^+|_R \cdot f_R)_R = 1$, so it has a component Γ^+ dominating Γ. By Theorem 2.23, the flopped curves of $Y \dashrightarrow Y^+$ are the strict transforms of the bisecant lines of Γ, and their images in V_5 are lines, so Γ^+ is not a flopped curve. Thus $Y \dashrightarrow Y^+$ is an isomorphism at the generic point of Γ^+, and the strict transform

Γ_Y of Γ^+ on Y is a curve contained in $E \cap R$. Hence Γ_Y is a component of R_0, so $\Gamma_Y = s$ or Γ_Y is a fiber.

Since $H^0(Y, h) = H^0(V_5, \mathcal{O}_{V_5}(1))$, the rational map of E defined by U is the composition of $E \dashrightarrow E^+$ and $E^+ \to V_5 \subset \mathbb{P}^6$. As U has no fixed curve, the rational map defined by U is defined at the generic point of Γ_Y and maps Γ_Y onto Γ. Let Γ_T be the strict transform of Γ_Y on T_E. Since the base point free linear system $p_E^*U - e_1 - e_2 - e_3 \subset |\Lambda|$ defines the composition of p_E with the rational map defined by U, the projection formula [Har77, Appendix A, §1, A4] implies $(\Lambda \cdot \Gamma_T) \geq \deg \Gamma = 5$. On the other hand, $(e_k \cdot \Gamma_T) \geq 0$ for $k = 1, 2, 3$, so

$$(\Lambda \cdot \Gamma_T) \leq (p_E^*(2s+3f) \cdot \Gamma_T) = ((2s+3f) \cdot \Gamma_Y) \leq \max\{((2s+3f) \cdot s), ((2s+3f) \cdot f)\} = \max\{1, 2\} = 2.$$

This is a contradiction. Hence $\mu_s \leq 2$.

Step 5. In this step we bound $\sum_{k \geq 1} \dim W_{m,j}^{\geq k}$ for each j.

Let $m \geq 1$ and $0 \leq j \leq 2m$, and let $w := \max\{0, j - m\}$. By Construction-Proposition 11.3(4), $W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m+j)f) \otimes I_{Z_U}^w)$, and $w \leq m$. By (11.13) with $(a, b) = (j, m + j)$,

$$\sum_{k \geq 1} \dim W_{m,j}^{\geq k} \leq \sum_{\ell=0}^{j} \ell\Big(m + \ell + 1 - \max\Big\{0, j - m - \frac{\ell}{3}\Big\}\Big). \tag{11.14}$$

Let $2m < j \leq 3m$ and $q := 3m - j$. By Construction-Proposition 11.3(4), $W_{m,j} = r_0^{j-2m} U_q$, and $\operatorname{ord}_s(r_0^{j-2m}\sigma) = (j - 2m)\mu_s + \operatorname{ord}_s \sigma$ for $\sigma \in U_q$. Since multiplication by r_0^{j-2m} is injective,

$$\sum_{k \geq 1} \dim W_{m,j}^{\geq k} = (j - 2m)\mu_s \dim W_{m,j} + \sum_{k \geq 1} \dim U_q^{\geq k}.$$

By Construction-Proposition 11.3(3), $U_q \subset H^0(E, \mathcal{O}_E(2qs + 3qf) \otimes I_{Z_U}^q)$. Since $q - \ell/3 \geq 0$ for $0 \leq \ell \leq 2q$, (11.13) with $(a, b, w) = (2q, 3q, q)$ implies

$$\sum_{k \geq 1} \dim U_q^{\geq k} \leq \sum_{\ell=0}^{2q} \ell\Big(\frac{4\ell}{3} + 1\Big). \tag{11.15}$$

Finally, $W_{m,j} = 0$ for $j > 3m$.

Step 6. In this step we compute the limit of the bounds of **Step 5** and conclude the proof.

For $t \in [0, 3]$ let

$$\omega(t) := \begin{cases} \displaystyle\int_0^t \theta\Big(1 + \theta - \max\Big\{0, t - 1 - \frac{\theta}{3}\Big\}\Big) d\theta & \text{if } 0 \leq t \leq 2, \\ \displaystyle\int_0^{2(3-t)} \frac{4}{3}\theta^2 d\theta & \text{if } 2 \leq t \leq 3. \end{cases}$$

With $\theta = \ell/m$ and $t = j/m$, the right sides of (11.14) and (11.15) are $m^3\omega(j/m) + O(m^2)$, uniformly in j, since they are Riemann sums of piecewise polynomial integrands over $O(m)$ rows. Summing over the $O(m)$ values of j changes the error to $O(m^3)$, and $\sum_j m^3\omega(j/m) = m^4 \int_0^3 \omega(t)dt + O(m^3)$. Since $H^3 = 22$, $N_m = \frac{11}{3}m^3 + O(m^2)$, so the errors are $o(mN_m)$. By Lemma 11.1(4), $\lim_m (mN_m)^{-1} \sum_{j > 2m} (j - 2m) \dim W_{m,j} = J_1 = 5/88$. Together with **Step 5**, we obtain

$$\limsup_m B_m(s) \leq \frac{3}{11} \int_0^3 \omega(t)dt + \frac{5\mu_s}{88}. \tag{11.16}$$

We compute $\int_0^3 \omega(t)dt$. For $0 \le t \le 2$ we have $2\int_0^t \theta(1+\theta)d\theta = t^2 + \frac{2}{3}t^3$. For $1 \le t \le 2$, the function $t - 1 - \theta/3$ is positive exactly for $\theta < 3(t-1)$, so

$$2\int_0^t \theta \max\left\{0, t-1-\frac{\theta}{3}\right\}d\theta = (t-1)\min\{t, 3(t-1)\}^2 - \frac{2}{9}\min\{t, 3(t-1)\}^3$$
$$= \begin{cases} 3(t-1)^3 & \text{if } 1 \le t \le \frac{3}{2}, \\ (t-1)t^2 - \frac{2}{9}t^3 & \text{if } \frac{3}{2} \le t \le 2, \end{cases}$$

and the left side is 0 for $0 \le t \le 1$. For $2 \le t \le 3$,

$$2\omega(t) = 2\int_0^{2(3-t)} \frac{4}{3}\theta^2 d\theta = \frac{64}{9}(3-t)^3.$$

Thus

$$\int_0^2 \left(t^2 + \frac{2}{3}t^3\right)dt = \frac{16}{3}, \qquad \int_2^3 \frac{64}{9}(3-t)^3 dt = \frac{16}{9},$$

$$\int_1^{3/2} 3(t-1)^3 dt + \int_{3/2}^2 \left((t-1)t^2 - \frac{2}{9}t^3\right)dt = \frac{3}{64} + \frac{337}{576} = \frac{91}{144},$$

and hence

$$2\int_0^3 \omega(t)dt = \frac{16}{3} - \frac{91}{144} + \frac{16}{9} = \frac{311}{48}. \tag{11.17}$$

By (11.16), (11.17), and $\mu_s \le 2$ (**Step 4**),

$$\limsup_m B_m(s) \le \frac{3}{22}\cdot\frac{311}{48} + 2\cdot\frac{5}{88} = \frac{311}{352} + \frac{40}{352} = \frac{351}{352} < 1.$$

Thus $S(W; \mathrm{ord}_s) = \lim_m B_m(s) \le 351/352$, so (11.10) holds, and **Step 1** implies the proposition. □

We now treat conics. For a conic of type (4) in Proposition 11.2, the prime divisors left open by Proposition 11.2 are those whose center on the blow-up is the negative section of the exceptional divisor $\mathbb{F}_2$. The rational map $Y \dashrightarrow Q$ of Theorem 2.24, restricted to the exceptional divisor, has base points away from the negative section, namely four simple base points or one double base point. We keep the vanishing conditions at two points over them, possibly infinitely near (two of the four simple base points, or the double base point and the point infinitely near to it in the fiber direction), and bound the dimensions of the refined pieces by h^0 on a toric surface, using the following lemma.

Lemma 11.5. *Let s be the negative section of $\mathbb{F}_2$, and let f be the class of a fiber of $\mathbb{F}_2 \to \mathbb{P}^1$. Let $p_1 \in \mathbb{F}_2 \backslash s$, let p_2 be a point of $\mathrm{Bl}_{p_1}\mathbb{F}_2$ not contained in the strict transform of s, and let T be the blow-up of $\mathrm{Bl}_{p_1}\mathbb{F}_2$ at p_2. We write s and f also for their pullbacks to T, and e_1 and e_2 for the total transforms on T of the exceptional curves over p_1 and p_2. Let $p_0 \in \mathbb{F}_2 \backslash s$, let p_0' be the point of the exceptional curve of $\mathrm{Bl}_{p_0}\mathbb{F}_2$ corresponding to the tangent direction at p_0 of the fiber through p_0, let T_0 be the surface T for $p_1 = p_0$ and $p_2 = p_0'$, and write s, f, e_1, e_2 for the corresponding classes on T_0. Let $f' := f - e_1 - e_2$ on T_0. Then the following hold.*

(1) For any integers $a \ge 0$, b, and $w \ge 0$,

$$h^0(T, as + bf - w(e_1+e_2)) \le h^0(T_0, as + bf - w(e_1+e_2)).$$

(2) T_0 is a projective toric surface, the torus-invariant curves of T_0 have classes s, f, $s + 2f - e_1$, $e_1 - e_2$, e_2, and f', and $\mathrm{NE}(T_0)$ is generated by s, f, $s + 2f - e_1$, $e_1 - e_2$, e_2, and f'.

(3) Let a, b, w be real numbers such that either $(a,b,w) = (t-v, t+2, t-1)$ for some $1 \le t \le 2$ and $0 \le v \le t$, or $(a,b,w) = (2r-v, 4r, r)$ for some $0 \le r \le 1$ and $0 \le v \le 2r$. Then

$$\mathrm{vol}_{T_0}(as+bf-w(e_1+e_2)) = 2ab - 2a^2 - 2w^2 + \frac{1}{2}\max\{2w-a,0\}^2.$$

Proof. (1) The complement $\mathbb{F}_2 \backslash s$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(2)$, and the ruling is the bundle projection. The following automorphisms of $\mathbb{F}_2$ preserve s and the ruling: the lifts of automorphisms of $\mathbb{P}^1$ to $\mathcal{O}_{\mathbb{P}^1}(2) \cong T_{\mathbb{P}^1}$, the multiplications by non-zero scalars on the fibers, and the translations by elements of $H^0(\mathbb{P}^1, \mathcal{O}(2))$. For distinct points $x_1, x_2 \in \mathbb{P}^1$ and any $x \in \mathbb{P}^1$, the restriction maps from $H^0(\mathbb{P}^1, \mathcal{O}(2))$ to the fibers of $\mathcal{O}(2)$ at x_1 and x_2 and to the first-order jets of $\mathcal{O}(2)$ at x are surjective. Thus translations move any two points of $\mathbb{F}_2 \backslash s$ on distinct fibers into the zero section, and move any point of $\mathbb{F}_2 \backslash s$ together with a tangent direction different from the fiber direction to a point of the zero section with the tangent direction of the zero section. Since $\mathrm{Aut}(\mathbb{P}^1)$ acts 2-transitively on $\mathbb{P}^1$ and the scalar multiplications act transitively on the non-zero points of a fiber, each of the following four sets of pairs (p_1, p_2) is a single orbit:

(i) $p_2 \in \mathbb{F}_2 \backslash s$ and p_2 is not on the fiber through p_1;
(ii) $p_2 \in \mathbb{F}_2 \backslash (s \cup \{p_1\})$ and p_2 is on the fiber through p_1;
(iii) p_2 is a point of the exceptional curve of $\mathrm{Bl}_{p_1} \mathbb{F}_2$ different from the fiber direction;
(iv) p_2 is the point of the exceptional curve of $\mathrm{Bl}_{p_1} \mathbb{F}_2$ corresponding to the fiber direction.

An automorphism of $\mathbb{F}_2$ preserving s and the ruling and mapping (p_1, p_2) to (p_1', p_2') lifts to an isomorphism of the blow-ups mapping s, f, e_1, e_2 to s, f, e_1, e_2. Hence the left side of the inequality in (1) is constant on any orbit, and T_0 is the surface T of a pair in (iv).

Let τ be the coordinate of $\mathbb{A}^1$, and consider the following families $p_2(\tau)$ with fixed p_1: for (i) to (ii), $p_2(\tau) \in \mathbb{F}_2 \backslash s$ is on the fiber over a point $x(\tau) \in \mathbb{P}^1$ different from the image of p_1 for $\tau \neq 0$, and $p_2(0) \neq p_1$ is on the fiber through p_1; for (ii) to (iv), $p_2(\tau)$ moves on the fiber through p_1 with non-zero velocity and $p_2(0) = p_1$; for (iii) to (iv), $p_2(\tau)$ moves on the exceptional curve of $\mathrm{Bl}_{p_1} \mathbb{F}_2$ and $p_2(0)$ is the fiber direction. In each case, blow up $\mathbb{F}_2 \times \mathbb{A}^1$ along $\{p_1\} \times \mathbb{A}^1$, and then blow up the closure of the strict transform of $\{(p_2(\tau), \tau) : \tau \neq 0\}$. The closure is a section over $\mathbb{A}^1$, and in the family from (ii) to (iv) it intersects the fiber over $\tau = 0$ at the fiber direction at p_1, since $p_2(\tau)$ moves along the fiber. Thus the total space is smooth and projective over $\mathbb{A}^1$, the fiber over τ is the surface T of $(p_1, p_2(\tau))$, and s, f, e_1, e_2 extend to Cartier divisors on the total space. By upper semicontinuity [Har77, Theorem III.12.8], the value of $h^0(T, as+bf-w(e_1+e_2))$ at $\tau = 0$ is at least the value at $\tau \neq 0$. Thus the values on (i) are at most those on (ii), the values on (ii) and (iii) are at most those on (iv), and (1) follows.

(2) By the orbit description in (1), we may take p_0 to be a torus-fixed point of $\mathbb{F}_2$. Let f_0 and f_∞ be the torus-invariant fibers and $s_\infty \in |s+2f|$ the torus-invariant section disjoint from s, and take $p_0 = f_0 \cap s_\infty$. The fiber direction at p_0 is the intersection of the exceptional curve with the strict transform of f_0, again a torus-fixed point. Blowups of torus-fixed points of smooth toric surfaces are toric [CLS11, Proposition 3.3.15]. The torus-invariant curves of T_0 are the strict transforms of s, f_∞, s_∞, of the first exceptional curve, and of f_0, together with the second exceptional curve. Since the second point is not on the strict transform of s_∞, their classes are s, f, $s+2f-e_1$, e_1-e_2, f', and e_2. The cone of curves of a projective toric surface is generated by the torus-invariant curves [CLS11, Theorem 6.3.20].

(3) On T_0 we have $s^2 = -2$, $s \cdot f = 1$, $f^2 = 0$, $e_1^2 = e_2^2 = -1$, and $e_i \cdot s = e_i \cdot f = e_1 \cdot e_2 = 0$ [Har77, Propositions V.2.3, V.2.9 and V.3.2]. Let $D := as + bf - w(e_1+e_2)$, let $c := \frac{1}{2}\max\{2w-a, 0\}$, and let $P := D - cf'$. Then $D^2 = 2ab - 2a^2 - 2w^2$, $f'^2 = -2$, $D \cdot f' = a - 2w$, and the intersection numbers with the six curves of (2) are

(11.18)

	s	f	$s+2f-e_1$	e_1-e_2	e_2	f'
D	$b-2a$	a	$b-w$	0	w	$a-2w$
P (if $a < 2w$)	$b-w-\frac{3}{2}a$	a	$b-w$	0	$\frac{a}{2}$	0

In the first range $b-2a=2-t+2v$, $a=t-v\le 2$, and $b-w=3$, and in the second range $b-2a=2v$, $a\le 2r$, and $b-w=3r$. Hence $b-w-\frac{3}{2}a\ge 0$ in both ranges.

Assume that $a\ge 2w$. Then all entries of the row of D in (11.18) are non-negative, so D is nef by (2), and $\mathrm{vol}(D)=D^2$.

Assume that $a<2w$. Since $b-w-\frac{3}{2}a\ge 0$, all entries of the row of P in (11.18) are non-negative, and P is nef by (2). Since $c>0$, $P\cdot f'=0$, and $f'^2=-2$, $D=P+cf'$ is the Zariski decomposition of D. Thus [Laz04a, Theorem 2.3.19 and Corollary 2.3.22] implies

$$\mathrm{vol}(D)=P^2=D^2-2c(a-2w)-2c^2=D^2+2c^2=D^2+\frac{1}{2}(2w-a)^2. \qquad \square$$

Proposition 11.6. *Let X be a smooth prime Fano threefold of genus 12 and $C\subset X$ an integral curve with $H\cdot C=2$, and let E be the exceptional divisor of $\mathrm{Bl}_C X\to X$. Then every prime divisor F over X with $\mathrm{center}_X F=C$ satisfies*

$$A_X(F)-S_X(F)\ge \frac{59}{88}\,\mathrm{mult}_F E>0.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the proposition to the bound $S(W;\mathrm{ord}_s)\le 1$, where W is the refinement by E and s is the negative section of $E\cong\mathbb{F}_2$.

Let $\pi: Y:=\mathrm{Bl}_C X\to X$, and write H also for π^*H. By Proposition 11.2, C is a smooth rational curve, and $A_X(F)-S_X(F)\ge\frac{59}{88}\,\mathrm{mult}_F E>0$ unless case (4) of Proposition 11.2 holds and $\mathrm{center}_Y F=s_E$. Thus we may assume that $N_{C/X}\cong\mathcal{O}(1)\oplus\mathcal{O}(-1)$, $E\cong\mathbb{F}_2$, and $\mathrm{center}_Y F=s_E$. Let s be the negative section of E. It is the curve s_E of Proposition 11.2. Let W be the refinement of $(H^0(X,mH))_m$ by E (Definition 2.8). Thus $W_{m,j}$ is the image of $H^0(Y,mH-jE)\to H^0(E,(mH-jE)|_E)$.

Since Y and E are smooth and $-E$ is π-ample, E is a prime divisor over X of plt type with plt extraction π and different $\Delta_E=0$, so $A_{E,\Delta_E}(\mathrm{ord}_s)=1$. The only prime divisor over E whose center on E contains the generic point of s is s. We have $A_X(E)=2$ [KM98, Lemma 2.29] and $S_X(E)=\frac{117}{88}$ (Lemma 11.1(2)), and $\mathrm{mult}_F E>0$ since $\mathrm{center}_Y F\subset E$. If $S(W;\mathrm{ord}_s)\le 1$, then Lemma 3.1, applied to the complete series of $H=-K_X$ with $\lambda=1$ and the generic point of s, implies

$$A_X(F)-S_X(F)\ge\big(A_X(E)-S_X(E)\big)\,\mathrm{mult}_F E=\Big(2-\frac{117}{88}\Big)\,\mathrm{mult}_F E=\frac{59}{88}\,\mathrm{mult}_F E>0,$$

since $s\subset\mathrm{center}_Y F$. Thus it suffices to prove

$$S(W;\mathrm{ord}_s)\le 1. \tag{11.19}$$

Step 2. In this step we show that the restriction of $|h|$ to E has finitely many base points, all away from s, with multiplicities b_i satisfying $\sum_i b_i^2=4$.

Let f be the class of a fiber of $E\to C$. Since C is a smooth conic, Theorem 2.24 holds for C. Let $Y\dashrightarrow Y^+$ be the flop, $Y^+\to Q$ the blow-up, and h and R the divisors given by Theorem 2.24, so that $h=H-2E$ and $R=2H-5E$ on Y.

Since $H|_E=2f$, $E|_E\cdot f=-1$, and $(E|_E)^2=E^3=0$ (Lemma 11.1(1)), we have $-E|_E=s+f$. Since $-K_Y=H-E$, we obtain, for any integers m and j,

$$-K_Y|_E=s+3f,\quad h|_E=2s+4f,\quad R|_E=5s+9f,\quad (mH-jE)|_E=js+(2m+j)f. \tag{11.20}$$

Since $(s+3f)\cdot s=1$ and $(s+3f)\cdot f=1$, $-K_Y|_E$ is ample [Har77, Proposition V.2.20]. By Theorem 2.24, $|-K_Y|$ is base point free and the morphism it defines contracts no divisor. Let $\phi: Y\to Y_0$ be its Stein factorization. Then Y_0 is normal, $\mathrm{Exc}(\phi)$ has codimension at least two, and $-K_Y\sim\phi^*A$ for an ample Cartier divisor A on Y_0. The flop $Y\dashrightarrow Y^+$ is the flop of ϕ [KM98, Definition 6.10]. Let $\phi^+: Y^+\to Y_0$ be the corresponding contraction, so that $\mathrm{Exc}(\phi^+)$ also has codimension at least two. Any curve C' contracted by ϕ satisfies $-K_Y\cdot C'=0$, so $C'\not\subset E$. Since ϕ contracts no divisor, $\mathrm{Exc}(\phi)$ is a union of finitely many curves, and hence $E\cap\mathrm{Exc}(\phi)$ is finite.

The flop maps $Y \setminus \mathrm{Exc}(\phi)$ isomorphically onto $Y^+ \setminus \mathrm{Exc}(\phi^+)$. Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one, strict transform is a bijection from $|h|$ on Y to $|h|$ on Y^+, and $|h|$ on Y^+ is base point free, being the pullback of $|\mathcal{O}_Q(1)|$. Thus $\mathrm{Bs}|h| \subset \mathrm{Exc}(\phi)$.

Let U be the image of $H^0(Y,h) \to H^0(E, h|_E)$. The base locus of U is $\mathrm{Bs}|h| \cap E$, so it is finite. We show that it is disjoint from s. By (11.20), $(h|_E \cdot s) = (2s+4f)\cdot s = 0$, so $h|_E|_s \cong \mathcal{O}_s$. If $x \in s$ is a base point of U, then for any $u \in U$, $u|_s$ is a section of $\mathcal{O}_s$ vanishing at x, so $u|_s = 0$. Thus s is contained in the base locus of U, a contradiction.

Let $p_E : T_E \to E$ be the composite of the blow-ups at the base points $p_1, p_2, \dots$ of U, proper or infinitely near. Let e_i be the total transform on T_E of the exceptional curve over p_i, and let $b_i \geq 1$ be the multiplicity of U at p_i, so that the fixed part of $p_E^* U$ is $\sum_i b_i e_i$ and the moving part of $p_E^* U$ is base point free of class

$$\Lambda := p_E^*(2s+4f) - \sum_i b_i e_i.$$

All p_i are over $E \setminus s$.

Since $Q \subset \mathbb{P}^4$ is a quadric hypersurface, the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^4}(-1) \to \mathcal{O}_{\mathbb{P}^4}(1) \to \mathcal{O}_Q(1) \to 0$ and [Har77, Theorem III.5.1] show that $H^0(\mathbb{P}^4, \mathcal{O}(1)) \to H^0(Q, \mathcal{O}_Q(1))$ is an isomorphism. Since $Y^+ \to Q$ is a birational morphism and Q is normal, pullback $H^0(Q, \mathcal{O}_Q(1)) \to H^0(Y^+, h)$ is an isomorphism [Har77, Proposition II.6.3A]. Together with the bijection of $|h|$ on Y and on Y^+ above, any member of $|h|$ on Y is the strict transform of the pullback to Y^+ of a hyperplane section of Q. The rational map $E \dashrightarrow \mathbb{P}^4$ defined by U is the restriction to E of the composite $Y \dashrightarrow Y^+ \to Q \subset \mathbb{P}^4$. By Theorem 2.24, the strict transform E^+ of E on Y^+ is the unique member of $|2h - R|$, so the image S_Q of E^+ in Q is an integral member of $|\mathcal{O}_Q(2)|$, and $E^+ \to S_Q$ is birational. Since $E \cap \mathrm{Exc}(\phi)$ is finite, $E \dashrightarrow E^+$ is birational. Thus the morphism $T_E \to \mathbb{P}^4$ defined by the moving part of $p_E^* U$ is birational onto S_Q, and $\deg S_Q = 2 \deg Q = 4$, since $S_Q \in |\mathcal{O}_Q(2)|$. Hence $\Lambda^2 = 4$ [Har77, Appendix A, §1, the definition of f_* and axiom A4]. Since $(2s+4f)^2 = 8$ and $e_i \cdot e_k = -\delta_{ik}$ [Har77, Proposition V.3.2], we obtain

$$\sum_i b_i^2 = (2s+4f)^2 - \Lambda^2 = 4. \tag{11.21}$$

Since the b_i are positive integers, either U has exactly four base points, all with $b_i = 1$, or U has exactly one base point p_1, with $b_1 = 2$.

Step 3. In this step we show that the sections in $W_{m,j}$ vanish to order $(j-m)b_i$ at the points p_i for $m \leq j \leq 2m$.

Recall that $W_{m,j}$ is the image of $H^0(Y, mH - jE) \to H^0(E, (mH - jE)|_E)$. For a non-zero section σ of a line bundle, we write $(\sigma = 0)$ for the divisor of zeros of σ.

Let $Y' \to Y$ be the composite of the blow-ups at the points $p_1, p_2, \dots$, in the order in which they are blown up in p_E. Since E is a smooth surface in the smooth threefold Y, the blow-up of Y at a point of E induces on the strict transform of E the blow-up of E at that point, so the strict transform of E on Y' is T_E. Let Z be a resolution of the graph of $Y' \dashrightarrow Y^+$, and let $\alpha : Z \to Y$ and $\beta : Z \to Y^+$ be the induced morphisms. Then the strict transform E_Z of E on Z admits a birational morphism $\gamma : E_Z \to T_E$ with $p_E \circ \gamma = \alpha|_{E_Z}$.

For a divisor D on Y with strict transform D^+ on Y^+, the divisor $\alpha^* D - \beta^* D^+$ is α-exceptional, since for a prime divisor Θ of Z not exceptional over Y, the coefficients of Θ in $\alpha^* D$ and in $\beta^* D^+$ are both equal to the coefficient of $\alpha_* \Theta$ in D. If $D' = D + (\xi)$ for a rational function ξ, then $D'^+ = D^+ + (\xi)$ on Y^+ and $\alpha^*(\xi) = (\xi) = \beta^*(\xi)$ on Z. Thus $\alpha^* D - \beta^* D^+$ depends only on the linear equivalence class of D. Moreover, fix a canonical divisor K_Z and let $K_Y := \alpha_* K_Z$, $K_{Y^+} := \beta_* K_Z$, and $K_{Y_0} := (\phi \circ \alpha)_* K_Z$, so that $K_{Y_0} = \phi_* K_Y = \phi_*^+ K_{Y^+}$. Write $K_Y = -\phi^* A + (\zeta)$ for a rational function ζ. Then $K_{Y_0} = -A + (\zeta)$ is Cartier and $K_Y = \phi^* K_{Y_0}$. The divisors K_{Y^+} and $\phi^{+*} K_{Y_0}$ coincide outside $\mathrm{Exc}(\phi^+)$, which has codimension at least two, so $K_{Y^+} = \phi^{+*} K_{Y_0}$.

Since $\phi\circ\alpha=\phi^+\circ\beta$, we obtain $\alpha^*K_Y=\beta^*K_{Y^+}$. Let

$$\Phi:=\alpha^*D_1-\beta^*D_1^+\quad\text{for } D_1\in|h|.$$

Choosing D_1 such that D_1^+ is a general member of the base point free system $|h|$ on Y^+, D_1^+ contains no center on Y^+ of a β-exceptional divisor, so $\beta^*D_1^+$ is the strict transform of D_1 on Z. Thus $\Phi\ge 0$, and the class of Φ is $\alpha^*h-\beta^*h$. Since Φ is α-exceptional, $E_Z\not\subset\operatorname{Supp}\Phi$.

We show that

$$\Phi|_{E_Z}=\gamma^*\Big(\sum_i b_ie_i\Big). \tag{11.22}$$

For $D_1\in|h|$ with $E\not\subset D_1$, we have $(\alpha^*D_1)|_{E_Z}=\gamma^*p_E^*(D_1|_E)$ and $(\alpha^*D_1)|_{E_Z}=\Phi|_{E_Z}+(\beta^*D_1^+)|_{E_Z}$. As D_1 varies, the divisors $(\beta^*D_1^+)|_{E_Z}$ form the restriction to E_Z of the base point free system $\beta^*|h|$, which has no fixed component, and the divisors $p_E^*(D_1|_E)$ form p_E^*U, whose fixed part is $\sum_i b_ie_i$ and whose moving part is base point free. Computing the fixed part of the system $\{(\alpha^*D_1)|_{E_Z}\}$ in both ways, we obtain (11.22).

Let $m\le j\le 2m$ be integers, and let $\sigma\in W_{m,j}$ be non-zero, the restriction of a section of $mH-jE$ with divisor D. Then $E\not\subset D$ and

$$mH-jE=(2m-j)(-K_Y)+(j-m)h.$$

Since $\alpha^*D-\beta^*D^+$ depends only on the class of D, $\alpha^*K_Y=\beta^*K_{Y^+}$, and strict transform is additive, we obtain $\alpha^*D-\beta^*D^+=(j-m)\Phi$. Since $\beta^*D^+\ge 0$ and $E_Z\not\subset\operatorname{Supp}\beta^*D^+$, restriction to E_Z and (11.22) imply $\gamma^*p_E^*(\sigma=0)\ge(j-m)\gamma^*(\sum_i b_ie_i)$. Applying γ_*, we obtain

$$p_E^*(\sigma=0)\ge(j-m)\sum_i b_ie_i\quad\text{for any integers } m\le j\le 2m \text{ and any nonzero } \sigma\in W_{m,j}. \tag{11.23}$$

Step 4. In this step we keep the vanishing conditions at two points over the base points of U and bound the dimensions of the pieces of $W_{m,j}$ by h^0 on the surface T_0 of Lemma 11.5.

If U has four base points with $b_i=1$, we number them so that $p_1\in E$, p_2 is either a point of E or a point of the exceptional curve over p_1, and p_1 and p_2 are blown up first in p_E. In this case, let p_2' be the point of $\operatorname{Bl}_{p_1}E$ over p_2. If U has one base point p_1 with $b_1=2$, let p_2' be the point of the exceptional curve of $\operatorname{Bl}_{p_1}E$ corresponding to the fiber direction at p_1. Let T be the blow-up of $\operatorname{Bl}_{p_1}E$ at p_2', and let e_1 and e_2 be the total transforms on T of the exceptional curves over p_1 and over p_2'. In the first case, p_E factors through T, and the pullbacks of e_1 and e_2 to T_E are the classes e_1 and e_2 of **Step 2**. In the second case, $T_E=\operatorname{Bl}_{p_1}E$, and the pullback of e_1 from T_E to T is e_1. Since all base points of U are over $E\backslash s$ (**Step 2**), $p_1\notin s$, and in both cases p_2' is over a point of $E\backslash s$. As $\operatorname{Bl}_{p_1}E\to E$ is an isomorphism over a neighborhood of s, p_2' is not contained in the strict transform of s. Thus the surface T satisfies the hypotheses of Lemma 11.5 for the points p_1 and p_2'. For a section u on E, let u_T be the pullback of u to T. We show that for integers $m\le j\le 2m$ and non-zero $u\in W_{m,j}$,

$$(u_T=0)\ge(j-m)(e_1+e_2). \tag{11.24}$$

In the first case, e_3 and e_4 are effective, so (11.24) follows from (11.23) by pushing forward to T. In the second case, $T_E=\operatorname{Bl}_{p_1}E$, and (11.23) implies

$$(u_T=0)\ge 2(j-m)e_1=(j-m)(e_1+e_2)+(j-m)(e_1-e_2)\ge(j-m)(e_1+e_2),$$

since e_1-e_2 is the strict transform of the exceptional curve over p_1.

For integers $k\ge 0$, let $W_{m,j}^{\ge k}:=\{u\in W_{m,j}:\operatorname{ord}_s(u)\ge k\}$, and let σ_s be a section defining s. Any $u\in W_{m,j}^{\ge k}$ is of the form $u=\sigma_s^k u'$ with $u'\in H^0(E,(j-k)s+(2m+j)f)$ by (11.20). Since the pullback of s to T is disjoint from the supports of e_1 and e_2, (11.24) implies $(u_T'=0)\ge(j-m)(e_1+e_2)$ for $m\le j\le 2m$. If $k>j$, then $((j-k)s+(2m+j)f)\cdot f<0$ and $W_{m,j}^{\ge k}=0$.

Thus Lemma 11.5(1) implies, for $0 \le k \le j$,

$$\dim W_{m,j}^{\ge k} \le \begin{cases} h^0(E, (j-k)s + (2m+j)f) & \text{if } 0 \le j \le m, \\ h^0(T_0, (j-k)s + (2m+j)f - (j-m)(e_1+e_2)) & \text{if } m \le j \le 2m. \end{cases} \tag{11.25}$$

Now let $2m \le j$, let $q := 5m - 2j$, and let U_q be the image of $H^0(Y, qh) \to H^0(E, qh|_E)$. By Lemma 11.1(3), $W_{m,j} = 0$ if $j > \frac{5}{2}m$, and if $2m \le j \le \frac{5}{2}m$, then

$$W_{m,j} = r_0^{j-2m} \cdot U_q,$$

where r_0 is the restriction to E of a section σ_R defining R, and multiplication by r_0^{j-2m} is injective. Since $qh = qH - 2qE$, we have $U_q = W_{q,2q}$. Let μ_s be the coefficient of s in $(r_0 = 0) = R|_E$. Since $R \ne E$, $R|_E - \mu_s s$ is effective and does not contain s, so (11.20) implies $5 - \mu_s = (R|_E - \mu_s s) \cdot f \ge 0$, that is, $\mu_s \le 5$. Since $\operatorname{ord}_s(r_0^{j-2m} u') = (j-2m)\mu_s + \operatorname{ord}_s(u')$ for any non-zero $u' \in U_q$, we obtain

$$\sum_{k \ge 1} \dim W_{m,j}^{\ge k} = (j-2m)\mu_s \dim W_{m,j} + \sum_{k \ge 1} \dim W_{q,2q}^{\ge k} \quad \text{for } 2m \le j \le \frac{5}{2}m, \tag{11.26}$$

and (11.25) with $(m, j) = (q, 2q)$ bounds $\dim W_{q,2q}^{\ge k}$ by $h^0(T_0, (2q-k)s + 4qf - q(e_1+e_2))$.

Step 5. In this step we show that $S(W; \operatorname{ord}_s) \le \frac{87}{88}$.

Let

$$B_m(s) := \frac{1}{mN_m} \sum_{j \ge 0} \sum_{k \ge 1} \dim W_{m,j}^{\ge k}.$$

The inner sum is the sum of the jumping numbers of ord_s on $W_{m,j}$, so $S(W; \operatorname{ord}_s) = \lim_m B_m(s)$, and the limit exists by [AZ22, Lemma 2.9].

We bound $B_m(s)$ by (11.25) and (11.26). Let $t := j/m$, $v := k/m$, and $r := q/m = 5 - 2t$. The divisors in (11.25) are m times $(t-v)s + (t+2)f$ and $(t-v)s + (t+2)f - (t-1)(e_1+e_2)$, and the divisor bounding $W_{q,2q}^{\ge k}$ is m times $(2r-v)s + 4rf - r(e_1+e_2)$. On T_0, $h^0(T_0, D)$ is the number of lattice points of the polygon of D [CLS11, Proposition 4.3.3], whose area is $\frac{1}{2}\operatorname{vol}(D)$. For divisors D with coefficients bounded by a constant times m, the boundary of the polygon is covered by $O(m)$ unit squares, so $h^0(T_0, D) = \frac{1}{2}\operatorname{vol}(D) + O(m)$ uniformly. On $E \cong \mathbb{F}_2$, $h^0(E, as + bf) = \sum_{i=0}^{a} h^0(\mathbb{P}^1, \mathcal{O}(b - 2i))$ for $a \ge 0$ [Har77, Exercise III.8.4(a) and Proposition V.2.8], which implies $h^0(E, D) = \frac{1}{2}\operatorname{vol}(D) + O(m)$ uniformly for divisors D on E with coefficients bounded by a constant times m. The number of pairs (j, k) is $O(m^2)$, so the lattice point errors and the errors of the Riemann sums contribute $O(m^3)$ in total, and $mN_m = \frac{11}{3}m^4 + O(m^3)$ since $H^3 = 22$. By Lemma 11.1(4), the first term on the right of (11.26) contributes at most $\mu_s J_2 \le \frac{5}{88}$ in the limit. Thus

$$\limsup_m B_m(s) \le \frac{3}{22}\Big(\int_0^2 \int_0^t V_1(t,v)\,dv\,dt + \int_2^{5/2} \int_0^{2r} V_2(r,v)\,dv\,dt \Big) + \frac{5}{88}, \tag{11.27}$$

where $V_1(t,v) := \operatorname{vol}_{\mathbb{F}_2}((t-v)s + (t+2)f)$ for $0 \le t \le 1$, $V_1(t,v) := \operatorname{vol}_{T_0}((t-v)s + (t+2)f - (t-1)(e_1+e_2))$ for $1 \le t \le 2$, and $V_2(r,v) := \operatorname{vol}_{T_0}((2r-v)s + 4rf - r(e_1+e_2))$.

We compute the integrals. On $\mathbb{F}_2$, $as + bf$ with $b \ge 2a \ge 0$ is nef with square $2ab - 2a^2$. For $0 \le v \le t \le 2$, $(t+2) - 2(t-v) = 2 - t + 2v \ge 0$, so

$$\int_0^t \operatorname{vol}_{\mathbb{F}_2}((t-v)s + (t+2)f)\,dv = \int_0^t 2(t-v)(2+v)\,dv = 2t^2 + \frac{t^3}{3}.$$

For $1 \le t \le 2$, let $w := t - 1$. Lemma 11.5(3) with $(a, b, w) = (t - v, t + 2, t - 1)$ and $2w - a = v - (2 - t)$ implies

$$\int_0^t V_1(t,v)\,dv = 2t^2 + \frac{t^3}{3} - \int_0^t \Big(2w^2 - \frac{1}{2}\max\{v - (2-t), 0\}^2\Big)\,dv = 2t^2 + \frac{t^3}{3} - 2w^2 - \frac{2}{3}w^3.$$

Lemma 11.5(3) with $(a,b,w)=(2r-v,4r,r)$ and $2w-a=v$ implies

$$\int_0^{2r} V_2(r,v)dv=\int_0^{2r}\Big(2(4r^2-v^2)-2r^2+\frac{v^2}{2}\Big)dv=8r^3.$$

Since $dt=-\frac{1}{2}dr$ on $[2,\frac{5}{2}]$, we obtain

$$\int_0^2\int_0^t V_1dvdt+\int_2^{5/2}\int_0^{2r}V_2dvdt=\frac{20}{3}-\int_0^1\Big(2w^2+\frac{2}{3}w^3\Big)dw+\int_0^1 4r^3dr=\frac{20}{3}-\frac{5}{6}+1=\frac{41}{6}.$$

By (11.27),

$$S(W;\mathrm{ord}_s)=\lim_m B_m(s)\le\frac{3}{22}\cdot\frac{41}{6}+\frac{5}{88}=\frac{87}{88}. \tag{11.28}$$

This implies (11.19). □

11.3. **Special lines.** In this subsection, we treat the prime divisors over X whose center on $\mathrm{Bl}_L X$ is the negative section s of $E\cong\mathbb{F}_3$ for a special line L. As for special lines in genus ten (Propositions 5.7 and 5.9), we extract the prime divisor P over s. It satisfies $S_X(P)\le 61/22<3=A_X(P)$, and the sections of $mH-kP$ with $4m\le k\le 5m$ are computed through the projection of V_5 from the inverse bisecant onto a quadric threefold, where they form a complete linear system.

Lemma 11.7. *Notation as in Set-up 10.10. Then the following hold.*

(1) $A_X(P)=3$, $K_T=(\pi p_T)^*K_X+E'+2P$, *and* $S_X(P)\le 61/22$.

(2) $\mathrm{mult}_s(R|_E)=2$ *and* $R_T\sim H-3E'-5P$. *If* $P\cong\mathbb{F}_0$, *then* $H|_P\sim f_P$, $-P|_P\sim u_P+f_P$, $E'|_P\sim u_P+2f_P$, *and* $R_T|_P\sim 2u_P$. *If* $P\cong\mathbb{F}_2$, *then* $H|_P\sim f_P$, $-P|_P\sim\sigma+2f_P$, $E'|_P\sim\sigma+3f_P$, *and* $R_T|_P\sim 2\sigma+2f_P$. *The curve* $E'\cap P$ *is smooth and irreducible, and* $E'|_P=E'\cap P$.

(3) The unique hyperplane section of V_5 *containing* Γ *is* S_M.

(4) For integers $m\ge 1$ *and* $k\ge 0$, *any element of* $H^0(T,mH-kP)$ *vanishes along* E' *to order at least* $\lceil k/2\rceil$, *and* $H^0(T,mH-kP)=0$ *if* $k>5m$. *For integers* $4m\le k\le 5m$, *any element of* $H^0(T,mH-kP)$ *vanishes along* E' *to order at least* $k-2m$ *and along* R_T *to order at least* $k-4m$,

$$mH-kP\sim(k-2m)E'+(k-4m)R_T+(5m-k)(H-2E'-4P),$$

and multiplication by $e'^{k-2m}r_T^{k-4m}$, *where* e' *and* r_T *are sections defining* E' *and* R_T, *is an isomorphism from* $H^0(Q,\mathcal{O}_Q(5m-k))$, *viewed as a space of rational sections on* T *through* ψ_M *and the birational map* $T\dashrightarrow\mathrm{Bl}_M V_5$, *onto* $H^0(T,mH-kP)$.

(5) Let $n_{m,k}:=h^0(T,mH-kP)-h^0(T,mH-(k+1)P)$. *Let* $a_{m,k}:=\lceil k/2\rceil$ *if* $k\le 4m$ *and* $a_{m,k}:=k-2m$ *if* $k>4m$, *and let*

$$a_m:=\frac{1}{mN_m}\sum_{k\ge 0}a_{m,k}n_{m,k},\qquad r_m:=\frac{1}{mN_m}\sum_{k>4m}(k-4m)n_{m,k}.$$

Then $\limsup_{m\to+\infty}a_m\le 123/88$ *and* $\lim_{m\to+\infty}r_m=1/44$.

Proof. The proof is long, so we divide it into several steps. Let $\phi: Y^+=\mathrm{Bl}_\Gamma V_5\to V_5$ be the blow-up of Theorem 2.23, let $M'\subset Y^+$ be the strict transform of M, which is the flopped curve of s (Set-up 10.10), and let E^+ be the strict transform of E on Y^+. Since $h=H-2E$ and $R=H-3E$, we have $E^+\sim h-R$. For a section τ of a line bundle on a smooth surface and a curve C on the surface, we write $\mathrm{ord}_C(\tau)$ for the order of vanishing of τ along C, and for $u\in H^0(X,mH)$ we write $\mathrm{mult}_F u$ for the coefficient of a prime divisor F over X in the pullback of $(u=0)$.

Step 1. In this step we compare multiplicities along s and M', and we compute K_T.

By Theorem 2.23, the flopping curves of $Y\dashrightarrow Y^+$ are pairwise disjoint, so s is a connected component of the flopping locus. By Set-up 10.10, $N_{s/Y}\cong\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{s/Y}\cong\mathcal{O}_{\mathbb{P}^1}\oplus\mathcal{O}_{\mathbb{P}^1}(-2)$,

and by Proposition 10.8, s satisfies the hypotheses of Definition-Proposition 2.21, so the flopping contraction maps s to an isolated singular point. Let $w(s)$ be the width of s, and let $p : W \to Y$, $q : W \to Y^+$, and $P_1, \dots, P_{w(s)}$ be as in Construction-Proposition 10.2 applied to $C = s$. By the construction in Construction-Proposition 10.2 and by its item (1), the first blow-up composing p is the blow-up of s and the first blow-up composing q is the blow-up of M', and the later blow-ups have curves as centers. Hence, for any divisor $D \geq 0$ on Y with strict transform D^+ on Y^+, the coefficient of P_1 in p^*D is $\operatorname{mult}_s D$ and the coefficient of P_1 in q^*D^+ is $\operatorname{mult}_{M'} D^+$. Comparing the coefficients of P_1 in Lemma 10.3, we obtain

$$\operatorname{mult}_s D = \operatorname{mult}_{M'} D^+ - (D \cdot s). \tag{11.29}$$

Moreover, $T_1 = \operatorname{Bl}_s Y = T$ in Construction-Proposition 10.2, so P_1 is the strict transform of P, and $\operatorname{mult}_P u = \operatorname{mult}_s(u = 0)$ for $u \in H^0(X, mH)$, where $(u = 0)$ is taken on Y.

By [KM98, Lemma 2.29], $K_Y = \pi^*K_X + E$ and $K_T = p_T^*K_Y + P$. Since E is smooth along s, $p_T^*E = E' + P$. Thus

$$K_T - (\pi p_T)^*K_X = E' + 2P,$$

and $A_X(P) = 3$.

Step 2. In this step we prove $\operatorname{mult}_s R = 2$ and $\operatorname{mult}_s(R|_E) = 2$.

We have $(E \cdot f) = -1$, $((E|_E)^2)_E = E^3 = 1$ by Lemma 11.1(1), and $H|_E \sim f$ by [IP99, Lemma 2.2.14]. Hence $-E|_E \sim s + 2f$, and

$$(H \cdot s) = 1, \quad (E \cdot s) = 1, \quad (h \cdot s) = -1, \quad (R \cdot s) = -2.$$

Since $\phi(M') = M$ and $\phi(R) = \Gamma$, $M' \not\subset R$. Applying (11.29) to $D = R$, we obtain $\operatorname{mult}_s R = 0 + 2 = 2$. Since $E \not\subset R$, the restriction $R|_E$ is a divisor on E with $R|_E \geq 0$ and $\operatorname{mult}_s(R|_E) \geq \operatorname{mult}_s R = 2$.

Suppose that $\operatorname{mult}_s(R|_E) \geq 3$. Since $R|_E \sim 3s + 7f$, we have $R|_E = 3s + \sum_i f_{x_i}$ for points $x_1, \dots, x_7 \in L$. Let $\ell \subset R$ be a general fiber of $R \to \Gamma$. Then $(E^+ \cdot \ell) = (h \cdot \ell) - (R \cdot \ell) = 1$, so $E^+ \cap \ell \neq \emptyset$. Hence $E^+ \cap R$ contains a curve C^+ with $\phi(C^+) = \Gamma$. By Theorem 2.23, the flopped curves are strict transforms of bisecant lines of Γ, so C^+ is not a flopped curve. Let $C \subset E \cap R$ be the strict transform of C^+ on Y. Then C is not a flopping curve, so $C \neq s$, and $C = f_{x_i}$ for some i. The rational map $Y \dashrightarrow V_5$ is defined by the linear system $|h|$ on Y, whose base locus is contained in the flopping locus, and it maps C onto Γ. Thus $5 = \deg \Gamma \leq (h \cdot C)$. However, $h|_E \sim 2s + 5f$ and $(h \cdot f_{x_i}) = 2$, a contradiction. Thus $\operatorname{mult}_s(R|_E) = 2$.

Step 3. In this step we prove (2).

By **Step 2**, $p_T^*R = R_T + 2P$. Since $p_T^*E = E' + P$, we obtain

$$R_T \sim H - 3(E' + P) - 2P = H - 3E' - 5P.$$

By [IP99, Lemma 2.2.14], $H|_P \sim (H \cdot s)f_P = f_P$, $(p_T^*E)|_P \sim (E \cdot s)f_P = f_P$, $(P \cdot f_P) = -1$, and $P^3 = -\deg N_{s/Y} = 2$. If $P \cong \mathbb{F}_0$, then $-P|_P \sim u_P + bf_P$ with $2b = 2$, so $-P|_P \sim u_P + f_P$. If $P \cong \mathbb{F}_2$, then $-P|_P \sim \sigma + bf_P$ with $-2 + 2b = 2$, so $-P|_P \sim \sigma + 2f_P$. Since $E'|_P = (p_T^*E)|_P - P|_P$ and $R_T|_P = (H - 3E' - 5P)|_P$, we obtain

$$E'|_P \sim u_P + 2f_P, \quad R_T|_P \sim f_P - 3(u_P + 2f_P) + 5(u_P + f_P) = 2u_P$$

if $P \cong \mathbb{F}_0$, and

$$E'|_P \sim \sigma + 3f_P, \quad R_T|_P \sim f_P - 3(\sigma + 3f_P) + 5(\sigma + 2f_P) = 2\sigma + 2f_P$$

if $P \cong \mathbb{F}_2$. Since s is a Cartier divisor on the smooth surface E, the morphism $E' \to E$ is an isomorphism, and $E' \cap P$ is the section of $P \to s$ given by the subbundle $N_{s/E} \subset N_{s/Y}$. Thus $E' \cap P$ is smooth and irreducible. Since $(E'|_P \cdot f_P) = 1$, $E'|_P = E' \cap P$.

Step 4. In this step we prove (3).

Let S_Γ be the unique hyperplane section of V_5 containing Γ. By Theorem 2.23, E^+ is the strict transform of S_Γ. Since $E^+ \sim h - R$, we have $\phi^*S_\Gamma = E^+ + R$. Applying (11.29) to $D = E$, we obtain $\operatorname{mult}_{M'} E^+ = \operatorname{mult}_s E + (E \cdot s) = 2$. Since $M' \not\subset R$ and ϕ is an isomorphism at the

generic point of M', $\operatorname{mult}_M S_\Gamma = \operatorname{mult}_{M'} \phi^* S_\Gamma = 2$. Hence S_Γ is a hyperplane section singular along M, and $S_\Gamma = S_M$ by Set-up 10.10.

Step 5. In this step we prove $S_X(P) \le 61/22$.

Let $V(t) := \operatorname{vol}(H - tE)$, given by Lemma 11.1(2). For integers $m \ge 1$ and $0 \le j \le 3m$, let $W_{m,j}$ be the image of $H^0(Y, mH - jE)$ in $H^0(E, (mH - jE)|_E)$, where $(mH - jE)|_E \sim js + (m+2j)f$, and let $c_{m,j} := \max(j - m, 0)$. For $0 \le t \le 3$ let

$$\Lambda(t) := \begin{cases} ts + (1+2t)f & \text{if } 0 \le t \le 1, \\ s + (1+2t)f & \text{if } 1 \le t \le 2, \\ (3-t)(s+5f) & \text{if } 2 \le t \le 3. \end{cases}$$

We claim that there exists an injective linear map $W_{m,j} \to H^0(E, m\Lambda(j/m))$, $\bar{u} \to \bar{u}'$, such that $\operatorname{ord}_s(\bar{u}) = c_{m,j} + \operatorname{ord}_s(\bar{u}')$. If $j \le m$, we let $\bar{u}' := \bar{u}$. If $m \le j \le 2m$, then $((mH - jE) \cdot s) = m - j$, so by (11.29) any member of $|mH - jE|$ has multiplicity at least $j - m$ along s. Thus $\bar{u}$ vanishes along s to order at least $j - m$, and we let $\bar{u}'$ be the quotient of $\bar{u}$ by the $(j-m)$-th power of a section defining $s \subset E$, so that $\bar{u}' \in H^0(E, ms + (m+2j)f) = H^0(E, m\Lambda(j/m))$. If $2m \le j \le 3m$, then by Lemma 11.1(3), $\bar{u} = r_0^{j-2m} \bar{u}_h$, where r_0 is the restriction to E of a section defining R and $\bar{u}_h$ is the restriction of an element of $H^0(Y, (3m-j)h)$. By (11.29) and $((3m-j)h \cdot s) = -(3m-j)$, $\operatorname{ord}_s(\bar{u}_h) \ge 3m - j$, and we let $\bar{u}'$ be the quotient of $\bar{u}_h$ by the $(3m-j)$-th power of a section defining s, so that $\bar{u}' \in H^0(E, (3m-j)(h|_E - s)) = H^0(E, (3m-j)(s+5f))$. Since $\operatorname{ord}_s(r_0) = 2$ by **Step 2**,

$$\operatorname{ord}_s(\bar{u}) = 2(j - 2m) + (3m - j) + \operatorname{ord}_s(\bar{u}') = c_{m,j} + \operatorname{ord}_s(\bar{u}').$$

This proves the claim.

By [AZ22, Lemma 3.1], there exists a basis $u_1, \dots, u_{N_m}$ of $H^0(X, mH)$ compatible with the filtrations by mult_E and by mult_P. Let $j_i := \operatorname{mult}_E u_i$, and let $\bar{u}_i \in W_{m,j_i}$ be the image of u_i. At a general point of s choose local coordinates (x, y, z) of Y with $E = (x = 0)$ and $s = (x = y = 0)$, and write a local function representing u_i as $x^{j_i} g$ with $g|_E$ representing $\bar{u}_i$. Since $g|_E$ vanishes along s to order $\operatorname{ord}_s(\bar{u}_i)$, g has order at most $\operatorname{ord}_s(\bar{u}_i)$ along $(x = y = 0)$, so

$$\operatorname{mult}_P u_i \le j_i + \operatorname{ord}_s(\bar{u}_i) = j_i + c_{m,j_i} + \operatorname{ord}_s(\bar{u}_i').$$

For each j, the elements $\bar{u}_i$ with $j_i = j$ form a basis of $W_{m,j}$, so the corresponding $\bar{u}_i'$ are linearly independent in $H^0(E, m\Lambda(j/m))$, and the number of such i with $\operatorname{ord}_s(\bar{u}_i') \ge l$ is at most $h^0(E, m\Lambda(j/m) - ls)$ for $l \ge 1$. Since $\dim W_{m,j} = h^0(Y, mH - jE) - h^0(Y, mH - (j+1)E)$, we have $\sum_j c_{m,j} \dim W_{m,j} = \sum_{j>m} h^0(Y, mH - jE)$. Summing over i, we obtain

$$S_m(P) \le S_m(E) + \frac{1}{mN_m} \sum_{j>m} h^0(Y, mH - jE) + \frac{1}{mN_m} \sum_{j=0}^{3m} \sum_{l \ge 1} h^0(E, m\Lambda(j/m) - ls).$$

The last two terms are Riemann sums with errors $O(1/m)$. By $S_X(F) = \lim_m S_m(F)$ [AZ22, Lemma 2.9], we obtain

$$S_X(P) \le S_X(E) + \frac{1}{22} \int_1^3 V(t)dt + \frac{3}{22} \int_0^3 \int_0^\infty \operatorname{vol}_E(\Lambda(t) - vs)dvdt.$$

By Lemma 11.1(2), $S_X(E) = 17/11$ and

$$\int_1^3 V(t)dt = \int_1^2 \big(18 - 9(t-1) - 6(t-1)^2 + 2(t-1)^3\big)dt + \int_2^3 5(3-t)^3 dt = 12 + \frac{5}{4} = \frac{53}{4}.$$

For integers $0 \le 3x \le y$, the class $(x - v)s + yf$ on $E \cong \mathbb{F}_3$ is nef for $0 \le v \le x$ and is not pseudo-effective for $v > x$ [Har77, Proposition V.2.20]. Thus, by continuity in x and y,

$$\int_0^\infty \operatorname{vol}_E(xs + yf - vs)dv = \int_0^x \big(2y(x-v) - 3(x-v)^2\big)dv = x^2 y - x^3$$

for real $0 \le 3x \le y$. For $\Lambda(t)$ the integral equals $t^2 + t^3$, $2t$, and $4(3-t)^3$ on the three intervals, whose integrals over t are $7/12$, 3, and 1. Hence

$$S_X(P) \le \frac{17}{11} + \frac{53}{88} + \frac{3}{22} \cdot \frac{55}{12} = \frac{136}{88} + \frac{53}{88} + \frac{55}{88} = \frac{61}{22}.$$

Step 6. In this step we prove (4).

First, let $o \in L$ be a general point. Choose local analytic coordinates (z, y_1, y_2) of X at o such that $L = (y_1 = y_2 = 0)$ and s is the section of $E \to L$ given by the normal direction of the surface $(y_2 = 0)$ along L. In the chart $y_1 = x_1$, $y_2 = x_1 y$ of Y, we have $E = (x_1 = 0)$ and $s = (x_1 = y = 0)$, and the monomial $y_1^a y_2^b$ pulls back to $x_1^{a+b} y^b$. Hence, for a local function g at o, $\operatorname{mult}_E g$ is the minimum of $a + b$ and $\operatorname{mult}_P g$ is the minimum of $a + 2b$ over the monomials $y_1^a y_2^b$ occurring in g. Thus $\operatorname{mult}_P \le 2 \operatorname{mult}_E$, and any $u \in H^0(T, mH - kP)$ satisfies $\operatorname{mult}_{E'} u = \operatorname{mult}_E u \ge \lceil k/2 \rceil$.

Next, by Theorem 2.23, $\pi^* H = 3h - 2R$, and Y and Y^+ are isomorphic in codimension one. Since $\phi_* \mathcal{O}_{Y^+}(-2mR) = I_\Gamma^{2m}$, we obtain

$$H^0(X, mH) \cong H^0(Y^+, 3mh - 2mR) \cong H^0(V_5, \mathcal{O}_{V_5}(3m) \otimes I_\Gamma^{2m}).$$

For $u \in H^0(X, mH)$ let $\Delta_u \in |\mathcal{O}_{V_5}(3m)|$ be the divisor of the corresponding section. Then the divisor of u on Y^+ is $\phi^* \Delta_u - 2mR$. Since $(mH \cdot s) = m$, $M' \not\subset R$, and ϕ is an isomorphism at the generic point of M', (11.29) implies

$$\operatorname{mult}_P u = \operatorname{mult}_M \Delta_u - m.$$

Thus $H^0(T, mH - kP)$ is identified with the space of sections of $\mathcal{O}_{V_5}(3m) \otimes I_\Gamma^{2m}$ vanishing to order at least $m + k$ along M.

Let $J \subset \operatorname{Bl}_M V_5$ be the exceptional divisor of ψ_M. By Set-up 10.10 and [Fuj26a, Theorem 3.6(4)–(6)], ψ_M is the blow-up of Q along a twisted cubic, $h - E_M = \psi_M^* \mathcal{O}_Q(1)$, $J \sim h - 2E_M$, and $\phi_M(J) = S_M$. Since J is ψ_M-exceptional and Q is normal, $\psi_{M*} \mathcal{O}(aJ) = \mathcal{O}_Q$ for $a \ge 0$ [Har77, Proposition II.6.3A]. Let $k \ge 2m$. Since

$$3mh - (m+k)E_M = (5m - k)(h - E_M) + (k - 2m)J,$$

the projection formula implies that multiplication by the $(k-2m)$-th power of a section defining J is an isomorphism

$$H^0(Q, \mathcal{O}_Q(5m - k)) \to H^0(\operatorname{Bl}_M V_5, 3mh - (m+k)E_M) = H^0(V_5, \mathcal{O}_{V_5}(3m) \otimes I_M^{m+k}).$$

In particular, both spaces are zero if $k > 5m$, and any section of $\mathcal{O}_{V_5}(3m) \otimes I_M^{m+k}$ is the product of the $(k-2m)$-th power of a linear form defining S_M with a section of $\mathcal{O}_{V_5}(5m - k)$. If moreover $k \ge 4m$, then $k - 2m \ge 2m$, and since $\Gamma \subset S_M$ by (3), the product is an element of $H^0(V_5, \mathcal{O}_{V_5}(3m) \otimes I_\Gamma^{2m})$. Hence, for $4m \le k$,

$$H^0(T, mH - kP) \cong H^0(V_5, \mathcal{O}_{V_5}(3m) \otimes I_M^{m+k}) \cong H^0(Q, \mathcal{O}_Q(5m - k)),$$

and for $u \in H^0(T, mH - kP)$ we have $\Delta_u \ge (k - 2m) S_M$. Since $\phi^* S_M = E^+ + R$ by **Step 4**, the divisor of u on Y^+ satisfies

$$\phi^* \Delta_u - 2mR \ge (k - 2m)E^+ + (k - 4m)R.$$

Since $T \dashrightarrow Y^+$ is an isomorphism at the generic points of E' and R_T, u vanishes along E' to order at least $k - 2m$ and along R_T to order at least $k - 4m$, and the factor $(k - 2m)S_M$ corresponds to the factor $e'^{k-2m} r_T^{k-4m}$ on T. This proves the isomorphism of (4) and $H^0(T, mH - kP) = 0$ for $k > 5m$. Finally, by (2),

$$(k - 2m)E' + (k - 4m)(H - 3E' - 5P) + (5m - k)(H - 2E' - 4P) = mH - kP.$$

Step 7. In this step we prove (5).

By (4), for $4m \le k \le 5m$,

$$n_{m,k} = h^0(Q, \mathcal{O}_Q(5m - k)) - h^0(Q, \mathcal{O}_Q(5m - k - 1)) = (5m - k + 1)^2,$$

where the last equality follows from $h^0(Q, \mathcal{O}_Q(q)) = \binom{q+4}{4} - \binom{q+2}{4}$ for $q \geq 0$ [Har77, Theorem III.5.1]. Since $N_m = 22m^3/6 + O(m^2)$, we obtain

$$\lim_{m\to+\infty} r_m = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{i=0}^{m} i(m-i+1)^2 = \frac{6}{22} \int_0^1 x(1-x)^2 dx = \frac{1}{44}.$$

For any k, $a_{m,k} \leq \frac{k}{2} + \frac{1}{2}\max(k-4m, 0) + \frac{1}{2}$. Since $\sum_k n_{m,k} = N_m$ and $\sum_k k n_{m,k} = mN_m S_m(P)$, we obtain

$$a_m \leq \frac{1}{2} S_m(P) + \frac{1}{2} r_m + \frac{1}{2m}.$$

By [AZ22, Lemma 2.9] and (1),

$$\limsup_{m\to+\infty} a_m \leq \frac{1}{2} \cdot \frac{61}{22} + \frac{1}{88} = \frac{123}{88}. \quad \square$$

Since $K_T = (\pi p_T)^* K_X + E' + 2P$ (Lemma 11.7(1)), the $\mathbb{R}$-divisors D_m and Ψ_m of Definition-Lemma 11.8 below satisfy

$$(\pi p_T)^*(K_X + D_m) = K_T + (S_m(P) - 2)P + (a_m - 1)E' + r_m R_T + \Psi_m.$$

The boundary Θ_m is the part of the right-hand side other than K_T and the multiple of P, with the coefficient $a_m - 1$ of E' replaced by $\max(a_m - 1, 0)$, so that $\Theta_m \geq 0$ by Definition-Lemma 11.8(1) and inversion of adjunction along P can be applied.

Definition-Lemma 11.8. Notation as in Set-up 10.10, with $a_{m,k}$, $n_{m,k}$, a_m, and r_m defined in Lemma 11.7(5). Let $C_0 := E' \cap P$. For $0 \leq t \leq 5$ let $\Lambda_P(t)$ be the $\mathbb{R}$-divisor class on P given by

$$\Lambda_P(t) := \begin{cases} (H - tP - \frac{t}{2}E')|_P & \text{if } 0 \leq t \leq 4, \\ (5-t)(H - 2E' - 4P)|_P & \text{if } 4 \leq t \leq 5, \end{cases}$$

and for a divisor D on P let

$$I_P(D) := \frac{3}{22} \int_0^5 \int_0^\infty \mathrm{vol}_P(\Lambda_P(t) - vD) dv dt.$$

Let $C \subset P$ be an irreducible curve with $p_T(C) = s$, and let F be a prime divisor over X with $\mathrm{center}_T F = C$. For any integer $m \geq 1$ let $u_1, \ldots, u_{N_m}$ be a basis of $H^0(X, mH)$ compatible with the filtrations by mult_P and by mult_F (such a basis exists by [AZ22, Lemma 3.1]), let

$$D_m := \frac{1}{mN_m} \sum_{i=1}^{N_m} (u_i = 0)$$

be the associated m-basis type divisor (Definition 2.8), and define Ψ_m by

$$(\pi p_T)^* D_m = S_m(P)P + a_m E' + r_m R_T + \Psi_m,$$

and let $\Theta_m := \max(a_m - 1, 0)E' + r_m R_T + \Psi_m$. Then the following hold.

(1) $\Psi_m \geq 0$ and $P \not\subset \mathrm{Supp}\, \Psi_m$.
(2) If D is a divisor on P such that $C - D$ is linearly equivalent to an effective divisor, then $\limsup_{m\to+\infty} \mathrm{mult}_C(\Psi_m|_P) \leq I_P(D)$.
(3) If $\limsup_{m\to+\infty} \mathrm{mult}_C(\Theta_m|_P) < 1$, then

$$A_X(F) - S_X(F) \geq (3 - S_X(P)) \,\mathrm{mult}_F P.$$

Proof. Since the basis is compatible with the filtrations by mult_P and by mult_F, we have $\mathrm{mult}_P D_m = S_m(P)$ and $\mathrm{mult}_F D_m = S_m(F)$ by Definition 2.8, and the number of i with $\mathrm{mult}_P u_i = k$ is $n_{m,k}$. Let $k_i := \mathrm{mult}_P u_i$.

(1) By Lemma 11.7(4),

$$\Psi(u_i) := (\pi p_T)^*(u_i = 0) - k_i P - a_{m,k_i} E' - \max(k_i - 4m, 0) R_T \geq 0,$$

and $P \not\subset \operatorname{Supp}\Psi(u_i)$. Since $\sum_i k_i = mN_mS_m(P)$, $\sum_i a_{m,k_i} = mN_m a_m$, and $\sum_i \max(k_i-4m,0) = mN_m r_m$, we have $\Psi_m = \frac{1}{mN_m}\sum_i \Psi(u_i)$. This proves (1).

(2) For an integer $k \geq 0$ let

$$D_{m,k} := \big(mH - kP - a_{m,k}E' - \max(k-4m,0)R_T\big)|_P.$$

Then $\Psi(u_i)|_P = (\tau_i = 0)$ for a section $\tau_i \in H^0(P, D_{m,k_i})$, which is the image of u_i in $H^0(P, (mH - k_iP)|_P)$ divided by a section defining $(a_{m,k_i}E' + \max(k_i-4m,0)R_T)|_P$. By compatibility of the basis with mult_P, for each k the sections τ_i with $k_i = k$ are linearly independent. Since $C - D$ is linearly equivalent to an effective divisor, the number of i with $k_i = k$ and $\operatorname{ord}_C(\tau_i) \geq l$ is at most $h^0(P, D_{m,k} - lC) \leq h^0(P, D_{m,k} - lD)$ for $l \geq 1$. Hence

$$\operatorname{mult}_C(\Psi_m|_P) = \frac{1}{mN_m}\sum_i \operatorname{ord}_C(\tau_i) \leq \frac{1}{mN_m}\sum_{k=0}^{5m}\sum_{l\geq 1} h^0(P, D_{m,k} - lD).$$

By Lemma 11.7(4), $D_{m,k} = (5m-k)(H-2E'-4P)|_P$ for $4m \leq k \leq 5m$, and $D_{m,k} - m\Lambda_P(k/m) = (\frac{k}{2} - \lceil \frac{k}{2} \rceil)E'|_P$ for $k \leq 4m$. By the formula $h^0(\mathbb{F}_n, xs_n + yf_n) = \sum_{i=0}^{x}\max(y - ni + 1, 0)$ for integers $x \geq 0$ and y, where s_n is the negative section and f_n a fiber [Har77, Exercise III.8.4(a) and Propositions V.2.8 and V.2.9], the right-hand side is a Riemann sum for $I_P(D)$ with error $O(1/m)$. This proves (2).

(3) Fix a real number $c < 1$ and an integer m such that $\operatorname{mult}_C(\Theta_m|_P) < c$. Since $P \not\subset \operatorname{Supp}\Theta_m$, $\Theta_m|_P \geq 0$ is an $\mathbb{R}$-divisor on P. Let $U \subset T$ be the complement of the singular locus of C and of the components of $\Theta_m|_P$ different from C. Then $U \cap C \neq \emptyset$, and $(P \cap U, \Theta_m|_{P\cap U})$ is klt. Since klt is an open condition, there exists a $\mathbb{Q}$-divisor $\Theta' \geq \Theta_m$ on T with $\operatorname{Supp}\Theta' = \operatorname{Supp}\Theta_m$ such that $(P \cap U, \Theta'|_{P\cap U})$ is klt. Since T is smooth, by [KM98, Theorem 5.50(1)], $(U, P + \Theta'|_U)$ is plt near $P \cap U$, and so is $(U, P + \Theta_m|_U)$ since $\Theta_m \leq \Theta'$. Since $\operatorname{center}_T F = C$ intersects U and F is exceptional over T,

$$A_T(F) - \operatorname{mult}_F P - \operatorname{mult}_F \Theta_m > 0.$$

By Lemma 11.7(1), $A_X(F) = A_T(F) + \operatorname{mult}_F E' + 2\operatorname{mult}_F P$. Since $a_m E' \leq E' + \max(a_m - 1, 0)E'$, we have $(\pi p_T)^* D_m \leq S_m(P)P + E' + \Theta_m$. Thus

$$S_m(F) = \operatorname{mult}_F(\pi p_T)^* D_m \leq S_m(P)\operatorname{mult}_F P + \operatorname{mult}_F E' + \operatorname{mult}_F \Theta_m,$$

and

$$\begin{aligned} A_X(F) - S_m(F) &> A_T(F) + \operatorname{mult}_F E' + 2\operatorname{mult}_F P - S_m(P)\operatorname{mult}_F P \\ &\quad - \operatorname{mult}_F E' - A_T(F) + \operatorname{mult}_F P = (3 - S_m(P))\operatorname{mult}_F P. \end{aligned}$$

Since the inequality holds for any sufficiently large m, [AZ22, Lemma 2.9] implies $A_X(F) - S_X(F) \geq (3 - S_X(P))\operatorname{mult}_F P$. □

The values of $I_P(D)$ needed below are listed in Table 13.

Lemma 11.9. *Notation as in Definition-Lemma 11.8. Then $\Lambda_P(t)$, the integral $\int_0^\infty \operatorname{vol}_P(\Lambda_P(t) - vD)dv$, and $I_P(D)$ are as in Table 13. In particular, if $P \cong \mathbb{F}_0$, then $I_P(u_P) = 19/22$ and $I_P(C_0) = 23/88$, and if $P \cong \mathbb{F}_2$, then $I_P(\sigma + 2f_P) = 181/352$.*

Proof of Lemma 11.9. By Lemma 11.7(2),

$$(H - tP - \frac{t}{2}E')|_P \sim \begin{cases} f_P + t(u_P + f_P) - \frac{t}{2}(u_P + 2f_P) = \frac{t}{2}u_P + f_P & \text{if } P \cong \mathbb{F}_0, \\ f_P + t(\sigma + 2f_P) - \frac{t}{2}(\sigma + 3f_P) = \frac{t}{2}\sigma + (1 + \frac{t}{2})f_P & \text{if } P \cong \mathbb{F}_2, \end{cases}$$

and $(H - 2E' - 4P)|_P$ equals $2u_P + f_P$ and $2\sigma + 3f_P$, respectively. Also $C_0 \sim u_P + 2f_P$ if $P \cong \mathbb{F}_0$.

On $\mathbb{F}_n$ with negative section s_n and fiber f_n, a class $xs_n + yf_n$ is pseudo-effective if and only if $x, y \geq 0$, and nef if and only if $x \geq 0$ and $y \geq nx$ [Har77, Proposition V.2.20]. We have $\operatorname{vol}(N) = N^2$ for a nef class N, and $\operatorname{vol}(N_+ + N_-) = N_+^2$ for a pseudo-effective class with Zariski decomposition $N_+ + N_-$, for rational classes by [Laz04a, Corollary 1.4.41] and [Laz04a,

TABLE 13. The classes $\Lambda_P(t)$ and the integrals defining $I_P(D)$ in Lemma 11.9. The sixth column is the integral of the fifth column over the interval of t in the third column, and $I_P(D)$ is $\frac{3}{22}$ times the sum of the sixth column over the rows of D.

P	D	t	$\Lambda_P(t)$	$\int_0^\infty \mathrm{vol}_P(\Lambda_P(t)-vD)dv$	$\int\cdot dt$	$I_P(D)$
$\mathbb{F}_0$	u_P	$[0,4]$	$\frac{t}{2}u_P+f_P$	$\frac{t^2}{4}$	$\frac{16}{3}$	$\frac{19}{22}$
		$[4,5]$	$(5-t)(2u_P+f_P)$	$4(5-t)^3$	1	
$\mathbb{F}_0$	C_0	$[0,1]$	$\frac{t}{2}u_P+f_P$	$\frac{t^2}{4}-\frac{t^3}{12}$	$\frac{1}{16}$	$\frac{23}{88}$
		$[1,4]$	$\frac{t}{2}u_P+f_P$	$\frac{t}{4}-\frac{1}{12}$	$\frac{13}{8}$	
		$[4,5]$	$(5-t)(2u_P+f_P)$	$\frac{11}{12}(5-t)^3$	$\frac{11}{48}$	
$\mathbb{F}_2$	$\sigma+2f_P$	$[0,2]$	$\frac{t}{2}\sigma+(1+\frac{t}{2})f_P$	$\frac{t^2}{4}-\frac{t^3}{24}$	$\frac{1}{2}$	$\frac{181}{352}$
		$[2,4]$	$\frac{t}{2}\sigma+(1+\frac{t}{2})f_P$	$\frac{(2+t)^3}{96}$	$\frac{65}{24}$	
		$[4,5]$	$(5-t)(2\sigma+3f_P)$	$\frac{9}{4}(5-t)^3$	$\frac{9}{16}$	

Theorem 2.3.19 and Corollary 2.3.22], and for real classes by continuity of the volume [Laz04a, Corollary 2.2.45]. Let $x,y\geq 0$.

(1) On $\mathbb{F}_0$, $(x-v)u_P+yf_P$ is nef for $v\leq x$ and not pseudo-effective for $v>x$, so

$$\int_0^\infty \mathrm{vol}(xu_P+yf_P-vu_P)dv=\int_0^x 2(x-v)ydv=x^2y.$$

(2) On $\mathbb{F}_0$, $(x-v)u_P+(y-2v)f_P$ is nef for $v\leq \min(x,y/2)$ and not pseudo-effective for $v>\min(x,y/2)$, so

$$\int_0^\infty \mathrm{vol}(xu_P+yf_P-vC_0)dv=\int_0^{\min(x,y/2)} 2(x-v)(y-2v)dv=\begin{cases}x^2y-\frac{2}{3}x^3 & \text{if } x\leq y/2,\\ \frac{1}{2}xy^2-\frac{1}{12}y^3 & \text{if } x\geq y/2.\end{cases}$$

(3) On $\mathbb{F}_2$, consider $(x-v)\sigma+(y-2v)f_P$. If $y\geq 2x$, then $(x-v)\sigma+(y-2v)f_P$ is nef for $v\leq x$ with volume $2(x-v)(y-x-v)$, and not pseudo-effective for $v>x$. If $y\leq 2x$, then for $v\leq y/2$ the positive part of the Zariski decomposition of $(x-v)\sigma+(y-2v)f_P$ is $\frac{y-2v}{2}(\sigma+2f_P)$, with self-intersection $\frac{1}{2}(y-2v)^2$, and for $v>y/2$ the class $(x-v)\sigma+(y-2v)f_P$ is not pseudo-effective. Thus

$$\int_0^\infty \mathrm{vol}(x\sigma+yf_P-v(\sigma+2f_P))dv=\begin{cases}x^2y-\frac{4}{3}x^3 & \text{if } y\geq 2x,\\ \frac{1}{12}y^3 & \text{if } y\leq 2x.\end{cases}$$

Substituting $\Lambda_P(t)$ into (1)–(3), we obtain the fifth column of Table 13: for $\frac{t}{2}\sigma+(1+\frac{t}{2})f_P$ we have $y\geq 2x$ if and only if $t\leq 2$, for $\frac{t}{2}u_P+f_P$ we have $x\leq y/2$ if and only if $t\leq 1$, and the classes on $[4,5]$ satisfy $x\geq y/2$ and $y\leq 2x$, respectively. Integrating over t and multiplying the sums $19/3$, $23/12$, and $181/48$ by $3/22$, we obtain the last column. □

Proposition 11.10. *Notation as in Set-up 10.10. Assume that $N_{s/Y}\cong\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Then any prime divisor F over X with $\mathrm{center}_Y F=s$ satisfies*

$$A_X(F)-S_X(F)\geq\frac{5}{22}\mathrm{mult}_F P>0.$$

Hence $\beta_X(F')>0$ for any prime divisor F' over X with $\mathrm{center}_X F'=L$.

Proof. By Lemma 11.7(1), $3-S_X(P)\geq 3-61/22=5/22$. If $F=P$, the inequality follows from $A_X(P)=3$. Otherwise, since $p_T(\mathrm{center}_T F)=s$ and $F\neq P$, $C:=\mathrm{center}_T F$ is an irreducible

curve in P with $p_T(C) = s$, and $\operatorname{mult}_F P > 0$. By Definition-Lemma 11.8(3), it suffices to show that $\limsup_m \operatorname{mult}_C(\Theta_m|_P) < 1$ for any choice of the bases.

Here $P \cong \mathbb{F}_0$, and $C \sim xu_P + yf_P$ with $x = (C \cdot f_P) \geq 1$ and $y \geq 0$. Recall that $E'|_P = C_0 \sim u_P + 2f_P$ and $R_T|_P \sim 2u_P$ (Lemma 11.7(2)), and that $\limsup_m a_m \leq 123/88$ and $\lim_m r_m = 1/44$ (Lemma 11.7(5)).

Suppose that $C \neq C_0$. Then $\operatorname{mult}_C(E'|_P) = 0$. Since $C - u_P \sim (x-1)u_P + yf_P$ is effective, Definition-Lemma 11.8(2) and Lemma 11.9 imply $\limsup_m \operatorname{mult}_C(\Psi_m|_P) \leq I_P(u_P) = 19/22$. If $c := \operatorname{mult}_C(R_T|_P)$, then $2u_P - cC \sim (2-cx)u_P - cyf_P$ is effective, so $c \leq 2$. Thus

$$\limsup_{m\to+\infty} \operatorname{mult}_C(\Theta_m|_P) \leq \frac{19}{22} + 2 \cdot \frac{1}{44} = \frac{10}{11} < 1.$$

Suppose that $C = C_0$. Then $\limsup_m \operatorname{mult}_C(\Psi_m|_P) \leq I_P(C_0) = 23/88$. Since $2u_P - C_0 \sim u_P - 2f_P$ is not effective, $\operatorname{mult}_{C_0}(R_T|_P) = 0$. Since $\operatorname{mult}_{C_0}(E'|_P) = 1$,

$$\limsup_{m\to+\infty} \operatorname{mult}_{C_0}(\Theta_m|_P) \leq \left(\frac{123}{88} - 1\right) + \frac{23}{88} = \frac{58}{88} < 1.$$

This proves the inequality.

Let F' be a prime divisor over X with $\operatorname{center}_X F' = L$. If $\operatorname{center}_Y F' \neq s$, then $\beta_X(F') > 0$ by Proposition 11.2. If $\operatorname{center}_Y F' = s$, then $\beta_X(F') \geq \frac{5}{22} \operatorname{mult}_{F'} P > 0$. □

Proposition 11.11. *Notation as in Set-up 10.10. Assume that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Then the width of s is 2 or 3, and any prime divisor F over X with $\operatorname{center}_Y F = s$ such that $F = P$ or $\operatorname{center}_T F \neq \sigma$ satisfies*

$$A_X(F) - S_X(F) \geq \frac{5}{22} \operatorname{mult}_F P > 0.$$

Proof. Let $p : W \to Y$, $q : W \to Y^+$, and $P_1, \ldots, P_{w(s)}$ be as in Construction-Proposition 10.2 applied to $C = s$. By Proposition 10.8, the widths of the flopping curves of $Y \dashrightarrow Y^+$ sum to three, so $w(s) \leq 3$. If $w(s) = 1$, then $W = T$ and $P = P_1 \cong \mathbb{F}_0$ by Construction-Proposition 10.2(1). Since $P \cong \mathbb{F}_2$, $w(s) \geq 2$. Thus $w(s) \in \{2, 3\}$.

By Lemma 11.7(1), $3 - S_X(P) \geq 5/22$. If $F = P$, the inequality follows from $A_X(P) = 3$. Otherwise $C := \operatorname{center}_T F$ is an irreducible curve in P with $p_T(C) = s$ and $C \neq \sigma$, and $\operatorname{mult}_F P > 0$. By Definition-Lemma 11.8(3), it suffices to show that $\limsup_m \operatorname{mult}_C(\Theta_m|_P) < 1$ for any choice of the bases.

Write $C \sim x\sigma + yf_P$. Then $x = (C \cdot f_P) \geq 1$, and $(C \cdot \sigma) = y - 2x \geq 0$ since $C \neq \sigma$. Hence $C - (\sigma + 2f_P) \sim (x-1)(\sigma + 2f_P) + (y-2x)f_P$ is effective, and Definition-Lemma 11.8(2) and Lemma 11.9 imply

$$\limsup_{m\to+\infty} \operatorname{mult}_C(\Psi_m|_P) \leq I_P(\sigma + 2f_P) = \frac{181}{352}.$$

By Lemma 11.7(2), $E'|_P = C_0 \sim \sigma + 3f_P$ and $R_T|_P \sim 2\sigma + 2f_P$. If $c := \operatorname{mult}_C(R_T|_P)$, then $2\sigma + 2f_P - cC \sim (2-cx)\sigma + (2-cy)f_P$ is effective. Since $y \geq 2x \geq 2$, we obtain $c \leq 1$.

Suppose that $C \neq C_0$. Then $\operatorname{mult}_C(E'|_P) = 0$, and by Lemma 11.7(5),

$$\limsup_{m\to+\infty} \operatorname{mult}_C(\Theta_m|_P) \leq \frac{181}{352} + \frac{1}{44} = \frac{189}{352} < 1.$$

Suppose that $C = C_0$. Since $2\sigma + 2f_P - C_0 \sim \sigma - f_P$ is not effective, $\operatorname{mult}_{C_0}(R_T|_P) = 0$. Since $\operatorname{mult}_{C_0}(E'|_P) = 1$ and $\limsup_m a_m \leq 123/88$ by Lemma 11.7(5),

$$\limsup_{m\to+\infty} \operatorname{mult}_{C_0}(\Theta_m|_P) \leq \left(\frac{123}{88} - 1\right) + \frac{181}{352} = \frac{321}{352} < 1.$$

This proves the inequality. □

It remains to treat the prime divisors over X whose center on T is the negative section σ of $P \cong \mathbb{F}_2$. If $M \cap \Gamma$ consists of two distinct points, then the upper bounds for $n_{m,k}$ in **Step 2** of the proof below sum to $\frac{11}{3}m^3 + O(m^2) = N_m + O(m^2)$, so they are asymptotically sharp, and

$S_X(P) = 117/44$ is an exact value (Proposition 11.12), and the estimate along σ has to keep the conditions at both points of $M \cap \Gamma$ (Remark 11.15). When $M \cap \Gamma$ is one point of length two, we do not prove such a local estimate, and this case is treated in Section 12.

Proposition 11.12. *Notation as in Set-up 10.10. Assume that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and that $M \cap \Gamma$ consists of two distinct points. Then $S_X(P) = 117/44$, and every prime divisor F over X with $\operatorname{center}_T F = \sigma$ satisfies*

$$A_X(F) - S_X(F) \geq \frac{15}{44} \operatorname{mult}_F P > 0.$$

Hence $\beta_X(F') > 0$ for every prime divisor F' over X with $\operatorname{center}_X F' = L$.

Proof. The proof is long, so we divide it into several steps. Recall that

$$K_T = (\pi p_T)^* K_X + E' + 2P, \qquad R_T \sim H - 3E' - 5P,$$

$$H|_P \sim f_P, \quad -P|_P \sim \sigma + 2f_P, \quad E'|_P \sim \sigma + 3f_P, \quad R_T|_P \sim 2\sigma + 2f_P$$

(Lemma 11.7(1)(2)), and that $(\sigma^2) = -2$, $(\sigma \cdot f_P) = 1$, and $(f_P^2) = 0$ on $P \cong \mathbb{F}_2$. Let $H_Q := H - 2E' - 4P$. Since $-K_X = H$, we have $-K_T = H - E' - 2P$, and

$$\begin{aligned} -K_T|_P &\sim f_P - (\sigma + 3f_P) + 2(\sigma + 2f_P) = \sigma + 2f_P, \\ H_Q|_P &\sim f_P - 2(\sigma + 3f_P) + 4(\sigma + 2f_P) = 2\sigma + 3f_P. \end{aligned} \tag{11.30}$$

In particular $(-K_T \cdot \sigma) = ((\sigma + 2f_P) \cdot \sigma) = 0$ and $(H_Q \cdot \sigma) = ((2\sigma + 3f_P) \cdot \sigma) = -1$.

Step 1. In this step, we compare T with the blow-up of $Y^+ = \operatorname{Bl}_\Gamma V_5$ along the strict transform of M and with the blow-up of $\operatorname{Bl}_M V_5$ along the strict transform of Γ. We show that there are points $p_1, p_2 \in P$ on distinct fibers of $P \to s$, at most one of which is a point of σ, and an integer $c \geq 0$, such that for integers $a, b \geq 0$ and any divisor $D \geq 0$ on T with $P \not\subset \operatorname{Supp} D$ and $D \sim a(-K_T) + bH_Q$ or $D \sim a(-K_T) + bH_Q + P$, we have $D|_P \geq (b-c)\sigma$ and $D|_P - (b-c)\sigma$ has multiplicity at least $b - c$ at p_1 and p_2, with $c = 0$ in the first case (Claim 11.13).

The blow-up of Y^+. Let $M' \subset Y^+$ be the strict transform of M. It is the flopped curve of s (Set-up 10.10). Let $T_a := \operatorname{Bl}_{M'} Y^+$ with exceptional divisor P_a. Let $W \to Y$ and $W \to Y^+$ be Reid's pagoda resolution of the flop $Y \dashrightarrow Y^+$ over a neighborhood of s, and let $P_1, P_2, \dots$ be the strict transforms on W of the exceptional divisors of the blow-ups composing $W \to Y$, in the order of the blow-ups. The width of s is 2 or 3 (Proposition 11.11). By the construction in Construction-Proposition 10.2 and by its item (1), $W \to Y$ factors through a morphism $p_W : W \to T$, $W \to Y^+$ factors through a morphism $q_W : W \to T_a$, and P_1 is the strict transform of P and of P_a. Over a neighborhood of P, the birational map $T \dashrightarrow T_a$ is the flop of σ, the morphisms p_W and q_W form Reid's pagoda resolution of the flop of σ, the width of σ is the width of s minus one, and $P_2, P_3, \dots$ are the exceptional divisors of p_W, with $\operatorname{center}_T P_2 = \sigma$ [BO95, Section 3]. Thus σ satisfies the hypotheses of Definition-Proposition 2.21, and we can use Lemma 10.3 for the flop $T \dashrightarrow T_a$ of σ over a neighborhood of P. Since σ is a Cartier divisor on P, $P_1 \cap P_2$ is the section of $P_2 \cong \mathbb{P}(N_{\sigma/T}) \to \sigma$ corresponding to the subbundle $N_{\sigma/P} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$ of $N_{\sigma/T}$. If σ has width one, then there is no P_3. If σ has width two, then $N_{\sigma/T} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ by Definition-Proposition 2.21, so $P_2 \cong \mathbb{F}_2$, and

$$((P_1 \cap P_2)^2)_{P_2} = \deg N_{\sigma/T} - 2 \deg N_{\sigma/P} = -2 + 4 = 2.$$

Hence $P_1 \cap P_2$ is disjoint from the negative section of P_2, which is the center of the next blow-up, and $P_1 \cap P_3 = \emptyset$. Since the width of s is at most 3, the width of σ is at most 2, and P_3 is the last of the divisors P_i. Hence $P_1 \cap P_i = \emptyset$ for $i \geq 3$. Since σ is a Cartier divisor on P, the morphisms p_W and q_W restrict to isomorphisms $P_1 \to P$ and $P_1 \to P_a$, and $P_2|_{P_1} = \sigma$ under the first isomorphism. Let $\iota : P \to P_a$ be the induced isomorphism. We write σ, f_P, and P also for $\iota(\sigma)$, the class $\iota_* f_P$, and P_a when we restrict divisors to P_a. Since $\mathbb{F}_2$ has only one ruling, ι maps the fibers of $P \to s$ to the fibers of $P_a \to M'$.

For a divisor D on T, we write D_a and D_b for the strict transforms of D on T_a and on the threefold T_b defined below. Since $(H_Q \cdot \sigma) = -1$ and $(-K_T \cdot \sigma) = 0$, Lemma 10.3 applied to $T \dashrightarrow T_a$ implies that, over a neighborhood of P,

$$p_W^* H_Q - q_W^* H_{Q,a} = \sum_{i \geq 2} (i-1) P_i, \qquad p_W^* K_T = q_W^* K_{T_a}. \tag{11.31}$$

Restricting the first equality to P_1, we obtain

$$H_{Q,a}|_{P_a} \sim (2\sigma + 3f_P) - \sigma = \sigma + 3f_P. \tag{11.32}$$

The blow-up of $\mathrm{Bl}_M V_5$ *and two Atiyah flops.* Let $\Gamma' \subset \mathrm{Bl}_M V_5$ be the strict transform of Γ, which is isomorphic to Γ since Γ is a smooth curve. Let $T_b := \mathrm{Bl}_{\Gamma'} \mathrm{Bl}_M V_5$, and let P_b be the strict transform of E_M. Since $\Gamma' \not\subset E_M$, the divisor P_b is the pullback of E_M. Let x_1, x_2 be the points of $M \cap \Gamma$. Since $M \cap \Gamma$ is reduced, M and Γ intersect transversally at x_1 and x_2. The threefold T_a is obtained by blowing up Γ and then the strict transform of M, and T_b by blowing up M and then the strict transform of Γ. Hence $T_b \dashrightarrow T_a$ is an isomorphism over $V_5 \backslash \{x_1, x_2\}$, and over a neighborhood of each x_i, where M and Γ meet transversally at one point, it is the flop of the strict transform ℓ_i of the fiber of $E_M \to M$ over x_i [Fuj26a, Example 2.10]. In particular P_b is the strict transform of P_a.

We compute near x_i. Choose local coordinates z_1, z_2, z_3 at x_i such that $M = (z_2 = z_3 = 0)$ and $\Gamma = (z_1 = z_2 = 0)$. On the chart of $\mathrm{Bl}_M V_5$ with coordinates z_1, z_2', z_3 and $z_2 = z_2' z_3$, we have $E_M = (z_3 = 0)$, the fiber of E_M over x_i is $(z_1 = z_3 = 0)$, and $\Gamma' = (z_1 = z_2' = 0)$. Thus Γ' intersects E_M transversally at the point o_i with coordinates $(0,0,0)$, which is a point of the fiber over x_i, and Γ' does not intersect the fiber elsewhere. Hence $P_b \to E_M$ is the blow-up of o_i near o_i, and

$$(\ell_i^2)_{P_b} = -1, \qquad (P_b \cdot \ell_i) = (E_M \cdot \ell_i) = -1.$$

Hence N_{ℓ_i/T_b} is an extension of $\mathcal{O}_{\ell_i}(-1)$ by $\mathcal{O}_{\ell_i}(-1)$, so $N_{\ell_i/T_b} \cong \mathcal{O}_{\ell_i}(-1)^{\oplus 2}$, and $T_b \dashrightarrow T_a$ is near each ℓ_i the flop of a $(-1,-1)$-curve, that is, an Atiyah flop [Fuj26a, Example 2.10]. Thus each ℓ_i satisfies the hypotheses of Definition-Proposition 2.21 and has width one, and we can use Lemma 10.3 for $T_b \dashrightarrow T_a$ at ℓ_1 and at ℓ_2, with width one. By adjunction, $(K_{T_b} \cdot \ell_i) = \deg K_{\ell_i} - \deg N_{\ell_i/T_b} = -2 - (-2) = 0$. Let $T_{ab} := \mathrm{Bl}_{\ell_1 \cup \ell_2} T_b$ with exceptional divisors $\Omega_1, \Omega_2 \cong \mathbb{P}^1 \times \mathbb{P}^1$, and let $\ell_1^+, \ell_2^+ \subset T_a$ be the flopped curves, so that $T_{ab} \to T_a$ is the blow-up of $\ell_1^+ \cup \ell_2^+$. Since ℓ_i is a Cartier divisor on P_b, the strict transform of P_b on T_{ab} is isomorphic to P_b, and it intersects Ω_i along a section $\ell_i \times \{y_i\}$, $y_i \in \mathbb{P}^1$, given by the subbundle $N_{\ell_i/P_b} \subset N_{\ell_i/T_b}$. The morphism $T_{ab} \to T_a$ contracts Ω_i along the curves $\ell_i \times \{y\}$, $y \in \mathbb{P}^1$. Hence $P_b \to P_a$ contracts ℓ_1 and ℓ_2 to points p_1, p_2, that is,

$$\gamma : P_b \cong \mathrm{Bl}_{p_1,p_2} P_a \to P_a \cong P,$$

with exceptional curves $e_i := \ell_i$. By Lemma 10.3, $(P_a \cdot \ell_i^+) = -(P_b \cdot \ell_i) = 1$, so ℓ_i^+ intersects P_a transversally at the single point p_i. Since $P_b \cong \mathrm{Bl}_{o_1,o_2} E_M$, the birational map $P \cong P_a \dashrightarrow E_M$ is the composite of the elementary transformations [Har77, Example V.5.7.1] with centers p_1 and p_2: blow up p_i, then contract the strict transform of the fiber through p_i, which is the exceptional curve of $P_b \to E_M$ over o_i. The points p_1 and p_2 are on the fibers of $P_a \to M'$ over $x_1 \neq x_2$, so on distinct fibers of $P \to s$.

The class H_Q *on* T_b. For a divisor D on Y with strict transform D^+ on Y^+, pushing forward the formula of Lemma 10.3 along q_W shows that the strict transform on T_a of $p_T^* D$ is linearly equivalent to the pullback of D^+ minus $(D \cdot s) P_a$. By Lemma 11.7(2), $(p_T^* H)|_P \sim f_P$ and $(p_T^* E)|_P = (E' + P)|_P \sim (\sigma + 3f_P) - (\sigma + 2f_P) = f_P$, so $(H \cdot s) = (E \cdot s) = 1$. The strict transforms of H and E on Y^+ are $3h - 2R$ and $h - R$ (Theorem 2.23). Since $E' = p_T^* E - P$, we obtain $H_Q = p_T^* H - 2p_T^* E - 2P$, whose strict transform on T_a is linearly equivalent to

$$(3h - 2R - P_a) - 2(h - R - P_a) - 2P_a = h - P_a,$$

where h and R also denote their pullbacks to T_a. On T_b the divisor P_b is the pullback of E_M, so $H_{Q,b}$ is the pullback of $h - E_M$, that is, of $\mathcal{O}_Q(1)$ by ψ_M. In particular $|H_{Q,b}|$ is base point free, and

$$(H_{Q,b} \cdot \ell_i) = ((h - E_M) \cdot \ell_i) = 0 + 1 = 1.$$

By Lemma 10.3 applied to the Atiyah flops $T_a \dashrightarrow T_b$, we have $(H_{Q,a} \cdot \ell_i^+) = -1$, and on T_{ab} the pullbacks satisfy

$$H_{Q,a} - H_{Q,b} = \Omega_1 + \Omega_2, \qquad K_{T_a} = K_{T_b}. \tag{11.33}$$

Restricting the first equality to the strict transform of P_b and using (11.32), we obtain

$$H_{Q,b}|_{P_b} \sim \gamma^*(\sigma + 3f_P) - e_1 - e_2, \qquad \big((\gamma^*(\sigma + 3f_P) - e_1 - e_2)^2\big) = 4 - 2 = 2. \tag{11.34}$$

If $p_1, p_2 \in \sigma$, then the strict transform of σ on P_b is $\gamma^*\sigma - e_1 - e_2$ and

$$\big((\gamma^*(\sigma + 3f_P) - e_1 - e_2) \cdot (\gamma^*\sigma - e_1 - e_2)\big) = 1 - 2 = -1,$$

which contradicts the base point freeness of $|H_{Q,b}|$. Hence at most one of p_1, p_2 is a point of σ.

The fixed part on a common resolution. The morphism $W \to Y$ is the composite of the blow-up of s and of blow-ups of curves over s (Construction-Proposition 10.2), so it is defined over all of Y, and $p_W : W \to T$ is a morphism, while q_W is a morphism outside the strict transforms of the flopping curves of $Y \dashrightarrow Y^+$ other than s, in particular over a neighborhood of P. The threefold $T_{ab} = \mathrm{Bl}_{\ell_1 \cup \ell_2} T_b$ is defined globally, and $T_{ab} \to T_a$ is the blow-up of $\ell_1^+ \cup \ell_2^+$, since T_a is the blow-up of $\mathrm{Bl}_\Gamma V_5$ along the strict transform of M, T_b is the blow-up of $\mathrm{Bl}_M V_5$ along the strict transform of Γ, and $T_a \dashrightarrow T_b$ is an isomorphism over $V_5 \backslash (M \cap \Gamma)$. Let $\tilde{T}$ be a resolution of the closure of the graph of the birational map $W \dashrightarrow T_{ab}$ [KM98, Theorem 0.2], and let $g : \tilde{T} \to T$ be the morphism through W, and $g_a : \tilde{T} \to T_a$ and $g_b : \tilde{T} \to T_b$ the morphisms through T_{ab}. Let $\tilde{P} \subset \tilde{T}$ be the strict transform of P, and let $g_P := g_b|_{\tilde{P}} : \tilde{P} \to P_b$, so that $g|_{\tilde{P}} = \gamma \circ g_P$. Let Υ be the sum of the pullbacks to $\tilde{T}$ of $\sum_{i \geq 2}(i-1)P_i$ and of $\Omega_1 + \Omega_2$.

Let $\Phi \subset T$ be the union of the strict transforms of the flopping curves of $Y \dashrightarrow Y^+$ other than s. The flopping curves of $Y \dashrightarrow Y^+$ other than s are the strict transforms of the lines on X meeting L, and they are disjoint from s (Theorem 2.23). Hence $\Phi \cap P = \emptyset$. Since $T \to Y$ is an isomorphism outside P and $T_a \to Y^+$ is an isomorphism outside P_a, the map $T \dashrightarrow T_a$ is an isomorphism outside $\sigma \cup \Phi$, and near Φ it is the flop $Y \dashrightarrow Y^+$. The map $T_a \dashrightarrow T_b$ is an isomorphism outside $\ell_1^+ \cup \ell_2^+$. Since $T \dashrightarrow T_b$ is an isomorphism in codimension one and $|H_{Q,b}|$ is base point free, we can fix $G \in |H_Q|$ with $P \not\subset G$. Applying (11.31) over a neighborhood of P and (11.33) to G and its strict transforms, we obtain

$$\begin{gathered} g^*G - g_b^*G_b = \Upsilon + \Upsilon', \qquad g^*(-K_T) = g_b^*(-K_{T_b}), \\ \Upsilon|_{\tilde{P}} = g_P^*(\gamma^*\sigma + e_1 + e_2), \qquad \Upsilon'|_{\tilde{P}} = 0, \end{gathered} \tag{11.35}$$

where Υ' is a g-exceptional divisor with $g(\operatorname{Supp}\Upsilon') \subset \Phi$. The second equality holds near P by (11.31) and (11.33), and near Φ since $Y \dashrightarrow Y^+$ is a flop. The third equality follows from $P_2|_{P_1} = \sigma$, $P_1 \cap P_i = \emptyset$ for $i \geq 3$, and $\Omega_i|_{P_b} = e_i$, and the fourth from $\Phi \cap P = \emptyset$.

Claim 11.13. *Let $a, b \geq 0$ be integers, and let $D \geq 0$ be a divisor on T with $P \not\subset \operatorname{Supp} D$. Assume that $D \sim a(-K_T) + bH_Q$ or $D \sim a(-K_T) + bH_Q + P$. Then there exists an integer $c \geq 0$, independent of a, b, and D, with $c = 0$ in the first case, such that $D|_P \geq (b-c)\sigma$, and*

$$\operatorname{mult}_{p_i}\big(D|_P - (b-c)\sigma\big) \geq b - c \qquad (i = 1, 2).$$

Proof. Let $\Xi := g^*P - g_b^*P_b$. Since $T \dashrightarrow T_b$ is an isomorphism in codimension one, the exceptional prime divisors of g and of g_b coincide, and Ξ is supported on them. By (11.35),

$$g^*D - g_b^*D_b \sim b(\Upsilon + \Upsilon') + \Xi$$

in the second case, and $g^*D - g_b^*D_b \sim b(\Upsilon + \Upsilon')$ in the first. Both sides are supported on the exceptional divisors of g. A divisor $\operatorname{div}(\varphi)$ supported on the exceptional divisors of g satisfies $g_* \operatorname{div}(\varphi) = 0$, so φ is a regular invertible function on the normal projective variety

T, hence a constant. Hence $g^*D - b\Upsilon' = g_b^*D_b + b\Upsilon + \Xi \geq b\Upsilon + \Xi$ in the second case, and $g^*D - b\Upsilon' = g_b^*D_b + b\Upsilon \geq b\Upsilon$ in the first. Over a neighborhood of P, the map $T \dashrightarrow T_a$ is an isomorphism outside σ, the map $T_a \dashrightarrow T_b$ is an isomorphism outside $\ell_1^+ \cup \ell_2^+$, and $\ell_i^+ \cap P_a = \{p_i\}$. Hence $\Xi|_{\tilde{P}}$ is supported on $(\gamma g_P)^{-1}(\sigma \cup \{p_1, p_2\})$, which is the support of $g_P^*(\gamma^*\sigma + e_1 + e_2)$, and in the second case there exists an integer $c \geq 0$ with $\Xi|_{\tilde{P}} \geq -cg_P^*(\gamma^*\sigma + e_1 + e_2)$. In the first case, let $c := 0$. Restricting to $\tilde{P}$, which is not contained in $\operatorname{Supp} g_b^*D_b$, and using $\Upsilon'|_{\tilde{P}} = 0$ and (11.35), we obtain

$$g_P^*\gamma^*(D|_P) \geq (b-c)g_P^*(\gamma^*\sigma + e_1 + e_2).$$

Since $g_{P*}g_P^*$ is the identity, $\gamma^*(D|_P) - (b-c)\gamma^*\sigma \geq (b-c)(e_1+e_2)$, which is the claim. □

Proof of Proposition 11.12 continued. For a divisor L' on P, integers $v, \mu \geq 0$, and points $x, x' \in P$, let $V(L', v, \mu; x, x') \subset H^0(P, L')$ be the subspace of sections vanishing to order at least v along σ and with multiplicity at least μ at x and x'.

Claim 11.14. *Assume that $p_1, p_2 \notin \sigma$. Then there exist points $p_1' \in \sigma$ and $p_2' \notin \sigma$ on distinct fibers of $P \to s$ such that*

$$\dim V(L', v, \mu; p_1, p_2) \leq \dim V(L', v, \mu; p_1', p_2')$$

for every divisor L' on P and all integers $v, \mu \geq 0$.

Proof. The surface $P\backslash\sigma$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(2)$ over $s \cong \mathbb{P}^1$. A translation in the fiber direction by a section of $\mathcal{O}_{\mathbb{P}^1}(2)$ moves p_2 to the zero section and keeps p_1 off the zero section. Automorphisms of $\mathbb{F}_2$ act trivially on $\operatorname{Pic}\mathbb{F}_2$ and preserve σ, so the translation does not change $\dim V(L', v, \mu; p_1, p_2)$. Thus we may assume that p_2 is a point of the zero section. The scaling of the fiber coordinate by τ^{-1}, $\tau \in \mathbb{C}^*$, fixes σ, the zero section, and p_2, preserves the fibers, and moves p_1 to a point $p_1(\tau)$ with $p_1(1) = p_1$ and $\lim_{\tau\to 0} p_1(\tau) =: p_1(0) \in \sigma$. Let $p_1' := p_1(0)$ and $p_2' := p_2$. Fix L', v, and μ, and let Z_τ be the union of the subschemes defined by the μ-th powers of the maximal ideals of p_2 and of $p_1(\tau)$. Then $(Z_\tau)_{\tau\in\mathbb{A}^1}$ is flat, so $h^0(P, \mathcal{O}_P(L' - v\sigma) \otimes I_{Z_\tau})$ is upper semicontinuous in τ [Har77, Theorem III.12.8], and it is constant for $\tau \neq 0$, since the scalings are automorphisms of P. For $\tau \neq 0$, we have $p_1(\tau), p_2 \notin \sigma$, so division by the v-th power of a section defining σ does not change the multiplicities at $p_1(\tau)$ and p_2, and

$$V(L', v, \mu; p_1(\tau), p_2) = \sigma^v \cdot H^0(P, \mathcal{O}_P(L' - v\sigma) \otimes I_{Z_\tau}).$$

For $\tau = 0$, we have

$$V(L', v, \mu; p_1', p_2') \supset \sigma^v \cdot H^0(P, \mathcal{O}_P(L' - v\sigma) \otimes I_{Z_0}).$$

Hence

$$\dim V(L', v, \mu; p_1, p_2) = h^0(P, \mathcal{O}_P(L'-v\sigma)\otimes I_{Z_1}) \leq h^0(P, \mathcal{O}_P(L'-v\sigma)\otimes I_{Z_0}) \leq \dim V(L', v, \mu; p_1', p_2'),$$

and p_1' and p_2' are points of the fibers through p_1 and p_2, which are distinct. □

Proof of Proposition 11.12 continued.

Step 2. In this step, we bound the ranks of the refinement by P and compute $S_X(P)$. For integers $m \geq 1$ and $k \geq 0$, let $V_{m,k} := H^0(T, mH - kP)$, let $W_{m,k}$ be the image of the restriction $V_{m,k} \to H^0(P, (mH-kP)|_P)$, and let $n_{m,k} := \dim W_{m,k} = h^0(T, mH-kP) - h^0(T, mH-(k+1)P)$. Since $V_{m,0} = H^0(X, mH)$ and $V_{m,k} = 0$ for $k > 5m$ (Lemma 11.7(4)),

$$\sum_{k\geq 0} n_{m,k} = N_m = \frac{11}{3}m^3 + O(m^2),$$

where the last equality follows from $(H^3) = 22$ (Lemma 11.1(1)) and asymptotic Riemann–Roch. Let $t := k/m$ and $j := \lceil k/2 \rceil$, so that $2j - k \in \{0, 1\}$. For $2 \leq t \leq 4$, let $w := t/2 - 1$, and for $4 \leq t \leq 5$, let $q := 5 - t$.

Let $0 \neq \bar{u} \in W_{m,k}$ be the restriction of $u \in V_{m,k} \backslash V_{m,k+1}$, and let $D_u \sim mH - kP$ be the divisor of u, so that $P \not\subset \operatorname{Supp} D_u$. By Lemma 11.7(4), $D_u \geq jE'$, and

$$D_u - jE' \sim mH - jE' - kP = (2m-j)(-K_T) + (j-m)H_Q + (2j-k)P. \tag{11.36}$$

Thus division by the j-th power of a section defining $E'|_P$ is an injective linear map from $W_{m,k}$ to $H^0(P, L_{m,k})$, where

$$L_{m,k} := (mH - jE' - kP)|_P \sim j\sigma + (m+j)f_P - (2j-k)(\sigma + 2f_P).$$

For a divisor B on P and an integer $\mu \geq 0$, sections of B with multiplicity at least μ at p_1 and p_2 are the sections of $\gamma^* B - \mu(e_1 + e_2)$ on P_b. For a divisor L' on P_b and a smooth rational curve $C \subset P_b$, the restriction sequence implies $0 \leq h^0(P_b, L') - h^0(P_b, L' - C) \leq \max\{0, (L' \cdot C) + 1\}$. Each of $\gamma^*\sigma$, $\gamma^* f_P$, e_1, and e_2 is linearly equivalent to a sum of at most two smooth rational curves, and the intersection numbers below are $O(m)$. Hence a change of the classes and multiplicities below by quantities bounded independently of m and k changes the dimensions by $O(m)$.

The range $0 \leq t \leq 2$. The class $L_{m,k}$ is nef, and $h^0(\mathbb{F}_2, \alpha\sigma + bf_P) = (\alpha+1)(b-\alpha+1)$ for $b \geq 2\alpha \geq 0$ [Har77, Exercise III.8.4(a), Propositions V.2.8 and V.2.9]. Hence $n_{m,k} \leq (j+1)(m+1) = m^2 \cdot t/2 + O(m)$.

The range $2 \leq t \leq 4$. By (11.36) and Claim 11.13 applied to $D_u - jE'$, the divisor of the image of $\bar{u}$ in $H^0(P, L_{m,k})$ contains $(j-m-c)\sigma$, and the residual divisor has multiplicity at least $j-m-c$ at p_1 and p_2. Hence

$$n_{m,k} \leq h^0\big(P_b, \gamma^*(m\sigma + (m+j)f_P) - (j-m)(e_1+e_2)\big) + O(m), \tag{11.37}$$

where $m\sigma + (m+j)f_P = m(\sigma + (2+w)f_P)$ and $j - m = mw$ up to $O(1)$.

The dimension in (11.37) is $\dim V(m\sigma + (m+j)f_P, 0, j-m; p_1, p_2)$, and the dimensions bounded in **Step 3** are also of the form $\dim V(L', v, \mu; p_1, p_2)$. By Claim 11.14, possibly renumbering p_1 and p_2, and possibly replacing p_1 and p_2 with the points p_1' and p_2' of Claim 11.14, we may assume that $p_1 \in \sigma$ and $p_2 \notin \sigma$, on distinct fibers. Moving the fibers through p_1 and p_2 to the fibers f_0 and f_∞ over $0, \infty \in \mathbb{P}^1$, and p_2 to the zero section $\sigma_0 \sim \sigma + 2f_P$, we may assume that $p_1 = \sigma \cap f_0$ and $p_2 = \sigma_0 \cap f_\infty$, and we replace P_b with the blow-up of P at p_1 and p_2. Then P_b is a smooth projective toric surface whose torus invariant curves are the six curves in Table 14, and the cone of curves of P_b is generated by them [CLS11, Theorem 6.3.20]. For real numbers $\xi \geq \zeta \geq 0$ and $0 \leq v \leq \xi$, let

$$D_v := \gamma^*\big((\xi - v)\sigma + (2\xi + \zeta)f_P\big) - (\zeta - v)e_1 - \zeta e_2. \tag{11.38}$$

TABLE 14. The torus invariant curves on P_b and their intersection numbers with D_v from (11.38).

curve C	$\gamma^*\sigma - e_1$	$\gamma^* f_P - e_1$	e_1	$\gamma^* f_P - e_2$	e_2	$\gamma^*(\sigma + 2f_P) - e_2$
$(D_v \cdot C)$	$3v$	$\xi - \zeta$	$\zeta - v$	$\xi - v - \zeta$	ζ	2ξ

For $\xi = 1$, $\zeta = w$, and $v = 0$, we have $D_0 = \gamma^*(\sigma + (2+w)f_P) - w(e_1 + e_2)$. By Table 14, D_0 is nef for $0 \leq w \leq 1$, and

$$(D_0^2) = -2 + 2(2+w) - 2w^2 = 2(1 + w - w^2).$$

For a nef divisor D on the fixed smooth projective toric surface P_b, $H^i(P_b, D) = 0$ for $i > 0$ [CLS11, Theorem 9.2.3], so $h^0(P_b, D) = \frac{1}{2}(D \cdot (D - K_{P_b})) + 1$ by Riemann–Roch [Har77, Theorem V.1.6]. The term $-\frac{1}{2}(D \cdot K_{P_b})$ is $O(m)$ uniformly for the classes in (11.37). Hence $n_{m,k} \leq m^2(1 + w - w^2) + O(m)$.

The range $4 \leq t \leq 5$. By Lemma 11.7(4), $h^0(T, mH - kP) = h^0(Q, \mathcal{O}_Q(5m-k))$ for $4m \leq k \leq 5m$, and $h^0(T, mH - (5m+1)P) = 0$. Since $Q \subset \mathbb{P}^4$ is a quadric hypersurface, $h^0(Q, \mathcal{O}_Q(r)) = \binom{r+4}{4} - \binom{r+2}{4}$ for $r \geq 0$ [Har77, Theorem III.5.1], so $h^0(Q, \mathcal{O}_Q(r)) - h^0(Q, \mathcal{O}_Q(r-1)) = (r+1)^2$ for $r \geq 0$. Hence $n_{m,k} = (5m - k + 1)^2 = m^2(5-t)^2 + O(m)$.

Let $\rho(t) := t/2$ for $0 \le t \le 2$, $\rho(t) := 1 + w - w^2$ for $2 \le t \le 4$, and $\rho(t) := (5-t)^2$ for $4 \le t \le 5$. We have shown that $n_{m,k} \le m^2\rho(k/m) + O(m)$ uniformly in $0 \le k \le 5m$. Moreover

$$\int_0^2 \frac{t}{2}dt = 1, \qquad \int_2^4 (1+w-w^2)dt = \frac{7}{3}, \qquad \int_4^5 (5-t)^2 dt = \frac{1}{3}, \tag{11.39}$$

and $1 + 7/3 + 1/3 = 11/3$. The Riemann sums of the piecewise polynomial functions in this proof differ from the integrals by $O(1/m)$. Hence $\sum_k (m^2\rho(k/m) - n_{m,k}) = O(m^2)$. Since each of the $5m+1$ summands is at least $-O(m)$, we obtain $\sum_k |m^2\rho(k/m) - n_{m,k}| = O(m^2)$. Therefore, for real numbers $\theta_{m,k} \ge 0$ with $\theta_{m,k} = O(m)$ uniformly in k,

$$\sum_k \theta_{m,k} n_{m,k} = \sum_k \theta_{m,k} m^2 \rho(k/m) + O(m^3). \tag{11.40}$$

By Definition 2.8, a basis compatible with the filtration by order along P computes $S_m(P) = \frac{1}{mN_m}\sum_k k n_{m,k}$. By (11.40) with $\theta_{m,k} = k$ and by [AZ22, Lemma 2.9],

$$S_X(P) = \frac{3}{11}\Big(\int_0^2 \frac{t^2}{2}dt + \int_2^4 t(1+w-w^2)dt + \int_4^5 t(5-t)^2 dt\Big) = \frac{3}{11}\Big(\frac{4}{3} + 7 + \frac{17}{12}\Big) = \frac{117}{44}. \tag{11.41}$$

Step 3. In this step, we bound the σ-coefficient of the restriction to P of an m-basis type divisor (Definition 2.8) compatible with P, keeping the point conditions at p_1 and p_2. Let $\bar{u}_1, \dots, \bar{u}_{n_{m,k}}$ be a basis of $W_{m,k}$, and let $W_{m,k}^{(v)} \subset W_{m,k}$ be the subspace of sections vanishing to order at least v along σ. For each $v \ge 1$, the sections $\bar{u}_i$ with $\operatorname{ord}_\sigma \bar{u}_i \ge v$ are linearly independent elements of $W_{m,k}^{(v)}$. Hence

$$\sum_i \operatorname{ord}_\sigma \bar{u}_i \le \sum_{v \ge 1} \dim W_{m,k}^{(v)}. \tag{11.42}$$

Since $E'|_P$ is an irreducible curve in $|\sigma + 3f_P|$ different from σ (Lemma 11.7(2)), $\sigma \not\subset E'$. If $R_T|_P - r\sigma$ is effective, then $r \le 2$, since $((R_T|_P - r\sigma)\cdot f_P) = 2 - r$ and f_P is nef. Thus $\operatorname{mult}_\sigma(R_T|_P) \le 2$.

Let $0 \ne \bar{u} \in W_{m,k}$. For $0 \le t \le 2$, let $c_{m,k} := 0$, and let $\bar{u}'$ be the image of $\bar{u}$ in $H^0(P, L_{m,k})$, so that $\operatorname{ord}_\sigma \bar{u} = \operatorname{ord}_\sigma \bar{u}'$. For $2 \le t \le 4$, by (11.36) and Claim 11.13, the image of $\bar{u}$ in $H^0(P, L_{m,k})$ is $\sigma^{c_{m,k}}\bar{u}'$, where $c_{m,k} := \max\{j - m - c, 0\}$ and $\bar{u}'$ is a section of the class in (11.44) below with multiplicity at least $c_{m,k}$ at p_1 and p_2, so that $\operatorname{ord}_\sigma \bar{u} = c_{m,k} + \operatorname{ord}_\sigma \bar{u}'$.

For $4 \le t \le 5$, $V_{m,k}$ consists of the products of $e'^{k-2m} r_T^{k-4m}$ with the sections of $(5m-k)H_Q$ (Lemma 11.7(4)), where e' and r_T are sections defining E' and R_T. Thus $\bar{u}$ is the restriction of $e'^{k-2m} r_T^{k-4m} u_0$ for a section u_0 of $(5m-k)H_Q$ whose divisor does not contain P. By Claim 11.13 in the first case, with $a = 0$ and $b = 5m - k$, $u_0|_P = \sigma^{5m-k}\bar{u}'$, where $\bar{u}'$ is a section of the class in (11.45) below with multiplicity at least $5m - k$ at p_1 and p_2. Since $\sigma \not\subset E'$,

$$\operatorname{ord}_\sigma \bar{u} = (k - 4m)\operatorname{mult}_\sigma(R_T|_P) + (5m - k) + \operatorname{ord}_\sigma \bar{u}'.$$

For $4 \le t \le 5$, let $c_{m,k} := 2(k-4m) + (5m-k) = m(t-3)$. Thus the class of $\bar{u}'$ on P and its multiplicities at p_1 and p_2, divided by m, are the following, up to $O(1/m)$:

$$\tfrac{t}{2}\sigma + (1 + \tfrac{t}{2})f_P, \text{ no point conditions}, \qquad 0 \le t \le 2, \tag{11.43}$$

$$\sigma + (2+w)f_P, \text{ multiplicity } w, \qquad 2 \le t \le 4, \tag{11.44}$$

$$q(\sigma + 3f_P), \text{ multiplicity } q, \qquad 4 \le t \le 5. \tag{11.45}$$

In each range, the linear map sending $\bar{u}$ to $\bar{u}'$ is injective, and $\operatorname{ord}_\sigma \bar{u} \ge c_{m,k} + v$ implies $\operatorname{ord}_\sigma \bar{u}' \ge v$, in the range $4 \le t \le 5$ since $\operatorname{mult}_\sigma(R_T|_P) \le 2$. Hence $\dim W_{m,k}^{(c_{m,k}+v)}$ is at most the

dimension of the space of sections $\bar{u}'$ with the point conditions above and $\operatorname{ord}_\sigma \bar{u}' \geq v$. Moreover $\dim W^{(v)}_{m,k} \leq n_{m,k}$ for $v \leq c_{m,k}$. Splitting the sum in (11.42) at $v = c_{m,k}$, we obtain

$$\sum_i \operatorname{ord}_\sigma \bar{u}_i \leq c_{m,k} n_{m,k} + \sum_{v \geq 1} h^0\big(P, L_{m,k} - v\sigma\big) + O(m^2)$$

in (11.43), and

$$\sum_i \operatorname{ord}_\sigma \bar{u}_i \leq c_{m,k} n_{m,k} + \sum_{v \geq 1} h^0\big(P_b, mD_{v/m}\big) + O(m^2)$$

in (11.44) and (11.45), with D_v from (11.38) for $(\xi, \zeta) = (1, w)$ and $(\xi, \zeta) = (q, q)$, respectively.

Indeed, by Claim 11.14, we keep the assumption $p_1 \in \sigma$ and $p_2 \notin \sigma$ of **Step 2**. Then $\gamma^*\sigma$ is the sum of the strict transform of σ and e_1. Hence, for integers $v \geq 0$, the sections of $m\xi\sigma + m(2\xi + \zeta)f_P$ which vanish to order at least v along σ and have multiplicity at least $m\zeta$ at p_1 and p_2 are, after division by σ^v, exactly the sections of $mD_{v/m}$ on P_b. The class and the multiplicities of $\bar{u}'$ differ from $m\xi\sigma + m(2\xi + \zeta)f_P$ and $m\zeta$ by $O(1)$, and $m\xi$ and $m\zeta$ differ from integers by $O(1)$, which changes the dimensions only by $O(m)$, as in **Step 2**.

By Table 14, the negative part of the Zariski decomposition of D_v is

$$(v - \zeta)_+ e_1 + (\zeta - \xi + v)_+ (\gamma^* f_P - e_2),$$

since e_1 and $\gamma^* f_P - e_2$ are disjoint (-1)-curves and subtracting the negative part leaves a divisor with non-negative intersection numbers with the six torus invariant curves, hence a nef divisor [Laz04a, Theorem 2.3.19]. For rational ξ, ζ, and v by [Laz04a, Corollary 2.3.22], and for real ξ, ζ, and v by continuity of the volume [Laz04a, Corollary 2.2.45], compared with $\operatorname{vol}((\xi - v)\sigma + (2\xi + \zeta)f_P)$, the volume of D_v decreases by

$$(\zeta - v)_+^2 + \zeta^2 - (\zeta - \xi + v)_+^2.$$

Removing the negative part first, the Riemann–Roch estimate of **Step 2** holds with error $O(m)$. For real numbers α, β with $\beta \geq 2\alpha \geq 0$, the class $(\alpha - v)\sigma + \beta f_P$ is nef for $0 \leq v \leq \alpha$, so its volume is its self-intersection number, for rational classes by [Laz04a, Corollary 1.4.41] and for real classes by continuity of the volume [Laz04a, Corollary 2.2.45], and

$$\int_0^\alpha \operatorname{vol}((\alpha - v)\sigma + \beta f_P) dv = \alpha^2\beta - \frac{2}{3}\alpha^3.$$

For (11.43), (11.44), and (11.45), ignoring the point conditions, we obtain

$$\int_0^2 \Big(\frac{t^2}{4} + \frac{t^3}{24}\Big) dt = \frac{5}{6}, \qquad \int_2^4 \Big(\frac{1}{3} + \frac{t}{2}\Big) dt = \frac{11}{3}, \qquad \int_4^5 \frac{7}{3} q^3 dt = \frac{7}{12}. \tag{11.46}$$

Integrating the decrease in v, we obtain w^2 in (11.44), where $\xi = 1$ and $\zeta = w$, and q^3 in (11.45), where $\xi = \zeta = q$. Hence

$$\int_2^4 \Big(\frac{t}{2} - 1\Big)^2 dt = \frac{2}{3}, \qquad \int_4^5 (5 - t)^3 dt = \frac{1}{4}. \tag{11.47}$$

The number of pairs (k, v) is $O(m^2)$, and each dimension is at most $\frac{m^2}{2}$ times the corresponding volume plus $O(m)$, so after division by $mN_m = \frac{11}{3}m^4 + O(m^3)$ the free parts contribute at most

$$\frac{3}{22}\Big(\frac{5}{6} + \frac{11}{3} + \frac{7}{12}\Big) - \frac{3}{22}\Big(\frac{2}{3} + \frac{1}{4}\Big) + o(1) = \frac{61}{88} - \frac{1}{8} + o(1) = \frac{25}{44} + o(1). \tag{11.48}$$

We have $c_{m,k} = mw + O(1)$ in (11.44) and $c_{m,k} = m(t-3)$ in (11.45). By (11.40) with $\theta_{m,k} = c_{m,k}$, the fixed parts contribute at most

$$\frac{3}{11}\int_2^4 w(1 + w - w^2) dt + \frac{3}{11}\int_4^5 (t - 3)(5 - t)^2 dt + o(1) = \frac{14}{44} + \frac{5}{44} + o(1) = \frac{19}{44} + o(1). \tag{11.49}$$

Let D_m be an m-basis type divisor compatible with the filtration by order along P, and write $(\pi p_T)^* D_m = S_m(P)P + \Delta_m$, where $\Delta_m \geq 0$ and $P \notin \operatorname{Supp} \Delta_m$. Then $\Delta_m|_P =$

$\frac{1}{mN_m}\sum_k\sum_i \operatorname{div}(\bar{u}_i)$ for bases $\bar{u}_i$ of the spaces $W_{m,k}$. By (11.42), (11.48), and (11.49), there exist $\epsilon_m \geq 0$ with $\lim_{m\to+\infty}\epsilon_m = 0$, independent of D_m, such that

$$\operatorname{mult}_\sigma(\Delta_m|_P) \leq \frac{25}{44} + \frac{19}{44} + \epsilon_m = 1 + \epsilon_m. \tag{11.50}$$

Step 4. In this step, we prove the inequality for F and the positivity of $\beta_X(F')$. Let F be a prime divisor over X with $\operatorname{center}_T F = \sigma$. By [AZ22, Lemma 3.1], there exists a basis of $H^0(X, mH)$ compatible with the filtrations by order along P and along F. Let D_m be the associated m-basis type divisor, so that $\operatorname{mult}_P D_m = S_m(P)$ and $\operatorname{mult}_F D_m = S_m(F)$, and write $(\pi p_T)^* D_m = S_m(P)P + \Delta_m$ as in **Step 3**. Let $\lambda_m := (1+\epsilon_m)^{-1}$. By (11.50), $(P, \lambda_m\Delta_m|_P)$ is lc at the generic point of σ. Let $U \subset T$ be the complement of the components of $\operatorname{Supp}(\Delta_m|_P)$ other than σ and of E'. For $0 < \delta < 1$ with $(1-\delta)\lambda_m \in \mathbb{Q}$, the pair $(P\cap U, (1-\delta)\lambda_m\Delta_m|_{P\cap U})$ is klt, so $(U, P + (1-\delta)\lambda_m\Delta_m)$ is plt near $P\cap U$ by inversion of adjunction [KM98, Theorem 5.50(1)]. Letting $\delta \to 0$, the pair $(U, P + \lambda_m\Delta_m)$ is lc near $P\cap U$, and $\operatorname{center}_T F \cap U \neq \emptyset$. Hence

$$A_T(F) \geq \operatorname{mult}_F P + \lambda_m \operatorname{mult}_F \Delta_m.$$

By Lemma 11.7(1), $A_X(F) = A_T(F) + \operatorname{mult}_F E' + 2\operatorname{mult}_F P$. Since $\sigma \not\subset E'$ (**Step 3**), $\operatorname{mult}_F E' = 0$. Moreover $\lambda_m S_m(F) = \lambda_m S_m(P)\operatorname{mult}_F P + \lambda_m \operatorname{mult}_F \Delta_m$. Thus

$$A_X(F) - \lambda_m S_m(F) \geq \operatorname{mult}_F P + \lambda_m \operatorname{mult}_F \Delta_m + 2\operatorname{mult}_F P - \lambda_m S_m(P)\operatorname{mult}_F P - \lambda_m \operatorname{mult}_F \Delta_m = (3 - \lambda_m S_m(P))$$

Let $m \to +\infty$. Since $S_m(P) \to S_X(P)$ and $S_m(F) \to S_X(F)$ [AZ22, Lemma 2.9], (11.41) implies

$$A_X(F) - S_X(F) \geq \left(3 - \frac{117}{44}\right)\operatorname{mult}_F P = \frac{15}{44}\operatorname{mult}_F P > 0, \tag{11.51}$$

where $\operatorname{mult}_F P > 0$ since $\operatorname{center}_T F \subset P$.

Let F' be a prime divisor over X with $\operatorname{center}_X F' = L$. If $\operatorname{center}_Y F' \neq s$, then $\beta_X(F') > 0$ by Proposition 11.2. If $\operatorname{center}_Y F' = s$, then either $F' = P$, or $\operatorname{center}_T F'$ is a curve on P dominating s. If $F' = P$ or $\operatorname{center}_T F' \neq \sigma$, then $\beta_X(F') > 0$ by Proposition 11.11. If $\operatorname{center}_T F' = \sigma$, then $\beta_X(F') > 0$ by (11.51) applied to $F = F'$. □

Remark 11.15. The estimate (11.50) has no margin: $25/44 + 19/44 = 1$, and the bound $3 - S_X(P) = 15/44$ in **Step 4** uses the coefficient bound 1 exactly. Without the conditions at p_1 and p_2, the free part in (11.48) would be bounded only by $61/88$, and the bound in (11.50) would become $61/88 + 19/44 = 99/88 > 1$. Also, the integrals in (11.39) sum to $11/3 = \lim_{m\to+\infty} N_m/m^3$, so the upper bounds for $n_{m,k}$ in **Step 2** are asymptotically equalities, and (11.41) is the value of $S_X(P)$ and not only an upper bound. A loss in the analysis of the base points of $H_Q|_P$ would increase the bound in (11.50) above 1, and the argument of **Step 4** would no longer apply.

12. Special lines and torus degenerations in genus twelve

In this section, we determine which configurations of special lines lead to smooth degenerations to members with a torus action, and we complete the treatment of curves of degree one and two (Corollary 12.8). The local estimates of Section 11 leave open one configuration of a special line L: in the notation of Set-up 10.10, $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, the center on T is σ, and $M \cap \Gamma$ is non-reduced of length two (Corollary 12.8). In that configuration, X degenerates smoothly to a member with a faithful $\mathbb{G}_m$-action, so X is K-semistable by openness. A smooth degeneration of X to a member with a faithful $\mathbb{G}_m$-action also arises from a special line which intersects no other line (the implication (ii)⇒(iii) of Theorem 1.3(2)), and it cannot occur for nets in Ω^{st} (Section 17).

12.1. **Quintic curves through the singular point of the conductor.** In this subsection, we prove that the threefold of genus twelve associated with a rational normal quintic curve whose strict transform in $\mathbb{F}_3$ contains the point $s_0 \cap f_0$ (Lemma 12.1) is K-semistable (Proposition 12.4).

Lemma 12.1 ([FT92, Theorem, part (B)], [DFK25, Proposition 5], [Fuj26a, Theorem 3.6])**.** *Let $M \subset V_5$ be a line with $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$. Then there exists a unique hyperplane section S_M of V_5 singular along M, and the normalization of S_M is isomorphic to $\mathbb{F}_3$. Let $\nu : \mathbb{F}_3 \to S_M$ be the normalization and s_0 the negative section of $\mathbb{F}_3$. Then $\nu^{-1}(M) = s_0 \cup f_0$ for a fiber f_0 of $\mathbb{F}_3$, the conductor of ν is $s_0 + f_0$, and $\nu^*\mathcal{O}_{V_5}(1) \cong \mathcal{O}_{\mathbb{F}_3}(s_0 + 4f_0)$. If $\Gamma \subset S_M$ is a smooth rational normal quintic curve, then the strict transform $\tilde{\Gamma} \subset \mathbb{F}_3$ of Γ is isomorphic to Γ, $\tilde{\Gamma} \in |s_0 + 4f_0|$, and $(\tilde{\Gamma} \cdot s_0) = (\tilde{\Gamma} \cdot f_0) = 1$.*

Proof. Let h be the pullback to $\mathrm{Bl}_M V_5$ of the hyperplane class of V_5 and E_M the exceptional divisor of $\mathrm{Bl}_M V_5 \to V_5$. The strict transform on $\mathrm{Bl}_M V_5$ of a hyperplane section of V_5 singular along M belongs to $|h - 2E_M|$. By [Fuj26a, Theorem 3.6(4)(5)], there exists a contraction $\psi_M : \mathrm{Bl}_M V_5 \to Q$ onto a smooth quadric threefold Q whose exceptional divisor J satisfies $J \sim h - 2E_M$. Since J is ψ_M-exceptional, $\psi_{M*}\mathcal{O}_{\mathrm{Bl}_M V_5}(J) = \mathcal{O}_Q$ [Har77, Proposition II.6.3A], so $h^0(\mathrm{Bl}_M V_5, \mathcal{O}_{\mathrm{Bl}_M V_5}(J)) = 1$. Thus J is the only member of $|h - 2E_M|$, and S_M is unique. The normalization of S_M is $\mathbb{F}_3$ by [FT92, Theorem, parts (B)(1),(2)]. By [DFK25, Proposition 5(b)–(d)], $\nu^{-1}(M) = s_0 \cup f_0$ for a fiber f_0, and

$$\nu^*\omega_{S_M}^{\vee} \cong \omega_{\mathbb{F}_3}^{\vee} \otimes \mathcal{O}_{\mathbb{F}_3}(-s_0 - f_0),$$

so the conductor of ν is $s_0 + f_0$ (see also [FT92, Theorem, part (B)(2)b]). Since $S_M \in |\mathcal{O}_{V_5}(1)|$ and $K_{V_5} \sim -2S_M$, adjunction implies $\omega_{S_M}^{\vee} \cong \mathcal{O}_{S_M}(1)$. Since $-K_{\mathbb{F}_3} \sim 2s_0 + 5f_0$, we obtain $\nu^*\mathcal{O}_{V_5}(1) \cong \mathcal{O}_{\mathbb{F}_3}(s_0 + 4f_0)$.

Let $\Gamma \subset S_M$ be a smooth rational normal quintic curve and $\tilde{\Gamma} \subset \mathbb{F}_3$ its strict transform. Since ν is finite and $\Gamma \not\subset M$, the morphism $\nu|_{\tilde{\Gamma}} : \tilde{\Gamma} \to \Gamma$ is finite and birational onto the smooth curve Γ, hence an isomorphism. Since

$$(s_0 \cdot (s_0 + 4f_0)) = (f_0 \cdot (s_0 + 4f_0)) = 1,$$

ν maps s_0 and any fiber of $\mathbb{F}_3$ onto lines. Thus $\tilde{\Gamma}$ is neither s_0 nor a fiber. Write $\tilde{\Gamma} \sim \alpha s_0 + \beta f_0$ with $\alpha, \beta \in \mathbb{Z}$. Then

$$\alpha = (\tilde{\Gamma} \cdot f_0) \geq 1, \qquad \beta - 3\alpha = (\tilde{\Gamma} \cdot s_0) \geq 0, \qquad 5 = (\tilde{\Gamma} \cdot (s_0 + 4f_0)) = \alpha + \beta.$$

Thus $(\alpha, \beta) = (1, 4)$, $\tilde{\Gamma} \in |s_0 + 4f_0|$, and $(\tilde{\Gamma} \cdot s_0) = (\tilde{\Gamma} \cdot f_0) = 1$ (cf. [DFK25, Lemma 10]). □

Set-up 12.2. Let M, S_M, ν, s_0, f_0, Γ, $\tilde{\Gamma}$, X, L, Y, and Y^+ be as follows.

(1) $M \subset V_5$ is a line with $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$.
(2) S_M is the hyperplane section of V_5 singular along M, $\nu : \mathbb{F}_3 \to S_M$ is the normalization, s_0 is the negative section of $\mathbb{F}_3$, and f_0 is the fiber of $\mathbb{F}_3$ with $\nu^{-1}(M) = s_0 \cup f_0$ (Lemma 12.1).
(3) $\Gamma \subset S_M$ is a smooth rational normal quintic curve whose strict transform $\tilde{\Gamma} \subset \mathbb{F}_3$ contains $s_0 \cap f_0$.
(4) X is the smooth prime Fano threefold of genus 12 associated with (V_5, Γ) by the inverse link of Theorem 2.23, $L \subset X$ is the line marked by the inverse link, $Y := \mathrm{Bl}_L X$, $Y^+ := \mathrm{Bl}_\Gamma V_5$, and $Y \dashrightarrow Y^+$ is the flop of Theorem 2.23.

The objects of Set-up 12.2 form the diagram

$$\begin{array}{ccc} Y = \mathrm{Bl}_L X & \dashrightarrow & Y^+ = \mathrm{Bl}_\Gamma V_5 \\ \downarrow & & \downarrow \\ X & & V_5, \end{array}$$

where the vertical arrows are the blow-ups of L and of Γ, and the horizontal arrow is the flop of Theorem 2.23. The condition that $\tilde{\Gamma}$ contains $s_0 \cap f_0$ is what allows a one-parameter subgroup of

$\mathrm{Aut}(V_5; M)$ to degenerate Γ to an invariant curve which is still a smooth rational normal quintic, so that the inverse link yields a smooth family of threefolds.

The following lemma is one of the key constructions of this paper. We degenerate (V_5, Γ) by a torus on the side of V_5, and we reconstruct the family of threefolds by the inverse link. The outcome is a smooth anti-canonical test configuration whose central fiber carries a faithful $\mathbb{G}_m$-action.

Lemma 12.3. *Notation as in Set-up 12.2. Then there exists a smooth projective morphism $\pi : \mathcal{X} \to \mathbb{A}^1$ with a $\mathbb{G}_m$-action covering the standard action on $\mathbb{A}^1$, such that $-K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier and π-ample, $(\mathcal{X}, -K_{\mathcal{X}/\mathbb{A}^1})|_{\pi^{-1}(\mathbb{A}^1\setminus\{0\})}$ is $\mathbb{G}_m$-equivariantly isomorphic to $(X, -K_X) \times (\mathbb{A}^1\setminus\{0\})$ with the trivial action on X, every fiber of π is a smooth prime Fano threefold of genus 12, and the induced $\mathbb{G}_m$-action on $\mathcal{X}_0$ is faithful. In particular π is an anti-canonical test configuration of X with smooth central fiber.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix coordinates on $\mathbb{F}_3$ and on the hyperplane spanned by S_M, and a one-parameter subgroup $\rho : \mathbb{G}_m \to \mathrm{Aut}(V_5; M)$.

Let $\langle S_M \rangle \cong \mathbb{P}^5$ be the hyperplane of $\mathbb{P}^6$ spanned by S_M. We write $\mathbb{F}_3$ as the quotient of $(\mathbb{A}^2\setminus\{0\}) \times (\mathbb{A}^2\setminus\{0\})$, with coordinates (r, t) and (a, b), by the action of $\mathbb{G}_m^2$ with weights

$$\deg r = \deg t = (1, 0), \qquad \deg a = (0, 1), \qquad \deg b = (-3, 1).$$

Then $s_0 = (b = 0)$, the zero locus of a non-zero polynomial of bidegree (x, y) is linearly equivalent to $ys_0 + (x+3y)f_0$, and $[r : t]$ is the coordinate on the base $\mathbb{P}^1$ of $\mathbb{F}_3$. By [DFK25, Example 15 and Lemma 10], we may choose the coordinates (r, t) with $f_0 = (t = 0)$ and homogeneous coordinates $[x_0 : \cdots : x_5]$ on $\langle S_M \rangle$ such that

$$\nu(r, t, a, b) = \left[ra : \frac{4}{5}(ta + r^4 b) : tr^3 b : t^2 r^2 b : t^3 r b : t^4 b\right], \tag{12.1}$$

and such that there exists a one-parameter subgroup $\tau : \mathbb{G}_m \to \mathrm{Aut}(V_5; M)$ for which $(r, t, a, b) \to (r, \lambda t, a, \lambda b)$ is the lift of $\tau(\lambda)$ through ν. Let $\rho(\lambda) := \tau(\lambda^{-1})$. Then $\rho(\lambda)$ lifts to the automorphism

$$(r, t, a, b) \to (r, \lambda^{-1} t, a, \lambda^{-1} b)$$

of $\mathbb{F}_3$. Since $\rho(\lambda)$ preserves M, it preserves S_M (Lemma 12.1) and $\langle S_M \rangle$, and (12.1) implies

$$\rho(\lambda)[x_0 : x_1 : \cdots : x_5] = [x_0 : \lambda^{-1} x_1 : \lambda^{-2} x_2 : \lambda^{-3} x_3 : \lambda^{-4} x_4 : \lambda^{-5} x_5].$$

In particular $\rho(\lambda)$ acts trivially on $\langle S_M \rangle$ only if $\lambda = 1$, so ρ is injective.

Step 2. In this step, we write an equation of $\tilde{\Gamma}$ and construct a family $\mathcal{C} \subset V_5 \times \mathbb{A}^1$ of rational normal quintic curves Γ_z with $\Gamma_1 = \Gamma$ and $\rho(\lambda)\Gamma_z = \Gamma_{\lambda z}$.

By Lemma 12.1, $\tilde{\Gamma} \cong \Gamma$ and $\tilde{\Gamma} \sim s_0 + 4f_0$. Thus $\tilde{\Gamma}$ is the zero locus of a polynomial $a\ell(r, t) + bp_4(r, t)$, where ℓ is linear and p_4 is homogeneous of degree four. The point $s_0 \cap f_0$ is $(t = b = 0)$, so $\ell(1, 0) = 0$. If $\ell = 0$, then $s_0 \subset \tilde{\Gamma}$, which is impossible. Thus, after rescaling the equation, $\ell = t$ and $\tilde{\Gamma} = (at + bp_4 = 0)$. Write

$$p_4 = cr^4 + c_1 r^3 t + c_2 r^2 t^2 + c_3 r t^3 + c_4 t^4.$$

If $c = 0$, then t divides $at + bp_4$ and $f_0 \subset \tilde{\Gamma}$, which is impossible. Thus $c \neq 0$.

For $z \in \mathbb{A}^1$, let

$$p_{4,z} := cr^4 + zc_1 r^3 t + z^2 c_2 r^2 t^2 + z^3 c_3 r t^3 + z^4 c_4 t^4,$$

so that $p_{4,1} = p_4$ and

$$p_{4,z}(r, \lambda t) = p_{4,\lambda z}(r, t) \quad \text{for any } \lambda \in \mathbb{G}_m. \tag{12.2}$$

Since $c \neq 0$, the forms t and $p_{4,z}$ have no common zero on $\mathbb{P}^1$. Thus, for any z, $[r : t] \to (r, t, -p_{4,z}(r, t), t)$ is an isomorphism from $\mathbb{P}^1$ onto the zero locus of $at + bp_{4,z}$ in $\mathbb{F}_3$. Let

$\gamma : \mathbb{P}^1 \times \mathbb{A}^1 \to \langle S_M \rangle \times \mathbb{A}^1$ be the composite of $([r : t], z) \to \big((r, t, -p_{4,z}(r, t), t), z\big)$ with $\nu \times \mathrm{id}_{\mathbb{A}^1}$. By (12.1),

$$\gamma([r : t], z) = \Big(\Big[-rp_{4,z} : \frac{4}{5}t(r^4 - p_{4,z}) : r^3t^2 : r^2t^3 : rt^4 : t^5\Big], z\Big). \tag{12.3}$$

Let $W \subset H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(5))$ be the span of r^3t^2, r^2t^3, rt^4, and t^5. Modulo $W \otimes \mathbb{C}[z]$, the first two forms in (12.3) are congruent to

$$-cr^5 - zc_1r^4t \qquad \text{and} \qquad \frac{4}{5}(1 - c)r^4t.$$

Thus the coefficient matrix of the six forms in (12.3) with respect to the basis $r^5, r^4t, \dots, t^5$ is upper triangular with diagonal entries $-c$, $\frac{4}{5}(1 - c)$, 1, 1, 1, 1. Hence the determinant of the coefficient matrix is $-\frac{4}{5}c(1 - c)$, independently of z. At $z = 1$, the map $[r : t] \to \gamma([r : t], 1)$ is the composite of the isomorphisms $\mathbb{P}^1 \to \tilde{\Gamma} \to \Gamma$. Since Γ is a rational normal quintic curve, it spans $\langle S_M \rangle$, so the six forms are linearly independent at $z = 1$. Thus $c(1 - c) \neq 0$, and the coefficient matrix belongs to $\mathrm{GL}_6(\mathbb{C}[z])$. Hence γ is the composite of the relative Veronese embedding of degree five $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{P}^5 \times \mathbb{A}^1$ with an automorphism of $\mathbb{P}^5 \times \mathbb{A}^1$ over $\mathbb{A}^1$, and γ is a closed immersion. Let

$$\mathcal{C} := \gamma(\mathbb{P}^1 \times \mathbb{A}^1) \subset S_M \times \mathbb{A}^1 \subset V_5 \times \mathbb{A}^1,$$

and let $\Gamma_z \subset V_5$ be the fiber of $\mathcal{C}$ over $z \in \mathbb{A}^1$. Then $\mathcal{C} \cong \mathbb{P}^1 \times \mathbb{A}^1$, any Γ_z is a rational normal quintic curve contained in S_M, and $\Gamma_1 = \Gamma$.

By **Step 1** and (12.2), $\rho(\lambda)$ maps the point $\big(r, t, -p_{4,z}(r, t), t\big)$ of $\mathbb{F}_3$ to

$$\big(r, \lambda^{-1}t, -p_{4,z}(r, t), \lambda^{-1}t\big) = \big(r, t', -p_{4,\lambda z}(r, t'), t'\big), \qquad t' := \lambda^{-1}t.$$

Thus $\rho(\lambda)\Gamma_z = \Gamma_{\lambda z}$ for any $\lambda \in \mathbb{G}_m$ and $z \in \mathbb{A}^1$. In particular $\Gamma_z = \rho(z)\Gamma$ for $z \neq 0$, and

$$\Gamma_0 = \Big\{\Big[-cr^5 : \frac{4}{5}(1 - c)r^4t : r^3t^2 : r^2t^3 : rt^4 : t^5\Big]\Big\}$$

is a ρ-invariant rational normal quintic curve. Moreover, $\mathcal{C}$ is invariant under the $\mathbb{G}_m$-action on $V_5 \times \mathbb{A}^1$ given by $\lambda \cdot (v, z) = (\rho(\lambda)v, \lambda z)$.

Step 3. In this step, we construct the blow-up $g : \mathcal{Y} \to \mathbb{A}^1$ of $V_5 \times \mathbb{A}^1$ along $\mathcal{C}$ and prove that the graded algebra $\bigoplus_m g_*\mathcal{O}_{\mathcal{Y}}(m(3h - 2R))$ has locally free pieces, base change, and generators in degrees at most three.

Let $\mathcal{Y} := \mathrm{Bl}_{\mathcal{C}}(V_5 \times \mathbb{A}^1)$, and let $g : \mathcal{Y} \to \mathbb{A}^1$ be the induced morphism. Let h be the pullback of $\mathcal{O}_{V_5}(1)$ to $\mathcal{Y}$, let R be the exceptional divisor of $\mathcal{Y} \to V_5 \times \mathbb{A}^1$, and let $D := 3h - 2R$. Since $\mathcal{C} \cong \mathbb{P}^1 \times \mathbb{A}^1$ is smooth, $\mathcal{Y}$ is a smooth fourfold, projective over $\mathbb{A}^1$. By [Ful98, Appendix B.6.7 and B.6.9], g is flat, and for $z \in \mathbb{A}^1$ the blow-up $\mathrm{Bl}_{\Gamma_z} V_5$ is a closed subscheme of the fiber $\mathcal{Y}_z$. The fiber $\mathcal{Y}_z$ is a Cartier divisor in the smooth fourfold $\mathcal{Y}$, so $\mathcal{Y}_z$ is Cohen–Macaulay of pure dimension three. Moreover, $\dim(R \cap \mathcal{Y}_z) = 2$, and $\mathcal{Y}_z \backslash R \cong V_5 \backslash \Gamma_z$. Thus $\mathcal{Y}_z$ is integral and $\mathcal{Y}_z = \mathrm{Bl}_{\Gamma_z} V_5$. Since $\mathcal{C}$ is flat over $\mathbb{A}^1$, the ideal of $\mathcal{C}$ restricts to the ideal of Γ_z in $V_5 \times \{z\}$, so $R|_{\mathcal{Y}_z}$ is the exceptional divisor of $\mathrm{Bl}_{\Gamma_z} V_5 \to V_5$. Thus

$$D|_{\mathcal{Y}_z} = 3h - 2R|_{\mathcal{Y}_z}.$$

Since Γ_z is a rational normal quintic curve in V_5, Theorem 2.23 implies that there exist a smooth prime Fano threefold X_z of genus 12, a line $L_z \subset X_z$, and a flop $\mathrm{Bl}_{L_z} X_z \dashrightarrow \mathrm{Bl}_{\Gamma_z} V_5$ such that $3h - 2R|_{\mathcal{Y}_z}$ is the strict transform on $\mathrm{Bl}_{\Gamma_z} V_5$ of the pullback of H to $\mathrm{Bl}_{L_z} X_z$. By Set-up 12.2(4) and $\Gamma_1 = \Gamma$, we have $(X_1, L_1) = (X, L)$. Since the flop is an isomorphism in codimension one between normal projective varieties, pullback of rational functions induces isomorphisms $H^0(\mathrm{Bl}_{\Gamma_z} V_5, mD|_{\mathcal{Y}_z}) \cong H^0(\mathrm{Bl}_{L_z} X_z, mH)$ compatible with multiplication. Since X_z is normal, $H^0(\mathrm{Bl}_{L_z} X_z, mH) = H^0(X_z, mH)$. Thus there is an isomorphism of graded $\mathbb{C}$-algebras

$$\bigoplus_{m \geq 0} H^0(\mathcal{Y}_z, mD|_{\mathcal{Y}_z}) \cong \bigoplus_{m \geq 0} H^0(X_z, mH). \tag{12.4}$$

Since X_z is a smooth prime Fano threefold of genus 12, Lemma 2.12(2) gives

$$h^0(X_z, mH) = N_m = \frac{11}{3}m^3 + \frac{11}{2}m^2 + \frac{23}{6}m + 1 \qquad \text{for any integer } m \geq 0,$$

independently of z. For $m \geq 0$, let $A_m := g_*\mathcal{O}_{\mathcal{Y}}(mD)$. Since g is flat and projective and $h^0(\mathcal{Y}_z, mD|_{\mathcal{Y}_z}) = N_m$ for any z, A_m is locally free of rank N_m and the natural map $A_m \otimes k(z) \to H^0(\mathcal{Y}_z, mD|_{\mathcal{Y}_z})$ is an isomorphism for any z by Grauert's theorem [Har77, Corollary III.12.9]. In particular $A_0 = \mathcal{O}_{\mathbb{A}^1}$.

Fix $z \in \mathbb{A}^1$. By Lemma 2.12(1), H is very ample on X_z. By Kodaira vanishing [Laz04a, Theorem 4.2.1], $H^i(X_z, mH) = 0$ for $i > 0$ and $m \geq 0$, since $mH - K_{X_z} = (m+1)H$ is ample. In particular

$$H^1(X_z, 2H) = H^2(X_z, H) = H^3(X_z, \mathcal{O}_{X_z}) = 0,$$

so $\mathcal{O}_{X_z}$ is 3-regular with respect to H. By Mumford's theorem [Laz04a, Theorem 1.8.5], the multiplication map

$$H^0(X_z, mH) \otimes H^0(X_z, H) \to H^0(X_z, (m+1)H)$$

is surjective for $m \geq 3$. By (12.4) and the base change isomorphisms above, the multiplication map $A_m \otimes A_1 \to A_{m+1}$ is surjective on the fiber over any z, hence surjective by Nakayama's lemma, for $m \geq 3$. Since the image of $A_1 \otimes A_1 \otimes A_1$ is contained in A_3, it follows that $A_{3k} \otimes A_3 \to A_{3k+3}$ is surjective for $k \geq 1$. Thus the graded $\mathcal{O}_{\mathbb{A}^1}$-algebra

$$\mathcal{A} := \bigoplus_{k \geq 0} A_{3k}$$

is generated by A_3, and $\mathcal{A} \otimes k(z) \cong \bigoplus_{k \geq 0} H^0(X_z, 3kH)$ is generated in degree one for any z.

Step 4. In this step, we define $\pi : \mathcal{X} \to \mathbb{A}^1$ and prove that π is smooth and projective, that $\mathcal{X}_z \cong X_z$ for any z, and that $-K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier and π-ample.

Let $\mathcal{X} := \operatorname{Proj}_{\mathbb{A}^1} \mathcal{A}$ and let $\pi : \mathcal{X} \to \mathbb{A}^1$ be the structure morphism. Since $\mathcal{A}$ is generated by A_3, $\mathcal{X}$ is a closed subscheme of the projective bundle $\operatorname{Proj}_{\mathbb{A}^1} \operatorname{Sym} A_3$ over $\mathbb{A}^1$, and $\mathcal{O}_{\mathcal{X}}(1)$ is very ample over $\mathbb{A}^1$. Each A_{3k} is a free $\mathbb{C}[z]$-module of finite rank. For a homogeneous element $u \in A_{3d}$ with $d \geq 1$, the coordinate ring of the open subset $D_+(u) \subset \mathcal{X}$ is the filtered colimit of the free $\mathbb{C}[z]$-modules A_{3dn} along multiplication by u, hence flat over $\mathbb{C}[z]$. Thus π is flat. Formation of Proj commutes with base change [Gro61, Proposition 3.5.3], so

$$\mathcal{X}_z \cong \operatorname{Proj} \bigoplus_{k \geq 0} H^0(X_z, 3kH) \cong X_z,$$

where the second isomorphism holds since $3H$ is very ample and $\bigoplus_k H^0(X_z, 3kH)$ is generated in degree one. Under the isomorphism $\mathcal{X}_z \cong X_z$, $\mathcal{O}_{\mathcal{X}}(1)|_{\mathcal{X}_z} \cong \mathcal{O}_{X_z}(3H)$. Since π is flat and any fiber is smooth, π is smooth [Har77, Theorem III.10.2], and any fiber of π is a smooth prime Fano threefold of genus 12.

Since π is smooth, $\Omega_{\mathcal{X}/\mathbb{A}^1}$ is locally free of rank 3. Thus $\omega_{\mathcal{X}/\mathbb{A}^1}$ is invertible and $\omega_{\mathcal{X}/\mathbb{A}^1}|_{\mathcal{X}_z} \cong \omega_{X_z}$ for any z [Har77, Proposition II.8.10]. Thus $K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier, and the line bundle $\mathcal{O}_{\mathcal{X}}(1) \otimes \omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes 3}$ is trivial on any fiber of π. Since any fiber is connected, $h^0(\mathcal{X}_z, \mathcal{O}_{\mathcal{X}_z}) = 1$ for any z. By [Har77, Corollary III.12.9], $\pi_*(\mathcal{O}_{\mathcal{X}}(1) \otimes \omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes 3})$ is a line bundle on $\mathbb{A}^1$, and the evaluation map

$$\pi^*\pi_*\big(\mathcal{O}_{\mathcal{X}}(1) \otimes \omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes 3}\big) \to \mathcal{O}_{\mathcal{X}}(1) \otimes \omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes 3}$$

is an isomorphism on any fiber, hence an isomorphism. Since $\operatorname{Pic}(\mathbb{A}^1) = 0$, we have

$$\mathcal{O}_{\mathcal{X}}(1) \otimes \omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes 3} \cong \mathcal{O}_{\mathcal{X}}.$$

Thus $\omega_{\mathcal{X}/\mathbb{A}^1}^{\otimes -3} \cong \mathcal{O}_{\mathcal{X}}(1)$, and $-K_{\mathcal{X}/\mathbb{A}^1}$ is π-ample.

Step 5. In this step, we construct the $\mathbb{G}_m$-action on $\mathcal{X}$, the equivariant trivialization over $\mathbb{A}^1 \backslash \{0\}$, and prove that the action on $\mathcal{X}_0$ is faithful.

Recall from **Step 2** that $\lambda \cdot (v, z) = (\rho(\lambda)v, \lambda z)$ defines a $\mathbb{G}_m$-action on $V_5 \times \mathbb{A}^1$ preserving $\mathcal{C}$. By the universal property of blowing up [Har77, Proposition II.7.14], the action lifts to $\mathcal{Y}$, commutes with g and the action $z \to \lambda z$ on $\mathbb{A}^1$, and preserves R. The preimage of $\rho(\mathbb{G}_m)$ in GL_7 is a two-dimensional torus, and the kernel of the surjection of the preimage onto $\rho(\mathbb{G}_m)$ is the connected group of scalar matrices. Hence ρ lifts to a one-parameter subgroup of GL_7, so $\mathcal{O}_{V_5}(1)$ is ρ-linearized. The ideal sheaf $\mathcal{O}_{\mathcal{Y}}(-R)$ is $\mathbb{G}_m$-invariant. Thus $\mathcal{O}_{\mathcal{Y}}(mD)$ is $\mathbb{G}_m$-linearized for any m, the multiplication maps $A_m \otimes A_{m'} \to A_{m+m'}$ are $\mathbb{G}_m$-equivariant, and $\mathbb{G}_m$ acts on $\mathcal{A}$ by graded automorphisms covering the action on $\mathbb{C}[z]$. Thus $\mathbb{G}_m$ acts on $\mathcal{X}$, and the action covers the standard action on $\mathbb{A}^1$. Since the action on $\mathcal{X}$ covers an action on $\mathbb{A}^1$, it induces a canonical linearization of $\omega^{\vee}_{\mathcal{X}/\mathbb{A}^1}$.

The isomorphism $V_5 \times \mathbb{G}_m \to V_5 \times \mathbb{G}_m$, $(v, z) \to (\rho(z)v, z)$, maps $\Gamma \times \mathbb{G}_m$ onto $\mathcal{C} \cap (V_5 \times \mathbb{G}_m)$, since $\Gamma_z = \rho(z)\Gamma$ for $z \neq 0$ (**Step 2**). It is $\mathbb{G}_m$-equivariant, where $\mathbb{G}_m$ acts on the first copy of $V_5 \times \mathbb{G}_m$ by $\lambda \cdot (v, z) = (v, \lambda z)$, since $(\rho(\lambda z)v, \lambda z) = (\rho(\lambda)\rho(z)v, \lambda z)$. Thus it induces a $\mathbb{G}_m$-equivariant isomorphism $\mathrm{Bl}_\Gamma V_5 \times \mathbb{G}_m \cong g^{-1}(\mathbb{G}_m)$ which preserves h and R, and hence, by (12.4) for $z = 1$, a $\mathbb{G}_m$-equivariant isomorphism

$$\pi^{-1}(\mathbb{A}^1 \backslash \{0\}) \cong X_1 \times (\mathbb{A}^1 \backslash \{0\}) = X \times (\mathbb{A}^1 \backslash \{0\})$$

over $\mathbb{A}^1 \backslash \{0\}$, with the trivial action on X. An isomorphism over the base preserves relative canonical bundles, so we obtain an equivariant isomorphism of $(\mathcal{X}, -K_{\mathcal{X}/\mathbb{A}^1})|_{\pi^{-1}(\mathbb{A}^1 \backslash \{0\})}$ with $(X, -K_X) \times (\mathbb{A}^1 \backslash \{0\})$.

Finally, $\mathbb{G}_m$ preserves $\mathcal{X}_0 \cong X_0$. Through (12.4) for $z = 0$, the automorphism of X_0 induced by $\lambda \in \mathbb{G}_m$ is the automorphism induced through the inverse link by $\rho(\lambda) \in \mathrm{Aut}(V_5; \Gamma_0)$. By Theorem 2.23 ([KPS18, Remark 5.2.6]), the inverse link induces an isomorphism $\mathrm{Aut}(V_5; \Gamma_0) \cong \mathrm{Aut}(X_0; L_0)$. Since ρ is injective by **Step 1**, the $\mathbb{G}_m$-action on $\mathcal{X}_0$ is faithful.

By **Step 4**, π is smooth and projective, and $-K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier and π-ample. Since π is smooth and $\mathbb{A}^1$ is smooth, $\mathcal{X}$ is smooth. The $\mathbb{G}_m$-action on $\mathcal{X}$ constructed above covers the standard action on $\mathbb{A}^1$, linearizes $-K_{\mathcal{X}/\mathbb{A}^1}$, and is compatible with the equivariant isomorphism over $\mathbb{A}^1 \backslash \{0\}$ above. Thus π is an anti-canonical test configuration of X (Definition 2.4) with smooth central fiber. □

Proposition 12.4. *Notation as in Set-up 12.2. Then X is K-semistable. In particular $A_X(F) \geq S_X(F)$ for every prime divisor F over X.*

Proof. By Lemma 12.3, there exists an anti-canonical test configuration $\pi : \mathcal{X} \to \mathbb{A}^1$ of X whose central fiber $\mathcal{X}_0$ is a smooth prime Fano threefold of genus 12 with a faithful $\mathbb{G}_m$-action. By Theorem 2.26, $\mathcal{X}_0$ is K-polystable, hence K-semistable. By Lemma 2.6, X is K-semistable. By Theorem 2.5(1), $A_X(F) \geq S_X(F)$ for every prime divisor F over X. □

12.2. **Inverse bisecants and curves of degree one and two.** In this subsection, we show that the inverse bisecant of a negative section s with $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ is a line of V_5 with normal bundle $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, and we prove $A_X(F) \geq S_X(F)$ for any prime divisor F over X whose center on X is a curve of degree one or two (Corollary 12.8).

Proposition 12.5 translates the condition that $M \cap \Gamma$ is non-reduced into the condition that the strict transform of Γ on the normalization $\mathbb{F}_3$ of S_M passes through the intersection point $s_0 \cap f_0$ of the conductor components, which is the situation of Lemma 12.3.

Proposition 12.5. *Notation as in Set-up 10.10. Assume that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Then the following hold.*

(1) $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, and the scheme $M \cap \Gamma$ has length two.
(2) The normalization of S_M is $\nu : \mathbb{F}_3 \to S_M$ with $\nu^* \mathcal{O}_{V_5}(1) \cong \mathcal{O}_{\mathbb{F}_3}(s_0 + 4f_0)$ and conductor $s_0 + f_0$, where s_0 is the negative section and f_0 is a fiber of $\mathbb{F}_3$.
(3) The strict transform $\tilde{\Gamma} \subset \mathbb{F}_3$ of Γ is a smooth curve with $\tilde{\Gamma} \sim s_0 + 4f_0$.
(4) The scheme $M \cap \Gamma$ is non-reduced of length two if and only if $\tilde{\Gamma}$ contains the point $s_0 \cap f_0$.

(5) If $M \cap \Gamma$ is non-reduced of length two, then X is K-semistable, and $A_X(F) \geq S_X(F)$ for any prime divisor F over X.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the strict transform of M on Y^+ has normal bundle $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, and that $M \cap \Gamma$ has length two.

Let $\phi_\Gamma : Y^+ = \mathrm{Bl}_\Gamma V_5 \to V_5$ be the blow-up, whose exceptional divisor is R, and let $M' \subset Y^+$ be the strict transform of M. By Set-up 10.10, M' is the flopped curve of s. Since $M \not\subset \Gamma$, the scheme $M \cap \Gamma$ is a Cartier divisor on M. Thus ϕ_Γ restricts to an isomorphism $M' \to M$, under which $\mathcal{O}_{Y^+}(-R)|_{M'}$ is the ideal sheaf of $M \cap \Gamma$ in M [Har77, Corollary II.7.15]. Hence $h^0(\mathcal{O}_{M\cap\Gamma}) = (R \cdot M')$. Since $Y \dashrightarrow Y^+$ is a flop, $(K_{Y^+} \cdot M') = 0$. Since $-K_{V_5} \sim 2h$, we have $K_{Y^+} \sim -2h + R$. Since $(h \cdot M') = (h \cdot M) = 1$, we obtain

$$h^0(\mathcal{O}_{M\cap\Gamma}) = (R \cdot M') = (K_{Y^+} \cdot M') + 2(h \cdot M') = 2.$$

This is the second part of (1).

By Proposition 11.11, $w(s) \in \{2, 3\}$. By Construction-Proposition 10.2(1), Reid's pagoda resolution of the flop near s is obtained on the side of Y^+ by $w(s)$ blow-ups of smooth rational curves starting with the blow-up of M', and, since $w(s) \geq 2$, the exceptional divisor $\mathbb{P}(N_{M'/Y^+})$ of the blow-up of M' is isomorphic to $\mathbb{F}_2$ (see also [BO95, Section 3]). Thus $N_{M'/Y^+} \cong \mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(a-2)$ for an integer a [Har77, Exercise V.2.6, Proposition V.2.8 and Example V.2.11.3]. Since

$$\deg N_{M'/Y^+} = -(K_{Y^+} \cdot M') - 2 = -2,$$

we have $a = 0$, that is, $N_{M'/Y^+} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

Step 2. In this step we compute the intersection of E_M with the exceptional divisor of ψ_M, and we show that the strict transform of Γ on $\mathrm{Bl}_M V_5$ is contained in that exceptional divisor.

Let $J \subset \mathrm{Bl}_M V_5$ be the exceptional divisor of ψ_M. By [Fuj26a, Theorem 3.6(4)], ψ_M is the blow-up of Q along a twisted cubic curve,

$$J \sim h - 2E_M, \qquad K_{\mathrm{Bl}_M V_5} \sim -2h + E_M.$$

By [Fuj26a, Theorem 3.6(5)], $J \cong \mathbb{F}_n$ with $n \in \{1, 3\}$, the morphism $\phi_M|_J : J \to S_M$ is the normalization, and $n = 3$ if and only if $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$. Let s_J be the negative section of J and f_J the class of a fiber of $J \to \psi_M(J)$. Since $(K_{\mathrm{Bl}_M V_5} \cdot f_J) = (J \cdot f_J) = -1$, we have

$$-2(h \cdot f_J) + (E_M \cdot f_J) = -1, \qquad (h \cdot f_J) - 2(E_M \cdot f_J) = -1,$$

so $(h \cdot f_J) = (E_M \cdot f_J) = 1$. Since $\phi_M(E_M) = M$ is a curve, $(h^2 \cdot E_M) = 0$, and

$$((h|_J)^2)_J = (h^2 \cdot J) = (h^3) - 2(h^2 \cdot E_M) = 5.$$

Since $(s_J^2)_J = -n$, this implies $h|_J \equiv s_J + \frac{n+5}{2} f_J$ [Har77, Propositions V.2.3 and V.2.9]. By adjunction, $K_J \sim (K_{\mathrm{Bl}_M V_5} + J)|_J \sim (-h - E_M)|_J$. Since $-K_J \equiv 2s_J + (n+2)f_J$ [Har77, Corollary V.2.11], we obtain

$$E_M|_J \equiv -K_J - h|_J \equiv s_J + \frac{n-1}{2} f_J.$$

Since $(E_M|_J \cdot s_J)_J = -n + \frac{n-1}{2} < 0$, the curve s_J is a component of $E_M|_J$, and $E_M|_J - s_J \in |\frac{n-1}{2} f_J|$. Thus the scheme $E_M \cap J$ is the curve s_J if $n = 1$, and $E_M \cap J = s_J + f_0$ for a fiber f_0 of J if $n = 3$.

By Lemma 11.7(3), $\Gamma \subset S_M$. Let $\tilde{\Gamma} \subset \mathrm{Bl}_M V_5$ be the strict transform of Γ. Since $M \cap \Gamma$ is a Cartier divisor on the smooth curve Γ, the morphism ϕ_M restricts to an isomorphism $\tilde{\Gamma} \to \Gamma$, which maps the scheme

$$W := \tilde{\Gamma} \cap E_M$$

isomorphically onto $M \cap \Gamma$ [Har77, Corollary II.7.15]. In particular $h^0(\mathcal{O}_W) = 2$ by **Step 1**. Since J is the strict transform of S_M and $\Gamma \not\subset M$, we have $\tilde{\Gamma} \subset J$. Hence $W = \tilde{\Gamma} \cap (E_M \cap J)$, and $W \subset E_M \cap J$.

Step 3. In this step we compare the normal bundles of M' and M.

Since ϕ_Γ restricts to an isomorphism $M' \to M$, the differential of ϕ_Γ induces a homomorphism $N_{M'/Y^+} \to N_{M/V_5}$ of $\mathcal{O}_M$-modules, which is an isomorphism outside $M \cap \Gamma$. For a nowhere vanishing section $v \in H^0(M, N_{M/V_5})$, let $C_v \subset E_M = \mathbb{P}(N_{M/V_5})$ be the section of $E_M \to M$ defined by the line subbundle $\mathcal{O}_M v \subset N_{M/V_5}$.

Claim 12.6. *Let $v \in H^0(M, N_{M/V_5})$ be a nowhere vanishing section. Then v is contained in the image of $H^0(M', N_{M'/Y^+})$ if and only if $W \subset C_v$.*

Proof. Let $p \in M \cap \Gamma$, and let $k \geq 1$ be the length of $M \cap \Gamma$ at p. Let (x, y, z) be local analytic coordinates of V_5 at p with $M = \{x = y = 0\}$. Replacing z with $z + \lambda x + \mu y$ for general $\lambda, \mu \in \mathbb{C}$ keeps $M = \{x = y = 0\}$, so we may assume that $z|_\Gamma$ is a local parameter of Γ at p. Then

$$\Gamma = \{x = \alpha(z),\ y = \beta(z)\}$$

near p with $\alpha(0) = \beta(0) = 0$, and the ideal of $M \cap \Gamma$ in $\mathcal{O}_{\Gamma,p} \cong \mathbb{C}\{z\}$ is $(\alpha, \beta) = (z^k)$. A linear change of the coordinates x, y also keeps $M = \{x = y = 0\}$, so we may assume that α has order k. Let $a := \beta/\alpha \in \mathbb{C}\{z\}$.

Near the point of M' over p, the threefold Y^+ has local coordinates (x, z, t) with $y - \beta(z) = t(x - \alpha(z))$, and $M' = \{x = 0,\ t = a(z)\}$. The differential of ϕ_Γ maps ∂_x to $\partial_x + t\partial_y$ and ∂_t to $(x - \alpha(z))\partial_y$, and it maps the tangent vector $\partial_z + a'\partial_t$ of M' to

$$\partial_z + (\beta' - a\alpha' - a'\alpha)\partial_y = \partial_z,$$

since $\beta = a\alpha$. Thus, in the basis ∂_x, ∂_y of N_{M/V_5} near p, the image of N_{M'/Y^+} near p is generated by $(1, a)$ and $(0, \alpha)$. In particular $N_{M'/Y^+} \to N_{M/V_5}$ is injective, and a section $v = (v_x, v_y)$ of N_{M/V_5} near p is a section of the image if and only if

$$v_y - a v_x \in (z^k).$$

Near the fiber of E_M over p, consider the chart of $\mathrm{Bl}_M V_5$ with coordinates (x, z, τ) and $y = x\tau$. On the chart, $E_M = \{x = 0\}$, $\tilde{\Gamma} = \{x = \alpha(z),\ \tau = a(z)\}$, and

$$W = \{x = 0,\ z^k = 0,\ \tau = a(z)\}$$

near p. If $v_x(0) \neq 0$, then $C_v = \{x = 0,\ \tau = v_y/v_x\}$ near the fiber over p, so $W \subset C_v$ near p if and only if $v_y/v_x - a \in (z^k)$, that is, $v_y - a v_x \in (z^k)$. If $v_x(0) = 0$, then $v_y(0) \neq 0$, so $v_y - a v_x \notin (z^k)$, and the point of C_v over p is not contained in the chart, while W is. Thus, near any point of $M \cap \Gamma$, v is a section of the image of N_{M'/Y^+} if and only if $W \subset C_v$. Since $N_{M'/Y^+} \to N_{M/V_5}$ is injective and an isomorphism outside $M \cap \Gamma$, the claim follows. □

Proof of Proposition 12.5 continued.

Step 4. In this step we prove the first part of (1).

The normal bundle of a line of V_5 is $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}$ or $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ [Fuj26a, Theorem 3.6(3)]. Suppose that $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}$. Then $n = 1$ in **Step 2**, so $E_M \cap J = s_J$ is an integral curve. Moreover $E_M \cong M \times \mathbb{P}^1$, any non-zero $v \in H^0(M, N_{M/V_5}) \cong \mathbb{C}^2$ is nowhere vanishing, and $C_v = M \times \{[v]\}$. Let f_M be the class of a fiber of $E_M \to M$. Then

$$(C_v^2)_{E_M} = 0, \qquad (C_v \cdot f_M)_{E_M} = 1, \qquad h|_{E_M} \equiv f_M.$$

By [IP99, Lemma 2.2.14], $(E_M^3) = -\deg N_{M/V_5} = 0$. Since $(E_M \cdot f_M) = -1$, this implies $-E_M|_{E_M} \equiv C_v$. Thus $J|_{E_M} \equiv 2C_v + f_M$, and

$$(C_v \cdot J|_{E_M})_{E_M} = (C_v \cdot (2C_v + f_M))_{E_M} = 1.$$

Since the integral curve $E_M \cap J$ has class $2C_v + f_M$ on E_M, we have $C_v \not\subset E_M \cap J$, and the scheme $C_v \cap (E_M \cap J)$ has length one. Since $W \subset E_M \cap J$ and $h^0(\mathcal{O}_W) = 2$ (**Step 2**), we have $W \not\subset C_v$ for any non-zero $v \in H^0(M, N_{M/V_5})$. By Claim 12.6, the image of $H^0(M', N_{M'/Y^+}) \to H^0(M, N_{M/V_5})$ is zero. Since $N_{M'/Y^+} \to N_{M/V_5}$ is injective, $H^0(M', N_{M'/Y^+}) = 0$. This contradicts **Step 1**, since $h^0(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)) = 1$. Thus $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$.

Step 5. In this step we prove (2), (3), (4), and (5).

By (1) and **Step 2**, $n = 3$, and $\nu := \phi_M|_J : J \cong \mathbb{F}_3 \to S_M$ is the normalization [Fuj26a, Theorem 3.6(4)(5)]. Let $s_0 := s_J$, and let f_0 be the fiber of **Step 2**, so that $E_M \cap J = s_0 + f_0$ and $\nu^{-1}(M) = s_0 \cup f_0$. By **Step 2**, $\nu^*\mathcal{O}_{V_5}(1) \cong \mathcal{O}_J(h|_J) \cong \mathcal{O}_{\mathbb{F}_3}(s_0 + 4f_0)$. By Lemma 12.1, the conductor of ν is $s_0 + f_0$ (see also [FT92, Theorem, parts (B)(1), (2)]). This proves (2).

By (1), M and $\Gamma \subset S_M$ satisfy the assumptions of Lemma 12.1, which proves (3).

We prove (4). By **Step 2**, $W = \tilde{\Gamma} \cap (s_0 \cup f_0)$, and ϕ_M maps W isomorphically onto $M \cap \Gamma$. Since $(\tilde{\Gamma} \cdot s_0)_J = (\tilde{\Gamma} \cdot f_0)_J = 1$ (Lemma 12.1), the schemes $\tilde{\Gamma} \cap s_0$ and $\tilde{\Gamma} \cap f_0$ are reduced points. Since $(s_0 \cdot f_0)_J = 1$, the intersection $s_0 \cap f_0$ is one point x_0. Suppose that $M \cap \Gamma$ is non-reduced of length two. Then $M \cap \Gamma$ is supported at one point, so W is supported at one point. Since $\tilde{\Gamma} \cap s_0$ and $\tilde{\Gamma} \cap f_0$ are non-empty subsets of the support of W, both are equal to the support of W, so $x_0 \in \tilde{\Gamma}$. Conversely, suppose that $x_0 \in \tilde{\Gamma}$. Then $\tilde{\Gamma} \cap s_0 = \tilde{\Gamma} \cap f_0 = \{x_0\}$, so W is supported at x_0. Since $h^0(\mathcal{O}_W) = 2$, the scheme $M \cap \Gamma \cong W$ is non-reduced of length two. This proves (4).

We prove (5). Assume that $M \cap \Gamma$ is non-reduced of length two. By (1), (2), (3), and (4), M, S_M, ν, s_0, f_0, Γ, and $\tilde{\Gamma}$ satisfy conditions (1), (2), and (3) of Set-up 12.2. That is, M and Γ are as in Set-up 12.2, and, since the line link and the inverse link of Theorem 2.23 are mutually inverse [KPS18, Theorem 5.2.2(ii) and the sentence after it], X is the threefold associated there with (V_5, Γ), and L is the line marked by the inverse link. By Proposition 12.4, X is K-semistable, and $A_X(F) \geq S_X(F)$ for any prime divisor F over X. This proves (5). □

If $w(s) = 3$, that is, L intersects no other line (Proposition 10.8), then $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ by Definition-Proposition 2.21 and Set-up 10.10, and $M \cap \Gamma$ is non-reduced of length two by Lemma 12.7, so the conclusion of Proposition 12.5(5) holds for X.

Lemma 12.7. *Notation as in Set-up 10.10. Assume that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$. Then the following hold.*

(1) If $M \cap \Gamma$ consists of two distinct points, then $w(s) = 2$.
(2) If $w(s) = 3$, then $M \cap \Gamma$ is non-reduced of length two, the strict transform of Γ in the normalization $\mathbb{F}_3$ of S_M contains $s_0 \cap f_0$, where $s_0 + f_0$ is the conductor, with s_0 the negative section and f_0 a fiber, and X is K-semistable.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we write the curve $E_M \cap J$ on E_M, where J is the exceptional divisor of ψ_M.

Let $J \subset \mathrm{Bl}_M V_5$ be the exceptional divisor of ψ_M, and let $\tilde{\Gamma} \subset \mathrm{Bl}_M V_5$ be the strict transform of Γ. By Proposition 12.5(1) and [Fuj26a, Theorem 3.6(4)(5)], we have $J \sim h - 2E_M$, $K_{\mathrm{Bl}_M V_5} \sim -2h + E_M$, and $\nu := \phi_M|_J : J \cong \mathbb{F}_3 \to S_M$ is the normalization. Let s_0 be the negative section of J, and let f_0 be the fiber of J with $\nu^{-1}(M) = s_0 \cup f_0$ (Lemma 12.1). By Proposition 12.5(2), $h|_J \sim s_0 + 4f_0$. By adjunction,

$$E_M|_J \sim -K_J - h|_J \sim (2s_0 + 5f_0) - (s_0 + 4f_0) = s_0 + f_0.$$

Since $E_M|_J$ is supported on $\nu^{-1}(M) = s_0 \cup f_0$, we have $E_M|_J = s_0 + f_0$. In particular, the scheme $E_M \cap J$ is the reduced curve $s_0 \cup f_0$. Since $J|_J \sim (h - 2E_M)|_J \sim -s_0 + 2f_0$,

$$(J \cdot s_0) = ((-s_0 + 2f_0) \cdot s_0)_J = 5, \qquad (J \cdot f_0) = ((-s_0 + 2f_0) \cdot f_0)_J = -1.$$

By Proposition 12.5(1), $E_M = \mathbb{P}(N_{M/V_5}) \cong \mathbb{F}_2$. Let σ_M be the negative section of E_M and f_M the class of a fiber of $E_M \to M$. Then $h|_{E_M} \sim f_M$. By [IP99, Lemma 2.2.14],

$(E_M^3) = -\deg N_{M/V_5} = 0$. Since $(E_M \cdot f_M) = -1$, this implies $-E_M|_{E_M} \sim \sigma_M + f_M$. Thus

$$J|_{E_M} \sim f_M + 2(\sigma_M + f_M) = 2\sigma_M + 3f_M.$$

Since $(h \cdot s_0) = ((s_0 + 4f_0) \cdot s_0)_J = 1$ and $(h \cdot f_0) = ((s_0 + 4f_0) \cdot f_0)_J = 1$, the morphism ϕ_M maps s_0 and f_0 isomorphically onto the line M. Thus s_0 and f_0 are sections of $E_M \to M$. A section $C \neq \sigma_M$ of $E_M \to M$ satisfies $C \equiv \sigma_M + cf_M$ with $c = (C \cdot \sigma_M)_{E_M} + 2 \geq 2$. Since $s_0 + f_0 = J|_{E_M} \sim 2\sigma_M + 3f_M$ as divisors on E_M, the classes of s_0 and f_0 on E_M are σ_M and $\sigma_M + 3f_M$ in some order. Since

$$(J \cdot \sigma_M) = ((2\sigma_M + 3f_M) \cdot \sigma_M)_{E_M} = -1,$$

the equalities $(J \cdot s_0) = 5$ and $(J \cdot f_0) = -1$ imply that $f_0 = \sigma_M$ as curves, and $s_0 \equiv \sigma_M + 3f_M$ on E_M.

Step 2. In this step we blow up $\tilde{\Gamma}$ and find the intersection points of J with the flopping curves.

Assume that $M \cap \Gamma$ consists of two distinct points p_1 and p_2. By Proposition 12.5(1), $M \cap \Gamma$ has length two, so $M \cap \Gamma$ is reduced, and M and Γ intersect transversally at p_1 and p_2. By [Har77, Corollary II.7.15], ϕ_M restricts to an isomorphism $\tilde{\Gamma} \to \Gamma$, and $W := \tilde{\Gamma} \cap E_M$ consists of two reduced points. Since $\tilde{\Gamma} \subset J$, we have $W \subset s_0 \cup f_0$. By Proposition 12.5(3), $(\tilde{\Gamma} \cdot s_0)_J = (\tilde{\Gamma} \cdot f_0)_J = 1$, and by Proposition 12.5(4), $s_0 \cap f_0 \not\subset \tilde{\Gamma}$. Thus, after renumbering p_1 and p_2,

$$\tilde{\Gamma} \cap f_0 = \{q_1\}, \qquad \tilde{\Gamma} \cap s_0 = \{q_2\}, \qquad \phi_M(q_i) = p_i,$$

with $q_1 \notin s_0$ and $q_2 \notin f_0$. Let $f_{M,i}$ be the fiber of $E_M \to M$ over p_i. Since $(\sigma_M \cdot f_M)_{E_M} = ((\sigma_M + 3f_M) \cdot f_M)_{E_M} = 1$ and $f_0 = \sigma_M$, we have $f_{M,1} \cap f_0 = \{q_1\}$ and $f_{M,2} \cap s_0 = \{q_2\}$. Let q_1' be the point of $f_{M,1} \cap s_0$ and q_2' the point of $f_{M,2} \cap f_0$. Then $q_1' \neq q_1$ since $q_1 \notin s_0$, and $q_2' \neq q_2$ since $q_2 \notin f_0$. Moreover $q_1' \notin f_0$ and $q_2' \notin s_0$, and $q_1', q_2' \notin \tilde{\Gamma}$ since $\tilde{\Gamma} \cap E_M = \{q_1, q_2\}$ and $f_{M,1} \cap f_{M,2} = \emptyset$.

Let $g_1 : Y_1 := \mathrm{Bl}_{\tilde{\Gamma}}(\mathrm{Bl}_M V_5) \to \mathrm{Bl}_M V_5$ be the blow-up with exceptional divisor R_1, and let $E_{M,1}$ and J_1 be the strict transforms of E_M and of J. Since W is reduced, $E_{M,1} = g_1^* E_M$ is the blow-up of E_M at q_1 and q_2 [Har77, Corollary II.7.15]. Let e_1 and e_2 be the exceptional curves of $E_{M,1} \to E_M$. Since $\tilde{\Gamma}$ is a Cartier divisor on the smooth surface J, the morphism g_1 maps J_1 isomorphically onto J, and $J_1 = g_1^* J - R_1$. Let $\ell_i \subset E_{M,1}$ be the strict transform of $f_{M,i}$. Then $\ell_i \equiv f_M - e_i$ on $E_{M,1}$ and $(\ell_i^2)_{E_{M,1}} = -1$. Since the scheme $f_{M,i} \cap \tilde{\Gamma} \subset W$ is the reduced point q_i, we have $(R_1 \cdot \ell_i) = 1$ [Har77, Corollary II.7.15]. Thus

$$(J_1 \cdot \ell_i) = (J \cdot f_{M,i}) - (R_1 \cdot \ell_i) = ((2\sigma_M + 3f_M) \cdot f_M)_{E_M} - 1 = 1.$$

Since $q_i' \notin \tilde{\Gamma}$, g_1 is an isomorphism near q_i', and ℓ_i intersects J_1 at the point over q_i', which we denote again by q_i'. Since $\ell_i \not\subset J_1$, the scheme $J_1 \cap \ell_i$ is the reduced point q_i'. Under the isomorphism $J_1 \cong J$, we have $q_1' \in s_0 \backslash f_0$ and $q_2' \in f_0 \backslash s_0$.

Step 3. In this step we show that the birational map from Y_1 to the blow-up Y_2 of the strict transform of M on Y^+ is the composite of two Atiyah flops, and we locate the negative section of the new exceptional surface.

Let $M' \subset Y^+$ be the strict transform of M, let $g_2 : Y_2 := \mathrm{Bl}_{M'} Y^+ \to Y^+$ be the blow-up, and let P^+ be the exceptional divisor of g_2. By Proposition 11.11, $w(s) \geq 2$, so $P^+ \cong \mathbb{F}_2$ by Construction-Proposition 10.2(1) (see also [BO95, Section 3]). Let σ^+ be the negative section of P^+ and f_{P^+} the class of a fiber of $P^+ \to M'$.

Since M and Γ intersect transversally at p_1 and p_2, the birational map $Y_1 \dashrightarrow Y_2$ over V_5 obtained by exchanging the order of the blow-ups of M and Γ is a flop whose flopping curves are ℓ_1 and ℓ_2 [Fuj26a, Example 2.10]. Since $E_{M,1}$ is smooth, $(\ell_i^2)_{E_{M,1}} = -1$, and $(E_{M,1} \cdot \ell_i) = (g_1^* E_M \cdot \ell_i) = (E_M \cdot f_{M,i}) = -1$, the exact sequence

$$0 \to N_{\ell_i/E_{M,1}} \to N_{\ell_i/Y_1} \to N_{E_{M,1}/Y_1}|_{\ell_i} \to 0$$

has both outer terms isomorphic to $\mathcal{O}_{\mathbb{P}^1}(-1)$. Since

$$\mathrm{Ext}^1(\mathcal{O}_{\mathbb{P}^1}(-1),\mathcal{O}_{\mathbb{P}^1}(-1)) = H^1(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}) = 0,$$

the sequence splits, and $N_{\ell_i/Y_1} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Let $\hat{g}_1 : \hat{Y} := \mathrm{Bl}_{\ell_1\cup\ell_2} Y_1 \to Y_1$ be the blow-up with exceptional divisors D_1 and D_2. By [Rei83, Definition (5.3) and the sentence after it, and (5.5)], $D_i \cong \ell_i \times \mathbb{P}^1$ with $\hat{g}_1|_{D_i}$ the first projection, and there exists a morphism $\hat{g}_2 : \hat{Y} \to Y_2$ which is an isomorphism outside $D_1 \cup D_2$ and which contracts D_i onto the flopped curve $\ell_i^+ \subset Y_2$ by the second projection $D_i \to \mathbb{P}^1 \cong \ell_i^+$.

Since $Y_1 \dashrightarrow Y_2$ is an isomorphism in codimension one, and $E_{M,1}$ and P^+ are the only prime divisors on Y_1 and on Y_2 whose images on V_5 are M, the divisor P^+ is the strict transform of $E_{M,1}$. Let $\hat{E} \subset \hat{Y}$ be the strict transform of $E_{M,1}$. Since ℓ_i is a Cartier divisor on $E_{M,1}$, $\hat{g}_1$ maps $\hat{E}$ isomorphically onto $E_{M,1}$, and

$$\hat{E} \cap D_i = \mathbb{P}(N_{\ell_i/E_{M,1}}) \subset \mathbb{P}(N_{\ell_i/Y_1}) = D_i.$$

Since $N_{\ell_i/E_{M,1}} \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ is a line subbundle of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, it is given by a constant vector, so $\hat{E} \cap D_i = \ell_i \times \{t_i\}$ for a point $t_i \in \mathbb{P}^1$, and $\hat{E} \cap D_i$ is contracted by $\hat{g}_2$. Thus $\hat{g}_2|_{\hat{E}} : \hat{E} \to P^+$ is a birational morphism of smooth projective surfaces which contracts exactly the curves $\hat{E} \cap D_1$ and $\hat{E} \cap D_2$. Under $\hat{E} \cong E_{M,1}$, it is the contraction of the (-1)-curves ℓ_1 and ℓ_2.

Let $\sigma_{M,1} \subset E_{M,1}$ be the strict transform of σ_M. Since $q_1 \in f_0 = \sigma_M$ and $q_2 \notin \sigma_M$, we have $\sigma_{M,1} \equiv \sigma_M - e_1$, and

$$(\sigma_{M,1}^2)_{E_{M,1}} = -3, \qquad (\sigma_{M,1}\cdot\ell_1)_{E_{M,1}} = ((\sigma_M - e_1)\cdot(f_M - e_1))_{E_{M,1}} = 0,$$

$$(\sigma_{M,1}\cdot\ell_2)_{E_{M,1}} = ((\sigma_M - e_1)\cdot(f_M - e_2))_{E_{M,1}} = 1.$$

By [Har77, Propositions V.3.2 and V.3.6], the image of $\sigma_{M,1}$ on P^+ has self-intersection $-3 + 0^2 + 1^2 = -2$. Since σ^+ is the only integral curve on $P^+ \cong \mathbb{F}_2$ with negative self-intersection, the image of $\sigma_{M,1}$ is σ^+. Since $\sigma_M = f_0$ (**Step 1**), σ^+ is the strict transform on Y_2 of the curve $f_0 \subset J_1$.

Step 4. In this step we show that the strict transform J_2 of J_1 on Y_2 is the blow-up of J_1 at q_1' and q_2'.

Let $J_2 \subset Y_2$ and $\hat{J} \subset \hat{Y}$ be the strict transforms of J_1. By **Step 2** and [Har77, Corollary II.7.15], $\hat{g}_1|_{\hat{J}} : \hat{J} \to J_1$ is the blow-up of J_1 at q_1' and q_2', and $\hat{J} \cap D_i = \{q_i'\} \times \mathbb{P}^1$ is the fiber of $D_i \to \ell_i$ over q_i'. By **Step 3**, $\hat{g}_2$ maps $\{q_i'\} \times \mathbb{P}^1$ isomorphically onto ℓ_i^+, and any curve contracted by $\hat{g}_2$ is contained in $D_1 \cup D_2$. Hence $\hat{g}_2|_{\hat{J}} : \hat{J} \to J_2$ is finite and birational, and $\ell_i^+ \subset J_2$. By Lemma 10.3 for the flop of ℓ_i, which has width one, and since $\ell_i \not\subset J_1$, we have

$$\hat{g}_2^* J_2 = \hat{g}_1^* J_1 + (J_1 \cdot \ell_i) D_i = \hat{J} + D_i$$

near D_i. Thus J_2 has multiplicity one along ℓ_i^+. Since $J_2 \backslash (\ell_1^+ \cup \ell_2^+) \cong \hat{J} \backslash (D_1 \cup D_2)$ is smooth, J_2 is smooth outside a finite subset of $\ell_1^+ \cup \ell_2^+$. Since J_2 is a Cartier divisor on the smooth threefold Y_2, J_2 is normal [Har77, Proposition II.8.23(b)]. Since $\hat{g}_2|_{\hat{J}}$ is finite and birational and J_2 is normal, $(\hat{g}_2|_{\hat{J}})_* \mathcal{O}_{\hat{J}} = \mathcal{O}_{J_2}$, so $\hat{g}_2|_{\hat{J}}$ is an isomorphism. Thus J_2 is smooth, and $J_2 \cong \mathrm{Bl}_{q_1',q_2'} J_1$. Under the isomorphism $J_2 \cong \mathrm{Bl}_{q_1',q_2'} J_1$, the curve σ^+ is the strict transform of f_0 by **Step 3**.

Step 5. In this step we compute N_{σ^+/Y_2} and prove (1).

Since $(f_0^2)_J = 0$, $q_2' \in f_0$, and $q_1' \notin f_0$, **Step 4** and [Har77, Propositions V.3.2 and V.3.6] imply

$$((\sigma^+)^2)_{J_2} = 0 - 1 = -1,$$

so $N_{\sigma^+/J_2} \cong \mathcal{O}_{\mathbb{P}^1}(-1)$. Since $Y \dashrightarrow Y^+$ is a flop, $(K_{Y^+}\cdot M') = 0$, so $\deg N_{M'/Y^+} = -2$ and $((P^+)^3) = -\deg N_{M'/Y^+} = 2$ [IP99, Lemma 2.2.14]. Since $(P^+ \cdot f_{P^+}) = -1$ and $(\sigma^+)^2 = -2$ on

P^+, this implies $-P^+|_{P^+} \equiv \sigma^+ + 2f_{P^+}$. Since $K_{Y_2} \sim g_2^* K_{Y^+} + P^+$ and g_2 maps σ^+ isomorphically onto M',

$$(K_{Y_2} \cdot \sigma^+) = (K_{Y^+} \cdot M') - ((\sigma^+ + 2f_{P^+}) \cdot \sigma^+)_{P^+} = 0 - (-2+2) = 0.$$

Thus $\deg N_{\sigma^+/Y_2} = -(K_{Y_2} \cdot \sigma^+) - 2 = -2$. Since J_2 is smooth along σ^+, there is an exact sequence

$$0 \to N_{\sigma^+/J_2} \to N_{\sigma^+/Y_2} \to N_{J_2/Y_2}|_{\sigma^+} \to 0,$$

in which $N_{\sigma^+/J_2} \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ and $\deg N_{J_2/Y_2}|_{\sigma^+} = -2 - (-1) = -1$. Since

$$\operatorname{Ext}^1(\mathcal{O}_{\mathbb{P}^1}(-1), \mathcal{O}_{\mathbb{P}^1}(-1)) = H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0,$$

the sequence splits, and

$$N_{\sigma^+/Y_2} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}.$$

By Construction-Proposition 10.2(1), the first blow-ups of Reid's pagoda resolution of the flop of s are $p_T : T \to Y$ on the side of Y and $g_2 : Y_2 \to Y^+$ on the side of Y^+, with common exceptional divisor $P \cong P^+$. By [BO95, Section 3], the birational map $T \dashrightarrow Y_2$ induced by $Y \dashrightarrow Y^+$ is, over a neighborhood of P, the flop of σ to σ^+, and $w(\sigma^+) = w(s) - 1 \geq 1$. Since $N_{\sigma^+/Y_2} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, [Rei83, (5.3), (5.9)] implies $w(\sigma^+) = 1$. Thus $w(s) = 2$. This proves (1).

Step 6. In this step we prove (2).

Assume that $w(s) = 3$. By (1), $M \cap \Gamma$ does not consist of two distinct points. By Proposition 12.5(1), $M \cap \Gamma$ has length two, so $M \cap \Gamma$ is non-reduced of length two. By Proposition 12.5(4), the strict transform of Γ in the normalization $\mathbb{F}_3$ of S_M contains $s_0 \cap f_0$, and by Proposition 12.5(5), X is K-semistable. This proves (2). □

Corollary 12.8 collects the bounds for curves of degree one and two proved in Section 11 and in this subsection.

Corollary 12.8. *Let X be a smooth prime Fano threefold of genus 12 and $C \subset X$ an integral curve with $H \cdot C \in \{1, 2\}$. Let E be the exceptional divisor of $\operatorname{Bl}_C X \to X$. Then any prime divisor F over X with $\operatorname{center}_X F = C$ satisfies $A_X(F) \geq S_X(F)$. More precisely, the following hold.*

(1) If $H \cdot C = 2$, then $A_X(F) - S_X(F) \geq \frac{59}{88} \operatorname{mult}_F E > 0$.
(2) If C is a non-special line, then $A_X(F) - S_X(F) \geq \frac{5}{11} \operatorname{mult}_F E > 0$.
(3) If $C = L$ is a special line and $A_X(F) = S_X(F)$, then, in the notation of Set-up 10.10, $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, $\operatorname{center}_T F = \sigma$, M is a line of V_5 with normal bundle $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, and $M \cap \Gamma$ is non-reduced of length two. Moreover, if $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and $M \cap \Gamma$ is non-reduced of length two, then X is K-semistable.

Proof. By Proposition 11.2, C is a smooth conic, a non-special line, or a special line. In the notation of Set-up 10.10 for a special line L, the cases are listed in Table 15.

The rows of Table 15 exhaust all prime divisors F over X with $\operatorname{center}_X F = C$. For a special line L, $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ or $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ by Set-up 10.10. If $\operatorname{center}_Y F = s$, then $F = P$ or $\operatorname{center}_T F \subset P$, so $\operatorname{center}_T F = \sigma$ or $\operatorname{center}_T F \neq \sigma$. If $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, then $M \cap \Gamma$ has length two by Proposition 12.5(1), so $M \cap \Gamma$ consists of two distinct points or is non-reduced of length two.

The first two rows of Table 15 prove (1) and (2). The bounds in the first six rows are positive, so $A_X(F) = S_X(F)$ implies that F is in the last row. In the last row, $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ by Proposition 12.5(1). This proves the first part of (3), and the second part of (3) is Proposition 12.5(5). □

Curve	Prime divisor F	Bound for $A_X(F) - S_X(F)$	Reference
$H \cdot C = 2$	$\operatorname{center}_X F = C$	$\geq \frac{59}{88} \operatorname{mult}_F E > 0$	Proposition 11.6
non-special line	$\operatorname{center}_X F = C$	$\geq \frac{5}{11} \operatorname{mult}_F E > 0$	Proposition 11.4
special line L	$F = E$ or $\operatorname{center}_Y F \neq s$	$\geq \frac{5}{11} \operatorname{mult}_F E > 0$	Proposition 11.2
L special, $N_{s/Y} \cong \mathcal{O}(-1)^{\oplus 2}$	$\operatorname{center}_Y F = s$	$\geq \frac{5}{22} \operatorname{mult}_F P > 0$	Proposition 11.10
L special, $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$	$\operatorname{center}_Y F = s$, and $F = P$ or $\operatorname{center}_T F \neq \sigma$	$\geq \frac{5}{22} \operatorname{mult}_F P > 0$	Proposition 11.11
L special, $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$, $M \cap \Gamma$ two distinct points	$\operatorname{center}_T F = \sigma$	$\geq \frac{15}{44} \operatorname{mult}_F P > 0$	Proposition 11.12
L special, $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$, $M \cap \Gamma$ non-reduced	$\operatorname{center}_T F = \sigma$	≥ 0, and X is K-semistable	Proposition 12.5(5)

TABLE 15. Bounds for $A_X(F) - S_X(F)$ for prime divisors F over curves of degree one and two in genus 12.

12.3. **Special lines meeting no other line, and pointed deformations.** In this subsection, we show that a special line which intersects no other line forces a smooth anti-canonical test configuration whose central fiber carries a faithful $\mathbb{G}_m$-action (Proposition 12.9), and we determine the points x of incidence type 3 on special lines, that is, the points x such that $\Sigma(X, x)$ is curvilinear of length three and supported at one line (Definition 2.11 and Proposition 12.10).

Lemma 12.3 starts from a quintic whose strict transform in $\mathbb{F}_3$ contains $s_0 \cap f_0$. Conversely, if $L \subset X$ is a special line which intersects no other line, then the line link of Theorem 2.23 maps (X, L) to (V_5, Γ) for such a quintic Γ, and hence X has a smooth degeneration to a member with a faithful $\mathbb{G}_m$-action (Proposition 12.9). Theorems 18.1 and 19.1 pass from the incidence of lines to degenerations through the following proposition.

Proposition 12.9. *Let X be a smooth prime Fano threefold of genus 12, and let $L \subset X$ be a special line which intersects no other line on X. Then the line link of Theorem 2.23 maps (X, L) to a pair (V_5, Γ) such that the unique hyperplane section of V_5 containing Γ is singular along a line $M \subset V_5$ with $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$. The normalization of the hyperplane section is $\mathbb{F}_3$ with conductor $s_0 + f_0$ for the negative section s_0 and a fiber f_0, and the strict transform of Γ in $\mathbb{F}_3$ belongs to $|s_0 + 4f_0|$ and contains $s_0 \cap f_0$. That is, M and Γ are as in Set-up 12.2, and X is the threefold associated there with (V_5, Γ). Hence X admits an anti-canonical test configuration $\pi : \mathcal{X} \to \mathbb{A}^1$ which is a smooth morphism, all of whose fibers are smooth prime Fano threefolds of genus 12, and whose central fiber carries a faithful induced $\mathbb{G}_m$-action.*

Proof. Let Y, s, P, M, and S_M be as in Set-up 10.10. By Proposition 10.8, the width of s is 3. By Construction-Proposition 10.2, in which $T_1 = \operatorname{Bl}_s Y$ and $P_1 = P$, we have $P \cong \mathbb{F}_2$, so $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ (Set-up 10.10). By Lemma 12.7, $M \cap \Gamma$ is non-reduced of length two. By Proposition 12.5(1)–(4), $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, the normalization of S_M is $\mathbb{F}_3$ with conductor $s_0 + f_0$, and the strict transform $\tilde{\Gamma} \in |s_0 + 4f_0|$ of Γ contains $s_0 \cap f_0$. Since $\Gamma \subset S_M$ (Lemma 11.7(3)) and the hyperplane section of V_5 containing Γ is unique (Theorem 2.23), it is S_M. Thus M and Γ satisfy (1)–(3) of Set-up 12.2. Since the line link and the inverse link of Theorem 2.23 are inverse to each other [KPS18, Theorem 5.2.2], (4) of Set-up 12.2 holds. The existence of π now follows from Lemma 12.3. □

Proposition 12.9 does not exclude that π is a product test configuration, and X may itself admit a faithful $\mathbb{G}_m$-action.

Next we locate the points of incidence type 3 on a special line L, that is, the points $x \in L$ such that $\Sigma(X, x)$ is curvilinear of length three and supported at L. The hypothesis of Theorem 14.1 excludes such points. By Proposition 12.10, L contains at most one such point, and, in the notation of Set-up 10.10, it is the image of the intersection point of σ with the curve $E' \cap P$.

Proposition 12.10. *Notation as in Set-up 10.1, with L special. Consider the exact sequence*

$$0 \to N_{s/E} \to N_{s/Y} \to N_{E/Y}|_s \to 0,$$

where $N_{s/E} \cong \mathcal{O}_{\mathbb{P}^1}(-3)$ and $N_{E/Y}|_s \cong \mathcal{O}_{\mathbb{P}^1}(1)$. Let $\epsilon \in H^1(\mathbb{P}^1, \mathcal{O}(-4))$ be the extension class of the sequence and $\partial_\epsilon : H^0(\mathcal{O}(1)) \to H^1(\mathcal{O}(-3))$ the connecting map. For $x \in L$ let $\ell_x := H^0(\mathcal{O}(1)(-x)) \subset H^0(\mathcal{O}(1))$. Then a non-zero first-order deformation of L containing x extends to a deformation of L over $\mathbb{C}[t]/(t^3)$ containing $x \times \operatorname{Spec} \mathbb{C}[t]/(t^3)$ if and only if $\partial_\epsilon(\ell_x) = 0$. Hence the following conditions are equivalent.

(1) $\Sigma(X, x)$ is curvilinear of length three and supported at L.
(2) $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ and x is the unique zero of the image in $\mathcal{O}(1)$ of a non-zero section of $N_{s/Y}$.

In case (2), let T, $P \cong \mathbb{F}_2$, σ, and E' be as in Set-up 10.10, and let $C_0 := E' \cap P$. Then C_0 and σ intersect transversally in one point q_0, which maps to x. A special line L with $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ has exactly one point as in (1), and a special line with $N_{s/Y} \cong \mathcal{O}(-1)^{\oplus 2}$ has none. For any point as in (1), the inverse bisecant M is a line of V_5 with normal bundle $\mathcal{O}(1) \oplus \mathcal{O}(-1)$. If $M \cap \Gamma$ is non-reduced or the width of s is three, then X is K-semistable. Otherwise the width of s is two and $M \cap \Gamma$ consists of two distinct points.

Proof. The proof is long, so we divide it into several steps. Identify L with $\mathbb{P}^1$ and s with L via $E \to L$. By Proposition 11.2, $N_{L/X} = N_+ \oplus N_-$ with $N_+ \cong \mathcal{O}(1)$ and $N_- \cong \mathcal{O}(-2)$. Then $N_{s/E} \cong N_- \otimes N_+^{-1}$ and $N_{E/Y}|_s \cong N_+$.

Step 1. In this step we express ϵ by the quadratic transition functions of coordinates of X along L.

The exact sequence $0 \to T_L \to T_X|_L \to N_{L/X} \to 0$ splits, since

$$\operatorname{Ext}^1(N_+ \oplus N_-, T_L) = H^1(L, \mathcal{O}(1) \oplus \mathcal{O}(4)) = 0.$$

Cover L by affine open subsets U_i, and pick formal coordinates (z_i, u_i, v_i) of X along U_i such that $L = (u_i = v_i = 0)$, $z_i|_{U_i}$ is a coordinate on U_i, and the linear parts of u_i and v_i are compatible with the splitting, u_i along N_+ and v_i along N_-. Then on $U_i \cap U_j$,

$$z_j = \varphi_{ij}(z_i) + O((u_i, v_i)^2), \qquad u_j = g_{ij} u_i + O((u_i, v_i)^2),$$

$$v_j = h_{ij} v_i + a_{ij} u_i^2 + O(u_i v_i, v_i^2) + O((u_i, v_i)^3),$$

where g_{ij} and h_{ij} are the transition functions of N_+ and N_-, and φ_{ij}, g_{ij}, h_{ij}, a_{ij} are functions of z_i. The functions a_{ij} form a Čech 1-cocycle of $\operatorname{Hom}(N_+^{\otimes 2}, N_-) \cong \mathcal{O}(-4)$, and replacing v_i by $v_i + c_i u_i^2$ changes the cocycle by a coboundary. Let $[a] \in H^1(L, \mathcal{O}(-4))$ be the class of (a_{ij}).

On the chart of Y where $v_i = u_i w_i$, we have $E = (u_i = 0)$ and $s = (u_i = w_i = 0)$, since s corresponds to the normal direction N_+. Modulo $(u_i, w_i)^2$,

$$u_j \equiv g_{ij} u_i, \qquad w_j \equiv \frac{h_{ij}}{g_{ij}} w_i + \frac{a_{ij}}{g_{ij}} u_i.$$

Here w_i is the coordinate along $N_{s/E} \cong N_- \otimes N_+^{-1}$ and u_i the coordinate along $N_{E/Y}|_s \cong N_+$. Hence, up to the sign convention of the Čech connecting map, $\epsilon = [a]$ under $\operatorname{Hom}(N_+^{\otimes 2}, N_-) \cong \operatorname{Hom}(N_+, N_- \otimes N_+^{-1})$, and ∂_ϵ is the cup product with $[a]$.

Step 2. In this step we show that a non-zero first-order deformation of L containing x extends to a $\mathbb{C}[t]/(t^3)$-point of $\Sigma(X,x)$, that is, to a deformation of L over $\mathbb{C}[t]/(t^3)$ containing $x \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$, if and only if $\partial_\epsilon(\ell_x) = 0$.

The tangent space of $\Sigma(X,x)$ at $[L]$ is $H^0(L, N_{L/X}(-x))$ (Proposition 4.1). Since $N_+(-x) \cong \mathcal{O}$ and $N_-(-x) \cong \mathcal{O}(-3)$,

$$H^0(L, N_{L/X}(-x)) = H^0(N_+(-x)) \oplus H^0(N_-(-x)) = \ell_x \oplus 0.$$

Let $l \in \ell_x$ be non-zero, and let l_i be the local expression of l on U_i. Since L is a local complete intersection, a deformation of L over $\mathbb{C}[t]/(t^3)$ extending the first-order deformation defined by l is glued from local extensions [Har10, Theorem 6.2 and Remark 6.2.1]. Thus it is given on U_i by a graph

$$u_i = tl_i + t^2 r_i, \qquad v_i = t^2 b_i,$$

where r_i and b_i are functions of z_i. Since $z_j - \varphi_{ij}(z_i) \in (t^2)$ on the graph, a change of the tangent coordinate changes u_j and v_j only by elements of (t^3). The deformation contains $x \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$ if and only if $r_i(x) = b_i(x) = 0$ for any i with $x \in U_i$. Substituting the graph into the transition formulas of **Step 1**, the graphs glue modulo t^3 if and only if

$$b_j - h_{ij} b_i = a_{ij} l_i^2 \qquad \text{and} \qquad r_j - g_{ij} r_i = c_{ij}$$

on $U_i \cap U_j$, where $c_{ij} := e_{ij} l_i^2$, and e_{ij} is the coefficient of u_i^2 in the expansion of u_j as a power series in u_i, v_i with coefficients functions of z_i. The cocycle condition for (c_{ij}) follows from the gluing of the u-coordinates on triple overlaps, and c_{ij} vanishes at x since it is quadratic in l_i. Thus (c_{ij}) is a cocycle of $N_+(-x)$. Since $H^1(N_+(-x)) = H^1(\mathcal{O}) = 0$, the second equation is solvable with r_i vanishing at x.

The first equation is solvable with b_i vanishing at x if and only if the class of $(a_{ij} l_i^2)$ vanishes in $H^1(N_-(-x))$. Since l is a section of $N_+ \cong \mathcal{O}(1)$ with divisor x, multiplication by l is an isomorphism

$$N_- \otimes N_+^{-1} \to N_-(-x),$$

which maps the class of $(a_{ij} l_i)$ to the class of $(a_{ij} l_i^2)$. By **Step 1**, the class of $(a_{ij} l_i)$ in $H^1(N_- \otimes N_+^{-1}) = H^1(\mathcal{O}(-3))$ is $\partial_\epsilon(l)$ up to sign. Thus the first equation is solvable with $b_i(x) = 0$ if and only if $\partial_\epsilon(l) = 0$.

Conversely, if both equations are solved with $r_i(x) = b_i(x) = 0$, the graphs glue to a closed subscheme of $X \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$ which is locally a graph over $L \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$. It is flat over $\mathbb{C}[t]/(t^3)$ with the Hilbert polynomial of L, and contains $x \times \operatorname{Spec}\mathbb{C}[t]/(t^3)$. Hence it is a $\mathbb{C}[t]/(t^3)$-point of $\Sigma(X,x)$ extending the first-order deformation defined by l.

Step 3. In this step we prove the equivalence of (1) and (2), and the number of points as in (1).

Since $H^0(\mathcal{O}(-3)) = 0$ and $H^1(\mathcal{O}(1)) = 0$, the exact sequence of the proposition induces

$$0 \to H^0(N_{s/Y}) \to H^0(\mathcal{O}(1)) \xrightarrow{\partial_\epsilon} H^1(\mathcal{O}(-3)) \to H^1(N_{s/Y}) \to 0,$$

and $h^0(\mathcal{O}(1)) = h^1(\mathcal{O}(-3)) = 2$. By Lemma 10.6, whose connecting map is ∂_ϵ, $\operatorname{rank}\partial_\epsilon = 2$ if $N_{s/Y} \cong \mathcal{O}(-1)^{\oplus 2}$ and $\operatorname{rank}\partial_\epsilon = 1$ if $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$.

If $N_{s/Y} \cong \mathcal{O}(-1)^{\oplus 2}$, then ∂_ϵ is an isomorphism, and $\partial_\epsilon(\ell_x) \neq 0$ for any $x \in L$. Suppose that $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$. Then $\operatorname{Ker}\partial_\epsilon$ is the image of $H^0(N_{s/Y})$, and $h^0(N_{s/Y}) = 1$. Let τ be a non-zero section of $N_{s/Y}$. Then τ is a section of the summand $\mathcal{O}$, and hence τ vanishes nowhere. Let $l_0 \in H^0(\mathcal{O}(1))$ be the image of τ. Since $H^0(N_{s/E}) = 0$, $l_0 \neq 0$, so l_0 vanishes at exactly one point $x_0 \in L$. Thus $\partial_\epsilon(\ell_x) = 0$ if and only if $\ell_x = \mathbb{C} l_0$, if and only if $x = x_0$.

Assume (2). Then $x = x_0$. Let $\mathcal{O}$ be the local ring of $\Sigma(X,x)$ at $[L]$. Since $\Sigma(X,x)$ is finite of length at most three [KP23, Corollary A.6], $\mathcal{O}$ is Artinian. By **Step 2**, the tangent space of $\mathcal{O}$ is ℓ_x, so $\mathcal{O} \cong \mathbb{C}[t]/(t^k)$ for some $k \geq 2$. By **Step 2**, there exists a local homomorphism $\mathcal{O} \to \mathbb{C}[t]/(t^3)$ with $t \to \lambda t + \mu t^2$ and $\lambda \neq 0$. Since

$$(\lambda t + \mu t^2)^2 = \lambda^2 t^2 \neq 0 \quad \text{in } \mathbb{C}[t]/(t^3),$$

we have $k \geq 3$. Since length $\Sigma(X,x) \leq 3$, it follows that $k = 3$ and $\Sigma(X,x) = \operatorname{Spec} \mathcal{O}$. Thus (1) holds.

Assume (1). The isomorphism $\operatorname{Spec} \mathbb{C}[t]/(t^3) \cong \Sigma(X,x)$ is a $\mathbb{C}[t]/(t^3)$-point of $\Sigma(X,x)$ extending a non-zero first-order deformation of L containing x. By **Step 2**, $\partial_\epsilon(\ell_x) = 0$. Hence $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ and $x = x_0$, which is (2).

Thus a special line L with $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ has exactly one point as in (1), namely x_0, and a special line with $N_{s/Y} \cong \mathcal{O}(-1)^{\oplus 2}$ has none.

Step 4. In this step we show that C_0 and σ intersect transversally in one point q_0, which maps to x.

Assume (2), so that $x = x_0$. The surface $P = \mathbb{P}(N_{s/Y})$ parametrizes the one-dimensional subspaces of the fibers of $N_{s/Y}$. Let $\mathcal{O} \subset N_{s/Y}$ be the trivial summand, which is spanned by τ. Then $\sigma = \mathbb{P}(\mathcal{O})$, since the normal bundle of $\mathbb{P}(\mathcal{O})$ in P is $\operatorname{Hom}(\mathcal{O}, \mathcal{O}(-2)) = \mathcal{O}(-2)$. Since E is smooth along s, $C_0 = E' \cap P$ is the section $\mathbb{P}(N_{s/E})$. The normal bundle of C_0 in P is

$$\operatorname{Hom}(N_{s/E}, N_{s/Y}/N_{s/E}) \cong \operatorname{Hom}(\mathcal{O}(-3), \mathcal{O}(1)) = \mathcal{O}(4),$$

so $(C_0^2)_P = 4$. Write $C_0 \equiv \sigma + bf_P$, with f_P as in Set-up 10.10. Then

$$(\sigma + bf_P)^2 = -2 + 2b = 4$$

implies $b = 3$, and hence

$$(C_0 \cdot \sigma) = (\sigma + 3f_P) \cdot \sigma = -2 + 3 = 1$$

[Har77, Propositions V.2.3 and V.2.9]. Since C_0 and σ are distinct smooth curves with $(C_0 \cdot \sigma) = 1$, they intersect transversally in one point q_0 [Har77, Proposition V.1.4].

For $y \in s$, the fiber of P over y contains a point of $\sigma \cap C_0$ if and only if $\mathbb{C}\tau(y) = N_{s/E,y}$ in $N_{s/Y,y}$, if and only if the image of $\tau(y)$ in $N_{E/Y,y}$ is zero, if and only if $l_0(y) = 0$. Thus q_0 is contained in the fiber of P over the point of s over x_0, and q_0 maps to $x_0 = x$.

Step 5. In this step we conclude the proof.

Let x be a point as in (1). By **Step 3**, $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$. By Proposition 12.5, M is a line with $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$, and X is K-semistable if $M \cap \Gamma$ is non-reduced. If the width of s is three, then X is K-semistable by Lemma 12.7. Suppose that $M \cap \Gamma$ is reduced and the width of s is not three. By Proposition 11.11, the width of s is two. Since length$(M \cap \Gamma) = 2$ (Proposition 12.5(1)), $M \cap \Gamma$ consists of two distinct points. □

When the width of s is two and $M \cap \Gamma$ consists of two distinct points, the point of incidence type 3 on L is treated in Section 15.

13. Points on non-special lines in genus twelve

In this section, we prove lower bounds for $\delta_x(X)$ at the points x of incidence type 1, $1+1$, and $1+1+1$ (Definition 2.11) of a smooth prime Fano threefold X of genus 12. For such points every local ring of $\Sigma(X,x)$ is isomorphic to $\mathbb{C}$, so the lines through them are non-special by Proposition 4.1. The three incidence types are treated in Construction-Theorem 13.2 and Corollary 13.3, in Theorem 13.21, and in Theorem 13.24, respectively.

Compared with genus ten, two changes occur. For a point on one line we use the log canonical threshold of a septic tangent cone, and for a point on two lines one configuration is treated by a weighted blow-up in place of the blow-up of the point. The technical reasons are explained before the corresponding results.

13.1. **Points on one line.** In this subsection, we treat the points x such that $\Sigma(X,x)$ consists of one reduced point.

Proposition 4.5 computes the volume of $\pi^*H - tE$ on the blow-up of the point only for t between 0 and 2, and the tail of the volume, for t between 3 and 7/2, is beyond the range of Proposition 4.5. For t between 2 and 3, only the total rank $h^0(Y, m\pi^*H - 2mE) - h^0(Y, m\pi^*H - 3mE)$ enters

the estimates of Construction-Theorem 13.2, and it is given by the values at the endpoints 2 and 3. To compute the tail, we pass first through the link of Theorem 2.23 along the line through x and then through the blow-up of a conic in V_5. After both links, the sections of $m\pi^*H - jE$ become the sections of a multiple of a base point free divisor on the blow-up of $\mathbb{P}^3$ along a conic, whose dimensions can be computed.

Lemma 13.1. *Let X be a smooth prime Fano threefold of genus 12, let $x \in X$ be a closed point, let $\pi : Y := \mathrm{Bl}_x X \to X$ be the blow-up with exceptional plane E, and let W be the refinement of $(H^0(X, mH))_m$ by E (Definition 2.8). Assume that $\Sigma(X, x)$ consists of one reduced point. Then the following hold.*

(1) The pseudo-effective threshold of E with respect to π^*H is $7/2$, and

$$\mathrm{vol}_Y(\pi^*H - tE) = 2(7 - 2t)^3 \qquad \text{for } 3 \le t \le \frac{7}{2}.$$

(2) There exists a smooth plane conic $T_x \subset \mathbb{P}^3$ such that, with $U_x := \mathrm{Bl}_{T_x} \mathbb{P}^3$, a the pullback of $\mathcal{O}_{\mathbb{P}^3}(1)$, R_x the exceptional divisor, and $L_x := 2a - R_x$, we have

$$h^0(Y, m\pi^*H - jE) = h^0(U_x, (7m - 2j)L_x)$$

for all integers $m \ge 1$ and $3m \le j \le 7m/2$, and $h^0(Y, m\pi^*H - jE) = 0$ for all integers $m \ge 1$ and $j > 7m/2$.

(3) We have

$$\dim W_{m,j} = 2(7m - 2j)^2 + O(m)$$

uniformly for $3m \le j < \lfloor 7m/2 \rfloor$. Moreover, $h^0(Y, m\pi^*H - 3mE) = m^3/3 + O(m^2)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the line L through x is non-special, and that the fiber of $\mathrm{Bl}_L X \to X$ over x becomes a smooth conic $C \subset V_5$ under the link of Theorem 2.23.

Let L be the line through x. Since $\Sigma(X, x)$ consists of one reduced point, its local ring at $[L]$ is isomorphic to $\mathbb{C}$, so L is non-special by Proposition 4.1.

Let $\pi_L : Y_L := \mathrm{Bl}_L X \to X$ be the blow-up with exceptional divisor J, and let $f_x := \pi_L^{-1}(x)$ be the fiber of $J \to L$ over x. By Theorem 2.23, applied to L with Y_L and J in place of Y and E, we obtain a flop $Y_L \dashrightarrow Y^+$ and the blow-up $\phi : Y^+ = \mathrm{Bl}_\Gamma V_5 \to V_5$ of a rational normal quintic curve Γ, with exceptional divisor R, such that under strict transform

$$h = \pi_L^*H - 2J, \qquad R = \pi_L^*H - 3J, \qquad \pi_L^*H = 3h - 2R. \tag{13.1}$$

Since L is non-special, the flopping curves of $Y_L \dashrightarrow Y^+$ are the strict transforms of the lines $M \ne L$ on X intersecting L. If the strict transform of such a line M intersects f_x, then $x \in M$, which is impossible since L is the only line through x. Since the flopping curves are pairwise disjoint, the flop $Y_L \dashrightarrow Y^+$ is an isomorphism on a neighborhood of f_x. We denote the image of f_x in Y^+ again by f_x. Since $(J \cdot f_x) = -1$ and $(\pi_L^*H \cdot f_x) = 0$, (13.1) implies

$$(h \cdot f_x) = 2, \qquad (R \cdot f_x) = 3.$$

The morphism $R \to \Gamma$ is a $\mathbb{P}^1$-bundle, so any curve in R is either a fiber, of h-degree 0, or maps onto Γ, with h-degree a positive multiple of $(h \cdot \Gamma) = 5$. Thus $f_x \not\subset R$. Since ϕ is an isomorphism outside R, ϕ maps f_x birationally onto an integral curve $C \subset V_5$ with $\deg C = (h \cdot f_x) = 2$. Thus C is a smooth conic and $f_x \to C$ is an isomorphism. Since C is a smooth curve, $I_\Gamma \cdot \mathcal{O}_C$ is invertible, f_x is the strict transform of C, and $R|_{f_x}$ is the divisor defined by $I_\Gamma \cdot \mathcal{O}_C$. Thus the scheme $C \cap \Gamma$ has length $(R \cdot f_x) = 3$.

Step 2. In this step we pass from V_5 to $\mathbb{P}^3$ through the blow-up of C, and we construct T_x.

Let $V' := \mathrm{Bl}_C V_5$ with exceptional divisor Q, and denote the pullback of h to V' again by h. By [IP99, Lemma 2.2.14],

$$(h^3) = 5, \qquad (h^2 \cdot Q) = 0, \qquad (h \cdot Q^2) = -\deg C = -2,$$
$$(Q^3) = -\deg N_{C/V_5} = -\big((2h \cdot C) - 2\big) = -2,$$

so $((h-Q)^3) = 5 - 6 + 2 = 1$. Since $h^0(V_5, \mathcal{O}_{V_5}(1)) = 7$ and C spans a plane, $h^0(V', h-Q) = 4$. By [PCS19, Remark 6.4], V' is a smooth Fano threefold and $|h-Q|$ is base point free. Thus $|h-Q|$ defines a birational morphism $V' \to \mathbb{P}^3$, which is not an isomorphism since V' has Picard number 2. Since moreover $((h-Q)\cdot \bar{f}) = 1$ for a fiber $\bar{f}$ of $Q \to C$, the morphism $V' \to \mathbb{P}^3$ defined by $|h-Q|$ is the contraction of the extremal ray of $\mathrm{NE}(V')$ not containing $[\bar{f}]$. By [MM81, Table 2, No. 22], which covers any smooth conic $C \subset V_5$, the morphism $V' \to \mathbb{P}^3$ is the blow-up of $\mathbb{P}^3$ along a smooth rational quartic curve C_4. Let a be the pullback of $\mathcal{O}_{\mathbb{P}^3}(1)$ to V', so that $a = h - Q$, and let R_7 be the exceptional divisor of $V' \to \mathbb{P}^3$. By [KM98, Lemma 2.29], $-K_{V'} = 2h - Q = 4a - R_7$, hence

$$h = 3a - R_7, \qquad Q = 2a - R_7. \tag{13.2}$$

Let $\Gamma' \subset V'$ be the strict transform of Γ. Since Γ is smooth and $\Gamma \not\subset C$, $I_C \cdot \mathcal{O}_\Gamma$ is invertible, so $\Gamma' \cong \Gamma$ and $(Q \cdot \Gamma')$ is the length of $C \cap \Gamma$, which is 3 by **Step 1**. Thus

$$(a \cdot \Gamma') = (h \cdot \Gamma') - (Q \cdot \Gamma') = 5 - 3 = 2.$$

Any curve in R_7 is either a fiber of $R_7 \to C_4$, of a-degree 0, or maps onto C_4, with a-degree a positive multiple of 4. Thus $\Gamma' \not\subset R_7$, and $V' \to \mathbb{P}^3$ maps Γ' birationally onto an integral curve $T_x \subset \mathbb{P}^3$ of degree 2. Thus T_x is a smooth plane conic and $\Gamma' \cong T_x$.

Step 3. In this step we show that

$$H^0(Y, m\pi^*H - jE) \cong H^0(U_x, (9m-2j)a - 2mR_x) \qquad \text{for all integers } m \geq 1 \text{ and } j \geq 3m. \tag{13.3}$$

Recall that $f_x \subset J$ is the fiber over x. Let $\mu_L : \widetilde{Y}_L := \mathrm{Bl}_{f_x} Y_L \to Y_L$ be the blow-up with exceptional divisor $\widetilde{E}$. Let (z,u,v) be local coordinates of X at x with $L = (u = v = 0)$. On the chart of Y_L with coordinates (z,u,w), where $v = uw$, we have $J = (u = 0)$ and $f_x = (z = u = 0)$, so $\mathfrak{m}_x \cdot \mathcal{O}_{Y_L} = (z, u, uw) = (z, u)$ is the ideal of f_x, and on the chart with coordinates (z, v, w'), where $u = vw'$, we have $\mathfrak{m}_x \cdot \mathcal{O}_{Y_L} = (z, vw', v) = (z, v)$, which is again the ideal of f_x. By [Har77, Proposition II.7.14], $\pi_L \circ \mu_L$ factors through π. On the chart of $\widetilde{Y}_L$ with coordinates (z, u', w), where $u = zu'$, we have $\widetilde{E} = (z = 0)$, and the induced morphism to the chart of Y with coordinates $(z, u/z, v/z)$ is $(z, u', w) \to (z, u', u'w)$, which maps $\widetilde{E}$ birationally onto E. Thus $\mathrm{ord}_{\widetilde{E}} = \mathrm{ord}_E$ as valuations of the function field of X, and

$$H^0(Y, m\pi^*H - jE) = \{s \in H^0(X, mH) : \mathrm{ord}_E(s) \geq j\} = H^0\big(\widetilde{Y}_L, m(\pi_L \circ \mu_L)^*H - j\widetilde{E}\big).$$

Next we introduce three more smooth projective threefolds. Let $\widetilde{Y}^+ := \mathrm{Bl}_{f_x} Y^+$, whose exceptional divisor we also denote by $\widetilde{E}$, since $Y_L \dashrightarrow Y^+$ is an isomorphism near f_x (**Step 1**), and let $V'' := \mathrm{Bl}_{\Gamma'} V'$ with exceptional divisor R'. Let R_x be the exceptional divisor of $U_x = \mathrm{Bl}_{T_x} \mathbb{P}^3 \to \mathbb{P}^3$, and let $C_4' \subset U_x$ be the strict transform of C_4. Since C_4 is smooth and $C_4 \not\subset T_x$, we have $C_4' \cong C_4$ and $C_4' \not\subset R_x$. Let $\rho : Y'' := \mathrm{Bl}_{C_4'} U_x \to U_x$ be the blow-up, and denote its exceptional divisor again by R_7. On $\widetilde{Y}^+$, V'', and Y'', we denote the pullbacks of h, a, R, Q, R_7, and R_x again by h, a, R, Q, R_7, and R_x. Since $f_x \not\subset R$, $\Gamma' \not\subset Q \cup R_7$, and $C_4' \not\subset R_x$, the pullbacks of R, Q, R_7, and R_x are prime divisors, equal to their strict transforms.

We show that $\widetilde{Y}_L$, $\widetilde{Y}^+$, V'', and Y'' are isomorphic in codimension one, with the following correspondences of divisor classes:

$$(\pi_L \circ \mu_L)^*H \leftrightarrow 3h - 2R \leftrightarrow 3h - 2R' \leftrightarrow 9a - 3R_7 - 2R_x, \qquad \widetilde{E} \leftrightarrow \widetilde{E} \leftrightarrow Q \leftrightarrow 2a - R_7. \tag{13.4}$$

Recall that if the sets of divisorial valuations defined by the prime divisors on two birational normal projective varieties are equal, then the birational map between them is an isomorphism at

the generic point of any prime divisor, hence an isomorphism in codimension one. Moreover, by [KM98, Lemmas 2.29 and 2.30, and Corollary 2.31(2)], a prime divisor over a smooth threefold whose center is a smooth curve and whose log discrepancy is 2 is the exceptional divisor of the blow-up of that curve.

First, $Y_L \dashrightarrow Y^+$ is an isomorphism in codimension one and an isomorphism on a neighborhood of f_x, so it lifts to a birational map $\widetilde{Y}_L \dashrightarrow \widetilde{Y}^+$ which is an isomorphism in codimension one and maps the exceptional divisor $\widetilde{E}$ of μ_L to that of $\widetilde{Y}^+ \to Y^+$. For a Weil divisor D on Y_L with strict transform D^+ on Y^+, we have $\operatorname{mult}_{f_x} D = \operatorname{mult}_{f_x} D^+$, so $\mu_L^* D$ corresponds to the pullback of D^+. By (13.1) and $f_x \notin R$, $(\pi_L \circ \mu_L)^* H$ corresponds to $3h - 2R$.

Second, consider $\widetilde{Y}^+$ and V'' over V_5. The prime divisors on $\widetilde{Y}^+$ exceptional over V_5 are R and $\widetilde{E}$, with centers Γ and C on V_5, and $A_{V_5}(\widetilde{E}) = 2 + \operatorname{mult}_{f_x} R = 2$. The prime divisors on V'' exceptional over V_5 are Q and R', with centers C and Γ, and $A_{V_5}(R') = 2 + \operatorname{mult}_{\Gamma'} Q = 2$. Thus both sets consist of the exceptional divisors of $\operatorname{Bl}_\Gamma V_5$ and $\operatorname{Bl}_C V_5$, and $\widetilde{Y}^+ \dashrightarrow V''$ is an isomorphism in codimension one with $R \leftrightarrow R'$ and $\widetilde{E} \leftrightarrow Q$. The pullbacks of a hyperplane section of V_5 not containing $C \cup \Gamma$ have no exceptional components on either side, so $h \leftrightarrow h$.

Third, consider V'' and Y'' over $\mathbb{P}^3$. The prime divisors on V'' exceptional over $\mathbb{P}^3$ are R_7 and R', with centers C_4 and T_x, and $A_{\mathbb{P}^3}(R') = 2 + \operatorname{mult}_{\Gamma'} R_7 = 2$. The prime divisors on Y'' exceptional over $\mathbb{P}^3$ are R_x and R_7, with centers T_x and C_4, and $A_{\mathbb{P}^3}(R_7) = 2 + \operatorname{mult}_{C_4'} R_x = 2$. Thus $V'' \dashrightarrow Y''$ is an isomorphism in codimension one with $R' \leftrightarrow R_x$, $R_7 \leftrightarrow R_7$, and $a \leftrightarrow a$. Since (13.2) holds on V'' after pullback, we have $3h - 2R' = 9a - 3R_7 - 2R'$ and $Q = 2a - R_7$ on V''. This proves (13.4).

For a Weil divisor D on a normal variety, $H^0(\mathcal{O}(D))$ consists of 0 and the rational functions φ with $\operatorname{div}\varphi + D \ge 0$, which is a condition at the prime divisors only. Thus, for an isomorphism in codimension one $\psi : Z \dashrightarrow Z'$ between normal varieties and a Weil divisor D on Z, we have $H^0(Z, \mathcal{O}_Z(D)) \cong H^0(Z', \mathcal{O}_{Z'}(\psi_* D))$, and (13.4) implies

$$H^0(Y, m\pi^* H - jE) \cong H^0\big(Y'', (9m - 2j)a + (j - 3m)R_7 - 2mR_x\big).$$

Assume that $j \ge 3m$. Since $\rho_* \mathcal{O}_{Y''}((j - 3m)R_7) = \mathcal{O}_{U_x}$ [Har77, Proposition II.6.3A] and a and R_x on Y'' are pullbacks from U_x, the projection formula implies (13.3).

Step 4. In this step we show that $h^0(Y, m\pi^* H - jE) = h^0(U_x, (7m - 2j)L_x)$ for all integers $m \ge 1$ and $j \ge 3m$, and that L_x is base point free with $(L_x^3) = 2$.

Let $S \subset U_x$ be the strict transform of the plane containing T_x. Then $S \sim a - R_x$, $S \cong \mathbb{P}^2$, $a|_S \cong \mathcal{O}_{\mathbb{P}^2}(1)$, and $R_x|_S \cong \mathcal{O}_{\mathbb{P}^2}(2)$ since $R_x \cap S$ is the conic $T_x \subset S$. Thus $S|_S \cong \mathcal{O}_{\mathbb{P}^2}(-1)$. Let k and n be integers with $n \ge 0$ and $2n \ge k$. For any integer $0 \le b < 2n - k$,

$$(ka - nR_x - bS)|_S \cong \mathcal{O}_{\mathbb{P}^2}(k - 2n + b), \qquad k - 2n + b < 0,$$

so the exact sequence

$$0 \to \mathcal{O}_{U_x}(ka - nR_x - (b+1)S) \to \mathcal{O}_{U_x}(ka - nR_x - bS) \to \mathcal{O}_S(k - 2n + b) \to 0$$

shows that multiplication by a section defining S is an isomorphism on global sections. Applying the isomorphism for $b = 0, 1, \dots, 2n - k - 1$, we obtain

$$H^0(U_x, ka - nR_x) \cong H^0\big(U_x, ka - nR_x - (2n - k)S\big) = H^0(U_x, (k - n)L_x). \tag{13.5}$$

For $j \ge 3m$, we take $k = 9m - 2j$ and $n = 2m$. Then $2n - k = 2j - 5m \ge m > 0$ and $k - n = 7m - 2j$, so (13.3) and (13.5) imply

$$h^0(Y, m\pi^* H - jE) = h^0(U_x, (7m - 2j)L_x) \qquad \text{for all integers } m \ge 1 \text{ and } j \ge 3m. \tag{13.6}$$

The ideal of T_x in $\mathbb{P}^3$ is generated by a linear form λ and a quadratic form κ, so the ideal sheaf of T_x twisted by $\mathcal{O}_{\mathbb{P}^3}(2)$ is generated by the global sections κ and the products of λ with a basis of the linear forms on $\mathbb{P}^3$. Since $I_{T_x} \cdot \mathcal{O}_{U_x} = \mathcal{O}_{U_x}(-R_x)$, $|L_x|$ is base point free. By [IP99,

Lemma 2.2.14], $(a^3) = 1$, $(a^2 \cdot R_x) = 0$, $(a \cdot R_x^2) = -\deg T_x = -2$, and $(R_x^3) = -\deg N_{T_x/\mathbb{P}^3} = -6$, hence

$$(L_x^3) = 8(a^3) - 12(a^2 \cdot R_x) + 6(a \cdot R_x^2) - (R_x^3) = 8 - 0 - 12 + 6 = 2.$$

If $j > 7m/2$ and $D \in |(7m - 2j)L_x|$, then $(D \cdot L_x^2) \geq 0$ since L_x is nef, whereas $(D \cdot L_x^2) = 2(7m - 2j) < 0$. Thus $h^0(U_x, (7m - 2j)L_x) = 0$ for $j > 7m/2$. Together with (13.6), this proves (2).

Step 5. In this step we prove (1) and (3).

By [KM98, Lemma 2.29], $-K_{U_x} = 4a - R_x = 2a + L_x$. For any integer $p \geq 0$, $pL_x - K_{U_x} = (p + 1)L_x + 2a$ is nef and big, so $H^i(U_x, pL_x) = 0$ for $i > 0$ by [KM98, Theorem 2.64]. By the Hirzebruch–Riemann–Roch theorem [Har77, Appendix A, Theorem 4.1], there exist rational numbers c_0, c_1, and c_2 such that, for all integers $p \geq 0$,

$$h^0(U_x, pL_x) = \chi(U_x, pL_x) = \frac{(L_x^3)}{6}p^3 + c_2p^2 + c_1p + c_0 = \frac{p^3}{3} + c_2p^2 + c_1p + c_0. \tag{13.7}$$

Since all smooth plane conics in $\mathbb{P}^3$ are projectively equivalent, c_0, c_1, and c_2 do not depend on X or x. Taking $j = 3m$ in (13.6), we obtain $h^0(Y, m\pi^*H - 3mE) = \frac{m^3}{3} + O(m^2)$.

We prove (1). Let $t \in [3, 7/2]$ be a rational number, and let $m \geq 1$ be an integer with $mt \in \mathbb{Z}$. By (13.6) and (13.7), for any integer $k \geq 1$,

$$h^0\big(Y, k(m\pi^*H - mtE)\big) = h^0\big(U_x, km(7 - 2t)L_x\big) = \frac{k^3m^3(7 - 2t)^3}{3} + O(k^2),$$

so $\mathrm{vol}_Y(\pi^*H - tE) = m^{-3}\,\mathrm{vol}_Y(m\pi^*H - mtE) = 2(7 - 2t)^3$. Since the volume is continuous on $N^1(Y)_{\mathbb{R}}$ [Laz04a, Theorem 2.2.44 and Corollary 2.2.45], the formula holds for any real number $t \in [3, 7/2]$. In particular, $\pi^*H - 3E$ is big. Since $h^0(Y, 2\pi^*H - 7E) = h^0(U_x, \mathcal{O}_{U_x}) = 1$ by (13.6), $\pi^*H - \frac{7}{2}E$ is pseudo-effective. Suppose that $\pi^*H - t_0E$ is pseudo-effective for some real number $t_0 > 7/2$. Pick a rational number $t \in (7/2, t_0)$. Then

$$\pi^*H - tE = \frac{t - 3}{t_0 - 3}(\pi^*H - t_0E) + \frac{t_0 - t}{t_0 - 3}(\pi^*H - 3E),$$

where both coefficients are positive, is big, whereas $h^0(Y, k(m\pi^*H - mtE)) = 0$ for any integers $k, m \geq 1$ with $mt \in \mathbb{Z}$ by (2), a contradiction. Thus the pseudo-effective threshold of E with respect to π^*H is $7/2$.

We prove (3). For any integers m and j, the exact sequence

$$0 \to \mathcal{O}_Y(m\pi^*H - (j + 1)E) \to \mathcal{O}_Y(m\pi^*H - jE) \to \mathcal{O}_E(j) \to 0$$

implies

$$\dim W_{m,j} = h^0(Y, m\pi^*H - jE) - h^0(Y, m\pi^*H - (j + 1)E).$$

Assume that $3m \leq j < \lfloor 7m/2 \rfloor$, and let $p := 7m - 2j$. Then $j + 1 \leq 7m/2$, so $2 \leq p \leq m$. By (13.6) and (13.7),

$$\begin{aligned}\dim W_{m,j} &= h^0(U_x, pL_x) - h^0(U_x, (p - 2)L_x) \\ &= 2p^2 + (4c_2 - 4)p + \frac{8}{3} - 4c_2 + 2c_1 = 2(7m - 2j)^2 + O(m),\end{aligned}$$

uniformly in j, since $0 \leq p \leq m$. This proves (3). □

Next we treat a point x which belongs to exactly one line. In the top range $3m \leq j \leq 7m/2$ of the refinement by the exceptional plane E, every graded piece $W_{m,j}$ equals $\tau^{j-3m}W_{n,3n}$ with $n = 7m - 2j$, so its fixed part contains $(j - 3m)B_7$. Here τ defines the plane septic $B_7 \subset E$ cut out by the tangent cone at x of the unique member of $|2H|$ of multiplicity seven at x. The estimate of $\delta_x(X)$ then reduces to the klt property of $(E, \frac{1}{3}B_7)$, which we prove by relating $(E, \frac{1}{3}B_7)$ to a pair on a blow-up of a smooth quadric surface. The bound $S(W; v) \leq \frac{85}{88}A_E(v) + \frac{1}{88}v(B_7)$ of Construction-Theorem 13.2 gives only $\delta_x(X) \geq 1$ when $(E, \frac{1}{3}B_7)$ is lc, and the strict inequality requires the klt property, that is, $\mathrm{lct}(E; B_7) > 1/3$ (Corollary 13.3).

Construction-Theorem 13.2. Let X be a smooth prime Fano threefold of genus 12 and $x \in X$ a closed point such that $\Sigma(X,x)$ consists of one reduced point. Let $\pi : Y := \mathrm{Bl}_x X \to X$ with exceptional plane E, and let W be the refinement of $(H^0(X,mH))_m$ by E. There is a unique section, up to scalar, of $2H$ vanishing to order at least seven at x, and the order of vanishing of the section at x is exactly seven. Let $B_7 \subset E \cong \mathbb{P}^2$ be the septic curve defined by the restriction to E of this section, viewed as a section of $2\pi^*H - 7E$. Equivalently, B_7 is defined by the tangent cone at x of the divisor of the section. Then $(E, \frac{1}{3}B_7)$ is klt, $S_X(E) \le \frac{119}{44}$, and

$$S(W;v) \le \frac{85}{88}A_E(v) + \frac{1}{88}v(B_7) \le A_E(v)$$

for every prime divisor v over E. In particular $A_X(F) \ge S_X(F)$ for every prime divisor F over X with $x \in \mathrm{center}_X F$, that is, $\delta_x(X) \ge 1$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the line through x is non-special, and we describe the line link near the fiber over x.

Let L be the line with $\Sigma(X,x) = \{[L]\}$. By Proposition 4.1, the Zariski tangent space of $\Sigma(X,x)$ at $[L]$ is $H^0(L, N_{L/X}(-x))$, and it is zero because $\Sigma(X,x)$ is a reduced point. We have $\deg N_{L/X} = -2 - (K_X \cdot L) = -1$ [IP99, Lemma 2.2.14], so $N_{L/X} \cong \mathcal{O}_L(c) \oplus \mathcal{O}_L(-1-c)$ for an integer $c \ge 0$ [Gro57, Théorème 2.1]. Since $h^0(L, \mathcal{O}_L(c-1)) = 0$, we have $c = 0$. Thus $N_{L/X} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1)$, that is, L is non-special (Definition 2.11).

Let $\pi_L : Y_L := \mathrm{Bl}_L X \to X$ be the blow-up of L. Since E denotes the exceptional plane of π, we write J for the exceptional divisor of π_L, which is denoted by E in Theorem 2.23. By Theorem 2.23 and [Fuj26a, Theorem 4.3(2)(3)(i)], there exists a flop $Y_L \dashrightarrow Y^+$ followed by the blow-up $Y^+ = \mathrm{Bl}_\Gamma V_5 \to V_5$ of a rational normal quintic curve $\Gamma \subset V_5$, with exceptional divisor R, such that

$$h = \pi_L^*H - 2J, \qquad R = \pi_L^*H - 3J, \qquad \pi_L^*H = 3h - 2R, \qquad J = h - R. \tag{13.8}$$

Let $f_x := \pi_L^{-1}(x)$ be the fiber of $J \to L$ over x. Since L is non-special, the flopping curves are the strict transforms of the lines $M \ne L$ intersecting L. Such a line M does not contain x because $\Sigma(X,x) = \{[L]\}$, so the strict transform of M is disjoint from f_x. Thus the flop is an isomorphism on a neighborhood of f_x, and we denote the image of f_x in Y^+ again by f_x. Since $(\pi_L^*H \cdot f_x) = 0$ and $(J \cdot f_x) = -1$, (13.8) implies

$$(h \cdot f_x) = 2, \qquad (R \cdot f_x) = 3.$$

Since $R \to \Gamma$ is a $\mathbb{P}^1$-bundle and $\deg \Gamma = 5$, we have $(h \cdot D) \in 5\mathbb{Z}_{\ge 0}$ for any curve $D \subset R$. Hence $f_x \not\subset R$. Since $Y^+ \to V_5$ is an isomorphism outside R, f_x maps birationally onto an integral curve $C \subset V_5$ of degree 2. Thus C is a smooth conic and $f_x \to C$ is an isomorphism. For $y \in C \cap \Gamma$ let $k_y := \mathrm{length}_y(C \cap \Gamma)$.

Step 2. In this step we fix local coordinates at the points of $C \cap \Gamma$ and show that $\sum_y k_y = 3$.

Let $V' := \mathrm{Bl}_C V_5$ with exceptional divisor Q, and let $\Gamma' \subset V'$ be the strict transform of Γ. Fix $y \in C \cap \Gamma$ and write $k := k_y$. Choose analytic coordinates (t, u_1, u_2) of V_5 at y such that $C = (u_1 = u_2 = 0)$ and dt is non-zero on the tangent lines of C and of Γ at y. Then $\Gamma = (u_1 - \gamma_1(t) = u_2 - \gamma_2(t) = 0)$ for holomorphic functions γ_1, γ_2 of t with $\min\{\mathrm{ord}_0 \gamma_1, \mathrm{ord}_0 \gamma_2\} = k$. After a linear change of (u_1, u_2) whose coefficients are holomorphic functions of t and whose determinant is a unit, we may assume that $(\gamma_1, \gamma_2) = (0, t^k)$, that is,

$$C = (u_1 = u_2 = 0), \qquad \Gamma = (u_1 = u_2 - t^k = 0).$$

Near the points of f_x over y, the threefold Y^+ has coordinates (t, z, u_2) with $u_1 = (u_2 - t^k)z$, in which $R = (u_2 - t^k = 0)$ and $f_x = (z = u_2 = 0)$. Near the points of $Q \cap \Gamma'$ over y, the threefold

V' has coordinates (t, w, u_2) with $u_1 = wu_2$, in which $Q = (u_2 = 0)$ and $\Gamma' = (w = u_2 - t^k = 0)$. Hence, near y,

$$R \cap f_x = (t^k = 0) \subset f_x, \qquad \Gamma' \cap Q = (w = t^k = 0) \subset Q. \tag{13.9}$$

In particular $\sum_y k_y = (R \cdot f_x) = 3$, and $\Xi := \Gamma' \cap Q$ is a curvilinear subscheme of Q of length $(Q \cdot \Gamma') = 3$.

Step 3. In this step we describe V' as a blow-up of $\mathbb{P}^3$ and the position of Ξ on the surface Q.

The linear system $|h - Q|$ is base point free [PCS19, Remark 6.4], and the morphism defined by it is the blow-up $V' \to \mathbb{P}^3$ of a smooth rational quartic curve $C_4 \subset \mathbb{P}^3$ [MM81, Table 2, No. 22], [PCS19, diagram (6.14) in the proof of Lemma 6.13]. Let a be the pullback of $\mathcal{O}_{\mathbb{P}^3}(1)$ and R_7 the exceptional divisor over C_4. Then $a = h - Q$, and $K_{V'} = -2h + Q = -4a + R_7$ implies

$$h = 3a - R_7, \qquad Q = 2a - R_7, \qquad a = h - Q, \qquad R_7 = 2h - 3Q. \tag{13.10}$$

By $Q = 2a - R_7$, the image Q_0 of Q in $\mathbb{P}^3$ is a quadric surface containing C_4, and Q_0 is smooth by [PCS19, Lemma 6.12]. Since C_4 is a Cartier divisor on Q_0, the morphism $Q \to Q_0$ is an isomorphism and $R_7|_Q = C_4$. A fiber $\bar{f}_y$ of $Q \to C$ satisfies $(Q \cdot \bar{f}_y) = -1$ and $(h \cdot \bar{f}_y) = 0$, so $(a \cdot \bar{f}_y) = 1$ and $(R_7 \cdot \bar{f}_y) = 3$. We fix an isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ under which $\bar{f}_y \in |\mathcal{O}_Q(1, 0)|$. Then $a|_Q \in |\mathcal{O}_Q(1, 1)|$ and $K_Q \in |\mathcal{O}_Q(-2, -2)|$, and $(R_7 \cdot \bar{f}_y) = 3$ and $(a \cdot C_4) = 4$ imply

$$C_4 = R_7|_Q \in |\mathcal{O}_Q(1, 3)|.$$

By **Step 2**, $(h \cdot \Gamma') = 5$ and $(Q \cdot \Gamma') = 3$, so $(a \cdot \Gamma') = 2$ and $(R_7 \cdot \Gamma') = 1$ by (13.10). Since $R_7 \to C_4$ is a $\mathbb{P}^1$-bundle and $\deg C_4 = 4$, we have $(a \cdot D) \in 4\mathbb{Z}_{\geq 0}$ for any curve $D \subset R_7$. Hence $\Gamma' \not\subset R_7$, and $\Gamma' \cong \Gamma$ maps isomorphically onto a smooth conic $T_x \subset \mathbb{P}^3$. The conic T_x agrees with the conic of Lemma 13.1, which is constructed there as the image of the strict transform of Γ under the morphism $\mathrm{Bl}_C V_5 \to \mathbb{P}^3$ defined by $|a|$. Let $P_T \in |a|$ be the pullback of the plane spanned by T_x. Since Q_0 is not a plane, $\Lambda := P_T|_Q$ is a member of $|\mathcal{O}_Q(1, 1)|$. Since $\Gamma' \subset P_T$ and $\Xi \cap C_4 = \Gamma' \cap Q \cap R_7 \subset \Gamma' \cap R_7$, we have

$$\Xi \subset \Lambda, \qquad \mathrm{length}(\Xi \cap C_4) \leq (R_7 \cdot \Gamma') = 1. \tag{13.11}$$

If Λ were not reduced, then $\Lambda \geq 2D$ for a prime divisor D of class $(\alpha, \beta) \neq (0, 0)$ with $\alpha, \beta \geq 0$, and $\Lambda - 2D \geq 0$ would be a divisor of class $(1 - 2\alpha, 1 - 2\beta)$, which is impossible. Thus Λ is reduced. Hence Λ is either a smooth conic or the union of a member of $|\mathcal{O}_Q(1, 0)|$ and a member of $|\mathcal{O}_Q(0, 1)|$, and in both cases Λ is a reduced snc divisor.

Step 4. In this step we show that the image D_x of R_7 in X is the unique member of $|2H|$ with multiplicity at least seven at x, and we compute B_7 on a blow-up of E.

Let $\mu_L : \widetilde{Y}_L := \mathrm{Bl}_{f_x} Y_L \to Y_L$ with exceptional divisor $\widetilde{E}$. Take local coordinates (z, u, v) of X at x with $L = (u = v = 0)$. On the chart of Y_L with $v = uv'$, the ideal $\mathfrak{m}_x \mathcal{O}_{Y_L} = (z, u, uv') = (z, u)$ is the ideal of f_x, and on the chart with $u = vu'$, the ideal $\mathfrak{m}_x \mathcal{O}_{Y_L} = (z, vu', v) = (z, v)$ is the ideal of f_x. Hence $\mathfrak{m}_x \mathcal{O}_{\widetilde{Y}_L} = \mathcal{O}_{\widetilde{Y}_L}(-\widetilde{E})$, and $\pi_L \circ \mu_L$ factors through a morphism $\mu : \widetilde{Y}_L \to Y$ with $\mu^* E = \widetilde{E}$ [Har77, Proposition II.7.14]. Thus $\widetilde{E}$ is the strict transform of E, and μ induces a birational morphism $b : \widetilde{E} \to E$. Since $\mu(J)$ is the strict transform of L, b contracts the curve $J \cap \widetilde{E}$. Since

$$N_{f_x/Y_L} \cong N_{f_x/J} \oplus N_{J/Y_L}|_{f_x} \cong \mathcal{O} \oplus \mathcal{O}(-1),$$

we have $\widetilde{E} \cong \mathbb{F}_1$. Since $\mathrm{Pic}(\widetilde{E}) \cong \mathbb{Z}^2$ and $\mathrm{Pic}(E) \cong \mathbb{Z}$, the birational morphism b is a single monoidal transformation, so b contracts exactly one curve, and the contracted curve is a (-1)-curve. On $\mathbb{F}_1$ the only irreducible curve of negative self-intersection is the negative section. Hence $J \cap \widetilde{E}$ is the negative section $\tilde{s}$ of $\widetilde{E}$, b is the blow-up of a point, and $b^* \mathcal{O}_E(1) = \tilde{s} + \tilde{f}$, where $\tilde{f}$ is the fiber class [Har77, Propositions V.2.3, V.2.20(a), V.3.2, V.3.6 and Corollary V.5.4]. Moreover $\mu_L^* J = J + \widetilde{E}$, $(J \cdot f_x) = -1$, and $\widetilde{E}|_{\widetilde{E}} = b^*(E|_E) \sim -\tilde{s} - \tilde{f}$, so $J|_{\widetilde{E}} \sim \tilde{s}$. Since $|\tilde{s}| = \{\tilde{s}\}$, we obtain $J|_{\widetilde{E}} = \tilde{s}$.

Let $\widetilde{Y}^+ := \mathrm{Bl}_{f_x} Y^+$, and let $V'' := \mathrm{Bl}_{\Gamma'} V'$ with exceptional divisor R'. Since the flop is an isomorphism near f_x, the induced birational map $\widetilde{Y}_L \dashrightarrow \widetilde{Y}^+$ is an isomorphism in codimension one and on a neighborhood of $\widetilde{E}$, and it maps $\mu_L^* h$ to the pullback of h. The exceptional divisors of $\widetilde{Y}^+ \to V_5$ are R and $\widetilde{E}$, and those of $V'' \to V_5$ are Q and R'. The prime divisors R and R' define the valuation ord_Γ, and $\widetilde{E}$ and Q define the valuation ord_C. Hence $\widetilde{Y}^+ \dashrightarrow V''$ is an isomorphism in codimension one, and the strict transform of $\widetilde{E}$ on V'' is Q. On V'', the pullbacks of R_7 and Q are their strict transforms because $\Gamma' \not\subset R_7 \cup Q$, so $R_7 \sim 2h - 3Q$ by (13.10). Write H also for the pullback of H to $\widetilde{Y}_L$. Since $\mu_L^* h = H - 2J - 2\widetilde{E}$ by (13.8), we obtain

$$R_7 \sim 2\mu_L^* h - 3\widetilde{E} \sim 2H - 4J - 7\widetilde{E} \quad \text{on } \widetilde{Y}_L. \tag{13.12}$$

Since $\mathrm{Pic}(\widetilde{Y}_L) = \mathbb{Z}H \oplus \mathbb{Z}J \oplus \mathbb{Z}\widetilde{E}$, the prime divisor R_7 is different from J and $\widetilde{E}$. Thus R_7 is the strict transform of a prime divisor $D_x \in |2H|$, and $(\pi_L \circ \mu_L)^* D_x = R_7 + 4J + 7\widetilde{E}$. Let $\widetilde{D}_x \subset Y$ be the strict transform of D_x. Since $\widetilde{E}$ is the strict transform of E and $\mu^* E = \widetilde{E}$, we have $\mathrm{mult}_x D_x = 7$, $\pi^* D_x = \widetilde{D}_x + 7E$, and $\mu^* \widetilde{D}_x = R_7 + 4J$.

By Lemma 13.1 with $(m, j) = (2, 7)$ and $(m, j) = (2, 8)$, we have $h^0(Y, 2\pi^* H - 7E) = h^0(U_x, \mathcal{O}_{U_x}) = 1$ and $h^0(Y, 2\pi^* H - 8E) = 0$. Thus the section of $2H$ defining D_x is the unique section up to scalar vanishing to order at least seven at x, and the order of vanishing at x is exactly seven. Let $\sigma_7 \in H^0(Y, 2\pi^* H - 7E)$ be a section with divisor $\widetilde{D}_x$ and $\tau := \sigma_7|_E \in H^0(E, \mathcal{O}_E(7))$. Then $B_7 = (\tau = 0) = \widetilde{D}_x|_E$ is the septic curve on $E \cong \mathbb{P}^2$ defined by the tangent cone of D_x at x. Restricting $\mu^* \widetilde{D}_x = R_7 + 4J$ to $\widetilde{E}$ and using $J|_{\widetilde{E}} = \tilde{s}$ and (13.12), we obtain

$$b^* B_7 = R_7|_{\widetilde{E}} + 4\tilde{s}, \qquad R_7|_{\widetilde{E}} \sim 3\tilde{s} + 7\tilde{f}. \tag{13.13}$$

Step 5. In this step we show that $W_{m,j} = \tau^{j-3m} W_{n,3n}$ with $n := 7m - 2j$ for integers $m \geq 1$ and $3m \leq j \leq 7m/2$.

Fix such m and j. Then $n \geq 0$, $n + 2(j - 3m) = m$, and $3n + 7(j - 3m) = j$. Hence multiplication by σ_7^{j-3m} is an injective linear map

$$H^0(Y, n\pi^* H - 3nE) \to H^0(Y, m\pi^* H - jE).$$

By Lemma 13.1, both spaces have dimension $h^0(U_x, nL_x)$. For $n = 0$ both spaces are one-dimensional. Thus the map is an isomorphism. Since restriction to E is multiplicative and $\sigma_7|_E = \tau$, we obtain

$$W_{m,j} = \tau^{j-3m} W_{n,3n}, \qquad \dim W_{n,3n} = \dim W_{m,j}. \tag{13.14}$$

Step 6. In this step we construct a smooth projective surface $\widetilde{Q}$ with birational morphisms $p_1 : \widetilde{Q} \to \widetilde{E}$ and $q : \widetilde{Q} \to Q$, and we compute the relative canonical divisors.

By **Step 4**, the prime divisors $\widetilde{E} \subset \widetilde{Y}^+$ and $Q \subset V'$ over V_5 both define the valuation ord_C. Hence the function fields of $\widetilde{E}$ and Q are both the residue field of ord_C, and the identity map of the residue field defines a birational map $g : \widetilde{E} \dashrightarrow Q$ over $f_x \cong C$. Since $Y^+ \to V_5$ is an isomorphism near the points of f_x outside Γ, g is an isomorphism over $C \backslash \Gamma$.

Fix $y \in C \cap \Gamma$, $k := k_y$, and the coordinates of **Step 2**. Over a neighborhood of y, the restrictions z' and $z'' = 1/z'$ of z/u_2 and u_2/z to $\widetilde{E}$ are fiber coordinates of $\widetilde{E} \to f_x$, and w is a fiber coordinate of $Q \to C$. On Y^+ we have $u_1/u_2 = z - t^k z/u_2$, and z vanishes on $\widetilde{E}$. Hence, as rational functions,

$$g^* w = -t^k z', \quad \text{that is,} \quad w z'' = -t^k. \tag{13.15}$$

Let $q : \widetilde{Q} \to Q$ be the composite of the following blow-ups over the points $y \in C \cap \Gamma$: the blow-up of $c_1 := (t = w = 0)$ with exceptional curve e_1, and then, for $1 \leq i \leq k - 1$, the blow-up of the point $c_{i+1} := (t = w_i = 0) \in e_i$ with exceptional curve e_{i+1}, where (t, w_i) are coordinates on the

chart of the i-th blow-up with $w = t^i w_i$ and $e_i = (t = 0)$. Let l be the strict transform of $\bar{f}_y$ on $\widetilde{Q}$. When y varies we write l_y and $e_{y,i}$. Then

$$\operatorname{ord}_l t = \operatorname{ord}_{e_i} t = 1, \qquad \operatorname{ord}_l w = 0, \qquad \operatorname{ord}_{e_i} w = i, \tag{13.16}$$

so $q^*\bar{f}_y = l + e_1 + \cdots + e_k$.

We show that $g \circ q$ extends to a birational morphism $p_1 : \widetilde{Q} \to \widetilde{E}$. Write $w_0 := w$. On Q near $\bar{f}_y \backslash \{c_1\}$, (13.15) gives $z'' = -t^k w^{-1}$, and $z'' = -t^k w'$ with $w' := 1/w$ near $w = +\infty$. On the chart (t, w_i) of the i-th blow-up, $z'' = -t^{k-i} w_i^{-1}$, which is regular off $c_{i+1} = (w_i = 0)$. On the other chart of the i-th blow-up, with coordinates (s, w_{i-1}), we have $t = s w_{i-1}$, $e_i = (w_{i-1} = 0)$, and

$$z'' = -s^{k-i+1} w_{i-1}^{k-i}.$$

On the chart (t, w_k), (13.15) implies $z' = -w_k$. Since (t, z''), resp. (t, z'), are coordinates of $\widetilde{E}$ near the point $(t, z'') = (0, 0)$, resp. near the fiber $\tilde{f}_y$ of $\widetilde{E}$ over y minus that point, $g \circ q$ is given on the charts above by (t, z'') or (t, z'), and it extends to a morphism near $q^{-1}(\bar{f}_y)$. Since g is an isomorphism over $C \backslash \Gamma$, we obtain a birational morphism $p_1 : \widetilde{Q} \to \widetilde{E}$. The functions t and z'' vanish on $l, e_1, \dots, e_{k-1}$, so p_1 contracts $l, e_1, \dots, e_{k-1}$ to $(t, z'') = (0, 0)$. On the chart (t, w_k), the curve $e_k = (t = 0)$ maps isomorphically onto $\tilde{f}_y$ minus that point. In particular $p_1^* \tilde{f}_y = q^* \bar{f}_y$.

On the chart (t, w_i), $dt \wedge dw = t^i dt \wedge dw_i$, so the coefficient of e_i in $K_{\widetilde{Q}/Q}$ is i. By (13.15), $dt \wedge dz'' = t^k w^{-2} dt \wedge dw$. The form $dt \wedge dz''$ generates $K_{\widetilde{E}}$ near $(t, z'') = (0, 0)$ and near a general point of $\tilde{f}_y$, and $dt \wedge dw$ generates K_Q near c_1 and near a general point of $\bar{f}_y$. Hence the coefficient of a curve $D \in \{l, e_1, \dots, e_k\}$ in $K_{\widetilde{Q}/\widetilde{E}} - K_{\widetilde{Q}/Q}$ is $k \operatorname{ord}_D t - 2 \operatorname{ord}_D w$. By (13.16) we obtain

$$K_{\widetilde{Q}/Q} = \sum_y \sum_{i=1}^{k_y} i e_{y,i}, \qquad K_{\widetilde{Q}/\widetilde{E}} = \sum_y \Big(k_y l_y + \sum_{i=1}^{k_y - 1} (k_y - i) e_{y,i} \Big). \tag{13.17}$$

Since $\sum_y k_y = 3$, we obtain

$$K_{\widetilde{Q}/\widetilde{E}} + K_{\widetilde{Q}/Q} = \sum_y k_y q^* \bar{f}_y \sim 3 q^* \mathcal{O}_Q(1, 0).$$

Since

$$p_1^* K_{\widetilde{E}} + K_{\widetilde{Q}/\widetilde{E}} = K_{\widetilde{Q}} = q^* K_Q + K_{\widetilde{Q}/Q},$$

$K_{\widetilde{E}} \sim -2\tilde{s} - 3\tilde{f}$, $K_Q \sim \mathcal{O}_Q(-2, -2)$, and $p_1^* \tilde{f} \sim q^* \mathcal{O}_Q(1, 0)$, we obtain the linear equivalence

$$2 p_1^* \tilde{s} \sim 2\big(q^* \mathcal{O}_Q(1, 1) - K_{\widetilde{Q}/Q} \big).$$

Since $q : \widetilde{Q} \to Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ is a composite of point blow-ups, $\operatorname{Pic}(\widetilde{Q})$ is torsion free [Har77, Proposition V.3.2], and therefore

$$p_1^* \tilde{s} \sim q^* \mathcal{O}_Q(1, 1) - K_{\widetilde{Q}/Q}. \tag{13.18}$$

Step 7. In this step we show that $(E, \frac{1}{3} B_7)$ is klt.

Let $D_0 := q^* \Lambda - K_{\widetilde{Q}/Q}$. We show that D_0 is a reduced snc divisor. Fix $y \in C \cap \Gamma$ and $k := k_y$, and write q near $\bar{f}_y$ as $Q_k \to \cdots \to Q_0 = Q$, where $Q_i \to Q_{i-1}$ is the blow-up of c_i. Let $D^{(0)} := \Lambda$, and let $D^{(i)}$ be the pullback of $D^{(i-1)}$ minus e_i. Since c_{i+1} belongs to e_i and to no other curve over $\bar{f}_y$, $K_{\widetilde{Q}/Q}$ is the sum of the total transforms of $e_1, \dots, e_k$ near $\bar{f}_y$ [Har77, Proposition V.3.3], so $D^{(k)} = D_0$ near $q^{-1}(\bar{f}_y)$. By (13.9) and (13.11), a local equation φ_Λ of Λ at c_1 belongs to the ideal (w, t^k). On the chart (t, w_i), $D^{(i)}$ has the local equation $t^{-i} \varphi_\Lambda(t, t^i w_i) \in (w_i, t^{k-i})$, so $c_{i+1} \in D^{(i)}$ for $i < k$. Assume that $D^{(i-1)}$ is a reduced snc divisor containing c_i. If c_i is a smooth point of $D^{(i-1)}$, then $D^{(i)}$ is the strict transform of $D^{(i-1)}$. If c_i is a node of $D^{(i-1)}$, then $D^{(i)}$ is the strict transform of $D^{(i-1)}$ plus e_i. In both cases $D^{(i)}$ is a reduced snc divisor. Since Λ is a reduced snc divisor by **Step 3**, induction on i shows that D_0 is a reduced snc divisor.

In particular $D_0 \geq 0$. Since $\Lambda \in |\mathcal{O}_Q(1,1)|$, (13.18) implies $D_0 \sim p_1^*\tilde{s}$. Since p_1 is birational, $h^0(\widetilde{Q}, p_1^*\tilde{s}) = h^0(\widetilde{E}, \tilde{s}) = 1$. Thus

$$D_0 = p_1^*\tilde{s}. \tag{13.19}$$

Next we show that q^*C_4 is a reduced snc divisor. Near the fibers $\bar{f}_y$ with $c_1 \notin C_4$, q^*C_4 is the strict transform of the smooth curve C_4. Assume that $c_1 \in C_4$ for some y, let $k := k_y$, and let φ_4 be a local equation of C_4 at c_1. If $k \geq 2$, then $\varphi_4 \notin (w, t^2)$, since otherwise $\operatorname{length}(\Xi \cap C_4) \geq \dim \mathcal{O}_{Q,c_1}/(w,t^2) = 2$, contrary to (13.11). Hence the tangent line of C_4 at c_1 is different from the tangent line of $(w=0)$, so the strict transform of C_4 does not contain c_2. Thus, near $q^{-1}(c_1)$,

$$q^*C_4 = C_4 + e_1 + \cdots + e_k,$$

where the strict transform C_4 intersects only e_1, transversally at a point outside e_2, and e_i intersects e_{i+1} transversally at one point. Hence q^*C_4 is a reduced snc divisor.

Let $R_V \subset V_5$ be the image of R_7. Since $\operatorname{Pic}(V') = \mathbb{Z}h \oplus \mathbb{Z}Q$, (13.10) implies that $R_V \in |2h|$ and that the pullback of R_V to V' is $R_7 + 3Q$. Fix $y \in C \cap \Gamma$ and $k := k_y$. A local equation φ of R_V at y belongs to $(u_1, u_2)^3$, and we write $\varphi \equiv \varphi_3(t; u_1, u_2)$ modulo $(u_1, u_2)^4$, where φ_3 is a cubic form in (u_1, u_2) whose coefficients are holomorphic functions of t. On the chart (t, w, u_2) of V', $\varphi = u_2^3(\varphi_3(t; w, 1) + u_2\varphi')$ for a holomorphic function φ', so $\varphi_3(t; w, 1)$ is a local equation of $R_7|_Q = C_4$ at c_1 and at a general point of $\bar{f}_y$. Since $\Gamma' \not\subset R_7$, $\Gamma \not\subset R_V$, so $R_7 \subset Y^+$ is defined by $\varphi(t, (u_2 - t^k)z, u_2) \equiv \varphi_3(t; -t^k z, u_2)$ modulo $(z, u_2)^4$. On the chart of $\widetilde{Y}^+$ with $u_2 = z''z$, the function $\varphi(t, (u_2 - t^k)z, u_2)$ equals $z^3(\varphi_3(t; -t^k, z'') + z\varphi'')$ for a holomorphic function φ'', so $\varphi_3(t; -t^k, z'')$ is a local equation of $R_7|_{\widetilde{E}}$ at $(t, z'') = (0,0)$ and at a general point of $\tilde{f}_y$. By (13.15), $-t^k/z'' = w$, so

$$\varphi_3(t; -t^k, z'') = z''^3\varphi_3(t; w, 1)$$

as rational functions on $\widetilde{Q}$. Let $D \in \{l, e_1, \ldots, e_k\}$. Then $p_1(D)$ is the point $(0,0)$ or the curve $\tilde{f}_y$, and $q(D)$ is the point c_1 or the curve $\bar{f}_y$. Hence the coefficients of D in $p_1^*(R_7|_{\widetilde{E}})$ and in q^*C_4 are $\operatorname{ord}_D \varphi_3(t; -t^k, z'')$ and $\operatorname{ord}_D \varphi_3(t; w, 1)$. By (13.15), (13.16), and (13.17), $\operatorname{ord}_D z'' = k \operatorname{ord}_D t - \operatorname{ord}_D w$ is the coefficient of D in $K_{\widetilde{Q}/\widetilde{E}}$. Over $C \backslash \Gamma$, and over the points $t \neq 0$ near y, the map g is an isomorphism mapping $R_7|_{\widetilde{E}}$ to C_4, by the equality $\varphi_3(t; -t^k z', 1) = \varphi_3(t; w, 1)$. Therefore

$$p_1^*(R_7|_{\widetilde{E}}) = q^*C_4 + 3K_{\widetilde{Q}/\widetilde{E}}. \tag{13.20}$$

Let $p := b \circ p_1 : \widetilde{Q} \to E$. Since $K_{\widetilde{E}} = b^*K_E + \tilde{s}$, the equalities (13.13), (13.19), and (13.20) imply

$$p^*B_7 = q^*C_4 + 3K_{\widetilde{Q}/\widetilde{E}} + 4D_0, \qquad K_{\widetilde{Q}} - p^*K_E = K_{\widetilde{Q}/\widetilde{E}} + D_0,$$

and hence

$$K_{\widetilde{Q}} + \frac{1}{3}\big(q^*C_4 + D_0\big) = p^*\Big(K_E + \frac{1}{3}B_7\Big). \tag{13.21}$$

Since q^*C_4 and D_0 are reduced snc divisors, the pairs $(\widetilde{Q}, q^*C_4)$ and $(\widetilde{Q}, D_0)$ are lc [KM98, Corollary 2.31(3)]. Thus $\operatorname{mult}_v(q^*C_4) \leq A_{\widetilde{Q}}(v)$ and $\operatorname{mult}_v D_0 \leq A_{\widetilde{Q}}(v)$ for any prime divisor v over $\widetilde{Q}$. By (13.21) and [KM98, Lemma 2.30], for any prime divisor v over E,

$$A_{E, \frac{1}{3}B_7}(v) = A_{\widetilde{Q}}(v) - \frac{1}{3}\operatorname{mult}_v\big(q^*C_4 + D_0\big) \geq \frac{1}{3}A_{\widetilde{Q}}(v) > 0.$$

Thus $(E, \frac{1}{3}B_7)$ is klt. In particular $v(B_7) < 3A_E(v)$ for any prime divisor v over E.

Step 8. In this step we show that $S(W; v) \leq \frac{85}{88}A_E(v) + \frac{1}{88}v(B_7)$.

Fix a prime divisor v over E and let $\alpha := A_E(v)$. For a linear subspace $U \subset H^0(E, \mathcal{O}_E(j))$, let $w_v(U)$ be the sum of the jumping numbers of the filtration induced by v on U, and let $r_{m,j} := \dim W_{m,j}$. By Definition 2.8 and [AZ22, Lemma 2.9],

$$S(W; v) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j\geq 0} w_v(W_{m,j}).$$

By Lemma 13.1, $W_{m,j} = 0$ for $j > 7m/2$. The jumping numbers of v on $H^0(E, \mathcal{O}_E(j))$ are at most a constant multiple of j depending on v. Hence the errors $o(m^3)$ in the sums of ranks in (13.22) below and $O(m)$ in the individual ranks $r_{m,j}$ for $j \geq 3m$, the errors $o(j^3)$ in Lemma 3.2, and the replacement of the sums over j by integrals contribute $o(m^4)$ to the sums below.

Since $H^3 = 22$, $N_m = \frac{11}{3}m^3 + O(m^2)$. By Proposition 4.5 with $k = 1$, $\mathrm{vol}_Y(\pi^*H - 2E) = \frac{29}{2}$, and by Lemma 13.1, $h^0(Y, m\pi^*H - 3mE) = \frac{m^3}{3} + O(m^2)$. Since $\sum_{j\geq j_0} r_{m,j} = h^0(Y, m\pi^*H - j_0E)$, we obtain

$$\begin{aligned} \sum_{j<2m} r_{m,j} &= \frac{5}{4}m^3 + o(m^3), \\ \sum_{2m\leq j<3m} r_{m,j} &= \frac{25}{12}m^3 + o(m^3), \\ \sum_{j\geq 3m} r_{m,j} &= \frac{1}{3}m^3 + O(m^2). \end{aligned} \tag{13.22}$$

For $j < 2m$, $w_v(W_{m,j}) \leq w_v(H^0(E, \mathcal{O}_E(j)))$, since $w_v(U) = \int_0^\infty \dim\{s \in U : v(s) \geq t\}dt$. By Lemma 3.2(1) with $\rho = 1$,

$$w_v(H^0(E, \mathcal{O}_E(j))) \leq \frac{\alpha}{3} j \dim H^0(E, \mathcal{O}_E(j)) + o(j^3) = \frac{\alpha}{6}j^3 + o(j^3).$$

Hence

$$\sum_{j<2m} w_v(W_{m,j}) \leq \frac{\alpha}{6} \sum_{j<2m} j^3 + o(m^4) = \frac{2\alpha}{3}m^4 + o(m^4).$$

For $2m \leq j < 3m$, Lemma 3.2(2), with $a = \alpha$, implies

$$w_v(W_{m,j}) \leq \alpha\Big(jr_{m,j} - \frac{2\sqrt{2}}{3}r_{m,j}^{3/2}\Big) + o(j^3).$$

For any real numbers $r \geq 0$ and $0 \leq c \leq j$,

$$jr - \frac{2\sqrt{2}}{3}r^{3/2} \leq cr + \frac{1}{6}(j-c)^3,$$

since the maximum of $(j-c)r - \frac{2\sqrt{2}}{3}r^{3/2}$ over $r \geq 0$ is $\frac{1}{6}(j-c)^3$, attained at $r = \frac{1}{2}(j-c)^2$. We take $c = \frac{m}{2}$, which is at most j for $2m \leq j < 3m$ and gives the constant $\frac{85}{88}$ below. By (13.22),

$$\sum_{2m\leq j<3m} w_v(W_{m,j}) \leq \alpha\Big(\frac{m}{2}\cdot\frac{25}{12}m^3 + \frac{m^4}{6}\int_2^3 \Big(t - \frac{1}{2}\Big)^3 dt\Big) + o(m^4) = \frac{59}{24}\alpha m^4 + o(m^4).$$

Since $\frac{2}{3} + \frac{59}{24} = \frac{75}{24}$ and $\frac{75}{24}\cdot\frac{3}{11} = \frac{75}{88}$, the grades $j < 3m$ contribute at most $\frac{75}{88}\alpha$ to $S(W; v)$.

For $3m \leq j \leq 7m/2$, let $n := 7m - 2j$. By (13.14), the filtration induced by v on $W_{m,j}$ is the filtration on $W_{n,3n}$ shifted by $(j-3m)v(B_7)$, so

$$w_v(W_{m,j}) = (j-3m)r_{m,j}v(B_7) + w_v(W_{n,3n}).$$

By Lemma 13.1 and (13.14), $\dim W_{n,3n} = r_{m,j} = 2n^2 + O(m)$ uniformly in j. The function $r \to 3nr - \frac{2\sqrt{2}}{3}r^{3/2}$ has derivative of absolute value $O(n)$ on $[0, \dim H^0(E, \mathcal{O}_E(3n))]$, and equals $\frac{10}{3}n^3$

at $r = 2n^2$. Hence Lemma 3.2(2) in degree $3n$, with $a = \alpha$, implies $w_v(W_{n,3n}) \le \frac{10}{3}\alpha n^3 + o(m^3)$. Since

$$\int_3^{7/2} \frac{10}{3}(7-2t)^3 dt = \frac{5}{12}, \qquad \int_3^{7/2} 2(t-3)(7-2t)^2 dt = \frac{1}{24},$$

we obtain

$$\sum_{3m \le j \le 7m/2} w_v(W_{m,j}) \le \Big(\frac{5}{12}\alpha + \frac{1}{24}v(B_7)\Big)m^4 + o(m^4).$$

Since $\frac{5}{12} \cdot \frac{3}{11} = \frac{10}{88}$ and $\frac{1}{24} \cdot \frac{3}{11} = \frac{1}{88}$, we conclude that

$$S(W; v) \le \frac{75}{88}\alpha + \frac{10}{88}\alpha + \frac{1}{88}v(B_7) = \frac{85}{88}A_E(v) + \frac{1}{88}v(B_7).$$

By **Step 7**, $v(B_7) < 3A_E(v)$, so $S(W; v) \le A_E(v)$.

Step 9. In this step we bound $S_X(E)$ and conclude the proof.

The jumping numbers of ord_E on $H^0(X, mH)$ are the integers j with multiplicities $r_{m,j}$. Hence, by [AZ22, Lemma 2.9], by $W_{m,j} = 0$ for $j > 7m/2$, and by (13.22),

$$S_X(E) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_j j r_{m,j} \le \frac{3}{11}\Big(2 \cdot \frac{5}{4} + 3 \cdot \frac{25}{12} + \frac{7}{2} \cdot \frac{1}{3}\Big) = \frac{119}{44}.$$

The pair (Y, E) is plt [KM98, Corollary 2.31(3)], the different of (Y, E) on E is zero [Har77, Proposition II.8.20], and $A_X(E) = 3$ [KM98, Lemma 2.29]. By **Step 8**, $\delta_{Z'}(E; W) \ge 1$ for any irreducible closed subset $Z' \subset E$. Theorem 2.9 with $Z = \{x\}$ implies

$$\delta_x(X) \ge \min\Big\{\frac{3}{S_X(E)}, 1\Big\} = 1,$$

since $3/S_X(E) \ge \frac{132}{119}$. Thus $A_X(F) \ge S_X(F)$ for any prime divisor F over X with $x \in \mathrm{center}_X F$. □

The klt property in Construction-Theorem 13.2 improves $\delta_x(X) \ge 1$ to a strict inequality, with a lower bound depending on x.

Corollary 13.3. *Let X be a smooth prime Fano threefold of genus 12 and $x \in X$ a closed point which belongs to exactly one line L, and assume that L is non-special. Let E and B_7 be as in Construction-Theorem 13.2 and $\ell_x := \mathrm{lct}(E; B_7)$. Then $\ell_x > 1/3$ and*

$$\delta_x(X) \ge \min\Big\{\frac{132}{119}, \frac{88}{85 + \ell_x^{-1}}\Big\} > 1.$$

Proof. We first show that $\Sigma(X, x)$ consists of one reduced point. Since x belongs to exactly one line, the support of $\Sigma(X, x)$ is $\{[L]\}$. Since $N_{L/X} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1)$, Proposition 4.1 shows that the Zariski tangent space of $\Sigma(X, x)$ at $[L]$ is

$$H^0(L, N_{L/X}(-x)) = H^0(L, \mathcal{O}_L(-1) \oplus \mathcal{O}_L(-2)) = 0.$$

Thus the local ring of $\Sigma(X, x)$ at $[L]$ is a zero-dimensional Noetherian local ring whose maximal ideal $\mathfrak{m}$ satisfies $\mathfrak{m} = \mathfrak{m}^2$, so $\mathfrak{m} = 0$ by Nakayama's lemma. Hence the hypotheses of Construction-Theorem 13.2 hold.

Let $\pi : Y = \mathrm{Bl}_x X \to X$ and the refinement W of $(H^0(X, mH))_m$ by E be as in Construction-Theorem 13.2. By Construction-Theorem 13.2, $(E, \frac{1}{3}B_7)$ is klt. Since the log canonical threshold and the klt property are read off from the coefficients of a log resolution of (E, B_7) [KM98, Corollary 2.32], we have $\ell_x > 1/3$, and $(E, \ell_x B_7)$ is lc. Thus $v(B_7) \le \ell_x^{-1} A_E(v)$ for any prime divisor v over E. By Construction-Theorem 13.2,

$$S(W; v) \le \frac{85}{88}A_E(v) + \frac{1}{88}v(B_7) \le c_x A_E(v), \qquad c_x := \frac{85 + \ell_x^{-1}}{88},$$

for any prime divisor v over E, and $c_x < 1$ since $\ell_x^{-1} < 3$. Hence $\delta_{Z'}(E;W) \geq c_x^{-1}$ for any irreducible closed subset $Z' \subset E$. Since $A_X(E) = 3$, the different of (Y,E) on E is zero, and $S_X(E) \leq \frac{119}{44}$ by Construction-Theorem 13.2, Theorem 2.9 with $Z = \{x\}$ implies

$$\delta_x(X) \geq \min\left\{\frac{3}{S_X(E)}, \frac{1}{c_x}\right\} \geq \min\left\{\frac{132}{119}, \frac{88}{85+\ell_x^{-1}}\right\}.$$

Both terms of the minimum are larger than 1, since $\ell_x^{-1} < 3$. □

13.2. **The divisor R_0 and the base points of U.** In this subsection, we study the divisor R_0 and the base points of U for a non-special line L. Lemma 13.5 bounds the coefficients of the fibers of $E \to L$ in R_0 and excludes three configurations of the base points of U. Propositions 13.10 and 13.11 prove $\delta_x(X) \geq 1$ when Z_U is the disjoint union of a subscheme of f_x of length two and a point of $E\backslash(s \cup f_x)$. Lemma 13.5 is used in Subsections 13.3 and 13.4, and Proposition 13.11, whose proof uses Proposition 13.10, is used in Subsection 13.4.

Set-up 13.4. Notation as in Construction-Proposition 11.3, so that L is non-special and $E \cong \mathbb{F}_1$. Let W be the refinement of the anti-canonical series of X by E (Definition 2.8), whose pieces are the spaces $W_{m,j}$. For $x \in L$ let W^{f_x} be the refinement of W by f_x. Write $p_E : T_E \to E$ (Construction-Proposition 11.3(2)) as the composite of three blow-ups

$$T_E = E^{(3)} \to E^{(2)} \to E^{(1)} \to E^{(0)} = E.$$

For a curve $C \subset E$, let k_C be the number of indices $k \in \{1,2,3\}$ such that the center of $E^{(k)} \to E^{(k-1)}$ is a point of the strict transform of C on $E^{(k-1)}$.

We use the following coordinates on E. Let z be an affine coordinate on $L \cong \mathbb{P}^1$. When a point $x \in L$ is fixed, we choose z with $f_x = (z = 0)$. The complement in E of a section of $E \to L$ disjoint from s is the total space of the normal bundle of s, and we let u be a linear coordinate on the fibers of the total space of the normal bundle of s over $(z \neq +\infty)$, so that $s = (u = 0)$. For integers $b \geq a \geq 0$, a basis of $H^0(E, \mathcal{O}_E(as+bf))$ is given by the monomials

$$z^i u^\ell, \qquad 0 \leq \ell \leq a, \qquad 0 \leq i \leq b - a + \ell$$

[Har77, Exercise III.8.4(a) and Propositions V.2.8 and V.2.9]. On $E\backslash s$ we use the fiber coordinate $u' := u^{-1}$. In the coordinates (z,u'), the monomial $z^i u^\ell$ corresponds to $z^i u'^{\ell'}$ with $\ell' := a - \ell$, and the conditions on the exponents become $0 \leq \ell' \leq a$ and $0 \leq i \leq b - \ell'$.

By Construction-Proposition 11.3(4)(b), the divisor R_0 is a fixed component of the pieces $W_{m,j}$ with $2m < j \leq 3m$.

The next lemma controls the components of R_0. All three exclusions in the lemma come from the fact that Γ spans a hyperplane of $\mathbb{P}^6$.

Lemma 13.5. *Notation as in Set-up 13.4. Then the following hold.*

(1) The divisor $D_R := p_E^* R_0 - 2(e_1+e_2+e_3)$ on T_E satisfies $D_R \geq 0$, and it has exactly one component C_Γ with $(\Lambda \cdot C_\Gamma) > 0$. The curve C_Γ is the strict transform of a component of R_0, $\operatorname{mult}_{C_\Gamma} D_R = 1$, the morphism $\varphi : T_E \to \mathbb{P}(U^\vee)$ defined by the linear system $p_E^* U - (e_1+e_2+e_3) \subset |\Lambda|$ maps C_Γ isomorphically onto Γ, and $(\Lambda \cdot C_\Gamma) = 5$. Here $\mathbb{P}(U^\vee)$ is identified with the hyperplane of $\mathbb{P}(H^0(Y,h)^\vee) \cong \mathbb{P}^6$ spanned by Γ. Any component $D' \neq C_\Gamma$ of D_R satisfies $(\Lambda \cdot D') = 0$.

(2) If $s \subset \operatorname{Supp} R_0$, then $k_s = 1$. If $f_x \subset \operatorname{Supp} R_0$ for some $x \in L$, then $k_{f_x} = 2$ and $f_{x'} \not\subset \operatorname{Supp} R_0$ for any $x' \in L\backslash\{x\}$. If $k_s = 0$, then $\operatorname{mult}_{f_x} R_0 \leq 1$ for any $x \in L$. Moreover, $\operatorname{mult}_{f_x} R_0 \leq 2$ for any $x \in L$.

(3) Let $x \in L$. The base points of U do not form a chain of three points starting at $s \cap f_x$, in which the second point is infinitely near the first in the direction of f_x, and the third point is infinitely near the second and does not belong to the strict transform of f_x. They do not consist of a subscheme of length two of f_x disjoint from s together with a point of

$s\backslash f_x$. They do not consist of the points $s\cap f_x$ and $q\in f_x\backslash s$ together with a point infinitely near $s\cap f_x$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we describe U and φ through the line link.

By Theorem 2.23, the link is a flop $Y \dashrightarrow Y^+ = \mathrm{Bl}_\Gamma V_5$, where $\Gamma \subset V_5 \subset \mathbb{P}^6$ is a rational normal quintic curve. Let $\rho : Y^+ \to V_5$ be the blow-up of Γ. As in Theorem 2.23, we write h, E, and R also for the strict transforms on Y^+. Then $h = \rho^*\mathcal{O}_{V_5}(1)$, R is the exceptional divisor of ρ, $\rho(E)$ is the unique hyperplane section of V_5 containing Γ, and any flopped curve $\ell \subset Y^+$ is the strict transform of a bisecant line of Γ, so that $(h\cdot\ell) = 1$ (see also [KPS18, Lemma 5.2.5]). Since $Y \dashrightarrow Y^+$ is an isomorphism in codimension one and R is ρ-exceptional, we have

$$h^0(Y,h) = h^0(Y^+,h) = h^0(V_5,\mathcal{O}_{V_5}(1)) = 7 \quad \text{and} \quad h^0(Y,R) = h^0(Y^+,R) = 1.$$

Since $h - E = R$, the exact sequence

$$0 \to \mathcal{O}_Y(R) \to \mathcal{O}_Y(h) \to \mathcal{O}_E(h|_E) \to 0$$

shows that the kernel of the restriction map $H^0(Y,h) \to U$ is spanned by the section of h with divisor $E + R$. Thus $\dim U = 6$.

On Y^+ we have $\rho^*\rho(E) = E + cR$, where $c \geq 1$ is the multiplicity of $\rho(E)$ along Γ. Since $E + R \sim h \sim E + cR$, we have $(c-1)R \sim 0$, and $c = 1$ because $R \neq 0$ is a divisor with $R \geq 0$ on the projective variety Y^+. Thus $E \sim h - R$ on Y^+, and under the identification $H^0(Y,h) = H^0(V_5,\mathcal{O}_{V_5}(1))$, the section with divisor $E+R$ defines the hyperplane of $\mathbb{P}^6$ containing $\rho(E) \supset \Gamma$. Since Γ is a rational normal quintic curve, Γ spans a $\mathbb{P}^5$ [Har92, Example 1.14], so the hyperplane defined by the section with divisor $E + R$ is the span of Γ. Since U is the quotient of $H^0(Y,h)$ by the line spanned by the section with divisor $E + R$, the span of Γ is equal to $\mathbb{P}(U^\vee)$. The rational map $E \dashrightarrow \mathbb{P}(U^\vee)$ defined by U is the restriction to E of the rational map $Y \dashrightarrow V_5 \subset \mathbb{P}^6$ defined by $|h|$, which is the composition of the flop with ρ.

By Construction-Proposition 11.3(2)(a), the base ideal of the linear system p_E^*U is $I_{Z_U}\cdot\mathcal{O}_{T_E} = \mathcal{O}_{T_E}(-e_1 - e_2 - e_3)$. Thus $p_E^*U - (e_1 + e_2 + e_3)$ is a base point free linear subsystem of $|\Lambda|$. It defines the morphism $\varphi : T_E \to \mathbb{P}(U^\vee)$, which extends the rational map $E \dashrightarrow \mathbb{P}(U^\vee)$ defined by U, and $\varphi^*\mathcal{O}_{\mathbb{P}(U^\vee)}(1) = \mathcal{O}_{T_E}(\Lambda)$. In particular, Λ is nef.

Step 2. In this step, we prove (1).

By Construction-Proposition 11.3(5), we have $D_R = p_E^*R_0 - 2(e_1 + e_2 + e_3) \geq 0$.

Let ℓ_R be the fiber of $R \to \Gamma$ over a general point of Γ. Then $(h\cdot\ell_R) = 0$ and $(R\cdot\ell_R) = -1$, so

$$(E\cdot\ell_R) = ((h-R)\cdot\ell_R) = 1.$$

Since $E \neq R$, the curve ℓ_R is not contained in E, and $E\cap\ell_R$ is one reduced point. Thus $E\cap R$ contains exactly one curve Γ^+ dominating Γ, and $\Gamma^+ \to \Gamma$ is birational. In particular, $(h\cdot\Gamma^+) = 5$. Since $(h\cdot\ell) = 1$ for any flopped curve ℓ, the curve Γ^+ is not a flopped curve, and the flop is an isomorphism at the generic point of Γ^+. Hence the strict transform of Γ^+ on Y is a curve contained in $E\cap R = \operatorname{Supp} R_0$, that is, a component of R_0. Let C_Γ be the strict transform on T_E of the strict transform of Γ^+ on Y. The morphism φ coincides with the composition of the flop with ρ on a neighborhood of the generic point of C_Γ, so $\varphi(C_\Gamma) = \rho(\Gamma^+) = \Gamma$. By the projection formula [Har77, Appendix A, §1, A4],

$$(\Lambda\cdot C_\Gamma) = \deg(\varphi|_{C_\Gamma} : C_\Gamma \to \Gamma)\cdot\deg\Gamma = 5\deg(\varphi|_{C_\Gamma} : C_\Gamma \to \Gamma) \geq 5.$$

On the other hand, $(p_E^*D\cdot e_i) = 0$ for any divisor D on E and $(e_i\cdot e_j) = -\delta_{ij}$ [Har77, Proposition V.3.2]. Since $R_0 \in |3s + 4f|$, $(s^2) = -1$, $(s\cdot f) = 1$, and $(f^2) = 0$, we obtain

$$(\Lambda\cdot D_R) = ((2s+3f)\cdot(3s+4f)) + 2\sum_{i=1}^{3}(e_i\cdot e_i) = 11 - 6 = 5.$$

Write $D_R = cC_\Gamma + D''$, where $c = \mathrm{mult}_{C_\Gamma} D_R$ and $D'' \geq 0$ does not contain C_Γ. Since C_Γ is the strict transform of a component of R_0 and C_Γ is not a component of any e_i, we have $c = \mathrm{mult}_{p_E(C_\Gamma)} R_0 \geq 1$. Since Λ is nef, $(\Lambda \cdot D'') \geq 0$, so

$$5 = c(\Lambda \cdot C_\Gamma) + (\Lambda \cdot D'') \geq 5c \geq 5.$$

Thus $c = 1$, $(\Lambda \cdot C_\Gamma) = 5$, and $(\Lambda \cdot D'') = 0$. Since Λ is nef, any component D' of D'' satisfies $(\Lambda \cdot D') = 0$. In particular, C_Γ is the only component of D_R with positive Λ-degree. Moreover, $\deg(\varphi|_{C_\Gamma} : C_\Gamma \to \Gamma) = 1$, so $\varphi|_{C_\Gamma}$ is a birational morphism of projective curves onto the smooth curve Γ. Thus $\varphi|_{C_\Gamma}$ is finite [Har77, Exercise III.11.2], and $(\varphi|_{C_\Gamma})_*\mathcal{O}_{C_\Gamma}$ is a coherent $\mathcal{O}_\Gamma$-algebra contained in the function field of Γ and integral over $\mathcal{O}_\Gamma$. Since Γ is normal, $(\varphi|_{C_\Gamma})_*\mathcal{O}_{C_\Gamma} = \mathcal{O}_\Gamma$, so $\varphi|_{C_\Gamma}$ is an isomorphism. This proves (1).

Step 3. In this step, we prove (2).

Let $C \subset E$ be a smooth curve and C_T the strict transform of C on T_E. Then

$$C_T = p_E^* C - \sum_{i \in K_C} e_i,$$

where $K_C \subset \{1, 2, 3\}$ is the set of the indices counted by k_C in Set-up 13.4, so that $\#K_C = k_C$ [Har77, Propositions V.3.2 and V.3.6]. Taking $C = s$ and $C = f_x$, we obtain

$$(\Lambda \cdot s_T) = ((2s + 3f) \cdot s) - k_s = 1 - k_s \quad \text{and} \quad (\Lambda \cdot (f_x)_T) = ((2s + 3f) \cdot f) - k_{f_x} = 2 - k_{f_x}.$$

Since Λ is nef, $k_s \leq 1$ and $k_{f_x} \leq 2$. In particular, $(\Lambda \cdot s_T) \leq 1$ and $(\Lambda \cdot (f_x)_T) \leq 2$, so neither s_T nor $(f_x)_T$ is C_Γ by (1).

Suppose that $s \subset \operatorname{Supp} R_0$. Then s_T is a component of D_R, because $\mathrm{mult}_{s_T} D_R = \mathrm{mult}_s R_0 \geq 1$. Since $s_T \neq C_\Gamma$, (1) implies that $1 - k_s = 0$, so $k_s = 1$. If $f_x \subset \operatorname{Supp} R_0$, then $(f_x)_T$ is a component of D_R, because $\mathrm{mult}_{(f_x)_T} D_R = \mathrm{mult}_{f_x} R_0 \geq 1$. Since $(f_x)_T \neq C_\Gamma$, (1) implies that $2 - k_{f_x} = 0$, so $k_{f_x} = 2$. For $x' \in L \backslash \{x\}$, the fibers f_x and $f_{x'}$ are disjoint, so their strict transforms on the blow-ups are disjoint and $K_{f_x} \cap K_{f_{x'}} = \emptyset$. Since there exist exactly three base points and $k_{f_x} = 2$, we have $k_{f_{x'}} \leq 1$, so $f_{x'} \not\subset \operatorname{Supp} R_0$.

Suppose that $k_s = 0$. Then $s \not\subset \operatorname{Supp} R_0$ by the first sentence of (2). Let $x \in L$ and $\mu := \mathrm{mult}_{f_x} R_0$, and suppose that $\mu \geq 2$. Then $R_0 - \mu f_x \geq 0$ does not contain s, and

$$((R_0 - \mu f_x) \cdot s) = ((3s + (4 - \mu)f) \cdot s) = 1 - \mu < 0.$$

This contradicts [Har77, Proposition V.1.4]. Thus $\mathrm{mult}_{f_x} R_0 \leq 1$.

Finally, let $x \in L$ and suppose that $\mu = \mathrm{mult}_{f_x} R_0 \geq 3$. By (1) and the first paragraph of this step, $p_E(C_\Gamma)$ is a component of $R_0 - \mu f_x$ different from s and from the fibers of $E \to L$. Any integral curve $D \neq s$ on E has class $as + bf$ with $a = (D \cdot f) \geq 0$ and $b - a = (D \cdot s) \geq 0$, while s has class s and a fiber has class f. Since $R_0 - \mu f_x$ has class $3s + (4 - \mu)f$ with $4 - \mu \leq 1$, the curve $p_E(C_\Gamma)$ has class $as + bf$ with $a \leq b \leq 1$. Since $C_\Gamma = p_E^* p_E(C_\Gamma) - \sum_i m_i e_i$ with $m_i \geq 0$, we obtain

$$(\Lambda \cdot C_\Gamma) = ((2s + 3f) \cdot (as + bf)) - \sum_i m_i \leq a + 2b \leq 3 < 5,$$

which contradicts (1). Thus $\mathrm{mult}_{f_x} R_0 \leq 2$. This proves (2).

Step 4. In this step, we prove (3). We first show the common reason for the three exclusions.

Claim 13.6. *Let $x \in L$. Then there exists no non-zero $\sigma \in U$ such that $R_0 = s + f_x + D_\sigma$, where D_σ is the divisor of σ.*

Proof. Suppose that such a section σ exists. By **Step 3**, $p_E(C_\Gamma)$ is neither s nor f_x, so $p_E(C_\Gamma)$ is a component of D_σ. Since $\sigma \in U$, the divisor $p_E^* D_\sigma - (e_1 + e_2 + e_3)$ is the member of the linear system $p_E^* U - (e_1 + e_2 + e_3)$ defined by σ, that is, φ^* of the hyperplane $(\sigma = 0)$ of $\mathbb{P}(U^\vee)$. The curve C_Γ is a component of $p_E^* D_\sigma - (e_1 + e_2 + e_3)$, so $\Gamma = \varphi(C_\Gamma)$ is contained in the hyperplane $(\sigma = 0)$. This contradicts the fact that Γ spans $\mathbb{P}(U^\vee)$ (**Step 1**). □

Proof of Lemma 13.5 continued. In the three configurations, we use the following facts. By Construction-Proposition 11.3(5), the local equations of R_0 belong to $I_{Z_U}^2$. By Construction-Proposition 11.3(3) with $q=1$, we have $U \subset H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U})$, and $\dim U = 6$ by **Step 1**. Thus, if the three conditions defined by I_{Z_U} on $H^0(E, \mathcal{O}_E(2s+3f))$ are linearly independent, then $\dim H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}) = 9-3 = 6$ and

$$U = H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}).$$

When x is fixed, we denote by zu the section of $\mathcal{O}_E(s+f)$ with expansion zu, whose divisor is $s+f_x$. In each configuration, we show that $r_0 = zu \cdot \sigma$ for some non-zero $\sigma \in H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}) = U$, which contradicts Claim 13.6.

The first configuration. Suppose that the base points of U form a chain of three points starting at $p := s \cap f_x$, in which the second point is infinitely near the first in the direction of f_x, and the third point is infinitely near the second and does not belong to the strict transform of f_x. Then Z_U is supported at p, and (z,u) are coordinates of E at p with $s = (u=0)$ and $f_x = (z=0)$. Since Z_U is curvilinear of length three (Construction-Proposition 11.3(1)), and the second base point is in the direction of f_x, the ideal of Z_U at p is $(z - cu^2, u^3)$ for some $c \in \mathbb{C}$. Since the third base point does not belong to the strict transform of f_x, the curve f_x does not contain Z_U, so $c \neq 0$. Replacing z with $c^{-1}z$, which does not change the monomial bases of Set-up 13.4, we may assume that $I_{Z_U} = (z-u^2, u^3)$ at p. Since Z_U is supported at p, a section belongs to $H^0(E, \mathcal{O}_E(as+bf) \otimes I_{Z_U}^w)$ if and only if the polynomial expansion of the section in z and u belongs to $(z-u^2, u^3)^w$.

The space $H^0(E, \mathcal{O}_E(2s+3f))$ has the basis of nine monomials

$$1,\ z;\quad u,\ zu,\ z^2u;\quad u^2,\ zu^2,\ z^2u^2,\ z^3u^2.$$

Modulo $(z-u^2, u^3)$, whose quotient has basis $1, u, u^2$, the monomial z maps to u^2, and the monomials zu, z^2u, zu^2, z^2u^2, z^3u^2 map to zero. Thus a linear combination of the nine monomials belongs to I_{Z_U} if and only if the coefficients of 1 and u vanish and the coefficients of z and u^2 have sum zero. The three conditions are linearly independent, and $H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}) = U$ has the basis

$$z-u^2,\ zu,\ z^2u,\ zu^2,\ z^2u^2,\ z^3u^2. \tag{13.23}$$

The space $H^0(E, \mathcal{O}_E(3s+4f))$ has the basis of fourteen monomials consisting of the nine monomials above and

$$u^3,\ zu^3,\ z^2u^3,\ z^3u^3,\ z^4u^3.$$

At p, the ideal $I_{Z_U}^2 = (z-u^2, u^3)^2$ is generated by $(z-u^2)^2$, $(z-u^2)u^3$, and u^6, and the quotient $\mathcal{O}_{E,p}/I_{Z_U}^2$ has the basis

$$1,\ u,\ u^2,\ u^3,\ u^4,\ u^5,\ z-u^2,\ (z-u^2)u,\ (z-u^2)u^2.$$

Writing $z = (z-u^2) + u^2$, the fourteen monomials have the following images, grouped by the power of u:

$$1 \to 1, \quad z \to (z-u^2)+u^2,$$
$$u \to u, \quad zu \to (z-u^2)u + u^3, \quad z^2u \to u^5,$$
$$u^2 \to u^2, \quad zu^2 \to (z-u^2)u^2 + u^4, \quad z^2u^2 \to 0, \quad z^3u^2 \to 0,$$
$$u^3 \to u^3, \quad zu^3 \to u^5, \quad z^2u^3 \to 0, \quad z^3u^3 \to 0, \quad z^4u^3 \to 0.$$

The images span an eight-dimensional space with basis

$$1,\ (z-u^2)+u^2,\ u,\ (z-u^2)u+u^3,\ u^2,\ (z-u^2)u^2+u^4,\ u^3,\ u^5.$$

Thus $\dim H^0(E, \mathcal{O}_E(3s+4f) \otimes I_{Z_U}^2) = 14-8 = 6$, and $H^0(E, \mathcal{O}_E(3s+4f) \otimes I_{Z_U}^2)$ has the basis

$$z^2u - zu^3,\ z^2u^2,\ z^3u^2,\ z^2u^3,\ z^3u^3,\ z^4u^3.$$

The six elements of the last basis are zu times the elements of (13.23), so

$$H^0(E, \mathcal{O}_E(3s+4f) \otimes I_{Z_U}^2) = zu \cdot H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}).$$

Since the local equations of R_0 belong to $I_{Z_U}^2$, we have $r_0 \in H^0(E, \mathcal{O}_E(3s+4f) \otimes I_{Z_U}^2)$. Thus $r_0 = zu \cdot \sigma$ for a non-zero $\sigma \in U$, that is, $R_0 = s + f_x + D_\sigma$. This contradicts Claim 13.6.

The second configuration. Suppose that the base points of U consist of a subscheme $Z_f \subset f_x$ of length two with $Z_f \cap s = \emptyset$, together with a point $r \in s \backslash f_x$. Then Z_U is the disjoint union of Z_f and the reduced point r. We choose z such that r belongs to the open subset of E over $(z \neq +\infty)$.

We first show that $f_x \subset \operatorname{Supp} R_0$. If Z_f consists of two points q_1, q_2, then the local equations of R_0 at q_1 and q_2 belong to the squares of the maximal ideals. If Z_f is supported at one point q, then in local coordinates (z, u_1) of E at q with $f_x = (z = 0)$ the ideal of Z_U at q is (z, u_1^2), and the local equation of R_0 at q belongs to $(z, u_1^2)^2 = (z^2, zu_1^2, u_1^4)$, whose restriction to f_x is (u_1^4). In both cases, if $f_x \not\subset \operatorname{Supp} R_0$, then $(R_0 \cdot f_x) \geq 4$ by [Har77, Proposition V.1.4], which contradicts $(R_0 \cdot f_x) = ((3s+4f) \cdot f) = 3$. Similarly, the local equation of R_0 at r belongs to the square of the maximal ideal, and if $s \not\subset \operatorname{Supp} R_0$, then $(R_0 \cdot s) \geq 2$, which contradicts $(R_0 \cdot s) = ((3s+4f) \cdot s) = 1$. Thus $R_0 - s - f_x \geq 0$, and $r_0 = zu \cdot \sigma$ for a non-zero section σ of $\mathcal{O}_E(2s+3f)$.

We show that $\sigma \in H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U})$. At a point q_i, the local equation of s is a unit and z is a member of a regular system of parameters, so $z\sigma$ belongs to the square of the maximal ideal only if $\sigma(q_i) = 0$. At r, the local equation z of f_x is a unit and u is a member of a regular system of parameters, so $u\sigma$ belongs to the square of the maximal ideal only if $\sigma(r) = 0$. If Z_f is supported at q, then $z\sigma \in (z^2, zu_1^2, u_1^4)$. Since (z^2, zu_1^2, u_1^4) is generated by monomials, and $z \cdot z^i u_1^k \in (z^2, zu_1^2, u_1^4)$ if and only if $i \geq 1$ or $k \geq 2$, we obtain $\sigma \in (z, u_1^2)$.

We show that the three conditions defined by I_{Z_U} are linearly independent. The monomials 1, u, u^2 restrict to a basis of $H^0(f_x, \mathcal{O}_E(2s+3f)|_{f_x})$, where $((2s+3f) \cdot f) = 2$, and the subscheme Z_f of length two of $f_x \cong \mathbb{P}^1$ defines two linearly independent conditions on $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2))$. The monomial z vanishes along f_x, hence on Z_f, and $z(r) \neq 0$ since $r \notin f_x$. Thus the three conditions are linearly independent, and $\sigma \in U$. This contradicts Claim 13.6.

The third configuration. Suppose that the base points of U are $p := s \cap f_x$, a point $q \in f_x \backslash s$, and a point infinitely near p. Then $k_{f_x} \geq 2$ and $k_s \geq 1$, so $k_{f_x} = 2$ and $k_s = 1$ by **Step 3**. Thus the third base point belongs to neither the strict transform of f_x nor that of s. In the coordinates (z, u) at p, the ideal of Z_U at p is

$$J := (z - cu, u^2)$$

for some $c \neq 0$, since the tangent direction of Z_U at p is different from those of $s = (u = 0)$ and $f_x = (z = 0)$. The ideal of Z_U at q is the maximal ideal $\mathfrak{m}_q$ of $\mathcal{O}_{E,q}$.

The local equations of R_0 at p and at q belong to the squares of the maximal ideals. If $f_x \not\subset \operatorname{Supp} R_0$, then $(R_0 \cdot f_x) \geq 4 > 3$, and if $s \not\subset \operatorname{Supp} R_0$, then $(R_0 \cdot s) \geq 2 > 1$, both contradicting [Har77, Proposition V.1.4]. Thus $r_0 = zu \cdot \sigma$ for a non-zero section σ of $\mathcal{O}_E(2s+3f)$. At q, the local equation of s is a unit and $z\sigma$ belongs to the square of the maximal ideal, so $\sigma(q) = 0$. At p, the quotient $\mathcal{O}_{E,p}/J^2$ has the basis

$$1,\ u,\ u^2,\ u^3,\ z - cu,\ (z - cu)u,$$

and $zu = (z - cu)u + cu^2 \in J$. Write $\sigma \equiv \sigma_0 + \sigma_1 u \pmod{J}$ with $\sigma_0, \sigma_1 \in \mathbb{C}$. Since $zu \cdot J \subset J^2$ and $(z - cu)u^2 \in J^2$, we have

$$zu \cdot \sigma \equiv c\sigma_0 u^2 + c\sigma_1 u^3 + \sigma_0(z - cu)u \pmod{J^2}.$$

Since $zu \cdot \sigma = r_0 \in J^2$ and $c \neq 0$, we obtain $\sigma_0 = \sigma_1 = 0$, that is, $\sigma \in J$. Thus

$$\sigma \in H^0(E, \mathcal{O}_E(2s+3f) \otimes I_{Z_U}).$$

We show that the three conditions defined by I_{Z_U} are linearly independent. The monomials 1 and u map to a basis of $\mathcal{O}_{E,p}/J$. The monomial u^2 belongs to J, and $u^2(q) \neq 0$, since the divisor of

u^2 is $2s$ plus three times the fiber over $(z = +\infty)$ and $q \notin s$. Thus the map $H^0(E, \mathcal{O}_E(2s+3f)) \to \mathcal{O}_{E,p}/J \oplus \mathcal{O}_{E,q}/\mathfrak{m}_q$ is surjective, the three conditions are linearly independent, and $\sigma \in U$. This contradicts Claim 13.6.

This proves (3). □

The next remark shows that for a fourth configuration, a chain of three base points as in the first configuration of Lemma 13.5(3) but starting at a point of $f_x \backslash s$, the flag consisting of E and f_x does not prove $\delta_x(X) \geq 1$.

Remark 13.7. Notation as in Set-up 13.4. Suppose that the base points of U form a chain of three points starting at a point of $f_x \backslash s$, in which the second point is infinitely near the first in the direction of f_x, and the third point is infinitely near the second and does not belong to the strict transform of f_x. Then $\operatorname{mult}_{f_x} R_0 = 1$, $S(W; f_x) = 47/44$, and $S(W^{f_x}; q) \leq 10/11$ for every $q \in f_x$. Hence the flag consisting of E and f_x does not prove $\delta_x(X) \geq 1$ for such a base point configuration. By Definition-Proposition 10.9, $\Sigma(X, x) \backslash \{[L]\} \cong Z_U \cap f_x$ has length two, so x has incidence type $2 + 1$. Such points are treated in Theorem 14.1.

We explain the values above. Let $p \in f_x \backslash s$ be the first base point, and choose z with $f_x = (z = 0)$. Let $u_1 := u' - u'(p)$, where u' is as in Set-up 13.4, so that (z, u_1) are coordinates at $p = (0, 0)$ and $s \cap f_x$ is the point $u_1 = +\infty$ of f_x. The span of the monomials $z^i u'^{\ell'}$ with $\ell' \leq a$ and $i + \ell' \leq b$ does not change under a translation of u', so for integers $b \geq a \geq 0$ a basis of $H^0(E, \mathcal{O}_E(as + bf))$ is given by the monomials $z^i u_1^\ell$ with $0 \leq \ell \leq a$ and $0 \leq i \leq b - \ell$. As in the first configuration of Lemma 13.5(3), after a scaling of z, the ideal of Z_U at p is $(z - u_1^2, u_1^3)$, and Z_U is supported at p.

The coefficient of f_x. No base point belongs to the strict transform of s, so $k_s = 0$, and $\operatorname{mult}_{f_x} R_0 \leq 1$ by Lemma 13.5(2). By Construction-Proposition 11.3(5), the local equation of R_0 at p belongs to $(z - u_1^2, u_1^3)^2$, whose restriction to $f_x = (z = 0)$ is (u_1^4). Since $4 > (R_0 \cdot f_x) = 3$, we have $f_x \subset \operatorname{Supp} R_0$ [Har77, Proposition V.1.4]. Thus $\operatorname{mult}_{f_x} R_0 = 1$.

The polygon. Let $b \geq a \geq 0$ and $w \geq 0$ be integers. We assign weight 2 to z and weight 1 to u_1. The ideal $(z - u_1^2, u_1^3)$ is generated by weighted homogeneous elements and $H^0(E, \mathcal{O}_E(as + bf))$ is spanned by monomials, so $H^0(E, \mathcal{O}_E(as + bf) \otimes I_{Z_U}^w)$ is a direct sum of weighted homogeneous parts.

Let $\theta := z/u_1^2$. Then $(z - u_1^2)^k u_1^{d-2k} = u_1^d(\theta - 1)^k$, and $u_1^d(\theta - 1)^k \in (z - u_1^2, u_1^3)^w$ if and only if $d + k \geq 3w$. Thus the part of weight d consists of the elements $u_1^d g(\theta)$, where g is a linear combination of θ^i with $i_0 \leq i \leq \lfloor d/2 \rfloor$, $i_0 := \max\{0, \lceil (d-a)/2 \rceil, d - b\}$, and g is divisible by $(\theta - 1)^\kappa$ with $\kappa := \max\{0, 3w - d\}$. Writing $g = \theta^{i_0}(\theta - 1)^\kappa g_0$ with $\deg g_0 \leq \lfloor d/2 \rfloor - i_0 - \kappa$, the lowest power of θ in g, which is the order of $u_1^d g(\theta)$ along f_x, takes exactly the values $i_0, \dots, \lfloor d/2 \rfloor - \kappa$. In the exponent $\ell = d - 2i$ of u_1, the bound $i \leq \lfloor d/2 \rfloor - \kappa$ is equivalent to $3\ell + 4i \geq 6w$. Therefore the graded pieces of $H^0(E, \mathcal{O}_E(as + bf) \otimes I_{Z_U}^w)$ for the order along f_x, restricted to f_x after division by the power of z, have the bases u_1^ℓ indexed by

$$0 \leq \ell \leq a, \qquad 0 \leq i \leq b - \ell, \qquad 3\ell + 4i \geq 6w, \tag{13.24}$$

where i is the order along f_x. For fixed i, the allowed ℓ are the integers from $\ell_{\min}$ to $\ell_{\max}$, where

$$\ell_{\min} := \max\{0, \lceil (6w - 4i)/3 \rceil\}, \qquad \ell_{\max} := \min\{a, b - i\}.$$

Thus the graded piece of order i is $u_1^{\ell_{\min}}$ times the complete linear system of degree $\ell_{\max} - \ell_{\min}$ on f_x, with fixed order $\ell_{\min}$ at p and $a - \ell_{\max}$ at $s \cap f_x$. In particular, for any $q \in f_x \backslash \{p, s \cap f_x\}$, the sum of the orders at q over a basis adapted to the order at q is at most the corresponding sums at p and at $s \cap f_x$. For a subspace, the sums of orders at a point can only decrease, by extending an adapted basis of the subspace to the containing space.

The moments. By Construction-Proposition 11.3(3) and (4)(a)(b), $W_{m,j}$ is contained in $H^0(E, \mathcal{O}_E(as+bf) \otimes I_{Z_U}^w)$ with $(a, b, w) = (j, m+j, 0)$ for $0 \leq j \leq m$ and $(a, b, w) = (j, m+j, j-m)$ for $m \leq j \leq 2m$, and $W_{m,j}$ is contained in r_0^{j-2m} times $H^0(E, \mathcal{O}_E(as + bf) \otimes I_{Z_U}^w)$ with

$(a,b,w) = (2(3m-j), 3(3m-j), 3m-j)$ for $2m \le j \le 3m$. We denote by $\widetilde{W}_{m,j}$ the space containing $W_{m,j}$ in each range. We call the space $\widetilde{W}_{m,j}$ for $j \le 2m$, and the space $H^0(E, \mathcal{O}_E(as+bf) \otimes I^w_{Z_U})$ with $\widetilde{W}_{m,j} = r_0^{j-2m} H^0(E, \mathcal{O}_E(as+bf) \otimes I^w_{Z_U})$ for $j > 2m$, the *free part* of $\widetilde{W}_{m,j}$. Likewise, the *free part* of $W_{m,j}$ is $W_{m,j}$ for $j \le 2m$ and U_{3m-j} for $j > 2m$. Write $\tau = j/m$, $\alpha = i/m$, and $\beta = \ell/m$. By (13.24), the normalized polygon of the free part of $\widetilde{W}_{m,j}$ is

$$P_\tau := \{(\beta,\alpha) : 0 \le \beta \le \tau,\ 0 \le \alpha \le 1+\tau-\beta\} \quad \text{for } 0 \le \tau \le 1,$$

$$P_\tau := \left\{(\beta,\alpha) : 0 \le \beta \le \tau,\ \max\left\{0, \frac{3}{2}(\tau-1) - \frac{3}{4}\beta\right\} \le \alpha \le 1+\tau-\beta\right\} \quad \text{for } 1 \le \tau \le 2,$$

and $P_\tau := (3-\tau)P_2$ for $2 \le \tau \le 3$. The areas of P_τ are $\tau + \tau^2/2$, $-\tau^2 + 4\tau - 3/2$, and $\frac{5}{2}(3-\tau)^2$ in the three ranges, with integrals $2/3$, $13/6$, and $5/6$ over the ranges, whose sum is $11/3$. The integrals of α over P_τ are

$$\frac{\tau}{2} + \frac{\tau^2}{2} + \frac{\tau^3}{6}, \qquad \frac{\tau}{2} + \frac{\tau^2}{2} + \frac{\tau^3}{6} - \frac{3}{4}(\tau-1)^3, \qquad \frac{43}{12}(3-\tau)^3,$$

and the integrals of β over P_τ are $\tau^2/2 + \tau^3/6$, $\tau^2/2 + \tau^3/6 - (\tau-1)^3$, and $\frac{7}{3}(3-\tau)^3$. The order at $s \cap f_x$ is $a - \ell$, normalized to $a/m - \beta$, where $a/m = \tau$ for $\tau \le 2$ and $a/m = 2(3-\tau)$ for $\tau \ge 2$. The integrals over the three ranges of τ are listed in Table 16.

TABLE 16. The integrals of α, β, and $a/m - \beta$ over the polygons P_τ in the three ranges of τ.

range of τ	order along f_x	order at p	order at $s \cap f_x$
$[0,1]$	$11/24$	$5/24$	$1/4$
$[1,2]$	$113/48$	$37/24$	$43/24$
$[2,3]$	$43/48$	$7/12$	$2/3$

Since $N_m = h^0(X, mH) = \frac{11}{3}m^3 + O(m^2)$, the sums over lattice points normalized by mN_m are $\frac{3}{11}$ times the integrals up to $O(1/m)$, and the limits are the invariants by [AZ22, Lemma 2.9]. Thus the free parts of $\widetilde{W}$ have moment $\frac{3}{11} \cdot \frac{89}{24} = \frac{89}{88}$ along f_x, moment $\frac{3}{11} \cdot \frac{7}{3} = \frac{7}{11}$ at p, and moment $\frac{3}{11} \cdot \frac{65}{24} = \frac{65}{88}$ at $s \cap f_x$. The total dimension of the spaces $\widetilde{W}_{m,j}$ is $\frac{11}{3}m^3 + O(m^2)$ by the areas above, and the total dimension of the spaces $W_{m,j}$ is N_m. Since $W_{m,j} \subset \widetilde{W}_{m,j}$, the total codimension is $O(m^2)$, and since the orders along f_x in $\widetilde{W}_{m,j}$ are at most $4m$, the moments along f_x of the free parts of W and $\widetilde{W}$ differ by $O(m^3)$, which vanishes after division by mN_m. Thus the free part of W has moment exactly $89/88$ along f_x.

In the range $2m < j \le 3m$, the factor r_0^{j-2m} adds $(j-2m)\operatorname{mult}_{f_x} R_0$ to the order along f_x. By Lemma 11.1(4),

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j>2m} (j-2m) \dim W_{m,j} = \frac{5}{88}.$$

Since $\operatorname{mult}_{f_x} R_0 = 1$, we obtain

$$S(W; f_x) = \frac{89}{88} + \frac{5}{88} = \frac{47}{44}.$$

For $q \in f_x$, the divisor $R_0 - f_x$ restricts to a divisor of degree $((3s+3f)\cdot f) = 3$ on f_x, so the factor r_0^{j-2m} adds at most $3(j-2m)$ to the order at q in W^{f_x}. Since the sums of orders at q are bounded by the sums at p and at $s \cap f_x$, the free part contributes at most $\max\{7/11, 65/88\} = 65/88$. Thus

$$S(W^{f_x}; q) \le \frac{65}{88} + 3 \cdot \frac{5}{88} = \frac{10}{11}.$$

In the bound of [AZ22, Theorem 3.4] along the flag consisting of E and f_x, the term of f_x is $1/S(W; f_x) = 44/47 < 1$.

The incidence type. The ideal of $Z_U \cap f_x$ at p is $(z-u_1^2, u_1^3)+(z) = (z, u_1^2)$, so $Z_U \cap f_x$ has length two and is supported at one point p. By Definition-Proposition 10.9, p corresponds to a line $L' \neq L$ through x, and the connected component of $\Sigma(X,x)$ at $[L']$ has length two. Since L is non-special, the local ring of $\Sigma(X,x)$ at $[L]$ is $\mathbb{C}$ by Proposition 4.1. Thus x has incidence type $2+1$ (Definition 2.11), and x satisfies the assumption of Theorem 14.1 with the line L'. By Theorem 14.1(1), L' is a special line.

The next set-up and lemma give a lattice point count. Propositions 13.10 and 13.11 use it to bound the expected vanishing orders along a fiber of E and at the points of that fiber.

Set-up 13.8. Let X be a smooth prime Fano threefold of genus 12, and let $N_m := h^0(X, -mK_X)$ for integers $m \geq 1$. For integers $m \geq 1$ and $0 \leq j \leq 3m$ let

$$(a_j, b_j, w_j) := \begin{cases} (j, m+j, 0) & \text{if } 0 \leq j \leq m, \\ (j, m+j, j-m) & \text{if } m < j \leq 2m, \\ (2(3m-j), 3(3m-j), 3m-j) & \text{if } 2m < j \leq 3m, \end{cases}$$

and let Δ_m be the set of triples (j,i,ℓ) of integers with $0 \leq j \leq 3m$, $0 \leq \ell \leq a_j$, $0 \leq i \leq \min\{b_j - a_j + \ell,\ b_j - w_j\}$, and $2i + \ell \geq 2w_j$.

In Proposition 13.10, a triple $(j,i,\ell) \in \Delta_m$ indexes the monomial $z^i u^\ell$ at $p = s \cap f_x$, with i the order along f_x and ℓ the order at p after division by z^i and restriction to f_x. Proposition 13.11 uses the counts over Δ_m as upper bounds.

Lemma 13.9. *Notation as in Set-up 13.8. Then*

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{(j,i,\ell)\in\Delta_m} i = \frac{39}{44} \quad \textit{and} \quad \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{(j,i,\ell)\in\Delta_m} \ell = \frac{73}{88}.$$

Proof. For a real number $\tau \in [0,3]$, let

$$(a(\tau), b(\tau), w(\tau)) := \begin{cases} (\tau, 1+\tau, 0) & \text{if } 0 \leq \tau \leq 1, \\ (\tau, 1+\tau, \tau-1) & \text{if } 1 < \tau \leq 2, \\ (2(3-\tau), 3(3-\tau), 3-\tau) & \text{if } 2 < \tau \leq 3, \end{cases}$$

and let

$$Q_\tau := \big\{(\alpha,\beta) \in \mathbb{R}^2 \;\big|\; 0 \leq \beta \leq a(\tau),\ 0 \leq \alpha \leq \min\{b(\tau)-a(\tau)+\beta,\ b(\tau)-w(\tau)\},\ 2\alpha+\beta \geq 2w(\tau)\big\}.$$

The inequalities defining Δ_m are linear and homogeneous in (j,i,ℓ,m), and $(a_j, b_j, w_j) = m \cdot (a(j/m), b(j/m), w(j/m))$. Thus, for $0 \leq j \leq 3m$, the pairs (i,ℓ) with $(j,i,\ell) \in \Delta_m$ are the lattice points of $mQ_{j/m}$, the coordinate i corresponding to α and ℓ to β. Let

$$I_f(\tau) := \int_{Q_\tau} \alpha d\alpha d\beta \quad \text{and} \quad I_p(\tau) := \int_{Q_\tau} \beta d\alpha d\beta.$$

We describe Q_τ on the three ranges of τ. If $0 \leq \tau \leq 1$, then $(a(\tau), b(\tau), w(\tau)) = (\tau, 1+\tau, 0)$. Since $\min\{1+\beta, 1+\tau\} = 1+\beta$ for $\beta \leq \tau$ and the inequality $2\alpha+\beta \geq 0$ holds for any $\alpha, \beta \geq 0$, we have

$$Q_\tau = \{(\alpha,\beta) \mid 0 \leq \beta \leq \tau,\ 0 \leq \alpha \leq 1+\beta\}. \tag{13.25}$$

If $1 \leq \tau \leq 2$, let $w := \tau - 1 \in [0,1]$. Then $(a(\tau), b(\tau), w(\tau)) = (\tau, 1+\tau, w)$ and $b(\tau) - w(\tau) = 2$, so

$$Q_\tau = \left\{(\alpha,\beta) \;\middle|\; 0 \leq \beta \leq \tau,\ \max\left\{0, w - \frac{\beta}{2}\right\} \leq \alpha \leq \min\{1+\beta, 2\}\right\}. \tag{13.26}$$

Here $w - \frac{\beta}{2} \leq 1 \leq \min\{1+\beta, 2\}$ for $\beta \geq 0$, so for any $\beta \in [0,\tau]$ the bounds on α define a non-empty interval. If $2 \leq \tau \leq 3$, then

$$(a(\tau), b(\tau), w(\tau)) = (3-\tau) \cdot (2,3,1) = (3-\tau) \cdot (a(2), b(2), w(2)),$$

so $Q_\tau = (3-\tau)Q_2$.

We now compute the moment functions. On the first range, (13.25) implies

$$I_f(\tau) = \frac{1}{2}\int_0^\tau (1+\beta)^2 d\beta \quad \text{and} \quad I_p(\tau) = \int_0^\tau \beta(1+\beta)d\beta.$$

On the second range, the lower bound $w - \frac{\beta}{2}$ in (13.26) is positive exactly for $\beta < 2w$, and $2w \le \tau$ since $\tau \le 2$. Splitting the upper bound at $\beta = 1$, we obtain

$$I_f(\tau) = \frac{1}{2}\Big(\int_0^1 (1+\beta)^2 d\beta + \int_1^\tau 4 d\beta - \int_0^{2w} \Big(w - \frac{\beta}{2}\Big)^2 d\beta\Big)$$

and

$$I_p(\tau) = \int_0^1 \beta(1+\beta)d\beta + \int_1^\tau 2\beta d\beta - \int_0^{2w} \beta\Big(w - \frac{\beta}{2}\Big)d\beta.$$

On the third range, since $Q_\tau = (3-\tau)Q_2$ and the integrands are linear, $I_f(\tau) = (3-\tau)^3 I_f(2)$ and $I_p(\tau) = (3-\tau)^3 I_p(2)$. A direct calculation of the integrals leads to Table 17.

TABLE 17. The moment functions I_f and I_p of the polygons Q_τ and their integrals over the three ranges of τ.

range of τ	$I_f(\tau)$	$\int I_f(\tau)d\tau$	$I_p(\tau)$	$\int I_p(\tau)d\tau$
$[0,1]$	$\frac{\tau}{2} + \frac{\tau^2}{2} + \frac{\tau^3}{6}$	$\frac{11}{24}$	$\frac{\tau^2}{2} + \frac{\tau^3}{3}$	$\frac{1}{4}$
$[1,2]$	$2\tau - \frac{5}{6} - \frac{(\tau-1)^3}{3}$	$\frac{25}{12}$	$\tau^2 - \frac{1}{6} - \frac{2(\tau-1)^3}{3}$	2
$[2,3]$	$\frac{17}{6}(3-\tau)^3$	$\frac{17}{24}$	$\frac{19}{6}(3-\tau)^3$	$\frac{19}{24}$

The formulas in Table 17 coincide at $\tau = 1$ and at $\tau = 2$, where I_f takes the values $\frac{7}{6}$ and $\frac{17}{6}$ and I_p takes the values $\frac{5}{6}$ and $\frac{19}{6}$. Thus I_f and I_p are continuous and piecewise polynomial on $[0,3]$, and

$$\int_0^3 I_f(\tau)d\tau = \frac{11}{24} + \frac{25}{12} + \frac{17}{24} = \frac{13}{4}, \qquad \int_0^3 I_p(\tau)d\tau = \frac{1}{4} + 2 + \frac{19}{24} = \frac{73}{24}. \tag{13.27}$$

For any j, the sum of i (resp. of ℓ) over the lattice points of $mQ_{j/m}$ differs from $m^3 I_f(j/m)$ (resp. $m^3 I_p(j/m)$) by $O(m^2)$ uniformly in j, since the polygons Q_τ have at most six edges and bounded size. Summing over the $3m+1$ values of j changes the numerator by $O(m^3)$, and $\sum_{j=0}^{3m} I_f(j/m) = m\int_0^3 I_f(\tau)d\tau + O(1)$ because I_f is continuous and piecewise polynomial. Similarly, $\sum_{j=0}^{3m} I_p(j/m) = m\int_0^3 I_p(\tau)d\tau + O(1)$. By Lemma 2.12(2), $mN_m = \frac{11}{3}m^4 + O(m^3)$. Thus (13.27) implies

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{(j,i,\ell)\in\Delta_m} i = \frac{3}{11}\cdot\frac{13}{4} = \frac{39}{44} \quad \text{and} \quad \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{(j,i,\ell)\in\Delta_m} \ell = \frac{3}{11}\cdot\frac{73}{24} = \frac{73}{88}.$$

□

In the configuration of the next proposition, two of the three base points of U are on f_x, both bounds for the flag $E \supset f_x$ equal 1, and the strict inequality for $\operatorname{center}_X F = \{x\}$ follows from the term $\frac{5}{11}\operatorname{mult}_F E$ in Lemma 3.1.

Proposition 13.10. *Notation as in Set-up 13.4. Let $x \in L$ and assume that the base points of U are $p := s \cap f_x$, the point infinitely near p in the direction of f_x, and a point $r \in E\backslash(s \cup f_x)$. Then $S(W; f_x) \le 1$ and $S(W^{f_x}; q) \le 1$ for any $q \in f_x$. Any prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies*

$$A_X(F) - S_X(F) \ge \frac{5}{11}\operatorname{mult}_F E \ge 0.$$

In particular $\delta_x(X) \geq 1$, *and* $A_X(F) > S_X(F)$ *if* $\operatorname{center}_X F = \{x\}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a torus action on E for which p and r are fixed points.

By [Har77, Example V.2.11.5], there exists a birational morphism $E \to \mathbb{P}^2$ contracting s to a point o. The fibers of the ruling of E are the strict transforms of the lines through o, and the members of $|s+f|$ not containing s are the strict transforms of the lines not through o. Let f_∞ be the fiber through r. Then $f_\infty \neq f_x$ since $r \notin f_x$. Since $r \notin s$, the image $\bar{r}$ of r in $\mathbb{P}^2$ is different from o. Choose homogeneous coordinates y_0, y_1, y_2 on $\mathbb{P}^2$ such that the images of f_x and f_∞ are the lines $(y_0 = 0)$ and $(y_1 = 0)$, and the line $(y_2 = 0)$ passes through $\bar{r}$ and not through o. Then $o = [0:0:1]$ and $\bar{r} = [1:0:0]$. Let $s_\infty \in |s+f|$ be the strict transform of $(y_2 = 0)$. The diagonal torus of PGL_3 fixes o and preserves the three lines $(y_0 = 0)$, $(y_1 = 0)$, and $(y_2 = 0)$, so it acts on E, and s, f_x, s_∞, and f_∞ are invariant curves. Thus $p = s \cap f_x$ and $r = s_\infty \cap f_\infty$ are fixed points, and the point infinitely near p in the direction of f_x is fixed as well.

Step 2. In this step we describe spaces spanned by monomials that contain the spaces $W_{m,j}$ and U_{3m-j}.

Let $(z,u) := (y_0/y_1, y_1/y_2)$. Then (z,u) are coordinates on the invariant affine chart $E \backslash (s_\infty \cup f_\infty) \cong \mathbb{A}^2$, the point p is the origin, $f_x = (z = 0)$, and $s = (u = 0)$. The coordinate u restricts to an affine coordinate on $f_x \cong \mathbb{P}^1$ vanishing at p. The coordinates (z,u) are those of Set-up 13.4, with s_∞ as the section disjoint from s and $f_\infty = (z = +\infty)$. Thus, for integers $b \geq a \geq 0$, Set-up 13.4 gives the basis

$$\{z^i u^\ell \mid 0 \leq \ell \leq a,\ 0 \leq i \leq b - a + \ell\} \tag{13.28}$$

of $H^0(E, \mathcal{O}_E(as+bf))$. The monomial $z^i u^\ell$ has order i along f_x. After division by z^i, the restriction to f_x is u^ℓ, whose order at p is ℓ. In terms of y_0, y_1, y_2 of **Step 1**, the monomial $z^i u^\ell$ corresponds to $y_0^i y_1^{b-a+\ell-i} y_2^{a-\ell}$ [Har77, Proposition V.3.6]. In the coordinates $(y_1/y_0, y_2/y_0)$ at r, the monomial $z^i u^\ell$ becomes $(y_1/y_0)^{b-a+\ell-i}(y_2/y_0)^{a-\ell}$, so the multiplicity of $z^i u^\ell$ at r is $(b-a+\ell-i)+(a-\ell) = b-i$.

We number the blow-ups in $p_E : T_E \to E$ so that e_1 and e_2 correspond to p and to the point infinitely near p, and e_3 corresponds to r. Let C_1, C_2, and C_3 be the strict transforms on T_E of the three exceptional curves, so that $e_1 = C_1 + C_2$, $e_2 = C_2$, and $e_3 = C_3$. Then ord_{C_1} is the monomial valuation with $\operatorname{ord}_{C_1}(z) = \operatorname{ord}_{C_1}(u) = 1$, and ord_{C_2} is the monomial valuation with $\operatorname{ord}_{C_2}(z) = 2$ and $\operatorname{ord}_{C_2}(u) = 1$, since C_2 is the exceptional curve over the point of C_1 in the direction of $f_x = (z = 0)$. The valuation mult_r is monomial in the coordinates at r. For a monomial valuation, the value of a section is the minimum of the values of the monomials in the expansion of the section. Let $w \geq 0$ be an integer. By Construction-Proposition 11.3(2)(c), a section σ of $\mathcal{O}_E(as+bf)$ belongs to $H^0(E, \mathcal{O}_E(as+bf) \otimes I^w_{Z_U})$ if and only if $p_E^* \operatorname{div}(\sigma) \geq w(e_1+e_2+e_3) = wC_1 + 2wC_2 + wC_3$, that is,

$$\operatorname{ord}_{C_1} \sigma \geq w, \quad \operatorname{ord}_{C_2} \sigma \geq 2w, \quad \text{and} \quad \operatorname{mult}_r \sigma \geq w.$$

For $\sigma = z^i u^\ell$, the three conditions read $i + \ell \geq w$, $2i + \ell \geq 2w$, and $i \leq b - w$, and the first follows from the second. Thus $H^0(E, \mathcal{O}_E(as+bf) \otimes I^w_{Z_U})$ is spanned by the monomials $z^i u^\ell$ in (13.28) satisfying

$$2i + \ell \geq 2w \quad \text{and} \quad i \leq b - w. \tag{13.29}$$

Let $m \geq 1$ be an integer, and let (a_j, b_j, w_j) and Δ_m be as in Set-up 13.8. For $0 \leq j \leq 3m$, let $\overline{W}_{m,j} \subset H^0(E, \mathcal{O}_E(a_j s + b_j f))$ be the span of the monomials $z^i u^\ell$ with $(j,i,\ell) \in \Delta_m$. By (13.29),

$$\overline{W}_{m,j} = H^0(E, \mathcal{O}_E(a_j s + b_j f) \otimes I^{w_j}_{Z_U}).$$

Thus the triple $(j,i,\ell) \in \Delta_m$ corresponds to the monomial $z^i u^\ell$ of $\overline{W}_{m,j}$, where i is the order along f_x and ℓ is the order at p after the refinement by f_x. By Construction-Proposition 11.3(3)

and (4),

$$\begin{aligned} &W_{m,j} \subset \overline{W}_{m,j} && \text{for } 0 \le j \le 2m, \\ &W_{m,j} = r_0^{j-2m} \cdot U_{3m-j}, \quad U_{3m-j} \subset \overline{W}_{m,j} && \text{for } 2m \le j \le 3m, \end{aligned} \tag{13.30}$$

and $W_{m,j} = 0$ for $j > 3m$. Here for $2m \le j \le 3m$ we use $(a_j, b_j, w_j) = (2n, 3n, n)$ with $n := 3m - j$, so that $\mathcal{O}_E(a_j s + b_j f) = \mathcal{O}_E(n(2s + 3f))$.

Step 3. In this step we prove $S(W; f_x) \le 1$ and $S(W^{f_x}; q) \le 1$ for any $q \in f_x$.

For a valuation v and a space V of sections with $\dim V < +\infty$, let $\mathcal{T}_v(V)$ be the sum of the values of v on a basis of V adapted to the filtration of V by v. We write $\mathcal{T}_{f_x}$ and $\mathcal{T}_q$ for $v = \mathrm{ord}_{f_x}$ and $v = \mathrm{ord}_q$ with $q \in f_x$. Since ord_{f_x} and ord_q take integer values,

$$\mathcal{T}_v(V) = \sum_{h \ge 1} \dim\{\sigma \in V \mid v(\sigma) \ge h\}.$$

For a divisor D on E, a subspace $W' \subset H^0(E, \mathcal{O}_E(D))$, and an integer i, let $W'_{\ge i} := \{\sigma \in W' \mid \mathrm{ord}_{f_x} \sigma \ge i\}$, so that $W'_{\ge i} = W'$ for $i \le 0$, and let $\mathrm{gr}^i W' \subset H^0(f_x, \mathcal{O}_{f_x}(D))$ be the image of $W'_{\ge i}$ under division by the i-th power of the section defining f_x followed by restriction to f_x. If $W' \subset W''$, then $W'_{\ge i} \subset W''_{\ge i}$ and $\mathrm{gr}^i W' \subset \mathrm{gr}^i W''$ for any i, hence

$$\mathcal{T}_{f_x}(W') \le \mathcal{T}_{f_x}(W'') \quad \text{and} \quad \sum_{i \ge 0} \mathcal{T}_q(\mathrm{gr}^i W') \le \sum_{i \ge 0} \mathcal{T}_q(\mathrm{gr}^i W''). \tag{13.31}$$

By definition, $W^{f_x}_{m,j,i} = \mathrm{gr}^i W_{m,j}$, and $\sum_j \dim W_{m,j} = N_m$. By Definition 2.8,

$$S_m(W; f_x) = \frac{1}{mN_m} \sum_{j=0}^{3m} \mathcal{T}_{f_x}(W_{m,j}) \quad \text{and} \quad S_m(W^{f_x}; q) = \frac{1}{mN_m} \sum_{j=0}^{3m} \sum_{i \ge 0} \mathcal{T}_q(\mathrm{gr}^i W_{m,j}),$$

and $S(W; f_x) = \lim_m S_m(W; f_x)$ and $S(W^{f_x}; q) = \lim_m S_m(W^{f_x}; q)$ by [AZ22, Lemma 2.9].

We first treat the spaces $\overline{W}_{m,j}$. Since ord_{f_x} of a section is the minimal exponent of z in the expansion, the monomials $z^i u^\ell$ with $(j, i, \ell) \in \Delta_m$ form a basis of $\overline{W}_{m,j}$ adapted to the filtration by ord_{f_x}, and $\mathrm{gr}^i \overline{W}_{m,j}$ is spanned by the u^ℓ with $(j, i, \ell) \in \Delta_m$. Hence

$$\mathcal{T}_{f_x}(\overline{W}_{m,j}) = \sum_{i,\ell:(j,i,\ell)\in\Delta_m} i. \tag{13.32}$$

Let $i \ge 0$ with $\mathrm{gr}^i \overline{W}_{m,j} \ne 0$. Then $i \le b_j - w_j$, and the integers ℓ with $(j, i, \ell) \in \Delta_m$ are those with $\ell_{j,i} \le \ell \le a_j$, where $\ell_{j,i} := \max\{0, i - b_j + a_j, 2w_j - 2i\}$. We identify $H^0(f_x, \mathcal{O}_{f_x}(a_j))$ with the polynomials in u of degree at most a_j. Then

$$\mathrm{gr}^i \overline{W}_{m,j} = u^{\ell_{j,i}} H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(a_j - \ell_{j,i})),$$

the space of sections of $\mathcal{O}_{f_x}(a_j)$ vanishing to order at least $\ell_{j,i}$ at p. At $q = p$, we have

$$\mathcal{T}_p(\mathrm{gr}^i \overline{W}_{m,j}) = \sum_{\ell=\ell_{j,i}}^{a_j} \ell.$$

At $q \ne p$, with $d := a_j - \ell_{j,i}$, the dimension of the subspace of sections vanishing to order at least h at q is $\max\{0, d + 1 - h\}$, so $\mathcal{T}_q(\mathrm{gr}^i \overline{W}_{m,j}) = \frac{d(d+1)}{2} \le \sum_{\ell=\ell_{j,i}}^{a_j} \ell$. Summing over i, we obtain

$$\sum_{i \ge 0} \mathcal{T}_q(\mathrm{gr}^i \overline{W}_{m,j}) \le \sum_{i \ge 0} \mathcal{T}_p(\mathrm{gr}^i \overline{W}_{m,j}) = \sum_{i,\ell:(j,i,\ell)\in\Delta_m} \ell \quad \text{for any } q \in f_x. \tag{13.33}$$

We next treat the factor r_0^{j-2m} for $2m < j \le 3m$. Let $\mu := \mathrm{mult}_{f_x} R_0$. Then $\mu \le 2$ by Lemma 13.5(2). The divisor $R_0 - \mu f_x$ satisfies $R_0 - \mu f_x \ge 0$ and does not contain f_x, and

$$((R_0 - \mu f_x) \cdot f_x) = ((3s + 4f) \cdot f) = 3$$

by Construction-Proposition 11.3(1). Let $\rho \in H^0(f_x, \mathcal{O}_{f_x}(3))$ be the restriction to f_x of r_0 divided by the μ-th power of the section defining f_x. Then $\rho \neq 0$, and $\operatorname{ord}_q \rho \leq 3$ for any $q \in f_x$. Let $n := 3m - j$. For $\sigma \in U_n$, we have $\operatorname{ord}_{f_x}(r_0^{j-2m}\sigma) = \mu(j-2m) + \operatorname{ord}_{f_x}\sigma$, and the leading term of $r_0^{j-2m}\sigma$ along f_x is ρ^{j-2m} times the leading term of σ. Thus, for any integer i,

$$(W_{m,j})_{\geq i} = r_0^{j-2m} \cdot (U_n)_{\geq i-\mu(j-2m)} \quad \text{and} \quad \operatorname{gr}^i W_{m,j} = \rho^{j-2m} \cdot \operatorname{gr}^{i-\mu(j-2m)} U_n.$$

Since multiplication by ρ^{j-2m} raises the order at q by $(j-2m)\operatorname{ord}_q \rho$, and $\dim W_{m,j} = \dim U_n = \sum_i \dim \operatorname{gr}^i U_n$, we obtain

$$\begin{aligned} \mathcal{T}_{f_x}(W_{m,j}) &= \mu(j-2m)\dim W_{m,j} + \mathcal{T}_{f_x}(U_n), \\ \sum_{i\geq 0} \mathcal{T}_q(\operatorname{gr}^i W_{m,j}) &= (j-2m)\operatorname{ord}_q \rho \cdot \dim W_{m,j} + \sum_{i\geq 0} \mathcal{T}_q(\operatorname{gr}^i U_n). \end{aligned} \tag{13.34}$$

By (13.30), (13.31), (13.32), (13.33), and (13.34),

$$mN_m S_m(W; f_x) \leq \sum_{(j,i,\ell)\in\Delta_m} i + \mu \sum_{j>2m} (j-2m)\dim W_{m,j}$$

and

$$mN_m S_m(W^{f_x}; q) \leq \sum_{(j,i,\ell)\in\Delta_m} \ell + \operatorname{ord}_q \rho \cdot \sum_{j>2m} (j-2m)\dim W_{m,j}.$$

By Lemma 13.9, Lemma 11.1(4) with $J_1 = \frac{5}{88}$, $\mu \leq 2$, and $\operatorname{ord}_q \rho \leq 3$, we obtain

$$S(W; f_x) \leq \frac{39}{44} + \frac{5\mu}{88} \leq \frac{78}{88} + \frac{10}{88} = 1 \quad \text{and} \quad S(W^{f_x}; q) \leq \frac{73}{88} + \frac{5}{88}\operatorname{ord}_q \rho \leq \frac{73}{88} + \frac{15}{88} = 1$$

for any $q \in f_x$.

Step 4. In this step we prove the inequality for $A_X(F) - S_X(F)$ and conclude the proof.

Recall that $f_x = \pi^{-1}(x)$ is the fiber of $E \to L$ over x. Since Y and E are smooth, (Y, E) is plt and the different of E is zero. Since E and f_x are smooth, f_x is a prime divisor on E of plt type with plt extraction the identity of E, and the different of f_x is zero. Let $q \in f_x$. The only irreducible closed subset of f_x with image $\{q\}$ is $\{q\}$, and the only prime divisor over the smooth curve f_x whose center contains q is q, with $A_{f_x}(q) = 1$. Thus $\delta_q(f_x; W^{f_x}) = 1/S(W^{f_x}; q)$ by Definition 2.8. By Theorem 2.9 applied to the klt pair $(E, 0)$, the series W, $Z = \{q\}$, and the plt type divisor f_x with $A_E(f_x) = 1$, together with **Step 3**, we have

$$\delta_q(E; W) \geq \min\left\{\frac{1}{S(W; f_x)}, \frac{1}{S(W^{f_x}; q)}\right\} \geq 1.$$

Let v be a prime divisor over E with $\operatorname{center}_E v \cap f_x \neq \emptyset$, and let $q \in \operatorname{center}_E v \cap f_x$. Since $\delta_q(E; W) \geq 1$ and $q \in \operatorname{center}_E v$, we have $S(W; v) \leq A_E(v)$.

By Lemma 3.1 with $\lambda = 1$ and $G = f_x$, any prime divisor F over X with $\operatorname{center}_Y F \cap f_x \neq \emptyset$ satisfies

$$A_X(F) \geq S_X(F) + (A_X(E) - S_X(E))\operatorname{mult}_F E.$$

Since $K_Y = \pi^* K_X + E$, we have $A_X(E) = 2$. By Lemma 11.1(2), $A_X(E) - S_X(E) = 2 - \frac{17}{11} = \frac{5}{11}$.

Let F be a prime divisor over X with $x \in \operatorname{center}_X F$. Since $\pi(\operatorname{center}_Y F) = \operatorname{center}_X F$, there exists a point of $\operatorname{center}_Y F$ in $\pi^{-1}(x) = f_x$, so $\operatorname{center}_Y F \cap f_x \neq \emptyset$. Since $E \geq 0$, $\operatorname{mult}_F E \geq 0$. Thus

$$A_X(F) - S_X(F) \geq \frac{5}{11}\operatorname{mult}_F E \geq 0.$$

Since $S_X(F) > 0$, this implies $A_X(F)/S_X(F) \geq 1$, hence $\delta_x(X) \geq 1$. If $\operatorname{center}_X F = \{x\}$, then $\operatorname{center}_Y F \subset \pi^{-1}(x) = f_x \subset E$, so $\operatorname{mult}_F E > 0$ and $A_X(F) > S_X(F)$. □

We now treat the configurations of the base points of U in which two base points belong to f_x and one base point belongs to $E \backslash (s \cup f_x)$.

Proposition 13.11. *Notation as in Set-up 13.4. Let $x \in L$ and assume that the base points of U consist of a subscheme $Z_f \subset f_x$ of length two, either two distinct points or a point together with the point infinitely near in the direction of f_x, and a point $r \in E\backslash(s \cup f_x)$. Then $S(W; f_x) \le 1$ and $S(W^{f_x}; q) \le 1$ for any $q \in f_x$. If $Z_f \cap s = \emptyset$, then $S(W; f_x) \le 83/88$ and $S(W^{f_x}; q) \le 39/44$ for any $q \in f_x$. Let $c := 88/83$ if $Z_f \cap s = \emptyset$ and $c := 1$ otherwise. Then any prime divisor F over X with $x \in \operatorname{center}_X F$ satisfies*

$$A_X(F) \ge cS_X(F) + \Big(2 - \frac{17c}{11}\Big) \operatorname{mult}_F E.$$

In particular $\delta_x(X) \ge c$, and $A_X(F) > S_X(F)$ if $\operatorname{center}_X F = \{x\}$.

Proof. The proof is long, so we divide it into several steps. By hypothesis, Z_U is the disjoint union of Z_f and the reduced point r. Let

$$R' := R_0 - (\operatorname{mult}_{f_x} R_0) f_x.$$

Since $f_x \not\subset \operatorname{Supp} R'$ and $(R' \cdot f_x) = ((3s + 4f) \cdot f) = 3$, we have $\operatorname{mult}_q(R'|_{f_x}) \le 3$ for any $q \in f_x$. By Lemma 13.5(2), $\operatorname{mult}_{f_x} R_0 \le 2$. If $Z_f \cap s = \emptyset$, then no base point of U belongs to s, so $k_s = 0$ and $\operatorname{mult}_{f_x} R_0 \le 1$ (Lemma 13.5(2)).

For integers $0 \le a \le b$, a subspace $W' \subset H^0(E, \mathcal{O}_E(as + bf))$, and an integer $i \ge 0$, let $\operatorname{gr}^i W' \subset H^0(f_x, \mathcal{O}_E(as + bf - if_x)|_{f_x}) = H^0(f_x, \mathcal{O}_{f_x}(a))$ be the image of $\{t \in W' \mid \operatorname{ord}_{f_x} t \ge i\}$. Then $W^{f_x}_{m,j,i} = \operatorname{gr}^i W_{m,j}$ is the piece of W^{f_x} of order i along f_x. For a valuation v and a vector space W'' of sections of a line bundle with $\dim W'' < +\infty$, let $\mathcal{T}_v(W'')$ be the sum of the values of v on a basis of W'' adapted to the filtration of W'' by v. Then $\mathcal{T}_v(W''') \le \mathcal{T}_v(W'')$ for any subspace $W''' \subset W''$, because a basis of W''' adapted to v extends to a basis of W'' adapted to v. Moreover $\mathcal{T}_{f_x}(W') = \sum_i i \dim \operatorname{gr}^i W'$, $\operatorname{gr}^i W''' \subset \operatorname{gr}^i W'$ for $W''' \subset W'$, and by Definition 2.8 and [AZ22, Lemma 2.9],

$$S(W; f_x) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_j \mathcal{T}_{f_x}(W_{m,j}) \quad \text{and} \quad S(W^{f_x}; q) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j,i} \mathcal{T}_q(W^{f_x}_{m,j,i}).$$

Step 1. In this step we separate the contribution of R_0 to $S(W; f_x)$ and $S(W^{f_x}; q)$.

Let $m \ge 1$, let (a_j, b_j, w_j) be as in Set-up 13.8 for $0 \le j \le 3m$, and let $W^\circ_{m,j} := W_{m,j}$ if $j \le 2m$ and $W^\circ_{m,j} := U_{3m-j}$ if $2m < j \le 3m$. By Construction-Proposition 11.3(3)(4),

$$W^\circ_{m,j} \subset H^0\big(E, \mathcal{O}_E(a_j s + b_j f) \otimes I^{w_j}_{Z_U}\big) \tag{13.35}$$

for $0 \le j \le 3m$, $W_{m,j} = r_0^{j-2m} \cdot W^\circ_{m,j}$ for $2m \le j \le 3m$, and $W_{m,j} = 0$ for $j > 3m$.

Let $2m < j \le 3m$ and $t \in W^\circ_{m,j}$. Then $\operatorname{ord}_{f_x}(r_0^{j-2m} t) = (j - 2m) \operatorname{mult}_{f_x} R_0 + \operatorname{ord}_{f_x} t$. Hence

$$\mathcal{T}_{f_x}(W_{m,j}) = (j - 2m)(\operatorname{mult}_{f_x} R_0) \dim W_{m,j} + \mathcal{T}_{f_x}(W^\circ_{m,j}).$$

Moreover, multiplication by the $(j - 2m)$-th power of a section of $\mathcal{O}_{f_x}(3)$ defining $R'|_{f_x}$ maps $\operatorname{gr}^{i-(j-2m)\operatorname{mult}_{f_x} R_0} W^\circ_{m,j}$ isomorphically onto $W^{f_x}_{m,j,i}$ for any i. Since $\sum_i \dim \operatorname{gr}^i W^\circ_{m,j} = \dim W_{m,j}$, we obtain

$$\sum_i \mathcal{T}_q(W^{f_x}_{m,j,i}) = (j - 2m) \operatorname{mult}_q(R'|_{f_x}) \dim W_{m,j} + \sum_i \mathcal{T}_q(\operatorname{gr}^i W^\circ_{m,j})$$

for any $q \in f_x$. By Lemma 11.1(4),

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{j>2m} (j - 2m) \dim W_{m,j} = \frac{5}{88}.$$

Since $\operatorname{mult}_q(R'|_{f_x}) \le 3$, we obtain

$$\begin{aligned} S(W; f_x) &\le \frac{5}{88} \operatorname{mult}_{f_x} R_0 + \limsup_{m\to+\infty} \frac{1}{mN_m} \sum_{j=0}^{3m} \mathcal{T}_{f_x}(W^\circ_{m,j}), \\ S(W^{f_x}; q) &\le \frac{15}{88} + \limsup_{m\to+\infty} \frac{1}{mN_m} \sum_{j=0}^{3m} \sum_i \mathcal{T}_q(\operatorname{gr}^i W^\circ_{m,j}). \end{aligned} \tag{13.36}$$

Step 2. In this step we assume that $Z_f \cap s = \emptyset$, and we describe the right-hand side of (13.35) in coordinates.

In **Steps 2–4**, we use the chart (z, u') of Set-up 13.4 on $E\backslash s$, with s_∞ a section through r disjoint from s, so that s is the locus where u' has a pole and the basis of $H^0(E, \mathcal{O}_E(as+bf))$ is $z^i u'^{\ell'}$, $0 \le \ell' \le a$, $0 \le i \le b-\ell'$. Such a section exists: since $E \cong \mathbb{F}_1$ is the blow-up of $\mathbb{P}^2$ at a point with exceptional curve s, and $s+f$ is the pullback of a line, there exists $s_\infty \in |s+f|$ with $r \in s_\infty$ and $s_\infty \cap s = \emptyset$. Let $\infty \in L$ be the image of r, let f_∞ be the fiber over ∞, and choose z with $z(x) = 0$ and with its pole at ∞. Then u' is a fiber coordinate on $E\backslash(s \cup f_\infty) \cong \mathbb{A}^2$, and the function $z^i u'^{\ell'}$ has a pole of order ℓ' along s and of order $i+\ell'$ along f_∞. For integers $0 \le a \le b$ and $i \ge 0$, let $d_i := \min\{a, b-i\}$. Identifying $\mathcal{O}_E(as+bf)$ with $\mathcal{O}_E(as+bf_\infty)$, any $t \in H^0(E, \mathcal{O}_E(as+bf))$ has a unique expansion

$$t = \sum_{i\ge 0} z^i g_i(u'), \qquad g_i \in \mathbb{C}[u'],\ \deg g_i \le d_i,$$

where $g_i = 0$ if $d_i < 0$ ([Har77, Exercise III.8.4(a) and Propositions V.2.8 and V.2.9]). Near r, the functions $z' := 1/z$ and $\bar{u} := u'/z$ are local coordinates centered at r, z' is a local equation of f_∞, and $z'^b z^i u'^{\ell'} = z'^{b-i-\ell'} \bar{u}^{\ell'}$. Thus distinct monomials $z^i u'^{\ell'}$ have distinct local expressions at r, of order $b-i$.

On $f_x\backslash s = \{z = 0\}$, the subscheme Z_f is defined by $(z, Q(u'))$ for a monic polynomial Q of degree two. For $w \ge 0$ let $\kappa_i := \max\{0, w-i\}$, and let $\mathbb{C}[u']_{\le e}$ be the space of polynomials of degree at most e, which is 0 if $e < 0$. Let

$$\mathcal{W}_Q(a,b,w) := \bigoplus_{0\le i\le b-w} z^i Q(u')^{\kappa_i} \cdot \mathbb{C}[u']_{\le d_i - 2\kappa_i},$$

and let $\Omega(a,b,w)$ be the set of pairs $(i,\ell') \in \mathbb{Z}^2$ with $0 \le \ell' \le a$, $0 \le i \le b-\ell'$, $i \le b-w$, and $2i+\ell' \ge 2w$. Then $H^0(E, \mathcal{O}_E(as+bf) \otimes I^w_{Z_U}) = \mathcal{W}_Q(a,b,w)$. To prove the equality, recall that membership in an ideal is a local property. At r, the ideal $I^w_{Z_U}$ is the w-th power of the maximal ideal, so by the description of the local expressions at r, the condition at r is $g_i = 0$ for $i > b-w$. Near Z_f, the ideal $I^w_{Z_U}$ is generated by $(z,Q)^w = \sum_{0\le i\le w} z^i Q^{w-i} \cdot \mathbb{C}[z,u']$, and a polynomial $\sum_i z^i g_i(u')$ belongs to $(z,Q)^w$ if and only if Q^{κ_i} divides g_i for any i.

We compute the orders of $\mathcal{W}_Q(a,b,w)$. The row $z^i Q^{\kappa_i} \mathbb{C}[u']_{\le d_i - 2\kappa_i}$ is contained in $\{\operatorname{ord}_{f_x} \ge i\}$, so $\operatorname{gr}^i \mathcal{W}_Q(a,b,w) = Q^{\kappa_i}\mathbb{C}[u']_{\le d_i-2\kappa_i}$, where a polynomial g with $\deg g \le a$ is regarded as a section of $\mathcal{O}_{f_x}(a)$ vanishing to order $a - \deg g$ at $s \cap f_x$. Let $n_i := \#\{\ell' \mid (i,\ell') \in \Omega(a,b,w)\}$. Since $(i,\ell') \in \Omega(a,b,w)$ if and only if $i \le b-w$ and $2\kappa_i \le \ell' \le d_i$, we have $\dim \operatorname{gr}^i \mathcal{W}_Q(a,b,w) = n_i$, and

$$\mathcal{T}_{f_x}(\mathcal{W}_Q(a,b,w)) = \sum_{(i,\ell')\in\Omega(a,b,w)} i. \tag{13.37}$$

If $n_i > 0$, then $\operatorname{gr}^i \mathcal{W}_Q(a,b,w)$ is the space of sections of $\mathcal{O}_{f_x}(a)$ vanishing on the divisor $D_i := \kappa_i Z_f + (a-d_i)(s \cap f_x)$, and $\deg D_i = a+1-n_i$. Hence the values of ord_q on an adapted basis are $\operatorname{mult}_q D_i, \operatorname{mult}_q D_i + 1, \dots, \operatorname{mult}_q D_i + n_i - 1$, and

$$\mathcal{T}_q(\operatorname{gr}^i \mathcal{W}_Q(a,b,w)) = n_i \operatorname{mult}_q D_i + \frac{n_i(n_i-1)}{2}.$$

Here $\operatorname{mult}_q D_i \le 2\kappa_i$ if $q \in \operatorname{Supp} Z_f$, $\operatorname{mult}_q D_i = a - d_i$ if $q = s \cap f_x$, and $\operatorname{mult}_q D_i = 0$ otherwise. On the other hand, $\sum_{\ell'} \ell' = 2\kappa_i n_i + \frac{n_i(n_i-1)}{2}$ and $\sum_{\ell'}(a-\ell') = (a-d_i)n_i + \frac{n_i(n_i-1)}{2}$, where ℓ' runs through the integers with $(i,\ell') \in \Omega(a,b,w)$. Summing over i, we obtain

$$\sum_i \mathcal{T}_q(\operatorname{gr}^i \mathcal{W}_Q(a,b,w)) \le \begin{cases} \sum_{(i,\ell')\in\Omega(a,b,w)} \ell' & \text{if } q \in \operatorname{Supp} Z_f, \\ \sum_{(i,\ell')\in\Omega(a,b,w)} (a-\ell') & \text{otherwise.} \end{cases} \tag{13.38}$$

For $(i,\ell') \in \Omega(a,b,w)$, the integers i, ℓ', and $a-\ell'$ are the order of $z^i u'^{\ell'}$ along f_x and the orders at the point $u' = 0$ of f_x and at $s \cap f_x$ of the restriction of $u'^{\ell'}$ to f_x, as a section of $\mathcal{O}_{f_x}(a)$, respectively.

Step 3. In this step we prove the bounds of the proposition when $Z_f \cap s = \emptyset$.

Let Ω_m be the set of triples (j,i,ℓ') with $0 \le j \le 3m$ and $(i,\ell') \in \Omega(a_j,b_j,w_j)$. By (13.35), the equality $H^0(E, \mathcal{O}_E(as+bf) \otimes I_{Z_U}^w) = \mathcal{W}_Q(a,b,w)$ of **Step 2**, (13.37), (13.38), and the monotonicity of $\mathcal{T}$,

$$\sum_{j=0}^{3m} \mathcal{T}_{f_x}(W^\circ_{m,j}) \le \sum_{(j,i,\ell')\in\Omega_m} i,$$

and for any $q \in f_x$, $\sum_{j,i} \mathcal{T}_q(\operatorname{gr}^i W^\circ_{m,j})$ is bounded by $\sum_{\Omega_m} \ell'$ if $q \in \operatorname{Supp} Z_f$ and by $\sum_{\Omega_m}(a_j - \ell')$ otherwise.

For $0 \le \tau \le 3$ let

$$(a(\tau), b(\tau), w(\tau)) := \begin{cases} (\tau, 1+\tau, 0) & \text{if } 0 \le \tau \le 1, \\ (\tau, 1+\tau, \tau-1) & \text{if } 1 \le \tau \le 2, \\ (2(3-\tau), 3(3-\tau), 3-\tau) & \text{if } 2 \le \tau \le 3, \end{cases}$$

so that $(a_j,b_j,w_j) = m\big(a(j/m), b(j/m), w(j/m)\big)$. Let $P_\tau \subset \mathbb{R}^2$ be the polygon of points (α,β) with $0 \le \beta \le a(\tau)$, $0 \le \alpha \le b(\tau) - \beta$, $\alpha \le b(\tau) - w(\tau)$, and $2\alpha + \beta \ge 2w(\tau)$, and let

$$I_f(\tau) := \int_{P_\tau} \alpha, \qquad I_0(\tau) := \int_{P_\tau} \beta, \qquad I_s(\tau) := \int_{P_\tau} (a(\tau) - \beta).$$

In the variables $(\tau,\alpha,\beta) = (j/m, i/m, \ell'/m)$, the set Ω_m is the set of lattice points of the dilation by m of the region $\{(\tau,\alpha,\beta) \mid 0 \le \tau \le 3,\ (\alpha,\beta) \in P_\tau\}$, which is a finite union of polytopes. The boundary of the dilation contains $O(m^2)$ lattice points, each contributing $O(m)$ to the sums, and $N_m = \frac{11}{3}m^3 + O(m^2)$ by Lemma 2.12(2). Thus

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{\Omega_m} i = \frac{3}{11}\int_0^3 I_f(\tau)d\tau, \qquad \lim_{m\to+\infty} \frac{1}{mN_m} \sum_{\Omega_m} \ell' = \frac{3}{11}\int_0^3 I_0(\tau)d\tau,$$

$$\lim_{m\to+\infty} \frac{1}{mN_m} \sum_{\Omega_m} (a_j - \ell') = \frac{3}{11}\int_0^3 I_s(\tau)d\tau.$$

For $2 \le \tau \le 3$ we have $P_\tau = (3-\tau)P_2$. A direct calculation shows the values in Table 18.

TABLE 18. The moment functions I_f, I_0, and I_s of the polygons P_τ and their integrals over the three ranges of τ. The functions I_f, I_0, I_s correspond to the orders along f_x, at the point $u' = 0$ of f_x, and at $s \cap f_x$.

range of τ	$I_f(\tau)$	$\int I_f(\tau)d\tau$	$I_0(\tau)$	$\int I_0(\tau)d\tau$	$I_s(\tau)$	$\int I_s(\tau)d\tau$
$[0,1]$	$\frac{\tau}{2} + \frac{\tau^2}{2} + \frac{\tau^3}{6}$	$\frac{11}{24}$	$\frac{\tau^2}{2} + \frac{\tau^3}{6}$	$\frac{5}{24}$	$\frac{\tau^2}{2} + \frac{\tau^3}{3}$	$\frac{1}{4}$
$[1,2]$	$2\tau - \frac{5}{6} - \frac{(\tau-1)^3}{3}$	$\frac{25}{12}$	$\tau^2 - \frac{\tau}{2} + \frac{1}{6} - \frac{2(\tau-1)^3}{3}$	$\frac{19}{12}$	$2\tau - \frac{7}{6} - \frac{(\tau-1)^3}{3}$	$\frac{7}{4}$
$[2,3]$	$\frac{17}{6}(3-\tau)^3$	$\frac{17}{24}$	$\frac{5}{2}(3-\tau)^3$	$\frac{5}{8}$	$\frac{5}{2}(3-\tau)^3$	$\frac{5}{8}$

The entries in the columns $I_f(\tau)$ and $\int I_f(\tau)d\tau$ are equal to the corresponding entries of Table 17.

Summing the three ranges and multiplying by 3/11 yields 39/44, 29/44, and 63/88. Since $\frac{29}{44} < \frac{63}{88}$ and $\operatorname{mult}_{f_x} R_0 \le 1$, (13.36) implies

$$S(W; f_x) \le \frac{5}{88} + \frac{39}{44} = \frac{83}{88}, \qquad S(W^{f_x}; q) \le \frac{15}{88} + \frac{63}{88} = \frac{39}{44}$$

for any $q \in f_x$. In particular $S(W; f_x) \le 1$ and $S(W^{f_x}; q) \le 1$.

Step 4. In this step we prove $S(W; f_x) \le 1$ and $S(W^{f_x}; q) \le 1$ for any $q \in f_x$ when $Z_f \cap s \ne \emptyset$.

Let $p := s \cap f_x$. If $\operatorname{Supp} Z_f = \{p\}$, then Z_f consists of p and the point infinitely near p in the direction of f_x, and Proposition 13.10 implies the proposition with $c = 1$, because $2 - \frac{17}{11} = \frac{5}{11}$. Thus we may assume that $Z_f = \{p, q_0\}$ with $q_0 \in f_x \backslash s$.

As in **Step 2**, there exists $s_\infty \in |s + f|$ with $r, q_0 \in s_\infty$ and $s_\infty \cap s = \emptyset$, since $r \notin f_x$. We define z, u', z', $\bar{u}$, and d_i as in **Step 2**, using s_∞ through r and q_0, so that $q_0 = \{z = u' = 0\}$. Near p, the functions z and $u = 1/u'$ of Set-up 13.4 are local coordinates centered at p, u is a local equation of s, and $u^a z^i u'^{\ell'} = z^i u^{a-\ell'}$. At each of the three reduced points p, q_0, r, the ideal $I^w_{Z_U}$ is the w-th power of the maximal ideal, and each monomial $z^i u'^{\ell'}$ has a monomial local expression at each of the three points. Thus $H^0(E, \mathcal{O}_E(as + bf) \otimes I^w_{Z_U})$ is spanned by the monomials $z^i u'^{\ell'}$ with $i \le b - w$ and

$$\max\{\kappa_i, a - d_i\} \le a - \ell' \le a - \kappa_i,$$

where $a - \ell'$ is the order at p of the restriction of $u'^{\ell'}$ to f_x, $a - \ell' \ge \kappa_i$ is the condition at p, $\ell' \ge \kappa_i$ is the condition at q_0, and $\ell' \le d_i$ is the bound $\deg g_i \le d_i$ in the expansion $t = \sum_i z^i g_i(u')$ of **Step 2**. Let $\bar{n}_i$ be the number of such ℓ'.

Let Δ_m be as in Set-up 13.8, fix j and i, and let $(a, b, w) = (a_j, b_j, w_j)$. Since $a - d_i = \max\{0, i - (b - a)\}$ and $2\kappa_i = \max\{0, 2(w - i)\}$, the integers ℓ with $(j, i, \ell) \in \Delta_m$ are the integers with $\max\{2\kappa_i, a - d_i\} \le \ell \le a$ if $i \le b - w$, and no such ℓ exists if $i > b - w$. Let n'_i be their number, so that $n'_i \le a + 1$ and the integers ℓ with $(j, i, \ell) \in \Delta_m$ are $a - n'_i + 1, \dots, a$. If $i \le b - w$, then the integers $a - \ell'$ above are the integers in $[\max\{\kappa_i, a - d_i\}, a - \kappa_i]$, and since

$$\kappa_i + \max\{\kappa_i, a - d_i\} \ge \max\{2\kappa_i, a - d_i\},$$

the interval $[\max\{\kappa_i, a - d_i\}, a - \kappa_i]$ contains at most n'_i integers. If $i > b - w$, then $\bar{n}_i = 0$. Hence $\bar{n}_i \le n'_i$ in both cases, and by (13.35),

$$\dim \operatorname{gr}^i W^\circ_{m,j} \le \bar{n}_i \le n'_i = \#\{\ell \mid (j, i, \ell) \in \Delta_m\}.$$

Let $q \in f_x$ and $e := \dim \operatorname{gr}^i W^\circ_{m,j}$. Since $\operatorname{gr}^i W^\circ_{m,j}$ is a subspace of $H^0(f_x, \mathcal{O}_{f_x}(a_j))$, the values of ord_q on an adapted basis are e distinct integers in $\{0, \dots, a_j\}$, so

$$\mathcal{T}_q(\operatorname{gr}^i W^\circ_{m,j}) \le \frac{e(2a_j - e + 1)}{2} \le \frac{n'_i(2a_j - n'_i + 1)}{2} = \sum_{\ell:(j,i,\ell)\in\Delta_m} \ell,$$

where the second inequality holds because $e \to e(2a_j - e + 1)/2$ is nondecreasing for $0 \le e \le a_j + 1$. Moreover $\mathcal{T}_{f_x}(W^\circ_{m,j}) = \sum_i i \dim \operatorname{gr}^i W^\circ_{m,j} \le \sum_{i,\ell:(j,i,\ell)\in\Delta_m} i$. Summing over j and i, we find by Lemma 13.9 that the limits superior in (13.36) are at most $\frac{39}{44}$ and $\frac{73}{88}$, respectively. Since $\operatorname{mult}_{f_x} R_0 \le 2$, we obtain

$$S(W; f_x) \le \frac{10}{88} + \frac{39}{44} = 1, \qquad S(W^{f_x}; q) \le \frac{15}{88} + \frac{73}{88} = 1$$

for any $q \in f_x$.

Step 5. In this step we prove the inequality for $A_X(F)$.

If Z_f is supported at $s \cap f_x$, then the inequality for $A_X(F)$ with $c = 1$ follows from Proposition 13.10, as shown at the beginning of **Step 4**. Otherwise, by **Steps 3 and 4**, $S(W; f_x) \le \frac{1}{c}$ and $S(W^{f_x}; q) \le \frac{1}{c}$ for any $q \in f_x$, where for $c = \frac{88}{83}$ we use $\frac{39}{44} < \frac{83}{88}$.

The pairs (Y,E) and (E,f_x) are smooth, so both differents are zero, and $A_E(f_x)=1$. Any prime divisor over the curve f_x with center q defines a positive multiple of ord_q, so $\delta_q(f_x;W^{f_x})=1/S(W^{f_x};q)$ by Definition 2.8. By Theorem 2.9 applied to the klt pair $(E,0)$, the series W, $Z=\{q\}$, and the plt type divisor f_x, we obtain

$$\delta_q(E;W)\geq\min\left\{\frac{A_E(f_x)}{S(W;f_x)},\frac{1}{S(W^{f_x};q)}\right\}\geq c$$

for any $q\in f_x$. Let v be a prime divisor over E with $\mathrm{center}_E v\cap f_x\neq\emptyset$, and pick $q\in\mathrm{center}_E v\cap f_x$. Then $cS(W;v)\leq A_E(v)$ because $\delta_q(E;W)\geq c$.

Thus Lemma 3.1, applied with $\lambda=c$ and $G=f_x$, implies

$$A_X(F)\geq cS_X(F)+\big(A_X(E)-cS_X(E)\big)\,\mathrm{mult}_F E$$

for any prime divisor F over X with $\mathrm{center}_Y F\cap f_x\neq\emptyset$. If $x\in\mathrm{center}_X F$, then $\pi(\mathrm{center}_Y F)=\mathrm{center}_X F$ contains x, so $\mathrm{center}_Y F\cap f_x\neq\emptyset$. By Lemma 11.1(2), $S_X(E)=\frac{17}{11}$, so $A_X(E)-cS_X(E)=2-\frac{17c}{11}$, and we obtain

$$A_X(F)\geq cS_X(F)+\left(2-\frac{17c}{11}\right)\mathrm{mult}_F E.$$

We have $2-\frac{17c}{11}=\frac{5}{11}$ if $c=1$ and $2-\frac{17c}{11}=\frac{30}{83}$ if $c=\frac{88}{83}$. In particular $A_X(F)\geq cS_X(F)$ for any F with $x\in\mathrm{center}_X F$, so $\delta_x(X)\geq c$. If $\mathrm{center}_X F=\{x\}$, then $\mathrm{center}_Y F\subset f_x\subset E$, so $\mathrm{mult}_F E>0$, and $A_X(F)>cS_X(F)\geq S_X(F)$. □

13.3. **Points on two lines.** In this subsection, we fix the notation for points of X through which exactly two or three lines pass, all of them reduced points of $\Sigma(X,p)$, and we prove the lower bounds of Proposition 13.15 for two lines by the refinement along the blow-up of one of the lines. The refinement covers all configurations of two lines except one, which Theorem 13.21 treats with a weighted extraction. Proposition 13.15(4) shows that the refinement along a line does not suffice there.

Set-up 13.12. Let X be a smooth prime Fano threefold of genus 12 and $p\in X$ a closed point such that $\Sigma(X,p)$ consists of $r\in\{2,3\}$ reduced points $[L_1],\dots,[L_r]$. For each i use the notation of Set-up 13.4 for L_i with an index i. For $i\neq j$ let $p_{ij}\in f_{i,p}$ be the point corresponding to L_j under Definition-Proposition 10.9, and let $n_{ij}\in\{1,2,3\}$ be the length of Z_{U_i} at p_{ij}.

By Proposition 4.1, a reduced point of $\Sigma(X,p)$ corresponds to a non-special line, so each L_i is non-special and the notation of Set-up 13.4 is defined for L_i. In particular, $\pi_i: Y_i=\mathrm{Bl}_{L_i}X\to X$ has exceptional divisor $E_i\cong\mathbb{F}_1$ with negative section s_i and fiber class f_i, $f_{i,p}$ is the fiber of E_i over p, and $A_X(E_i)=2$.

Lemma 13.13. *Notation as in Set-up 13.12. Then for any j, the point $s_j\cap f_{j,p}$ of $f_{j,p}=\mathbb{P}(N_{L_j/X}\otimes k(p))$ is the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$ of $N_{L_j/X}\cong\mathcal{O}_{\mathbb{P}^1}\oplus\mathcal{O}_{\mathbb{P}^1}(-1)$. Moreover, for $i\neq j$, $n_{ij}>1$ if and only if $p_{ji}\in s_j$.*

Proof. Since L_j is non-special, $N_{L_j/X}\cong\mathcal{O}_{\mathbb{P}^1}\oplus\mathcal{O}_{\mathbb{P}^1}(-1)$ (Lemma 2.12(3)), and the summand $\mathcal{O}_{\mathbb{P}^1}$ is the image of the evaluation map $H^0(L_j,N_{L_j/X})\otimes\mathcal{O}_{L_j}\to N_{L_j/X}$. The subbundle $\mathcal{O}_{\mathbb{P}^1}\subset N_{L_j/X}$ defines a section C_0 of $E_j=\mathbb{P}(N_{L_j/X})\to L_j$ with normal bundle $\mathcal{H}om(\mathcal{O}_{\mathbb{P}^1},\mathcal{O}_{\mathbb{P}^1}(-1))\cong\mathcal{O}_{\mathbb{P}^1}(-1)$. Thus $(C_0^2)_{E_j}=-1$, and $C_0=s_j$, because s_j is the only curve on $E_j\cong\mathbb{F}_1$ with negative self-intersection. Taking the fiber over p yields the first statement.

Let $i\neq j$. By Definition-Proposition 10.9 for $L=L_i$, the completed local ring of Inc_{L_i} at (L_j,p) is isomorphic to $\mathbb{C}[t]/(t^{n_{ij}})$. Thus the Zariski tangent space T of Inc_{L_i} at (L_j,p) satisfies $\dim T=1$ if $n_{ij}>1$, and $T=0$ if $n_{ij}=1$.

We compute T. Let $\mathcal{L}\to\Sigma(X)$ be the universal line with evaluation morphism $\phi:\mathcal{L}\to X$, and let $\xi_{L_i}\in T_pL_i$ and $\xi_{L_j}\in T_pL_j$ be non-zero vectors. Since L_i and L_j are distinct lines through p in the embedding of Lemma 2.12(1), $\xi_{L_i}\notin\mathbb{C}\xi_{L_j}$. By [KPS18, Corollary 2.1.6(ii)], $\Sigma(X)$ is smooth of dimension one at $[L_j]$, and $T_{[L_j]}\Sigma(X)=H^0(L_j,N_{L_j/X})$ by [KPS18, Lemma 2.1.4 and

the paragraph following it]. Hence, after choosing a splitting of the tangent space of $\mathcal{L}$ at (L_j, p), a tangent vector of $\mathcal{L}$ at (L_j, p) is a pair (σ, a) with $\sigma \in H^0(L_j, N_{L_j/X})$ and $a \in \mathbb{C}$, and the differential of ϕ sends (σ, a) to

$$\mathrm{ev}_p(\sigma) + a\xi_{L_j} \in T_pX,$$

where $\mathrm{ev}_p(\sigma)$ is a lift of $\sigma(p) \in N_{L_j/X} \otimes k(p) = T_pX/\mathbb{C}\xi_{L_j}$ [KPS18, Corollary 2.1.6(ii) and the map (2.2.2)]. Since Inc_{L_i} is open in $\mathcal{L} \times_X L_i$,

$$T = \left\{(\sigma, a, b) \in H^0(L_j, N_{L_j/X}) \times \mathbb{C} \times \mathbb{C} \;\middle|\; \mathrm{ev}_p(\sigma) + a\xi_{L_j} = b\xi_{L_i}\right\}.$$

If $(0, a, b) \in T$, then $a\xi_{L_j} = b\xi_{L_i}$, hence $a = b = 0$. Let σ_0 be a generator of $H^0(L_j, N_{L_j/X}) \cong \mathbb{C}$. Then $T \neq 0$ if and only if $\sigma_0(p) \in \mathbb{C}\bar{\xi}_{L_i}$, where $\bar{\xi}_{L_i}$ is the image of ξ_{L_i} in $T_pX/\mathbb{C}\xi_{L_j}$. Since $\sigma_0(p) \neq 0$ spans the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$ and $\bar{\xi}_{L_i} \neq 0$, $T \neq 0$ if and only if the normal direction of L_i at p relative to L_j is the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$. By the first statement, this holds if and only if the point of $f_{j,p}$ given by the normal direction of L_i belongs to s_j. By Definition-Proposition 10.9 for $L = L_j$, that point is p_{ji}. Thus $n_{ij} > 1$ if and only if $p_{ji} \in s_j$. □

Lemma 13.14. *Notation as in Set-up 13.12 with $r = 2$. Let $\{i, j\} = \{1, 2\}$, let W_i be the refinement of the complete series of H by E_i, and assume that either $p_{ij} \notin s_i$, or $p_{ij} \in s_i$ and $n_{ij} \in \{2, 3\}$. Define*

$$\lambda_i := \begin{cases} 44/43 & \text{if } p_{ij} \notin s_i, \\ 88/83 & \text{if } p_{ij} \in s_i \text{ and } n_{ij} = 2, \\ 176/181 & \text{if } p_{ij} \in s_i \text{ and } n_{ij} = 3. \end{cases}$$

Then any prime divisor v over E_i with $\mathrm{center}_{E_i} v \cap f_{i,p} \neq \emptyset$ satisfies

$$\lambda_i S(W_i; v) \leq A_{E_i}(v). \tag{13.39}$$

Moreover, if $p_{ij} \in s_i$ and $n_{ij} = 3$, then $S(W_i; \mathrm{ord}_{p_{ij}}) = 181/88$, where $\mathrm{ord}_{p_{ij}}$ is the order of vanishing at the point p_{ij} of E_i.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe Z near f and the restriction of R_0 to f.

We drop the index i: we write Y, E, s, W, $Z := Z_{U_i}$, $I := I_{Z_{U_i}}$, $f := f_{i,p}$, $n := n_{ij}$, and $q := s \cap f$. By Set-up 13.4, W^f is the refinement of W by f.

By Definition-Proposition 10.9, $\Sigma(X, p) \backslash \{[L_i]\} \cong Z \cap f$. Since $\Sigma(X, p)$ consists of two reduced points, $Z \cap f = \{p_{ij}\}$ is a reduced point. Hence the residual part of Z, of length $3 - n$, is supported outside f, and if $n \geq 2$, then the curvilinear scheme Z is transverse to f at p_{ij}. Thus exactly one of the three base points of U belongs to the strict transforms of f, that is, $k_f = 1$.

Since Λ is base point free, $(\Lambda \cdot \tilde{s}) \geq 0$ for the strict transform $\tilde{s}$ of s on T_E. Since $(s^2)_E = -1$ and $(s \cdot f)_E = 1$,

$$(\Lambda \cdot \tilde{s}) = ((2s + 3f) \cdot s)_E - k_s = 1 - k_s \geq 0.$$

Thus at most one of the three base points of U belongs to the strict transforms of s. In particular, if $p_{ij} = q$ and $n \geq 2$, then Z is transverse to s at q.

Since $k_f = 1$, Lemma 13.5(2) implies $f \not\subset \mathrm{Supp}\, R_0$. By Lemma 13.5(1), $p_E^* R_0 - 2(e_1 + e_2 + e_3) \geq 0$, so the local equations of R_0 belong to I^2 by Construction-Proposition 11.3(2). Since $Z \cap f = \{p_{ij}\}$ is reduced, $R_0|_f$ is a divisor on f with

$$\deg R_0|_f = ((3s + 4f) \cdot f)_E = 3, \qquad R_0|_f \geq 2p_{ij}. \tag{13.40}$$

Step 2. In this step we fix the containing spaces of W, the normalization, and the expected coefficient of the fixed part.

For integers $a, b, w \geq 0$ write

$$\mathcal{C}(a, b, w) := H^0(E, \mathcal{O}_E(as + bf) \otimes I^w).$$

By Construction-Proposition 11.3(3)(4), for any $m \geq 1$,

$$W_{m,j} \subset \begin{cases} \mathcal{C}(j, m+j, 0) & 0 \leq j \leq m, \\ \mathcal{C}(j, m+j, j-m) & m \leq j \leq 2m, \\ r_0^{j-2m} \cdot \mathcal{C}(2(3m-j), 3(3m-j), 3m-j) & 2m \leq j \leq 3m, \end{cases} \tag{13.41}$$

where $W_{m,j} = r_0^{j-2m} \cdot U_{3m-j}$ for $2m \leq j \leq 3m$, and $W_{m,j} = 0$ for $j > 3m$.

On $E \backslash s$ let z be a coordinate on the base with $f = (z = 0)$, and let u be a fiber coordinate of $\mathcal{O}_{\mathbb{P}^1}(1)$. For $b \geq a \geq 0$, the space $H^0(E, \mathcal{O}_E(as + bf))$ has the basis

$$z^k u^\ell, \qquad 0 \leq \ell \leq a,\ 0 \leq k \leq b - \ell. \tag{13.42}$$

Near s let $\nu := 1/u$. Here u and ν are the coordinates u' and u of Set-up 13.4. In the corresponding local generator, the basis (13.42) is

$$z^k \nu^\ell, \qquad 0 \leq \ell \leq a,\ 0 \leq k \leq b - a + \ell, \tag{13.43}$$

and $z^k \nu^\ell$ vanishes to order ℓ along s and to order k along f.

We use the scaled variables $w := j/m - 1 \in [0, 1]$ in the middle range, so that $a/m = 1 + w$ and $b/m = 2 + w$, $\eta := \ell/m$, and $\theta := (3m - j)/m \in [0, 1]$ in the tail range $2m \leq j \leq 3m$. In the middle range $m \leq j \leq 2m$, the scaled w is the exponent $w = j - m$ of I in $\mathcal{C}(a, b, w)$ divided by m, and we keep the letter w after scaling, as for d in **Step 5**. By Lemma 2.12(2), $N_m = \frac{11}{3}m^3 + O(m^2)$, so $mN_m = \frac{11}{3}m^4 + O(m^3)$, and a sum over j of moments of order m^3 per j contributes $\frac{3}{11}$ times the corresponding integral to a normalized expected order. All lattice regions below have fixed rational facets and bounded scaled support. Thus rounding the boundaries changes a rank by $O(m)$ and a moment by $O(m^2)$ for each j, and changes normalized expected orders by $O(1/m)$, uniformly in the point and the basis. The limits $S = \lim_m S_m$ exist by [AZ22, Lemma 2.9].

By Construction-Proposition 11.3(2), $\mathcal{C}(a, b, w) = H^0(T_E, p_E^*(as + bf) - w(e_1 + e_2 + e_3))$. For the three ranges in (13.41), the divisor $p_E^*(as + bf) - w(e_1 + e_2 + e_3)$ equals

$$m\big(\tfrac{j}{m} p_E^*(s + f) + p_E^* f\big), \qquad m\big(w\Lambda + (1 - w)p_E^*(s + 2f)\big), \qquad m\theta\Lambda,$$

respectively. Each is nef, as a non-negative combination of the base point free divisors $p_E^*(s + f)$, $p_E^* f$, $p_E^*(s + 2f)$, and Λ. Since $(\Lambda^2)_{T_E} = 5$, $(\Lambda \cdot p_E^*(s + 2f))_{T_E} = ((2s + 3f) \cdot (s + 2f))_E = 5$, $((s + 2f)^2)_E = 3$, and $((s + f)^2)_E = 1$, the self-intersections divided by $2m^2$ are

$$\frac{j}{m} + \frac{1}{2}\Big(\frac{j}{m}\Big)^2, \qquad \frac{3 + 4w - 2w^2}{2}, \qquad \frac{5}{2}\theta^2.$$

For a nef divisor D on the smooth projective surface T_E that is a non-negative integral combination of the four divisors above with coefficients at most $3m$, $h^0(T_E, D) = \frac{1}{2}(D^2) + O(m)$ by the Riemann–Roch theorem, since $h^2(T_E, D) = 0$ and $h^1(T_E, D) = O(m)$ (add the base point free summands one curve at a time and use the exact sequences of restriction to the curves). Thus

$$\lim_{m \to +\infty} \frac{1}{m^3} \sum_{j=0}^{3m} \dim(\text{containing space of } W_{m,j}) = \frac{2}{3} + \frac{13}{6} + \frac{5}{6} = \frac{11}{3} = \lim_{m \to +\infty} \frac{N_m}{m^3}. \tag{13.44}$$

Since $\sum_j \dim W_{m,j} = N_m$, the codimensions of $W_{m,j}$ in the containing spaces of (13.41), summed over j, are $o(m^3)$. Since all vanishing orders under consideration are $O(m)$, replacing $W_{m,j}$ by the containing spaces changes any sum of vanishing orders over adapted bases by $o(m^4)$, which is $o(mN_m)$. By Lemma 11.1(4), the expected coefficient of the fixed factor r_0 is

$$\lim_{m \to +\infty} \frac{1}{mN_m} \sum_{j > 2m} (j - 2m) \dim W_{m,j} = J_1 = \frac{5}{88}. \tag{13.45}$$

Step 3. In this step we assume that $p_{ij} \notin s$, and we prove $S(W; f) \leq 83/88$.

Since the automorphisms $u \to u + \beta_1$ of E preserve s and f, possibly composing with one of them, we may assume that $p_{ij} = (z, u) = (0, 0)$, and we choose the point $z = +\infty$ of the base outside the image of $\operatorname{Supp} Z$. Since $Z \cap f$ is reduced, the ideal of Z at p_{ij} is $(u - P(z), z^n)$ with $P \in \mathbb{C}[z]$ and $P(0) = 0$. Let μ be a positive integer, and for $\tau \in \mathbb{G}_m$ let ψ_τ be the automorphism $(z, u) \to (\tau^{-1}z, \tau^\mu u)$ of E, which preserves s and f. Let I_τ be the ideal of $\psi_\tau(Z)$. The coefficient of z^k in the transformed polynomial P acquires the factor $\tau^{\mu+k}$, so the ideal of $\psi_\tau(Z)$ at p_{ij} tends to (u, z^n) as $\tau \to 0$.

At infinity let $z_\infty := 1/z$, $u_\infty := u/z$, and $\nu_\infty := 1/u_\infty$, let $y_\infty := (z_\infty, u_\infty) = (0, 0)$, and let $q_\infty := s \cap f_\infty$, where $f_\infty = (z_\infty = 0)$. The automorphism ψ_τ multiplies z_∞ and u_∞ by τ and $\tau^{\mu+1}$. Thus the residual points of Z outside s tend to y_∞, and the residual points on s tend to q_∞. Two distinct residual points outside s with distinct base coordinates, and a residual double point outside s with tangent direction different from the fiber direction, tend to the horizontal double point (u_∞, z_∞^2) at y_∞, because the slopes of their interpolating graphs acquire the factor τ^μ. Two residual points with equal base coordinates, and a residual double point tangent to the fiber, tend to the vertical double point (z_∞, u_∞^2) at y_∞. By **Step 1**, at most one base point of U belongs to the strict transforms of s. Hence Z has at most one residual point on s, and a residual double point supported on s is transverse to s. Such a double point is the graph $z_\infty = z_\infty^0 + \omega\nu_\infty$ modulo ν_∞^2, and the transformed slope is $\omega\tau^{\mu+2}$, so it tends to (z_∞, ν_∞^2) at q_∞. Thus the limit Z_0 of $\psi_\tau(Z)$ consists of the point (u, z^n) at p_{ij} and one of the following residual parts:

- $n = 3$: none;
- $n = 2$: the reduced point y_∞, or the reduced point q_∞;
- $n = 1$: (u_∞, z_∞^2) at y_∞, (z_∞, u_∞^2) at y_∞, (z_∞, ν_∞^2) at q_∞, or the reduced points y_∞ and q_∞.

Near each limit point the ideal I_τ is generated by x_1 and $Q(x_2)$, where x_2 is a coordinate, x_1 is a transverse coordinate defined by a polynomial graph with coefficients regular at $\tau = 0$ (linear at the residual points, where for two points with distinct base coordinates the slope has the factor τ^μ, and for two points with equal base coordinates we take $x_2 = u_\infty$), and Q is a monic polynomial in x_2 with $\deg Q \le 3$. The quotient by $(x_1, Q(x_2))^w$ is free with basis $x_1^{i'}x_2^{j'}$, $0 \le i' < w$, $0 \le j' < (w - i')\deg Q$, and distinct limit points are treated on disjoint neighborhoods. Hence the powers I_τ^w form flat families over $\mathbb{A}^1$.

For integers $a, b, w, k \ge 0$, the subspace of $\mathcal{C}(a, b, w)$ of sections vanishing to order at least k along f is the kernel of the composition of multiplication by z^k and restriction,

$$H^0(E, \mathcal{O}_E(as + (b - k)f)) \to H^0(E, \mathcal{O}_E(as + bf) \otimes \mathcal{O}_E/I_\tau^w).$$

The domain and the codomain are locally free over $\mathbb{A}^1$, so the kernel dimension at τ equals the rank of the domain minus the rank of the codomain plus the fiber dimension of the cokernel, which is upper semicontinuous [Har77, Exercise II.5.8(a)]. Since ψ_τ is an automorphism preserving f for $\tau \ne 0$, the dimension of the subspace of $\mathcal{C}(a, b, w)$ of sections vanishing to order at least k along f is at most the corresponding dimension for Z_0. The space defined by Z_0 is spanned by monomials. In the coordinates at infinity, $z^k u^\ell$ is $z_\infty^{b-k-\ell}u_\infty^\ell = z_\infty^{b-k-\ell}\nu_\infty^{a-\ell}$ times a local generator. Hence $z^k u^\ell$ belongs to the space defined by Z_0 if and only if $k \ge n(w - \ell)$ and, for each point of the residual part of Z_0, the corresponding condition among the following holds: $b - k \ge w$ for y_∞, $a + b - k - 2\ell \ge w$ for q_∞, $b - k + \ell \ge 2w$ for (u_∞, z_∞^2), $2(b - k) - \ell \ge 2w$ for (z_∞, u_∞^2), and $2b + a - 2k - 3\ell \ge 2w$ for (z_∞, ν_∞^2).

In the middle range, for fixed w and η, the allowed values of k/m form an interval with lower end $\max(0, n(w - \eta))$ and upper end the minimum of $2 + w - \eta$ and the bounds of Table 19, and the vanishing order of $z^k u^\ell$ along f is k. Thus the sum of the orders is m^2 times half the difference of the squares of the upper and lower ends, up to $O(m)$, and the middle integral is the integral over $0 \le w \le 1$ and $0 \le \eta \le 1 + w$ of $\frac{1}{2}(\text{upper end}^2 - \text{lower end}^2)$, over the set where the interval is non-empty. The range $0 \le j \le m$ has no conditions and contributes $11/24$. By homogeneity in θ, the tail integral is $\int_0^1 \theta^3 d\theta = \frac{1}{4}$ times the middle integrand at $w = 1$, integrated

in η. Since $f \not\subset \operatorname{Supp} R_0$, the factors r_0^{j-2m} do not change orders along f. A direct calculation, splitting the (w,η)-region along the lines where two of the affine bounds coincide, yields the middle and tail integrals listed in Table 99 of Appendix B, and hence the last column of Table 19.

TABLE 19. Bounds for $S(W;f)$ when $p_{ij} \notin s$, where the last column is $\frac{3}{11}$ times the sum of $\frac{11}{24}$ and the middle and tail integrals of Table 99.

n	residual part	extra bounds on k/m	$S(W;f) \le$
3	none	none	10/11
2	y_∞	2	37/44
2	q_∞	$3+w-2\eta$	83/88
1	(u_∞, z_∞^2)	$2-w+\eta$	17/22
1	(z_∞, u_∞^2)	$2-\eta/2$	17/22
1	(z_∞, ν_∞^2)	$5/2+w/2-3\eta/2$	39/44
1	y_∞ and q_∞	2 and $3+w-2\eta$	73/88

By (13.41), the subspaces of $W_{m,j}$ of sections vanishing to order at least k along f have dimensions at most those for $\mathcal{C}$, hence at most those for Z_0. Since the largest entry of the last column of Table 19 is 83/88,

$$S(W;f) \le \frac{83}{88}. \tag{13.46}$$

Step 4. In this step we bound $S(W^f;y)$ for $y \in f$, and we prove (13.39) when $p_{ij} \notin s$.

For $j \ge 2m$ write the elements of $W_{m,j}$ as $r_0^{j-2m} g'$ with $g' \in U_{3m-j}$, and set $(a,b) := (2(3m-j), 3(3m-j))$. For $j \le 2m$ set $(a,b) := (j, m+j)$. Since $f \not\subset \operatorname{Supp} R_0$, the order along f of $r_0^{j-2m} g'$ equals the order of g'. For $k \ge 0$, division by z^k and restriction to f embed the quotient of the subspaces of $W_{m,j}$ of orders at least k and at least $k+1$ along f into $H^0(f, \mathcal{O}_f(j))$, and by (13.42) the image of an element is $(r_0|_f)^{\max(0,j-2m)}$ times a polynomial in u of degree at most $\min(a, b-k)$. The images of the quotients form W^f.

Let $y \in f \backslash \{q\}$. The space of polynomials in u of degree at most d' has vanishing sequence $0, 1, \dots, d'$ at y, so any basis of a subspace has total order at most $d'(d'+1)/2$ at y. For fixed (a,b), summing over k yields $\frac{1}{2}a^2 b - \frac{1}{3}a^3$ up to lower order terms. In the following display, χ stands for j/m, $0 \le \chi \le 2$, in the range $j \le 2m$. Since

$$\int_0^2 \Big(\frac{\chi^2(1+\chi)}{2} - \frac{\chi^3}{3} \Big) d\chi + \int_0^1 \theta^3 \Big(\frac{4 \cdot 3}{2} - \frac{8}{3} \Big) d\theta = 2 + \frac{5}{6} = \frac{17}{6},$$

the polynomial factors contribute at most $\frac{3}{11} \cdot \frac{17}{6} = \frac{17}{22}$. By (13.40), $\operatorname{mult}_y R_0|_f \le 3$, so the factors $(r_0|_f)^{j-2m}$ contribute at most $3J_1$. By (13.45), for any configuration,

$$S(W^f;y) \le \frac{17}{22} + 3J_1 = \frac{83}{88} \quad \text{for } y \in f \backslash \{q\}. \tag{13.47}$$

Assume now that $p_{ij} \notin s$, and let $y = q$. By (13.41) and since $I \subset \mathfrak{m}_{p_{ij}}$, any $g \in W_{m,j}$ with $j \le 2m$, and the factor g' of any element of $W_{m,j}$ with $j \ge 2m$, vanishes at $p_{ij} = (0,0)$ to order at least the exponent w of I in the containing space of $W_{m,j}$ in (13.41), that is, $\max(0, j-m)$ for $j \le 2m$, and $3m-j$ for the factor g' if $j \ge 2m$. Thus the polynomial factor of the image of g vanishes to order at least $\max(0, w-k)$ at $u=0$, and it belongs to the span of $u^{\max(0,w-k)}, \dots, u^{\min(a,b-k)}$. Since u^ℓ vanishes to order $a-\ell$ at q, the total order at q of a basis of a subspace is at most $\sum_{\ell=\max(0,w-k)}^{\min(a,b-k)} (a-\ell)$. Integrating over k and η yields $\frac{1}{4}$ for $0 \le j \le m$, $\int_0^1 (w^2 + 2w + \frac{5}{6}) dw = \frac{13}{6}$ for $m \le j \le 2m$, and $\frac{23}{24}$ for $2m \le j \le 3m$. Since $R_0|_f \ge 2p_{ij}$ has degree three and $p_{ij} \ne q$, $\operatorname{mult}_q R_0|_f \le 1$. Thus

$$S(W^f;q) \le \frac{3}{11} \Big(\frac{1}{4} + \frac{13}{6} + \frac{23}{24} \Big) + J_1 = \frac{81}{88} + \frac{5}{88} = \frac{43}{44}. \tag{13.48}$$

We now use the flag $f \subset E$. The pair (E, f) is plt with zero different, since E and f are smooth, and $A_f(\mathrm{ord}_y) = 1$ and $A_E(f) = 1$. Let $y \in f$ and $\lambda > 0$. Lemma 3.1 for the pair $(E, 0)$, the series W, the divisor f, and $G = \{y\}$ shows the following: if $\lambda S(W^f; y) \le 1$, then $A_E(v) \ge \lambda S(W; v) + (1 - \lambda S(W; f))\,\mathrm{mult}_v f$ for any prime divisor v over E with $y \in \mathrm{center}_E v$. Hence, if $\lambda S(W^f; y) \le 1$ and $\lambda S(W; f) \le 1$, then any prime divisor v over E with $y \in \mathrm{center}_E v$ satisfies

$$\lambda S(W; v) \le A_E(v). \tag{13.49}$$

Let $\lambda = 44/43$. By (13.46), $\lambda S(W; f) \le \frac{44}{43} \cdot \frac{83}{88} = \frac{83}{86} < 1$. By (13.47) and (13.48), $\lambda S(W^f; y) \le 1$ for any $y \in f$. Since any prime divisor v over E with $\mathrm{center}_E v \cap f \ne \emptyset$ satisfies $y \in \mathrm{center}_E v$ for some $y \in f$, (13.49) implies (13.39) when $p_{ij} \notin s$.

Step 5. In this step we assume that $p_{ij} = q$ and $n \in \{2, 3\}$, and we describe the containing spaces $\mathcal{C}(a, b, w)$ by homogeneous blocks at q.

By **Step 1**, Z is transverse to f and to s at q. Thus the ideal of Z at q is $(\nu - \gamma_1 z - \gamma_2 z^2, z^n)$ with $\gamma_1 \ne 0$. If $n = 2$, the residual part of Z is a reduced point outside f, and outside s since $q \in s$ and at most one base point of U belongs to the strict transforms of s. We choose the base coordinate z so that the residual point is over $z = +\infty$. The automorphisms $u \to \alpha u + \beta_1 + \beta_2 z$ of E, with $\alpha \ne 0$, preserve s and f, and act near q by $\nu \to \nu/(\alpha + (\beta_1 + \beta_2 z)\nu)$. Take $\alpha = \gamma_1$. If $n = 3$, take $\beta_1 = \gamma_2/\gamma_1$. If $n = 2$, choose β_2 so that the residual point becomes $y_\infty = (z_\infty, u_\infty) = (0, 0)$, with $z_\infty = 1/z$ and $u_\infty = u/z$. Possibly replacing the coordinates by their images, we may assume that the ideal of Z at q is

$$I_q = (\nu - z, z^n),$$

and, if $n = 2$, that the residual part of Z is the reduced point y_∞.

The automorphisms $(z, \nu) \to (\tau z, \tau\nu)$, $\tau \in \mathbb{G}_m$, of E fix q and y_∞, and they preserve I_q and the ideal of y_∞. Hence $\mathcal{C}(a, b, w)$ is the direct sum of the subspaces $\mathcal{C}(a, b, w)_d$ of forms $\sum_{k+\ell=d} c_{k\ell} z^k \nu^\ell$ homogeneous of degree d in the basis (13.43). By (13.43), $z^{d-\ell}\nu^\ell$ belongs to the basis (13.43) and, if $n = 2$, vanishes to order at least w at y_∞, if and only if $\ell_- \le \ell \le \ell_+$, where

$$\begin{aligned}
\ell_+ &:= \min(a, d),\\
\ell_- &:= \max\left(0, \left\lceil \frac{d - b + a}{2} \right\rceil\right) \text{ if } n = 3,\\
\ell_- &:= \max\left(0, \left\lceil \frac{d - b + a}{2} \right\rceil, w + d - b\right) \text{ if } n = 2.
\end{aligned}$$

For $n = 2$, the last term is the condition at y_∞, since $z^k\nu^\ell$ vanishes at y_∞ to order $(b - a + \ell - k) + (a - \ell) = b - d + \ell$. In the coordinates $\nu - z$ and z, the ideal I_q^w is generated by monomials, and $(\nu - z)^h z^{d-h} \in I_q^w$ if and only if $d + (n - 1)h \ge nw$. Let

$$\kappa := \max\left(0, \left\lceil \frac{nw - d}{n - 1} \right\rceil\right).$$

Since $\nu^{\ell_-} z^{d-\ell_+}$ is prime to $\nu - z$,

$$\mathcal{C}(a, b, w)_d = \nu^{\ell_-} z^{d-\ell_+} (\nu - z)^\kappa \cdot \mathbb{C}[\nu, z]_{\ell_+ - \ell_- - \kappa} \tag{13.50}$$

if $\ell_+ - \ell_- - \kappa \ge 0$, and $\mathcal{C}(a, b, w)_d = 0$ otherwise. In particular, $\dim \mathcal{C}(a, b, w)_d = \ell_+ - \ell_- - \kappa + 1$.

Here a *line through* q is a curve on E defined near q by a non-zero linear form in z and ν, such as $f = (z = 0)$, $s = (\nu = 0)$, and the curves C_1 and C_t below. For a non-zero element g of (13.50) of order k along f, its *leading coefficient along* f is the restriction of $z^{-k}g$ to f, a polynomial in ν, as in **Step 4**. For any integer $d' \ge 0$, the binary forms of degree d' have vanishing sequence $0, 1, \dots, d'$ along any line through q. Hence an adapted basis of (13.50) has the following sums

of orders:

$$\begin{aligned}
\text{along } f=(z=0):&\ (d-\ell_+)(\ell_+-\ell_--\kappa+1)+\tbinom{\ell_+-\ell_--\kappa+1}{2},\\
\text{along } s=(\nu=0):&\ \ell_-(\ell_+-\ell_--\kappa+1)+\tbinom{\ell_+-\ell_--\kappa+1}{2},\\
\text{along } C_1:=(\nu=z):&\ \kappa(\ell_+-\ell_--\kappa+1)+\tbinom{\ell_+-\ell_--\kappa+1}{2},\\
\text{along } C_t:=(\nu=tz),\ t\neq 0,1:&\ \tbinom{\ell_+-\ell_--\kappa+1}{2}.
\end{aligned}$$

Each element of (13.50) has multiplicity d at q. For a basis of (13.50) adapted to ord_f, the orders at q of the leading coefficients along f add up to $\ell_+(\ell_+-\ell_--\kappa+1)-\binom{\ell_+-\ell_--\kappa+1}{2}$. For any basis of any subspace of (13.50), the sums of the orders along f, s, C_1, and C_t are at most the corresponding sums for the complete block, since for each $t'\geq 1$ the number of basis elements of order at least t' along a line through q is at most the dimension of the subspace of elements of order at least t', which is at most the corresponding dimension for the block. The sum of the multiplicities at q of a basis of a subspace is d times its dimension, hence at most the sum for the block. For a basis of a subspace adapted to ord_f, the orders at q of the leading coefficients along f are the numbers $d-\operatorname{ord}_f\geq 0$, and the values of ord_f on such a basis are distinct values of ord_f on the block, since the elements of the block of order exactly k along f have leading term a multiple of $z^k\nu^{d-k}$. Thus the sum of the orders at q of the leading coefficients along f of such a basis is also at most the corresponding sum for the complete block.

We scale by m as in **Step 2**: in the middle range $\ell_+=\min(1+w,d)$, $\ell_-=\max(0,\frac{d-1}{2},d-2)$ for $n=2$ and $\ell_-=\max(0,\frac{d-1}{2})$ for $n=3$, and $\kappa=\max(0,\frac{nw-d}{n-1})$, where d now denotes the scaled degree. We integrate the leading terms of the sums above over the region $0\leq w\leq 1$, $d\geq 0$, $\ell_+-\ell_--\kappa\geq 0$, together with the ranges $0\leq j\leq m$ and $2m\leq j\leq 3m$ (the tail moments are $\frac14$ of the corresponding moments at $w=1$, by homogeneity). The cells of the integration region on which all integrands are polynomials of degree at most two, and the integrals over them, are listed in Tables 100 and 101 of Appendix B, and their normalized sums are the values in Table 20. The rank integrals are $\frac23$, $\frac{13}{6}$, and $\frac56$ again.

TABLE 20. Normalized sums of orders of adapted bases of the blocks (13.50) at $q=p_{ij}$, without the factors r_0^{j-2m}.

n	f	leading coeff. along f, at q	s	C_1	C_t	mult_q
2	299/352	305/352	73/88	39/88	13/32	151/88
3	125/132	31/33	299/352	39/88	373/1056	83/44

Since $f\not\subset\operatorname{Supp}R_0$, the first column of Table 20 and (13.41) imply

$$S(W;f)\leq\frac{299}{352}\text{ if }n=2,\qquad S(W;f)\leq\frac{125}{132}\text{ if }n=3. \tag{13.51}$$

Step 6. In this step we assume that $p_{ij}=q$ and $n\in\{2,3\}$, and we prove $\lambda_iS(W;v)\leq A_E(v)$ for any prime divisor v over E with $q\in\operatorname{center}_E v$.

We first compute the lowest homogeneous form of r_0 at q. By Construction-Proposition 11.3(1) and **Step 1**, $r_0\in\mathcal{C}(3,4,2)$. By (13.50) with $(a,b,w)=(3,4,2)$, $\mathcal{C}(3,4,2)_d=0$ for $d<3$, while $\mathcal{C}(3,4,2)_3=\nu(\nu-z)\cdot\mathbb{C}[\nu,z]_1$ if $n=2$ and $\mathcal{C}(3,4,2)_3=\mathbb{C}\nu(\nu-z)^2$ if $n=3$. By (13.43), the restriction of a section of $\mathcal{O}_E(3s+4f)$ to $f=(z=0)$ is a polynomial in ν of degree at most three, so $r_0|_f$ is the coefficient of ν^3 in the degree-three component of r_0. Since $f\not\subset\operatorname{Supp}R_0$, the coefficient of ν^3 is non-zero. Thus the lowest homogeneous form of r_0 at q is

$$\nu(\nu-z)(\epsilon\nu+\epsilon'z)\text{ with }\epsilon\neq 0\text{ if }n=2,\qquad \epsilon\nu(\nu-z)^2\text{ with }\epsilon\neq 0\text{ if }n=3. \tag{13.52}$$

In particular, $\operatorname{mult}_qR_0=3$. The coefficients in (13.52) of f, s, C_1, and any other line through q are at most 0, 2, 2, and 1 if $n=2$, and equal 0, 1, 2, and 0 if $n=3$.

Let $\rho : \mathbb{G}_m \to \mathrm{Aut}(E)$ be the one-parameter subgroup with $\rho(\tau)^* z = \tau^{-1} z$ and $\rho(\tau)^* \nu = \tau^{-1}\nu$, so that, for the action on sections in Lemma 3.6, the monomial $z^k \nu^\ell$ has weight $k + \ell$ up to the character of the linearization. It is contained in the torus of the toric surface $E \cong \mathbb{F}_1$, it fixes q, and the sheaves $\mathcal{O}_E(as + bf)$ carry ρ-linearizations. For any m and j, the limit $\mathrm{in}_\rho W_{m,j}$ is spanned by the lowest homogeneous forms at q of the elements of $W_{m,j}$. Since the lowest form of a product is the product of the lowest forms, (13.41) and (13.52) imply that

$$\mathrm{in}_\rho W_{m,j} \subset \begin{cases} \bigoplus_d \mathcal{C}(j, m+j, \max(0, j-m))_d & j \le 2m, \\ \mathrm{in}(r_0)^{j-2m} \cdot \bigoplus_d \mathcal{C}(2(3m-j), 3(3m-j), 3m-j)_d & j \ge 2m, \end{cases}$$

where $\mathrm{in}(r_0)$ is (13.52). Let D be an eigen-basis divisor at level m in the sense of Lemma 3.6, applied with the one-parameter subgroup ρ at each level m. Near q, D is a sum of lines through q. By **Step 5**, Table 20, (13.45), and (13.52), the coefficients of D and $\mathrm{mult}_q D$ are bounded, up to $O(1/m)$ uniformly in D, as follows:

$$\begin{aligned} n = 2: \quad & f: \tfrac{299}{352}, \quad s: \tfrac{73}{88} + 2J_1 = \tfrac{83}{88}, \quad C_1: \tfrac{39}{88} + 2J_1 = \tfrac{49}{88}, \\ & C_t: \tfrac{13}{32} + J_1, \quad \mathrm{mult}_q: \tfrac{151}{88} + 3J_1 = \tfrac{83}{44}; \\ n = 3: \quad & f: \tfrac{125}{132}, \quad s: \tfrac{299}{352} + J_1 = \tfrac{319}{352}, \quad C_1: \tfrac{39}{88} + 2J_1 = \tfrac{49}{88}, \\ & C_t: \tfrac{373}{1056}, \quad \mathrm{mult}_q: \tfrac{83}{44} + 3J_1 = \tfrac{181}{88}. \end{aligned}$$

Let B be a $\mathbb{Q}$-divisor on E whose components through q are distinct lines through q, with coefficients less than one and with sum of the coefficients of the components through q less than two. Then (E, B) is klt at q, since on the blow-up of q the strict transforms of the lines are disjoint and intersect the exceptional curve transversely, the exceptional curve has coefficient $\mathrm{mult}_q B - 1 < 1$ in the crepant pullback, where $\mathrm{mult}_q B$ is the sum of the coefficients of the components of B through q, and the pullback has simple normal crossing support with all coefficients less than one [KM98, Lemma 2.29, Lemma 2.30, and Corollary 2.31(3)].

Let $0 < c < 88/83$ if $n = 2$, and $0 < c < 176/181$ if $n = 3$. If $n = 2$, then $c \cdot \frac{83}{88} < 1$ and $c \cdot \frac{83}{44} < 2$, and the other coefficients are smaller than $\frac{83}{88}$. If $n = 3$, then $c \cdot \frac{181}{88} < 2$, $c \cdot \frac{319}{352} < \frac{319}{362} < 1$, $c \cdot \frac{125}{132} < \frac{500}{543} < 1$, and the other coefficients are smaller than $\frac{319}{352}$. Thus (E, cD) is klt at q for any $m \gg 0$ and any eigen-basis divisor D at level m. By Lemma 3.6(2) with $c_m = c$, $cS(W; v) \le A_E(v)$ for any prime divisor v over E with $q \in \mathrm{center}_E v$. Letting c tend to $88/83$ (resp. $176/181$) yields $\lambda_i S(W; v) \le A_E(v)$ for such v.

Step 7. In this step we finish the proof of (13.39) when $p_{ij} = q$, and we prove $S(W; \mathrm{ord}_q) = 181/88$ when $n = 3$.

Assume that $p_{ij} = q$ and $n \in \{2, 3\}$. Let $y \in f \backslash \{q\}$. Since $\lambda_i \le 88/83$ (it equals $88/83$ if $n = 2$ and $176/181$ if $n = 3$), (13.47) gives $\lambda_i S(W^f; y) \le \frac{88}{83} \cdot \frac{83}{88} = 1$. By (13.51), $\lambda_i S(W; f) \le \frac{88}{83} \cdot \frac{299}{352} = \frac{299}{332} < 1$ if $n = 2$, and $\lambda_i S(W; f) \le \frac{176}{181} \cdot \frac{125}{132} = \frac{500}{543} < 1$ if $n = 3$. Thus (13.49) implies $\lambda_i S(W; v) \le A_E(v)$ for any prime divisor v over E with $y \in \mathrm{center}_E v$. Together with **Step 6**, this proves (13.39).

Assume now that $n = 3$. Then Z is supported at q, so (13.50) describes the containing spaces of (13.41) completely, and an adapted basis of $\mathcal{C}(a, b, w)$ for ord_q consists of homogeneous forms. Thus the sum of the orders at q over an adapted basis of the containing space of $W_{m,j}$, without the factor r_0^{j-2m}, is $\sum_d d \dim \mathcal{C}(a, b, w)_d$, and the normalized limit of the sums is $\frac{125}{132} + \frac{31}{33} = \frac{83}{44}$ by Table 20. By **Step 2**, replacing $W_{m,j}$ by the containing spaces changes the sums of orders over adapted bases by $o(mN_m)$. Since $\mathrm{ord}_q(r_0^{j-2m} g') = 3(j - 2m) + \mathrm{ord}_q(g')$ by (13.52),

$$S(W; \mathrm{ord}_q) = \frac{83}{44} + 3J_1 = \frac{83}{44} + \frac{15}{88} = \frac{181}{88}.$$

Since $q = p_{ij}$ and $W = W_i$, this proves $S(W_i; \mathrm{ord}_{p_{ij}}) = 181/88$. □

Proposition 13.15. *Notation as in Set-up 13.12 with $r = 2$. Then the following hold.*

(1) $\delta_p(X) \geq 176/181$.
(2) If $p_{ij} \notin s_i$ *for some* i, *then* $\delta_p(X) \geq 44/43$.
(3) If $p_{12} \in s_1$, $p_{21} \in s_2$, *and* $2 \in \{n_{12}, n_{21}\}$, *then* $\delta_p(X) \geq 88/83$.
(4) If neither the assumption of (2) nor the assumption of (3) holds, then $n_{12} = n_{21} = 3$, $p_{12} \in s_1$, $p_{21} \in s_2$, *and any prime divisor* F *over* X *with* $p \in \operatorname{center}_X F$ *satisfies*

$$A_X(F) \geq \frac{176}{181} S_X(F) + \frac{90}{181} \operatorname{mult}_F E_i$$

for $i = 1, 2$. *Moreover,* $S(W_i; \operatorname{ord}_{p_{ij}}) = 181/88 > 2 = A_{E_i}(\operatorname{ord}_{p_{ij}})$ *for* $\{i, j\} = \{1, 2\}$, *where* $\operatorname{ord}_{p_{ij}}$ *is the order of vanishing at the point* p_{ij} *of* E_i.

Proof. Assume that neither the assumption of (2) nor the assumption of (3) holds. Then $p_{12} \in s_1$ and $p_{21} \in s_2$. By Lemma 13.13, $n_{21} > 1$ and $n_{12} > 1$, so $n_{12}, n_{21} \in \{2, 3\}$. Since $2 \notin \{n_{12}, n_{21}\}$, $n_{12} = n_{21} = 3$.

Let $\{i, j\} = \{1, 2\}$, and assume that either $p_{ij} \notin s_i$, or $p_{ij} \in s_i$ and $n_{ij} \in \{2, 3\}$. Let λ_i and W_i be as in Lemma 13.14. The pair (Y_i, E_i) is plt with zero different because Y_i and E_i are smooth. By Lemma 11.1(2), $S_X(E_i) = \frac{17}{11}$, so $A_X(E_i) - \lambda_i S_X(E_i) = 2 - \frac{17\lambda_i}{11}$. Let F be a prime divisor over X with $p \in \operatorname{center}_X F$. Since $\pi_i^{-1}(p) = f_{i,p}$ and $\pi_i(\operatorname{center}_{Y_i} F) = \operatorname{center}_X F$, $\operatorname{center}_{Y_i} F \cap f_{i,p} \neq \emptyset$. By Lemma 13.14 and Lemma 3.1 with $E = E_i$, $G = f_{i,p}$, and $\lambda = \lambda_i$,

$$A_X(F) \geq \lambda_i S_X(F) + \left(2 - \frac{17}{11}\lambda_i\right) \operatorname{mult}_F E_i \geq \lambda_i S_X(F), \tag{13.53}$$

where $2 - \frac{17}{11}\lambda_i > 0$ because $\lambda_i < 22/17$. If $p_{ij} \notin s_i$ for some i, then the assumption above holds for such i, and (13.53) for such i yields (2). If $p_{12} \in s_1$, $p_{21} \in s_2$, and $n_{ij} = 2$ for some $\{i, j\} = \{1, 2\}$, then the assumption above holds for such i, and (13.53) for such i yields (3). In the configuration of (4), the assumption above holds for $i = 1, 2$, so (13.53) holds for $i = 1, 2$ with $\lambda_i = 176/181$, and $2 - \frac{17}{11} \cdot \frac{176}{181} = \frac{90}{181}$. Since $88/83 > 44/43 > 176/181$ and $\operatorname{mult}_F E_i \geq 0$, (1) follows from (2)–(4).

In the configuration of (4), Lemma 13.14 yields $S(W_i; \operatorname{ord}_{p_{ij}}) = 181/88$ for $\{i, j\} = \{1, 2\}$. Since $\operatorname{ord}_{p_{ij}}$ is the exceptional divisor of the blow-up of the smooth point p_{ij} of E_i, $A_{E_i}(\operatorname{ord}_{p_{ij}}) = 2$ [KM98, Lemma 2.29]. This proves the last statement of (4). □

We now treat the remaining case of Proposition 13.15(4), where both lengths n_{12}, n_{21} are three and both points p_{12}, p_{21} are on the negative sections. In this case, Construction-Proposition 13.19 below computes the associated graded algebra of the anti-canonical ring of X for a $(1, 1, 2)$-weighted monomial valuation at p, and Theorem 13.21 is deduced from it. We first write the canonical Mukai net in coordinates adapted to p.

Set-up 13.16. Let X be a smooth prime Fano threefold of genus 12 and $p \in X$ a closed point. By Definition-Theorem 2.20, we identify X with the scheme of three-dimensional subspaces of V_X isotropic for any form in Π_X, and H with the restriction of the Plücker class. Let $V_3 \subset V_X$ be the subspace with $p = [V_3]$, let $u_1, \ldots, u_7$ be a basis of V_X with $V_3 = \langle u_1, u_2, u_3 \rangle$, and let $V_4 := \langle u_4, \ldots, u_7 \rangle$.

(1) For a 3×4 matrix $M = (M_{ij})$, let $V_M \subset V_X$ be the span of the vectors $u_i + \sum_{j=1}^4 M_{ij} u_{3+j}$, $i = 1, 2, 3$.
(2) For a set $I \subset \{1, \ldots, 7\}$ of three indices, let $\sigma_I \in H^0(\operatorname{Gr}(3, V_X), \mathcal{O}(1))$ be the Plücker coordinate of the columns I with respect to $u_1, \ldots, u_7$, and let $\sigma_0 := \sigma_{\{1,2,3\}}$.
(3) The map $M \to [V_M]$ is an isomorphism from the affine space of 3×4 matrices onto $(\sigma_0 \neq 0) \subset \operatorname{Gr}(3, V_X)$, and $(\sigma_I/\sigma_0)([V_M])$ is the minor of the 3×7 matrix $(I_3|M)$ with columns I, and is, up to sign, the minor of M with rows $\{1, 2, 3\} \backslash I$ and columns $\{j : 3 + j \in I\}$.
(4) We write $\Delta_{I',J'}$ for the minor of M with rows I' and columns J', and J_{ij} for the 3×4 matrix with entry 1 at (i, j) and 0 elsewhere.

(5) For $1 \le i \le 3$ and $1 \le j, l \le 4$, let $\ell_{ij}, d_{jl} \in \Pi_X^\vee$ be the linear forms $\omega \to \omega(u_i, u_{3+j})$ and $\omega \to \omega(u_{3+j}, u_{3+l})$, and let $C := (\ell_{ij})$ and $D := (d_{jl})$.
(6) Since V_3 is isotropic for any $\omega \in \Pi_X$, the value of ω on the i-th and the i'-th vectors spanning V_M is the entry (i, i') of $Q(M)(\omega)$, where

$$Q(M) := CM^t - MC^t + MDM^t. \tag{13.54}$$

(7) The intersection $X \cap (\sigma_0 \ne 0)$ is the closed subscheme $Q(M) = 0$ of the affine space of 3×4 matrices.
(8) The point p is $[V_0]$, and T_pX is the space of matrices M with $CM^t - MC^t = 0$.
(9) Choosing a basis of $\Pi_X^\vee$, the coefficients of the entries $(1,2)$, $(1,3)$, and $(2,3)$ of $Q(M)$ are nine polynomial equations in the entries of M.

Lemma 13.17. *Notation as in Set-ups 13.12 and 13.16 with $r = 2$, where X and p in Set-up 13.16 are X and p of Set-up 13.12. Assume that $p_{12} \in s_1$ and $p_{21} \in s_2$. Then there exist a basis $u_1, \dots, u_7$ of V_X as in Set-up 13.16 and a basis a, b, c of $\Pi_X^\vee$ with the following properties.*

(1) L_1 and L_2 are the closures of $\{[V_{\tau J_{11}}] : \tau \in \mathbb{C}\}$ and $\{[V_{\tau J_{22}}] : \tau \in \mathbb{C}\}$.
(2) There exist $\kappa_1, \kappa_2, \beta, \gamma \in \mathbb{C}$ with $\kappa_1\kappa_2 \ne 1$ such that

$$C = \begin{pmatrix} a & 0 & c & \gamma b \\ 0 & b & \beta a & c \\ 0 & 0 & \kappa_1 a + b & a + \kappa_2 b \end{pmatrix}, \tag{13.55}$$

and the coefficient κ_0 of c in d_{12} is non-zero.
(3) $n_{12} = 3$ if and only if $\kappa_1 = 0$, and in this case $\beta \ne 0$. Moreover, $n_{21} = 3$ if and only if $\kappa_2 = 0$, and in this case $\gamma \ne 0$.

Proof. The proof is long, so we divide it into several steps. We write $\mathbf{e}_1, \mathbf{e}_2, \dots$ for the standard basis vectors of $\mathbb{C}^3$ and of $\mathbb{C}^4$, so that $J_{ij} = \mathbf{e}_i \mathbf{e}_j^t$.

Step 1. In this step we find a basis with property (1), and we show that then $\ell_{21} = \ell_{31} = \ell_{12} = \ell_{32} = 0$.

Let M' be a 3×4 matrix. By [Har92, Exercise 6.9(i)], the lines of Plücker degree one in $\mathrm{Gr}(3, V_X)$ through $[V_{M'}]$ are the sets $\{[U_3] : U_2 \subset U_3 \subset U_4\}$ with $U_2 \subset V_{M'} \subset U_4$ and $\dim U_i = i$. In the chart of Set-up 13.16, such a line is the closure of $\{[V_{M' + \tau\boldsymbol{\alpha}\mathbf{u}^t}] : \tau \in \mathbb{C}\}$ for non-zero vectors $\boldsymbol{\alpha} \in \mathbb{C}^3$ and $\mathbf{u} \in \mathbb{C}^4$, determined up to scalars, and, by Lemma 2.19, closures of families $\{[V_{M' + \tau\boldsymbol{\alpha}\mathbf{u}^t}]\}$ also describe the families of lines over a local Artinian $\mathbb{C}$-algebra. Since H is the restriction of the Plücker class, the lines of X are the lines of Plücker degree one contained in X. Since $\mathbf{u}^t D \mathbf{u} = 0$, we have

$$Q(\tau\boldsymbol{\alpha}\mathbf{u}^t) = \tau\big(C\mathbf{u}\boldsymbol{\alpha}^t - \boldsymbol{\alpha}(C\mathbf{u})^t\big).$$

Thus the line through p with direction $\boldsymbol{\alpha}\mathbf{u}^t$ is contained in X if and only if

$$C\mathbf{u} = \lambda\boldsymbol{\alpha} \quad \text{for some } \lambda \in \Pi_X^\vee. \tag{13.56}$$

Let $\boldsymbol{\alpha}^{(i)}(\mathbf{u}^{(i)})^t$ be the direction of L_i for $i = 1, 2$. Suppose that $\boldsymbol{\alpha}^{(1)}$ and $\boldsymbol{\alpha}^{(2)}$ are proportional. Then (13.56) holds for $\boldsymbol{\alpha}^{(1)}$ and any linear combination of $\mathbf{u}^{(1)}$ and $\mathbf{u}^{(2)}$, so infinitely many lines of X pass through p. This contradicts $\Sigma(X, p) = \{[L_1], [L_2]\}$. Suppose that $\mathbf{u}^{(1)}$ and $\mathbf{u}^{(2)}$ are proportional. Then $C\mathbf{u}^{(1)}$ is a multiple of both $\boldsymbol{\alpha}^{(1)}$ and $\boldsymbol{\alpha}^{(2)}$, hence $C\mathbf{u}^{(1)} = 0$, and (13.56) holds for $\mathbf{u}^{(1)}$ and any $\boldsymbol{\alpha}$, again a contradiction. Thus $\boldsymbol{\alpha}^{(1)}, \boldsymbol{\alpha}^{(2)}$ are linearly independent, and so are $\mathbf{u}^{(1)}, \mathbf{u}^{(2)}$. Replacing u_1, u_2, u_3 and $u_4, \dots, u_7$ with other bases of V_3 and V_4 replaces M with hMg for invertible matrices h and g, and hence replaces $\boldsymbol{\alpha}\mathbf{u}^t$ with $(h\boldsymbol{\alpha})(g^t\mathbf{u})^t$. Possibly replacing the bases in this way, we may assume that $\boldsymbol{\alpha}^{(i)} = \mathbf{u}^{(i)} = \mathbf{e}_i$ for $i = 1, 2$. Then (1) holds, and (13.56) for $\mathbf{e}_1$ and $\mathbf{e}_2$ says $\ell_{21} = \ell_{31} = 0$ and $\ell_{12} = \ell_{32} = 0$. Let $a := \ell_{11}$ and $b := \ell_{22}$. Then

$$C = \begin{pmatrix} a & 0 & \ell_{13} & \ell_{14} \\ 0 & b & \ell_{23} & \ell_{24} \\ 0 & 0 & \ell_{33} & \ell_{34} \end{pmatrix}. \tag{13.57}$$

Step 2. In this step we show that a, b, d_{12} are linearly independent, and that ℓ_{33}, ℓ_{34} are linearly independent.

Suppose that a non-zero $\omega \in \Pi_X$ satisfies $a(\omega) = b(\omega) = d_{12}(\omega) = 0$. By (13.57), the five-dimensional subspace $V_3 + \langle u_4, u_5 \rangle$ is isotropic for ω. Hence $\operatorname{rank} \omega \leq 2(7-5) = 4$. This contradicts the non-degeneracy of Π_X (Definition-Theorem 2.20), by which any non-zero form in Π_X has rank six.

Suppose that ℓ_{33} and ℓ_{34} are linearly dependent. Then there exists a two-dimensional subspace $\Pi' \subset \Pi_X$ on which ℓ_{33} and ℓ_{34} vanish. By (13.57), $u_3 \in \operatorname{Ker} \omega$ for any $\omega \in \Pi'$. The Pfaffian of the form induced by ω on $V_X/\langle u_3 \rangle$ is a cubic form on Π', so it vanishes at some non-zero $\omega \in \Pi'$. Then $\operatorname{rank} \omega \leq 4$, a contradiction.

Step 3. In this step we show that $\operatorname{Span}(\ell_{33}, \ell_{34}) = \operatorname{Span}(a, b)$, using $p_{12} \in s_1$ and $p_{21} \in s_2$.

The tangent line $T_p L_1$ is spanned by J_{11}. By Definition-Proposition 10.9, p_{12} is the image of $T_p L_2 = \mathbb{C} J_{22}$ in $N_{L_1/X} \otimes k(p) = T_p X / \mathbb{C} J_{11}$. By Lemma 13.13, $s_1 \cap f_{1,p}$ is the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$ of $N_{L_1/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$. Thus $H^0(L_1, N_{L_1/X}) \cong \mathbb{C}$ is spanned by a section σ such that $\sigma(p)$ spans the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$. Since $p_{12} \in s_1$, after rescaling σ, the vector $\sigma(p)$ is the image of J_{22}. By [KPS18, the paragraph following Lemma 2.1.4, Corollary 2.1.6(ii), and the map (2.2.2)], since L_1 is non-special, $T_{[L_1]} \Sigma(X) = H^0(L_1, N_{L_1/X})$, its elements are the lines of X over $\mathbb{C}[\epsilon]/(\epsilon^2)$ restricting to L_1, and for such a line L^ϵ and any $\mathbb{C}[\epsilon]/(\epsilon^2)$-point of L^ϵ restricting to p, the class in $N_{L_1/X} \otimes k(p)$ of the tangent vector at p is the value at p of the corresponding section. Thus σ is a line L_1^ϵ of X over $\mathbb{C}[\epsilon]/(\epsilon^2)$ restricting to L_1. The line L_1^ϵ contains a $\mathbb{C}[\epsilon]/(\epsilon^2)$-point $[V_{\epsilon M_1}]$ restricting to p, where $M_1 \in T_p X$ has image $\sigma(p)$ in $N_{L_1/X} \otimes k(p)$, so $M_1 = J_{22} + \nu J_{11}$ for some $\nu \in \mathbb{C}$. By **Step 1**, L_1^ϵ has a direction $J_{11} + \epsilon \Xi$ of the form $(\mathbf{e}_1 + \epsilon \boldsymbol{\alpha}')(\mathbf{e}_1 + \epsilon \mathbf{u}')^t$ with $\boldsymbol{\alpha}' \in \mathbb{C}^3$ and $\mathbf{u}' \in \mathbb{C}^4$. Changing the parameter τ into $\tau - \epsilon\nu$, the line L_1^ϵ is the closure of the family $[V_M]$ with

$$M = \epsilon J_{22} + \tau(J_{11} + \epsilon \Xi), \qquad \epsilon^2 = 0, \tag{13.58}$$

where $\Xi = (\xi_{ij})$ satisfies $\xi_{ij} = 0$ whenever $i \neq 1$ and $j \neq 1$. Since $Q(\tau J_{11}) = Q(\epsilon J_{22}) = 0$, the coefficient of $\epsilon\tau$ in $Q(M) = 0$ is

$$C \Xi^t - \Xi C^t + J_{11} D J_{22}^t + J_{22} D J_{11}^t = 0.$$

By (13.57), the entries $(1,3)$ and $(1,2)$ of the last equation are

$$a\xi_{31} = \xi_{13}\ell_{33} + \xi_{14}\ell_{34}, \tag{13.59}$$

$$0 = a\xi_{21} - b\xi_{12} - \xi_{13}\ell_{23} - \xi_{14}\ell_{24} + d_{12}. \tag{13.60}$$

Suppose that a, ℓ_{33}, ℓ_{34} are linearly independent. Then (13.59) implies $\xi_{31} = \xi_{13} = \xi_{14} = 0$, and (13.60) implies $d_{12} = b\xi_{12} - a\xi_{21} \in \operatorname{Span}(a, b)$, contrary to **Step 2**. Since ℓ_{33}, ℓ_{34} are linearly independent by **Step 2**, we have $a \in \operatorname{Span}(\ell_{33}, \ell_{34})$.

Exchanging the roles of L_1, J_{11}, s_1, p_{12} and L_2, J_{22}, s_2, p_{21} in the proof of (13.59) and (13.60), Lemma 13.13 for L_2 and $p_{21} \in s_2$ imply that there exists $\Xi' = (\xi'_{ij})$ with $\xi'_{ij} = 0$ whenever $i \neq 2$ and $j \neq 2$ such that $M = \epsilon J_{11} + \tau(J_{22} + \epsilon \Xi')$ is a line of X over $\mathbb{C}[\epsilon]/(\epsilon^2)$. The entries $(2,3)$ and $(1,2)$ of the coefficient of $\epsilon\tau$ in $Q(M) = 0$ are

$$b\xi'_{32} = \xi'_{23}\ell_{33} + \xi'_{24}\ell_{34}, \tag{13.61}$$

$$0 = a\xi'_{21} - b\xi'_{12} + \xi'_{23}\ell_{13} + \xi'_{24}\ell_{14} + d_{12}. \tag{13.62}$$

Suppose that b, ℓ_{33}, ℓ_{34} are linearly independent. Then (13.61) implies $\xi'_{32} = \xi'_{23} = \xi'_{24} = 0$, and (13.62) implies $d_{12} = b\xi'_{12} - a\xi'_{21} \in \operatorname{Span}(a, b)$, contrary to **Step 2**. Hence $b \in \operatorname{Span}(\ell_{33}, \ell_{34})$. Since a, b are linearly independent by **Step 2**, $\operatorname{Span}(\ell_{33}, \ell_{34}) = \operatorname{Span}(a, b)$.

Step 4. In this step we define c, show $\kappa_0 \neq 0$, and normalize ℓ_{33}, ℓ_{34} and the coefficients of c in $\ell_{13}, \ell_{14}, \ell_{23}, \ell_{24}$.

Let $\omega_0 \in \Pi_X$ be non-zero with $a(\omega_0) = b(\omega_0) = 0$, and let $c \in \Pi_X^\vee$ with $c(\omega_0) = 1$. Then a, b, c is a basis of $\Pi_X^\vee$, and the coefficient of c in any $\ell \in \Pi_X^\vee$ is $\ell(\omega_0)$. By **Step 3**, $\ell_{33}(\omega_0) = \ell_{34}(\omega_0) = 0$,

so $u_3 \in \operatorname{Ker}\omega_0$. Since $\operatorname{rank}\omega_0 = 6$, the form induced by ω_0 on $V_X/\langle u_3\rangle$ is non-degenerate. In the basis of $V_X/\langle u_3\rangle$ given by the images of $u_1, u_2, u_4, u_5, u_6, u_7$, the vectors u_1, u_2 pair trivially with u_1, u_2, u_4, u_5 by (13.57). Expanding the Pfaffian, we obtain

$$\mathrm{Pf} = \pm \det G \cdot d_{12}(\omega_0), \qquad G := \begin{pmatrix} \ell_{13}(\omega_0) & \ell_{14}(\omega_0) \\ \ell_{23}(\omega_0) & \ell_{24}(\omega_0) \end{pmatrix}.$$

Hence $\det G \neq 0$ and $\kappa_0 := d_{12}(\omega_0) \neq 0$. Replacing (u_6, u_7) with $(u_6, u_7)G^{-1}$ multiplies the last two columns of C by G^{-1} from the right, and keeps (1) and (13.57). Possibly replacing (u_6, u_7) in this way, we may assume that $G = I_2$.

Write $\ell_{33} = \kappa_{11}a + \kappa_{12}b$ and $\ell_{34} = \kappa_{21}a + \kappa_{22}b$. By **Step 2**, $\kappa_{11}\kappa_{22} - \kappa_{12}\kappa_{21} \neq 0$. Evaluating (13.60) at ω_0 and using $G = I_2$, we obtain $\xi_{14} = \kappa_0 \neq 0$. The coefficient of b in (13.59) is $0 = \kappa_{12}\xi_{13} + \kappa_{22}\xi_{14}$. If $\kappa_{12} = 0$, then $\kappa_{22} = 0$, a contradiction. Thus $\kappa_{12} \neq 0$. Likewise, evaluating (13.62) at ω_0, we obtain $\xi'_{23} = -\kappa_0 \neq 0$, and the coefficient of a in (13.61) is $0 = \kappa_{11}\xi'_{23} + \kappa_{21}\xi'_{24}$. Thus $\kappa_{21} \neq 0$. Replacing u_4, u_5, a, b with $\kappa_{21}u_4, \kappa_{12}u_5, \kappa_{21}a, \kappa_{12}b$ keeps (1), (13.57), ω_0, c, and G, and multiplies κ_0 by $\kappa_{12}\kappa_{21}$. Possibly replacing u_4, u_5, a, b in this way, we may assume that $\kappa_{12} = \kappa_{21} = 1$. Let $\kappa_1 := \kappa_{11}$ and $\kappa_2 := \kappa_{22}$. Then

$$\ell_{33} = \kappa_1 a + b, \qquad \ell_{34} = a + \kappa_2 b, \qquad \kappa_1\kappa_2 \neq 1.$$

Step 5. In this step we remove the coefficients of a and b in $\ell_{13}, \ell_{14}, \ell_{23}, \ell_{24}$, except the coefficient of b in ℓ_{14} and the coefficient of a in ℓ_{23}.

Let λ_1 be the coefficient of b in ℓ_{13}, and λ_2 the coefficient of a in ℓ_{24}. Replacing u_1 and u_2 with $u_1 - \lambda_1 u_3$ and $u_2 - \lambda_2 u_3$ subtracts λ_1 times the third row of C from the first row and λ_2 times the third row from the second row. Since $\ell_{33} = \kappa_1 a + b$ and $\ell_{34} = a + \kappa_2 b$, after the replacement the coefficient of b in ℓ_{13} and the coefficient of a in ℓ_{24} vanish. The first two columns of C, the third row of C, and G do not change. Next, let $\lambda_3, \lambda_4, \lambda_5, \lambda_6$ be the coefficients of a in ℓ_{13}, of b in ℓ_{23}, of a in ℓ_{14}, and of b in ℓ_{24}. Replacing u_6 and u_7 with $u_6 - \lambda_3 u_4 - \lambda_4 u_5$ and $u_7 - \lambda_5 u_4 - \lambda_6 u_5$ replaces $\ell_{13}, \ell_{23}, \ell_{14}, \ell_{24}$ with $\ell_{13} - \lambda_3 a$, $\ell_{23} - \lambda_4 b$, $\ell_{14} - \lambda_5 a$, $\ell_{24} - \lambda_6 b$, and keeps the other entries of C. Both replacements change M into hM and Mg with $hJ_{ii} = J_{ii}g = J_{ii}$ for $i = 1, 2$, so (1) still holds. They keep u_4, u_5, and ω_0, so $\kappa_0 = d_{12}(\omega_0)$ does not change. Let β be the coefficient of a in ℓ_{23}, and γ the coefficient of b in ℓ_{14}. Then C is given by (13.55). This proves (1) and (2).

Step 6. In this step we compute the length of the incidence scheme Inc_{L_1} at (L_2, p), and prove (3).

By Definition-Proposition 10.9, n_{12} is the length at (L_2, p) of the incidence scheme Inc_{L_1}. The points of L_1 near p are $[V_{\epsilon J_{11}}]$, $\epsilon \in \mathbb{C}$. By **Step 1**, the pairs consisting of a point of L_1 near p and a line of $\mathrm{Gr}(3, V_X)$ through it near L_2 are parametrized by an open subset of $\mathbb{A}^6$, with coordinates $(\epsilon, \alpha_1, \alpha_3, \mu_1, \mu_3, \mu_4)$ centered at (L_2, p), through the lines

$$M = \epsilon J_{11} + \tau \boldsymbol{\alpha}\mathbf{u}^t, \qquad \boldsymbol{\alpha} = (\alpha_1, 1, \alpha_3)^t, \qquad \mathbf{u} = (\mu_1, 1, \mu_3, \mu_4)^t,$$

and Inc_{L_1} is the closed subscheme where the line is contained in X. Since $Q(\epsilon J_{11}) = 0$ and $\mathbf{u}^t D \mathbf{u} = 0$, we have

$$Q(\epsilon J_{11} + \tau \boldsymbol{\alpha}\mathbf{u}^t) = \tau\big(\boldsymbol{\eta}\boldsymbol{\alpha}^t - \boldsymbol{\alpha}\boldsymbol{\eta}^t\big), \qquad \boldsymbol{\eta} = (\eta_1, \eta_2, \eta_3)^t := C\mathbf{u} + \epsilon(D\mathbf{u})_1 \mathbf{e}_1.$$

Thus Inc_{L_1} is defined by $\eta_1 = \alpha_1\eta_2$ and $\eta_3 = \alpha_3\eta_2$, which are equalities in $\Pi_X^\vee$ with coefficients in the coordinate ring of the chart. Write $D = aD_a + bD_b + cD_c$ with constant alternating matrices D_a, D_b, D_c, and $q_v := (D_v)_{12} + (D_v)_{13}\mu_3 + (D_v)_{14}\mu_4$ for $v \in \{a, b, c\}$. Then $(D_c)_{12} = \kappa_0$. By (13.55), the coefficients of a, b, c in $\eta_3 = \alpha_3\eta_2$ and in $\eta_1 = \alpha_1\eta_2$ are

$$\kappa_1\mu_3 + \mu_4 = \beta\alpha_3\mu_3, \qquad \mu_3 + \kappa_2\mu_4 = \alpha_3, \qquad \alpha_3\mu_4 = 0, \tag{13.63}$$

$$\mu_1 + \epsilon q_a = \beta\alpha_1\mu_3, \qquad \gamma\mu_4 + \epsilon q_b = \alpha_1, \qquad \mu_3 + \epsilon q_c = \alpha_1\mu_4. \tag{13.64}$$

The linear parts in $(\mu_1, \alpha_1, \mu_3, \mu_4, \alpha_3)$ of the five equations other than $\alpha_3\mu_4 = 0$ are μ_1, $\gamma\mu_4 - \alpha_1$, μ_3, $\kappa_1\mu_3 + \mu_4$, and $\mu_3 + \kappa_2\mu_4 - \alpha_3$, which are linearly independent. By the formal implicit

function theorem, the five equations define near the origin a smooth curve with coordinate ϵ, on which $\mu_1, \alpha_1, \mu_3, \mu_4, \alpha_3$ are power series in ϵ. Solving (13.64) and (13.63) modulo ϵ^2, we obtain

$$\mu_3 = -\kappa_0\epsilon + O(\epsilon^2), \qquad \mu_4 = \kappa_1\kappa_0\epsilon + O(\epsilon^2), \qquad \alpha_3 = -(1-\kappa_1\kappa_2)\kappa_0\epsilon + O(\epsilon^2).$$

Hence the completed local ring of Inc_{L_1} at (L_2, p) is $\mathbb{C}[[\epsilon]]/(\alpha_3\mu_4)$.

Suppose that $\kappa_1 \neq 0$. Since $\kappa_0 \neq 0$ and $\kappa_1\kappa_2 \neq 1$, the order of $\alpha_3\mu_4$ in ϵ is 2, so $n_{12} = 2$. Suppose that $\kappa_1 = 0$. Then the first equation of (13.63) reads $\mu_4 = \beta\alpha_3\mu_3$. If $\beta = 0$, then $\mu_4 = 0$, and the completed local ring of Inc_{L_1} at (L_2, p) is $\mathbb{C}[[\epsilon]]$. This contradicts Definition-Proposition 10.9, by which the completed local ring of Inc_{L_1} at (L_2, p) is $\mathbb{C}[t]/(t^{n_{12}})$. Thus $\beta \neq 0$, and

$$\mu_4 = \beta\kappa_0^2\epsilon^2 + O(\epsilon^3), \qquad \alpha_3\mu_4 = -\beta\kappa_0^3\epsilon^3 + O(\epsilon^4).$$

So $n_{12} = 3$. This proves the first half of (3).

Exchanging u_1 with u_2, u_4 with u_5, u_6 with u_7, and a with b exchanges L_1 with L_2, keeps the shape (13.55) with (κ_1, β) and (κ_2, γ) exchanged, and replaces κ_0 with $-\kappa_0$. Thus the second half of (3) follows from the first half. □

The next lemma computes the lowest-weight parts at p of the entries and minors of M used in Construction-Proposition 13.19, for the normal form of Lemma 13.17 with $\kappa_1 = \kappa_2 = 0$. The parts it computes depend on D only through the coefficient κ_0 of c in d_{12}.

Lemma 13.18. *Notation as in Set-up 13.16. Assume that there exist a basis a, b, c of $\Pi_X^\vee$ and $\beta, \gamma \in \mathbb{C}$ such that*

$$C = \begin{pmatrix} a & 0 & c & \gamma b \\ 0 & b & \beta a & c \\ 0 & 0 & b & a \end{pmatrix}$$

and the coefficient κ_0 of c in d_{12} is non-zero. Then the following hold.

(1) The restrictions to X of $x := M_{11}$, $y := M_{22}$, and $z := \kappa_0^{-1}M_{14}$ are regular parameters of X at p.

(2) Let $\zeta := z - xy$, let v_{w} be the monomial valuation on $\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x, y, z]]$ with weights $(1, 1, 2)$ on (x, y, z), and for $f \neq 0$ let $\mathrm{in}(f)$ be the part of f of weight $v_{\mathrm{w}}(f)$. Then

$$\begin{gathered} v_{\mathrm{w}}(M_{12}),\ v_{\mathrm{w}}(M_{21}) \geq 2, \qquad v_{\mathrm{w}}(M_{13}),\ v_{\mathrm{w}}(M_{24}) \geq 3, \\ v_{\mathrm{w}}(M_{23} - \kappa_0\zeta),\ v_{\mathrm{w}}(M_{31} - \kappa_0 z),\ v_{\mathrm{w}}(M_{32} - \kappa_0\zeta) \geq 3, \\ v_{\mathrm{w}}(M_{33} + \kappa_0^2 x\zeta),\ v_{\mathrm{w}}(M_{34} - \kappa_0^2 yz) \geq 4. \end{gathered}$$

(3) The functions $M_{14} - M_{23}$, M_{33}, M_{34}, $\Delta_{13,13}$, $\Delta_{23,24}$, $\Delta_{13,24}$, $\Delta_{13,34}$, $\Delta_{23,34}$, $\Delta_{123,134}$, and $\Delta_{123,234}$ satisfy

$$\begin{gathered} \mathrm{in}(M_{14} - M_{23}) = \kappa_0 xy, \quad \mathrm{in}(M_{33}) = -\kappa_0^2 x\zeta, \quad \mathrm{in}(M_{34}) = \kappa_0^2 yz, \\ \mathrm{in}(\Delta_{13,13}) = -\kappa_0^2 x^2\zeta, \quad \mathrm{in}(\Delta_{23,24}) = \kappa_0^2 y^2 z, \quad \mathrm{in}(\Delta_{13,24}) = -\kappa_0^2 z\zeta, \\ \mathrm{in}(\Delta_{13,34}) = \kappa_0^3 xz\zeta, \quad \mathrm{in}(\Delta_{23,34}) = \kappa_0^3 yz\zeta, \\ \mathrm{in}(\Delta_{123,134}) = -\kappa_0^3 z\zeta^2, \quad \mathrm{in}(\Delta_{123,234}) = -\kappa_0^3 z^2\zeta. \end{gathered} \tag{13.65}$$

Proof. For (1), a direct calculation from the nine linear equations $CM^t - MC^t = 0$ shows that T_pX consists of the matrices

$$M_0 = \begin{pmatrix} x_0 & 0 & 0 & z_0 \\ 0 & y_0 & z_0 & 0 \\ z_0 & z_0 & 0 & 0 \end{pmatrix}, \qquad x_0, y_0, z_0 \in \mathbb{C}. \tag{13.66}$$

Thus the differentials of M_{11}, M_{22}, M_{14} at p are linearly independent on T_pX. Since X is a smooth threefold, x, y, z are regular parameters at p. We regard the entries of M as elements of the maximal ideal $\mathfrak{m}$ of $\mathbb{C}[[x, y, z]]$.

For (2), write $D = aD_a + bD_b + cD_c$ with constant alternating matrices D_a, D_b, D_c, and let $\Psi_v := MD_vM^t$ for $v \in \{a,b,c\}$. Then $(D_c)_{12} = \kappa_0$. Taking the coefficients of a, b, c in the entries $(1,2)$, $(1,3)$, and $(2,3)$ of $Q(M) = 0$, a direct calculation shows that the nine equations are equivalent to

$$\begin{aligned} &M_{21} = \beta M_{13} - (\Psi_a)_{12}, \quad M_{12} = \gamma M_{24} + (\Psi_b)_{12}, \quad M_{23} = M_{14} - (\Psi_c)_{12},\\ &M_{31} = M_{14} - (\Psi_a)_{13}, \quad M_{13} = \gamma M_{34} + (\Psi_b)_{13}, \quad M_{33} = -(\Psi_c)_{13},\\ &M_{24} = \beta M_{33} + (\Psi_a)_{23}, \quad M_{32} = M_{23} - (\Psi_b)_{23}, \quad M_{34} = -(\Psi_c)_{23}. \end{aligned} \tag{13.67}$$

By (13.66) and the formal implicit function theorem, (13.67) has a unique solution with entries in $\mathfrak{m}$ and with $M_{11} = x$, $M_{22} = y$, $M_{14} = \kappa_0 z$, namely the matrix M restricted to X.

Let $e_{11} = e_{22} = 1$, let $e_{ij} = 3$ for $(i,j) \in \{(1,3),(2,4),(3,3),(3,4)\}$, and let $e_{ij} = 2$ for the remaining six pairs (i,j). Let $\mathcal{N}$ be the set of 3×4 matrices $\tilde{M}$ with entries in $\mathfrak{m}$ such that $\tilde{M}_{11} = x$, $\tilde{M}_{22} = y$, $\tilde{M}_{14} = \kappa_0 z$, and $v_{\mathrm{w}}(\tilde{M}_{ij}) \geq e_{ij}$ for all (i,j). For $\tilde{M} \in \mathcal{N}$ and $v \in \{a,b,c\}$,

$$v_{\mathrm{w}}\big((\tilde{M}D_v\tilde{M}^t)_{rs}\big) \geq \min_{i \neq l}(e_{ri} + e_{sl}),$$

which is 2 for $(r,s) = (1,2)$ and 3 for $(r,s) = (1,3),(2,3)$. Moreover, the minimum is attained only at $(i,l) = (1,2)$, $(1,2)$, and $(2,1)$, respectively. Hence the parts of weight 2, 3, and 3 of $(\tilde{M}D_v\tilde{M}^t)_{12}$, $(\tilde{M}D_v\tilde{M}^t)_{13}$, and $(\tilde{M}D_v\tilde{M}^t)_{23}$ are

$$(D_v)_{12}xy, \qquad (D_v)_{12}x[\tilde{M}_{32}]_2, \qquad -(D_v)_{12}y[\tilde{M}_{31}]_2, \tag{13.68}$$

where $[f]_2$ denotes the part of weight 2 of f. Let $\Theta(\tilde{M})$ be the matrix with the entries $x, y, \kappa_0 z$ at $(1,1),(2,2),(1,4)$ and with the right-hand sides of (13.67), evaluated at $\tilde{M}$, as the other nine entries. Comparing each term of the right-hand sides of (13.67) with the numbers e_{ij}, we obtain $\Theta(\mathcal{N}) \subset \mathcal{N}$. If $\tilde{M}, \tilde{M}' \in \mathcal{N}$ and $\tilde{M} - \tilde{M}'$ has entries in $\mathfrak{m}^k$, then $\Theta(\tilde{M}) - \Theta(\tilde{M}')$ is the sum of a matrix with entries in $\mathfrak{m}^{k+1}$, coming from Ψ_v, and of the terms $\beta(\tilde{M}_{13} - \tilde{M}'_{13})$, $\gamma(\tilde{M}_{24} - \tilde{M}'_{24})$, $\gamma(\tilde{M}_{34} - \tilde{M}'_{34})$, $\beta(\tilde{M}_{33} - \tilde{M}'_{33})$, and $\tilde{M}_{23} - \tilde{M}'_{23}$. The linear terms express the differences of the entries $(2,1),(1,2),(1,3),(2,4),(3,2)$ through the differences of the entries $(1,3),(2,4),(3,4),(3,3),(2,3)$, and the equations for the entries $(3,4),(3,3),(2,3),(3,1)$ have no linear terms in the other dependent entries. The longest chain of such dependences is $(3,4) \to (1,3) \to (2,1)$. Hence $\Theta^3(\tilde{M}) - \Theta^3(\tilde{M}')$ has entries in $\mathfrak{m}^{k+1}$. Thus, for $\tilde{M} \in \mathcal{N}$, the iterates $\Theta(\tilde{M}), \Theta^2(\tilde{M}), \dots$ converge $\mathfrak{m}$-adically to a solution of (13.67), which is M by uniqueness. Since $\mathcal{N}$ is closed in the $\mathfrak{m}$-adic topology, $M \in \mathcal{N}$.

This proves the first line of (2). By (13.67) and (13.68) with $\tilde{M} = M$ and $(D_c)_{12} = \kappa_0$,

$$\begin{aligned} &M_{23} = \kappa_0 z - \kappa_0 xy + O_{\mathrm{w}}(3) = \kappa_0\zeta + O_{\mathrm{w}}(3), \quad M_{31} = \kappa_0 z + O_{\mathrm{w}}(3),\\ &M_{32} = M_{23} + O_{\mathrm{w}}(3) = \kappa_0\zeta + O_{\mathrm{w}}(3), \quad M_{33} = -\kappa_0 x[M_{32}]_2 + O_{\mathrm{w}}(4) = -\kappa_0^2 x\zeta + O_{\mathrm{w}}(4),\\ &M_{34} = \kappa_0 y[M_{31}]_2 + O_{\mathrm{w}}(4) = \kappa_0^2 yz + O_{\mathrm{w}}(4), \end{aligned}$$

where $O_{\mathrm{w}}(d)$ denotes an element f with $v_{\mathrm{w}}(f) \geq d$. This proves (2).

For (3), the first three equalities of (13.65) follow from (2). For the minors of size two, (2) implies

$$\begin{aligned} \Delta_{13,13} &= M_{11}M_{33} - M_{13}M_{31} = -\kappa_0^2 x^2\zeta + O_{\mathrm{w}}(5),\\ \Delta_{23,24} &= M_{22}M_{34} - M_{24}M_{32} = \kappa_0^2 y^2 z + O_{\mathrm{w}}(5),\\ \Delta_{13,24} &= M_{12}M_{34} - M_{14}M_{32} = -\kappa_0^2 z\zeta + O_{\mathrm{w}}(5),\\ \Delta_{13,34} &= M_{13}M_{34} - M_{14}M_{33} = \kappa_0^3 xz\zeta + O_{\mathrm{w}}(6),\\ \Delta_{23,34} &= M_{23}M_{34} - M_{24}M_{33} = \kappa_0^3 yz\zeta + O_{\mathrm{w}}(6), \end{aligned}$$

since $v_{\mathrm{w}} \geq 5$ on each of $M_{13}M_{31}$, $M_{24}M_{32}$, and $M_{12}M_{34}$, and $v_{\mathrm{w}} \geq 6$ on $M_{13}M_{34}$ and $M_{24}M_{33}$. For $\Delta_{123,134}$, by (2) the products in the expansion of the determinant other than $M_{11}M_{23}M_{34}$

and $-M_{14}M_{23}M_{31}$ have weight at least 7. Hence

$$\begin{aligned}\Delta_{123,134} &= x\cdot\kappa_0\zeta\cdot\kappa_0^2yz-\kappa_0z\cdot\kappa_0\zeta\cdot\kappa_0z+O_{\mathrm{w}}(7)\\ &= \kappa_0^3z\zeta(xy-z)+O_{\mathrm{w}}(7)=-\kappa_0^3z\zeta^2+O_{\mathrm{w}}(7).\end{aligned}$$

For $\Delta_{123,234}$, the products in the expansion other than $M_{14}M_{22}M_{33}$ and $-M_{14}M_{23}M_{32}$ have weight at least 7, and

$$\begin{aligned}\Delta_{123,234} &= \kappa_0z\cdot y\cdot(-\kappa_0^2x\zeta)-\kappa_0z\cdot\kappa_0\zeta\cdot\kappa_0\zeta+O_{\mathrm{w}}(7)\\ &= -\kappa_0^3z\zeta(xy+\zeta)+O_{\mathrm{w}}(7)=-\kappa_0^3z^2\zeta+O_{\mathrm{w}}(7).\end{aligned}$$

This proves (3). □

Construction-Proposition 13.19. Notation as in Set-up 13.12 with $r=2$. Assume that $n_{12}=n_{21}=3$, $p_{12}\in s_1$, and $p_{21}\in s_2$. Then there exist regular parameters x,y,z of X at p and a local trivialization of H at p with the following properties, where $\zeta:=z-xy$ and v_{w} is the monomial valuation with weights $(1,1,2)$ on (x,y,z).

(1) With T of degree one, the associated graded algebra of $\bigoplus_{m\ge0}H^0(X,mH)$ for v_{w} is the subalgebra

$$\mathcal{R}:=\mathbb{C}[T,Tx,Ty,Tz,Txy,Tx\zeta,Tyz,Tz\zeta,Txz\zeta,Tyz\zeta,Tx^2\zeta,Ty^2z,Tz\zeta^2,Tz^2\zeta]$$

of $\mathbb{C}[T,x,y,z]$.

(2) Let $\mathcal{R}_m\subset\mathbb{C}[x,y,z]$ be the coefficient space of T^m in $\mathcal{R}$, and let $\mathcal{P}\subset\mathbb{R}^3$ be the polytope defined by $i_1,i_2,i_3\ge0$, $i_1-i_3\le1$, $i_3-i_2\le1$, $i_1+i_3\le3$, $i_2+i_3\le3$, and $-2i_1+i_2+i_3\le1$. For all integers $m,k\ge0$, the image of $\mathcal{R}_m\cap\zeta^k\mathbb{C}[x,y,z]$ in $\zeta^k\mathbb{C}[x,y,z]/\zeta^{k+1}\mathbb{C}[x,y,z]\cong\mathbb{C}[x,y]\zeta^k$ is spanned by the monomials $x^{i_1}y^{i_2}\zeta^k$ with $(i_1,i_2,k)\in m\mathcal{P}$.

(3) $\dim\mathcal{R}_m=\#(m\mathcal{P}\cap\mathbb{Z}^3)=N_m$ for any $m\ge0$.

Proof. By Lemma 13.17, there exist a basis $u_1,\dots,u_7$ of V_X as in Set-up 13.16 and a basis a,b,c of $\Pi_X^\vee$ such that C is given by (13.55) with $\kappa_0\ne0$. Since $n_{12}=n_{21}=3$, Lemma 13.17(3) implies $\kappa_1=\kappa_2=0$. Thus the assumptions of Lemma 13.18 hold. Let $x:=M_{11}$, $y:=M_{22}$, and $z:=\kappa_0^{-1}M_{14}$ on X. By Lemma 13.18(1), x,y,z are regular parameters of X at p. Since $p=[V_0]$, the section σ_0 trivializes H near p.

We first identify the associated graded algebra. For $s\in H^0(X,mH)$, let $f_s:=s/\sigma_0^m\in\mathbb{C}[[x,y,z]]$, and let $\mathcal{F}^jH^0(X,mH)$ be the subspace of the sections s with $v_{\mathrm{w}}(f_s)\ge j$. Since X is integral, $s\to f_s$ is injective. Thus $\mathcal{F}^0H^0(X,mH)=H^0(X,mH)$ and $\mathcal{F}^jH^0(X,mH)=0$ for $j\gg0$, so

$$\sum_{j\ge0}\dim\left(\mathcal{F}^jH^0(X,mH)/\mathcal{F}^{j+1}H^0(X,mH)\right)=N_m.$$

Sending the class of $s\in\mathcal{F}^jH^0(X,mH)\backslash\mathcal{F}^{j+1}H^0(X,mH)$ to $T^m\,\mathrm{in}(f_s)$, where $\mathrm{in}(f)$ is the part of f of weight $v_{\mathrm{w}}(f)$, defines an injective homomorphism of algebras from the associated graded algebra to $\mathbb{C}[T,x,y,z]$, since $\mathrm{in}(fg)=\mathrm{in}(f)\,\mathrm{in}(g)$. We identify the associated graded algebra with the image, and we write $\mathrm{gr}_m\subset\mathbb{C}[x,y,z]$ for the coefficient space of T^m. Then $\dim\mathrm{gr}_m=N_m$, and $N_m=\frac{11}{3}m^3+\frac{11}{2}m^2+\frac{23}{6}m+1$ by Lemma 2.12(2).

By Set-up 13.16(3), the Plücker coordinates restrict to sections of H, and any entry or minor of M is, up to sign, the quotient σ_I/σ_0 for some I. This holds in particular for 1, M_{11}, M_{22}, M_{14}, M_{23}, M_{33}, M_{34}, $\Delta_{13,13}$, $\Delta_{23,24}$, $\Delta_{13,24}$, $\Delta_{13,34}$, $\Delta_{23,34}$, $\Delta_{123,134}$, and $\Delta_{123,234}$. Thus σ_0, σ_0M_{11}, σ_0M_{22}, σ_0M_{14}, $\sigma_0(M_{14}-M_{23})$, and the sections $\sigma_0M_{33},\dots,\sigma_0\Delta_{123,234}$ belong to $H^0(X,H)$. By Lemma 13.18(3), their images in gr_1 are non-zero multiples of

$$1,\ x,\ y,\ z,\ xy,\ x\zeta,\ yz,\ x^2\zeta,\ y^2z,\ z\zeta,\ xz\zeta,\ yz\zeta,\ z\zeta^2,\ z^2\zeta.$$

Since the associated graded algebra is closed under products, $\mathcal{R}$ is contained in it. In particular $\mathcal{R}_m\subset\mathrm{gr}_m$ and $\dim\mathcal{R}_m\le N_m$ for any $m\ge0$.

Next we compute the image of $\mathcal{R}_m$ under the order along ζ. We have $\mathbb{C}[x,y,z]=\mathbb{C}[x,y,\zeta]$. For $f\neq 0$, write $f=\sum_{k\geq k_0} f_k(x,y)\zeta^k$ with $f_{k_0}\neq 0$, and let $\mathrm{in}_\zeta(f):=f_{k_0}\zeta^{k_0}$ be the lowest term of f in ζ. Then

$$\mathrm{in}_\zeta(fg)=\mathrm{in}_\zeta(f)\,\mathrm{in}_\zeta(g).$$

For a subspace $W\subset\mathbb{C}[x,y,z]$ with $\dim W<+\infty$, let $\mathrm{in}_\zeta(W)$ be the direct sum over $k\geq 0$ of the images of $W\cap\zeta^k\mathbb{C}[x,y,z]$ in $\zeta^k\mathbb{C}[x,y,z]/\zeta^{k+1}\mathbb{C}[x,y,z]\cong\mathbb{C}[x,y]\zeta^k$. Then $\dim\mathrm{in}_\zeta(W)=\dim W$. The coefficient space $\mathcal{R}_1$ is spanned by the fourteen polynomials listed above. Since $z=\zeta+xy$, we have $\mathrm{Span}(z,xy)=\mathrm{Span}(\zeta,xy)$, and $\mathcal{R}_1$ contains the fourteen polynomials

$$1,\ x,\ y,\ xy,\ \zeta,\ x\zeta,\ yz,\ z\zeta,\ xz\zeta,\ yz\zeta,\ x^2\zeta,\ y^2z,\ z^2\zeta,\ z\zeta^2,$$

whose images under in_ζ are the monomials $x^{i_1}y^{i_2}\zeta^{i_3}$ with $(i_1,i_2,i_3)=\mathbf{w}_0,\dots,\mathbf{w}_{13}$, where

$$\begin{aligned}&\mathbf{w}_0,\dots,\mathbf{w}_6=(0,0,0),(1,0,0),(0,1,0),(1,1,0),(0,0,1),(1,0,1),(1,2,0),\\&\mathbf{w}_7,\dots,\mathbf{w}_{13}=(1,1,1),(2,1,1),(1,2,1),(2,0,1),(1,3,0),(2,2,1),(1,1,2).\end{aligned}\tag{13.69}$$

For example, $\mathrm{in}_\zeta(yz)=xy^2$, $\mathrm{in}_\zeta(y^2z)=xy^3$, $\mathrm{in}_\zeta(z^2\zeta)=x^2y^2\zeta$, and $\mathrm{in}_\zeta(z\zeta^2)=xy\zeta^2$. Let $\mathcal{S}_m$ be the set of sums of m vectors among $\mathbf{w}_0,\dots,\mathbf{w}_{13}$, with $\mathcal{S}_0=\{0\}$. Taking products of m of the fourteen polynomials, we see that $\mathrm{in}_\zeta(\mathcal{R}_m)$ contains the monomials $x^{i_1}y^{i_2}\zeta^{i_3}$ with $(i_1,i_2,i_3)\in\mathcal{S}_m$. Hence

$$\#\mathcal{S}_m\leq\dim\mathrm{in}_\zeta(\mathcal{R}_m)=\dim\mathcal{R}_m\leq N_m.\tag{13.70}$$

By Appendix B, the twenty-two tetrahedra listed in (B.4), with the vertices of Table 102, form a unimodular triangulation of $\mathcal{P}$ with 14, 46, 55, 22 faces of dimensions $0,1,2,3$, and $\#(m\mathcal{P}\cap\mathbb{Z}^3)=N_m$. The vertices of the triangulation are $\mathbf{w}_0,\dots,\mathbf{w}_{13}$.

We show that $\mathcal{S}_m=m\mathcal{P}\cap\mathbb{Z}^3$. Let $m\geq 1$ and $\mathbf{i}\in m\mathcal{P}\cap\mathbb{Z}^3$. Then $\mathbf{i}\in m\mathcal{T}$ for a tetrahedron $\mathcal{T}$ of the triangulation with vertices $\mathbf{w}_{j_0},\dots,\mathbf{w}_{j_3}$, and $(m,\mathbf{i})=\sum_t\lambda_t(1,\mathbf{w}_{j_t})$ with $\lambda_t\geq 0$. Since $\mathcal{T}$ is unimodular, the vectors $(1,\mathbf{w}_{j_t})$ form a basis of $\mathbb{Z}^4$, so $\lambda_t\in\mathbb{Z}$. Thus $\mathbf{i}\in\mathcal{S}_m$. Conversely, $\mathcal{S}_m\subset m\mathcal{P}\cap\mathbb{Z}^3$, since $\mathbf{w}_0,\dots,\mathbf{w}_{13}\in\mathcal{P}$ and $\mathcal{P}$ is convex. Hence $\mathcal{S}_m=m\mathcal{P}\cap\mathbb{Z}^3$ for any $m\geq 0$, and $\#\mathcal{S}_m=N_m$.

By (13.70), $N_m=\#\mathcal{S}_m\leq\dim\mathcal{R}_m\leq N_m$. Thus $\dim\mathcal{R}_m=N_m=\dim\mathrm{gr}_m$, so $\mathcal{R}_m=\mathrm{gr}_m$ for any $m\geq 0$. This proves (1), and (3) follows since $\#(m\mathcal{P}\cap\mathbb{Z}^3)=N_m$. Moreover, $\mathrm{in}_\zeta(\mathcal{R}_m)$ contains the span of the monomials $x^{i_1}y^{i_2}\zeta^{i_3}$ with $(i_1,i_2,i_3)\in m\mathcal{P}\cap\mathbb{Z}^3$, and both spaces have dimension N_m. Since both spaces are direct sums of their parts in $\mathbb{C}[x,y]\zeta^k$, they coincide in any degree k in ζ. This proves (2). □

We first prove a lemma on $\mathbb{P}(1,1,2)$, used in the proof of Theorem 13.21 below. It bounds the divisors associated with bases consisting of eigenvectors of a one-parameter subgroup.

Lemma 13.20. *Let $E:=\mathbb{P}(1,1,2)$ with weighted homogeneous coordinates x,y,z of weights $1,1,2$, let $\zeta:=z-xy$, and let $\lambda:\mathbb{G}_m\to\mathrm{Aut}(E)$ be the one-parameter subgroup given by*

$$\lambda(t)\cdot[x:y:z]=[tx:y:tz].$$

Let $\mathcal{P}\subset\mathbb{R}^3$ be the polytope of Construction-Proposition 13.19(2). For integers $d_1,d_2\geq 0$ let $\mathbb{C}[x,y,z]_{(d_1,d_2)}$ be the span of the monomials $x^{i_1}y^{i_2}z^{i_3}$ with $i_1+i_3=d_1$ and $i_2+i_3=d_2$, for an integer $j\geq 0$ let $\mathbb{C}[x,y,z]_j=H^0(E,\mathcal{O}_E(j))$ be the space of weighted homogeneous polynomials of degree j, and let ι be the automorphism of $\mathbb{C}[x,y,z]$ with $\iota(x)=y$, $\iota(y)=x$, and $\iota(z)=xy-z$. For any integer $m\geq 0$ let $W_m\subset\mathbb{C}[x,y,z]$ be a linear subspace with $\dim W_m<+\infty$, and let $W_m^{d_1,d_2}:=W_m\cap\mathbb{C}[x,y,z]_{(d_1,d_2)}$. Assume that the following hold for any m.

(i) $W_m=\bigoplus_{d_1,d_2\geq 0}W_m^{d_1,d_2}$.

(ii) $\iota(W_m)=W_m$.

(iii) For any integer $k\geq 0$, the image of $W_m\cap\zeta^k\mathbb{C}[x,y,z]$ in

$$\zeta^k\mathbb{C}[x,y,z]/\zeta^{k+1}\mathbb{C}[x,y,z]\cong\mathbb{C}[x,y]\zeta^k$$

is spanned by the monomials $x^{i_1}y^{i_2}\zeta^k$ with $(i_1,i_2,k)\in m\mathcal{P}$.

Then the following hold.

(1) $\dim W_m = \#(m\mathcal{P}\cap\mathbb{Z}^3)$, $\mathrm{vol}(\mathcal{P}) = 11/3$, *the barycenter of* $\mathcal{P}$ *is* $(87/88, 87/88, 8/11)$, *and*

$$\frac{1}{m\dim W_m}\sum_{u\in m\mathcal{P}\cap\mathbb{Z}^3} u = \Big(\frac{87}{88},\frac{87}{88},\frac{8}{11}\Big) + O\Big(\frac{1}{m}\Big).$$

(2) Let $m, d_1, d_2 \ge 0$ *be integers. If* $\max\{d_1,d_2\} > 3m$, *then* $W_m^{d_1,d_2} = 0$. *If* $d_1, d_2 \le 3m$, *let*

$$d_0 := \min\{d_1,d_2\},\quad k_\zeta := \max\Big\{0, \Big\lceil\frac{d_1-m}{2}\Big\rceil\Big\},\quad k_z := \max\Big\{0,\Big\lceil\frac{d_2-m}{2}\Big\rceil\Big\},\quad \nu := d_0-k_z-k_\zeta.$$

Then

$$W_m^{d_1,d_2} = x^{d_1-d_0}y^{d_2-d_0}z^{k_z}\zeta^{k_\zeta}\mathbb{C}[z,xy]_\nu,$$

where $\mathbb{C}[z,xy]_\nu$ *is the span of* $z^k(xy)^{\nu-k}$, $0\le k\le\nu$, *and* $\mathbb{C}[z,xy]_\nu = 0$ *if* $\nu<0$.

(3) For any real number $0<\theta<88/87$ *there exists an integer* m_θ *such that* $(E,\theta D_m)$ *is klt for any integer* $m\ge m_\theta$ *and any* $\mathbb{Q}$*-divisor*

$$D_m = \frac{1}{m\dim W_m}\sum_j\sum_i \mathrm{div}(s_{j,i}),$$

where for each j, $(s_{j,i})_i$ *is a basis of* $W_m\cap\mathbb{C}[x,y,z]_j$ *consisting of* λ*-eigenvectors.*

Proof. We prove (1). Since $\dim W_m < +\infty$ and $\bigcap_k \zeta^k\mathbb{C}[x,y,z] = 0$, we have $W_m\cap\zeta^k\mathbb{C}[x,y,z] = 0$ for $k\gg 0$. Thus $\dim W_m$ is the sum over k of the dimensions of the images in (iii), which is $\#(m\mathcal{P}\cap\mathbb{Z}^3)$. The polytope $\mathcal{P}$ has a unimodular triangulation into 22 lattice tetrahedra, see (B.4) and Table 102. Thus $\mathrm{vol}(\mathcal{P}) = 22/6 = 11/3$, and the barycenter of $\mathcal{P}$ is the average of the barycenters of the 22 tetrahedra. A direct calculation with the vertices in Table 102 shows that the sum over the 22 tetrahedra of the sums of their four vertices is $(87,87,64)$. Thus the barycenter of $\mathcal{P}$ is $(87/88,87/88,8/11)$. Finally, we compare $\sum_{u\in m\mathcal{P}\cap\mathbb{Z}^3} u$ and $\dim W_m$ with the integrals of u and 1 over $m\mathcal{P}$, unit cube by unit cube. The $O(m^2)$ unit cubes intersecting the boundary of $m\mathcal{P}$ contribute $O(m^3)$ to the first difference and $O(m^2)$ to the second. Since $\mathrm{vol}(m\mathcal{P}) = \frac{11}{3}m^3$, the last formula of (1) follows.

We prove (2). Any monomial $x^{i_1}y^{i_2}z^{i_3}$ in $\mathbb{C}[x,y,z]_{(d_1,d_2)}$ satisfies $i_3\le d_0$ and equals $x^{d_1-d_0}y^{d_2-d_0}(xy)^{d_0-i_3}z^{i_3}$. Thus

$$\mathbb{C}[x,y,z]_{(d_1,d_2)} = x^{d_1-d_0}y^{d_2-d_0}\mathbb{C}[z,xy]_{d_0}. \tag{13.71}$$

Since ζ belongs to $\mathbb{C}[x,y,z]_{(1,1)}$, (i) implies that the image in (iii) is the direct sum over (d_1,d_2) of the images of $W_m^{d_1,d_2}\cap\zeta^k\mathbb{C}[x,y,z]$. If $f\in W_m^{d_1,d_2}$ is non-zero with ζ-order k, then the image of f in $\mathbb{C}[x,y]\zeta^k$ is a non-zero multiple of $x^{d_1-k}y^{d_2-k}\zeta^k$. Hence, by (iii), the set of ζ-orders of non-zero elements of $W_m^{d_1,d_2}$ is

$$\mathcal{K} := \{k\in\mathbb{Z} : (d_1-k, d_2-k, k)\in m\mathcal{P}\},$$

and $\dim W_m^{d_1,d_2} = \#\mathcal{K}$. Substituting $(i_1,i_2,i_3) = (d_1-k,d_2-k,k)$ into the inequalities defining $m\mathcal{P}$, we see that $k\in\mathcal{K}$ if and only if $d_1,d_2\le 3m$ and

$$\max\Big\{0,\Big\lceil\frac{d_1-m}{2}\Big\rceil\Big\}\le k\le\min\Big\{d_1,d_2,\Big\lfloor\frac{d_2+m}{2}\Big\rfloor,\Big\lfloor\frac{m+2d_1-d_2}{2}\Big\rfloor\Big\}.$$

In particular $W_m^{d_1,d_2} = 0$ if $\max\{d_1,d_2\} > 3m$. Assume that $d_1,d_2\le 3m$. If $d_1\ge d_2$, then $\lfloor(m+2d_1-d_2)/2\rfloor \ge \lfloor(d_2+m)/2\rfloor$ and the minimum is $\min\{d_2,\lfloor(d_2+m)/2\rfloor\} = d_0-k_z$. If $d_1\le d_2$, then $\lfloor(m+2d_1-d_2)/2\rfloor\le\lfloor(d_2+m)/2\rfloor$ and the minimum is $\min\{d_1,\lfloor(m+2d_1-d_2)/2\rfloor\} = d_0-k_z$. Thus

$$\mathcal{K} = \{k\in\mathbb{Z} : k_\zeta\le k\le d_0-k_z\},\qquad \dim W_m^{d_1,d_2} = \max\{0,\nu+1\}.$$

In particular any non-zero $f\in W_m^{d_1,d_2}$ has ζ-order at least k_ζ. Since $\iota(\zeta) = -z$ and ι maps $\mathbb{C}[x,y,z]_{(d_1,d_2)}$ onto $\mathbb{C}[x,y,z]_{(d_2,d_1)}$, (ii) and the lower bound $\max\{0,\lceil(d_2-m)/2\rceil\}$ on the ζ-orders of non-zero elements of $W_m^{d_2,d_1}$ imply that any non-zero $f\in W_m^{d_1,d_2}$ has z-order at least k_z. By

(13.71), write $f = x^{d_1-d_0}y^{d_2-d_0}g$ with $g \in \mathbb{C}[z, xy]_{d_0}$. The binary form g is a product of linear forms in z and xy, and z and $\zeta = z - xy$ are non-proportional linear forms, each a prime element of $\mathbb{C}[x, y, z]$ not dividing x or y. Thus $g \in z^{k_z}\zeta^{k_\zeta}\mathbb{C}[z, xy]_\nu$, that is, $W_m^{d_1,d_2}$ is contained in the right-hand side of (2). Both sides have dimension $\max\{0, \nu + 1\}$. Thus they are equal.

We prove (3). Since $\lambda(t)$ acts on $\mathbb{C}[x, y, z]_{(d_1,d_2)}$ by t^{d_1} and

$$\mathbb{C}[x, y, z]_j = \bigoplus_{d_1+d_2=j} \mathbb{C}[x, y, z]_{(d_1,d_2)},$$

(i) implies that any λ-eigenvector in $W_m \cap \mathbb{C}[x, y, z]_j$ belongs to $W_m^{d_1,d_2}$ for some $d_1 + d_2 = j$. Thus $\bigcup_j (s_{j,i})_i$ is a union of bases of the non-zero spaces $W_m^{d_1,d_2}$. By (2), a basis of $W_m^{d_1,d_2}$ has the form $x^{d_1-d_0}y^{d_2-d_0}z^{k_z}\zeta^{k_\zeta}g_l$, $1 \le l \le \nu + 1$, where $(g_l)_l$ is a basis of $\mathbb{C}[z, xy]_\nu$. Each g_l is, up to a constant, a product of the forms xy and $z - \mu xy$ with $\mu \in \mathbb{C}$. Let D_x, D_y, D_z, D_ζ be the curves $(x = 0)$, $(y = 0)$, $(z = 0)$, $(\zeta = 0)$, and let $\Gamma_\mu := (z = \mu xy)$ for $\mu \in \mathbb{C}\backslash\{0, 1\}$. Then $\operatorname{div}(xy) = D_x + D_y$, $\operatorname{div}(z - \mu xy) = \Gamma_\mu$, and $\operatorname{Supp} D_m$ is contained in the union of D_x, D_y, D_z, D_ζ, and finitely many Γ_μ.

Let ℓ be one of the forms xy and $z - \mu xy$, $\mu \in \mathbb{C}$. The forms in $\mathbb{C}[z, xy]_\nu$ divisible by ℓ^k form a subspace of codimension k. Thus at most $\nu + 1 - k$ of the g_l have ℓ-order at least k, and

$$\sum_{l=1}^{\nu+1} \operatorname{ord}_\ell g_l \le \sum_{k=1}^{\nu} (\nu + 1 - k) = \frac{\nu(\nu + 1)}{2}. \tag{13.72}$$

For the block $W_m^{d_1,d_2}$ with $\nu \ge 0$, the elements of $\mathcal{K}$ are $k_\zeta, \dots, k_\zeta + \nu$. Thus the lattice points $u = (d_1 - k, d_2 - k, k)$, $k \in \mathcal{K}$, satisfy

$$\sum_{k\in\mathcal{K}} u_1 = (\nu + 1)\Big(d_1 - k_\zeta - \frac{\nu}{2}\Big), \quad \sum_{k\in\mathcal{K}} u_2 = (\nu + 1)\Big(d_2 - k_\zeta - \frac{\nu}{2}\Big), \quad \sum_{k\in\mathcal{K}} u_3 = (\nu + 1)\Big(k_\zeta + \frac{\nu}{2}\Big).$$

By (13.72), the sum over the basis of $W_m^{d_1,d_2}$ of the coefficients of D_x in the divisors is at most

$$(\nu + 1)\Big(d_1 - d_0 + \frac{\nu}{2}\Big) = \sum_{k\in\mathcal{K}} u_1 - (\nu + 1)k_z \le \sum_{k\in\mathcal{K}} u_1,$$

where we use $d_0 - k_\zeta - \nu = k_z$. The corresponding sums for D_y, D_ζ, and Γ_μ are at most

$$(\nu + 1)\Big(d_2 - d_0 + \frac{\nu}{2}\Big) = \sum_{k\in\mathcal{K}} u_2 - (\nu + 1)k_z,$$

$$(\nu + 1)k_\zeta + \frac{\nu(\nu + 1)}{2} = \sum_{k\in\mathcal{K}} u_3,$$

$$\frac{\nu(\nu + 1)}{2} \le \sum_{k\in\mathcal{K}} u_3,$$

respectively. The corresponding sum for D_z is at most $(\nu + 1)(k_z + \frac{\nu}{2})$. Exchanging d_1 and d_2 exchanges k_ζ and k_z and preserves ν, so the sum of $(\nu + 1)(k_z + \frac{\nu}{2})$ over all (d_1, d_2) equals the sum of $(\nu + 1)(k_\zeta + \frac{\nu}{2})$ over all (d_1, d_2). Since $m\mathcal{P} \cap \mathbb{Z}^3$ is the disjoint union over (d_1, d_2) of the sets $\{(d_1 - k, d_2 - k, k) : k \in \mathcal{K}\}$, (1) implies that there exists a sequence $\epsilon_m \to 0$, independent of the bases, such that

$$\operatorname{mult}_{D_x} D_m,\ \operatorname{mult}_{D_y} D_m \le \frac{87}{88} + \epsilon_m, \qquad \operatorname{mult}_{D_z} D_m,\ \operatorname{mult}_{D_\zeta} D_m,\ \operatorname{mult}_{\Gamma_\mu} D_m \le \frac{8}{11} + \epsilon_m$$

for any μ. On the chart $x = 1$, the point $[1 : 0 : 0]$ is smooth with local coordinates y, z, and any element of $W_m^{d_1,d_2}$ is a non-zero homogeneous polynomial of degree $d_2 = u_2 + u_3$ in y, z. On the

chart $y = 1$, the point $[0 : 1 : 0]$ is smooth with local coordinates x, z, and any element of $W_m^{d_1,d_2}$ is homogeneous of degree $d_1 = u_1 + u_3$ in x, z. Thus, by (1),

$$\operatorname{mult}_{[1:0:0]} D_m,\ \operatorname{mult}_{[0:1:0]} D_m \le \frac{87}{88} + \frac{8}{11} + \epsilon_m = \frac{151}{88} + \epsilon_m,$$

after possibly enlarging ϵ_m.

Let m_θ be an integer such that $\theta(87/88 + \epsilon_m) < 1$ and $\theta(151/88 + \epsilon_m) < 2$ for any $m \ge m_\theta$. It exists, since $\theta \cdot 87/88 < 1$ and $\theta \cdot 151/88 < 151/87 < 2$. Let $m \ge m_\theta$. Any two of the curves $(z = \mu xy)$, $\mu \in \mathbb{C}$, intersect only at $[1 : 0 : 0]$ and $[0 : 1 : 0]$, D_x intersects them only at $[0 : 1 : 0]$, D_y intersects them only at $[1 : 0 : 0]$, and $D_x \cap D_y = \{[0 : 0 : 1]\}$. The curves D_x, D_y, and $(z = \mu xy)$, $\mu \in \mathbb{C}$, are smooth. Thus, at any point $q \notin \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\}$, at most one component of $\operatorname{Supp} D_m$ passes through q, and $(E, \theta D_m)$ is klt near q by [KM98, Corollary 2.31(3)].

Let $h : E' \to E$ be the blow-up of $[1 : 0 : 0]$ with exceptional curve F, and let D'_m be the strict transform of D_m. Then

$$h^*(K_E + \theta D_m) = K_{E'} + \theta D'_m + \big(\theta \operatorname{mult}_{[1:0:0]} D_m - 1\big)F, \qquad \theta \operatorname{mult}_{[1:0:0]} D_m - 1 < 1.$$

In the coordinates y, z, the components of $\operatorname{Supp} D_m$ through $[1 : 0 : 0]$ are among the lines $y = 0$ and $z = \mu y$, $\mu \in \mathbb{C}$, which have pairwise distinct tangent directions. Thus the strict transforms of the components are pairwise disjoint near F and intersect F transversally. All coefficients of $\theta D'_m$ are less than 1, so $(E, \theta D_m)$ is klt near $[1 : 0 : 0]$ by [KM98, Lemma 2.30 and Corollary 2.31(3)]. Exchanging the roles of x and y, we see that $(E, \theta D_m)$ is klt near $[0 : 1 : 0]$. Near $[0 : 0 : 1]$, the surface E is the quotient of $\mathbb{C}^2$ with coordinates x, y by μ_2 acting by $(x, y) \to (-x, -y)$, and the quotient morphism is étale in codimension one. Only D_x and D_y pass through $[0 : 0 : 1]$, and the pullback of θD_m is supported on the coordinate axes with coefficients less than 1. Thus $(E, \theta D_m)$ is klt near $[0 : 0 : 1]$ by [KM98, Corollary 2.31(3) and Proposition 5.20(4)]. □

The following theorem gives $\delta_p(X) \ge 88/87$ for every point p as in Set-up 13.12 with $r = 2$. In the configuration left open by Proposition 13.15(4), its proof uses the algebra $\mathcal{R}$ of Construction-Proposition 13.19(1).

Theorem 13.21. *Notation as in Set-up 13.12 with $r = 2$. Then $\delta_p(X) \ge 88/87$. If moreover $n_{12} = n_{21} = 3$, $p_{12} \in s_1$, and $p_{21} \in s_2$, then there exist regular parameters x, y, z at p for which the weighted blow-up $\pi : Y \to X$ with weights $(1, 1, 2)$ is a plt extraction with exceptional divisor $E \cong \mathbb{P}(1, 1, 2)$, $\operatorname{Diff}_E(0) = 0$, $A_X(E) = 4$, and $S_X(E) = 151/44$. For the refinement W of $(H^0(X, mH))_{m\ge0}$ by E, any prime divisor v over E satisfies $S(W; v) \le \frac{87}{88} A_E(v)$, and any prime divisor F over X with $p \in \operatorname{center}_X F$ satisfies*

$$A_X(F) - S_X(F) \ge \frac{1}{88} A_X(F) + \frac{23}{44} \operatorname{mult}_F E.$$

Proof of Theorem 13.21. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the theorem to the case $n_{12} = n_{21} = 3$, $p_{12} \in s_1$, and $p_{21} \in s_2$. If the assumption of Proposition 13.15(2) holds, then $\delta_p(X) \ge 44/43 > 88/87$. If the assumption of Proposition 13.15(3) holds, then $\delta_p(X) \ge 88/83 > 88/87$. Otherwise Proposition 13.15(4) implies that $n_{12} = n_{21} = 3$, $p_{12} \in s_1$, and $p_{21} \in s_2$. In this case the last inequality of the theorem and $\operatorname{mult}_F E \ge 0$ imply that $\frac{87}{88} A_X(F) \ge S_X(F)$ for any prime divisor F over X with $p \in \operatorname{center}_X F$. Thus $\delta_p(X) \ge 88/87$. From now on we assume that $n_{12} = n_{21} = 3$, $p_{12} \in s_1$, and $p_{21} \in s_2$.

Step 2. In this step we construct the extraction. Let x, y, z be regular parameters of X at p, together with a local trivialization of H at p, satisfying (1) and (2) of Construction-Proposition 13.19, and let $\zeta := z - xy$. Let $\pi : Y \to X$ be the weighted blow-up with weights $(1, 1, 2)$ on (x, y, z), and let E be the exceptional divisor. Then $Y = \operatorname{Proj}_X \bigoplus_{j\ge0} \mathfrak{a}_j$, where $\mathfrak{a}_j$ is the ideal of functions of weight at least j, and $E \cong \mathbb{P}(1, 1, 2)$ with weighted homogeneous coordinates x, y, z. Moreover, $-E$ is π-ample and $\operatorname{ord}_E = v_{\mathrm{w}}$.

The variety Y is covered over a neighborhood of p by three charts. The chart $x=\rho$, $y=\rho y'$, $z=\rho^2 z'$ is smooth, and so is the chart obtained by exchanging x and y. The third chart is the quotient of $\mathbb{C}^3$ with coordinates x', y', ρ by μ_2 acting by $(x', y', \rho) \to (-x', -y', -\rho)$, where $x=\rho x'$, $y=\rho y'$, $z=\rho^2$, and E is the image of $(\rho=0)$. The quotient morphism $\varphi : \mathbb{C}^3 \to Y$ onto the chart is étale in codimension one, since μ_2 acts freely outside the origin. A direct calculation in the chart $x=\rho x'$, $y=\rho y'$, $z=\rho^2$, where

$$dx \wedge dy \wedge dz = 2\rho^3 dx' \wedge dy' \wedge d\rho,$$

and in the other charts, where $dx \wedge dy \wedge dz = \rho^3 d\rho \wedge dy' \wedge dz'$ up to exchanging x and y, shows that

$$K_Y + E = \pi^* K_X + 4E.$$

Thus $A_X(E)=4$. The pair $(\mathbb{C}^3, (\rho=0))$ is plt, and $K_{\mathbb{C}^3} + (\rho=0) = \varphi^*(K_Y+E)$. Hence (Y,E) is plt near the singular point of Y by [KM98, Proposition 5.20(2)] and [KM98, Definition 2.34]. Outside the singular point, Y is smooth and E is smooth, so (Y,E) is plt. Thus π is a plt extraction with exceptional divisor E. Since the singular locus of Y near E is a single point, Y is smooth at the generic point of every curve in E, and hence $\operatorname{Diff}_E(0)=0$.

Step 3. In this step we identify the refinement W and compute $S_X(E)$. Consider the algebra $\mathcal{R}$ of Construction-Proposition 13.19(1) and the spaces $\mathcal{R}_m$ of Construction-Proposition 13.19(2). The restriction $-E|_E$ in the sense of [AZ22, Lemma 2.1] is $\mathcal{O}_E(1)$, and $\pi^*H|_E$ is trivial. Since $\operatorname{ord}_E = v_{\mathrm{w}}$, the space $W_{m,j} \subset H^0(E, \mathcal{O}_E(j)) = \mathbb{C}[x,y,z]_j$ is the degree j part of the associated graded space of $H^0(X, mH)$ for v_{w}, which is the weighted homogeneous part of degree j of $\mathcal{R}_m$ by Construction-Proposition 13.19(1).

We check that $W_m := \mathcal{R}_m$ satisfies (i), (ii), and (iii) of Lemma 13.20. With $\deg x = (1,0)$, $\deg y = (0,1)$, and $\deg z = \deg \zeta = (1,1)$, each of the fourteen generators of $\mathcal{R}$ is a monomial in T, x, y, z, ζ, hence bihomogeneous. Thus (i) holds. Extend ι by $\iota(T)=T$. Then $\iota(\zeta) = -z$, $\iota(Tz) = Txy - Tz$, and ι maps the other thirteen generators to generators up to sign:

$$Tx \leftrightarrow Ty, \quad Tx\zeta \leftrightarrow -Tyz, \quad Tx^2\zeta \leftrightarrow -Ty^2 z, \quad Txz\zeta \leftrightarrow Tyz\zeta, \quad Tz\zeta^2 \leftrightarrow -Tz^2\zeta,$$

while T, Txy, and $Tz\zeta$ are fixed. Thus $\iota(\mathcal{R}) = \mathcal{R}$, and (ii) holds. Condition (iii) is Construction-Proposition 13.19(2), and $\dim \mathcal{R}_m = N_m$ by Construction-Proposition 13.19(3). Thus $\dim W_m = N_m$, and $W_m \cap \mathbb{C}[x,y,z]_j = W_{m,j}$ for any j.

By (iii), the ζ-adic filtration of $W_{m,j}$ has graded pieces spanned by the monomials $x^{i_1} y^{i_2} \zeta^k$ with $(i_1, i_2, k) \in m\mathcal{P}$ and $i_1 + i_2 + 2k = j$. Thus

$$\sum_j j \dim W_{m,j} = \sum_{u \in m\mathcal{P} \cap \mathbb{Z}^3} (u_1 + u_2 + 2u_3).$$

By [AZ22, Lemma 2.9] and Lemma 13.20(1),

$$S_X(E) = \lim_{m\to+\infty} \frac{1}{mN_m} \sum_j j \dim W_{m,j} = \frac{87}{88} + \frac{87}{88} + 2 \cdot \frac{8}{11} = \frac{151}{44}.$$

Step 4. In this step we prove that $S(W;v) \le \frac{87}{88} A_E(v)$ for any prime divisor v over E. Let λ be the one-parameter subgroup of Lemma 13.20. The sheaves $\mathcal{O}_E(j)$ carry the λ-linearizations given by the action of λ on x, y, z. By (i), each $W_{m,j}$ is λ-invariant, so $\operatorname{in}_\lambda W_{m,j} = W_{m,j}$. Thus the eigen-basis divisors at level m of Lemma 3.6, for $\lambda_m := \lambda$ and $\Delta_E := \operatorname{Diff}_E(0) = 0$, are the $\mathbb{Q}$-divisors D_m of Lemma 13.20(3).

Choose real numbers $0 < \theta_1 < \theta_2 < \cdots$ with $\lim_k \theta_k = 88/87$. By Lemma 13.20(3), there exist integers $m_1 < m_2 < \cdots$ such that $(E, \theta_k D_m)$ is klt for any $m \ge m_k$ and any $\mathbb{Q}$-divisor D_m associated with a basis consisting of λ-eigenvectors at level m. Let $c_m := \theta_k$ for $m_k \le m < m_{k+1}$, and $c_m := \theta_1$ for $m < m_1$. Then $\lim_m c_m = 88/87$, and $(E, c_m D_m)$ is klt for any $m \ge m_1$ and any such D_m. By Lemma 3.6(1), $\frac{88}{87} S(W;v) \le A_E(v)$ for any prime divisor v over E.

Step 5. In this step we prove the last inequality of the theorem. Let F be a prime divisor over X with $p \in \operatorname{center}_X F$. Since $\pi(\operatorname{center}_Y F) = \operatorname{center}_X F$ and $\pi^{-1}(p) = E$, we have $\operatorname{center}_Y F \cap E \neq \emptyset$. By **Step 4**, the assumption of Lemma 3.1 holds for $(X, 0)$, the complete series of H, the plt extraction π, $G := E$, and the real number $88/87$ in place of the constant λ of the lemma. Thus, by Lemma 3.1, $A_X(E) = 4$ (**Step 2**), and $S_X(E) = 151/44$ (**Step 3**), we have

$$A_X(F) \geq \frac{88}{87} S_X(F) + \left(4 - \frac{88}{87} \cdot \frac{151}{44}\right) \operatorname{mult}_F E = \frac{88}{87} S_X(F) + \frac{46}{87} \operatorname{mult}_F E.$$

Multiplying by $87/88$, we obtain $\frac{87}{88} A_X(F) \geq S_X(F) + \frac{23}{44} \operatorname{mult}_F E$, that is,

$$A_X(F) - S_X(F) \geq \frac{1}{88} A_X(F) + \frac{23}{44} \operatorname{mult}_F E. \quad \square$$

13.4. **Points on three lines.** In this subsection, we treat the points $p \in X$ such that $\Sigma(X, p)$ consists of three reduced points, and we prove Theorem 13.24. The widths of the strict transforms of the lines intersecting a non-special line sum to three, and the switching identity of Lemma 13.13 relates the lengths n_{ij} and n_{ji}. Thus only two cases remain: either there exists an index i with $n_{ij} = n_{ik} = 1$, where we use Proposition 13.11, or $n_{12} = n_{23} = n_{31} = 2$ after relabelling.

Proposition 13.22. *Notation as in Set-up 13.12 with $r = 3$. Then exactly one of the following holds.*

(1) There exists an index i with $n_{ij} = n_{ik} = 1$, where $\{j, k\} = \{1, 2, 3\} \backslash \{i\}$.
(2) After relabelling, $n_{12} = n_{23} = n_{31} = 2$ and $n_{21} = n_{32} = n_{13} = 1$.

Moreover, if i is an index as in (1), then Z_{U_i} consists of p_{ij}, p_{ik}, and a point outside $s_i \cup f_{i,p}$, and any prime divisor F over X with $p \in \operatorname{center}_X F$ satisfies $A_X(F) \geq S_X(F)$, with strict inequality if $\operatorname{center}_X F = \{p\}$.

Proof. **Step 1.** In this step, we show that Z_{U_i} is a curvilinear scheme of length three with $Z_{U_i} \cap f_{i,p} = \{p_{ij}, p_{ik}\}$, and that $(n_{ij}, n_{ik}) \in \{(1, 1), (1, 2), (2, 1)\}$.

Fix $i \in \{1, 2, 3\}$ and let $\{j, k\} = \{1, 2, 3\} \backslash \{i\}$. Since E_i is the union of the fibers $f_{i,x}$ with $x \in L_i$, Definition-Proposition 10.9 implies that the points of Z_{U_i} correspond to the pairs (L', y), where $L' \neq L_i$ is a line and $y \in L' \cap L_i$, that the local ring of Z_{U_i} at the point of (L', y) is $\mathbb{C}[t]/(t^n)$, where n is the width of the strict transform of L' in Y_i, and that the widths of the strict transforms of all lines $L' \neq L_i$ intersecting L_i sum to three. Thus Z_{U_i} is a curvilinear scheme of length three (see also Construction-Proposition 11.3(1)). By Construction-Proposition 11.3(2), the base points of U_i are resolved by three blow-ups $p_{E_i} : T_{E_i} \to E_i$ with $I_{Z_{U_i}} \cdot \mathcal{O}_{T_{E_i}} = \mathcal{O}_{T_{E_i}}(-e_1 - e_2 - e_3)$. Hence the three base points of U_i are the points of Z_{U_i}, proper or infinitely near. By Definition-Proposition 10.9 again,

$$Z_{U_i} \cap f_{i,p} \cong \Sigma(X, p) \backslash \{[L_i]\} = \{[L_j], [L_k]\},$$

and the right-hand side is reduced. Hence $Z_{U_i} \cap f_{i,p} = \{p_{ij}, p_{ik}\}$ with $p_{ij} \neq p_{ik}$. Since $n_{ij} \geq 1$, $n_{ik} \geq 1$, and the length of Z_{U_i} is three, we have

$$n_{ij} + n_{ik} \leq 3, \qquad \text{hence} \qquad (n_{ij}, n_{ik}) \in \{(1, 1), (1, 2), (2, 1)\}. \tag{13.73}$$

Step 2. In this step, we prove the last sentence of the proposition. Let i be an index with $n_{ij} = n_{ik} = 1$.

By **Step 1**, Z_{U_i} is reduced at p_{ij} and at p_{ik}, and Z_{U_i} has exactly one further point z_i, which is a proper point of E_i with $z_i \neq p_{ij}, p_{ik}$. Since $Z_{U_i} \cap f_{i,p} = \{p_{ij}, p_{ik}\}$, we have $z_i \notin f_{i,p}$. Thus the base points of U_i are the three proper points p_{ij}, p_{ik}, and z_i.

We show that $z_i \notin s_i$. Suppose that $z_i \in s_i$. If $p_{ij} \notin s_i$ and $p_{ik} \notin s_i$, then the base points of U_i consist of the subscheme $\{p_{ij}, p_{ik}\}$ of length two of $f_{i,p}$, which is disjoint from s_i, together with the point $z_i \in s_i \backslash f_{i,p}$. This is the second configuration excluded in Lemma 13.5(3), a contradiction. Otherwise $s_i \cap f_{i,p} \in \{p_{ij}, p_{ik}\}$, and s_i contains two distinct proper base points $s_i \cap f_{i,p}$ and z_i of U_i. Let $R_{i,0}$ be the divisor $R_0 = R|_E$ of Construction-Proposition 11.3 for L_i, where $R \in |\pi_i^* H - 3E_i|$.

Since $(H \cdot L_i) = 1$, we have $\pi_i^* H|_{E_i} \equiv f_i$. Since $U_i \subset H^0(E_i, \mathcal{O}_{E_i}(2s_i + 3f_i))$ is the restriction of $H^0(Y_i, \pi_i^* H - 2E_i)$, we have $(\pi_i^* H - 2E_i)|_{E_i} \sim 2s_i + 3f_i$. Hence $-E_i|_{E_i} \equiv s_i + f_i$ and

$$R_{i,0} \equiv (\pi_i^* H - 3E_i)|_{E_i} \equiv 3s_i + 4f_i, \qquad (R_{i,0} \cdot s_i) = 3(s_i^2) + 4(f_i \cdot s_i) = -3 + 4 = 1.$$

By Lemma 13.5(1), $p_{E_i}^* R_{i,0} - 2(e_1 + e_2 + e_3) \geq 0$, so $\operatorname{mult}_q R_{i,0} \geq 2$ for any proper base point q of U_i. If $s_i \not\subset \operatorname{Supp} R_{i,0}$, then

$$1 = (R_{i,0} \cdot s_i) \geq \operatorname{mult}_{s_i \cap f_{i,p}} R_{i,0} + \operatorname{mult}_{z_i} R_{i,0} \geq 4,$$

a contradiction. Hence $s_i \subset \operatorname{Supp} R_{i,0}$, and $k_{s_i} = 1$ by Lemma 13.5(2), where k_{s_i} is the number of base points of U_i on the strict transforms of s_i (Set-up 13.4). However, s_i contains the proper base points $s_i \cap f_{i,p}$ and z_i of U_i, so $k_{s_i} \geq 2$, a contradiction. Therefore $z_i \notin s_i$.

Thus Z_{U_i} consists of p_{ij}, p_{ik}, and the point $z_i \notin s_i \cup f_{i,p}$. We use Proposition 13.11 with $x := p$, $Z_f := \{p_{ij}, p_{ik}\}$, and the point z_i in place of r. Let $c \in \{1, 88/83\}$ be the constant of Proposition 13.11. Then

$$2 - \frac{17c}{11} \geq 2 - \frac{17}{11} \cdot \frac{88}{83} = \frac{30}{83} > 0.$$

Since $c \geq 1$, $S_X(F) \geq 0$, and $\operatorname{mult}_F E_i \geq 0$, Proposition 13.11 implies that

$$A_X(F) \geq cS_X(F) + \left(2 - \frac{17c}{11}\right) \operatorname{mult}_F E_i \geq S_X(F)$$

for any prime divisor F over X with $p \in \operatorname{center}_X F$, and that $A_X(F) > S_X(F)$ if $\operatorname{center}_X F = \{p\}$.

Step 3. In this step, we show that exactly one of (1) and (2) holds.

In case (2), any i has an index j with $n_{ij} = 2$, so (1) and (2) do not hold simultaneously. Suppose that (1) does not hold. By (13.73), for any i there exists a unique $\sigma(i) \in \{1, 2, 3\} \backslash \{i\}$ with $n_{i\sigma(i)} = 2$, and $n_{ik} = 1$ for the remaining index k.

We show that $\sigma(\sigma(i)) \neq i$ for any i. Suppose that $j := \sigma(i)$ satisfies $\sigma(j) = i$, and let k be the remaining index. Since $n_{ji} = 2 > 1$, Lemma 13.13 implies that $p_{ij} \in s_i$, that is, $p_{ij} = s_i \cap f_{i,p}$. By **Step 1**, Z_{U_i} has length two at p_{ij} and length one at p_{ik}. Since the local ring of Z_{U_i} at p_{ij} is $\mathbb{C}[t]/(t^2)$, the base points of U_i are $s_i \cap f_{i,p}$, the point $p_{ik} \in f_{i,p} \backslash s_i$, and a point infinitely near $s_i \cap f_{i,p}$. This is the third configuration excluded in Lemma 13.5(3), a contradiction.

Thus σ is a map from $\{1, 2, 3\}$ to itself without fixed points such that $\sigma \circ \sigma$ has no fixed points. After relabelling, $\sigma(1) = 2$. Since $\sigma(2) \neq 2$ and $\sigma(2) \neq 1$, we have $\sigma(2) = 3$. Since $\sigma(3) \neq 3$ and $\sigma(3) \neq 2$, we have $\sigma(3) = 1$. Hence $n_{12} = n_{23} = n_{31} = 2$ and $n_{13} = n_{21} = n_{32} = 1$, which is (2). □

Remark 13.23. Let $V = V_1 \oplus V_2$ with bases (v_1, v_2, v_3) of V_1 and $(v_4, \dots, v_7)$ of V_2, and let $\Pi_{\mathrm{Kl}} \subset \Lambda^2 V^\vee$ be the net of alternating forms whose block matrices are

$$\begin{pmatrix} 0 & C_{\mathrm{Kl}} \\ -C_{\mathrm{Kl}}^t & D_{\mathrm{Kl}} \end{pmatrix},$$

where $(a, b, c) \in \mathbb{C}^3$ are linear coordinates on Π_{Kl} and

$$C_{\mathrm{Kl}} = \begin{pmatrix} a & 0 & 0 & c \\ 0 & b & 0 & a \\ 0 & 0 & c & b \end{pmatrix}, \qquad D_{\mathrm{Kl}} = \begin{pmatrix} 0 & -c & b & 0 \\ c & 0 & -a & 0 \\ -b & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then $X_{\Pi_{\mathrm{Kl}}}$ is a smooth prime Fano threefold of genus 12, and $\Sigma(X_{\Pi_{\mathrm{Kl}}}, [V_1])$ consists of three reduced points. With $X := X_{\Pi_{\mathrm{Kl}}}$ and $p := [V_1]$ in Set-up 13.12, the integers n_{ij} satisfy (2) of Proposition 13.22.

Proof of Remark 13.23. Let $(w_1, \dots, w_7)$ be the basis of $\mathbb{C}^7$ and y_0, y_1, y_2 the coordinates on the net in [MS01, the net η_{klein} following Theorem 4.3]. The substitution

$$(v_1, \dots, v_7) = (w_1, w_2, w_3, -w_6, -w_5, -w_4, w_7), \qquad (a, b, c) = (y_1, y_2, y_0)$$

transforms the net of [MS01, the net η_{klein} following Theorem 4.3] into Π_{Kl}. For example, the diagonal of the first three columns of C_{Kl} is (y_1, y_2, y_0), the last column of C_{Kl} is (y_0, y_1, y_2), and the entries of D_{Kl} above the diagonal in the first three rows are $-y_0$, y_2, and $-y_1$. Thus $X := X_{\Pi_{\text{Kl}}}$ is a smooth prime Fano threefold of genus 12 by [MS01, Theorems 4.3 and 4.4 and the net η_{klein} following Theorem 4.3]. Let $p := [V_1]$. Since the upper left block of any form in Π_{Kl} is zero, $p \in X$.

We work in the affine chart of $\mathrm{Gr}(3, V)$ around p. For $\Phi \in \mathrm{Hom}(V_1, V_2) = V_1^\vee \otimes V_2$, let $\Phi = (\phi_{ij})$ also denote the 3×4 matrix such that ϕ_{ij} is the coefficient of v_{3+j} in $\Phi(v_i)$, and let $V_\Phi := \{v + \Phi(v) : v \in V_1\}$. Then $\Phi \to [V_\Phi]$ is an open immersion of $\mathrm{Hom}(V_1, V_2)$ into $\mathrm{Gr}(3, V)$ sending 0 to p [Har92, Example 16.1]. Let $(v_1^\vee, v_2^\vee, v_3^\vee)$ be the dual basis of $V_1^\vee$, so that the matrix unit in position (i, j) is $v_i^\vee \otimes v_{3+j}$. Since V_Φ is the row space of $(I_3|\Phi)$, the subspace V_Φ is isotropic for the form with coordinates (a, b, c) if and only if

$$(I_3|\Phi)\begin{pmatrix} 0 & C_{\text{Kl}} \\ -C_{\text{Kl}}^t & D_{\text{Kl}} \end{pmatrix}\begin{pmatrix} I_3 \\ \Phi^t \end{pmatrix} = \Psi(\Phi) + \Phi D_{\text{Kl}} \Phi^t = 0, \qquad \Psi(\Phi) := C_{\text{Kl}}\Phi^t - \Phi C_{\text{Kl}}^t. \tag{13.74}$$

The left-hand side of (13.74) is an alternating 3×3 matrix, and the entries of $\Psi(\Phi)$ above the diagonal are

$$\begin{aligned} \Psi(\Phi)_{12} &= a(\phi_{21} - \phi_{14}) - b\phi_{12} + c\phi_{24}, \\ \Psi(\Phi)_{13} &= a\phi_{31} - b\phi_{14} + c(\phi_{34} - \phi_{13}), \\ \Psi(\Phi)_{23} &= a\phi_{34} + b(\phi_{32} - \phi_{24}) - c\phi_{23}. \end{aligned}$$

The equations of X in the chart are the coefficients of a, b, and c in the entries above the diagonal of (13.74). The linear parts of the nine equations vanish if and only if $\phi_{ij} = 0$ for any $(i, j) \notin \{(1,1), (2,2), (3,3)\}$. Hence

$$T_p X = \mathbb{C}v_1^\vee \otimes v_4 \oplus \mathbb{C}v_2^\vee \otimes v_5 \oplus \mathbb{C}v_3^\vee \otimes v_6 \subset \mathrm{Hom}(V_1, V_2).$$

By [Har92, Exercise 6.9(i)], the lines of $\mathrm{Gr}(3, V)$ in the Plücker embedding are the sets of three-dimensional subspaces containing a fixed two-dimensional subspace and contained in a fixed four-dimensional subspace. By Lemma 2.19, the scheme of lines of $\mathrm{Gr}(3, V)$ through p is $\mathbb{P}(V_1^\vee) \times \mathbb{P}(V_2)$: to $([u], [w])$ corresponds the line $\ell_{u\otimes w}$ of subspaces containing $\operatorname{Ker} u$ and contained in $V_1 + \mathbb{C}w$, and $\ell_{u\otimes w}$ is the closure of $\{[V_{tu\otimes w}] \mid t \in \mathbb{C}\}$. In matrix form $u \otimes w = uw^t$ with column vectors u and w, and $w^t D_{\text{Kl}} w = 0$ since D_{Kl} is alternating. Hence

$$tu \otimes w D_{\text{Kl}} (tu \otimes w)^t = t^2 (w^t D_{\text{Kl}} w) u u^t = 0,$$

over any $\mathbb{C}$-algebra, and $V_{tu\otimes w}$ is isotropic for Π_{Kl} if and only if $t\Psi(u \otimes w) = 0$. Since X is closed, $\ell_{u\otimes w} \subset X$ if and only if $u \otimes w \in T_p X$. Let $\ell_i := \ell_{v_i^\vee \otimes v_{3+i}}$ for $i = 1, 2, 3$. Since $\mathrm{Pic}(X) = \mathbb{Z}H$, the Plücker class restricts to dH for an integer $d \geq 1$, and $1 = d(H \cdot \ell_1)$ implies $d = 1$. Thus the lines of X are the lines of $\mathrm{Gr}(3, V)$ contained in X, and $\Sigma(X, p)$ is the intersection of the Segre embedding of $\mathbb{P}(V_1^\vee) \times \mathbb{P}(V_2)$ in $\mathbb{P}(\mathrm{Hom}(V_1, V_2))$ with $\mathbb{P}(T_p X)$. In the coordinates $[x_1 : x_2 : x_3]$ of $\mathbb{P}(T_p X)$ given by the point $x_1 v_1^\vee \otimes v_4 + x_2 v_2^\vee \otimes v_5 + x_3 v_3^\vee \otimes v_6$, the 2×2 minors of the matrix Φ generate the ideal

$$(x_1x_2, x_1x_3, x_2x_3),$$

which defines the three reduced points $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$. Thus $\Sigma(X, p) = \{[\ell_1], [\ell_2], [\ell_3]\}$ consists of three reduced points, and we use Set-up 13.12 with $r = 3$ and $L_i := \ell_i$. The tangent direction of L_i at p is $v_i^\vee \otimes v_{3+i}$.

We construct a first-order deformation of each L_i inside X. Over $\mathbb{C}[\epsilon]/(\epsilon^2)$, let

$$\begin{aligned} \Phi_1(t) &= \epsilon v_3^\vee \otimes v_6 + t(v_1^\vee + \epsilon v_2^\vee) \otimes (v_4 + \epsilon v_7), \\ \Phi_2(t) &= \epsilon v_1^\vee \otimes v_4 + t(v_2^\vee + \epsilon v_3^\vee) \otimes (v_5 + \epsilon v_7), \\ \Phi_3(t) &= \epsilon v_2^\vee \otimes v_5 + t(v_3^\vee + \epsilon v_1^\vee) \otimes (v_6 + \epsilon v_7). \end{aligned}$$

Write $\Phi_i(t) = \Phi_i(0) + tu_i \otimes w_i$. The vectors u_i and w_i are non-zero modulo ϵ, so $V_{\Phi_i(t)}$ contains the free submodule $\{v + \Phi_i(0)(v) \mid v \in \operatorname{Ker} u_i\}$ of rank two and is contained in the free submodule

$V_{\Phi_i(0)} + (\mathbb{C}[\epsilon]/(\epsilon^2))w_i$ of rank four, neither depending on t. Hence the closure of $\{[V_{\Phi_i(t)}]\}_t$ is a family $\mathcal{L}_i \cong \mathbb{P}^1 \times \operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^2)$ of lines of $\operatorname{Gr}(3, V)$ whose reduction modulo ϵ is L_i. We substitute $\Phi_i(t)$ into (13.74) modulo ϵ^2. For $\Phi_1(t)$ the only non-zero entries above the diagonal are

$$\Psi(\Phi_1(t))_{13} = -\epsilon bt, \qquad (\Phi_1(t) D_{\mathrm{Kl}} \Phi_1(t)^t)_{13} = \epsilon bt.$$

For $\Phi_2(t)$ they are

$$\Psi(\Phi_2(t))_{12} = \epsilon ct, \qquad (\Phi_2(t) D_{\mathrm{Kl}} \Phi_2(t)^t)_{12} = -\epsilon ct.$$

For $\Phi_3(t)$ they are

$$\Psi(\Phi_3(t))_{23} = \epsilon at, \qquad (\Phi_3(t) D_{\mathrm{Kl}} \Phi_3(t)^t)_{23} = -\epsilon at.$$

Thus (13.74) holds for $\Phi_i(t)$ for any t and any (a, b, c). Since the complement of the section at infinity is schematically dense in $\mathcal{L}_i$, we have $\mathcal{L}_i \subset X \times \operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^2)$. Let $\theta_i \in T_{[L_i]}\Sigma(X)$ be the tangent vector defined by $\mathcal{L}_i$.

By [KPS18, the paragraph following Lemma 2.1.4, Corollary 2.1.6(ii), and the map (2.2.2)], $T_{[L_i]}\Sigma(X) = H^0(L_i, N_{L_i/X})$, and the value $\theta_i(p) \in N_{L_i/X} \otimes k(p) = T_pX/T_pL_i$ is the class of the tangent vector at p of the section $t = 0$ of $\mathcal{L}_i$. That tangent vector is $v_3^\vee \otimes v_6$, $v_1^\vee \otimes v_4$, and $v_2^\vee \otimes v_5$ for $i = 1, 2, 3$, respectively. Hence $\theta_1(p)$, $\theta_2(p)$, and $\theta_3(p)$ are the classes of the tangent directions of L_3, L_1, and L_2, respectively, and they are non-zero since T_pX is the direct sum of the tangent lines of L_1, L_2, and L_3. In particular $\theta_i \neq 0$. Since $N_{L_i/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ (Lemma 2.12(3)) and $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) = 0$, the section θ_i spans $H^0(L_i, N_{L_i/X})$ and $\theta_i(p)$ spans the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$. By Lemma 13.13, $s_i \cap f_{i,p}$ is the point $[\theta_i(p)]$ of $f_{i,p} = \mathbb{P}(N_{L_i/X} \otimes k(p))$. By Definition-Proposition 10.9, p_{ij} is the normal direction of L_j at p relative to L_i. Therefore

$$p_{13} = s_1 \cap f_{1,p}, \qquad p_{21} = s_2 \cap f_{2,p}, \qquad p_{32} = s_3 \cap f_{3,p},$$

and Lemma 13.13 implies that $n_{31} > 1$, $n_{12} > 1$, and $n_{23} > 1$. Since n_{ij} is the width of the strict transform of L_j in Y_i and the widths of the strict transforms of the lines intersecting L_i sum to three (Definition-Proposition 10.9), we have $n_{ij} + n_{ik} \leq 3$ for $\{i, j, k\} = \{1, 2, 3\}$. Hence $n_{12} = n_{23} = n_{31} = 2$ and $n_{21} = n_{32} = n_{13} = 1$, which is (2) of Proposition 13.22. □

By Remark 13.23, case (2) of Proposition 13.22 occurs, so the estimate of Proposition 13.11 does not cover all points on three reduced lines.

We now treat case (2) of Proposition 13.22. In that case the canonical Mukai net has a normal form at p for which the associated graded ring of the anti-canonical ring, for the blow-up of p, is generated by fourteen monomials, so it can be computed exactly.

Theorem 13.24. *Notation as in Set-up 13.12 with $r = 3$. Then the following hold.*

(1) $A_X(F) \geq S_X(F)$ for any prime divisor F over X with $p \in \operatorname{center}_X F$, with strict inequality if $\operatorname{center}_X F = \{p\}$.

(2) Assume that (2) of Proposition 13.22 holds. Let $\pi : Y := \operatorname{Bl}_p X \to X$ be the blow-up of the point p with exceptional plane E, and let W be the refinement of $(H^0(X, mH))_m$ by E. Then $A_X(E) = 3$, $S_X(E) = 219/88$, $\delta_p(X) = A_X(E)/S_X(E) = 88/73$, $A_X(F) - S_X(F) \geq \frac{15}{88} A_X(F)$ for any F as in (1), and $S(W; v) \leq \frac{73}{88} A_E(v)$ for any prime divisor v over E.

(3) Under the assumption of (2), there exist regular parameters x, y, z at p and a local trivialization of H such that, with T of degree one, the associated graded ring of $\bigoplus_m H^0(X, mH)$ for ord_E is

$$\mathbb{C}[T, Tx, Ty, Tz, Txy, Txz, Tyz, Tx^2y, Ty^2z, Tz^2x, Txyz, Tx^2yz, Txy^2z, Txyz^2].$$

By (2), E computes $\delta_p(X)$, so the constant $15/88$ in (2) is optimal at such points.

Proof of Theorem 13.24. The proof is long, so we divide it into several steps. Until **Step 9** we assume that (2) of Proposition 13.22 holds, and after relabelling we assume that $n_{12} = n_{23} = n_{31} = 2$.

Step 1. In this step we write X near p in a chart of a Grassmannian.

Let $V := V_X$ and $\Pi := \Pi_X$. By Definition-Theorem 2.20, X is the scheme of three-dimensional subspaces of V isotropic for any $\omega \in \Pi$, any non-zero $\omega \in \Pi$ has rank six, and H is the restriction of the Plücker class. Let $V_1 \subset V$ be the subspace with $p = [V_1]$, and fix a complement V_2, so that $V = V_1 \oplus V_2$, $\dim V_1 = 3$, and $\dim V_2 = 4$. For a basis (v_1, v_2, v_3) of V_1 and a basis $(v_4, \dots, v_7)$ of V_2, we write the matrix of $\omega \in \Pi$ as

$$\omega = \begin{pmatrix} 0 & C \\ -C^t & D \end{pmatrix}, \qquad C_{ik} = \omega(v_i, v_{3+k}), \qquad D_{jk} = \omega(v_{3+j}, v_{3+k}),$$

where the upper left block vanishes because $p \in X$. The entries of C and D are linear forms on Π. We also view C as the linear map $V_2 \to V_1^\vee$, $w \to \omega(\cdot, w)|_{V_1}$.

For $\Phi \in \operatorname{Hom}(V_1, V_2)$ with matrix (ϕ_{ik}), that is, $\Phi(v_i) = \sum_k \phi_{ik} v_{3+k}$, let $V_\Phi := \{v + \Phi(v) \mid v \in V_1\}$. The subspaces V_Φ form an affine chart of $\operatorname{Gr}(3, V)$ around p with coordinates ϕ_{ik}, and $T_p \operatorname{Gr}(3, V) = \operatorname{Hom}(V_1, V_2)$ [Har92, Example 16.1]. Since $\omega(v_i + \Phi(v_i), v_j + \Phi(v_j))$ is the (i, j) entry of

$$Q_\omega(\Phi) := C\Phi^t - \Phi C^t + \Phi D \Phi^t, \tag{13.75}$$

X is defined in the chart by the $(1,2)$, $(1,3)$, and $(2,3)$ entries of $Q_\omega(\Phi)$, where ω runs through a basis of Π. In particular, $T_p X = \{\Phi \mid C\Phi^t = \Phi C^t \text{ for any } \omega \in \Pi\}$.

We will use the following rank formula for $\omega \in \Pi$:

$$\operatorname{rank} \omega = 2 \operatorname{rank} C + \operatorname{rank} \left(D|_{\operatorname{Ker} C}\right). \tag{13.76}$$

To prove (13.76), let $V_1'' \subset V_1$ be the common zero locus of the image of C, and fix complements $V_1 = V_1' \oplus V_1''$ and $V_2 = K' \oplus \operatorname{Ker} C$. Then $V_1'' \subset \operatorname{Ker} \omega$, $\omega|_{V_1'} = 0$, and ω pairs V_1' and K' perfectly. Let $h : \operatorname{Ker} C \to V_1'$, $w \to h(w)$, where $h(w)$ is the unique vector with $\omega(w + h(w), K') = 0$. Since $\omega(V_1', \operatorname{Ker} C) = 0$, we have $\omega(w + h(w), w' + h(w')) = \omega(w, w')$ for $w, w' \in \operatorname{Ker} C$, and $\omega(w + h(w), V_1) = 0$. Thus, with respect to $V = V_1'' \oplus (V_1' \oplus K') \oplus \{w + h(w) \mid w \in \operatorname{Ker} C\}$, the form ω is block diagonal with blocks 0, a block of rank $2 \operatorname{rank} C$, and $D|_{\operatorname{Ker} C}$. Since $\dim \operatorname{Ker} C = 4 - \operatorname{rank} C$, (13.76) implies

$$\operatorname{rank} C \le 1 \implies \operatorname{rank} \omega \le 4. \tag{13.77}$$

Let $\xi \in \operatorname{Hom}(V_1, V_2)$ be of rank one. Then $\xi D \xi^t = 0$, since D is alternating. Thus, if $\xi \in T_p X$, then $Q_\omega(\tau\xi) = \tau(C\xi^t - \xi C^t) = 0$ for any $\tau \in \mathbb{C}$, and the closure of $\{V_{\tau\xi} \mid \tau \in \mathbb{C}\}$ is a line on X through p. Conversely, a line L' of $\operatorname{Gr}(3, V)$ through p is the set of V' with $W_2 \subset V' \subset W_4$ for subspaces $W_2 \subset V_1 \subset W_4$ of dimensions two and four by Lemma 2.19, and the tangent direction of L' at p is the rank one map $V_1 \to V_1/W_2 \to W_4/V_1$. Hence the points of $\Sigma(X, p)$ correspond to the points of $\mathbb{P}(T_p X)$ of rank one, by taking the tangent direction. For $i = 1, 2, 3$ let $\xi_i = \alpha_i \otimes \beta_i$, $\alpha_i \in V_1^\vee$, $\beta_i \in V_2$, be a tangent vector of L_i at p, that is, $\xi_i(v) = \alpha_i(v)\beta_i$. By Proposition 4.1, $[\xi_1]$, $[\xi_2]$, $[\xi_3]$ are not collinear, so (ξ_1, ξ_2, ξ_3) is a basis of $T_p X$.

Step 2. In this step we show that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent and that $\beta_1, \beta_2, \beta_3$ are linearly independent.

In matrix form $\xi_i = \alpha_i \beta_i^t$, so $C\xi_i^t - \xi_i C^t = (C\beta_i)\alpha_i^t - \alpha_i (C\beta_i)^t$. Thus $\xi_i \in T_p X$ means that for any $\omega \in \Pi$ there exists $\mu_i(\omega) \in \mathbb{C}$ with

$$C\beta_i = \mu_i(\omega)\alpha_i, \tag{13.78}$$

and μ_i is a linear form on Π. If α_1 and α_2 were proportional, then $\alpha_1 \otimes \beta \in T_p X$ would have rank at most one for any $\beta \in \operatorname{Span}(\beta_1, \beta_2)$, and $\Sigma(X, p)$ would be infinite by **Step 1**. If β_1 and β_2 were proportional, then $\alpha \otimes \beta_1 \in T_p X$ would have rank at most one for any $\alpha \in \operatorname{Span}(\alpha_1, \alpha_2)$, and $\Sigma(X, p)$ would again be infinite. Thus no two of the α_i and no two of the β_i are proportional, and $\dim \operatorname{Span}(\alpha_i)$, $\dim \operatorname{Span}(\beta_i) \in \{2, 3\}$.

Case 1. $\dim \operatorname{Span}(\alpha_i) = \dim \operatorname{Span}(\beta_i) = 2$. Then $T_p X \subset \operatorname{Span}(\alpha_i) \otimes \operatorname{Span}(\beta_i)$, and $\dim T_p X = 3$. The locus of rank at most one in $\mathbb{P}(\operatorname{Span}(\alpha_i) \otimes \operatorname{Span}(\beta_i)) \cong \mathbb{P}^3$ is a quadric surface, which intersects the plane $\mathbb{P}(T_p X)$ in a curve. Thus $\Sigma(X, p)$ is infinite, a contradiction.

Case 2. $\dim \mathrm{Span}(\alpha_i) = 2$ and $\dim \mathrm{Span}(\beta_i) = 3$. Let $0 \neq u_0 \in V_1$ with $\alpha_i(u_0) = 0$ for $i = 1, 2, 3$. By (13.78), $\omega(u_0, \beta_i) = \mu_i(\omega)\alpha_i(u_0) = 0$, and $\omega(u_0, V_1) = 0$. Let $w \in V_2 \setminus \mathrm{Span}(\beta_i)$. The linear form $\omega \to \omega(u_0, w)$ vanishes on a subspace $\Pi' \subset \Pi$ with $\dim \Pi' \geq 2$, and $u_0 \in \mathrm{Ker}\, \omega$ for $\omega \in \Pi'$. Thus any $\omega \in \Pi'$ induces an alternating form $\bar{\omega}$ on $V/\mathbb{C}u_0$ with $\mathrm{rank}\, \bar{\omega} = \mathrm{rank}\, \omega$. The Pfaffian of $\bar{\omega}$ is a cubic form on Π', so it vanishes at some $0 \neq \omega_0 \in \Pi'$, and $\mathrm{rank}\, \omega_0 \leq 4$, a contradiction.

Case 3. $\dim \mathrm{Span}(\alpha_i) = 3$ and $\dim \mathrm{Span}(\beta_i) = 2$. Write $\beta_3 = \eta_1\beta_1 + \eta_2\beta_2$. Then $\eta_1\eta_2 \neq 0$, and possibly replacing (α_i, β_i) with $(\alpha_i/\eta_i, \eta_i\beta_i)$ for $i = 1, 2$, we may assume that $\beta_3 = \beta_1 + \beta_2$. By (13.78), $\mu_3(\omega)\alpha_3 = \mu_1(\omega)\alpha_1 + \mu_2(\omega)\alpha_2$, so $\mu_1 = \mu_2 = \mu_3 = 0$ and $\mathrm{Span}(\beta_1, \beta_2) \subset \mathrm{Ker}\, C$ for any $\omega \in \Pi$. Let $0 \neq \omega \in \Pi$ with $\omega(\beta_1, \beta_2) = 0$. Then $\mathrm{rank}\, C \leq 2$. If $\mathrm{rank}\, C = 2$, then $\mathrm{Ker}\, C = \mathrm{Span}(\beta_1, \beta_2)$ and $D|_{\mathrm{Ker}\, C} = 0$, so $\mathrm{rank}\, \omega = 4$ by (13.76). If $\mathrm{rank}\, C \leq 1$, then $\mathrm{rank}\, \omega \leq 4$ by (13.77). Both contradict $\mathrm{rank}\, \omega = 6$.
Thus $\dim \mathrm{Span}(\alpha_i) = \dim \mathrm{Span}(\beta_i) = 3$.

Step 3. In this step we fix the bases of V_1 and V_2 and show three non-vanishing conditions on D.

By **Step 2**, we may let (v_1, v_2, v_3) be the basis of V_1 dual to $(\alpha_1, \alpha_2, \alpha_3)$ and $v_{3+i} := \beta_i$ for $i = 1, 2, 3$. Then $\xi_i = v_i^\vee \otimes v_{3+i}$, the matrix of ξ_i has a single non-zero entry 1 at (i, i), and (13.78) says that the first three columns of C form the matrix $\mathrm{diag}(a, b, c)$, where $a := \mu_1$, $b := \mu_2$, and $c := \mu_3$. If a, b, c were linearly dependent, then some $0 \neq \omega \in \Pi$ would have $a(\omega) = b(\omega) = c(\omega) = 0$, hence $\mathrm{rank}\, C \leq 1$, contradicting (13.77). Thus (a, b, c) are linear coordinates on Π. Replacing v_7 with $v_7 - \chi_1 v_4 - \chi_2 v_5 - \chi_3 v_6$ subtracts $(\chi_1 a, \chi_2 b, \chi_3 c)^t$ from the fourth column of C. Possibly replacing v_7 in this way, we may assume that

$$C = \begin{pmatrix} a & 0 & 0 & \sigma_1 \\ 0 & b & 0 & \sigma_2 \\ 0 & 0 & c & \sigma_3 \end{pmatrix}, \qquad \begin{aligned} \sigma_1 &= \kappa_{12} b + \kappa_{13} c, \\ \sigma_2 &= \kappa_{21} a + \kappa_{23} c, \\ \sigma_3 &= \kappa_{31} a + \kappa_{32} b, \end{aligned} \tag{13.79}$$

with $\kappa_{ij} \in \mathbb{C}$. Write $D_{jk} = d_{jk}^{(1)} a + d_{jk}^{(2)} b + d_{jk}^{(3)} c$ with $d_{jk}^{(l)} \in \mathbb{C}$. We show that

$$d_{23}^{(1)} \neq 0, \qquad d_{13}^{(2)} \neq 0, \qquad d_{12}^{(3)} \neq 0. \tag{13.80}$$

Let $\omega \in \Pi$ with $(a, b, c) = (1, 0, 0)$. The non-zero columns of C are the first column $(1, 0, 0)^t$ and the fourth column $(0, \kappa_{21}, \kappa_{31})^t$. By (13.77), $\mathrm{rank}\, C = 2$, so $\mathrm{Ker}\, C = \mathrm{Span}(v_5, v_6)$, and (13.76) implies that $\mathrm{rank}\, \omega = 6$ if and only if $D_{23}(\omega) = d_{23}^{(1)} \neq 0$. The computation for $(a, b, c) = (1, 0, 0)$, applied to the forms with $(a, b, c) = (0, 1, 0)$ and $(0, 0, 1)$, which have $\mathrm{Ker}\, C = \mathrm{Span}(v_4, v_6)$ and $\mathrm{Ker}\, C = \mathrm{Span}(v_4, v_5)$, respectively, implies $d_{13}^{(2)} \neq 0$ and $d_{12}^{(3)} \neq 0$.

Step 4. In this step we compute the normal direction at p of the first-order deformations of L_1, L_2, and L_3, and we show that $\kappa_{12} = \kappa_{23} = \kappa_{31} = 0$ and $\kappa_{13}\kappa_{21}\kappa_{32} \neq 0$.

Since L_1 is non-special, $N_{L_1/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, so $\dim H^0(N_{L_1/X}) = 1$ and the value at p of a non-zero section spans the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$. The first-order deformations of L_1 in X are the elements of $H^0(N_{L_1/X})$, and the value at p of a first-order deformation is the image in $N_{L_1/X} \otimes k(p) = T_p X/\mathbb{C}\xi_1$ of the first-order displacement of p [KPS18, Lemma 2.1.4 and the paragraph following it, Corollary 2.1.6(ii), and the map (2.2.2)]. By Lemma 13.13, the fiber at p of the summand $\mathcal{O}_{\mathbb{P}^1}$ is the point $s_1 \cap f_{1,p}$, and $n_{31} = 2 > 1$ implies $p_{13} \in s_1$. By Definition-Proposition 10.9, p_{13} is the normal direction of L_3 at p relative to L_1, that is, the image of ξ_3. Thus the value at p of a non-zero first-order deformation of L_1 in X is a non-zero multiple of the image of ξ_3. Replacing (L_1, L_3, n_{31}) by (L_2, L_1, n_{12}) and by (L_3, L_2, n_{23}) in the argument for L_1, we see that the value at p of a non-zero first-order deformation of L_2 (resp. L_3) in X is a non-zero multiple of the image of ξ_1 (resp. ξ_2).

We compute the first-order deformations of L_1 in X. Let $\mathbb{C}[\epsilon]$ denote the ring of dual numbers $\mathbb{C}[\epsilon]/(\epsilon^2)$. By Lemma 2.19, after a reparametrization $\tau \to \tau + \epsilon(\tau_1\tau + \tau_0)$ with $\tau_0, \tau_1 \in \mathbb{C}$, a first-order deformation of L_1 in $\mathrm{Gr}(3, V)$ is the closure of the family $V_{\Phi(\tau)}$ with

$$\Phi(\tau) = \epsilon\rho + \tau(\xi_1 + \epsilon\Phi'),$$

where $\rho \in \mathrm{Hom}(V_1, V_2)$ has vanishing $(1,1)$ entry and the matrix $\Phi' = (\phi'_{ik})$ has $\phi'_{11} = 0$ by the reparametrization, and $\phi'_{ik} = 0$ for $i \neq 1$ and $k \neq 1$ since the 2×2 minors of $\xi_1 + \epsilon\Phi'$ vanish. The value at p of such a deformation is the image of ρ. Since the affine part of the family is schematically dense in the closure, the deformation is contained in X if and only if $Q_\omega(\Phi(\tau)) = 0$ in $\mathbb{C}[\epsilon][\tau]$ for any $\omega \in \Pi$. Over $\mathbb{C}[\epsilon]$ we have

$$\xi_1 + \epsilon\Phi' = \big(v_1^\vee + \epsilon \sum_{i>1} \phi'_{i1} v_i^\vee\big) \otimes \big(v_4 + \epsilon \sum_{k>1} \phi'_{1k} v_{3+k}\big),$$

so the coefficient of τ^2 in $Q_\omega(\Phi(\tau))$ vanishes as D is alternating. The coefficient of τ modulo ϵ is $C\xi_1^t - \xi_1 C^t = 0$. The coefficient of ϵ is $C\rho^t - \rho C^t$, so $\rho \in T_pX$ and $\rho = \rho_2\xi_2 + \rho_3\xi_3$ with $\rho_2, \rho_3 \in \mathbb{C}$. The coefficient of $\epsilon\tau$ is $Q'_\omega := C\Phi'^t - \Phi' C^t + \rho D \xi_1^t + \xi_1 D \rho^t$. By (13.79), the $(2,3)$ entry of Q'_ω vanishes identically, and the $(1,2)$ and $(1,3)$ entries of Q'_ω are

$$a\phi'_{21} - b\phi'_{12} - \sigma_2\phi'_{14} + \rho_2 D_{12}, \qquad a\phi'_{31} - c\phi'_{13} - \sigma_3\phi'_{14} + \rho_3 D_{13}.$$

Both vanish for any $\omega \in \Pi$ if and only if their coefficients of a, b, and c vanish. The coefficients of a and b in the first one and of a and c in the second one determine ϕ'_{21}, ϕ'_{12}, ϕ'_{31}, and ϕ'_{13} from $(\rho_2, \rho_3, \phi'_{14})$, and the remaining coefficients, of c in the first entry and of b in the second entry, are

$$d_{12}^{(3)}\rho_2 = \kappa_{23}\phi'_{14}, \qquad d_{13}^{(2)}\rho_3 = \kappa_{32}\phi'_{14}.$$

By (13.80), the first-order deformations of L_1 in X form a line spanned by the deformation with $\phi'_{14} = 1$, whose value at p is the image of

$$\frac{\kappa_{23}}{d_{12}^{(3)}}\xi_2 + \frac{\kappa_{32}}{d_{13}^{(2)}}\xi_3.$$

Since the value at p of a non-zero first-order deformation of L_1 in X is a non-zero multiple of the image of ξ_3, we obtain $\kappa_{23} = 0$ and $\kappa_{32} \neq 0$.

The form (13.79) of C is preserved if we permute (v_1, v_2, v_3), (v_4, v_5, v_6), (a, b, c), and the indices of the κ_{ij} simultaneously by one permutation of $\{1,2,3\}$. Applying the cyclic permutation $i \to i+1$ (resp. $i \to i+2$) of $\{1,2,3\}$ to the computation above, the value at p of a non-zero first-order deformation of L_2 (resp. L_3) is a non-zero multiple of the image of $\kappa_{13}\xi_1/d_{21}^{(3)} + \kappa_{31}\xi_3/d_{23}^{(1)}$ (resp. $\kappa_{12}\xi_1/d_{31}^{(2)} + \kappa_{21}\xi_2/d_{32}^{(1)}$). Since D is alternating, $d_{kj}^{(l)} = -d_{jk}^{(l)}$, so $d_{21}^{(3)}$, $d_{23}^{(1)}$, $d_{31}^{(2)}$, and $d_{32}^{(1)}$ are non-zero by (13.80). Since the values for L_2 and L_3 are non-zero multiples of the images of ξ_1 and ξ_2, respectively, we obtain $\kappa_{31} = \kappa_{12} = 0$ and $\kappa_{13}\kappa_{21} \neq 0$.

Thus $\sigma_1 = \kappa_{13}c$, $\sigma_2 = \kappa_{21}a$, and $\sigma_3 = \kappa_{32}b$. Replacing v_4, v_5, v_6, a, b, and c with $\kappa_{21}v_4$, $\kappa_{32}v_5$, $\kappa_{13}v_6$, $\kappa_{21}a$, $\kappa_{32}b$, and $\kappa_{13}c$, respectively, we obtain

$$C = \begin{pmatrix} a & 0 & 0 & c \\ 0 & b & 0 & a \\ 0 & 0 & c & b \end{pmatrix}, \tag{13.81}$$

and (13.80) still holds, because each $d_{jk}^{(l)}$ is multiplied by a non-zero constant. The tangent vectors ξ_i are multiplied by non-zero constants, so T_pX is still the space of matrices whose only non-zero entries are at $(1,1)$, $(2,2)$, and $(3,3)$.

Step 5. In this step we find regular parameters at p and fourteen sections of H whose leading forms are non-zero multiples of monomials.

By (13.81), the linear parts of the $(1,2)$, $(1,3)$, and $(2,3)$ entries of $Q_\omega(\Phi)$ are

$$\begin{aligned} &a(\phi_{21} - \phi_{14}) - b\phi_{12} + c\phi_{24}, \\ &a\phi_{31} - b\phi_{14} + c(\phi_{34} - \phi_{13}), \\ &a\phi_{34} + b(\phi_{32} - \phi_{24}) - c\phi_{23}. \end{aligned} \tag{13.82}$$

Their nine coefficients of a, b, and c are linear forms in the nine entries ϕ_{ik}, $i \neq k$, of rank nine, and they do not involve $\phi_{11}, \phi_{22}, \phi_{33}$. Thus $x := \phi_{11}|_X$, $y := \phi_{22}|_X$, and $z := \phi_{33}|_X$ are

regular parameters of X at p, and, by the implicit function theorem for formal power series, $\phi_{ik}|_X \in \mathfrak{m}_p^2$ for $i \neq k$, where $\mathfrak{m}_p \subset \widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x,y,z]]$ is the maximal ideal. Hence, writing $(x_1,x_2,x_3) := (x,y,z)$, we have $(\Phi D \Phi^t)_{jk} \equiv x_j x_k D_{jk}$ modulo $\mathfrak{m}_p^3$. Taking the coefficient of c in the first entry of (13.82), of b in the second, and of a in the third, together with the corresponding coefficients of xyD_{12}, xzD_{13}, and yzD_{23}, we obtain

$$\phi_{24} \equiv -d_{12}^{(3)} xy, \qquad \phi_{14} \equiv d_{13}^{(2)} xz, \qquad \phi_{34} \equiv -d_{23}^{(1)} yz \qquad \text{modulo } \mathfrak{m}_p^3. \tag{13.83}$$

Let ϖ_0 be the Plücker coordinate dual to $v_1 \wedge v_2 \wedge v_3$. Then $\varpi_0(p) \neq 0$, so $\varpi_0|_X \in H^0(X,H)$ trivializes H near p. On the chart, the Plücker coordinates divided by ϖ_0 are the maximal minors of the 3×7 matrix $(I_3 \mid \Phi)$, that is, up to sign, the minors of Φ of all sizes. For $I \subset \{1,2,3\}$ and $J \subset \{1,2,3,4\}$ with $|I| = |J|$, let $\Phi_{I,J}$ be the minor of Φ with rows I and columns J. For a non-zero $s \in H^0(X, mH)$ write $s = f\varpi_0^m$ near p with $f \in \mathbb{C}[[x,y,z]]$, and let $\operatorname{ord}_E(s)$ and $\operatorname{lf}(s)$ be the order and the leading form of f. Table 21 lists fourteen sections of H and their leading forms. Each entry follows from (13.83): an off-diagonal entry of Φ has order at least two and a diagonal entry has order one, so in each minor the product indicated in the table is the only term of the smallest order. For example, the columns $1,2,4$ contain no diagonal entry in the third row, so any term of $\Phi_{123,124}$ has order at least four, with equality only for $\phi_{11}\phi_{22}\phi_{34}$. By (13.80), all the constants in Table 21 are non-zero, so the leading forms are non-zero multiples of

$$1,\ x,\ y,\ z,\ xy,\ xz,\ yz,\ x^2y,\ y^2z,\ z^2x,\ xyz,\ x^2yz,\ xy^2z,\ xyz^2. \tag{13.84}$$

TABLE 21. Fourteen sections of H near p, the products of entries of Φ of the smallest order in their expansions, and their leading forms.

section/ϖ_0	leading product	leading form
1		1
$\phi_{11},\ \phi_{22},\ \phi_{33}$		$x,\ y,\ z$
$\phi_{24},\ \phi_{14},\ \phi_{34}$		$-d_{12}^{(3)}xy,\ d_{13}^{(2)}xz,\ -d_{23}^{(1)}yz$
$\Phi_{12,14} = \phi_{11}\phi_{24} - \phi_{14}\phi_{21}$	$\phi_{11}\phi_{24}$	$-d_{12}^{(3)}x^2y$
$\Phi_{23,24} = \phi_{22}\phi_{34} - \phi_{24}\phi_{32}$	$\phi_{22}\phi_{34}$	$-d_{23}^{(1)}y^2z$
$\Phi_{13,34} = \phi_{13}\phi_{34} - \phi_{14}\phi_{33}$	$-\phi_{14}\phi_{33}$	$-d_{13}^{(2)}xz^2$
$\Phi_{123,123}$	$\phi_{11}\phi_{22}\phi_{33}$	xyz
$\Phi_{123,134}$	$-\phi_{11}\phi_{24}\phi_{33}$	$d_{12}^{(3)}x^2yz$
$\Phi_{123,124}$	$\phi_{11}\phi_{22}\phi_{34}$	$-d_{23}^{(1)}xy^2z$
$\Phi_{123,234}$	$\phi_{14}\phi_{22}\phi_{33}$	$d_{13}^{(2)}xyz^2$

Step 6. In this step we show that, for any $m \geq 1$, $H^0(X,mH)$ has a basis compatible with ord_E consisting of products of m of the fourteen sections of **Step 5**, one for each $u \in m\mathcal{P} \cap \mathbb{Z}^3$ with $\mathcal{P}$ as in (13.85) below, whose leading form is a non-zero multiple of $x^{u_1}y^{u_2}z^{u_3}$. This proves (3).

Let $\mathcal{P} \subset \mathbb{R}^3$ be the polytope of $u = (u_1,u_2,u_3)$ with

$$\begin{aligned} &u_1,u_2,u_3 \geq 0, \qquad u_1+u_2+u_3 \leq 4, \\ &u_1+u_2,\ u_2+u_3,\ u_3+u_1 \leq 3, \qquad u_1-u_2,\ u_2-u_3,\ u_3-u_1 \leq 1. \end{aligned} \tag{13.85}$$

The inequalities (13.85) are invariant under the cyclic permutation $(u_1,u_2,u_3) \to (u_3,u_1,u_2)$, and so is $\mathcal{P}$. We write $u_1u_2u_3$ for the point (u_1,u_2,u_3) when $0 \leq u_i \leq 9$. A direct calculation, intersecting any three of the ten planes in (13.85) and testing the remaining inequalities, shows that the vertices of $\mathcal{P}$ are

$$000,\ 100,\ 010,\ 001,\ 210,\ 021,\ 102,\ 211,\ 121,\ 112.$$

Since $u_1 \leq u_2 + 1$ and $u_1 + u_2 \leq 3$ imply $u_1 \leq 2$, any $u \in \mathcal{P}$ satisfies $0 \leq u_i \leq 2$, and testing the 27 points of $\{0,1,2\}^3$ shows that $\mathcal{P} \cap \mathbb{Z}^3$ consists of the fourteen exponent vectors of (13.84).

In particular, $\mathcal{P}$ is the convex hull of $\mathcal{P}\cap\mathbb{Z}^3$, and $u_* := (1,1,1)$ is the only lattice point in the interior of $\mathcal{P}$. Each of the ten facets of $\mathcal{P}$ is defined by a primitive integral linear form ℓ with $|\ell(\text{facet}) - \ell(u_*)| = 1$.

We triangulate the four facets in Table 22 using all lattice points of the facet as listed there, and the other six facets by the images under the cyclic permutation. Each edge of $\mathcal{P}$ is subdivided by all lattice points on it, so the triangulations of two facets coincide on their common edge. Let Θ be a triangle of the triangulation, contained in the facet $\{\ell = \ell_\Theta\}$ of $\mathcal{P}$ with ℓ primitive and $\ell_\Theta \in \mathbb{Z}$. Then Θ contains no lattice points other than the vertices of Θ, so by Pick's theorem [BR15, Theorem 2.8] the area of Θ is $1/2$ for the lattice $\ell^{-1}(0)\cap\mathbb{Z}^3$, that is, the edge vectors of Θ form a basis of $\ell^{-1}(0)\cap\mathbb{Z}^3$. Since $|\ell_\Theta - \ell(u_*)| = 1$, the edge vectors of the tetrahedron with vertices Θ and u_* form a basis of $\mathbb{Z}^3$. Thus the 22 tetrahedra with vertices Θ and u_* form a triangulation of $\mathcal{P}$ into unimodular tetrahedra, and $\operatorname{vol}(\mathcal{P}) = 22/6 = 11/3$. The boundary triangulation is a triangulated two-sphere with 13 vertices and 22 triangles, so it has 33 edges by Euler's formula, and the numbers of vertices, edges, triangles, and tetrahedra of the triangulation of $\mathcal{P}$ are

$$(f_0, f_1, f_2, f_3) = (14, 46, 55, 22). \tag{13.86}$$

TABLE 22. Four facets of $\mathcal{P}$, the other six being their images under the cyclic permutation of coordinates, with their triangulations using all lattice points of the facet, and the sum over the triangles Θ of the sum s_Θ of the coordinates of the barycenter of the tetrahedron with vertices Θ and u_*.

facet	triangles	$\sum s_\Theta$
$u_3 = 0$	$(000,100,110)$, $(100,210,110)$, $(000,110,010)$	$21/4$
$u_1 - u_2 = 1$	$(100,210,101)$, $(101,210,211)$, $(101,211,102)$	$33/4$
$u_1 + u_2 = 3$	$(210,211,121)$	$7/2$
$u_1 + u_2 + u_3 = 4$	$(211,121,112)$	$15/4$

Let $m \ge 1$ and $u \in m\mathcal{P}\cap\mathbb{Z}^3$. Let Δ be a tetrahedron of the triangulation with $u/m \in \Delta$, with vertices $w_0,\dots,w_3$, and write $u/m = \sum_i t_i w_i$ with $t_i \ge 0$ and $\sum_i t_i = 1$. Since $u - mw_0 = \sum_{i\ge1} mt_i(w_i - w_0)$ and $(w_1 - w_0, w_2 - w_0, w_3 - w_0)$ is a basis of $\mathbb{Z}^3$, the numbers mt_i are non-negative integers with sum m. Thus u is a sum of m of the fourteen exponent vectors. Moreover, the relative interior of $m\Delta'$ for a face Δ' of the triangulation contains $\binom{m-1}{\dim\Delta'}$ lattice points. Summing over the faces and using (13.86), we obtain

$$\begin{aligned}\#(m\mathcal{P}\cap\mathbb{Z}^3) &= 14 + 46(m-1) + 55\binom{m-1}{2} + 22\binom{m-1}{3}\\ &= \frac{11}{3}m^3 + \frac{11}{2}m^2 + \frac{23}{6}m + 1.\end{aligned}$$

By Lemma 2.12(2) with $g = 12$, $N_m = \dim H^0(X, mH) = \frac{22}{12}m(m+1)(2m+1) + 2m + 1 = \frac{11}{3}m^3 + \frac{11}{2}m^2 + \frac{23}{6}m + 1$, and hence

$$\#(m\mathcal{P}\cap\mathbb{Z}^3) = N_m. \tag{13.87}$$

Let $F^jH^0(X,mH) := \{s \mid \operatorname{ord}_E(s) \ge j\}$. The map $s \to T^m \operatorname{lf}(s)$ induces injective linear maps $F^jH^0(X,mH)/F^{j+1}H^0(X,mH) \to T^m\mathbb{C}[x,y,z]_j$, compatible with products, so the associated graded ring of $\bigoplus_m H^0(X,mH)$ for ord_E is isomorphic to the span of the $T^m \operatorname{lf}(s)$, and the dimension of the part of degree m is $\dim H^0(X,mH) = N_m$. For each $u \in m\mathcal{P}\cap\mathbb{Z}^3$, choose a product $s_u \in H^0(X,mH)$ of m of the sections in Table 21 whose exponent vectors sum to u, which exists by the decomposition above. The leading form of s_u is a non-zero multiple of $x^{u_1}y^{u_2}z^{u_3}$. The leading forms of the sections s_u are non-zero multiples of pairwise distinct monomials, so any non-zero linear combination of the sections s_u has ord_E equal to the smallest $u_1 + u_2 + u_3$ with non-zero coefficient. Thus the sections s_u are linearly independent, and by

(13.87) they form a basis of $H^0(X,mH)$ compatible with ord_E. Hence the N_m distinct monomials $T^m x^{u_1}y^{u_2}z^{u_3}$ with $u\in m\mathcal{P}\cap\mathbb{Z}^3$ span the part of degree m. Thus the associated graded ring is the ring in (3), where T corresponds to ϖ_0. Moreover, $E\cong\mathbb{P}^2$ has homogeneous coordinates x,y,z, $\mathcal{O}_E(-E)\cong\mathcal{O}_{\mathbb{P}^2}(1)$, and $W_{m,j}\subset H^0(E,\mathcal{O}_E(j))$ is the image of $F^jH^0(X,mH)$, so

$$W_{m,j}=\mathrm{Span}\left\{x^{u_1}y^{u_2}z^{u_3}\;\middle|\;u\in m\mathcal{P}\cap\mathbb{Z}^3,\ u_1+u_2+u_3=j\right\}. \tag{13.88}$$

Step 7. In this step we compute $S_X(E)$.

The basis $(s_u)_{u\in m\mathcal{P}\cap\mathbb{Z}^3}$ of $H^0(X,mH)$ of **Step 6** is compatible with the filtration by ord_E, and $\mathrm{ord}_E(s_u)=u_1+u_2+u_3$, so

$$S_m(E)=\frac{1}{mN_m}\sum_{u\in m\mathcal{P}\cap\mathbb{Z}^3}(u_1+u_2+u_3)=3b_m,\qquad b_m:=\frac{1}{mN_m}\sum_{u\in m\mathcal{P}\cap\mathbb{Z}^3}u_1,$$

where the second equality follows from the cyclic symmetry of $\mathcal{P}$. The tetrahedra of **Step 6** have volume $1/6$, and the integral of $u_1+u_2+u_3$ over a tetrahedron Δ is $\mathrm{vol}(\Delta)$ times the sum of the coordinates of the barycenter of Δ. By Table 22 and the cyclic symmetry,

$$\int_{\mathcal{P}}(u_1+u_2+u_3)du=\frac{1}{6}\Big(3\cdot\frac{21}{4}+3\cdot\frac{33}{4}+3\cdot\frac{7}{2}+\frac{15}{4}\Big)=\frac{73}{8}.$$

Since $\#(m\mathcal{P}\cap\mathbb{Z}^3)=\mathrm{vol}(\mathcal{P})m^3+O(m^2)$ and the sums above are Riemann sums of $\mathcal{P}$ with error $O(1/m)$, we obtain $\lim_m b_m=\frac{73}{8}\cdot\frac{3}{11}\cdot\frac{1}{3}=\frac{73}{88}$, and, by [AZ22, Lemma 2.9],

$$S_X(E)=\lim_m S_m(E)=\frac{219}{88}.$$

Since E is the exceptional divisor of the blow-up of a smooth point of a threefold, $A_X(E)=3$ [KM98, Lemma 2.29].

Step 8. In this step we show that $S(W;v)\le\frac{73}{88}A_E(v)$ for any prime divisor v over E.

Since Y is smooth and E is a smooth divisor, $\Delta_E=\mathrm{Diff}_E(0)=0$ and $A_{E,\Delta_E}(v)=A_E(v)$. The series W has bounded support and contains an ample series [AZ22, Example 2.6] (see [AZ22, Section 2.5] for the definitions). For $m\ge1$ let $\lambda_m:\mathbb{G}_m\to\mathrm{Aut}(E)$ be the one-parameter subgroup

$$\lambda_m(t)\cdot[x:y:z]=[tx:t^{2m+1}y:z],$$

with the induced linearizations of $\mathcal{O}_E(j)$. By (13.88), $W_{m,j}$ is spanned by λ_m-eigenvectors, so $\mathrm{in}_{\lambda_m}W_{m,j}=W_{m,j}$. The weight of $x^{u_1}y^{u_2}z^{u_3}$ is $u_1+(2m+1)u_2$. For $u\in m\mathcal{P}$ we have $0\le u_1\le 2m$ by **Step 6**, so the weight determines (u_1,u_2), and $u_3=j-u_1-u_2$ on $W_{m,j}$. Thus the monomials in $W_{m,j}$ have distinct weights, any basis of $W_{m,j}$ consisting of λ_m-eigenvectors consists of multiples of monomials, and the only eigen-basis divisor at level m is

$$\frac{1}{mN_m}\sum_{u\in m\mathcal{P}\cap\mathbb{Z}^3}(u_1\Gamma_x+u_2\Gamma_y+u_3\Gamma_z)=b_m(\Gamma_x+\Gamma_y+\Gamma_z),$$

where $\Gamma_x,\Gamma_y,\Gamma_z\subset E$ are the lines $(x=0),(y=0),(z=0)$, and the equality follows from the cyclic symmetry of $\mathcal{P}$. Let $\gamma_m:=1/b_m$. Then $\big(E,\gamma_m b_m(\Gamma_x+\Gamma_y+\Gamma_z)\big)=(E,\Gamma_x+\Gamma_y+\Gamma_z)$ is log canonical [KM98, Corollary 2.31(3)], and $\lim_m\gamma_m=88/73$ by **Step 7**. By Lemma 3.6(1) with $c_m:=\gamma_m$, we have $\frac{88}{73}S(W;v)\le A_E(v)$ for any prime divisor v over E.

Step 9. In this step we prove (2) and (1).

By **Step 8**, the assumption of Lemma 3.1 holds for the complete series of H, the plt type divisor E with plt extraction π, $\lambda=88/73$, and $G=E$. Since

$$A_X(E)-\frac{88}{73}S_X(E)=3-\frac{88}{73}\cdot\frac{219}{88}=0$$

by **Step 7**, Lemma 3.1 implies that $A_X(F)\ge\frac{88}{73}S_X(F)$ for any prime divisor F over X with $\mathrm{center}_Y F\cap E\ne\emptyset$, that is, with $p\in\mathrm{center}_X F$. Thus $\delta_p(X)\ge88/73$. Since $A_X(E)/S_X(E)=$

$3 \cdot \frac{88}{219} = \frac{88}{73}$, we obtain $\delta_p(X) = 88/73$, and $A_X(F) - S_X(F) \geq (1 - \frac{73}{88})A_X(F) = \frac{15}{88}A_X(F)$ for any F with $p \in \operatorname{center}_X F$. Together with **Steps 6–8**, this proves (2).

We prove (1) without assuming (2) of Proposition 13.22. By Proposition 13.22, either (1) or (2) of Proposition 13.22 holds. If (1) of Proposition 13.22 holds, then Theorem 13.24(1) is the last statement of Proposition 13.22. If (2) of Proposition 13.22 holds, then $A_X(F) - S_X(F) \geq \frac{15}{88}A_X(F) > 0$ for any F with $p \in \operatorname{center}_X F$, since $A_X(F) > 0$ as X is smooth. This proves Theorem 13.24(1). □

14. Points on a doubled special line in genus twelve

In this section, we prove Theorem 14.1, which treats the points of incidence type 2 and $2+1$ together. For such a point the divisor extracted over X is the prime divisor P over the special line studied in Section 11. When $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, the inverse bisecant M of s (Set-up 10.10) satisfies $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, and $M \cap \Gamma$ is non-reduced of length two, X is K-semistable by the degeneration of Section 12 (Theorem 14.1(3)), and the rest of this section does not treat this case. We first prove two lemmas on surface series, normalized as in Definition 7.3. The proof of Theorem 14.1 is given in Subsection 14.2.

Theorem 14.1. *Let X be a smooth prime Fano threefold of genus 12 and $x \in X$ a closed point. Assume that the connected component of $\Sigma(X, x)$ at some point $[L]$ has length exactly two, that is, x has incidence type 2 or $2+1$. Then the following hold.*

1. *The line L is special, and $A_X(F) \geq S_X(F)$ for any prime divisor F over X with $\operatorname{center}_X F = \{x\}$.*
2. *In the notation of Set-up 10.10, the prime divisor P, viewed over X, is the reduced exceptional divisor of the blow-up $\hat{\pi} : \hat{X} \to X$ of the non-reduced conic supported on L. The pair $(\hat{X}, P)$ is plt and $A_X(P) = 3$. The surface P is isomorphic to $\mathbb{F}_0$ or to $\mathbb{F}_2$, and the different of $(\hat{X}, P)$ on P is $\frac{1}{2}B$ for a smooth section B of P over L.*
3. *If $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, the inverse bisecant M of s satisfies $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, and $M \cap \Gamma$ is non-reduced of length two, then X is K-semistable.*
4. *If the assumption of (3) does not hold, then $S_X(P) = 117/44$, and any prime divisor F over X with $\operatorname{center}_X F = \{x\}$ satisfies*
$$A_X(F) - S_X(F) \geq \frac{15}{44} \operatorname{mult}_F P > 0,$$
where P is regarded as the $\mathbb{Q}$-Cartier divisor on $\hat{X}$.

14.1. **Surface series.** In this subsection, we prove two inequalities for surface series. Lemmas 14.3 and 14.5 bound $S(R; v)$ for the series R of Set-ups 14.2 and 14.4, which arise from $P \cong \mathbb{F}_0$ and $P \cong \mathbb{F}_2$ in the proof of Theorem 14.1. On $\mathbb{F}_0$ the inequality holds for any prime divisor over the surface, and on $\mathbb{F}_2$ for prime divisors whose center is not contained in the fiber through $\sigma \cap B$.

Set-up 14.2. Let S, σ, f, B, Δ_B, ι, R_Δ, and $R_{m,k}$ be as follows.

1. $S := \mathbb{P}^1 \times \mathbb{P}^1$ with ruling classes σ and f, and $B \subset S$ is a smooth curve of class $\sigma + 2f$.
2. $\Delta_B \geq 0$ is a divisor of degree two on B on which both ruling projections restrict to closed immersions, and $I_{\Delta_B} \subset \mathcal{O}_S$ is the ideal sheaf of Δ_B as a subscheme of S.
3. ι is the involution of the double cover $B \to \mathbb{P}^1$ given by the ruling with fibers of class σ, and R_Δ is the sum, with multiplicities, of the fibers of class σ through Δ_B, so that $R_\Delta|_B = \Delta_B + \iota(\Delta_B)$.
4. For integers $m \geq 1$ and $0 \leq k \leq \lfloor 5m/2 \rfloor$, $R_{m,k} \subset H^0(S, \mathcal{O}_S(k\sigma + mf))$ is the subspace $H^0(S, \mathcal{O}_S(k\sigma + mf))$ if $k \leq m$, the subspace $H^0(S, \mathcal{O}_S(k\sigma + mf) \otimes I_{\Delta_B}^{k-m})$ if $m < k \leq 2m$, and the subspace
$$s_B^{k-2m} s_{R_\Delta}^{2k-4m} H^0(S, \mathcal{O}_S(\ell(2\sigma + f)) \otimes I_{\Delta_B}^{\ell}), \qquad \ell := 5m - 2k,$$
if $2m < k \leq \lfloor 5m/2 \rfloor$, where s_B and s_{R_Δ} are defining sections of B and R_Δ.

At $k = m$ the first two formulas in Set-up 14.2(4) coincide, since $I^0_{\Delta_B} = \mathcal{O}_S$, and at $k = 2m$ the last two coincide, since then the exponents of s_B and s_{R_Δ} vanish and $\ell = m$. In the proof of Theorem 14.1, one ruling projection maps Δ_B isomorphically onto the scheme $M \cap \Gamma \subset M$ of length two, where M is the inverse bisecant of Set-up 10.10, and the other ruling projection restricts on B to the isomorphism $B \to L$. Thus both ruling projections restrict to closed immersions on Δ_B, and condition (2) of Set-up 14.2 excludes no configuration that occurs there, also when $M \cap \Gamma$ is non-reduced.

Lemma 14.3. *Notation as in Set-up 14.2, and let n_m, $S(R; v)$, and $A_{S,B/2}(v)$ be as in Definition 7.3. Then $n_m = \frac{11}{6}m^3 + O(m^2)$, and*

$$S(R; v) \le A_{S,B/2}(v)$$

for any prime divisor v over S.

Proof. The proof is long, so we divide it into several steps.

In the proof we write a for the ratio k/m and also for the corresponding real variable of integration, $\xi_+ := \max\{\xi, 0\}$ for a real number ξ, and T_Δ for the union of the fibers of class f through $\operatorname{Supp}\Delta_B$. Since $(B \cdot f) = 1$, we have $T_\Delta \cap B = \operatorname{Supp}\Delta_B$. For a closed point $z \in S$, C_z denotes the exceptional curve of the blow-up of z.

Step 1. In this step we collect facts on basis type divisors and log canonicity.

The sections s_B and s_{R_Δ} vanish on the subscheme Δ_B, that is, they are sections of $\mathcal{O}_S(B)\otimes I_{\Delta_B}$ and of $\mathcal{O}_S(R_\Delta)\otimes I_{\Delta_B}$. For $2m < k$ the exponents of I_{Δ_B} add up to $(k-2m)+(2k-4m)+\ell = k-m$, so

$$R_{m,k} \subset H^0(S, \mathcal{O}_S(k\sigma + mf) \otimes I^{(k-m)_+}_{\Delta_B})$$

for any k, with equality for $k \le 2m$. We check that $R_{m,k}\cdot R_{m',k'} \subset R_{m'',k''}$, where $(m'', k'') := (m+m', k+k')$. If $k'' \le 2m''$, this follows from $I^a_{\Delta_B} I^b_{\Delta_B} \subset I^{a+b}_{\Delta_B}$ and $(k-m)_+ + (k'-m')_+ \ge (k''-m'')_+$. If $k'' > 2m''$, we may assume $k' > 2m'$. If moreover $k > 2m$, the claim holds since the exponents of s_B, of s_{R_Δ}, and of I_{Δ_B}, and the integers ℓ are additive. If $k \le 2m$, let $g \in R_{m,k}$ and let

$$s_B^{k'-2m'} s_{R_\Delta}^{2k'-4m'} g' \in R_{m',k'}, \qquad g' \in H^0(S, \mathcal{O}_S(\ell'(2\sigma + f)) \otimes I^{\ell'}_{\Delta_B}), \quad \ell' := 5m' - 2k'.$$

Their product is

$$s_B^{k''-2m''} s_{R_\Delta}^{2k''-4m''} \cdot g g' s_B^{2m-k} s_{R_\Delta}^{4m-2k}.$$

Here $g \in I^{(k-m)_+}_{\Delta_B}$, $g' \in I^{\ell'}_{\Delta_B}$, $s_B^{2m-k} s_{R_\Delta}^{4m-2k} \in I^{6m-3k}_{\Delta_B}$, and $(k-m)_+ + 6m - 3k \ge 5m - 2k$. Since $B \sim \sigma + 2f$ and $R_\Delta \sim 2\sigma$, the factor $gg' s_B^{2m-k} s_{R_\Delta}^{4m-2k}$ belongs to $H^0(S, \mathcal{O}_S(\ell''(2\sigma+f))\otimes I^{\ell''}_{\Delta_B})$ with $\ell'' := 5m'' - 2k''$. Thus R is a multigraded linear series on S, graded by (m, k). It has bounded support since $k \le \lfloor 5m/2 \rfloor$, and it contains an ample series since $R_{m,k} = H^0(S, \mathcal{O}_S(k\sigma + mf))$ for $k \le m$. Hence the klt pair $(S, B/2)$ and the series R satisfy the hypotheses of Theorem 2.9 and [AZ22, Lemma 2.9], and below we refine R by fibers of class f, by B, and by exceptional curves of blow-ups and weighted blow-ups of points.

(a) For $m \ge 1$ let $S_m(R; v)$ and $S_m(R^C; q)$ be the normalized sums at level m whose lim sup as $m \to +\infty$ are $S(R; v)$ and $S(R^C; q)$ in Definition 7.3. By [AZ22, Lemma 3.1], for prime divisors v and v' over S each $R_{m,k}$ has a basis compatible with both v and v'. Let D_m be the m-basis type divisor of such bases (Definition 2.8), normalized by mn_m. Then

$$v(D_m) = S_m(R; v), \qquad v'(D_m) = S_m(R; v'),$$

and $v''(D) \le S_m(R; v'')$ for any m-basis type divisor D of R and any prime divisor v'' over S. If $v' = \operatorname{ord}_C$ for a curve $C \subset S$, then $D_m = S_m(R; \operatorname{ord}_C)C + \Omega_m$ with $C \not\subset \operatorname{Supp}\Omega_m$ and $\operatorname{mult}_q(\Omega_m|_C) \le S_m(R^C; q)$ for any $q \in C$, since the basis elements of $R_{m,k}$ of C-order j, divided by the j-th power of a defining section of C and restricted to C, form a basis of $R^C_{m,k,j}$.

(b) Let $z \in S$ be a closed point, v a prime divisor over S with $z \in \operatorname{center}_S v$, D_m as in (a) for v and some v', and $\lambda \in (0,1) \cap \mathbb{Q}$. If $(S, B/2 + \lambda D_m)$ is lc at z for any $m \gg 0$, then

$$0 \le A_S(v) - \operatorname{mult}_v \left(\frac{1}{2}B + \lambda D_m\right) = A_{S,B/2}(v) - \lambda S_m(R; v)$$

for any $m \gg 0$, hence $\lambda S(R; v) \le A_{S,B/2}(v)$.

(c) Let h and C be as in Definition 7.3, with C a curve on S or the exceptional curve of the blow-up or of a weighted blow-up of a closed point of S, and let $q \in C$ be a closed point near which the pair $(S', \frac{1}{2}B'_C + C)$ of Lemma 7.4 is plt, with different Δ_C on C. If $S(R; \operatorname{ord}_C) \le A_{S,B/2}(\operatorname{ord}_C)$ and $S(R^C; q) \le 1 - \operatorname{mult}_q \Delta_C$, then $S(R; v) \le A_{S,B/2}(v)$ for any prime divisor v over S whose center on the surface containing C contains q, by Lemma 7.4 with $\lambda = 1$.

(d) Let $z \in B$, let $\mu \in [0,1]$ be rational, and let $\Omega \ge 0$ be a $\mathbb{Q}$-divisor on S with $B \not\subset \operatorname{Supp} \Omega$, $\operatorname{mult}_z \Omega \le 1$, and $(B \cdot \Omega)_z \le 2 - \mu$. Then $(S, \mu B + \Omega)$ is lc at z by [CRZ19, Proposition 3.3].

(e) Let $V \subset W$ be spaces of sections of a line bundle, or of a $\mathbb{Q}$-divisor class on a curve, and v a valuation that is non-negative on W. Since $\dim\{s \in V : v(s) \ge t\} \le \dim\{s \in W : v(s) \ge t\}$ for any $t \ge 0$, the sum of the jumping numbers of v on V is at most the sum on W. If $W = H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(e))$ and v is the order at a point, then $\dim\{s \in W : v(s) \ge i\} = e - i + 1$ for $0 \le i \le e$, so the sum on V is at most $e(e+1)/2$.

Step 2. In this step we resolve Δ_B and prove $n_m = \frac{11}{6}m^3 + O(m^2)$.

Let $\psi : S' \to S$ be the blow-up of $\operatorname{Supp} \Delta_B$ if Δ_B is reduced. If $\Delta_B = 2z$, let ψ be the blow-up of z followed by the blow-up of the point of C_z corresponding to the tangent direction of B at z. Let e_1 and e_2 be the total transforms on S' of the exceptional curves of the first and of the second blow-up. In the first case the curves with negative self-intersection on S' are the strict transforms of the fibers through $\operatorname{Supp} \Delta_B$ and the exceptional curves, of classes $\sigma - e_i$, $f - e_i$, and e_i. In the second case they have classes $\sigma - e_1$, $f - e_1$, $e_1 - e_2$, and e_2, since the tangent direction of Δ_B at z differs from both ruling directions. The listed classes generate the cone of curves of S', because the class $\alpha\sigma + \beta f - \epsilon_1 e_1 - \epsilon_2 e_2$ of any other irreducible curve satisfies $0 \le \epsilon_i \le \min\{\alpha, \beta\}$, by intersecting with the strict transforms of the fibers through the points of $\operatorname{Supp} \Delta_B$ and with e_i, respectively with $e_1 - e_2$ and e_2, and

$$\alpha\sigma + \beta f - \epsilon_1 e_1 - \epsilon_2 e_2 = \epsilon_1(\sigma - e_1) + \epsilon_2(f - e_2) + (\alpha - \epsilon_1)\sigma + (\beta - \epsilon_2)f,$$

where σ, f, and $f - e_2$ are non-negative combinations of the listed classes. The intersection numbers of $-K_{S'} = 2\sigma + 2f - e_1 - e_2$ are $1, 1, 1$ with $\sigma - e_i$, $f - e_i$, e_i in the first case, and $1, 1, 0, 1$ with $\sigma - e_1$, $f - e_1$, $e_1 - e_2$, e_2 in the second case, and $(K^2_{S'}) = 6$. Thus $-K_{S'}$ is nef and big. By Kawamata–Viehweg vanishing [KM98, Theorem 2.70], the Riemann–Roch theorem [Har77, Theorem V.1.6], and the Zariski decomposition $\eta = \eta^+ + \eta^-$ of a big $\mathbb{Q}$-divisor class η on S' [Laz04a, Theorem 2.3.19, Proposition 2.3.21, and Corollary 2.3.22], any member of $|m\eta|$ contains $m\eta^-$ and

$$h^0(S', \mathcal{O}_{S'}(m\eta)) = \frac{m^2}{2}((\eta^+)^2) + O(m)$$

when $m\eta$ is integral, uniformly for η in a bounded set of the classes below. For each class η below we give η^- explicitly. It is supported on curves with negative definite intersection matrix, and $\eta^+ := \eta - \eta^-$ is nef and satisfies $(\eta^+ \cdot E) = 0$ for every component E of η^-, as one checks by intersecting η^+ with the generators listed above. By [Laz04a, Theorem 2.3.19], $\eta = \eta^+ + \eta^-$ is then the Zariski decomposition.

At a point of $\operatorname{Supp} \Delta_B$, the ideal of Δ_B is (u_1, u_2^r), where u_1 is a local equation of B, u_2 restricts to a uniformizer of B, and r is the multiplicity of Δ_B at the point. The ideal (u_1, u_2^r) is complete, and so are the powers of (u_1, u_2^r) by [ZS60, Appendix 5, Theorem 2']. Thus $I^t_{\Delta_B} = \psi_* \mathcal{O}_{S'}(-t(e_1 + e_2))$ and

$$H^0(S, \mathcal{O}_S(\Theta) \otimes I^t_{\Delta_B}) = H^0(S', \mathcal{O}_{S'}(\psi^*\Theta - t(e_1 + e_2)))$$

for any divisor Θ on S and any $t \geq 0$. Moreover, $\psi^*B = B' + e_1 + e_2$ for the strict transform B' of B, so $s_B^j g$ is a section of $\mathcal{O}_S(\Theta) \otimes I_{\Delta_B}^t$ if and only if g is a section of $\mathcal{O}_S(\Theta - jB) \otimes I_{\Delta_B}^{(t-j)_+}$.

For $k \leq 2m$ we obtain $\psi^* R_{m,k} = H^0(S', \mathcal{O}_{S'}(m\eta_a))$, where $\eta_a := a\sigma + f - (a-1)_+(e_1+e_2)$ is nef for $0 \leq a \leq 2$ and $(\eta_a^2) = 2a - 2(a-1)_+^2$. For $k > 2m$, multiplication by $s_B^{k-2m} s_{R_\Delta}^{2k-4m}$ is an isomorphism onto $R_{m,k}$ from $H^0(S', \mathcal{O}_{S'}(\ell(2\sigma + f - e_1 - e_2)))$, where $2\sigma + f - e_1 - e_2$ is nef and $((2\sigma + f - e_1 - e_2)^2) = 2$. Hence $\dim R_{m,k} = \rho(a)m^2 + O(m)$ with

$$\rho(a) = \begin{cases} a, & 0 \leq a \leq 1, \\ a - (a-1)^2, & 1 \leq a \leq 2, \\ (5-2a)^2, & 2 \leq a \leq \frac{5}{2}, \end{cases} \tag{14.1}$$

and $n_m = m^3 \int_0^{5/2} \rho(a)da + O(m^2) = \frac{11}{6}m^3 + O(m^2)$. In particular, $mn_m = \frac{11}{6}m^4 + O(m^3)$, and all normalized integrals below carry the factor $6/11$.

Here and below each sum over k and over orders along a curve is a Riemann sum for an integral of a piecewise polynomial function over finitely many polygons with rational vertices. Rounding changes each rank by $O(1)$ and each order by $O(m)$, and the index pairs within bounded distance of the walls of the polygons are $O(m)$ in number and contribute $O(m^3)$ in total. Thus all errors are $O(1/m)$ after division by mn_m, uniformly in the point and in the basis.

Step 3. In this step we prove $S(R; v) \leq A_{S,B/2}(v)$ for any prime divisor v over S whose center contains a closed point $z \notin B$.

Let C be the fiber of class f through z. For $k \leq 2m$, an element of $R_{m,k}$ vanishes along C to order at most m, since $((k\sigma + mf) \cdot \sigma) = m$. For $k > 2m$, the curve C is not a component of B or of R_Δ, and an element of $H^0(S, \mathcal{O}_S(\ell(2\sigma + f)))$ vanishes along C to order at most ℓ. Thus

$$S(R; \mathrm{ord}_C) \leq \frac{6}{11}\Big(\int_0^1 ada + \int_1^2 (a - (a-1)^2)da + \int_2^{5/2} (5-2a)^3 da \Big) = \frac{6}{11}\Big(\frac{1}{2} + \frac{7}{6} + \frac{1}{8}\Big) = \frac{43}{44}.$$

Since $(C^2) = 0$, for $k \leq 2m$ the space $R^C_{m,k,j}$ is contained in $H^0(C, \mathcal{O}_C(k))$, and j takes at most $m+1$ values. For $k > 2m$, $R^C_{m,k,j}$ is contained in $(s_B^{k-2m} s_{R_\Delta}^{2k-4m})|_C \cdot H^0(C, \mathcal{O}_C(2\ell))$, and j takes at most $\ell + 1$ values. The section s_B does not vanish at z, and $s_{R_\Delta}|_C$ vanishes at z to order $\mathrm{mult}_z(R_\Delta|_C) \leq (R_\Delta \cdot f) = 2$. By **Step 1**(e) and (14.1),

$$\begin{aligned} S(R^C; z) &\leq \frac{6}{11}\Big(\int_0^2 \frac{a^2}{2} da + \int_2^{5/2} 2(5-2a)^3 da + \mathrm{mult}_z(R_\Delta|_C) \int_2^{5/2} (2a-4)(5-2a)^2 da \Big) \\ &= \frac{19}{22} + \frac{\mathrm{mult}_z(R_\Delta|_C)}{44} \leq \frac{10}{11}. \end{aligned}$$

Since $z \notin B$, the pair $(S, \frac{1}{2}B + C)$ is plt near z and the different on C vanishes near z, while $A_{S,B/2}(\mathrm{ord}_C) = 1$. Thus **Step 1**(c) with $q = z$ proves the claim. The argument of this step holds for any fiber of class f, in particular for the fibers contained in T_Δ.

Step 4. In this step we prove $S(R; \mathrm{ord}_B) = 19/88$ and, for $z \in B \setminus \operatorname{Supp} \Delta_B$,

$$S(R^B; z) \leq \frac{94 + 5\,\mathrm{mult}_z\, \iota(\Delta_B)}{88}. \tag{14.2}$$

Let $j \geq 0$ and $u := j/m$. For $k \leq 2m$, by **Step 2** the subspace $\{s \in R_{m,k} : \mathrm{ord}_B(s) \geq j\}$ is identified with s_B^j times the sections of $m\eta_{a,u}$, where

$$\eta_{a,u} := (a-u)\sigma + (1-2u)f - (a-1-u)_+(e_1+e_2), \qquad 0 \leq u \leq \min\Big\{a, \frac{1}{2}\Big\},$$

and $\{s \in R_{m,k} : \mathrm{ord}_B(s) \geq j\} = 0$ for larger u. The negative part of $\eta_{a,u}$ is $\eta^-_{a,u} = (a+u-2)_+(2\sigma - e_1 - e_2)$. If Δ_B is reduced, $2\sigma - e_1 - e_2$ is the sum of the disjoint (-1)-curves $\sigma - e_1$ and $\sigma - e_2$, and if $\Delta_B = 2z$, then $2\sigma - e_1 - e_2 = 2(\sigma - e_1) + (e_1 - e_2)$ is supported on a chain

with negative definite intersection matrix. For instance, if $0 \le a \le 2$ and $a+u>2$, then $a-1-u \ge 1-2u \ge 0$ and

$$\eta^+_{a,u} = (4-a-3u)\sigma + (1-2u)(f-e_1-e_2),$$

whence $(\eta^+_{a,u}\cdot(\sigma-e_i))=0$, $(\eta^+_{a,u}\cdot(e_1-e_2))=0$, $(\eta^+_{a,u}\cdot e_i)=1-2u\ge 0$, and $(\eta^+_{a,u}\cdot(f-e_i))=3-a-u>0$. For $k>2m$, any element of $R_{m,k}$ has B-order at least $k-2m$, and $\{s\in R_{m,k}:\operatorname{ord}_B(s)\ge k-2m+j\}$ is identified with $s_B^{k-2m+j}s_{R_\Delta}^{2k-4m}$ times the sections of $m\eta_{a,u}$, where now

$$\eta_{a,u} := (2(5-2a)-u)\sigma + ((5-2a)-2u)f - ((5-2a)-u)(e_1+e_2), \qquad 0\le u\le \frac{5-2a}{2},$$

with $\eta^-_{a,u}=u(2\sigma-e_1-e_2)$. In both cases let $n:=(\eta^+_{a,u}\cdot B')$, where $B'\sim\sigma+2f-e_1-e_2$. Then

$$n=\begin{cases} 2a+1-4u-2(a-1-u)_+-2(a+u-2)_+, & 0\le a\le 2,\\ 3(5-2a)-4u, & 2<a\le\frac52,\end{cases}$$

and the derivative of $\frac12((\eta^+_{a,u})^2)$ in u is $-n$.

Since $(\sigma-e_i)\cdot B'=1$, the strict transform of the fiber of class σ through a point $z_1\in\operatorname{Supp}\Delta_B$ intersects B' over $\iota(z_1)$. Moreover, $(e_1+e_2)|_{B'}$ maps to Δ_B, and $R_\Delta|_B=\Delta_B+\iota(\Delta_B)$. Thus any element of $R^B_{m,k,j}$ vanishes, up to $O(1)$, on

$$\begin{cases} m\big((a-1-u)_+\Delta_B+(a+u-2)_+\iota(\Delta_B)\big), & 0\le a\le 2,\\ m\big((1-u)\Delta_B+(2a-4+u)\iota(\Delta_B)\big), & 2<a\le\frac52,\end{cases}$$

and the residual sections belong to $H^0(B,\mathcal{O}_B(mn+O(1)))$. The chambers are listed in Remark B.6, and the totals of the integrals over them are

$$\iint n\,du\,da=\frac{11}{6},\qquad \iint (u+(a-2)_+)n\,du\,da=\frac{19}{48},\qquad \iint \frac{n^2}{2}\,du\,da=\frac{47}{24},$$

$$\int_0^2\int (a+u-2)_+n\,du\,da+\int_2^{5/2}\int (2a-4+u)n\,du\,da=\frac{5}{48},$$

where u runs over the ranges above. Since

$$\sum_{j\ge1}\dim\{s\in R_{m,k}:\operatorname{ord}_B(s)\ge j\}=m^3\int (u+(a-2)_+)n\,du+O(m^2)$$

by integration by parts in u, we obtain $S(R;\operatorname{ord}_B)=\frac{6}{11}\cdot\frac{19}{48}=\frac{19}{88}$, and $S_m(R;\operatorname{ord}_B)=\frac{19}{88}+O(1/m)$. For $z\in B\setminus\operatorname{Supp}\Delta_B$, **Step 1**(e) bounds the sum of the jumping numbers of ord_z on $R^B_{m,k,j}$ by the free part $\frac12(mn+O(1))^2$ plus the vanishing order at z of the fixed part times $mn+O(1)$. Thus

$$S(R^B;z)\le\frac{6}{11}\Big(\frac{47}{24}+\frac{5}{48}\operatorname{mult}_z\iota(\Delta_B)\Big)=\frac{94+5\operatorname{mult}_z\iota(\Delta_B)}{88}.$$

Step 5. In this step we let $z\in B\setminus\operatorname{Supp}\Delta_B$ and prove

$$S_m(R;\operatorname{ord}_{C_z})\le\frac{87+10\operatorname{mult}_z\iota(\Delta_B)}{88}+O(1/m). \tag{14.3}$$

Let C be the fiber of class f through z. Since $(B\cdot f)=1$ and $z\notin\operatorname{Supp}\Delta_B$, the fiber C is disjoint from $\operatorname{Supp}\Delta_B$, so the strict transform of C on S' has class f and is disjoint from the curves contracted by ψ. Let s_C be a defining section of C, $j\ge0$, and $u:=j/m$. For $k\le2m$, $\{s\in R_{m,k}:\operatorname{ord}_C(s)\ge j\}$ is identified with s_C^j times the sections of $m\theta_{a,u}$, where

$$\theta_{a,u}:=a\sigma+(1-u)f-(a-1)_+(e_1+e_2),\qquad \theta^-_{a,u}=(a+u-2)_+(2\sigma-e_1-e_2),\qquad 0\le u\le1.$$

For $k > 2m$, it is identified with $s_B^{k-2m} s_{R_\Delta}^{2k-4m} s_C^j$ times the sections of $m\theta_{a,u}$, where

$$\theta_{a,u} := 2(5-2a)\sigma + (5-2a-u)f - (5-2a)(e_1+e_2),$$
$$\theta^-_{a,u} = u(2\sigma - e_1 - e_2), \qquad 0 \le u \le 5-2a.$$

Let $n := (\theta^+_{a,u} \cdot f)$, so that $n = a - 2(a+u-2)_+$ for $0 \le a \le 2$ and $n = 2(5-2a-u)$ for $2 < a \le 5/2$. The strict transforms of the fibers of class σ through $\operatorname{Supp}\Delta_B$ intersect C at the points of $R_\Delta|_C$, so any element of $R^C_{m,k,j}$ vanishes, up to $O(1)$, on

$$m(a+u-2)_+ R_\Delta|_C \quad (0 \le a \le 2), \qquad m\big((2a-4+u)R_\Delta|_C + (a-2)B|_C\big) \quad \Big(2 < a \le \frac{5}{2}\Big).$$

The chambers are listed in Remark B.6, and the totals of the integrals over them are

$$\iint n\,du\,da = \frac{11}{6}, \qquad \iint un\,du\,da = \frac{19}{24}, \qquad \iint \frac{n^2}{2}\,du\,da = 1,$$
$$\int_0^2\!\!\int (a+u-2)_+ n\,du\,da + \int_2^{5/2}\!\!\int (2a-4+u)n\,du\,da = \frac{5}{24}, \qquad \int_2^{5/2}\!\!\int (a-2)n\,du\,da = \frac{1}{48}.$$

Thus $S(R;\operatorname{ord}_C) = \frac{19}{44}$, and by **Step 1**(e)

$$S(R^C;z) \le \frac{6}{11} + \frac{5}{44}\operatorname{mult}_z(R_\Delta|_C) + \frac{1}{88}\operatorname{mult}_z(B|_C).$$

Here $\operatorname{mult}_z(B|_C) = 1$. The fiber of class σ through z intersects B at z and $\iota(z)$, it is a component of R_Δ of multiplicity $\operatorname{mult}_{\iota(z)}\Delta_B = \operatorname{mult}_z \iota(\Delta_B)$, and it intersects C transversally at z, while no other component of R_Δ passes through z. Hence $\operatorname{mult}_z(R_\Delta|_C) = \operatorname{mult}_z \iota(\Delta_B)$ and

$$S(R^C;z) \le \frac{49 + 10\operatorname{mult}_z \iota(\Delta_B)}{88}.$$

By **Step 1**(a), there exists a basis of each $R_{m,k}$ compatible with ord_{C_z} and ord_C. For an element s of such a basis with $\operatorname{ord}_C(s) = j$,

$$\operatorname{ord}_z(s) \le j + \operatorname{ord}_z\big((s/s_C^j)|_C\big),$$

and the restrictions $(s/s_C^j)|_C$ of the basis elements of C-order j form a basis of $R^C_{m,k,j}$. Summing over the basis, we obtain $S_m(R;\operatorname{ord}_{C_z}) \le S_m(R;\operatorname{ord}_C) + S_m(R^C;z)$. By **Step 2** the quantities $S_m(R;\operatorname{ord}_C)$ and $S_m(R^C;z)$ exceed $19/44$ and $(49+10\operatorname{mult}_z\iota(\Delta_B))/88$, respectively, by at most $O(1/m)$. Since

$$\frac{19}{44} + \frac{49 + 10\operatorname{mult}_z \iota(\Delta_B)}{88} = \frac{87 + 10\operatorname{mult}_z \iota(\Delta_B)}{88},$$

this proves (14.3).

Step 6. In this step we prove $S(R;v) \le A_{S,B/2}(v)$ for any prime divisor v over S whose center contains a point $z \in B \setminus \operatorname{Supp}\Delta_B$.

The bound (14.2) exceeds 1, so **Step 1**(c) along B does not suffice, and we keep the coefficient of B in the pair. Let D_m be an m-basis type divisor compatible with v and ord_B as in **Step 1**(a), and write $D_m = b_m B + \Omega_m$ with $b_m := S_m(R;\operatorname{ord}_B)$ and $B \not\subset \operatorname{Supp}\Omega_m$. By **Step 4** and **Step 1**(a),

$$b_m = \frac{19}{88} + O(1/m), \qquad (B\cdot\Omega_m)_z \le S_m(R^B;z) \le \frac{94 + 5\operatorname{mult}_z \iota(\Delta_B)}{88} + O(1/m).$$

Since B is smooth at z, $\operatorname{mult}_z D_m = b_m + \operatorname{mult}_z \Omega_m$, and $\operatorname{mult}_z D_m = \operatorname{mult}_{C_z} D_m \le S_m(R;\operatorname{ord}_{C_z})$ by **Step 1**(a). By (14.3) and $\operatorname{mult}_z \iota(\Delta_B) \le \deg \Delta_B = 2$,

$$\begin{aligned}\operatorname{mult}_z \Omega_m &\le \frac{87 + 10\operatorname{mult}_z \iota(\Delta_B)}{88} - b_m + O(1/m)\\ &= \frac{68 + 10\operatorname{mult}_z \iota(\Delta_B)}{88} + O(1/m) \le 1 + O(1/m).\end{aligned}$$

Fix $\lambda \in (0,1) \cap \mathbb{Q}$ and let $\mu_m := \frac{1}{2} + \lambda b_m$, so that $\frac{1}{2}B + \lambda D_m = \mu_m B + \lambda \Omega_m$. For $m \gg 0$ we have $\mu_m \le 1$, $\lambda \operatorname{mult}_z \Omega_m \le 1$, and

$$\lambda(B \cdot \Omega_m)_z + \mu_m \le \frac{1}{2} + \lambda \cdot \frac{113 + 5 \operatorname{mult}_z \iota(\Delta_B)}{88} + O(1/m) \le \frac{1}{2} + \frac{123}{88} + O(1/m) < 2.$$

By **Step 1**(d), $(S, \frac{1}{2}B + \lambda D_m)$ is lc at z for $m \gg 0$. By **Step 1**(b), $\lambda S(R;v) \le A_{S,B/2}(v)$, and letting $\lambda \to 1$ proves the claim.

Step 7. In this step we let $z \in \operatorname{Supp} \Delta_B$ be a point at which the double cover $B \to \mathbb{P}^1$ of Set-up 14.2(3) is unramified, and we prove $S(R;v) \le A_{S,B/2}(v)$ for any prime divisor v over S with $z \in \operatorname{center}_S v$.

Let $r := \operatorname{mult}_z \Delta_B \in \{1,2\}$. Choose affine coordinates (x,y) on S centered at z such that the fiber of class f through z is $\{x=0\}$ and the fiber of class σ through z is $\{y=0\}$. Then $H^0(S, \mathcal{O}_S(k\sigma + mf))$ consists of the polynomials of degree at most m in x and at most k in y. The tangent line of B at z is $\{y = cx\}$ with $c \ne 0$, since $(B \cdot f) = 1$ and z is not a ramification point. Since the projection given by the ruling with fibers of class σ restricts to a closed immersion on Δ_B, no point of $\operatorname{Supp} \Delta_B$ other than z belongs to $\{y=0\}$, so $R_\Delta = r\{y=0\}$ near z. If $r=2$, the tangent direction of Δ_B is that of B. Let q_f, q_σ, and q_B be the intersection points of C_z with the strict transforms of $\{x=0\}$, $\{y=0\}$, and B. The points q_f, q_σ, and q_B are distinct. We have $A_{S,B/2}(\operatorname{ord}_{C_z}) = 2 - \frac{1}{2} = \frac{3}{2}$, the strict transform of $\frac{1}{2}B$ plus C_z is plt near C_z, and the different on C_z is $\frac{1}{2}q_B$. Thus the bounds of **Step 1**(c) on C_z are 1 at q_f, 1 at q_σ, $\frac{1}{2}$ at q_B, and 1 at any other point.

Set $(\alpha_1, \alpha_2, t) := (1, a, 0)$ for $0 \le a \le 1$, $(1, a, a-1)$ for $1 < a \le 2$, and $(5-2a, 2(5-2a), 5-2a)$ for $2 < a \le 5/2$. Any element of $R_{m,k}$ is g if $k \le 2m$ and $s_B^{k-2m} s_{R_\Delta}^{2k-4m} g$ if $k > 2m$, where g is a polynomial of degree at most $\alpha_1 m$ in x and at most $\alpha_2 m$ in y which belongs to $I_{\Delta_B}^{tm}$. Let dm be the order of g at z. Since $I_{\Delta_B} \subset \mathfrak{m}_z$, we have $d \ge t$. The homogeneous part of degree dm of g is a binary form in the span of the monomials $x^i y^{dm-i}$ with $mi_- \le i \le mi_+$, where

$$i_- := \max\{0, d - \alpha_2\}, \qquad i_+ := \min\{\alpha_1, d\}.$$

If $r=2$, the completed local ideal of Δ_B is $(y - \varphi(x), x^2)$ with $\varphi'(0) = c$, and $I_{\Delta_B}^{tm}$ pulls back with multiplicity tm at z and at q_B. Thus, after removing dm times C_z, the strict transform of $\{g=0\}$ passes through q_B with multiplicity at least $(2t-d)_+ m$, and the homogeneous part of degree dm of g is divisible by $(y - cx)^{\kappa m}$ with $\kappa := (2t-d)_+$. If $r=1$, let $\kappa := 0$. Since $y - cx$ is coprime to x and y, the binary forms in the span above divisible by $(y-cx)^{\kappa m}$ are $x^{mi_-} y^{dm - mi_+} (y - cx)^{\kappa m}$ times the forms of degree $m(i_+ - i_- - \kappa)$, and the space of such products has rank $mn + O(1)$ with $n := (i_+ - i_- - \kappa)_+$. On C_z with homogeneous coordinates $[x:y]$, the monomial $x^i y^{dm-i}$ vanishes to order i at q_f and to order $dm - i$ at q_σ, and $y - cx$ vanishes only at q_B. For $k > 2m$, the factor s_B^{k-2m} has order $(a-2)m$ at z with tangent direction q_B, and $s_{R_\Delta}^{2k-4m}$ has order $2r(a-2)m$ at z with tangent direction q_σ, the direction of $\{y=0\}$. By **Step 1**(e), the rank, the first order, and the secondary orders at q_B, q_f, and q_σ are bounded, after division by m^2, by the integrands

$$n, \quad \big(d + (1+2r)(a-2)_+\big)n, \quad \big(\kappa + (a-2)_+\big)n + \frac{n^2}{2}, \quad i_- n + \frac{n^2}{2}, \quad \big(d - i_+ + 2r(a-2)_+\big)n + \frac{n^2}{2},$$

integrated over $d \ge t$ with $n > 0$. At any other point of C_z the secondary integrand is at most $\frac{n^2}{2}$. The integrals over the ranges $0 \le a \le 1$, $1 \le a \le 2$, and $2 \le a \le \frac{5}{2}$ are

$$\Big(\frac{1}{2}, \frac{5}{12}, \frac{1}{8}, \frac{1}{4}, \frac{1}{6}\Big), \quad \Big(\frac{4}{3}, \frac{11}{6}, \frac{13}{24}, \frac{17}{24}, \frac{9}{8}\Big), \quad \Big(\frac{1}{4}, \frac{41}{96}, \frac{11}{96}, \frac{5}{48}, \frac{7}{24}\Big)$$

if $r=1$, and

$$\Big(\frac{1}{2}, \frac{5}{12}, \frac{1}{8}, \frac{1}{4}, \frac{1}{6}\Big), \quad \Big(\frac{7}{6}, \frac{5}{3}, \frac{1}{2}, \frac{29}{48}, 1\Big), \quad \Big(\frac{1}{6}, \frac{17}{48}, \frac{1}{12}, \frac{1}{16}, \frac{1}{4}\Big)$$

if $r = 2$. After multiplication by $6/11$,

$$r = 1: \quad S(R; \mathrm{ord}_{C_z}) \le \frac{257}{176}, \quad S(R^{C_z}; q_B) \le \frac{75}{176}, \quad S(R^{C_z}; q_f) \le \frac{51}{88}, \quad S(R^{C_z}; q_\sigma) \le \frac{19}{22},$$
$$r = 2: \quad S(R; \mathrm{ord}_{C_z}) \le \frac{117}{88}, \quad S(R^{C_z}; q_B) \le \frac{17}{44}, \quad S(R^{C_z}; q_f) \le \frac{1}{2}, \quad S(R^{C_z}; q_\sigma) \le \frac{17}{22}.$$

The first bounds are less than $\frac{3}{2}$, the bounds at q_B are less than $\frac{1}{2}$, the bounds at q_f and q_σ are less than 1, and at any other point of C_z the bound is at most the bound at q_B. The center on the blow-up of z of any prime divisor v over S with $z \in \mathrm{center}_S\, v$ contains a point of C_z. Thus **Step 1**(c) at the points of C_z proves the claim.

Step 8. In this step we let $z \in \mathrm{Supp}\, \Delta_B$ be a ramification point of the double cover $B \to \mathbb{P}^1$ of Set-up 14.2(3), and we bound the first order of R along the exceptional curve C'_z of a weighted blow-up of z.

The curve B is tangent at z to the fiber of class σ through z. Since the projection given by the ruling with fibers of class σ restricts to a closed immersion on Δ_B, we have $\mathrm{mult}_z\, \Delta_B = 1$ and $\Delta_B = z + z'$ with $z' \ne z$. Since $(B \cdot f) = 1$ and $(B \cdot \sigma) = 2$, the curve B is the graph of a morphism $\mathbb{P}^1 \to \mathbb{P}^1$ of degree two. Moving the ramification points of $B \to \mathbb{P}^1$ and their images to 0 and ∞ and rescaling, we choose affine coordinates (x, y) on S with $\{x = 0\}$ the fiber of class f through z such that $B = \{y = x^2\}$, $z = (0, 0)$, and the other ramification point of B is the point z_∞ at which x and y are both ∞.

Let C'_z be the exceptional curve of the weighted blow-up of z with weights $(1, 2)$ for (x, y). Then $A_S(\mathrm{ord}_{C'_z}) = 3$ [JM12, Proposition 5.1] and $\mathrm{mult}_{C'_z} B = 2$, so $A_{S,B/2}(\mathrm{ord}_{C'_z}) = 2$. The function $w := y/x^2$ is a coordinate on C'_z. The toric charts of the weighted blow-up show that the blown-up surface is smooth along C'_z except for one singular point of type $\frac{1}{2}(1, 1)$ at the point where w is ∞, and that the strict transform of B is $\{w = 1\}$ near C'_z, transversal to C'_z. Let q'_0, q'_B, and q'_∞ be the points of C'_z where w is 0, 1, and ∞. The strict transform of $\frac{1}{2}B$ plus C'_z is plt near C'_z, and the different on C'_z is $\frac{1}{2}(q'_\infty + q'_B)$ by [Pro01, Theorem 2.2.4(i)]. Thus the bounds of **Step 1**(c) on C'_z are 1 at q'_0, $\frac{1}{2}$ at q'_B and at q'_∞, and 1 at any other point.

We bound $S(R; \mathrm{ord}_{C'_z})$ by moving z' to z_∞. For $\tau \in \mathbb{C}^*$ let g_τ be the automorphism $(x, y) \to (\tau^{-1}x, \tau^{-2}y)$ of S. It preserves both rulings, B, z, z_∞, and $\mathrm{ord}_{C'_z}$. Let $\Delta_\tau := z + g_\tau(z')$ for $\tau \in \mathbb{C}^*$ and $\Delta_0 := z + z_\infty$. For any $\tau \in \mathbb{C}$, both ruling projections restrict to closed immersions on Δ_τ, and we let $R(\tau)$ be the series of Set-up 14.2 defined with Δ_τ in place of Δ_B. For $\tau \ne 0$, g_τ^* maps $R(\tau)_{m,k}$ isomorphically onto $R(1)_{m,k} = R_{m,k}$ and preserves the order along C'_z. As $\tau \to 0$, $g_\tau(z') \to z_\infty$. Near $\tau = 0$ the support of Δ_τ consists of two distinct points moving in a family, so the subschemes defined by $I^t_{\Delta_\tau}$ form a flat family of finite subschemes, and $R(\tau)_{m,k}$ for $k \le 2m$ is the kernel of a family of linear maps from $H^0(S, \mathcal{O}_S(k\sigma + mf))$. For $k > 2m$, $R(\tau)_{m,k}$ is $s_B^{k-2m} s_{R_{\Delta_\tau}}^{2k-4m}$ times such a kernel in $H^0(S, \mathcal{O}_S(\ell(2\sigma + f)))$, and the weighted order at z of $s_B^{k-2m} s_{R_{\Delta_\tau}}^{2k-4m}$ is $6(k - 2m)$ for any τ, since B and the fiber $\{y = 0\}$ both have weighted order 2 at z. Fix $e \ge 0$ and restrict the maps above to the fixed subspace of sections of weighted order at least e. The rank of a family of linear maps is lower semicontinuous, so the dimension of the subspace of $R(\tau)_{m,k}$ of weighted order at least e is upper semicontinuous in τ. In other words, the dimension at $\tau = 0$ is at least the dimension at $\tau \ne 0$, so the fixed weighted-order subspaces can only become larger in the limit. Since the sum of the jumping numbers of $\mathrm{ord}_{C'_z}$ on a space V of sections is $\sum_{e \ge 1} \dim\{s \in V : \mathrm{ord}_{C'_z}(s) \ge e\}$, we obtain

$$S_m(R; \mathrm{ord}_{C'_z}) \le S_m(R(0); \mathrm{ord}_{C'_z}),$$

where both sides are normalized by mn_m. Thus the limit series $R(0)$ bounds the first order of R from above.

With (α_1, α_2, t) as in **Step 7**, write each element of $R(0)_{m,k}$ as g if $k \le 2m$ and as $s_B^{k-2m} s_{R_{\Delta_0}}^{2k-4m} g$ if $k > 2m$. Since the ideals of z and z_∞ are monomial, the space of the factors g is spanned by

the monomials $x^{i_1}y^{i_2}$ with

$$0 \le i_1 \le \alpha_1 m, \qquad 0 \le i_2 \le \alpha_2 m, \qquad i_1 + i_2 \ge tm, \qquad (\alpha_1 m - i_1) + (\alpha_2 m - i_2) \ge tm,$$

where the last condition is the condition at z_∞ in the coordinates (x^{-1}, y^{-1}). The weighted order at z of $x^{i_1}y^{i_2}$ is $i_1 + 2i_2$, and for $k > 2m$ the fixed factor adds $6(a-2)m$. Integrating $i_1 + 2i_2 + 6(a-2)_+$ over the scaled polygons yields $\frac{7}{12}$, $\frac{7}{3}$, and $\frac{7}{16}$ on the three ranges of a. Hence

$$S(R; \mathrm{ord}_{C'_z}) \le \frac{6}{11}\Big(\frac{7}{12} + \frac{7}{3} + \frac{7}{16}\Big) = \frac{161}{88} < 2 = A_{S,B/2}(\mathrm{ord}_{C'_z}).$$

Step 9. In this step we bound the secondary orders on C'_z at the point z of **Step 8** and prove $S(R; v) \le A_{S,B/2}(v)$ for any prime divisor v over S with $z \in \mathrm{center}_S\, v$.

We bound the images of R itself, discarding only the conditions at z'. The local ideal of Δ_B at z is $\mathfrak{m}_z$. Any element of $R_{m,k}$ is g if $k \le 2m$ and $s_B^{k-2m} s_{R_\Delta}^{2k-4m} g$ if $k > 2m$, where g has degree at most $\alpha_1 m$ in x and at most $\alpha_2 m$ in y and belongs to $\mathfrak{m}_z^{tm}$. Since $\mathfrak{m}_z^{tm}$ is a monomial ideal, the weighted homogeneous part of g of weighted degree dm is spanned by the monomials $x^{dm-2i}y^i$ with $mi_- \le i \le mi_+$, where

$$i_- := \max\Big\{0, \frac{d-\alpha_1}{2}\Big\}, \qquad i_+ := \min\Big\{\alpha_2, \frac{d}{2}, d-t\Big\}, \qquad n := (i_+ - i_-)_+.$$

On C'_z the monomial $x^{dm-2i}y^i$ restricts to w^i, whose $\mathbb{Q}$-divisor is $iq'_0 + (\frac{dm}{2} - i)q'_\infty$. For $k > 2m$, the fixed factor s_B^{k-2m} contributes $(a-2)m$ at q'_B, and $s_{R_\Delta}^{2k-4m}$ contributes $2(a-2)m$ at q'_0, since near z the curve R_Δ is the fiber $\{y = 0\}$, whose strict transform is $\{w = 0\}$. By **Step 1**(e), the secondary orders at q'_B, q'_∞, and q'_0 are bounded, after division by m^2, by the integrands

$$(a-2)_+ n + \frac{n^2}{2}, \qquad \Big(\frac{d}{2} - i_+\Big)n + \frac{n^2}{2}, \qquad \big(i_- + 2(a-2)_+\big)n + \frac{n^2}{2},$$

integrated over $d \ge t$ with $n > 0$, and at any other point of C'_z by $\frac{n^2}{2}$. Here $\frac{d}{2} - i_+$ includes the fractional order at the point q'_∞ of index two. The integrals over the ranges $0 \le a \le 1$, $1 \le a \le 2$, and $2 \le a \le \frac{5}{2}$ are

$$\Big(\frac{17}{192}, \frac{1}{8}, \frac{1}{6}\Big), \quad \Big(\frac{19}{64}, \frac{17}{48}, \frac{9}{8}\Big), \quad \Big(\frac{7}{96}, \frac{5}{96}, \frac{7}{24}\Big).$$

After multiplication by $6/11$,

$$S(R^{C'_z}; q'_B) \le \frac{1}{4} < \frac{1}{2}, \qquad S(R^{C'_z}; q'_\infty) \le \frac{51}{176} < \frac{1}{2}, \qquad S(R^{C'_z}; q'_0) \le \frac{19}{22} < 1,$$

and at any other point of C'_z the bound is at most $\frac{1}{4}$. Together with the bound of **Step 8**, **Step 1**(c) at the points of C'_z proves the claim, since the center on the weighted blow-up of any prime divisor v over S with $z \in \mathrm{center}_S\, v$ contains a point of C'_z.

Step 10. In this step we conclude the proof.

Let v be a prime divisor over S and z a closed point of $\mathrm{center}_S\, v$. If $z \notin B$, then **Step 3** applied to the fiber of class f through z proves $S(R; v) \le A_{S,B/2}(v)$. If $z \in B \setminus \mathrm{Supp}\, \Delta_B = B \setminus T_\Delta$, then **Step 6** proves it. If $z \in \mathrm{Supp}\, \Delta_B = B \cap T_\Delta$, then **Step 7** or **Step 9** proves it, according as the double cover $B \to \mathbb{P}^1$ is unramified or ramified at z. Together with **Step 2**, the four cases above prove the lemma. □

We next treat a surface series on the Hirzebruch surface $\mathbb{F}_2$.

Set-up 14.4. Let S, σ, f, B, q_σ, q_B, C_σ, C_B, Δ_2, q_0, C_{q_0}, and $R_{m,k}$ be as follows.

(1) $S := \mathbb{F}_2$ with negative section σ and fiber class f, and $B \subset S$ is a smooth curve of class $\sigma + 3f$.
(2) $q_\sigma \in \sigma \setminus B$ and $q_B \in B \setminus \sigma$ are points on distinct fibers C_σ and C_B, $\Delta_2 := q_\sigma + q_B$, and I_{Δ_2} is the ideal sheaf of $\{q_\sigma, q_B\}$.
(3) $q_0 := \sigma \cap B$, and C_{q_0} is the fiber through q_0.

(4) For a divisor $D \geq 0$ on S, s_D is the canonical section of $\mathcal{O}_S(D)$.
(5) For integers $m \geq 1$ and $0 \leq k \leq \lfloor 5m/2 \rfloor$, $R_{m,k} \subset H^0(S, \mathcal{O}_S(k\sigma + (m+k)f))$ is the subspace

$$R_{m,k} := \begin{cases} H^0(S, \mathcal{O}_S(k\sigma + (m+k)f)), & k \leq m, \\ s_\sigma^{k-m} H^0(S, \mathcal{O}_S(m\sigma + (m+k)f) \otimes I_{\Delta_2}^{k-m}), & m < k \leq 2m, \\ s_\sigma^{2k-3m} s_B^{k-2m} s_{C_\sigma + C_B}^{2k-4m} H^0(S, \mathcal{O}_S(\ell(\sigma + 3f)) \otimes I_{\Delta_2}^{\ell}), & 2m < k \leq \lfloor 5m/2 \rfloor, \end{cases}$$

where $\ell := 5m - 2k$.

The set-up is used in the proof of Theorem 14.1, where q_σ and q_B are the images of the points of $M \cap \Gamma$. The points q_σ and q_B are on distinct fibers unless $M \cap \Gamma$ is non-reduced, which is case (3) of Theorem 14.1 and is treated there by degeneration. The fiber C_{q_0} is the fiber over the point of incidence type 3 on L (Proposition 12.10), so the application there does not need Lemma 14.5(1) for prime divisors whose center is contained in C_{q_0}.

Lemma 14.5. *Notation as in Set-up 14.4. Let n_m, $S(R; v)$, and $A_{S,B/2}(v)$, and for a curve $C \subset S$ and a point $p \in C$ the secondary expected vanishing order $S(R^C; p)$, be as in Definition 7.3. Then the following hold.*

(1) $n_m = \frac{11}{6}m^3 + O(m^2)$, and $S(R; v) \leq A_{S,B/2}(v)$ for every prime divisor v over S with $\operatorname{center}_S v \not\subset C_{q_0}$.
(2) Let $C \neq C_{q_0}$ be a fiber. Then $S(R; \operatorname{ord}_C) = 59/88$ if $C \notin \{C_\sigma, C_B\}$, $S(R; \operatorname{ord}_C) = 17/22$ if $C = C_\sigma$, and $S(R; \operatorname{ord}_C) = 39/44$ if $C = C_B$. Moreover, $S(R^C; C \cap \sigma) \leq 1$, $S(R^C; C \cap B) \leq 29/88$, and $S(R^C; p) \leq 7/22$ (resp. $S(R^C; p) \leq 3/11$) for every point $p \in C \backslash (\sigma \cup B)$ if $C \neq C_B$ (resp. $C = C_B$).

Proof. **Step 1.** In this step we blow up q_σ and q_B, describe the effective cone of the blow-up, and compute n_m.

By [Har77, Propositions V.2.3 and V.2.9 and Lemma V.2.10], $(\sigma^2) = -2$, $(\sigma \cdot f) = 1$, $(f^2) = 0$, and $K_S \sim -2\sigma - 4f$. Thus $(\sigma \cdot B) = 1$ and $(f \cdot B) = 1$. Since B is smooth, the irreducible components of B are pairwise disjoint. Since $(f \cdot (B - f)) = 1$ and $(\sigma \cdot (B - \sigma)) = 3$, neither a fiber nor σ is a component of B. Hence any fiber C intersects σ transversally in one point and B transversally in one point, and $C \cap \sigma = C \cap B$ if and only if $C = C_{q_0}$.

Let $g : \tilde{S} \to S$ be the blow-up of q_σ and q_B, with exceptional curve e_1 over q_σ and e_2 over q_B. We write σ and f also for $g^*\sigma$ and g^*f. Since $q_\sigma \in \sigma \cap C_\sigma$, $q_B \in C_B \backslash \sigma$, and $C_\sigma \neq C_B$, the strict transforms of σ, C_σ, and C_B are

$$\tilde{\sigma} = \sigma - e_1, \qquad \tilde{C}_\sigma = f - e_1, \qquad \tilde{C}_B = f - e_2,$$

and the intersection matrix of $(\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2)$ is

$$\Theta = \begin{pmatrix} -3 & 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}.$$

We show that the effective cone of $\tilde{S}$ is generated by $\tilde{\sigma}$, $\tilde{C}_\sigma$, e_1, $\tilde{C}_B$, and e_2. Let $C' \subset \tilde{S}$ be an irreducible curve different from $\tilde{\sigma}$, $\tilde{C}_\sigma$, e_1, $\tilde{C}_B$, and e_2. Then $g(C')$ is a curve different from σ, C_σ, and C_B. If $g(C')$ is a fiber, then $C' \equiv f = \tilde{C}_\sigma + e_1$. Otherwise, $g(C') \equiv \alpha\sigma + \beta f$ with $\alpha = (g(C') \cdot f) \geq 1$ and $\beta - 2\alpha = (g(C') \cdot \sigma) \geq 0$, and $C' \equiv \alpha\sigma + \beta f - \mu_1 e_1 - \mu_2 e_2$, where μ_1 and μ_2 are the multiplicities of $g(C')$ at q_σ and q_B [Har77, Propositions V.3.2 and V.3.6]. Since $C' \neq \tilde{C}_\sigma, \tilde{C}_B$, we have $(C' \cdot \tilde{C}_\sigma) = \alpha - \mu_1 \geq 0$ and $(C' \cdot \tilde{C}_B) = \alpha - \mu_2 \geq 0$. Thus

$$C' \equiv \alpha\tilde{\sigma} + \mu_2\tilde{C}_B + (\beta - \mu_2)\tilde{C}_\sigma + (\alpha - \mu_1 + \beta - \mu_2)e_1,$$

and all coefficients are non-negative since $\beta - \mu_2 \geq 2\alpha - \alpha \geq 0$. Hence an $\mathbb{R}$-divisor D on $\tilde{S}$ is nef if and only if $(D \cdot \Gamma) \geq 0$ for $\Gamma \in \{\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2\}$.

By [Har77, Proposition V.3.3], $K_{\tilde{S}} \sim -2\sigma - 4f + e_1 + e_2$, so

$$-\Big(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma}\Big) \equiv \frac{3}{2}\sigma + 4f - \frac{1}{2}e_1 - e_2.$$

The intersection numbers of $-(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma})$ with $\tilde{\sigma}$, $\tilde{C}_\sigma$, e_1, $\tilde{C}_B$, and e_2 are $\frac{1}{2}$, 1, $\frac{1}{2}$, $\frac{1}{2}$, and 1, and $((K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma})^2) = \frac{25}{4} > 0$. Thus $-(K_{\tilde{S}} + \frac{1}{2}\tilde{\sigma})$ is nef and big. Since $\tilde{\sigma}$ is a smooth curve, $(\tilde{S}, \frac{1}{2}\tilde{\sigma})$ is klt [KM98, Corollary 2.31(3)]. Therefore, for any nef divisor L on $\tilde{S}$, Kawamata–Viehweg vanishing [KM98, Theorem 2.70] implies $H^i(\tilde{S}, \mathcal{O}_{\tilde{S}}(L)) = 0$ for $i > 0$, and the Riemann–Roch theorem [Har77, Theorem V.1.6] implies

$$h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(L)) = \frac{1}{2}(L \cdot (L - K_{\tilde{S}})) + 1.$$

For $m \geq 1$ and $0 \leq k \leq \lfloor 5m/2 \rfloor$ let

$$\tilde{L}_{m,k} := \begin{cases} k\sigma + (m+k)f & \text{if } 0 \leq k \leq m, \\ m\sigma + (m+k)f - (k-m)(e_1+e_2) & \text{if } m < k \leq 2m, \\ \ell(\sigma + 3f - e_1 - e_2) & \text{if } 2m < k \leq \lfloor 5m/2 \rfloor, \end{cases}$$

with $\ell = 5m - 2k$. For a regular function y near q_σ (resp. q_B), $\operatorname{ord}_{e_1}(g^*y)$ (resp. $\operatorname{ord}_{e_2}(g^*y)$) is the multiplicity of $\operatorname{div}(y)$ at q_σ (resp. q_B) [Har77, Proposition V.3.6]. Thus, for $w \geq 0$, $g_*\mathcal{O}_{\tilde{S}}(-w(e_1+e_2))$ is $\mathfrak{m}_{q_\sigma}^w$ near q_σ and $\mathfrak{m}_{q_B}^w$ near q_B, that is, $g_*\mathcal{O}_{\tilde{S}}(-w(e_1+e_2)) = I_{\Delta_2}^w$. Since $g_*\mathcal{O}_{\tilde{S}}(-w(e_1+e_2)) = I_{\Delta_2}^w$ and multiplication by s_σ^{k-m} and by $s_\sigma^{2k-3m} s_B^{k-2m} s_{C_\sigma + C_B}^{2k-4m}$ is injective, $\dim R_{m,k} = h^0(\tilde{S}, \mathcal{O}_{\tilde{S}}(\tilde{L}_{m,k}))$. The intersection numbers of $\tilde{L}_{m,k}$ with $(\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2)$ are

$$(m-k, k, 0, k, 0), \qquad (0, 2m-k, k-m, 2m-k, k-m), \qquad (0, 0, \ell, 0, \ell)$$

on the three ranges of k, so $\tilde{L}_{m,k}$ is nef. Moreover, $(\tilde{L}_{m,k}^2) = 2m^2\rho(k/m)$, where

$$\rho(a) := \begin{cases} a & \text{if } 0 \leq a \leq 1, \\ a - (a-1)^2 & \text{if } 1 < a \leq 2, \\ (5-2a)^2 & \text{if } 2 < a \leq 5/2, \end{cases}$$

and $(\tilde{L}_{m,k} \cdot K_{\tilde{S}}) = O(m)$. Thus $\dim R_{m,k} = m^2\rho(k/m) + O(m)$ uniformly in k. Since ρ is continuous and piecewise polynomial and

$$\int_0^{5/2} \rho(a)da = \frac{1}{2} + \frac{7}{6} + \frac{1}{6} = \frac{11}{6},$$

we obtain $n_m = \frac{11}{6}m^3 + O(m^2)$.

Step 2. In this step we fix a fiber $C \neq C_{q_0}$ and describe the filtration of $R_{m,k}$ by the order along C.

Let $\tilde{C} \subset \tilde{S}$ be the strict transform of C. If $C = C_\sigma$ (resp. $C = C_B$), let $e_C := e_1$ (resp. $e_C := e_2$). If $C \notin \{C_\sigma, C_B\}$, then $C \cap \{q_\sigma, q_B\} = \emptyset$, $\tilde{C} \equiv f$, and we set $e_C := 0$. Let $\tilde{p}_\sigma$ and $\tilde{p}_B$ be the points of $\tilde{C}$ over $C \cap \sigma$ and $C \cap B$. For real numbers $0 \leq a \leq 5/2$ and $u \geq 0$ let

$$\epsilon_{a,u} := \begin{cases} 0 & \text{if } 0 \leq a \leq 1, \\ \max(a - 1 - u, 0) & \text{if } 1 < a \leq 2, \\ \max(5 - 2a - u, 0) & \text{if } 2 < a \leq 5/2, \end{cases}$$

and let $D_{a,u}$ be the $\mathbb{R}$-divisor class

$$D_{a,u} := \begin{cases} a\sigma + (a+1-u)f & \text{if } 0 \leq a \leq 1, \\ \sigma + (a+1-u)f - (a-1)(e_1+e_2) + (a - 1 - \epsilon_{a,u})e_C & \text{if } 1 < a \leq 2, \\ (5-2a)(\sigma + 3f - e_1 - e_2) - uf + (5 - 2a - \epsilon_{a,u})e_C & \text{if } 2 < a \leq 5/2. \end{cases}$$

Thus the coefficient of e_C in $D_{a,u}$ is $-\epsilon_{a,u}$ when $C \in \{C_\sigma, C_B\}$ and $a > 1$. On the three ranges of a, let $\gamma_\sigma(a)$ be 0, $a-1$, and $2a-3$, and let $\gamma_B(a)$ be 0, 0, and $a-2$. Let $\kappa(a) := 2a-4$ if $C \in \{C_\sigma, C_B\}$ and $2 < a \le 5/2$, and $\kappa(a) := 0$ otherwise. We set $\tau := u + \kappa(a)$.

Fix integers $m \ge 1$ and $0 \le k \le \lfloor 5m/2 \rfloor$, and let $a := k/m$. Let

$$(\varphi_{m,k}, w) := \begin{cases} (1,0) & \text{if } 0 \le k \le m, \\ (s_\sigma^{k-m}, k-m) & \text{if } m < k \le 2m, \\ (s_\sigma^{2k-3m} s_B^{k-2m} s_{C_\sigma + C_B}^{2k-4m}, \ell) & \text{if } 2m < k \le \lfloor 5m/2 \rfloor, \end{cases}$$

and let $\mathcal{L}_{m,k} := \mathcal{O}_S(k\sigma + (m+k)f)$ be the line bundle of Definition 7.3. Thus $R_{m,k} = \varphi_{m,k} H^0(S, \mathcal{L}_{m,k}(-\operatorname{div}\varphi_{m,k}) \otimes I_{\Delta_2}^w)$ and $\operatorname{ord}_C(\varphi_{m,k}) = m\kappa(a)$. For an integer $j \ge 0$ let

$$F^j R_{m,k} := \{r \in R_{m,k} : \operatorname{ord}_C(r) \ge j\}.$$

Then $F^j R_{m,k} = R_{m,k}$ if $j \le m\kappa(a)$. Let $j \ge m\kappa(a)$ and $i := j - m\kappa(a)$. If $C \in \{C_\sigma, C_B\}$, let $q_C := C \cap \{q_\sigma, q_B\}$ and let t be a local equation of C at q_C. Since the associated graded ring of $\mathcal{O}_{S,q_C}$ with respect to $\mathfrak{m}_{q_C}$ is a polynomial ring, $\operatorname{ord}_{q_C}(t^i y) = i + \operatorname{ord}_{q_C}(y)$ for $y \in \mathcal{O}_{S,q_C}$, hence

$$(\mathfrak{m}_{q_C}^w : t^i) = \mathfrak{m}_{q_C}^{\max(w-i,0)}, \qquad \text{and} \qquad \max(w-i,0) = m\epsilon_{a,i/m}.$$

Thus the elements of $F^j R_{m,k}$ are exactly the sections $r = \varphi_{m,k} s_C^i r_1$, where r_1 is a section of $\mathcal{L}_{m,k}(-\operatorname{div}\varphi_{m,k} - iC)$ vanishing to order at least $m\epsilon_{a,i/m}$ at q_C (if $C \in \{C_\sigma, C_B\}$) and to order at least w at the points of $\{q_\sigma, q_B\} \backslash C$. For such r let

$$\Gamma_r := g^* \operatorname{div}(r_1) - m\epsilon_{a,i/m} e_C - w(e_1 + e_2 - e_C).$$

Then $\Gamma_r \ge 0$ is a divisor in the class $mD_{a,i/m}$, and $r \to \Gamma_r$ is a bijection from the set of non-zero elements of $F^j R_{m,k}$ up to scalars onto the linear system $|mD_{a,i/m}|$. If $\operatorname{ord}_C(r) = j$, then $\tilde{C} \not\subset \operatorname{Supp}\Gamma_r$, and since $\tilde{C} \cap (e_1 + e_2 - e_C) = \emptyset$, the image of r in $R^C_{m,k,j}$ has divisor

$$\operatorname{div}\left((\varphi_{m,k} s_C^{-m\kappa(a)})|_C\right) + \left(\Gamma_r + m\epsilon_{a,i/m} e_C\right)|_{\tilde{C}} \tag{14.4}$$

on $C \cong \tilde{C}$. Since C intersects σ and B transversally and $C_\sigma \cap C_B = \emptyset$, the first term of (14.4) is $m\gamma_\sigma(a)[C \cap \sigma] + m\gamma_B(a)[C \cap B]$, where $[x]$ denotes the divisor of a point $x \in C$.

Step 3. In this step we compute the Zariski decomposition of $D_{a,u}$.

When $D_{a,u}$ is pseudo-effective, let $D_{a,u} = P_{a,u} + N_{a,u}$ be the Zariski decomposition [Laz04a, Theorem 2.3.19], and let

$$\begin{aligned} d &:= (P_{a,u} \cdot \tilde{C}), \\ \zeta &:= \gamma_\sigma(a) + \operatorname{mult}_{\tilde{p}_\sigma}\left((N_{a,u} + \epsilon_{a,u} e_C)|_{\tilde{C}}\right), \\ b &:= \gamma_B(a) + \operatorname{mult}_{\tilde{p}_B}\left((N_{a,u} + \epsilon_{a,u} e_C)|_{\tilde{C}}\right). \end{aligned}$$

Table 23 lists, for each type of fiber, triangles in the (a,u)-plane, and for each triangle a set $I \subset \{\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2\}$ and the functions d, ζ, and b on the triangle. In each row, let Θ_I be the intersection matrix of the curves in I, a principal submatrix of Θ, and let

$$N_{a,u} := \sum_{\Gamma \in I} x_\Gamma \Gamma, \qquad (x_\Gamma)_{\Gamma \in I} := \Theta_I^{-1}\left((D_{a,u} \cdot \Gamma)\right)_{\Gamma \in I}.$$

Each Θ_I is negative definite. The coefficients x_Γ and the intersection numbers of $D_{a,u} - N_{a,u}$ with $\tilde{\sigma}$, $\tilde{C}_\sigma$, e_1, $\tilde{C}_B$, and e_2 are affine functions of (a,u) on each triangle. The coefficients x_Γ are listed in Table 23, and a direct calculation at the three vertices shows that $x_\Gamma \ge 0$ for $\Gamma \in I$ and $((D_{a,u} - N_{a,u}) \cdot \Gamma) \ge 0$ for $\Gamma \in \{\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2\}$. Hence $D_{a,u} - N_{a,u}$ is nef by **Step 1**, $((D_{a,u} - N_{a,u}) \cdot \Gamma) = 0$ for $\Gamma \in I$, and so $N_{a,u}$ is the negative part of $D_{a,u}$ [Laz04a, Theorem 2.3.19]. In the rows with $0 \le a \le 1$, $N_{a,u} = \left(\frac{a+u-1}{2}\right)_+ g^*\sigma$ is the pullback of the negative part on S. Substituting $N_{a,u}$ into the definitions of d, ζ, and b, a direct calculation yields the last three

columns of Table 23. For instance, for $C \notin \{C_\sigma, C_B\}$ and the triangle with vertices $(1,0)$, $(2,0)$, $(\frac{3}{2},\frac{3}{2})$, we have $(D_{a,u}\cdot\tilde{\sigma}) = -u$, so $N_{a,u} = \frac{u}{3}\tilde{\sigma}$, $d = 1-\frac{u}{3}$, and $\zeta = (a-1)+\frac{u}{3}$.

TABLE 23. The negative part $N_{a,u} = \sum_{\Gamma\in I} x_\Gamma \Gamma$ of $D_{a,u}$, with the coefficients x_Γ of $N_{a,u}$ in the order of I, and the functions d, ζ, and b on the triangles of the (a,u)-plane, for each type of fiber $C \neq C_{q_0}$.

vertices in (a,u)	I	x_Γ, $\Gamma\in I$	d	ζ	b
any fiber $C \neq C_{q_0}$					
$(0,0),(1,0),(0,1)$	$\emptyset$		a	0	0
$(1,0),(1,2),(0,1)$	$\tilde{\sigma}, e_1$	$\frac{a+u-1}{2}, \frac{a+u-1}{2}$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$	0
$C \notin \{C_\sigma, C_B\}$					
$(1,0),(2,0),(\frac{3}{2},\frac{3}{2})$	$\tilde{\sigma}$	$\frac{u}{3}$	$1-\frac{u}{3}$	$a+\frac{u}{3}-1$	0
$(1,0),(\frac{3}{2},\frac{3}{2}),(1,2)$	$\tilde{\sigma}, e_1$	$\frac{u-a+1}{2}, \frac{u-3a+3}{2}$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$	0
$(2,0),(2,2),(\frac{3}{2},\frac{3}{2})$	$\tilde{\sigma}, \tilde{C}_B$	$\frac{a+u-2}{2}, \frac{3a+u-6}{2}$	$2-\frac{a+u}{2}$	$\frac{3a+u}{2}-2$	0
$(2,2),(1,2),(\frac{3}{2},\frac{3}{2})$	$\tilde{\sigma}, e_1, \tilde{C}_B$	$u-1, u-a, a+u-3$	$2-u$	$a+u-2$	0
$(2,0),(\frac{5}{2},0),(2,2)$	$\tilde{\sigma}, \tilde{C}_B$	$\frac{u}{2}, \frac{u}{2}$	$5-2a-\frac{u}{2}$	$2a+\frac{u}{2}-3$	$a-2$
$C = C_\sigma$					
$(1,0),(2,0),(2,1)$	$\emptyset$		$2-a+u$	$2a-u-2$	0
$(1,0),(2,1),(1,2)$	$\tilde{\sigma}, e_1$	$\frac{u-a+1}{2}, \frac{u-a+1}{2}$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$	0
$(2,1),(2,2),(1,2)$	$\tilde{\sigma}, e_1, \tilde{C}_B$	$u-1, u-1, a+u-3$	$2-u$	$a+u-2$	0
$(2,0),(\frac{5}{2},0),(2,1)$	$\emptyset$		u	$2-u$	$a-2$
$(\frac{5}{2},0),(2,2),(2,1)$	$\tilde{\sigma}, e_1, \tilde{C}_B$	$2a+u-5, 2a+u-5, 2a+u-5$	$10-4a-u$	$4a+u-8$	$a-2$
$C = C_B$					
$(1,0),(2,0),(2,1)$	$\tilde{\sigma}$	$\frac{u}{3}$	$2-a+\frac{2u}{3}$	$a+\frac{u}{3}-1$	$a-u-1$
$(1,0),(2,1),(2,3)$	$\tilde{\sigma}$	$\frac{u}{3}$	$1-\frac{u}{3}$	$a+\frac{u}{3}-1$	0
$(1,0),(2,3),(1,2)$	$\tilde{\sigma}, e_1$	$\frac{u-a+1}{2}, \frac{u-3a+3}{2}$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$	0
$(2,0),(\frac{5}{2},0),(2,1)$	$\tilde{\sigma}$	$\frac{u}{3}$	$\frac{2u}{3}$	$2a+\frac{u}{3}-3$	$3-a-u$
$(\frac{5}{2},0),(2,3),(2,1)$	$\tilde{\sigma}$	$\frac{u}{3}$	$5-2a-\frac{u}{3}$	$2a+\frac{u}{3}-3$	$a-2$

For each type of fiber and each $0 \le a \le 5/2$, the points (a,u) of the triangles of Table 23 form the interval $0 \le u \le u_{\max}(a)$, where

$$u_{\max}(a) := \begin{cases} a+1 & \text{if } 0 \le a \le 1, \\ 2 \ (\text{resp. } a+1) & \text{if } 1 < a \le 2 \text{ and } C \neq C_B \ (\text{resp. } C = C_B), \\ 10-4a \ (\text{resp. } 15-6a) & \text{if } 2 < a \le 5/2 \text{ and } C \neq C_B \ (\text{resp. } C = C_B). \end{cases}$$

We show that $D_{a,u}$ is not pseudo-effective for $u > u_{\max}(a)$. The intersection numbers of $g^*(\sigma+2f)$ and $g^*(\sigma+2f)-e_2$ with $(\tilde{\sigma}, \tilde{C}_\sigma, e_1, \tilde{C}_B, e_2)$ are $(0,1,0,1,0)$ and $(0,1,0,0,1)$, so both classes are nef by **Step 1**. Let $M := g^*(\sigma+2f)$ if $a \le 1$ or $C = C_B$, and $M := g^*(\sigma+2f)-e_2$ otherwise. Then $(\sigma\cdot M) = 0$, $(f\cdot M) = 1$, $(e_1\cdot M) = 0$, and $(e_C\cdot M) = 0$, while $(e_2\cdot M) = 0$ if $M = g^*(\sigma+2f)$ and $(e_2\cdot M) = 1$ otherwise. For $u \ge u_{\max}(a)$ we have $\epsilon_{a,u} = 0$, and thus

$$(D_{a,u}\cdot M) = u_{\max}(a) - u.$$

Since $(D\cdot M) \ge 0$ for any pseudo-effective $\mathbb{R}$-divisor class D on $\tilde{S}$, $D_{a,u}$ is not pseudo-effective for $u > u_{\max}(a)$. Moreover, d is continuous on the union of the triangles, $d = 0$ for $u = u_{\max}(a)$, and a direct calculation shows

$$\int_0^{u_{\max}(a)} d(a,u)\,du = \rho(a) \qquad \text{for } 0 \le a \le 5/2. \tag{14.5}$$

Step 4. In this step we prove (2).

Recall that $C \neq C_{q_0}$ is the fiber fixed in **Step 2**. Let $\Omega \subset \mathbb{R}^2$ be the image of the union of the triangles of Table 23 for the type of C under the map $\psi(a,u) := (a, u+\kappa(a)) = (a,\tau)$. Each triangle is contained in $\{a \le 2\}$ or in $\{a \ge 2\}$, so ψ is affine with determinant one on each triangle, and ψ is injective. Thus Ω is the union of the image triangles, which have pairwise disjoint interiors, and

$$\int_\Omega G da d\tau = \int G \circ \psi \, da du$$

for any integrable function G on Ω, where the right-hand side is the integral over the triangles. We regard d, ζ, and b as functions on Ω through ψ. They are affine on each image triangle, $d \ge 0$ since $P_{a,u}$ is nef, and $\zeta, b \ge 0$ since $\gamma_\sigma, \gamma_B \ge 0$ and $N_{a,u} + \epsilon_{a,u} e_C \ge 0$.

We check the three hypotheses of Lemma 7.6 for h the identity, the fiber C, $\gamma = \frac{11}{6}$, $c = \frac{5}{2}$, Ω, $\varphi = d$, $q_1 = C \cap \sigma$ with $\mu_1 = \zeta$, and $q_2 = C \cap B$ with $\mu_2 = b$.

(1) S is smooth, $C \cong \mathbb{P}^1$, and $n_m = \frac{11}{6} m^3 + O(m^2)$ by **Step 1**. By (14.5) and **Step 1**,

$$\int_\Omega d da d\tau = \int_0^{5/2} \int_0^{u_{\max}(a)} d(a,u) du da = \int_0^{5/2} \rho(a) da = \frac{11}{6}.$$

(2) Let $R^C_{m,k,j} \neq 0$, and let $a := k/m$. Since every element of $R_{m,k}$ is divisible by $\varphi_{m,k}$ and $\operatorname{ord}_C(\varphi_{m,k}) = m\kappa(a)$, we have $F^j R_{m,k} = F^{j+1} R_{m,k} = R_{m,k}$ for $j < m\kappa(a)$, hence $R^C_{m,k,j} = 0$ for such j. Thus $j \ge m\kappa(a)$. Let $u := j/m - \kappa(a)$. By **Step 2**, $F^j R_{m,k} \neq 0$ implies $|mD_{a,u}| \neq \emptyset$, so $D_{a,u}$ is pseudo-effective and $u \le u_{\max}(a)$ by **Step 3**. Thus (a,u) is contained in a triangle of Table 23, and $(k/m, j/m) = (a, u+\kappa(a)) \in \Omega$. Moreover, since $(C^2) = 0$,

$$\deg\big((\mathcal{L}_{m,k} - jC)|_C\big) = \big((k\sigma + (m+k)f - jC) \cdot C\big) = k \le \frac{5}{2} m.$$

(3) Let $(a,t) := (k/m, j/m)$ be contained in the interior of an image triangle, and let $u := t - \kappa(a)$, so that (a,u) is contained in the interior of a triangle of Table 23. Every row of Table 23 has $d + \zeta + b = a$, so

$$\deg\big((\mathcal{L}_{m,k} - jC)|_C\big) = k = m\big(d(a,u) + \zeta(a,u) + b(a,u)\big).$$

A non-zero element of $R^C_{m,k,j}$ is the image of some $r \in F^j R_{m,k}$ with $\operatorname{ord}_C(r) = j$. Since $\Gamma_r \ge 0$ and $\Gamma_r \equiv mD_{a,u}$, we have $\Gamma_r \ge mN_{a,u}$ [Laz04a, Theorem 2.3.19 and Proposition 2.3.21]. Since $\tilde{C} \not\subset \operatorname{Supp} \Gamma_r$, $\tilde{C} \not\subset \operatorname{Supp}(\Gamma_r - mN_{a,u})$, and $\tilde{C}$ is not one of the curves in I by Table 23. Hence

$$\big(\Gamma_r + m\epsilon_{a,u} e_C\big)|_{\tilde{C}} \ge m\big(N_{a,u} + \epsilon_{a,u} e_C\big)|_{\tilde{C}},$$

and by the definitions of ζ and b in **Step 3**, the right-hand side has multiplicity $m(\zeta(a,u) - \gamma_\sigma(a))$ at $\tilde{p}_\sigma$ and $m(b(a,u) - \gamma_B(a))$ at $\tilde{p}_B$. By (14.4) and **Step 2**, the divisor of the image of r in $R^C_{m,k,j}$ is at least $m\zeta(a,u)[C \cap \sigma] + mb(a,u)[C \cap B]$.

Thus Lemma 7.6 gives

$$S(R; \operatorname{ord}_C) = \frac{6}{11} \int \tau d, \quad S(R^C; C \cap \sigma) \le \frac{6}{11} \int \Big(\frac{d^2}{2} + d\zeta\Big), \quad S(R^C; C \cap B) \le \frac{6}{11} \int \Big(\frac{d^2}{2} + db\Big),$$

and $S(R^C; p) \le \frac{6}{11} \int \frac{d^2}{2}$ for any point $p \in C \backslash (\sigma \cup B)$, where all integrals without limits are over the union of the triangles of Table 23 for the type of C, with respect to $da du$.

A direct calculation over the triangles shows that, for $C \neq C_B$,

$$\int d = \frac{11}{6}, \quad \int \frac{d^2}{2} = \frac{7}{12}, \quad \int d\zeta = \frac{5}{4}, \quad \int db = \frac{1}{48}, \quad \int \tau d = \begin{cases} \frac{59}{48} & \text{if } C \notin \{C_\sigma, C_B\}, \\ \frac{17}{12} & \text{if } C = C_\sigma, \end{cases} \tag{14.6}$$

and, for $C = C_B$,

$$\int d = \frac{11}{6}, \quad \int \frac{d^2}{2} = \frac{1}{2}, \quad \int d\zeta = \frac{4}{3}, \quad \int db = \frac{5}{48}, \quad \int \tau d = \frac{13}{8}.$$

The integrals over the individual triangles are listed in Remark B.7.

By (14.6), $S(R;\operatorname{ord}_C) = \frac{6}{11}\cdot\frac{59}{48} = \frac{59}{88}$ if $C \notin \{C_\sigma, C_B\}$, $S(R;\operatorname{ord}_C) = \frac{6}{11}\cdot\frac{17}{12} = \frac{17}{22}$ if $C = C_\sigma$, and $S(R;\operatorname{ord}_C) = \frac{6}{11}\cdot\frac{13}{8} = \frac{39}{44}$ if $C = C_B$. By (14.6), the three upper bounds above are

$$\begin{aligned} &\frac{6}{11}\Big(\frac{5}{4}+\frac{7}{12}\Big) = \frac{15}{22}+\frac{7}{22} = 1, \quad \frac{6}{11}\Big(\frac{1}{48}+\frac{7}{12}\Big) = \frac{29}{88}, \quad \frac{6}{11}\cdot\frac{7}{12} = \frac{7}{22} && \text{if } C \neq C_B,\\ &\frac{6}{11}\Big(\frac{4}{3}+\frac{1}{2}\Big) = \frac{8}{11}+\frac{3}{11} = 1, \quad \frac{6}{11}\Big(\frac{5}{48}+\frac{1}{2}\Big) = \frac{29}{88}, \quad \frac{6}{11}\cdot\frac{1}{2} = \frac{3}{11} && \text{if } C = C_B, \end{aligned} \tag{14.7}$$

for $p = C\cap\sigma$, $p = C\cap B$, and $p \in C\backslash(\sigma\cup B)$, respectively. This proves (2).

Step 5. In this step we prove the inequality of (1).

Let v be a prime divisor over S with $\operatorname{center}_S v \not\subset C_{q_0}$, let $q \in \operatorname{center}_S v\backslash C_{q_0}$ be a closed point, and let C be the fiber through q. Then $C \neq C_{q_0}$. We check the hypotheses of Lemma 7.4 for h the identity, the fiber C, the point q, and $\lambda = 1$. Since C is not a component of B, $B'_C = B$. By **Step 1**, C and B are smooth and intersect transversally in one point, so $(S, \frac{1}{2}B + C)$ is plt near C [KM98, Corollary 2.31(3)] and $\Delta_C = \frac{1}{2}[C\cap B]$. By (2),

$$A_{S,B/2}(\operatorname{ord}_C) = 1 \geq \frac{39}{44} \geq S(R;\operatorname{ord}_C).$$

Since $C \neq C_{q_0}$, $C\cap\sigma \notin B$. Thus, by (2) and $\frac{3}{11} \leq \frac{7}{22}$,

$$S(R^C;q) \leq \begin{cases} 1 = 1-\operatorname{mult}_q\Delta_C & \text{if } q = C\cap\sigma,\\ \frac{29}{88} \leq \frac{1}{2} = 1-\operatorname{mult}_q\Delta_C & \text{if } q = C\cap B,\\ \frac{7}{22} \leq 1 = 1-\operatorname{mult}_q\Delta_C & \text{otherwise.}\end{cases}$$

Since $q \in \operatorname{center}_S v$, Lemma 7.4 gives $A_{S,B/2}(v) \geq S(R;v)$. Together with **Step 1**, this proves (1). □

14.2. **Points of incidence type 2 and 2 + 1.** In this subsection, we prove Theorem 14.1 by the surface estimates of Lemmas 14.3 and 14.5 for the plt extraction of the non-reduced conic supported on L.

Proof of Theorem 14.1. The proof is long, so we divide it into several steps.

Step 1. In this step we construct the plt extraction $\hat\pi : \hat X \to X$, prove (2), and describe the refinement by P.

We use the notation of Set-ups 10.1 and 10.10. By Proposition 4.1, a local ring of length two of $\Sigma(X,x)$ corresponds to a special line. Thus L is special, $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1)\oplus\mathcal{O}_{\mathbb{P}^1}(-2)$, and $E \cong \mathbb{F}_3$. Let (u,v) be parameters transverse to L near a point of L such that the class of v generates the summand $\mathcal{O}_L(2)$ of $I_L/I_L^2 \cong N^\vee_{L/X}$. The non-reduced conic C_L supported on L with $I_L/I_{C_L} \cong \mathcal{O}_L(-1)$ has ideal $I_{C_L} = (v,u^2)$. Let $\hat\pi : \hat X \to X$ be the blow-up of I_{C_L}. It is covered by the charts $v = u^2t$ and $u^2 = vt'$. In the first chart $\hat X$ is smooth and $I_{C_L}\mathcal{O}_{\hat X} = (u^2)$. In the second chart $\hat X$ is the product of L with the A_1 surface singularity $u^2 = vt'$, and $I_{C_L}\mathcal{O}_{\hat X} = (v)$ defines twice the surface $u = v = 0$. Hence the reduced exceptional divisor of $\hat\pi$ is the divisor of weights $(1,2)$ in (u,v), and twice the reduced exceptional divisor is Cartier.

On Y, in the chart $v = uv'$ of π, we have $E = (u = 0)$ and $s = (u = v' = 0)$, since s corresponds to the summand $\mathcal{O}(1)$ of $N_{L/X}$. The exceptional divisor P of p_T is the valuation with $\operatorname{ord}(u) = \operatorname{ord}(v') = 1$, that is, $\operatorname{ord}(u) = 1$ and $\operatorname{ord}(v) = 2$. Thus P is the reduced exceptional divisor of $\hat\pi$. Since the multiplicities of u and v along E' are 1,

$$I_{C_L}\mathcal{O}_T = \mathcal{O}_T(-E' - 2P),$$

so πp_T factors through a morphism $\mu : T \to \hat X$ [Har77, Proposition II.7.14], which contracts E' onto the singular curve of $\hat X$ and maps P isomorphically onto the reduced exceptional divisor of

$\hat\pi$, also denoted P. In particular

$$\mu^*P = P + \tfrac{1}{2}E'. \tag{14.8}$$

Since P is smooth and $\hat X$ has transversal A_1 singularities along the curve $B := E' \cap P \subset P$, the pair $(\hat X, P)$ is plt and the different of $(\hat X, P)$ on P is $\frac{1}{2}B$ [Pro01, Theorem 2.2.4(i)], as near B the threefold $\hat X$ is locally the product of an A_1 surface singularity and a curve, so a general surface section transverse to B reduces the computation to the surface case. The curve B is smooth and irreducible, and $A_X(P) = 3$ (Lemma 11.7). If $P \cong \mathbb{F}_0$, we set $\sigma := u_P$, and in both cases we set $f := f_P$. By Lemma 11.7(2),

$$\begin{aligned} &P \cong \mathbb{F}_0: \quad H|_P \sim f, \quad -P|_P \sim \sigma + f, \quad B = E'|_P \sim \sigma + 2f, \quad R_T|_P \sim 2\sigma;\\ &P \cong \mathbb{F}_2: \quad H|_P \sim f, \quad -P|_P \sim \sigma + 2f, \quad B = E'|_P \sim \sigma + 3f, \quad R_T|_P \sim 2\sigma + 2f. \end{aligned} \tag{14.9}$$

Thus $(B \cdot f) = 1$, and B is a section of P over L. This proves (2). Let P_x be the fiber of $P \to s \cong L$ over x. Then $\hat\pi^{-1}(x) = P_x$, so $\operatorname{center}_{\hat X} F \subset P_x$ for any prime divisor F over X with $\operatorname{center}_X F = \{x\}$.

Let $W_{\bullet,\bullet}$ be the refinement by P of the complete series of H (Definition 2.8). By (14.8), $H^0(\hat X, mH - jP) = H^0(T, mH - jP - \lceil j/2\rceil E')$, and by Lemma 11.7(4), multiplication by $e'^{\lceil j/2\rceil}$, where e' is a section defining E', is an isomorphism from $H^0(T, mH - jP - \lceil j/2\rceil E')$ onto $H^0(T, mH - jP)$. Let

$$W^\circ_{m,j} \subset H^0\big(P, (mH - jP - \lceil j/2\rceil E')|_P\big)$$

be the image of $H^0(T, mH - jP - \lceil j/2\rceil E')$ under restriction. By (14.8) and [AZ22, Lemma 2.1], the $\mathbb{Q}$-divisors on P of the elements of $W_{m,j}$ are the $\mathbb{Q}$-divisors $D + (\lceil j/2\rceil - j/2)B$, where D runs through the divisors of the elements of $W^\circ_{m,j}$, and $\dim W_{m,j} = \dim W^\circ_{m,j}$. By (14.9), for $j = 2k$ the class of $W^\circ_{m,2k}$ is

$$\xi_{m,k} := k\sigma + mf \quad (P \cong \mathbb{F}_0), \qquad \xi_{m,k} := k\sigma + (m+k)f \quad (P \cong \mathbb{F}_2), \tag{14.10}$$

and for $j = 2k+1$ the class of $W^\circ_{m,2k+1}$ is $\xi_{m,k} + (-P - E')|_P = \xi_{m,k} - f$.

Step 2. In this step we compare the normal bundles of s and of the inverse bisecant M, describe the contact of M with Γ, and prove (3).

By Lemma 11.7(3), $\Gamma \subset S_M$. Let $V_5' := \operatorname{Bl}_M V_5$, so that $\phi_M : V_5' \to V_5$ has exceptional divisor E_M, and let $J \subset V_5'$ be the exceptional divisor of ψ_M. Then $J \sim h - 2E_M$ and $\phi_M(J) = S_M$ (Set-up 10.10). Let $\Gamma_1 \subset J$ be the strict transform of Γ. Since Γ is a smooth curve, ϕ_M maps Γ_1 isomorphically onto Γ, and

$$Z_0 := \Gamma_1 \cap E_M$$

maps isomorphically onto the subscheme $M \cap \Gamma$ of M. Since M is a bisecant of Γ, $\operatorname{length}(Z_0) = 2$, and the projection $Z_0 \to M$ is a closed immersion.

By [Fuj26a, Theorem 3.6(3)–(6)], $N_{M/V_5} \cong \mathcal{O}^{\oplus 2}$ or $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$, so $E_M \cong \mathbb{F}_0$ or $E_M \cong \mathbb{F}_2$. Let f_M be the class of a fiber of $E_M \to M$, and let σ_M be the class of the other ruling if $E_M \cong \mathbb{F}_0$ and the negative section if $E_M \cong \mathbb{F}_2$. Then $h|_{E_M} \sim f_M$, and

$$\begin{aligned} &N_{M/V_5} \cong \mathcal{O}^{\oplus 2}: \quad -E_M|_{E_M} \sim \sigma_M, \quad J|_{E_M} \sim 2\sigma_M + f_M;\\ &N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1): \quad -E_M|_{E_M} \sim \sigma_M + f_M, \quad J|_{E_M} \sim 2\sigma_M + 3f_M. \end{aligned} \tag{14.11}$$

Assume that $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$ until (14.12). Then $J \cong \mathbb{F}_3$ is the normalization of S_M [Fuj26a, Theorem 3.6(3)–(6)]. Let s_0 and f_0 be the classes of the negative section and of a fiber of $J \cong \mathbb{F}_3$. Since $((h - E_M) \cdot \sigma_M) = ((\sigma_M + 2f_M) \cdot \sigma_M)_{E_M} = 0$, the curve σ_M is contracted by ψ_M, so $\sigma_M \subset J$ is a fiber of J over a point of the twisted cubic $\psi_M(J)$, and $(h - E_M)|_J \sim 3f_0$. Write $E_M|_J \sim \alpha s_0 + \beta f_0$. The equations

$$((h|_J)^2)_J = (h^2 \cdot (h - 2E_M)) = 5, \qquad (h|_J \cdot E_M|_J)_J = (f_M \cdot (2\sigma_M + 3f_M))_{E_M} = 2,$$

with $h|_J \sim E_M|_J + 3f_0$, imply $\alpha = \beta = 1$. Thus $J \cap E_M = \sigma_M + \Upsilon$, where Υ is the residual curve. On J we have $\sigma_M = f_0$ and $\Upsilon = s_0$, and $\Upsilon \in |\sigma_M + 3f_M|$ on E_M. Write $\Gamma_1 \sim \alpha_1 s_0 + \beta_1 f_0$

on J. Then $(h \cdot \Gamma_1) = \alpha_1 + \beta_1 = 5$ and $(E_M \cdot \Gamma_1) = \beta_1 - 2\alpha_1 = \mathrm{length}(Z_0) = 2$, so $\Gamma_1 \sim s_0 + 4f_0$. Since $Z_0 = \Gamma_1 \cap (J \cap E_M)$, we obtain

$$\mathrm{length}(Z_0 \cap \sigma_M) = (\Gamma_1 \cdot f_0)_J = 1, \qquad \mathrm{length}(Z_0 \cap \Upsilon) = (\Gamma_1 \cdot s_0)_J = 1. \tag{14.12}$$

We compute directly instead of quoting Proposition 12.5(2) and (3), since Proposition 12.5 assumes $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$, whereas here we assume $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$, and the equivalence of $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ and $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$ is proved only at the end of this step.

Let $\tilde{M} \subset Y^+ = \mathrm{Bl}_\Gamma V_5$ be the strict transform of M. By [KPS18, Lemma 5.2.5 and the paragraph following it], $N_{s/Y}$ is isomorphic to $\mathcal{O}(-1)^{\oplus 2}$ or to $\mathcal{O} \oplus \mathcal{O}(-2)$. The flopping curves s and $\tilde{M}$ satisfy $w(s) = w(\tilde{M})$ by the symmetry of the pagoda [Rei83, (5.8) and Remark (5.13)(b)]. Hence, by Definition-Proposition 2.21 and [Rei83, (5.3), (5.9)], the normal bundles $N_{s/Y}$ and $N_{\tilde{M}/Y^+}$ are both isomorphic to $\mathcal{O}(-1)^{\oplus 2}$ or both isomorphic to $\mathcal{O} \oplus \mathcal{O}(-2)$. The differential of $Y^+ \to V_5$ is an injection $N_{\tilde{M}/Y^+} \to N_{M/V_5}$, which is an isomorphism away from the image of Z_0 in M. We determine the image near the image in M of a point of Z_0. Let (y_1, y_2, t) be local coordinates on V_5 with $M = (y_1 = y_2 = 0)$ and $\Gamma = (y_1 = \gamma_1(t),\ y_2 = \gamma_2(t))$, where $\mathrm{ord}_t \gamma_1 \le \mathrm{ord}_t \gamma_2$, so that $\mathrm{ord}_t \gamma_1$ is the length of $M \cap \Gamma$ at $t = 0$. Near $\tilde{M}$, the blow-up of Γ has coordinates (a, w, t) with

$$y_1 = \gamma_1(t) + a, \qquad y_2 = \gamma_2(t) + wa,$$

and $\tilde{M} = (a = -\gamma_1(t),\ w = \gamma_2(t)/\gamma_1(t))$. A normal vector $(\delta a, \delta w)$ to $\tilde{M}$ maps to $(\delta y_1, \delta y_2) = (\delta a, (\gamma_2/\gamma_1)\delta a - \gamma_1 \delta w)$. Hence a section (v_1, v_2) of N_{M/V_5} near $t = 0$ is contained in the image of $N_{\tilde{M}/Y^+}$ if and only if

$$\gamma_1 \text{ divides } v_2 - (\gamma_2/\gamma_1)v_1. \tag{14.13}$$

The point of Z_0 over $t = 0$ is the direction $[1 : (\gamma_2/\gamma_1)(0)]$ in $E_M = \mathbb{P}(N_{M/V_5})$. If $\mathrm{ord}_t \gamma_1 = 2$, then Z_0 is defined by $(w - (\gamma_2/\gamma_1)(t), t^2)$, where w also denotes the fiber coordinate v_2/v_1 of E_M.

Assume that $N_{M/V_5} \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$. The curve σ_M is the direction of the summand $\mathcal{O}(1)$. By (14.12), if Γ_1 contains $z_0 := \sigma_M \cap \Upsilon$, then Z_0 is non-reduced and supported at z_0, and otherwise $Z_0 = z_\sigma + z_\Upsilon$ with $z_\sigma \in \sigma_M \backslash \Upsilon$ and $z_\Upsilon \in \Upsilon \backslash \sigma_M$. In particular Z_0 is non-reduced if and only if $z_0 \in \Gamma_1$. If Z_0 is reduced, let θ be a non-zero section of the summand $\mathcal{O}(1)$ vanishing at the image of z_Υ in M. At z_Υ we have $\mathrm{ord}_t \gamma_1 = 1$ and $\theta(0) = 0$, and at z_σ we have $\mathrm{ord}_t \gamma_1 = 1$ and $\theta(0) \ne 0$ has direction $z_\sigma \in \sigma_M$, since $Z_0 \to M$ is a closed immersion. In both cases (14.13) holds. If Z_0 is non-reduced, let $\theta = t\theta_1$ be a non-zero section of $\mathcal{O}(1)$ vanishing at the image of z_0. Since $\theta_1(0)$ is a non-zero vector of the summand $\mathcal{O}(1)$, the direction of $\theta_1(0)$ is $z_0 \in \sigma_M$, so t divides $\theta_{1,2} - (\gamma_2/\gamma_1)\theta_{1,1}$, and (14.13) holds with $\mathrm{ord}_t \gamma_1 = 2$. In both cases $H^0(\tilde{M}, N_{\tilde{M}/Y^+}) \ne 0$. Since $\deg N_{\tilde{M}/Y^+} = -2$, we obtain $N_{\tilde{M}/Y^+} \cong \mathcal{O} \oplus \mathcal{O}(-2)$, and hence $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$.

Together with Proposition 12.5(1), this shows that $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ if and only if $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, that is, $E_M \cong P$. Thus the first two conditions in the assumption of (3) are equivalent. If $N_{s/Y} \cong \mathcal{O} \oplus \mathcal{O}(-2)$ and $M \cap \Gamma$ is non-reduced, then X is K-semistable and $A_X(F) \ge S_X(F)$ for any prime divisor F over X by Proposition 12.5. This proves (3) and (1) in this case. From now on we assume that the assumption of (3) does not hold. Then either

(i) $P \cong E_M \cong \mathbb{F}_0$ and Z_0 is subject to no condition beyond $\mathrm{length}(Z_0) = 2$ and the closed immersion $Z_0 \to M$, or
(ii) $P \cong E_M \cong \mathbb{F}_2$ and $Z_0 = z_\sigma + z_\Upsilon$ with $z_\sigma \in \sigma_M \backslash \Upsilon$ and $z_\Upsilon \in \Upsilon \backslash \sigma_M$.

Step 3. In this step we construct a model V_5'' isomorphic to T in codimension one and compare the restrictions of sections to P on both sides.

Let $V_5'' := \mathrm{Bl}_{\Gamma_1} V_5'$ with exceptional divisor R''. Since $\Gamma_1 \not\subset E_M$, the pullback of E_M to V_5'' is the strict transform $\tilde{E}_M$ of E_M. We write h for the pullback of the hyperplane class of V_5, $H_Q := h - \tilde{E}_M$ for the pullback of the hyperplane class of Q, and J' for the strict transform of J.

Then

$$J' \sim h - 2\tilde{E}_M - R'', \qquad -K_{V_5''} \sim 2h - \tilde{E}_M - R''. \tag{14.14}$$

The prime divisors of T are the strict transforms of the prime divisors of V_5 together with R_T and P, since $Y \dashrightarrow Y^+$ is small. Over Y^+ the divisor $\tilde{E}_M$ has center $\tilde{M}$ and $A_{Y^+}(\tilde{E}_M) = A_{V_5}(E_M) = 2$, as $M \not\subset \Gamma$. Since $Y \dashrightarrow Y^+$ is a flop of s to $\tilde{M}$, we have $A_Y(\tilde{E}_M) = 2$ and $\operatorname{center}_Y(\tilde{E}_M) \subset s$. Thus $\operatorname{center}_Y(\tilde{E}_M) = s$ and $\tilde{E}_M$ is the valuation P [KM98, Lemma 2.29, Lemma 2.30 and Corollary 2.31(2)]. Since ϕ_M is an isomorphism over $V_5 \backslash M$ and $\Gamma \not\subset M$, R'' is the valuation R. Moreover $E \sim h - R$ on Y^+ (Set-up 10.1), so E is the strict transform of S_M by Lemma 11.7(3), and E' is the valuation J'. Hence the birational map $T \dashrightarrow V_5''$ is an isomorphism in codimension one, with E', P, and R_T corresponding to J', $\tilde{E}_M$, and R''. For a divisor D on T we write D'' for the corresponding divisor on V_5'', so that $H^0(T, D) = H^0(V_5'', D'')$. By Lemma 11.7(1) and (14.14),

$$H'' \sim -K_{V_5''} + J' + 2\tilde{E}_M \sim 3h - \tilde{E}_M - 2R'',$$

and a direct calculation shows that, for integers k and m,

$$(mH - 2kP - kE')'' \sim (2m-k)(-K_{V_5''}) + (k-m)H_Q, \qquad (-P-E')'' \sim -(h - \tilde{E}_M - R''). \tag{14.15}$$

Let $\tilde{P} \to E_M$ be the blow-up of the points of Z_0, where in the non-reduced case the second point is the point of the first exceptional curve determined by Z_0. Let e_1 and e_2 be the total transforms of the exceptional curves, and $e_Z := e_1 + e_2$. Then $I_{Z_0}\mathcal{O}_{\tilde{P}} = \mathcal{O}_{\tilde{P}}(-e_Z)$, $\tilde{E}_M$ is the blow-up of E_M along $Z_0 = \Gamma_1 \cap E_M$, and $\tilde{P}$ is the minimal resolution of $\tilde{E}_M$. Restricting to $\tilde{P}$ we obtain

$$\begin{aligned} &h|_{\tilde{P}} \sim f_M, \qquad R''|_{\tilde{P}} = e_Z, \\ &-\tilde{E}_M|_{\tilde{P}} \sim \sigma_M \text{ (case (i))}, \qquad -\tilde{E}_M|_{\tilde{P}} \sim \sigma_M + f_M \text{ (case (ii))}. \end{aligned} \tag{14.16}$$

Let $b : \tilde{P} \to P^+$ be the contraction of the strict transforms in $\tilde{P}$ of the fibers of $E_M \to M$ through Z_0, and in the non-reduced case also of the strict transform of the first exceptional curve, so that $E_M \dashrightarrow P^+$ is the elementary transformation of E_M at Z_0 [Har77, Example V.5.7.1]. The curves contracted by b have classes $f_M - e_1$ and $f_M - e_2$ if Z_0 is reduced, and $f_M - e_1$ and $e_1 - e_2$ if Z_0 is non-reduced. In the non-reduced case the second point is not on the strict transform of the fiber through the first point, since $Z_0 \to M$ is a closed immersion, so the strict transform of the fiber through the first point has class $f_M - e_1$. After the contraction of the (-1)-curve of class $f_M - e_1$, the image of the curve of class $e_1 - e_2$ is a (-1)-curve. The surface P^+ is smooth, since $Z_0 \to M$ is a closed immersion. Let $e_\Delta := 2f_M - e_Z$, the sum of the total transforms of the exceptional curves of b, and let $\Delta \subset P^+$ be the subscheme defined by $I_\Delta := b_*\mathcal{O}_{\tilde{P}}(-e_\Delta)$. Then Δ has length two, b is the composite of the blow-ups of the points of Δ (infinitely near in the non-reduced case), and $I_\Delta\mathcal{O}_{\tilde{P}} = \mathcal{O}_{\tilde{P}}(-e_\Delta)$. Hence the sheaf $b_*\mathcal{O}_{\tilde{P}}(-ce_\Delta)$ is the integral closure of I_Δ^c for any integer $c \geq 0$. At a point q of Δ, the ideal I_Δ is the maximal ideal (w, t) of $\mathcal{O}_{P^+,q}$ or equal to (w, t^2) for local parameters (w, t) at q. The ideal (w, t^2) is integrally closed, since the image in the discrete valuation ring $\mathcal{O}_{P^+,q}/(w)$ of an element integral over (w, t^2) is integral over $(\bar{t}^2)$, and hence is contained in $(\bar{t}^2)$. Since products of integrally closed ideals in a two-dimensional regular local ring are integrally closed [HS06, Theorem 14.4.4], I_Δ^c is integrally closed. Thus

$$H^0(\tilde{P}, b^*D^+ - ce_\Delta) = H^0(P^+, \mathcal{O}_{P^+}(D^+) \otimes I_\Delta^c) \tag{14.17}$$

for any divisor D^+ on P^+ and any integer $c \geq 0$.

The birational map $T \dashrightarrow V_5''$ is the composite of the flop $Y \dashrightarrow Y^+$, whose blow-ups of s and of $\tilde{M}$ both extract P [Rei83, (5.3), (5.5), and (5.8)], and of the exchange of the order of the blow-ups of Γ and M, which is the inverse of the flop $\chi : V_5'' \dashrightarrow \operatorname{Bl}_{\tilde{M}} Y^+$ of the strict transforms in V_5'' of the fibers of $E_M \to M$ through the points of the support of Z_0 [Fuj26a, Example 2.10]. If Z_0 is non-reduced, there is only one such curve. On V_5'' we have $\tilde{E}_M = \operatorname{Bl}_{Z_0} E_M$. The flopping

curves of χ are contained in $\tilde{E}_M$. The strict transform of $\tilde{E}_M$ under χ is the exceptional divisor $\mathbb{P}(N_{\tilde{M}/Y^+})$ of the blow-up of Y^+ along $\tilde{M}$, which is P, since the blow-ups of s and of $\tilde{M}$ both extract P. In case (ii), the width of s is at least two (Definition-Proposition 2.21), and by Construction-Proposition 10.2(1), applied with $T_1 = T$ and $P_1 = P$, over a neighborhood of P the map $T \dashrightarrow \mathrm{Bl}_{\tilde{M}} Y^+$ is the flop of the negative section σ of P. The flop of σ maps P isomorphically onto the strict transform of P, since $(\sigma^2)_P = -2$ [Rei83, (5.13)(a)]. The flopped curves of χ are the strict transforms of the fibers of $R \to \Gamma$ over the points of the support of $M \cap \Gamma$ [Fuj26a, Example 2.10], and they map onto curves in Y^+ different from $\tilde{M}$, so no flopped curve of χ is contained in P. Thus the inverse of the induced map $P \dashrightarrow \tilde{P}$ contracts on $\tilde{P}$ exactly the curves contracted by b, where in the non-reduced case the strict transform of the first exceptional curve is first contracted by the minimal resolution $\tilde{P} \to \tilde{E}_M$. Since P and P^+ are smooth surfaces, b composed with $P \dashrightarrow \tilde{P}$ is an isomorphism $P \to P^+$, and we identify P^+ with P. In case (i) the flop of s is an Atiyah flop, which exchanges the rulings [Rei83, (5.5)], so $b^*\sigma = f_M$. In case (ii), $\mathbb{F}_2$ has one ruling, so $b^*f = f_M$. Let D be a divisor on T, let r be a rational section of $\mathcal{O}_T(D)$ not vanishing identically on P, and let r'' be the corresponding section of $\mathcal{O}_{V_5''}(D'')$. Set $D_P := D|_P$ and $D_P^+ := b_*(D''|_{\tilde{P}})$. In case (i), $P \cong \mathbb{P}^1 \times \mathbb{P}^1$ is the exceptional divisor of both blow-ups of the Atiyah flop of s, and no flop through P occurs between T and the Y^+ side [Rei83, (5.5)]. Thus $D_P^+ = D_P$, and the divisors $\operatorname{div}(r|_P)$ and $b_* \operatorname{div}(r''|_{\tilde{P}})$ coincide away from the finite set Δ, hence everywhere. In case (ii), $D_P^+ = D_P + (D_P \cdot \sigma)\sigma$, the divisors $\operatorname{div}(r|_P)$ and $b_* \operatorname{div}(r''|_{\tilde{P}})$ coincide away from σ [Rei83, (6.1), Theorem (6.3), (6.4), Claim (6.5)], and $(D_P^+ \cdot \sigma) = -(D_P \cdot \sigma)$, since $(\sigma^2) = -2$. Thus we obtain

$$\text{(14.18)}\quad \operatorname{div}(r|_P) = b_* \operatorname{div}(r''|_{\tilde{P}}) \ \text{ (case (i))}, \qquad \operatorname{div}(r|_P) = b_* \operatorname{div}(r''|_{\tilde{P}}) + (D_P^+ \cdot \sigma)\sigma \ \text{ (case (ii))}.$$

The classes $b^*\sigma$ and b^*f are determined by their intersection numbers with each other and with the curves contracted by b. By (14.11), (14.14), and (14.16), a direct calculation shows

$$\text{(14.19)}\quad \begin{aligned} &\text{(i)}: \ b^*f = \sigma_M + f_M - e_Z, \quad -K_{V_5''}|_{\tilde{P}} \sim b^*(\sigma + f), \\ &\qquad H_Q|_{\tilde{P}} \sim b^*(2\sigma + f) - e_\Delta, \quad J'|_{\tilde{P}} \sim b^*(\sigma + 2f) - e_\Delta; \\ &\text{(ii)}: \ b^*\sigma = \sigma_M + f_M - e_Z, \quad -K_{V_5''}|_{\tilde{P}} \sim b^*(\sigma + 2f), \\ &\qquad H_Q|_{\tilde{P}} \sim b^*(\sigma + 3f) - e_\Delta, \quad J'|_{\tilde{P}} \sim b^*(2\sigma + 3f) - e_\Delta, \end{aligned}$$

and $(h - \tilde{E}_M - R'')|_{\tilde{P}} \sim b^*f$ in case (i), $(h - \tilde{E}_M - R'')|_{\tilde{P}} \sim b^*(\sigma + f)$ in case (ii). For example, in case (i),

$$(b^*f \cdot (f_M - e_1)) = ((\sigma_M + f_M - e_1 - e_2) \cdot (f_M - e_1)) = 1 - 1 = 0,$$

and if moreover Z_0 is non-reduced, then $(b^*f \cdot (e_1 - e_2)) = ((\sigma_M + f_M - e_1 - e_2) \cdot (e_1 - e_2)) = 1 - 1 = 0$. The other equalities follow similarly from (14.11), (14.14), and (14.16).

In case (i), $J'|_{\tilde{P}} \sim b^*B - e_\Delta$ and $b_*(J'|_{\tilde{P}}) = B$ by (14.18), so $J'|_{\tilde{P}}$ is the strict transform of B, and $\Delta_B := \Delta$ is a subscheme of B. The projection of Δ_B along the fibers of class σ is the projection of Z_0 to M, and the projection along the fibers of class f is the isomorphism $B \to L$. Both are closed immersions. Moreover $R_T|_P = b_*(e_Z)$ is the sum R_Δ of the fibers of class σ through Δ_B, with multiplicities. Thus P, σ, f, B, and Δ_B satisfy the conditions of Set-up 14.2.

In case (ii), let q_σ and q_B be the images under b of the strict transforms of the fibers of E_M through z_Υ and through z_σ, respectively. By (14.19), the strict transform of σ_M is $b^*\sigma$ minus the curve over q_σ, and the strict transform of Υ is $b^*(\sigma + 3f)$ minus the curve over q_B. Hence b maps the strict transforms of σ_M and Υ onto σ and onto a curve $B^+ \in |\sigma + 3f|$, with $q_\sigma \in \sigma$ and $q_B \in B^+$. Applying (14.18) to e', with $D_P^+ = 2\sigma + 3f$ and $(D_P^+ \cdot \sigma) = -1$, we obtain $B = \sigma + B^+ - \sigma = B^+$. Let C_σ and C_B be the fibers of P through q_σ and q_B. The curves C_σ and C_B are the fibers over the images of z_Υ and z_σ in M, which are distinct, since $b^*f = f_M$. The point $q_0 := \sigma \cap B$ is the image of $\sigma_M \cap \Upsilon$, and the fiber of E_M through $\sigma_M \cap \Upsilon$ contains neither z_σ nor z_Υ, since $(f_M \cdot \sigma_M) = (f_M \cdot \Upsilon) = 1$. Thus the fiber C_{q_0} through q_0 is different

from C_σ and C_B. Since $(\sigma \cdot B) = 1$, we obtain $q_\sigma \notin B$ and $q_B \notin \sigma$. Moreover $b_*(e_Z) = C_\sigma + C_B$, and (14.18) with $D_P^+ = 2f$ implies

$$R_T|_P = 2\sigma + C_\sigma + C_B. \tag{14.20}$$

Thus P, σ, f, B, q_σ, and q_B satisfy the conditions of Set-up 14.4, with $\Delta_2 := q_\sigma + q_B = \Delta$. By Proposition 12.10, the point q_0 of P maps to the point of incidence type 3 on L. Since x has incidence type 2 or $2+1$, x is not the point of incidence type 3, and $P_x \neq C_{q_0}$.

Step 4. In this step we prove that both parities of $W_{\bullet,\bullet}$ embed into the series $R_{\bullet,\bullet}$ of Set-up 14.2 in case (i), and of Set-up 14.4 in case (ii).

Fix a point $y \in L$ with $y \neq x$, and let s_{P_y} be a section defining the fiber P_y. We show that for all integers $m \geq 1$ and $k \geq 0$,

$$W^\circ_{m,2k} \subset R_{m,k} \quad \text{if } 2k \leq 5m, \qquad s_{P_y} W^\circ_{m,2k+1} \subset R_{m,k} \quad \text{if } 2k+1 \leq 5m, \tag{14.21}$$

and $W^\circ_{m,j} = 0$ if $j > 5m$. The vanishing for $j > 5m$ holds by Lemma 11.7(4). Let $\ell := 5m - 2k$.

If $k \leq m$, then $R_{m,k} = H^0(P, \xi_{m,k})$, and (14.21) follows from (14.10), since $W^\circ_{m,2k+1} \subset H^0(P, \xi_{m,k} - f)$.

Let $m < k \leq 2m$. By (14.15) and (14.19), the class of $(mH - 2kP - kE')''|_{\tilde{P}}$ is $b^* D_P^+ - (k-m)e_\Delta$, where $D_P^+ = k\sigma + mf$ in case (i) and $D_P^+ = m\sigma + (m+k)f$ in case (ii). By (14.17) and (14.18), with $(D_P^+ \cdot \sigma) = k - m$ in case (ii), every element of $W^\circ_{m,2k}$ is contained in

$$\begin{aligned} &\text{(i)}: \ H^0\big(P, \mathcal{O}_P(k\sigma + mf) \otimes I_{\Delta_B}^{k-m}\big), \\ &\text{(ii)}: \ s_\sigma^{k-m} H^0\big(P, \mathcal{O}_P(m\sigma + (m+k)f) \otimes I_{\Delta_2}^{k-m}\big), \end{aligned} \tag{14.22}$$

which is $R_{m,k}$. If moreover $k < 2m$, then by (14.15) and (14.19) the class of $(mH - (2k+1)P - (k+1)E')''|_{\tilde{P}}$ is $b^*(D_P^+ - f) - (k-m)e_\Delta$ in case (i) and $b^*(D_P^+ - \sigma - f) - (k-m)e_\Delta$ in case (ii). In case (ii), $((D_P^+ - \sigma - f) \cdot \sigma) = k - m + 1$. Hence, by (14.17) and (14.18),

$$\begin{aligned} &\text{(i)}: W^\circ_{m,2k+1} \subset H^0\big(P, \mathcal{O}_P(k\sigma + (m-1)f) \otimes I_{\Delta_B}^{k-m}\big), \\ &\text{(ii)}: W^\circ_{m,2k+1} \subset s_\sigma^{k-m+1} H^0\big(P, \mathcal{O}_P((m-1)\sigma + (m+k-1)f) \otimes I_{\Delta_2}^{k-m}\big). \end{aligned}$$

Multiplying by s_{P_y}, and in case (ii) using $s_\sigma s_{P_y} \in H^0(P, \mathcal{O}_P(\sigma + f))$, we obtain $s_{P_y} W^\circ_{m,2k+1} \subset R_{m,k}$ by (14.22).

Let $4m \leq j \leq 5m$. By Lemma 11.7(4), with j in place of k, $H^0(T, mH - jP) = e'^{j-2m} r_T^{j-4m} \mathcal{H}_{5m-j}$, where r_T defines R_T and $\mathcal{H}_c \subset H^0(T, c(H - 2E' - 4P))$ is the image of $H^0(Q, \mathcal{O}_Q(c))$. Since $(H - 2E' - 4P)'' \sim H_Q$, by (14.19), (14.17), and (14.18), with $(c(\sigma + 3f) \cdot \sigma) = c$ in case (ii), the restriction of $\mathcal{H}_c$ to P is contained in

$$\text{(i)}: \ H^0\big(P, \mathcal{O}_P(c(2\sigma + f)) \otimes I_{\Delta_B}^c\big), \qquad \text{(ii)}: \ s_\sigma^c H^0\big(P, \mathcal{O}_P(c(\sigma + 3f)) \otimes I_{\Delta_2}^c\big). \tag{14.23}$$

Moreover $e'|_P = s_B$, and $r_T|_P = s_{R_\Delta}$ in case (i) and $r_T|_P = s_\sigma^2 s_{C_\sigma + C_B}$ in case (ii) by (14.20). For $j = 2k$ we divide by e'^k. The restriction is

$$W^\circ_{m,2k} = s_B^{k-2m} (r_T|_P)^{2k-4m} \mathcal{H}_\ell|_P,$$

and (14.23) with $c = \ell$ implies $W^\circ_{m,2k} \subset R_{m,k}$. In case (ii) the exponent of s_σ is $2(2k-4m) + \ell = 2k - 3m$. For $j = 2k+1$ we divide by e'^{k+1}, and

$$s_{P_y} W^\circ_{m,2k+1} = s_B^{k-2m} (r_T|_P)^{2k-4m} \cdot \big(s_{P_y} r_T|_P\big) \cdot \mathcal{H}_{\ell-1}|_P.$$

In case (i), the section $s_{P_y} s_{R_\Delta}$ of $\mathcal{O}_P(2\sigma + f)$ is an element of $H^0(P, \mathcal{O}_P(2\sigma + f) \otimes I_{\Delta_B})$, since the projection $P \to \mathbb{P}^1$ along the fibers of class σ maps Δ_B isomorphically onto a subscheme $\Delta' \subset \mathbb{P}^1$, and R_Δ is the pullback of Δ'. In case (ii), $s_{P_y} r_T|_P = s_\sigma \cdot s_\sigma s_{C_\sigma + C_B} s_{P_y}$, and $s_\sigma s_{C_\sigma + C_B} s_{P_y} \in H^0(P, \mathcal{O}_P(\sigma + 3f) \otimes I_{\Delta_2})$, since $q_\sigma \in C_\sigma$ and $q_B \in C_B$. By (14.23) for $c = \ell - 1$, we obtain $s_{P_y} W^\circ_{m,2k+1} \subset R_{m,k}$. This proves (14.21).

Step 5. In this step we prove $S_X(P) = 117/44$.

Let $n_m := \sum_k \dim R_{m,k}$ (Definition 7.3). By Lemma 14.3 and Lemma 14.5(1), $n_m = \frac{11}{6}m^3 + O(m^2)$. By Lemma 2.12(2), $N_m = h^0(X, mH) = \frac{11}{3}m^3 + O(m^2)$, and $\sum_j \dim W_{m,j} = N_m$ (Definition 2.8). By (14.21), $\dim R_{m,k} - \dim W_{m,2k} \geq 0$ and $\dim R_{m,k} - \dim W_{m,2k+1} \geq 0$, and

$$\sum_k \left(2\dim R_{m,k} - \dim W_{m,2k} - \dim W_{m,2k+1}\right) = 2n_m - N_m = O(m^2). \tag{14.24}$$

Let

$$\rho(a) := a \ \ (0 \leq a \leq 1), \qquad \rho(a) := a - (a-1)^2 \ \ (1 \leq a \leq 2), \qquad \rho(a) := (5-2a)^2 \ \ (2 \leq a \leq 5/2).$$

For $a, c \geq 0$ and $e \in \{0, 2\}$ with $c \geq ae - 1$, we have $h^0(\mathbb{F}_e, a\sigma + cf) = \sum_{i=0}^{a}(c - ie + 1)$ [Har77, Exercise III.8.4(a) and Propositions V.2.8 and V.2.9], and the colength of $I^c_{\Delta_B}$ and of $I^c_{\Delta_2}$ is $c(c+1)$. In both cases a direct count shows

$$\begin{aligned}
&\dim R_{m,k} \geq (k+1)(m+1) && (k \leq m),\\
&\dim R_{m,k} \geq (k+1)(m+1) - (k-m)(k-m+1) && (m < k \leq 2m),\\
&\dim R_{m,k} \geq (\ell+1)^2 && (2m < k \leq \lfloor 5m/2 \rfloor),
\end{aligned}$$

so $\dim R_{m,k} \geq m^2\rho(k/m) - O(m)$ uniformly in k. Since $\int_0^{5/2} \rho(a)da = \frac{11}{6}$ and ρ is Lipschitz, $\sum_k m^2\rho(k/m) = \frac{11}{6}m^3 + O(m^2) = n_m + O(m^2)$. Hence $\sum_k |\dim R_{m,k} - m^2\rho(k/m)| = O(m^2)$. The jumping numbers of P on $H^0(X, mH)$ are the integers j, each counted $\dim W_{m,j}$ times, and $j \leq 5m$. By (14.24),

$$\sum_j j \dim W_{m,j} = \sum_k (4k+1)\dim R_{m,k} + O(m^3) = 4m^4 \int_0^{5/2} a\rho(a)da + O(m^3).$$

Dividing by mN_m and letting $m \to +\infty$ [AZ22, Lemma 2.9], we obtain

$$S_X(P) = \frac{6}{11}\int_0^{5/2} 2a\rho(a)da = \frac{6}{11}\left(\frac{2}{3} + \frac{7}{2} + \frac{17}{24}\right) = \frac{117}{44}. \tag{14.25}$$

Step 6. In this step we prove (1) and (4) by Lemma 3.1.

The data for Lemma 3.1 are the complete series of H, the plt extraction $\hat{\pi}$ of P with different $\frac{1}{2}B$, $G := P_x$, and $\lambda := 1$. Let v be a prime divisor over P with $\operatorname{center}_P v \cap P_x \neq \emptyset$. Since distinct fibers of $P \to L$ are disjoint, $\operatorname{center}_P v \not\subset P_y$, so $v(s_{P_y}) = 0$. Moreover $P_x \neq C_{q_0}$ in case (ii) by **Step 3**, so $\operatorname{center}_P v \not\subset C_{q_0}$.

For a subspace U of a space U' of sections, the sum of the jumping numbers of v on U is at most that on U', since a basis of U compatible with v extends to a basis of U' and v takes non-negative values on sections. By **Step 1**, an element of $W_{m,2k+1}$ has divisor $D + \frac{1}{2}B$ with D the divisor of an element of $W^{\circ}_{m,2k+1}$, and $v(D) = v(D + P_y)$. Hence, by (14.21), the sum of the jumping numbers of v on $W_{m,2k}$ and on $W_{m,2k+1}$ is at most twice the sum on $R_{m,k}$ plus $\frac{1}{2}v(B)\dim W_{m,2k+1}$. Summing over k and dividing by mN_m, we obtain

$$S_m(W; v) \leq \frac{2n_m}{N_m} \cdot \frac{1}{mn_m} \sum_k \left(\text{sum of the jumping numbers of } v \text{ on } R_{m,k}\right) + \frac{v(B)}{2m}.$$

Since (P, B) is log canonical, $v(B) \leq A_P(v)$, that is, $v(B) \leq 2A_{P,B/2}(v)$, so the last term is at most $A_{P,B/2}(v)/m$. Since $2n_m/N_m \to 1$, letting $m \to +\infty$ we obtain $S(W; v) \leq S(R; v)$. By Lemma 14.3 in case (i) and Lemma 14.5(1) in case (ii),

$$S(W; v) \leq S(R; v) \leq A_{P,B/2}(v).$$

Thus Lemma 3.1, with $A_X(P) = 3$ and (14.25), implies

$$A_X(F) \geq S_X(F) + \left(3 - \frac{117}{44}\right)\operatorname{mult}_F P = S_X(F) + \frac{15}{44}\operatorname{mult}_F P$$

for any prime divisor F over X with $\operatorname{center}_{\hat{X}} F \cap P_x \neq \emptyset$, where P is regarded as the $\mathbb{Q}$-Cartier divisor on $\hat{X}$. Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$. By **Step 1**, $\operatorname{center}_{\hat{X}} F \subset P_x \subset P$, so the inequality holds for F, and $\operatorname{mult}_F P > 0$ since $2P$ is Cartier and contains $\operatorname{center}_{\hat{X}} F$. This proves (4), and (1) in the remaining case. □

15. Curvilinear triple points in genus twelve

In this section, we study the points of incidence type 3 on a smooth prime Fano threefold X of genus 12, the last incidence type in genus twelve. Exactly one or exactly two conics on X contain such a point (Lemma 15.3), and the number of conics determines the method: if exactly one conic contains the point, then X degenerates smoothly to a member with a faithful $\mathbb{G}_m$-action (Proposition 15.2), which is K-polystable by Theorem 2.26, so X is K-semistable by Lemma 2.6, and if two conics contain the point, we obtain a strict inequality from a weighted extraction (Theorem 15.6).

15.1. **The normal form and one conic.** In this subsection, we write the canonical Mukai net in a normal form adapted to a point of incidence type 3, and we prove that X is K-semistable when exactly one conic contains the point.

Construction-Proposition 15.1. Let X be a smooth prime Fano threefold of genus 12 and let $p \in X$ be a closed point. Assume that $\Sigma(X,p)$ is curvilinear of length three and supported at one line, that is, p has incidence type 3. Then there exist a basis $e_0,\dots,e_6$ of V_X and a basis $\omega_0,\omega_1,\omega_2$ of Π_X with the following properties, where $U_3 := \langle e_0,e_1,e_2\rangle$ and $Q_4 := \langle e_3,\dots,e_6\rangle$.

(1) $p = [U_3]$, and the Gram matrix of ω_j in the basis $e_0,\dots,e_6$ is $\begin{pmatrix} 0 & C_j \\ -C_j^t & B_j \end{pmatrix}$, $j = 0,1,2$, with B_j alternating of size 4 and

$$C_0 = \begin{pmatrix} 1 & 0 & 0 & \alpha_0 \\ 0 & 1 & 0 & \beta_0 \\ 0 & 0 & 1 & \gamma_0 \end{pmatrix}, \quad C_1 = \begin{pmatrix} 0 & 1 & 0 & \alpha_1 \\ 0 & 0 & 1 & \beta_1 \\ 0 & 0 & 0 & \gamma_1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} 0 & 0 & 1 & \alpha_2 \\ 0 & 0 & 0 & \beta_2 \\ 0 & 0 & 0 & \gamma_2 \end{pmatrix}.$$

(2) $\det J \neq 0$ for

$$J := \begin{pmatrix} \gamma_1-\beta_2 & \alpha_2-\gamma_0 & \beta_0-\alpha_1 \\ -\gamma_2 & \gamma_1 & \alpha_2-\beta_1 \\ 0 & -\gamma_2 & \beta_2 \end{pmatrix}.$$

(3) For $M \in \operatorname{Mat}_{3\times 4}(\mathbb{C})$ let $[M] \in \operatorname{Gr}(3,V_X)$ be the span of the rows of $(I_3|M)$ in the basis $e_0,\dots,e_6$. In the chart $M \to [M]$ of $\operatorname{Gr}(3,V_X)$ at p, the local equations of X are $C_jM^t - MC_j^t + MB_jM^t = 0$, $j = 0,1,2$.

(4) The functions $x := M_{11}$, $y := M_{12}$, $z := M_{13}$ are regular parameters of X at p.

(5) For $a,b,c \in \mathbb{C}$ let

$$\Theta(a,b,c) := \begin{pmatrix} a & b & c & 0 \\ b & c & 0 & 0 \\ c & 0 & 0 & 0 \end{pmatrix}.$$

Then $T_pX = \{\Theta(a,b,c) : a,b,c \in \mathbb{C}\}$.

(6) Any entry of the matrix $M - \Theta(x,y,z)$ over $\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x,y,z]]$ belongs to the ideal $(xz-y^2, yz, z^2)$.

(7) $\Sigma(X,p) \subset \mathbb{P}(T_pX)$ is the subscheme with homogeneous ideal $(ac-b^2, bc, c^2)$ in the coordinates a,b,c.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we write the local equations of X at p in an arbitrary adapted basis and show that $\Sigma(X,p)$ is the intersection of a Segre variety with $\mathbb{P}(T_pX)$.

By Definition-Theorem 2.20, we view X as the scheme of three-dimensional subspaces of V_X isotropic for any form in Π_X, with H the restriction of the Plücker class. Let $U_3 \subset V_X$ be the

subspace with $p = [U_3]$, let Q_4 be a complement of U_3 in V_X, and let e_0, e_1, e_2 and $e_3, \dots, e_6$ be bases of U_3 and Q_4. For $\omega \in \Pi_X$ let $(\omega(e_i, e_k))_{0 \le i,k \le 6}$ be the Gram matrix. Since $\omega|_{U_3} = 0$, it has the form

$$\begin{pmatrix} 0 & C_\omega \\ -C_\omega^t & B_\omega \end{pmatrix}$$

with $C_\omega \in \mathrm{Mat}_{3\times 4}(\mathbb{C})$, the *cross block* of ω, and B_ω alternating of size 4. The Gram matrix of ω on the rows of $(I_3|M)$ is

$$(I_3|M) \begin{pmatrix} 0 & C_\omega \\ -C_\omega^t & B_\omega \end{pmatrix} (I_3|M)^t = C_\omega M^t - M C_\omega^t + M B_\omega M^t. \tag{15.1}$$

The map $M \to [M]$ is an open immersion of $\mathrm{Mat}_{3\times 4}(\mathbb{C}) \cong \mathbb{A}^{12}$ into $\mathrm{Gr}(3, V_X)$ sending 0 to p. Thus, in the chart, X is cut out by the entries of (15.1) for ω in a basis of Π_X, that is, by nine equations. Since X is smooth and $\dim X = 3$, the linear parts of the nine equations have rank 9, and

$$T_p X = \{N \in \mathrm{Mat}_{3\times 4}(\mathbb{C}) : C_\omega N^t = N C_\omega^t \text{ for any } \omega \in \Pi_X\}. \tag{15.2}$$

The Grassmannian $\mathrm{Gr}(3, V_X) \subset \mathbb{P}(\Lambda^3 V_X)$ is cut out by quadrics. By [KP23, Lemma 5.6], $\Sigma(\mathrm{Gr}(3, V_X), p) \subset \mathbb{P}(T_p \mathrm{Gr}(3, V_X)) = \mathbb{P}(\mathrm{Mat}_{3\times 4}(\mathbb{C}))$ is the projectivization of the cone $\mathrm{Gr}(3, V_X) \cap T_p \mathrm{Gr}(3, V_X)$. In the chart, the Plücker coordinates of $[M]$ are 1 and the minors of M of all orders, and the embedded tangent space is cut out by the minors of order at least 2. Since the 3×3 minors belong to the ideal of the 2×2 minors, the cone is the locus of matrices of rank at most one, and $\Sigma(\mathrm{Gr}(3, V_X), p)$ is the Segre variety $\mathbb{P}(U_3^\vee) \times \mathbb{P}(Q_4)$ cut out by the 2×2 minors [Har92, Example 2.11]. A point of the Segre variety is $[uv^t]$ with $u \in \mathbb{C}^3$ the coordinates of an element of $U_3^\vee$ and $v \in \mathbb{C}^4$ the coordinates of an element of Q_4.

Let N_0 be a matrix of rank one. The line of $\mathrm{Gr}(3, V_X)$ through p with direction $[N_0]$ intersects the chart in the dense open subset $\{[sN_0] : s \in \mathbb{C}\}$. Similarly, a family of lines through p over a base scheme S is, locally on S, given by a 3×4 matrix N_0 with entries in $\mathcal{O}_S$ that generate the unit ideal and with vanishing 2×2 minors, and the line intersects the chart in $\{[sN_0]\}$. We use families to compare scheme structures in (15.4). By (15.1), the equations of X on $\{[sN_0] : s \in \mathbb{C}\}$ are

$$s(C_\omega N_0^t - N_0 C_\omega^t) + s^2 N_0 B_\omega N_0^t = 0.$$

For any $N \in \mathrm{Mat}_{3\times 4}(\mathbb{C})$ and any alternating $B = (B_{ab})$ of size 4,

$$(NBN^t)_{ik} = \sum_{1 \le a < b \le 4} B_{ab}(N_{ia}N_{kb} - N_{ib}N_{ka}), \tag{15.3}$$

so the entries of $N_0 B_\omega N_0^t$ belong to the ideal of the 2×2 minors. Hence, over $\Sigma(\mathrm{Gr}(3, V_X), p)$, the line is contained in X if and only if $C_\omega N_0^t - N_0 C_\omega^t = 0$ for any $\omega \in \Pi_X$, that is, $[N_0] \in \mathbb{P}(T_p X)$ by (15.2), and

$$\Sigma(X, p) = \big(\mathbb{P}(U_3^\vee) \times \mathbb{P}(Q_4)\big) \cap \mathbb{P}(T_p X) \tag{15.4}$$

as schemes.

Step 2. In this step, we show that, after changing the bases of U_3 and Q_4, $T_p X$ is one of three explicit planes.

Let $Z := \Sigma(X, p)$. By assumption $Z \cong \operatorname{Spec} \mathbb{C}[t]/(t^3)$. By Proposition 4.1, Z is not contained in a line of $\mathbb{P}(T_p X) \cong \mathbb{P}^2$, so Z spans $\mathbb{P}(T_p X)$. By (15.4), the embedding of Z is $t \to [u(t)v(t)^t]$ for jets

$$u(t) = u_0 + tu_1 + t^2 u_2 \in \mathbb{C}^3 \otimes \mathbb{C}[t]/(t^3), \qquad v(t) = v_0 + tv_1 + t^2 v_2 \in \mathbb{C}^4 \otimes \mathbb{C}[t]/(t^3),$$

determined up to units of $\mathbb{C}[t]/(t^3)$.

We claim that u_0, u_1 are linearly independent, and so are v_0, v_1. Suppose that $u_1 \in \mathbb{C}u_0$. Then the subscheme $Z_2 := \operatorname{Spec} \mathbb{C}[t]/(t^2)$ of Z is contained in the linear subspace $\{[u_0]\} \times \mathbb{P}(Q_4)$ of $\mathbb{P}(\mathrm{Mat}_{3\times 4}(\mathbb{C}))$, which is contained in the Segre variety. The line spanned by Z_2 is thus contained

in the Segre variety and in $\mathbb{P}(T_pX)$, so it is contained in $\Sigma(X,p)$ by (15.4). This contradicts the finiteness of $\Sigma(X,p)$ (Proposition 4.1). If $v_1 \in \mathbb{C}v_0$, then Z_2 is contained in the linear subspace $\mathbb{P}(U_3^\vee) \times \{[v_0]\}$ of the Segre variety, and the line spanned by Z_2 is contained in $\Sigma(X,p)$ by (15.4), which again contradicts Proposition 4.1.

Next we normalize u. If u_0, u_1, u_2 are linearly independent, we take them as the new coordinate basis, so that $u(t) = (1, t, t^2)$. Otherwise $u_2 = a'u_0 + b'u_1$ for some $a', b' \in \mathbb{C}$, and modulo t^3,

$$(1 + a't^2)^{-1}u(t) \equiv u_0 + (t + b't^2)u_1 \equiv (1 - b't)^{-1}\big(u_0 + t(u_1 - b'u_0)\big).$$

Taking u_0, $u_1 - b'u_0$, and a third vector as the new coordinate basis, we obtain $u(t) = (1, t, 0)$ up to a unit. For v, if v_0, v_1, v_2 are linearly independent, we extend them to a coordinate basis of $\mathbb{C}^4$, so that $v(t) = (1, t, t^2, 0)$. Otherwise $v_2 = a''v_0 + b''v_1$ for some $a'', b'' \in \mathbb{C}$, the congruence above with a'', b'', v in place of a', b', u holds, and with v_0, $v_1 - b''v_0$, and two further vectors as the new coordinate basis we obtain $v(t) = (1, t, 0, 0)$ up to a unit. The parameter t is not changed. Thus, after changing the bases of U_3 and Q_4, we may assume that

$$u(t) = (1, t, \epsilon_u t^2), \qquad v(t) = (1, t, \epsilon_v t^2, 0), \qquad \epsilon_u, \epsilon_v \in \{0, 1\}.$$

The coefficients of $u(t)v(t)^t$ modulo t^3 are E_{11}, $E_{12} + E_{21}$, and $\epsilon_v E_{13} + E_{22} + \epsilon_u E_{31}$, where E_{ia} are the matrix units. Since Z spans $\mathbb{P}(T_pX)$ and $\dim T_pX = 3$,

$$T_pX = \{M_\epsilon(a,b,c) : a, b, c \in \mathbb{C}\}, \qquad M_\epsilon(a,b,c) := \begin{pmatrix} a & b & \epsilon_v c & 0 \\ b & c & 0 & 0 \\ \epsilon_u c & 0 & 0 & 0 \end{pmatrix}.$$

If $\epsilon_u = \epsilon_v = 0$, then $\operatorname{rank} M_\epsilon(a,b,c) \le 1$ if and only if $ac - b^2 = 0$, so (15.4) implies that $\Sigma(X,p)$ contains a conic of $\mathbb{P}(T_pX)$, a contradiction. Thus $(\epsilon_u, \epsilon_v) \in \{(1,0), (0,1), (1,1)\}$.

Step 3. In this step, we exclude $(\epsilon_u, \epsilon_v) = (1,0)$ by the smoothness of X and $(\epsilon_u, \epsilon_v) = (0,1)$ by the non-degeneracy of Π_X.

Let $C = (C_{ia})$ be the cross block of some $\omega \in \Pi_X$. By (15.2), $CM_\epsilon(a,b,c)^t$ is symmetric for any a, b, c.

Suppose that $(\epsilon_u, \epsilon_v) = (1,0)$. The entries (i,k) of $CM_\epsilon(a,b,c)^t$ are $C_{i1}a + C_{i2}b$, $C_{i1}b + C_{i2}c$, and $C_{i1}c$ for $k = 1, 2, 3$. Comparing the coefficients of a, b, c in the pairs $(1,2)$, $(1,3)$, and $(2,3)$, we obtain $C_{21} = C_{12} = 0$, $C_{22} = C_{11}$, $C_{31} = C_{32} = C_{11} = 0$. Thus the first two columns of the cross block of any $\omega \in \Pi_X$ vanish. Then $C_\omega N^t - NC_\omega^t$ depends only on the last two columns of N, and by (15.2) T_pX contains all matrices whose last two columns vanish. Hence $\dim T_pX \ge 6$, a contradiction.

Suppose that $(\epsilon_u, \epsilon_v) = (0,1)$. The entries (i,k) of $CM_\epsilon(a,b,c)^t$ are $C_{i1}a + C_{i2}b + C_{i3}c$, $C_{i1}b + C_{i2}c$, and 0 for $k = 1, 2, 3$. Comparing the coefficients of a, b, c in the pairs $(1,2)$, $(1,3)$, and $(2,3)$, we obtain

$$C = \begin{pmatrix} c_0 & c_1 & c_2 & d_1 \\ 0 & c_0 & c_1 & d_2 \\ 0 & 0 & 0 & d_3 \end{pmatrix}.$$

Thus $\omega(e_2, e_k) = 0$ for $k \ne 6$, and $\omega(e_2, e_6) = d_3(\omega)$, where $d_3 : \Pi_X \to \mathbb{C}$ is linear. Let $\omega, \omega' \in \operatorname{Ker} d_3$ be linearly independent. Then $e_2 \in \operatorname{Ker}(s\omega + s'\omega')$ for any $s, s' \in \mathbb{C}$, and $s\omega + s'\omega'$ induces an alternating form on the six-dimensional space $V_X/\mathbb{C}e_2$. The Pfaffian of the induced form is a homogeneous cubic in (s, s'), so it vanishes at some $[s : s'] \in \mathbb{P}^1$. For such $[s : s']$, the non-zero form $s\omega + s'\omega'$ has rank at most 4 on $V_X/\mathbb{C}e_2$, hence on V_X. This contradicts the non-degeneracy of Π_X (Definition-Theorem 2.20).

Thus $(\epsilon_u, \epsilon_v) = (1,1)$ and $T_pX = \{\Theta(a,b,c)\}$. The 2×2 minors of $\Theta(a,b,c)$ are 0, $\pm(ac - b^2)$, $\pm bc$, and $\pm c^2$, so by (15.4) the homogeneous ideal of $\Sigma(X,p) \subset \mathbb{P}(T_pX)$ is $(ac - b^2, bc, c^2)$. This proves (7).

Step 4. In this step, we normalize the cross blocks, compute $\det J \ne 0$, and prove that x, y, z are regular parameters.

The entries (i,k) of $C\Theta(a,b,c)^t$ are $C_{i1}a+C_{i2}b+C_{i3}c$, $C_{i1}b+C_{i2}c$, and $C_{i1}c$ for $k=1,2,3$. Comparing the coefficients of a,b,c in the pairs $(1,2)$, $(1,3)$, and $(2,3)$, we obtain $C_{21}=0$, $C_{22}=C_{11}$, $C_{23}=C_{12}$, $C_{31}=C_{32}=0$, and $C_{33}=C_{11}$, that is,

$$C=\begin{pmatrix} c_0 & c_1 & c_2 & d_1\\ 0 & c_0 & c_1 & d_2\\ 0 & 0 & c_0 & d_3\end{pmatrix}.$$

We claim that the linear map $\Pi_X\to\mathbb{C}^3$, $\omega\to(c_0,c_1,c_2)$, is injective. Otherwise some non-zero $\omega\in\Pi_X$ has cross block $C_\omega=df_4^t$, where $d=(d_1,d_2,d_3)^t$ and $f_4=(0,0,0,1)^t$. Then $C_\omega N^t-NC_\omega^t=dw^t-wd^t$, where w is the fourth column of N, and the linear map $N\to dw^t-wd^t$ has rank at most 2. The other two forms of a basis of Π_X contribute at most 3 linear equations each, so the nine linear parts in **Step 1** have rank at most 8, a contradiction. Thus the map is an isomorphism, and we take $\omega_0,\omega_1,\omega_2$ to be the preimages of the standard basis. Their cross blocks are the matrices C_0,C_1,C_2 of (1), with $\alpha_j,\beta_j,\gamma_j$ the entries d_1,d_2,d_3 of C_{ω_j}, and $B_j:=B_{\omega_j}$. This proves (1), and (15.1) implies the local equations in (3).

The linear parts of the equations of ω_0 in the pairs $(1,2)$, $(1,3)$, of ω_1 in the pairs $(1,2)$, $(1,3)$, $(2,3)$, and of ω_2 in the pair $(1,2)$ are equivalent to

$$\begin{aligned} &M_{21}=y+\beta_0M_{14}-\alpha_0M_{24}, && M_{31}=z+\gamma_0M_{14}-\alpha_0M_{34}, && M_{22}=z+\beta_1M_{14}-\alpha_1M_{24},\\ &M_{32}=\gamma_1M_{14}-\alpha_1M_{34}, && M_{33}=\gamma_1M_{24}-\beta_1M_{34}, && M_{23}=\beta_2M_{14}-\alpha_2M_{24}.\end{aligned}\tag{15.5}$$

Substituting (15.5) into the linear parts of the three remaining equations, namely ω_0 in the pair $(2,3)$ and ω_2 in the pairs $(1,3)$ and $(2,3)$, we obtain $J(M_{14},M_{24},M_{34})^t=0$, and x,y,z do not occur. Hence the space of solutions of the nine linear equations is the direct sum of the coordinate space of (x,y,z) and $\operatorname{Ker}J$. Since $\dim T_pX=3$, $\operatorname{Ker}J=0$, that is, $\det J\neq 0$. Moreover $M_{14}=M_{24}=M_{34}=0$ on T_pX, and (15.5) implies $T_pX=\{\Theta(x,y,z)\}$ in the coordinates x,y,z.

Since the linear part of the nine equations in the nine entries other than x,y,z is invertible, the cotangent space of X at p is spanned by the differentials of x,y,z. As X is smooth of dimension 3 at p, x,y,z are regular parameters and $\widehat{\mathcal{O}}_{X,p}=\mathbb{C}[[x,y,z]]$. This proves (2), (4), and (5).

Step 5. In this step, we prove that the entries of $M-\Theta(x,y,z)$ belong to $I:=(xz-y^2,yz,z^2)\subset\mathbb{C}[[x,y,z]]$.

Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[x,y,z]]$, and let Mat° be the space of matrices in $\mathrm{Mat}_{3\times4}$ with vanishing entries $(1,1)$, $(1,2)$, $(1,3)$. Let $\ell(N):=(C_jN^t-NC_j^t)_{j=0,1,2}$. By **Step 4**, ℓ restricts to an isomorphism from $\mathrm{Mat}^\circ(\mathbb{C})$ onto the space of triples of alternating 3×3 matrices. Write $M=\Theta+R$ with $\Theta:=\Theta(x,y,z)$ and $R\in\mathrm{Mat}^\circ(\mathfrak{m})$. Since $\ell(\Theta)=0$, the equations of (3) read

$$R=-\ell^{-1}\Big(\big(\Theta B_j\Theta^t+\Theta B_jR^t+RB_j\Theta^t+RB_jR^t\big)_{j=0,1,2}\Big).\tag{15.6}$$

By (15.3), the entries of $\Theta B_j\Theta^t$ are linear combinations of 2×2 minors of Θ, which belong to I. If two solutions $R,R'\in\mathrm{Mat}^\circ(\mathfrak{m})$ of (15.6) satisfy $R-R'\in\mathrm{Mat}^\circ(\mathfrak{m}^k)$, then (15.6) implies $R-R'\in\mathrm{Mat}^\circ(\mathfrak{m}^{k+1})$, since the entries of Θ, R, and R' belong to $\mathfrak{m}$. Thus (15.6) has a unique solution in $\mathrm{Mat}^\circ(\mathfrak{m})$. Let $R^{(0)}:=0$ and let $R^{(n+1)}$ be the right-hand side of (15.6) at $R^{(n)}$. The right-hand side of (15.6) changes by an element of $\mathrm{Mat}^\circ(\mathfrak{m}^{k+1})$ when $R\in\mathrm{Mat}^\circ(\mathfrak{m})$ changes by an element of $\mathrm{Mat}^\circ(\mathfrak{m}^k)$, so $R^{(n+1)}-R^{(n)}\in\mathrm{Mat}^\circ(\mathfrak{m}^{n+1})$, and $R^{(n)}$ converges $\mathfrak{m}$-adically. If $R\in\mathrm{Mat}^\circ(I)$, then any entry of ΘB_jR^t, $RB_j\Theta^t$, and RB_jR^t belongs to I, and any entry of ℓ^{-1} of a triple is a $\mathbb{C}$-linear combination of the entries of the triple, so the right-hand side of (15.6) belongs to $\mathrm{Mat}^\circ(I)$. By induction on n, $R^{(n)}\in\mathrm{Mat}^\circ(I)$ for any n. The limit of $R^{(n)}$ is a solution of (15.6) in $\mathrm{Mat}^\circ(\mathfrak{m})$, and so is $M-\Theta(x,y,z)$ by (3), so they are equal by uniqueness. Since I is closed in $\mathbb{C}[[x,y,z]]$, the entries of $M-\Theta(x,y,z)$ belong to I. This proves (6). □

By Lemma 15.3 below, $\gamma_2=0$ if and only if exactly one conic on X contains p. If $\gamma_2=0$, the weight filtration of a diagonal one-parameter subgroup of $\mathrm{SL}(V_X)$ degenerates Π_X to an explicit $\mathbb{G}_m$-invariant net. The isotropic scheme of the limit net is smooth, since the locus of

points where it fails to be smooth is invariant under the torus and the four torus-fixed points are smooth points.

Proposition 15.2. *Notation and conditions as in Construction-Proposition 15.1. Assume that* $\gamma_2 = 0$. *Then* $\beta_2\gamma_1(\gamma_1 - \beta_2) \neq 0$ *and* $(B_2)_{12} \neq 0$. *Let* $\tau := \beta_2/\gamma_1$, *so that* $\tau \notin \{0, 1\}$. *There exists a smooth projective morphism* $\pi : \mathcal{X} \to \mathbb{A}^1$ *with Cartier relatively ample anti-canonical bundle and with a* $\mathbb{G}_m$*-action covering the standard action on* $\mathbb{A}^1$, *such that* $\pi^{-1}(\mathbb{A}^1\backslash\{0\})$ *is* $\mathbb{G}_m$*-equivariantly isomorphic to* $X \times (\mathbb{A}^1\backslash\{0\})$ *with the trivial action on* X, *and the central fiber is isomorphic to* X_{Π_τ}, *where* $\Pi_\tau \subset \Lambda^2 V_X^\vee$ *is the net spanned by* $e'_{03} + e'_{14} + e'_{25}$, $e'_{04} + e'_{15} + e'_{26}$, *and* $e'_{05} + \tau e'_{16} + e'_{34}$. *Here* $e'_{ij} := e_i'^\vee \wedge e_j'^\vee$ *for a basis* $e'_0, \dots, e'_6$ *of* V_X *with* $e'_i \in \mathbb{C}^* e_i$ *for any* i. *The central fiber is a smooth prime Fano threefold of genus* 12 *and admits a faithful* $\mathbb{G}_m$*-action. In particular,* X *is K-semistable.*

Proof. The proof is long, so we divide it into several steps. For $0 \leq i < k \leq 6$ let $e_{ik} := e_i^\vee \wedge e_k^\vee$. A form with Gram matrix (G_{ik}) is $\sum_{i<k} G_{ik} e_{ik}$, so by Construction-Proposition 15.1(1),

$$\begin{aligned}
\omega_0 &= e_{03} + e_{14} + e_{25} + \alpha_0 e_{06} + \beta_0 e_{16} + \gamma_0 e_{26} + \textstyle\sum_{3\leq a<b\leq 6} (B_0)_{a-2,b-2} e_{ab},\\
\omega_1 &= e_{04} + e_{15} + \alpha_1 e_{06} + \beta_1 e_{16} + \gamma_1 e_{26} + \textstyle\sum_{3\leq a<b\leq 6} (B_1)_{a-2,b-2} e_{ab},\\
\omega_2 &= e_{05} + \alpha_2 e_{06} + \beta_2 e_{16} + \textstyle\sum_{3\leq a<b\leq 6} (B_2)_{a-2,b-2} e_{ab}.
\end{aligned} \tag{15.7}$$

Step 1. In this step, we prove $\beta_2\gamma_1(\gamma_1 - \beta_2) \neq 0$ and $b_2 := (B_2)_{12} \neq 0$.

Since $\gamma_2 = 0$, the matrix J is upper triangular and $\det J = \beta_2\gamma_1(\gamma_1 - \beta_2)$, which is non-zero by Construction-Proposition 15.1(2). The third row of C_2 vanishes, so $e_2 \in \operatorname{Ker}\omega_2$. Since $\operatorname{rank}\omega_2 = 6$ (Definition-Theorem 2.20), ω_2 induces a non-degenerate form on $V_X/\mathbb{C}e_2$, with basis the images of $e_0, e_1, e_3, e_4, e_5, e_6$. By (15.7), e_1 pairs non-trivially only with e_6, with $\omega_2(e_1, e_6) = \beta_2$, and e_0 pairs non-trivially only with e_5 and e_6. Thus the only perfect matching contributing to the Pfaffian matches e_1 with e_6, e_0 with e_5, and e_3 with e_4, and the Pfaffian equals $\pm\beta_2 b_2$. Hence $b_2 \neq 0$. Moreover $\tau = \beta_2/\gamma_1 \notin \{0, 1\}$.

Step 2. In this step, we construct the family $\mathcal{X}$ and compute the central fiber.

Let $\lambda : \mathbb{G}_m \to \mathrm{SL}(V_X)$ be the one-parameter subgroup with $\lambda(s)e_i = s^{w_i} e_i$ and $(w_0, \dots, w_6) = (-1, -2, -3, 0, 1, 2, 3)$. Then $\lambda(s) \cdot e_{ik} = s^{-(w_i+w_k)} e_{ik}$, and we call $-(w_i + w_k)$ the *weight* of e_{ik}. The weights of e_{03}, e_{14}, e_{25} are 1, those of e_{04}, e_{15}, e_{26} are 0, those of e_{05}, e_{16}, e_{34} are -1, the weight of e_{06} is -2, and the weights of $e_{35}, e_{36}, e_{45}, e_{46}, e_{56}$ are at most -2. Let $(d_0, d_1, d_2) := (1, 0, -1)$. By (15.7), any term of ω_j has weight at most d_j, and the terms of weight d_j are

$$\omega_0^{\mathrm{top}} = e_{03} + e_{14} + e_{25}, \qquad \omega_1^{\mathrm{top}} = e_{04} + e_{15} + \gamma_1 e_{26}, \qquad \omega_2^{\mathrm{top}} = e_{05} + \beta_2 e_{16} + b_2 e_{34}.$$

For $t \neq 0$ let $\omega_j(t) := t^{d_j} \lambda(t^{-1}) \cdot \omega_j$. The coefficient of e_{ik} in ω_j is multiplied by t to the power d_j minus the weight of e_{ik}, which is a non-negative integer. Thus $\omega_j(t)$ extends to a polynomial family of forms on $\mathbb{A}^1$ with $\omega_j(0) = \omega_j^{\mathrm{top}}$. The coefficients of e_{03}, e_{04}, e_{05} in $(\omega_0(t), \omega_1(t), \omega_2(t))$ form the identity matrix for any t, so the three forms are linearly independent for any $t \in \mathbb{A}^1$.

Let $\mathcal{X} \subset \mathrm{Gr}(3, V_X) \times \mathbb{A}^1$ be the closed subscheme of the points (U, t) with $\omega_j(t)|_U = 0$ for $j = 0, 1, 2$, and let $\pi : \mathcal{X} \to \mathbb{A}^1$ be the projection. Since $(g \cdot \omega)|_U = 0$ if and only if $\omega|_{g^{-1}U} = 0$, the fiber over $t \neq 0$ is $\lambda(t^{-1})X$. For $s \in \mathbb{G}_m$ we have

$$\omega_j(st) = s^{d_j} \lambda(s^{-1}) \cdot \omega_j(t)$$

for $t \neq 0$, hence for any t. Thus $s \cdot (U, t) := (\lambda(s^{-1})U, st)$ is a $\mathbb{G}_m$-action on $\mathcal{X}$ covering the standard action on $\mathbb{A}^1$. The isomorphism $X \times (\mathbb{A}^1\backslash\{0\}) \to \pi^{-1}(\mathbb{A}^1\backslash\{0\})$, $(U, t) \to (\lambda(t^{-1})U, t)$, is $\mathbb{G}_m$-equivariant for the trivial action on X, since $s \cdot (\lambda(t^{-1})U, t) = (\lambda((st)^{-1})U, st)$.

The central fiber is X_{Π_0}, where Π_0 is spanned by $\omega_0^{\mathrm{top}}, \omega_1^{\mathrm{top}}, \omega_2^{\mathrm{top}}$. Let $e'_i := m_i e_i$ with

$$(m_0, \dots, m_6) := (1, b_2^{-1}, b_2^{-2}, 1, b_2, b_2^2, b_2^3\gamma_1^{-1}).$$

Then $e_{ik} = m_i m_k e'_{ik}$, and

$$\omega_0^{\mathrm{top}} = e'_{03} + e'_{14} + e'_{25}, \qquad \omega_1^{\mathrm{top}} = b_2(e'_{04} + e'_{15} + e'_{26}), \qquad \omega_2^{\mathrm{top}} = b_2^2(e'_{05} + \tau e'_{16} + e'_{34}).$$

Hence $\Pi_0 = \Pi_\tau$ and the central fiber is X_{Π_τ}.

Step 3. In this step, we prove that X_{Π_τ} is smooth of dimension 3.

Each generator of Π_τ is an eigenvector of λ, so $\lambda(\mathbb{G}_m)$ preserves X_{Π_τ}. The weights w_i are pairwise distinct, so the λ-fixed points of $\mathrm{Gr}(3, V_X)$ are the coordinate subspaces $\langle e'_i, e'_j, e'_k\rangle$. The nine pairs occurring in the generators of Π_τ are

$$\{0,3\},\ \{1,4\},\ \{2,5\},\ \{0,4\},\ \{1,5\},\ \{2,6\},\ \{0,5\},\ \{1,6\},\ \{3,4\},$$

each in exactly one generator and with non-zero coefficient, as $\tau \neq 0$. Thus $\langle e'_i, e'_j, e'_k\rangle \in X_{\Pi_\tau}$ if and only if $\{i,j,k\}$ contains none of the nine pairs. Suppose that $\{i,j,k\}$ contains none of the nine pairs. If $0 \in \{i,j,k\}$, then $3,4,5 \notin \{i,j,k\}$. If moreover $6 \in \{i,j,k\}$, then $1,2 \notin \{i,j,k\}$ because of $\{1,6\}$ and $\{2,6\}$, which is impossible, so $\{i,j,k\} = \{0,1,2\}$. If $1 \in \{i,j,k\}$ and $0 \notin \{i,j,k\}$, then $4,5,6 \notin \{i,j,k\}$, so $\{i,j,k\} = \{1,2,3\}$. If $2 \in \{i,j,k\}$ and $0,1 \notin \{i,j,k\}$, then $5,6 \notin \{i,j,k\}$, so $\{i,j,k\} = \{2,3,4\}$, which contains $\{3,4\}$. Thus this case does not occur. If $\{i,j,k\} \subset \{3,4,5,6\}$, then $\{3,4\} \not\subset \{i,j,k\}$, so $\{i,j,k\}$ is $\{3,5,6\}$ or $\{4,5,6\}$. Conversely, none of the four triples contains one of the nine pairs. Hence the λ-fixed points of X_{Π_τ} are

$$\langle e'_0, e'_1, e'_2\rangle, \quad \langle e'_1, e'_2, e'_3\rangle, \quad \langle e'_3, e'_5, e'_6\rangle, \quad \langle e'_4, e'_5, e'_6\rangle. \tag{15.8}$$

At a coordinate subspace $\langle e'_{i_1}, e'_{i_2}, e'_{i_3}\rangle$ with $i_1 < i_2 < i_3$ and complement $k_1 < \cdots < k_4$, we use the chart $N = (n_{ab}) \to$ the span of the rows of $(I_3|N)$ in the basis $e'_{i_1}, e'_{i_2}, e'_{i_3}, e'_{k_1}, \ldots, e'_{k_4}$. We list the linear parts of the nine equations, generator by generator and in the pairs $(1,2)$, $(1,3)$, $(2,3)$ for each generator. They are computed directly from (15.1).

$$\begin{aligned}
\langle e'_0, e'_1, e'_2\rangle :&\ n_{21} - n_{12},\ n_{31} - n_{13},\ n_{32} - n_{23};\ n_{22} - n_{13},\ n_{32} - n_{14},\ n_{33} - n_{24};\\
&\ n_{23} - \tau n_{14},\ n_{33},\ \tau n_{34}.\\
\langle e'_1, e'_2, e'_3\rangle :&\ n_{22} - n_{13},\ n_{11} + n_{32},\ n_{21} + n_{33};\ n_{23} - n_{14},\ n_{33},\ n_{34};\\
&\ \tau n_{24},\ \tau n_{34} - n_{12},\ -n_{22}.\\
\langle e'_3, e'_5, e'_6\rangle :&\ n_{13} - n_{21},\ -n_{31},\ -n_{33};\ n_{12},\ n_{13},\ n_{23} - n_{32};\\
&\ n_{11} + n_{24},\ \tau n_{12} + n_{34},\ \tau n_{22} - n_{31}.\\
\langle e'_4, e'_5, e'_6\rangle :&\ n_{13} - n_{22},\ -n_{32},\ -n_{33};\ n_{12} - n_{21},\ n_{13} - n_{31},\ n_{23} - n_{32};\\
&\ n_{11} - n_{24},\ \tau n_{12} - n_{34},\ \tau n_{22} - n_{31}.
\end{aligned} \tag{15.9}$$

At $\langle e'_0, e'_1, e'_2\rangle$, the equations imply $n_{33} = n_{34} = n_{24} = 0$ and $n_{32} = n_{14} = n_{23} = \tau n_{14}$, so $(\tau - 1)n_{14} = 0$ and $n_{14} = n_{23} = n_{32} = 0$, while $n_{21} = n_{12}$ and $n_{31} = n_{22} = n_{13}$. At $\langle e'_1, e'_2, e'_3\rangle$, they imply $n_{33} = n_{34} = n_{24} = n_{12} = n_{22} = n_{13} = n_{21} = 0$, $n_{32} = -n_{11}$, and $n_{23} = n_{14}$. At $\langle e'_3, e'_5, e'_6\rangle$, they imply $n_{31} = n_{33} = n_{12} = n_{13} = n_{21} = n_{34} = n_{22} = 0$, $n_{23} = n_{32}$, and $n_{11} = -n_{24}$. At $\langle e'_4, e'_5, e'_6\rangle$, they imply $n_{32} = n_{33} = n_{23} = 0$ and $n_{22} = n_{31} = n_{13} = \tau n_{22}$, so $(\tau - 1)n_{22} = 0$ and $n_{13} = n_{22} = n_{31} = 0$, while $n_{21} = n_{12}$, $n_{34} = \tau n_{12}$, and $n_{24} = n_{11}$. In each case the solution space is three-dimensional, so the nine linear parts have rank 9.

Let $\mathcal{E} := (\Lambda^2 \mathcal{U}^\vee)^{\oplus 3}$ on $\mathrm{Gr}(3, V_X)$, of rank 9, and let σ_τ be the section of $\mathcal{E}$ defined by the generators of Π_τ, so that X_{Π_τ} is the zero scheme of σ_τ. The locus of points of X_{Π_τ} where the differential of σ_τ has rank less than 9 is closed in the projective scheme X_{Π_τ} and λ-invariant. If it were non-empty, it would contain a λ-fixed point by the Borel fixed point theorem [Bor91, Theorem III.10.4], which contradicts (15.9). Thus X_{Π_τ} is smooth and $\dim X_{\Pi_\tau} = 12 - 9 = 3$.

Step 4. In this step, we prove that π is smooth with relatively ample anti-canonical bundle and that X_{Π_τ} is a smooth prime Fano threefold of genus 12.

The scheme $\mathcal{X}$ is the zero scheme of the section of the pullback of $\mathcal{E}$ to $\mathrm{Gr}(3, V_X) \times \mathbb{A}^1$ defined by $\omega_0(t), \omega_1(t), \omega_2(t)$. Over $t \neq 0$ the fiber is isomorphic to the smooth threefold X, and over $t = 0$ it is X_{Π_τ}, smooth of dimension 3 by **Step 3**. Thus the differential in the fiber direction has

rank 9 at any point of $\mathcal{X}$. By the relative Jacobian criterion [BLR90, Section 2.2, Proposition 7], π is smooth of relative dimension 3, and π is projective since $\mathcal{X}$ is closed in $\mathrm{Gr}(3, V_X) \times \mathbb{A}^1$.

Let $H_{\mathrm{Pl}} := \det \mathcal{U}^\vee$ be the Plücker line bundle. By [Har92, Example 16.1], $T_{\mathrm{Gr}(3,V_X)} \cong \mathcal{H}om(\mathcal{U}, V_X/\mathcal{U})$, so

$$\det T_{\mathrm{Gr}(3,V_X)} \cong (\det \mathcal{U}^\vee)^{\otimes 4} \otimes \det(V_X/\mathcal{U})^{\otimes 3} \cong H_{\mathrm{Pl}}^7,$$

since $\det(V_X/\mathcal{U}) \cong \det \mathcal{U}^\vee$. Thus $\omega_{\mathrm{Gr}(3,V_X)} \cong H_{\mathrm{Pl}}^{-7}$. Moreover $\det \mathcal{E} \cong (\det \Lambda^2 \mathcal{U}^\vee)^{\otimes 3} \cong H_{\mathrm{Pl}}^6$. Since $\mathcal{X}$ is the smooth zero scheme of a section of the pullback of $\mathcal{E}$ with surjective differential, the normal bundle of $\mathcal{X}$ is the restriction of $\mathcal{E}$, and adjunction [Har77, Proposition II.8.20] implies

$$\omega_{\mathcal{X}/\mathbb{A}^1} \cong H_{\mathrm{Pl}}^{-1}|_{\mathcal{X}}.$$

Thus the relative anti-canonical bundle is Cartier and relatively ample, and any fiber of π is a smooth Fano threefold.

By Ehresmann's theorem [Voi02, Theorem 9.3], the fibers of π are diffeomorphic by diffeomorphisms preserving the classes $c_1(H_{\mathrm{Pl}})$. Thus X_{Π_τ} is connected, $H^2(X_{\Pi_\tau}, \mathbb{Z}) \cong H^2(X, \mathbb{Z})$ with $c_1(-K_{X_{\Pi_\tau}})$ corresponding to $c_1(-K_X)$, and $(-K_{X_{\Pi_\tau}})^3 = (-K_X)^3 = 22$. For a smooth Fano threefold F, Kodaira vanishing [Har77, Remark III.7.15] implies $H^1(F, \mathcal{O}_F) = H^2(F, \mathcal{O}_F) = 0$, so the exponential sequence implies $\mathrm{Pic}(F) \cong H^2(F, \mathbb{Z})$. Applying the isomorphism to X with $\mathrm{Pic}(X) = \mathbb{Z}[-K_X]$, we obtain $H^2(X_{\Pi_\tau}, \mathbb{Z}) = \mathbb{Z}c_1(-K_{X_{\Pi_\tau}})$, and applying it to X_{Π_τ}, we obtain $\mathrm{Pic}(X_{\Pi_\tau}) = \mathbb{Z}[-K_{X_{\Pi_\tau}}]$. Hence X_{Π_τ} is a smooth prime Fano threefold of genus 12.

Step 5. In this step, we conclude the proof.

The $\mathbb{G}_m$-action on $\mathcal{X}$ restricts to the action $s \cdot U = \lambda(s^{-1})U$ on X_{Π_τ}. The action is non-trivial: otherwise any point of the threefold X_{Π_τ} would be λ-fixed, whereas the λ-fixed points are the four points of (15.8). Thus the kernel of the action is a proper closed subgroup of $\mathbb{G}_m$, that is, μ_k for some $k \geq 1$, and $\mathbb{G}_m/\mu_k \cong \mathbb{G}_m$ acts faithfully on X_{Π_τ}. By Theorem 2.26, X_{Π_τ} is K-polystable, in particular K-semistable. By Lemma 2.6 applied to π, $\pi^{-1}(t)$ is K-semistable for any t in a non-empty open subset of $\mathbb{A}^1$. Since $\pi^{-1}(t) \cong X$ for any $t \neq 0$, X is K-semistable. □

Lemma 15.3. *Notation as in Construction-Proposition 15.1, and let L be the line with $\Sigma(X, p)$ supported at $[L]$. For $c = [c_0 : c_1 : c_2] \in \mathbb{P}^2$ let $\omega_c := c_0\omega_0 + c_1\omega_1 + c_2\omega_2$ and $\mathcal{C}_c := \{U' \in X : \mathrm{Ker}\,\omega_c \subset U'\}$. Then exactly one or exactly two conics on X contain p, and each of them contains L. There exists exactly one such conic if and only if $\gamma_2 = 0$. If $\gamma_2 \neq 0$, the conics through p are $\mathcal{C}_{[0:0:1]}$ and $\mathcal{C}_{[0:\gamma_2:-\gamma_1]}$. If there exists exactly one conic through p, then X is K-semistable.*

Proof. Here a conic is a closed subscheme of X with Hilbert polynomial $2t+1$ with respect to H, as in Definition 2.11, with no smoothness or reducedness condition. Since Π_X is non-degenerate (Definition-Theorem 2.20), $\mathrm{rank}\,\omega_c = 6$ and $\mathrm{Ker}\,\omega_c$ is one-dimensional for any $c \in \mathbb{P}^2$. By [BKM26b, Lemmas B.1–B.4], the map $\mathbb{P}^2 \to \mathbb{P}(V_X)$, $c \to \mathrm{Ker}\,\omega_c$, is a closed embedding, $\mathcal{C}_c$ is a conic for any c, and any conic on X is $\mathcal{C}_c$ for some c (cf. [KPS18, Proposition 2.3.6(v)], where the Hilbert scheme of conics on X is shown to be isomorphic to $\mathbb{P}^2$).

We claim that $\mathcal{C}_c \neq \mathcal{C}_{c'}$ for $c \neq c'$. Since $c \to \mathrm{Ker}\,\omega_c$ is a closed embedding, $\mathrm{Ker}\,\omega_c \neq \mathrm{Ker}\,\omega_{c'}$, so $W := \mathrm{Ker}\,\omega_c + \mathrm{Ker}\,\omega_{c'}$ is two-dimensional. If $\mathcal{C}_c = \mathcal{C}_{c'}$, then $\mathcal{C}_c = \mathcal{C}_c \cap \mathcal{C}_{c'}$ is the scheme of points $U' \in X$ with $W \subset U'$. The subspaces $U' \supset W$ form the linear subspace $\mathbb{P}(V_X/W) \cong \mathbb{P}^4$ of $\mathbb{P}(\Lambda^3 V_X)$, and for $U' = W + \mathbb{C}v$ the isotropy conditions for a form ω are $\omega|_W = 0$ and $\omega(W, v) = 0$, which are constant or linear in v. Hence $\mathcal{C}_c$ is a linear subspace of $\mathbb{P}(\Lambda^3 V_X)$. The Hilbert polynomial of a linear subspace of dimension r is $\binom{t+r}{r}$, which is never $2t+1$, a contradiction.

The point $p = [U_3]$ belongs to $\mathcal{C}_c$ if and only if $\mathrm{Ker}\,\omega_c \subset U_3$. Write $C(c) := c_0C_0 + c_1C_1 + c_2C_2$. For $u = \xi_0 e_0 + \xi_1 e_1 + \xi_2 e_2 \in U_3$ we have $\omega_c(u, U_3) = 0$ and $(\omega_c(u, e_3), \dots, \omega_c(u, e_6)) = (\xi_0, \xi_1, \xi_2)C(c)$. Thus $p \in \mathcal{C}_c$ if and only if $C(c)$ has a non-zero left kernel, that is, $\mathrm{rank}\,C(c) \leq 2$.

We have

$$C(c) = \begin{pmatrix} c_0 & c_1 & c_2 & \delta_1 \\ 0 & c_0 & c_1 & \delta_2 \\ 0 & 0 & c_0 & \delta_3 \end{pmatrix}, \qquad \delta_1 = \sum_j \alpha_j c_j, \quad \delta_2 = \sum_j \beta_j c_j, \quad \delta_3 = \sum_j \gamma_j c_j.$$

The maximal minors of $C(c)$ in the columns $\{1,2,3\}$, $\{1,2,4\}$, $\{1,3,4\}$, $\{2,3,4\}$ are

$$c_0^3, \qquad c_0^2\delta_3, \qquad c_0(c_1\delta_3 - c_0\delta_2), \qquad c_1^2\delta_3 - c_0c_2\delta_3 - c_0c_1\delta_2 + c_0^2\delta_1.$$

Thus $\operatorname{rank} C(c) \le 2$ if and only if $c_0 = 0$ and

$$c_1^2(\gamma_1 c_1 + \gamma_2 c_2) = 0. \tag{15.10}$$

If $\gamma_2 = 0$, then $\gamma_1 \neq 0$ by Proposition 15.2, and (15.10) has the unique solution $[0:0:1]$. If $\gamma_2 \neq 0$, then (15.10) has exactly two distinct solutions, $[0:0:1]$ and $[0:\gamma_2:-\gamma_1]$. Since $c \to \mathcal{C}_c$ is injective, exactly one conic contains p if $\gamma_2 = 0$, and the conics through p are exactly $\mathcal{C}_{[0:0:1]}$ and $\mathcal{C}_{[0:\gamma_2:-\gamma_1]}$ if $\gamma_2 \neq 0$.

By Construction-Proposition 15.1(7), $\Sigma(X,p)$ is supported at the direction $[\Theta(1,0,0)]$, so the points of L in the chart are $\langle e_0 + xe_3, e_1, e_2\rangle$, $x \in \mathbb{C}$. Thus L is the line of subspaces U' with $\langle e_1, e_2\rangle \subset U' \subset \langle e_0, e_1, e_2, e_3\rangle$. If $\operatorname{Ker}\omega_c \subset \langle e_1, e_2\rangle$, then L is contained in $\{U' : \operatorname{Ker}\omega_c \subset U'\}$ and in X, hence in $\mathcal{C}_c$, as L is reduced. For $c = [0:0:1]$, $C(c) = C_2$ has third column $(1,0,0)^t$, so any left kernel vector has $\xi_0 = 0$ and $\operatorname{Ker}\omega_c \subset \langle e_1, e_2\rangle$. For $c = [0:\gamma_2:-\gamma_1]$ with $\gamma_2 \neq 0$, $C(c) = \gamma_2 C_1 - \gamma_1 C_2$ has second column $(\gamma_2, 0, 0)^t$ and third column $(-\gamma_1, \gamma_2, 0)^t$, so any left kernel vector has $\xi_0 = \xi_1 = 0$ and $\operatorname{Ker}\omega_c = \mathbb{C}e_2$. Thus any conic through p contains L.

Finally, if exactly one conic contains p, then $\gamma_2 = 0$, and X is K-semistable by Proposition 15.2. □

15.2. **Two conics.** In this subsection, we compute the associated graded ring of the anti-canonical ring of X for a $(1,3,5)$-weighted valuation at a point p of incidence type 3 through which exactly two conics pass. We normalize two forms of Π_X simultaneously. Then the leading forms of fourteen Plücker coordinates are linearly independent, and the symbols of their products of degree m along a curve in the exceptional divisor are exactly the N_m lattice points of the m-th dilate of a polytope. Thus the associated graded ring is generated by the fourteen leading forms, and all the S-invariants below are moments of one polytope.

Construction-Proposition 15.4. Let X be a smooth prime Fano threefold of genus 12 and $p \in X$ a closed point such that $\Sigma(X,p)$ is curvilinear of length three and supported at one line. Assume that exactly two conics on X contain p. Then there exist regular parameters x, y, z at p and $\lambda \in \mathbb{C}^*$ with the following properties, where $D_2 := y^2 - xz$, $G_3 := yz + \lambda x^2 D_2$, $J_3 := z^2 + \lambda xyD_2$, and w is the monomial valuation with weights $(1,3,5)$ on (x,y,z), so that D_2, G_3, J_3 have weights $6, 8, 10$.

(1) The weighted blow-up $\pi : Y \to X$ with weights $(1,3,5)$ on (x,y,z) is a plt extraction with exceptional divisor $E \cong \mathbb{P}(1,3,5)$, $\operatorname{Diff}_E(0) = 0$, and $A_X(E) = 9$.
(2) Let T be the leading form of a section of H not vanishing at p. The associated graded ring of $\bigoplus_m H^0(X, mH)$ for w is

$$\mathbb{C}[T, Tx, Ty, Tz, TD_2, TxD_2, TG_3, TyD_2, TJ_3, TzD_2, TD_2^2, TxD_2^2, TD_2G_3, TyD_2^2] \subset \mathbb{C}[x,y,z,T].$$

(3) Let W be the refinement of $(H^0(X,mH))_m$ by E (Definition 2.8), and let $C := (D_2 = 0) \subset E$, parametrized by $x = u^2$, $y = uv$, $z = v^2$. The associated graded algebra of W for the order along C has basis $T^m u^\alpha v^{2r-\alpha} h_C^c$, where h_C is the leading form of D_2 and m, r, α, c are integers with $0 \le r \le 2m$, $0 \le \alpha \le 2r$, $0 \le c \le 2m$, $r + \alpha \le 3m$, $r + c \le 3m$, and $r - \alpha + c \le 2m$. The basis element $T^m u^\alpha v^{2r-\alpha} h_C^c$ belongs to the associated graded space of $W_{m,j}$ for the order along C, where $j = 5r - 2\alpha + 6c$.
(4) $S_X(E) = 733/88$ and $S(W; \operatorname{ord}_C) = 9/11$.
(5) The coarse curve of C is $\mathbb{P}^1$, and the different of (E,C) on C is $\frac{4}{5}P_z$, where $P_x := [1:0:0]$ and $P_z := [0:0:1]$.

(6) Let W^C be the refinement of W by C (Definition 2.8). Then $S(W^C; P_x) = 53/44$, $S(W^C; P_z) = 9/55$, and $S(W^C; q) \leq 9/55$ for any point $q \in C \backslash \{P_x, P_z\}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we make $\alpha_0 = \alpha_1 = \alpha_2 = 0$ and compute the weight-six part of the graph coordinates.

We use the bases $e_0, \dots, e_6$ of V_X and $\omega_0, \omega_1, \omega_2$ of Π_X, the constants $\alpha_j, \beta_j, \gamma_j$, the matrices B_j, J, $\Theta(a,b,c)$, and the chart $M \to [M]$ of Construction-Proposition 15.1. Let $e_0^\vee, \dots, e_6^\vee$ be the dual basis of $V_X^\vee$. Then

$$\omega_j = \sum_{i=0}^{2} \sum_{l=1}^{4} (C_j)_{i+1,l} e_i^\vee \wedge e_{l+2}^\vee + \sum_{1 \leq p < q \leq 4} (B_j)_{pq} e_{p+2}^\vee \wedge e_{q+2}^\vee,$$

and the subspaces $U' \in X$ near p are the subspaces U' with $\omega_j|_{U'} = 0$ for $j = 0, 1, 2$ (Construction-Proposition 15.1(3)). Let $x_0 := M_{11}$, $y_0 := M_{12}$, $z_0 := M_{13}$, which are regular parameters at p, let $D_0 := y_0^2 - x_0 z_0$, and let $I := (D_0, y_0 z_0, z_0^2) \subset \widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x_0, y_0, z_0]]$.

Replace e_6 with $e_6' := e_6 - \alpha_0 e_3 - \alpha_1 e_4 - \alpha_2 e_5$. In the basis $e_0, \dots, e_5, e_6'$ the Gram matrices of ω_j have the form of Construction-Proposition 15.1(1) and (2) with $\alpha_0 = \alpha_1 = \alpha_2 = 0$, with $\beta_0, \gamma_0, \beta_1$ replaced by $\beta_0 - \alpha_1, \gamma_0 - \alpha_2, \beta_1 - \alpha_2$, while $\gamma_1, \gamma_2, \beta_2$ and $(B_j)_{12}$ keep their values. In particular, the matrix J does not change. The new graph coordinates are $M_{il} + \alpha_{l-1} M_{i4}$ for $l \leq 3$ and M_{i4} for $l = 4$. Since $M_{14} \in I$, the new parameters differ from x_0, y_0, z_0 by elements of I. Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{X,p}$. Then $D_0, y_0 z_0, z_0^2$ with x_0, y_0, z_0 replaced by the new parameters differ from $D_0, y_0 z_0, z_0^2$ by elements of $\mathfrak{m} I$, so they generate an ideal $I' \subset I$ with $I = I' + \mathfrak{m} I$, and $I' = I$ by Nakayama's lemma. Moreover, any entry of the new matrix minus Θ of the new parameters belongs to I. Thus we may assume that $\alpha_0 = \alpha_1 = \alpha_2 = 0$, and properties (1)–(7) of Construction-Proposition 15.1 still hold.

Since exactly two conics contain p, $\gamma_2 \neq 0$ (Lemma 15.3). Let $k := (B_2)_{12}$. The vector $e_1 - \frac{\beta_2}{\gamma_2} e_2$ belongs to the kernel of ω_2. Since Π_X is non-degenerate (Definition-Theorem 2.20), ω_2 has rank 6, so the restriction of ω_2 to $\langle e_0, e_2, e_3, e_4, e_5, e_6 \rangle$ is non-degenerate. On $\langle e_0, e_2, e_3, e_4, e_5, e_6 \rangle$ the only non-zero values of ω_2 on pairs containing e_0 or e_2 are $\omega_2(e_0, e_5) = 1$ and $\omega_2(e_2, e_6) = \gamma_2$, so the Pfaffian of the restriction is $\pm \gamma_2 k$. Thus $k \neq 0$. Let

$$\theta := \frac{\gamma_1 \beta_2 - \gamma_2 \beta_1}{\gamma_2^2}, \qquad \eta := \frac{k \gamma_2^2}{\det J}.$$

Here

$$\det J = (\gamma_1 - \beta_2)(\gamma_1 \beta_2 - \gamma_2 \beta_1) - \gamma_0 \gamma_2 \beta_2 + \beta_0 \gamma_2^2 \neq 0,$$

so $\eta \neq 0$.

Let w be the monomial valuation with weights $(1, 3, 5)$ on (x_0, y_0, z_0). For $f \in \widehat{\mathcal{O}}_{X,p}$ we call the weighted homogeneous part of f of weight j the *weight-j part* of f. Any $f \in I$ has $w(f) \geq 6$, and the weight-six part of f is a constant multiple of D_0, because $y_0 z_0$ and z_0^2 have weights 8 and 10. By Construction-Proposition 15.1(6), there exists a matrix $A = (a_{il}) \in \mathrm{Mat}_{3 \times 4}(\mathbb{C})$ with $a_{11} = a_{12} = a_{13} = 0$ such that

$$w\big(M_{il} - \Theta(x_0, y_0, z_0)_{il} - a_{il} D_0\big) \geq 7 \qquad \text{for all } i, l.$$

Since the local equations of X at p are $C_j M^t - M C_j^t + M B_j M^t = 0$ (Construction-Proposition 15.1(3)), $T_p X$ is the common kernel of their linear parts $C_j N^t - N C_j^t$, so Construction-Proposition 15.1(5) implies $C_j \Theta(a,b,c)^t = \Theta(a,b,c) C_j^t$ for any $a, b, c \in \mathbb{C}$. Hence $C_j \Theta^t = \Theta C_j^t$ for $\Theta = \Theta(x_0, y_0, z_0)$, so the weight-six part of $C_j M^t - M C_j^t$ is $(C_j A^t - A C_j^t) D_0$. Since the entries of Θ have $w \geq 1$, the weight-six part of $M B_j M^t$ equals the weight-six part of $\Theta B_j \Theta^t$. The 2×2 minors of Θ in rows $1, 2$ are $-D_0$, $-y_0 z_0$, $-z_0^2$, and 0, and the 2×2 minors in rows $1, 3$ and in rows $2, 3$ belong to $\{-y_0 z_0, -z_0^2, 0\}$. Thus the weight-six parts of the entries $(1,2)$, $(1,3)$, $(2,3)$ of $\Theta B_j \Theta^t$ are $-(B_j)_{12} D_0$, 0, 0. Taking the weight-six parts of six of the nine equations

$C_jM^t - MC_j^t + MB_jM^t = 0$, namely the entries $(2,3)$ and $(1,3)$ for $j=2$, $(2,3)$ and $(1,3)$ for $j=1$, $(1,2)$ for $j=2$, and $(2,3)$ for $j=0$, we obtain

$$\beta_2 a_{34} - \gamma_2 a_{24} = 0, \quad a_{33} - \gamma_2 a_{14} = 0, \quad a_{33} + \beta_1 a_{34} - \gamma_1 a_{24} = 0,$$
$$a_{32} - \gamma_1 a_{14} = 0, \quad a_{23} - \beta_2 a_{14} = k, \quad a_{32} - a_{23} + \beta_0 a_{34} - \gamma_0 a_{24} = 0.$$

The first five equations imply $a_{24} = \frac{\beta_2}{\gamma_2} a_{34}$, $a_{33} = \gamma_2\theta a_{34}$, $a_{14} = \theta a_{34}$, $a_{32} = \gamma_1\theta a_{34}$, and $a_{23} = \beta_2\theta a_{34} + k$. Substituting them into the last equation, we obtain $\frac{\det J}{\gamma_2^2} a_{34} = k$, that is, $a_{34} = \eta$. In particular,

$$w(M_{34} - \eta D_0) \geq 7. \tag{15.11}$$

Step 2. In this step, we normalize $\omega_1 - \frac{\gamma_1}{\gamma_2}\omega_2$ and ω_2 simultaneously, and we define x, y, z and λ. Let

$$\epsilon_2 := e_2^\vee + \theta e_0^\vee + \frac{\beta_2}{\gamma_2} e_1^\vee, \quad \epsilon_3 := e_3^\vee, \quad \epsilon_4 := e_4^\vee - \frac{\gamma_1}{\gamma_2} e_5^\vee, \quad \epsilon_5 := e_5^\vee - \gamma_2\theta e_6^\vee, \quad \epsilon_6 := e_6^\vee.$$

Since $\gamma_1\beta_2/\gamma_2 - \beta_1 = \gamma_2\theta$, a direct calculation from the matrices C_1 and C_2 (with $\alpha_j = 0$) shows

$$\omega_1 - \frac{\gamma_1}{\gamma_2}\omega_2 = e_0^\vee \wedge \epsilon_4 + e_1^\vee \wedge \epsilon_5 + \sigma_1, \qquad \omega_2 = e_0^\vee \wedge \epsilon_5 + \gamma_2 \epsilon_2 \wedge \epsilon_6 + \sigma_2,$$

with $\sigma_1, \sigma_2 \in \Lambda^2\langle \epsilon_3, \epsilon_4, \epsilon_5, \epsilon_6\rangle$. Substituting $e_4^\vee = \epsilon_4 + \frac{\gamma_1}{\gamma_2} e_5^\vee$ and $e_5^\vee = \epsilon_5 + \gamma_2\theta\epsilon_6$ into $\sum_{p<q} (B_2)_{pq} e_{p+2}^\vee \wedge e_{q+2}^\vee$ shows that the coefficient of $\epsilon_3 \wedge \epsilon_4$ in σ_2 is k. Since $\epsilon_3 \wedge \epsilon_4, \epsilon_3 \wedge \epsilon_5, \epsilon_3 \wedge \epsilon_6, \epsilon_4 \wedge \epsilon_5, \epsilon_4 \wedge \epsilon_6, \epsilon_5 \wedge \epsilon_6$ form a basis of $\Lambda^2\langle \epsilon_3, \dots, \epsilon_6\rangle$, there exist unique $\ell_0 \in \langle \epsilon_3, \epsilon_6\rangle$, $\ell_1 \in \langle \epsilon_3, \epsilon_4, \epsilon_6\rangle$, and $\kappa \in \mathbb{C}$ with $\sigma_1 = \ell_0 \wedge \epsilon_4 + \ell_1 \wedge \epsilon_5 + \kappa\epsilon_3 \wedge \epsilon_6$. Let $\epsilon_0 := e_0^\vee + \ell_0$ and $\epsilon_1 := e_1^\vee + \ell_1$. Then $\epsilon_0, \dots, \epsilon_6$ is a basis of $V_X^\vee$, and

$$\omega_1 - \frac{\gamma_1}{\gamma_2}\omega_2 = \epsilon_0 \wedge \epsilon_4 + \epsilon_1 \wedge \epsilon_5 + \kappa\epsilon_3 \wedge \epsilon_6, \qquad \omega_2 = \epsilon_0 \wedge \epsilon_5 + \gamma_2 \epsilon_2 \wedge \epsilon_6 + \sum_{1 \leq p < q \leq 4} (B_2')_{pq} \epsilon_{p+2} \wedge \epsilon_{q+2} \tag{15.12}$$

for an alternating matrix B_2'. Substituting $e_0^\vee = \epsilon_0 - \ell_0$ into ω_2 changes σ_2 by $-\ell_0 \wedge \epsilon_5$, which has no $\epsilon_3 \wedge \epsilon_4$ term since $\ell_0 \in \langle \epsilon_3, \epsilon_6\rangle$. Hence $(B_2')_{12} = k$. The form $\omega_1 - \frac{\gamma_1}{\gamma_2}\omega_2$ is a non-zero element of Π_X, so it has rank 6 since Π_X is non-degenerate (Definition-Theorem 2.20). By (15.12), the rank of $\omega_1 - \frac{\gamma_1}{\gamma_2}\omega_2$ would be 4 if $\kappa = 0$. Thus $\kappa \neq 0$. Let $\lambda := \kappa\eta \neq 0$.

Let $f_0, \dots, f_6$ be the basis of V_X dual to $\epsilon_0, \dots, \epsilon_6$. Since $\langle \epsilon_3, \dots, \epsilon_6\rangle = \langle e_3^\vee, \dots, e_6^\vee\rangle$, we have $\langle f_0, f_1, f_2\rangle = U_3$, so $p = [U_3]$ is the origin of the chart $M' \to [M']$ of $\mathrm{Gr}(3, V_X)$ defined by the basis $f_0, \dots, f_6$, where $[M']$ is the span of the rows of $(I_3|M')$. We compute M' in terms of M. Let r_1, r_2, r_3 be the rows of $(I_3|M)$, viewed as vectors of V_X in the basis $e_0, \dots, e_6$, and let $r_1' := r_1 - \theta r_3$, $r_2' := r_2 - \frac{\beta_2}{\gamma_2} r_3$, $r_3' := r_3$. Then $e_0^\vee, e_1^\vee, \epsilon_2$ take the values $\delta_{i1}, \delta_{i2}, \delta_{i3}$ on r_i'. Let $\widetilde{M}$ be the 3×4 matrix with $\widetilde{M}_{il} := \epsilon_{l+2}(r_i')$, and let $\Xi \in \mathrm{Mat}_{4\times 3}(\mathbb{C})$ with $\ell_0 = \sum_l \Xi_{l1}\epsilon_{l+2}$, $\ell_1 = \sum_l \Xi_{l2}\epsilon_{l+2}$, and $\Xi_{l3} = 0$. Then $\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3, \dots, \epsilon_6$ take the values given by the rows of $(I_3 + \widetilde{M}\Xi|\widetilde{M})$ on r_1', r_2', r_3', so

$$M' = (I_3 + \widetilde{M}\Xi)^{-1}\widetilde{M} = \widetilde{M} + \Delta, \qquad \Delta := \sum_{n \geq 1} (-\widetilde{M}\Xi)^n \widetilde{M}.$$

The entries of M' are regular functions near p, since $I_3 + \widetilde{M}\Xi = I_3$ at p. From the definitions of $\epsilon_3, \dots, \epsilon_6$ and **Step 1**, the entries of

$$\widetilde{M} - \begin{pmatrix} x_0 - \theta z_0 & y_0 - \frac{\gamma_1}{\gamma_2} z_0 & z_0 & 0 \\ y_0 - \frac{\beta_2}{\gamma_2} z_0 & z_0 & 0 & 0 \\ z_0 & 0 & 0 & \eta D_0 \end{pmatrix}$$

have $w \geq 6$, and the $(3,4)$ entry has $w \geq 7$ by (15.11). In particular $w(\widetilde{M}_{il}) \geq \rho_i$ with $(\rho_1, \rho_2, \rho_3) = (1, 3, 5)$, and for $j \in \{1, 2\}$ we have $w(\widetilde{M}_{jl}) \geq \rho_l'$ with $(\rho_1', \rho_2', \rho_3', \rho_4') = (1, 3, 5, 6)$. Since $\Xi_{l3} = 0$,

the (i,l) entry of $(\widetilde{M}\Xi)^n\widetilde{M}$ is a sum of products $(\widetilde{M}\Xi)_{ij_1}(\widetilde{M}\Xi)_{j_1j_2}\cdots(\widetilde{M}\Xi)_{j_{n-1}j_n}\widetilde{M}_{j_nl}$ with $j_1,\dots,j_n\in\{1,2\}$. Since $w((\widetilde{M}\Xi)_{ij})\ge\rho_i$, we obtain

$$w\big(\Delta_{il}\big)\ge\rho_i+\rho'_l,\qquad w\big(\Delta_{il}+(\widetilde{M}\Xi\widetilde{M})_{il}\big)\ge\rho_i+\rho'_l+1. \tag{15.13}$$

Let $x:=M'_{11}$, $y:=M'_{12}$, $z:=M'_{13}$. By (15.13), $w(x-x_0)\ge2$, $w(y-y_0)\ge4$, and $w(z-z_0)\ge6$. Thus x,y,z are regular parameters at p with leading forms x_0,y_0,z_0 for w. Hence a weighted homogeneous polynomial of weight j in x,y,z has w exactly j, so w is also the monomial valuation with weights $(1,3,5)$ on (x,y,z). Moreover $w(D_2-D_0)\ge7$ for $D_2=y^2-xz$, since $w((y-y_0)(y+y_0))\ge7$ and $w((x-x_0)z+x_0(z-z_0))\ge7$.

We bound the entry $M'_{21}-y=(\widetilde{M}_{21}-\widetilde{M}_{12})+(\Delta_{21}-\Delta_{12})$. First, $\widetilde{M}_{21}-\widetilde{M}_{12}=\frac{\gamma_1-\beta_2}{\gamma_2}z_0+(w\ge6)$, so $w(\widetilde{M}_{21}-\widetilde{M}_{12})\ge5$. Second, by (15.13) it suffices to bound $(\widetilde{M}\Xi\widetilde{M})_{21}-(\widetilde{M}\Xi\widetilde{M})_{12}$. The terms of $(\widetilde{M}\Xi\widetilde{M})_{21}=\sum_{j=1}^2(\widetilde{M}\Xi)_{2j}\widetilde{M}_{j1}$ and $(\widetilde{M}\Xi\widetilde{M})_{12}=\sum_{j=1}^2(\widetilde{M}\Xi)_{1j}\widetilde{M}_{j2}$ with $j=2$ have $w\ge6$. Since $w(\widetilde{M}_{2l})\ge5$ and $w(\widetilde{M}_{1l})\ge3$ for $l\ge2$, we have $w((\widetilde{M}\Xi)_{21}-\Xi_{11}\widetilde{M}_{21})\ge5$ and $w((\widetilde{M}\Xi)_{11}-\Xi_{11}\widetilde{M}_{11})\ge3$. Hence

$$(\widetilde{M}\Xi)_{21}\widetilde{M}_{11}-(\widetilde{M}\Xi)_{11}\widetilde{M}_{12}=\Xi_{11}\widetilde{M}_{11}(\widetilde{M}_{21}-\widetilde{M}_{12})+(w\ge6),$$

and $w(M'_{21}-y)\ge5$. The other bounds below follow directly from (15.13) and the leading terms of $\widetilde{M}$.

From now on we write M for M' and $e_0^\vee,\dots,e_6^\vee$ are no longer used. We have shown

$$\begin{gathered}w(M_{21}-y)\ge5,\quad w(M_{22}-z)\ge6,\quad w(M_{31}-z)\ge6,\quad w(M_{34}-\eta D_2)\ge7,\\ w(M_{14}),\ w(M_{24}),\ w(M_{23}),\ w(M_{32}),\ w(M_{33})\ge6.\end{gathered} \tag{15.14}$$

In the basis $f_0,\dots,f_6$, the condition $\omega|_{[M]}=0$ for the forms $\omega_1-\frac{\gamma_1}{\gamma_2}\omega_2$ and ω_2 of (15.12) reads $C'M^t-MC'^t+MB'M^t=0$, where (C',B') is the pair of blocks of the Gram matrix of the form. Writing out the entries $(1,2)$, $(1,3)$, $(2,3)$, we obtain

$$\begin{aligned}M_{22}&=z-\kappa(xM_{24}-M_{14}M_{21}),&\qquad M_{32}&=-\kappa(xM_{34}-M_{14}M_{31}),\\ M_{33}&=-\kappa(M_{21}M_{34}-M_{24}M_{31}),&\qquad M_{23}+(MB'_2M^t)_{12}&=0,\\ M_{33}-\gamma_2M_{14}+(MB'_2M^t)_{13}&=0,&\qquad -\gamma_2M_{24}+(MB'_2M^t)_{23}&=0,\end{aligned} \tag{15.15}$$

where $(MB'_2M^t)_{ab}=\sum_{p<q}(B'_2)_{pq}(M_{ap}M_{bq}-M_{aq}M_{bp})$.

Claim 15.5. *The entries of M satisfy*

$$\begin{gathered}w(M_{21}-y)\ge5,\quad w(M_{31}-z)\ge6,\quad w(M_{22}-z)\ge10,\quad w(M_{14})\ge7,\quad w(M_{24})\ge9,\\ w(M_{23}-kD_2)\ge7,\quad w(M_{34}-\eta D_2)\ge7,\quad w(M_{32}+\lambda xD_2)\ge8,\quad w(M_{33}+\lambda yD_2)\ge10.\end{gathered}$$

In particular $w(M_{32})\ge7$ and $w(M_{33})\ge9$.

Proof. The first, second, and seventh bounds are in (15.14). By the second equation of (15.15),

$$M_{32}+\lambda xD_2=-\kappa x(M_{34}-\eta D_2)+\kappa M_{14}M_{31},$$

where the terms have $w\ge1+7$ and $w\ge6+5$ by (15.14). By the third equation of (15.15),

$$M_{33}+\lambda yD_2=-\kappa(M_{21}-y)M_{34}-\kappa y(M_{34}-\eta D_2)+\kappa M_{24}M_{31},$$

where the terms have $w\ge5+6$, $w\ge3+7$, and $w\ge6+5$. This proves the last two bounds.

Next we use the last three equations of (15.15), which come from ω_2. By (15.14) and the bounds for M_{32} and M_{33} just proved, the entries of the three rows of M have w at least $(1,3,5,6)$, $(3,5,6,6)$, and $(5,7,9,6)$. Hence each minor $M_{1p}M_{3q}-M_{1q}M_{3p}$ has $w\ge7$. The minimum 7 is attained only by xM_{34}, from the minor with $(p,q)=(1,4)$. By the fifth equation of (15.15), $\gamma_2M_{14}=M_{33}+(MB'_2M^t)_{13}$ has $w\ge7$, so $w(M_{14})\ge7$. Each minor $M_{2p}M_{3q}-M_{2q}M_{3p}$ has $w\ge9$: the smallest products are $M_{21}M_{34}$ of weight ≥9, $M_{21}M_{32}$ and $M_{22}M_{31}$ of weight ≥10,

and the others have weight ≥ 11. By the sixth equation of (15.15), $w(M_{24}) \geq 9$. Each minor $M_{1p}M_{2q} - M_{1q}M_{2p}$ with $(p,q) \neq (1,2)$ has $w \geq 7$, and

$$xM_{22} - yM_{21} = -D_2 + x(M_{22} - z) - y(M_{21} - y)$$

with $w(x(M_{22} - z)) \geq 7$ and $w(y(M_{21} - y)) \geq 8$. Since $(B_2')_{12} = k$, the fourth equation of (15.15) implies $M_{23} = kD_2 + (w \geq 7)$. Finally, by the first equation of (15.15), $M_{22} - z = -\kappa(xM_{24} - M_{14}M_{21})$ has $w \geq \min\{1+9, 7+3\} = 10$. □

Proof of Construction-Proposition 15.4 continued.

Step 3. In this step, we compute the leading forms of fourteen Plücker coordinates.

For $0 \leq i < j < l \leq 6$ let $P_{ijl} := \epsilon_i \wedge \epsilon_j \wedge \epsilon_l$, viewed as a linear form on $\Lambda^3 V_X$. Since the Plücker line bundle of $\mathrm{Gr}(3, V_X)$ restricts to H on X (Definition-Theorem 2.20), $P_{ijl}|_X \in H^0(X, H)$. Let $s_0 := P_{012}|_X$. Then $s_0(p) \neq 0$, and on the chart $M \to [M]$ the function P_{ijl}/s_0 is the 3×3 minor of $(I_3|M)$ in the columns i, j, l, where the columns are numbered $0, \dots, 6$. Let $R_m := H^0(X, mH)$ and $F^j R_m := \{s \in R_m : w(s/s_0^m) \geq j\}$. For $0 \neq s \in R_m$ with $w(s/s_0^m) = j$, let $\mathrm{in}(s) := fT^m$, where $f \in \mathbb{C}[x,y,z]$ is the weight-j part of s/s_0^m. Since w is a valuation, $\mathrm{in}(ss') = \mathrm{in}(s)\,\mathrm{in}(s')$, so in embeds the associated graded ring

$$\mathrm{gr}_w R := \bigoplus_{m,j \geq 0} F^j R_m / F^{j+1} R_m$$

into $\mathbb{C}[x,y,z,T]$, and we regard $\mathrm{gr}_w R$ as a subring of $\mathbb{C}[x,y,z,T]$. Here $T = \mathrm{in}(s_0)$ is the leading form of a section of H not vanishing at p, as in (2).

Expanding the minors, we obtain $P_{012}/s_0 = 1$, $P_{123}/s_0 = x$, $P_{124}/s_0 = y$, $P_{125}/s_0 = z$, $P_{016}/s_0 = M_{34}$,

$$P_{136}/s_0 = -xM_{34} + M_{14}M_{31}, \quad P_{134}/s_0 = yM_{31} - xM_{32}, \quad P_{146}/s_0 = -yM_{34} + M_{14}M_{32},$$
$$P_{135}/s_0 = zM_{31} - xM_{33}, \quad P_{156}/s_0 = -zM_{34} + M_{14}M_{33},$$

and

$$\begin{aligned}
P_{346}/s_0 &= M_{34}(xM_{22} - yM_{21}) - M_{24}(xM_{32} - yM_{31}) + M_{14}(M_{21}M_{32} - M_{22}M_{31}),\\
P_{356}/s_0 &= x(M_{23}M_{34} - M_{24}M_{33}) - z(M_{21}M_{34} - M_{24}M_{31}) + M_{14}(M_{21}M_{33} - M_{23}M_{31}),\\
P_{345}/s_0 &= M_{23}(yM_{31} - xM_{32}) + M_{33}(xM_{22} - yM_{21}) + z(M_{21}M_{32} - M_{22}M_{31}),\\
P_{456}/s_0 &= y(M_{23}M_{34} - M_{24}M_{33}) - z(M_{22}M_{34} - M_{24}M_{32}) + M_{14}(M_{22}M_{33} - M_{23}M_{32}).
\end{aligned} \tag{15.16}$$

Table 24 lists, for each of the fourteen functions, the lowest term, the weight of the lowest term, and lower bounds for the weights of the remaining terms. Each bound in the last column is the sum of the bounds of Claim 15.5 for the factors, together with $w(x) = 1$, $w(y) = 3$, $w(z) = 5$, $w(M_{21}) = 3$, $w(M_{22}) = w(M_{31}) = 5$, $w(M_{23}) = w(M_{34}) = 6$, and the following three consequences of Claim 15.5:

$$w\big(yM_{31} - xM_{32} - G_3\big) \geq 9, \qquad w\big(xM_{22} - yM_{21} + D_2\big) \geq 8, \qquad w\big(M_{21}M_{32} - M_{22}M_{31}\big) \geq 10.$$

For the first, $yM_{31} - xM_{32} - G_3 = y(M_{31} - z) - x(M_{32} + \lambda x D_2)$. For the second, $xM_{22} - yM_{21} + D_2 = x(M_{22} - z) - y(M_{21} - y)$. For the third, $w(M_{21}M_{32}) \geq 3 + 7$ and $w(M_{22}M_{31}) \geq 5 + 5$. In the rows P_{356}, P_{345}, and P_{456} the lowest term is the lowest term of $xM_{23}M_{34}$, $M_{23}(yM_{31} - xM_{32})$, and $yM_{23}M_{34}$, respectively, and the bounds for the differences use

$$M_{23}M_{34} - k\eta D_2^2 = (M_{23} - kD_2)M_{34} + kD_2(M_{34} - \eta D_2), \qquad w \geq 13,$$

and

$$\begin{aligned}
M_{23}(yM_{31} - xM_{32}) - kD_2G_3 = {} & (M_{23} - kD_2)(yM_{31} - xM_{32})\\
& + kD_2(yM_{31} - xM_{32} - G_3), \qquad w \geq 15.
\end{aligned}$$

The fourteen weights $0, 1, 3, 5, 6, \dots, 15$ are distinct, and $k, \eta, \lambda \neq 0$. Hence, after multiplication by non-zero constants, the leading forms $\mathrm{in}(P_{ijl}|_X)$ are the fourteen generators

$$T,\ Tx,\ Ty,\ Tz,\ TD_2,\ TxD_2,\ TG_3,\ TyD_2,\ TJ_3,\ TzD_2,\ TD_2^2,\ TxD_2^2,\ TD_2G_3,\ TyD_2^2$$

TABLE 24. Lowest terms of P_{ijl}/s_0 for w, with lower bounds for the weights of the remaining terms.

P_{ijl}	lowest term	weight	remaining terms
P_{012}	1	0	none
P_{123}	x	1	none
P_{124}	y	3	none
P_{125}	z	5	none
P_{016}	ηD_2	6	$M_{34}-\eta D_2$: ≥ 7
P_{136}	$-\eta x D_2$	7	$-x(M_{34}-\eta D_2)$: ≥ 8; $M_{14}M_{31}$: ≥ 12
P_{134}	G_3	8	$y(M_{31}-z)$: ≥ 9; $-x(M_{32}+\lambda x D_2)$: ≥ 9
P_{146}	$-\eta y D_2$	9	$-y(M_{34}-\eta D_2)$: ≥ 10; $M_{14}M_{32}$: ≥ 14
P_{135}	J_3	10	$z(M_{31}-z)$: ≥ 11; $-x(M_{33}+\lambda y D_2)$: ≥ 11
P_{156}	$-\eta z D_2$	11	$-z(M_{34}-\eta D_2)$: ≥ 12; $M_{14}M_{33}$: ≥ 16
P_{346}	$-\eta D_2^2$	12	$M_{34}(xM_{22}-yM_{21})+\eta D_2^2$: ≥ 13; $M_{24}(xM_{32}-yM_{31})$: ≥ 17; $M_{14}(M_{21}M_{32}-M_{22}M_{31})$: ≥ 17
P_{356}	$k\eta x D_2^2$	13	$x(M_{23}M_{34}-k\eta D_2^2)$: ≥ 14; $xM_{24}M_{33}$: ≥ 19; $zM_{21}M_{34}$: ≥ 14; $zM_{24}M_{31}$: ≥ 19; $M_{14}M_{21}M_{33}$: ≥ 19; $M_{14}M_{23}M_{31}$: ≥ 18
P_{345}	kD_2G_3	14	$M_{23}(yM_{31}-xM_{32})-kD_2G_3$: ≥ 15; $M_{33}(xM_{22}-yM_{21})$: ≥ 15; $z(M_{21}M_{32}-M_{22}M_{31})$: ≥ 15
P_{456}	$k\eta y D_2^2$	15	$y(M_{23}M_{34}-k\eta D_2^2)$: ≥ 16; $yM_{24}M_{33}$: ≥ 21; $zM_{22}M_{34}$: ≥ 16; $zM_{24}M_{32}$: ≥ 21; $M_{14}M_{22}M_{33}$: ≥ 21; $M_{14}M_{23}M_{32}$: ≥ 20

of the algebra in (2), which we denote by $\mathcal{B}$. Since in is multiplicative, $\mathcal{B} \subset \operatorname{gr}_w R$. For $m, j \geq 0$ let $\mathcal{B}_{m,j} := \mathcal{B} \cap \mathbb{C}[x,y,z]_j T^m$, where $\mathbb{C}[x,y,z]_j$ is the space of weighted homogeneous polynomials of weight j.

Step 4. In this step, we prove (2) and determine the order along D_2 on $\mathcal{B}$.

The polynomial D_2 is prime, and $\mathbb{C}[x,y,z]/(D_2) \cong \mathbb{C}[u^2, uv, v^2] \subset \mathbb{C}[u,v]$ by $x \to u^2$, $y \to uv$, $z \to v^2$. Thus the associated graded ring of $\mathbb{C}[x,y,z]$ for the order along D_2 is $\mathbb{C}[u^2,uv,v^2][h_C]$, where h_C is the class of D_2, and it is a domain. Let $\operatorname{sym}(f)$ denote the class in $\mathbb{C}[u^2,uv,v^2][h_C]$ of a non-zero $f \in \mathbb{C}[x,y,z]$. Then $\operatorname{sym}(ff') = \operatorname{sym}(f)\operatorname{sym}(f')$. We write $T^m u^\alpha v^{2r-\alpha} h_C^c$ as the lattice point (r,α,c). Since $G_3 \equiv yz$ and $J_3 \equiv z^2$ modulo D_2, the symbols of the fourteen generators of $\mathcal{B}$ (with T of degree $m=1$) are the points

$$\begin{aligned} c=0: &\quad (r,\alpha) = (0,0),(1,0),(1,1),(1,2),(2,0),(2,1),\\ c=1: &\quad (r,\alpha) = (0,0),(1,0),(1,1),(1,2),(2,1),\\ c=2: &\quad (r,\alpha) = (0,0),(1,1),(1,2). \end{aligned} \tag{15.17}$$

The points in the three rows are, in the order listed, the symbols of $T, Tz, Ty, Tx, TJ_3, TG_3$, of $TD_2, TzD_2, TyD_2, TxD_2, TD_2G_3$, and of TD_2^2, TyD_2^2, TxD_2^2. For $m \geq 0$ let Λ_m be the set of $(r,\alpha,c) \in \mathbb{Z}^3$ with

$$\begin{gathered} 0 \leq r \leq 2m, \quad 0 \leq \alpha \leq 2r, \quad 0 \leq c \leq 2m,\\ r+\alpha \leq 3m, \quad r+c \leq 3m, \quad r-\alpha+c \leq 2m. \end{gathered} \tag{15.18}$$

A direct check shows that Λ_1 is exactly the set of fourteen points in (15.17): for $r=0$ the conditions are $\alpha=0$ and $c \leq 2$; for $r=1$ they are $\alpha \leq 2$ and $c \leq \min\{2, 1+\alpha\}$; for $r=2$ they are $\alpha \leq 1$ and $c \leq \alpha$. Since the inequalities (15.18) are linear and homogeneous in (m,r,α,c), a sum of m points of Λ_1 belongs to Λ_m.

We prove the converse: any $(r,\alpha,c) \in \Lambda_m$ is a sum $b_1 + \cdots + b_m$ of points $b_i = (r_i,\alpha_i,c_i) \in \Lambda_1$. By the description of Λ_1, if $r_i = 0$ then $\alpha_i = 0$ and $c_i \leq 2$; if $r_i = 1$ then $\alpha_i \leq 2$, and $c_i \leq 1$ for $\alpha_i = 0$ and $c_i \leq 2$ for $\alpha_i \geq 1$; if $r_i = 2$ then $\alpha_i \leq 1$ and $c_i \leq \alpha_i$. Conversely any (r_i,α_i,c_i) with $\alpha_i, c_i \geq 0$ satisfying the conditions just listed belongs to Λ_1.

Suppose $r \leq m$. Then (15.18) is equivalent to $\alpha \leq 2r$ and $c \leq \min\{2m, 2m-r+\alpha\}$. Let $r_i = 1$ for $i \leq r$ and $r_i = 0$ for $i > r$. Let $\alpha_i = 1$ for $i \leq \min\{\alpha, r\}$, add a second unit to α_i for

$i \le \max\{0, \alpha - r\}$, and let $\alpha_i = 0$ otherwise. Exactly $\max\{0, r-\alpha\}$ indices $i \le r$ have $\alpha_i = 0$, so the sum of the largest admitted values of c_i is

$$2(m-r) + 2r - \max\{0, r-\alpha\} = \min\{2m, 2m - r + \alpha\}.$$

Lowering the values c_i one unit at a time realizes any smaller $c \ge 0$.

Suppose $m \le r \le 2m$. Then (15.18) is equivalent to $\alpha \le 3m-r$ and $c \le \min\{3m-r, 2m-r+\alpha\}$. Let $r_i = 1$ for $i \le 2m - r$ and $r_i = 2$ for $i > 2m - r$. Let $\alpha_i = 1$ for $\min\{\alpha, m\}$ indices i, and if $\alpha > m$, add a second unit to α_i for $\alpha - m$ indices $i \le 2m - r$. This is possible since $\alpha - m \le 2m - r$. The largest admitted c_i is α_i if $r_i = 2$, and it is 1 or 2 if $r_i = 1$, according as $\alpha_i = 0$ or $\alpha_i \ge 1$. Hence the sum of the largest admitted values of c_i is $(2m - r) + \min\{\alpha, m\} = \min\{3m - r, 2m - r + \alpha\}$, and any smaller $c \ge 0$ is realized by lowering the values c_i one unit at a time.

Next we count Λ_m. For $r \le m$, α runs from 0 to $2r$ and c from 0 to $2m - \max\{0, r-\alpha\}$, so there are $(2r+1)(2m+1) - \frac{r(r+1)}{2}$ points. For $m \le r \le 2m$, α runs from 0 to $3m - r$ and c from 0 to $3m - r - \max\{0, m-\alpha\}$, so there are $(3m - r + 1)^2 - \frac{m(m+1)}{2}$ points. Summing the first number over $0 \le r \le m$ and the second over $m + 1 \le r \le 2m$, we obtain

$$\begin{aligned}\#\Lambda_m &= (2m+1)(m+1)^2 - \frac{m(m+1)(m+2)}{6} + \frac{m(2m+1)(7m+1)}{6} - \frac{m^2(m+1)}{2} \\ &= \frac{11}{3}m^3 + \frac{11}{2}m^2 + \frac{23}{6}m + 1,\end{aligned}$$

which equals N_m by Lemma 2.12(2) with $g = 12$.

Fix $m, j \ge 0$. The products of m generators of $\mathcal{B}$ belong to $\mathcal{B}_{m,j}$ when their weights sum to j, and their symbols are $T^m u^\alpha v^{2r-\alpha} h_C^c$ with (r, α, c) the sums of the corresponding points of (15.17). With the weights $(1, 5)$ on (u, v), the monomial $u^\alpha v^{2r-\alpha}$ has weight $\alpha + 5(2r - \alpha) = 2(5r - 2\alpha)$, twice the weight in $\mathbb{C}[x, y, z]$, and h_C has weight 6. Thus the point (r, α, c) has weight $5r - 2\alpha + 6c$. By the converse proved above, any $(r, \alpha, c) \in \Lambda_m$ with $5r - 2\alpha + 6c = j$ is the symbol of a product in $\mathcal{B}_{m,j}$. Distinct monomials are linearly independent. The order along D_2 defines a finite filtration of $\mathcal{B}_{m,j}$, so $\mathcal{B}_{m,j}$ and its associated graded space have equal dimensions. Hence

$$\dim \mathcal{B}_{m,j} \ge \#\{(r, \alpha, c) \in \Lambda_m : 5r - 2\alpha + 6c = j\}.$$

The filtration $F^\bullet R_m$ is finite, since $\dim R_m = N_m < +\infty$ and $w(f) < +\infty$ for $f \ne 0$. Thus $\dim(\mathrm{gr}_w R)_m = N_m$, and summing over j,

$$N_m = \#\Lambda_m \le \sum_j \dim \mathcal{B}_{m,j} \le \dim(\mathrm{gr}_w R)_m = N_m.$$

Hence all inequalities are equalities. In particular $\mathcal{B} = \mathrm{gr}_w R$, which is (2), and for any (m, j) the associated graded space of $\mathcal{B}_{m,j}$ for the order along D_2 has basis $T^m u^\alpha v^{2r-\alpha} h_C^c$, $(r, \alpha, c) \in \Lambda_m$, $5r - 2\alpha + 6c = j$.

Step 5. In this step, we prove (1), (3), and (4).

Let $\mathfrak{a}_j \subset \mathcal{O}_X$ be the ideal of functions f with $w(f) \ge j$ near p, and $\mathfrak{a}_j = \mathcal{O}_X$ away from p, and let $\pi : Y := \mathrm{Proj}_X \bigoplus_{j \ge 0} \mathfrak{a}_j \to X$ be the weighted blow-up. Since x, y, z are regular parameters, étale locally over p the variety Y is covered by three charts: the chart $x = x_1$, $y = x_1^3 y_1$, $z = x_1^5 z_1$, which is smooth; a chart $\mathbb{A}^3/\mu_3$ of type $\frac{1}{3}(1, 2, 1)$; and a chart $\mathbb{A}^3/\mu_5$ of type $\frac{1}{5}(1, 4, 2)$, where in the last two the first coordinate defines the exceptional divisor on the cover. The exceptional divisor is $E = \mathrm{Proj}\, \mathbb{C}[x, y, z] \cong \mathbb{P}(1, 3, 5)$, and $-E$ is π-ample since $\mathfrak{a}_{15j}\mathcal{O}_Y = \mathcal{O}_Y(-15jE)$ for $j \ge 0$. On the first chart $\mathrm{ord}_E(x^a y^b z^c) = a + 3b + 5c$, so $\mathrm{ord}_E = w$. On the first chart,

$$dx \wedge dy \wedge dz = x_1^8 dx_1 \wedge dy_1 \wedge dz_1,$$

so $K_Y = \pi^* K_X + 8E$ and $A_X(E) = 1 + 3 + 5 = 9$. Near E, the variety Y has exactly two singular points, both isolated cyclic quotient singularities. Hence Y is smooth at the generic point of any curve in E, so E is Cartier there and $\mathrm{Diff}_E(0) = 0$. The surface $E \cong \mathbb{P}(1, 3, 5)$ has only cyclic quotient singularities, so $(E, 0)$ is klt [KM98, Proposition 5.20(4)]. Since Y has only quotient

singularities, $K_Y + E$ is $\mathbb{Q}$-Cartier, and by inversion of adjunction [KM98, Theorem 5.50], (Y, E) is plt near E. Thus π is the plt extraction of E, which proves (1).

Since s_0 trivializes π^*H near E, and $\mathfrak{a}_j/\mathfrak{a}_{j+1} = \mathbb{C}[x,y,z]_j = H^0(E, (-jE)|_E)$, the restriction map sends $s \in F^j R_m$ to the weight-j part of s/s_0^m. Hence $W_{m,j} = (\mathrm{gr}_w R)_{m,j} = \mathcal{B}_{m,j}$ by Definition 2.8 and **Step 4**. The curve $C = (D_2 = 0) \subset E$ is reduced and irreducible, and D_2 defines C at a general point of C, so the order along C of $f \in \mathbb{C}[x,y,z]_j$ is the largest c with $D_2^c \mid f$. Thus (3) follows from **Step 4**.

Let $\mathcal{P} \subset \mathbb{R}^3$ be the polytope defined by (15.18) with $m = 1$, so that $\Lambda_m = m\mathcal{P} \cap \mathbb{Z}^3$. The jumping numbers of ord_E on R_m are the numbers $5r - 2\alpha + 6c$, $(r, \alpha, c) \in \Lambda_m$, and the jumping numbers of ord_C on the spaces $W_{m,j}$ are the numbers c, $(r, \alpha, c) \in \Lambda_m$. The slice of $\mathcal{P}$ at $r \in [0,1]$ is $\{0 \le \alpha \le r,\ 0 \le c \le 2 - r + \alpha\} \cup \{r \le \alpha \le 2r,\ 0 \le c \le 2\}$, and the slice at $r \in [1,2]$ is $\{0 \le \alpha \le 1,\ 0 \le c \le 2 - r + \alpha\} \cup \{1 \le \alpha \le 3 - r,\ 0 \le c \le 3 - r\}$. A direct calculation over the four regions in the slices shows that the integrals of 1, r, α, and c over the part of $\mathcal{P}$ with $r \in [0,1]$ are $\frac{11}{6}$, $\frac{29}{24}$, $\frac{31}{24}$, and $\frac{41}{24}$, and over the part with $r \in [1,2]$ they are $\frac{11}{6}$, $\frac{5}{2}$, $\frac{41}{24}$, and $\frac{31}{24}$. Summing, we obtain

$$\int_{\mathcal{P}} 1 = \frac{11}{3}, \qquad \int_{\mathcal{P}} r = \frac{89}{24}, \qquad \int_{\mathcal{P}} \alpha = 3, \qquad \int_{\mathcal{P}} c = 3. \tag{15.19}$$

The limits $S = \lim_m S_m$ exist [AZ22, Lemma 2.9]. Since Λ_m is the set of lattice points of $m\mathcal{P}$ and the weights are linear, the sums defining S_m are Riemann sums of $\mathcal{P}$, with errors $O(1/m)$ after normalization by mN_m. Thus, by Definition 2.8,

$$S_X(E) = \frac{5 \cdot \frac{89}{24} - 2 \cdot 3 + 6 \cdot 3}{11/3} = \frac{733}{88}, \qquad S(W; \mathrm{ord}_C) = \frac{3}{11/3} = \frac{9}{11},$$

which is (4).

Step 6. In this step, we prove (5) and (6).

The curve C does not contain $P_y := [0:1:0]$, so C is covered by the charts $x \ne 0$ and $z \ne 0$ of E. On the chart $x \ne 0$, which is $\mathbb{A}^2$ with coordinates y, z, the curve C is $z = y^2$, which is smooth and contains P_x at $y = 0$. The chart $z \ne 0$ is $\mathbb{A}^2/\mu_5$ with coordinates x, y of weights $(1,3)$, and the preimage of C is $x = y^2$, which is smooth and parametrized by y. The group μ_5 acts faithfully on y, and freely outside the origin of $\mathbb{A}^2$. Let $g : \mathbb{A}^2 \to \mathbb{A}^2/\mu_5$ be the quotient map. Since g is étale in codimension one and $g^{-1}C$ is the smooth curve $x = y^2$, we have $K_{\mathbb{A}^2} + g^{-1}C = g^*(K_E + C)$ and $\mathrm{discrep}(\mathbb{A}^2, g^{-1}C) > -1$. Thus $\mathrm{discrep}(E, C) > -1$ near P_z [KM98, Proposition 5.20(2)], that is, (E, C) is plt near P_z [KM98, Definition 2.34]. Since E and C are smooth along $C \backslash \{P_z\}$, the pair (E, C) is plt. The different of (E, C) on C is $\frac{4}{5}P_z$ [Pro01, Theorem 2.2.4(i)]. The map $[u:v] \to [u^2 : uv : v^2]$ from $\mathbb{P}(1,5)$ onto C is bijective, so the coarse curve of C is $\mathbb{P}^1$. This proves (5). Since (E, C) is plt and E is $\mathbb{Q}$-factorial, C is of plt type over $(E, 0)$ with plt extraction id_E.

By (3), the refinement $W^C_{m,j,c}$ has basis the restrictions to C of the residual monomials $u^\alpha v^{2r-\alpha}$, $(r, \alpha, c) \in \Lambda_m$, $5r - 2\alpha + 6c = j$, viewed as sections of the $\mathbb{Q}$-divisor class $(m\pi^*H - jE)|_E - cC$ restricted to C [AZ22, Lemma 2.1]. The zero divisor of $u^\alpha v^{2r-\alpha}$ on C is $(2r - \alpha)P_x + \frac{\alpha}{5}P_z$. At P_x we set $u = 1$, and $v = y$ is a coordinate on C. At P_z we set $v = 1$, and $u = y$ is the coordinate of the preimage of C in the μ_5-cover, of which u^5 is a coordinate of the coarse curve. For fixed (m, j, c), distinct basis elements have distinct α, so they have distinct orders at P_x and at P_z. Hence the jumping numbers of ord_{P_x} and ord_{P_z} on $W^C_{m,j,c}$ are the orders of the basis elements. By the Riemann sums of **Step 5** and (15.19),

$$S(W^C; P_x) = \frac{2 \cdot \frac{89}{24} - 3}{11/3} = \frac{53}{44}, \qquad S(W^C; P_z) = \frac{3/5}{11/3} = \frac{9}{55}.$$

Let $q \in C \backslash \{P_x, P_z\}$, and fix (m, j, c). The points (r, α) with $5r - 2\alpha = j - 6c$ form an arithmetic progression with step $(2, 5)$, and the inequalities (15.18) cut out n consecutive terms of it, say with $\alpha = \alpha_0, \alpha_0 + 5, \dots, \alpha_0 + 5(n-1)$. The quotient of consecutive basis elements is $\zeta := u^5/v$,

a coordinate on $C\backslash\{P_x,P_z\}\cong\mathbb{G}_m$. Thus $W^C_{m,j,c}=h\cdot\langle 1,\zeta,\dots,\zeta^{n-1}\rangle$, where h has neither zeros nor poles on $C\backslash\{P_x,P_z\}$. The basis $h(\zeta-\zeta(q))^i$, $0\le i\le n-1$, has the orders $0,1,\dots,n-1$ at q, so the sum of the jumping numbers of ord_q is $\frac{n(n-1)}{2}$. The sum of the jumping numbers of ord_{P_z} is

$$\sum_{i=0}^{n-1}\frac{\alpha_0+5i}{5}=\frac{n\alpha_0}{5}+\frac{n(n-1)}{2}\ge\frac{n(n-1)}{2}.$$

Summing over (j,c), we obtain $S_m(W^C;q)\le S_m(W^C;P_z)$ for any m. Thus $S(W^C;q)\le S(W^C;P_z)=\frac{9}{55}$, which proves (6). □

We now bound $S(W;v)$ for any prime divisor v over E. The weights $(0,1,2)$ on (x,y,z) degenerate W to a series whose eigenspaces are explicit. The divisor associated with a basis consisting of eigenvectors then has explicit coefficient bounds, which show that, for large level, the pair on E with that divisor multiplied by a number c_m is log canonical, where c_m tends to $\frac{132}{125}$, and Lemma 3.6 carries the bound back to W.

Theorem 15.6. *Let X be a smooth prime Fano threefold of genus 12 and $p\in X$ a closed point. Assume that $\Sigma(X,p)$ is curvilinear of length three and supported at one line, and that exactly two conics contain p. Let E and W be as in Construction-Proposition 15.4, so that $\Delta_E=\mathrm{Diff}_E(0)=0$. Then:*

(1) $S(W;v)\le\frac{125}{132}A_{E,\Delta_E}(v)$ for any prime divisor v over E;

(2) any prime divisor F over X with $\mathrm{center}_X F=\{p\}$ satisfies

$$A_X(F)-S_X(F)\ge\frac{7}{132}A_X(F)+\frac{17}{88}\,\mathrm{mult}_F E>0.$$

Proof. The proof is long, so we divide it into several steps. We use the regular parameters x,y,z, the constant λ, and the polynomials D_2, G_3, J_3 of Construction-Proposition 15.4, and we write $\mathcal{B}\subset\mathbb{C}[x,y,z,T]$ for the ring in Construction-Proposition 15.4(2).

Step 1. In this step we degenerate each $W_{m,j}$ by the weights $(0,1,2)$ on (x,y,z).

For $j\ge 0$ write $\mathbb{C}[x,y,z]_j$ for the space of polynomials of degree j for the weights $(1,3,5)$, so that $H^0(E,\mathcal{O}_E(j))=\mathbb{C}[x,y,z]_j$ on $E\cong\mathbb{P}(1,3,5)$. Since π^*H is trivialized near E by a section of H not vanishing at p, since ord_E restricted to $\widehat{\mathcal{O}}_{X,p}$ is the monomial valuation w, and since $H^0(E,(-jE)|_E)=\mathbb{C}[x,y,z]_j$, Definition 2.8 shows that $W_{m,j}$ is the degree-(m,j) part of the associated graded ring of $\bigoplus_m H^0(X,mH)$ for w. By Construction-Proposition 15.4(2), that associated graded ring is $\mathcal{B}$, so

$$W_{m,j}=\{f\in\mathbb{C}[x,y,z]_j:\ T^m f\in\mathcal{B}\}.$$

For a non-zero polynomial f, write $\mathrm{in}_{(0,1,2)}(f)$ for the sum of the terms of f of largest degree for the weights $(0,1,2)$ on (x,y,z), and $\deg_{(0,1,2)} f$ for that degree. Let $W^0_{m,j}$ be the span of the $\mathrm{in}_{(0,1,2)}(f)$ with $0\ne f\in W_{m,j}$, and let $W^0_{m,j,\ell}$ be the subspace of elements of $W^0_{m,j}$ homogeneous of degree ℓ for the weights $(0,1,2)$. Then

$$W^0_{m,j}=\bigoplus_\ell W^0_{m,j,\ell},\qquad W^0_{m,j,\ell}=\{\mathrm{in}_{(0,1,2)}(f):\ f\in W_{m,j},\ \deg_{(0,1,2)} f=\ell\}\cup\{0\}.$$

Choose a basis $s_1,\dots,s_n$ of $W_{m,j}$ such that, for any d, the s_i with $\deg_{(0,1,2)} s_i\le d$ form a basis of the subspace of $W_{m,j}$ of elements of $\deg_{(0,1,2)}$ at most d. Then $\mathrm{in}_{(0,1,2)}(s_1),\dots,\mathrm{in}_{(0,1,2)}(s_n)$ form a basis of $W^0_{m,j}$. In particular, $\dim W^0_{m,j}=\dim W_{m,j}$. Let $d_i:=\deg_{(0,1,2)} s_i$ and

$$s_{i,t}(x,y,z):=t^{d_i}s_i(x,t^{-1}y,t^{-2}z),\qquad t\in\mathbb{C}.$$

Each $s_{i,t}$ is a polynomial in t with $s_{i,0}=\mathrm{in}_{(0,1,2)}(s_i)$, and for $t\ne 0$ the $s_{i,t}$ span the image of $W_{m,j}$ under the automorphism $(x:y:z)\to(x:ty:t^2z)$ of E, acting on $\mathcal{O}_E(j)$ by $f\to f(x,t^{-1}y,t^{-2}z)$. Thus $W^0_{m,j}$ is the limit of the images $\{f(x,t^{-1}y,t^{-2}z):f\in W_{m,j}\}$ as $t\to 0$ in the Grassmannian,

which is the limit in Lemma 3.6 for the one-parameter subgroup $t \to [(x:y:z) \to (x:ty:t^2z)]$ of $\mathrm{Aut}(E)$. The one-parameter subgroup does not depend on m, and it fixes $\Delta_E = 0$. An element of $\mathbb{C}[x,y,z]_j$ is an eigenvector of $f \to f(x,t^{-1}y,t^{-2}z)$ if and only if it is homogeneous for the weights $(0,1,2)$. Hence a basis of $W^0_{m,j}$ consisting of eigenvectors is the union of bases of the $W^0_{m,j,\ell}$.

Step 2. In this step we determine every $W^0_{m,j,\ell}$.

Let $\mathcal{Q} \subset \mathbb{R}^3$ be the polytope of points (r,α,c) satisfying the inequalities of Construction-Proposition 15.4(3) with $m=1$, and let $I_m := m\mathcal{Q} \cap \mathbb{Z}^3$, which is the set of (r,α,c) satisfying those inequalities. By Construction-Proposition 15.4(3), the elements indexed by $(r,\alpha,c) \in I_m$ form a basis of the associated graded space of $\bigoplus_j W_{m,j}$ for the order along C, and $\dim \bigoplus_j W_{m,j} = N_m$. Hence $\#I_m = N_m$. The element indexed by (r,α,c) has order c along C and degree $j = 5r-2\alpha+6c$ for the weights $(1,3,5)$. The weights $(0,1,2)$ multiply D_2 by a scalar, so they act on C. For any $a_1,a_2,a_3 \geq 0$ with $2a_1+a_2 = \alpha$ and $a_2+2a_3 = 2r-\alpha$, the element indexed by (r,α,c) is the leading form along C of $T^m x^{a_1} y^{a_2} z^{a_3} D_2^c$. Since $x^{a_1}y^{a_2}z^{a_3}$ and D_2 are homogeneous of degrees a_2+2a_3 and 2 for the weights $(0,1,2)$, the element indexed by (r,α,c) is homogeneous of degree $\ell = a_2+2a_3+2c = 2r-\alpha+2c$ for them. At fixed (j,ℓ,c) we obtain

$$r = j-2\ell-2c, \qquad \alpha = 2j-5\ell-2c, \qquad 2r-\alpha = \ell-2c, \qquad j-3\ell = \alpha-r. \tag{15.20}$$

Substituting (15.20) into the inequalities of Construction-Proposition 15.4(3), the inequalities $c \geq 0$, $r \leq 2m$, $r+\alpha \leq 3m$, $r+c \leq 3m$ bound c from below, and the inequalities $c \leq 2m$, $2r-\alpha \geq 0$, $\alpha \geq 0$, $r \geq 0$, $r-\alpha+c \leq 2m$ bound c from above by $2m$, $\frac{\ell}{2}$, $\frac{2j-5\ell}{2}$, $\frac{j-2\ell}{2}$, $2m+j-3\ell$. Let

$$\begin{aligned} &e_x := \max(0, j-3\ell), \qquad e_z := \max(0, 3\ell-j), \qquad i_{\max} := \min\left(\lfloor \tfrac{\ell}{2} \rfloor, 2m\right), \\ &c_{\min} := \left\lceil \max\left(0, \frac{j-2\ell-2m}{2}, \frac{3j-7\ell-3m}{4}, j-2\ell-3m\right) \right\rceil, \qquad c_{\max} := i_{\max} - e_z. \end{aligned} \tag{15.21}$$

If $j \geq 3\ell$, the integer part of the minimum of the upper bounds is $\min(2m, \lfloor \frac{\ell}{2} \rfloor)$. If $j < 3\ell$, it is $\min(2m, \lfloor \frac{\ell}{2} \rfloor) - (3\ell - j)$. In both cases it is $c_{\max}$. Thus the points of I_m with weights (j,ℓ) correspond to the integers c with $c_{\min} \leq c \leq c_{\max}$, and their number is $n_{j,\ell} := \max(0, c_{\max} - c_{\min} + 1)$. Since each term in the maximum defining $c_{\min}$ is non-increasing in ℓ, the integer $c_{\min}$ is a non-increasing function of ℓ for fixed j and m.

We claim that

$$W^0_{m,j,\ell} = x^{e_x} z^{e_z} y^{\ell-2i_{\max}} D_2^{c_{\min}} \cdot \mathbb{C}[y^2, xz]_{c_{\max}-c_{\min}}, \tag{15.22}$$

where $\mathbb{C}[y^2,xz]_d$ denotes the space of homogeneous binary forms of degree d in y^2 and xz, and is zero for $d<0$. Each of the fourteen generators of $\mathcal{B}$ is T times a polynomial of degree at most 2 in z, so any element of $W_{m,j}$, and any element of $W^0_{m,j}$, has degree at most $2m$ in z. A monomial of degree j for $(1,3,5)$ and degree ℓ for $(0,1,2)$ is $x^{j-3\ell+i}y^{\ell-2i}z^i$, and

$$x^{j-3\ell+i}y^{\ell-2i}z^i = x^{e_x}z^{e_z}y^{\ell-2i_{\max}}(xz)^{i-e_z}(y^2)^{i_{\max}-i}$$

for $e_z \leq i \leq i_{\max}$. Hence any element of $W^0_{m,j,\ell}$ is $x^{e_x}z^{e_z}y^{\ell-2i_{\max}}g(y^2,xz)$ with g a binary form of degree $c_{\max}$. Let $0 \neq f \in W_{m,j}$ with $\deg_{(0,1,2)} f = \ell$, and write $f = D_2^{c_f} g'$ with $D_2 \nmid g'$, so that $c_f := \mathrm{ord}_C f$ and $\mathrm{in}_{(0,1,2)}(f) = D_2^{c_f}\,\mathrm{in}_{(0,1,2)}(g')$. The leading form of f along C is a non-zero combination of the elements indexed by the points $(r,\alpha,c_f) \in I_m$ with $5r-2\alpha+6c_f = j$. Since D_2 is homogeneous for the weights $(0,1,2)$, the leading form of f along C is the sum of the leading forms of the homogeneous components of f of order c_f along C, each of degree at most ℓ. Since the elements indexed by I_m are linearly independent and homogeneous, some $(r,\alpha,c_f) \in I_m$ with $5r-2\alpha+6c_f = j$ satisfies $\ell' := 2r-\alpha+2c_f \leq \ell$. Thus

$$c_f \geq c_{\min}(j,\ell') \geq c_{\min}(j,\ell),$$

so $D_2^{c_{\min}}$ divides $\mathrm{in}_{(0,1,2)}(f)$. Since $D_2 = y^2 - xz$ is coprime to x, y, and z, the form g is divisible by $(y^2 - xz)^{c_{\min}}$. Therefore $W^0_{m,j,\ell}$ is contained in the right-hand side of (15.22), which is of dimension $n_{j,\ell}$. Summing over ℓ,

$$\sum_\ell \dim W^0_{m,j,\ell} \le \sum_\ell n_{j,\ell} = \#\{(r,\alpha,c) \in I_m :\ 5r - 2\alpha + 6c = j\} = \dim W_{m,j} = \dim W^0_{m,j},$$

where the middle equality is Construction-Proposition 15.4(3). Hence equality holds for every ℓ, which proves (15.22).

For $s \in \mathbb{C}^*$ let $C_s := (xz - sy^2 = 0) \subset E$, so that $C_1 = C$. A binary form in y^2 and xz is a product of linear forms in y^2 and xz. By (15.22), the zero divisor of any eigenvector in $W^0_{m,j}$ is supported on $(x = 0)$, $(y = 0)$, $(z = 0)$, and finitely many curves C_s.

Step 3. In this step we bound the coefficients of the divisor associated with a basis consisting of eigenvectors.

Fix m, and for each j fix a basis of $W^0_{m,j}$ consisting of eigenvectors. Let D be the associated divisor, that is, $\frac{1}{mN_m}$ times the sum of the zero divisors of all basis elements. By definition, D is an eigen-basis divisor at level m in the sense of Lemma 3.6. Since E is well formed, the multiplicity of the zero divisor of a polynomial along $(x = 0)$, $(y = 0)$, $(z = 0)$, or C_s is the order of the polynomial at the generic point of that curve.

Let $n \ge 1$, let $\mathbb{C}[Y,Z]_{n-1}$ be the space of binary forms of degree $n-1$, and let $q \in \mathbb{P}^1$. For $1 \le k \le n$ the forms vanishing to order at least k at q form a subspace of codimension k. Hence for any basis $g_1, \dots, g_n$ of $\mathbb{C}[Y,Z]_{n-1}$,

$$\sum_{i=1}^{n} \mathrm{ord}_q\, g_i = \sum_{k \ge 1} \#\{i : \mathrm{ord}_q\, g_i \ge k\} \le \sum_{k=1}^{n-1} (n-k) = \frac{n(n-1)}{2}.$$

We apply the inequality to (15.22) with $Y = y^2$, $Z = xz$, and $n = n_{j,\ell} \ge 1$. The factor $D_2^{c_{\min}}$ does not vanish along $(x = 0)$, $(y = 0)$, $(z = 0)$, and C_s for $s \ne 1$. The factor $x^{e_x} z^{e_z} y^{\ell - 2i_{\max}}$ does not vanish along any C_s. The point $Y = 0$ of $\mathbb{P}^1$ contributes order two along $(y = 0)$. Thus, for any basis of $W^0_{m,j,\ell}$, the average multiplicities of the zero divisors of the basis elements are at most

$$\begin{aligned} &\text{along } (x=0):\ e_x + \tfrac{n-1}{2}, \qquad \text{along } (z=0):\ e_z + \tfrac{n-1}{2}, \\ &\text{along } (y=0):\ \ell - 2e_z - c_{\max} - c_{\min}, \\ &\text{along } C:\ \tfrac{c_{\max}+c_{\min}}{2}, \qquad \text{along } C_s,\ s \ne 1:\ \tfrac{c_{\max}-c_{\min}}{2}, \end{aligned} \tag{15.23}$$

where we use $i_{\max} = c_{\max} + e_z$ for the bound along $(y = 0)$. The bounds (15.23) do not depend on the basis or on s.

Next we sum (15.23) over (j,ℓ). By (15.20), $e_x = (\alpha - r)_+$ and $e_z = (r - \alpha)_+$ at any point $(r,\alpha,c) \in I_m$ of weights (j,ℓ). Moreover $\sum_{c=c_{\min}}^{c_{\max}} (c - c_{\min}) = \frac{n(n-1)}{2}$ and $\sum_{c=c_{\min}}^{c_{\max}} c = \frac{n(c_{\max}+c_{\min})}{2}$.

Writing $c_{\min}$ for $c_{\min}(j,\ell)$ at the weights (j,ℓ) of (r,α,c), we obtain

$$\begin{aligned}
\operatorname{mult}_{(x=0)} D &\le \frac{1}{mN_m}\sum_{I_m}\big((\alpha-r)_+ + c - c_{\min}\big),\\
\operatorname{mult}_{(z=0)} D &\le \frac{1}{mN_m}\sum_{I_m}\big((r-\alpha)_+ + c - c_{\min}\big),\\
\operatorname{mult}_{(y=0)} D &\le \frac{1}{mN_m}\sum_{I_m}\big(\ell - 2(r-\alpha)_+ - 2c\big),\\
\operatorname{mult}_{C} D &\le \frac{1}{mN_m}\sum_{I_m} c,\\
\operatorname{mult}_{C_s} D &\le \frac{1}{mN_m}\sum_{I_m}(c - c_{\min}) \quad (s\neq 1).
\end{aligned} \tag{15.24}$$

By (15.20) and (15.21),

$$c - c_{\min} = \Big\lfloor \min\Big(c,\ m-\frac{r}{2},\ \frac{3m-r-\alpha}{4},\ 3m-r-c\Big)\Big\rfloor.$$

Now we compute the limits. For a function φ_0 on $\mathcal{Q}$ write $\langle\varphi_0\rangle := \operatorname{vol}(\mathcal{Q})^{-1}\int_{\mathcal{Q}}\varphi_0$. The image $\mathcal{Q}'$ of $\mathcal{Q}$ in the (r,α)-plane is the polygon with vertices $(0,0)$, $(2,0)$, $(2,1)$, $(1,2)$, and over $(r,\alpha)\in\mathcal{Q}'$ the coordinate c runs through $[0,\varphi(r,\alpha)]$, where

$$\varphi(r,\alpha) := 3-r-\psi(r,\alpha), \qquad \psi(r,\alpha) := \max(0,1-r,1-\alpha).$$

A direct calculation over $\mathcal{Q}'$ shows that

$$\operatorname{vol}(\mathcal{Q}) = \frac{11}{3},\quad \langle r\rangle = \frac{89}{88},\quad \langle\alpha\rangle = \frac{9}{11},\quad \langle c\rangle = \frac{9}{11},$$
$$\int_{\mathcal{Q}'}(\alpha-r)_+\varphi\,dr\,d\alpha = \frac{47}{96},\quad \int_{\mathcal{Q}'}(r-\alpha)_+\varphi\,dr\,d\alpha = \frac{115}{96}.$$

By linearity, $\langle j\rangle = 5\langle r\rangle - 2\langle\alpha\rangle + 6\langle c\rangle = \frac{733}{88}$. Hence

$$\begin{aligned}
\langle\ell\rangle &= 2\langle r\rangle - \langle\alpha\rangle + 2\langle c\rangle = \frac{125}{44},\\
\langle(\alpha-r)_+\rangle &= \frac{47}{352}, \qquad \langle(r-\alpha)_+\rangle = \frac{115}{352}.
\end{aligned} \tag{15.25}$$

For the term $c-c_{\min}$, let $\sigma := \min(1-\frac{r}{2}, \frac{3-r-\alpha}{4})$. On $\mathcal{Q}'$ we have $0\le\sigma\le\frac{3-r}{2}$ and $\sigma\le\varphi$, and $3-r-c\ge\psi$ for $c\le\varphi$. Integrating $\min(c,\sigma,3-r-c)$ in c over $[0,\varphi]$ yields

$$\int_0^{\varphi(r,\alpha)}\min(c,\sigma,3-r-c)\,dc = \begin{cases}\sigma(3-r) - \frac{\sigma^2}{2} - \sigma\psi & \text{if } \psi\ge\sigma,\\ \sigma(3-r) - \sigma^2 - \frac{\psi^2}{2} & \text{if } \psi\le\sigma.\end{cases} \tag{15.26}$$

The polygon $\mathcal{Q}'$ is the union of six polygons, overlapping only along edges, on each of which σ and ψ are affine and one case of (15.26) holds. The integrals of (15.26) over the six polygons are listed in Table 106, and their sum is $\frac{11}{12}$. Dividing by $\operatorname{vol}(\mathcal{Q}) = \frac{11}{3}$,

$$\big\langle \min(c,\sigma,3-r-c)\big\rangle = \frac{1}{4}. \tag{15.27}$$

Each summand in (15.24) differs by at most one from the value at (r,α,c) of a function on $m\mathcal{Q}$ which is m times a fixed function on $\mathcal{Q}$, piecewise affine with respect to finitely many rational hyperplanes, and of absolute value $O(m)$ on $m\mathcal{Q}$. The unit cubes meeting a facet of $m\mathcal{Q}$ or one of the scaled hyperplanes contain $O(m^2)$ lattice points, and on any other unit cube the function is affine with gradient $O(1)$. Hence the sum differs from m^4 times the integral over $\mathcal{Q}$ by $O(m^3)$. Dividing by $mN_m = \frac{11}{3}m^4 + O(m^3)$, we see that the sums in (15.24) converge to the corresponding means with error $O(\frac{1}{m})$. The bounds (15.23) are maxima over all bases and do

not depend on s, so the error is uniform in D and in s. By (15.24)–(15.27), using $\frac{47}{352}+\frac{1}{4}=\frac{135}{352}$, $\frac{115}{352}+\frac{1}{4}=\frac{203}{352}$, and $\frac{125}{44}-2\cdot\frac{115}{352}-2\cdot\frac{9}{11}=\frac{97}{176}$,

$$\begin{aligned} \operatorname{mult}_{(x=0)} D &\le \frac{135}{352}+O\Big(\frac{1}{m}\Big), & \operatorname{mult}_{(z=0)} D &\le \frac{203}{352}+O\Big(\frac{1}{m}\Big),\\ \operatorname{mult}_{(y=0)} D &\le \frac{97}{176}+O\Big(\frac{1}{m}\Big), & \operatorname{mult}_{C} D &\le \frac{9}{11}+O\Big(\frac{1}{m}\Big),\\ \operatorname{mult}_{C_s} D &\le \frac{1}{4}+O\Big(\frac{1}{m}\Big)\ (s\neq 1). \end{aligned} \tag{15.28}$$

Two further orders do not depend on the basis. The chart $(x\neq 0)$ of E is $\mathbb{A}^2$ with coordinates (y,z). Let ν_x be the exceptional divisor of the weighted blow-up of P_x with weights $(1,2)$ on (y,z), so that $A_{E,\Delta_E}(\nu_x)=3$. The chart $(z\neq 0)$ is the quotient of $\mathbb{A}^2$ with coordinates (x,y) by μ_5 acting with weights $(1,3)$. Let $h_z:\mathbb{A}^2\to E$ be the quotient map onto the chart, and let ν_z be the exceptional divisor of the weighted blow-up of $0\in\mathbb{A}^2$ with weights $(2,1)$ on (x,y), so that $A_{\mathbb{A}^2}(\nu_z)=3$. By (15.22), any element f of $W^0_{m,j,\ell}$ satisfies $\nu_x(f(1,y,z))=\ell$ and $\nu_z(f(x,y,1))=2(j-3\ell+i)+(\ell-2i)=2j-5\ell$. Therefore $\nu_x(D)=\frac{1}{mN_m}\sum_{I_m}\ell$ and $\nu_z(h_z^*D)=\frac{1}{mN_m}\sum_{I_m}(2j-5\ell)$, and by the value of $\langle j\rangle$ above, (15.25), and the lattice-point count above,

$$\nu_x(D)=\frac{125}{44}+O\Big(\frac{1}{m}\Big),\qquad \nu_z(h_z^*D)=2\cdot\frac{733}{88}-5\cdot\frac{125}{44}+O\Big(\frac{1}{m}\Big)=\frac{27}{11}+O\Big(\frac{1}{m}\Big). \tag{15.29}$$

Step 4. In this step we prove (1).

The largest of the ratios $\frac{135}{352}$, $\frac{97}{176}$, $\frac{203}{352}$, $\frac{9}{11}$, $\frac{1}{4}$, $\frac{1}{3}\cdot\frac{125}{44}=\frac{125}{132}$, and $\frac{1}{3}\cdot\frac{27}{11}=\frac{9}{11}$ is $\frac{125}{132}$. By (15.28) and (15.29), there exist a constant $\kappa'>0$ and an integer m_0 such that, for $m\ge m_0$ and any eigen-basis divisor D at level m, the five bounds of (15.28) and the bounds for $\frac{1}{3}\nu_x(D)$ and $\frac{1}{3}\nu_z(h_z^*D)$ hold with $O(\frac{1}{m})$ replaced by $\frac{\kappa'}{m}$. Let

$$c_m:=\Big(\frac{125}{132}+\frac{\kappa'}{m}\Big)^{-1},$$

so that $\lim_m c_m=\frac{132}{125}$. Then for $m\ge m_0$ and any such D, every coefficient of c_mD is at most one, $c_m\nu_x(D)\le 3$, and $c_m\nu_z(h_z^*D)\le 3$. We show that (E,c_mD) is lc.

Let $P_y:=[0:1:0]$. The curves $(x=0)$, $(y=0)$, $(z=0)$ intersect pairwise only at P_x, P_y, P_z. On the chart $(x\neq 0)$ the curve C_s is $(z=sy^2)$, on the cover $\mathbb{A}^2$ of the chart $(z\neq 0)$ it is $(x=sy^2)$, and on the chart $(y\neq 0)$ it is the image of $(xz=s)$. Therefore C_s is smooth away from P_x and P_z, does not contain P_y, and intersects $(x=0)$, $(y=0)$, $(z=0)$, and $C_{s'}$ for $s'\neq s$ only at P_x and P_z. Hence away from P_x, P_y, P_z the support of D is a disjoint union of smooth curves on the smooth part of E, and (E,c_mD) is lc there. Near P_y, the chart $(y\neq 0)$ is the quotient of $\mathbb{A}^2$ with coordinates (x,z) by μ_3, the quotient map is étale in codimension one, and the pullback of D is supported on the coordinate axes $(x=0)$ and $(z=0)$, with the coefficients of $(x=0)$ and $(z=0)$ in D. Hence the pullback pair is lc by [KM98, Corollary 2.31(3)], and (E,c_mD) is lc near P_y by [KM98, Proposition 5.20(4)].

Near P_x we use the coordinates $(a,b):=(y,z)$ on the chart $(x\neq 0)$, the $\mathbb{Q}$-divisor $D':=D$, and $\nu:=\nu_x$. Near P_z we use the coordinates $(a,b):=(y,x)$ on the cover $\mathbb{A}^2$ of the chart $(z\neq 0)$, the $\mathbb{Q}$-divisor $D':=h_z^*D$, and $\nu:=\nu_z$. Since h_z is étale in codimension one, (E,c_mD) is lc near P_z if $(\mathbb{A}^2,c_mD')$ is lc near 0 [KM98, Proposition 5.20(4)]. In both cases D' is supported near 0 on $(a=0)$, $(b=0)$, and curves $(b=sa^2)$ with $s\in\mathbb{C}^*$, and ν is the exceptional divisor Γ of the weighted blow-up $\pi':S'\to\mathbb{A}^2$ of 0 with weights $(1,2)$ on (a,b). Since the log discrepancy of Γ is 3,

$$K_{S'}+c_m\pi'^{-1}_*D'+(c_m\nu(D')-2)\Gamma=\pi'^*(K_{\mathbb{A}^2}+c_mD'),$$

and $c_m\nu(D')-2\le 1$. On the chart $b=a^2w$ of S', which is $\mathbb{A}^2$ with coordinates (a,w), the curve Γ is $(a=0)$, the strict transform of $(b=0)$ is $(w=0)$, and the strict transform of $(b=sa^2)$ is $(w=s)$. In particular, the strict transforms of $(b=0)$ and of the curves $(b=sa^2)$ intersect

Γ transversally at distinct points. The other chart of S' is the quotient of $\mathbb{A}^2$ by μ_2, étale in codimension one, and on that cover the preimages of Γ and of the strict transform of $(a = 0)$ are the coordinate axes. Thus on S', and on the μ_2-cover, the support of the crepant pullback is simple normal crossing with all coefficients at most one. By [KM98, Corollary 2.31(3) and Proposition 5.20(4)], the pair $(S', c_m\pi'^{-1}_* D' + (c_m\nu(D') - 2)\Gamma)$ is sub-lc, and hence $(\mathbb{A}^2, c_m D')$ is lc near 0 [KM98, Lemma 2.30].

Therefore $(E, \Delta_E + c_m D) = (E, c_m D)$ is lc for $m \geq m_0$ and any eigen-basis divisor D at level m. By **Step 1**, the assumption of Lemma 3.6(1) holds for the one-parameter subgroup $t \to [(x : y : z) \to (x : ty : t^2 z)]$ and the numbers c_m, whose limit is $\frac{132}{125}$. Thus $\frac{132}{125} S(W; v) \leq A_{E,\Delta_E}(v)$ for any prime divisor v over E, which is (1).

Step 5. In this step we prove (2).

Let F be a prime divisor over X with $\operatorname{center}_X F = \{p\}$. Since $\pi^{-1}(p) = E$, we have $\operatorname{center}_Y F \subset E$. By (1), the assumption of Lemma 3.1, applied with $G = E$ and the constant $\frac{132}{125}$, holds. Since $A_X(E) = 9$ and $S_X(E) = \frac{733}{88}$ by Construction-Proposition 15.4(1) and (4), Lemma 3.1 implies

$$A_X(F) \geq \frac{132}{125} S_X(F) + \left(9 - \frac{132}{125} \cdot \frac{733}{88}\right) \operatorname{mult}_F E = \frac{132}{125} S_X(F) + \frac{51}{250} \operatorname{mult}_F E.$$

Multiplying by $\frac{125}{132}$,

$$S_X(F) \leq \frac{125}{132} A_X(F) - \frac{17}{88} \operatorname{mult}_F E, \qquad \text{that is,} \qquad A_X(F) - S_X(F) \geq \frac{7}{132} A_X(F) + \frac{17}{88} \operatorname{mult}_F E.$$

Since X is smooth, $A_X(F) > 0$, and since $\operatorname{center}_Y F \subset E$ and E is $\mathbb{Q}$-Cartier, $\operatorname{mult}_F E > 0$. This proves (2). □

16. Nets of alternating forms and plane quartics

In this section, we prove the properties of the GIT of nets and of apolar quartics that are used in Sections 17–20. From now on we pass from estimates at points of a fixed threefold of genus 12 to the moduli of smooth prime Fano threefolds of genus 12. Two GIT problems enter, Mukai nets under $\mathrm{SL}(V)$ and plane quartics under SL_3, together with two dictionaries: non-degenerate nets versus non-degenerate quartics (Lemma 16.3), and the apolar quartic f_X versus the quartic $Q_X = (\mathrm{Sc}(f_X) = 0)$ of Definition-Proposition 2.28, which is isomorphic to $\Sigma(X)$ (Theorem 16.11).

16.1. **Nets and GIT.** In this subsection, we show that the non-degenerate nets form an affine open subset of $\mathbb{B}$ consisting of GIT-semistable points, we relate non-degenerate nets to non-degenerate quartics, and we show that Ω is the non-vanishing locus of an $\mathrm{SL}(V)$-invariant function on $\mathbb{B}^{\mathrm{nd}}$.

Lemma 16.1. *Let A be a commutative $\mathbb{C}$-algebra, let $\epsilon_1, \dots, \epsilon_7$ be the standard basis of A^7 with dual basis $\epsilon_1^*, \dots, \epsilon_7^*$, and let M be an alternating 7×7 matrix with entries in A. Let $\omega_M := \sum_{a<b} M_{ab} \epsilon_a^* \wedge \epsilon_b^*$, and let $p(M) \in A^7$ be the vector such that*

$$\frac{1}{6} \omega_M^{\wedge 3} = \iota_{p(M)}(\epsilon_1^* \wedge \cdots \wedge \epsilon_7^*).$$

Then the following hold.

1. *$p(M)_a = (-1)^{a+1} \operatorname{Pf}(M_{\hat{a}})$ for $1 \leq a \leq 7$, where $M_{\hat{a}}$ is the principal 6×6 submatrix of M obtained by deleting the a-th row and column.*
2. *$Mp(M) = 0$.*
3. *$p(g^t M g) = \det(g) g^{-1} p(M)$ for any $g \in \mathrm{GL}_7(A)$.*
4. *If A is a field, then $\operatorname{rank} M \leq 4$ if and only if $p(M) = 0$, and if $\operatorname{rank} M = 6$, then $\operatorname{Ker} M = A \cdot p(M)$.*

Proof. Write $\mathrm{vol} := \epsilon_1^* \wedge \cdots \wedge \epsilon_7^*$. (1) For an alternating 6×6 matrix N, the top power $\frac{1}{6}\omega_N^{\wedge 3}$ is $\mathrm{Pf}(N)$ times the wedge product of the six dual basis vectors. Thus the coefficient of $\epsilon_1^* \wedge \cdots \wedge \widehat{\epsilon_a^*} \wedge \cdots \wedge \epsilon_7^*$ in $\frac{1}{6}\omega_M^{\wedge 3}$ is $\mathrm{Pf}(M_{\hat{a}})$, and (1) follows from

$$\iota_{\epsilon_a} \mathrm{vol} = (-1)^{a+1} \epsilon_1^* \wedge \cdots \wedge \widehat{\epsilon_a^*} \wedge \cdots \wedge \epsilon_7^*.$$

(2) For a linear form α on A^7 we have $\alpha \wedge \iota_{p(M)} \mathrm{vol} = \alpha(p(M))\,\mathrm{vol}$, since $\iota_{p(M)}(\alpha \wedge \mathrm{vol}) = 0$. For $x \in A^7$, $\omega_M^{\wedge 4} = 0$ implies

$$0 = \iota_x(\omega_M^{\wedge 4}) = 4(\iota_x \omega_M) \wedge \omega_M^{\wedge 3} = 24\omega_M(x, p(M))\,\mathrm{vol}\,.$$

Taking $x = \epsilon_a$ implies $(Mp(M))_a = 0$.

(3) The matrix $g^t M g$ is the matrix of the form $g^*\omega_M$, $(g^*\omega_M)(x,y) = \omega_M(gx, gy)$. Since $g^*(\iota_v \beta) = \iota_{g^{-1}v}(g^*\beta)$ for $v \in A^7$ and $\beta \in \Lambda^7(A^7)^\vee$, and $g^* \mathrm{vol} = \det(g)\,\mathrm{vol}$, we obtain

$$\frac{1}{6}(g^*\omega_M)^{\wedge 3} = g^*\big(\iota_{p(M)}\,\mathrm{vol}\big) = \det(g)\iota_{g^{-1}p(M)}\,\mathrm{vol}\,.$$

(4) If $\mathrm{rank}\, M = 2r$, then in some basis $\epsilon_1', \ldots, \epsilon_7'$ of A^7 we have $\omega_M = \sum_{j=1}^r \epsilon_{2j-1}'^* \wedge \epsilon_{2j}'^*$, so $\omega_M^{\wedge 3} = 0$ if and only if $r \le 2$. If $\mathrm{rank}\, M = 6$, then $\dim \mathrm{Ker}\, M = 1$, and $0 \ne p(M) \in \mathrm{Ker}\, M$ by (2). □

Definition-Proposition 16.2. Let $\mathcal{A} := \det \mathcal{S}^\vee$ be the Plücker line bundle on $\mathbb{B} = \mathrm{Gr}(3, \Lambda^2 V^\vee)$, where $\mathcal{S}$ is the tautological subbundle. Let $Z_4 \subset \mathbb{P}(\Lambda^2 V^\vee)$ be the reduced locus of forms of rank at most four, and let $D_{\mathrm{Pf}} := \{\Pi \in \mathbb{B} : \mathbb{P}(\Pi) \cap Z_4 \ne \emptyset\}$ with the reduced structure. Then Z_4 is irreducible of codimension three and degree 14, D_{Pf} is a non-zero irreducible divisor with $\mathcal{O}_{\mathbb{B}}(D_{\mathrm{Pf}}) \cong \mathcal{A}^{14}$, and there exists an $\mathrm{SL}(V)$-invariant section $c_{\mathrm{Pf}} \in H^0(\mathbb{B}, \mathcal{A}^{14})$ with divisor D_{Pf}, unique up to scalar among sections with divisor D_{Pf}. Hence $\mathbb{B}^{\mathrm{nd}} = \mathbb{B} \backslash D_{\mathrm{Pf}}$ is a non-empty affine $\mathrm{SL}(V)$-invariant open subset, and any point of $\mathbb{B}^{\mathrm{nd}}$ is GIT-semistable for $\mathcal{A}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that Z_4 is irreducible of codimension three and degree 14.

Let $R := \mathrm{Sym}(\Lambda^2 V)$ be the coordinate ring of $\Lambda^2 V^\vee$, fix a basis $v_1, \ldots, v_7$ of V, let M be the alternating matrix with entries $M_{ab} \in \Lambda^2 V \subset R$, $M_{ab}(\omega) = \omega(v_a, v_b)$, and let $I \subset R$ be the ideal generated by the entries of $p(M)$, that is, by the seven principal 6×6 Pfaffians of M (Lemma 16.1(1)). By Lemma 16.1(4), the zero set of I in $\Lambda^2 V^\vee$ is the affine cone over Z_4.

A form has rank at most four if and only if the radical of the form contains a three-dimensional subspace. Thus Z_4 is the image of the $\mathbb{P}^5$-bundle $\{(V_3, [\omega]) : \omega \in \Lambda^2(V/V_3)^\vee\}$ over $\mathrm{Gr}(3, V)$, which is irreducible of dimension $12 + 5 = 17$. Over a form of rank four the radical is the unique such V_3, and the forms of rank four are dense in Z_4, since the forms of rank two form the image of $\mathrm{Gr}(2, V^\vee)$, of dimension 10. Hence Z_4 is irreducible, $\dim Z_4 = 17$, and I has height 3 in the polynomial ring R in 21 variables.

Let ω_0 be a form of rank four, and choose $v_1, \ldots, v_4$ spanning a subspace V' with $\omega_0|_{V'}$ non-degenerate. Write

$$M = \begin{pmatrix} M_1 & M_2 \\ -M_2^t & M_3 \end{pmatrix}$$

with M_1 the $V' \times V'$ block. On the open subset where $\mathrm{Pf}(M_1) \ne 0$, the matrix

$$g := \begin{pmatrix} 1 & -M_1^{-1} M_2 \\ 0 & 1 \end{pmatrix}$$

satisfies $\det(g) = 1$ and

$$g^t M g = \begin{pmatrix} M_1 & 0 \\ 0 & M_3 + M_2^t M_1^{-1} M_2 \end{pmatrix}.$$

The principal 6×6 Pfaffians of $g^t M g$ vanish when a row of M_1 is deleted, and equal $\pm \mathrm{Pf}(M_1)$ times an entry of the alternating 3×3 matrix $M_3 + M_2^t M_1^{-1} M_2$ otherwise. By Lemma 16.1(3), on

the open subset $\{\mathrm{Pf}(M_1) \neq 0\}$ the ideal I is generated by the three entries of $M_3 + M_2^t M_1^{-1} M_2$ above the diagonal, which together with the entries of M_1 and M_2 form a coordinate system. Thus $\operatorname{Spec} R/I$ is smooth at ω_0.

Since I has height 3, the Pfaffian complex

$$0 \to R(-7) \xrightarrow{p(M)} R(-4)^{\oplus 7} \xrightarrow{M} R(-3)^{\oplus 7} \xrightarrow{p(M)^t} R \to R/I \to 0$$

is exact and R/I is Gorenstein by the theorem of Buchsbaum and Eisenbud [BE77, Theorem 2.1], see also [Kan12, Section 2.1, Theorem 2.1]. In particular R/I is Cohen–Macaulay, hence satisfies (S_1). Since $\operatorname{Spec} R/I$ is irreducible and smooth at ω_0, the ring R/I satisfies (R_0). Hence R/I is reduced [Mat89, Section 23, p. 183], and I is the homogeneous ideal of Z_4. Since

$$(1-t)^3(1+3t+6t^2+3t^3+t^4) = 1-7t^3+7t^4-t^7,$$

the Hilbert series of R/I is

$$\frac{1-7t^3+7t^4-t^7}{(1-t)^{21}} = \frac{1+3t+6t^2+3t^3+t^4}{(1-t)^{18}}.$$

Thus $\deg Z_4 = 1+3+6+3+1 = 14$.

Step 2. In this step we show that D_{Pf} is an irreducible divisor with $\mathcal{O}_{\mathbb{B}}(D_{\mathrm{Pf}}) \cong \mathcal{A}^{14}$.

Let $\mathcal{I}_Z := \{(z, \Pi) \in Z_4 \times \mathbb{B} : z \in \mathbb{P}(\Pi)\}$. The projection $\mathcal{I}_Z \to Z_4$ is a $\mathrm{Gr}(2,20)$-bundle, so $\mathcal{I}_Z$ is irreducible and $\dim \mathcal{I}_Z = 17 + 36 = 53$. The projection $\mathcal{I}_Z \to \mathbb{B}$ is proper with image D_{Pf}. Thus D_{Pf} is closed and irreducible, $\dim D_{\mathrm{Pf}} \leq 53$, and the fiber of $\mathcal{I}_Z \to D_{\mathrm{Pf}}$ over Π is $\mathbb{P}(\Pi) \cap Z_4$.

Let $T \subset \Lambda^2 V^\vee$ be a general four-dimensional subspace. By [Ful98, Example 8.4.12(c)], $\mathbb{P}(T) \cong \mathbb{P}^3$ intersects Z_4 transversally in $\deg Z_4 = 14$ points $z_1, \dots, z_{14}$, all smooth points of Z_4. Let $U \subset T$ be a general two-dimensional subspace. Then $z_i \notin \mathbb{P}(U)$, and the line through z_i and z_j does not intersect $\mathbb{P}(U)$ for $i \neq j$. Let $\Pi_i := U + \mathbb{C}z_i$, where z_i also denotes a vector spanning it. Then $\Pi_1, \dots, \Pi_{14}$ are distinct and $\mathbb{P}(\Pi_i) \cap Z_4 = \mathbb{P}(\Pi_i) \cap \mathbb{P}(T) \cap Z_4 = \{z_i\}$ as sets. The nets Π with $U \subset \Pi \subset T$ form a line $\ell \cong \mathbb{P}(T/U)$ of $\mathbb{B}$ in the Plücker embedding of $\mathbb{B}$ [Har92, Exercise 6.9(i)], so $\deg \mathcal{A}|_\ell = 1$. A net $\Pi \in \ell$ belongs to D_{Pf} if and only if $\mathbb{P}(\Pi)$ contains some z_i, that is, $\ell \cap D_{\mathrm{Pf}} = \{\Pi_1, \dots, \Pi_{14}\}$ as sets.

We compute the length of $\ell \cap D_{\mathrm{Pf}}$ at Π_i. Since z_i is a smooth point of Z_4, $\mathcal{I}_Z$ is smooth at (z_i, Π_i). The fiber $\mathbb{P}(\Pi_i) \cap Z_4$ of $\mathcal{I}_Z \to D_{\mathrm{Pf}}$ over Π_i is a closed subscheme of $\mathbb{P}(T) \cap Z_4$ supported at z_i, and $\mathbb{P}(T) \cap Z_4$ is reduced at z_i, so $\mathbb{P}(\Pi_i) \cap Z_4$ is one reduced point. Since $\mathcal{I}_Z \to D_{\mathrm{Pf}}$ is proper with a finite fiber over Π_i, it is finite over a neighborhood of Π_i, and since $\mathbb{P}(\Pi_i) \cap Z_4$ is one reduced point, it is a closed immersion over a smaller neighborhood by Nakayama's lemma. As $\mathcal{I}_Z$ is integral and surjects onto the reduced scheme D_{Pf}, the morphism $\mathcal{I}_Z \to D_{\mathrm{Pf}}$ is an isomorphism over a neighborhood of Π_i. In particular $\dim D_{\mathrm{Pf}} = 53 = \dim \mathbb{B} - 1$, and D_{Pf} is smooth at Π_i. Thus the length of $\ell \cap D_{\mathrm{Pf}}$ at Π_i is the length of $\ell \times_{\mathbb{B}} \mathcal{I}_Z$ at (z_i, Π_i). The morphism $z \to (z, U + \mathbb{C}z)$ is an isomorphism from $(\mathbb{P}(T) \setminus \mathbb{P}(U)) \cap Z_4$ onto an open subscheme of $\ell \times_{\mathbb{B}} \mathcal{I}_Z$ containing (z_i, Π_i), with inverse $(z, \Pi) \to z$. Hence the length of $\ell \cap D_{\mathrm{Pf}}$ at Π_i equals the length of $\mathbb{P}(T) \cap Z_4$ at z_i, which is 1. Therefore

$$\deg \mathcal{O}_{\mathbb{B}}(D_{\mathrm{Pf}})|_\ell = 14.$$

Since $\mathrm{Pic}(\mathbb{B}) = \mathbb{Z}[\mathcal{A}]$ [Ful98, Section 14.7] and $\deg \mathcal{A}|_\ell = 1$, we have $\mathcal{O}_{\mathbb{B}}(D_{\mathrm{Pf}}) \cong \mathcal{A}^{14}$. Moreover $D_{\mathrm{Pf}} \neq \mathbb{B}$ since $\ell \not\subset D_{\mathrm{Pf}}$, and $D_{\mathrm{Pf}} \neq \emptyset$.

Step 3. In this step we construct c_{Pf} and conclude the proof.

Let $c \in H^0(\mathbb{B}, \mathcal{A}^{14})$ be a section with divisor D_{Pf}. Two such sections differ by a nowhere vanishing regular function on the connected projective variety $\mathbb{B}$, which is constant, so c is unique up to scalar. The group $\mathrm{SL}(V)$ preserves the rank of forms, hence preserves Z_4 and D_{Pf}. For the natural $\mathrm{SL}(V)$-linearization of $\mathcal{A}$ and $g \in \mathrm{SL}(V)$, the section $g \cdot c$ has divisor $g(D_{\mathrm{Pf}}) = D_{\mathrm{Pf}}$, so $g \cdot c = \chi(g)c$ for a character χ of $\mathrm{SL}(V)$. Since $\mathrm{SL}(V)$ has no non-trivial characters, $c_{\mathrm{Pf}} := c$ is $\mathrm{SL}(V)$-invariant.

Any non-zero $\omega \in \Lambda^2 V^\vee$ has rank 2, 4, or 6, so $\mathbb{B}^{\mathrm{nd}} = \mathbb{B} \setminus D_{\mathrm{Pf}}$, which is non-empty by **Step 2** and $\mathrm{SL}(V)$-invariant. The line bundle $\mathcal{A}$ is very ample, so in the embedding of $\mathbb{B}$ defined by $\mathcal{A}^{14}$ the divisor D_{Pf} is a hyperplane section, and $\mathbb{B}^{\mathrm{nd}}$ is a closed subvariety of the complement of a hyperplane. Hence $\mathbb{B}^{\mathrm{nd}}$ is affine [Har77, Exercises I.3.5 and II.3.11(b)]. Finally, c_{Pf} is an invariant section of a positive power of $\mathcal{A}$ which does not vanish at any point of $\mathbb{B}^{\mathrm{nd}}$, so any point of $\mathbb{B}^{\mathrm{nd}}$ is GIT-semistable [MFK94, Definition 1.7]. □

Canonical Mukai nets belong to $\mathbb{B}^{\mathrm{nd}}$ (Definition-Theorem 2.20). Since $\mathbb{B}^{\mathrm{nd}}$ is affine, the closure of any $\mathrm{SL}(V)$-orbit in $\mathbb{B}^{\mathrm{nd}}$ contains exactly one closed orbit [BGM19, Section 2.2].

The next lemma relates non-degenerate nets to non-degenerate quartics, for all non-degenerate nets and in families over $\operatorname{Spec}\mathbb{C}[[\tau]]$.

Lemma 16.3. *Let $\Pi \in \mathbb{B}^{\mathrm{nd}}$, and identify $\operatorname{Sym}\Pi^\vee$ with $\mathbb{C}[\partial]$ and $\operatorname{Sym}\Pi$ with $\mathbb{C}[x,y,z]$ by a basis of Π as in Definition-Proposition 2.28. Then the following hold.*

(1) The ideal $I_\Pi \subset \mathbb{C}[\partial]$ of the seven principal 6×6 Pfaffians of the alternating matrix of Π has height three, the Pfaffian complex

$$0 \to \mathbb{C}[\partial](-7) \to \mathbb{C}[\partial](-4)^{\oplus 7} \to \mathbb{C}[\partial](-3)^{\oplus 7} \to \mathbb{C}[\partial] \to \mathbb{C}[\partial]/I_\Pi \to 0$$

is exact, and $\mathbb{C}[\partial]/I_\Pi$ has Hilbert function $(1,3,6,3,1)$. In particular $[f_\Pi] \in U^{\mathrm{cat}}$.

(2) For any $[f] \in U^{\mathrm{cat}}$ there exist $\Pi' \in \mathbb{B}^{\mathrm{nd}}$ and a basis of Π' in which $f_{\Pi'}$ is a non-zero multiple of f, and $\mathrm{VPS}(f,6) \cong X_{\Pi'}$.

(3) Let $\Pi' \in \mathbb{B}^{\mathrm{nd}}$. Then f_Π and $f_{\Pi'}$, written in bases of Π and Π', are projectively equivalent if and only if $\Pi' \in \mathrm{PGL}(V)\cdot\Pi$. The natural map $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \to \mathrm{Stab}_{\mathrm{PGL}(\Pi^\vee)}([f_\Pi])$ is an isomorphism of algebraic groups.

(4) Let $c : \operatorname{Spec}\mathbb{C}[[\tau]] \to \mathbb{B}^{\mathrm{nd}}$ be a morphism, fix a basis $\omega_1,\omega_2,\omega_3$ of the free $\mathbb{C}[[\tau]]$-module $H^0(\operatorname{Spec}\mathbb{C}[[\tau]], c^*\mathcal{S})$, and identify the symmetric algebras of the dual module and of the module with $\mathbb{C}[[\tau]][\partial]$ and $\mathbb{C}[[\tau]][x,y,z]$ accordingly. Then there exists $f_c \in \mathbb{C}[[\tau]][x,y,z]_4$ such that $\det \mathrm{Cat}_2(f_c)$ is a unit of $\mathbb{C}[[\tau]]$ and the annihilator of f_c in $\mathbb{C}[[\tau]][\partial]$ is the ideal of the seven principal 6×6 Pfaffians of the alternating matrix of c. In particular the reduction of f_c modulo τ is a non-zero multiple of $f_{c(0)}$ in the basis given by the reductions of $\omega_1,\omega_2,\omega_3$.

(5) Let $\Pi_0,\Pi_1 \in \mathbb{B}^{\mathrm{nd}}$, and assume that Π_0 belongs to the closure of $\mathrm{PGL}(V)\cdot\Pi_1$ in $\mathbb{B}^{\mathrm{nd}}$. Then $[f_{\Pi_0}]$ belongs to the closure in U^{cat} of the PGL_3-orbit of $[f_{\Pi_1}]$.

Proof. Fix a basis $v_1,\dots,v_7$ of V. For a commutative $\mathbb{C}$-algebra A and $\omega \in \Lambda^2 V^\vee \otimes A$, let $M(\omega) := (\omega(v_a,v_b))_{a,b}$. For a net with basis $\omega_1,\omega_2,\omega_3$, the alternating matrix is $\partial_x M(\omega_1) + \partial_y M(\omega_2) + \partial_z M(\omega_3)$, whose value at $\omega = c_1\omega_1 + c_2\omega_2 + c_3\omega_3$ is $M(\omega)$. By Lemma 16.1(1), the Pfaffian ideal is generated by the entries of p of the alternating matrix, and by Lemma 16.1(3) it does not depend on the basis of V. Let M_Π be the alternating matrix of Π in the given basis $\omega_1,\omega_2,\omega_3$ of Π. For a non-zero $f \in \mathbb{C}[x,y,z]_4$, the map $\mathbb{C}[\partial]_4 \to \mathbb{C}$, $P \to P(f)$, is non-zero with kernel $(f^\perp)_4$, so the quartics annihilated by $(f^\perp)_4$ form the line $\mathbb{C}f$. In particular, if non-zero quartics f, f' satisfy $f^\perp = f'^\perp$, then $f' \in \mathbb{C}^* f$.

(1) By Definition-Proposition 2.28, $I_\Pi = f_\Pi^\perp$ is a Gorenstein ideal of codimension three, hence of height three, and f_Π is non-degenerate, that is, $[f_\Pi] \in U^{\mathrm{cat}}$. Hence the Pfaffian complex, with the maps $p(M_\Pi)$, M_Π, and $p(M_\Pi)^t$, is exact [BE77, Theorem 2.1], and the Hilbert series of $\mathbb{C}[\partial]/I_\Pi$ is

$$\frac{1 - 7t^3 + 7t^4 - t^7}{(1-t)^3} = 1 + 3t + 6t^2 + 3t^3 + t^4.$$

In particular $(I_\Pi)_2 = 0$ and $\dim(I_\Pi)_3 = 7$.

(2) Let $[f] \in U^{\mathrm{cat}}$. By [BE77, Theorem 2.1] and [Sch01, Theorems 2.1 and 2.6], there exists an alternating 7×7 matrix $M_f = \partial_x M_x + \partial_y M_y + \partial_z M_z$ with $M_x, M_y, M_z \in \mathrm{M}_7(\mathbb{C})$ alternating such that the entries of $p(M_f)$ generate $f^\perp$, and $\mathrm{VPS}(f,6)$ is isomorphic to the scheme of three-dimensional subspaces isotropic for the net spanned by the forms with matrices M_x, M_y, M_z.

Let $\omega'_x, \omega'_y, \omega'_z \in \Lambda^2 V^\vee$ be the forms with $M(\omega'_x) = M_x$, $M(\omega'_y) = M_y$, and $M(\omega'_z) = M_z$, and let Π' be their span. Since $\mathbb{C}[\partial]/f^\perp$ has finite length, the zero set of $f^\perp$ in $\mathbb{C}^3$ is $\{0\}$. Thus $p(M(c_1\omega'_x + c_2\omega'_y + c_3\omega'_z)) \neq 0$ for $(c_1, c_2, c_3) \neq 0$, so any non-zero linear combination of $\omega'_x, \omega'_y, \omega'_z$ has rank 6 (Lemma 16.1(4)). Hence $\omega'_x, \omega'_y, \omega'_z$ are linearly independent, $\Pi' \in \mathbb{B}^{\mathrm{nd}}$, and M_f is the alternating matrix of Π' in the basis $\omega'_x, \omega'_y, \omega'_z$. Thus $f^\perp_{\Pi'} = I_{\Pi'} = f^\perp$, so $f_{\Pi'}$ is a non-zero multiple of f by the opening paragraph, and $\mathrm{VPS}(f, 6) \cong X_{\Pi'}$.

(3) We first show the following statement (†): if $\Pi'' \in \mathbb{B}^{\mathrm{nd}}$ and an endomorphism ϕ of V maps the radical of any non-zero $\omega \in \Pi''$ into itself, then ϕ is a scalar multiple of the identity. For non-zero $\omega \in \Pi''$, the radical of ω is the line spanned by the vector with coordinates $p(M(\omega))$ (Lemma 16.1(4)). Thus the image of the morphism $\kappa'' : \mathbb{P}(\Pi'') \to \mathbb{P}(V)$, $[\omega] \to [p(M(\omega))]$, consists of eigenlines of ϕ. The morphism κ'' is defined by the entries of $p(M_{\Pi''})$, which are linearly independent since they span the seven-dimensional space $(I_{\Pi''})_3$ by (1). Hence the image of κ'' is irreducible and not contained in a hyperplane. It is contained in the finite union of the linear subspaces $\mathbb{P}(\mathrm{Ker}(\phi - \lambda))$, λ an eigenvalue of ϕ, hence in one of them, and then $\mathrm{Ker}(\phi - \lambda) = V$.

Assume that $\Pi' = g \cdot \Pi$ for some $g \in \mathrm{GL}(V)$, where $(g \cdot \omega)(u, u') = \omega(g^{-1}u, g^{-1}u')$, and denote also by g the matrix of g in the basis $v_1, \dots, v_7$. In the basis $(g \cdot \omega_k)_k$ of Π', the alternating matrix of Π' is $g^{-t} M_\Pi g^{-1}$. By Lemma 16.1(3), $p(g^{-t} M_\Pi g^{-1}) = \det(g)^{-1} g p(M_\Pi)$, so the Pfaffian ideal of Π' in the basis $(g \cdot \omega_k)_k$ is I_Π. Thus $f_{\Pi'}$ in the basis $(g \cdot \omega_k)_k$ has apolar ideal $I_\Pi = f^\perp_\Pi$, so it is a non-zero multiple of f_Π by the opening paragraph, and in any other basis of Π' it is projectively equivalent to f_Π.

Conversely, assume that f_Π and $f_{\Pi'}$ are projectively equivalent. Changing the basis $\omega'_1, \omega'_2, \omega'_3$ of Π', we may assume that $f_{\Pi'}$ is a non-zero multiple of f_Π. Then $I_{\Pi'} = f^\perp_{\Pi'} = f^\perp_\Pi = I_\Pi$ as ideals of $\mathbb{C}[\partial]$. By (1), the entries of $p(M_\Pi)$ and those of $p(M_{\Pi'})$ are two bases of the seven-dimensional space $(I_\Pi)_3$, so $p(M_{\Pi'}) = a p(M_\Pi)$ for some $a \in \mathrm{GL}_7(\mathbb{C})$. Let $\widetilde{M} := a^{-t} M_\Pi a^{-1}$. By Lemma 16.1(3),

$$p(\widetilde{M}) = \det(a)^{-1} a p(M_\Pi) = \det(a)^{-1} p(M_{\Pi'}).$$

By Lemma 16.1(2), $p(\widetilde{M})^t \widetilde{M} = -(\widetilde{M} p(\widetilde{M}))^t = 0$, so the columns of $\widetilde{M}$ are homogeneous elements of degree four of the kernel of $p(M_{\Pi'})^t : \mathbb{C}[\partial](-3)^{\oplus 7} \to \mathbb{C}[\partial]$. By the exactness in (1) for Π', they are $\mathbb{C}$-linear combinations of the columns of $M_{\Pi'}$, that is, $\widetilde{M} = M_{\Pi'} C$ for some $C \in \mathrm{M}_7(\mathbb{C})$. Since $\widetilde{M}$ and $M_{\Pi'}$ are alternating,

$$M_{\Pi'} C = \widetilde{M} = -\widetilde{M}^t = C^t M_{\Pi'},$$

and hence $M(\omega')C = C^t M(\omega')$ for any $\omega' \in \Pi'$. If $M(\omega')u = 0$, then $M(\omega')Cu = C^t M(\omega')u = 0$. Thus the endomorphism of V with matrix C maps the radical of any non-zero $\omega' \in \Pi'$ into itself, and $C = \lambda$ is a scalar matrix by (†). Moreover $\lambda \neq 0$ since $\widetilde{M} \neq 0$. Evaluating $a^{-t} M_\Pi a^{-1} = \lambda M_{\Pi'}$ at $(c_1, c_2, c_3) \in \mathbb{C}^3$ implies

$$c_1\omega'_1 + c_2\omega'_2 + c_3\omega'_3 = \lambda^{-1} g \cdot (c_1\omega_1 + c_2\omega_2 + c_3\omega_3),$$

where $g \in \mathrm{GL}(V)$ has matrix a. Thus $\Pi' = g \cdot \Pi$.

For $g \in \mathrm{GL}(V)$ with $g \cdot \Pi = \Pi$, the case $\Pi' = \Pi$ of the computation of $p(g^{-t} M_\Pi g^{-1})$ above shows that the linear automorphism $g|_\Pi$ of Π preserves I_Π, hence fixes $[f_\Pi]$, and scalar matrices act on Π by scalars. Thus we obtain the natural homomorphism $\rho : \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \to \mathrm{Stab}_{\mathrm{PGL}(\Pi^\vee)}([f_\Pi])$, where $\mathrm{PGL}(\Pi) = \mathrm{PGL}(\Pi^\vee)$. Let $h \in \mathrm{GL}(\Pi)$ fix $[f_\Pi]$. The converse direction, applied to $\Pi' = \Pi$ with the basis $h(\omega_1), h(\omega_2), h(\omega_3)$, shows that there exist $g \in \mathrm{GL}(V)$ and $\lambda \in \mathbb{C}^*$ with $h(\omega) = \lambda^{-1} g \cdot \omega$ for $\omega \in \Pi$, so ρ is surjective. If $g \in \mathrm{GL}(V)$ acts on Π by a scalar, then g maps the radical of any non-zero $\omega \in \Pi$ onto the radical of $g \cdot \omega$, which is the radical of ω, so g is a scalar matrix by (†). Thus ρ is injective. Similarly, let $\delta \in \mathfrak{gl}(V)$ preserve Π and act on Π by a scalar c, that is, $-\omega(\delta u, u') - \omega(u, \delta u') = c\omega(u, u')$ for $\omega \in \Pi$ and $u, u' \in V$. If u belongs to the radical of a non-zero $\omega \in \Pi$, then $\omega(\delta u, u') = 0$ for any $u' \in V$, so δu belongs to the radical of ω, and δ is a scalar matrix by (†). Thus the differential of ρ is injective. Thus the kernel of ρ is an affine group scheme with one point and zero Lie algebra, hence trivial, so ρ is a closed

immersion [Mil17, Theorem 5.34]. Since ρ is surjective and $\mathrm{Stab}_{\mathrm{PGL}(\Pi^\vee)}([f_\Pi])$ is smooth [Mil17, Theorem 3.23], ρ is an isomorphism.

(4) Let M_c be the alternating matrix of c in the basis $\omega_1, \omega_2, \omega_3$, let $I_c \subset \mathbb{C}[[\tau]][\partial]$ be the ideal generated by the entries of $p(M_c)$, and let $A_c := \mathbb{C}[[\tau]][\partial]/I_c$. Let $F_\bullet$ be the complex

$$0 \to \mathbb{C}[[\tau]][\partial](-7) \xrightarrow{p(M_c)} \mathbb{C}[[\tau]][\partial](-4)^{\oplus 7} \xrightarrow{M_c} \mathbb{C}[[\tau]][\partial](-3)^{\oplus 7} \xrightarrow{p(M_c)^t} \mathbb{C}[[\tau]][\partial],$$

which is a complex by Lemma 16.1(2). The reductions $\bar\omega_1, \bar\omega_2, \bar\omega_3$ of $\omega_1, \omega_2, \omega_3$ modulo τ form a basis of $c(0)$, and $F_\bullet/\tau F_\bullet$ is the Pfaffian complex of $c(0)$ in the basis $\bar\omega_1, \bar\omega_2, \bar\omega_3$, which is exact in positive homological degrees by (1). The short exact sequence

$$0 \to F_\bullet \xrightarrow{\tau} F_\bullet \to F_\bullet/\tau F_\bullet \to 0$$

implies that multiplication by τ is surjective on $H_i(F_\bullet)$ for $i \geq 1$ and injective on $H_0(F_\bullet) = A_c$. Each graded piece of $H_i(F_\bullet)$ is a finitely generated $\mathbb{C}[[\tau]]$-module, so $H_i(F_\bullet) = 0$ for $i \geq 1$ by Nakayama's lemma, and each graded piece $(A_c)_d$ is a finitely generated torsion free, hence free, $\mathbb{C}[[\tau]]$-module. Since $A_c/\tau A_c = \mathbb{C}[\partial]/I_{c(0)}$, (1) implies that $(A_c)_d$ is free of rank $1, 3, 6, 3, 1$ for $d = 0, \dots, 4$ and $(A_c)_d = 0$ for $d \geq 5$. Moreover $(A_c)_2 = \mathbb{C}[[\tau]][\partial]_2$ since I_c is generated in degree 3.

Fix an isomorphism $\mu : (A_c)_4 \to \mathbb{C}[[\tau]]$. The pairing

$$\mathbb{C}[[\tau]][\partial]_4 \times \mathbb{C}[[\tau]][x, y, z]_4 \to \mathbb{C}[[\tau]], \qquad (P, f) \to P(f),$$

is perfect, since $\partial^\alpha(x^\beta) = \alpha!\delta_{\alpha\beta}$ for multi-indices α, β of degree four. Hence there exists a unique $f_c \in \mathbb{C}[[\tau]][x, y, z]_4$ with $P(f_c) = \mu(\bar P)$ for $P \in \mathbb{C}[[\tau]][\partial]_4$, where $\bar P$ is the class of P in $(A_c)_4$. For $0 \leq d \leq 4$ consider the pairing

$$(A_c)_d \times (A_c)_{4-d} \to \mathbb{C}[[\tau]], \qquad (\bar P, \bar Q) \to \mu(\overline{PQ}).$$

Modulo τ it is, up to a non-zero scalar, the pairing $(\bar P, \bar Q) \to (PQ)(f_{c(0)})$ on $\mathbb{C}[\partial]/I_{c(0)} = \mathbb{C}[\partial]/f_{c(0)}^\perp$, which is perfect: if $(PQ)(f_{c(0)}) = 0$ for all Q of degree $4 - d$, then $P(f_{c(0)}) = 0$. Since $(A_c)_d$ and $(A_c)_{4-d}$ are free, the determinant of the pairing in bases is a unit of $\mathbb{C}[[\tau]]$, so the pairing is perfect. For P homogeneous of degree $d \leq 4$, $P(f_c) = 0$ if and only if $(QP)(f_c) = 0$ for all Q of degree $4 - d$, that is, $\mu(\overline{QP}) = 0$ for all such Q, that is, $P \in I_c$. In degrees at least five both I_c and the annihilator of f_c contain all elements. Thus the annihilator of f_c is I_c. For a basis $P_1, \dots, P_6$ of $\mathbb{C}[\partial]_2$, the matrix $((P_iP_j)(f_c))$ is the matrix of the perfect pairing on $(A_c)_2 = \mathbb{C}[[\tau]][\partial]_2$, so $\det \mathrm{Cat}_2(f_c)$ is a unit. Finally, the reduction $\bar f_c$ of f_c modulo τ is non-zero, and $(I_{c(0)})_4$ annihilates it, so $\bar f_c$ is a non-zero multiple of $f_{c(0)}$ by the opening paragraph, since $(I_{c(0)})_4 = (f_{c(0)}^\perp)_4$.

(5) The orbit $O := \mathrm{PGL}(V) \cdot \Pi_1$ is locally closed [Bor91, Proposition I.1.8]. By [Mum74, §6], there exists an irreducible curve C_0 in the closure of O which contains Π_0 and intersects O. Let $\widetilde C$ be the normalization of C_0, let $y \in \widetilde C$ be a point over Π_0, and let $c : \mathrm{Spec}\, \widehat{\mathcal{O}}_{\widetilde C, y} \to \mathbb{B}$ be the induced morphism, where $\widehat{\mathcal{O}}_{\widetilde C, y} \cong \mathbb{C}[[\tau]]$ [Ser79, Chapter II, §4, Theorem 2]. Then c maps the closed point to Π_0 and the generic point η to O, so c factors through $\mathbb{B}^{\mathrm{nd}}$. Let $\mathbb{K} := \mathbb{C}((\tau))$. The fiber over $c(\eta)$ of the orbit morphism $\mathrm{GL}(V) \to O$, $g \to g \cdot \Pi_1$, is a non-empty $\mathbb{K}$-scheme of finite type, so it has a point over a finite extension $\mathbb{K}'$ of $\mathbb{K}$. The normalization of $\mathbb{C}[[\tau]]$ in $\mathbb{K}'$ is a complete discrete valuation ring with residue field $\mathbb{C}$, hence isomorphic to $\mathbb{C}[[\tau']]$ [Ser79, Chapter II, §2, Proposition 3 and §4, Theorem 2]. Possibly replacing $\mathbb{C}[[\tau]]$ with $\mathbb{C}[[\tau']]$, we may assume that $c(\eta) = g \cdot (\Pi_1 \otimes \mathbb{K})$ for some $g \in \mathrm{GL}(V)(\mathbb{K})$.

Fix a basis $\omega_1, \omega_2, \omega_3$ of $H^0(c^*\mathcal{S})$ and a basis $\omega_1'', \omega_2'', \omega_3''$ of Π_1, and let f_c be as in (4). Since $\mathbb{K}$ is flat over $\mathbb{C}[[\tau]]$, the annihilator of $f_c \otimes 1$ in $\mathbb{K}[\partial]$ is $I_c \otimes \mathbb{K}$, the Pfaffian ideal of $c(\eta)$ in the basis $\omega_1, \omega_2, \omega_3$. Let M'' be the alternating matrix of Π_1 in the basis $\omega_1'', \omega_2'', \omega_3''$. The alternating matrix of $c(\eta)$ in the basis $g \cdot \omega_1'', g \cdot \omega_2'', g \cdot \omega_3''$ is $g^{-t}M''g^{-1}$, and $p(g^{-t}M''g^{-1}) = \det(g)^{-1}gp(M'')$ by Lemma 16.1(3) over $\mathbb{K}$. Hence the Pfaffian ideal of $c(\eta)$ in the basis $(g \cdot \omega_k'')_k$ is $I_{\Pi_1} \otimes \mathbb{K}$. The bases $(\omega_k)_k$ and $(g \cdot \omega_k'')_k$ of $c(\eta)$ differ by some $h \in \mathrm{GL}_3(\mathbb{K})$, so $f_c \otimes 1$ is a non-zero multiple

of the image of f_{Π_1} under h. Hence the morphism $\gamma : \operatorname{Spec}\mathbb{C}[[\tau]] \to \mathbb{P}(\mathbb{C}[x,y,z]_4)$ defined by f_c, which is defined since $\bar{f}_c \neq 0$, maps η into the orbit $\mathrm{PGL}_3 \cdot [f_{\Pi_1}]$. Thus γ^{-1} of the closure of $\mathrm{PGL}_3 \cdot [f_{\Pi_1}]$ is a closed subset containing η, so it contains the closed point. By (4), γ maps the closed point to $[f_{\Pi_0}]$ in the basis $\bar{\omega}_1, \bar{\omega}_2, \bar{\omega}_3$, and $[f_{\Pi_0}] \in U^{\mathrm{cat}}$ by (1). Since the closure of the PGL_3-orbit of $[f_{\Pi_1}]$ is PGL_3-invariant, the conclusion does not depend on the bases. □

The next lemma exhibits a non-degenerate net whose isotropic scheme has an ordinary double point.

Lemma 16.4. *Let $e_1, e_2, e_3, g_1, g_2, g_3, g_4$ be a basis of V with dual basis $e_1^*, e_2^*, e_3^*, g_1^*, g_2^*, g_3^*, g_4^*$, let*

$$\omega_1 := e_1^* \wedge g_1^* + (e_2^* + e_3^*) \wedge g_4^* + g_2^* \wedge g_3^*,$$
$$\omega_2 := e_2^* \wedge g_2^* + (e_1^* + e_3^*) \wedge g_4^* + g_3^* \wedge g_1^*,$$
$$\omega_3 := e_3^* \wedge g_3^* + (e_1^* + e_2^*) \wedge g_4^* + g_1^* \wedge g_2^*,$$

let $\Pi_0 := \mathbb{C}\omega_1 + \mathbb{C}\omega_2 + \mathbb{C}\omega_3$ and $V_3 := \mathbb{C}e_1 + \mathbb{C}e_2 + \mathbb{C}e_3$, and let s_{Π_0} be the section of $\Lambda^2\mathcal{U}^\vee \otimes \Pi_0^\vee$ on $\mathrm{Gr}(3,V)$ induced by restriction of forms. Then $\Pi_0 \in \mathbb{B}^{\mathrm{nd}}$, $V_3 \in X_{\Pi_0}$, the differential of s_{Π_0} at V_3 has rank 8, and the germ of X_{Π_0} at V_3 is analytically isomorphic to the germ at the origin of

$$\{w_1w_2 + w_1w_3 + w_2w_3 - 3w_4^2 = 0\} \subset \mathbb{C}^4.$$

In particular $\dim_{V_3} X_{\Pi_0} = 3$ and V_3 is an isolated singular point of X_{Π_0}.

Proof. Let $V_4 := \mathbb{C}g_1 + \mathbb{C}g_2 + \mathbb{C}g_3 + \mathbb{C}g_4$. We first show that a form $\omega \in \Lambda^2 V^\vee$ with $\omega|_{V_3} = 0$ has rank 6 if either the $V_3 \times V_4$ block of ω, that is, the linear map $V_4 \to V_3^\vee$, $q \to \omega(\cdot, q)|_{V_3}$, has rank 3, or it has rank 2 with kernel V_2 and $\omega|_{V_2} \neq 0$. Let $u + q$, $u \in V_3$, $q \in V_4$, belong to the radical of ω. Pairing with V_3 shows that q belongs to the kernel of the $V_3 \times V_4$ block. In the first case the kernel of the block is a line, and the vectors u with $\omega(u, V_4) = 0$ form the zero space, so the radical is at most a line. In the second case $q \in V_2$ and $0 = \omega(u + q, q') = \omega(q, q')$ for $q' \in V_2$, so $q = 0$ since $\omega|_{V_2}$ is a non-zero form on the plane V_2. Then u belongs to the line of vectors u with $\omega(u, V_4) = 0$. In both cases the radical is at most a line, so $\operatorname{rank}\omega = 6$.

No ω_i involves a term $e_a^* \wedge e_b^*$, so $\omega_i|_{V_3} = 0$ and $V_3 \in X_{\Pi_0}$. Let $\omega := a_1\omega_1 + a_2\omega_2 + a_3\omega_3$ with $(a_1, a_2, a_3) \neq 0$. In the bases e_1, e_2, e_3 and $g_1, \dots, g_4$, the $V_3 \times V_4$ block of ω and the restriction $\omega|_{V_4}$ are

$$\begin{pmatrix} a_1 & 0 & 0 & a_2 + a_3 \\ 0 & a_2 & 0 & a_1 + a_3 \\ 0 & 0 & a_3 & a_1 + a_2 \end{pmatrix} \quad \text{and} \quad a_1 g_2^* \wedge g_3^* + a_2 g_3^* \wedge g_1^* + a_3 g_1^* \wedge g_2^*.$$

The automorphism of V with $e_1 \to e_2 \to e_3 \to e_1$, $g_1 \to g_2 \to g_3 \to g_1$, and $g_4 \to g_4$ permutes $\omega_1, \omega_2, \omega_3$ cyclically, so we may permute a_1, a_2, a_3 cyclically. If $a_1a_2a_3 \neq 0$, the block has rank 3. If $a_3 = 0$ and $a_1a_2(a_1 + a_2) \neq 0$, the columns of g_1, g_2, g_4 form a matrix with determinant $a_1a_2(a_1 + a_2)$, so the block has rank 3. If $a_3 = 0$ and $a_2 = -a_1 \neq 0$, the block has rank 2 with kernel $V_2 = \mathbb{C}g_3 + \mathbb{C}(g_1 + g_2 + g_4)$, and $\omega|_{V_4} = a_1(g_1^* + g_2^*) \wedge g_3^*$ satisfies $\omega(g_1 + g_2 + g_4, g_3) = 2a_1 \neq 0$. If $a_2 = a_3 = 0$, the block has rank 2 with kernel $V_2 = \mathbb{C}g_2 + \mathbb{C}g_3$, and $\omega(g_2, g_3) = a_1 \neq 0$. Thus $\operatorname{rank}\omega = 6$ in all cases, $\omega_1, \omega_2, \omega_3$ are linearly independent, and $\Pi_0 \in \mathbb{B}^{\mathrm{nd}}$.

Consider the affine chart $\mathrm{Hom}(V_3, V_4) \to \mathrm{Gr}(3, V)$, $\theta \to \{u + \theta(u) : u \in V_3\}$, with $\mathcal{U}$ trivialized by $u \to u + \theta(u)$. Write $\theta(e_b) = \sum_{j=1}^4 y_{jb} g_j$ and $y_j := (y_{j1}, y_{j2}, y_{j3})$. Since $\omega_i|_{V_3} = 0$, the scheme X_{Π_0} is defined in the chart by the nine equations

$$F_{i,ab} := \omega_i(e_a + \theta e_a, e_b + \theta e_b) = \omega_i(e_a, \theta e_b) + \omega_i(\theta e_a, e_b) + \omega_i(\theta e_a, \theta e_b)$$

for $i \in \{1, 2, 3\}$ and $(a, b) \in \{(2, 3), (3, 1), (1, 2)\}$. Their linear parts $L_{i,ab}$ are

$$\begin{array}{lll} L_{1,23} = y_{43} - y_{42}, & L_{1,31} = y_{41} - y_{13}, & L_{1,12} = y_{12} - y_{41}, \\ L_{2,23} = y_{23} - y_{42}, & L_{2,31} = y_{41} - y_{43}, & L_{2,12} = y_{42} - y_{21}, \\ L_{3,23} = y_{43} - y_{32}, & L_{3,31} = y_{31} - y_{43}, & L_{3,12} = y_{42} - y_{41}. \end{array}$$

Each of $y_{12}, y_{13}, y_{21}, y_{23}, y_{31}, y_{32}$ appears in exactly one of them, and $L_{1,23} + L_{2,31} + L_{3,12} = 0$. Hence the linear parts span a space of dimension 8, which is the rank of the differential of s_{Π_0} at V_3, and their common kernel is the four-dimensional space $\mathcal{T}$ of

$$y_1 = (w_1, w_4, w_4), \quad y_2 = (w_4, w_2, w_4), \quad y_3 = (w_4, w_4, w_3), \quad y_4 = (w_4, w_4, w_4), \qquad w \in \mathbb{C}^4.$$

The ideal of X_{Π_0} in the chart is generated by the eight equations $F_{i,ab}$ with $(i, ab) \neq (3, 12)$ and by

$$\begin{aligned} F_{1,23} + F_{2,31} + F_{3,12} &= \omega_1(\theta e_2, \theta e_3) + \omega_2(\theta e_3, \theta e_1) + \omega_3(\theta e_1, \theta e_2) \\ &= (y_{22}y_{33} - y_{32}y_{23}) + (y_{33}y_{11} - y_{13}y_{31}) + (y_{11}y_{22} - y_{21}y_{12}). \end{aligned}$$

The eight equations have linearly independent linear parts, so by the implicit function theorem they define a smooth germ Y of dimension four at 0, with tangent space $\mathcal{T}$, parametrized by $w \in (\mathbb{C}^4, 0)$ with $y = \ell(w) + O(|w|^2)$, where ℓ is the linear parametrization of $\mathcal{T}$ above. Thus the restriction of $F_{1,23} + F_{2,31} + F_{3,12}$ to Y is

$$(w_2w_3 - w_4^2) + (w_3w_1 - w_4^2) + (w_1w_2 - w_4^2) + O(|w|^3) = w_1w_2 + w_1w_3 + w_2w_3 - 3w_4^2 + O(|w|^3).$$

The quadratic form $w_1w_2 + w_1w_3 + w_2w_3 - 3w_4^2$ is non-degenerate, since the symmetric matrix of $w_1w_2 + w_1w_3 + w_2w_3$ has eigenvalues $1, -\frac{1}{2}, -\frac{1}{2}$. The holomorphic Morse lemma [AGV85, Section 6.2, Theorem] completes the proof. □

The next proposition describes Ω inside $\mathbb{B}^{\mathrm{nd}}$ and shows that Ω is saturated for the affine quotient. Together with Theorem 16.12(2), it shows that degenerations in $\mathbb{B}^{\mathrm{nd}}$ of orbits in Ω stay in Ω, and that if $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$ for $\Pi \in \Omega$, then a non-trivial torus acts faithfully on X_{Π_0} for Π_0 in the closed orbit in the closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$.

Proposition 16.5. *Let $\mathcal{X} \subset \mathrm{Gr}(3, V) \times \mathbb{B}$ be the zero scheme of the section of $\Lambda^2\mathcal{U}^\vee \boxtimes \mathcal{S}^\vee$ induced by restriction of forms, let $\pi : \mathcal{X} \to \mathbb{B}$ be the projection, and let $\mathcal{X}_\Omega := \pi^{-1}(\Omega)$. Then the following hold.*

(1) A net $\Pi \in \mathbb{B}^{\mathrm{nd}}$ belongs to Ω if and only if X_Π is smooth of dimension three.

(2) The morphism $\pi|_{\mathcal{X}_\Omega} : \mathcal{X}_\Omega \to \Omega$ is smooth and projective, the fiber of $\pi|_{\mathcal{X}_\Omega}$ over Π is X_Π, and

$$\omega_{\mathcal{X}_\Omega/\Omega}^{-1} \cong \big(\mathcal{O}_{\mathrm{Gr}(3,V)}(1) \boxtimes \mathcal{A}^{-3}\big)|_{\mathcal{X}_\Omega}.$$

In particular the Plücker line bundle $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ restricts to $\omega_{X_\Pi}^{-1}$ for $\Pi \in \Omega$.

(3) The closed subset $\mathbb{B}^{\mathrm{nd}} \backslash \Omega$ is a non-empty irreducible divisor, and there exists an $\mathrm{SL}(V)$-invariant regular function r_Ω on $\mathbb{B}^{\mathrm{nd}}$ with $\Omega = \{r_\Omega \neq 0\}$.

(4) Ω is saturated for the affine quotient $\mathbb{B}^{\mathrm{nd}} \to \mathbb{B}^{\mathrm{nd}} /\!\!/ \mathrm{SL}(V)$, that is, Ω is the preimage of the image of Ω under the quotient morphism. In particular, for $\Pi \in \Omega$, any orbit contained in the closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$ is contained in Ω.

(5) Let $\Pi \in \Omega$. If $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, then the stabilizer in $\mathrm{SL}(V)$ of a point of the unique closed orbit in the closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$ is reductive of positive dimension and contains a non-trivial torus.

Proof. The proof is long, so we divide it into several steps. Recall that $\mathcal{O}_{\mathrm{Gr}(3,V)}(1) = \det \mathcal{U}^\vee$. For $\Pi \in \mathbb{B}$, let s_Π be the section of $\mathcal{E}_\Pi := \Lambda^2\mathcal{U}^\vee \otimes \Pi^\vee$ defining X_Π. We have $\operatorname{rank} \mathcal{E}_\Pi = 9$, $\dim \mathrm{Gr}(3, V) = 12$, and $T_x X_\Pi = \operatorname{Ker} ds_\Pi(x)$ for $x \in X_\Pi$.

Step 1. In this step we show (1) and (2).

Let $\mathbb{B}^{\mathrm{tr}} \subset \mathbb{B}$ be the set of Π such that $ds_\Pi(x)$ is surjective for any $x \in X_\Pi$. By the Jacobian criterion, $\Pi \in \mathbb{B}^{\mathrm{tr}}$ if and only if X_Π is empty or smooth of dimension three. The set of $(x, \Pi) \in \mathcal{X}$ with $\operatorname{rank} ds_\Pi(x) < 9$ is closed and π is proper, so $\mathbb{B}^{\mathrm{tr}}$ is open. For a general $\Pi \in \mathbb{B}$, X_Π is a smooth prime Fano threefold of genus 12 by a theorem of Mukai, see [Sch01, Theorem 1.1, Corollary 2.7], and conversely any such threefold is isomorphic to some X_Π [Muk92, Section 3, Theorem 3]. Thus $\mathbb{B}^{\mathrm{tr}}$ is non-empty, hence irreducible.

Let $\mathcal{X}^{\mathrm{tr}} := \pi^{-1}(\mathbb{B}^{\mathrm{tr}})$, the zero scheme of the section of $\mathcal{E} := \Lambda^2\mathcal{U}^\vee \boxtimes \mathcal{S}^\vee$ on $\mathrm{Gr}(3,V)\times\mathbb{B}^{\mathrm{tr}}$. At any point of $\mathcal{X}^{\mathrm{tr}}$ the differential of the section of $\mathcal{E}$ is surjective already on the tangent vectors along $\mathrm{Gr}(3,V)$. Hence $\mathcal{X}^{\mathrm{tr}}$ is smooth, and any tangent vector of $\mathbb{B}^{\mathrm{tr}}$ lifts to a tangent vector of $\mathcal{X}^{\mathrm{tr}}$ after adding a tangent vector along $\mathrm{Gr}(3,V)$, so the differential of $\pi|_{\mathcal{X}^{\mathrm{tr}}}$ is surjective at any point, so $\pi|_{\mathcal{X}^{\mathrm{tr}}}$ is smooth and projective of relative dimension three. The image of $\pi|_{\mathcal{X}^{\mathrm{tr}}}$ is open, as $\pi|_{\mathcal{X}^{\mathrm{tr}}}$ is flat [Har77, Exercise III.9.1], and closed, as π is proper, so it is the connected space $\mathbb{B}^{\mathrm{tr}}$. Thus X_Π is non-empty for $\Pi\in\mathbb{B}^{\mathrm{tr}}$. Since $\det\Lambda^2\mathcal{U}^\vee \cong \mathcal{O}_{\mathrm{Gr}(3,V)}(2)$, we have $\det\mathcal{E}\cong\mathcal{O}_{\mathrm{Gr}(3,V)}(6)\boxtimes\mathcal{A}^3$, and $\omega_{\mathrm{Gr}(3,V)}\cong\mathcal{O}_{\mathrm{Gr}(3,V)}(-7)$ since $T_{\mathrm{Gr}(3,V)}\cong\mathcal{U}^\vee\otimes\mathcal{Q}$ [Ful98, Appendix B.5.8]. The conormal sequence of $\mathcal{X}^{\mathrm{tr}}$ in $\mathrm{Gr}(3,V)\times\mathbb{B}^{\mathrm{tr}}$ relative to $\mathbb{B}^{\mathrm{tr}}$ implies

$$\omega_{\mathcal{X}^{\mathrm{tr}}/\mathbb{B}^{\mathrm{tr}}} \cong \big(\mathcal{O}_{\mathrm{Gr}(3,V)}(-1)\boxtimes\mathcal{A}^3\big)|_{\mathcal{X}^{\mathrm{tr}}}.$$

In particular $\omega_{X_\Pi}^{-1}\cong\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_{X_\Pi}$ is ample for $\Pi\in\mathbb{B}^{\mathrm{tr}}$.

For a general $\Pi\in\mathbb{B}^{\mathrm{tr}}$, X_Π is connected, $H^2(X_\Pi,\mathbb{Z}) = \mathrm{Pic}(X_\Pi) = \mathbb{Z}[-K_{X_\Pi}]$ [IP99, Proposition 2.1.2], and $(-K_{X_\Pi})^3 = 22$. By Ehresmann's theorem [Voi02, Theorem 9.3], applied over contractible neighborhoods of points of the connected manifold $\mathbb{B}^{\mathrm{tr}}$, the fibers of the smooth projective morphism $\pi|_{\mathcal{X}^{\mathrm{tr}}}$ over the connected base $\mathbb{B}^{\mathrm{tr}}$ are diffeomorphic, compatibly with the restrictions of $c_1(\mathcal{O}_{\mathrm{Gr}(3,V)}(1))$, which are the classes $-K_{X_\Pi}$. Thus for any $\Pi\in\mathbb{B}^{\mathrm{tr}}$, X_Π is connected, $H^2(X_\Pi,\mathbb{Z}) = \mathbb{Z}[-K_{X_\Pi}]$, and $(-K_{X_\Pi})^3 = 22$, and $\mathrm{Pic}(X_\Pi) = H^2(X_\Pi,\mathbb{Z})$ [IP99, Proposition 2.1.2]. Since genus 12 means $(-K_{X_\Pi})^3 = 22$ and $\mathrm{Pic}(X_\Pi) = \mathbb{Z}[-K_{X_\Pi}]$, X_Π is a smooth prime Fano threefold of genus 12 for any $\Pi\in\mathbb{B}^{\mathrm{tr}}$. Hence $\Omega = \mathbb{B}^{\mathrm{nd}}\cap\mathbb{B}^{\mathrm{tr}}$, which implies (1), and (2) follows from the formula for $\omega_{\mathcal{X}^{\mathrm{tr}}/\mathbb{B}^{\mathrm{tr}}}$ since $\mathcal{X}_\Omega = \mathcal{X}^{\mathrm{tr}}\times_{\mathbb{B}^{\mathrm{tr}}}\Omega$.

Step 2. In this step we describe $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ by an incidence variety $\mathcal{I}$ with $\dim\mathcal{I} = 62$.

Let $P^{\mathrm{nd}}\subset(\Lambda^2V^\vee)^{\oplus3}$ be the open subset of linearly independent triples $\underline{\omega} = (\omega_1,\omega_2,\omega_3)$ whose span belongs to $\mathbb{B}^{\mathrm{nd}}$, so that $P^{\mathrm{nd}}\to\mathbb{B}^{\mathrm{nd}}$ is the principal GL_3-bundle whose fiber over Π is the set of ordered bases of Π, and $\dim P^{\mathrm{nd}} = 63$. Let $V_3\in\mathrm{Gr}(3,V)$, let V_4 be a complement of V_3, and consider the affine chart $\mathrm{Hom}(V_3,V_4)\to\mathrm{Gr}(3,V)$, $\theta\to\{u+\theta(u) : u\in V_3\}$, with $\mathcal{U}$ trivialized by $u\to u+\theta(u)$. For a triple $\underline{\omega}$ spanning Π, the section s_Π is $\theta\to(\omega_i(u+\theta u, u'+\theta u'))_{i=1,2,3}$, as a function of $u,u'\in V_3$. Thus $V_3\in X_\Pi$ if and only if $\omega_i|_{V_3} = 0$ for $i=1,2,3$, which are nine linear equations on $\underline{\omega}$, and then

$$ds_\Pi(V_3)(\theta)_i(u,u') = \omega_i(u,\theta u') + \omega_i(\theta u, u'),\qquad u,u'\in V_3.$$

The dual of the cokernel of $ds_\Pi(V_3)$ is the space of $\xi = (\xi_1,\xi_2,\xi_3)\in(\Lambda^2V_3)^{\oplus3}$ annihilating the image of $ds_\Pi(V_3)$. For $\alpha\in V_3^\vee$ let $\iota_\alpha:\Lambda^2V_3\to V_3$ be the contraction, $\iota_\alpha(u\wedge u') = \alpha(u)u'-\alpha(u')u$. For $\theta=\alpha\otimes q$, that is, $\theta(u)=\alpha(u)q$ with $q\in V_4$, we have $ds_\Pi(V_3)(\theta)_i(u,u') = -\omega_i(\iota_\alpha(u\wedge u'),q)$. Since such θ span $\mathrm{Hom}(V_3,V_4)$, and $\beta(\iota_\alpha\xi_i) = -\alpha(\iota_\beta\xi_i)$ for $\alpha,\beta\in V_3^\vee$, the element ξ annihilates the image if and only if

$$(*)\qquad \sum_{i=1}^{3}\iota_{\omega_i(\cdot,q)|_{V_3}}\xi_i = 0\in V_3\quad\text{for any } q\in V,$$

where the condition for $q\in V_3$ holds since $\omega_i|_{V_3}=0$. For $0\neq\xi_i\in\Lambda^2V_3$, the map $\alpha\to\iota_\alpha\xi_i$ has rank 2 with image the plane $W_i\subset V_3$ with $\xi_i\in\Lambda^2W_i$. If $\dim(\mathbb{C}\xi_1+\mathbb{C}\xi_2+\mathbb{C}\xi_3)\geq2$, two of the planes W_i are distinct and span V_3, so the linear map $(V_3^\vee)^{\oplus3}\to V_3$, $(\alpha_1,\alpha_2,\alpha_3)\to\sum_i\iota_{\alpha_i}\xi_i$, is surjective. For such ξ, letting q run over a basis of V_4, $(*)$ is a system of $3\cdot4=12$ linearly independent linear equations on the $V_3\times V_4$ blocks of $\omega_1,\omega_2,\omega_3$.

Let $\mathcal{F}$ be the variety of pairs $(V_3,[\xi])$ with $V_3\in\mathrm{Gr}(3,V)$, $[\xi]\in\mathbb{P}((\Lambda^2V_3)^{\oplus3})$, and $\dim(\mathbb{C}\xi_1+\mathbb{C}\xi_2+\mathbb{C}\xi_3)\geq2$. It is an open subset of a $\mathbb{P}^8$-bundle over $\mathrm{Gr}(3,V)$, so it is smooth and irreducible, and $\dim\mathcal{F} = 12+8=20$. Let $\mathcal{I}\subset(\Lambda^2V^\vee)^{\oplus3}\times\mathcal{F}$ be the set of $(\underline{\omega},V_3,[\xi])$ with $\omega_i|_{V_3}=0$ for $i=1,2,3$ and $(*)$. The fibers of $\mathcal{I}\to\mathcal{F}$ are linear subspaces of codimension $9+12=21$ in $(\Lambda^2V^\vee)^{\oplus3}$, so $\mathcal{I}$ is a vector bundle of rank 42 over $\mathcal{F}$, irreducible, and $\dim\mathcal{I}=62$.

Let $\underline{\omega}\in P^{\mathrm{nd}}$ span Π, and let $V_3\in X_\Pi$ be a point where $ds_\Pi(V_3)$ is not surjective. We claim that any non-zero ξ annihilating the image satisfies $\dim(\mathbb{C}\xi_1+\mathbb{C}\xi_2+\mathbb{C}\xi_3)\geq2$. Otherwise $\xi_i = a_i\xi_0$

with $(a_1, a_2, a_3) \neq 0$ and $\xi_0 = u \wedge u' \neq 0$. Then $\omega := a_1\omega_1 + a_2\omega_2 + a_3\omega_3 \neq 0$, and $(*)$ reads $\omega(u, q)u' - \omega(u', q)u = 0$ for $q \in V$, so u and u' belong to the radical of ω. Since $\dim V$ is odd, the radical of ω has odd dimension, so the radical is at least three-dimensional and $\operatorname{rank}\omega \leq 4$, which contradicts $\Pi \in \mathbb{B}^{\mathrm{nd}}$.

Let $\mathcal{I}^{\mathrm{nd}} := \mathcal{I} \cap (P^{\mathrm{nd}} \times \mathcal{F})$, an open subset of $\mathcal{I}$. By **Step 1** and the claim, the image of $\mathcal{I}^{\mathrm{nd}}$ in P^{nd} is the preimage of $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$, which is closed. Thus the preimage of $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ in P^{nd} is either empty or irreducible of dimension at most 62.

Step 3. In this step we show that $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ is a non-empty irreducible divisor.

By Lemma 16.4, there exist $\Pi' \in \mathbb{B}^{\mathrm{nd}}$ and $V_3 \in X_{\Pi'}$ such that $ds_{\Pi'}(V_3)$ has rank 8 and the germ of $X_{\Pi'}$ at V_3 is a hypersurface germ in a smooth germ of dimension four with an isolated singular point at V_3. Let $\underline{\omega}$ be a basis of Π', and let ξ^0 span the dual of the cokernel of $ds_{\Pi'}(V_3)$. Then $z_0 := (\underline{\omega}, V_3, [\xi^0]) \in \mathcal{I}^{\mathrm{nd}}$. If $(\underline{\omega}, V_3', [\xi]) \in \mathcal{I}^{\mathrm{nd}}$ with V_3' near V_3, then $\dim T_{V_3'}X_{\Pi'} = 12 - \operatorname{rank} ds_{\Pi'}(V_3') \geq 4$, so $V_3' = V_3$, and $[\xi] = [\xi^0]$. Thus z_0 is an isolated point of the fiber of $\mathcal{I}^{\mathrm{nd}} \to P^{\mathrm{nd}}$ containing z_0. Since $\mathcal{I}^{\mathrm{nd}}$ is irreducible and $\dim \mathcal{I}^{\mathrm{nd}} = 62$, the closure of the image of $\mathcal{I}^{\mathrm{nd}}$ in P^{nd} is at least 62-dimensional [Har77, Exercise II.3.22(b)]. By **Step 2**, the preimage of $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ in P^{nd}, which contains $\underline{\omega}$, is a non-empty irreducible closed subset of dimension $62 = \dim P^{\mathrm{nd}} - 1$. Since $P^{\mathrm{nd}} \to \mathbb{B}^{\mathrm{nd}}$ is Zariski locally isomorphic to the projection $\mathbb{B}^{\mathrm{nd}} \times \mathrm{GL}_3 \to \mathbb{B}^{\mathrm{nd}}$, $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ is irreducible of dimension $62 - 9 = 53 = \dim \mathbb{B} - 1$, and it contains Π'.

Step 4. In this step we construct r_Ω and show (3) and (4).

Let $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega}$ be the closure of $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ in $\mathbb{B}$. By **Step 3** it is a prime divisor, and since $\mathbb{B}^{\mathrm{nd}}\backslash\Omega$ is closed in $\mathbb{B}^{\mathrm{nd}}$, we have $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega} \cap \mathbb{B}^{\mathrm{nd}} = \mathbb{B}^{\mathrm{nd}}\backslash\Omega$. Since $\mathrm{Pic}(\mathbb{B}) = \mathbb{Z}[\mathcal{A}]$ [Ful98, Section 14.7], there exist an integer $d > 0$ and a section $\sigma_\Omega \in H^0(\mathbb{B}, \mathcal{A}^d)$ with divisor $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega}$. Since $X_{g\cdot\Pi} = g(X_\Pi)$ for $g \in \mathrm{SL}(V)$, the subsets Ω and $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega}$ are $\mathrm{SL}(V)$-invariant. Thus $g \cdot \sigma_\Omega = \chi(g)\sigma_\Omega$ for a character χ of $\mathrm{SL}(V)$, as any section of $\mathcal{A}^d$ with divisor $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega}$ is a non-zero constant multiple of σ_Ω, and χ is trivial since $\mathrm{SL}(V)$ has no non-trivial characters. Let

$$r_\Omega := \frac{\sigma_\Omega^{14}}{c_{\mathrm{Pf}}^d},$$

which is an $\mathrm{SL}(V)$-invariant regular function on $\mathbb{B}^{\mathrm{nd}}$ since c_{Pf} does not vanish on $\mathbb{B}^{\mathrm{nd}}$ (Definition-Proposition 16.2). The zero set of r_Ω is $\overline{\mathbb{B}^{\mathrm{nd}}\backslash\Omega} \cap \mathbb{B}^{\mathrm{nd}} = \mathbb{B}^{\mathrm{nd}}\backslash\Omega$, so $\Omega = \{r_\Omega \neq 0\}$, and (3) holds.

Let $\psi_{\mathrm{nd}} : \mathbb{B}^{\mathrm{nd}} \to \mathbb{B}^{\mathrm{nd}} /\!\!/ \mathrm{SL}(V)$ be the affine quotient, which is surjective, and any fiber of which contains exactly one closed orbit [MFK94, Theorem 1.1 and Corollary 1.2], [BGM19, Section 2.2]. As an invariant regular function, $r_\Omega = \bar{r} \circ \psi_{\mathrm{nd}}$ for a regular function $\bar{r}$ on $\mathbb{B}^{\mathrm{nd}} /\!\!/ \mathrm{SL}(V)$. Thus $\Omega = \psi_{\mathrm{nd}}^{-1}(\{\bar{r} \neq 0\})$ and $\psi_{\mathrm{nd}}(\Omega) = \{\bar{r} \neq 0\}$, so $\Omega = \psi_{\mathrm{nd}}^{-1}(\psi_{\mathrm{nd}}(\Omega))$. For $\Pi \in \Omega$, the invariant function r_Ω is constant on the closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$ with value $r_\Omega(\Pi) \neq 0$, so the closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$ is contained in Ω. This proves (4).

Step 5. In this step we show (5).

Let $\Pi \in \Omega$. The closure of $\mathrm{SL}(V) \cdot \Pi$ in $\mathbb{B}^{\mathrm{nd}}$ is contained in the fiber $\psi_{\mathrm{nd}}^{-1}(\psi_{\mathrm{nd}}(\Pi))$, and an orbit of minimal dimension in the closure of $\mathrm{SL}(V) \cdot \Pi$ is closed [Bor91, Proposition I.1.8]. Thus the closure contains a unique closed orbit $\mathrm{SL}(V) \cdot \Pi_0$ [MFK94, Theorem 1.1 and Corollary 1.2], [BGM19, Section 2.2], and $\Pi_0 \in \Omega$ by (4). The orbit $\mathrm{SL}(V) \cdot \Pi_0 \cong \mathrm{SL}(V)/\operatorname{Stab}_{\mathrm{SL}(V)}(\Pi_0)$ is closed in the affine variety $\mathbb{B}^{\mathrm{nd}}$, hence affine. By Matsushima's criterion [Arz08, Theorem 2.1], $\operatorname{Stab}_{\mathrm{SL}(V)}(\Pi_0)$ is reductive. If $\mathrm{SL}(V)\cdot\Pi$ is not closed, then $\mathrm{SL}(V)\cdot\Pi_0$ is contained in the boundary of $\mathrm{SL}(V) \cdot \Pi$, so $\dim \mathrm{SL}(V) \cdot \Pi_0 < \dim \mathrm{SL}(V) \cdot \Pi$ [Bor91, Proposition I.1.8], and

$$\dim \operatorname{Stab}_{\mathrm{SL}(V)}(\Pi_0) = 48 - \dim \mathrm{SL}(V) \cdot \Pi_0 > 48 - \dim \mathrm{SL}(V) \cdot \Pi = \dim \operatorname{Stab}_{\mathrm{SL}(V)}(\Pi) \geq 0.$$

The identity component of $\operatorname{Stab}_{\mathrm{SL}(V)}(\Pi_0)$ is a connected reductive group of positive dimension, so it is not unipotent and contains a non-trivial torus [Bor91, Corollary IV.11.5(2)]. □

Proposition 17.4 uses Proposition 16.7 to turn a non-closed orbit of a canonical Mukai net into a non-product test configuration with vanishing Donaldson–Futaki invariant, after verifying that the limit net is non-degenerate with three-dimensional isotropic scheme and that the total space is normal. The CM line bundle goes back to [KM76, Theorem 4] and [PT06, Definition 1 and Theorem 1], and we use the normalization of [FR06, Theorem 2.1 and Definition 2.2].

Lemma 16.6. *Let $\mathcal{U}$ be the tautological subbundle on $\mathrm{Gr}(3,V)$, and let $h := c_1(\mathcal{O}_{\mathrm{Gr}(3,V)}(1))$. Then*

$$\int_{\mathrm{Gr}(3,V)} h^3 c_3(\Lambda^2\mathcal{U}^\vee)^3 = 22, \qquad \int_{\mathrm{Gr}(3,V)} h^4 c_2(\Lambda^2\mathcal{U}^\vee) c_3(\Lambda^2\mathcal{U}^\vee)^2 = 89.$$

Proof. Let x, y, and z be the Chern roots of $\mathcal{U}^\vee$, and let $e_2 := xy + xz + yz$, so that $h = c_1(\det\mathcal{U}^\vee) = x + y + z$. The Chern roots of $\Lambda^2\mathcal{U}^\vee$ are $x+y$, $x+z$, and $y+z$. Hence

$$c_3(\Lambda^2\mathcal{U}^\vee) = (x+y)(x+z)(y+z), \qquad c_2(\Lambda^2\mathcal{U}^\vee) = h^2 + e_2.$$

For a symmetric homogeneous polynomial $P(x,y,z)$ of degree 12, let $[P]_{ijk}$ be the coefficient of $x^i y^j z^k$ in P. For a partition $\mu = (\mu_1,\mu_2,\mu_3)$ of 12 with at most three parts, let $s_\mu(x,y,z)$ be the Schur polynomial. Since $c(-\mathcal{U}) = c(\mathcal{U})^{-1} = (1-x)^{-1}(1-y)^{-1}(1-z)^{-1}$, the class $c_k(-\mathcal{U})$ is the complete homogeneous symmetric polynomial of degree k in x, y, and z, so the Jacobi–Trudi formula [Ful98, Lemma A.9.3] gives

$$\Delta_\mu(c(-\mathcal{U})) = \det\left(c_{\mu_i+j-i}(-\mathcal{U})\right)_{1\le i,j\le 3} = s_\mu(x,y,z).$$

By [Ful98, Example 14.2.2] applied with X a point, $E = V$, $F = 0$, $d = 3$, and $n = 7$, we have $\int_{\mathrm{Gr}(3,V)} s_\mu(x,y,z) = \Delta_{\mu-(4,4,4)}(1)$. The determinant $\Delta_{\mu-(4,4,4)}(1)$ is non-zero only if $i \to i - \mu_i + 4$ is a permutation of $\{1,2,3\}$. Since $i \to i-\mu_i+4$ is strictly increasing, it is a permutation only if it is the identity, that is, $\mu = (4,4,4)$, and $\Delta_{(0,0,0)}(1) = 1$. Thus $\int_{\mathrm{Gr}(3,V)} s_\mu(x,y,z) = 1$ if $\mu = (4,4,4)$ and $\int_{\mathrm{Gr}(3,V)} s_\mu(x,y,z) = 0$ otherwise. By the bialternant formula [Ful98, Lemma A.9.3], $(x-y)(x-z)(y-z)s_\mu(x,y,z)$ is the alternating sum of the monomials obtained by permuting the exponents of $x^{\mu_1+2}y^{\mu_2+1}z^{\mu_3}$, so its coefficient of $x^6y^5z^4$ is 1 if $\mu = (4,4,4)$ and 0 otherwise. Since the Schur polynomials s_μ form a basis of the symmetric polynomials of degree 12 in x, y, and z, it follows that $\int_{\mathrm{Gr}(3,V)} P$ is the coefficient of $x^6y^5z^4$ in $(x-y)(x-z)(y-z)P$. Since $(x-y)(x-z)(y-z) = x^2y - x^2z - xy^2 + xz^2 + y^2z - yz^2$ and P is symmetric,

$$\int_{\mathrm{Gr}(3,V)} P = [P]_{444} - 2[P]_{543} + [P]_{552} + [P]_{633} - [P]_{642}. \tag{16.1}$$

The coefficients of $P = h^3c_3(\Lambda^2\mathcal{U}^\vee)^3$ and $P = h^4(h^2+e_2)c_3(\Lambda^2\mathcal{U}^\vee)^2$ needed in (16.1) are listed in Table 25. They are obtained by expanding $h = x+y+z$, h^2+e_2, and $c_3(\Lambda^2\mathcal{U}^\vee) = (x+y)(x+z)(y+z)$ by the binomial theorem.

TABLE 25. The five coefficients in (16.1) for $P = h^3c_3(\Lambda^2\mathcal{U}^\vee)^3$ and $P = h^4(h^2+e_2)c_3(\Lambda^2\mathcal{U}^\vee)^2$.

P	$[P]_{444}$	$[P]_{543}$	$[P]_{552}$	$[P]_{633}$	$[P]_{642}$
$h^3c_3(\Lambda^2\mathcal{U}^\vee)^3$	1092	834	468	490	360
$h^4(h^2+e_2)c_3(\Lambda^2\mathcal{U}^\vee)^2$	4716	3636	2058	2208	1621

By (16.1) and Table 25,

$$\int_{\mathrm{Gr}(3,V)} h^3c_3(\Lambda^2\mathcal{U}^\vee)^3 = 1092 - 2\cdot 834 + 468 + 490 - 360 = 22,$$

$$\int_{\mathrm{Gr}(3,V)} h^4c_2(\Lambda^2\mathcal{U}^\vee)c_3(\Lambda^2\mathcal{U}^\vee)^2 = 4716 - 2\cdot 3636 + 2058 + 2208 - 1621 = 89. \qquad \square$$

Proposition 16.7. *Let $\mathcal{A}$, $\mathcal{S}$, and D_{Pf} be as in Definition-Proposition 16.2, let $\mathcal{U}$ be the tautological subbundle on $\mathrm{Gr}(3,V)$ and $\mathcal{H} := \det \mathcal{U}^\vee$, and let $\mathcal{X} \subset \mathrm{Gr}(3,V) \times \mathbb{B}$ be the universal isotropic scheme, the zero scheme of the section of $\Lambda^2\mathcal{U}^\vee \boxtimes \mathcal{S}^\vee$ given by restriction of alternating forms. Let $\mathbb{B}^{(3)} \subset \mathbb{B}$ be the set of Π with $\dim X_\Pi = 3$, and let $p : \mathcal{X}|_{\mathbb{B}^{(3)}} \to \mathbb{B}^{(3)}$ be the projection. Let $\widetilde{\mathcal{L}} := \mathcal{H} \boxtimes \mathcal{A}^{-3} \otimes (\det V)^3$, let $\mathcal{K}^\bullet$ be the Koszul complex of the section above, let $q : \mathrm{Gr}(3,V) \times \mathbb{B} \to \mathbb{B}$ be the projection, and let $\mathcal{D}_m := \det Rq_*(\mathcal{K}^\bullet \otimes \widetilde{\mathcal{L}}^m)$ for $m \in \mathbb{Z}$. For $k \in \{3,4\}$ let*

$$\mathcal{M}_k^{\mathrm{Kos}} := \bigotimes_{j=0}^{k} \mathcal{D}_j^{(-1)^{k-j}\binom{k}{j}}, \qquad \lambda_{\mathrm{CM}}^{\mathrm{Kos}} := (\mathcal{M}_4^{\mathrm{Kos}})^{15} \otimes (\mathcal{M}_3^{\mathrm{Kos}})^{-8}.$$

Then the following hold.

(1) $\mathbb{B}^{(3)}$ is a non-empty open subset, the morphism p is flat and Gorenstein with relative anti-canonical bundle $\mathcal{L} := \widetilde{\mathcal{L}}|_{\mathcal{X}|_{\mathbb{B}^{(3)}}}$, and the fibers have anti-canonical degree 22. Write

$$\det Rp_*(\mathcal{L}^m) = \bigotimes_{i=0}^{4} \mathcal{M}_i^{\binom{m}{i}}$$

and $\lambda_{\mathrm{CM}} := \mathcal{M}_4^{15} \otimes \mathcal{M}_3^{-8}$. The exponents $15 = 3 \cdot 4 + 3$ and $-8 = -2 \cdot 4$ are those of [FR06, Definition 2.2] for a family of threefolds whose fibers have Hilbert polynomial $a_0m^3 + a_1m^2 + O(m)$ with $2a_1/a_0 = 3$, as for $\mathcal{L}$. Then $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}|_{\mathbb{B}^{(3)}} \cong \lambda_{\mathrm{CM}}$, and $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}$ is $\mathrm{GL}(V)$-equivariantly isomorphic to $\mathcal{A}^{175} \otimes (\det V)^{-150}$.

(2) D_{Pf} is a non-zero irreducible $\mathrm{GL}(V)$-invariant divisor, and $\Pi_X \in \mathbb{B}^{(3)} \setminus D_{\mathrm{Pf}}$ for any smooth prime Fano threefold X of genus 12. In particular Π_X is GIT-semistable for $\mathcal{A}$.

(3) Let X be a smooth prime Fano threefold of genus 12, and let $\lambda : \mathbb{G}_m \to \mathrm{GL}(V)$ be a one-parameter subgroup with $\Pi_0 := \lim_{t\to 0} \lambda(t) \cdot \Pi_X \in \mathbb{B}^{(3)}$. Let $\mathcal{X}_\lambda \to \mathbb{A}^1$ and $\mathcal{L}_\lambda$ be the pullbacks of p and $\mathcal{L}$ along the morphism $\mathbb{A}^1 \to \mathbb{B}^{(3)}$ given by $t \to \lambda(t) \cdot \Pi_X$ for $t \neq 0$ and $0 \to \Pi_0$. Then $\mathcal{X}_\lambda \to \mathbb{A}^1$ is flat, polarized by $\mathcal{L}_\lambda$, and isotrivial over $\mathbb{A}^1 \setminus \{0\}$. Let $a_0m^3 + a_1m^2 + O(m)$ be the Hilbert polynomial of the central fiber, so that $a_0 = 11/3$ and $a_1 = 11/2$, let $b_0m^4 + b_1m^3 + O(m^2)$ be the total weight of the induced $\mathbb{G}_m$-action on H^0 of the central fiber polarized by $\mathcal{L}_\lambda^m$, and let $\lambda(t)$ act on the fiber of $\mathcal{A}$ at Π_0 by $t^{\mu_{\mathcal{A}}}$ and on $\det V$ by t^{d_λ}. Here the total weight is the integer w such that $\lambda(t)$ acts on $\det H^0$ of the central fiber by t^w. Then

$$\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) := \frac{2(a_1b_0 - a_0b_1)}{a_0^2} = \frac{175\mu_{\mathcal{A}} - 150d_\lambda}{88} \geq 0,$$

with strict inequality if $\Pi_0 \in D_{\mathrm{Pf}}$ and equality if $\Pi_0 \notin D_{\mathrm{Pf}}$. If $\mathcal{X}_\lambda$ is normal, then $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a test configuration of X, and $\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is the Donaldson–Futaki invariant of $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ in the sense of Definition 2.4.

Proof. For a smooth prime Fano threefold X of genus 12 we fix a linear isomorphism $H^0(X, \mathcal{U}_X^\vee)^\vee \cong V$, so that $\Pi_X \in \mathbb{B}$, and Π_X is determined by X up to the action of $\mathrm{GL}(V)$. Statement (2) is invariant under $\mathrm{GL}(V)$ (**Steps 1 and 7**), and λ in (3) is arbitrary, so the proposition does not depend on the isomorphism.

Let $\mathcal{E} := \Lambda^2\mathcal{U}^\vee \boxtimes \mathcal{S}^\vee$, a vector bundle of rank 9 on $\mathrm{Gr}(3,V) \times \mathbb{B}$, so that $\mathcal{X}$ is the zero scheme of a section of $\mathcal{E}$ and $\mathcal{K}^\bullet = \mathcal{K}^\bullet(\mathcal{E}^\vee)$. All vector bundles below carry their natural $\mathrm{GL}(V)$-linearizations, and $\det V = \Lambda^7 V$ is viewed as the trivial line bundle with $\mathrm{GL}(V)$ acting by the determinant.

The proof is long, so we divide it into several steps.

Step 1. In this step we describe $\mathcal{X}$, and we show that $\mathbb{B}^{(3)}$ is a non-empty open subset and that p is flat and Gorenstein.

Let $\mathcal{K}$ be the kernel of the surjective morphism of vector bundles $\Lambda^2 V^\vee \otimes \mathcal{O}_{\mathrm{Gr}(3,V)} \to \Lambda^2\mathcal{U}^\vee$ given by restriction of alternating forms, so that $\operatorname{rank} \mathcal{K} = 21 - 3 = 18$. The section of $\mathcal{E}$ defining

$\mathcal{X}$ is the composite $\mathcal{S} \to \Lambda^2 V^\vee \otimes \mathcal{O} \to \Lambda^2 \mathcal{U}^\vee$, so $\mathcal{X}$ is the scheme of pairs (U_0, Π) with $\Pi \subset \mathcal{K}_{U_0}$. Thus

$$\mathcal{X} = \mathrm{Gr}_{\mathrm{Gr}(3,V)}(3, \mathcal{K})$$

is a Grassmann bundle over $\mathrm{Gr}(3, V)$. Hence $\mathcal{X}$ is smooth and irreducible, and

$$\dim \mathcal{X} = 12 + 3(18 - 3) = 57, \qquad \dim \mathbb{B} = 3(21 - 3) = 54.$$

In particular $\operatorname{codim}(\mathcal{X}, \mathrm{Gr}(3, V) \times \mathbb{B}) = 66 - 57 = 9 = \operatorname{rank} \mathcal{E}$.

For $\Pi \in \mathbb{B}$, the scheme X_Π is the fiber of $q|_{\mathcal{X}}$ over Π, the zero scheme of a section of the rank-nine bundle $\Lambda^2 \mathcal{U}^\vee \otimes \Pi^\vee$ on the twelve-dimensional smooth variety $\mathrm{Gr}(3, V)$. By Krull's height theorem [Mat89, Theorem 13.5], any irreducible component of X_Π is of dimension at least 3. Let $W \subset \mathbb{B}$ be the set of Π with $X_\Pi = \emptyset$ or $\dim X_\Pi \leq 3$. The morphism $\mathcal{X} \to q(\mathcal{X})$ is a dominant morphism of integral schemes, since $\mathcal{X}$ is a smooth irreducible Grassmann bundle. Thus the set of points of $\mathcal{X}$ lying on an irreducible component of dimension at least 4 of their fiber is closed by upper semicontinuity of fiber dimension [Har77, Exercise II.3.22(d)]. Since $q|_{\mathcal{X}}$ is proper, its image in $\mathbb{B}$ is closed, so W is open. By Definition-Theorem 2.20, $X_{\Pi_X} \cong X$ for any smooth prime Fano threefold X of genus 12, so $\Pi_X \in W$. Hence the general fiber of $\mathcal{X} \to q(\mathcal{X})$ is of dimension at most 3, and since any irreducible component Z of the fiber over Π_X satisfies $\dim Z \geq \dim \mathcal{X} - \dim q(\mathcal{X})$ [Har77, Exercise II.3.22(b)], we have

$$\dim q(\mathcal{X}) \geq \dim \mathcal{X} - 3 = 54 = \dim \mathbb{B}.$$

Since $q(\mathcal{X})$ is closed and $\mathbb{B}$ is irreducible, $q(\mathcal{X}) = \mathbb{B}$. Thus $X_\Pi \neq \emptyset$ for any $\Pi \in \mathbb{B}$, and $\mathbb{B}^{(3)} = W$ is a non-empty open subset. It is $\mathrm{GL}(V)$-invariant, since $X_{g\Pi} = g \cdot X_\Pi$ for $g \in \mathrm{GL}(V)$.

Let $\Pi \in \mathbb{B}^{(3)}$ and $U_0 \in X_\Pi$. In the twelve-dimensional regular local ring $\mathcal{O}_{\mathrm{Gr}(3,V),U_0}$, the scheme X_Π is defined by nine elements generating an ideal of height 9, so the nine elements form a regular sequence [Mat89, Theorem 17.4(i) and (iii)]. Thus X_Π is a local complete intersection of pure dimension 3, and in particular X_Π is Gorenstein [Mat89, Theorems 21.2(ii) and 21.3]. By [Mat89, Theorem 17.4(i) and (iii)], since $\mathcal{X}|_{\mathbb{B}^{(3)}}$ is of codimension 9 in the smooth variety $\mathrm{Gr}(3, V) \times \mathbb{B}^{(3)}$, the section of $\mathcal{E}$ is regular on $\mathrm{Gr}(3, V) \times \mathbb{B}^{(3)}$. The varieties $\mathcal{X}|_{\mathbb{B}^{(3)}}$ and $\mathbb{B}^{(3)}$ are smooth, and all fibers of p are of dimension $3 = 57 - 54$. Hence p is flat by [Mat89, Theorem 23.1]. Since the fibers of p are Gorenstein, p is a Gorenstein morphism.

Step 2. In this step we prove that $\mathcal{L}$ is the relative anti-canonical bundle of p, compute the weights of the scalar matrices, and compute the Hilbert polynomial of the fibers of p.

Let $\mathcal{Q} := (V \otimes \mathcal{O}_{\mathrm{Gr}(3,V)})/\mathcal{U}$. The tangent bundle of $\mathrm{Gr}(3, V)$ is $\mathrm{GL}(V)$-equivariantly isomorphic to $\mathcal{U}^\vee \otimes \mathcal{Q}$ [Ful98, Appendix B.5.8], and $\det \mathcal{Q} \cong \det V \otimes \mathcal{H}$, where $\mathcal{H} = \det \mathcal{U}^\vee$, that is, $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ with the natural $\mathrm{GL}(V)$-linearization. Since $\operatorname{rank} \mathcal{U}^\vee = 3$ and $\operatorname{rank} \mathcal{Q} = 4$, we have $\det(\mathcal{U}^\vee \otimes \mathcal{Q}) \cong \mathcal{H}^4 \otimes (\det V \otimes \mathcal{H})^3$, that is,

$$\omega_{\mathrm{Gr}(3,V)} \cong \mathcal{H}^{-7} \otimes (\det V)^{-3}.$$

Since $\operatorname{rank} \mathcal{U} = 3$, we have $\det \Lambda^2 \mathcal{U}^\vee \cong \mathcal{H}^2$, so $\det \mathcal{E} \cong \mathcal{H}^6 \boxtimes \mathcal{A}^3$. The normal bundle of $\mathcal{X}|_{\mathbb{B}^{(3)}}$ in $\mathrm{Gr}(3, V) \times \mathbb{B}^{(3)}$ is $\mathcal{E}|_{\mathcal{X}|_{\mathbb{B}^{(3)}}}$, so the adjunction formula [Har77, Proposition II.8.20 and Exercise II.8.3] implies

$$\omega_{\mathcal{X}|_{\mathbb{B}^{(3)}}/\mathbb{B}^{(3)}} \cong (\omega_{\mathrm{Gr}(3,V)} \otimes \det \mathcal{E})|_{\mathcal{X}|_{\mathbb{B}^{(3)}}} \cong \left(\mathcal{H}^{-1} \boxtimes \mathcal{A}^3 \otimes (\det V)^{-3}\right)|_{\mathcal{X}|_{\mathbb{B}^{(3)}}} = \mathcal{L}^{-1}.$$

For $\Pi \in \mathbb{B}^{(3)}$ the restriction $\mathcal{L}|_{X_\Pi} \cong \mathcal{H}|_{X_\Pi}$ is the Plücker line bundle, which is ample. Hence $\mathcal{L}$ is p-ample.

For $c \in \mathbb{C}^*$, the scalar matrix $c \cdot \mathrm{id}_V$ acts trivially on $\mathrm{Gr}(3, V)$ and on $\mathbb{B}$, and it acts on the fibers of $\mathcal{U}$, $\Lambda^2 \mathcal{U}^\vee$, and $\mathcal{S}^\vee$ by c, c^{-2}, and c^2, respectively. Thus $c \cdot \mathrm{id}_V$ acts on the fibers of $\mathcal{H}$, $\mathcal{A}$, and $\det V$ by c^{-3}, c^6, and c^7, respectively, and it acts trivially on $\mathcal{E}$ and on $\widetilde{\mathcal{L}}$, since

$$-3 - 3 \cdot 6 + 3 \cdot 7 = 0.$$

Since p is flat, $\mathbb{B}^{(3)}$ is integral (an open subset of $\mathbb{B}$), and $\mathcal{L}|_{X_\Pi} \cong \mathcal{H}|_{X_\Pi}$ is the restriction of $\mathcal{O}(1)$ under the Plücker embedding of $\mathrm{Gr}(3,V)$, the Hilbert polynomial $\chi(X_\Pi, \mathcal{L}^m)$ does not depend on $\Pi \in \mathbb{B}^{(3)}$ [Har77, Theorem III.9.9]. By Definition-Theorem 2.20, $\mathcal{L}|_{X_{\Pi_X}} = \mathcal{H}|_{X_{\Pi_X}}$ corresponds to $-K_X$ under $X_{\Pi_X} \cong X$, and $(-K_X)^3 = 22$. By Riemann–Roch [Har77, Appendix A, Exercise 6.7],

$$\chi(X_\Pi, \mathcal{L}^m) = \chi(X, -mK_X) = \frac{22}{6}m^3 + \frac{22}{4}m^2 + O(m) = \frac{11}{3}m^3 + \frac{11}{2}m^2 + O(m) \tag{16.2}$$

for any $\Pi \in \mathbb{B}^{(3)}$. In particular $(\mathcal{L}|_{X_\Pi})^3 = 22$, and the coefficients $a_0 = 11/3$ and $a_1 = 11/2$ of m^3 and m^2 satisfy $2a_1/a_0 = 3$. Forgetting the linearizations, the formula for $\omega_{\mathcal{X}|_{\mathbb{B}^{(3)}}/\mathbb{B}^{(3)}}$ over Ω is Proposition 16.5(2).

Step 3. In this step we prove that $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}|_{\mathbb{B}^{(3)}} \cong \lambda_{\mathrm{CM}}$.

The complex $\mathcal{K}^\bullet$ has the term $\Lambda^j\mathcal{E}^\vee$ in degree $-j$ for $0 \le j \le 9$. It is a $\mathrm{GL}(V)$-equivariant complex of vector bundles. Since q is smooth and projective, $Rq_*(\mathcal{K}^\bullet \otimes \widetilde{\mathcal{L}}^m)$ is a perfect complex on $\mathbb{B}$, and $\mathcal{D}_m$ is a $\mathrm{GL}(V)$-linearized line bundle on $\mathbb{B}$ for any $m \in \mathbb{Z}$. By **Step 1**, the section of $\mathcal{E}$ is regular on $\mathrm{Gr}(3,V) \times \mathbb{B}^{(3)}$, so $\mathcal{K}^\bullet$ restricts to a resolution of $\mathcal{O}_{\mathcal{X}|_{\mathbb{B}^{(3)}}}$ [Mat89, Theorem 16.5(i)]. Hence, $\mathrm{GL}(V)$-equivariantly,

$$\mathcal{D}_m|_{\mathbb{B}^{(3)}} \cong \det Rp_*(\mathcal{L}^m) \cong \bigotimes_{i=0}^{4} \mathcal{M}_i^{\binom{m}{i}}$$

for any $m \in \mathbb{Z}$, where the second isomorphism is the Knudsen–Mumford expansion [KM76, Theorem 4], [FR06, Theorem 2.1 and equation (2.2)]. Let $\mathcal{M}_k^{\mathrm{Kos}}$ for $k \in \{3,4\}$ be as in the statement. For integers $i, k \ge 0$, the k-th forward difference of $m \to \binom{m}{i}$ at $m = 0$ is

$$\sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j}\binom{j}{i} = \binom{0}{i-k} = \begin{cases} 1 & \text{if } i = k, \\ 0 & \text{if } i \ne k. \end{cases}$$

Thus $\mathcal{M}_k^{\mathrm{Kos}}|_{\mathbb{B}^{(3)}} \cong \mathcal{M}_k$ for $k \in \{3,4\}$, and $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}|_{\mathbb{B}^{(3)}} \cong \lambda_{\mathrm{CM}}$.

Step 4. In this step we compute $c_1(\lambda_{\mathrm{CM}}^{\mathrm{Kos}})$ in terms of a push-forward from $\mathrm{Gr}(3,V) \times \mathbb{B}$.

We work in the rational Chow rings, where $c_1(\det V) = 0$, and we suppress pullbacks. Let $h := c_1(\mathcal{H})$ and $l := c_1(\widetilde{\mathcal{L}}) = h - 3c_1(\mathcal{A})$. By [Ful98, Example 3.2.5],

$$\mathrm{ch}(\mathcal{K}^\bullet) = \sum_{j=0}^{9} (-1)^j \,\mathrm{ch}(\Lambda^j\mathcal{E}^\vee) = c_9(\mathcal{E})\,\mathrm{td}(\mathcal{E})^{-1}.$$

The relative tangent bundle of q is the pullback of the tangent bundle $T_{\mathrm{Gr}(3,V)}$, so the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2] implies

$$\mathrm{ch}\left(Rq_*(\mathcal{K}^\bullet \otimes \widetilde{\mathcal{L}}^m)\right) = q_*\left(e^{ml} c_9(\mathcal{E})\,\mathrm{td}(T_{\mathrm{Gr}(3,V)})\,\mathrm{td}(\mathcal{E})^{-1}\right).$$

By **Step 2**, $c_1(T_{\mathrm{Gr}(3,V)}) - c_1(\mathcal{E}) = 7h - (6h + 3c_1(\mathcal{A})) = l$, so

$$\mathrm{td}(T_{\mathrm{Gr}(3,V)})\,\mathrm{td}(\mathcal{E})^{-1} = 1 + \frac{l}{2} + (\text{classes of degree at least } 2).$$

The class $c_1(\mathcal{D}_m)$ is the degree-one part of the left-hand side, that is, the push-forward of the degree-13 part of $e^{ml} c_9(\mathcal{E})\,\mathrm{td}(T_{\mathrm{Gr}(3,V)})\,\mathrm{td}(\mathcal{E})^{-1}$. Since $c_9(\mathcal{E})$ has degree 9, letting

$$\gamma := q_*\left(l^4 c_9(\mathcal{E})\right) \in CH^1(\mathbb{B})_{\mathbb{Q}},$$

we obtain, for any $m \in \mathbb{Z}$,

$$c_1(\mathcal{D}_m) = \frac{\gamma}{24}m^4 + \frac{\gamma}{12}m^3 + (\text{terms of degree at most } 2 \text{ in } m).$$

Let $P(m)$ be the right-hand side, a polynomial of degree at most 4 in m. Then $c_1(\mathcal{M}_k^{\mathrm{Kos}})$ is the k-th forward difference of P at $m=0$, and the Newton interpolation formula implies

$$P(m)=\sum_{k=0}^{4}\Big(\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}P(j)\Big)\binom{m}{k},$$

where the inner sums for $k=3,4$ are $c_1(\mathcal{M}_3^{\mathrm{Kos}})$ and $c_1(\mathcal{M}_4^{\mathrm{Kos}})$. Since $\binom{m}{4}=\frac{1}{24}(m^4-6m^3)+O(m^2)$ and $\binom{m}{3}=\frac{1}{6}m^3+O(m^2)$, comparing the coefficients of m^4 and m^3 implies

$$\frac{c_1(\mathcal{M}_4^{\mathrm{Kos}})}{24}=\frac{\gamma}{24},\qquad -\frac{c_1(\mathcal{M}_4^{\mathrm{Kos}})}{4}+\frac{c_1(\mathcal{M}_3^{\mathrm{Kos}})}{6}=\frac{\gamma}{12}.$$

Thus $c_1(\mathcal{M}_4^{\mathrm{Kos}})=\gamma$, $c_1(\mathcal{M}_3^{\mathrm{Kos}})=2\gamma$, and

$$c_1(\lambda_{\mathrm{CM}}^{\mathrm{Kos}})=15\gamma-8\cdot 2\gamma=-\gamma. \tag{16.3}$$

On $\mathbb{B}^{(3)}$, this is the formula for the first Chern class of the CM line bundle [PT06, Definition 1 and equation (1.3)], in the form of [FR06, Definition 2.2 and equation (2.7)], specialized to the relative anti-canonical polarization.

Step 5. In this step we compute $\gamma=-175c_1(\mathcal{A})$ by two integrals on $\mathrm{Gr}(3,V)$.

Let w_1, w_2, and w_3 be the Chern roots of $\mathcal{S}^\vee$, so that $w_1+w_2+w_3=c_1(\mathcal{A})$. For each j, the product of $(r+w_j)$ over the Chern roots r of $\Lambda^2\mathcal{U}^\vee$ equals $c_3(\Lambda^2\mathcal{U}^\vee)+w_jc_2(\Lambda^2\mathcal{U}^\vee)+w_j^2c_1(\Lambda^2\mathcal{U}^\vee)+w_j^3$. Taking the product over $j=1,2,3$, we obtain

$$\begin{aligned}c_9(\mathcal{E})&=c_3(\Lambda^2\mathcal{U}^\vee)^3+c_1(\mathcal{A})c_2(\Lambda^2\mathcal{U}^\vee)c_3(\Lambda^2\mathcal{U}^\vee)^2\\&\quad+(\text{terms of degree at least 2 in } w_1,w_2,w_3),\end{aligned}$$

and

$$l^4=h^4-12h^3c_1(\mathcal{A})+(\text{terms of degree at least 2 in } c_1(\mathcal{A})).$$

The class $l^4c_9(\mathcal{E})$ is a polynomial in Chern classes pulled back from $\mathrm{Gr}(3,V)$ and $\mathbb{B}$, the pull-back product of $\alpha\in CH^*(\mathrm{Gr}(3,V))$ and $\beta\in CH^*(\mathbb{B})$ is the exterior product $\alpha\times\beta$, and $q_*(\alpha\times\beta)=(\int_{\mathrm{Gr}(3,V)}\alpha)\beta$ [Ful98, Proposition 1.10(b)(i) and Example 8.3.7]. A term of $l^4c_9(\mathcal{E})$ contributes to $CH^1(\mathbb{B})$ only if it has degree 12 in the classes from $\mathrm{Gr}(3,V)$ and degree 1 in the classes from $\mathbb{B}$. Thus

$$\gamma=\Big(\int_{\mathrm{Gr}(3,V)}h^4c_2(\Lambda^2\mathcal{U}^\vee)c_3(\Lambda^2\mathcal{U}^\vee)^2-12\int_{\mathrm{Gr}(3,V)}h^3c_3(\Lambda^2\mathcal{U}^\vee)^3\Big)c_1(\mathcal{A}). \tag{16.4}$$

By Lemma 16.6, $\gamma=(89-12\cdot 22)c_1(\mathcal{A})=-175c_1(\mathcal{A})$. Thus, by (16.3),

$$c_1(\lambda_{\mathrm{CM}}^{\mathrm{Kos}})=175c_1(\mathcal{A}).$$

Step 6. In this step we determine the $\mathrm{GL}(V)$-linearization of $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}$ and conclude the proof of (1).

Fix a subspace $N\subset\Lambda^2V^\vee$ of dimension 18. The Schubert divisor $\{\Pi:\Pi\cap N\neq 0\}$ is an irreducible hyperplane section of $\mathbb{B}$ in the Plücker embedding, and its complement is an affine space of dimension 54. Since $\mathbb{B}$ is smooth, $\mathrm{Pic}(\mathbb{B})=\mathbb{Z}[\mathcal{A}]$ [Har77, Proposition II.6.2, Proposition II.6.5(c), and Corollary II.6.16], so **Step 5** implies $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}\cong\mathcal{A}^{175}$ as line bundles. Two $\mathrm{GL}(V)$-linearizations of a line bundle on the projective variety $\mathbb{B}$ differ by a character of $\mathrm{GL}(V)$, and the characters of $\mathrm{GL}(V)$ are the integral powers of det. Thus there exists an integer η such that

$$\lambda_{\mathrm{CM}}^{\mathrm{Kos}}\cong\mathcal{A}^{175}\otimes(\det V)^\eta$$

as $\mathrm{GL}(V)$-linearized line bundles. By **Step 2**, the scalar matrices act trivially on $\mathrm{Gr}(3,V)\times\mathbb{B}$, on $\mathcal{E}$, and on $\widetilde{\mathcal{L}}$. Hence they act trivially on any term of $\mathcal{K}^\bullet\otimes\widetilde{\mathcal{L}}^m$, on $Rq_*(\mathcal{K}^\bullet\otimes\widetilde{\mathcal{L}}^m)$, on $\mathcal{D}_m$, and on $\lambda_{\mathrm{CM}}^{\mathrm{Kos}}$. On the other hand, $c\cdot\mathrm{id}_V$ acts on the fibers of $\mathcal{A}^{175}\otimes(\det V)^\eta$ by $c^{6\cdot 175+7\eta}$. Thus $6\cdot 175+7\eta=0$, that is, $\eta=-150$. Together with **Steps 1–3**, this proves (1).

Step 7. In this step we prove (2).

By Definition-Proposition 16.2, D_{Pf} is a non-zero irreducible divisor. It is $\mathrm{GL}(V)$-invariant, since $g \in \mathrm{GL}(V)$ preserves the rank of alternating forms, so $g \cdot Z_4 = Z_4$. Let X be a smooth prime Fano threefold of genus 12. By Definition-Theorem 2.20, Π_X is non-degenerate and $X_{\Pi_X} \cong X$. Thus $\Pi_X \in \mathbb{B}^{\mathrm{nd}} = \mathbb{B}\backslash D_{\mathrm{Pf}}$ and $\dim X_{\Pi_X} = 3$, that is, $\Pi_X \in \mathbb{B}^{(3)}\backslash D_{\mathrm{Pf}}$. By Definition-Proposition 16.2, Π_X is GIT-semistable for $\mathcal{A}$. The divisor D_{Pf} is of the type studied by the incidence method for Grassmannian GIT in [GLT14, Section 2.7].

Step 8. In this step we prove that $\mathcal{X}_\lambda \to \mathbb{A}^1$ is flat, polarized by $\mathcal{L}_\lambda$, and isotrivial over $\mathbb{A}^1\backslash\{0\}$, and that $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a test configuration of X if $\mathcal{X}_\lambda$ is normal.

Let $\varphi : \mathbb{A}^1 \to \mathbb{B}^{(3)}$ be the morphism $t \to \lambda(t) \cdot \Pi_X$ for $t \neq 0$ and $0 \to \Pi_0$. For $s \in \mathbb{G}_m$ we have $\lambda(s) \cdot \Pi_0 = \lim_{t\to 0} \lambda(st) \cdot \Pi_X = \Pi_0$, so Π_0 is fixed by λ. Moreover $\varphi(st) = \lambda(s) \cdot \varphi(t)$ for $s, t \in \mathbb{G}_m$, hence for $s \in \mathbb{G}_m$ and $t \in \mathbb{A}^1$, since $\mathbb{B}$ is separated. Thus $\mathbb{G}_m$ acts on

$$\mathcal{X}_\lambda = \mathcal{X} \times_{\mathbb{B}} \mathbb{A}^1 \subset \mathrm{Gr}(3, V) \times \mathbb{A}^1$$

by $s \cdot (U_0, t) = (\lambda(s)U_0, st)$, covering the standard action on $\mathbb{A}^1$, and the $\mathrm{GL}(V)$-linearization of $\widetilde{\mathcal{L}}$ induces a $\mathbb{G}_m$-linearization of $\mathcal{L}_\lambda$. By **Step 1** and **Step 2**, $\mathcal{X}_\lambda \to \mathbb{A}^1$ is flat and projective, and $\mathcal{L}_\lambda$ is relatively ample. The morphism

$$X_{\Pi_X} \times \mathbb{G}_m \to \mathcal{X}_\lambda|_{\mathbb{G}_m}, \qquad (U_0, t) \to (\lambda(t)U_0, t), \tag{16.5}$$

is a $\mathbb{G}_m$-equivariant isomorphism, where $\mathbb{G}_m$ acts trivially on X_{Π_X}. The linearization of $\widetilde{\mathcal{L}}$ maps the fiber at U_0 to the fiber at $\lambda(t)U_0$. Thus the pullback of $\mathcal{L}_\lambda$ under the isomorphism (16.5) is the pullback of $\mathcal{L}|_{X_{\Pi_X}}$ with the trivial $\mathbb{G}_m$-action. Since $X_{\Pi_X} \cong X$ and $\mathcal{L}|_{X_{\Pi_X}}$ corresponds to $-K_X$ (**Step 2**), the family $\mathcal{X}_\lambda$ is isotrivial over $\mathbb{A}^1\backslash\{0\}$. If $\mathcal{X}_\lambda$ is normal, then $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ satisfies all the conditions of Definition 2.4, since X is smooth.

Step 9. In this step we prove the formula for $\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$.

For a one-dimensional vector space ℓ on which $\mathbb{G}_m$ acts through λ, the weight of ℓ under λ is the integer w with $\lambda(t) \cdot v = t^w v$ for $v \in \ell$. The total weight $b_0 m^4 + b_1 m^3 + O(m^2)$ is the weight of $\det H^0(X_{\Pi_0}, \mathcal{L}^m)$ under λ. The central fiber of $\mathcal{X}_\lambda$ is X_{Π_0}, so (16.2) implies $a_0 = 11/3$ and $a_1 = 11/2$.

Since Π_0 is fixed by λ, the group $\mathbb{G}_m$ acts through λ on the fiber at Π_0 of any $\mathrm{GL}(V)$-linearized line bundle on $\mathbb{B}^{(3)}$. For $0 \leq i \leq 4$, let ν_i be the weight of the fiber of $\mathcal{M}_i$ at Π_0 under λ. For $m \gg 0$ we have $H^j(X_{\Pi_0}, \mathcal{L}^m) = 0$ for $j > 0$. By semicontinuity [Har77, Theorem III.12.8], $H^j(X_\Pi, \mathcal{L}^m) = 0$ for $j > 0$ and Π in an open neighborhood U_m of Π_0 in $\mathbb{B}^{(3)}$, so $h^0(X_\Pi, \mathcal{L}^m) = \chi(X_\Pi, \mathcal{L}^m)$ is constant on U_m. By [Har77, Corollary III.12.9], $R^j p_* \mathcal{L}^m = 0$ for $j > 0$ on U_m, and $p_* \mathcal{L}^m$ is locally free on U_m with fiber $H^0(X_{\Pi_0}, \mathcal{L}^m)$ at Π_0. Hence $\det Rp_*(\mathcal{L}^m)|_{U_m} = \det p_* \mathcal{L}^m$ has fiber $\det H^0(X_{\Pi_0}, \mathcal{L}^m)$ at Π_0, λ-equivariantly. By the Knudsen–Mumford expansion [KM76, Theorem 4], [FR06, Theorem 2.1 and equation (2.2)], we obtain

$$b_0 m^4 + b_1 m^3 + O(m^2) = \sum_{i=0}^{4} \nu_i \binom{m}{i}$$

for $m \gg 0$. Comparing the coefficients of m^4 and m^3, we obtain $\nu_4 = 24b_0$ and $-\frac{\nu_4}{4} + \frac{\nu_3}{6} = b_1$, that is,

$$\nu_4 = 24b_0, \qquad \nu_3 = 6b_1 + 36b_0.$$

Thus the weight of the fiber of λ_{CM} at Π_0 under λ is

$$15\nu_4 - 8\nu_3 = 72b_0 - 48b_1.$$

On the other hand, since $a_0 = 11/3$ and $a_1 = 11/2$,

$$\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) = \frac{2(a_1 b_0 - a_0 b_1)}{a_0^2} = \frac{9b_0 - 6b_1}{11},$$

so the weight of the fiber of λ_{CM} at Π_0 under λ is $88\,\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda)$. By (1), $\lambda_{\mathrm{CM}} \cong \mathcal{A}^{175}\otimes(\det V)^{-150}$ on $\mathbb{B}^{(3)}$, $\mathrm{GL}(V)$-equivariantly, so the weight of the fiber of λ_{CM} at Π_0 under λ is also $175\mu_{\mathcal{A}}-150d_\lambda$. Therefore

$$\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda) = \frac{175\mu_{\mathcal{A}} - 150d_\lambda}{88}. \tag{16.6}$$

The derivation of (16.6) uses only the flatness of $\mathcal{X}_\lambda$ and the vanishing of higher cohomology on the central fiber. Assume that $\mathcal{X}_\lambda$ is normal. The normalization of the closure of the graph of the $\mathbb{G}_m$-equivariant birational map $\mathcal{X}_\lambda \dashrightarrow X\times\mathbb{A}^1$ given by the inverse of (16.5) and $X_{\Pi_X}\cong X$ admits $\mathbb{G}_m$-equivariant projective birational morphisms to $\mathcal{X}_\lambda$ and to $X\times\mathbb{A}^1$, and pulling back $\mathcal{L}_\lambda^m$ to it does not change H^0, since $\mathcal{X}_\lambda$ is normal. By [BHJ17, Lemmas 2.13 and 2.14], for $m\gg 0$ the function $w(m)$ in [Fuj19, Proposition 2.12], defined by the filtration of the test configuration, is the total weight of $\mathbb{G}_m$ on $H^0(X_{\Pi_0},\mathcal{L}^m)$, where X_{Π_0} is the central fiber of $\mathcal{X}_\lambda$, with the convention $\lambda(t)\cdot v = t^w v$. Thus [Fuj19, Proposition 2.12(2)(iii)] computes the Donaldson–Futaki invariant of Definition 2.4 as $-2F_1$, where F_1 is the coefficient of m^{-1} in the expansion of $(b_0m^4+b_1m^3+O(m^2))/(m(a_0m^3+a_1m^2+O(m)))$. Since $F_1=(a_0b_1-a_1b_0)/a_0^2$, it coincides with $\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda)$.

Step 10. In this step we prove the sign of $\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda)$, and conclude the proof of (3).

Let $c_{\mathrm{Pf}}\in H^0(\mathbb{B},\mathcal{A}^{14})$ be the $\mathrm{SL}(V)$-invariant section with divisor D_{Pf} (Definition-Proposition 16.2). For $c\in\mathbb{C}^*$, the scalar matrix $c\cdot\mathrm{id}_V$ acts trivially on $\mathbb{B}$ and by c^6 on the fibers of $\mathcal{A}$ (**Step 2**), so it multiplies c_{Pf} by c^{84}. Since $\mathrm{GL}(V)$ is generated by $\mathrm{SL}(V)$ and the scalar matrices, the line $\mathbb{C}c_{\mathrm{Pf}}$ is $\mathrm{GL}(V)$-stable, and $\mathrm{GL}(V)$ acts on it by a character trivial on $\mathrm{SL}(V)$, that is, by $\det^k$ for an integer k with $7k=84$. Thus $k=12$, and c_{Pf} is a $\mathrm{GL}(V)$-invariant section of $\mathcal{A}^{14}\otimes(\det V)^{-12}$. In particular

$$\mathcal{O}_{\mathbb{B}}(D_{\mathrm{Pf}}) \cong \mathcal{A}^{14}\otimes(\det V)^{-12}$$

as $\mathrm{GL}(V)$-linearized line bundles, and the weight of the fiber of $\mathcal{A}^{14}\otimes(\det V)^{-12}$ at Π_0 under λ is $14\mu_{\mathcal{A}}-12d_\lambda$.

The pullback φ^*c_{Pf} is a $\mathbb{G}_m$-invariant section of the $\mathbb{G}_m$-linearized line bundle $\varphi^*(\mathcal{A}^{14}\otimes(\det V)^{-12})$ on $\mathbb{A}^1$. By **Step 7** and the $\mathrm{GL}(V)$-invariance of D_{Pf}, we have $\lambda(t)\cdot\Pi_X\notin D_{\mathrm{Pf}}$ for $t\neq 0$, so φ^*c_{Pf} does not vanish on $\mathbb{A}^1\backslash\{0\}$. Let $m_{\mathrm{Pf}}\geq 0$ be the order of vanishing of φ^*c_{Pf} at 0. Then $m_{\mathrm{Pf}}>0$ if and only if $\Pi_0\in D_{\mathrm{Pf}}$. Since $\mathrm{Pic}(\mathbb{A}^1)=0$, there exists a nowhere vanishing section e of $\varphi^*(\mathcal{A}^{14}\otimes(\det V)^{-12})$. Write $s\cdot e(t)=\chi(s,t)e(st)$ for $s\in\mathbb{G}_m$ and $t\in\mathbb{A}^1$, where χ is an invertible function on $\mathbb{G}_m\times\mathbb{A}^1$ with $\chi(1,t)=1$. Then $\chi(s,t)=s^w$ for an integer w, and evaluating at $t=0$ shows that $w=14\mu_{\mathcal{A}}-12d_\lambda$. Write $\varphi^*c_{\mathrm{Pf}}=r(t)e$ with $r\in\mathbb{C}[t]$, so that $r(t)=t^{m_{\mathrm{Pf}}}u(t)$ with $u(0)\neq 0$. The invariance of φ^*c_{Pf} means $r(st)=s^wr(t)$, so r is a non-zero multiple of t^w. Thus

$$14\mu_{\mathcal{A}}-12d_\lambda = m_{\mathrm{Pf}}\geq 0.$$

By (16.6),

$$\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda) = \frac{175\mu_{\mathcal{A}}-150d_\lambda}{88} = \frac{25}{2}\cdot\frac{14\mu_{\mathcal{A}}-12d_\lambda}{88} = \frac{25m_{\mathrm{Pf}}}{176}.$$

Hence $\mathrm{DF}(\mathcal{X}_\lambda,\mathcal{L}_\lambda)\geq 0$, with strict inequality if $\Pi_0\in D_{\mathrm{Pf}}$ and equality if $\Pi_0\notin D_{\mathrm{Pf}}$. Together with **Steps 8 and 9**, this proves (3). □

16.2. **Apolar quartics and lines.** In this subsection, we prove that the smooth members X_Π, $\Pi\in\Omega$, are exactly the varieties of apolar hexagons of non-degenerate plane quartics (Proposition 16.10), and that the Hilbert scheme of lines of X_Π is the Scorza quartic of f_Π, with scheme structure and in families over Ω (Theorem 16.11), and we compare the stabilizers of Π and $[f_\Pi]$ and the stability of Π, $[f_\Pi]$, and $[\mathrm{Sc}(f_\Pi)]$ (Theorem 16.12). We first fix notation for the Pfaffians of a net.

Let $\Pi\in\mathbb{B}^{\mathrm{nd}}$, identify $\mathrm{Sym}\,\Pi^\vee$ with $\mathbb{C}[\partial]$ by a basis of Π, and let $e_1,\dots,e_7$ be a basis of V with dual basis $e_1^\vee,\dots,e_7^\vee$. Let M_Π be the alternating matrix of Π in the basis $e_1,\dots,e_7$ and the basis of Π fixed above, whose entry $(M_\Pi)_{ij}\in\mathbb{C}[\partial]_1=\Pi^\vee$ is the linear function $\omega\to\omega(e_i,e_j)$,

and for $v, w \in V$ let $M_\Pi(v,w) := \sum_{i,j} e_i^\vee(v) e_j^\vee(w) (M_\Pi)_{ij}$. Let $\mathrm{Pf}_i \in \mathbb{C}[\partial]_3$ be the Pfaffian of the principal 6×6 submatrix of M_Π obtained by deleting the i-th row and column, and let $I_\Pi \subset \mathbb{C}[\partial]$ be the ideal generated by $\mathrm{Pf}_1, \dots, \mathrm{Pf}_7$. By Lemma 16.1(1), we regard the Pfaffian vector of M_Π as

$$p(M_\Pi) = \sum_{i=1}^{7} (-1)^{i+1} \mathrm{Pf}_i \, e_i \in V \otimes \mathbb{C}[\partial]_3, \qquad \lambda(p(M_\Pi)) := \sum_{i=1}^{7} (-1)^{i+1} \lambda(e_i) \, \mathrm{Pf}_i \quad \text{for } \lambda \in V^\vee.$$

By Lemma 16.1(2) and (4), $M_\Pi(v, p(M_\Pi)) = 0$ for any $v \in V$, and for any non-zero $\omega \in \Pi$, which has rank six, the vector $p(M_\Pi)(\omega)$ spans the kernel k_ω of ω, and the morphism $\kappa : \mathbb{P}(\Pi) \to \mathbb{P}(V)$, $[\omega] \to [k_\omega]$, is a closed immersion by [BKM26b, Lemma B.1].

Lemma 16.8. *Let $\Pi \in \mathbb{B}^{\mathrm{nd}}$, and use the notation above. Then the following hold.*

(1) The map $V^\vee \to (I_\Pi)_3$, $\lambda \to \lambda(p(M_\Pi))$, is an isomorphism.

(2) Let $\mu : \Pi^\vee \otimes V^\vee \to \mathbb{C}[\partial]_4$ be the linear map $l \otimes \lambda \to l \cdot \lambda(p(M_\Pi))$, and let $s : V \to \Pi^\vee \otimes V^\vee$ be the linear map $s(v) := \sum_j M_\Pi(v, e_j) \otimes e_j^\vee$. Then s is an isomorphism onto $\operatorname{Ker} \mu$.

Proof. (1) The cubics $\mathrm{Pf}_1, \dots, \mathrm{Pf}_7$ generate I_Π, so the map is surjective. By Lemma 16.3(1), $\dim (I_\Pi)_3 = \dim \mathbb{C}[\partial]_3 - 3 = 7 = \dim V^\vee$.

(2) For $v \in V$ we have $\mu(s(v)) = \sum_j M_\Pi(v, e_j) e_j^\vee(p(M_\Pi)) = M_\Pi(v, p(M_\Pi)) = 0$. If $s(v) = 0$, then $v \in k_\omega$ for any non-zero $\omega \in \Pi$. Since κ is injective, two such kernels are distinct lines, so $v = 0$. By (1), $\operatorname{Ker} \mu$ is the degree-four part of the module of syzygies of $\mathrm{Pf}_1, \dots, \mathrm{Pf}_7$. By the exactness of the Pfaffian complex of Lemma 16.3(1), the module of syzygies is the image of $\mathbb{C}[\partial](-4)^{\oplus 7}$, and the degree-four part $\mathbb{C}^7$ of $\mathbb{C}[\partial](-4)^{\oplus 7}$ maps injectively to it because $\mathbb{C}[\partial](-7)$ is zero in degree four. Thus $\dim \operatorname{Ker} \mu = 7 = \dim V$, and s is an isomorphism onto $\operatorname{Ker} \mu$. □

Lemma 16.9. *Let $g \in \mathbb{C}[x,y,z]_3$. Then the following hold.*

(1) If $c(g) \neq 0$ for any non-zero $c \in \mathbb{C}[\partial]_1$, then $\dim (g^\perp)_2 = 3$.

(2) If $c(g) \neq 0$ for any non-zero $c \in \mathbb{C}[\partial]_1$ and $\mathfrak{A}(g) = 0$, then there exists a 3×2 matrix $L = (L_{kj})$ with entries in $\mathbb{C}[\partial]_1$ with the following property. Let $L_{(k)}$ be the 2×2 submatrix of L obtained by deleting the k-th row, and let $q_k := (-1)^{k+1} \det L_{(k)}$ for $k = 1, 2, 3$, that is, (q_1, q_2, q_3) is the cross product of the first and the second column of L. Then q_1, q_2, q_3 form a basis of $(g^\perp)_2$, and the columns of L form a basis of the space of linear syzygies

$$\Big\{ (l_1, l_2, l_3) \in \mathbb{C}[\partial]_1^{\oplus 3} : \sum_k l_k q_k = 0 \Big\}.$$

(3) If $g = b(f)$ for a non-degenerate $f \in \mathbb{C}[x,y,z]_4$ and a non-zero $b \in \mathbb{C}[\partial]_1$, then $c(g) \neq 0$ for any non-zero $c \in \mathbb{C}[\partial]_1$.

Proof. Let $T := \mathbb{C}[\partial]$.

(1) The image of $T_2 \to \mathbb{C}[x,y,z]_1$, $q \to q(g)$, is annihilated by $c \in T_1$ if and only if $(qc)(g) = q(c(g)) = 0$ for any $q \in T_2$, that is, if and only if $c(g) = 0$. Hence the image is three-dimensional and $\dim (g^\perp)_2 = 6 - 3 = 3$.

(2) Since $(g^\perp)_1 = 0$ and $g \neq 0$, the Artinian Gorenstein algebra $T/g^\perp$ [IK99, Lemma 2.12] has Hilbert function $(1,3,3,1)$ by (1). By [BE77, Theorem 2.1], $g^\perp$ is minimally generated by the $2k+1$ principal $2k \times 2k$ Pfaffians of an alternating $(2k+1) \times (2k+1)$ matrix Υ with entries in $T_{\geq 1}$, and if $a_1, \dots, a_{2k+1}$ are the degrees of the principal Pfaffians of Υ, then the minimal free resolution of $T/g^\perp$ is

$$0 \to T(-6) \to \bigoplus_i T(a_i - 6) \xrightarrow{\Upsilon} \bigoplus_i T(-a_i) \to T,$$

so that Υ_{ij} is homogeneous of degree $6 - a_i - a_j$. Since $(g^\perp)_1 = 0$ and $(g^\perp)_2$ is three-dimensional, exactly three a_i are equal to 2 and the others are at least 3. Since $T_4 \subset g^\perp$, no minimal generator

has degree at least 5. The Hilbert series of $T/g^\perp$ is $(1+t)^3$, so comparing it with the resolution yields

$$1-\sum_i t^{a_i}+\sum_i t^{6-a_i}-t^6=(1-t^2)^3=1-3t^2+3t^4-t^6.$$

Thus the multiset of the a_i different from 2, which belong to $\{3,4\}$, is invariant under $a\to 6-a$, so all of them are equal to 3, and there are $2k-2$ of them. The entries Υ_{ij} with $a_i=a_j=3$ have degree 0, so they vanish by minimality. If $k\geq 3$, then deleting the row and the column of a quadratic generator leaves a $2k\times 2k$ alternating matrix with a zero principal block of size $2k-2>k$, whose Pfaffian is zero. This contradicts the fact that the deleted generator is non-zero. If $k=1$, then $g^\perp$ is generated by three quadrics and has height three, so $T/g^\perp$ is a complete intersection of three quadrics. This contradicts $\mathfrak{A}(g)=0$ by [Sch01, Lemma 3.1], since the zero locus of $\mathfrak{A}$ is the closure of the PGL_3-orbit of $[x^3+y^3+z^3]$ [Sch01, Section 3, pp. 304–305]. Hence $k=2$, and ordering the generators so that $a_1=a_2=a_3=2$ and $a_4=a_5=3$, we may write

$$\Upsilon=\begin{pmatrix}\Upsilon' & L\\ -L^t & 0\end{pmatrix},$$

where Υ' is a 3×3 alternating matrix of quadrics and L is a 3×2 matrix of linear forms. Let $\mathrm{Pf}_i(\Upsilon)$ be the principal Pfaffian of Υ obtained by deleting index i. In the resolution, the map $\bigoplus_i T(-a_i)\to T$ is given by the entries $p(\Upsilon)_i:=(-1)^{i+1}\mathrm{Pf}_i(\Upsilon)$ [BE77, Theorem 2.1]. For $k\leq 3$, since $\Upsilon_{45}=0$, we have $\mathrm{Pf}_k(\Upsilon)=-\det L_{(k)}$, so

$$q_k=(-1)^{k+1}\det L_{(k)}=-p(\Upsilon)_k\qquad(k=1,2,3).$$

Thus q_1,q_2,q_3 form a basis of $(g^\perp)_2$. The syzygies of degree three in the resolution form a two-dimensional space spanned by the last two rows $(-L^t,0)$ of Υ, and since $q_k=-p(\Upsilon)_k$, the rows $(-L^t,0)$ give the syzygies $\sum_k L_{kj}q_k=0$, $j=1,2$. A linear syzygy among q_1,q_2,q_3 is a syzygy of degree three of the generators of $g^\perp$. Hence the columns of L form a basis of the space of linear syzygies among q_1,q_2,q_3.

(3) If $g=b(f)$ and $c(g)=0$ for some non-zero $c\in T_1$, then the non-zero quadric cb belongs to $(f^\perp)_2=0$, which is impossible. □

The following proposition is essentially contained in [Muk92, Section 6, Theorem 11], [Sch01, Theorems 2.3 and 2.6, Corollary 2.7], [KPS18, Proposition B.4.1, Lemmas 2.3.3 and 2.3.4], and [BKM26b, Lemmas B.1–B.4, Proposition B.6]. See also [Muk04, Section 5]. We give a complete proof of (1)–(4), since Theorem 16.11 uses the scheme-theoretic form of (4).

Proposition 16.10. *Let $f\in\mathbb{C}[x,y,z]_4$ be non-degenerate. Then the following hold.*

(1) If $\mathrm{VPS}(f,6)$ is smooth, then $\mathrm{VPS}(f,6)$ is a smooth prime Fano threefold of genus 12 and $\mathrm{VSP}(f,6)=\mathrm{VPS}(f,6)$ as schemes.

(2) If $\mathrm{VSP}(f,6)$ is a smooth prime Fano threefold of genus 12, then $\mathrm{VSP}(f,6)=\mathrm{VPS}(f,6)$.

(3) Every smooth prime Fano threefold X of genus 12 is isomorphic to $\mathrm{VSP}(f_X,6)$.

(4) Assume that $X:=\mathrm{VPS}(f,6)$ is smooth, and let $\Pi\in\mathbb{B}^{\mathrm{nd}}$ with a basis $\omega_1,\omega_2,\omega_3$ in which f_Π is a non-zero multiple of f (Lemma 16.3(2)), and identify $\mathbb{P}(\mathbb{C}[x,y,z]_1)$ with $\mathbb{P}(\Pi)$ through the basis $\omega_1,\omega_2,\omega_3$. For $[\omega]\in\mathbb{P}(\Pi)$ let $k_\omega\subset V$ be the kernel of ω, let $\kappa:\mathbb{P}(\Pi)\to\mathbb{P}(V)$ be the morphism $[\omega]\to[k_\omega]$, and for $U\in X_\Pi$ let $\Gamma_U:=\kappa^{-1}(\mathbb{P}(U))$. Then $U\to\Gamma_U$ is an isomorphism from X_Π onto X. For any $[\omega]\in\mathbb{P}(\Pi)$ the scheme $\mathcal{C}_\omega:=\{U\in X_\Pi : k_\omega\subset U\}$ is a conic, and the morphism $[\omega]\to\mathcal{C}_\omega$ is an isomorphism from $\mathbb{P}(\Pi)$ onto the Hilbert scheme of conics of X_Π. The universal family $\{(U,[\omega]) : k_\omega\subset U\}\subset X_\Pi\times\mathbb{P}(\Pi)$ is finite and flat of degree six over X_Π and generically étale. In particular a general point of X_Π is contained in six distinct conics of X_Π.

Proof of Proposition 16.10. We first prove that $\mathrm{VPS}(f,6)$ is a smooth prime Fano threefold of genus 12 if it is smooth. By Lemma 16.3(2), there exists $\Pi'\in\mathbb{B}^{\mathrm{nd}}$ such that $f_{\Pi'}$ is a non-zero multiple of f in a basis of Π' and $\mathrm{VPS}(f,6)\cong X_{\Pi'}$. By [BKM26b, Proposition B.6], $X_{\Pi'}$ is a

three-dimensional local complete intersection scheme. If $\mathrm{VPS}(f,6)$ is smooth, then $X_{\Pi'}$ is smooth of dimension three, so $\Pi' \in \Omega$ by Proposition 16.5. Thus $\mathrm{VPS}(f,6) \cong X_{\Pi'}$ is a smooth prime Fano threefold of genus 12.

We next prove (4). Let Π be as in (4). Since f_Π and $f_{\Pi'}$ are both non-zero multiples of f, Lemma 16.3(3) implies $\Pi \in \mathrm{PGL}(V) \cdot \Pi'$. Hence $X_\Pi \cong X_{\Pi'} \cong X$, and X_Π is a smooth prime Fano threefold, in particular integral. We use the notation introduced before Lemma 16.8, so that κ is the closed immersion $[\omega] \to [p(M_\Pi)(\omega)]$. By Definition-Proposition 2.28, $f^\perp = f_\Pi^\perp = I_\Pi$.

Let $U \in X_\Pi$ be a closed point, and choose the basis of V so that $U = \mathrm{Span}\{e_5, e_6, e_7\}$. Since any form of Π vanishes on $\Lambda^2 U$,

$$M_\Pi = \begin{pmatrix} M' & \Psi \\ -\Psi^t & 0 \end{pmatrix},$$

where M' is an alternating 4×4 matrix and $\Psi = ((M_\Pi)_{ij})_{1 \le i \le 4 < j}$ is a 4×3 matrix with entries in $\mathbb{C}[\partial]_1$. For $i \le 4$, the principal 6×6 submatrix of M_Π obtained by deleting index i has a 3×3 zero block, so $\mathrm{Pf}_i = \pm \det \Psi_{(i)}$, where $\Psi_{(i)}$ is the 3×3 submatrix of Ψ formed by the rows of index different from i. The subscheme Γ_U is cut out by the pullbacks $e_i^\vee(p(M_\Pi)) = \pm \mathrm{Pf}_i$, $i \le 4$, of the linear equations of $\mathbb{P}(U)$. Hence the ideal of Γ_U is generated by

$$J_U := (\mathrm{Pf}_1, \dots, \mathrm{Pf}_4) = I_3(\Psi) \subset I_\Pi = f^\perp,$$

the ideal of maximal minors of Ψ. By Lemma 16.8(1), $(J_U)_3$ is the image of $U^\perp := \{\lambda \in V^\vee : \lambda|_U = 0\}$ under $\lambda \to \lambda(p(M_\Pi))$, and $\mathrm{Pf}_1, \dots, \mathrm{Pf}_4$ are linearly independent.

We show that J_U has height at least two. Suppose that Γ_U contains an integral curve $(r = 0)$ with $r \in \mathbb{C}[\partial]$ irreducible of degree d. Then r divides the four linearly independent cubics $\mathrm{Pf}_1, \dots, \mathrm{Pf}_4$. If $d \ge 2$, the cubics divisible by r form a space of dimension at most 3, which is impossible. If $d = 1$, write $\mathrm{Pf}_i = r\tilde{q}_i$. Then $\tilde{q}_i(r(f)) = (r\tilde{q}_i)(f) = 0$ for the four linearly independent quadrics $\tilde{q}_1, \dots, \tilde{q}_4$, which contradicts $\dim(r(f)^\perp)_2 = 3$ (Lemma 16.9(1) and (3)). Thus Γ_U is finite and J_U has height at least two. By the Hilbert–Burch theorem [Eis95, Theorem 20.15(b)], the complex

$$0 \to \mathbb{C}[\partial](-4)^{\oplus 3} \xrightarrow{\Psi} \mathbb{C}[\partial](-3)^{\oplus 4} \to \mathbb{C}[\partial] \to \mathbb{C}[\partial]/J_U \to 0,$$

where $\mathbb{C}[\partial](-3)^{\oplus 4} \to \mathbb{C}[\partial]$ is given by the signed maximal minors of Ψ, is exact. Hence $\mathbb{C}[\partial]/J_U$ has depth one, so J_U is saturated, and the Hilbert polynomial of Γ_U is

$$\binom{t+2}{2} - 4\binom{t-1}{2} + 3\binom{t-2}{2} = 6.$$

Thus Γ_U is a subscheme of length six of $\mathbb{P}(\Pi)$ whose homogeneous ideal J_U is contained in $f^\perp$, that is, $\Gamma_U \in \mathrm{VPS}(f,6)$. Moreover $(I_{\Gamma_U})_3 = (J_U)_3$ is the image of $U^\perp$, so U is determined by Γ_U.

Let $\mathcal{Z} := \{(U, [\omega]) : k_\omega \subset U\} \subset X_\Pi \times \mathbb{P}(\Pi)$, the preimage under $\mathrm{id}_{X_\Pi} \times \kappa$ of the incidence subscheme $\{(U, [k]) : k \subset U\}$ of $\mathrm{Gr}(3, V) \times \mathbb{P}(V)$. The fiber of $\mathcal{Z}$ over a closed point U is Γ_U, whose Hilbert polynomial is 6. Since X_Π is integral, $\mathcal{Z}$ is flat over X_Π [Har77, Theorem III.9.9], so $U \to \Gamma_U$ is a morphism $\phi : X_\Pi \to \mathrm{Hilb}^6(\mathbb{P}(\Pi))$, which factors through the closed subscheme $\mathrm{VPS}(f,6)$ because X_Π is reduced. By the previous paragraph, ϕ is injective on closed points. Hence ϕ is quasi-finite and proper, so finite, and $\phi(X_\Pi) = X$ since both are irreducible of dimension three. By generic smoothness [Har77, Corollary III.10.7], applied to the restriction of ϕ to the smooth locus of X_Π, there is a dense open subset of X over which ϕ is étale. Since ϕ is injective, its fibers there are reduced points, so the finite morphism ϕ has degree one and is birational. Since X is normal, $\phi_* \mathcal{O}_{X_\Pi} = \mathcal{O}_X$ (Zariski's Main Theorem [Har77, Corollary III.11.4] and its proof), so the finite morphism ϕ is an isomorphism.

For $[\omega] \in \mathbb{P}(\Pi)$, the fiber $\mathcal{C}_\omega$ of $\mathcal{Z}$ over $[\omega]$ is a conic [BKM26b, Lemma B.4]. Since $\mathbb{P}(\Pi)$ is integral and the Hilbert polynomial $2t + 1$ of $\mathcal{C}_\omega$ is constant, $\mathcal{Z}$ is flat over $\mathbb{P}(\Pi)$ [Har77, Theorem III.9.9], and $[\omega] \to \mathcal{C}_\omega$ is a morphism ψ from $\mathbb{P}(\Pi)$ to the Hilbert scheme of conics of X_Π, which is isomorphic to $\mathbb{P}^2$ [KPS18, Proposition B.4.1]. Suppose that $\mathcal{C}_\omega = \mathcal{C}_{\omega'}$ for distinct $[\omega], [\omega']$. Since κ is injective, $k_\omega + k_{\omega'}$ is a two-dimensional subspace A, and $\mathcal{C}_\omega$ is contained in

$\{U \in \mathrm{Gr}(3,V) : A \subset U\} \cong \mathbb{P}(V/A)$, which is a linear subspace in the Plücker embedding. For $U = A + \mathbb{C}u$, the conditions $\omega''(\Lambda^2 U) = 0$, $\omega'' \in \Pi$, are linear in u, so $X_\Pi \cap \mathbb{P}(V/A)$ is a linear subspace of $\mathbb{P}(V/A)$. It contains the conic $\mathcal{C}_\omega$, hence the plane spanned by $\mathcal{C}_\omega$, which contradicts the fact that X_Π contains no plane [BKM26b, Lemma B.3]. Thus ψ is injective on closed points, hence quasi-finite and proper, so finite. The image $\psi(\mathbb{P}(\Pi))$ is closed of dimension two in the integral surface $\mathbb{P}^2$, so ψ is surjective. Since $\mathbb{P}(\Pi)$ and $\mathbb{P}^2$ are smooth, generic smoothness [Har77, Corollary III.10.7] and Zariski's Main Theorem [Har77, Corollary III.11.4] show, as for ϕ, that ψ is an isomorphism.

By construction, $\mathcal{Z}$ is the universal family of ϕ and of ψ. It is finite and flat of degree six over X_Π. By [KPS18, Lemma 2.3.3 and the last paragraph of the proof of Lemma 2.3.4], a general conic of X_Π is smooth, so the generic fiber of $\mathcal{Z} \to \mathbb{P}(\Pi)$ is reduced. Since $\mathcal{Z}$ is flat over the integral scheme $\mathbb{P}(\Pi)$, non-zero local functions on $\mathbb{P}(\Pi)$ are nonzerodivisors on $\mathcal{Z}$, so the local rings of $\mathcal{Z}$ embed into localizations of the generic fiber, and $\mathcal{Z}$ is reduced. Hence the generic fiber of $\mathcal{Z} \to X_\Pi$ is a reduced finite scheme over a field of characteristic zero, and the finite flat morphism $\mathcal{Z} \to X_\Pi$ is étale over a dense open subset $X^\circ \subset X_\Pi$. For $U \in X^\circ$, Γ_U consists of six distinct reduced points $[\omega^{(1)}], \dots, [\omega^{(6)}]$, and U is contained in the six conics $\mathcal{C}_{\omega^{(1)}}, \dots, \mathcal{C}_{\omega^{(6)}}$, which are distinct since ψ is injective. This proves (4).

We finish the proof of (1). For $U \in X^\circ$, write $\Gamma_U = \{[\ell_1], \dots, [\ell_6]\}$ with $\ell_i \in \mathbb{C}[x,y,z]_1$. Since $I_{\Gamma_U} \subset f^\perp$, the apolarity lemma [IK99, Lemma 1.15] implies $f \in \mathbb{C}\ell_1^4 + \cdots + \mathbb{C}\ell_6^4$. Thus $\phi(X^\circ)$ is contained in the locus of polar hexagons, and $\mathrm{VSP}(f,6)$ contains the closure of the dense open subset $\phi(X^\circ)$ of the integral scheme X. Hence $\mathrm{VSP}(f,6) = X = \mathrm{VPS}(f,6)$ as schemes.

(2) Let $Y := \mathrm{VSP}(f,6)$ and let Π' be as in the first paragraph, so that Y is a closed integral subscheme of dimension three of $X_{\Pi'} \cong \mathrm{VPS}(f,6)$. By [BKM26b, Proposition B.6], $X_{\Pi'}$ is a local complete intersection, hence Cohen–Macaulay and of pure dimension three, and the Plücker degree of $X_{\Pi'}$ is 22. Thus Y is an irreducible component of $X_{\Pi'}$. Since $\mathrm{Pic}(Y) = \mathbb{Z}[-K_Y]$ and the Plücker line bundle $\mathcal{O}(1)$ is ample, $\mathcal{O}(1)|_Y \cong \mathcal{O}_Y(-dK_Y)$ for an integer $d \geq 1$. The Plücker degree of $X_{\Pi'}$ is the sum of the degrees of the irreducible components of $X_{\Pi'}$ weighted by their generic multiplicities, so

$$22d^3 = (-dK_Y)^3 \leq 22.$$

Hence $d = 1$, Y is the only irreducible component of $X_{\Pi'}$, and the generic multiplicity of $X_{\Pi'}$ along Y is one. A generically reduced Cohen–Macaulay scheme is reduced [Mat89, Section 23, p. 183], so $X_{\Pi'}$ is reduced, and hence $X_{\Pi'} = Y$. Thus $\mathrm{VSP}(f,6) = \mathrm{VPS}(f,6)$.

(3) By Definition-Theorem 2.20, $X \cong X_{\Pi_X}$ with $\Pi_X \in \Omega$. By Lemma 16.3(2), $\mathrm{VPS}(f_X,6) \cong X_{\Pi''}$ for some $\Pi'' \in \mathbb{B}^{\mathrm{nd}}$ such that $f_{\Pi''}$ is a non-zero multiple of f_X, and $\Pi'' \in \mathrm{PGL}(V) \cdot \Pi_X$ by Lemma 16.3(3). Hence $\mathrm{VPS}(f_X,6) \cong X$ is smooth, and $\mathrm{VSP}(f_X,6) = \mathrm{VPS}(f_X,6) \cong X$ by (1). □

Theorem 19.1 reads off the singularities of the Hilbert scheme of lines at special lines, so a statement for a general member would not suffice. For a general X, an isomorphism $\Sigma(X) \cong Q_X$ is [Sch01, Theorem 6.1, (c)⇔(d)]. By [KPS18, Remark 5.4.8], no complete proof of the isomorphism of Theorem 16.11(2) for every smooth X has been published.

Theorem 16.11. *Let X be a smooth prime Fano threefold of genus 12, let V and $\Pi_X \subset \Lambda^2 V^\vee$ be as in Definition-Theorem 2.20, and identify X with $X_{\Pi_X} \subset \mathrm{Gr}(3,V)$ by Definition-Theorem 2.20. Let $\mathcal{S}$ be the tautological subbundle on $\mathbb{B}$. Then the following hold.*

(1) $\mathrm{Cat}_2(f_X)$ *is invertible and* $\mathrm{Sc}(f_X) \neq 0$.

(2) Let $A \subset B \subset V$ be subspaces of dimensions 2 and 4 such that the line $\{U \in \mathrm{Gr}(3,V) : A \subset U \subset B\}$ is contained in X. Then the image of $\Lambda^2 B$ under the dual $\Lambda^2 V \to \Pi_X^\vee$ of the inclusion $\Pi_X \subset \Lambda^2 V^\vee$ is one-dimensional. Sending such a line to the point of $\mathbb{P}(\Pi_X^\vee)$ given by the image of $\Lambda^2 B$ defines a closed immersion of the scheme $\Sigma(X)$ into $\mathbb{P}(\Pi_X^\vee)$ with image Q_X. This is the canonical embedding of the Gorenstein curve $\Sigma(X)$, so the plane $\mathbb{P}(\Pi_X^\vee)$ and the projective equivalence class of Q_X are intrinsic to X.

(3) The Hilbert scheme of conics of X is $\mathbb{P}(\Pi_X)$, the plane dual to $\mathbb{P}(\Pi_X^\vee)$.
(4) For any non-degenerate $f \in \mathbb{C}[x,y,z]_4$ such that $\mathrm{VPS}(f,6)$ is a smooth prime Fano threefold, $\Sigma(\mathrm{VPS}(f,6)) \cong (\mathrm{Sc}(f)=0)$ in the plane dual to the plane of conics.
(5) Over Ω, the isomorphisms in (2) and (4) are isomorphisms of families between the relative Hilbert scheme of lines of $\{X_\Pi\}_{\Pi\in\Omega}$ and the relative Scorza quartic $\{(\mathrm{Sc}(f_\Pi)=0)\}_{\Pi\in\Omega} \subset \mathbb{P}(\mathcal{S}^\vee|_\Omega)$, and they commute with arbitrary base change.

Proof of Theorem 16.11. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1). By Definition-Proposition 2.28, f_X is non-degenerate, that is, $\mathrm{Cat}_2(f_X)$ is invertible. For any non-degenerate quartic f, $\mathrm{Sc}(f) \neq 0$ by [DK93, Definition (6.2.1) and Corollary (6.6.3)(iv)], as recalled in [Sch01, Section 3, p. 305].

Step 2. In this step we show that $\Sigma(\mathcal{X}_\Omega/\Omega)$ and a relative Scorza quartic $\mathcal{Q}_\Omega \subset \mathbb{P}(\mathcal{S}^\vee|_\Omega)$ are flat over Ω with reduced total spaces.

Let $\mathcal{X}_\Omega \subset \mathrm{Gr}(3,V)\times\Omega$ be as in Proposition 16.5, let $\Sigma(\mathcal{X}_\Omega/\Omega)$ be the relative Hilbert scheme of lines of $\mathcal{X}_\Omega \to \Omega$ with respect to the Plücker polarization, and let $\mathrm{Fl}(2,4;V)$ be the variety of flags $A \subset B \subset V$ with $\dim A = 2$ and $\dim B = 4$, with tautological bundles $A \subset B$. By Proposition 16.5(2), $\mathcal{X}_\Omega \to \Omega$ is smooth and projective with fibers X_Π, and $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ restricts to $\omega_{X_\Pi}^{-1}$ on each fiber. Thus the fiber of $\Sigma(\mathcal{X}_\Omega/\Omega)$ over Π is $\Sigma(X_\Pi)$. By Lemma 2.19, the Hilbert scheme of lines of $\mathrm{Gr}(3,V)$ in the Plücker embedding is $\mathrm{Fl}(2,4;V)$ as a scheme, a flag $A \subset B$ corresponding to the line $\{U : A \subset U \subset B\}$, so the identification of $\mathrm{Fl}(2,4;V)$ with the Hilbert scheme of lines of $\mathrm{Gr}(3,V)$ is compatible with base change. For $U = A + \mathbb{C}u$ with $u \in B$, the space $\Lambda^2 U$ is spanned by $\Lambda^2 A$ and $A \wedge u$, and the rank-five subbundle $A \wedge B \subset \Lambda^2 V$ is spanned by the subspaces $\Lambda^2 U$ with $U = A + \mathbb{C}u$, $u \in B$. Thus the line of a flag is contained in X_Π if and only if the map $A \wedge B \to \Pi^\vee$ induced by $\Lambda^2 V \to \Pi^\vee$ vanishes. Over an arbitrary base $T \to \mathrm{Fl}(2,4;V)\times\Omega$, the family of lines given by the pulled-back flag is contained in $\mathcal{X}_\Omega \times_\Omega T$ if and only if the pullback of the map $A \wedge B \to \mathcal{S}^\vee$ vanishes. Hence $\Sigma(\mathcal{X}_\Omega/\Omega)$ is the zero scheme in $\mathrm{Fl}(2,4;V)\times\Omega$ of the induced section of the rank-15 bundle $\mathrm{Hom}(A\wedge B, \mathcal{S}^\vee)$.

Let x be a closed point of $\Sigma(\mathcal{X}_\Omega/\Omega)$ over Π. Since $\dim \mathrm{Fl}(2,4;V) = 16$ and $\dim_x \Sigma(X_\Pi) = 1$ (Lemma 2.12(4)), the fifteen local equations form a regular sequence in the regular local ring of $\mathrm{Fl}(2,4;V)$ at x. By [Mat89, Theorem 22.5], applied fifteen times, the local ring of $\Sigma(\mathcal{X}_\Omega/\Omega)$ at x is flat over the local ring of Ω at Π. Thus $\Sigma(\mathcal{X}_\Omega/\Omega) \to \Omega$ is flat.

Let $W \subset \Omega$ be an open subset over which $\mathcal{S}$ has a basis $\omega_1,\omega_2,\omega_3$, and identify $\mathrm{Sym}\,\mathcal{S}^\vee|_W$ and $\mathrm{Sym}\,\mathcal{S}|_W$ with $\mathcal{O}_W[\partial]$ and $\mathcal{O}_W[x,y,z]$. By Lemma 16.3(1), the degree-four part of the ideal generated by the seven principal Pfaffians of the alternating matrix of $\mathcal{S}|_W$ has rank 14 at every point of the reduced scheme W, so it is a subbundle $\mathcal{I}_W \subset \mathcal{O}_W[\partial]_4$. The annihilator of $\mathcal{I}_W$ under the apolarity pairing $\mathcal{O}_W[\partial]_4 \times \mathcal{O}_W[x,y,z]_4 \to \mathcal{O}_W$ is a line subbundle. Shrinking W, it is spanned by a quartic f_W whose value at $\Pi \in W$ is a non-zero multiple of f_Π. By **Step 1**, the quartic $\mathrm{Sc}(f_W) \in \mathcal{O}(W)[x,y,z]_4$ is non-zero at every point of W, so $(\mathrm{Sc}(f_W)=0) \subset \mathbb{P}(\mathcal{S}^\vee|_W)$ is a relative effective Cartier divisor, flat over W [Mat89, Theorem 22.5]. A change of the basis $\omega_1,\omega_2,\omega_3$ multiplies f_W by a unit and $\mathrm{Sc}(f_W)$ by a unit, since $\mathfrak{A}$ is an SL_3-invariant. Hence the subschemes $(\mathrm{Sc}(f_W)=0)$ glue to a relative effective Cartier divisor $\mathcal{Q}_\Omega \subset \mathbb{P}(\mathcal{S}^\vee|_\Omega)$, flat over Ω, whose fiber over Π is $(\mathrm{Sc}(f_\Pi)=0)$.

By Proposition 16.5, Ω is a dense open subset of the irreducible variety $\mathbb{B}^{\mathrm{nd}}$. Let $O \subset U^{\mathrm{cat}}$ be a dense open subset. Replacing O by $\mathrm{PGL}_3 \cdot O$, we may assume that O is PGL_3-invariant, so that the subset $\{\Pi \in \mathbb{B}^{\mathrm{nd}} : [f_\Pi] \in O\}$ does not depend on the choice of bases of Π. It is open in $\mathbb{B}^{\mathrm{nd}}$, since locally it is the preimage of O under $\Pi \to [f_W(\Pi)]$, where f_W spans the annihilator of $\mathcal{I}_W$ for an open subset $W \subset \mathbb{B}^{\mathrm{nd}}$ over which $\mathcal{S}$ has a basis (Lemma 16.3(1) holds on $\mathbb{B}^{\mathrm{nd}}$), and it is non-empty by Lemma 16.3(2). Hence its intersection with Ω is a dense open subset of Ω. The properties of a general quartic used below are PGL_3-invariant, so they hold for f_Π with $\Pi \in \Omega$ general. For a general f, the quartic $(\mathrm{Sc}(f)=0)$ is smooth [DK93, (6.8.1)], see also [Sch01, Lemma 3.2]. By [Sch01, Corollaries 6.3 and 6.4], for a general f there is a general

net of quadrics q such that the threefold $H(q)$ associated with q in [Sch01, Section 5] is the scheme $G(3, V_q, \eta_q)$ of three-dimensional subspaces of a seven-dimensional space V_q isotropic for a non-degenerate net η_q of alternating forms on V_q [Sch01, Theorem 5.1 and Proposition 5.2], and the apolar quartic of η_q is a non-zero multiple of f. Let $\Pi \in \Omega$ be general, and take $f = f_\Pi$. After an identification $V_q \cong V$, Lemma 16.3(3) implies that $\eta_q \in \mathrm{PGL}(V) \cdot \Pi$. Hence $\Sigma(X_\Pi) \cong \Sigma(H(q)) \cong (\mathrm{Sc}(f_\Pi) = 0)$ [Sch01, Theorem 6.1], which is a smooth plane quartic [Sch01, Lemma 3.2]. In particular the generic fibers of $\Sigma(\mathcal{X}_\Omega/\Omega)$ and $\mathcal{Q}_\Omega$ over the integral scheme Ω are reduced. By flatness, non-zero local functions on Ω are nonzerodivisors on both schemes, so their local rings embed into localizations of the generic fibers. Thus $\Sigma(\mathcal{X}_\Omega/\Omega)$ and $\mathcal{Q}_\Omega$ are reduced.

Step 3. In this step we construct a morphism $h : \Sigma(\mathcal{X}_\Omega/\Omega) \to \mathcal{Q}_\Omega$ over Ω and compute it at closed points. We follow the argument of [Sch01, Section 6], carried out at every closed point instead of a general one.

On $\Sigma(\mathcal{X}_\Omega/\Omega)$, the map $\Lambda^2 B \to \mathcal{S}^\vee$ vanishes on $A \wedge B$, so it factors through the line bundle $\Lambda^2 B/(A \wedge B) \cong \Lambda^2(B/A)$. The induced map $\Lambda^2(B/A) \to \mathcal{S}^\vee$ is non-zero at every point: otherwise B would be isotropic for any form of Π, which is impossible by [BKM26b, Lemma B.2]. Hence the image of $\Lambda^2 B$ is a line subbundle of $\mathcal{S}^\vee$, which defines a morphism $h : \Sigma(\mathcal{X}_\Omega/\Omega) \to \mathbb{P}(\mathcal{S}^\vee|_\Omega)$ over Ω. This proves that the image of $\Lambda^2 B$ is one-dimensional, as claimed in (2).

Let $\Pi \in \Omega$ and let $A_0 \subset B_0$ be a flag whose line is contained in X_Π. Choose a basis $e_1, \dots, e_7$ of V with $A_0 = \mathrm{Span}\{e_1, e_2\}$ and $B_0 = \mathrm{Span}\{e_1, \dots, e_4\}$, and use the notation introduced before Lemma 16.8. Since Π kills $A_0 \wedge B_0$, we have $(M_\Pi)_{ij} = 0$ for $i \in \{1, 2\}$ and $j \le 4$, so the image of $\Lambda^2 B_0$ is spanned by $a := (M_\Pi)_{34} \neq 0$, and $h(A_0 \subset B_0) = [a]$. Let $\Xi := ((M_\Pi)_{ij})_{i\in\{1,2\}, 5\le j\le 7}$, and for $5 \le k \le 7$ let ξ_k be the 2×2 minor of Ξ obtained by deleting the column of index k. In the principal 6×6 submatrix of M_Π obtained by deleting index $k \ge 5$, the rows 1 and 2 have non-zero entries only in the columns of indices in $\{5, 6, 7\}\backslash\{k\}$, so

$$\mathrm{Pf}_k = \pm(M_\Pi)_{34}\xi_k = \pm a\xi_k \quad (5 \le k \le 7).$$

By Lemma 16.8(1), $\mathrm{Pf}_5, \mathrm{Pf}_6, \mathrm{Pf}_7$, hence ξ_5, ξ_6, ξ_7, are linearly independent. Let $g := a(f_\Pi)$. Then $\xi_k(g) = (a\xi_k)(f_\Pi) = 0$ since $a\xi_k \in I_\Pi = f_\Pi^\perp$, so $(g^\perp)_2 = \mathrm{Span}\{\xi_5, \xi_6, \xi_7\}$ by Lemma 16.9(1) and (3). For $r = 1, 2$, the expansion of the vanishing 3×3 determinant with rows $((M_\Pi)_{r5}, (M_\Pi)_{r6}, (M_\Pi)_{r7})$, $((M_\Pi)_{15}, (M_\Pi)_{16}, (M_\Pi)_{17})$, and $((M_\Pi)_{25}, (M_\Pi)_{26}, (M_\Pi)_{27})$ is a linear syzygy $\sum_k \pm(M_\Pi)_{rk}\xi_k = 0$. The syzygies for $r = 1$ and $r = 2$ are linearly independent: otherwise a non-trivial constant combination of the rows of Ξ would vanish, and all ξ_k would vanish. Hence $(g^\perp)_2$ has two linearly independent linear syzygies, so the three quadrics spanning $(g^\perp)_2$ do not form a regular sequence and $\mathbb{C}[\partial]/g^\perp$ is not a complete intersection of three quadrics. By [Sch01, Lemma 3.1] and the description of the zero locus of $\mathfrak{A}$ in [Sch01, Section 3, pp. 304–305], $\mathrm{Sc}(f_\Pi)(a) = \mathfrak{A}(g) = 0$, that is, $h(A_0 \subset B_0) \in \mathcal{Q}_\Omega$. Moreover, $e_k^\vee(p(M_\Pi)) = \pm\,\mathrm{Pf}_k$ and $s(e_r) = \sum_{k=5}^7 (M_\Pi)_{rk} \otimes e_k^\vee$ for $r = 1, 2$. Letting $B_0^\perp \subset V^\vee$ be the annihilator of B_0, we obtain

$$a \cdot (g^\perp)_2 = \{\lambda(p(M_\Pi)) : \lambda \in B_0^\perp\} \quad \text{and} \quad s(A_0) \subset \Pi^\vee \otimes B_0^\perp. \tag{16.7}$$

Since h maps every closed point of $\Sigma(\mathcal{X}_\Omega/\Omega)$ into the closed subscheme $\mathcal{Q}_\Omega$, and $\Sigma(\mathcal{X}_\Omega/\Omega)$ is reduced of finite type over $\mathbb{C}$ (**Step 2**), h factors through $\mathcal{Q}_\Omega$.

Step 4. In this step we construct bundles $\mathcal{N}$, $\mathcal{R}$, $\mathcal{E}$, A', and B' on $\mathcal{Q}_\Omega$.

Work locally on $\mathcal{Q}_\Omega$, over an open subset W as in **Step 2** and with a trivialization of the tautological line bundle of $\mathbb{P}(\mathcal{S}^\vee|_W)$, so that f_W and the tautological point determine a quartic f and a linear form $a \in \mathcal{O}[\partial]_1$, and let $g := a(f)$. The constructions below change by units under a change of W, of the basis $\omega_1, \omega_2, \omega_3$, and of the trivialization, so they glue over $\mathcal{Q}_\Omega$. Let $\mathcal{N}$ be the kernel of $\mathcal{O}[\partial]_2 \to \mathcal{O}[x, y, z]_1$, $q \to q(g)$. By Lemma 16.9(1) and (3), the map $q \to q(g)$ is surjective at every point, so $\mathcal{N}$ is a subbundle of rank 3 whose fiber at a closed point is $(g^\perp)_2$. Let $\mathcal{R}$ be the kernel of the multiplication map $\mathcal{O}[\partial]_1 \otimes \mathcal{N} \to \mathcal{O}[\partial]_3$. At a closed point of $\mathcal{Q}_\Omega$, $\mathfrak{A}(g) = 0$, so the fiber of the kernel is two-dimensional by Lemma 16.9(2) and (3), and the map

has rank 7. Since $\mathcal{Q}_\Omega$ is reduced, the cokernel is locally free [Har77, Exercise II.5.8(c)], and $\mathcal{R}$ is a subbundle of rank 2.

Since $(aq)(f) = q(g) = 0$ for $q \in \mathcal{N}$, multiplication by a maps $\mathcal{N}$ injectively into the degree-three part of the Pfaffian ideal. The map $\lambda \to \lambda(p(M_\Pi))$ of Lemma 16.8(1) is defined in families and is injective at every point, so it is an isomorphism from $V^\vee \otimes \mathcal{O}$ onto the degree-three part of the Pfaffian ideal. Let $\mathcal{E} \subset V^\vee \otimes \mathcal{O}$ be the rank-three subbundle corresponding to $a\mathcal{N}$, and let $B' \subset V \otimes \mathcal{O}$ be the annihilator of $\mathcal{E}$, of rank 4. Multiplication by a maps $\mathcal{R}$ injectively into $\mathcal{O}[\partial]_1 \otimes \mathcal{E} \subset \mathcal{O}[\partial]_1 \otimes V^\vee$, and the image is contained in the kernel of the map μ of Lemma 16.8(2), defined in families, because $\sum_k l_k \cdot aq_k = a\sum_k l_k q_k$. By Lemma 16.8(2), applied at every point, s is an isomorphism from $V \otimes \mathcal{O}$ onto the subbundle $\operatorname{Ker}\mu$. Let $A' \subset V \otimes \mathcal{O}$ be the rank-two subbundle with $s(A') = a\mathcal{R}$. At a closed point,

$$A' = \{v \in V : s(v) \in \Pi^\vee \otimes \mathcal{E}\}, \tag{16.8}$$

since a linear syzygy among elements of $\mathcal{E}$, viewed in $\mathbb{C}[\partial]_3$, is a times a linear syzygy among elements of $\mathcal{N}$.

Step 5. In this step we show that $A' \subset B'$ and that Π kills $A' \wedge B'$, which defines a morphism $j : \mathcal{Q}_\Omega \to \Sigma(\mathcal{X}_\Omega/\Omega)$ over Ω.

For a local section v of A', we have $s(v) \in \mathcal{O}[\partial]_1 \otimes \mathcal{E}$, so contracting $s(v)$ with a local section w of B' yields $M_\Pi(v, w) = 0$. Thus Π kills $A' \wedge B'$.

We show that $A' \subset B'$ at a closed point $[a]$ of $\mathcal{Q}_\Omega$ over Π. Let $g := a(f_\Pi)$, so that $\mathfrak{A}(g) = \operatorname{Sc}(f_\Pi)(a) = 0$. By Lemma 16.9(2) and (3), there exists a 3×2 matrix $L = (L_{kj})$ with entries in $\mathbb{C}[\partial]_1$ such that the signed 2×2 minors $q_k = (-1)^{k+1} \det L_{(k)}$, $k = 1, 2, 3$, of L form a basis of $(g^\perp)_2$ and the columns of L form a basis of the space of linear syzygies among q_1, q_2, q_3. Let $\epsilon_1, \epsilon_2, \epsilon_3 \in \mathcal{E}$ be given by $\epsilon_k(p(M_\Pi)) = aq_k$. Since $\sum_k L_{kj} \cdot aq_k = 0$, Lemma 16.8(2) and (16.8) yield a basis v_1, v_2 of A' with $s(v_j) = \sum_k L_{kj} \otimes \epsilon_k$, that is,

$$M_\Pi(v_j, w) = \sum_k L_{kj}\epsilon_k(w) \quad \text{for any } w \in V.$$

Let Θ be the 3×2 matrix with entries $\Theta_{ki} := \epsilon_k(v_i) \in \mathbb{C}$. Then $M_\Pi(v_j, v_i) = (\Theta^t L)_{ij}$, and since M_Π is alternating,

$$\Theta^t L = \begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix} \quad \text{for some } c \in \mathbb{C}[\partial]_1.$$

For $G \in \mathrm{GL}_3(\mathbb{C})$, the vector of signed 2×2 minors of GL is $\det(G)(G^t)^{-1}$ times the vector (q_1, q_2, q_3), so the signed minors of GL also span $(g^\perp)_2$. Suppose that Θ has rank one, and write $\Theta = t\beta^t$ with non-zero $t \in \mathbb{C}^3$ and $\beta \in \mathbb{C}^2$. Then $\Theta^t L = \beta(t^t L)$ has rank at most one, so $c^2 = \det(\Theta^t L) = 0$ and $\beta(t^t L) = 0$. Hence $t^t L = 0$, and for G with first row t^t the matrix GL has a zero row, so at most one of the minors of GL is non-zero, a contradiction. Suppose that Θ has rank two, let t_1, t_2 be the columns of Θ, complete them to a basis t_1, t_2, t_3 of $\mathbb{C}^3$, and let G be the matrix with rows t_1^t, t_2^t, t_3^t. Then the rows of GL are $(0, c)$, $(-c, 0)$, and (u, w) for some $u, w \in \mathbb{C}[\partial]_1$, and the signed minors of GL are $-cw$, cu, and c^2. Hence $(g^\perp)_2 \subset c\mathbb{C}[\partial]_1$, and comparing dimensions, $c \neq 0$ and $(g^\perp)_2 = c\mathbb{C}[\partial]_1$. Thus $b(c(g)) = 0$ for any $b \in \mathbb{C}[\partial]_1$, so $c(g) = 0$, which contradicts Lemma 16.9(3) for $g = a(f_\Pi)$. Therefore $\Theta = 0$, that is, A' is contained in the annihilator B' of $\mathcal{E}$.

Hence the bundle map $A' \to (V \otimes \mathcal{O})/B'$ vanishes at every closed point of the reduced scheme $\mathcal{Q}_\Omega$, so it vanishes. Thus $A' \subset B'$ defines a morphism $\mathcal{Q}_\Omega \to \mathrm{Fl}(2, 4; V) \times \Omega$ over Ω, and it factors through $\Sigma(\mathcal{X}_\Omega/\Omega)$ by **Step 2**, since Π kills $A' \wedge B'$. We denote the morphism by j.

Step 6. In this step we show that h and j are mutually inverse, and we conclude the proof.

Let $\Pi \in \Omega$, let $A_0 \subset B_0$ be a closed point of $\Sigma(X_\Pi)$, and let $[a] := h(A_0 \subset B_0)$. By (16.7), the fiber of $\mathcal{E}$ at $[a]$ is $B_0^\perp$, so the fiber of B' is B_0, and $s(A_0) \subset \Pi^\vee \otimes B_0^\perp$, so A_0 is contained in the fiber of A' by (16.8). Since both are two-dimensional, $j(h(A_0 \subset B_0)) = (A_0 \subset B_0)$.

Let $[a] \neq [a']$ be closed points of $\mathcal{Q}_\Omega$ over Π with $j([a]) = j([a'])$, and let $g := a(f_\Pi)$ and $g' := a'(f_\Pi)$. Then the fibers of B', hence of $\mathcal{E}$, coincide, so $a(g^\perp)_2 = a'(g'^\perp)_2$ in $\mathbb{C}[\partial]_3$. The three-dimensional space $a(g^\perp)_2$ consists of cubics divisible by a and by a', so it equals $aa'\mathbb{C}[\partial]_1$, and $(g^\perp)_2 = a'\mathbb{C}[\partial]_1$. Thus $a'(g) = 0$, which contradicts Lemma 16.9(3) for $g = a(f_\Pi)$. Hence j is injective on closed points. For a closed point $[a]$ of $\mathcal{Q}_\Omega$, we have $j(h(j([a]))) = j([a])$, so $h(j([a])) = [a]$.

Thus $h \circ j$ and $j \circ h$ coincide with the identities at every closed point. Since $\mathcal{Q}_\Omega$ and $\Sigma(\mathcal{X}_\Omega/\Omega)$ are reduced of finite type over $\mathbb{C}$ and separated, the locus where $h \circ j$ (resp. $j \circ h$) coincides with the identity is a closed subscheme containing every closed point, hence is everything. Therefore $h : \Sigma(\mathcal{X}_\Omega/\Omega) \to \mathcal{Q}_\Omega$ is an isomorphism over Ω. Since the formation of the relative Hilbert scheme of lines and of the relative effective Cartier divisor $\mathcal{Q}_\Omega$ commutes with base change, so does h. This proves (5).

By Definition-Theorem 2.20, $\Pi_X \in \Omega$, and restricting h to the fiber over Π_X yields the closed immersion $\Sigma(X) \to \mathbb{P}(\Pi_X^\vee)$ of (2) with image Q_X. Since Q_X is the zero scheme of a non-zero quartic form on $\mathbb{P}(\Pi_X^\vee)$, it is Gorenstein with $\omega_{Q_X} \cong \mathcal{O}_{Q_X}(1)$ [Har77, Theorem III.7.11], and the restriction $H^0(\mathbb{P}(\Pi_X^\vee), \mathcal{O}(1)) \to H^0(Q_X, \mathcal{O}_{Q_X}(1))$ is an isomorphism because H^0 and H^1 of $\mathcal{O}_{\mathbb{P}^2}(-3)$ vanish [Har77, Theorem III.5.1]. Hence the embedding is the canonical embedding of $\Sigma(X)$, and (2) follows.

By Proposition 16.10(3) and (2), $\mathrm{VPS}(f_X, 6) = \mathrm{VSP}(f_X, 6) \cong X$ is smooth, so Proposition 16.10(4) with $f = f_X$ and $\Pi = \Pi_X$ implies that $[\omega] \to \mathcal{C}_\omega$ is an isomorphism from $\mathbb{P}(\Pi_X)$ onto the Hilbert scheme of conics of X. This proves (3).

Let f be as in (4). By Lemma 16.3(2), there exists $\Pi \in \mathbb{B}^{\mathrm{nd}}$ with a basis in which f_Π is a non-zero multiple of f, and $X_\Pi \cong \mathrm{VPS}(f, 6)$, so $\Pi \in \Omega$ by Proposition 16.5. Since $\mathrm{Sc}(f_\Pi)$ is a non-zero multiple of $\mathrm{Sc}(f)$, restricting h to the fiber over Π yields

$$\Sigma(\mathrm{VPS}(f, 6)) \cong \Sigma(X_\Pi) \cong (\mathrm{Sc}(f) = 0) \subset \mathbb{P}(\Pi^\vee),$$

and $\mathbb{P}(\Pi)$ is the Hilbert scheme of conics of X_Π by Proposition 16.10(4). This proves (4). □

Theorem 16.12 shows that $\mathrm{Aut}(X_\Pi)$ is isomorphic to the stabilizers of Π in $\mathrm{PGL}(V)$ and of $[f_\Pi]$ in $\mathrm{PGL}(\Pi^\vee)$, and it relates the condition $\Pi \in \Omega^{\mathrm{st}}$ to $\mathbb{G}_m$-actions on X_Π and to the GIT-stability of $[f_\Pi]$ and $[\mathrm{Sc}(f_\Pi)]$. It is used in particular in the proof of Theorem 19.1. Item (5) follows from the SL_3-equivariance of Sc and the Hilbert–Mumford criterion, (6) from Lemma 16.3(1), (3), and (5) and the GIT-semistability of the points of U^{cat}, and (4) from (5) and (6). The converse of (4) is part of Theorem 19.1(1).

Theorem 16.12. *Let $\Pi \in \Omega$, and let $\mathcal{U}$ be the tautological subbundle on $\mathrm{Gr}(3, V)$. Then the following hold.*

(1) The restriction $\mathcal{U}|_{X_\Pi}$ is the Mukai bundle of X_Π, the restriction map $V^\vee \to H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$ is an isomorphism, and under the isomorphism $V^\vee \cong H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$ the net Π is the canonical Mukai net of X_Π.

(2) The natural maps induce isomorphisms of algebraic groups

$$\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \cong \mathrm{Aut}(X_\Pi) \cong \mathrm{Stab}_{\mathrm{PGL}(\Pi^\vee)}([f_\Pi]),$$

and the sequence

$$1 \to \mu_7 \to \mathrm{Stab}_{\mathrm{SL}(V)}(\Pi) \to \mathrm{Aut}(X_\Pi) \to 1$$

is exact.

(3) $\Pi \notin \Omega^{\mathrm{st}}$ if and only if $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$ or X_Π admits a faithful $\mathbb{G}_m$-action.

(4) If Q_{X_Π} is GIT-stable, then $\Pi \in \Omega^{\mathrm{st}}$.

(5) Let $f \in \mathbb{C}[x, y, z]_4$ be non-zero with $\mathrm{Sc}(f) \neq 0$. If $[\mathrm{Sc}(f)]$ is GIT-stable, then $[f]$ is GIT-stable.

(6) Let $\Pi' \in \mathbb{B}^{\mathrm{nd}}$. If $[f_{\Pi'}]$ is GIT-stable, then $\mathrm{SL}(V) \cdot \Pi'$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and the stabilizer of Π' in $\mathrm{SL}(V)$ is finite.

Proof. The proof is long, so we divide it into several steps. Let $\mathcal{O}(1) := \det \mathcal{U}^\vee$ be the Plücker line bundle on $\mathrm{Gr}(3, V)$. Recall that X_Π is the zero scheme of the section s_Π of $\Lambda^2\mathcal{U}^\vee \otimes \Pi^\vee$ obtained by restricting the forms in Π to the fibers of $\mathcal{U}$.

Step 1. In this step we compute the cohomology on $\mathrm{Gr}(3, V)$ of the bundles

$$\mathcal{G} \otimes \Lambda^p(\Lambda^2\mathcal{U} \otimes \Pi), \qquad \mathcal{G} \in \{\mathcal{U},\ \mathcal{U}^\vee,\ \Lambda^2\mathcal{U}^\vee,\ \mathcal{E}nd(\mathcal{U})\}, \qquad 0 \le p \le 9.$$

We show that all the cohomology groups of the bundles above vanish, except the following groups, which are all in degree zero:

$$H^0(\mathrm{Gr}(3, V), \mathcal{U}^\vee) = V^\vee, \quad H^0(\mathrm{Gr}(3, V), \Lambda^2\mathcal{U}^\vee) = \Lambda^2 V^\vee, \quad H^0(\mathrm{Gr}(3, V), \mathcal{E}nd(\mathcal{U})) = \mathbb{C} \cdot \mathrm{id}_\mathcal{U}$$

for $p = 0$, and

$$H^0(\mathrm{Gr}(3, V), \Lambda^2\mathcal{U}^\vee \otimes \Lambda^2\mathcal{U} \otimes \Pi) = \{\mathrm{id}_{\Lambda^2\mathcal{U}} \otimes \omega : \omega \in \Pi\} \cong \Pi$$

for $p = 1$.

For a nonincreasing $\mu = (\mu_1, \mu_2, \mu_3) \in \mathbb{Z}^3$, let $\Sigma^\mu\mathcal{U}^\vee$ be the associated Schur functor of $\mathcal{U}^\vee$. Thus $\Sigma^{(1,0,0)}\mathcal{U}^\vee = \mathcal{U}^\vee$, $\Sigma^{(1,1,0)}\mathcal{U}^\vee = \Lambda^2\mathcal{U}^\vee$, $\Sigma^{(0,0,-1)}\mathcal{U}^\vee = \mathcal{U}$, and $\Sigma^{\mu+(k,k,k)}\mathcal{U}^\vee = \Sigma^\mu\mathcal{U}^\vee \otimes \mathcal{O}(k)$ for $k \in \mathbb{Z}$. Let

$$\tau(\mu) := (\mu_1 + 6,\ \mu_2 + 5,\ \mu_3 + 4,\ 3,\ 2,\ 1,\ 0).$$

We use the Borel–Weil–Bott theorem on $\mathrm{Gr}(3, V)$ [Wey03, Corollary 4.1.9], with the conventions of [BKM26b, Appendix A, pp. 155–156]: if two entries of $\tau(\mu)$ coincide, then $H^\bullet(\mathrm{Gr}(3, V), \Sigma^\mu\mathcal{U}^\vee) = 0$, and if $\mu_3 \ge 0$, then $H^{>0}(\mathrm{Gr}(3, V), \Sigma^\mu\mathcal{U}^\vee) = 0$ and $H^0(\mathrm{Gr}(3, V), \Sigma^\mu\mathcal{U}^\vee) = \Sigma^{(\mu_1,\mu_2,\mu_3,0,0,0,0)}V^\vee$.

Since $\Lambda^2\mathcal{U} \cong \mathcal{U}^\vee \otimes \mathcal{O}(-1)$ and $\dim \Pi = 3$, the Cauchy formula [Wey03, Theorem 2.3.2] implies

$$\Lambda^p(\Lambda^2\mathcal{U} \otimes \Pi) \cong \bigoplus_\rho \Sigma^\rho\mathcal{U}^\vee \otimes \mathcal{O}(-p) \otimes \Sigma^{\rho'}\Pi,$$

where ρ runs through the partitions with $3 \ge \rho_1 \ge \rho_2 \ge \rho_3 \ge 0$ and $\rho_1 + \rho_2 + \rho_3 = p$, ρ' is the conjugate partition, and $\Sigma^{\rho'}\Pi$ is a vector space. Let

$$D(\mathcal{U}) := \{-e_i\}, \quad D(\mathcal{U}^\vee) := \{e_i\},$$
$$D(\Lambda^2\mathcal{U}^\vee) := \{e_i + e_j : i < j\}, \quad D(\mathcal{E}nd(\mathcal{U})) := \{e_i - e_j : 1 \le i, j \le 3\},$$

where e_1, e_2, e_3 is the standard basis of $\mathbb{Z}^3$. In particular $0 = e_1 - e_1 \in D(\mathcal{E}nd(\mathcal{U}))$. By the Pieri rule, $\mathcal{G} \otimes \Sigma^\rho\mathcal{U}^\vee$ is a direct sum of bundles $\Sigma^{\rho+\delta}\mathcal{U}^\vee$ with $\delta \in D(\mathcal{G})$ and $\rho + \delta$ nonincreasing. Hence $\mathcal{G} \otimes \Lambda^p(\Lambda^2\mathcal{U} \otimes \Pi)$ is a direct sum of bundles $\Sigma^\mu\mathcal{U}^\vee$ tensored with vector spaces, where $\mu = \rho + \delta - (p, p, p)$ with $3 \ge \rho_1 \ge \rho_2 \ge \rho_3 \ge 0$, $\rho_1 + \rho_2 + \rho_3 = p$, and $\delta \in D(\mathcal{G})$. For such μ we have

$$\tau(\mu) = (\nu_1 + \delta_1,\ \nu_2 + \delta_2,\ \nu_3 + \delta_3,\ 3,\ 2,\ 1,\ 0),$$
$$\nu = (\nu_1, \nu_2, \nu_3) := (6 - \rho_2 - \rho_3,\ 5 - \rho_1 - \rho_3,\ 4 - \rho_1 - \rho_2).$$

Any $\delta \in D(\mathcal{G})$ satisfies $\delta_i \in \{-1, 0, 1\}$ for $i = 1, 2, 3$. Thus if $\nu_i \in \{1, 2\}$ for some i, then $\nu_i + \delta_i \in \{0, 1, 2, 3\}$ coincides with one of the last four entries of $\tau(\mu)$, and $\Sigma^\mu\mathcal{U}^\vee$ has no cohomology.

The case $p \ge 2$. We show that $\nu_i \in \{1, 2\}$ for some i unless $\rho = (2, 2, 0)$ or $\rho = (3, 3, 3)$. Since $p \ge 2$ and ρ is nonincreasing, $\rho_1 + \rho_2 \ge 2$, so $\nu_3 \le 2$. Assume that $\nu_3 \notin \{1, 2\}$. Then $\rho_1 + \rho_2 \ge 4$. If $\rho_1 + \rho_3 \le 2$, then $\rho_1 \ge 2$ implies $\rho_3 = 0$ and $\rho_1 = 2$, hence $\rho = (2, 2, 0)$. Otherwise $\nu_2 = 5 - \rho_1 - \rho_3 \le 2$, and assuming that $\nu_2 \notin \{1, 2\}$ we obtain $\rho_1 + \rho_3 \ge 5$, so $\rho_1 = 3$ and $\rho_3 \ge 2$. Then $\nu_1 = 6 - \rho_2 - \rho_3 \le 2$, and assuming that $\nu_1 \notin \{1, 2\}$ we obtain $\rho_2 + \rho_3 = 6$, so $\rho = (3, 3, 3)$.

Let $\rho = (2, 2, 0)$, so $p = 4$ and $\nu = (4, 3, 0)$. If $\tau(\mu)$ has distinct entries, then $3 + \delta_2 \notin \{0, 1, 2, 3\}$ and $0 + \delta_3 \notin \{0, 1, 2, 3\}$, so $\delta_2 = 1$ and $\delta_3 = -1$. The second entry is then 4, and $4 + \delta_1 \notin \{0, 1, 2, 3, 4\}$ forces $\delta_1 = 1$. Thus $\delta = (1, 1, -1)$, which is not of the form 0, $\pm e_i$, $e_i + e_j$, or $e_i - e_j$. Hence $\delta \notin D(\mathcal{G})$ for any $\mathcal{G}$, and the summands with $\rho = (2, 2, 0)$ have no cohomology.

Let $\rho = (3, 3, 3)$, so $p = 9$ and $\nu = (0, -1, -2)$. If $\tau(\mu)$ has distinct entries, then $\delta_1 \notin \{0, 1, 2, 3\}$ forces $\delta_1 = -1$. The second entry $-1 + \delta_2$ must differ from -1 and 0, so $\delta_2 = -1$, and the third

entry $-2+\delta_3$ must differ from -2, -1, and 0, so $\delta_3=-1$. Thus $\delta=(-1,-1,-1)\notin D(\mathcal{G})$ for any $\mathcal{G}$, and the summands with $\rho=(3,3,3)$ have no cohomology.

The case $p=1$. Here $\rho=(1,0,0)$, $\rho'=(1)$, and $\nu=(6,4,3)$. If $\tau(\mu)$ has distinct entries, then $3+\delta_3\notin\{0,1,2,3\}$ forces $\delta_3=1$, then $4+\delta_2\notin\{0,1,2,3,4\}$ forces $\delta_2=1$, and $6+\delta_1\notin\{4,5\}$ forces $\delta_1\in\{0,1\}$. Since $(1,1,1)$ belongs to no $D(\mathcal{G})$, the only possibility is $\mathcal{G}=\Lambda^2\mathcal{U}^\vee$ and $\delta=(0,1,1)$, with $\mu=(1,1,1)-(1,1,1)=0$ and multiplicity space $\Sigma^{(1)}\Pi=\Pi$. The trivial summand of $\Lambda^2\mathcal{U}^\vee\otimes\Lambda^2\mathcal{U}=\mathcal{E}nd(\Lambda^2\mathcal{U})$ is spanned by $\mathrm{id}_{\Lambda^2\mathcal{U}}$. Thus $H^\bullet(\mathrm{Gr}(3,V),\Lambda^2\mathcal{U}^\vee\otimes\Lambda^2\mathcal{U}\otimes\Pi)$ is concentrated in degree zero and equals $\{\mathrm{id}_{\Lambda^2\mathcal{U}}\otimes\omega:\omega\in\Pi\}$.

The case $p=0$. Here $\Lambda^0(\Lambda^2\mathcal{U}\otimes\Pi)=\mathcal{O}_{\mathrm{Gr}(3,V)}$, and

$$\mathcal{U}=\Sigma^{(0,0,-1)}\mathcal{U}^\vee,\quad \mathcal{U}^\vee=\Sigma^{(1,0,0)}\mathcal{U}^\vee,\quad \Lambda^2\mathcal{U}^\vee=\Sigma^{(1,1,0)}\mathcal{U}^\vee,\quad \mathcal{E}nd(\mathcal{U})=\mathcal{O}_{\mathrm{Gr}(3,V)}\oplus\Sigma^{(1,0,-1)}\mathcal{U}^\vee.$$

Since $\tau(0,0,-1)=(6,5,3,3,2,1,0)$ and $\tau(1,0,-1)=(7,5,3,3,2,1,0)$, the bundles $\mathcal{U}$ and $\Sigma^{(1,0,-1)}\mathcal{U}^\vee$ have no cohomology. The remaining bundles have $\mu_3\geq 0$, and the Borel–Weil–Bott theorem implies $H^0(\mathrm{Gr}(3,V),\mathcal{U}^\vee)=V^\vee$, $H^0(\mathrm{Gr}(3,V),\Lambda^2\mathcal{U}^\vee)=\Lambda^2V^\vee$, and $H^0(\mathrm{Gr}(3,V),\mathcal{O}_{\mathrm{Gr}(3,V)})=\mathbb{C}$, with no higher cohomology. The isomorphisms for $\mathcal{U}^\vee$ and $\Lambda^2\mathcal{U}^\vee$ are the natural ones, given by regarding a linear or alternating form on V as a section of $\mathcal{U}^\vee$ or $\Lambda^2\mathcal{U}^\vee$ by restriction to the fibers of $\mathcal{U}$.

Step 2. In this step we prove (1).

Since $\mathrm{Gr}(3,V)$ is smooth of dimension 12, $\dim X_\Pi=3$, and $\mathrm{rank}(\Lambda^2\mathcal{U}^\vee\otimes\Pi^\vee)=9$, the section s_Π is locally given by a regular sequence. Hence the Koszul complex

$$0\to\Lambda^9(\Lambda^2\mathcal{U}\otimes\Pi)\to\cdots\to\Lambda^2\mathcal{U}\otimes\Pi\to\mathcal{O}_{\mathrm{Gr}(3,V)}\to\mathcal{O}_{X_\Pi}\to 0,$$

whose differentials are the contractions with s_Π, is exact. By adjunction,

$$\omega_{X_\Pi}\cong\big(\omega_{\mathrm{Gr}(3,V)}\otimes\det(\Lambda^2\mathcal{U}^\vee\otimes\Pi^\vee)\big)|_{X_\Pi}\cong\big(\mathcal{O}(-7)\otimes\mathcal{O}(2)^{\otimes 3}\big)|_{X_\Pi}=\mathcal{O}(-1)|_{X_\Pi}=\det(\mathcal{U}|_{X_\Pi}),$$

where we use $\omega_{\mathrm{Gr}(3,V)}\cong\mathcal{O}(-7)$ and $\det\Lambda^2\mathcal{U}^\vee\cong\det(\mathcal{U}\otimes\mathcal{O}(1))\cong\mathcal{O}(2)$.

Let $\mathcal{G}$ be one of the four bundles of **Step 1**. Tensoring the Koszul complex with $\mathcal{G}$, we obtain a spectral sequence

$$E_1^{-p,q}=H^q\big(\mathrm{Gr}(3,V),\mathcal{G}\otimes\Lambda^p(\Lambda^2\mathcal{U}\otimes\Pi)\big)\Rightarrow H^{q-p}(X_\Pi,\mathcal{G}|_{X_\Pi}).$$

By **Step 1**, the only non-zero terms are $E_1^{0,0}$ for $\mathcal{G}\neq\mathcal{U}$, and $E_1^{-1,0}\cong\Pi$ for $\mathcal{G}=\Lambda^2\mathcal{U}^\vee$. For $\mathcal{G}=\Lambda^2\mathcal{U}^\vee$, the differential $d_1:E_1^{-1,0}\to E_1^{0,0}$ is induced by contraction with s_Π, so it sends $\mathrm{id}_{\Lambda^2\mathcal{U}}\otimes\omega$ to the restriction of ω to the fibers of $\mathcal{U}$, that is, it is the inclusion $\Pi\subset\Lambda^2V^\vee$. Since all other terms vanish, the spectral sequence degenerates at E_2, and we obtain

$$H^\bullet(X_\Pi,\mathcal{U}|_{X_\Pi})=0,\qquad H^\bullet(X_\Pi,\mathcal{E}nd(\mathcal{U}|_{X_\Pi}))=H^0(X_\Pi,\mathcal{E}nd(\mathcal{U}|_{X_\Pi}))=\mathbb{C},$$
$$H^{>0}(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})=0,$$
$$V^\vee\xrightarrow{\cong}H^0(X_\Pi,\mathcal{U}^\vee|_{X_\Pi}),\tag{16.9}$$

and an exact sequence

$$0\to\Pi\to\Lambda^2V^\vee\to H^0(X_\Pi,\Lambda^2\mathcal{U}^\vee|_{X_\Pi})\to 0,\tag{16.10}$$

where the maps from $V^\vee$ and $\Lambda^2V^\vee$ are the restriction maps.

Thus $\mathcal{U}|_{X_\Pi}$ is a vector bundle of rank 3 with $\det(\mathcal{U}|_{X_\Pi})\cong\mathcal{O}_{X_\Pi}(K_{X_\Pi})$ and $H^\bullet(X_\Pi,\mathcal{U}|_{X_\Pi})=0$. It is exceptional, because $\mathrm{Ext}^i(\mathcal{U}|_{X_\Pi},\mathcal{U}|_{X_\Pi})=H^i(X_\Pi,\mathcal{E}nd(\mathcal{U}|_{X_\Pi}))$. The dual $\mathcal{U}^\vee|_{X_\Pi}$ is a quotient of $V^\vee\otimes\mathcal{O}_{X_\Pi}$, hence globally generated, and $h^0(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})=7$ and $H^{>0}(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})=0$. Since $\Pi\in\Omega$, X_Π is a smooth prime Fano threefold of genus $12=3\cdot 4$. The properties of $\mathcal{U}|_{X_\Pi}$ above are the conditions of [BKM26a, Theorem 1.1] for genus $3\cdot 4$ and rank 3, with $3+4=7$ global sections of the dual, so $\mathcal{U}|_{X_\Pi}$ is the Mukai bundle of X_Π by the uniqueness in [BKM26a, Theorem 1.1]. The restriction map $V^\vee\to H^0(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})$ is an isomorphism by (16.9), and (16.10) shows that, under $V^\vee\cong H^0(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})$, Π is the kernel of $\Lambda^2H^0(X_\Pi,\mathcal{U}^\vee|_{X_\Pi})\to H^0(X_\Pi,\Lambda^2\mathcal{U}^\vee|_{X_\Pi})$, that is, the canonical Mukai net of X_Π (Definition-Theorem 2.20). This proves (1).

Since X_Π is reduced, a section of $\Lambda^2\mathcal{U}^\vee|_{X_\Pi}$ vanishes if and only if it vanishes at all closed points. Thus (1) implies

$$\Pi = \{\omega \in \Lambda^2 V^\vee : \omega|_W = 0 \text{ for any } [W] \in X_\Pi\}. \tag{16.11}$$

Step 3. In this step we prove (2).

By adjunction (**Step 2**), $-K_{X_\Pi} = \mathcal{O}(1)|_{X_\Pi}$ is very ample, so $\mathrm{Aut}(X_\Pi)$ is a closed subgroup of $\mathrm{PGL}(H^0(X_\Pi, -K_{X_\Pi}))$, in particular an algebraic group. Any $g \in \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi)$ acts on $\mathrm{Gr}(3,V)$ and maps X_Π onto X_Π. Hence restriction defines a homomorphism of algebraic groups

$$r : \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \to \mathrm{Aut}(X_\Pi).$$

We show that r is surjective. This is the $\mathrm{Aut}(X_\Pi)$-equivariance of the embedding $X_\Pi \to \mathrm{Gr}(3,V)$ defined by the Mukai bundle [KP20, Corollary 5.6]. We prove the $\mathrm{Aut}(X_\Pi)$-equivariance of $X_\Pi \subset \mathrm{Gr}(3,V)$ in the present notation. Let $\phi \in \mathrm{Aut}(X_\Pi)$. The bundle $\phi^*(\mathcal{U}|_{X_\Pi})$ satisfies all the conditions in Definition-Theorem 2.20, so by the uniqueness in [BKM26a, Theorem 1.1] there exists an isomorphism $\alpha : \phi^*(\mathcal{U}|_{X_\Pi}) \to \mathcal{U}|_{X_\Pi}$. Since $H^0(X_\Pi, \mathcal{E}nd(\mathcal{U}|_{X_\Pi})) = \mathbb{C}$ by **Step 2**, α is unique up to a non-zero scalar. Let $\psi \in \mathrm{GL}(V^\vee)$ be the composition

$$V^\vee \cong H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi}) \xrightarrow{\phi^*} H^0(X_\Pi, \phi^*(\mathcal{U}^\vee|_{X_\Pi})) \xrightarrow{(\alpha^\vee)^{-1}} H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi}) \cong V^\vee,$$

where the outer isomorphisms are the restriction maps of (1). Then ψ is determined by ϕ up to a non-zero scalar. Let $[W] \in X_\Pi$, and let $[W'] := \phi([W])$. For $\sigma \in V^\vee$, the value of $\phi^*\sigma$ at $[W]$ is $\sigma|_{W'}$, and $(\alpha^\vee)^{-1}$ is an isomorphism on fibers, so $\psi(\sigma)|_W = 0$ if and only if $\sigma|_{W'} = 0$. Let $h \in \mathrm{PGL}(V)$ be the class of the transpose $\psi^t \in \mathrm{GL}(V)$. Then $\psi^t(W) = W'$, so h induces ϕ on X_Π. By (16.11), h preserves Π. Thus $h \in \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi)$ and $r(h) = \phi$.

We show that r is injective. Let $g \in \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi)$ with $r(g) = \mathrm{id}_{X_\Pi}$, and let $\tilde{g} \in \mathrm{GL}(V)$ be a representative of g. Then $\tilde{g}(W) = W$ for any $[W] \in X_\Pi$, so $\tilde{g}$ induces an automorphism of the vector bundle $\mathcal{U}|_{X_\Pi}$ over id_{X_Π}. Since $H^0(X_\Pi, \mathcal{E}nd(\mathcal{U}|_{X_\Pi})) = \mathbb{C}$, there exists $c \in \mathbb{C}^*$ with $\tilde{g}|_W = c \cdot \mathrm{id}_W$ for any $[W] \in X_\Pi$. Hence for $\sigma \in V^\vee$, the form $\sigma \circ \tilde{g} - c\sigma$ restricts to zero in $H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$, so $\sigma \circ \tilde{g} = c\sigma$ by (1). Thus $\tilde{g} = c \cdot \mathrm{id}_V$ and $g = 1$.

Thus r is bijective on $\mathbb{C}$-points. The group schemes $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi)$ and $\mathrm{Aut}(X_\Pi)$ are closed subgroup schemes of $\mathrm{PGL}(V)$ and of $\mathrm{PGL}(H^0(X_\Pi, -K_{X_\Pi}))$, hence affine, and affine group schemes of finite type over $\mathbb{C}$ are smooth by Cartier's theorem [Mil17, Theorem 3.23]. Hence $\mathrm{Ker}\, r$ is reduced, so it is trivial and r is a closed immersion. Since $\mathrm{Aut}(X_\Pi)$ is reduced and r is surjective on $\mathbb{C}$-points, r is an isomorphism of algebraic groups. By Lemma 16.3(3), the natural map $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \to \mathrm{Stab}_{\mathrm{PGL}(\Pi^\vee)}([f_\Pi])$ is an isomorphism of algebraic groups. This proves the isomorphisms in (2).

The scalars in $\mathrm{SL}(V)$ preserve Π, and $\mathrm{SL}(V) \to \mathrm{PGL}(V)$ is surjective with kernel μ_7. Hence $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is the inverse image of $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi)$, and

$$1 \to \mu_7 \to \mathrm{Stab}_{\mathrm{SL}(V)}(\Pi) \to \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi) \to 1$$

is exact. Composing with r, we obtain the exact sequence in (2).

Step 4. In this step we prove (3). Let $\Gamma := \mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$.

Suppose that $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$. Then $\Pi \notin \Omega^{\mathrm{st}}$ by the definition of Ω^{st}. Suppose that X_Π admits a faithful $\mathbb{G}_m$-action. Then $\dim \mathrm{Aut}(X_\Pi) \geq 1$. By (2), $\Gamma \to \mathrm{Aut}(X_\Pi)$ is surjective with finite kernel μ_7, so $\dim \Gamma = \dim \mathrm{Aut}(X_\Pi) \geq 1$, and $\Pi \notin \Omega^{\mathrm{st}}$.

Conversely, assume that $\Pi \notin \Omega^{\mathrm{st}}$ and that $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$. Then Γ is not finite, and since Γ is an algebraic group, $\dim \Gamma > 0$. By Definition-Proposition 16.2, $\mathbb{B}^{\mathrm{nd}}$ is an affine $\mathrm{SL}(V)$-invariant open subset of $\mathbb{B}$. Since $\mathrm{SL}(V)$ is linearly reductive, the morphism from $[\mathbb{B}^{\mathrm{nd}}/\mathrm{SL}(V)]$ to the spectrum of the ring of invariants of $\mathbb{B}^{\mathrm{nd}}$ is a good moduli space [Alp13, Theorem 13.2]. The closed orbit $\mathrm{SL}(V) \cdot \Pi$ is a closed point of $[\mathbb{B}^{\mathrm{nd}}/\mathrm{SL}(V)]$ with stabilizer Γ, so Γ is linearly reductive [Alp13, Proposition 12.14]. Since we work in characteristic zero, Γ is reductive. Let Γ^0 be the identity component of Γ. Then Γ^0 is a connected reductive group with $\dim \Gamma^0 > 0$,

so the unipotent radical of Γ^0 is trivial [Bor91, IV.11.21]. If a Borel subgroup of Γ^0 contained no non-trivial torus, then Γ^0 would be unipotent [Bor91, Corollary IV.11.5(2)], hence equal to the unipotent radical of Γ^0, which contradicts $\dim \Gamma^0 > 0$. Thus Γ^0 contains a torus T with $\dim T \geq 1$. Since $\Gamma \to \mathrm{Aut}(X_\Pi)$ has finite kernel μ_7 by (2), the image of T in $\mathrm{Aut}(X_\Pi)$ is a torus of dimension $\dim T \geq 1$. Hence $\mathrm{Aut}(X_\Pi)$ contains a closed subgroup isomorphic to $\mathbb{G}_m$, that is, X_Π admits a faithful $\mathbb{G}_m$-action. This proves (3).

Step 5. In this step we prove (5).

We use the action of SL_3 and the form of the Hilbert–Mumford criterion fixed in Section 2.6. For a one-parameter subgroup $\lambda : \mathbb{G}_m \to \mathrm{SL}_3$ and $0 \neq u \in \mathbb{C}[x,y,z]_4$, write $u = \sum_{k\in\mathbb{Z}} u_k$ with $\lambda(t)\cdot u_k = t^k u_k$. Then $[u]$ is GIT-stable if and only if for any non-trivial λ there exists $k < 0$ with $u_k \neq 0$ [MFK94, Theorem 2.1 and Proposition 2.3], that is, if and only if for any non-trivial λ the limit $\lim_{t\to 0} \lambda(t)\cdot u$ does not exist in $\mathbb{C}[x,y,z]_4$.

For $a \in \mathbb{C}^3$ and $f \in \mathbb{C}[x,y,z]_4$ let $\partial_a f := \sum_i a_i \partial_i f$. Under the identification of a with $\sum_i a_i\partial_i \in \mathbb{C}[\partial]_1$, we have $\partial_a f = a(f)$ for the apolarity action of Definition 2.27, which is the notation $a(f)$ used for Theorem 16.11, and $\mathrm{Sc}(f)(a) = \mathfrak{A}(\partial_a f)$. For $g \in \mathrm{SL}_3$, the chain rule implies $\partial_a(g\cdot f) = g\cdot(\partial_{g^{-1}a} f)$. Since $\mathfrak{A}$ is SL_3-invariant,

$$\mathrm{Sc}(g\cdot f)(a) = \mathfrak{A}\big(g\cdot(\partial_{g^{-1}a} f)\big) = \mathfrak{A}(\partial_{g^{-1}a} f) = \mathrm{Sc}(f)(g^{-1}a).$$

Since we regard $\mathrm{Sc}(f)$ as a quartic in (x,y,z) by renaming a, this means

$$\mathrm{Sc}(g\cdot f) = g\cdot \mathrm{Sc}(f) \qquad \text{for any } g \in \mathrm{SL}_3\,.$$

Moreover, $f \to \partial_a f$ is linear and $\mathfrak{A}$ is a form of degree four, so, unlike the projective Scorza map, $\mathrm{Sc} : \mathbb{C}[x,y,z]_4 \to \mathbb{C}[x,y,z]_4$ is a homogeneous polynomial map of degree four defined on all of $\mathbb{C}[x,y,z]_4$.

Let f be as in (5), and suppose that $[f]$ is not GIT-stable. By the Hilbert–Mumford criterion there exists a non-trivial one-parameter subgroup λ such that $t \to \lambda(t)\cdot f$ extends to a morphism $\mathbb{A}^1 \to \mathbb{C}[x,y,z]_4$. Composing with the polynomial map Sc, the morphism $t \to \mathrm{Sc}(\lambda(t)\cdot f) = \lambda(t)\cdot\mathrm{Sc}(f)$ also extends to $\mathbb{A}^1$, so $\lim_{t\to 0}\lambda(t)\cdot\mathrm{Sc}(f)$ exists. Since $\mathrm{Sc}(f) \neq 0$, the Hilbert–Mumford criterion implies that $[\mathrm{Sc}(f)]$ is not GIT-stable. This proves (5).

Step 6. In this step we prove (6).

We first show that any point of U^{cat} is GIT-semistable. Fix a basis $(P_1,\dots,P_6)$ of $\mathbb{C}[\partial]_2$. By Definition 2.27, $\det \mathrm{Cat}_2(f)$ is the Gram determinant of the bilinear form $(P,P') \to (PP')(f)$ on $\mathbb{C}[\partial]_2$ with respect to $(P_1,\dots,P_6)$, and the matrix entries $(P_iP_j)(f)$ are linear in f. For $g \in \mathrm{SL}_3$, the Gram matrix of $g\cdot f$ is $A^t\big((P_iP_j)(f)\big)A$, where A is the matrix of the action of g^{-1} on $\mathbb{C}[\partial]_2 \cong \mathrm{Sym}^2\mathbb{C}^3$, and $\det A = \det(g^{-1})^4 = 1$. Hence $\det\mathrm{Cat}_2$ is an SL_3-invariant form of degree six on $\mathbb{C}[x,y,z]_4$, and it does not vanish on U^{cat}. Thus U^{cat} is an SL_3-invariant open subset of the GIT-semistable locus $\mathbb{P}(\mathbb{C}[x,y,z]_4)^{\mathrm{ss}}$.

Let Π' be as in (6), and identify $\mathrm{Sym}\,\Pi'$ with $\mathbb{C}[x,y,z]$ by a basis of Π'. By Lemma 16.3(1), $[f_{\Pi'}] \in U^{\mathrm{cat}}$. Since $[f_{\Pi'}]$ is GIT-stable, the orbit $\mathrm{PGL}_3\cdot[f_{\Pi'}] = \mathrm{SL}_3\cdot[f_{\Pi'}]$ is closed in $\mathbb{P}(\mathbb{C}[x,y,z]_4)^{\mathrm{ss}}$ [MFK94, Amplification 1.11], hence closed in U^{cat}.

Let Π_0 belong to the closure of $\mathrm{SL}(V)\cdot\Pi'$ in $\mathbb{B}^{\mathrm{nd}}$. Since $\mathrm{SL}(V) \to \mathrm{PGL}(V)$ is surjective, $\mathrm{SL}(V)\cdot\Pi' = \mathrm{PGL}(V)\cdot\Pi'$. By Lemma 16.3(5), $[f_{\Pi_0}]$ belongs to the closure in U^{cat} of $\mathrm{PGL}_3\cdot[f_{\Pi'}]$, which is $\mathrm{PGL}_3\cdot[f_{\Pi'}]$. Thus f_{Π_0} and $f_{\Pi'}$ are projectively equivalent, and Lemma 16.3(3) implies $\Pi_0 \in \mathrm{PGL}(V)\cdot\Pi' = \mathrm{SL}(V)\cdot\Pi'$. Hence $\mathrm{SL}(V)\cdot\Pi'$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

Since $[f_{\Pi'}]$ is GIT-stable, the stabilizer of $[f_{\Pi'}]$ in SL_3 is finite, and so is the stabilizer in $\mathrm{PGL}_3 = \mathrm{SL}_3/\mu_3$. By Lemma 16.3(3), $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi') \cong \mathrm{Stab}_{\mathrm{PGL}(\Pi'^\vee)}([f_{\Pi'}])$ is finite. Since $\mathrm{SL}(V) \to \mathrm{PGL}(V)$ has finite kernel μ_7, $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi')$ is finite. This proves (6).

Step 7. In this step we prove (4).

By (1), Π is the canonical Mukai net of X_Π under $V^\vee \cong H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$, so we may take $\Pi_{X_\Pi} = \Pi$ and $f_{X_\Pi} = f_\Pi$, and $Q_{X_\Pi} = (\mathrm{Sc}(f_\Pi) = 0)$. By Theorem 16.11(1), $\mathrm{Sc}(f_\Pi) \neq 0$. Assume that Q_{X_Π} is GIT-stable, that is, $[\mathrm{Sc}(f_\Pi)]$ is GIT-stable. Then $[f_\Pi]$ is GIT-stable by (5), and (6)

applied to $\Pi' = \Pi$ implies that $\mathrm{SL}(V)\cdot\Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite. Since $\Pi \in \Omega$, we obtain $\Pi \in \Omega^{\mathrm{st}}$. This proves (4). □

16.3. **Degenerations and torus actions.** In this subsection, we relate degenerations of prime Fano threefolds of genus 12 to the GIT of nets, and we determine the non-degenerate nets whose stabilizer contains a non-trivial torus. Proposition 16.13(4) shows that if the geometric generic fiber of such a degeneration is isomorphic to X_Π over $\overline{\mathbb{C}((\tau))}$ for some $\Pi \in \Omega$ whose orbit is closed in $\mathbb{B}^{\mathrm{nd}}$, then the central fiber is isomorphic to X_Π. It is used in the proofs of Lemma 17.3 and Theorem 19.1.

Let Y be a prime Fano threefold of genus 12 with factorial terminal singularities. A Mukai sheaf of type $(3,4)$ on Y [BKM26a, Definition 5.1] is a maximal Cohen–Macaulay sheaf $\mathcal{U}_Y$ on Y with $\mathrm{rank}\,\mathcal{U}_Y = 3$, $c_1(\mathcal{U}_Y) = K_Y$, $H^i(Y,\mathcal{U}_Y) = 0$ for all i, $\mathcal{U}_Y^\vee$ globally generated, $h^0(Y,\mathcal{U}_Y^\vee) = 7$, and $H^i(Y,\mathcal{U}_Y^\vee) = 0$ for $i > 0$. Such a sheaf exists and is unique [BKM26a, Theorem 5.3], and it is locally free and exceptional [BKM26b, Corollary 6.10], [BKM26a, Corollary 5.9].

Proposition 16.13. *Let $p : \mathcal{X} \to \mathrm{Spec}\,\mathbb{C}[[\tau]]$ be a flat projective Gorenstein morphism. Assume that any geometric fiber Y of p is a normal prime Fano threefold with factorial terminal singularities, $\mathrm{Pic}(Y) = \mathbb{Z}[-K_Y]$, and $(-K_Y)^3 = 22$, and that the geometric generic fiber of p is smooth. Then the following hold.*

(1) There exists a vector bundle $\mathcal{U}_\mathcal{X}$ of rank three on $\mathcal{X}$ such that $p_*(\mathcal{U}_\mathcal{X}^\vee)$ is a free $\mathbb{C}[[\tau]]$-module of rank seven and, for any geometric fiber Y, the restriction $\mathcal{U}_Y := \mathcal{U}_\mathcal{X}|_Y$ satisfies $\det\mathcal{U}_Y \cong \mathcal{O}_Y(K_Y)$, $H^i(Y,\mathcal{U}_Y) = 0$ for all i, $h^0(Y,\mathcal{U}_Y^\vee) = 7$, $H^i(Y,\mathcal{U}_Y^\vee) = 0$ for $i > 0$, and $\mathcal{U}_Y^\vee$ is globally generated. In particular, $\mathcal{U}_Y$ is the unique locally free Mukai sheaf of type $(3,4)$ on Y.

(2) Fix a basis of $p_*(\mathcal{U}_\mathcal{X}^\vee)$ and write $p_*(\mathcal{U}_\mathcal{X}^\vee) = V^\vee\otimes_\mathbb{C}\mathbb{C}[[\tau]]$. Then the module $\mathrm{Ker}\,\big(\Lambda^2 p_*(\mathcal{U}_\mathcal{X}^\vee) \to p_*(\Lambda^2\mathcal{U}_\mathcal{X}^\vee)\big)$ is a direct summand of rank three whose fiber at any geometric point is $\mathrm{Ker}\,\big(\Lambda^2 H^0(Y,\mathcal{U}_Y^\vee) \to H^0(Y,\Lambda^2\mathcal{U}_Y^\vee)\big)$, where Y is the fiber of p over the geometric point, and the classifying morphism $c : \mathrm{Spec}\,\mathbb{C}[[\tau]] \to \mathbb{B}$ of the summand factors through $\mathbb{B}^{\mathrm{nd}}$.

(3) The evaluation map induces an isomorphism j from $\mathcal{X}$ onto the closed subscheme of $\mathrm{Gr}(3,V)\times\mathrm{Spec}\,\mathbb{C}[[\tau]]$ of three-dimensional subspaces isotropic for any form of the summand in (2), and $\omega_{\mathcal{X}/\mathbb{C}[[\tau]]}^{-1} \cong j^*\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$, where $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ is the Plücker line bundle of $\mathrm{Gr}(3,V)$.

(4) Let $\Pi \in \Omega$. Assume that $\mathrm{SL}(V)\cdot\Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and that $\mathcal{X}_{\overline{\mathbb{C}((\tau))}} \cong (X_\Pi)_{\overline{\mathbb{C}((\tau))}}$. Then the central fiber of $\mathcal{X}$ is isomorphic to X_Π.

Proof. The proof is long, so we divide it into several steps. Let $\mathcal{X}_0$ be the central fiber of p, let $\mathbb{K} := \mathbb{C}((\tau))$, and let $\overline{\mathbb{K}}$ be an algebraic closure of $\mathbb{K}$.

Step 1. In this step, we show that on a geometric fiber Y of p, the kernel of the restriction map of 2-forms on the sections of the dual of the Mukai bundle is the net defining Y.

Let Y be a geometric fiber of p over an algebraically closed field $\mathbf{k} \in \{\mathbb{C}, \overline{\mathbb{K}}\}$. The results of [BKM26b] and [BKM26a] hold over any algebraically closed field of characteristic zero [BKM26b, Section 1.4], [BKM26a, Section 1.6], so they apply to Y. By [BKM26a, Theorem 5.3], [BKM26b, Corollary 6.10], and [BKM26a, Corollary 5.9], Y carries a unique Mukai sheaf $\mathcal{U}_Y$ of type $(3,4)$, and $\mathcal{U}_Y$ is locally free and exceptional. Let $W_Y := H^0(Y,\mathcal{U}_Y^\vee)$, so $\dim W_Y = 7$. By [BKM26b, Corollary 6.10, Lemma 4.2, and Definition 4.1], there exists a three-dimensional subspace $\Pi_Y \subset \Lambda^2 W_Y$ whose non-zero elements have rank 6 such that the evaluation map of $\mathcal{U}_Y^\vee$ induces an isomorphism of Y onto the scheme of three-dimensional subspaces of $W_Y^\vee$ isotropic for any form of Π_Y. Moreover, $H^i(Y,\mathcal{O}_Y) = 0$ for $i > 0$ by [BKM26b, Lemma 4.2]. Let

$$\rho_Y : \Lambda^2 W_Y \to H^0(Y,\Lambda^2\mathcal{U}_Y^\vee)$$

be the restriction map. We show that $\mathrm{Ker}\,\rho_Y = \Pi_Y$.

Since Y is contained in the zero scheme of any form of Π_Y, $\Pi_Y \subset \operatorname{Ker}\rho_Y$. By [BKM26a, Lemma 5.2], there exists a smooth K3 surface $S \in |-K_Y|$ with $\operatorname{Pic}(S) = \mathbb{Z}\cdot(-K_Y)|_S$ such that $\mathcal{U}_Y|_S$ is a Mukai bundle on S in the sense of [BKM26b, Definition 3.1]. Since $\mathcal{O}_Y(-S) \cong \omega_Y$, there is an exact sequence

$$0 \to \mathcal{U}_Y^\vee \otimes \omega_Y \to \mathcal{U}_Y^\vee \to \mathcal{U}_Y^\vee|_S \to 0,$$

and $H^i(Y, \mathcal{U}_Y^\vee \otimes \omega_Y) \cong H^{3-i}(Y, \mathcal{U}_Y)^\vee = 0$ for all i by Serre duality on the Gorenstein threefold Y [Har77, Theorem III.7.6 and Corollary III.7.7]. Thus the restriction map $W_Y \to H^0(S, \mathcal{U}_Y^\vee|_S)$ is an isomorphism, and the composition $S \to Y \to \operatorname{Gr}(3, W_Y^\vee)$ is the morphism defined by the evaluation map $W_Y \otimes \mathcal{O}_S \to \mathcal{U}_Y^\vee|_S$, that is, the Gushel embedding of S [BKM26b, (3.7) and Lemma 3.4].

Let $\omega \in \operatorname{Ker}\rho_Y$ be non-zero, and suppose that $\operatorname{rank}\omega \le 4$. Then $\omega = f_1 \wedge f_2 + f_3 \wedge f_4$ for some linearly independent $f_1, f_2, f_3, f_4 \in W_Y$, or $\omega = f_1 \wedge f_2$ for some $f_1, f_2 \in W_Y$, and in the second case we pick $f_3 \in W_Y \setminus \operatorname{Span}(f_1, f_2)$. In both cases

$$\omega \wedge f_3 = f_1 \wedge f_2 \wedge f_3 \neq 0.$$

Let $U \in S$, viewed as a three-dimensional subspace of $W_Y^\vee$. Since $\omega|_U = 0$, we have $(f_1 \wedge f_2 \wedge f_3)|_U = (\omega \wedge f_3)|_U = 0$, so $f_1|_U, f_2|_U, f_3|_U$ are linearly dependent, that is, $U \cap \{f_1 = f_2 = f_3 = 0\} \neq 0$. Since $\dim\{f_1 = f_2 = f_3 = 0\} = 4$, the subspaces $U \in \operatorname{Gr}(3, W_Y^\vee)$ with $U \cap \{f_1 = f_2 = f_3 = 0\} \neq 0$ form a Schubert divisor of $\operatorname{Gr}(3, W_Y^\vee)$. Thus S is contained in a Schubert divisor, which contradicts [BKM26b, Lemma 3.5] applied with $r = 3$ and $s = 4$, since $(S, (-K_Y)|_S)$ is a prime K3 surface of genus $12 = rs$. Therefore any non-zero element of $\operatorname{Ker}\rho_Y$ has rank 6. The forms of rank at most four form a closed subset of codimension three of $\mathbb{P}(\Lambda^2 W_Y)$ (Definition-Proposition 16.2), so any linear subspace of dimension three of $\mathbb{P}(\Lambda^2 W_Y)$ intersects it [Har77, Theorem I.7.2]. Hence $\dim\operatorname{Ker}\rho_Y \le 3$, and $\operatorname{Ker}\rho_Y = \Pi_Y$.

Step 2. In this step, we extend $\mathcal{U}_{\mathcal{X}_0}$ to a vector bundle $\mathcal{U}_{\mathcal{X}}$ of rank three on $\mathcal{X}$.

For $n \ge 0$, let $\mathcal{X}_n := \mathcal{X} \times_{\operatorname{Spec}\mathbb{C}[[\tau]]} \operatorname{Spec}\mathbb{C}[\tau]/(\tau^{n+1})$. Since $\mathcal{U}_{\mathcal{X}_0}$ is locally free and exceptional,

$$H^i(\mathcal{X}_0, \mathcal{E}nd(\mathcal{U}_{\mathcal{X}_0})) = \operatorname{Ext}^i(\mathcal{U}_{\mathcal{X}_0}, \mathcal{U}_{\mathcal{X}_0}) = 0 \text{ for } i > 0.$$

Let $\mathcal{U}_n$ be a vector bundle on $\mathcal{X}_n$ with $\mathcal{U}_n|_{\mathcal{X}_0} \cong \mathcal{U}_{\mathcal{X}_0}$. The kernel of $\mathbb{C}[\tau]/(\tau^{n+2}) \to \mathbb{C}[\tau]/(\tau^{n+1})$ is $(\tau^{n+1})/(\tau^{n+2}) \cong \mathbb{C}$, and by [Har10, Theorem 7.1] the obstruction to extending $\mathcal{U}_n$ to a vector bundle on $\mathcal{X}_{n+1}$ belongs to $H^2(\mathcal{X}_0, \mathcal{E}nd(\mathcal{U}_{\mathcal{X}_0})) = 0$. Thus there exist vector bundles $\mathcal{U}_n$ on $\mathcal{X}_n$ with $\mathcal{U}_{n+1}|_{\mathcal{X}_n} \cong \mathcal{U}_n$ for all $n \ge 0$. By the Grothendieck existence theorem [Sta, Tag 088E and Tag 0883], there exists a coherent sheaf $\mathcal{U}_{\mathcal{X}}$ on $\mathcal{X}$ with $\mathcal{U}_{\mathcal{X}}|_{\mathcal{X}_n} \cong \mathcal{U}_n$ for all n.

Let $\mathcal{T}$ be a scheme proper over $\operatorname{Spec}\mathbb{C}[[\tau]]$, such as $\mathcal{X}$ or a closed subscheme of $\operatorname{Gr}(3, V) \times \operatorname{Spec}\mathbb{C}[[\tau]]$. Since the image of a closed subset of $\mathcal{T}$ in $\operatorname{Spec}\mathbb{C}[[\tau]]$ is closed, any non-empty closed subset of $\mathcal{T}$ intersects the closed fiber of $\mathcal{T}$. In particular, by Nakayama's lemma, a coherent sheaf $\mathcal{F}$ on $\mathcal{T}$ with $\mathcal{F} \otimes_{\mathbb{C}[[\tau]]} \mathbb{C} = 0$ is zero.

Let $z \in \mathcal{X}_0$, and let M be the stalk of $\mathcal{U}_{\mathcal{X}}$ at z. For any n, $M/\tau^{n+1}M$ is a free $\mathcal{O}_{\mathcal{X},z}/(\tau^{n+1})$-module of rank three, and bases of the modules $M/\tau^{n+1}M$ lift compatibly by Nakayama's lemma. Thus the τ-adic completion of M is free of rank three over the τ-adic completion of $\mathcal{O}_{\mathcal{X},z}$, which is faithfully flat over $\mathcal{O}_{\mathcal{X},z}$ [Mat89, Theorem 8.14], so M is free of rank three. Since the set of points at which $\mathcal{U}_{\mathcal{X}}$ is not locally free of rank three is closed [Har77, Exercise II.5.7(a)] and does not intersect $\mathcal{X}_0$, $\mathcal{U}_{\mathcal{X}}$ is a vector bundle of rank three.

Step 3. In this step, we prove (1) and construct j.

Since p is projective and flat with Gorenstein, hence Cohen–Macaulay, fibers, the relative dualizing sheaf $\omega_{\mathcal{X}/\mathbb{C}[[\tau]]}$ is flat over $\mathbb{C}[[\tau]]$ and commutes with base change [Kol23, 2.68.2]. Its restriction to any fiber is the dualizing sheaf of the fiber, which is invertible. Thus $\omega_{\mathcal{X}/\mathbb{C}[[\tau]]}$ is invertible at the points of $\mathcal{X}_0$ by Nakayama's lemma and flatness over $\mathbb{C}[[\tau]]$, hence invertible by **Step 2**. Let $L := \det\mathcal{U}_{\mathcal{X}} \otimes \omega_{\mathcal{X}/\mathbb{C}[[\tau]]}^{-1}$. Then $L|_{\mathcal{X}_0} \cong \mathcal{O}_{\mathcal{X}_0}$, since $\det\mathcal{U}_{\mathcal{X}_0} \cong \omega_{\mathcal{X}_0}$. For $n \ge 0$, there is an exact sequence

$$0 \to \mathcal{O}_{\mathcal{X}_0} \to \mathcal{O}_{\mathcal{X}_{n+1}}^* \to \mathcal{O}_{\mathcal{X}_n}^* \to 1,$$

where the first map is $a \to 1+\tau^{n+1}a$. Since $H^1(\mathcal{X}_0, \mathcal{O}_{\mathcal{X}_0}) = 0$ (**Step 1**), $\operatorname{Pic}(\mathcal{X}_{n+1}) \to \operatorname{Pic}(\mathcal{X}_n)$ is injective and $H^0(\mathcal{X}_{n+1}, \mathcal{O}^*_{\mathcal{X}_{n+1}}) \to H^0(\mathcal{X}_n, \mathcal{O}^*_{\mathcal{X}_n})$ is surjective. Thus $L|_{\mathcal{X}_n}$ has trivializations for all n that are compatible under restriction. Since completion is fully faithful on coherent sheaves on $\mathcal{X}$ [Sta, Tag 0883], $L \cong \mathcal{O}_{\mathcal{X}}$, that is,

$$\det \mathcal{U}_{\mathcal{X}} \cong \omega_{\mathcal{X}/\mathbb{C}[[\tau]]}.$$

Since $\mathcal{U}_{\mathcal{X}}$ is locally free and $\mathcal{X}$ is flat over $\mathbb{C}[[\tau]]$, the sheaves $\mathcal{U}_{\mathcal{X}}$ and $\mathcal{U}_{\mathcal{X}}^\vee$ are flat over $\mathbb{C}[[\tau]]$. By **Step 1** and the definition of a Mukai sheaf, $H^i(\mathcal{X}_0, \mathcal{U}_{\mathcal{X}_0}) = 0$ for all i, $H^i(\mathcal{X}_0, \mathcal{U}_{\mathcal{X}_0}^\vee) = 0$ for $i > 0$, and $h^0(\mathcal{X}_0, \mathcal{U}_{\mathcal{X}_0}^\vee) = 7$. By cohomology and base change [Har77, Theorem III.12.11],

$$R^i p_* \mathcal{U}_{\mathcal{X}} = 0 \text{ for all } i, \quad R^i p_*(\mathcal{U}_{\mathcal{X}}^\vee) = 0 \text{ for } i > 0,$$

and $p_*(\mathcal{U}_{\mathcal{X}}^\vee)$ is a free $\mathbb{C}[[\tau]]$-module of rank seven whose formation commutes with base change. In particular, $H^i(Y, \mathcal{U}_Y) = 0$ for all i, $H^i(Y, \mathcal{U}_Y^\vee) = 0$ for $i > 0$, and $h^0(Y, \mathcal{U}_Y^\vee) = 7$ for the geometric generic fiber Y. The cokernel of the evaluation map $p^* p_*(\mathcal{U}_{\mathcal{X}}^\vee) \to \mathcal{U}_{\mathcal{X}}^\vee$ restricts on $\mathcal{X}_0$ to the cokernel of $H^0(\mathcal{X}_0, \mathcal{U}_{\mathcal{X}_0}^\vee) \otimes \mathcal{O}_{\mathcal{X}_0} \to \mathcal{U}_{\mathcal{X}_0}^\vee$, which is zero, so the evaluation map is surjective (**Step 2**). Hence for any geometric fiber Y, $\mathcal{U}_Y$ satisfies the properties in (1), where $\det \mathcal{U}_Y \cong \mathcal{O}_Y(K_Y)$ follows from $\det \mathcal{U}_{\mathcal{X}} \cong \omega_{\mathcal{X}/\mathbb{C}[[\tau]]}$. Since Y is Gorenstein and $\mathcal{U}_Y$ is locally free, $\mathcal{U}_Y$ is maximal Cohen–Macaulay, so $\mathcal{U}_Y$ is a Mukai sheaf of type $(3, 4)$ [BKM26a, Definition 5.1], and it is the unique one by [BKM26a, Theorem 5.3]. This proves (1).

Fix a basis as in (2), and write $p_*(\mathcal{U}_{\mathcal{X}}^\vee) = V^\vee \otimes_{\mathbb{C}} \mathbb{C}[[\tau]]$. Since the evaluation map $V^\vee \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{X}} \to \mathcal{U}_{\mathcal{X}}^\vee$ is surjective, the dual of the evaluation map realizes $\mathcal{U}_{\mathcal{X}}$ as a subbundle of rank three of $V \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{X}}$. Thus there exists a morphism $j : \mathcal{X} \to \operatorname{Gr}(3, V) \times \operatorname{Spec} \mathbb{C}[[\tau]]$ over $\operatorname{Spec} \mathbb{C}[[\tau]]$ such that the pullback of the tautological subbundle $\mathcal{U}$ is $\mathcal{U}_{\mathcal{X}}$, and

$$j^* \mathcal{O}_{\operatorname{Gr}(3,V)}(1) \cong \det \mathcal{U}_{\mathcal{X}}^\vee \cong \omega_{\mathcal{X}/\mathbb{C}[[\tau]]}^{-1}.$$

On a geometric fiber Y, the restriction of j is the morphism of **Step 1** defined by the evaluation map of $\mathcal{U}_Y^\vee$ and the basis of $W_Y = V^\vee \otimes_{\mathbb{C}} \mathbf{k}$.

Step 4. In this step, we prove (2) and (3).

Let

$$\rho : \Lambda^2 p_*(\mathcal{U}_{\mathcal{X}}^\vee) \to p_*(\Lambda^2 \mathcal{U}_{\mathcal{X}}^\vee) = H^0(\mathcal{X}, \Lambda^2 \mathcal{U}_{\mathcal{X}}^\vee)$$

be the restriction map, and let $\Pi_{\mathcal{X}} := \operatorname{Ker} \rho$. Since $\mathcal{X}$ is flat over $\mathbb{C}[[\tau]]$, multiplication by τ is injective on the vector bundle $\Lambda^2 \mathcal{U}_{\mathcal{X}}^\vee$, hence on $H^0(\mathcal{X}, \Lambda^2 \mathcal{U}_{\mathcal{X}}^\vee)$. Thus $\Lambda^2 p_*(\mathcal{U}_{\mathcal{X}}^\vee)/\Pi_{\mathcal{X}}$ is a finitely generated torsion-free module over the principal ideal domain $\mathbb{C}[[\tau]]$, hence free [Lan02, Theorem III.7.3], and $\Pi_{\mathcal{X}}$ is a direct summand of $\Lambda^2 p_*(\mathcal{U}_{\mathcal{X}}^\vee)$.

Let $Y := \mathcal{X}_{\overline{\mathbb{K}}}$. Since $\overline{\mathbb{K}}$ is flat over $\mathbb{C}[[\tau]]$, we have $\Pi_{\mathcal{X}} \otimes \overline{\mathbb{K}} = \operatorname{Ker}(\rho \otimes \overline{\mathbb{K}})$, and $\rho \otimes \overline{\mathbb{K}} = \rho_Y$ by flat base change [Har77, Proposition III.9.3] and **Step 3**. By **Step 1**, $\Pi_{\mathcal{X}}$ is of rank three, and the fiber of $\Pi_{\mathcal{X}}$ at the geometric generic point is $\operatorname{Ker} \rho_Y$. Since $\Pi_{\mathcal{X}}$ is a direct summand, $\Pi_{\mathcal{X}} \otimes_{\mathbb{C}[[\tau]]} \mathbb{C} \to \Lambda^2 W_{\mathcal{X}_0}$ is injective, and the image of the map is contained in $\operatorname{Ker} \rho_{\mathcal{X}_0}$, since $\rho_{\mathcal{X}_0}$ is the reduction of ρ modulo τ followed by restriction to $\mathcal{X}_0$. Since $\dim \operatorname{Ker} \rho_{\mathcal{X}_0} = 3$ by **Step 1**, the fiber of $\Pi_{\mathcal{X}}$ at the closed point is $\operatorname{Ker} \rho_{\mathcal{X}_0} = \Pi_{\mathcal{X}_0}$. Let $c : \operatorname{Spec} \mathbb{C}[[\tau]] \to \mathbb{B}$ be the classifying morphism of $\Pi_{\mathcal{X}} \subset \Lambda^2 V^\vee \otimes_{\mathbb{C}} \mathbb{C}[[\tau]]$. By **Step 1**, the image of the closed point is $\Pi_{\mathcal{X}_0} \in \mathbb{B}^{\mathrm{nd}}$. Since $c^{-1}(\mathbb{B}^{\mathrm{nd}})$ is open and contains the closed point, $c^{-1}(\mathbb{B}^{\mathrm{nd}}) = \operatorname{Spec} \mathbb{C}[[\tau]]$. This proves (2).

Let $\mathcal{Z} \subset \operatorname{Gr}(3, V) \times \operatorname{Spec} \mathbb{C}[[\tau]]$ be the closed subscheme of three-dimensional subspaces isotropic for any form of $\Pi_{\mathcal{X}}$. Since $\rho(\Pi_{\mathcal{X}}) = 0$ and $j^* \mathcal{U} = \mathcal{U}_{\mathcal{X}}$, j factors through a morphism $j : \mathcal{X} \to \mathcal{Z}$. By **Step 1** and **Step 3** and the computation of the fibers of $\Pi_{\mathcal{X}}$, the restrictions $j_0 : \mathcal{X}_0 \to \mathcal{Z}_0$ and $j_{\overline{\mathbb{K}}} : \mathcal{X}_{\overline{\mathbb{K}}} \to \mathcal{Z}_{\overline{\mathbb{K}}}$ of j are isomorphisms. Thus j is quasi-finite and projective, hence finite [Har77, Exercise III.11.2]. Let Q be the cokernel and I the kernel of $\mathcal{O}_{\mathcal{Z}} \to j_* \mathcal{O}_{\mathcal{X}}$. Since j is affine and $\mathcal{X}$ is flat over $\mathbb{C}[[\tau]]$, $j_* \mathcal{O}_{\mathcal{X}}$ is flat over $\mathbb{C}[[\tau]]$ and $(j_* \mathcal{O}_{\mathcal{X}}) \otimes_{\mathbb{C}[[\tau]]} \mathbb{C} = j_{0*} \mathcal{O}_{\mathcal{X}_0}$. Since j_0 is an isomorphism, $Q \otimes_{\mathbb{C}[[\tau]]} \mathbb{C} = 0$, so $Q = 0$ (**Step 2**). Since $j_* \mathcal{O}_{\mathcal{X}}$ is flat over $\mathbb{C}[[\tau]]$, the exact sequence $0 \to I \to \mathcal{O}_{\mathcal{Z}} \to j_* \mathcal{O}_{\mathcal{X}} \to 0$ remains exact after $\otimes_{\mathbb{C}[[\tau]]} \mathbb{C}$, so $I \otimes_{\mathbb{C}[[\tau]]} \mathbb{C} = 0$, and $I = 0$

(**Step 2**). Hence $j : \mathcal{X} \to \mathcal{Z}$ is an isomorphism, and $\omega^{-1}_{\mathcal{X}/\mathbb{C}[[\tau]]} \cong j^*\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ by **Step 3**. This proves (3).

Step 5. In this step, we prove (4).

Let $\Pi \in \Omega$ satisfy the assumptions of (4), and let $Y := \mathcal{X}_{\overline{\mathbb{K}}}$. By Theorem 16.12(1), $\mathcal{U}|_{X_\Pi}$ is the Mukai bundle of X_Π, $V^\vee \to H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$ is an isomorphism, and Π is the kernel of $\Lambda^2 V^\vee \to H^0(X_\Pi, \Lambda^2\mathcal{U}^\vee|_{X_\Pi})$. By flat base change, $\mathcal{U}|_{(X_\Pi)_{\overline{\mathbb{K}}}}$ is the Mukai bundle of $(X_\Pi)_{\overline{\mathbb{K}}}$, the map $V^\vee \otimes_{\mathbb{C}} \overline{\mathbb{K}} \to H^0((X_\Pi)_{\overline{\mathbb{K}}}, \mathcal{U}^\vee)$ is an isomorphism, and $\Pi \otimes_{\mathbb{C}} \overline{\mathbb{K}}$ is the kernel of $\Lambda^2 V^\vee \otimes_{\mathbb{C}} \overline{\mathbb{K}} \to H^0((X_\Pi)_{\overline{\mathbb{K}}}, \Lambda^2\mathcal{U}^\vee)$. Let $\phi : Y \to (X_\Pi)_{\overline{\mathbb{K}}}$ be an isomorphism. By the uniqueness of the Mukai sheaf, $\phi^*(\mathcal{U}|_{(X_\Pi)_{\overline{\mathbb{K}}}}) \cong \mathcal{U}_Y$, and an isomorphism $\phi^*(\mathcal{U}|_{(X_\Pi)_{\overline{\mathbb{K}}}}) \cong \mathcal{U}_Y$ induces a linear isomorphism $H^0((X_\Pi)_{\overline{\mathbb{K}}}, \mathcal{U}^\vee) \to H^0(Y, \mathcal{U}_Y^\vee)$ that maps the kernel of the restriction map of 2-forms onto $\operatorname{Ker}\rho_Y$. In the bases of **Step 3**, the linear isomorphism is given by an element $g \in \mathrm{GL}(V)(\overline{\mathbb{K}})$ with $g \cdot \Pi = c(\overline{\eta})$, where $\overline{\eta}$ is the geometric generic point of $\operatorname{Spec}\mathbb{C}[[\tau]]$ (**Step 4**). Since scalar matrices act trivially on $\mathbb{B}$ and $\overline{\mathbb{K}}$ is algebraically closed, we may assume that $g \in \mathrm{SL}(V)(\overline{\mathbb{K}})$.

We regard $\mathrm{SL}(V) \cdot \Pi$ as a closed subscheme of $\mathbb{B}^{\mathrm{nd}}$ with the reduced structure. Since $\mathrm{SL}(V)$ is reduced, the orbit morphism $\mathrm{SL}(V) \to \mathbb{B}^{\mathrm{nd}}$, $g' \to g' \cdot \Pi$, factors through $\mathrm{SL}(V) \cdot \Pi$, so $c(\overline{\eta}) = g \cdot \Pi$ is a $\overline{\mathbb{K}}$-point of $\mathrm{SL}(V) \cdot \Pi$. Let $\mathfrak{a} \subset \mathbb{C}[[\tau]]$ be the ideal of the closed subscheme $c^{-1}(\mathrm{SL}(V) \cdot \Pi)$. Then the image of $\mathfrak{a}$ in $\overline{\mathbb{K}}$ is zero, and since $\mathbb{C}[[\tau]] \to \overline{\mathbb{K}}$ is injective, $\mathfrak{a} = 0$. Thus c factors through $\mathrm{SL}(V) \cdot \Pi$, so c maps the closed point of $\operatorname{Spec}\mathbb{C}[[\tau]]$ to $g_0 \cdot \Pi$ for some $g_0 \in \mathrm{SL}(V)$. By (3), $\mathcal{X}_0 \cong X_{g_0 \cdot \Pi} = g_0 \cdot X_\Pi \cong X_\Pi$. □

Next we determine the non-degenerate nets whose stabilizer contains a non-trivial torus, compute their apolar quartics, and decide which of the nets define smooth threefolds. The normalization in Definition-Proposition 16.14(6) is the one we use later.

Definition-Proposition 16.14. Fix a basis $e_0, \dots, e_6$ of V with dual basis $e_0^\vee, \dots, e_6^\vee$. For $h \in \mathbb{C}\backslash\{1\}$, let $k := h - 1$, and let $\Pi_h \subset \Lambda^2 V^\vee$ be the span of

$$\begin{aligned} x &:= k(e_0^\vee \wedge e_1^\vee + e_2^\vee \wedge e_3^\vee) + e_5^\vee \wedge e_6^\vee, \\ y &:= -ke_0^\vee \wedge e_3^\vee + ke_1^\vee \wedge e_5^\vee - e_4^\vee \wedge e_6^\vee, \\ z &:= -ke_0^\vee \wedge e_6^\vee + ke_1^\vee \wedge e_2^\vee + e_4^\vee \wedge e_5^\vee. \end{aligned}$$

Then the following hold.

(1) For any $h \in \mathbb{C}\backslash\{1\}$, $\Pi_h \in \mathbb{B}^{\mathrm{nd}}$, x, y, z is a basis of Π_h, and in the basis x, y, z the apolar quartic of Π_h is the quartic $F_h = \frac{h}{6}x^4 + 2x^2yz + y^2z^2$ of Definition 2.27, viewed in $\operatorname{Sym}^4 \Pi_h$.

(2) Any $\Pi \in \mathbb{B}^{\mathrm{nd}}$ whose stabilizer in $\mathrm{PGL}(V)$ contains a non-trivial torus is $\mathrm{PGL}(V)$-conjugate to Π_h for exactly one $h \in \mathbb{C}\backslash\{1\}$.

(3) The projective stabilizers $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi_h)$ and $\mathrm{Stab}_{\mathrm{PGL}(\Pi_h^\vee)}([F_h])$ are both isomorphic to $\mathbb{G}_m \rtimes \mu_2$, with the involution acting by inversion, if $h \neq 6$, and to PGL_2 if $h = 6$.

(4) The scheme X_{Π_h} is smooth if and only if $h \neq 2$. For $h \notin \{1, 2\}$, X_{Π_h} is a smooth prime Fano threefold of genus 12, and $X_{\Pi_6} \cong X^{\mathrm{MU}}$.

(5) The scheme X_{Π_2} has an affine open subscheme isomorphic to $\operatorname{Spec}\mathbb{C}[a, b, c, d]/(bd - a^2)$, and X_{Π_2} is singular.

(6) There exists a basis $e'_{-3}, \dots, e'_3$ of V in which the net Π_h is spanned by

$$\begin{aligned} &e'^\vee_{-3} \wedge e'^\vee_2 + e'^\vee_{-2} \wedge e'^\vee_1 + e'^\vee_{-1} \wedge e'^\vee_0, \\ &e'^\vee_{-3} \wedge e'^\vee_3 + e'^\vee_{-2} \wedge e'^\vee_2 + ve'^\vee_{-1} \wedge e'^\vee_1, \\ &e'^\vee_{-2} \wedge e'^\vee_3 + e'^\vee_{-1} \wedge e'^\vee_2 + e'^\vee_0 \wedge e'^\vee_1, \end{aligned}$$

where $v = 1/(h-1)$. Here v ranges over $\mathbb{C}\backslash\{0\}$, X_{Π_h} is smooth if and only if $v \neq 1$, and $X_{\Pi_h} \cong X^{\mathrm{MU}}$ for $v = 1/5$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

Let $\partial_x, \partial_y, \partial_z$ be the basis of $\Pi_h^\vee$ dual to x, y, z, so that $\operatorname{Sym}\Pi_h^\vee = \mathbb{C}[\partial]$ and $\operatorname{Sym}\Pi_h = \mathbb{C}[x, y, z]$. The forms x, y, z are linearly independent, since no $e_i^\vee \wedge e_j^\vee$ occurs in two of them. The alternating matrix of Π_h, with entries in $\Pi_h^\vee$, is

$$M = \begin{pmatrix} 0 & k\partial_x & 0 & -k\partial_y & 0 & 0 & -k\partial_z \\ -k\partial_x & 0 & k\partial_z & 0 & 0 & k\partial_y & 0 \\ 0 & -k\partial_z & 0 & k\partial_x & 0 & 0 & 0 \\ k\partial_y & 0 & -k\partial_x & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \partial_z & -\partial_y \\ 0 & -k\partial_y & 0 & 0 & -\partial_z & 0 & \partial_x \\ k\partial_z & 0 & 0 & 0 & \partial_y & -\partial_x & 0 \end{pmatrix},$$

with rows and columns numbered from 0 to 6. Let

$$g = (g_0, \dots, g_6)^T := \left(\partial_x\partial_y^2,\ \partial_x\partial_z^2,\ \partial_y^3,\ \partial_z^3,\ \partial_x^3 - h\partial_x\partial_y\partial_z,\ \partial_x^2\partial_y - \partial_y^2\partial_z,\ \partial_x^2\partial_z - \partial_y\partial_z^2\right)^T.$$

A direct calculation shows $Mg = 0$, and the signed principal Pfaffians $(-1)^i \operatorname{Pf}(M_i)$ of M are $k^2 g_i$, where M_i denotes the 6×6 submatrix of M with the i-th row and the i-th column deleted.

An alternating 7×7 matrix has rank at most four if and only if the seven principal 6×6 Pfaffians of the matrix vanish. Let $\omega = a_x x + a_y y + a_z z \in \Pi_h$ with $g_i(a_x, a_y, a_z) = 0$ for all i. Then $g_2 = \partial_y^3$ and $g_3 = \partial_z^3$ imply $a_y = a_z = 0$, and g_4 implies $a_x^3 = 0$, so $\omega = 0$. Since $k \neq 0$, any non-zero $\omega \in \Pi_h$ has rank 6, and $\Pi_h \in \mathbb{B}^{\mathrm{nd}}$.

Each g_i annihilates F_h. For example,

$$(\partial_x^3 - h\partial_x\partial_y\partial_z)(F_h) = 4hx - 4hx = 0, \quad (\partial_x^2\partial_y - \partial_y^2\partial_z)(F_h) = 4z - 4z = 0.$$

By Lemma 16.3(1), the ideal I_{Π_h} generated by $k^2 g_0, \dots, k^2 g_6$ satisfies $\dim(\mathbb{C}[\partial]/I_{\Pi_h})_4 = 1$. Thus the quartics annihilated by $(I_{\Pi_h})_4$ form a one-dimensional space, which contains F_h and, by Definition-Proposition 2.28, is spanned by f_{Π_h}. Hence f_{Π_h} is a non-zero multiple of F_h.

Step 2. In this step, we prove (2) and (3).

Let $\Pi \in \mathbb{B}^{\mathrm{nd}}$ whose stabilizer in $\mathrm{PGL}(V)$ contains a non-trivial torus. By Lemma 16.3(3), the stabilizer of $[f_\Pi]$ in $\mathrm{PGL}(\Pi^\vee)$, which acts on $\operatorname{Sym}\Pi$ through the contragredient action, contains a non-trivial torus. A non-trivial one-parameter subgroup of the torus lifts, after replacing it with a positive power, to a one-parameter subgroup λ of $\mathrm{SL}(\Pi)$, and λ is diagonalizable. Thus there exist a basis ℓ_1, ℓ_2, ℓ_3 of Π, integers $n_1 \geq n_2 \geq n_3$, not all zero, with $n_1 + n_2 + n_3 = 0$, and $q \in \mathbb{Z}$ such that $\lambda(t)\ell_i = t^{n_i}\ell_i$ and $\lambda(t) f_\Pi = t^q f_\Pi$ for $t \in \mathbb{G}_m$.

Let λ act on $\mathbb{C}[\partial] = \operatorname{Sym}\Pi^\vee$ through the contragredient action, so that $(\lambda(t)P)(\lambda(t)f) = \lambda(t)(P(f))$ for $P \in \mathbb{C}[\partial]$ and $f \in \operatorname{Sym}\Pi$. For $P, P' \in \mathbb{C}[\partial]_2$, $(PP')(f_\Pi)$ is a constant, so

$$\big((\lambda(t)P)(\lambda(t)P')\big)(f_\Pi) = t^{-q}(PP')(f_\Pi).$$

If P and P' are monomials in the basis of $\Pi^\vee$ dual to ℓ_1, ℓ_2, ℓ_3, of weights w and w', then the left-hand side is $t^{w+w'}(PP')(f_\Pi)$. Hence $(PP')(f_\Pi) = 0$ unless $w + w' = -q$. Since f_Π is non-degenerate (Definition-Proposition 2.28), the bilinear form $(P, P') \to (PP')(f_\Pi)$ on $\mathbb{C}[\partial]_2$ is non-degenerate, so $w \to -q - w$ preserves the multiset of weights

$$\{-2n_1, -2n_2, -2n_3, -n_1 - n_2, -n_1 - n_3, -n_2 - n_3\}.$$

The sum of the six weights is $-4(n_1 + n_2 + n_3) = 0$ and also equals $-3q$, so $q = 0$. The largest weight is $-2n_3$ and the smallest is $-2n_1$, so $-2n_3 - 2n_1 = -q = 0$. Thus $n_3 = -n_1$, $n_2 = 0$, and $n := n_1 > 0$. Let $(x', y', z') := (\ell_2, \ell_1, \ell_3)$, of weights $0, n, -n$. Since $q = 0$, f_Π is a linear combination of the monomials $x'^a y'^b z'^c$ with $a + b + c = 4$ and $n(b - c) = 0$, that is,

$$f_\Pi = Ax'^4 + Bx'^2 y' z' + Cy'^2 z'^2$$

for some $A, B, C \in \mathbb{C}$. In the basis $(\partial_{x'}^2, \partial_{y'}^2, \partial_{z'}^2, \partial_{x'}\partial_{y'}, \partial_{x'}\partial_{z'}, \partial_{y'}\partial_{z'})$, where $\partial_{x'}, \partial_{y'}, \partial_{z'}$ is the basis of $\Pi^\vee$ dual to x', y', z', the non-zero entries of the symmetric matrix $((P_iP_j)(f_\Pi))$ are

$$(1,1) : 24A, \quad (1,6),(6,1) : 2B, \quad (2,3),(3,2) : 4C, \quad (4,5),(5,4) : 2B, \quad (6,6) : 4C,$$

so

$$\det \mathrm{Cat}_2(f_\Pi) = 256B^2C^2(24AC - B^2).$$

Since $\det \mathrm{Cat}_2(f_\Pi) \neq 0$, we have $B \neq 0$ and $C \neq 0$. The substitution $y' \to \alpha y'$ changes (A, B, C) to $(A, \alpha B, \alpha^2 C)$, the substitution $x' \to \beta x'$ changes (A, B, C) to $(\beta^4 A, \beta^2 B, C)$, and $24AC/B^2$ is invariant under both substitutions and under scaling f_Π. Possibly rescaling x' and y' and multiplying f_Π by a non-zero constant, we may assume that $C = 1$ and $B = 2$. Then $A = h/6$ for $h := 24AC/B^2$, and $\det \mathrm{Cat}_2(f_\Pi) = 4096(h-1) \neq 0$, so $h \neq 1$ and $f_\Pi = F_h(x', y', z')$. By **Step 1** and Lemma 16.3(3), $\Pi \in \mathrm{PGL}(V) \cdot \Pi_h$. If $\Pi_h \in \mathrm{PGL}(V) \cdot \Pi_{h'}$ for some $h' \in \mathbb{C}\backslash\{1\}$, then F_h and $F_{h'}$ are projectively equivalent by Lemma 16.3(3) and **Step 1**, so $h = h'$ by Proposition 16.16. This proves (2).

By **Step 1** and Lemma 16.3(3),

$$\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi_h) \cong \mathrm{Stab}_{\mathrm{PGL}(\Pi_h^\vee)}([F_h]).$$

By Proposition 16.16, $\mathrm{Stab}_{\mathrm{PGL}(\Pi_h^\vee)}([F_h])$ is PGL_2 if $h = 6$, and for $h \neq 6$ it is $\mathbb{G}_m \rtimes \mu_2$, where $\mathbb{G}_m$ acts by $\mathrm{diag}(1, s, s^{-1})$ and the involution exchanges y and z, hence acts on $\mathbb{G}_m$ by $s \to s^{-1}$. This proves (3).

Step 3. In this step, we show that X_{Π_h} is smooth for $h \notin \{1, 2\}$.

Let $\mathbb{G}_m$ act on V by $e_i \to t^{w_i} e_i$ with

$$(w_0, \dots, w_6) = (2, -2, 3, -3, 0, 1, -1).$$

Then $\mathbb{G}_m$ acts on $e_i^\vee \wedge e_j^\vee$ with weight $-w_i - w_j$, and x, y, z are weight vectors of weights $0, 1, -1$. Thus $\mathbb{G}_m$ preserves Π_h and X_{Π_h}. Since $w_0, \dots, w_6$ are distinct, the $\mathbb{G}_m$-fixed points of $\mathrm{Gr}(3, V)$ are the subspaces $\langle e_{i_0}, e_{i_1}, e_{i_2}\rangle$, $i_0 < i_1 < i_2$. Since $k \neq 0$, such a subspace belongs to X_{Π_h} if and only if $\{i_0, i_1, i_2\}$ contains none of the pairs

$$\{0,1\}, \{2,3\}, \{5,6\}, \{0,3\}, \{1,5\}, \{4,6\}, \{0,6\}, \{1,2\}, \{4,5\},$$

that is, if and only if $(i_0, i_1, i_2) \in \{(0,2,4), (0,2,5), (1,3,4), (1,3,6)\}$.

For $(i_0, i_1, i_2) \in \{(0,2,4), (0,2,5), (1,3,4), (1,3,6)\}$, consider the affine chart of $\mathrm{Gr}(3, V)$ with coordinates $\xi_{\alpha,j}$, $\alpha \in \{0,1,2\}$, $j \notin \{i_0, i_1, i_2\}$, whose points are the spans of

$$r_\alpha := e_{i_\alpha} + \sum_{j \notin \{i_0, i_1, i_2\}} \xi_{\alpha,j} e_j, \quad \alpha = 0, 1, 2.$$

In the chart, X_{Π_h} is defined by the nine equations $\omega(r_\alpha, r_\beta) = 0$, $\omega \in \{x, y, z\}$, $0 \leq \alpha < \beta \leq 2$. We order the twelve columns of the Jacobian matrix by α and then by increasing j, numbered from 0 to 11, and the nine rows by ω in the order x, y, z and then by (α, β) in the order $(0,1), (0,2), (1,2)$. At the origin of the chart,

$$\frac{\partial \omega(r_\alpha, r_\beta)}{\partial \xi_{\gamma,j}} = \delta_{\alpha\gamma}\omega(e_j, e_{i_\beta}) + \delta_{\beta\gamma}\omega(e_{i_\alpha}, e_j). \tag{16.12}$$

The 9×9 minors in Table 26 are computed directly from (16.12). For $h \notin \{1, 2\}$, they are non-zero, so the Jacobian matrix has rank 9 at the four $\mathbb{G}_m$-fixed points of X_{Π_h}.

Since X_{Π_h} is locally defined by nine equations in a smooth variety of dimension 12, $\dim_p X_{\Pi_h} \geq 3$ for any $p \in X_{\Pi_h}$. Hence X_{Π_h} is smooth of dimension three at p if and only if $\dim T_p X_{\Pi_h} = 3$. The set of $p \in X_{\Pi_h}$ with $\dim T_p X_{\Pi_h} > 3$ is closed and $\mathbb{G}_m$-invariant. If the set is non-empty, then it contains a $\mathbb{G}_m$-fixed point by the Borel fixed point theorem [Bor91, Theorem III.10.4], since X_{Π_h} is projective. For $h \notin \{1, 2\}$, $\dim T_p X_{\Pi_h} = 12 - 9 = 3$ at the four $\mathbb{G}_m$-fixed points by the Jacobian criterion. Thus X_{Π_h} is smooth of dimension three, and $\Pi_h \in \Omega$ by Proposition 16.5. That is, X_{Π_h} is a smooth prime Fano threefold of genus 12.

TABLE 26. The 9×9 minors of the Jacobian matrix of X_{Π_h} at the four torus-fixed points, with the chart columns numbered from 0.

(i_0,i_1,i_2)	columns	determinant
$(0,2,4)$	$0,1,2,3,5,6,7,8,9$	$-k^5$
$(0,2,5)$	$0,1,2,3,4,5,6,7,9$	$-(h-2)k^5$
$(1,3,4)$	$0,1,2,3,5,6,7,8,9$	$-k^5$
$(1,3,6)$	$0,1,2,3,4,5,6,7,9$	$-(h-2)k^5$

Step 4. In this step, we prove (5).

Let $h=2$, and let $\xi_0,\dots,\xi_{11}$ be the coordinates of the chart of **Step 3** for $(i_0,i_1,i_2)=(0,2,5)$, numbered in the column order of **Step 3**. An explicit calculation of the nine equations $\omega(r_\alpha,r_\beta)=0$ shows that the nine equations of X_{Π_2} in the chart are equivalent to the eight relations

$$\xi_0=\xi_9=\xi_7,\quad \xi_6=\xi_8=\xi_3,\quad \xi_{11}=\xi_2,\quad \xi_1=\xi_4=\xi_2\xi_3-\xi_7\xi_{10},\quad \xi_5=\xi_3^2-\xi_2\xi_7,$$

together with

$$\xi_3\xi_{10}-\xi_2^2=0.$$

Let $a:=\xi_2$, $b:=\xi_3$, $c:=\xi_7$, and $d:=\xi_{10}$. Then the intersection of X_{Π_2} with the chart is isomorphic to $\operatorname{Spec}\mathbb{C}[a,b,c,d]/(bd-a^2)$, which is singular along $(a=b=d=0)$. This proves (5). With **Step 3**, X_{Π_h} is smooth if and only if $h\neq 2$.

Step 5. In this step, we prove $X_{\Pi_6}\cong X^{\mathrm{MU}}$ and (6).

By **Step 3**, $\Pi_6\in\Omega$, and by Theorem 16.12(1), Π_6 is the canonical Mukai net of X_{Π_6}. Thus $f_{X_{\Pi_6}}$ is a non-zero multiple of $F_6=(x^2+yz)^2$ by **Step 1**, and $X_{\Pi_6}\cong \mathrm{VSP}(F_6,6)=\mathrm{VPS}(F_6,6)$ by Proposition 16.10. Since F_6 is a double smooth conic, $\mathrm{VSP}(F_6,6)\cong X^{\mathrm{MU}}$ by [Muk92, §7, Theorem 14] and [DKK15, Section 2.1]. This completes the proof of (4).

Let

$$(e'_{-3},e'_{-2},e'_{-1},e'_0,e'_1,e'_2,e'_3):=(e_3,e_1,e_6,ke_4,e_5,e_0,e_2).$$

Then $e'^{\vee}_0=k^{-1}e^{\vee}_4$, and $e'^{\vee}_m$ is the corresponding $e^{\vee}_i$ for $m\neq 0$. A direct calculation shows that $k^{-1}y$, $-k^{-1}x$, and $k^{-1}z$ are the three forms in (6) with $v=1/k=1/(h-1)$. Since $h\in\mathbb{C}\backslash\{1\}$, we have $v\in\mathbb{C}\backslash\{0\}$, $h=2$ if and only if $v=1$, and $h=6$ if and only if $v=1/5$. With (4), this proves (6). □

Remark 16.15. By Proposition 16.16(2) and (3), the quartics F_h, $h\in\mathbb{C}\backslash\{1\}$, are, up to projective equivalence and non-zero constants, the GIT-polystable points of U^{cat} that are not GIT-stable. The smooth prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action are classified in [KPS18, Theorem 5.3.10 and Proposition 5.4.6] and [KP18, Theorem 1.3]. Their plane quartics are described in [DKK15, Section 2.1]. In [KP18, Theorem 1.3], the prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action are parametrized by $u\in\mathbb{P}^1\backslash\{0,1,\infty\}$. The parameter u in [CS23, Introduction and Theorem 1.6] is the parameter of [KP18]. Both u and our $h\in\mathbb{C}\backslash\{1,2\}$ (Definition-Proposition 16.14(2)–(4), Definition-Theorem 2.20, and Theorem 16.12(2)) parametrize the isomorphism classes of the smooth prime Fano threefolds of genus 12 with a faithful $\mathbb{G}_m$-action, and X^{MU} corresponds to $u=-1/4$ and to $h=6$, that is, $v=1/5$. We do not use the relation between u and h for other values.

16.4. **GIT of quartics.** In this subsection, we study the GIT of plane quartics with invertible middle catalecticant and of plane quartics of multiplicity at most two, and we apply the second study to the quartics Q_{X_Π} for $\Pi\in\Omega$. As in Theorem 2.29, GIT-stability of a point of $\mathbb{P}(\mathbb{C}[x,y,z]_4)$ refers to the action of SL_3 with the linearization $\mathcal{O}(1)$.

The next proposition shows that the points of U^{cat} which are GIT-polystable but not GIT-stable are exactly the points of the orbits of $[F_h]$, $h\in\mathbb{C}\backslash\{1\}$, and that any point of U^{cat} which is not GIT-stable degenerates to the orbit of $[F_h]$ for some $h\in\mathbb{C}\backslash\{1\}$.

Proposition 16.16. *Let F_h, $h \in \mathbb{C}\backslash\{1\}$, be the quartics of Definition 2.27. Then the following hold.*

(1) Any point of U^{cat} is GIT-semistable.

(2) A point $[f] \in U^{\mathrm{cat}}$ is not GIT-stable if and only if, after a projective change of coordinates and multiplication by a non-zero constant,

$$f = F_h + a_1x^3y + a_2x^2y^2 + a_3xy^3 + a_4y^4 + a_5xy^2z + a_6y^3z$$

for some $h \in \mathbb{C}\backslash\{1\}$ and $a_1, \dots, a_6 \in \mathbb{C}$.

(3) For f as in (2), the quartics

$$\begin{aligned} f_t(x,y,z) &:= f(x,ty,t^{-1}z) \\ &= F_h + t(a_1x^3y + a_5xy^2z) + t^2(a_2x^2y^2 + a_6y^3z) + t^3a_3xy^3 + t^4a_4y^4 \end{aligned}$$

satisfy $[f_t] \in U^{\mathrm{cat}}$ for any $t \in \mathbb{C}$, with $\det \mathrm{Cat}_2(f_t) = 4096(h-1)$ in the basis $(\partial_x^2, \partial_y^2, \partial_z^2, \partial_x\partial_y, \partial_x\partial_z, \partial_y\partial_z)$. The orbit of $[F_h]$ is the unique closed orbit in the closure of $\mathrm{SL}_3 \cdot [f]$ in the GIT-semistable locus, and $[F_h]$ is GIT-polystable and not GIT-stable. Hence $[f]$ is GIT-polystable if and only if f is projectively equivalent to F_h.

(4) For $h, h' \in \mathbb{C}\backslash\{1\}$, the quartics F_h and $F_{h'}$ are projectively equivalent if and only if $h = h'$.

(5) The quartic $F_6 = (x^2 + yz)^2$ is a double smooth conic with projective stabilizer PGL_2, the quartic $F_0 = yz(2x^2 + yz)$ is a smooth conic together with two tangent lines, and for $h \notin \{0, 1, 6\}$ the quartic F_h is the union of two smooth conics tangent at two points. For $h \neq 1, 6$ the projective stabilizer of $[F_h]$ is $\mathbb{G}_m \rtimes \mu_2$, where $\mathbb{G}_m$ acts by $\mathrm{diag}(1, s, s^{-1})$ and the involution exchanges y and z.

(6) Let λ be a non-trivial one-parameter subgroup of SL_3, and assume that $\lim_{t\to 0} \lambda(t) \cdot f$ exists in $\mathbb{C}[x,y,z]_4$ for some f with $[f] \in U^{\mathrm{cat}}$. Then the weights of λ are $(0, n, -n)$ for some $n > 0$, up to order and sign.

Proof. We use the action of GL_3 on $\mathbb{C}[x,y,z]$ and $\mathbb{C}[\partial]$ and the form of the Hilbert–Mumford criterion fixed in Section 2.6.

Step 1. In this step we show that $\det \mathrm{Cat}_2$ is an SL_3-invariant and prove (1).

Let $g \in \mathrm{SL}_3$, let $f \in \mathbb{C}[x,y,z]_4$, and let $P, P' \in \mathbb{C}[\partial]_2$. Since $(PP')(f)$ is a constant, it is fixed by g, so

$$(PP')(g \cdot f) = g \cdot \Big(\big((g^{-1} \cdot P)(g^{-1} \cdot P')\big)(f)\Big) = \big((g^{-1} \cdot P)(g^{-1} \cdot P')\big)(f).$$

Thus $\det \mathrm{Cat}_2(g \cdot f) = \det(M)^2 \det \mathrm{Cat}_2(f)$, where M is the matrix of the action of g^{-1} on $\mathbb{C}[\partial]_2$ in the basis of Definition 2.27. If g^{-1} acts on $\mathbb{C}[\partial]_1$ diagonally with eigenvalues α, β, γ, then the eigenvalues of g^{-1} on $\mathbb{C}[\partial]_2$ are $\alpha^2, \beta^2, \gamma^2, \alpha\beta, \alpha\gamma, \beta\gamma$, so

$$\det M = (\alpha\beta\gamma)^4 = \det(g)^{-4} = 1.$$

Since diagonalizable elements are dense in SL_3, $\det M = 1$ for any $g \in \mathrm{SL}_3$. Thus $\det \mathrm{Cat}_2$ is an SL_3-invariant homogeneous polynomial of degree six on $\mathbb{C}[x,y,z]_4$. For $[f] \in U^{\mathrm{cat}}$ we have $\det \mathrm{Cat}_2(f) \neq 0$ (Definition 2.27), so $[f]$ is GIT-semistable. This proves (1).

Step 2. In this step we prove (6).

Let λ and f be as in (6), and let $f_0 := \lim_{t\to 0} \lambda(t) \cdot f$. By **Step 1**, $\det \mathrm{Cat}_2(\lambda(t) \cdot f) = \det \mathrm{Cat}_2(f)$ for any $t \in \mathbb{C}^*$, so by continuity

$$\det \mathrm{Cat}_2(f_0) = \det \mathrm{Cat}_2(f) \neq 0.$$

Moreover, $\lambda(s) \cdot f_0 = \lim_{t\to 0} \lambda(st) \cdot f = f_0$ for any $s \in \mathbb{C}^*$. Choose coordinates such that λ acts diagonally on $\mathbb{C}[x,y,z]_1$ with weights $w_1 \geq w_2 \geq w_3$. Since λ is a non-trivial one-parameter subgroup of SL_3, we have $w_1 + w_2 + w_3 = 0$ and $w_1 > 0$. The weights of λ on $\mathbb{C}[\partial]_2$ are $-(w_i + w_j)$ for $1 \leq i \leq j \leq 3$. Let $P, P' \in \mathbb{C}[\partial]_2$ be weight vectors of weights u and u'. For any $t \in \mathbb{C}^*$,

$$(PP')(f_0) = \lambda(t) \cdot \big((PP')(f_0)\big) = \big((\lambda(t) \cdot P)(\lambda(t) \cdot P')\big)(\lambda(t) \cdot f_0) = t^{u+u'}(PP')(f_0).$$

Thus $(PP')(f_0) = 0$ unless $u + u' = 0$. Since the bilinear form $\mathrm{Cat}_2(f_0)$ is non-degenerate, for any $u \in \mathbb{Z}$ it restricts to a perfect pairing between the weight-u and the weight-$(-u)$ subspaces of $\mathbb{C}[\partial]_2$. Thus the multiset of weights of λ on $\mathbb{C}[\partial]_2$ is symmetric under $u \to -u$. The largest weight is $-2w_3$ and the smallest weight is $-2w_1$, so $-2w_3 = 2w_1$. Thus $w_3 = -w_1$ and $w_2 = 0$, and the weights of λ on $\mathbb{C}[x, y, z]_1$ are $(n, 0, -n)$ with $n := w_1 > 0$. Thus the weights of λ on $\mathbb{C}^3$ are $(-n, 0, n)$. This proves (6).

Step 3. In this step we prove (2) and the first sentence of (3).

Suppose that $[f] \in U^{\mathrm{cat}}$ is not GIT-stable. By the Hilbert–Mumford criterion, there exists a non-trivial one-parameter subgroup λ of SL_3 such that $f_0 := \lim_{t\to 0} \lambda(t) \cdot f$ exists. By **Step 2**, after a linear change of coordinates,

$$\lambda(t) \cdot x = x, \quad \lambda(t) \cdot y = t^n y, \quad \lambda(t) \cdot z = t^{-n} z$$

for some $n > 0$. Then $\lambda(t) \cdot (x^i y^j z^k) = t^{n(j-k)} x^i y^j z^k$, so any monomial $x^i y^j z^k$ with non-zero coefficient in f satisfies $j \geq k$. The monomials of degree four with $j \geq k$ are

$$x^4,\ x^2yz,\ y^2z^2,\ x^3y,\ x^2y^2,\ xy^3,\ y^4,\ xy^2z,\ y^3z,$$

and f_0 is the part of f of weight zero:

$$f_0 = b_0 x^4 + b_1 x^2 yz + b_2 y^2 z^2$$

for some $b_0, b_1, b_2 \in \mathbb{C}$. The operators ∂_x^2 and $\partial_y \partial_z$ have weight 0, the operators ∂_y^2 and ∂_z^2 have weights $-2n$ and $2n$, and the operators $\partial_x \partial_y$ and $\partial_x \partial_z$ have weights $-n$ and n. By **Step 2** and a direct calculation, in the ordered basis $(\partial_x^2, \partial_y \partial_z, \partial_y^2, \partial_z^2, \partial_x \partial_y, \partial_x \partial_z)$ the matrix $((P_i P_j)(f_0))$ is block diagonal with blocks

$$\begin{pmatrix} 24b_0 & 2b_1 \\ 2b_1 & 4b_2 \end{pmatrix}, \quad \begin{pmatrix} 0 & 4b_2 \\ 4b_2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 2b_1 \\ 2b_1 & 0 \end{pmatrix}.$$

Reordering a basis does not change the determinant of a Gram matrix, so in the basis of (3),

$$\det \mathrm{Cat}_2(f_0) = 256 b_1^2 b_2^2 (24 b_0 b_2 - b_1^2). \tag{16.13}$$

Since $\det \mathrm{Cat}_2(f_0) \neq 0$ (**Step 2**), we have $b_1 b_2 \neq 0$ and $24 b_0 b_2 \neq b_1^2$. Let $\kappa \in \mathbb{C}$ with $\kappa^2 = 2b_2/b_1$. Replacing x by κx and dividing f by b_2, the part of f of weight zero becomes

$$\frac{4b_0 b_2}{b_1^2} x^4 + 2x^2 yz + y^2 z^2 = F_h, \qquad h := \frac{24 b_0 b_2}{b_1^2} \neq 1.$$

The substitution $x \to \kappa x$ multiplies each of the nine monomials above by a non-zero constant, so f has the form in (2).

Conversely, let f be as in (2). Let λ_0 be the one-parameter subgroup of SL_3 such that $(\lambda_0(t) \cdot q)(x, y, z) = q(x, ty, t^{-1}z)$ for any $q \in \mathbb{C}[x, y, z]$ and $t \in \mathbb{C}^*$. Then $f_t = \lambda_0(t) \cdot f$ for $t \in \mathbb{C}^*$, the formula for f_t in (3) is polynomial in t, and $f_t = F_h$ for $t = 0$. By **Step 1**, the polynomial $t \to \det \mathrm{Cat}_2(f_t)$ is constant on $\mathbb{C}^*$, hence constant on $\mathbb{C}$. By (16.13) with $(b_0, b_1, b_2) = (h/6, 2, 1)$,

$$\det \mathrm{Cat}_2(f_t) = \det \mathrm{Cat}_2(F_h) = 256 \cdot 4 \cdot (4h - 4) = 4096(h - 1) \neq 0$$

for any $t \in \mathbb{C}$. Thus $[f_t] \in U^{\mathrm{cat}}$ for any $t \in \mathbb{C}$, and in particular $[f] \in U^{\mathrm{cat}}$. Since $\lim_{t\to 0} \lambda_0(t) \cdot f = F_h$ exists, $[f]$ is not GIT-stable by the Hilbert–Mumford criterion. This proves (2) and the first sentence of (3).

Step 4. In this step we describe the curves $(F_h = 0)$ and prove the rest of (3).

Let $c_1, c_2 \in \mathbb{C}$ with $c_1 + c_2 = 2$ and $c_1 c_2 = h/6$. Then

$$F_h = (yz + c_1 x^2)(yz + c_2 x^2).$$

If $h = 6$, then $c_1 = c_2 = 1$ and $F_6 = (x^2 + yz)^2$ is a double smooth conic. If $h = 0$, then $\{c_1, c_2\} = \{0, 2\}$ and $F_0 = yz(2x^2 + yz)$. The line $(y = 0)$ intersects the smooth conic $(2x^2 + yz = 0)$ only at $[0 : 0 : 1]$, where in the chart $z = 1$ the intersection has ideal $(y, 2x^2 + y) = (y, x^2)$ of length two. Exchanging y and z, the line $(z = 0)$ is tangent to the conic at $[0 : 1 : 0]$. If

$h \notin \{0,6\}$, then $c_1c_2 \neq 0$, and $c_1 \neq c_2$ since $c_1 = c_2$ if and only if $c_1 = c_2 = 1$. Thus the conics $D_i := (yz + c_ix^2 = 0)$, $i = 1, 2$, are smooth and distinct. A common point of D_1 and D_2 satisfies $(c_1 - c_2)x^2 = 0$ and $yz = 0$, so $D_1 \cap D_2 = \{[0:0:1], [0:1:0]\}$. In the chart $z = 1$ the intersection has ideal $(y + c_1x^2, (c_1 - c_2)x^2) = (y + c_1x^2, x^2)$ of length two, and exchanging y and z shows that $D_1 \cap D_2$ also has length two at $[0:1:0]$. Thus D_1 and D_2 are tangent at two points. This proves the first sentence of (5).

Substituting $(y, x, -z)$ for (x, y, z),

$$F_h(y, x, -z) = (xz - c_1y^2)(xz - c_2y^2), \qquad (c_1, c_2) \neq (0, 0).$$

By the classification of GIT-polystable plane quartics (Theorem 2.29(3), see also [CDGF+25, Theorem 5.1 and Theorem 1.12(2)–(4)]), $[F_h]$ is GIT-polystable and not GIT-stable. In particular, the orbit of $[F_h]$ is closed in the semistable locus.

Let f be as in (2). For $t \in \mathbb{C}^*$ we have $[f_t] \in \mathrm{SL}_3 \cdot [f]$, and $[f_t] \to [F_h]$ as $t \to 0$. Thus $[F_h]$ is contained in the closure of $\mathrm{SL}_3 \cdot [f]$ in the semistable locus. The closure in the semistable locus of the orbit of a GIT-semistable point contains exactly one closed orbit [MFK94, Theorem 1.10, Corollary 1.2, Amplification 1.11] (see also [Alp13, Theorem 13.6 and Theorem 4.16(iv)]). Thus the orbit of $[F_h]$ is the unique closed orbit in the closure of $\mathrm{SL}_3 \cdot [f]$ in the GIT-semistable locus. If $[f]$ is GIT-polystable, then $\mathrm{SL}_3 \cdot [f]$ is closed in the semistable locus, so $\mathrm{SL}_3 \cdot [f] = \mathrm{SL}_3 \cdot [F_h]$ and f is projectively equivalent to F_h. Conversely, if f is projectively equivalent to F_h, then $[f]$ is GIT-polystable. This proves (3).

Step 5. In this step we compute the projective stabilizers and prove (4) and the rest of (5).

For $h \in \mathbb{C}\backslash\{1\}$ let $G_h := \mathrm{Stab}_{\mathrm{PGL}_3}([F_h])$, and let G_h^0 be the identity component of G_h. Let $T \subset \mathrm{PGL}_3$ be the image of $\{\mathrm{diag}(1, s, s^{-1}) : s \in \mathbb{C}^*\}$, so that $T \cong \mathbb{G}_m$, and let $\sigma \in \mathrm{PGL}_3$ be the class of the transposition exchanging y and z. Since $F_h(x, sy, s^{-1}z) = F_h(x, y, z) = F_h(x, z, y)$, we have $T \subset G_h^0$ and $\sigma \in G_h$.

We first show that $G_h^0 = T$ for $h \neq 6$. Suppose that $h \notin \{0, 6\}$. The connected group G_h^0 permutes the finite sets $\{D_1, D_2\}$ and $D_1 \cap D_2$, so it fixes D_1 and the points $[0:0:1]$ and $[0:1:0]$. Thus restriction defines a homomorphism from G_h^0 to the group of automorphisms of $D_1 \cong \mathbb{P}^1$ fixing two points, which is isomorphic to $\mathbb{G}_m$. The homomorphism is injective: an element of PGL_3 fixing D_1 pointwise fixes four points of D_1, no three of which are collinear since a line intersects D_1 in at most two points, so it is the identity. Thus $\dim G_h^0 \leq 1$. Since $T \subset G_h^0$ is closed, connected, and of dimension one, $G_h^0 = T$. Suppose that $h = 0$. Then G_0^0 fixes each of the lines $(y = 0)$ and $(z = 0)$ and the conic $(2x^2 + yz = 0)$, hence fixes the points $[1:0:0]$, $[0:0:1]$, and $[0:1:0]$. Thus any element of G_0^0 is the class of a diagonal matrix $\mathrm{diag}(\alpha, \beta, \gamma)$. Since

$$F_0(\alpha x, \beta y, \gamma z) = 2\alpha^2\beta\gamma x^2yz + \beta^2\gamma^2y^2z^2$$

is a multiple of F_0 if and only if $\alpha^2 = \beta\gamma$, the class of $\mathrm{diag}(\alpha, \beta, \gamma)$ is in G_0 if and only if it is in T. Thus $G_0^0 = T$.

Next we show that any element of PGL_3 normalizing T is the class of a diagonal matrix or of a diagonal matrix composed with the transposition of y and z. Let $g \in \mathrm{GL}_3$ represent an element normalizing T. Conjugation by g induces an automorphism $s \to s^\epsilon$ of $T \cong \mathbb{G}_m$ with $\epsilon \in \{1, -1\}$, so there exists a character $s \to s^m$ of $\mathbb{G}_m$ such that

$$g\,\mathrm{diag}(1, s, s^{-1})g^{-1} = s^m\,\mathrm{diag}(1, s^\epsilon, s^{-\epsilon})$$

for any $s \in \mathbb{C}^*$. Comparing determinants, $1 = s^{3m}$, so $m = 0$. For $s \in \mathbb{C}^*$ with $1, s, s^{-1}$ pairwise distinct and for $\mu \in \{1, s, s^{-1}\}$, g maps the μ-eigenline of $\mathrm{diag}(1, s, s^{-1})$ to the μ-eigenline of $\mathrm{diag}(1, s^\epsilon, s^{-\epsilon})$. Thus g is diagonal if $\epsilon = 1$, and g is a diagonal matrix composed with the transposition of y and z if $\epsilon = -1$.

Let $h \neq 6$. Since G_h normalizes $G_h^0 = T$ and $\sigma \in G_h$, any element of G_h is of the form δ or $\delta\sigma$, where $\delta \in G_h$ is the class of a diagonal matrix $\mathrm{diag}(\alpha, \beta, \gamma)$. Since

$$F_h(\alpha x, \beta y, \gamma z) = \frac{h}{6}\alpha^4 x^4 + 2\alpha^2\beta\gamma x^2 yz + \beta^2\gamma^2 y^2 z^2$$

and the coefficients of x^2yz and y^2z^2 in F_h are non-zero, $\delta \in G_h$ implies $\alpha^2 = \beta\gamma$, hence $\delta \in T$. Moreover, $\sigma \notin T$, $\sigma^2 = 1$, and $\sigma t \sigma^{-1} = t^{-1}$ for $t \in T$. Thus $G_h = T \rtimes \{1, \sigma\} \cong \mathbb{G}_m \rtimes \mu_2$, where μ_2 acts on $\mathbb{G}_m$ by $t \to t^{-1}$.

Let $h = 6$. Since $\mathbb{C}[x, y, z]$ is a unique factorization domain, an element of PGL_3 fixes $[(x^2+yz)^2]$ if and only if it fixes $[x^2 + yz]$, that is, if and only if it preserves the conic $D := (x^2 + yz = 0)$. Restriction defines a homomorphism $G_6 \to \mathrm{Aut}(D) \cong \mathrm{PGL}_2$. It is injective, since an element of PGL_3 fixing D pointwise is the identity as shown above. It is surjective, since D is the image of $\mathbb{P}^1$ under the morphism defined by the complete linear system $|\mathcal{O}_{\mathbb{P}^1}(2)|$, and any automorphism of $\mathbb{P}^1$ acts linearly on $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2))$. Thus $G_6 \cong \mathrm{PGL}_2$. This proves (5).

Finally, let $h, h' \in \mathbb{C}\backslash\{1\}$, and suppose that $g \in \mathrm{PGL}_3$ maps $[F_h]$ to $[F_{h'}]$. By **Step 4**, F_6 is non-reduced and $F_{h''}$ is reduced for $h'' \neq 1, 6$, so $h = 6$ if and only if $h' = 6$. Suppose that $h, h' \neq 6$. Then $gG_hg^{-1} = G_{h'}$, so $gTg^{-1} = T$, and g is the class of a diagonal matrix or of a diagonal matrix composed with the transposition of y and z. For a quartic $b_0x^4 + b_1x^2yz + b_2y^2z^2$ with $b_1b_2 \neq 0$, the substitution $(x, y, z) \to (\alpha x, \beta y, \gamma z)$ replaces (b_0, b_1, b_2) by $(\alpha^4 b_0, \alpha^2\beta\gamma b_1, \beta^2\gamma^2 b_2)$. Thus the number $24b_0b_2/b_1^2$ is invariant under diagonal substitutions, under the transposition of y and z, and under multiplication of the quartic by a non-zero constant. The number $24b_0b_2/b_1^2$ is h for F_h and h' for $F_{h'}$, so $h = h'$. The converse holds since $F_h = F_{h'}$ if $h = h'$. This proves (4). □

In the next proposition, we show that, for $\Pi \in \Omega\backslash\Omega^{\mathrm{st}}$, the threefold X_Π contains a special line which intersects no other line. The proof reads off the GIT-stability of Q_{X_Π} from the types of the singular points of Q_{X_Π}.

Proposition 16.17. *Let $f \in \mathbb{C}[x, y, z]_4$ be non-zero. Assume that the plane quartic $(f = 0)$ has multiplicity at most two at any point. Then the following hold.*

(1) If $(f = 0)$ is non-reduced, then it is a double smooth conic. If $(f = 0)$ is reduced, then any singular point of $(f = 0)$ is of type A_k with $1 \leq k \leq 7$, where A_k means a completed local equation $a^2 + b^{k+1} = 0$. A singular point of type A_7 occurs if and only if $(f = 0)$ is the union of two smooth conics with one common point, and then it is the only singular point. A singular point of type A_6 occurs only on irreducible quartics.

(2) $[f]$ is GIT-unstable if and only if $(f = 0) = C \cup L$ for an irreducible cubic C and a line L such that C is smooth at the unique point $p \in C \cap L$ and $(C \cdot L)_p = 3$, and then the singular points of $(f = 0)$ are $\{A_5\}$, $\{A_5, A_1\}$, or $\{A_5, A_2\}$. In particular, if $(f = 0)$ contains no line, then $[f]$ is GIT-semistable. Moreover $[f]$ is GIT-stable if and only if $(f = 0)$ is smooth or has only singular points of types A_1 and A_2.

Let $\Pi \in \Omega$. Then the following hold.

(3) The quartic Q_{X_Π} has multiplicity at most two at any point, and Q_{X_Π} is GIT-stable if and only if any special line on X_Π intersects another line.

(4) If $\Pi \notin \Omega^{\mathrm{st}}$, then X_Π contains a special line which intersects no other line.

Proof. Let $Q := (f = 0) \subset \mathbb{P}^2$.

Step 1. In this step we prove (1).

Suppose that an irreducible curve $D \subset \mathbb{P}^2$ appears in Q with multiplicity $m \geq 2$. Since $m \deg D \leq 4$, we have $\deg D \in \{1, 2\}$. Suppose that $D = (\ell = 0)$ is a line, and write $f = \ell^m f_1$ with $\ell \nmid f_1$. If $m \geq 3$, then Q has multiplicity at least three at any point of D. If $m = 2$, then f_1 is a quadratic form whose restriction to D is a non-zero binary quadratic form, so it vanishes at some point $q \in D$, and $\mathrm{mult}_q Q \geq 3$. Both cases contradict the assumption. Thus D is an

irreducible conic, hence smooth, and $m \deg D = 4$ implies $Q = 2D$. This proves the first sentence of (1).

Suppose that Q is reduced, and let $q \in Q$ be a singular point. Then $\operatorname{mult}_q Q = 2$. If the quadratic part of a local equation of Q at q is not a square, then q is a node, that is, of type A_1. Otherwise, choose local coordinates a, b at q such that the quadratic part is a^2. By the Weierstrass preparation theorem and completing the square, the completed local ring of Q at q is $\mathbb{C}[[a,b]]/(a^2 + \varphi(b))$ with $\varphi \in b^3\mathbb{C}[[b]]$. Since Q is reduced, the completed local ring is reduced [Mat89, Corollary to Theorem 23.9, and Remark 1 after Theorem 32.6], so $\varphi \neq 0$. Writing $\varphi = b^{k+1}u(b)$ with a unit u and replacing b by $bu(b)^{1/(k+1)}$, the point q is of type A_k with $k \geq 2$ [GLS07, Chapter I, Theorem 1.6 and Section 2.4].

For a point q of a reduced curve Γ, let δ_q be the length of $\widetilde{\mathcal{O}}_{\Gamma,q}/\mathcal{O}_{\Gamma,q}$, where $\widetilde{\mathcal{O}}_{\Gamma,q}$ is the normalization of $\mathcal{O}_{\Gamma,q}$. If q is a point of type A_k of Q, then $\delta_q = \lfloor (k+1)/2 \rfloor$. If $k = 2e-1$, then $a^2 + b^{2e} = (a + \sqrt{-1}b^e)(a - \sqrt{-1}b^e)$, so there exist two smooth branches with intersection multiplicity e, and $\widetilde{\mathcal{O}}_{Q,q}/\mathcal{O}_{Q,q}$ has length e. If $k = 2e$, then the branch has the parametrization $b = t^2$, $a = ct^{2e+1}$ with $c \neq 0$, and the missing powers $t, t^3, \dots, t^{2e-1}$ show that $\widetilde{\mathcal{O}}_{Q,q}/\mathcal{O}_{Q,q}$ has length e. Conversely, two smooth branches with intersection multiplicity e at q have local equation $a(a - \psi(b))$ with $\operatorname{ord}\psi = e$, and completing the square shows that q is of type A_{2e-1}.

Suppose that Q is irreducible, and let $\widetilde{Q}$ be the normalization of Q. Since $p_a(Q) = 3$ [Har77, Exercise I.7.2(b)],

$$\sum_{q \in Q} \delta_q = p_a(Q) - p_a(\widetilde{Q}) \leq 3$$

[Har77, Exercise IV.1.8(a)]. Thus $\lfloor (k+1)/2 \rfloor \leq 3$ for any singular point of type A_k, that is, $k \leq 6$.

Suppose that Q is reduced and reducible. Let $q \in Q_1 \cap Q_2$ for irreducible components $Q_1 \neq Q_2$ of Q. The assumption $\operatorname{mult}_q Q \leq 2$ implies that no other component of Q passes through q and that Q_1 and Q_2 are smooth at q. Thus q is of type A_{2e-1}, where $e := (Q_1 \cdot Q_2)_q$. The degrees of the irreducible components of Q form one of the partitions $3+1$, $2+2$, $2+1+1$, and $1+1+1+1$.

Partition $3+1$. Let C be the cubic and L the line. Since $\sum_{q \in C} \delta_q \leq p_a(C) = 1$ [Har77, Exercise IV.1.8(a)], the curve C has at most one singular point, of type A_1 or A_2. By Bézout's theorem, $e \leq 3$ at any point of $C \cap L$. Thus $k \in \{1, 2, 3, 5\}$.

Partition $2+2$. Both conics are smooth, and $e \leq 4$ by Bézout's theorem. Thus $k \in \{1, 3, 5, 7\}$.

Partition $2+1+1$. The conic is smooth, $e \leq 2$ at a common point of the conic and a line, and $e = 1$ at the common point of the lines. Thus $k \in \{1, 3\}$.

Partition $1+1+1+1$. All singular points are common points of two lines with $e = 1$. Thus $k = 1$.

Thus $k \leq 7$ in all cases, and A_6 occurs only on irreducible quartics. A point of type A_7 occurs exactly in the case $2+2$ with $e = 4$. In this case, by Bézout's theorem, q is the only common point of the conics, and since the conics are smooth, q is the only singular point of Q. Conversely, two smooth conics with one common point q have intersection multiplicity 4 at q by Bézout's theorem, so q is of type A_7. This proves (1).

Step 2. In this step we prove (2).

By Theorem 2.29(2), $[f]$ is GIT-unstable if and only if Q has a point of multiplicity at least three, or $Q = C \cup L$ for a cubic C and a line L tangent to C at an inflection point p, that is, C is smooth at $p \in L$ and $(C \cdot L)_p \geq 3$. The first alternative is excluded by the assumption. Suppose that $Q = C \cup L$ as in the second alternative. If $L \subset C$, then Q contains the double line $2L$, which is impossible by (1). Thus $(C \cdot L) = 3$ by Bézout's theorem, so $(C \cdot L)_p = 3$ and $C \cap L = \{p\}$. If C is not irreducible, then C contains at least two irreducible components counted with multiplicity, each of them intersects L by Bézout's theorem, hence passes through p, and $\operatorname{mult}_p C \geq 2$, a contradiction. Thus C is an irreducible cubic. Conversely, if C is an irreducible

cubic smooth at the unique point $p \in C \cap L$ with $(C \cdot L)_p = 3$, then L is tangent to C at the inflection point p, and $[f]$ is GIT-unstable by Theorem 2.29(2).

In this case, C and L are smooth branches at p with intersection multiplicity 3, so p is of type A_5 by **Step 1**. Since $p_a(C) = 1$, the curve C is smooth or has exactly one singular point, of type A_1 or A_2 (**Step 1**), and a singular point of C is not on L since $C \cap L = \{p\}$ and C is smooth at p. Thus the singular points of Q are $\{A_5\}$, $\{A_5, A_1\}$, or $\{A_5, A_2\}$. In particular, if Q contains no line, then $[f]$ is GIT-semistable.

By Theorem 2.29(1), $[f]$ is GIT-stable if and only if Q is reduced and has only singular points of types A_1 and A_2 (see also [Che17, Example 1.11]). If Q is non-reduced, then Q is a double smooth conic by (1), whose completed local ring at any point is $\mathbb{C}[[a,b]]/(a^2)$, so Q has singular points not of type A_k for any k, and $[f]$ is not GIT-stable. Thus $[f]$ is GIT-stable if and only if Q is smooth or has only singular points of types A_1 and A_2. This proves (2).

Step 3. In this step we prove (3) and (4).

Let $\Pi \in \Omega$. By Theorem 16.12(1), Π is the canonical Mukai net of X_Π, so Theorem 16.11(2) applied to X_Π implies that $\Sigma(X_\Pi) \cong Q_{X_\Pi}$. Let $L \subset X_\Pi$ be a line, and let $q \in Q_{X_\Pi}$ be the image of $[L]$. Since X_Π is a smooth prime Fano threefold of genus 12, the hypothesis of Set-up 10.1 holds for X_Π and L. By Proposition 10.8, the completed local ring of $\Sigma(X_\Pi)$ at $[L]$ is $\mathbb{C}[[t]]$ if L is non-special, and is $\mathbb{C}[[a,b]]/(g)$ with

$$g \in \{ab,\ a^2 + b^3\} \cup \{a^2 + \varphi(b) : \varphi \in b^4\mathbb{C}[[b]]\}$$

if L is special. In the first case Q_{X_Π} is smooth at q. In the second case Q_{X_Π} is singular at q, and the completed local ring of Q_{X_Π} at q is $\mathbb{C}[[u,v]]/(f_q)$ for a local equation f_q of order at least two. By [GLS07, Chapter I, Lemma 1.23], the isomorphism $\mathbb{C}[[u,v]]/(f_q) \cong \mathbb{C}[[a,b]]/(g)$ lifts to an isomorphism $\mathbb{C}[[u,v]] \cong \mathbb{C}[[a,b]]$ mapping (f_q) onto (g). Thus $\operatorname{mult}_q Q_{X_\Pi} = \operatorname{ord} g = 2$, and the type of Q_{X_Π} at q is the type of the germ $(g = 0)$. Thus Q_{X_Π} has multiplicity at most two at any point, and (2) holds for Q_{X_Π}. By (2), Q_{X_Π} is GIT-stable if and only if Q_{X_Π} is smooth or has only singular points of types A_1 and A_2, that is, if and only if $\Sigma(X_\Pi)$ has only smooth points and singular points of types A_1 and A_2. By Proposition 10.8, the condition on $\Sigma(X_\Pi)$ holds if and only if any special line on X_Π intersects another line. This proves (3).

Suppose that $\Pi \notin \Omega^{\mathrm{st}}$. By Theorem 16.12(4), Q_{X_Π} is not GIT-stable. By (3), there exists a special line on X_Π which intersects no other line. This proves (4). □

17. The stability criterion in genus twelve

In this section, we prove Theorems 17.1 and 17.5. Theorem 17.1 combines the estimates of Sections 4–15, which include the special-line configuration of Corollary 12.8(3) treated by a torus degeneration, with one further configuration in which K-semistability follows from a torus degeneration, and Theorem 17.5 uses the GIT of nets to decide when the strict inequality $A_X(F) > S_X(F)$ holds for any prime divisor F over X.

Theorem 17.1. *Every smooth prime Fano threefold of genus* 12 *is K-semistable.*

Proof. Let X be a smooth prime Fano threefold of genus 12. By Theorem 2.5(1), it suffices to show that $A_X(F) \geq S_X(F)$ for any prime divisor F over X.

We first treat one configuration. Suppose that X contains a closed point p of incidence type 3 such that exactly one conic on X contains p. Then X is K-semistable by Lemma 15.3, applied in the notation of Construction-Proposition 15.1 for (X, p).

Thus we may assume that X contains no such point. Let F be a prime divisor over X. Then $\operatorname{center}_X F$ is a prime divisor, an integral curve, or a closed point.

Case 1. $\operatorname{center}_X F$ is a prime divisor. Then $F = \operatorname{center}_X F$ is a prime divisor on X, and $A_X(F)/S_X(F) \geq 4$ by Lemma 4.2.

Case 2. $C := \operatorname{center}_X F$ is an integral curve. If $H \cdot C \geq 3$, then $A_X(F) > S_X(F)$ by Proposition 4.6. If $H \cdot C \in \{1, 2\}$, then $A_X(F) \geq S_X(F)$ by Corollary 12.8.

Case 3. $\operatorname{center}_X F = \{p\}$ for a closed point $p \in X$. By Proposition 4.1, the incidence type of p is one of $\emptyset$, 1, $1+1$, $1+1+1$, 2, $2+1$, and 3.

- If the incidence type is $\emptyset$, then $\delta_p(X) \geq 33/32$ by Proposition 4.4, so $A_X(F) \geq \frac{33}{32} S_X(F)$.
- If the incidence type is 1, then $\Sigma(X,p)$ consists of one reduced point, and $\delta_p(X) \geq 1$ by Construction-Theorem 13.2.
- If the incidence type is $1+1$, then we are in the situation of Set-up 13.12 with $r=2$, and $\delta_p(X) \geq 88/87$ by Theorem 13.21.
- If the incidence type is $1+1+1$, then we are in the situation of Set-up 13.12 with $r=3$, and $A_X(F) > S_X(F)$ by Theorem 13.24(1).
- If the incidence type is 2 or $2+1$, then $A_X(F) \geq S_X(F)$ by Theorem 14.1(1).
- If the incidence type is 3, then, by Proposition 4.1, $\Sigma(X,p)$ is curvilinear of length three and supported at one line. By Lemma 15.3, exactly one or exactly two conics on X contain p. Since X contains no such point, exactly two conics contain p. Thus Theorem 15.6(2) implies

$$A_X(F) - S_X(F) \geq \frac{7}{132} A_X(F) + \frac{17}{88} \operatorname{mult}_F E > 0,$$

where E is as in Construction-Proposition 15.4.

In all cases $A_X(F) \geq S_X(F)$. By Theorem 2.5(1), X is K-semistable. □

Remark 17.2. Theorem 17.1 is proved by bounding $A_X(F) - S_X(F)$ for each prime divisor F over X separately, except when X contains a special line as in the last sentence of Corollary 12.8(3), which also occurs in Theorem 14.1(1), or a point of incidence type 3 on exactly one conic (Lemma 15.3). In both cases K-semistability follows from a torus degeneration, by Theorem 2.26 and Lemma 2.6. Apart from the results quoted in Theorem 2.26 and Lemma 2.6, the proof does not use that, when $\delta(X) < 1$, the threshold $\delta(X)$ is computed by a prime divisor over X [LXZ22, Theorem 5.1].

The next lemma excludes degenerations of X_Π, for $\Pi \in \Omega^{\mathrm{st}}$, by smooth families whose central fiber admits a faithful $\mathbb{G}_m$-action. In the proof of Theorem 17.5, it is used together with Lemma 12.3 and Proposition 15.2, which construct such degenerations from the configurations of Proposition 12.5 and Lemma 15.3.

Lemma 17.3. *Let $\Pi \in \Omega^{\mathrm{st}}$. Then there exists no smooth projective morphism $\mathcal{X} \to \mathbb{A}^1$, all of whose fibers are smooth prime Fano threefolds of genus 12, whose geometric generic fiber is isomorphic to X_Π over an algebraic closure of $\mathbb{C}(t)$, and whose central fiber admits a faithful $\mathbb{G}_m$-action.*

Proof. Suppose that such a morphism $\mathcal{X} \to \mathbb{A}^1$ exists, and let $\mathcal{X}_0$ be the fiber over 0. Let $\widehat{\mathcal{X}} := \mathcal{X} \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[[t]]$. Since smoothness and projectivity are preserved under base change, $\widehat{\mathcal{X}} \to \operatorname{Spec} \mathbb{C}[[t]]$ is smooth and projective. After fixing an embedding $\overline{\mathbb{C}(t)} \subset \overline{\mathbb{C}((t))}$, the geometric generic fiber of $\widehat{\mathcal{X}}$ is the base change of $(X_\Pi)_{\overline{\mathbb{C}(t)}}$. Smoothness and the equality $(-K)^3 = 22$ are preserved under extensions of algebraically closed fields, and so is the condition $\operatorname{Pic} = \mathbb{Z}[-K]$, since the Picard scheme of a smooth projective variety X' with $H^1(X', \mathcal{O}_{X'}) = 0$ is étale over the base field [Kle05, Theorem 9.5.11 and Corollary 9.5.14]. The central fiber of $\widehat{\mathcal{X}}$ is $\mathcal{X}_0$. Thus $\widehat{\mathcal{X}} \to \operatorname{Spec} \mathbb{C}[[t]]$ satisfies the hypotheses of Proposition 16.13 with $\tau = t$.

Since $\Pi \in \Omega^{\mathrm{st}}$, the orbit $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$. By Proposition 16.13(4), $\mathcal{X}_0 \cong X_\Pi$. Since $\Pi \in \Omega^{\mathrm{st}}$, the group $\operatorname{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite, so $\operatorname{Aut}(X_\Pi)$ is finite by the exact sequence

$$1 \to \mu_7 \to \operatorname{Stab}_{\mathrm{SL}(V)}(\Pi) \to \operatorname{Aut}(X_\Pi) \to 1$$

of Theorem 16.12(2). On the other hand, a faithful $\mathbb{G}_m$-action on $\mathcal{X}_0 \cong X_\Pi$ is an injective homomorphism $\mathbb{G}_m \to \operatorname{Aut}(X_\Pi)$, which is impossible since $\mathbb{G}_m$ is infinite. □

The next proposition describes X_Π for the nets Π outside Ω^{st}, and it is used in the proof of Theorem 17.5. Test configurations, products, Donaldson–Futaki invariants, K-semistability, and K-polystability are as in Definition 2.4.

Proposition 17.4. *Let $\Pi \in \Omega$. Then the following hold.*

(1) X_Π is K-semistable.

(2) Let $\lambda : \mathbb{G}_m \to \mathrm{SL}(V)$ be a one-parameter subgroup with $\Pi_0 := \lim_{t\to 0} \lambda(t)\cdot\Pi \in \mathbb{B}^{\mathrm{nd}}$. Then the family $\mathcal{X}_\lambda \to \mathbb{A}^1$ of Proposition 16.7(3), that is, the pullback of the universal isotropic scheme along the morphism $\mathbb{A}^1 \to \mathbb{B}^{\mathrm{nd}}$ given by $t \to \lambda(t)\cdot\Pi$ for $t \neq 0$ and $0 \to \Pi_0$, is an integral normal Gorenstein anti-canonical test configuration of X_Π with Donaldson–Futaki invariant zero, and it is not a product if Π_0 is not $\mathrm{SL}(V)$-conjugate to Π.

(3) If $\mathrm{SL}(V)\cdot\Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, then there exists a one-parameter subgroup λ as in (2) such that Π_0 is not $\mathrm{SL}(V)$-conjugate to Π, and X_Π is not K-polystable.

(4) If $\Pi \notin \Omega^{\mathrm{st}}$, then $\delta(X_\Pi) = 1$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1) and introduce the family in (2).

Since $\Pi \in \Omega$, X_Π is a smooth prime Fano threefold of genus 12, so (1) follows from Theorem 17.1.

By Definition 2.17, $\Omega \subset \mathbb{B}^{\mathrm{nd}}$. By [BKM26b, Proposition B.6], the zero scheme of any non-degenerate net is a three-dimensional local complete intersection, so $\mathbb{B}^{\mathrm{nd}} \subset \mathbb{B}^{(3)}$. By Theorem 16.12(1), Π is the canonical Mukai net of X_Π under the restriction isomorphism $V^\vee \cong H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$, that is, $\Pi = \Pi_{X_\Pi}$ after identifying V with V_{X_Π}. Hence Proposition 16.7(3) holds for X_Π and for any one-parameter subgroup λ as in (2).

Let λ be as in (2), let $\pi : \mathcal{X}_\lambda \to \mathbb{A}^1$ and $\mathcal{L}_\lambda$ be as in (2) and Proposition 16.7(3), and let $\mathcal{X}_{\lambda,0} := \pi^{-1}(0)$. The fiber of π over $t \neq 0$ is $X_{\lambda(t)\cdot\Pi} = \lambda(t)X_\Pi$, and $\mathcal{X}_{\lambda,0} = X_{\Pi_0}$. The group $\mathbb{G}_m$ acts on $\mathcal{X}_\lambda \subset \mathrm{Gr}(3, V) \times \mathbb{A}^1$ by $a\cdot(U', t) = (\lambda(a)U', at)$, the morphism π is equivariant, and the morphism

$$X_\Pi \times (\mathbb{A}^1\backslash\{0\}) \to \pi^{-1}(\mathbb{A}^1\backslash\{0\}), \quad (U', t) \to (\lambda(t)U', t)$$

is an equivariant isomorphism. By Proposition 16.7(1), the universal isotropic scheme $\mathcal{X}|_{\mathbb{B}^{(3)}} \to \mathbb{B}^{(3)}$ is flat and Gorenstein with relative anti-canonical bundle $\mathcal{L}$. Since π is a pullback of $\mathcal{X}|_{\mathbb{B}^{(3)}} \to \mathbb{B}^{(3)}$ and the relative dualizing sheaf of a flat projective Gorenstein morphism is compatible with base change, π is flat and Gorenstein and $\mathcal{L}_\lambda \cong \omega^{-1}_{\mathcal{X}_\lambda/\mathbb{A}^1}$. By Proposition 16.7(3), π is polarized by $\mathcal{L}_\lambda$. Until normality is proved in **Step 4**, we regard $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ only as a flat polarized family over $\mathbb{A}^1$ with a $\mathbb{G}_m$-action.

Step 2. In this step we show that $\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) = 0$, where DF is defined by the formula in Proposition 16.7(3), which does not require $\mathcal{X}_\lambda$ to be normal.

By Definition-Proposition 16.2, $\mathbb{B}^{\mathrm{nd}} = \mathbb{B}\backslash D_{\mathrm{Pf}}$, so $\Pi_0 \notin D_{\mathrm{Pf}}$, and Proposition 16.7(3) gives $\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) = 0$.

Step 3. In this step we show that $\mathcal{X}_\lambda$ is integral and Gorenstein.

Since π is flat and Gorenstein and $\mathbb{A}^1$ is regular, $\mathcal{X}_\lambda$ is Gorenstein by [Mat89, Theorem 23.4]. In particular, $\mathcal{X}_\lambda$ is Cohen–Macaulay, so $\mathcal{X}_\lambda$ satisfies S_2, and any associated point of $\mathcal{X}_\lambda$ is a generic point of an irreducible component of $\mathcal{X}_\lambda$. Since π is flat, t is a nonzerodivisor on $\mathcal{X}_\lambda$, so no associated point of $\mathcal{X}_\lambda$ is contained in $\mathcal{X}_{\lambda,0}$. Since $\pi^{-1}(\mathbb{A}^1\backslash\{0\}) \cong X_\Pi \times (\mathbb{A}^1\backslash\{0\})$ is integral, $\mathcal{X}_\lambda$ has exactly one associated point, and $\mathcal{X}_\lambda$ is generically reduced. A Noetherian scheme satisfying R_0 and S_1 is reduced [Mat89, Section 23, p. 183], so $\mathcal{X}_\lambda$ is integral.

Step 4. In this step we show that $\mathcal{X}_\lambda$ is normal. Thus $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a test configuration of X_Π with Donaldson–Futaki invariant zero.

Let $\nu : \mathcal{X}^\nu_\lambda \to \mathcal{X}_\lambda$ be the normalization. Since $\mathcal{X}_\lambda$ is of finite type over $\mathbb{C}$, ν is finite. Since $\pi^{-1}(\mathbb{A}^1\backslash\{0\})$ is smooth, ν is an isomorphism over $\mathbb{A}^1\backslash\{0\}$. The $\mathbb{G}_m$-action on $\mathcal{X}_\lambda$ lifts to $\mathcal{X}^\nu_\lambda$ by the universal property of normalization. Since $\mathcal{X}^\nu_\lambda$ is integral and dominates $\mathbb{A}^1$, $\mathcal{O}_{\mathcal{X}^\nu_\lambda}$ is

torsion-free over $\mathbb{C}[t]$, so $\pi\circ\nu$ is flat. Since ν is finite, $\nu^*\mathcal{L}_\lambda$ is relatively ample over $\mathbb{A}^1$. Therefore $(\mathcal{X}_\lambda^\nu, \nu^*\mathcal{L}_\lambda)$ is a test configuration of X_Π.

Suppose that $\mathcal{X}_\lambda$ is not normal. Since $\mathcal{X}_\lambda$ satisfies S_2 (**Step 3**), Serre's criterion [Mat89, Theorem 23.8] implies that $\mathcal{X}_\lambda$ does not satisfy R_1. Let $\eta\in\mathcal{X}_\lambda$ be a point of codimension one such that $\mathcal{O}_{\mathcal{X}_\lambda,\eta}$ is not regular. Since $\pi^{-1}(\mathbb{A}^1\backslash\{0\})$ is smooth, $\eta\in\mathcal{X}_{\lambda,0}$. A one-dimensional Noetherian local domain is normal if and only if it is regular, so $\mathcal{O}_{\mathcal{X}_\lambda,\eta}$ is not normal and ν is not an isomorphism over η. Hence ν is a finite equivariant morphism, an isomorphism over $\mathbb{A}^1\backslash\{0\}$, which is not an isomorphism in codimension one. By [LX14, Section 8.2],

$$\mathrm{DF}(\mathcal{X}_\lambda^\nu, \nu^*\mathcal{L}_\lambda) < \mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) = 0.$$

Here only the signs of Donaldson–Futaki invariants and their vanishing are used, so the positive constant relating the normalization of [LX14, Definition 2 and Section 8.1] to the one in Proposition 16.7(3) plays no role. On the other hand, X_Π is K-semistable by (1), so any test configuration of X_Π has non-negative Donaldson–Futaki invariant (Definition 2.4). This is a contradiction, so $\mathcal{X}_\lambda$ is normal.

By Proposition 16.7(3), $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a test configuration of X_Π, and $\mathrm{DF}(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is the Donaldson–Futaki invariant of $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ in the sense of Definition 2.4. By **Steps 1–3**, $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is an integral normal Gorenstein anti-canonical test configuration of X_Π with Donaldson–Futaki invariant zero.

Step 5. In this step we show that $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is not a product if $\Pi_0\notin \mathrm{SL}(V)\cdot\Pi$. This finishes the proof of (2).

Suppose that $\Pi_0\notin\mathrm{SL}(V)\cdot\Pi$ and that $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a product. Then $X_{\Pi_0} = \mathcal{X}_{\lambda,0}\cong X_\Pi$. Thus X_{Π_0} is a smooth prime Fano threefold of genus 12. Since $\Pi_0\in\mathbb{B}^{\mathrm{nd}}$, we have $\Pi_0\in\Omega$.

By Theorem 16.12(1), applied to Π and to Π_0, the restrictions $\mathcal{U}|_{X_\Pi}$ and $\mathcal{U}|_{X_{\Pi_0}}$ are the Mukai bundles $\mathcal{U}_{X_\Pi}$ and $\mathcal{U}_{X_{\Pi_0}}$, the restriction maps $V^\vee\to H^0(X_\Pi, \mathcal{U}^\vee|_{X_\Pi})$ and $V^\vee\to H^0(X_{\Pi_0}, \mathcal{U}^\vee|_{X_{\Pi_0}})$ are isomorphisms, and under the restriction isomorphisms Π and Π_0 are the canonical Mukai nets of X_Π and X_{Π_0}, in the sense of Definition-Theorem 2.20.

Let $\varphi : X_{\Pi_0}\to X_\Pi$ be an isomorphism. Then $\varphi^*\mathcal{U}_{X_\Pi}$ satisfies the characterization of the Mukai bundle of X_{Π_0}, so $\varphi^*\mathcal{U}_{X_\Pi}\cong\mathcal{U}_{X_{\Pi_0}}$ by the uniqueness in Definition-Theorem 2.20. Fix such an isomorphism. Together with φ, it induces a linear isomorphism

$$H^0(X_\Pi, \mathcal{U}_{X_\Pi}^\vee)\to H^0(X_{\Pi_0}, \mathcal{U}_{X_{\Pi_0}}^\vee)$$

which carries the canonical Mukai net of X_Π onto that of X_{Π_0}, since, by Definition-Theorem 2.20, the canonical Mukai net of a smooth prime Fano threefold X' of genus 12 is the kernel of the map $\Lambda^2 V_{X'}^\vee\to H^0(X', \Lambda^2\mathcal{U}_{X'}^\vee)$, which is defined from $\mathcal{U}_{X'}$ alone. Composing with the restriction isomorphisms from $V^\vee$, we obtain $g\in\mathrm{GL}(V)$ such that $g\cdot\Pi=\Pi_0$. Scalar matrices act trivially on $\mathbb{B}$, so, replacing g with cg for a complex number c with $c^7\det g = 1$, we may assume that $g\in\mathrm{SL}(V)$. Thus $\Pi_0\in\mathrm{SL}(V)\cdot\Pi$, a contradiction.

Step 6. In this step we prove (3).

Suppose that $\mathrm{SL}(V)\cdot\Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$. By Definition-Proposition 16.2, $\mathbb{B}^{\mathrm{nd}}$ is an affine variety with an action of the reductive group $\mathrm{SL}(V)$. By the Hilbert–Mumford criterion in the form of [BHMR17, Section 1], for a reductive group acting on an affine variety, a non-closed orbit has a one-parameter subgroup whose limit exists and is not contained in the orbit. Hence there exists a one-parameter subgroup $\lambda:\mathbb{G}_m\to\mathrm{SL}(V)$ such that $\Pi_0 := \lim_{t\to0}\lambda(t)\cdot\Pi$ exists in $\mathbb{B}^{\mathrm{nd}}$ and $\Pi_0\notin\mathrm{SL}(V)\cdot\Pi$. By (2), $(\mathcal{X}_\lambda, \mathcal{L}_\lambda)$ is a test configuration of X_Π which is not a product and has Donaldson–Futaki invariant zero. Thus X_Π is not K-polystable (Definition 2.4).

Step 7. In this step we prove (4).

Suppose that $\Pi\notin\Omega^{\mathrm{st}}$. By Theorem 16.12(3), either $\mathrm{SL}(V)\cdot\Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, or X_Π admits a faithful $\mathbb{G}_m$-action. In the first case, **Step 6** implies that X_Π has a test configuration which is not a product and has Donaldson–Futaki invariant zero. Since a test configuration which is not a product is non-trivial, X_Π is not K-stable (Definition 2.4). In the second case, $\mathrm{Aut}(X_\Pi)$ is infinite, so X_Π is not K-stable by Theorem 2.7. In both cases, X_Π is not uniformly

K-stable by Theorem 2.5(4), so $\delta(X_\Pi) \leq 1$ by Theorem 2.5(3). Since X_Π is K-semistable by (1), $\delta(X_\Pi) \geq 1$ by Theorem 2.5(1). Therefore $\delta(X_\Pi) = 1$. □

Theorem 17.5. *Let $\Pi \in \Omega$ and $X := X_\Pi$. Then the following hold.*

(1) X is K-semistable, and the following conditions are equivalent:
  (i) $\delta(X) > 1$,
  (ii) X is K-stable,
  (iii) X is uniformly K-stable,
  (iv) $\Pi \in \Omega^{\mathrm{st}}$, and
  (v) $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Aut}(X)$ is finite.

(2) If $\Pi \in \Omega^{\mathrm{st}}$, then $A_X(F) > S_X(F)$ for any prime divisor F over X.

(3) X is K-polystable if and only if $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

(4) If $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and the stabilizer of Π in $\mathrm{SL}(V)$ is of positive dimension, then X admits a faithful $\mathbb{G}_m$-action, X is K-polystable and not K-stable, and $\delta(X) = 1$.

(5) If $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, then X is K-semistable, not K-polystable, and $\delta(X) = 1$.

Proof. Recall that X is a smooth prime Fano threefold of genus 12 (Definition 2.17). In **Steps 1–5** we assume that $\Pi \in \Omega^{\mathrm{st}}$. **Steps 1 and 2** exclude two configurations on X by means of Lemma 17.3, **Steps 3 and 4** prove (2), and **Step 5** deduces the K-stability of X. **Step 6** treats an arbitrary $\Pi \in \Omega$.

Step 1. In this step, we show that no special line on X satisfies the condition $(*)$ below. We say that a special line $L \subset X$ satisfies $(*)$ if, in the notation of Set-up 10.10, $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and $M \cap \Gamma$ is non-reduced of length two. By Proposition 12.5(1), $(*)$ implies $N_{M/V_5} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, so $(*)$ is the exceptional configuration of Corollary 12.8(3) and the assumption of Theorem 14.1(3).

Suppose that a special line $L \subset X$ satisfies $(*)$, and use the notation of Set-up 10.10. Since $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, Proposition 12.5(2) and (3) show that the normalization $\nu : \mathbb{F}_3 \to S_M$ has conductor $s_0 + f_0$ and that the strict transform $\tilde{\Gamma} \subset \mathbb{F}_3$ of Γ is a smooth curve of class $s_0 + 4f_0$. Since $M \cap \Gamma$ is non-reduced of length two, Proposition 12.5(4) implies that $s_0 \cap f_0 \in \tilde{\Gamma}$. Since $\tilde{\Gamma}$ is irreducible of class $s_0 + 4f_0$, we have $(\tilde{\Gamma} \cdot s_0) = -3 + 4 = 1$ and $(\tilde{\Gamma} \cdot f_0) = 1$, and since $s_0 \cap f_0 \in \tilde{\Gamma}$, it follows that $\tilde{\Gamma} \cap s_0 = \tilde{\Gamma} \cap f_0 = s_0 \cap f_0$.

Since moreover $\nu^{-1}(M) = s_0 \cup f_0$ (Lemma 12.1) and $\Gamma \subset S_M$ (Lemma 11.7(3)), M, S_M, ν, s_0, f_0, Γ, and $\tilde{\Gamma}$ satisfy conditions (1)–(3) of Set-up 12.2. Since $\Gamma \subset V_5$ is the rational normal quintic attached to L by Theorem 2.23, the pair (X, L) is the image of (V_5, Γ) under the inverse link of Theorem 2.23, so condition (4) of Set-up 12.2 also holds. By Lemma 12.3, there exist a smooth projective morphism $\mathcal{X} \to \mathbb{A}^1$ and a $\mathbb{G}_m$-action on $\mathcal{X}$ covering the standard action on $\mathbb{A}^1$ such that

- $\mathcal{X}|_{\mathbb{A}^1 \setminus \{0\}}$ is $\mathbb{G}_m$-equivariantly isomorphic to $X \times (\mathbb{A}^1 \setminus \{0\})$, with the trivial action on X,
- any fiber of $\mathcal{X} \to \mathbb{A}^1$ is a smooth prime Fano threefold of genus 12, and
- the induced $\mathbb{G}_m$-action on $\mathcal{X}_0$ is faithful.

In particular, the geometric generic fiber of $\mathcal{X} \to \mathbb{A}^1$ is isomorphic to $X \times_{\operatorname{Spec} \mathbb{C}} \operatorname{Spec} \overline{\mathbb{C}(t)}$. This contradicts Lemma 17.3, since $\Pi \in \Omega^{\mathrm{st}}$. Therefore no special line on X satisfies $(*)$.

Step 2. In this step, we show that there exists no closed point $p \in X$ of incidence type 3 (Definition 2.11) such that exactly one conic on X contains p.

Suppose that $p \in X$ is such a point. Then X and p satisfy the hypotheses of Construction-Proposition 15.1, and Lemma 15.3 implies that the coefficient γ_2 of the normal form of Construction-Proposition 15.1(1) vanishes. By Proposition 15.2, there exist a smooth projective morphism $\mathcal{X} \to \mathbb{A}^1$ and a $\mathbb{G}_m$-action on $\mathcal{X}$ covering the standard action on $\mathbb{A}^1$ such that $\mathcal{X}|_{\mathbb{A}^1 \setminus \{0\}}$ is $\mathbb{G}_m$-equivariantly isomorphic to $X \times (\mathbb{A}^1 \setminus \{0\})$ with the trivial action on X, and such that $\mathcal{X}_0$ is isomorphic to X_{Π_τ}, where Π_τ is the net of Proposition 15.2, and is a smooth prime Fano threefold of genus 12 admitting a faithful $\mathbb{G}_m$-action. Thus any fiber of $\mathcal{X} \to \mathbb{A}^1$ is a smooth

prime Fano threefold of genus 12, and the geometric generic fiber of $\mathcal{X} \to \mathbb{A}^1$ is isomorphic to $X \times_{\operatorname{Spec}\mathbb{C}} \operatorname{Spec}\overline{\mathbb{C}(t)}$. This contradicts Lemma 17.3.

Step 3. In this step, we show that $A_X(F) > S_X(F)$ for any prime divisor F over X with $\dim \operatorname{center}_X F \geq 1$.

Since F is a prime divisor over X, $\operatorname{center}_X F$ is an irreducible closed subset of X with $\dim \operatorname{center}_X F \leq 2$.

Suppose that $\dim \operatorname{center}_X F = 2$. Since the local ring of X at the generic point of $\operatorname{center}_X F$ is a discrete valuation ring, F is the prime divisor $\operatorname{center}_X F$ on X. By Lemma 4.2,

$$\frac{A_X(F)}{S_X(F)} \geq 4.$$

Suppose that $C := \operatorname{center}_X F$ is a curve. If $H \cdot C \geq 3$, then $A_X(F) > S_X(F)$ by Proposition 4.6. Assume that $H \cdot C \in \{1, 2\}$, and let E be the exceptional divisor of $\operatorname{Bl}_C X \to X$. By Corollary 12.8,

$$A_X(F) - S_X(F) \geq \frac{59}{88} \operatorname{mult}_F E > 0 \qquad \text{if } H \cdot C = 2,$$

$$A_X(F) - S_X(F) \geq \frac{5}{11} \operatorname{mult}_F E > 0 \qquad \text{if } C \text{ is a non-special line.}$$

If C is a special line, then Corollary 12.8 implies that $A_X(F) = S_X(F)$ is possible only if C satisfies $(*)$ of **Step 1**. By **Step 1**, C does not satisfy $(*)$, hence $A_X(F) > S_X(F)$.

Step 4. In this step, we show that $A_X(F) > S_X(F)$ for any prime divisor F over X with $\operatorname{center}_X F = \{p\}$ for a closed point $p \in X$.

By Proposition 4.1, the incidence type of p is one of $\emptyset$, 1, $1+1$, $1+1+1$, 2, $2+1$, and 3, a reduced point of $\Sigma(X, p)$ corresponds to a non-special line, and a point of $\Sigma(X, p)$ of length two or three corresponds to a special line. We treat the seven incidence types in turn. Recall that $\delta_p(X)$ is the infimum of $A_X(F')/S_X(F')$ over the prime divisors F' over X with $p \in \operatorname{center}_X F'$, so a lower bound $\delta_p(X) > 1$ implies $A_X(F) > S_X(F)$.

(a) Incidence type $\emptyset$. By Proposition 4.4, $\delta_p(X) \geq 33/32 > 1$.

(b) Incidence type 1. Then p belongs to exactly one line L, and L is non-special. By Corollary 13.3, $\delta_p(X) > 1$. The lower bound of Corollary 13.3 depends on p, and only $\delta_p(X) > 1$ is used here.

(c) Incidence type $1+1$. Then p belongs to exactly two lines, both non-special, and X and p satisfy the hypotheses of Set-up 13.12 with $r = 2$. By Theorem 13.21, $\delta_p(X) \geq 88/87 > 1$.

(d) Incidence type $1+1+1$. Then p belongs to exactly three lines, all non-special, and X and p satisfy the hypotheses of Set-up 13.12 with $r = 3$. Since $\operatorname{center}_X F = \{p\}$, Theorem 13.24(1) implies that $A_X(F) > S_X(F)$.

(e) Incidence type 2 *or* $2+1$. Let L be the line through p with $[L]$ of length two in $\Sigma(X, p)$. By Theorem 14.1(1), L is special. By **Step 1**, L does not satisfy $(*)$, so the assumption of Theorem 14.1(3) does not hold. Let $\hat{\pi} : \hat{X} \to X$ be the blow-up of the non-reduced conic supported on L, with reduced exceptional divisor P (Theorem 14.1(2)). By Theorem 14.1(4),

$$A_X(F) - S_X(F) \geq \frac{15}{44} \operatorname{mult}_F P.$$

We show that $\operatorname{mult}_F P > 0$. Since $\hat{\pi}$ is an isomorphism over $X \setminus L$ and $\hat{\pi}^{-1}(L) = P$ as sets, we have

$$\operatorname{center}_{\hat{X}} F \subset \hat{\pi}^{-1}(p) \subset P.$$

Let k be a positive integer such that kP is Cartier, and let h be a local equation of kP at the generic point of $\operatorname{center}_{\hat{X}} F$. Then h belongs to the maximal ideal of the local ring of $\hat{X}$ at the generic point of $\operatorname{center}_{\hat{X}} F$, so $\operatorname{ord}_F(h) > 0$. Thus $\operatorname{mult}_F P > 0$, and $A_X(F) > S_X(F)$.

(f) Incidence type 3. Then $\Sigma(X, p)$ is curvilinear of length three and supported at one line, since $\Sigma(X, p) \cong \operatorname{Spec}\mathbb{C}[t]/(t^3)$ (Proposition 4.1). By Lemma 15.3, exactly one or exactly two conics on X contain p, and since p has incidence type 3, **Step 2** excludes exactly one. Hence

exactly two conics on X contain p. Let E be the exceptional divisor of the plt extraction of Construction-Proposition 15.4. By Theorem 15.6(2),

$$A_X(F) - S_X(F) \geq \frac{7}{132} A_X(F) + \frac{17}{88} \operatorname{mult}_F E > 0.$$

The seven incidence types exhaust the closed points of X, which proves the claim of **Step 4**. Together with **Step 3**, we obtain $A_X(F) > S_X(F)$ for any prime divisor F over X. This proves (2).

Step 5. In this step, we show that (iv) implies (i), (ii), and (iii). By **Step 4**, $A_X(F) > S_X(F)$ for any prime divisor F over X. By Theorem 2.5(2), X is K-stable. By Theorem 2.5(4), X is uniformly K-stable, and by Theorem 2.5(3), $\delta(X) > 1$.

Step 6. In this step, we drop the assumption $\Pi \in \Omega^{\mathrm{st}}$ and conclude the proof. By Proposition 17.4, X is K-semistable (see also Theorem 17.1).

By Theorem 2.5(3) and (4), conditions (i), (ii), and (iii) are equivalent. By **Step 5**, (iv) implies (ii). Suppose that $\Pi \notin \Omega^{\mathrm{st}}$. By Proposition 17.4, $\delta(X) = 1$, so (i) fails. Hence (i) implies (iv). By Definition 2.17, $\Pi \in \Omega^{\mathrm{st}}$ if and only if $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite. By the exact sequence

$$1 \to \mu_7 \to \mathrm{Stab}_{\mathrm{SL}(V)}(\Pi) \to \mathrm{Aut}(X) \to 1$$

of Theorem 16.12(2), $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite if and only if $\mathrm{Aut}(X)$ is finite. Thus (iv) and (v) are equivalent, which proves (1).

Suppose that $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$. If $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite, then $\Pi \in \Omega^{\mathrm{st}}$, so X is K-stable by (1), hence K-polystable. Assume that $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is of positive dimension. Then $\Pi \notin \Omega^{\mathrm{st}}$ by Definition 2.17. By Theorem 16.12(3), either $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$ or X admits a faithful $\mathbb{G}_m$-action. Since $\mathrm{SL}(V) \cdot \Pi$ is closed, X admits a faithful $\mathbb{G}_m$-action. By Theorem 2.26, X is K-polystable. Since $\Pi \notin \Omega^{\mathrm{st}}$, $\delta(X) = 1$ by Proposition 17.4, and X is not K-stable by (1). This proves (4). Hence X is K-polystable whenever $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

Suppose that $\mathrm{SL}(V) \cdot \Pi$ is not closed in $\mathbb{B}^{\mathrm{nd}}$. Then $\Pi \notin \Omega^{\mathrm{st}}$, and Proposition 17.4 implies that X is K-semistable, not K-polystable, and $\delta(X) = 1$. This proves (5). Together with the case of a closed orbit, this proves (3). □

18. Special lines meeting no other line and the simple elliptic germ

In this section, we prove Theorem 18.1, the equivalence of (i), (ii), and (iii) in Theorem 1.3(2). Since $H = -K_X$, conditions (1), (2), and (3) of Theorem 18.1 are conditions (i), (ii), and (iii) of Theorem 1.3(2). The simple elliptic germ [Sai74, Section 1] of Conjecture 1.1 appears at a point of a line on X exactly when X contains a special line which intersects no other line on X, and so condition (i) of Theorem 1.3(2) is tied to conditions (ii) and (iii).

Theorem 18.1. *Let X be a smooth prime Fano threefold of genus 12. Then the following conditions are equivalent.*

(1) There exist $D \in |H|$ and a point $p \in D$ contained in a line on X such that $(D \ni p) \cong (\{y^2 = x^3 + t^4 x\} \ni 0)$ analytically.
(2) X contains a special line which intersects no other line on X.
(3) X admits an anti-canonical test configuration (Definition 2.4) with smooth central fiber on which the induced $\mathbb{G}_m$-action is non-trivial, where product test configurations are allowed.

Moreover, in (3) the central fiber is a smooth prime Fano threefold of genus 12, and in (1) the line through p is unique, special, and intersects no other line on X. In particular, any smooth prime Fano threefold of genus 12 with a faithful $\mathbb{G}_m$-action satisfies (1) and (2).

We start with the implication from (1) to (2). If the germ appears at a point p of a line L, then the divisor contains L, and the first normal jet of the divisor along L is a non-zero section of $N_{L/X}^\vee \otimes \mathcal{O}_L(H)$ vanishing to order three at p. This is possible only if L is special in the sense of

Definition 2.11. A calculation with widths (Definition 2.21) then shows that the negative section s of Set-up 10.1 is the only flopping curve of the line link (Theorem 2.23 and Proposition 10.8), so L intersects no other line.

Proposition 18.2. *Let X be a smooth prime Fano threefold of genus 12, let $D \in |H|$, let $p \in D$ be a point such that the germ $(D \ni p)$ is analytically isomorphic to $(\{y^2 = x^3 + t^4x\} \ni 0)$, let $L \subset X$ be a line with $p \in L$, and let I_L be the ideal sheaf of L. Then the following hold.*

(1) L is the unique line through p, L is special, and L intersects no other line on X.
(2) In the notation of Set-up 10.1, the flopping locus of $Y \dashrightarrow Y^+$ is s, and s has width three.
(3) $L \subset D$, and the first normal jet of D along L, that is, the image of a non-zero section of $\mathcal{O}_X(H)$ with divisor D in $H^0(L, I_L/I_L^2 \otimes \mathcal{O}_L(H)) = H^0(L, N_{L/X}^\vee \otimes \mathcal{O}_L(H))$, is a non-zero section of the summand $\mathcal{O}_{\mathbb{P}^1}(3)$ of $N_{L/X}^\vee \otimes \mathcal{O}_L(H) \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(3)$ whose zero divisor is $3p$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $L \subset D$, that L is the unique line through p, that L is special, and that (3) holds.

Since H is very ample (Lemma 2.12), any line on X is a line in the projective space of the embedding by $|H|$. In particular $L \cong \mathbb{P}^1$. The germ $y^2 - x^3 - t^4x$ has order two, so $\operatorname{mult}_p D = 2$. If a line L' through p is not contained in D, then

$$1 = (D \cdot L') \geq \operatorname{mult}_p D \cdot \operatorname{mult}_p L' = 2,$$

a contradiction. Thus any line through p is contained in D, and in particular $L \subset D$.

An analytic isomorphism of hypersurface germs of order at least two is induced by an analytic change of ambient coordinates, and the local equations then differ by a unit [GLS07, Chapter I, Lemma 1.23]. Hence there exist analytic coordinates (t, x, y) on X at p such that

$$\varphi := y^2 - x^3 - t^4x$$

is a local equation of D at p. Let τ be a local parameter of L at p, and let o_t, o_x, and o_y be the orders in τ of the restrictions of t, x, and y to L. Since L is smooth at p, $\min\{o_t, o_x, o_y\} = 1$. On L we have $y^2 = x^3 + t^4x$. If $o_y = 1$, the left side has order 2 and the right side has order at least 3. If $o_y \geq 2$ and $o_x = 1$, the left side has order at least 4 and the right side has order 3. Thus $o_x \geq 2$, $o_y \geq 2$, and $o_t = 1$. So $t|_L$ is a local parameter of L at p, and near p

$$L = \{x = A(t),\ y = B(t)\}$$

for convergent power series $A, B \in \mathbb{C}\{t\}$ with $\operatorname{ord}_t A \geq 2$. Since $B^2 = A^3 + t^4A$ and $\operatorname{ord}_t(A^3 + t^4A) \geq 6$, we have $\operatorname{ord}_t B \geq 3$. In particular the tangent line of L at p is the t-axis. The tangent line at p of any line through p is also the t-axis, since any such line is contained in D. Two distinct lines of a projective space through p have distinct tangent lines at p, so L is the unique line through p.

Let $x_1 := x - A(t)$ and $y_1 := y - B(t)$. Then (t, x_1, y_1) are analytic coordinates at p with $L = \{x_1 = y_1 = 0\}$, and, using $B^2 = A^3 + t^4A$,

$$\varphi = y_1^2 + 2B(t)y_1 - x_1^3 - 3A(t)x_1^2 - \big(3A(t)^2 + t^4\big)x_1. \tag{18.1}$$

Let $j \in H^0(L, N_{L/X}^\vee \otimes \mathcal{O}_L(H))$ be the first normal jet of D along L. Near p, the classes $\bar{x}_1, \bar{y}_1$ of x_1, y_1 form a basis of I_L/I_L^2, and by (18.1) the image of φ in I_L/I_L^2 is

$$2B(t)\bar{y}_1 - \big(3A(t)^2 + t^4\big)\bar{x}_1.$$

Since $\operatorname{ord}_t(2B) \geq 3$ and $\operatorname{ord}_t(3A^2 + t^4) \geq 4$, the section j vanishes to order at least three at p. Multiplying φ by a unit, or changing the local generator of $\mathcal{O}_X(H)$, does not change the order of j at p. If $j = 0$, then a local equation of D is contained in I_L^2 near any point of L, so D is singular along L. However, the partial derivatives $-4t^3x$, $-(3x^2 + t^4)$, and $2y$ of φ vanish simultaneously only at $t = x = y = 0$, so p is an isolated singular point of D. Thus $j \neq 0$.

By Definition 2.11 and Lemma 2.12(3), either $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, or $N_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ and L is special. In the first case

$$N^\vee_{L/X} \otimes \mathcal{O}_L(H) \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(2),$$

and no non-zero section of $\mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(2)$ vanishes to order three at a point. Therefore L is special, and

$$N^\vee_{L/X} \otimes \mathcal{O}_L(H) \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(3).$$

The image of j in the quotient $\mathcal{O}_{\mathbb{P}^1}$ by the summand $\mathcal{O}_{\mathbb{P}^1}(3)$ is a constant which vanishes at p, hence zero. So j is a non-zero section of the summand $\mathcal{O}_{\mathbb{P}^1}(3)$ vanishing to order at least three at p, and the zero divisor of j is $3p$. This proves (3).

Since the order of j at p is exactly 3, $\min\{\operatorname{ord}_t(2B), \operatorname{ord}_t(3A^2 + t^4)\} = 3$, and so $\operatorname{ord}_t B = 3$. Moreover, at a point $q \in L\backslash\{p\}$ we have $j(q) \neq 0$, so a local equation of D at q has non-zero differential at q. Thus D is smooth along $L\backslash\{p\}$, and $\operatorname{mult}_L D = 1$.

Step 2. In this step we show the following obstruction. Let Z be a smooth threefold, let $C \subset Z$ be a smooth rational curve with $N_{C/Z} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, and let $D_Z \in |-K_Z|$ be a divisor which contains C and is smooth at the generic point of C. Then the image j_Z of a section of $\mathcal{O}_Z(-K_Z)$ with divisor D_Z in $H^0(C, I_C/I_C^2 \otimes \mathcal{O}_C(-K_Z))$, where I_C is the ideal sheaf of C, vanishes to order at most one at any point of C, and $\operatorname{mult}_q D_Z \leq 2$ for any $q \in C$.

Since $\deg T_Z|_C = \deg T_C + \deg N_{C/Z} = 0$, we have $(-K_Z \cdot C) = 0$ and

$$j_Z \in H^0(C, N^\vee_{C/Z}) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}).$$

If $j_Z = 0$, then a local equation of D_Z is contained in I_C^2 near any point of C, and D_Z is singular along C. Thus $j_Z \neq 0$, and a non-zero section of $\mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$ vanishes to order at most one at any point. Suppose that $\operatorname{mult}_q D_Z \geq 3$ for some $q \in C$. Let (w, u_1, u_2) be analytic coordinates at q with $C = \{u_1 = u_2 = 0\}$, and let ψ be a local equation of D_Z at q. Any monomial of ψ has degree at least three and is divisible by u_1 or u_2, so $\psi = u_1\psi_1 + u_2\psi_2$ with $\psi_1, \psi_2 \in \mathfrak{m}_q^2$. In the basis $\bar{u}_1, \bar{u}_2$ of I_C/I_C^2 near q, the section j_Z is $(\psi_1|_C, \psi_2|_C)$, which vanishes to order at least two at q. This contradicts the previous sentence, so $\operatorname{mult}_q D_Z \leq 2$.

Step 3. In this step we show that the width w_L of s (Proposition 10.8) satisfies $w_L \geq 2$, and that $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

Since L is special, s is a flopping curve of $Y \dashrightarrow Y^+$ (Theorem 2.23), so $(K_Y \cdot s) = 0$, and s has finite width $w_L \in \{1, 2, 3\}$ (Proposition 10.8). By Definition 2.21 and [Rei83, (5.3) and (5.9)], a flopping curve of finite width has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ if the width is one, and $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ if the width is at least two.

Since $\operatorname{mult}_L D = 1$ (**Step 1**), we have $\pi^* D = D_1 + E$ for the strict transform D_1 of D on Y. As $K_Y = \pi^* K_X + E$, we have $D_1 \in |\pi^* H - E| = |-K_Y|$. The exceptional divisor E is the projectivization of the normal directions of L [Har77, Theorem II.8.24]. On the chart $y_1 = x_1 z$ of Y with coordinates (t, x_1, z), we have $E = \{x_1 = 0\}$, the point $(t, 0, z)$ is the normal direction of $\partial_{x_1} + z\partial_{y_1}$ at $(t, 0, 0) \in L$, and (18.1) implies $\varphi = x_1 g$ with

$$g = -x_1^2 + x_1\big(z^2 - 3A(t)\big) + 2B(t)z - 3A(t)^2 - t^4.$$

So $D_1 = \{g = 0\}$ on the chart.

Let $N^+ \subset N_{L/X}$ be the subbundle isomorphic to $\mathcal{O}_{\mathbb{P}^1}(1)$. It is unique, since any homomorphism $\mathcal{O}_{\mathbb{P}^1}(1) \to \mathcal{O}_{\mathbb{P}^1}(-2)$ is zero. We view j as a homomorphism $N_{L/X} \to \mathcal{O}_L(H)$. The restriction of j to N^+ is a homomorphism $\mathcal{O}_{\mathbb{P}^1}(1) \to \mathcal{O}_{\mathbb{P}^1}(1)$ vanishing at p, hence zero. Since $j(q) \neq 0$ for $q \in L\backslash\{p\}$, the kernel of $j(q)$ is the fiber of N^+ at q. The section of E given by N^+ has self-intersection

$$\deg(N_{L/X}/N^+) - \deg N^+ = -3,$$

so it is s. In the chart, the kernel of $j(q)$ at $q = (t, 0, 0)$ with $t \neq 0$ is given by $2B(t)z - 3A(t)^2 - t^4 = 0$. Let

$$\rho(t) := \frac{3A(t)^2 + t^4}{2B(t)}.$$

Since $\operatorname{ord}_t(3A^2 + t^4) \geq 4$ and $\operatorname{ord}_t B = 3$, we have $\rho \in \mathbb{C}\{t\}$ and $\operatorname{ord}_t \rho \geq 1$. Thus s and the curve $\{x_1 = 0,\ z = \rho(t)\}$ coincide over $t \neq 0$, hence near the point $p_s := (0, 0, 0)$, which is the point of s over p. Let $z_1 := z - \rho(t)$ and $\alpha(t) := \rho(t)^2 - 3A(t)$. Then

$$g = -x_1^2 + x_1\big(z_1^2 + 2\rho(t)z_1 + \alpha(t)\big) + 2B(t)z_1, \tag{18.2}$$

and $s = \{x_1 = z_1 = 0\}$ near p_s. We have $\operatorname{ord}_t \alpha \geq 2$ and $\operatorname{ord}_t B = 3$. The terms $-x_1^2$, $x_1 z_1^2$, and $2\rho x_1 z_1$ of (18.2) are contained in I_s^2, so the first normal jet j_s of D_1 along s, that is, the image of g in I_s/I_s^2, is

$$j_s = \alpha(t)\bar{x}_1 + 2B(t)\bar{z}_1$$

near p_s. Thus j_s vanishes to order at least two at p_s. At a point $(t, x_1, z_1) = (t, 0, 0)$ with $t \neq 0$ small, $\partial g/\partial z_1 = 2B(t) \neq 0$, so D_1 is smooth at the generic point of s.

If $w_L = 1$, then $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, and **Step 2** applied to (Y, s, D_1) contradicts the order of j_s at p_s. Thus $w_L \geq 2$ and $N_{s/Y} \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

Step 4. In this step we show that $w_L \neq 2$.

Suppose that $w_L = 2$. Since $(K_Y \cdot s) = 0$, j_s is a non-zero section of $N_{s/Y}^\vee \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2)$. Let $N_s^+ \subset N_{s/Y}$ be the subbundle isomorphic to $\mathcal{O}_{\mathbb{P}^1}$, unique since any homomorphism $\mathcal{O}_{\mathbb{P}^1} \to \mathcal{O}_{\mathbb{P}^1}(-2)$ is zero. The restriction of j_s to N_s^+ is a constant vanishing at p_s, hence zero. So j_s is a non-zero section of $(N_{s/Y}/N_s^+)^\vee \cong \mathcal{O}_{\mathbb{P}^1}(2)$ vanishing to order at least two at p_s, and the zero divisor of j_s is $2p_s$. As $\operatorname{ord}_t B = 3$, the formula for j_s in **Step 3** implies $\operatorname{ord}_t \alpha = 2$. Let

$$\beta(t) := -\frac{2B(t)}{\alpha(t)} \in \mathbb{C}\{t\}.$$

Then $\operatorname{ord}_t \beta = 1$, and $\alpha\beta + 2B = 0$. So the normal vector $\beta(t)\partial_{x_1} + \partial_{z_1}$ is in the kernel of j_s, and N_s^+ is spanned by $\beta(t)\partial_{x_1} + \partial_{z_1}$ near p_s.

Let $Y' := \operatorname{Bl}_s Y$ with exceptional divisor E_s. Then E_s is the projectivization of $N_{s/Y}$, so $E_s \cong \mathbb{F}_2$, and the negative section s_1 of E_s is given by N_s^+. Consider the chart $x_1 = z_1(z_2 + \beta(t))$ of Y' with coordinates (t, z_1, z_2). There $E_s = \{z_1 = 0\}$, the point $(t, 0, z_2)$ is the normal direction of $(z_2 + \beta(t))\partial_{x_1} + \partial_{z_1}$, and so $s_1 = \{z_1 = z_2 = 0\}$. Since $j_s \neq 0$, $\operatorname{mult}_s D_1 = 1$, and the strict transform D_2 of D_1 satisfies $D_2 \in |-K_{Y'}|$, as $K_{Y'}$ is the pullback of K_Y plus E_s. By (18.2) and $\alpha\beta + 2B = 0$, the local equation of D_2 is

$$\frac{g\big(t, z_1(z_2 + \beta(t)), z_1\big)}{z_1} = \alpha(t)z_2 - z_1\big(z_2 + \beta(t)\big)^2 + z_1^2\big(z_2 + \beta(t)\big) + 2\rho(t)z_1\big(z_2 + \beta(t)\big), \tag{18.3}$$

where we write g as a function of (t, x_1, z_1). Each term of (18.3) has order at least three at $p_{s_1} := (0, 0, 0) \in s_1$, because $\operatorname{ord}_t \alpha = 2$, $\operatorname{ord}_t \beta = 1$, and $\operatorname{ord}_t \rho \geq 1$. Hence $\operatorname{mult}_{p_{s_1}} D_2 \geq 3$. Each term of (18.3) is divisible by z_1 or z_2, so $s_1 \subset D_2$. At a point $(t, 0, 0)$ with $t \neq 0$ small, the derivative of (18.3) with respect to z_2 is $\alpha(t) \neq 0$, so D_2 is smooth at the generic point of s_1.

By [Rei83, Theorem (5.4)], s_1 has width $w_L - 1 = 1$ on Y', so $N_{s_1/Y'} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ [Rei83, (5.3) and (5.9)]. **Step 2** applied to (Y', s_1, D_2) implies $\operatorname{mult}_{p_{s_1}} D_2 \leq 2$, a contradiction. Thus $w_L \neq 2$.

Step 5. In this step we conclude the proof.

By **Steps 3 and 4**, $w_L \geq 3$. By Proposition 10.8,

$$w_L + \sum_{L' \neq L,\ L' \cap L \neq \emptyset} w_L(L') = 3.$$

Thus $w_L = 3$, and no line $L' \neq L$ intersects L. By Proposition 10.8, the flopping locus of $Y \dashrightarrow Y^+$ is s. This proves (2), and together with **Step 1** it proves (1). □

For the implication from (3) to (1) of Theorem 18.1, we write down, on each member with a torus action, an explicit invariant Plücker section which has the germ of Proposition 18.2 at the torus-fixed point p_- of a line L_-. This section is the member $P_{-2,-1,3} + cP_{-3,0,3}$ of the pencil spanned by two Plücker coordinates, and the parameter c is fixed by the vanishing of the seminvariant $2\alpha_2^3 - 9\alpha_3\alpha_2\alpha_1 + 27\alpha_3^2\alpha_0$ of the binary cubic ϕ in the proof below, that is, $c = r/v$ for a root r of the polynomial T_v below. The local equation is weighted homogeneous of degree six for the torus weights, and this is why the germ survives the degeneration of Lemma 18.4.

Lemma 18.3. *Let V have basis $e_{-3}, \dots, e_3$ with dual basis $e_{-3}^\vee, \dots, e_3^\vee$, let $e_{ij} := e_i^\vee \wedge e_j^\vee$, and let P_{ijk}, $i < j < k$, be the Plücker coordinates of $\mathrm{Gr}(3, V)$ for the basis $e_{-3}, \dots, e_3$. For $v \in \mathbb{C}\setminus\{0, 1\}$ let Π'_v be spanned by*

$$\omega_{-1} := e_{-3,2} + e_{-2,1} + e_{-1,0}, \quad \omega_0 := e_{-3,3} + e_{-2,2} + ve_{-1,1}, \quad \omega_1 := e_{-2,3} + e_{-1,2} + e_{0,1},$$

let $X'_v := X_{\Pi'_v}$, let $p_- := \langle e_{-3}, e_{-2}, e_{-1}\rangle$, let $U_- := \{P_{-3,-2,-1} \neq 0\} \subset \mathrm{Gr}(3, V)$, and let $L_- \subset \mathrm{Gr}(3, V)$ be the set of three-dimensional subspaces of V containing $\langle e_{-3}, e_{-2}\rangle$ and contained in $\langle e_{-3}, e_{-2}, e_{-1}, e_0\rangle$. For $q \in U_-$, let $m_{i,j}(q)$, $0 \le i \le 3$, $-3 \le j \le -1$, be the unique numbers such that the vectors $e_j + \sum_{i=0}^3 m_{i,j}(q)e_i$, $-3 \le j \le -1$, span q, and let $t := m_{0,-1}$, $x := m_{1,-1}$, and $y := m_{2,-1}$. Let $a := v - 1$ and, for a variable r, let

$$\ell_v(r) := 1 + (a-1)r, \qquad \kappa_v(r) := \ell_v(r)^2 + 4ar, \qquad W_v(r) := ar(1-r),$$
$$I_v(r) := \kappa_v(r)^2 - 24W_v(r)\ell_v(r), \qquad T_v(r) := \kappa_v(r)^3 - 36W_v(r)\kappa_v(r)\ell_v(r) + 216W_v(r)^2.$$

Then the following hold.

1. *Under the identification of e_i with the basis vector e'_i of Definition-Proposition 16.14 for $-3 \le i \le 3$, the net Π'_v is the net Π_h of Definition-Proposition 16.14 with $v = 1/(h-1)$. In particular, X'_v is a smooth prime Fano threefold of genus 12, and $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_{X'_v} \cong \mathcal{O}_{X'_v}(-K_{X'_v})$.*
2. *$U_- \cap X'_v \cong \mathbb{A}^3$ with coordinates (t, x, y) and origin p_-, and L_- is a line on X'_v such that $L_- \cap U_-$ is the t-axis.*
3. *T_v has degree six if $v \neq 2$ and degree four if $v = 2$. It has a root r with $I_v(r) \neq 0$, and any root r of T_v with $I_v(r) \neq 0$ satisfies $r \neq 1$. The only common roots of I_v and T_v with $v \notin \{0, 1\}$ are $(v, r) = (-3, 1)$ and $(1/5, -5/27)$, of multiplicities two and one as roots of T_v.*
4. *Let r be a root of T_v with $I_v(r) \neq 0$, let $c := r/v$, and let $\sigma_c \in H^0(X'_v, -K_{X'_v})$ be the restriction of $P_{-2,-1,3} + cP_{-3,0,3}$, with divisor D_c. On $U_- \cap X'_v$ the expression of σ_c in the local generator $(dt \wedge dx \wedge dy)^{-1}$ of $-K_{X'_v}$ is a weighted homogeneous polynomial of degree six for the weights $(1, 2, 3)$ of (t, x, y), and an invertible weight-preserving polynomial change of coordinates followed by multiplication by a non-zero constant carries it to $y^2 - x^3 - t^4x$. In particular, $(D_c \ni p_-) \cong (\{y^2 = x^3 + t^4x\} \ni 0)$ analytically.*
5. *Let k be a non-zero integer and let $\lambda \in \mathbb{G}_m$ act on V by $\lambda \cdot e_i := \lambda^{ki}e_i$. Then the action preserves Π'_v, p_-, U_-, and L_-. For $q \in U_- \cap X'_v$ we have*

$$t(\lambda \cdot q) = \lambda^k t(q), \quad x(\lambda \cdot q) = \lambda^{2k}x(q), \quad y(\lambda \cdot q) = \lambda^{3k}y(q),$$

and σ_c is invariant for the $\mathbb{G}_m$-linearization of $-K_{X'_v}$ induced by the action on X'_v.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

The spanning forms of Π'_v are the forms of Definition-Proposition 16.14 in the basis $e'_{-3}, \dots, e'_3$ with $v = 1/(h-1)$, that is, $h = 1 + 1/v$. Since $v \in \mathbb{C}\setminus\{0, 1\}$, we have $h \notin \{1, 2\}$. By Definition-Proposition 16.14, $X'_v = X_{\Pi_h}$ is a smooth prime Fano threefold of genus 12 and $\Pi'_v \in \Omega$. By Theorem 16.12(1), the restriction $\mathcal{U}|_{X'_v}$ of the tautological subbundle is the Mukai bundle of X'_v, so $\det(\mathcal{U}|_{X'_v}) \cong \mathcal{O}_{X'_v}(K_{X'_v})$ (Definition-Theorem 2.20). Since $\det\mathcal{U}^\vee \cong \mathcal{O}_{\mathrm{Gr}(3,V)}(1)$, we obtain $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_{X'_v} \cong \mathcal{O}_{X'_v}(-K_{X'_v})$.

Step 2. In this step we prove (2) and compute the Plücker coordinates $P_{-2,-1,3}$ and $P_{-3,0,3}$ on the chart.

Write $m_{i,j}$ for the functions $m_{i,j}(q)$. They are coordinates on $U_- \cong \mathbb{A}^{12}$, and the vectors $b_j := e_j + \sum_{i=0}^{3} m_{i,j} e_i$, $-3 \le j \le -1$, form a basis of $\mathcal{U}|_{U_-}$. In the basis b_{-3}, b_{-2}, b_{-1}, the section of $\Lambda^2\mathcal{U}^\vee \otimes \Pi_v'^\vee$ defining X_v' (Definition 2.17) has the nine components $\omega(b_j, b_{j'})$, $\omega \in \{\omega_{-1}, \omega_0, \omega_1\}$, $j < j'$. From $e_{ij}(\xi, \xi') = \xi_i\xi_j' - \xi_j\xi_i'$, where ξ_i is the coefficient of e_i in ξ, we obtain the nine equations of $U_- \cap X_v'$ in U_-:

$$\begin{gathered} m_{2,-2} = m_{1,-3}, \qquad m_{0,-3} = y, \qquad m_{0,-2} = x, \\ m_{3,-2} = m_{2,-3}, \qquad m_{3,-1} = v m_{1,-3}, \qquad v m_{1,-2} = y, \\ m_{3,-3} = m_{0,-3}m_{1,-2} - m_{0,-2}m_{1,-3}, \qquad m_{2,-3} = m_{0,-3}x - t m_{1,-3}, \\ m_{2,-2} - m_{3,-1} = m_{0,-2}x - t m_{1,-2}, \end{gathered} \tag{18.4}$$

where the first, second, and last three equations come from ω_{-1}, ω_0, and ω_1, and in each group of three the equations come from the pairs $(j, j') = (-3, -2)$, $(-3, -1)$, and $(-2, -1)$, in the order listed. The first six equations express $m_{0,-3}$, $m_{0,-2}$, $m_{1,-2}$, $m_{2,-2}$, $m_{3,-1}$, and $m_{3,-2}$ through t, x, y, $m_{1,-3}$, and $m_{2,-3}$. Substituting them into the last equation of (18.4), we obtain $x^2 - ty/v + (v-1)m_{1,-3} = 0$, and the seventh and eighth equations express $m_{3,-3}$ and $m_{2,-3}$ through t, x, y, and $m_{1,-3}$. As $v(v-1) \ne 0$, the nine equations are equivalent to

$$\begin{gathered} m_{0,-3} = y, \quad m_{0,-2} = x, \quad m_{1,-2} = \frac{y}{v}, \quad m_{1,-3} = m_{2,-2} = \mu, \quad m_{3,-1} = v\mu, \\ m_{2,-3} = m_{3,-2} = xy - t\mu, \quad m_{3,-3} = \frac{y^2}{v} - x\mu, \qquad \text{where } \mu := \frac{ty - vx^2}{v(v-1)}. \end{gathered} \tag{18.5}$$

Hence the coordinate ring of $U_- \cap X_v'$ is $\mathbb{C}[t, x, y]$, that is, $U_- \cap X_v' \cong \mathbb{A}^3$ with coordinates (t, x, y), and p_- is the origin.

At $x = y = 0$, (18.5) implies that all $m_{i,j}$ except $m_{0,-1} = t$ vanish, so the point is $\langle e_{-3}, e_{-2}, e_{-1} + te_0 \rangle$. Thus the t-axis is $L_- \cap U_-$. By definition, L_- consists of the three-dimensional subspaces containing $\langle e_{-3}, e_{-2} \rangle$ and contained in $\langle e_{-3}, e_{-2}, e_{-1}, e_0 \rangle$, so L_- is a line of $\mathrm{Gr}(3, V)$ in the Plücker embedding [Har92, Exercise 6.9(i)]. As X_v' is closed and contains the dense subset $L_- \cap U_-$ of L_-, we have $L_- \subset X_v'$. By (1), $(-K_{X_v'} \cdot L_-) = 1$, so L_- is a line on X_v'. This proves (2).

On U_- we have $P_{-3,-2,-1} = 1$, $P_{-2,-1,3} = m_{3,-3}$, and $P_{-3,0,3} = m_{0,-2}m_{3,-1} - t m_{3,-2}$. By (18.5), on $U_- \cap X_v'$

$$\begin{aligned} f_1 &:= v(v-1)P_{-2,-1,3} = vx^3 + (v-1)y^2 - txy, \\ f_2 &:= v(v-1)P_{-3,0,3} = (vx + t^2)(ty - vx^2) - v(v-1)txy \\ &\quad = t^3 y - vt^2x^2 - v(v-2)txy - v^2x^3. \end{aligned} \tag{18.6}$$

Step 3. In this step we prove (3).

If $a \ne 1$, that is, $v \ne 2$, then ℓ_v, κ_v, and W_v have degrees 1, 2, and 2 with leading coefficients $a-1$, $(a-1)^2$, and $-a$. So κ_v^3 has degree six with leading coefficient $(a-1)^6$, while $W_v\kappa_v\ell_v$ and W_v^2 have degrees at most five and four. Thus $\deg T_v = 6$. If $v = 2$, then a direct calculation shows

$$T_2(r) = 216r^4 - 224r^3 + 156r^2 - 24r + 1,$$

so $\deg T_2 = 4$. At $r = 1$ we have $W_v(1) = 0$, so $T_v(1) = \kappa_v(1)^3$ and $I_v(1) = \kappa_v(1)^2$ vanish together. Hence any root r of T_v with $I_v(r) \ne 0$ satisfies $r \ne 1$.

We determine the common roots. Since $v \notin \{0, 1\}$ and $a = v - 1$, we have $a \notin \{0, -1\}$. Suppose that $I_v(r) = T_v(r) = 0$, and write ℓ, κ, and W for $\ell_v(r)$, $\kappa_v(r)$, and $W_v(r)$. If $W = 0$, then $r \in \{0, 1\}$. At $r = 0$ we have $\kappa = 1$ and $I_v(0) = 1$, which is impossible. At $r = 1$ we have $\ell = a$ and $\kappa = a(a+4)$, so $I_v(1) = a^2(a+4)^2 = 0$ implies $a = -4$, that is, $v = -3$. Suppose now that

$W \neq 0$. From $I_v(r) = 0$ we obtain $\kappa^2 = 24W\ell$, and substituting $\kappa^2 = 24W\ell$ into $T_v(r) = 0$ we obtain $\kappa^3 = 432W^2$. So $\kappa \neq 0$ and $\ell \neq 0$, and dividing $\kappa^3 = 432W^2$ by $\kappa^2 = 24W\ell$ we obtain

$$\kappa\ell = 18W, \qquad \kappa = \frac{4}{3}\ell^2, \qquad W = \frac{2}{27}\ell^3.$$

Since $\kappa = \ell^2 + 4ar$ and $W = ar(1-r)$, the identities $\kappa = \frac{4}{3}\ell^2$ and $W = \frac{2}{27}\ell^3$ are equivalent to

$$ar = \frac{\ell^2}{12}, \qquad 1 - r = \frac{8}{9}\ell.$$

Together with $\ell = 1 + (a-1)r$, the second identity implies $(8a+1)r = 1$. Hence $8a + 1 \neq 0$, $r = 1/(8a+1)$, and $\ell = 9a/(8a+1)$. Substituting into $ar = \ell^2/12$ and using $a \neq 0$, we obtain $12(8a+1) = 81a$, so $a = -4/5$, $v = 1/5$, and $r = -5/27$. Conversely, a direct calculation shows that $I_v(r) = T_v(r) = 0$ for $(v, r) = (-3, 1)$ and $(1/5, -5/27)$.

A direct calculation shows

$$T_{-3}(1+\epsilon) = 17280\epsilon^2 + O(\epsilon^3), \qquad T'_{1/5}\Big(-\frac{5}{27}\Big) = \frac{4096}{3645} \neq 0.$$

So the root $r = 1$ of T_{-3} has multiplicity two, and the root $r = -5/27$ of $T_{1/5}$ has multiplicity one. For any $v \in \mathbb{C}\backslash\{0, 1\}$, T_v has at most one common root with I_v, of multiplicity at most two, and $\deg T_v \geq 4$. Therefore T_v has a root r with $I_v(r) \neq 0$. This proves (3).

Step 4. In this step we prove (4).

By (1) and (2), $P_{-3,-2,-1}|_{U_- \cap X'_v}$ and $(dt \wedge dx \wedge dy)^{-1}$ are nowhere vanishing sections of $\mathcal{O}_{X'_v}(-K_{X'_v})$ on $U_- \cap X'_v \cong \mathbb{A}^3$. Their ratio is a nowhere vanishing regular function on $\mathbb{A}^3$, hence a constant $\gamma \in \mathbb{C}^*$. By (18.6), the expression of σ_c in the local generator $(dt \wedge dx \wedge dy)^{-1}$ is

$$\frac{\gamma}{v(v-1)}\Phi, \qquad \Phi := f_1 + cf_2.$$

Each monomial of f_1 and f_2 has weighted degree six for the weights $(1, 2, 3)$ of (t, x, y), so Φ is weighted homogeneous of degree six. With $r = vc$, $a = v - 1$, and $\ell := \ell_v(r)$, $\kappa := \kappa_v(r)$, (18.6) implies

$$\Phi = ay^2 + (ct^3 - \ell tx)y + v(1-r)x^3 - rt^2x^2.$$

Let

$$\bar{x} := vx, \qquad \bar{y} := y + \frac{ct^3 - \ell tx}{2a}.$$

A direct calculation shows

$$4av^2\Phi = (2av\bar{y})^2 + \phi(\bar{x}, t), \qquad \phi(\bar{x}, t) := \alpha_3\bar{x}^3 + \alpha_2 t^2\bar{x}^2 + \alpha_1 t^4\bar{x} + \alpha_0 t^6, \tag{18.7}$$

with $\alpha_3 = 4a(1-r)$, $\alpha_2 = -\kappa$, $\alpha_1 = 2r\ell$, and $\alpha_0 = -r^2$. For the binary cubic ϕ, a direct calculation shows

$$\alpha_2^2 - 3\alpha_3\alpha_1 = I_v(r), \qquad 2\alpha_2^3 - 9\alpha_3\alpha_2\alpha_1 + 27\alpha_3^2\alpha_0 = -2T_v(r) = 0.$$

By (3), $r \neq 1$, so $\alpha_3 \neq 0$. Let $\hat{x} := \bar{x} + \alpha_2 t^2/(3\alpha_3)$. Since the coefficient of t^6 in ϕ after the substitution $\bar{x} = \hat{x} - \alpha_2 t^2/(3\alpha_3)$ is

$$\frac{2\alpha_2^3 - 9\alpha_3\alpha_2\alpha_1 + 27\alpha_3^2\alpha_0}{27\alpha_3^2} = 0,$$

we obtain

$$\phi(\bar{x}, t) = \alpha_3\hat{x}^3 - \frac{I_v(r)}{3\alpha_3}t^4\hat{x}.$$

Choose $\zeta_1, \zeta_2 \in \mathbb{C}^*$ with $\zeta_1^3 = -1/\alpha_3$ and $\zeta_2^4 = 3\alpha_3/(I_v(r)\zeta_1)$, which is possible since $I_v(r) \neq 0$, and let

$$t' := \zeta_2^{-1}t, \qquad x' := \zeta_1^{-1}\hat{x}, \qquad y' := 2av\bar{y}.$$

Then $4av^2\Phi = y'^2 - x'^3 - t'^4x'$ by (18.7). The coordinates t', x', and y' are weighted homogeneous polynomials in (t, x, y) of weights 1, 2, and 3, and the linear part of $(t, x, y) \to (t', x', y')$ is

invertible, since a, v, ζ_1, and ζ_2 are non-zero. Hence $(t,x,y)\to(t',x',y')$ is an automorphism of $\mathbb{A}^3$ fixing the origin, and it carries the expression of σ_c to $\gamma(4av^3(v-1))^{-1}(y'^2-x'^3-t'^4x')$. This proves (4).

Step 5. In this step we prove (5).

The action of λ on $\Lambda^2V^\vee$ sends e_{ij} to $\lambda^{-k(i+j)}e_{ij}$. Each term of ω_m has $i+j=m$, so λ sends ω_m to $\lambda^{-km}\omega_m$ for $m\in\{-1,0,1\}$, and the action preserves Π'_v, hence X'_v. It fixes p_-, multiplies $P_{-3,-2,-1}$ by a non-zero constant, so it preserves U_-, and it preserves the subspaces $\langle e_{-3},e_{-2}\rangle$ and $\langle e_{-3},e_{-2},e_{-1},e_0\rangle$, so it preserves L_-. For $q\in U_-$, the vectors

$$\lambda^{-kj}\Big(\lambda\cdot\big(e_j+\textstyle\sum_{i=0}^3 m_{i,j}(q)e_i\big)\Big)=e_j+\sum_{i=0}^3\lambda^{k(i-j)}m_{i,j}(q)e_i,\qquad -3\le j\le -1,$$

span $\lambda\cdot q$, so $m_{i,j}(\lambda\cdot q)=\lambda^{k(i-j)}m_{i,j}(q)$. With $(i,j)=(0,-1)$, $(1,-1)$, and $(2,-1)$, we obtain the formulas for t, x, and y.

Let g_λ be the automorphism of X'_v given by λ. The induced linearization sends a section σ of $-K_{X'_v}$ to $(g_\lambda^{-1})^*\sigma$. By **Step 4**, $\sigma_c=\gamma'\Phi(t,x,y)(dt\wedge dx\wedge dy)^{-1}$ on $U_-\cap X'_v$ for some $\gamma'\in\mathbb{C}^*$. As Φ is weighted homogeneous of degree six,

$$\Phi\circ g_\lambda^{-1}=\lambda^{-6k}\Phi,\qquad (g_\lambda^{-1})^*(dt\wedge dx\wedge dy)=\lambda^{-6k}dt\wedge dx\wedge dy.$$

Thus $(g_\lambda^{-1})^*\sigma_c=\sigma_c$ on the dense open subset $U_-\cap X'_v$, hence on X'_v. This proves (5). □

The following lemma carries a weighted homogeneous germ over from the central fiber of a test configuration to the general fiber: the invariant lift of the central section differs from it only by terms of higher weight, and by the normal form theorem of Arnold for semiquasihomogeneous functions the terms of higher weight do not change the analytic type of the germ, provided that the family of lines and the fixed point lift equivariantly.

Let $\mathbb{G}_m$ act on a scheme. A function ϕ on a $\mathbb{G}_m$-stable open subset, or an element ϕ of the completed local ring at a fixed point, has *character* $j\in\mathbb{Z}$ if $\phi(\lambda\cdot q)=\lambda^j\phi(q)$ for any $\lambda\in\mathbb{G}_m$, and a section ψ of a $\mathbb{G}_m$-linearized sheaf on a $\mathbb{G}_m$-stable open subset has character j if $\lambda^*\psi=\lambda^j\psi$ for any $\lambda\in\mathbb{G}_m$. A rational representation of $\mathbb{G}_m$ is the direct sum of the character spaces, so taking the character-j part is an exact functor. We use the exactness without further mention. For a $\mathbb{C}$-vector space W_0 of finite dimension, an element ξ of $W_0\otimes_{\mathbb{C}}\mathbb{C}((\tau))$ invariant under $\tau\to\lambda\tau$ for any $\lambda\in\mathbb{C}^*$ belongs to W_0, since the coordinates of ξ in a basis of W_0 are Laurent series $\sum_k a_k\tau^k$ with $a_k\lambda^k=a_k$ for any λ. Thus a $\mathbb{C}((\tau))$-point of a $\mathbb{C}$-scheme of finite type which is invariant under $\tau\to\lambda\tau$ for any $\lambda\in\mathbb{C}^*$ is the base change of a $\mathbb{C}$-point: take an affine open subset containing the image of the point, fix a closed embedding of the affine open subset into $\mathbb{A}^N$, and apply the case $W_0=\mathbb{C}$ to the coordinates.

Lemma 18.4. *Let X be a smooth Fano threefold, and let $\pi:\mathcal{X}\to\mathbb{A}^1$ be a smooth projective morphism whose fibers are connected smooth Fano threefolds, with a $\mathbb{G}_m$-action covering the standard action on $\mathbb{A}^1$ and a $\mathbb{G}_m$-equivariant isomorphism $\mathcal{X}|_{\mathbb{A}^1\setminus\{0\}}\cong X\times(\mathbb{A}^1\setminus\{0\})$, where $\mathbb{G}_m$ acts trivially on X. Let $\mathcal{L}\subset\mathcal{X}$ be a $\mathbb{G}_m$-invariant closed subscheme, smooth and proper of relative dimension one over $\mathbb{A}^1$, all of whose fibers are lines with respect to $-K_{\mathcal{X}/\mathbb{A}^1}$, let $Y:=\mathcal{X}_0$ and $L:=\mathcal{L}_1\subset X$, let $p_0\in\mathcal{L}_0$ be a $\mathbb{G}_m$-fixed point, and let r be a positive integer. Assume that there exist equivariant formal regular parameters (t,x,y) at p_0 on Y with*

$$t(\lambda\cdot q)=\lambda^{-r}t(q),\quad x(\lambda\cdot q)=\lambda^{-2r}x(q),\quad y(\lambda\cdot q)=\lambda^{-3r}y(q)\quad \text{for } \lambda\in\mathbb{G}_m,$$

and a non-zero section $\sigma_0\in H^0(Y,-K_Y)$, invariant for the natural $\mathbb{G}_m$-linearization of $-K_Y$, whose expression in the local generator $(dt\wedge dx\wedge dy)^{-1}$ of $-K_Y$ is carried to $y^2-x^3-t^4x$ by an invertible weight-preserving polynomial change of coordinates, for the weights $(1,2,3)$ of (t,x,y), and multiplication by a non-zero constant. Then there exist $p\in L$ and a non-zero $\sigma\in H^0(X,-K_X)$ whose divisor D satisfies

$$(D\ni p)\cong(\{y^2=x^3+t^4x\}\ni 0)$$

analytically.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we change the coordinates and multiply σ_0 by a non-zero constant so that the expression of σ_0 in $(dt \wedge dx \wedge dy)^{-1}$ is $y^2 - x^3 - t^4 x$.

Let $f_0 := y^2 - x^3 - t^4 x \in \mathbb{C}[t, x, y]$, and for $b = (b_1, b_2, b_3) \in \mathbb{Z}^3_{\geq 0}$ let $w(b) := b_1 + 2b_2 + 3b_3$ be the weight of $t^{b_1} x^{b_2} y^{b_3}$. The monomial $t^{b_1} x^{b_2} y^{b_3}$ has character $-rw(b)$, and $(dt \wedge dx \wedge dy)^{-1}$ has character $6r$. Since σ_0 is invariant, the expression ϕ_0 of σ_0 in $(dt \wedge dx \wedge dy)^{-1}$ has character $-6r$. Thus ϕ_0 is a weighted homogeneous polynomial of degree six. By assumption, there exist $c_0 \in \mathbb{C}^*$ and an invertible weight-preserving polynomial change of coordinates (t', x', y') of (t, x, y) with $\phi_0 = c_0 f_0(t', x', y')$. Then t', x', y' are equivariant formal regular parameters at p_0 of characters $-r, -2r, -3r$. The Jacobian determinant $\det\big(\partial(t', x', y')/\partial(t, x, y)\big)$ is weighted homogeneous of degree $(1 + 2 + 3) - (1 + 2 + 3) = 0$, hence equal to a constant j_0, and $j_0 \neq 0$ since the change of coordinates is invertible. Thus

$$\sigma_0 = \phi_0 (dt \wedge dx \wedge dy)^{-1} = c_0 j_0 f_0(t', x', y')(dt' \wedge dx' \wedge dy')^{-1}.$$

Possibly replacing σ_0 with $(c_0 j_0)^{-1}\sigma_0$, which is again invariant, and (t, x, y) with (t', x', y'), we may assume that the expression of σ_0 in $(dt \wedge dx \wedge dy)^{-1}$ is f_0.

Step 2. In this step we construct equivariant parameters of $\mathcal{X}$ at p_0 and the point $p \in L$.

The coordinate τ of $\mathbb{A}^1$ has character 1. Since π is smooth at p_0, the restriction map $\widehat{\mathcal{O}}_{\mathcal{X},p_0} \to \widehat{\mathcal{O}}_{Y,p_0}$ is surjective with kernel $\tau\widehat{\mathcal{O}}_{\mathcal{X},p_0}$. For any $n \geq 1$ and $j \in \mathbb{Z}$, the restriction map from the character-j part of $\mathcal{O}_{\mathcal{X},p_0}/\mathfrak{m}^n_{p_0}$ to the character-j part of $\mathcal{O}_{Y,p_0}/\mathfrak{m}^n_{p_0}$ is a surjection of vector spaces of finite dimension, and an inverse limit of such surjections is surjective. Thus there exist $z_1, z_2, z_3 \in \widehat{\mathcal{O}}_{\mathcal{X},p_0}$ of characters $-r, -2r, -3r$ which restrict to t, x, y. Since $\tau \notin \mathfrak{m}^2_{p_0}$ and z_1, z_2, z_3 restrict to regular parameters of Y at p_0,

$$\widehat{\mathcal{O}}_{\mathcal{X},p_0} = \mathbb{C}[[\tau, z_1, z_2, z_3]].$$

An equivariant formal section of π through p_0 is a $\mathbb{G}_m$-equivariant local $\mathbb{C}[[\tau]]$-algebra homomorphism $\mathbb{C}[[\tau, z_1, z_2, z_3]] \to \mathbb{C}[[\tau]]$. It sends z_i to a power series of character $-ri < 0$. Since τ^k has character $k \geq 0$, it sends z_i to 0. Thus there exists exactly one equivariant formal section of π through p_0, given by $z_1 = z_2 = z_3 = 0$. Composing $\mathcal{O}_{\mathcal{X},p_0} \to \widehat{\mathcal{O}}_{\mathcal{X},p_0}$ with the homomorphism $z_i \to 0$, we obtain an equivariant morphism $\varsigma : \operatorname{Spec}\mathbb{C}[[\tau]] \to \mathcal{X}$ over $\mathbb{A}^1$ which sends the closed point to p_0.

The cotangent space of $\mathcal{L}_0$ at p_0 is a quotient of the cotangent space of Y at p_0, so it has character $-r$, $-2r$, or $-3r$. Since $\mathcal{L} \to \mathbb{A}^1$ is smooth at p_0, the argument above giving $\widehat{\mathcal{O}}_{\mathcal{X},p_0} = \mathbb{C}[[\tau, z_1, z_2, z_3]]$ shows that $\widehat{\mathcal{O}}_{\mathcal{L},p_0} = \mathbb{C}[[\tau, l]]$ for some l of negative character, and $l = 0$ defines an equivariant formal section of $\mathcal{L} \to \mathbb{A}^1$ through p_0. It is an equivariant formal section of π through p_0, hence it is ς by uniqueness. Thus ς factors through $\mathcal{L}$.

Let $p_\eta : \operatorname{Spec}\mathbb{C}((\tau)) \to X$ be the composition of $\varsigma|_{\operatorname{Spec}\mathbb{C}((\tau))}$ with the isomorphism $\mathcal{X}|_{\mathbb{A}^1\setminus\{0\}} \cong X \times (\mathbb{A}^1\setminus\{0\})$ and the projection to X. Since ς is equivariant and $\mathbb{G}_m$ acts trivially on X, p_η is invariant under the substitutions $\tau \to \lambda\tau$, $\lambda \in \mathbb{C}^*$. By the paragraph before Lemma 18.4, p_η is the base change of a point $p \in X$. Since $\mathcal{L}$ is invariant and $\mathbb{G}_m$ acts trivially on X, the isomorphism $\mathcal{X}|_{\mathbb{A}^1\setminus\{0\}} \cong X \times (\mathbb{A}^1\setminus\{0\})$ maps $\mathcal{L}|_{\mathbb{A}^1\setminus\{0\}}$ onto $L \times (\mathbb{A}^1\setminus\{0\})$. Since ς factors through $\mathcal{L}$, we have $p \in L$.

Step 3. In this step we lift σ_0 to an invariant relative section and expand the lift in invariant parameters.

For any $s \in \mathbb{A}^1$, the fiber $\mathcal{X}_s$ is a smooth Fano threefold and $-K_{\mathcal{X}_s} = K_{\mathcal{X}_s} + (-2K_{\mathcal{X}_s})$, so $H^i(\mathcal{X}_s, -K_{\mathcal{X}_s}) = 0$ for $i > 0$ by Kodaira vanishing [Har77, Remark III.7.15]. By cohomology and base change [Har77, Theorem III.12.11], $\pi_*\omega^{-1}_{\mathcal{X}/\mathbb{A}^1}$ is locally free and the restriction map

$$H^0(\mathcal{X}, \omega^{-1}_{\mathcal{X}/\mathbb{A}^1}) \to H^0(Y, -K_Y)$$

is surjective. The restriction map is equivariant for the natural linearizations. Thus there exists an invariant $\tilde{\sigma} \in H^0(\mathcal{X}, \omega^{-1}_{\mathcal{X}/\mathbb{A}^1})$ with $\tilde{\sigma}|_Y = \sigma_0$.

Let $\theta := (dz_1 \wedge dz_2 \wedge dz_3)^{-1}$, a local generator of $\omega^{-1}_{\mathcal{X}/\mathbb{A}^1}$ at p_0 of character $6r$, and write $z^b := z_1^{b_1} z_2^{b_2} z_3^{b_3}$ and

$$\tilde{\sigma} = \sum_{k,b} c_{k,b} \tau^k z^b \theta, \qquad c_{k,b} \in \mathbb{C}.$$

The term $\tau^k z^b \theta$ has character $k - rw(b) + 6r$. Since $\tilde{\sigma}$ is invariant, $c_{k,b} \neq 0$ implies

$$k = r\big(w(b) - 6\big). \tag{18.8}$$

In particular $c_{k,b} \neq 0$ implies $w(b) \geq 6$, with $k = 0$ if and only if $w(b) = 6$. Since $\theta|_Y = (dt \wedge dx \wedge dy)^{-1}$ and σ_0 has expression f_0, we obtain $\sum_b c_{0,b} z^b = f_0(z_1, z_2, z_3)$.

Let $\mathcal{X}_{\mathbb{C}[[\tau]]} := \mathcal{X} \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[[\tau]]$, and let I be the ideal of the section of $\mathcal{X}_{\mathbb{C}[[\tau]]} \to \operatorname{Spec} \mathbb{C}[[\tau]]$ defined by ς. For $n \geq 1$, the closed subscheme defined by I^n is finite over $\mathbb{C}[[\tau]]$ and has the single closed point p_0. Thus the ring $\mathcal{O}_{\mathcal{X}_{\mathbb{C}[[\tau]]}}/I^n$ is local and complete, hence equal to the quotient of $\widehat{\mathcal{O}}_{\mathcal{X},p_0}$ by the ideal of the section:

$$\mathcal{O}_{\mathcal{X}_{\mathbb{C}[[\tau]]}}/I^n \cong \mathbb{C}[[\tau]][z_1, z_2, z_3]/(z_1, z_2, z_3)^n.$$

Let $u_i := \tau^{ri} z_i$ for $i = 1, 2, 3$. They have character 0. Through the isomorphism $\mathcal{X}|_{\mathbb{A}^1 \setminus \{0\}} \cong X \times (\mathbb{A}^1 \setminus \{0\})$, the generic fiber of $\mathcal{X}_{\mathbb{C}[[\tau]]}$ is $X \times \operatorname{Spec} \mathbb{C}((\tau))$, and the section is the point $p \times \operatorname{Spec} \mathbb{C}((\tau))$ by **Step 2**. Localizing at τ, we obtain

$$(\mathcal{O}_{X,p}/\mathfrak{m}_p^n) \otimes_{\mathbb{C}} \mathbb{C}((\tau)) \cong \mathbb{C}((\tau))[z_1, z_2, z_3]/(z_1, z_2, z_3)^n = \mathbb{C}((\tau))[u_1, u_2, u_3]/(u_1, u_2, u_3)^n.$$

The group $\mathbb{G}_m$ acts on the left-hand side through the factor $\mathbb{C}((\tau))$ only. By the paragraph before Lemma 18.4, the invariant part of the left-hand side is $\mathcal{O}_{X,p}/\mathfrak{m}_p^n$. Thus the image of u_i is contained in $\mathcal{O}_{X,p}/\mathfrak{m}_p^n$ for any n, and $u_i \in \widehat{\mathcal{O}}_{X,p}$. Since u_1, u_2, u_3 form a $\mathbb{C}((\tau))$-basis of $(\mathfrak{m}_p/\mathfrak{m}_p^2) \otimes_{\mathbb{C}} \mathbb{C}((\tau))$, they form a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$, and

$$\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[u_1, u_2, u_3]].$$

Since $H^0(X \times \mathbb{G}_m, \operatorname{pr}_1^* \omega_X^{-1}) = H^0(X, -K_X) \otimes \mathbb{C}[\tau, \tau^{-1}]$ and τ^k has character k, there exists a unique $\sigma \in H^0(X, -K_X)$ with $\tilde{\sigma}|_{X \times \mathbb{G}_m} = \operatorname{pr}_1^* \sigma$. Let $(du)^{-1} := (du_1 \wedge du_2 \wedge du_3)^{-1}$. Since $du_i = \tau^{ri} dz_i$ in the relative differentials, $\theta = \tau^{6r}(du)^{-1}$. By (18.8), $\tau^k z^b \theta = \tau^{k - rw(b) + 6r} u^b (du)^{-1} = u^b (du)^{-1}$ whenever $c_{k,b} \neq 0$. For b fixed, (18.8) determines k, so

$$\varphi := \sum_b c_{r(w(b)-6),b} u^b \in \mathbb{C}[[u_1, u_2, u_3]]$$

is well defined, and modulo $(u_1, u_2, u_3)^n$ for any n we have $\sigma = \varphi(du)^{-1}$. Thus φ is the expression of σ in the local generator $(du)^{-1}$ of $-K_X$ at p. By (18.8) and $\sum_b c_{0,b} z^b = f_0(z_1, z_2, z_3)$, any monomial of φ has weight at least 6 for the weights $(1, 2, 3)$ of (u_1, u_2, u_3), and the part of φ of weight 6 is $f_0(u_1, u_2, u_3)$.

Step 4. In this step we pass to a convergent germ and conclude.

Let $U_1, U_2, U_3 \in \mathcal{O}_{X,p}$ with $U_i - u_i \in \widehat{\mathfrak{m}}_p^8$. They are regular parameters at p, hence holomorphic coordinates near p. Let G be the expression of σ in the local generator $(dU_1 \wedge dU_2 \wedge dU_3)^{-1}$ of $-K_X$, a convergent power series in U_1, U_2, U_3. Let $J := \det(\partial U_i / \partial u_j)$. Then $J - 1 \in \widehat{\mathfrak{m}}_p^7$ and $G(U_1, U_2, U_3) = J\varphi$. Since $\varphi \in \widehat{\mathfrak{m}}_p^2$ and $U_i - u_i \in \widehat{\mathfrak{m}}_p^8$,

$$G(u_1, u_2, u_3) \equiv G(U_1, U_2, U_3) = J\varphi \equiv \varphi \mod \widehat{\mathfrak{m}}_p^8.$$

A monomial of weight at most 6 has total degree at most 6. Thus the monomials of weight at most 6 in G coincide with those in φ, and

$$G = f_0 + R,$$

where R is a convergent power series all of whose monomials have weight greater than 6.

We apply the normal form theorem for semiquasihomogeneous functions [AGV85, Section 12.1, Theorem]. The polynomial f_0 is quasihomogeneous of degree 6 for the weights $(1,2,3)$. The Jacobian ideal of f_0 is $(t^3x, 3x^2+t^4, y)$, and it contains

$$t^7 = t^3(3x^2+t^4) - 3x\cdot t^3x.$$

Modulo the Jacobian ideal we have $y\equiv 0$, $x^2\equiv -t^4/3$, $t^3x\equiv 0$, and $t^7\equiv 0$. Thus the Milnor algebra of f_0 is spanned by

$$1,\ t,\ t^2,\ t^3,\ t^4,\ t^5,\ t^6,\ x,\ tx,\ t^2x,$$

which have weights at most 6. Hence 0 is an isolated critical point of f_0, and the Milnor algebra has a monomial basis without monomials of weight greater than 6. By [AGV85, Section 12.1, Theorem], f_0+R is right equivalent to f_0 by a local biholomorphism at 0. Thus

$$(D\ni p)\cong(\{f_0=0\}\ni 0)=(\{y^2=x^3+t^4x\}\ni 0)$$

analytically, and $\sigma\neq 0$ since $G\neq 0$. Since $p\in L$ by **Step 2**, Lemma 18.4 follows. □

In the proof of the implication (3)⇒(1) of Theorem 18.1, the line on the central fiber is lifted as follows: the torus characters on H^0 and H^1 of the normal bundle of the line are positive, as is the character of τ, so the line lifts uniquely and equivariantly order by order, and after algebraization it descends to a line on X.

Proof of Theorem 18.1. (1)⇒(2). Let $D\in|H|$ and $p\in D$ be such that

$$(D\ni p)\cong(\{y^2=x^3+t^4x\}\ni 0)$$

analytically, and let $L\subset X$ be a line with $p\in L$. By Proposition 18.2(1), L is the unique line through p, L is special, and L intersects no other line on X. Thus (2) holds. By Proposition 18.2(1), the statement on the line through p in (1) also holds.

(2)⇒(3). By Proposition 12.9, X admits an anti-canonical test configuration $\pi:\mathcal{X}\to\mathbb{A}^1$ which is a smooth morphism, all of whose fibers are smooth prime Fano threefolds of genus 12, and whose central fiber carries a faithful induced $\mathbb{G}_m$-action. A faithful $\mathbb{G}_m$-action is non-trivial. Thus (3) holds.

(3)⇒(1). The proof is long, so we divide it into several steps. Let $\pi:\mathcal{X}\to\mathbb{A}^1$ be an anti-canonical test configuration of X such that $Y:=\mathcal{X}_0$ is smooth and the induced $\mathbb{G}_m$-action on Y is non-trivial.

Step 1. In this step we show that π is smooth and that Y is a smooth prime Fano threefold of genus 12.

The fibers of π over $\mathbb{A}^1\backslash\{0\}$ are isomorphic to X, and Y is smooth. Since π is flat and all of the fibers of π are smooth of dimension three, π is smooth [Har77, Theorem III.10.2]. Thus $\mathcal{X}$ is smooth, $\omega_{\mathcal{X}/\mathbb{A}^1}|_Y=\omega_Y$, and $-K_Y=-K_{\mathcal{X}/\mathbb{A}^1}|_Y$ is ample. Hence Y is a smooth Fano threefold, so $H^i(Y,\mathcal{O}_Y)=0$ for $i>0$ and $\mathrm{Pic}(Y)\cong H^2(Y,\mathbb{Z})$ [IP99, Proposition 2.1.2]. Since π is proper and smooth and $\mathbb{C}$ is contractible, Ehresmann's theorem [Voi02, Theorem 9.3] implies that Y is connected and that the restriction maps

$$H^2(Y,\mathbb{Z})\leftarrow H^2(\mathcal{X},\mathbb{Z})\to H^2(X,\mathbb{Z})$$

are isomorphisms, where $X=\mathcal{X}_1$. They map $c_1(\omega^{-1}_{\mathcal{X}/\mathbb{A}^1})$ to $c_1(-K_Y)$ and to $c_1(-K_X)$. Since $H^2(X,\mathbb{Z})\cong\mathrm{Pic}(X)=\mathbb{Z}[-K_X]$, we obtain $\mathrm{Pic}(Y)=\mathbb{Z}[-K_Y]$. Since $-K_{\mathcal{X}/\mathbb{A}^1}$ is ample over $\mathbb{A}^1$, there exists $m_0\geq 1$ such that $-m_0K_{\mathcal{X}/\mathbb{A}^1}$ gives a closed embedding $\mathcal{X}\subset\mathbb{P}^N_{\mathbb{A}^1}$ over $\mathbb{A}^1$. Since π is flat, the Hilbert polynomial $n\to\chi(\mathcal{X}_s,-nm_0K_{\mathcal{X}_s})$ does not depend on $s\in\mathbb{A}^1$ [Har77, Theorem III.9.9], so $(-K_Y)^3=(-K_X)^3=22$. Thus Y is a smooth prime Fano threefold of genus 12, which is the statement on the central fiber in (3).

Step 2. In this step we construct an equivariant isomorphism of Y with some X'_v for which the action of $\mathbb{G}_m$ has a normal form.

By **Step 1** and Definition-Theorem 2.20, $Y \cong X_{\Pi_Y}$ with $\Pi_Y \in \Omega$. By Theorem 16.12(2), $\mathrm{Aut}(X_{\Pi_Y}) \cong \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi_Y)$, so the stabilizer of Π_Y in $\mathrm{PGL}(V)$ contains the image of the non-trivial $\mathbb{G}_m$-action, a non-trivial torus. By Definition-Proposition 16.14, Π_Y is $\mathrm{PGL}(V)$-conjugate to Π_h for some $h \in \mathbb{C}\backslash\{1\}$, and $h \neq 2$ since X_{Π_Y} is smooth. Let $v := 1/(h-1) \in \mathbb{C}\backslash\{0,1\}$. By Lemma 18.3(1), we obtain an isomorphism $f : Y \to X'_v$ induced by a linear automorphism of V. By Theorem 16.12(2) for Π'_v, the $\mathbb{G}_m$-action on Y corresponds under f to a homomorphism $\rho : \mathbb{G}_m \to \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v)$ with non-trivial image.

For $\lambda \in \mathbb{G}_m$, let $d_\lambda \in \mathrm{GL}(V)$ be given by $d_\lambda e_i = \lambda^i e_i$. Then $d_\lambda e_{ij} = \lambda^{-(i+j)} e_{ij}$, so

$$d_\lambda \omega_{-1} = \lambda \omega_{-1}, \qquad d_\lambda \omega_0 = \omega_0, \qquad d_\lambda \omega_1 = \lambda^{-1} \omega_1.$$

Thus $d_\lambda \Pi'_v = \Pi'_v$. Since d_λ is a scalar only for $\lambda = 1$, the image $\bar{T}$ of $\lambda \to d_\lambda$ in $\mathrm{PGL}(V)$ is a one-dimensional torus contained in $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v)$. The image of ρ is a one-dimensional torus as well. In both cases below, $\bar{T}$ is a maximal torus of $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v)$, after replacing f the image of ρ is $\bar{T}$, and there exists $g' \in \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v)$ which acts on $\bar{T}$ by inversion.

If $v \neq 1/5$, then $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v) \cong \mathbb{G}_m \rtimes \mu_2$, with the involution acting by inversion, by Definition-Proposition 16.14. Thus $\bar{T}$ is the identity component of the stabilizer, the image of ρ is contained in $\bar{T}$, hence equal to $\bar{T}$, and any g' in the non-identity component acts on $\bar{T}$ by inversion.

If $v = 1/5$, then $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v) \cong \mathrm{PGL}_2$ by Definition-Proposition 16.14, and $\bar{T}$ is a maximal torus of the stabilizer. Since maximal tori of PGL_2 are conjugate [Bor91, Corollary IV.11.3(1)], there exists $g \in \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v)$ with $g\rho(\mathbb{G}_m)g^{-1} = \bar{T}$, and we replace f with $g \circ f$. By the conjugacy of maximal tori of PGL_2 [Bor91, Corollary IV.11.3(1)], there exists an isomorphism $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi'_v) \cong \mathrm{PGL}_2$ which carries $\bar{T}$ to the diagonal torus. Let g' be the class of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ under the isomorphism. Conjugation by g' sends the class of $\mathrm{diag}(\alpha, \beta)$ to that of $\mathrm{diag}(\beta, \alpha)$, so g' acts on $\bar{T}$ by inversion.

In both cases, $\rho(\lambda)$ is the class of d_{λ^m} for some non-zero integer m. If $m > 0$, then $g'\rho(\lambda)g'^{-1} = \rho(\lambda)^{-1}$, and we replace f with $g' \circ f$. Thus we may assume that there exists a positive integer r such that, under f, the $\mathbb{G}_m$-action on Y is induced by

$$\lambda \cdot e_i = \lambda^{-ri} e_i, \qquad -3 \leq i \leq 3.$$

The action is the case $k = -r$ of Lemma 18.3(5). By that lemma, the action preserves p_-, U_-, and L_-, and the coordinates (t, x, y) of $U_- \cap X'_v$ satisfy

$$t(\lambda \cdot q) = \lambda^{-r} t(q), \qquad x(\lambda \cdot q) = \lambda^{-2r} x(q), \qquad y(\lambda \cdot q) = \lambda^{-3r} y(q).$$

Let $p_0 := f^{-1}(p_-)$ and $L_0 := f^{-1}(L_-)$, and denote the pullbacks of t, x, y under f again by t, x, y. By Lemma 18.3(3), T_v has a root at which I_v does not vanish. Let $\sigma_c \in H^0(X'_v, -K_{X'_v})$ be the invariant Plücker section that Lemma 18.3(4) attaches to such a root, and let $\sigma_0 := f^*\sigma_c \in H^0(Y, -K_Y)$. By Lemma 18.3(5), σ_0 is invariant for the natural $\mathbb{G}_m$-linearization of $-K_Y$.

Step 3. In this step we compute the characters of $\mathbb{G}_m$ on $H^0(L_0, N_{L_0/Y})$ and $H^1(L_0, N_{L_0/Y})$.

With the notion of character fixed before Lemma 18.4, t, x, y have characters $-r, -2r, -3r$. Since $\partial_x x = 1$ and $\partial_y y = 1$, the vector fields ∂_x and ∂_y have characters $2r$ and $3r$.

By Lemma 18.3(4), $(D_c \ni p_-) \cong (\{y^2 = x^3 + t^4 x\} \ni 0)$ analytically, and $p_- \in L_-$. By Proposition 18.2(1) for $X'_v \cong Y$, the line L_0 is special, that is,

$$N_{L_0/Y} \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(-2).$$

Since $h^0(L_0, N_{L_0/Y} \otimes \mathcal{O}_{\mathbb{P}^1}(-1)) = 1$, there exists a unique subbundle $N^+ \subset N_{L_0/Y}$ isomorphic to $\mathcal{O}_{\mathbb{P}^1}(1)$. Thus N^+ is $\mathbb{G}_m$-stable, and so is the quotient $N^- := N_{L_0/Y}/N^+ \cong \mathcal{O}_{\mathbb{P}^1}(-2)$. Let $L_0^\circ := L_0 \cap f^{-1}(U_-)$ and $p_\infty := L_0 \backslash L_0^\circ$. By Lemma 18.3(2), L_0° is the t-axis, so the fiber of $N_{L_0/Y}$ at p_0 is spanned by the images of ∂_x and ∂_y. Since $2r \neq 3r$, the $\mathbb{G}_m$-stable line $N^+(p_0)$ is spanned by the image of ∂_x or by the image of ∂_y. Let $j_1 \in \{2r, 3r\}$ be the character of the vector field whose image spans $N^+(p_0)$, and let j_2 be the other element of $\{2r, 3r\}$.

There exists a section e of N^+ over L_0° of character j_1 with $e(p_0) \neq 0$. The zero locus of e is a finite $\mathbb{G}_m$-stable subset of $L_0^\circ \backslash \{p_0\}$. Since t has non-zero character, the $\mathbb{G}_m$-orbits in $L_0^\circ \backslash \{p_0\}$ are infinite, so e vanishes nowhere on L_0°. Since $\deg N^+ = 1$, the divisor of e on L_0 is p_∞, and

$$H^0(L_0, N^+) = \{(\alpha + \beta t)e : \alpha, \beta \in \mathbb{C}\},$$

whose characters are j_1 and $j_1 - r$. The fiber $N^-(p_0)$ is spanned by the image of the vector field of character j_2, so there exists a section e' of N^- over L_0° of character j_2 with $e'(p_0) \neq 0$. Since the $\mathbb{G}_m$-orbits in $L_0^\circ \backslash \{p_0\}$ are infinite, e' vanishes nowhere on L_0°. Since $\deg N^- = -2$, the divisor of e' on L_0 is $-2p_\infty$. For the cover of L_0 by L_0° and $L_0 \backslash \{p_0\}$, the sections of N^- over the intersection are $\mathbb{C}[t, t^{-1}]e'$, over L_0° they are $\mathbb{C}[t]e'$, and over $L_0 \backslash \{p_0\}$ they are $t^{-2}\mathbb{C}[t^{-1}]e'$. Thus $H^1(L_0, N^-)$ is spanned by the class of $t^{-1}e'$, of character $j_2 + r$. Since $H^0(L_0, N^-) = H^1(L_0, N^+) = 0$, the exact sequence $0 \to N^+ \to N_{L_0/Y} \to N^- \to 0$ implies $\mathbb{G}_m$-equivariant isomorphisms $H^0(L_0, N_{L_0/Y}) \cong H^0(L_0, N^+)$ and $H^1(L_0, N_{L_0/Y}) \cong H^1(L_0, N^-)$. Hence the characters are

$$\begin{array}{lll} j_1 = 2r: & H^0(L_0, N_{L_0/Y}): \ 2r,\ r, & H^1(L_0, N_{L_0/Y}): \ 4r, \\ j_1 = 3r: & H^0(L_0, N_{L_0/Y}): \ 3r,\ 2r, & H^1(L_0, N_{L_0/Y}): \ 3r. \end{array} \tag{18.9}$$

In both cases all of the characters are at least $r > 0$.

Step 4. In this step we lift L_0 equivariantly to any finite order.

For $n \geq 0$ let $A_n := \mathbb{C}[\tau]/(\tau^{n+1})$ and $\mathcal{X}_n := \mathcal{X} \times_{\mathbb{A}^1} \operatorname{Spec} A_n$. The group $\mathbb{G}_m$ acts on $\mathcal{X}_n$ covering $\tau \to \lambda\tau$. We construct $\mathbb{G}_m$-invariant closed subschemes $L_n \subset \mathcal{X}_n$, flat over A_n, with $L_n \cap \mathcal{X}_{n-1} = L_{n-1}$ for $n \geq 1$.

Let $n \geq 1$, and let $L_{n-1} \subset \mathcal{X}_{n-1}$ be invariant and flat over A_{n-1} with $L_{n-1} \cap Y = L_0$. The scheme $\mathcal{X}_n$ is a flat extension of $\mathcal{X}_{n-1}$ over the small extension $A_n \to A_{n-1}$ with kernel (τ^n). This is the situation of [Har10, Theorem 6.2 and Remark 6.2.1], where the ambient deformation need not be a product. Since $L_0 \subset Y$ is a local complete intersection, local extensions of L_{n-1} exist [Har10, Remark 6.2.1], there exists an obstruction class in $H^1(L_0, N_{L_0/Y}) \otimes (\tau^n)$ which vanishes if and only if L_{n-1} extends to a closed subscheme of $\mathcal{X}_n$ flat over A_n, and the extensions form a torsor under $H^0(L_0, N_{L_0/Y}) \otimes (\tau^n)$ [Har10, Theorem 6.2]. The obstruction class and the torsor of extensions are natural for automorphisms of $\mathcal{X}_n$ preserving L_{n-1}. Thus the obstruction class is $\mathbb{G}_m$-invariant, and $\mathbb{G}_m$ acts algebraically on the torsor of extensions by affine transformations whose linear part is the action on $H^0(L_0, N_{L_0/Y}) \otimes (\tau^n)$, that is, $\lambda \cdot (L' + \xi) = \lambda \cdot L' + \lambda \cdot \xi$.

The element τ^n has character n, so the characters of $H^i(L_0, N_{L_0/Y}) \otimes (\tau^n)$ are those of $H^i(L_0, N_{L_0/Y})$ plus n. By (18.9), they are at least $r + n > 0$. Thus neither $H^0(L_0, N_{L_0/Y}) \otimes (\tau^n)$ nor $H^1(L_0, N_{L_0/Y}) \otimes (\tau^n)$ contains a non-zero vector of character 0. Hence the obstruction class vanishes. The torsor of extensions is the preimage of 1 under a $\mathbb{G}_m$-equivariant linear surjection from a rational representation of $\mathbb{G}_m$ onto the trivial representation $\mathbb{C}$, with kernel $H^0(L_0, N_{L_0/Y}) \otimes (\tau^n)$. Since taking the character-zero part is exact, the torsor contains a $\mathbb{G}_m$-fixed point, and the fixed point is unique since $H^0(L_0, N_{L_0/Y}) \otimes (\tau^n)$ has no non-zero vector of character 0. Let L_n be the fixed point. By induction on n, we obtain the invariant subschemes L_n for all $n \geq 0$.

Step 5. In this step we algebraize the subschemes L_n, descend them to a line L on X, and construct a family of lines $\mathcal{L} \subset \mathcal{X}$.

Fix $m \geq 1$ such that $-mK_{\mathcal{X}/\mathbb{A}^1}$ is very ample over $\mathbb{A}^1$, and let $\mathcal{H} \to \mathbb{A}^1$ be the relative Hilbert scheme of closed subschemes of $\mathcal{X}$ with Hilbert polynomial $P(n) = mn + 1$, which is projective over $\mathbb{A}^1$ [Kol96, Theorem I.1.4]. The group $\mathbb{G}_m$ acts on $\mathcal{H}$ over $\mathbb{A}^1$. The compatible points $[L_n] \in \mathcal{H}(A_n)$ define a local homomorphism $\mathcal{O}_{\mathcal{H},[L_0]} \to \varprojlim A_n = \mathbb{C}[[\tau]]$, hence a morphism $\kappa : \operatorname{Spec} \mathbb{C}[[\tau]] \to \mathcal{H}$ over $\mathbb{A}^1$. Since any L_n is invariant, κ commutes with the substitutions $\tau \to \lambda\tau$ and the action of λ on $\mathcal{H}$ for any $\lambda \in \mathbb{C}^*$. Let $\mathcal{U} \to \mathcal{H}$ be the universal family, and let $\mathcal{C} \subset \mathcal{X} \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[[\tau]]$ be its pullback by κ. For a point $\gamma \in \mathcal{H}$ let $\mathcal{U}_\gamma$ be the fiber over γ. The morphism $\mathcal{U} \to \mathcal{H}$ is flat and projective, and $\dim \mathcal{U}_\gamma = 1$ for any $\gamma \in \mathcal{H}$. By the upper semicontinuity of $h^0(\mathcal{O})$ and $h^1(\mathcal{O})$ on the fibers [Har77, Theorem III.12.8], the points $\gamma \in \mathcal{H}$

with $h^0(\mathcal{U}_\gamma, \mathcal{O}_{\mathcal{U}_\gamma}) \leq 1$ and $h^1(\mathcal{U}_\gamma, \mathcal{O}_{\mathcal{U}_\gamma}) = 0$ form an open subset $\mathcal{H}^\circ$. The set Σ of points $q \in \mathcal{U}$ with $\dim_{k(q)} \Omega_{\mathcal{U}/\mathcal{H}} \otimes k(q) > 1$ is closed [Har77, Exercise II.5.8(a)], and its image in $\mathcal{H}$ is closed since $\mathcal{U} \to \mathcal{H}$ is proper. Let $\gamma \in \mathcal{H}^\circ$ be a point outside the image of Σ. The fiber $\mathcal{U}_\gamma$ is non-empty, since its Hilbert polynomial is $mn + 1$, so $H^0(\mathcal{U}_\gamma, \mathcal{O}_{\mathcal{U}_\gamma}) = k(\gamma)$, and $\mathcal{U}_\gamma$ is connected. Since $\mathcal{U}_\gamma$ is connected of dimension one, it has no isolated point, so $\dim Z = 1$ for any irreducible component Z of $\mathcal{U}_\gamma$, and $\mathcal{U}_\gamma$ is smooth [Har77, Theorem III.10.2]. Conversely, if $\gamma' \in \mathcal{H}$ and $\mathcal{U}_{\gamma'}$ is smooth and connected of genus zero, then $\gamma' \in \mathcal{H}^\circ$ and γ' is outside the image of Σ. Thus the points of $\mathcal{H}$ over which the universal family has smooth connected fibers of genus zero form the open subset $\mathcal{H}^\circ$ minus the image of Σ, which contains $[L_0]$. The preimage of the open subset in $\operatorname{Spec} \mathbb{C}[[\tau]]$ contains the closed point, hence is equal to $\operatorname{Spec} \mathbb{C}[[\tau]]$. Thus the generic fiber $\mathcal{C}_\eta$ is a smooth connected curve of genus zero.

Through the isomorphism $\mathcal{X}|_{\mathbb{A}^1\setminus\{0\}} \cong X \times (\mathbb{A}^1\setminus\{0\})$, $\mathcal{H}|_{\mathbb{A}^1\setminus\{0\}} \cong \operatorname{Hilb}^P(X) \times (\mathbb{A}^1\setminus\{0\})$ equivariantly, where $\mathbb{G}_m$ acts trivially on $\operatorname{Hilb}^P(X)$ and P is taken for $-mK_X$. Let $h_\eta : \operatorname{Spec} \mathbb{C}((\tau)) \to \operatorname{Hilb}^P(X)$ be the composition of $\kappa|_{\operatorname{Spec} \mathbb{C}((\tau))}$ with the projection. Then h_η is invariant under $\tau \to \lambda\tau$ for any $\lambda \in \mathbb{C}^*$. Since $\operatorname{Hilb}^P(X)$ is a $\mathbb{C}$-scheme of finite type, by the paragraph before Lemma 18.4, h_η is the base change of a point $[L] \in \operatorname{Hilb}^P(X)$, and $L \times \operatorname{Spec} \mathbb{C}((\tau)) \cong \mathcal{C}_\eta$. Hence L is a smooth connected curve of genus zero with $(-K_X \cdot L) = 1$, that is, L is a line on X.

The family $L \times (\mathbb{A}^1\setminus\{0\}) \subset \mathcal{X}|_{\mathbb{A}^1\setminus\{0\}}$ defines a section κ° of $\mathcal{H} \to \mathbb{A}^1$ over $\mathbb{A}^1\setminus\{0\}$. Since $\mathcal{H}$ is proper over $\mathbb{A}^1$, κ° extends to a section $\bar{\kappa}$ of $\mathcal{H} \to \mathbb{A}^1$ by the valuative criterion of properness, and the extension is unique by the valuative criterion of separatedness [Har77, Theorems II.4.3 and II.4.7]. The restrictions of $\bar{\kappa}$ and κ to $\operatorname{Spec} \mathbb{C}((\tau))$ coincide, since both correspond to $L \times \operatorname{Spec} \mathbb{C}((\tau))$. Since $\mathcal{H}$ is separated over $\mathbb{A}^1$, $\bar{\kappa}|_{\operatorname{Spec} \mathbb{C}[[\tau]]} = \kappa$, and in particular $\bar{\kappa}(0) = [L_0]$. For $\lambda \in \mathbb{G}_m$, the section $\lambda \circ \bar{\kappa} \circ \lambda^{-1}$ extends κ° as well, so it is equal to $\bar{\kappa}$. Let $\mathcal{L} \subset \mathcal{X}$ be the pullback of the universal family by $\bar{\kappa}$. Then $\mathcal{L}$ is $\mathbb{G}_m$-invariant and flat and proper over $\mathbb{A}^1$, with $\mathcal{L}_0 = L_0$ and $\mathcal{L}_s = L \times \{s\}$ for $s \neq 0$. All of the fibers of $\mathcal{L} \to \mathbb{A}^1$ are smooth, so $\mathcal{L} \to \mathbb{A}^1$ is smooth [Har77, Theorem III.10.2], and all of the fibers are lines with respect to $-K_{\mathcal{X}/\mathbb{A}^1}$.

Step 6. In this step we prove (1) by Lemma 18.4.

By **Step 1**, π is a smooth projective morphism whose fibers are connected smooth Fano threefolds, and by Definition 2.4 there exists a $\mathbb{G}_m$-equivariant isomorphism $\mathcal{X}|_{\mathbb{A}^1\setminus\{0\}} \cong X \times (\mathbb{A}^1\setminus\{0\})$ with $\mathbb{G}_m$ acting trivially on X. By **Step 5**, $\mathcal{L}$ is a $\mathbb{G}_m$-invariant closed subscheme, smooth and proper of relative dimension one over $\mathbb{A}^1$, all of whose fibers are lines, with $\mathcal{L}_0 = L_0$ and $\mathcal{L}_1 = L$. By **Step 2**, $p_0 \in L_0$ is a $\mathbb{G}_m$-fixed point, (t, x, y) are equivariant regular parameters at p_0 of characters $-r, -2r, -3r$, and σ_0 is invariant for the natural linearization of $-K_Y$. By Lemma 18.3(4), the expression of σ_0 in $(dt \wedge dx \wedge dy)^{-1}$ is a weighted homogeneous polynomial of degree six for the weights $(1, 2, 3)$, carried to $y^2 - x^3 - t^4x$ by an invertible weight-preserving polynomial change of coordinates and multiplication by a non-zero constant. This is the assumption of Lemma 18.4 on σ_0. By Lemma 18.4, there exist $p \in L$ and a non-zero $\sigma \in H^0(X, -K_X)$ whose divisor $D \in |H|$ satisfies $(D \ni p) \cong (\{y^2 = x^3 + t^4x\} \ni 0)$ analytically. Since L is a line on X, (1) holds.

Finally, we prove the last sentence of Theorem 18.1. Assume that X admits a faithful $\mathbb{G}_m$-action $\mu : \mathbb{G}_m \to \operatorname{Aut}(X)$. Let $\mathbb{G}_m$ act on $X \times \mathbb{A}^1$ by $\lambda \cdot (q, s) := (\mu(\lambda)q, \lambda s)$. The isomorphism $(q, s) \to (\mu(s)^{-1}q, s)$ of $X \times (\mathbb{A}^1\setminus\{0\})$ sends $(\mu(\lambda)q, \lambda s)$ to $(\mu(\lambda s)^{-1}\mu(\lambda)q, \lambda s) = (\mu(s)^{-1}q, \lambda s)$, so it is equivariant if $\mathbb{G}_m$ acts on the image by $\lambda \cdot (q, s) = (q, \lambda s)$. Thus $X \times \mathbb{A}^1$ with polarization $-K_{X\times\mathbb{A}^1/\mathbb{A}^1}$ is a product anti-canonical test configuration of X (Definition 2.4), with central fiber X and induced action μ. Hence X satisfies (3), and (1) and (2) follow from the equivalence of (1), (2), and (3). □

19. The plane-quartic dictionary

In this section, we prove that for a smooth prime Fano threefold X of genus 12, the GIT of the apolar quartic f_X, the GIT of the Hilbert quartic Q_X of lines, the $\mathrm{SL}(V)$-orbit of the canonical Mukai net Π_X, and the incidence of lines on X determine each other. The arguments are geometric and do not use K-stability, which enters only in the proof of Theorem 1.5, through Theorem 17.5.

To pass from polystability of the Hilbert quartic to polystability of the apolar quartic, we compute the differential of the Scorza map at the torus quartics (Lemma B.10).

Theorem 19.1. *Let X be a smooth prime Fano threefold of genus 12. All GIT notions for quartic forms on $\Pi_X^\vee$ refer to the action of $\mathrm{SL}(\Pi_X^\vee) \cong \mathrm{SL}_3$ with the linearization $\mathcal{O}(1)$. Then the following hold.*

1. *f_X and Q_X are GIT-semistable, and the following conditions are equivalent: f_X is GIT-stable, Q_X is GIT-stable, $\Pi_X \in \Omega^{\mathrm{st}}$, and every special line on X intersects another line.*
2. *The following conditions are equivalent: f_X is GIT-polystable, Q_X is GIT-polystable, $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$, and every special line on X intersects another line or X admits a faithful $\mathbb{G}_m$-action.*
3. *Among the quartics Q_X, the GIT-polystable but not GIT-stable ones are exactly the double smooth conics and the unions of two distinct smooth conics tangent at two distinct points. The quartic Q_X is a double conic, equivalently non-reduced, if and only if $X \cong X^{\mathrm{MU}}$, and it is the union of two smooth conics tangent at two points if and only if X admits a faithful $\mathbb{G}_m$-action and $X \ncong X^{\mathrm{MU}}$. The quartic Q_X has a singular point of type A_7 if and only if $X \cong X^a$, equivalently Q_X is the union of two smooth conics with one common point, of intersection multiplicity four. Every reduced Q_X has only singular points of type A_k with $k \leq 7$. Neither a smooth conic together with two distinct tangent lines nor a cubic together with an inflectional tangent line is of the form Q_X.*
4. *If Q_X is not GIT-stable, then the projective Scorza map is étale at $[f_X]$, and the identity components of the stabilizers of $[f_X]$ and $[\mathrm{Sc}(f_X)]$ in $\mathrm{PGL}(\Pi_X^\vee)$ coincide; here f_X and $\mathrm{Sc}(f_X)$ are both quartic forms on $\Pi_X^\vee$.*
5. *For $h \in \mathbb{C}\backslash\{1,2\}$, $\mathrm{Sc}(F_h)$ is a non-zero multiple of $x^4 + (h-4)x^2yz + y^2z^2$. The quartics F_6 and $\mathrm{Sc}(F_6)$ are double conics, F_0 is a smooth conic together with two tangent lines, and $\mathrm{Sc}(F_0)$ is the union of two smooth conics tangent at two points. For X^a one may take $f_{X^a} = (z^2-xy)^2 + x^4$, and then $\mathrm{Sc}(f_{X^a})$ is a non-zero multiple of $(z^2-xy)^2 - 6x^4$.*
6. *Assume that f_X is not GIT-polystable, and let $h \in \mathbb{C}\backslash\{1,2\}$ be the unique value such that $[F_h]$ represents the unique closed orbit in the closure of $\mathrm{SL}_3\cdot[f_X]$ in the GIT-semistable locus. Then, in coordinates in which f_X, after multiplication by a non-zero constant, has the normal form of Proposition 16.16(2),*

$$\lim_{t\to 0} \mathrm{diag}(1, t^{-1}, t)\cdot \mathrm{Sc}(f_X) = \mathrm{Sc}(F_h),$$

where $(g\cdot p)(w) = p(g^{-1}w)$ for $g \in \mathrm{SL}_3$ and $p \in \mathbb{C}[x,y,z]_4$, and the orbit of $[\mathrm{Sc}(F_h)]$ is the unique closed orbit in the closure of $\mathrm{SL}_3\cdot[\mathrm{Sc}(f_X)]$ in the GIT-semistable locus. The orbit of $[\mathrm{Sc}(F_h)]$ is also the orbit of $[\mathrm{Sc}(f_Y)]$ for the central fiber Y of every anti-canonical test configuration of X whose central fiber is smooth and carries a faithful induced $\mathbb{G}_m$-action.

The apolar quartic of X^a is determined by an additive symmetry. We need the following lemma on quartics with such a symmetry.

Lemma 19.2. *For $s \in \mathbb{C}$, let ϕ_s be the linear automorphism of $\mathbb{C}^3$ given by $\phi_s(x,y,z) = (x, y+2sz+s^2x, z+sx)$. Let $f \in \mathbb{C}[x,y,z]_4$ be a quartic such that $\mathrm{Cat}_2(f)$ is invertible. Assume that the stabilizer of $[f]$ in PGL_3 contains a closed subgroup Γ isomorphic to $\mathbb{G}_a$. Then the following hold.*

(1) After a projective change of coordinates, $\Gamma = \{[\phi_s] : s \in \mathbb{C}\}$ *and* $f \circ \phi_s = f$ *for any* $s \in \mathbb{C}$.
(2) After a projective change of coordinates and multiplication by a non-zero constant, $f = (z^2 - xy)^2 + cx^4$ *for some* $c \in \mathbb{C}$.
(3) For any $u \in \mathbb{C}$, $\mathrm{Sc}((z^2 - xy)^2 + ux^4)$ *is a non-zero multiple of* $(z^2 - xy)^2 - 6ux^4$.

Proof. (1) Since the base field has characteristic zero, $\mathrm{Lie}(\mathrm{PGL}_3) = \mathfrak{sl}_3$. Let $N \in \mathfrak{sl}_3$ span $\mathrm{Lie}(\Gamma)$ and let $\mathcal{A}(N) \subset \mathrm{SL}_3$ be the smallest closed subgroup whose Lie algebra contains N. The image of $\mathcal{A}(N)$ in PGL_3 is the smallest closed subgroup whose Lie algebra contains N, so it is contained in Γ and, being connected of positive dimension, equals Γ [Bor91, §II.7.1 and Proposition II.7.2]. Write $N = N_s + N_n$ for the Jordan decomposition. Then $\mathcal{A}(N) = \mathcal{A}(N_s) \cdot \mathcal{A}(N_n)$, and $\mathcal{A}(N_s)$ is a torus, non-trivial if $N_s \neq 0$, whose image in $\Gamma \cong \mathbb{G}_a$ would be a non-trivial torus. Hence N is nilpotent and $\mathcal{A}(N) = \{\exp(sN) : s \in \mathbb{C}\} \cong \mathbb{G}_a$ [Bor91, §II.7.3(1)–(3)]. Since $\mathbb{G}_a$ has no non-trivial character, $f \circ \exp(sN) = f$ for any $s \in \mathbb{C}$. The Jordan type of N is $(2,1)$ or (3).

Suppose that the Jordan type of N is $(2,1)$. Then after a change of coordinates $\exp(sN)(x,y,z) = (x, y + sx, z)$, so $f(x, y + sx, z) = f(x,y,z)$ for any s. Differentiating at $s = 0$, we obtain $x\partial_y f = 0$, hence $\partial_y f = 0$ and ∂_y^2 annihilates f. This contradicts the invertibility of $\mathrm{Cat}_2(f)$.

Thus the Jordan type of N is (3). The endomorphism $N_0(x,y,z) = (0, 2z, x)$ satisfies $N_0^2(x,y,z) = (0,2x,0)$ and $N_0^3 = 0$, so N_0 is nilpotent of Jordan type (3), and

$$\exp(sN_0)(x,y,z) = (x, y + 2sz + s^2x, z + sx) = \phi_s(x,y,z).$$

Since two nilpotent endomorphisms with equal Jordan types are conjugate, after a change of coordinates $N = N_0$. This implies (1).

(2) We use the coordinates of (1). Write $f(x,y,0) = \sum_{i=0}^4 b_i x^{4-i} y^i$. On the open subset $(x \neq 0)$, invariance under $\phi_{-z/x}$ implies

$$f(x,y,z) = f\Big(x, y - \frac{z^2}{x}, 0\Big) = \sum_{i=0}^4 b_i x^{4-2i} (xy - z^2)^i.$$

Let i_0 be the largest index with $b_{i_0} \neq 0$. As a polynomial in z with coefficients in $\mathbb{C}(x,y)$, the right-hand side has leading coefficient $(-1)^{i_0} b_{i_0} x^{4-2i_0}$ in degree $2i_0$. Since f is a polynomial, $i_0 \leq 2$, so

$$f = b_2(z^2 - xy)^2 - b_1 x^2 (z^2 - xy) + b_0 x^4.$$

If $b_2 = 0$, then ∂_y^2 annihilates f, which contradicts the invertibility of $\mathrm{Cat}_2(f)$. Thus $b_2 \neq 0$ and

$$\frac{1}{b_2} f = \Big(z^2 - xy - \frac{b_1}{2b_2} x^2\Big)^2 + \Big(\frac{b_0}{b_2} - \frac{b_1^2}{4b_2^2}\Big) x^4.$$

Replacing y with $y - \frac{b_1}{2b_2} x$ turns $z^2 - xy - \frac{b_1}{2b_2} x^2$ into $z^2 - xy$. This implies (2).

(3) We use the notation of Subsection B.2: $M(\cdot)$ and Sc_0 are as in Lemma B.10, and Pfaffians are computed with (B.12). By Lemma B.10(1), Sc is a non-zero multiple of Sc_0. The partial derivatives of $f = (z^2 - xy)^2 + ux^4$ are

$$f_x = 4ux^3 + 2xy^2 - 2yz^2, \qquad f_y = 2x^2 y - 2xz^2, \qquad f_z = 4z^3 - 4xyz,$$

from which $M(f)$ is formed. Computing the Pfaffian with (B.12) as an exact polynomial in u, we obtain

$$\mathrm{Sc}_0\big((z^2 - xy)^2 + ux^4\big) = \mathrm{Pf}\big(M((z^2 - xy)^2 + ux^4)_{0,\dots,7}\big)/256 = (z^2 - xy)^2 - 6ux^4. \tag{19.1}$$

This implies (3). □

Proof of Theorem 19.1. The proof is long, so we divide it into several steps.

By Definition-Theorem 2.20, $\Pi_X \in \Omega$ and $X \cong X_{\Pi_X}$. Let $G := \mathrm{PGL}_3$, let $\mathbb{P} := \mathbb{P}(\mathbb{C}[x,y,z]_4)$, and let $\mathbb{P}^{\mathrm{ss}} \subset \mathbb{P}$ be the open subset of GIT-semistable points. For $[f] \in \mathbb{P}$, let $G_{[f]}$ be the stabilizer of $[f]$ in G, and let $G^0_{[f]}$ be the identity component of $G_{[f]}$. For a smooth prime Fano threefold

X' of genus 12, we identify $\mathrm{Sym}^4 \Pi_{X'}$ with $\mathbb{C}[x,y,z]_4$ by a basis of $\Pi_{X'}$. The orbits $G \cdot [f_{X'}]$ and $G \cdot [\mathrm{Sc}(f_{X'})]$ do not depend on the basis (Definition-Proposition 2.28). By Lemma B.10(1), Sc is a non-zero multiple of Sc_0. Since (1)–(6) do not change if Sc is multiplied by a non-zero constant, we may assume that $\mathrm{Sc} = \mathrm{Sc}_0$. Let $U^{\mathrm{Sc}} \subset \mathbb{P}$ be the open subset of the points $[f]$ with $\mathrm{Sc}(f) \neq 0$, and let $\sigma : U^{\mathrm{Sc}} \to \mathbb{P}$, $[f] \to [\mathrm{Sc}(f)]$, be the projective Scorza map (Definition 2.27). Since the Aronhold invariant is SL_3-invariant (Definition 2.27), $\mathrm{Sc}(g \cdot f) = g \cdot \mathrm{Sc}(f)$ for any $g \in \mathrm{SL}_3$. Thus U^{Sc} is G-invariant and σ is G-equivariant.

Step 1. In this step, we describe the smooth members with a faithful $\mathbb{G}_m$-action, prove (5) except for the last sentence, and show that σ is étale along $G \cdot [F_h]$ for $h \in \mathbb{C}\backslash\{1,2\}$.

Let X' be a smooth prime Fano threefold of genus 12 which admits a faithful $\mathbb{G}_m$-action. We show that there exists $h \in \mathbb{C}\backslash\{1,2\}$ such that

$$X' \cong X_{\Pi_h}, \quad [f_{X'}] \in G \cdot [F_h], \quad [\mathrm{Sc}(f_{X'})] \in G \cdot [\mathrm{Sc}(F_h)], \tag{19.2}$$

and that $h = 6$ if and only if $X' \cong X^{\mathrm{MU}}$. By Theorem 16.12(2), the stabilizer of $\Pi_{X'}$ in $\mathrm{PGL}(V)$ is isomorphic to $\mathrm{Aut}(X')$, so it contains a non-trivial torus. By Definition-Proposition 16.14, $\Pi_{X'} \in \mathrm{PGL}(V) \cdot \Pi_h$ for some $h \in \mathbb{C}\backslash\{1\}$, hence $X' \cong X_{\Pi_{X'}} \cong X_{\Pi_h}$. Since X' is smooth, $h \neq 2$. Since the apolar quartic of Π_h is F_h (Definition-Proposition 16.14), Lemma 16.3(3) implies $[f_{X'}] \in G \cdot [F_h]$, and the G-equivariance of σ implies $[\mathrm{Sc}(f_{X'})] \in G \cdot [\mathrm{Sc}(F_h)]$. If $h = 6$, then $X' \cong X^{\mathrm{MU}}$ by Definition-Proposition 16.14. If $X' \cong X^{\mathrm{MU}}$, then $X_{\Pi_h} \cong X_{\Pi_6}$. By Theorem 16.12(1), Π_h and Π_6 are the canonical Mukai nets of isomorphic threefolds, so $\Pi_h \in \mathrm{PGL}(V) \cdot \Pi_6$ by Definition-Theorem 2.20, and $h = 6$ by the uniqueness in Definition-Proposition 16.14. Moreover, for any $h \in \mathbb{C}\backslash\{1,2\}$, the threefold X_{Π_h} is a smooth prime Fano threefold of genus 12 which admits a faithful $\mathbb{G}_m$-action, since the stabilizer of Π_h in $\mathrm{PGL}(V)$ contains $\mathbb{G}_m$ (Definition-Proposition 16.14 and Theorem 16.12(2)).

Let $h \in \mathbb{C}\backslash\{1,2\}$. By Lemma B.10(2),

$$\mathrm{Sc}(F_h) = x^4 + (h-4)x^2yz + y^2z^2 = (yz + rx^2)(yz + r^{-1}x^2),$$

where r is a root of $r^2 - (h-4)r + 1$. In particular, $r \neq 0$. The conics $C_r := (yz + rx^2 = 0)$ and $C_{r^{-1}} := (yz + r^{-1}x^2 = 0)$ are smooth, and $C_r = C_{r^{-1}}$ if and only if $r = \pm 1$, that is, $(h-4)^2 = 4$. Since $h \neq 2$, this holds if and only if $h = 6$, and then $\mathrm{Sc}(F_6) = (x^2 + yz)^2 = F_6$ is a double smooth conic. Assume that $h \neq 6$. Then $C_r \cap C_{r^{-1}} = \{[0:1:0], [0:0:1]\}$, since $(r - r^{-1})x^2 = 0$ on $C_r \cap C_{r^{-1}}$. Both conics have the tangent line $(z = 0)$ at $[0:1:0]$ and the tangent line $(y = 0)$ at $[0:0:1]$, so both local intersection numbers are at least 2. Since $(C_r \cdot C_{r^{-1}}) = 4$, both are equal to 2, and $(\mathrm{Sc}(F_h) = 0)$ is the union of two distinct smooth conics tangent at two distinct points. In particular, this holds for $\mathrm{Sc}(F_0) = x^4 - 4x^2yz + y^2z^2$. The quartic $F_0 = yz(2x^2 + yz)$ is the union of the smooth conic $(2x^2 + yz = 0)$ and the lines $(y = 0)$ and $(z = 0)$, which are tangent to the conic at $[0:0:1]$ and $[0:1:0]$ respectively. This proves (5) except for the last sentence. Exchanging x and y, the quartic $\mathrm{Sc}(F_h)$ becomes $(xz + ry^2)(xz + r^{-1}y^2)$. Thus, by Theorem 2.29(3),

$$[\mathrm{Sc}(F_h)] \text{ is GIT-polystable but not GIT-stable for any } h \in \mathbb{C}\backslash\{1,2\}. \tag{19.3}$$

By Lemma B.10(3), the differential $d\,\mathrm{Sc}_{F_h}$ of Sc at F_h satisfies

$$\det(d\,\mathrm{Sc}_{F_h}) = -\frac{19683}{2}(h-2)^6 \neq 0.$$

Since Sc is homogeneous of degree four, $d\,\mathrm{Sc}_{F_h}(F_h) = 4\,\mathrm{Sc}(F_h)$. Thus the isomorphism $d\,\mathrm{Sc}_{F_h}$ maps $\mathbb{C}F_h$ onto $\mathbb{C}\,\mathrm{Sc}(F_h)$ and induces an isomorphism

$$\mathbb{C}[x,y,z]_4/\mathbb{C}F_h \to \mathbb{C}[x,y,z]_4/\mathbb{C}\,\mathrm{Sc}(F_h),$$

which is the differential of σ at $[F_h]$ under the standard identifications $T_{[F_h]}\mathbb{P} \cong \mathbb{C}[x,y,z]_4/\mathbb{C}F_h$ and $T_{[\mathrm{Sc}(F_h)]}\mathbb{P} \cong \mathbb{C}[x,y,z]_4/\mathbb{C}\,\mathrm{Sc}(F_h)$. Let $W \subset U^{\mathrm{Sc}}$ be the set of points at which the differential of σ is an isomorphism. Since U^{Sc} and $\mathbb{P}$ are smooth of dimension 14, W is open, being locally the non-vanishing locus of the Jacobian determinant, and $\sigma|_W : W \to \mathbb{P}$ is smooth of relative

dimension 0, that is, étale [Har77, Proposition III.10.4 and Exercise III.10.3]. By the computation above, $[F_h] \in W$. Since σ is G-equivariant, W is G-invariant. Thus

$$G \cdot [F_h] \subset W \text{ for any } h \in \mathbb{C}\backslash\{1,2\}. \tag{19.4}$$

Step 2. In this step, we fix an anti-canonical test configuration $\pi : \mathcal{X} \to \mathbb{A}^1$ of X whose central fiber Y is smooth and carries a faithful induced $\mathbb{G}_m$-action, and we show that

$$G \cdot [f_Y] \subset \overline{G \cdot [f_X]}, \quad G \cdot [\mathrm{Sc}(f_Y)] \subset \overline{G \cdot [\mathrm{Sc}(f_X)]}, \quad [f_X] \in W, \quad [\mathrm{Sc}(f_X)] \in \mathbb{P}^{\mathrm{ss}}, \tag{19.5}$$

where the closures are taken in $\mathbb{P}$. We also show that such π exists if Q_X is not GIT-stable.

Since the induced $\mathbb{G}_m$-action on Y is faithful, hence non-trivial, π is a test configuration as in Theorem 18.1(3), so Y is a smooth prime Fano threefold of genus 12 by Theorem 18.1. Since π is flat and all fibers of π are smooth, π is smooth [Har77, Theorem III.10.2].

Let τ be a coordinate of $\mathbb{A}^1 = \operatorname{Spec}\mathbb{C}[\tau]$, let $R := \mathbb{C}[[\tau]]$, let $\mathbb{K} := \mathbb{C}((\tau))$, and let $\overline{\mathbb{K}}$ be an algebraic closure of $\mathbb{K}$. Let $p : \mathcal{X}_R \to \operatorname{Spec} R$ be the base change of π along the morphism $\operatorname{Spec} R \to \mathbb{A}^1$ induced by the inclusion $\mathbb{C}[\tau] \subset R$. Then p is smooth and projective, the closed fiber of p is Y, and the generic fiber of p is $X_{\mathbb{K}}$. Since $H^1(X, \mathcal{O}_X) = 0$, $\mathrm{Pic}_{X/\mathbb{C}}$ is étale [Kle05, Theorem 9.5.11 and Corollary 9.5.14], so $\mathrm{Pic}(X_{\overline{\mathbb{K}}}) = \mathrm{Pic}(X)$ and $X_{\overline{\mathbb{K}}}$ is a smooth prime Fano threefold of genus 12. Thus p satisfies the assumptions of Proposition 16.13. Let $c : \operatorname{Spec} R \to \mathbb{B}^{\mathrm{nd}}$ be the classifying morphism of Proposition 16.13(2). By Proposition 16.13(1), the restrictions of $\mathcal{U}_{\mathcal{X}_R}$ to Y and to $X_{\overline{\mathbb{K}}}$ are the Mukai bundles. Thus $c(0) \in \mathrm{SL}(V) \cdot \Pi_Y$ by Definition-Theorem 2.20, and the base change of c to $\overline{\mathbb{K}}$ is the image of $\Pi_X \otimes \overline{\mathbb{K}}$ under a linear isomorphism $V \otimes \overline{\mathbb{K}} \to V \otimes \overline{\mathbb{K}}$, since the formation of $H^0(X, \mathcal{U}_X^\vee)$ and of Π_X commutes with flat base change [Har77, Proposition III.9.3]. Let $f_c \in R[x,y,z]_4$ be the quartic of Lemma 16.3(4). Then $\det \mathrm{Cat}_2(f_c)$ is a unit of R, the reduction of f_c modulo τ is a non-zero multiple of $f_{c(0)}$, and $[f_{c(0)}] \in G \cdot [f_Y]$ by Lemma 16.3(3). Over $\overline{\mathbb{K}}$, the quartic f_c is annihilated by the Pfaffian ideal of $c \otimes \overline{\mathbb{K}}$. Since the Pfaffian ideal and the inverse system of the quotient in degree four are compatible with linear isomorphisms of V, the point $[f_c \otimes \overline{\mathbb{K}}]$ belongs to $G(\overline{\mathbb{K}}) \cdot [f_X]$.

The closure $Z := \overline{G \cdot [f_X]}$ is a closed G-invariant subset of $\mathbb{P}$, and the point $[f_c \otimes \overline{\mathbb{K}}]$ belongs to $Z(\overline{\mathbb{K}})$. Thus the morphism $[f_c] : \operatorname{Spec} R \to \mathbb{P}$ maps the generic point of $\operatorname{Spec} R$ into Z. Since R is an integral domain, $[f_c]$ factors through Z, so $[f_{c(0)}] \in Z$. Since Z is G-invariant, $G \cdot [f_Y] \subset Z$. Since Sc is a polynomial map, the reduction of $\mathrm{Sc}(f_c)$ modulo τ is a non-zero multiple of $\mathrm{Sc}(f_{c(0)})$, which is non-zero by Theorem 16.11(1) and the equivariance of Sc. Thus $[\mathrm{Sc}(f_c)]$ is an R-point of $\mathbb{P}$ and $[f_c]$ is an R-point of U^{Sc}. Possibly replacing f_c with $\mathrm{Sc}(f_c)$ and Z with $\overline{G \cdot [\mathrm{Sc}(f_X)]}$, the argument above implies $G \cdot [\mathrm{Sc}(f_Y)] \subset \overline{G \cdot [\mathrm{Sc}(f_X)]}$.

By **Step 1**, $[f_Y] \in G \cdot [F_h]$ for some $h \in \mathbb{C}\backslash\{1,2\}$. By (19.4), $[f_{c(0)}] \in W$. Since W is open, the inverse image of W under $[f_c]$ is an open subset of $\operatorname{Spec} R$ which contains the closed point, so it is $\operatorname{Spec} R$. Thus $[f_c \otimes \overline{\mathbb{K}}] \in W(\overline{\mathbb{K}})$. Since W is G-invariant and $[f_c \otimes \overline{\mathbb{K}}] \in G(\overline{\mathbb{K}}) \cdot [f_X]$, we have $[f_X] \in W$. By **Step 1** and (19.3), $[\mathrm{Sc}(f_{c(0)})] \in G \cdot [\mathrm{Sc}(F_h)] \subset \mathbb{P}^{\mathrm{ss}}$. Since $\mathbb{P}^{\mathrm{ss}}$ is open and G-invariant, possibly replacing f_c with $\mathrm{Sc}(f_c)$ and W with $\mathbb{P}^{\mathrm{ss}}$, the argument above implies $[\mathrm{Sc}(f_X)] \in \mathbb{P}^{\mathrm{ss}}$. This proves (19.5).

Finally, assume that Q_X is not GIT-stable. By Proposition 16.17(3), X contains a special line which intersects no other line on X. By Proposition 12.9, X admits an anti-canonical test configuration whose central fiber is smooth and carries a faithful induced $\mathbb{G}_m$-action. In particular, by (19.5), if Q_X is not GIT-stable, then σ is étale at $[f_X]$ and $[\mathrm{Sc}(f_X)] \in \mathbb{P}^{\mathrm{ss}}$.

Step 3. In this step, we prove (1), and we prove that f_X is GIT-polystable if and only if $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

By Lemma 16.3(1), $[f_X] \in U^{\mathrm{cat}}$, so $[f_X]$ is GIT-semistable by Proposition 16.16(1). By **Step 2**, Q_X is GIT-semistable if it is not GIT-stable, so Q_X is GIT-semistable.

By Proposition 16.17(3), Q_X is GIT-stable if and only if every special line on X intersects another line. By Theorem 16.11(1), $\mathrm{Sc}(f_X) \neq 0$. If Q_X is GIT-stable, then f_X is GIT-stable and $\Pi_X \in \Omega^{\mathrm{st}}$ by Theorem 16.12(5) and (4).

Assume that f_X is GIT-stable and Q_X is not GIT-stable. Let Y be the central fiber of a test configuration as in **Step 2**. By (19.5), $[f_Y]$ belongs to the closure of $G\cdot[f_X]$, and $[f_Y] \in U^{\mathrm{cat}} \subset \mathbb{P}^{\mathrm{ss}}$ by Lemma 16.3(1) and Proposition 16.16(1). Since $[f_X]$ is GIT-stable, $G\cdot[f_X]$ is closed in $\mathbb{P}^{\mathrm{ss}}$ and $G_{[f_X]}$ is finite [MFK94, Definitions 1.7 and 1.8, Theorem 1.10, and Amplification 1.11]. Thus $[f_Y] \in G\cdot[f_X]$. However, by **Step 1**, $[f_Y] \in G\cdot[F_h]$ for some $h \in \mathbb{C}\backslash\{1,2\}$, and $G_{[F_h]}$ contains the torus $\{[\mathrm{diag}(1,s,s^{-1})] : s \in \mathbb{C}^*\}$, since F_h is a polynomial in x^2 and yz. This contradicts the finiteness of $G_{[f_X]}$. Thus, if f_X is GIT-stable, then Q_X is GIT-stable.

Assume that $\Pi_X \in \Omega^{\mathrm{st}}$ and Q_X is not GIT-stable. Let $p : \mathcal{X}_R \to \operatorname{Spec} R$ be the family over $R = \mathbb{C}[[\tau]]$ of **Step 2**, with closed fiber Y. Since $\Pi_X \in \Omega^{\mathrm{st}}$, the orbit $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ by the definition of Ω^{st}. Since the geometric generic fiber of p is $X_{\overline{\mathbb{K}}} \cong (X_{\Pi_X})_{\overline{\mathbb{K}}}$, Proposition 16.13(4) implies $Y \cong X_{\Pi_X} \cong X$. Thus X admits a faithful $\mathbb{G}_m$-action. By the exact sequence of Theorem 16.12(2), the stabilizer of Π_X in $\mathrm{SL}(V)$ is infinite, which contradicts $\Pi_X \in \Omega^{\mathrm{st}}$. Thus, if $\Pi_X \in \Omega^{\mathrm{st}}$, then Q_X is GIT-stable. This proves (1).

Assume that $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$. If the stabilizer of Π_X in $\mathrm{SL}(V)$ is finite, then $\Pi_X \in \Omega^{\mathrm{st}}$, so f_X is GIT-stable by (1). Otherwise $\Pi_X \notin \Omega^{\mathrm{st}}$, so X admits a faithful $\mathbb{G}_m$-action by Theorem 16.12(3). By (19.2), $[f_X] \in G\cdot[F_h]$ for some $h \in \mathbb{C}\backslash\{1,2\}$, so f_X is GIT-polystable by Proposition 16.16(3).

Conversely, assume that f_X is GIT-polystable, and let Π' be a point of the closure of $\mathrm{SL}(V)\cdot\Pi_X$ in $\mathbb{B}^{\mathrm{nd}}$. Since $\mathrm{SL}(V)\cdot\Pi_X = \mathrm{PGL}(V)\cdot\Pi_X$, Lemma 16.3(5) implies that $[f_{\Pi'}]$ belongs to the closure of $G\cdot[f_X]$ in U^{cat}. Since $U^{\mathrm{cat}} \subset \mathbb{P}^{\mathrm{ss}}$ (Proposition 16.16(1)) and $G\cdot[f_X]$ is closed in $\mathbb{P}^{\mathrm{ss}}$, we have $[f_{\Pi'}] \in G\cdot[f_X]$. By Lemma 16.3(3), $\Pi' \in \mathrm{PGL}(V)\cdot\Pi_X = \mathrm{SL}(V)\cdot\Pi_X$. Since Π' is arbitrary, $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

Step 4. In this step, we prove (4) and (2), and we prove that if Q_X is GIT-polystable but not GIT-stable, then X admits a faithful $\mathbb{G}_m$-action.

Let $\xi := [f_X]$ and $\xi' := [\mathrm{Sc}(f_X)]$, and assume that Q_X is not GIT-stable. By **Step 2**, $\xi \in W$, so $d\sigma_\xi$ is an isomorphism. By the G-equivariance of σ, $G_\xi \subset G_{\xi'}$. For $o \in \{\xi,\xi'\}$, let $a_o : G \to \mathbb{P}$, $g \to g\cdot o$, be the orbit map. Then $\sigma\circ a_\xi = a_{\xi'}$, so $d\sigma_\xi \circ (da_\xi)_e = (da_{\xi'})_e$, where $e \in G$ is the identity. Since the orbit maps are separable in characteristic zero, $\mathrm{Lie}(G_o) = \mathrm{Ker}(da_o)_e$ for $o \in \{\xi,\xi'\}$ [Bor91, Proposition II.6.7]. If $A \in \mathrm{Lie}(G_{\xi'})$, then $d\sigma_\xi((da_\xi)_e(A)) = (da_{\xi'})_e(A) = 0$, so $(da_\xi)_e(A) = 0$ and $A \in \mathrm{Lie}(G_\xi)$. Thus $\mathrm{Lie}(G_\xi) = \mathrm{Lie}(G_{\xi'})$. Since algebraic groups over $\mathbb{C}$ are smooth, G^0_ξ and $G^0_{\xi'}$ are connected algebraic groups with $\dim G^0_\xi = \dim G^0_{\xi'}$ and $G^0_\xi \subset G^0_{\xi'}$, hence

$$G^0_{[f_X]} = G^0_{[\mathrm{Sc}(f_X)]} \text{ if } Q_X \text{ is not GIT-stable.} \tag{19.6}$$

This proves (4).

Assume that f_X is GIT-polystable. If f_X is GIT-stable, then Q_X is GIT-stable by (1). Otherwise, by Proposition 16.16(2) and (3), $[f_X] \in G\cdot[F_h]$ for some $h \in \mathbb{C}\backslash\{1\}$. By Lemma 16.3(3), $\Pi_X \in \mathrm{PGL}(V)\cdot\Pi_h$, so $X_{\Pi_h} \cong X$ is smooth and $h \neq 2$ by Definition-Proposition 16.14. Since $[\mathrm{Sc}(f_X)] \in G\cdot[\mathrm{Sc}(F_h)]$, Q_X is GIT-polystable by (19.3).

Assume that Q_X is GIT-polystable. If Q_X is GIT-stable, then f_X is GIT-stable by (1). Otherwise, by Theorem 2.29(3), after a projective change of coordinates $\mathrm{Sc}(f_X)$ is a non-zero multiple of $(xz-ay^2)(xz-by^2)$ for some $(a,b) \in \mathbb{C}^2\backslash\{(0,0)\}$. The torus $T := \{[\mathrm{diag}(t,1,t^{-1})] : t \in \mathbb{C}^*\}$ fixes xz and y^2, so $T \subset G^0_{[\mathrm{Sc}(f_X)]}$. By (19.6), $T \subset G^0_{[f_X]}$. By Theorem 16.12(2), $\mathrm{Aut}(X) \cong G_{[f_X]}$ contains a non-trivial torus, so X admits a faithful $\mathbb{G}_m$-action. By (19.2), $[f_X] \in G\cdot[F_h]$ and $[\mathrm{Sc}(f_X)] \in G\cdot[\mathrm{Sc}(F_h)]$ for some $h \in \mathbb{C}\backslash\{1,2\}$, so f_X is GIT-polystable by Proposition 16.16(3). Thus f_X is GIT-polystable if and only if Q_X is GIT-polystable.

By **Step 3**, f_X is GIT-polystable if and only if $\mathrm{SL}(V)\cdot\Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$. If every special line on X intersects another line, then f_X is GIT-stable by (1). If X admits a faithful $\mathbb{G}_m$-action, then $[f_X] \in G\cdot[F_h]$ for some $h \in \mathbb{C}\backslash\{1,2\}$ by (19.2), so f_X is GIT-polystable by Proposition 16.16(3). Conversely, assume that f_X is GIT-polystable. If f_X is GIT-stable, then every special line on X intersects another line by (1). Otherwise Q_X is GIT-polystable but not GIT-stable by (1) and the equivalence above, so X admits a faithful $\mathbb{G}_m$-action. This proves (2).

Step 5. In this step, we prove (3) and the last sentence of (5).

Let X' be a smooth prime Fano threefold of genus 12 which admits a faithful $\mathbb{G}_m$-action, and let $h \in \mathbb{C}\backslash\{1,2\}$ be as in (19.2). By **Step 1**, $Q_{X'}$ is a double smooth conic if $h = 6$, that is, if $X' \cong X^{\mathrm{MU}}$, and $Q_{X'}$ is the union of two distinct smooth conics tangent at two distinct points if $h \neq 6$, that is, if $X' \ncong X^{\mathrm{MU}}$. Since $\mathrm{Aut}(X^{\mathrm{MU}}) \cong \mathrm{PGL}_2$, the threefold X^{MU} admits a faithful $\mathbb{G}_m$-action. By **Step 4**, if Q_X is GIT-polystable but not GIT-stable, then X admits a faithful $\mathbb{G}_m$-action. Conversely, double smooth conics and unions of two smooth conics tangent at two points are GIT-polystable but not GIT-stable by Theorem 2.29(3), since any smooth conic is projectively equivalent to $(xz - y^2 = 0)$, and two smooth conics tangent at two points P_1, P_2 become $(xz - ay^2)(xz - by^2)$ with $ab \neq 0$, $a \neq b$ after taking $P_1 = [1 : 0 : 0]$, $P_2 = [0 : 0 : 1]$ and the common tangent lines $(z = 0)$, $(x = 0)$. Thus a quartic Q_X is GIT-polystable but not GIT-stable if and only if Q_X is a double smooth conic or the union of two distinct smooth conics tangent at two distinct points, Q_X is a double smooth conic if and only if $X \cong X^{\mathrm{MU}}$, and Q_X is the union of two smooth conics tangent at two points if and only if X admits a faithful $\mathbb{G}_m$-action and $X \ncong X^{\mathrm{MU}}$. This proves the first two sentences of (3), except for the equivalence between double conics and non-reduced quartics Q_X, which is proved below.

A smooth conic together with two distinct tangent lines is GIT-polystable but not GIT-stable by Theorem 2.29(3) and contains a line, whereas a GIT-polystable but not GIT-stable Q_X contains no line. Hence a smooth conic together with two distinct tangent lines is not of the form Q_X. A cubic together with an inflectional tangent line is GIT-unstable by Theorem 2.29(2), so it is not of the form Q_X by (1).

By Proposition 16.17(3), Q_X has multiplicity at most two at any point. Thus, by Proposition 16.17(1), Q_X is non-reduced if and only if Q_X is a double smooth conic, every reduced Q_X has only singular points of type A_k with $k \leq 7$, and Q_X has a singular point of type A_7 if and only if Q_X is the union of two smooth conics with one common point. In the last case, the intersection multiplicity at the common point is 4 by Bézout's theorem.

Assume that $X \cong X^a$. By Theorem 16.11(2), $Q_X \cong \Sigma(X)$, and by [KPS18, Proposition 5.4.4], $\Sigma(X)$ is the union of two rational curves C_1 and C_2 glued at a point P, with $\mathrm{Sing}\,\Sigma(X) = \{P\}$. Since the support of a double smooth conic is irreducible, Q_X is reduced, so P is a singular point of type A_k for some $1 \leq k \leq 7$, that is, in local analytic coordinates (w_1, w_2) at P the germ of Q_X is $(w_1^2 + w_2^{k+1} = 0)$. Since $P \in C_1 \cap C_2$, the germ of Q_X at P has at least two branches. If k is even, then $w_1^2 + w_2^{k+1}$ is irreducible in $\mathbb{C}[[w_1, w_2]]$, so the germ has one branch. Thus k is odd and

$$w_1^2 + w_2^{k+1} = \big(w_1 - iw_2^{(k+1)/2}\big)\big(w_1 + iw_2^{(k+1)/2}\big),$$

so the germ of Q_X at P consists of two smooth branches, one on C_1 and one on C_2. Since Q_X is smooth away from P, C_1 and C_2 are smooth rational plane curves. A smooth plane curve of degree d has genus $(d-1)(d-2)/2$, so $\deg C_1, \deg C_2 \leq 2$. Since $\deg C_1 + \deg C_2 = 4$, C_1 and C_2 are smooth conics with one common point. Thus Q_X has a singular point of type A_7.

Conversely, assume that Q_X has a singular point P of type A_7, so $Q_X = C_1 \cup C_2$ for smooth conics C_1 and C_2 with $C_1 \cap C_2 = \{P\}$. Choose coordinates such that $C_1 = (z^2 - xy = 0)$ and $P = [0 : 1 : 0]$. The conic C_1 is the image of $[u : v] \to [u^2 : v^2 : uv]$, and P is the image of $[0 : 1]$. Let $C_2 = (g = 0)$. Since $(C_1 \cdot C_2)_P = 4$, the restriction of g to C_1 is a non-zero multiple of u^4, which is the restriction of x^2. Since the quadrics vanishing on C_1 are the multiples of $z^2 - xy$, we have $g = \alpha(z^2 - xy) + \beta x^2$ for some $\alpha, \beta \in \mathbb{C}$. Thus $\mathrm{Sc}(f_X)$ is a non-zero multiple of $(z^2 - xy)(\alpha(z^2 - xy) + \beta x^2)$, which is invariant under ϕ_s of Lemma 19.2 for any $s \in \mathbb{C}$, since ϕ_s fixes x and $z^2 - xy$. The subgroup $\{[\phi_s] : s \in \mathbb{C}\} \subset G^0_{[\mathrm{Sc}(f_X)]}$ is isomorphic to $\mathbb{G}_a$. Since Q_X is not GIT-stable by Theorem 2.29(1), (19.6) implies that $G_{[f_X]} \cong \mathrm{Aut}(X)$ contains a subgroup isomorphic to $\mathbb{G}_a$ (Theorem 16.12(2)). By [KPS18, Theorem 5.3.10 and Proposition 5.4.6], a smooth prime Fano threefold of genus 12 with infinite automorphism group admits a faithful $\mathbb{G}_m$-action or is isomorphic to X^a. If X admits a faithful $\mathbb{G}_m$-action, then by (19.2) and **Step 1**,

Q_X is non-reduced or has two singular points, which is not the case. Thus $X \cong X^a$. This proves (3).

Finally, by [KPS18, Proposition 5.4.6], $\mathrm{Aut}(X^a) \cong \mathbb{G}_a \rtimes \mu_4$, so by Theorem 16.12(2), $G_{[f_{X^a}]}$ contains a closed subgroup isomorphic to $\mathbb{G}_a$. Since $\mathrm{Cat}_2(f_{X^a})$ is invertible (Theorem 16.11(1)), Lemma 19.2(2) implies that, after a projective change of coordinates and multiplication by a non-zero constant, $f_{X^a} = (z^2 - xy)^2 + cx^4$ for some $c \in \mathbb{C}$. If $c = 0$, then f_{X^a} is the square of a smooth conic, so $[f_{X^a}] \in G \cdot [F_6]$ since $F_6 = (x^2 + yz)^2$. By Lemma 16.3(3) and Definition-Proposition 16.14, $X^a \cong X_{\Pi_6} \cong X^{\mathrm{MU}}$, which is impossible since $\mathrm{Aut}(X^{\mathrm{MU}}) \cong \mathrm{PGL}_2$ and $\dim \mathrm{Aut}(X^a) = 1$. Thus $c \neq 0$. Replacing x with $c^{-1/4}x$ and y with $c^{1/4}y$ fixes $z^2 - xy$ and turns cx^4 into x^4, so we may take $f_{X^a} = (z^2 - xy)^2 + x^4$. By Lemma 19.2(3), $\mathrm{Sc}(f_{X^a})$ is a non-zero multiple of $(z^2 - xy)^2 - 6x^4$. This proves (5).

Step 6. In this step, we prove (6).

Assume that f_X is not GIT-polystable. Then f_X is not GIT-stable, so Q_X is not GIT-stable by (1). By Proposition 16.16(2) and (3), after a projective change of coordinates and multiplication by a non-zero constant,

$$f_X = F_h + a_1x^3y + a_2x^2y^2 + a_3xy^3 + a_4y^4 + a_5xy^2z + a_6y^3z$$

for some $h \in \mathbb{C}\backslash\{1\}$ and $a_1, \dots, a_6 \in \mathbb{C}$, and $G \cdot [F_h]$ is the unique closed orbit in the closure of $G \cdot [f_X]$ in $\mathbb{P}^{\mathrm{ss}}$. Let Y be the central fiber of a test configuration as in **Step 2**, which exists since Q_X is not GIT-stable. By (19.5) and **Step 1**, $G \cdot [f_Y] = G \cdot [F_{h'}]$ is contained in the closure of $G \cdot [f_X]$ for some $h' \in \mathbb{C}\backslash\{1, 2\}$. Since $[F_{h'}]$ is GIT-polystable (Proposition 16.16(3)), $G \cdot [F_{h'}] = G \cdot [F_h]$, so $h = h' \neq 2$ by Proposition 16.16(4). If $[F_{h''}]$ also represents the unique closed orbit in the closure of $G \cdot [f_X]$ in $\mathbb{P}^{\mathrm{ss}}$ for some $h'' \in \mathbb{C}\backslash\{1\}$, then $G \cdot [F_{h''}] = G \cdot [F_h]$, so $h'' = h$ by Proposition 16.16(4). Thus h is unique.

For $t \in \mathbb{C}^*$, let $g_t := \mathrm{diag}(1, t^{-1}, t) \in \mathrm{SL}_3$. Then $g_t \cdot f_X = f_t$, where f_t is the quartic of Proposition 16.16(3), $f_t(x, y, z) = f_X(x, ty, t^{-1}z)$. By the SL_3-equivariance of Sc, $g_t \cdot \mathrm{Sc}(f_X) = \mathrm{Sc}(f_t)$ for $t \neq 0$. The coefficients of f_t are polynomials in t and $f_0 = F_h$, so the coefficients of $\mathrm{Sc}(f_t)$ are polynomials in t, and

$$\lim_{t \to 0} \mathrm{diag}(1, t^{-1}, t) \cdot \mathrm{Sc}(f_X) = \mathrm{Sc}(F_h) \neq 0.$$

Thus $[\mathrm{Sc}(F_h)]$ belongs to the closure of $G \cdot [\mathrm{Sc}(f_X)]$ in $\mathbb{P}^{\mathrm{ss}}$. Since $[\mathrm{Sc}(F_h)]$ is GIT-polystable by (19.3), $G \cdot [\mathrm{Sc}(F_h)]$ is the unique closed orbit in the closure of $G \cdot [\mathrm{Sc}(f_X)]$ in $\mathbb{P}^{\mathrm{ss}}$ [MFK94, Theorem 1.10 and Corollary 1.2].

Let Y' be the central fiber of an anti-canonical test configuration of X such that Y' is smooth and carries a faithful induced $\mathbb{G}_m$-action. By (19.5) for Y' in place of Y, $G \cdot [\mathrm{Sc}(f_{Y'})]$ is contained in the closure of $G \cdot [\mathrm{Sc}(f_X)]$. By (19.2) and (19.3), $[\mathrm{Sc}(f_{Y'})]$ is GIT-polystable, hence GIT-semistable. Thus $G \cdot [\mathrm{Sc}(f_{Y'})]$ is the unique closed orbit in the closure of $G \cdot [\mathrm{Sc}(f_X)]$ in $\mathbb{P}^{\mathrm{ss}}$, that is, $G \cdot [\mathrm{Sc}(f_{Y'})] = G \cdot [\mathrm{Sc}(F_h)]$. This proves (6). □

20. The harmonic slice

In this section, we prove that the orbit of any GIT-polystable quartic $[f]$ with $\det \mathrm{Cat}_2(f) \neq 0$ intersects $\{[f_b] : b \in \mathrm{Harm}_4\}$ (Proposition 20.6), and we construct the family of Theorem 20.14, which is the main input for Theorem 1.7(3). Definition-Proposition 20.1 and Proposition 20.6 are written in coordinates x_1, x_2, x_3 adapted to the conic $q_0 = x_1^2 + x_2^2 + x_3^2$, while Theorem 1.7 uses coordinates x, y, z and the conic $q = xz - y^2$. The linear substitution relating them is Definition-Proposition 20.1(7).

20.1. **The metric criterion.** In this subsection, we turn the condition that the orbit of a quartic $[f]$ intersects $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$ into an equation on symmetric matrices (Definition-Proposition 20.1(2)). A compactness argument implies that the equation has a non-zero solution, but the solution may be degenerate, and if the operator C_f of Definition-Proposition 20.1 is

invertible, degenerate solutions come from singular points of $(f = 0)$ or of $(f^\vee = 0)$, where $f^\vee$ is the quartic defined there.

Definition-Proposition 20.1. Let $q_0 := x_1^2 + x_2^2 + x_3^2$, let

$$\mathrm{Harm}_4(q_0) := \{b \in \mathbb{C}[x_1, x_2, x_3]_4 : (\partial_1^2 + \partial_2^2 + \partial_3^2)b = 0\},$$

and let $\mathrm{SO}(q_0) \subset \mathrm{SL}_3$ be the special orthogonal group of q_0. Let $f = \sum F_{ijkl}x_ix_jx_kx_l$ be a non-zero quartic with fully symmetric coefficients. For a symmetric 3×3 matrix A let $C_f(A)_{ij} := \sum_{k,l} F_{ijkl}A_{kl}$ and $\Delta_A := \sum_{i,j} A_{ij}\partial_i\partial_j$, and if A is invertible let $q_A(x) := x^tA^{-1}x$. If C_f is invertible, let $f^\vee(w) := \mathrm{tr}\left(ww^tC_f^{-1}(ww^t)\right)$. Then the following hold.

(1) The group $\mathrm{SO}(q_0)$ maps isomorphically onto $\mathrm{Stab}_{\mathrm{PGL}_3}([q_0^2])$, and the morphism $\mathrm{PGL}_3 \times^{\mathrm{SO}(q_0)} \mathrm{Harm}_4(q_0) \to \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$ given by $[g, b] \to g \cdot [q_0^2 + b]$ is étale along the zero section. In the coordinates x, y, z of (7), for $\gamma \in \mathrm{SL}_2$ let $\sigma(\gamma)$ be the linear map with $\sigma(\gamma)(u^2, uv, v^2) = (u'^2, u'v', v'^2)$, where $(u', v') = \gamma(u, v)$. Then $\sigma : \mathrm{SL}_2 \to \mathrm{SO}(q_0)$ is a homomorphism, and $b \to b(u^2, uv, v^2)$ is an SL_2-equivariant isomorphism from $\mathrm{Harm}_4(q_0)$, with SL_2 acting through σ, onto the binary octics $\mathbb{C}[u, v]_8 \cong \mathrm{Sym}^8 \mathbb{C}^2$.

(2) The orbit of $[f]$ intersects $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$ if and only if there exist an invertible symmetric matrix A and $t \in \mathbb{C}^*$ with $AC_f(A) = tI_3$.

(3) For an invertible symmetric matrix A and $t \in \mathbb{C}^*$, we have $AC_f(A) = tI_3$ if and only if $\Delta_A f = 12tq_A$, and then $f = \frac{3t}{5}q_A^2 + b_A$ with $\Delta_A b_A = 0$.

(4) There exists a non-zero symmetric matrix A such that $AC_f(A)$ is scalar.

(5) Assume that C_f is invertible, and let A be a symmetric matrix such that $AC_f(A)$ is scalar. If $\mathrm{rank}\, A = 1$, then the point of $\mathbb{P}^2$ spanned by the image of A is a singular point of $(f = 0)$. If $\mathrm{rank}\, A = 2$, then the point of $\mathbb{P}^2$ spanned by $\ker A$ is a singular point of $(f^\vee = 0)$. If $(f = 0)$ and $(f^\vee = 0)$ are smooth, then the orbit of $[f]$ intersects $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$.

(6) Under the identifications $A \leftrightarrow \Delta_A$ of symmetric 3×3 matrices with $\mathbb{C}[\partial_1, \partial_2, \partial_3]_2$ and $S \leftrightarrow x^tSx$ of symmetric 3×3 matrices with $\mathbb{C}[x_1, x_2, x_3]_2$, we have $\mathrm{Cat}_2(f) = 12C_f$. In particular C_f is invertible if and only if $\mathrm{Cat}_2(f)$ is invertible.

(7) Under the linear substitution $x = x_1 + ix_2$, $z = x_1 - ix_2$, $y = ix_3$, we have $xz - y^2 = q_0$ and $\partial_1^2 + \partial_2^2 + \partial_3^2 = 4\partial_x\partial_z - \partial_y^2$. In particular $b \in \mathrm{Harm}_4(q_0)$ if and only if $(4\partial_x\partial_z - \partial_y^2)b = 0$ in the coordinates x, y, z.

Proof. The proof is long, so we divide it into several steps. For $g \in \mathrm{GL}_3$ and a form $\phi \in \mathbb{C}[x_1, x_2, x_3]$ we write $g \cdot \phi := \phi \circ g^{-1}$. The action $\phi \to g \cdot \phi$ of GL_3 induces an action of PGL_3 on $\mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$, and the orbits of PGL_3 and of SL_3 coincide, since any invertible matrix is a scalar times a matrix of determinant one. Let $\Delta := \partial_1^2 + \partial_2^2 + \partial_3^2$, and let $\mathcal{S}$ be the space of symmetric 3×3 matrices, with $\langle A, B\rangle := \mathrm{tr}(AB)$. We have $\dim \mathcal{S} = 6$.

Step 1. In this step we prove (6), (7), and the formula

$$\Delta_A f = 12x^tC_f(A)x \qquad \text{for any } A \in \mathcal{S}. \tag{20.1}$$

Since the coefficients F_{ijkl} are fully symmetric,

$$\partial_i f = 4\sum_{j,k,l} F_{ijkl}x_jx_kx_l \quad \text{and} \quad \partial_i\partial_j f = 12\sum_{k,l} F_{ijkl}x_kx_l.$$

Multiplying the second formula by A_{ij} and summing over i, j, we obtain

$$\Delta_A f = 12\sum_{k,l}\Big(\sum_{i,j} F_{klij}A_{ij}\Big)x_kx_l = 12\sum_{k,l} C_f(A)_{kl}x_kx_l = 12x^tC_f(A)x,$$

which is (20.1). By definition $\mathrm{Cat}_2(f)(\Delta_A) = \Delta_A f$, so (20.1) says that $\mathrm{Cat}_2(f)$ corresponds to $12C_f$ under the identifications of (6). Both identifications are linear isomorphisms, so C_f is invertible if and only if $\mathrm{Cat}_2(f)$ is invertible. This proves (6).

For (7), we have $xz = (x_1 + ix_2)(x_1 - ix_2) = x_1^2 + x_2^2$ and $y^2 = -x_3^2$, so $xz - y^2 = q_0$. By the chain rule,

$$\partial_1 = \partial_x + \partial_z, \qquad \partial_2 = i\partial_x - i\partial_z, \qquad \partial_3 = i\partial_y.$$

Thus

$$\partial_1^2 + \partial_2^2 + \partial_3^2 = (\partial_x + \partial_z)^2 - (\partial_x - \partial_z)^2 - \partial_y^2 = 4\partial_x\partial_z - \partial_y^2,$$

which proves (7).

Step 2. In this step we decompose quartics with respect to Δ and compare Δ_A with Δ.

For $d \in \{2, 4\}$ let

$$\mathrm{Harm}_d(q_0) := \{b \in \mathbb{C}[x_1, x_2, x_3]_d : \Delta b = 0\}.$$

Direct differentiation, together with the Euler formula $\sum_i x_i\partial_i b_2 = 2b_2$, shows that

$$\Delta q_0 = 6, \qquad \Delta(q_0^2) = 20q_0, \qquad \Delta(q_0 b_2) = 6b_2 + 4\sum_i x_i\partial_i b_2 = 14b_2 \quad \text{for } b_2 \in \mathrm{Harm}_2(q_0).$$

We claim that any $f \in \mathbb{C}[x_1, x_2, x_3]_4$ has a unique decomposition

$$f = b_4 + q_0 b_2 + cq_0^2, \qquad b_4 \in \mathrm{Harm}_4(q_0),\ b_2 \in \mathrm{Harm}_2(q_0),\ c \in \mathbb{C}. \tag{20.2}$$

For existence, let

$$c := \frac{\Delta^2 f}{120}, \qquad b_2 := \frac{\Delta f - 20cq_0}{14}, \qquad b_4 := f - q_0 b_2 - cq_0^2.$$

Then $\Delta b_2 = (\Delta^2 f - 120c)/14 = 0$ and $\Delta b_4 = \Delta f - 14b_2 - 20cq_0 = 0$. For uniqueness, if $b_4 + q_0 b_2 + cq_0^2 = 0$, then applying Δ^2 implies $120c = 0$, and then applying Δ implies $14b_2 = 0$, so $b_4 = 0$. Since $\Delta : \mathbb{C}[x_1, x_2, x_3]_2 \to \mathbb{C}$ is non-zero, $\dim \mathrm{Harm}_2(q_0) = 5$, and (20.2) implies $\dim \mathrm{Harm}_4(q_0) = 15 - 5 - 1 = 9$.

Next let $h \in \mathrm{GL}_3$ and $A := hh^t$. For a form ϕ, the chain rule implies that the Hessian matrix of $\phi \circ h^{-1}$ at x is $h^{-t}\,\mathrm{Hess}(\phi)(h^{-1}x)h^{-1}$. Since $\Delta_A\phi = \mathrm{tr}(A\,\mathrm{Hess}(\phi))$, we obtain

$$\Delta_A(\phi \circ h^{-1}) = \mathrm{tr}\left(hh^t h^{-t}\,\mathrm{Hess}(\phi)h^{-1}\right) \circ h^{-1} = (\Delta\phi) \circ h^{-1}, \qquad q_A = q_0 \circ h^{-1}. \tag{20.3}$$

The second formula holds because $x^t(hh^t)^{-1}x = (h^{-1}x)^t(h^{-1}x)$. Any invertible $A \in \mathcal{S}$ is of the form hh^t with $h \in \mathrm{GL}_3$, since any non-degenerate quadratic form is equivalent to q_0.

Step 3. In this step we prove that $\mathrm{SO}(q_0)$ maps isomorphically onto $\mathrm{Stab}_{\mathrm{PGL}_3}([q_0^2])$ and that the morphism of (1) is étale along the zero section.

Let $g \in \mathrm{GL}_3$ with $g \cdot q_0^2 = \lambda q_0^2$ for some $\lambda \in \mathbb{C}^*$. Then $(q_0 \circ g^{-1})^2 = \lambda q_0^2$, and since q_0 is irreducible, unique factorization implies $q_0 \circ g^{-1} = \lambda' q_0$ for some $\lambda' \in \mathbb{C}^*$. Choose $\mu \in \mathbb{C}^*$ with $\mu^{-2}\lambda' = 1$. Then $q_0 \circ (\mu g)^{-1} = q_0$, so $\det(\mu g) = \pm 1$, and replacing μ with $-\mu$ if necessary we may assume $\det(\mu g) = 1$. Thus the class of g in PGL_3 has a representative $\mu g \in \mathrm{SO}(q_0)$. If g' and $\mu'' g'$ both belong to $\mathrm{SO}(q_0)$ for some $\mu'' \in \mathbb{C}^*$, then $\mu''^2 = 1$ and $\mu''^3 = 1$, so $\mu'' = 1$. Conversely any element of $\mathrm{SO}(q_0)$ fixes q_0^2. Hence the homomorphism $\mathrm{SO}(q_0) \to \mathrm{Stab}_{\mathrm{PGL}_3}([q_0^2])$ is bijective. The differential of the homomorphism is the inclusion $\mathfrak{so}(q_0) \subset \mathfrak{sl}_3$, which is injective, and a bijective homomorphism of algebraic groups over $\mathbb{C}$ with injective differential is an isomorphism. Thus $\mathrm{SO}(q_0) \to \mathrm{Stab}_{\mathrm{PGL}_3}([q_0^2])$ is an isomorphism.

By (20.3) with $h \in \mathrm{SO}(q_0)$, so that $hh^t = I_3$, the group $\mathrm{SO}(q_0)$ preserves $\mathrm{Harm}_4(q_0)$. Since $h \cdot (q_0^2 + b) = q_0^2 + h \cdot b$ for $h \in \mathrm{SO}(q_0)$, the morphism

$$\mathrm{PGL}_3 \times^{\mathrm{SO}(q_0)} \mathrm{Harm}_4(q_0) \to \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4), \qquad [g, b] \to g \cdot [q_0^2 + b]$$

is well defined. Here $q_0^2 + b \neq 0$ by (20.2). The variety $\mathrm{PGL}_3 \times^{\mathrm{SO}(q_0)} \mathrm{Harm}_4(q_0)$ is a vector bundle over $\mathrm{PGL}_3 / \mathrm{SO}(q_0)$, so it is smooth with $\dim \mathrm{PGL}_3 \times^{\mathrm{SO}(q_0)} \mathrm{Harm}_4(q_0) = 8 - 3 + 9 = 14 = \dim \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$.

We compute the differential at $[1,0]$. The tangent space of $\mathbb{P}(\mathbb{C}[x_1,x_2,x_3]_4)$ at $[q_0^2]$ is $\mathbb{C}[x_1,x_2,x_3]_4/\mathbb{C}q_0^2$, and the tangent space of $\mathrm{PGL}_3 \times^{\mathrm{SO}(q_0)} \mathrm{Harm}_4(q_0)$ at $[1,0]$ is $(\mathfrak{sl}_3/\mathfrak{so}(q_0)) \oplus \mathrm{Harm}_4(q_0)$, where $\mathfrak{so}(q_0) = \{M : M + M^t = 0\}$. For $M \in \mathfrak{sl}_3$,

$$q_0\big((I_3 - \epsilon M)x\big)^2 = q_0^2 - 4\epsilon q_0 x^t M x + O(\epsilon^2).$$

Thus the differential at $[1,0]$ is

$$(M,b) \to -4q_0 x^t M x + b \mod \mathbb{C}q_0^2.$$

The kernel of the map $\mathfrak{sl}_3 \to \mathbb{C}[x_1,x_2,x_3]_2$, $M \to x^t M x$, is $\mathfrak{so}(q_0)$, and the image is $\{x^t S x : S \in \mathcal{S},\ \mathrm{tr}\, S = 0\} = \mathrm{Harm}_2(q_0)$, since $\Delta(x^t S x) = 2\,\mathrm{tr}\, S$. Hence the differential maps $(\mathfrak{sl}_3/\mathfrak{so}(q_0)) \oplus \mathrm{Harm}_4(q_0)$ injectively onto the image of $q_0\,\mathrm{Harm}_2(q_0) \oplus \mathrm{Harm}_4(q_0)$ in $\mathbb{C}[x_1,x_2,x_3]_4/\mathbb{C}q_0^2$, which is all of $\mathbb{C}[x_1,x_2,x_3]_4/\mathbb{C}q_0^2$ by (20.2). Since both varieties are smooth of dimension 14 and the differential at $[1,0]$ is an isomorphism, the morphism is étale at $[1,0]$. The morphism is PGL_3-equivariant and PGL_3 acts transitively on the zero section, so it is étale along the zero section. In particular, since étale morphisms are open and the étale locus is open, the image of the morphism contains a non-empty open subset of $\mathbb{P}(\mathbb{C}[x_1,x_2,x_3]_4)$.

Step 4. In this step we prove the last two sentences of (1).

Let σ be as in (1). In the coordinates x,y,z of (7) we have $q_0 = xz - y^2$, and $\sigma(\gamma) = \mathrm{Sym}^2\gamma$ in the basis u^2, uv, v^2. Any point of the cone $\{q_0 = 0\} \subset \mathbb{C}^3$ is of the form (u^2, uv, v^2), so $q_0 \circ \sigma(\gamma)$ vanishes on $\{q_0 = 0\}$. Since q_0 is irreducible, $q_0 \circ \sigma(\gamma) = \chi(\gamma) q_0$ for a character χ of SL_2. Since SL_2 has no non-trivial characters, $\chi = 1$ and $\det\sigma(\gamma) = 1$. Thus $\sigma : \mathrm{SL}_2 \to \mathrm{SO}(q_0)$ is a homomorphism, and $\mathrm{Harm}_4(q_0)$ is an SL_2-representation through σ.

Consider the linear map

$$\mathrm{Harm}_4(q_0) \to \mathbb{C}[u,v]_8, \qquad b \to b(u^2, uv, v^2).$$

For $\gamma \in \mathrm{SL}_2$ we have $(\sigma(\gamma)\cdot b)(u^2,uv,v^2) = b\big(\sigma(\gamma^{-1})(u^2,uv,v^2)\big)$, which is b evaluated at the point $(u'^2, u'v', v'^2)$ with $(u',v') = \gamma^{-1}(u,v)$, so the map is SL_2-equivariant. If $b(u^2,uv,v^2) = 0$, then b vanishes on the conic $(q_0 = 0)$, so $b \in q_0\mathbb{C}[x_1,x_2,x_3]_2$ because q_0 is irreducible. Since $\Delta q_0 = 6 \neq 0$, we have $\mathbb{C}[x_1,x_2,x_3]_2 = \mathrm{Harm}_2(q_0) \oplus \mathbb{C}q_0$, so $b = q_0(b_2 + cq_0)$ with $b_2 \in \mathrm{Harm}_2(q_0)$ and $c \in \mathbb{C}$. Comparing with the decomposition $b = b + q_0 \cdot 0 + 0 \cdot q_0^2$, the uniqueness in (20.2) implies $b = 0$. Thus the map is injective, and $\dim \mathrm{Harm}_4(q_0) = 9 = \dim \mathbb{C}[u,v]_8$, so it is an isomorphism of SL_2-representations. Finally $\mathbb{C}[u,v]_8 = \mathrm{Sym}^8(\mathbb{C}^2)^* \cong \mathrm{Sym}^8\mathbb{C}^2$, since $\mathbb{C}^2 \cong (\mathbb{C}^2)^*$ as SL_2-representations through the invariant alternating form.

Step 5. In this step we prove (3).

Let $A \in \mathcal{S}$ be invertible and $t \in \mathbb{C}^*$. Since A is invertible, $AC_f(A) = tI_3$ if and only if $C_f(A) = tA^{-1}$, which by the injectivity of $S \to x^t S x$ is equivalent to $x^t C_f(A) x = t q_A$. By (20.1), this is equivalent to $\Delta_A f = 12 t q_A$. Write $A = hh^t$ with $h \in \mathrm{GL}_3$ (**Step 2**). By (20.3),

$$\Delta_A(q_A^2) = \Delta_A(q_0^2 \circ h^{-1}) = (\Delta q_0^2) \circ h^{-1} = 20 q_0 \circ h^{-1} = 20 q_A.$$

Hence, if $\Delta_A f = 12 t q_A$, then $b_A := f - \frac{3t}{5} q_A^2$ satisfies

$$\Delta_A b_A = 12 t q_A - \frac{3t}{5} \cdot 20 q_A = 0.$$

This proves (3).

Step 6. In this step we prove (2).

Assume that $AC_f(A) = tI_3$ for an invertible $A \in \mathcal{S}$ and $t \in \mathbb{C}^*$, and write $A = hh^t$ with $h \in \mathrm{GL}_3$. By (3), $f = \frac{3t}{5} q_A^2 + b_A$ with $\Delta_A b_A = 0$. By (20.3) applied to $\phi := b_A \circ h$, we have $\Delta(b_A \circ h) \circ h^{-1} = \Delta_A b_A = 0$, so $b_A \circ h \in \mathrm{Harm}_4(q_0)$. Since $q_A \circ h = q_0$,

$$h^{-1} \cdot f = f \circ h = \frac{3t}{5}\Big(q_0^2 + \frac{5}{3t} b_A \circ h\Big).$$

Thus $h^{-1} \cdot [f] = [q_0^2 + b]$ with $b = \frac{5}{3t} b_A \circ h \in \mathrm{Harm}_4(q_0)$.

Conversely, assume that $g \cdot [f] = [q_0^2 + b]$ for some $g \in \mathrm{GL}_3$ and $b \in \mathrm{Harm}_4(q_0)$, that is, $f \circ g^{-1} = c(q_0^2 + b)$ for some $c \in \mathbb{C}^*$. Let $h := g^{-1}$ and $A := hh^t$. Then $f = c(q_0^2 + b) \circ h^{-1}$, and (20.3) implies

$$\Delta_A f = c\big(\Delta(q_0^2 + b)\big) \circ h^{-1} = 20cq_0 \circ h^{-1} = 20cq_A.$$

By (3) with $t := \frac{5c}{3} \neq 0$, we obtain $AC_f(A) = tI_3$, and A is invertible. This proves (2).

Step 7. In this step we prove (4) by a compactness argument.

Let $\mathcal{Z} \subset \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4) \times \mathbb{P}(\mathcal{S})$ be the subset defined by

$$AC_f(A) - \frac{1}{3} \operatorname{tr}\big(AC_f(A)\big) I_3 = 0.$$

Since the entries of the left-hand side are bihomogeneous of degree one in f and degree two in A, $\mathcal{Z}$ is closed, and $([f], [A]) \in \mathcal{Z}$ if and only if $AC_f(A)$ is scalar. Let $p : \mathcal{Z} \to \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$ be the projection. Since p is projective, p is proper [Har77, Theorem II.4.9], so $p(\mathcal{Z})$ is closed. By **Step 6**, for any $[f]$ in the PGL_3-orbit of a point of $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$ there exists an invertible $A \in \mathcal{S}$ with $AC_f(A) = tI_3$ for some $t \in \mathbb{C}^*$, so $[f] \in p(\mathcal{Z})$. The union of the PGL_3-orbits of the points of $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$ is the image of the morphism of (1), which contains a non-empty open subset by **Step 3**. Hence $p(\mathcal{Z})$ is a closed subset containing a dense subset of $\mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$, so $p(\mathcal{Z}) = \mathbb{P}(\mathbb{C}[x_1, x_2, x_3]_4)$. This proves (4).

Step 8. In this step we prove (5).

Assume that C_f is invertible, and let $A \in \mathcal{S}$ be non-zero with $AC_f(A) = tI_3$ for some $t \in \mathbb{C}$. If $\operatorname{rank} A \leq 2$, then $t^3 = \det(AC_f(A)) = 0$, so $t = 0$.

Assume that $\operatorname{rank} A = 1$, and write $A = avv^t$ with $a \in \mathbb{C}^*$ and $0 \neq v \in \mathbb{C}^3$. Then $0 = AC_f(A) = a^2 v\big(v^t C_f(vv^t)\big)$, so $v^t C_f(vv^t) = 0$. Since $C_f(vv^t)$ is symmetric, $C_f(vv^t)v = 0$. By **Step 1**,

$$\partial_i f(v) = 4 \sum_{j,k,l} F_{ijkl} v_j v_k v_l = 4\big(C_f(vv^t)v\big)_i = 0 \quad \text{for } i = 1, 2, 3,$$

and the Euler formula implies $4f(v) = \sum_i v_i \partial_i f(v) = 0$. Thus $[v]$, the point spanned by the image of A, is a singular point of $(f = 0)$.

Assume that $\operatorname{rank} A = 2$, and let v span $\ker A$. Since $t = 0$, the image of $C_f(A)$ is contained in $\ker A = \mathbb{C}v$. A symmetric matrix with image in $\mathbb{C}v$ is of the form avv^t, and $C_f(A) \neq 0$ because C_f is invertible. Thus $C_f(A) = avv^t$ with $a \in \mathbb{C}^*$, so $A = aC_f^{-1}(vv^t)$ and $C_f^{-1}(vv^t)v = a^{-1}Av = 0$. For $A', B \in \mathcal{S}$,

$$\langle B, C_f(A') \rangle = \sum_{i,j,k,l} F_{ijkl} B_{ij} A'_{kl} = \langle A', C_f(B) \rangle,$$

so C_f is self-adjoint for the non-degenerate pairing $\langle\ ,\ \rangle$ on $\mathcal{S}$, and hence so is C_f^{-1}. For $u \in \mathbb{C}^3$ let $\beta := uv^t + vu^t$. Then

$$\frac{d}{d\epsilon}\Big|_{\epsilon=0} f^\vee(v + \epsilon u) = \langle \beta, C_f^{-1}(vv^t) \rangle + \langle vv^t, C_f^{-1}(\beta) \rangle = 2\langle \beta, C_f^{-1}(vv^t) \rangle = 4u^t C_f^{-1}(vv^t)v = 0.$$

Thus all first derivatives of $f^\vee$ vanish at v, and the Euler formula implies $f^\vee(v) = 0$. Hence $[v]$, the point spanned by the kernel of A, is a singular point of $(f^\vee = 0)$.

Finally assume that $(f = 0)$ and $(f^\vee = 0)$ are smooth. By (4) there exists a non-zero $A \in \mathcal{S}$ with $AC_f(A) = tI_3$ for some $t \in \mathbb{C}$. By the cases $\operatorname{rank} A = 1$ and $\operatorname{rank} A = 2$ above, $\operatorname{rank} A = 3$. If $t = 0$, then $C_f(A) = A^{-1} \cdot 0 = 0$, which contradicts the invertibility of C_f. Thus $t \neq 0$, and (2) implies that the orbit of $[f]$ intersects $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$. This proves (5). □

Since for a symmetric matrix A of rank two with $AC_f(A)$ scalar the point spanned by $\ker A$ is a singular point of $(f^\vee = 0)$ (Definition-Proposition 20.1(5)), the count in Subsection 20.2 needs control of the singular points of $(f^\vee = 0)$. Lemma 20.5 below bounds the sum of the Milnor numbers of the singular points of $(f^\vee = 0)$ when $(f^\vee = 0)$ is reduced. Its proof uses the following coordinates on the space of symmetric 3×3 matrices. In Definition-Lemma 20.2

and Lemmas 20.3–20.5, x, y, z stand for x_1, x_2, x_3, not for the coordinates x, y, z of Definition-Proposition 20.1(7).

Definition-Lemma 20.2. Let $\mathcal{S}$ be the space of symmetric 3×3 matrices, with the pairing $\langle A, B\rangle := \operatorname{tr}(AB)$. Let $f = \sum_{i,j,k,l} F_{ijkl} x_i x_j x_k x_l$ be a quartic with fully symmetric coefficients such that C_f is invertible, where $(x_1, x_2, x_3) = (x, y, z)$. For $A \in \mathcal{S}$ let

$$\operatorname{vec}(A) := (A_{11}, A_{22}, A_{33}, A_{12}, A_{13}, A_{23})^t,$$

index the rows and columns of 6×6 matrices by the pairs $(11), (22), (33), (12), (13), (23)$ in the order of $\operatorname{vec}(A)$, denote also by C_f the 6×6 matrix with $\operatorname{vec}(C_f(A)) = C_f \operatorname{vec}(A)$, and let

$$D := \operatorname{diag}(1, 1, 1, 2, 2, 2), \qquad G := DC_f^{-1}.$$

For $g \in \mathrm{GL}_3$ write $g^{-t} := (g^{-1})^t$. Then the following hold.

(1) For $A, A' \in \mathcal{S}$, we have $\langle A, A'\rangle = \operatorname{tr}(AA') = \operatorname{vec}(A)^t D \operatorname{vec}(A')$, $\langle A', C_f(A)\rangle = \langle A, C_f(A')\rangle$, and $\langle A, C_f^{-1}(A)\rangle = \operatorname{vec}(A)^t G \operatorname{vec}(A)$.

(2) The matrix G is symmetric and invertible, and $G^{-1}_{(ij),(kl)} = F_{ijkl}$ for $i \le j$ and $k \le l$. In particular

$$\begin{aligned} &G^{-1}_{(11),(22)} = G^{-1}_{(12),(12)}, \quad G^{-1}_{(11),(33)} = G^{-1}_{(13),(13)}, \quad G^{-1}_{(22),(33)} = G^{-1}_{(23),(23)}, \\ &G^{-1}_{(11),(23)} = G^{-1}_{(12),(13)}, \quad G^{-1}_{(22),(13)} = G^{-1}_{(12),(23)}, \quad G^{-1}_{(33),(12)} = G^{-1}_{(13),(23)}. \end{aligned} \tag{20.4}$$

(3) For $w = (x, y, z)^t$ we have $\operatorname{vec}(ww^t) = (x^2, y^2, z^2, xy, xz, yz)^t$ and

$$f(w) = \langle ww^t, C_f(ww^t)\rangle, \qquad f^\vee(w) = \operatorname{vec}(ww^t)^t G \operatorname{vec}(ww^t),$$

and $\nabla f^\vee(w) = 4C_f^{-1}(ww^t)w$.

(4) Let $g \in \mathrm{GL}_3$ and $f'(w) := f(gw)$. Then $C_{f'}(A) = g^t C_f(gAg^t)g$, $C_{f'}$ is invertible,

$$C_{f'}^{-1}(A) = g^{-1} C_f^{-1}(g^{-t} A g^{-1}) g^{-t}, \qquad \operatorname{tr}\left(AC_{f'}^{-1}(A)\right) = \operatorname{tr}\left(PAP^t C_f^{-1}(PAP^t)\right),$$

where $P := g^{-t}$, and $f'^\vee(w) = f^\vee(g^{-t}w)$.

Proof. (1) Since A and A' are symmetric,

$$\langle A, A'\rangle = \operatorname{tr}(AA') = \sum_{i,j} A_{ij}A'_{ij} = \sum_i A_{ii}A'_{ii} + 2\sum_{i<j} A_{ij}A'_{ij} = \operatorname{vec}(A)^t D \operatorname{vec}(A').$$

Moreover $\langle A', C_f(A)\rangle = \sum_{i,j,k,l} A'_{ij} F_{ijkl} A_{kl}$ is symmetric in A and A' because $F_{ijkl} = F_{klij}$. The third identity follows from the first one with $A' = C_f^{-1}(A)$.

(2) Let $\hat{F}$ be the symmetric 6×6 matrix with $\hat{F}_{(ij),(kl)} = F_{ijkl}$. Since A is symmetric, $C_f(A)_{ij} = \sum_{k\le l} D_{(kl),(kl)} F_{ijkl} A_{kl}$, that is, $C_f = \hat{F}D$ as 6×6 matrices. Since C_f is invertible, so is $\hat{F}$, and $G = D(\hat{F}D)^{-1} = \hat{F}^{-1}$. Thus G is symmetric and $G^{-1} = \hat{F}$. The relations (20.4) are the identities $F_{1122} = F_{1212}$, $F_{1133} = F_{1313}$, $F_{2233} = F_{2323}$, $F_{1123} = F_{1213}$, $F_{2213} = F_{1223}$, and $F_{3312} = F_{1323}$ of the fully symmetric coefficients.

(3) The first identity is

$$f(w) = \sum_{i,j} w_i w_j \sum_{k,l} F_{ijkl} w_k w_l = \sum_{i,j} (ww^t)_{ij} C_f(ww^t)_{ij},$$

and the second one follows from (1). By (1), C_f, hence also C_f^{-1}, is self-adjoint for $\langle\ ,\ \rangle$. Thus the derivative of $A \to \langle A, C_f^{-1}(A)\rangle$ at A in the direction A' is $2\langle A', C_f^{-1}(A)\rangle$. For $u \in \mathbb{C}^3$ the derivative of $w \to ww^t$ in the direction u is $uw^t + wu^t$, and $C_f^{-1}(ww^t)$ is symmetric. Hence the derivative of $f^\vee$ at w in the direction u is

$$2\langle uw^t + wu^t, C_f^{-1}(ww^t)\rangle = 4u^t C_f^{-1}(ww^t)w.$$

(4) The quartic f' has the fully symmetric coefficients $F'_{abcd} = \sum_{i,j,k,l} F_{ijkl} g_{ia} g_{jb} g_{kc} g_{ld}$. Hence

$$C_{f'}(A)_{ab} = \sum_{i,j} g_{ia} g_{jb} \sum_{k,l} F_{ijkl} (gAg^t)_{kl} = \left(g^t C_f(gAg^t) g\right)_{ab}.$$

The map $A \to g^t C_f(gAg^t)g$ is a composite of invertible linear maps, and the inverse map is $A \to g^{-1} C_f^{-1}(g^{-t} A g^{-1}) g^{-t}$. Since the trace is invariant under cyclic permutations,

$$\operatorname{tr}(A C_{f'}^{-1}(A)) = \operatorname{tr}(g^{-t} A g^{-1} C_f^{-1}(g^{-t} A g^{-1})),$$

which is the identity in (4) because $P^t = g^{-1}$. Taking $A = ww^t$ and using $Pww^tP^t = (g^{-t}w)(g^{-t}w)^t$, we obtain $f'^{\vee}(w) = f^{\vee}(g^{-t}w)$. □

The next two lemmas treat a multiple line and a multiple conic in $(f^{\vee} = 0)$.

Lemma 20.3. *Let f be a quartic such that C_f is invertible. Assume that x^2 divides $f^{\vee}$. Then f is projectively equivalent to $x^4 - 6x^2(y^2+z^2) + (y^2+z^2)^2$.*

Proof. The proof is long, so we divide it into several steps. We use the notation of Definition-Lemma 20.2.

Step 1. In this step we determine three columns of G and show that a constant k defined below is non-zero.

For $(y_0, z_0) \in \mathbb{C}^2$ let $w_0 := (0, y_0, z_0)^t$ and $\Phi(y_0, z_0) := C_f^{-1}(w_0 w_0^t)$. The entries of $\Phi(y_0, z_0)$ are quadratic forms in (y_0, z_0). Since x^2 divides $f^{\vee}$, $\nabla f^{\vee}(w_0) = 0$, hence $\Phi(y_0, z_0) w_0 = 0$ by Definition-Lemma 20.2(3). The first row of $\Phi(y_0, z_0) w_0 = 0$ reads $\Phi_{12} y_0 + \Phi_{13} z_0 = 0$, so $\Phi_{12} = -z_0 \ell$ and $\Phi_{13} = y_0 \ell$ for a linear form $\ell = d_1 y_0 + d_2 z_0$. The second row reads $\Phi_{22} y_0 + \Phi_{23} z_0 = 0$, so $\Phi_{22} = z_0 m$ and $\Phi_{23} = -y_0 m$ for a linear form m, and then the third row reads $-y_0^2 m + \Phi_{33} z_0 = 0$, so $m = k z_0$ for a constant k. Thus there exist constants $a_1, a_2, a_3, d_1, d_2, k$ with

$$\Phi(y_0, z_0) = \begin{pmatrix} a_1 y_0^2 + 2a_2 y_0 z_0 + a_3 z_0^2 & -z_0 \ell & y_0 \ell \\ -z_0 \ell & k z_0^2 & -k y_0 z_0 \\ y_0 \ell & -k y_0 z_0 & k y_0^2 \end{pmatrix}, \qquad \ell = d_1 y_0 + d_2 z_0. \tag{20.5}$$

By the definition of G in Definition-Lemma 20.2, $G \operatorname{vec}(A) = D \operatorname{vec}(C_f^{-1}(A))$. We take $A = w_0 w_0^t$, for which

$$\operatorname{vec}(w_0 w_0^t) = (0, y_0^2, z_0^2, 0, 0, y_0 z_0)^t,$$

and take the coefficients of y_0^2, z_0^2, and $y_0 z_0$ in $G \operatorname{vec}(w_0 w_0^t) = D \operatorname{vec}(\Phi(y_0, z_0))$. We obtain the columns (22), (33), (23) of G. Order the rows and columns of G as (11), (12), (13), (22), (33), (23) and write

$$G = \begin{pmatrix} G_1 & C' \\ C'^t & G_2 \end{pmatrix}.$$

Then

$$C' = \begin{pmatrix} a_1 & a_3 & 2a_2 \\ 0 & -2d_2 & -2d_1 \\ 2d_1 & 0 & 2d_2 \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 0 & k & 0 \\ k & 0 & 0 \\ 0 & 0 & -2k \end{pmatrix}. \tag{20.6}$$

Suppose that $k = 0$. Then $G_2 = 0$. Since G is invertible, C' is invertible: if $C'v = 0$ for some $v \neq 0$, then $G(0, v)^t = 0$. Multiplying by G shows

$$G^{-1} = \begin{pmatrix} 0 & C'^{-t} \\ C'^{-1} & -C'^{-1} G_1 C'^{-t} \end{pmatrix}.$$

Thus the block of G^{-1} with rows and columns (11), (12), (13) is zero. By (20.4), also $G^{-1}_{(11),(22)} = G^{-1}_{(12),(12)} = 0$, $G^{-1}_{(11),(33)} = G^{-1}_{(13),(13)} = 0$, and $G^{-1}_{(11),(23)} = G^{-1}_{(12),(13)} = 0$. Hence the row (11) of G^{-1} is zero, a contradiction. Therefore $k \neq 0$.

Step 2. In this step we reduce to the case $\ell = 0$.

Let $\mu := d_2/k$, $\nu := -d_1/k$, let $g \in \mathrm{GL}_3$ be the matrix with

$$g^{-t} = \begin{pmatrix} 1 & 0 & 0 \\ \mu & 1 & 0 \\ \nu & 0 & 1 \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} 1 & \mu & \nu \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and let $f'(w) := f(gw)$. By Definition-Lemma 20.2(4), $C_{f'}$ is invertible and $f'^{\vee}(w) = f^{\vee}(g^{-t}w)$. The first coordinate of $g^{-t}w$ is x, so x^2 divides $f'^{\vee}$. Since $g^{-t}w_0 = w_0$, Definition-Lemma 20.2(4) also implies $C_{f'}^{-1}(w_0 w_0^t) = g^{-1}\Phi(y_0, z_0)g^{-t}$. The entries of $g^{-1}\Phi(y_0, z_0)g^{-t}$ in the rows and columns $2, 3$ are those of $\Phi(y_0, z_0)$, and

$$\begin{aligned} \left(g^{-1}\Phi g^{-t}\right)_{12} &= \Phi_{12} + \mu\Phi_{22} + \nu\Phi_{32} = -z_0(\ell - k\mu z_0 + k\nu y_0) = 0, \\ \left(g^{-1}\Phi g^{-t}\right)_{13} &= \Phi_{13} + \mu\Phi_{23} + \nu\Phi_{33} = y_0(\ell - k\mu z_0 + k\nu y_0) = 0. \end{aligned}$$

Thus (20.5) holds for f' with the constant k of f and with $\ell = 0$. Since f' is projectively equivalent to f, possibly replacing f with f', we may assume that $d_1 = d_2 = 0$.

Step 3. In this step we compute G^{-1} and conclude.

By (20.6) with $d_1 = d_2 = 0$, the only non-zero row of C' is $(a_1, a_3, 2a_2)$, and

$$G_2^{-1} = \begin{pmatrix} 0 & 1/k & 0 \\ 1/k & 0 & 0 \\ 0 & 0 & -1/(2k) \end{pmatrix}.$$

Let $\alpha := a_3/k$, $\beta := a_1/k$, and $\gamma := -a_2/k$. Then the first row of $C'G_2^{-1}$ is (α, β, γ) and the other rows are zero. Since G and G_2 are invertible, the Schur complement $G_1 - C'G_2^{-1}C'^t$ is invertible. Let $J = (j_{ab})_{1\le a,b\le 3}$ be the inverse of $G_1 - C'G_2^{-1}C'^t$, so that $j_{ab} = G^{-1}_{(1a),(1b)}$. The block inverse formula

$$G^{-1} = \begin{pmatrix} J & -JC'G_2^{-1} \\ -(C'G_2^{-1})^t J & G_2^{-1} + (C'G_2^{-1})^t JC'G_2^{-1} \end{pmatrix}$$

implies, for $a = 1, 2, 3$,

$$\begin{aligned} &G^{-1}_{(1a),(22)} = -j_{1a}\alpha, && G^{-1}_{(1a),(33)} = -j_{1a}\beta, && G^{-1}_{(1a),(23)} = -j_{1a}\gamma, \\ &G^{-1}_{(22),(22)} = j_{11}\alpha^2, && G^{-1}_{(33),(33)} = j_{11}\beta^2, && G^{-1}_{(22),(33)} = \tfrac{1}{k} + j_{11}\alpha\beta, \\ &G^{-1}_{(22),(23)} = j_{11}\alpha\gamma, && G^{-1}_{(33),(23)} = j_{11}\beta\gamma, && G^{-1}_{(23),(23)} = -\tfrac{1}{2k} + j_{11}\gamma^2. \end{aligned} \tag{20.7}$$

Substituting (20.7) into the six relations (20.4), we obtain

$$\begin{aligned} &j_{22} = -j_{11}\alpha, && j_{33} = -j_{11}\beta, && j_{23} = -j_{11}\gamma, \\ &j_{13}\alpha = j_{12}\gamma, && j_{12}\beta = j_{13}\gamma, && j_{11}(\gamma^2 - \alpha\beta) = \frac{3}{2k}. \end{aligned} \tag{20.8}$$

The last relation implies $j_{11} \neq 0$ and $\gamma^2 - \alpha\beta \neq 0$. The fourth and fifth relations form a linear system in (j_{12}, j_{13}) with determinant $\alpha\beta - \gamma^2 \neq 0$, hence $j_{12} = j_{13} = 0$. Let

$$\rho(y, z) := \alpha y^2 + 2\gamma yz + \beta z^2,$$

a non-degenerate binary quadratic form. By Definition-Lemma 20.2(2), $F_{ijkl} = G^{-1}_{(ij),(kl)}$, and (20.7) and (20.8) imply

$$F_{1111} = j_{11}, \quad F_{1122} = -j_{11}\alpha, \quad F_{1133} = -j_{11}\beta, \quad F_{1123} = -j_{11}\gamma,$$

$$F_{2222} = j_{11}\alpha^2, \quad F_{2223} = j_{11}\alpha\gamma, \quad F_{2233} = \tfrac{1}{k} + j_{11}\alpha\beta, \quad F_{2333} = j_{11}\beta\gamma, \quad F_{3333} = j_{11}\beta^2,$$

and $F_{ijkl} = 0$ when exactly one or exactly three of the indices are 1. Thus

$$f = j_{11}\left(x^4 - 6x^2\rho(y, z) + \alpha^2 y^4 + 4\alpha\gamma y^3 z + 4\beta\gamma yz^3 + \beta^2 z^4\right) + 6\left(\frac{1}{k} + j_{11}\alpha\beta\right)y^2 z^2.$$

The coefficient of y^2z^2 in $j_{11}\rho^2$ is $j_{11}(2\alpha\beta+4\gamma^2)$, and by the last relation of (20.8),

$$6\Big(\frac{1}{k}+j_{11}\alpha\beta\Big)-j_{11}(2\alpha\beta+4\gamma^2)=\frac{6}{k}-4j_{11}(\gamma^2-\alpha\beta)=0.$$

Hence

$$f=j_{11}\big(x^4-6x^2\rho(y,z)+\rho(y,z)^2\big). \tag{20.9}$$

Over $\mathbb{C}$, a linear change of the variables (y,z) takes the non-degenerate form ρ to y^2+z^2. Since $j_{11}\neq 0$, f is projectively equivalent to $x^4-6x^2(y^2+z^2)+(y^2+z^2)^2$. □

Lemma 20.4. *Let f be a quartic such that C_f is invertible. Assume that $f^\vee=cr^2$ for some $c\in\mathbb{C}^*$ and some irreducible quadratic form r. Then f is projectively equivalent to $(x^2+y^2+z^2)^2$.*

Proof. We use the notation of Definition-Lemma 20.2, and write $w=(x,y,z)^t$. Replacing r with $c^{1/2}r$, we may assume that $c=1$. Let S be the symmetric matrix with $r(w)=w^tSw$.

We first show that any quadratic form on $\mathcal{S}$ which vanishes on $\{ww^t : w\in\mathbb{C}^3\}$ is a linear combination of the six 2×2 minors $A_{22}A_{33}-A_{23}^2$, $A_{11}A_{33}-A_{13}^2$, $A_{11}A_{22}-A_{12}^2$, $A_{12}A_{33}-A_{13}A_{23}$, $A_{11}A_{23}-A_{12}A_{13}$, and $A_{12}A_{23}-A_{13}A_{22}$. Let $\mathcal{Q}$ be the space of quadratic forms on $\mathcal{S}$, so that $\dim\mathcal{Q}=21$. The restriction map $\mathcal{Q}\to\mathbb{C}[x,y,z]_4$, $Q\to Q(ww^t)$, is surjective, since any monomial of degree four is a product of two entries of ww^t, and $\dim\mathbb{C}[x,y,z]_4=15$. Hence the kernel $\mathcal{Q}_0$ of the restriction map satisfies $\dim\mathcal{Q}_0=21-15=6$. The six minors vanish on matrices of rank at most one, and they are linearly independent because no monomial appears in two of them. Hence $\mathcal{Q}_0$ is spanned by the six minors. The six minors are, up to sign, the entries of $\operatorname{adj}A$, so a linear combination of them is $\operatorname{tr}(T\operatorname{adj}A)$ for a symmetric matrix T. Since

$$\operatorname{tr}(ww^tC_f^{-1}(ww^t))=f^\vee(w)=r(w)^2=\operatorname{tr}(Sww^t)^2,$$

there exists a symmetric matrix T with

$$\operatorname{tr}\big(AC_f^{-1}(A)\big)=\operatorname{tr}(SA)^2+\operatorname{tr}(T\operatorname{adj}A)\quad\text{for any } A\in\mathcal{S}. \tag{20.10}$$

Next we show that T is invertible. For $P\in\mathrm{GL}_3$ we have $\operatorname{adj}(PAP^t)=\operatorname{adj}(P)^t\operatorname{adj}(A)\operatorname{adj}(P)$, hence $\operatorname{tr}(T\operatorname{adj}(PAP^t))=\operatorname{tr}(\operatorname{adj}(P)T\operatorname{adj}(P)^t\operatorname{adj}A)$. Since $A\to PAP^t$ is a linear automorphism and $\operatorname{adj}(P)=\det(P)P^{-1}$ runs over GL_3 as P does, the rank of the quadratic form $A\to\operatorname{tr}(T\operatorname{adj}A)$ depends only on $\operatorname{rank}T$. For $T=\operatorname{diag}(1,1,0)$ the form is $A_{33}(A_{11}+A_{22})-A_{13}^2-A_{23}^2$, of rank 4, for $T=\operatorname{diag}(1,0,0)$ it is $A_{22}A_{33}-A_{23}^2$, of rank 3, and for $T=0$ it is zero. By Definition-Lemma 20.2(1)(2), the left side of (20.10) has rank 6, while $\operatorname{tr}(SA)^2$ has rank at most one. Hence $A\to\operatorname{tr}(T\operatorname{adj}A)$ has rank at least 5, and $\operatorname{rank}T=3$.

Since T is a non-degenerate symmetric matrix and $\mathbb{C}$ is algebraically closed, there exists $P\in\mathrm{GL}_3$ with $\operatorname{adj}(P)T\operatorname{adj}(P)^t=I_3$. Let $g:=P^{-t}$ and $f'(w):=f(gw)$. By Definition-Lemma 20.2(4) and (20.10),

$$\operatorname{tr}\big(AC_{f'}^{-1}(A)\big)=\operatorname{tr}(SPAP^t)^2+\operatorname{tr}(T\operatorname{adj}(PAP^t))=\operatorname{tr}(P^tSPA)^2+\operatorname{tr}(\operatorname{adj}A),$$

and $f'^\vee(w)=f^\vee(Pw)=r(Pw)^2$, where $r(Pw)=w^tP^tSPw$ is irreducible. Since f' is projectively equivalent to f, possibly replacing f with f' and S with P^tSP, we may assume that $T=I_3$.

Let $\mathcal{L}(A):=(\operatorname{tr}(A)I_3-A)/2$, and for $A',A''\in\mathcal{S}$ let $(A'\otimes A'')(A):=\operatorname{tr}(A''A)A'$. Then $\mathcal{L}$ and $S\otimes S$ are self-adjoint for $\langle\ ,\ \rangle$, $\operatorname{tr}(A\mathcal{L}(A))=(\operatorname{tr}(A)^2-\operatorname{tr}(A^2))/2=\operatorname{tr}(\operatorname{adj}A)$, and $\operatorname{tr}(A(S\otimes S)(A))=\operatorname{tr}(SA)^2$. Since C_f^{-1} is self-adjoint (Definition-Lemma 20.2(1)) and a self-adjoint operator $\mathcal{T}$ is determined by the quadratic form $A\to\operatorname{tr}(A\mathcal{T}(A))$, (20.10) with $T=I_3$ implies

$$C_f^{-1}=\mathcal{L}+S\otimes S.$$

The operator $\mathcal{L}$ is invertible with $\mathcal{L}^{-1}(A)=\operatorname{tr}(A)I_3-2A$, since

$$\mathcal{L}(\operatorname{tr}(A)I_3-2A)=\big((3-2)\operatorname{tr}(A)I_3-\operatorname{tr}(A)I_3+2A\big)/2=A.$$

Let

$$E:=\mathcal{L}^{-1}(S),\qquad \lambda:=1+\operatorname{tr}(SE).$$

Then $(\mathcal{L}+S\otimes S)(E)=S+\operatorname{tr}(SE)S=\lambda S$, and $S\neq 0$. Since $\mathcal{L}+S\otimes S$ is invertible, $\lambda\neq 0$. This is the matrix determinant lemma $\det(\mathcal{L}+S\otimes S)=\lambda\det\mathcal{L}$ in the present case. The Sherman–Morrison formula reads

$$C_f(A)=\mathcal{L}^{-1}(A)-\frac{\operatorname{tr}(EA)}{\lambda}E, \tag{20.11}$$

which is verified by multiplying by $\mathcal{L}+S\otimes S$: since $\operatorname{tr}(S\mathcal{L}^{-1}(A))=\operatorname{tr}(EA)$ by the self-adjointness of $\mathcal{L}^{-1}$,

$$(\mathcal{L}+S\otimes S)\Big(\mathcal{L}^{-1}(A)-\frac{\operatorname{tr}(EA)}{\lambda}E\Big)=A+\operatorname{tr}(EA)S-\frac{\operatorname{tr}(EA)}{\lambda}\big(S+\operatorname{tr}(SE)S\big)=A.$$

By Definition-Lemma 20.2(3) and (20.11), with $\mathcal{L}^{-1}(ww^t)=(w^tw)I_3-2ww^t$,

$$f(w)=\operatorname{tr}\big(ww^tC_f(ww^t)\big)=-(w^tw)^2-\frac{(w^tEw)^2}{\lambda}. \tag{20.12}$$

For a symmetric matrix E', the fully symmetric coefficients of $(w^tE'w)^2$ are $(E'_{ij}E'_{kl}+E'_{ik}E'_{jl}+E'_{il}E'_{jk})/3$, so $C_{(w^tE'w)^2}(A)=(\operatorname{tr}(E'A)E'+2E'AE')/3$. Since $f\to C_f$ is linear, (20.12) implies

$$C_f(A)=-\frac{\operatorname{tr}(A)I_3+2A}{3}-\frac{\operatorname{tr}(EA)E+2EAE}{3\lambda}.$$

Equating the right side of the last identity with the right side of (20.11) and multiplying by $3\lambda/2$, we obtain

$$2\lambda\big(\operatorname{tr}(A)I_3-A\big)-\operatorname{tr}(EA)E+EAE=0\quad\text{for any } A\in\mathcal{S}. \tag{20.13}$$

For $A=\operatorname{diag}(1,0,0)$, the (i,j) entry of (20.13) with $i,j\in\{2,3\}$ reads $E_{11}E_{ij}-E_{i1}E_{1j}=2\lambda\delta_{ij}$, that is,

$$E_{11}E_{22}-E_{12}^2=2\lambda,\qquad E_{11}E_{33}-E_{13}^2=2\lambda,\qquad E_{11}E_{23}-E_{12}E_{13}=0.$$

Taking $A=\operatorname{diag}(0,1,0)$ and $A=\operatorname{diag}(0,0,1)$ in (20.13) determines the remaining cofactors of E, and we obtain $\operatorname{adj}E=2\lambda I_3$. Hence $(\det E)^2=\det(\operatorname{adj}E)=8\lambda^3\neq 0$, and $E\operatorname{adj}E=(\det E)I_3$ implies $E=\frac{\det E}{2\lambda}I_3$. By (20.12), f is a scalar multiple of $(w^tw)^2=(x^2+y^2+z^2)^2$, and $f\neq 0$ because C_f is invertible. Thus f is projectively equivalent to $(x^2+y^2+z^2)^2$. □

Lemma 20.5. *Let f be a ternary quartic with C_f invertible, and let $f^\vee$ be as in Definition-Proposition 20.1. Here the Milnor number of an isolated plane curve singularity $g(y,z)=0$ is $\dim_{\mathbb{C}}\mathbb{C}[[y,z]]/(g_y,g_z)$. Then the following hold.*

1. *$f^\vee\neq 0$, and $f^\vee$ is not a polynomial in two variables after any invertible linear change of coordinates.*
2. *If $f^\vee$ has a repeated linear factor, then f is projectively equivalent to $x^4-6x^2(y^2+z^2)+(y^2+z^2)^2$. If $f^\vee$ is a scalar multiple of the square of an irreducible quadratic form, then f is projectively equivalent to $(x^2+y^2+z^2)^2$. In particular, if $(f^\vee=0)$ is non-reduced, then f is projectively equivalent to $x^4-6x^2(y^2+z^2)+(y^2+z^2)^2$ or to $(x^2+y^2+z^2)^2$.*
3. *If $(f^\vee=0)$ is reduced, then the singular points of $(f^\vee=0)$ are isolated and the sum of their Milnor numbers is at most 7.*
4. *If $f=c_1x^2y^2+c_2xyz^2+c_3z^4$ with $c_1\neq 0$ and $(c_2,c_3)\neq(0,0)$, then $AC_f(A)=tI_3$ for some invertible symmetric matrix A and $t\neq 0$. Both quartics in (2) are projectively equivalent to quartics $c_1x^2y^2+c_2xyz^2+c_3z^4$ with $c_1\neq 0$ and $(c_2,c_3)\neq(0,0)$.*

Proof. We use the notation of Definition-Lemma 20.2, and write $w=(x,y,z)^t$.

(1) Suppose that (1) fails. Then there exists $g\in\mathrm{GL}_3$ such that $f^\vee(g^{-t}w)$ is a polynomial in y and z (when $f^\vee=0$, take $g=I_3$). By Definition-Lemma 20.2(4), possibly replacing f with $f(gw)$, we may assume that $f^\vee$ is a polynomial in y and z. By Definition-Lemma 20.2(3),

$$f^\vee(w)=\sum_{p,p'}G_{p,p'}\operatorname{vec}(ww^t)_p\operatorname{vec}(ww^t)_{p'},$$

where p, p' run over the six index pairs. The coefficients of x^4, x^3y, x^3z, xy^3, xz^3, x^2y^2, x^2z^2, x^2yz, xy^2z, and xyz^2 in $f^\vee$ are zero, that is,

$$\begin{aligned} &G_{(11),(11)} = G_{(11),(12)} = G_{(11),(13)} = G_{(22),(12)} = G_{(33),(13)} = 0,\\ &G_{(12),(12)} = -2G_{(11),(22)}, \qquad G_{(13),(13)} = -2G_{(11),(33)}, \qquad G_{(12),(13)} = -G_{(11),(23)},\\ &G_{(22),(13)} = -G_{(12),(23)}, \qquad G_{(33),(12)} = -G_{(13),(23)}. \end{aligned} \tag{20.14}$$

By (20.14), the $((11),(11))$ entry of $GG^{-1} = I_6$ reads

$$G_{(11),(22)}G^{-1}_{(22),(11)} + G_{(11),(33)}G^{-1}_{(33),(11)} + G_{(11),(23)}G^{-1}_{(23),(11)} = 1, \tag{20.15}$$

and, by (20.14), the sum of the $((12),(12))$ and $((13),(13))$ entries of $GG^{-1} = I_6$ reads

$$\begin{aligned} 2 = &-2\big(G_{(11),(22)}G^{-1}_{(12),(12)} + G_{(11),(33)}G^{-1}_{(13),(13)} + G_{(11),(23)}G^{-1}_{(12),(13)}\big)\\ &+ G_{(12),(23)}\big(G^{-1}_{(23),(12)} - G^{-1}_{(22),(13)}\big) + G_{(13),(23)}\big(G^{-1}_{(23),(13)} - G^{-1}_{(33),(12)}\big). \end{aligned}$$

By (20.4) and (20.15), the right side is -2, a contradiction. This proves (1).

(2) Assume that $f^\vee$ has a repeated linear factor ℓ_1. Choose $g \in \mathrm{GL}_3$ with $\ell_1(g^{-t}w) = x$, and let $f'(w) := f(gw)$. By Definition-Lemma 20.2(4), $C_{f'}$ is invertible and x^2 divides $f'^\vee(w) = f^\vee(g^{-t}w)$. By Lemma 20.3, f', hence f, is projectively equivalent to $x^4 - 6x^2(y^2+z^2) + (y^2+z^2)^2$. If $f^\vee$ is a scalar multiple of the square of an irreducible quadratic form, then f is projectively equivalent to $(x^2+y^2+z^2)^2$ by Lemma 20.4. If $(f^\vee = 0)$ is non-reduced, then, since $f^\vee \neq 0$ is of degree four by (1), there exists an irreducible form h of degree 1 or 2 with $h^2 \mid f^\vee$. If $\deg h = 1$, then $f^\vee$ has a repeated linear factor, and if $\deg h = 2$, then $f^\vee$ is a non-zero scalar multiple of h^2. This proves (2).

(3) Let $C := (f^\vee = 0) \subset \mathbb{P}^2$, a reduced curve of degree 4 by (1). The singular locus of a reduced curve is finite. Let $C_1, \dots, C_r$ be the irreducible components of C, let g_i be the geometric genus of C_i, and let $\nu : \tilde{C} \to C$ be the normalization. For $P \in C$ let $\delta_P := \dim_{\mathbb{C}}(\nu_*\mathcal{O}_{\tilde{C}})_P/\mathcal{O}_{C,P}$ [GBP22, Section 1], let b_P be the number of analytic branches of C at P, and let μ_P be the Milnor number of C at P, so that $\delta_P = \mu_P = b_P - 1 = 0$ at smooth points. Taking Euler characteristics in the exact sequence

$$0 \to \mathcal{O}_C \to \nu_*\mathcal{O}_{\tilde{C}} \to \bigoplus_{P\in C}(\nu_*\mathcal{O}_{\tilde{C}})_P/\mathcal{O}_{C,P} \to 0,$$

and using $\chi(\nu_*\mathcal{O}_{\tilde{C}}) = \sum_{i=1}^r(1-g_i)$ and $\chi(\mathcal{O}_C) = -2$, which follows from the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-4) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ and [Har77, Theorem III.5.1], we obtain, as in [Har77, Exercise IV.1.8], whose argument does not use the irreducibility of the curve,

$$\sum_{P\in C}\delta_P = 2 + r - \sum_{i=1}^r g_i.$$

By the Milnor formula $\mu_P = 2\delta_P - b_P + 1$ [Mil68, Theorem 10.5], [GBP22, Section 1],

$$\sum_{P\in C}\mu_P = 4 + 2r - 2\sum_{i=1}^r g_i - \sum_{P\in C}(b_P - 1) \le 4 + 2r - \sum_{P\in C}(b_P - 1). \tag{20.16}$$

Moreover b_P is at least the number of components C_i through P, with equality when the components through P are smooth at P. Since $\deg C = 4$, we have $r \le 4$. We show $\sum_P \mu_P \le 7$ case by case.

If $r = 1$, then (20.16) implies $\sum_P \mu_P \le 6$.

If $r = 2$, then $C_1 \cap C_2 \neq \emptyset$ by Bézout's theorem [Har77, Corollary I.7.8], and $b_P \ge 2$ for $P \in C_1 \cap C_2$. Thus (20.16) implies $\sum_P \mu_P \le 8 - 1 = 7$.

If $r = 3$, then C consists of a smooth conic C_1 and two distinct lines C_2, C_3. All g_i are zero and b_P is the number of components through P, so (20.16) reads $\sum_P \mu_P = 10 - \sum_P(b_P - 1)$. Suppose that $\sum_P(b_P - 1) \le 2$. Any two components intersect by Bézout's theorem. If no point is on all three components, then $C_1 \cap C_2$, $C_1 \cap C_3$, and $C_2 \cap C_3$ contain three distinct points

with $b_P = 2$, and $\sum_P (b_P - 1) \geq 3$. Hence there exists a point P_0 on all three components, with $b_{P_0} - 1 = 2$, and any other point is on at most one component. Thus $C_1 \cap C_i = \{P_0\}$ and, by Bézout's theorem, $(C_1 \cdot C_i)_{P_0} = 2$ for $i = 2, 3$. Hence C_2 and C_3 are both the tangent line of C_1 at P_0, a contradiction. Therefore $\sum_P (b_P - 1) \geq 3$ and $\sum_P \mu_P \leq 7$.

If $r = 4$, then C consists of four distinct lines, and (20.16) reads $\sum_P \mu_P = 12 - \sum_P (b_P - 1)$ with b_P the number of lines through P. If the four lines pass through a common point P_1, then after a linear change of coordinates taking P_1 to $(1 : 0 : 0)$, $f^\vee$ is a polynomial in y and z, which contradicts (1). Hence $b_P \leq 3$ for any P. If $b_P \leq 2$ for any P, then the six pairs of lines intersect in six distinct points, and $\sum_P (b_P - 1) = 6$. If $b_{P_0} = 3$ for some P_0, on the lines C_1, C_2, C_3, then C_4 intersects C_1, C_2, C_3 in three distinct points, each on exactly two lines, since two distinct lines intersect in exactly one point. Thus $\sum_P (b_P - 1) = 2 + 3 = 5$. In both cases $\sum_P \mu_P \leq 7$. This proves (3).

(4) The fully symmetric coefficients of f are $F_{1122} = c_1/6$, $F_{1233} = c_2/12$, $F_{3333} = c_3$, together with their permutations, and the other coefficients are zero. Let

$$A = \begin{pmatrix} 0 & u & 0 \\ u & 0 & 0 \\ 0 & 0 & v \end{pmatrix}, \qquad u, v \in \mathbb{C}.$$

Then $C_f(A)_{ij} = 2uF_{ij12} + vF_{ij33}$, and a direct calculation shows

$$C_f(A) = \begin{pmatrix} 0 & \frac{c_1 u}{3} + \frac{c_2 v}{12} & 0 \\ \frac{c_1 u}{3} + \frac{c_2 v}{12} & 0 & 0 \\ 0 & 0 & \frac{c_2 u}{6} + c_3 v \end{pmatrix},$$

$$AC_f(A) = \operatorname{diag}\Big(\frac{c_1 u^2}{3} + \frac{c_2 uv}{12},\ \frac{c_1 u^2}{3} + \frac{c_2 uv}{12},\ \frac{c_2 uv}{6} + c_3 v^2\Big).$$

Hence $AC_f(A) = tI_3$ if and only if

$$4c_1 u^2 - c_2 uv - 12c_3 v^2 = 0, \qquad \text{and then} \quad t = \frac{c_1 u^2}{3} + \frac{c_2 uv}{12} = \frac{u}{12}(4c_1 u + c_2 v). \tag{20.17}$$

Since $\det A = -u^2 v$, it suffices to find a solution of (20.17) with $uvt \neq 0$. Let $v = 1$. If $c_3 \neq 0$, the product of the roots of $4c_1 u^2 - c_2 u - 12c_3$ is $-3c_3/c_1 \neq 0$, so both roots are non-zero, and a root u has $t = 0$ if and only if $4c_1 u + c_2 = 0$. If both roots satisfy $4c_1 u + c_2 = 0$, they coincide, and their sum $c_2/(4c_1)$ equals $-c_2/(2c_1)$. Then $c_2 = 0$, hence $u = 0$, a contradiction. Thus one of the roots has $t \neq 0$. If $c_3 = 0$, then $c_2 \neq 0$, and $u = c_2/(4c_1)$ solves (20.17) with $t = c_2^2/(24c_1) \neq 0$.

For the second sentence of (4), the substitution $(x, y, z) \to (x + iy, x - iy, iz)$ in $(xy - z^2)^2 = x^2y^2 - 2xyz^2 + z^4$ yields $(x^2 + y^2 + z^2)^2$. Exchanging x and z in $x^4 - 6x^2(y^2 + z^2) + (y^2 + z^2)^2$ yields $z^4 - 6z^2(x^2 + y^2) + (x^2 + y^2)^2$, and the substitution $(x, y) \to ((x + y)/2, (x - y)/(2i))$ takes $x^2 + y^2$ to xy, hence takes $z^4 - 6z^2(x^2 + y^2) + (x^2 + y^2)^2$ to $x^2y^2 - 6xyz^2 + z^4$. Both $x^2y^2 - 2xyz^2 + z^4$ and $x^2y^2 - 6xyz^2 + z^4$ are of the form in (4) with $c_1 = c_3 = 1$. This proves (4). □

20.2. **Coverage of polystable quartics.** In this subsection, we prove that the PGL_3-orbit of any GIT-polystable quartic with invertible middle catalecticant intersects the harmonic slice $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$. The proof is an intersection count in $\mathbb{P}^5 \times \mathbb{P}^5$, where $\mathbb{P}^5$ is the projective space of symmetric 3×3 matrices: the variety of complete conics intersects the graph of C_f, and the intersection points over singular matrices cannot account for the intersection number.

Proposition 20.6. *Let q_0, $\mathrm{Harm}_4(q_0)$, C_f, q_A, and Δ_A be as in Definition-Proposition 20.1, and let f be a non-zero ternary quartic with invertible middle catalecticant. Assume that $[f]$ is GIT-polystable, or that $(f = 0)$ is reduced and has only singular points of types A_1 and A_2. Then the following hold.*

(1) The PGL_3*-orbit of* $[f]$ *intersects* $\{[q_0^2+b] : b \in \mathrm{Harm}_4(q_0)\}$*, that is, there exist an invertible symmetric* 3×3 *matrix* A *and* $t\in\mathbb{C}^*$ *with* $AC_f(A) = tI_3$*.*

(2) For any invertible symmetric 3×3 *matrix* M*, the* PGL_3*-orbit of* $[f]$ *intersects*

$$\{[q_M^2 + b] : b \in \mathbb{C}[x_1, x_2, x_3]_4,\ \Delta_M b = 0\}.$$

We fix notation for the space of complete conics.

Construction-Proposition 20.7. Let $\mathcal{S}$ be the space of symmetric 3×3 matrices with the symmetric bilinear form $\langle A, B\rangle := \mathrm{tr}(AB)$, and identify $\mathbb{P}(\mathcal{S})$ with $\mathbb{P}^5$. Let

$$V_2 := \{[\xi\xi^t] : \xi \in \mathbb{C}^3\backslash\{0\}\} \subset \mathbb{P}(\mathcal{S}),$$

let $\Gamma \subset \mathbb{P}(\mathcal{S})\times\mathbb{P}(\mathcal{S})$ be the closure of the graph of the rational map $[A] \to [\mathrm{adj}\, A]$, where $\mathrm{adj}\, A$ is the adjugate matrix, let $p_1, p_2 : \Gamma \to \mathbb{P}(\mathcal{S})$ be the first and second projections, and for $i = 1, 2$ let h_i be the pullback by p_i of the hyperplane class of $\mathbb{P}(\mathcal{S})$. For $g \in \mathrm{GL}_3$, let ι_g be the automorphism $([A], [B]) \to ([gAg^t], [g^{-t}Bg^{-1}])$ of $\mathbb{P}(\mathcal{S})\times\mathbb{P}(\mathcal{S})$. For a linear map $\Psi : \mathcal{S} \to \mathcal{S}$ with $\langle A, \Psi(B)\rangle = \langle\Psi(A), B\rangle$ for all A, B, let $\varphi_\Psi(w) := \mathrm{tr}(ww^t\Psi(ww^t))$ for $w \in \mathbb{C}^3$, and if Ψ is invertible, let $\Gamma_\Psi \subset \mathbb{P}(\mathcal{S})\times\mathbb{P}(\mathcal{S})$ be the graph of $[A] \to [\Psi(A)]$. Write $\nabla := (\partial_1, \partial_2, \partial_3)^t$ for the gradient. Then the following hold.

(1) p_1 is the blow-up of $\mathbb{P}(\mathcal{S})$ along the smooth surface V_2, so Γ is a smooth projective fivefold. The involution $\sigma : ([A], [B]) \to ([B], [A])$ maps Γ onto itself.
(2) $\int_\Gamma h_1^{5-i}h_2^i = 1, 2, 4, 4, 2, 1$ for $i = 0, 1, 2, 3, 4, 5$.
(3) Let $([A], [B]) \in \Gamma$. Then $AB = BA$ is a scalar matrix. If A is invertible, then $[B] = [\mathrm{adj}\, A]$. If A is singular, then $AB = 0$, B is singular, and $\mathrm{rank}\, A + \mathrm{rank}\, B \le 3$.
(4) $\iota_g(\Gamma) = \Gamma$ for any $g \in \mathrm{GL}_3$.
(5) $\nabla\varphi_\Psi(w) = 4\Psi(ww^t)w$ for any $w \in \mathbb{C}^3$.
(6) Let $f = \sum F_{ijkl}x_ix_jx_kx_l$ be a ternary quartic with fully symmetric coefficients, and let $f^\vee$ be as in Definition-Proposition 20.1. Then $\langle A, C_f(B)\rangle = \langle C_f(A), B\rangle$ for all A, B, $\varphi_{C_f} = f$,

$$\partial_k\partial_l f = 12\sum_{i,j} F_{ijkl}x_ix_j, \qquad \Delta_A f = 12x^tC_f(A)x \quad \text{for } A \in \mathcal{S},$$

and $C_{f\circ g}(A) = g^tC_f(gAg^t)g$ for $g \in \mathrm{GL}_3$ and $A \in \mathcal{S}$. If C_f is invertible, then $\langle A, C_f^{-1}(B)\rangle = \langle C_f^{-1}(A), B\rangle$ for all A, B and $\varphi_{C_f^{-1}} = f^\vee$.

Proof. (1) Up to sign, the entries of $\mathrm{adj}\, A$ are the 2×2 minors of A, and for symmetric A there are six of them. Let $\mathcal{I}$ be the ideal sheaf generated by the six minors. By [Har77, Example II.7.17.3], the rational map $[A] \to [\mathrm{adj}\, A]$ extends to a morphism $\mathrm{Bl}_{\mathcal{I}}\mathbb{P}(\mathcal{S}) \to \mathbb{P}^5$. The surjection $\mathcal{O}_{\mathbb{P}(\mathcal{S})}[T_0, \dots, T_5] \to \bigoplus_{d\ge 0}\mathcal{I}^d(2d)$ sending the T_i to the six minors embeds $\mathrm{Bl}_{\mathcal{I}}\mathbb{P}(\mathcal{S}) = \mathrm{Proj}\bigoplus_{d\ge 0}\mathcal{I}^d$ as a closed subscheme Γ' of $\mathbb{P}(\mathcal{S})\times\mathbb{P}^5$. The subscheme Γ' is integral and agrees with the graph of $[A] \to [\mathrm{adj}\, A]$ over the open set where some minor is non-zero, so $\Gamma' = \Gamma$. Thus p_1 is the blow-up of $\mathbb{P}(\mathcal{S})$ along $\mathcal{I}$. We show that $\mathcal{I}$ is the ideal sheaf of V_2. A matrix of rank at least two has a non-zero 2×2 minor, so $\mathcal{I} = \mathcal{O}_{\mathbb{P}(\mathcal{S})}$ outside V_2. Any non-zero $\xi\xi^t$ has a non-zero diagonal entry ξ_i^2, so the affine charts $\{A_{ii} \ne 0\}$, $i = 1, 2, 3$, cover V_2. On the chart $\{A_{11} \ne 0\}$, where we set $A_{11} = 1$, the three minors

$$A_{22} - A_{12}^2, \qquad A_{23} - A_{12}A_{13}, \qquad A_{33} - A_{13}^2$$

generate the ideal of the graph of $(A_{12}, A_{13}) \to (A_{12}^2, A_{12}A_{13}, A_{13}^2)$. The graph is a smooth surface whose points are the matrices $\xi\xi^t$ with $\xi = (1, A_{12}, A_{13})^t$, so the graph is $V_2 \cap \{A_{11} \ne 0\}$. All 2×2 minors vanish on V_2. Thus, on $\{A_{11} \ne 0\}$, $\mathcal{I}$ is contained in the ideal of V_2 and contains the three minors above, and $\mathcal{I}$ is the ideal of V_2. The other two charts are obtained by permuting the coordinates. Hence p_1 is the blow-up of $\mathbb{P}^5$ along the smooth surface V_2, and Γ is smooth by [Har77, Theorem II.8.24]. For invertible A we have $\mathrm{adj}(\mathrm{adj}\, A) = (\det A)A$, so σ maps the graph

of $[A] \to [\operatorname{adj} A]$ over the invertible matrices into Γ. Since the graph of $[A] \to [\operatorname{adj} A]$ over the invertible matrices is dense in Γ and $\sigma^2 = \mathrm{id}_\Gamma$, we have $\sigma(\Gamma) = \Gamma$.

(2) Let E be the exceptional divisor of p_1. The morphism p_2 is defined by the six minors, which are sections of $p_1^*\mathcal{O}(2)$ generating $p_1^{-1}\mathcal{I} \cdot \mathcal{O}_\Gamma = \mathcal{O}_\Gamma(-E)$. Hence $h_2 = 2h_1 - E$. Since p_1 is birational, $\int_\Gamma h_1^5 = 1$. Since $\dim V_2 = 2$, a general plane $\Lambda \subset \mathbb{P}^5$ is disjoint from V_2, so $p_1^{-1}(\Lambda)$, which represents h_1^3, is disjoint from E. Hence $h_1^3 \cdot E = 0$, and

$$\int_\Gamma h_1^4 h_2 = \int_\Gamma h_1^4(2h_1 - E) = 2, \qquad \int_\Gamma h_1^3 h_2^2 = \int_\Gamma h_1^3(2h_1 - E)^2 = 4.$$

By (1), $\sigma|_\Gamma$ is an automorphism of Γ with $(\sigma|_\Gamma)^* h_1 = h_2$, so $\int_\Gamma h_1^{5-i} h_2^i = \int_\Gamma h_1^i h_2^{5-i}$, and (2) follows. The six numbers in (2) are the characteristic numbers of complete conics, see [Ful98, Section 10.4].

(3) The condition that AB is a scalar matrix is given by the bihomogeneous equations $(AB)_{ij} = 0$ for $i \neq j$ and $(AB)_{11} = (AB)_{22} = (AB)_{33}$, so it defines a closed subset of $\mathbb{P}(\mathcal{S}) \times \mathbb{P}(\mathcal{S})$. It holds on the graph of $[A] \to [\operatorname{adj} A]$, since $A \operatorname{adj} A = (\det A) I_3$, so it holds on Γ. Moreover, $BA = (AB)^t = AB$. If A is invertible, then $[A] \notin V_2$, and by (1) the fiber $p_1^{-1}([A])$ is the single point $([A], [\operatorname{adj} A])$. If A is singular and $AB = sI_3$, then $s^3 = \det A \det B = 0$, so $AB = 0$. Then the image of B is contained in the kernel of A, and $\operatorname{rank} B \leq 3 - \operatorname{rank} A \leq 2$ as $A \neq 0$.

(4) For invertible A we have $\operatorname{adj}(gAg^t) = (\det g)^2 g^{-t}(\operatorname{adj} A) g^{-1}$. Thus ι_g maps the graph of $[A] \to [\operatorname{adj} A]$ over the invertible matrices onto itself, and hence maps Γ onto Γ.

(5) For $e \in \mathbb{C}^3$, since $\langle A, \Psi(B)\rangle = \langle \Psi(A), B\rangle$ and $\Psi(ww^t)$ is symmetric, we have

$$\frac{d}{d\epsilon}\Big|_{\epsilon=0} \varphi_\Psi(w + \epsilon e) = 2 \operatorname{tr}\left((ew^t + we^t)\Psi(ww^t)\right) = 4e^t \Psi(ww^t) w.$$

(6) We have $\langle A, C_f(B)\rangle = \sum_{i,j,k,l} F_{ijkl} A_{ij} B_{kl}$, which is symmetric in A and B because $F_{ijkl} = F_{klij}$, and $\varphi_{C_f}(w) = \sum F_{ijkl} w_i w_j w_k w_l = f(w)$. Differentiating f twice and using the symmetry of (F_{ijkl}) implies $\partial_k \partial_l f = 12 \sum_{i,j} F_{ijkl} x_i x_j$, hence $\Delta_A f = \sum_{k,l} A_{kl} \partial_k \partial_l f = 12 x^t C_f(A) x$. In particular $\Delta_A f = \operatorname{tr}(A \operatorname{Hess}(f))$, where $\operatorname{Hess}(f) := (\partial_k \partial_l f)_{k,l}$. For $g \in \mathrm{GL}_3$, the chain rule implies $\operatorname{Hess}(f \circ g)(x) = g^t \operatorname{Hess}(f)(gx) g$, so

$$\Delta_A (f \circ g)(x) = \operatorname{tr}\left(gAg^t \operatorname{Hess}(f)(gx)\right) = (\Delta_{gAg^t} f)(gx) = 12 x^t g^t C_f(gAg^t) g x.$$

Since a symmetric matrix M is determined by the quadratic form $x^t M x$, $C_{f \circ g}(A) = g^t C_f(gAg^t) g$. Finally, if C_f is invertible, then

$$\langle A, C_f^{-1}(B)\rangle = \langle C_f(C_f^{-1} A), C_f^{-1} B\rangle = \langle C_f^{-1}(A), B\rangle,$$

and $\varphi_{C_f^{-1}} = f^\vee$ is the definition of $f^\vee$. □

The next two lemmas compute the local intersection multiplicities of Γ with the graph of a linear map at points $([A], [B])$ with A singular.

Lemma 20.8. *Notation as in Construction-Proposition 20.7. Let $\Psi : \mathcal{S} \to \mathcal{S}$ be an invertible linear map with $\langle A, \Psi(B)\rangle = \langle \Psi(A), B\rangle$ for all A, B, and let $\xi_0 \in \mathbb{C}^3 \backslash \{0\}$. Assume that $\Psi(\xi_0 \xi_0^t)\xi_0 = 0$, that $\operatorname{rank} \Psi(\xi_0 \xi_0^t) = 2$, and that $[\xi_0]$ is an isolated singular point of $(\varphi_\Psi = 0)$. Let μ be the Milnor number of $(\varphi_\Psi = 0)$ at $[\xi_0]$ in the sense of Lemma 20.5. Then $p := ([\xi_0 \xi_0^t], [\Psi(\xi_0 \xi_0^t)])$ is an isolated point of $\Gamma \cap \Gamma_\Psi$, and the length of $\mathcal{O}_{\Gamma \cap \Gamma_\Psi, p}$ is μ.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case $\xi_0 = e_1$.

Let $g \in \mathrm{GL}_3$ with $g\xi_0 = e_1$, and let $\Psi'(A) := g^{-t} \Psi(g^{-1} A g^{-t}) g^{-1}$. Then

$$\langle A, \Psi'(B)\rangle = \langle g^{-1} A g^{-t}, \Psi(g^{-1} B g^{-t})\rangle,$$

so $\langle A, \Psi'(B)\rangle = \langle \Psi'(A), B\rangle$, and $\varphi_{\Psi'}(w) = \varphi_\Psi(g^{-1} w)$. By Construction-Proposition 20.7(4), $\iota_g(\Gamma) = \Gamma$. Moreover, $\iota_g(\Gamma_\Psi) = \Gamma_{\Psi'}$ and $\iota_g(p) = ([e_1 e_1^t], [\Psi'(e_1 e_1^t)])$. The matrix $\Psi'(e_1 e_1^t) = g^{-t} \Psi(\xi_0 \xi_0^t) g^{-1}$ has rank two and annihilates $e_1 = g\xi_0$, and $(\varphi_{\Psi'} = 0)$ is the image of $(\varphi_\Psi = 0)$

under $[w] \to [gw]$, so $[e_1]$ is an isolated singular point of $(\varphi_{\Psi'} = 0)$ with Milnor number μ. Possibly replacing Ψ with Ψ' and ξ_0 with e_1, we may assume that $\xi_0 = e_1$.

Step 2. In this step we write an analytic chart of Γ at p.

Let $\mathcal{S}_2$ be the space of symmetric 2×2 matrices, and for $X \in \mathcal{S}$ let $X_{\mathrm{low}} \in \mathcal{S}_2$ be the lower right 2×2 block. Since $B_0 := \Psi(e_1 e_1^t)$ is symmetric, has rank two, and satisfies $B_0 e_1 = 0$, we have $B_0 = \operatorname{diag}(0, D_0)$ with $D_0 \in \mathcal{S}_2$ invertible. On $\mathcal{S}_2$ the adjugate is a linear involution, and we let $Z_0 := \operatorname{adj} D_0$, so $\operatorname{adj} Z_0 = D_0$ and $\det Z_0 \neq 0$. Fix a linear form l on $\mathcal{S}_2$ with $l(Z_0) = 1$, and let $U := \{Z \in \mathcal{S}_2 : l(Z) = 1\}$. For $u = (u_1, u_2)^t \in \mathbb{C}^2$, $r \in \mathbb{C}$, and $Z \in U$, let $\xi := (1, u_1, u_2)^t$, $W := \operatorname{diag}(0, Z)$, and

$$A := \xi\xi^t + rW = \begin{pmatrix} 1 & u^t \\ u & uu^t + rZ \end{pmatrix}, \qquad B := \begin{pmatrix} r \det Z + u^t (\operatorname{adj} Z) u & -u^t \operatorname{adj} Z \\ -(\operatorname{adj} Z) u & \operatorname{adj} Z \end{pmatrix}.$$

Writing $A = g_u \operatorname{diag}(1, rZ) g_u^t$ with $g_u := \left(\begin{smallmatrix} 1 & 0 \\ u & I_2 \end{smallmatrix}\right)$, a direct calculation shows that $\operatorname{adj} A = rB$ and $B\xi = r(\det Z) e_1$. Let $\kappa(u, r, Z) := ([A], [B])$. If $r \det Z \neq 0$, then A is invertible and $\kappa(u, r, Z) = ([A], [\operatorname{adj} A]) \in \Gamma$. Since Γ is closed, κ maps a neighborhood of $(0, 0, Z_0)$ into Γ, and $\kappa(0, 0, Z_0) = ([e_1 e_1^t], [B_0]) = p$. The differential of κ at $(0, 0, Z_0)$ is injective: the derivatives in u and r are detected by the entries A_{12}, A_{13} and by the block $A_{\mathrm{low}} = uu^t + rZ$ of $[A]$ in the chart $A_{11} = 1$, while the derivatives in Z have zero component in $[A]$ and are detected by $B_{\mathrm{low}} = \operatorname{adj} Z$, as $Z \to \operatorname{adj} Z$ is a linear isomorphism and $T_{Z_0} U = \ker l$ does not contain Z_0. Since Γ is smooth of dimension five (Construction-Proposition 20.7(1)), κ is a local analytic isomorphism from a neighborhood of $(0, 0, Z_0)$ onto a neighborhood of p in Γ by the inverse function theorem.

Step 3. In this step we show that the analytic local ring of $\Gamma \cap \Gamma_\Psi$ at p is $\mathbb{C}\{u_1, u_2\}/I$, where

$$I = (\partial_1 \bar\varphi + \bar\varphi a_1,\ \partial_2 \bar\varphi + \bar\varphi a_2), \qquad \bar\varphi(u) := \varphi_\Psi(1, u_1, u_2),$$

for some convergent power series $a_1, a_2 \in \mathbb{C}\{u_1, u_2\}$.

For $X \in \mathcal{S}$ let $\tilde l(X) := l(\operatorname{adj} X_{\mathrm{low}})$. Since the adjugate on $\mathcal{S}_2$ is linear, $\tilde l$ is a linear form on $\mathcal{S}$, and $\tilde l(B) = l(Z) = 1$. Hence, near p, $[\Psi(A)] = [B]$ if and only if $\Psi(A) = \tilde l(\Psi(A)) B$. Introducing an additional coordinate λ near 1, the analytic local ring of $\Gamma \cap \Gamma_\Psi$ at p is the quotient of the ring of convergent power series in u, r, $\lambda - 1$, and affine coordinates of U centered at Z_0 by the entries of $\Psi(A) - \lambda B$, because applying $\tilde l$ to $\Psi(A) - \lambda B$ yields $\tilde l(\Psi(A)) - \lambda$. Let D_ξ be the lower right block of $\Psi(\xi\xi^t)$, a function of u with $D_\xi = D_0$ at $u = 0$. The lower right block of $\Psi(A) = \lambda B$ is

$$D_\xi + r\Psi(W)_{\mathrm{low}} = \lambda \operatorname{adj} Z.$$

The differential of $(Z, \lambda) \to \lambda \operatorname{adj} Z$ at $(Z_0, 1)$ is the map

$$\ker l \times \mathbb{C} \to \mathcal{S}_2, \qquad (\dot Z, \dot\lambda) \to \operatorname{adj} \dot Z + \dot\lambda D_0,$$

which is injective, since $\operatorname{adj} \dot Z = -\dot\lambda \operatorname{adj} Z_0$ implies $\dot Z = -\dot\lambda Z_0$ and then $l(\dot Z) = -\dot\lambda = 0$. Hence it is an isomorphism, since $\dim(\ker l \times \mathbb{C}) = 3 = \dim \mathcal{S}_2$. By the implicit function theorem, the lower block equation determines Z and λ as analytic functions of (u, r), with $Z(0, 0) = Z_0$ and $\lambda(0, 0) = 1$. Substituting them, $Q := \Psi(A) - \lambda B$ has $Q_{\mathrm{low}} = 0$, and the analytic local ring is $\mathbb{C}\{u, r\}$ modulo the entries of Qe_1. Write

$$Q = \begin{pmatrix} q_{11} & q^t \\ q & 0 \end{pmatrix}.$$

Then $Q\xi = (q_{11} + q^t u, q)^t$ and $\xi^t Q \xi = q_{11} + 2q^t u$, so the ideal of $\mathbb{C}\{u, r\}$ generated by the entries of Qe_1 equals the ideal generated by $\xi^t Q\xi$, $(Q\xi)_2$, $(Q\xi)_3$.

We compute $\xi^t Q \xi$. We have $\xi^t \Psi(\xi\xi^t)\xi = \bar\varphi(u)$, $\xi^t B \xi = r \det Z$, and

$$\xi^t \Psi(W) \xi = \langle \xi\xi^t, \Psi(W) \rangle = \langle \Psi(\xi\xi^t), W \rangle = \operatorname{tr}(D_\xi Z) = 2\lambda \det Z - r \operatorname{tr}(W \Psi(W)),$$

where the last equality uses the lower block equation and $\operatorname{tr}(Z \operatorname{adj} Z) = 2 \det Z$. Hence

$$\xi^t Q \xi = \bar\varphi(u) + r c(u, r), \qquad c(u, r) := \lambda \det Z - r \operatorname{tr}(W \Psi(W)),$$

with $c(0,0) = \det Z_0 \neq 0$. By the implicit function theorem, $(\xi^t Q\xi) = (r - \rho(u))$ in $\mathbb{C}\{u, r\}$ for a convergent power series ρ with $\rho(u)c(u, \rho(u)) = -\bar{\varphi}(u)$. Since $c(u, \rho(u))$ is a unit, $\rho = \bar{\varphi}\theta$ for some $\theta \in \mathbb{C}\{u\}$. Since $B\xi = r(\det Z)e_1$, for $i = 1, 2$ we have

$$(Q\xi)_{i+1} = (\Psi(\xi\xi^t)\xi)_{i+1} + r(\Psi(W)\xi)_{i+1}.$$

By Construction-Proposition 20.7(5), $\partial_i\bar{\varphi} = 4(\Psi(\xi\xi^t)\xi)_{i+1}$. Substituting $r = \rho(u) = \bar{\varphi}\theta$ and letting $a_i := 4\theta(\Psi(W)\xi)_{i+1}|_{r=\rho(u)}$, we obtain $4(Q\xi)_{i+1} = \partial_i\bar{\varphi} + \bar{\varphi}a_i$, which proves the claim of this step.

Step 4. In this step we show that $\ell(\mathbb{C}\{u\}/I) = \mu$, which proves the lemma.

Let $R := \mathbb{C}\{u_1, u_2\}$, let $\mathfrak{m}$ be the maximal ideal of R, and let $J := (\partial_1\bar{\varphi}, \partial_2\bar{\varphi})$. Since $[e_1]$ is an isolated singular point of $(\varphi_\Psi = 0)$, we have $\bar{\varphi}(0) = 0$, J is $\mathfrak{m}$-primary, and $\mu = \ell(R/J)$. Since R/J is Artinian, $\ell(R/J) = \ell(\mathbb{C}[[u_1, u_2]]/J\mathbb{C}[[u_1, u_2]])$. By [HS06, Corollary 7.1.4], $\bar{\varphi} \in \overline{\mathfrak{m}J}$. By [HS06, Theorem 7.1.7], there exist a neighborhood N of 0 and $C > 0$ such that

$$|\bar{\varphi}(u)| \leq C\|u\| \max_j |\partial_j\bar{\varphi}(u)| \qquad \text{for } u \in N.$$

Let $C' := C \sup_N \max_i |a_i|$, and shrink N so that $C'\|u\| \leq \frac{1}{2}$ on N. Then for $u \in N$ and $i = 1, 2$,

$$|\partial_i\bar{\varphi}| \leq |\partial_i\bar{\varphi} + \bar{\varphi}a_i| + \tfrac{1}{2}\max_j |\partial_j\bar{\varphi}|, \qquad |\partial_i\bar{\varphi} + \bar{\varphi}a_i| \leq \tfrac{3}{2}\max_j |\partial_j\bar{\varphi}|.$$

Thus

$$\max_j |\partial_j\bar{\varphi}| \leq 2\max_j |\partial_j\bar{\varphi} + \bar{\varphi}a_j| \qquad \text{on } N,$$

and by [HS06, Theorem 7.1.7] again, $\overline{I} = \overline{J}$. Since the radical of an ideal contains the integral closure of the ideal, $\sqrt{I} = \sqrt{J} = \mathfrak{m}$, so I is $\mathfrak{m}$-primary and p is an isolated point of $\Gamma \cap \Gamma_\Psi$. The ideals I and J are $\mathfrak{m}$-primary ideals generated by two elements in the two-dimensional regular local ring R, so $\ell(R/I) = e(I)$ and $\ell(R/J) = e(J)$ by [HS06, Proposition 11.1.10(2)], see also [Mat89, Theorem 17.11]. Since $\overline{I} = \overline{J}$, we have $e(I) = e(J)$ by [HS06, Proposition 11.2.1]. Hence $\ell(R/I) = \ell(R/J) = \mu$. Since the algebraic and the analytic local rings of $\Gamma \cap \Gamma_\Psi$ at p have isomorphic completions, the length of $\mathcal{O}_{\Gamma\cap\Gamma_\Psi,p}$ is μ. □

Lemma 20.9. *Notation as in Construction-Proposition 20.7. Let $f = \sum F_{ijkl}x_ix_jx_kx_l$ be a ternary quartic with fully symmetric coefficients and C_f invertible, and let $\xi_0 \in \mathbb{C}^3\backslash\{0\}$. Assume that $C_f(\xi_0\xi_0^t)\xi_0 = 0$, that $\operatorname{rank} C_f(\xi_0\xi_0^t) = 1$, and that $[\xi_0]$ is a singular point of type A_2 of $(f = 0)$. Then $p := ([\xi_0\xi_0^t], [C_f(\xi_0\xi_0^t)])$ is an isolated point of $\Gamma \cap \Gamma_{C_f}$, and the length of $\mathcal{O}_{\Gamma\cap\Gamma_{C_f},p}$ is 3.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case $\xi_0 = e_1$ and $C_f(e_1e_1^t) = e_3e_3^t$.

Write $C_f(\xi_0\xi_0^t) = \eta\eta^t$ with $\eta \in \mathbb{C}^3\backslash\{0\}$. Since $\eta(\eta^t\xi_0) = 0$, we have $\eta^t\xi_0 = 0$. Thus there exists $g \in \mathrm{GL}_3$ whose third row is η^t and with $g\xi_0 = e_1$, and then $g^{-t}\eta = e_3$. Let $f' := f \circ g^{-1}$. By Construction-Proposition 20.7(6), $C_{f'}(A) = g^{-t}C_f(g^{-1}Ag^{-t})g^{-1}$, so $C_{f'}$ is invertible, $C_{f'}(e_1e_1^t) = e_3e_3^t$, and ι_g maps Γ_{C_f} onto $\Gamma_{C_{f'}}$ and p to $([e_1e_1^t], [e_3e_3^t])$. By Construction-Proposition 20.7(4), $\iota_g(\Gamma) = \Gamma$. The curve $(f' = 0)$ is the image of $(f = 0)$ under $[x] \to [gx]$, so $[e_1]$ is a singular point of type A_2 of $(f' = 0)$. Possibly replacing f with f' and ξ_0 with e_1, we may assume that $\xi_0 = e_1$ and $C_f(e_1e_1^t) = e_3e_3^t$.

Since $C_f(e_1e_1^t)_{ij} = F_{11ij}$, we obtain $F_{1111} = F_{1112} = F_{1113} = F_{1122} = F_{1123} = 0$ and $F_{1133} = 1$. Thus

$$f(1, x_2, x_3) = 6x_3^2 + 4\big(F_{1222}x_2^3 + 3F_{1223}x_2^2x_3 + 3F_{1233}x_2x_3^2 + F_{1333}x_3^3\big) + f(0, x_2, x_3).$$

Since the quadratic part is $6x_3^2$, solving $\partial_3 f(1, x_2, x_3) = 0$ for $x_3 = O(x_2^2)$ and substituting shows that the germ is of type A_2 if and only if $F_{1222} \neq 0$. Hence $F_{1222} \neq 0$.

Step 2. In this step we write an analytic chart of Γ at p.

For $(u,v,w,r,s)\in\mathbb{C}^5$, let

$$A:=\begin{pmatrix}1 & u & v\\ u & u^2+r & uv+rw\\ v & uv+rw & v^2+rw^2+rs\end{pmatrix},$$

$$B:=\begin{pmatrix}rs+su^2+(uw-v)^2 & -su-w(uw-v) & uw-v\\ -su-w(uw-v) & s+w^2 & -w\\ uw-v & -w & 1\end{pmatrix},$$

and $\kappa(u,v,w,r,s):=([A],[B])$. Let

$$g_{u,v,w}:=\begin{pmatrix}1 & 0 & 0\\ u & 1 & 0\\ v & w & 1\end{pmatrix}.$$

Then

$$A=g_{u,v,w}\operatorname{diag}(1,r,rs)g_{u,v,w}^t,\qquad B=g_{u,v,w}^{-t}\operatorname{diag}(rs,s,1)g_{u,v,w}^{-1},$$

so $\operatorname{adj}A=rB$. If $rs\neq 0$, then A is invertible and $\kappa(u,v,w,r,s)=([A],[\operatorname{adj}A])\in\Gamma$. Since Γ is closed, κ maps a neighborhood of 0 into Γ, and $\kappa(0)=([e_1e_1^t],[e_3e_3^t])=p$. At 0, the linear parts of the coordinates A_{12},A_{13},A_{22} of $[A]$ in the chart $A_{11}=1$ are u,v,r, and the linear parts of the coordinates B_{22},B_{23} of $[B]$ in the chart $B_{33}=1$ are $s,-w$. Thus the differential of κ at 0 has rank five. Since Γ is smooth of dimension five (Construction-Proposition 20.7(1)), κ is a local analytic isomorphism from a neighborhood of 0 onto a neighborhood of p in Γ by the inverse function theorem.

Step 3. In this step we compute the length.

Let $c:=C_f(A)_{33}$, so $c=1+O(u,v,r)$. In particular c is a unit near p. In the chart $B_{33}=1$, the condition $[C_f(A)]=[B]$ is $C_f(A)=cB$. Thus the analytic local ring of $\Gamma\cap\Gamma_{C_f}$ at p is $\mathbb{C}\{u,v,w,r,s\}/(E_{11},E_{12},E_{13},E_{22},E_{23})$, where $E_{ij}:=C_f(A)_{ij}-cB_{ij}$, and the entry $E_{33}:=C_f(A)_{33}-cB_{33}$ vanishes identically, since $B_{33}=1$ and $c=C_f(A)_{33}$. Using the vanishing of F_{11ij} for $(i,j)\neq(3,3)$, we have

$$C_f(A)_{11}=v^2+rw^2+rs,$$

$$C_f(A)_{12}=F_{1222}(u^2+r)+2F_{1223}(uv+rw)+F_{1233}(v^2+rw^2+rs),$$

$$C_f(A)_{13}=2v+F_{1223}(u^2+r)+2F_{1233}(uv+rw)+F_{1333}(v^2+rw^2+rs),$$

and the linear parts of $E_{12},E_{13},E_{22},E_{23}$ are

$$F_{1222}r,\qquad F_{1223}r+3v,\qquad F_{2222}r+2F_{1223}v+2F_{1222}u-s,\qquad F_{2223}r+2F_{1233}v+2F_{1223}u+w.$$

The Jacobian determinant of the four linear forms above with respect to (r,v,s,w) is $-3F_{1222}\neq 0$. By the implicit function theorem, $E_{12}=E_{13}=E_{22}=E_{23}=0$ determines $r=r(u)$, $v=v(u)$, $s=s(u)$, $w=w(u)$ as analytic functions of u, and the analytic local ring is $\mathbb{C}\{u\}/(E_{11}(u))$, where $E_{11}(u)$ is E_{11} evaluated on $r(u),v(u),s(u),w(u)$. We solve $E_{12}=E_{13}=E_{22}=E_{23}=0$ to second order in u. Since $c=1+O(u)$, $B_{12}=-su-w(uw-v)$, and $B_{13}=uw-v$, we obtain

$$\begin{aligned}
&\text{linear parts of } E_{12},E_{13}: && r=O(u^2),\ v=O(u^2),\\
&E_{22}=0: && s=2F_{1222}u+O(u^2),\\
&E_{23}=0: && w=-2F_{1223}u+O(u^2),\\
&E_{12}=0:\ F_{1222}(u^2+r)+su=O(u^3), && r=-3u^2+O(u^3),\\
&E_{13}=0:\ 3v+F_{1223}(u^2+r)-uw=O(u^3), && v=O(u^3),
\end{aligned}$$

where in the last line we substitute $r=-3u^2+O(u^3)$ and $w=-2F_{1223}u+O(u^2)$. Hence $rs=-6F_{1222}u^3+O(u^4)$, $su^2=2F_{1222}u^3+O(u^4)$, and v^2, rw^2, $(uw-v)^2$ are $O(u^4)$, so

$$E_{11}(u)=(v^2+rw^2+rs)-c\big(rs+su^2+(uw-v)^2\big)=-2F_{1222}u^3+O(u^4).$$

Since $F_{1222} \neq 0$, the analytic local ring is $\mathbb{C}\{u\}/(u^3)$. Thus p is an isolated point of $\Gamma \cap \Gamma_{C_f}$, and, since the algebraic and the analytic local rings have isomorphic completions, the length of $\mathcal{O}_{\Gamma\cap\Gamma_{C_f},p}$ is 3. □

Proof of Proposition 20.6. We use the notation of Construction-Proposition 20.7. Write $f = \sum F_{ijkl}x_ix_jx_kx_l$ with fully symmetric coefficients. By Construction-Proposition 20.7(6),

$$\partial_k\partial_l f = 12\sum_{i,j} F_{ijkl}x_ix_j, \qquad \Delta_A f = 12x^tC_f(A)x \quad \text{for any } A \in \mathcal{S}.$$

Since $A \to \Delta_A$ and $B \to x^tBx$ are linear isomorphisms from $\mathcal{S}$ onto $\mathbb{C}[\partial_1,\partial_2,\partial_3]_2$ and $\mathbb{C}[x_1,x_2,x_3]_2$, the middle catalecticant $\mathrm{Cat}_2(f)$ (Definition 2.27) is invertible if and only if C_f is invertible, see Definition-Proposition 20.1(6). Thus C_f is invertible. By Definition-Proposition 20.1(2), the condition on the PGL_3-orbit of $[f]$ in (1) is equivalent to the existence of A and t in (1). In particular, the existence of A and t in (1) depends only on the PGL_3-orbit of $[f]$. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1) when $(f = 0)$ is reduced with only singular points of types A_1 and A_2, and $(f^\vee = 0)$ is non-reduced.

By Lemma 20.5(2), f is projectively equivalent to one of the quartics in Lemma 20.5(2), and by Lemma 20.5(4), each of them is projectively equivalent to a quartic f' for which there exist an invertible symmetric A and $t \neq 0$ with $AC_{f'}(A) = tI_3$. Thus (1) holds for f.

Step 2. In this step we show $[\Gamma]\cdot[\Gamma_{C_f}] = 14$ in $\mathbb{P}(\mathcal{S}) \times \mathbb{P}(\mathcal{S})$.

We also write h_1, h_2 for the pullbacks of the hyperplane class to $\mathbb{P}(\mathcal{S}) \times \mathbb{P}(\mathcal{S})$ by the first and second projections, so that their restrictions to Γ are the classes h_1, h_2 of Construction-Proposition 20.7. The Chow ring of $\mathbb{P}^5 \times \mathbb{P}^5$ is $\mathbb{Z}[h_1,h_2]/(h_1^6, h_2^6)$. Write the class of the diagonal as $\sum_i a_i h_1^i h_2^{5-i}$. Intersecting with $h_1^{5-j}h_2^j$ and using $\int_{\mathbb{P}^5} c_1(\mathcal{O}_{\mathbb{P}^5}(1))^5 = 1$ implies $a_j = 1$ for any j. The automorphism $\mathrm{id}_{\mathbb{P}^5} \times [C_f]$ maps the diagonal onto Γ_{C_f} and fixes h_1 and h_2, so $[\Gamma_{C_f}] = \sum_{i=0}^5 h_1^i h_2^{5-i}$. By Construction-Proposition 20.7(2),

$$[\Gamma]\cdot[\Gamma_{C_f}] = \sum_{i=0}^5 \int_\Gamma h_1^i h_2^{5-i} = 1+2+4+4+2+1 = 14.$$

Step 3. In this step, assuming that $(f = 0)$ is reduced with only singular points of types A_1 and A_2 and that $(f^\vee = 0)$ is reduced, we describe the points $([A],[B]) \in \Gamma \cap \Gamma_{C_f}$ with A singular.

Let $([A],[B]) \in \Gamma \cap \Gamma_{C_f}$ with A singular. Then $[B] = [C_f(A)]$, $AB = BA = 0$, and $(\operatorname{rank} A, \operatorname{rank} B) \in \{(1,1),(1,2),(2,1)\}$ by Construction-Proposition 20.7(3). Let P_{11}, P_{12}, and P_{21} be the corresponding sets of points.

Assume that $\operatorname{rank} A = 1$, and write $A = \xi\xi^t$. Then $\xi\xi^tC_f(\xi\xi^t) = 0$, so $C_f(\xi\xi^t)\xi = 0$, and $\nabla f(\xi) = 4C_f(\xi\xi^t)\xi = 0$ by Construction-Proposition 20.7(5) and (6). Thus $[\xi]$ is a singular point of $(f = 0)$, and the point $([\xi\xi^t],[C_f(\xi\xi^t)])$ is determined by $[\xi]$. Assume that $\operatorname{rank} B = 1$, and write $B = ww^t$. Then $[A] = [C_f^{-1}(ww^t)]$ and $ww^tC_f^{-1}(ww^t) = 0$, so $\nabla f^\vee(w) = 4C_f^{-1}(ww^t)w = 0$ by Construction-Proposition 20.7(5) and (6). Thus $[w]$ is a singular point of $(f^\vee = 0)$, and the point $([C_f^{-1}(ww^t)],[ww^t])$ is determined by $[w]$. Hence there exist injective maps

$$\alpha : P_{11} \cup P_{12} \to \mathrm{Sing}(f = 0), \qquad \beta : P_{11} \cup P_{21} \to \mathrm{Sing}(f^\vee = 0)$$

sending a point to the point of $\mathbb{P}^2$ defined by the rank one matrix. Since $(f = 0)$ is reduced, $\mathrm{Sing}(f = 0)$ is finite, and $\mathrm{Sing}(f^\vee = 0)$ is finite by Lemma 20.5(3). Hence $P_{11} \cup P_{12} \cup P_{21}$ is finite.

Let $p = ([\xi\xi^t],[C_f(\xi\xi^t)]) \in P_{11}$. By Construction-Proposition 20.7(6), the Hessian matrix of f at ξ is $12C_f(\xi\xi^t)$, which has rank one. By the Euler formula, $12C_f(\xi\xi^t)\xi = 3\nabla f(\xi) = 0$, so the quadratic part of the affine germ of f at $[\xi]$ has rank one. Thus $[\xi]$ is not of type A_1, and $[\xi]$ is a singular point of type A_2 of $(f = 0)$.

Step 4. In this step, under the assumptions of **Step 3**, we bound the sum of the lengths of $\mathcal{O}_{\Gamma\cap\Gamma_{C_f},p}$ over $p \in P_{11} \cup P_{12} \cup P_{21}$.

For a singular point x of $(f = 0)$ (resp. $(f^\vee = 0)$), let $\mu_x(f)$ (resp. $\mu_x(f^\vee)$) be its Milnor number (Lemma 20.5), and for $p \in \Gamma \cap \Gamma_{C_f}$ let ℓ_p be the length of $\mathcal{O}_{\Gamma\cap\Gamma_{C_f},p}$. Let $p \in P_{12}$. By Construction-Proposition 20.7(6), C_f is self-adjoint and $\varphi_{C_f} = f$, so Lemma 20.8 with $\Psi = C_f$ implies $\ell_p = \mu_{\alpha(p)}(f)$. Let $p \in P_{21}$. The involution σ maps Γ onto Γ (Construction-Proposition 20.7(1)) and Γ_{C_f} onto $\Gamma_{C_f^{-1}}$, and C_f^{-1} is self-adjoint with $\varphi_{C_f^{-1}} = f^\vee$. Writing $p = ([C_f^{-1}(ww^t)], [ww^t])$, the matrix $C_f^{-1}(ww^t)$ has rank two and annihilates w, so Lemma 20.8 with $\Psi = C_f^{-1}$, applied at $\sigma(p)$, implies $\ell_p = \mu_{\beta(p)}(f^\vee)$, because σ maps $\Gamma \cap \Gamma_{C_f}$ isomorphically onto $\Gamma \cap \Gamma_{C_f^{-1}}$. Let $p \in P_{11}$. By **Step 3** and Lemma 20.9, $\ell_p = 3$. Since $\alpha(p)$ is of type A_2, $\mu_{\alpha(p)}(f) = 2$, and $\mu_{\beta(p)}(f^\vee) \geq 1$ as $\beta(p)$ is a singular point. Thus $\ell_p \leq \mu_{\alpha(p)}(f) + \mu_{\beta(p)}(f^\vee)$. Since α and β are injective, any singular point of $(f = 0)$ or of $(f^\vee = 0)$ is counted at most once in the following sum, and

$$\begin{aligned}\sum_{p\in P_{11}\cup P_{12}\cup P_{21}} \ell_p &\leq \sum_{p\in P_{11}\cup P_{12}} \mu_{\alpha(p)}(f) + \sum_{p\in P_{11}\cup P_{21}} \mu_{\beta(p)}(f^\vee)\\ &\leq \sum_{x\in \mathrm{Sing}(f=0)} \mu_x(f) + \sum_{x\in \mathrm{Sing}(f^\vee=0)} \mu_x(f^\vee).\end{aligned}$$

By Lemma 20.5(3), the second sum on the right, $\sum_{x\in\mathrm{Sing}(f^\vee=0)} \mu_x(f^\vee)$, is at most 7.

Step 5. In this step we show $\sum_{x\in\mathrm{Sing}(f=0)} \mu_x(f) \leq 6$ when $(f = 0)$ is reduced with only singular points of types A_1 and A_2.

Let $C_1, \ldots, C_k$ be the irreducible components of $(f = 0)$, of degrees $d_1, \ldots, d_k$ with $\sum_i d_i = 4$. A singular point of type A_1 or A_2 has multiplicity two. Thus a singular point of C_i does not belong to C_j for $j \neq i$, and a point of $C_i \cap C_j$ with $i \neq j$ is a smooth point of both curves. Since a singular point of type A_2 is unibranch and a tangency of two smooth branches is not of type A_1, any point of $C_i \cap C_j$ is of type A_1 with local intersection number one, and by Bézout's theorem [Har77, Corollary I.7.8], $\#(C_i \cap C_j) = d_i d_j$. For a singular point of type A_1 or A_2, the delta invariant is $\delta = 1$ by Milnor's formula $\mu = 2\delta - b + 1$, where b is the number of branches [Mil68, Theorem 10.5], [GBP22, Section 1], and $\mu \leq 2 = 2\delta$. The sum of the delta invariants of the singular points of C_i is at most the arithmetic genus $\frac{(d_i-1)(d_i-2)}{2}$ [Har77, Exercise IV.1.8 and Exercise I.7.2(b)]. Hence

$$\sum_{x\in\mathrm{Sing}(f=0)} \mu_x(f) \leq \sum_i (d_i - 1)(d_i - 2) + \sum_{i<j} d_i d_j.$$

For $(d_1, \ldots, d_k)$ equal to (4), $(3, 1)$, $(2, 2)$, $(2, 1, 1)$, $(1, 1, 1, 1)$, the right-hand side is 6, 5, 4, 5, 6, respectively.

Step 6. In this step we prove (1) when $(f = 0)$ is reduced with only singular points of types A_1 and A_2.

By **Step 1**, we may assume that $(f^\vee = 0)$ is reduced. Suppose that no point $([A], [B]) \in \Gamma\cap\Gamma_{C_f}$ has A invertible. Then $\Gamma\cap\Gamma_{C_f} = P_{11}\cup P_{12}\cup P_{21}$ is finite by **Step 3**. Thus Γ and Γ_{C_f} are smooth fivefolds intersecting properly in the smooth tenfold $\mathbb{P}^5 \times \mathbb{P}^5$, and by [Ful98, Proposition 7.1 and Proposition 8.2],

$$[\Gamma] \cdot [\Gamma_{C_f}] = \sum_{p\in\Gamma\cap\Gamma_{C_f}} \ell_p \leq 6 + 7 = 13$$

by **Steps 4 and 5**. This contradicts **Step 2**. Hence there exists $([A], [B]) \in \Gamma \cap \Gamma_{C_f}$ with A invertible. By Construction-Proposition 20.7(3), $[B] = [\operatorname{adj} A]$, so $C_f(A) = \lambda \operatorname{adj} A$ for some $\lambda \in \mathbb{C}$, and $\lambda \neq 0$ since C_f is invertible. Thus $AC_f(A) = tI_3$ with $t := \lambda \det A \neq 0$.

Step 7. In this step we prove (1) when $[f]$ is GIT-polystable.

By Theorem 2.29(1), the quartics in **Step 6** are exactly the GIT-stable quartics. Thus we may assume that $[f]$ is GIT-polystable but not GIT-stable. By Theorem 2.29(3), f is projectively equivalent to

$$(xz - ay^2)(xz - by^2) = x^2z^2 - (a+b)xy^2z + aby^4$$

for some $(a,b) \in \mathbb{C}^2 \backslash \{(0,0)\}$. Exchanging y and z, f is projectively equivalent to

$$f' := c_1x^2y^2 + c_2xyz^2 + c_3z^4, \qquad c_1 = 1,\ c_2 = -(a+b),\ c_3 = ab.$$

If $c_2 = c_3 = 0$, then $a + b = ab = 0$, so $a = b = 0$, which is not the case. Thus $(c_2, c_3) \neq (0,0)$. Since the existence of A and t in (1) depends only on the PGL_3-orbit of $[f]$, it suffices to prove (1) for f', which is of the form in Lemma 20.5(4) with $c_1 = 1$. Since C_f is invertible and f' is projectively equivalent to f, $C_{f'}$ is invertible by the transformation formula of Construction-Proposition 20.7(6). By Lemma 20.5(4), there exist an invertible symmetric A and $t \neq 0$ with $AC_{f'}(A) = tI_3$. (For $a = b$, f is a double smooth conic. Then $c_3 = c_2^2/4 \neq 0$, so the argument above covers this case, and the orbit of $[f]$ contains $[q_0^2]$ by Definition-Proposition 20.1(7).) Thus (1) holds for f.

Step 8. In this step we prove (2).

By (1), there exists $b \in \mathrm{Harm}_4(q_0)$ such that $[q_0^2 + b]$ is in the PGL_3-orbit of $[f]$. Let $g \in \mathrm{GL}_3$ with $M = gg^t$. Then

$$q_0(g^{-1}x) = x^t g^{-t} g^{-1} x = x^t M^{-1} x = q_M(x).$$

The Hessian matrix of $b \circ g^{-1}$ at x is $g^{-t}\,\mathrm{Hess}(b)(g^{-1}x)g^{-1}$, so

$$\Delta_M(b \circ g^{-1})(x) = \mathrm{tr}\left(gg^t g^{-t}\,\mathrm{Hess}(b)(g^{-1}x)g^{-1}\right) = (\Delta_{I_3} b)(g^{-1}x) = 0.$$

Thus g maps $\{[q_0^2 + b] : b \in \mathrm{Harm}_4(q_0)\}$ into $\{[q_M^2 + b'] : \Delta_M b' = 0\}$, and $[(q_0^2 + b) \circ g^{-1}] = [q_M^2 + b \circ g^{-1}]$ is in the PGL_3-orbit of $[f]$. □

20.3. **The harmonic apolar family.** In this subsection, we compare orbit closures of non-degenerate nets with orbit closures of their apolar quartics, and we construct the family of threefolds over the harmonic slice at q^2.

The next lemma turns a family of non-degenerate quartics, together with a basis of the cubic apolar operators, into a morphism to $\mathbb{B}^{\mathrm{nd}}$.

Lemma 20.10. *Let $e_1, \dots, e_7$ be a basis of V with dual basis $e_1^\vee, \dots, e_7^\vee$. For an alternating 7×7 matrix M over a ring R, let $\omega(M) := \sum_{i<j} M_{ij} e_i^\vee \wedge e_j^\vee \in R \otimes \Lambda^2 V^\vee$, and write an alternating 7×7 matrix A with entries in $R \otimes \mathbb{C}[\partial]_1$ as $A = \partial_x A_x + \partial_y A_y + \partial_z A_z$ with alternating matrices A_x, A_y, A_z over R. Let T be a reduced scheme of finite type over $\mathbb{C}$, let $f_T \in H^0(T, \mathcal{O}_T) \otimes \mathbb{C}[x,y,z]_4$, and let $P_1, \dots, P_7 \in H^0(T, \mathcal{O}_T) \otimes \mathbb{C}[\partial]_3$. For a closed point $t \in T$, let f_t and $P_{i,t}$ be the values of f_T and P_i at t, and let $P_t := (P_{1,t}, \dots, P_{7,t})$. Assume that $\det \mathrm{Cat}_2(f_t) \neq 0$ and P_t is a basis of $(f_t^\perp)_3$ for any closed point $t \in T$. Then the following hold.*

1. *Let $t \in T$ be a closed point. Then the alternating 7×7 matrices A with entries in $\mathbb{C}[\partial]_1$ and $P_t A = 0$ form a one-dimensional vector space. For any non-zero such A, the forms $\omega(A_x)$, $\omega(A_y)$, $\omega(A_z)$ are a basis of a net $\Pi \in \mathbb{B}^{\mathrm{nd}}$, the apolar quartic f_Π written in the basis $(\omega(A_x), \omega(A_y), \omega(A_z))$ is a non-zero multiple of f_t, and $\mathrm{VPS}(f_t, 6) \cong X_\Pi$.*
2. *There exists a morphism $\varphi : T \to \mathbb{B}^{\mathrm{nd}}$ such that $\varphi(t)$ is the net Π of (1) for any closed point $t \in T$.*
3. *Any point of U^{cat} has an open neighborhood $W \subset U^{\mathrm{cat}}$ with a morphism $\varphi_W : W \to \mathbb{B}^{\mathrm{nd}}$ such that for any closed point $[f] \in W$, $f_{\varphi_W([f])}$ is a non-zero multiple of f in some basis of $\varphi_W([f])$, and $\mathrm{VPS}(f, 6) \cong X_{\varphi_W([f])}$.*

Proof. (1) By Lemma 16.3(2), there exist $\Pi' \in \mathbb{B}^{\mathrm{nd}}$ and a basis $\omega_x', \omega_y', \omega_z'$ of Π' in which $f_{\Pi'}$ is a non-zero multiple of f_t, and $\mathrm{VPS}(f_t, 6) \cong X_{\Pi'}$. Let A' be the alternating matrix of Π' with respect to $e_1, \dots, e_7$ and $\omega_x', \omega_y', \omega_z'$, with entries in $\mathbb{C}[\partial]_1$, and let $p' = (p_1', \dots, p_7')$ be the row of

signed principal 6×6 Pfaffians of A'. By Definition-Proposition 2.28, $f_t^\perp = f_{\Pi'}^\perp$ is generated by $p'_1,\dots,p'_7$, and by Lemma 16.3(1) the Pfaffian complex

$$0\to \mathbb{C}[\partial](-7)\xrightarrow{p'^t}\mathbb{C}[\partial](-4)^{\oplus 7}\xrightarrow{A'}\mathbb{C}[\partial](-3)^{\oplus 7}\xrightarrow{p'}\mathbb{C}[\partial]$$

is exact, and $\mathbb{C}[\partial]/f_t^\perp$ has Hilbert function $(1,3,6,3,1)$. Thus $\dim(f_t^\perp)_3=7$, and $p'_1,\dots,p'_7$ form a basis of $(f_t^\perp)_3$. Let $G\in \mathrm{GL}_7(\mathbb{C})$ with $P_t = p'G$, and let $A_0 := G^{-1}A'G^{-t}$. Since $p'G = P_t$, $A'G^{-t} = GA_0$, and $G^{-t}P_t^t = p'^t$, multiplication by G^{-t} on $\mathbb{C}[\partial](-4)^{\oplus 7}$ and by G on $\mathbb{C}[\partial](-3)^{\oplus 7}$ define an isomorphism from the complex

$$(*)\qquad 0\to \mathbb{C}[\partial](-7)\xrightarrow{P_t^t}\mathbb{C}[\partial](-4)^{\oplus 7}\xrightarrow{A_0}\mathbb{C}[\partial](-3)^{\oplus 7}\xrightarrow{P_t}\mathbb{C}[\partial]$$

to the Pfaffian complex of A'. Hence $(*)$ is exact.

Let A be an alternating matrix with entries in $\mathbb{C}[\partial]_1$ and $P_tA=0$. The columns of A belong to the degree-four part of $\ker P_t$, which is $A_0\cdot\mathbb{C}^7$ by exactness of $(*)$. Thus $A = A_0C$ for some $C\in M_7(\mathbb{C})$. Since A and A_0 are alternating, $A_0C = -C^tA_0^t = C^tA_0$. Since $A_0P_t^t = -(P_tA_0)^t = 0$, we obtain

$$A_0CP_t^t = C^tA_0P_t^t = 0.$$

The entries of CP_t^t are cubics, so exactness of $(*)$ at $\mathbb{C}[\partial](-4)^{\oplus 7}$ in degree seven implies $CP_t^t = cP_t^t$ for some $c\in\mathbb{C}$. Since $P_{1,t},\dots,P_{7,t}$ are linearly independent, $C = cI_7$, and hence $A = cA_0$. As $A_0\neq 0$, the space of such matrices is $\mathbb{C}A_0$.

Let $A = cA_0$ with $c\neq 0$. For $g\in\mathrm{GL}_7(\mathbb{C})$, let $h_g\in\mathrm{GL}(V^\vee)$ be the automorphism with $h_g(e_k^\vee) = \sum_i g_{ki}e_i^\vee$, so that, for an alternating 7×7 matrix M over a $\mathbb{C}$-algebra,

$$\omega(g^tMg) = \frac{1}{2}\sum_{k,l} M_{kl}h_g(e_k^\vee)\wedge h_g(e_l^\vee) = (\Lambda^2 h_g)(\omega(M)).$$

Let $h := h_{G^{-t}}$. Since $(G^{-t})^tMG^{-t} = G^{-1}MG^{-t}$, we have $\omega(G^{-1}MG^{-t}) = (\Lambda^2h)(\omega(M))$. Thus $\omega(A_x),\omega(A_y),\omega(A_z)$ are the images of $\omega'_x,\omega'_y,\omega'_z$ under $c\Lambda^2h$, and $\Pi = (\Lambda^2h)(\Pi')$ belongs to $\mathrm{GL}(V)\cdot\Pi'\subset\mathbb{B}^{\mathrm{nd}}$. For an alternating 7×7 matrix M with entries in $\mathbb{C}[\partial]$, let $p(M)$ be the row of signed principal 6×6 Pfaffians of M. Up to a sign independent of M, $p(M)$ is the row of coefficients of $\frac{1}{6}\omega(M)^{\wedge 3}$ in the basis $((-1)^{k-1}e_1^\vee\wedge\cdots\wedge\widehat{e_k^\vee}\wedge\cdots\wedge e_7^\vee)_k$ of $\Lambda^6V^\vee$, and comparing the coefficients of $\frac{1}{6}\omega(g^tMg)^{\wedge 3} = (\Lambda^6h_g)(\frac{1}{6}\omega(M)^{\wedge 3})$ gives

$$p(g^tMg)^t = \det(g)g^{-1}p(M)^t.$$

For $g = G^{-t}$ we have $g^tA'g = A_0$, so $p(A_0) = \det(G)^{-1}p'G = \det(G)^{-1}P_t$. The principal 6×6 Pfaffians are homogeneous of degree three in the entries, so $p(A) = c^3\det(G)^{-1}P_t$, and the Pfaffian ideal of A is generated by $P_{1,t},\dots,P_{7,t}$. Since P_t and p' are bases of $(f_t^\perp)_3$ and $f_t^\perp = f_{\Pi'}^\perp$ is generated by $p'_1,\dots,p'_7$ (Definition-Proposition 2.28), the Pfaffian ideal of A is $f_t^\perp$. The matrix A is the alternating matrix of Π with respect to $e_1,\dots,e_7$ and $\omega(A_x),\omega(A_y),\omega(A_z)$. Thus, by Definition-Proposition 2.28, the apolar ideal of f_Π written in the basis $\omega(A_x),\omega(A_y),\omega(A_z)$ is $f_t^\perp$. Since the pairing $\mathbb{C}[\partial]_4\times\mathbb{C}[x,y,z]_4\to\mathbb{C}$, $(P,f)\to P(f)$, is perfect, a quartic f is determined up to a non-zero scalar by $(f^\perp)_4$. Hence f_Π is a non-zero multiple of f_t. Moreover $X_\Pi\cong X_{\Pi'}\cong\mathrm{VPS}(f_t,6)$.

(2) Let $\mathfrak{a}$ be the space of alternating 7×7 matrices with entries in $\mathbb{C}[\partial]_1$, so that $\dim\mathfrak{a}=63$, and let $\mathcal{L}$ be the kernel of the $\mathcal{O}_T$-linear map

$$\mathcal{O}_T\otimes\mathfrak{a}\to\mathcal{O}_T\otimes(\mathbb{C}[\partial]_4)^{\oplus 7},\quad A\to(P_1,\dots,P_7)A.$$

By (1), the map has rank 62 at any closed point. Since T is reduced, $\mathcal{L}$ is a line subbundle of $\mathcal{O}_T\otimes\mathfrak{a}$ whose fiber at any closed point t is the kernel of the map at t [Har77, Exercise II.5.8(c)]. Let $T'\subset T$ be an open subset on which $\mathcal{L}$ is generated by a section A. Then $A_s\neq 0$ for any closed point $s\in T'$, so by (1) the sections $\omega(A_x),\omega(A_y),\omega(A_z)$ of $\mathcal{O}_{T'}\otimes\Lambda^2V^\vee$ are linearly independent at any closed point. Thus they span a subbundle of rank three [Har77, Exercise II.5.8(c)], which defines a morphism $T'\to\mathbb{B} = \mathrm{Gr}(3,\Lambda^2V^\vee)$ sending a closed point s to the net Π of (1). The

preimage of the open subset $\mathbb{B}^{\mathrm{nd}}$ is an open subset of T' containing any closed point, hence equal to T'. Another generator of $\mathcal{L}$ on an open subset T'' is uA with $u \in H^0(T' \cap T'', \mathcal{O}_T)^\times$ on $T' \cap T''$, and $\omega(uA_x), \omega(uA_y), \omega(uA_z)$ span the subbundle spanned by $\omega(A_x), \omega(A_y), \omega(A_z)$ on $T' \cap T''$. Thus the morphisms glue to φ.

(3) Let $[f_1] \in U^{\mathrm{cat}}$, let m be a monomial of degree four whose coefficient $c_m(f_1)$ in f_1 is non-zero, and let $W_m := \{[f] \in U^{\mathrm{cat}} : c_m(f) \neq 0\}$. Then W_m is smooth, and $[f] \to f/c_m(f)$ defines $f_{W_m} \in H^0(W_m, \mathcal{O}_{W_m}) \otimes \mathbb{C}[x,y,z]_4$. For any closed point $[f] \in W_m$, $\mathbb{C}[\partial]/f^\perp$ has Hilbert function $(1,3,6,3,1)$ by Lemma 16.3(1) and (2) and Definition-Proposition 2.28, so $\dim(f^\perp)_3 = 7$. Thus the kernel $\mathcal{E}$ of $\mathcal{O}_{W_m} \otimes \mathbb{C}[\partial]_3 \to \mathcal{O}_{W_m} \otimes \mathbb{C}[x,y,z]_1$, $P \to P(f_{W_m})$, is a subbundle of rank seven whose fiber at $[f]$ is $(f^\perp)_3$ [Har77, Exercise II.5.8(c)]. Let $W \subset W_m$ be an open neighborhood of $[f_1]$ on which $\mathcal{E}$ is free with a basis $P_1, \dots, P_7$. Then (1) and (2) for $T = W$ imply (3). □

Proposition 20.11. *Let $\Pi, \Pi' \in \mathbb{B}^{\mathrm{nd}}$. Regard $[f_\Pi]$ and $[f_{\Pi'}]$ as points of $\mathbb{P}(\mathbb{C}[x,y,z]_4)$ by choosing linear isomorphisms $\Pi \to \mathbb{C}[x,y,z]_1$ and $\Pi' \to \mathbb{C}[x,y,z]_1$. Other choices change $[f_\Pi]$ and $[f_{\Pi'}]$ by elements of PGL_3. Then Π belongs to the closure of $\mathrm{SL}(V) \cdot \Pi'$ in $\mathbb{B}^{\mathrm{nd}}$ if and only if $[f_\Pi]$ belongs to the closure of $\mathrm{PGL}_3 \cdot [f_{\Pi'}]$ in U^{cat}. Hence, for $\Pi \in \mathbb{B}^{\mathrm{nd}}$, $[f_\Pi]$ is GIT-polystable if and only if $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$, and for $\Pi \in \Omega$, $[f_\Pi]$ is GIT-stable if and only if $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Aut}(X_\Pi)$ is finite.*

Proof. A scalar $\lambda \in \mathrm{GL}(V)$ acts on $\Lambda^2 V^\vee$ by λ^{-2}, hence trivially on $\mathbb{B}$. Thus $\mathrm{SL}(V) \cdot \Pi_1 = \mathrm{PGL}(V) \cdot \Pi_1$ for any $\Pi_1 \in \mathbb{B}$. Closures of PGL_3-orbits are PGL_3-invariant, so the first equivalence does not depend on the choices of the isomorphisms.

If Π belongs to the closure of $\mathrm{SL}(V) \cdot \Pi'$, then $[f_\Pi]$ belongs to the closure of $\mathrm{PGL}_3 \cdot [f_{\Pi'}]$ in U^{cat} by Lemma 16.3(5). Conversely, assume that $[f_\Pi]$ belongs to the closure of $O' := \mathrm{PGL}_3 \cdot [f_{\Pi'}]$ in U^{cat}. By Lemma 20.10(3), $[f_\Pi]$ has an open neighborhood $W \subset U^{\mathrm{cat}}$ with a morphism $\varphi_W : W \to \mathbb{B}^{\mathrm{nd}}$ such that for any closed point $[f] \in W$, f is a non-zero multiple of $f_{\varphi_W([f])}$ in some basis of $\varphi_W([f])$. Thus Lemma 16.3(3) implies that $\varphi_W([f]) \in \mathrm{SL}(V) \cdot \Pi'$ if $[f] \in O'$, and that $\varphi_W([f_\Pi]) \in \mathrm{SL}(V) \cdot \Pi$. Since W is an open neighborhood of $[f_\Pi]$, the point $[f_\Pi]$ belongs to the closure of the set of closed points of $W \cap O'$. Since φ_W is continuous, $\varphi_W([f_\Pi])$ belongs to the closure of $\mathrm{SL}(V) \cdot \Pi'$. The closure of $\mathrm{SL}(V) \cdot \Pi'$ is $\mathrm{SL}(V)$-invariant, so it contains Π.

For $g \in \mathrm{GL}_3$, the symmetric bilinear form $(P, P') \to (PP')(f \circ g^{-1})$ on $\mathbb{C}[\partial]_2$ is the pullback of $(P, P') \to (PP')(f)$ by a linear automorphism of $\mathbb{C}[\partial]_2$ of determinant $(\det g)^{-4}$. Thus

$$\det \mathrm{Cat}_2(f \circ g^{-1}) = (\det g)^{-8} \det \mathrm{Cat}_2(f),$$

and $\det \mathrm{Cat}_2$ is an SL_3-invariant homogeneous polynomial of degree six. By [MFK94, Theorem 1.10] and [Alp13, Theorem 13.6], U^{cat} is a saturated invariant open subset of the GIT-semistable locus, namely the preimage of the affine open subset $D_+(\det \mathrm{Cat}_2)$ of the GIT quotient of plane quartics. In particular, an orbit in U^{cat} is closed in the GIT-semistable locus if and only if it is closed in U^{cat}.

Assume that $\mathrm{SL}(V) \cdot \Pi$ is closed, and let $[f_1]$ be a point of the closure of $\mathrm{PGL}_3 \cdot [f_\Pi]$ in U^{cat}. By Lemma 16.3(2), $[f_1] = [f_{\Pi_1}]$ for some $\Pi_1 \in \mathbb{B}^{\mathrm{nd}}$ and some basis of Π_1. By the first equivalence, $\Pi_1 \in \mathrm{SL}(V) \cdot \Pi$, so $[f_1] \in \mathrm{PGL}_3 \cdot [f_\Pi]$ by Lemma 16.3(3). Conversely, assume that $\mathrm{PGL}_3 \cdot [f_\Pi]$ is closed in U^{cat}, and let Π_1 be a point of the closure of $\mathrm{SL}(V) \cdot \Pi$. By the first equivalence, $[f_{\Pi_1}] \in \mathrm{PGL}_3 \cdot [f_\Pi]$, so $\Pi_1 \in \mathrm{SL}(V) \cdot \Pi$ by Lemma 16.3(3). Thus $[f_\Pi]$ is GIT-polystable if and only if $\mathrm{SL}(V) \cdot \Pi$ is closed.

By definition, $[f_\Pi]$ is GIT-stable if and only if it is GIT-polystable and the stabilizer of $[f_\Pi]$ in SL_3 is finite. The kernel of $\mathrm{SL}_3 \to \mathrm{PGL}_3$ is finite, and for $\Pi \in \Omega$ the stabilizer of $[f_\Pi]$ in PGL_3 is isomorphic to $\mathrm{Aut}(X_\Pi)$ by Theorem 16.12(2). Thus the last equivalence follows from the second one. □

Set-up 20.12. Let $q := xz - y^2$, let $\Delta_q := 4\partial_x\partial_z - \partial_y^2 \in \mathbb{C}[\partial]_2$, and let $\mathrm{Harm}_4 := \{b \in \mathbb{C}[x,y,z]_4 : \Delta_q b = 0\}$ with the action of $\mathrm{SO}(q) := \{g \in \mathrm{SL}_3 : q \circ g = q\}$ by $b \to b \circ g^{-1}$. For $b \in \mathrm{Harm}_4$ let

$f_b := q^2 + b$, let

$$\mathrm{Harm}_4^\circ := \{b \in \mathrm{Harm}_4 : \det \mathrm{Cat}_2(f_b) \neq 0 \text{ and } \mathrm{VPS}(f_b, 6) \text{ is smooth}\},$$

for $b \in \mathrm{Harm}_4^\circ$ let $X_b := \mathrm{VPS}(f_b, 6)$, and let $\varpi : \mathrm{Harm}_4 \to \mathrm{Harm}_4 /\!\!/ \mathrm{SO}(q)$ be the affine quotient.

Lemma 20.13. *Notation as in Set-up 20.12. Then the following hold.*

(1) $\Delta_q(b \circ g^{-1}) = (\Delta_q b) \circ g^{-1}$ *for* $g \in \mathrm{SO}(q)$ *and* $b \in \mathbb{C}[x, y, z]$, *and* $\Delta_q(q^2) = 20q$.

(2) The natural map $\mathrm{SO}(q) \to \mathrm{PGL}_3$ *is injective with image* $\mathrm{Stab}_{\mathrm{PGL}_3}([q^2])$, *and* $\mathrm{SO}(q) \cong \mathrm{PSL}_2$.

(3) Restriction to the conic $(x, y, z) = (u^2, uv, v^2)$ *induces an* $\mathrm{SO}(q)$*-equivariant isomorphism* $\mathrm{Harm}_4 \cong \mathrm{Sym}^8 \mathbb{C}^2$.

Proof. (1) Under the substitution of Definition-Proposition 20.1(7), q, Δ_q, and $\mathrm{SO}(q)$ correspond to q_0, $\partial_1^2 + \partial_2^2 + \partial_3^2$, and $\mathrm{SO}(q_0)$. Since orthogonal matrices commute with the Laplacian, $\Delta_q(b \circ g^{-1}) = (\Delta_q b) \circ g^{-1}$ for any $g \in \mathrm{SO}(q)$ and $b \in \mathbb{C}[x, y, z]$. In the coordinates of Definition-Proposition 20.1(7), the Euler formula implies

$$\Delta_q(qh) = (4k + 6)h + q\Delta_q h \quad \text{for } h \in \mathbb{C}[x, y, z]_k. \tag{20.18}$$

In particular $\Delta_q q = 6$ and $\Delta_q(q^2) = 20q$.

(2) A scalar matrix $\lambda I_3 \in \mathrm{SO}(q)$ satisfies $\lambda^2 = \lambda^3 = 1$, so $\lambda = 1$, and $\mathrm{SO}(q) \to \mathrm{PGL}_3$ is injective. If $g \in \mathrm{PGL}_3$ fixes $[q^2]$, then g fixes $[q]$ by unique factorization, and a lift of g to GL_3, rescaled by a constant, preserves q. Thus $\mathrm{Stab}_{\mathrm{PGL}_3}([q^2]) = \mathrm{Stab}_{\mathrm{PGL}_3}([q]) = \mathrm{O}(q)/\{\pm 1\}$. Since $\det(-I_3) = -1$, $\mathrm{O}(q) = \mathrm{SO}(q) \times \{\pm 1\}$, and the image of $\mathrm{SO}(q)$ is $\mathrm{Stab}_{\mathrm{PGL}_3}([q^2])$.

The action of SL_2 on (u, v) induces a linear action on the span of u^2, uv, v^2, hence, through the coordinates $(x, y, z) = (u^2, uv, v^2)$, a homomorphism $s : \mathrm{SL}_2 \to \mathrm{GL}_3$ with kernel $\{\pm 1\}$. Since $s(\sigma)$ preserves the cone $(q = 0)$, the image of $(u, v) \to (u^2, uv, v^2)$, we have $q \circ s(\sigma) = \chi(\sigma)q$ for a character χ of SL_2. As SL_2 has no non-trivial characters, s takes values in $\mathrm{SO}(q)$. Let $g \in \mathrm{SO}(q)$. Then g preserves the conic $C = (q = 0)$, the image of the isomorphism $\mathbb{P}^1 \to C$, $(u : v) \to (u^2 : uv : v^2)$, so $g|_C$ corresponds under the isomorphism to an automorphism of $\mathbb{P}^1$, hence induced by some $\sigma \in \mathrm{SL}_2$ [Har77, Example II.7.1.1]. Then $s(\sigma)^{-1}g$ fixes every point of C. Since C contains four points of $\mathbb{P}^2$ in general position, $s(\sigma)^{-1}g$ is a scalar matrix [Har77, Example II.7.1.1], hence equal to I_3 by the previous paragraph. Thus $s(\mathrm{SL}_2) = \mathrm{SO}(q)$. Hence s induces a bijective homomorphism $\bar{s} : \mathrm{PSL}_2 \to \mathrm{SO}(q)$ between smooth varieties, and $\mathrm{SO}(q) = s(\mathrm{SL}_2)$ is irreducible as the image of SL_2. By [Bor91, AG.18.2, Theorem], the function field of PSL_2 is purely inseparable over that of $\mathrm{SO}(q)$, hence equal to it in characteristic zero, so $\bar{s}$ is birational, and by [Bor91, AG.18.2, Theorem] again $\bar{s}$ is an isomorphism.

(3) The restriction $r : \mathbb{C}[x, y, z]_4 \to \mathbb{C}[u, v]_8$, $b \to b(u^2, uv, v^2)$, is SL_2-equivariant, where SL_2 acts on $\mathbb{C}[x, y, z]_4$ through s, and r is surjective. Moreover $\ker r = q \cdot \mathbb{C}[x, y, z]_2$, since q is irreducible. Let $h \in \mathbb{C}[x, y, z]_2$ with $\Delta_q(qh) = 0$, and let $h_2 := h - \frac{\Delta_q h}{6} q$, so that $\Delta_q h_2 = 0$ since $\Delta_q q = 6$. Then (20.18) with $k = 2$ gives

$$0 = \Delta_q(qh) = 14h_2 + \frac{10\Delta_q h}{3} q.$$

Applying Δ_q implies $\Delta_q h = 0$, hence $h_2 = 0$ and $h = 0$. Thus $\mathrm{Harm}_4 \cap \ker r = 0$. Since $\dim \mathrm{Harm}_4 \geq 15 - 6 = 9 = \dim \mathbb{C}[u, v]_8$ and $s(\mathrm{SL}_2) = \mathrm{SO}(q)$ by (2), r restricts to an $\mathrm{SO}(q)$-equivariant isomorphism $\mathrm{Harm}_4 \cong \mathrm{Sym}^8 \mathbb{C}^2$. □

The next theorem turns the coverage of single quartics in Proposition 20.6 into a family: by Lemma 20.10, the apolar nets vary algebraically along the harmonic slice.

Theorem 20.14. *Notation as in Set-up 20.12. Then the following hold.*

(1) U^{sm} *is a saturated invariant open subset of the GIT-semistable locus of* $\mathbb{P}(\mathbb{C}[x, y, z]_4)$, *and the image* $U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$ *of* U^{sm} *in the GIT quotient of plane quartics is an open subscheme of the affine chart* $U^{\mathrm{cat}} /\!\!/ \mathrm{PGL}_3$. *The set* Harm_4° *is an* $\mathrm{SO}(q)$*-invariant open*

subset containing 0 *with* $\mathrm{Harm}_4^\circ = \varpi^{-1}(\varpi(\mathrm{Harm}_4^\circ))$, $\varpi(\mathrm{Harm}_4^\circ)$ *is open, and* ϖ *restricts to a good quotient* $\mathrm{Harm}_4^\circ \to \varpi(\mathrm{Harm}_4^\circ)$, *so that* $\varpi(\mathrm{Harm}_4^\circ) = \mathrm{Harm}_4^\circ /\!\!/ \mathrm{SO}(q)$.

(2) *There exists an* $\mathrm{SO}(q)$*-equivariant smooth projective family over* Harm_4° *with fibers* $X_b = \mathrm{VPS}(f_b, 6) = \mathrm{VSP}(f_b, 6)$, *which are smooth prime Fano threefolds of genus* 12, *polarized by the relative anti-canonical bundle, with* $X_0 \cong X^{\mathrm{MU}}$. *The morphism* $\mathrm{PGL}_3 \times^{\mathrm{SO}(q)} \mathrm{Harm}_4 \to \mathbb{P}(\mathbb{C}[x,y,z]_4)$ *given by* $[g, b] \to g \cdot [f_b]$ *is étale along the zero section.*

(3) *Sending* b *to* $[f_b]$ *induces a surjective morphism* $\rho : \varpi(\mathrm{Harm}_4^\circ) \to U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$. *For any GIT-polystable* $[f] \in U^{\mathrm{sm}}$ *there exists* $b \in \mathrm{Harm}_4^\circ$ *whose* $\mathrm{SO}(q)$*-orbit is closed in* Harm_4 *such that* $[f_b]$ *is projectively equivalent to* $[f]$. *For any* $b \in \mathrm{Harm}_4^\circ$, $\rho(\varpi(b))$ *is the unique GIT-polystable orbit in the closure of* $\mathrm{PGL}_3 \cdot [f_b]$, *and the GIT-polystable orbit is contained in* U^{sm}.

Proof. The proof is long, so we divide it into several steps. We regard PGL_3-orbits in $\mathbb{P}(\mathbb{C}[x,y,z]_4)$ as SL_3-orbits, and quotients by PGL_3 are GIT quotients by SL_3.

Step 1. In this step, we prove the first sentence of (1). Since $\det \mathrm{Cat}_2$ is an SL_3-invariant homogeneous polynomial of degree six, U^{cat} is a saturated invariant open subset of the GIT-semistable locus, namely the preimage of $D_+(\det \mathrm{Cat}_2)$, and the quotient morphism $\psi : U^{\mathrm{cat}} \to U^{\mathrm{cat}} /\!\!/ \mathrm{PGL}_3 = D_+(\det \mathrm{Cat}_2)$ is a good quotient onto an affine scheme [MFK94, Theorem 1.10], [Alp13, Theorem 13.6].

By Lemma 20.10(3), U^{cat} is covered by open subsets W with morphisms $\varphi_W : W \to \mathbb{B}^{\mathrm{nd}}$ such that $\mathrm{VPS}(f, 6) \cong X_{\varphi_W([f])}$ for any closed point $[f] \in W$. Fix such W and φ_W, and let $[f] \in W$ be a closed point. If $\mathrm{VPS}(f, 6)$ is smooth, then it is a smooth prime Fano threefold of genus 12 by Proposition 16.10(1), so $\varphi_W([f]) \in \Omega$ by Proposition 16.5. Conversely, $\varphi_W([f]) \in \Omega$ implies that $\mathrm{VPS}(f, 6)$ is smooth. Thus $U^{\mathrm{sm}} \cap W = \varphi_W^{-1}(\Omega)$, which is open since $\Omega = \{r_\Omega \neq 0\}$ is open (Proposition 16.5). Hence U^{sm} is open. It is PGL_3-invariant, since $\mathrm{VPS}(f \circ g^{-1}, 6) \cong \mathrm{VPS}(f, 6)$ for $g \in \mathrm{GL}_3$.

Let $[f] \in U^{\mathrm{sm}}$, and let $\mathrm{PGL}_3 \cdot [f_1]$ be the closed orbit in the closure of $\mathrm{PGL}_3 \cdot [f]$ in U^{cat}. By Lemma 16.3(2), there exist $\Pi, \Pi_1 \in \mathbb{B}^{\mathrm{nd}}$ such that $[f] = [f_\Pi]$ and $[f_1] = [f_{\Pi_1}]$ in some bases, and $X_\Pi \cong \mathrm{VPS}(f, 6)$ is smooth, so $\Pi \in \Omega$. By Proposition 20.11, Π_1 belongs to the closure of $\mathrm{SL}(V) \cdot \Pi$, so $\Pi_1 \in \Omega$ by Proposition 16.5. Thus $[f_1] \in U^{\mathrm{sm}}$.

Let $Z := U^{\mathrm{cat}} \backslash U^{\mathrm{sm}}$, a closed invariant subset. By [Alp13, Theorem 4.16(ii)], $\psi(Z)$ is closed. If $\psi(z) = \psi(u)$ for points $z \in Z$ and $u \in U^{\mathrm{sm}}$, then the orbit closures of z and u in U^{cat} contain a common closed orbit [Alp13, Theorem 4.16(iv)], which is contained in Z since Z is closed and invariant, and in U^{sm} by the previous paragraph. This is a contradiction. Thus $U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3 = \psi(U^{\mathrm{sm}})$ is the open complement of $\psi(Z)$ in $U^{\mathrm{cat}} /\!\!/ \mathrm{PGL}_3$, and

$$U^{\mathrm{sm}} = \psi^{-1}(U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3).$$

By [Alp13, Remark 6.2], ψ restricts to a good quotient $U^{\mathrm{sm}} \to U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$. Since U^{cat} is saturated in the GIT-semistable locus, so is U^{sm}.

Step 2. In this step, we prove the second sentence of (1), construct ρ, and prove $X_0 \cong X^{\mathrm{MU}}$. For $g \in \mathrm{SO}(q)$ and $b \in \mathrm{Harm}_4$, $q \circ g^{-1} = q$ implies $f_{b \circ g^{-1}} = f_b \circ g^{-1}$. Thus $c(b) := \det \mathrm{Cat}_2(f_b)$ is an $\mathrm{SO}(q)$-invariant polynomial function on Harm_4, and

$$\mathrm{Harm}_4^{\mathrm{cat}} := \{b \in \mathrm{Harm}_4 : c(b) \neq 0\}$$

is the preimage under ϖ of the principal open subset $D(c)$ of $\mathrm{Harm}_4 /\!\!/ \mathrm{SO}(q)$. In particular ϖ restricts to a good quotient $\mathrm{Harm}_4^{\mathrm{cat}} \to D(c)$. Let $\iota : \mathrm{Harm}_4^{\mathrm{cat}} \to U^{\mathrm{cat}}$ send b to $[f_b]$. Then $\iota(b \circ g^{-1}) = g \cdot \iota(b)$ for $g \in \mathrm{SO}(q)$, so $\psi \circ \iota$ is $\mathrm{SO}(q)$-invariant, and by **Step 1**,

$$\mathrm{Harm}_4^\circ = \iota^{-1}(U^{\mathrm{sm}}) = (\psi \circ \iota)^{-1}(U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3).$$

By [Alp13, Lemma 6.3], Harm_4° is saturated for $\mathrm{Harm}_4^{\mathrm{cat}} \to D(c)$, hence $\mathrm{Harm}_4^\circ = \varpi^{-1}(\varpi(\mathrm{Harm}_4^\circ))$. In particular Harm_4° is open and $\mathrm{SO}(q)$-invariant. By [Alp13, Remark 6.2], $\varpi(\mathrm{Harm}_4^\circ)$ is open

and ϖ restricts to a good quotient $\mathrm{Harm}_4^\circ \to \varpi(\mathrm{Harm}_4^\circ)$. By [Alp13, Theorem 4.16(vi)], the invariant morphism $\psi \circ \iota|_{\mathrm{Harm}_4^\circ}$ descends to a morphism

$$\rho : \varpi(\mathrm{Harm}_4^\circ) \to U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3, \quad \rho \circ \varpi = \psi \circ \iota.$$

The conics $(q = 0)$ and $(x^2 + yz = 0)$ are smooth, so $q = (x^2 + yz) \circ g_0$ for some $g_0 \in \mathrm{GL}_3$, and $q^2 = F_6 \circ g_0$ with $F_6 = (x^2 + yz)^2$. By Proposition 16.16, $[F_6] \in U^{\mathrm{cat}}$, so $0 \in \mathrm{Harm}_4^{\mathrm{cat}}$. By Definition-Proposition 16.14, F_6 is the apolar quartic of Π_6 in a basis, so Lemma 16.3(2) and (3) imply

$$X_0 = \mathrm{VPS}(q^2, 6) \cong \mathrm{VPS}(F_6, 6) \cong X_{\Pi_6} \cong X^{\mathrm{MU}}.$$

In particular X_0 is smooth and $0 \in \mathrm{Harm}_4^\circ$.

Step 3. In this step, we construct the family of (2). For $l \in \mathbb{C}[\partial]_1$, $(\Delta_q l)(q^2) = l(\Delta_q(q^2)) = 20l(q)$ by Lemma 20.13(1). The map $l \to l(q)$ is an isomorphism $\mathbb{C}[\partial]_1 \to \mathbb{C}[x, y, z]_1$, since $\partial_x q = z$, $\partial_y q = -2y$, and $\partial_z q = x$. Let $\lambda : \mathbb{C}[x, y, z]_1 \to \mathbb{C}[\partial]_1$ be the inverse isomorphism. Then $P \to P(q^2)$ maps $\mathbb{C}[\partial]_3$ onto $\mathbb{C}[x, y, z]_1$, and $\Delta_q \cdot \mathbb{C}[\partial]_1 \cap (q^2)_3^\perp = 0$. Thus $\dim(q^2)_3^\perp = 7$ and

$$\mathbb{C}[\partial]_3 = (q^2)_3^\perp \oplus \Delta_q \cdot \mathbb{C}[\partial]_1.$$

Let $b \in \mathrm{Harm}_4$, $P \in (q^2)_3^\perp$, and $l \in \mathbb{C}[\partial]_1$. Since $P(q^2) = 0$ and $(\Delta_q l)(b) = l(\Delta_q b) = 0$, we have $(P + \Delta_q l)(f_b) = P(b) + 20l(q)$. Thus

$$(f_b^\perp)_3 = \left\{ P - \tfrac{1}{20}\Delta_q \lambda(P(b)) : P \in (q^2)_3^\perp \right\},$$

and the projection to $(q^2)_3^\perp$ along $\Delta_q \cdot \mathbb{C}[\partial]_1$ maps $(f_b^\perp)_3$ isomorphically onto $(q^2)_3^\perp$. Fix a basis $e_1, \dots, e_7$ of V with dual basis $e_1^\vee, \dots, e_7^\vee$ and a linear isomorphism $V^\vee \cong (q^2)_3^\perp$, let $Q_1, \dots, Q_7$ be the images of $e_1^\vee, \dots, e_7^\vee$, and let $P_i(b) := Q_i - \frac{1}{20}\Delta_q \lambda(Q_i(b))$. Lemma 20.10(2) for $T = \mathrm{Harm}_4^{\mathrm{cat}}$, $f_T = f_b$, and $P_1, \dots, P_7$ implies a morphism

$$\mathrm{Harm}_4^{\mathrm{cat}} \to \mathbb{B}^{\mathrm{nd}}, \quad b \to \Pi(b),$$

and $\mathrm{VPS}(f_b, 6) \cong X_{\Pi(b)}$ for any $b \in \mathrm{Harm}_4^{\mathrm{cat}}$ by Lemma 20.10(1).

We use the actions of Section 2.6. For $g \in \mathrm{SO}(q)$, $g \cdot q^2 = q^2$, and $g \cdot \Delta_q = \Delta_q$ by Lemma 20.13(1), so g preserves $(q^2)_3^\perp$ and $\Delta_q \cdot \mathbb{C}[\partial]_1$, and maps $(f_b^\perp)_3$ onto $(f_{g \cdot b}^\perp)_3$ compatibly with the projections to $(q^2)_3^\perp \cong V^\vee$. Through $V^\vee \cong (q^2)_3^\perp$, $\mathrm{SO}(q)$ acts on $V^\vee$, and on V by the contragredient action. For $b \in \mathrm{Harm}_4^{\mathrm{cat}}$ let $\pi_b : V^\vee \to (f_b^\perp)_3$ be the inverse of the projection $(f_b^\perp)_3 \to (q^2)_3^\perp \cong V^\vee$, so that $\pi_b(e_i^\vee) = P_i(b)$, and let $\mu_b : \mathbb{C}[\partial]_1 \otimes \Lambda^2 V^\vee \to \mathbb{C}[\partial]_4 \otimes V^\vee$ be the linear map $l \otimes \alpha \wedge \beta \to l\pi_b(\alpha) \otimes \beta - l\pi_b(\beta) \otimes \alpha$. An alternating 7×7 matrix A with entries in $\mathbb{C}[\partial]_1$ satisfies $(P_1(b), \dots, P_7(b))A = 0$ if and only if the corresponding element of $\mathbb{C}[\partial]_1 \otimes \Lambda^2 V^\vee$ belongs to $\ker \mu_b$, and for $A \neq 0$ in $\ker \mu_b$, $\Pi(b)$ is the image of the map $\mathbb{C}[\partial]_1^\vee \to \Lambda^2 V^\vee$ induced by A (Lemma 20.10(1)). For $g \in \mathrm{SO}(q)$, the compatibility with the projections gives $\pi_{g \cdot b}(g \cdot \alpha) = g \cdot \pi_b(\alpha)$, and multiplication in $\mathbb{C}[\partial]$ is g-equivariant, so $\mu_{g \cdot b} \circ (g \otimes \Lambda^2 g) = (g \otimes g) \circ \mu_b$. Hence $g \otimes \Lambda^2 g$ maps $\ker \mu_b$ onto $\ker \mu_{g \cdot b}$, and $\Pi(g \cdot b) = g \cdot \Pi(b)$ for any closed point b. As $\mathrm{Harm}_4^{\mathrm{cat}}$ is reduced, $b \to \Pi(b)$ is $\mathrm{SO}(q)$-equivariant.

For $b \in \mathrm{Harm}_4^{\mathrm{cat}}$, $\mathrm{VPS}(f_b, 6)$ is smooth if and only if $\Pi(b) \in \Omega$, by Propositions 16.10(1) and 16.5. Thus $b \to \Pi(b)$ restricts to an $\mathrm{SO}(q)$-equivariant morphism $\mathrm{Harm}_4^\circ \to \Omega$. By Proposition 16.5, the universal isotropic scheme over Ω is a smooth projective morphism with fiber X_Π over Π, and $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ restricts to $\omega_{X_\Pi}^{-1}$ on each fiber (Proposition 16.5(2)). The base change of the universal isotropic scheme along $\mathrm{Harm}_4^\circ \to \Omega$ is a smooth projective family with fiber $X_{\Pi(b)} \cong \mathrm{VPS}(f_b, 6) = X_b$ over b, and it carries the $\mathrm{SO}(q)$-action induced by the action on V. By Proposition 16.10(1), any X_b is a smooth prime Fano threefold of genus 12 and $\mathrm{VPS}(f_b, 6) = \mathrm{VSP}(f_b, 6)$. Since the fibers X_b are smooth Fano threefolds, the family over Harm_4° is polarized by its relative anti-canonical bundle, which is relatively ample. By **Step 2**, $X_0 \cong X^{\mathrm{MU}}$.

Under the substitution of Definition-Proposition 20.1(7), q, Harm_4, and $\mathrm{SO}(q)$ correspond to q_0, $\mathrm{Harm}_4(q_0)$, and $\mathrm{SO}(q_0)$, and the morphism $\mathrm{PGL}_3 \times^{\mathrm{SO}(q)} \mathrm{Harm}_4 \to \mathbb{P}(\mathbb{C}[x, y, z]_4)$ in (2)

corresponds to the morphism of Definition-Proposition 20.1(1). Thus it is étale along the zero section. This proves (2).

Step 4. In this step, we prove (3). Let $[f] \in U^{\mathrm{sm}}$ be GIT-polystable. Since $\mathrm{Cat}_2(f)$ is invertible, Proposition 20.6 and the substitution of Definition-Proposition 20.1(7) imply that $[f]$ is projectively equivalent to $[f_{b_1}]$ for some $b_1 \in \mathrm{Harm}_4$. Since projective equivalence preserves the invertibility of the middle catalecticant and the isomorphism class of $\mathrm{VPS}(\cdot, 6)$, $b_1 \in \mathrm{Harm}_4^\circ$. Let b be a point of the unique closed $\mathrm{SO}(q)$-orbit in the closure of $\mathrm{SO}(q) \cdot b_1$ in Harm_4. Then $\varpi(b) = \varpi(b_1)$, so $b \in \mathrm{Harm}_4^\circ$ by **Step 2**. Since ι is continuous and $\iota(b_1 \circ g^{-1}) = g \cdot \iota(b_1)$ for $g \in \mathrm{SO}(q)$, $[f_b]$ belongs to the closure of $\mathrm{PGL}_3 \cdot [f_{b_1}] = \mathrm{PGL}_3 \cdot [f]$ in U^{cat}, and $\mathrm{PGL}_3 \cdot [f]$ is closed in U^{cat}, since $[f]$ is GIT-polystable. Thus $[f_b]$ is projectively equivalent to $[f]$, and the $\mathrm{SO}(q)$-orbit of b is closed in Harm_4.

By [Alp13, Theorem 4.16(iv)], the closed points of $U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$ correspond to the closed orbits in U^{sm}, which are the GIT-polystable orbits in U^{sm} by **Step 1**. By the previous paragraph, for a GIT-polystable $[f] \in U^{\mathrm{sm}}$ there exists $b \in \mathrm{Harm}_4^\circ$ with $\rho(\varpi(b)) = \psi([f_b]) = \psi([f])$. Thus the image of ρ contains any closed point. The image of ρ is constructible by Chevalley's theorem [Har77, Exercise II.3.19], so the complement of the image is a constructible subset of $U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$ without closed points, hence empty. Thus ρ is surjective.

Let $b \in \mathrm{Harm}_4^\circ$. By [Alp13, Theorem 4.16(iv)], $\rho(\varpi(b)) = \psi([f_b])$ corresponds to the unique closed orbit in the closure of $\mathrm{PGL}_3 \cdot [f_b]$ in U^{cat}. Since U^{cat} is saturated in the GIT-semistable locus, the closure of $\mathrm{PGL}_3 \cdot [f_b]$ in the GIT-semistable locus is contained in U^{cat}, and the closed orbit is the unique GIT-polystable orbit in the closure. Since $U^{\mathrm{sm}} = \psi^{-1}(U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3)$ by **Step 1** and $[f_b] \in U^{\mathrm{sm}}$, the GIT-polystable orbit is contained in U^{sm}. This proves (3). □

21. An explicit counterexample

In this section, we construct the divisor D and the point p of Theorem 1.8 explicitly, and we prove the geometric properties of the explicit net. The proof of Theorem 1.8 is in Section 22.

The data of this section are the data of the counterexample to Donaldson's formulation, Conjecture 1.1 (see Section 1.3). Let $\Pi \in \Omega$. If X_Π is not K-polystable, then by Theorem 1.3(3), X_Π admits no faithful $\mathbb{G}_m$-action, hence no non-trivial $\mathbb{C}^*$-action (see the explanation after Conjecture 1.1), and X_Π satisfies condition (2)(i) of Theorem 1.3, so $\Pi \in \Delta'$ (Definition 2.17). Thus a counterexample is a K-polystable X_Π without non-trivial $\mathbb{C}^*$-action with Π in the closure of Δ'. Such an X_Π has reductive automorphism group (Theorem 2.7) and no faithful $\mathbb{G}_m$-action, so $\mathrm{Aut}(X_\Pi)$ is finite, and X_Π is K-stable (Definition 2.4). Since Π is the canonical Mukai net of X_Π (Theorem 16.12(1)), $\Pi \in \Omega^{\mathrm{st}}$ by Theorem 1.3(1), and by the equivalence of (i) and (v) in Theorem 1.3(2), a singular point of a divisor in $|-K_{X_\Pi}|$ with the germ in the definition of Δ' is not contained in any line on X_Π. Thus the point p of the example below has to be found outside the lines.

The quartic f_1 of Construction-Proposition 21.1 is a member of the pencil f_s of Theorem 1.8. The K-stability of X follows from the smoothness of f_1 (Theorem 2.29, Proposition 20.11, and Theorem 17.5), and not from the local picture near X^{MU}. The divisor D belongs to a pencil of Plücker sections ℓ_λ. At the rational point p we compute the weighted jets of order six of $\ell_\lambda|_X$, and the parameter λ_* is the zero in the interval $(1/102, 1/101)$ of an explicit expression J in the coefficients of the jets. The parameter λ_* is a root of the irreducible sextic Φ of Construction-Proposition 21.1, so λ_* is an algebraic number of degree six. Moreover, D is defined over $\mathbb{Q}(\lambda_*)$, and X and p are defined over $\mathbb{Q}$.

The proof of Construction-Proposition 21.1 uses only exact polynomial arithmetic over $\mathbb{Q}$, over the field $\mathbb{F}_3$ with 3 elements, and over the field $\mathbb{F}_{19}$ with 19 elements. The computations are a fixed-point iteration for the jets of weight at most 6 with seven iterations, one unit ideal identity over $\mathbb{F}_3$ for each of eight standard charts of the Grassmannian (the 35 standard charts reduce to the eight charts by Lemma C.2), Frobenius remainder identities and Bézout identities over $\mathbb{F}_{19}$, and a Sturm chain obtained by Euclidean division, each step multiplied by a positive

rational number. Each identity is printed, in the proof below or in Appendix C, and can be checked by expanding both sides. No conclusion depends on the output of a program.

Construction-Proposition 21.1. Let $V = \mathbb{C}^7$ with basis $e_0, \dots, e_6$, let $e_{ij} := e_i^\vee \wedge e_j^\vee$, let

$$\begin{aligned}\omega_1 &= 6e_{02} - 2e_{05} + 10e_{14} - 47e_{23} + 6e_{35},\\ \omega_2 &= 6e_{03} - 2e_{06} + 6e_{12} + 8e_{15} - 37e_{24} + 6e_{36} + 6e_{45},\\ \omega_3 &= 6e_{13} - 2e_{16} + 10e_{25} - 47e_{34} + 6e_{46},\end{aligned}$$

let $\Pi := \mathbb{C}\omega_1 + \mathbb{C}\omega_2 + \mathbb{C}\omega_3$, and let $X := X_\Pi \subset \mathrm{Gr}(3, V)$. Let $p \in \mathrm{Gr}(3, V)$ be the span of $(160, 0, 0, -68, 0, 0, 901)$, $(0, 1, 0, 0, 1, 0, 0)$, and $(0, 0, 14, 0, 0, -43, 0)$, and for $0 \le i < j < k \le 6$ let P_{ijk} be the Plücker coordinate given by the minor on the rows i, j, k. Let

$$\begin{aligned}\Phi(\lambda) = {}& 530057373590412142215787805512787949638890501147\lambda^6\\ &+ 771307305838978057623657135612022018743577802 4\lambda^5\\ &- 154665962353594740748558617724644296479084080\lambda^4\\ &- 221223045679408775242229109781136919803008 0\lambda^3\\ &+ 748164874832917583624196999734876385408 0\lambda^2\\ &+ 126022506304470028296257354535937665024\lambda\\ &+ 396012070987274205768039017056841728\end{aligned}$$

and for $\lambda \in \mathbb{C}$ let

$$\begin{aligned}\ell_\lambda := {}& -\tfrac{1752746171\lambda - 60374628}{246928000}P_{012} - \tfrac{4459(19349\lambda - 732)}{185196000}P_{015} - \tfrac{3653(1463\lambda + 9516)}{148156800}P_{024}\\ &- \tfrac{91(53849\lambda - 732)}{2806000}P_{036} + P_{123} + \lambda P_{126}.\end{aligned}$$

Then Φ is irreducible over $\mathbb{Q}$, and Φ has exactly one root in the open interval $(1/102, 1/101)$, denoted by λ_*. The root λ_* is simple, and λ_* has degree six over $\mathbb{Q}$. Let $D := X \cap \{\ell_{\lambda_*} = 0\}$. Then the following hold.

(1) $\Pi \in \mathbb{B}^{\mathrm{nd}}$, in the basis $(x, y, z) := (\omega_1/6, \omega_2/12, \omega_3/6)$ of Π the apolar quartic f_Π is a non-zero multiple of $f_1 = (xz - y^2)^2 + y(x^3 + z^3)$, the quartic f_1 is smooth, and $\det \mathrm{Cat}_2(f_1) = 216320$ in the basis $(\partial_x^2, \partial_x\partial_y, \partial_x\partial_z, \partial_y^2, \partial_y\partial_z, \partial_z^2)$.

(2) X is a smooth prime Fano threefold of genus 12 with $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_X \cong \mathcal{O}_X(-K_X)$, and the scheme defined by $\omega_1, \omega_2, \omega_3$ over $\mathbb{Z}_{(3)}$ is smooth and projective of relative dimension three.

(3) $D \in |-K_X|$, $p \in D$, and $(D \ni p) \cong (\{y^2 = x^3 + t^4 x\} \ni 0)$ analytically, a simple elliptic singularity of type $\tilde{E}_8$ with j-invariant 1728 and Milnor number 10.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

Let $(x, y, z) := (\omega_1/6, \omega_2/12, \omega_3/6)$. As in Definition-Proposition 2.28, we identify $\mathrm{Sym}\,\Pi$ with $\mathbb{C}[x, y, z]$ and $\mathrm{Sym}\,\Pi^\vee$ with $\mathbb{C}[\partial]$, so that $\partial_x, \partial_y, \partial_z$ is the basis of $\Pi^\vee$ dual to x, y, z. Regard ω_k as the alternating 7×7 matrix, with rows and columns indexed by $0, \dots, 6$, whose (i, j) entry for $i < j$ is the coefficient of e_{ij}. Then the alternating matrix of Π with entries in $\Pi^\vee$ is

$$M := \tfrac{1}{6}\partial_x\omega_1 + \tfrac{1}{12}\partial_y\omega_2 + \tfrac{1}{6}\partial_z\omega_3,$$

and the non-zero entries M_{ij} with $i < j$ are

$$\begin{aligned}&M_{02} = \partial_x, \quad M_{03} = \tfrac{1}{2}\partial_y, \quad M_{05} = -\tfrac{1}{3}\partial_x, \quad M_{06} = -\tfrac{1}{6}\partial_y, \quad M_{12} = \tfrac{1}{2}\partial_y, \quad M_{13} = \partial_z,\\ &M_{14} = \tfrac{5}{3}\partial_x, \quad M_{15} = \tfrac{2}{3}\partial_y, \quad M_{16} = -\tfrac{1}{3}\partial_z, \quad M_{23} = -\tfrac{47}{6}\partial_x, \quad M_{24} = -\tfrac{37}{12}\partial_y, \quad M_{25} = \tfrac{5}{3}\partial_z,\\ &M_{34} = -\tfrac{47}{6}\partial_z, \quad M_{35} = \partial_x, \quad M_{36} = \tfrac{1}{2}\partial_y, \quad M_{45} = \tfrac{1}{2}\partial_y, \quad M_{46} = \partial_z.\end{aligned}$$

Let

$$
\begin{aligned}
q_0 &= -50\partial_z^3 - \tfrac{15}{2}\partial_y(4\partial_x\partial_z - \partial_y^2), & q_1 &= 25\partial_y\partial_z^2 - \tfrac{15}{2}\partial_x(4\partial_x\partial_z - \partial_y^2),\\
q_2 &= -10\partial_x\partial_z^2 - 10\partial_y^2\partial_z, & q_3 &= 15\partial_x\partial_y\partial_z + \tfrac{5}{2}\partial_y^3,\\
q_4 &= -10\partial_x^2\partial_z - 10\partial_x\partial_y^2, & q_5 &= 25\partial_x^2\partial_y - \tfrac{15}{2}\partial_z(4\partial_x\partial_z - \partial_y^2),\\
q_6 &= -50\partial_x^3 - \tfrac{15}{2}\partial_y(4\partial_x\partial_z - \partial_y^2).
\end{aligned}
\tag{21.1}
$$

For $0 \le i \le 6$ let M_{ii} be the alternating 6×6 submatrix of M with rows and columns indexed by $\{0,\dots,6\}\backslash\{i\}$. For an alternating $2m\times 2m$ matrix B with rows and columns indexed by $0,\dots,2m-1$, the Pfaffian satisfies

$$\operatorname{Pf}(B) = \sum_{j=1}^{2m-1} (-1)^{j+1} B_{0j} \operatorname{Pf}(B^{(0,j)}),$$

where $B^{(0,j)}$ is the submatrix of B with rows and columns indexed by $\{1,\dots,2m-1\}\backslash\{j\}$, re-indexed by $0,\dots,2m-3$, and the Pfaffian of the empty matrix is 1. A direct calculation with the recurrence shows

$$(-1)^i \operatorname{Pf}(M_{ii}) = -\tfrac{13}{108} q_i, \qquad 0 \le i \le 6. \tag{21.2}$$

We show that $\Pi \in \mathbb{B}^{\mathrm{nd}}$. Let $(\xi_1,\xi_2,\xi_3) \in \mathbb{C}^3\backslash\{0\}$. The matrix of the form $\xi_1 x + \xi_2 y + \xi_3 z \in \Pi$ is the image of M under the substitution $(\partial_x,\partial_y,\partial_z) \to (\xi_1,\xi_2,\xi_3)$. Since an alternating 7×7 matrix has rank 6 if and only if some principal 6×6 Pfaffian of it is non-zero, by (21.2) it suffices to show that $q_0,\dots,q_6$ have no common zero $(\xi_1,\xi_2,\xi_3) \ne 0$. We have $q_2(\xi_1,\xi_2,\xi_3) = -10\xi_3(\xi_1\xi_3 + \xi_2^2)$ and $q_4(\xi_1,\xi_2,\xi_3) = -10\xi_1(\xi_1\xi_3 + \xi_2^2)$. Suppose that all q_i vanish at (ξ_1,ξ_2,ξ_3). Then $q_2 = q_4 = 0$ gives $\xi_1 = \xi_3 = 0$ or $\xi_1\xi_3 = -\xi_2^2$. If $\xi_1 = \xi_3 = 0$, then $q_3(\xi_1,\xi_2,\xi_3) = \frac{5}{2}\xi_2^3$, so $\xi_2 = 0$. If $\xi_1\xi_3 = -\xi_2^2$, then $q_3(\xi_1,\xi_2,\xi_3) = -15\xi_2^3 + \frac{5}{2}\xi_2^3 = -\frac{25}{2}\xi_2^3$, so $\xi_2 = 0$ and $\xi_1\xi_3 = 0$. If moreover $\xi_1 = 0$, then $q_0(\xi_1,\xi_2,\xi_3) = -50\xi_3^3$, so $\xi_3 = 0$. If moreover $\xi_3 = 0$, then $q_6(\xi_1,\xi_2,\xi_3) = -50\xi_1^3$, so $\xi_1 = 0$. Thus $\Pi \in \mathbb{B}^{\mathrm{nd}}$.

We show that $f_1 \in \mathbb{C} f_\Pi$. We have $f_1 = x^2z^2 - 2xy^2z + y^4 + x^3y + yz^3$, and direct differentiation shows $q_i(f_1) = 0$ for $0 \le i \le 6$. For example, $\partial_x\partial_y\partial_z f_1 = -4y$ and $\partial_y^3 f_1 = 24y$, so $q_3(f_1) = -60y + 60y = 0$. By Definition-Proposition 2.28 and (21.2), the apolar ideal $f_\Pi^\perp$ is the ideal generated by $q_0,\dots,q_6$. Thus $f_\Pi^\perp \subset f_1^\perp$, and in particular $(f_\Pi^\perp)_4 \subset (f_1^\perp)_4$. Both $(f_\Pi^\perp)_4$ and $(f_1^\perp)_4$ are kernels of non-zero linear forms on $\mathbb{C}[\partial]_4$, namely $P \to P(f_\Pi)$ and $P \to P(f_1)$, so $(f_\Pi^\perp)_4 = (f_1^\perp)_4$. Since the pairing $\mathbb{C}[\partial]_4 \times \mathbb{C}[x,y,z]_4 \to \mathbb{C}$, $(P,f) \to P(f)$, is perfect, f_1 is a non-zero multiple of f_Π.

We compute $\det \mathrm{Cat}_2(f_1)$. In the basis $(\partial_x^2, \partial_x\partial_y, \partial_x\partial_z, \partial_y^2, \partial_y\partial_z, \partial_z^2)$ of $\mathbb{C}[\partial]_2$, the symmetric matrix $\big((\partial^\alpha\partial^\beta)(f_1)\big)$ of Definition 2.27, where ∂^α and ∂^β run through the basis, has entries the fourth derivatives of f_1. The non-zero fourth derivatives of f_1 are $\partial_x^2\partial_z^2 f_1 = 4$, $\partial_x\partial_y^2\partial_z f_1 = -4$, $\partial_y^4 f_1 = 24$, $\partial_x^3\partial_y f_1 = 6$, and $\partial_y\partial_z^3 f_1 = 6$, so the matrix is

$$\begin{pmatrix} 0 & 6 & 0 & 0 & 0 & 4\\ 6 & 0 & 0 & 0 & -4 & 0\\ 0 & 0 & 4 & -4 & 0 & 0\\ 0 & 0 & -4 & 24 & 0 & 0\\ 0 & -4 & 0 & 0 & 0 & 6\\ 4 & 0 & 0 & 0 & 6 & 0 \end{pmatrix}. \tag{21.3}$$

After the simultaneous permutation of rows and columns that orders the basis as $\partial_x\partial_z$, ∂_y^2, ∂_x^2, $\partial_y\partial_z$, $\partial_x\partial_y$, ∂_z^2, the matrix (21.3) becomes the direct sum

$$\begin{pmatrix} 4 & -4\\ -4 & 24 \end{pmatrix} \oplus \begin{pmatrix} 0 & K\\ K^t & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 6 & 4\\ -4 & 6 \end{pmatrix}.$$

Thus

$$\det \mathrm{Cat}_2(f_1) = 80 \cdot (\det K)^2 = 80 \cdot 52^2 = 216320.$$

We show that the plane quartic $(f_1 = 0)$ is smooth. Write $f_1 = (xz - y^2)^2 + y(x^3 + z^3)$. Then

$$\partial_x f_1 = 2z(xz - y^2) + 3x^2 y, \qquad \partial_z f_1 = 2x(xz - y^2) + 3yz^2, \qquad \partial_y f_1 = -4y(xz - y^2) + x^3 + z^3.$$

Suppose that $(x, y, z) \neq 0$ is a common zero of the three partial derivatives. If $y = 0$, then $2xz^2 = 2x^2 z = x^3 + z^3 = 0$, so $x = z = 0$, a contradiction. Thus we may assume that $y = 1$. If $x = 0$, then $\partial_x f_1 = -2z$ and $\partial_y f_1 = 4 + z^3$ cannot both vanish. Similarly $z \neq 0$. Then $xz - 1 \neq 0$, since otherwise $\partial_x f_1 = 3x^2 \neq 0$. The quotient of the first two equations implies $z/x = x^2/z^2$, so $z = rx$ with $r^3 = 1$. The first equation becomes $2rx(rx^2 - 1) = -3x^2$, that is, $rx^2 - 1 = -3x/(2r)$. The third equation becomes $2x^3 = 4(rx^2 - 1) = -6x/r$, that is, $x^2 = -3/r$. Hence $rx^2 - 1 = -4$, so $x = 8r/3$ and $x^2 = 64r^2/9 = -3/r$. Thus $64r^3 = -27$, that is, $64 = -27$, a contradiction. This proves (1).

Step 2. In this step we define the scheme $X_{\mathbb{Z}_{(3)}}$ and the rank-drop locus $\mathcal{R}$ of $X_{\mathbb{Z}_{(3)}}$, and we show that the fiber $\mathcal{R}_{\mathbb{F}_3}$ is empty.

Let $\omega_1' := 2\omega_1$, $\omega_2' := \omega_2$, and $\omega_3' := 2\omega_3$. We use $\omega_1', \omega_2', \omega_3'$ because their reductions modulo 3 are the forms used in Appendix C. The Grassmannian $\mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)$ is smooth and projective over $\mathbb{Z}_{(3)}$ of relative dimension 12. Let $\mathcal{U}$ be the tautological subbundle of rank three on $\mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)$, let $\mathcal{E} := (\Lambda^2 \mathcal{U}^\vee)^{\oplus 3}$, a vector bundle of rank 9, and let $s := (\omega_1'|_{\mathcal{U}}, \omega_2'|_{\mathcal{U}}, \omega_3'|_{\mathcal{U}})$, a section of $\mathcal{E}$. Let $X_{\mathbb{Z}_{(3)}} \subset \mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)$ be the zero scheme of s. Then $X_{\mathbb{Z}_{(3)}}$ is projective over $\mathbb{Z}_{(3)}$. Since 2 is a unit in $\mathbb{Z}_{(3)}$, $X_{\mathbb{Z}_{(3)}}$ is also the zero scheme of $(\omega_1|_{\mathcal{U}}, \omega_2|_{\mathcal{U}}, \omega_3|_{\mathcal{U}})$, and $X_{\mathbb{Z}_{(3)}} \times_{\mathbb{Z}_{(3)}} \mathbb{C} = X$.

On $X_{\mathbb{Z}_{(3)}}$ the relative derivative

$$ds : T_{\mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)/\mathbb{Z}_{(3)}}|_{X_{\mathbb{Z}_{(3)}}} \to \mathcal{E}|_{X_{\mathbb{Z}_{(3)}}}$$

does not depend on the choice of a local trivialization of $\mathcal{E}$: a change of trivialization changes the derivative by a matrix multiplied by s, which vanishes on $X_{\mathbb{Z}_{(3)}}$. Let $\mathcal{R} \subset X_{\mathbb{Z}_{(3)}}$ be the closed subscheme where $\operatorname{rank} ds \leq 8$, defined by the 9×9 minors of ds. On each standard chart $\{P_I \neq 0\}$, the scheme $\mathcal{R}$ is defined by the nine local equations of $X_{\mathbb{Z}_{(3)}}$ and the maximal minors of their Jacobian matrix (see Appendix C). Let $\mathcal{R}_{\mathbb{F}_3} := \mathcal{R} \times_{\mathbb{Z}_{(3)}} \mathbb{F}_3$. By Lemma C.2 and the identities over $\mathbb{F}_3$ in Appendix C, $\mathcal{R}_{\mathbb{F}_3} = \emptyset$.

Step 3. In this step we show that $X_{\mathbb{Z}_{(3)}}$ is smooth over $\mathbb{Z}_{(3)}$ of relative dimension three and that X is non-empty.

The scheme $\mathcal{R}$ is a closed subscheme of $X_{\mathbb{Z}_{(3)}}$, so $\mathcal{R}$ is projective over $\mathbb{Z}_{(3)}$. A projective morphism of noetherian schemes is proper [Har77, Theorem II.4.9], so the image of $\mathcal{R}$ in $\operatorname{Spec} \mathbb{Z}_{(3)}$ is closed. By **Step 2**, the image of $\mathcal{R}$ does not contain the closed point. Since the closure of the generic point of $\operatorname{Spec} \mathbb{Z}_{(3)}$ contains the closed point, the image does not contain the generic point either. Thus $\mathcal{R} = \emptyset$, that is, ds has rank 9 at any point of $X_{\mathbb{Z}_{(3)}}$. Locally $X_{\mathbb{Z}_{(3)}}$ is defined by nine equations in $\mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)$, which is smooth of relative dimension 12, and the Jacobian matrix of the nine equations has rank 9 at any point of $X_{\mathbb{Z}_{(3)}}$. By the relative Jacobian criterion [BLR90, Section 2.2, Proposition 7], $X_{\mathbb{Z}_{(3)}}$ is smooth over $\mathbb{Z}_{(3)}$ of relative dimension $12 - 9 = 3$.

Dividing the first and the third spanning vectors of p by 160 and 14, the point p is spanned by the columns of the 7×3 matrix $U(p)$ with rows $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(-\frac{17}{40}, 0, 0)$, $(0, 1, 0)$, $(0, 0, -\frac{43}{14})$, and $(\frac{901}{160}, 0, 0)$. All entries of $U(p)$ belong to $\mathbb{Z}_{(3)}$, and a direct calculation shows $U(p)^t \omega_k U(p) = 0$ for $k = 1, 2, 3$. Thus p defines a $\mathbb{Z}_{(3)}$-point of $X_{\mathbb{Z}_{(3)}}$ with $P_{012}(p) = 1$. In particular X is a non-empty smooth threefold. This proves the second half of (2).

Step 4. In this step we prove the first half of (2).

By **Steps 1 and 3**, $\Pi \in \mathbb{B}^{\mathrm{nd}}$ and $X = X_\Pi$ is smooth of dimension three. By Proposition 16.5, $\Pi \in \Omega$, that is, X is a smooth prime Fano threefold of genus 12. In particular X is integral.

Since $\Pi \in \Omega$, Proposition 16.5(2) gives $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_X \cong \omega_X^{-1} = \mathcal{O}_X(-K_X)$. This proves (2).

Step 5. In this step we write the local equations of X at p and the local coordinates (t,x,y) of X at p. From now on (t,x,y) denote the local coordinates of X at p, and not the variables of f_1.

Any point of the chart $\{P_{012}\neq 0\}$ of $\mathrm{Gr}(3,V)$ is the span of the columns of a unique 7×3 matrix whose rows $0,1,2$ form the identity matrix. For $w=(w_0,\dots,w_{11})\in\mathbb{C}^{12}$ let

$$U(w):=\begin{pmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ -\frac{17}{40}+w_0 & w_1 & w_2\\ w_3 & 1+w_4 & w_5\\ w_6 & w_7 & -\frac{43}{14}+w_8\\ \frac{901}{160}+w_9 & w_{10} & w_{11}\end{pmatrix}. \tag{21.4}$$

Then $w\to[U(w)]$ is an isomorphism from $\mathbb{C}^{12}$ onto $\{P_{012}\neq 0\}$ sending 0 to p (**Step 3**). The coordinates $w_0,\dots,w_{11}$ are centered at p. For $k=1,2,3$ let $\rho_{3k-2}(w),\rho_{3k-1}(w),\rho_{3k}(w)$ be the entries $(0,1),(0,2),(1,2)$ of $U(w)^t\omega'_kU(w)$. Then $X\cap\{P_{012}\neq 0\}$ is defined by $\rho_1,\dots,\rho_9$, and a direct calculation shows

$$\begin{aligned}
\rho_1&=-20w_3-\tfrac{91}{10}w_7+12(w_0w_7-w_1w_6),\\
\rho_2&=\tfrac{400}{7}w_0-\tfrac{91}{10}w_8+12(w_0w_8-w_2w_6),\\
\rho_3&=\tfrac{400}{7}w_1+20w_5+12(w_1w_8-w_2w_7),\\
\rho_4&=-\tfrac{2223}{80}w_1-14w_6-\tfrac{91}{20}w_{10}+6(w_0w_{10}-w_1w_9+w_3w_7-w_4w_6),\\
\rho_5&=-\tfrac{2223}{80}w_2+\tfrac{130}{7}w_3-\tfrac{91}{20}w_{11}+6(w_0w_{11}-w_2w_9+w_3w_8-w_5w_6),\\
\rho_6&=\tfrac{130}{7}w_4+14w_8+6(w_1w_{11}-w_2w_{10}+w_4w_8-w_5w_7),\\
\rho_7&=-106w_0-\tfrac{221}{8}w_4-8w_9-94w_0w_4+94w_1w_3+12w_3w_{10}-12w_4w_9,\\
\rho_8&=-\tfrac{221}{8}w_5-20w_6-94w_0w_5+94w_2w_3+12w_3w_{11}-12w_5w_9,\\
\rho_9&=106w_2-20w_7+8w_{11}-94w_1w_5+94w_2w_4-12w_5w_{10}+12w_4w_{11}.
\end{aligned} \tag{21.5}$$

We write (21.5) as $(\rho_1,\dots,\rho_9)^t=Lw+\mathcal{Q}(w)$, where L is a 9×12 matrix with entries in $\mathbb{Q}$ and the entries of $\mathcal{Q}(w)$ are the quadratic parts of $\rho_1,\dots,\rho_9$. Let $L=(L_d\mid L_f)$, where L_d consists of the columns of $w_0,\dots,w_8$ and L_f consists of the columns of w_9,w_{10},w_{11}. The only non-zero entries of L_f are $-\frac{91}{20}$ (row 4, column w_{10}), $-\frac{91}{20}$ (row 5, column w_{11}), -8 (row 7, column w_9), and 8 (row 9, column w_{11}). After grouping the columns of L_d as (w_0,w_4,w_8), (w_1,w_5,w_6), (w_2,w_3,w_7) and the rows as $(2,6,7)$, $(3,4,8)$, $(1,5,9)$, the matrix L_d becomes block diagonal with 3×3 blocks of determinants 4186, -33215, and 29029, and the signs of the row permutation and the column permutation cancel. Thus

$$\det L_d=4186\cdot(-33215)\cdot 29029=-4036133811710\neq 0.$$

By the holomorphic implicit function theorem, there exist unique convergent power series $w_0,\dots,w_8$ in

$$t:=w_{11},\qquad x:=w_{10},\qquad y:=w_9,$$

vanishing at 0, such that near p the threefold X is the image of $(t,x,y)\to[U(w(t,x,y))]$. Thus (t,x,y) are local coordinates of X at p. The expansion of the minors of (21.4) on the rows i,j,k implies the following identities of analytic functions on X near p:

$$\begin{aligned}
&P_{012}=1,\qquad P_{015}=-\tfrac{43}{14}+w_8,\qquad P_{024}=-1-w_4,\\
&P_{036}=w_1t-w_2x,\qquad P_{123}=-\tfrac{17}{40}+w_0,\qquad P_{126}=\tfrac{901}{160}+y.
\end{aligned} \tag{21.6}$$

For $\lambda\in\mathbb{C}$ let $g_\lambda(t,x,y)$ be the value of ℓ_λ at $U(w(t,x,y))$. Since $P_{012}(U(w))=1$, g_λ is the local function of the section $\ell_\lambda|_X$ in the trivialization of $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ by P_{012} near p. In particular near p the divisor D is defined by $g_{\lambda_*}=0$. By (21.6), g_λ is a convergent power series in (t,x,y) depending linearly on λ.

Step 6. In this step we compute the terms of weight at most 6 of g_λ.

We assign to t, x, y the weights $1, 2, 3$. For a power series h in (t, x, y), let $\tau(h)$ be the sum of the terms of h of weight at most 6, and say that h has weight at least k if any monomial of h has weight at least k. We apply τ and the weight to vectors entrywise. Let $w_d := (w_0, \dots, w_8)^t$ and $w_f := (w_9, w_{10}, w_{11})^t = (y, x, t)^t$, where $w_0, \dots, w_8$ are the power series of **Step 5**. By (21.5),

$$w_d = -L_d^{-1}\big(L_f w_f + \mathcal{Q}(w_d, w_f)\big).$$

We show that $\tau(w_d)$ is the unique vector T of polynomials of weight at most 6 with zero constant terms such that

$$\tau\big(L_d T + L_f w_f + \mathcal{Q}(T, w_f)\big) = 0.$$

Since $\mathcal{Q}$ is quadratic and the entries of w_d have weight at least 1, the terms of weight greater than 6 of w_d contribute to $\mathcal{Q}(w_d, w_f)$ only in weight greater than 6, so $T = \tau(w_d)$ satisfies the equation. Let T and T' be two solutions, and assume that $T - T'$ has weight at least $k \geq 1$. Since $\mathcal{Q}(T, w_f) - \mathcal{Q}(T', w_f)$ is bilinear in $T - T'$ and $(T + T', 2w_f)$, whose entries have weight at least 1, it has weight at least $k + 1$. Thus $T - T' = -L_d^{-1}\tau\big(\mathcal{Q}(T, w_f) - \mathcal{Q}(T', w_f)\big)$ has weight at least $k + 1$. By induction, $T - T'$ has weight at least 7, so $T = T'$.

Applying the weight estimate for $\mathcal{Q}(T, w_f) - \mathcal{Q}(T', w_f)$ to $T = w^{(n)}$ and $T' = \tau(w_d)$, induction on n shows that the fixed-point iteration $w^{(0)} := -L_d^{-1} L_f w_f$ and

$$w^{(n+1)} := \tau\Big(- L_d^{-1}\big(L_f w_f + \mathcal{Q}(w^{(n)}, w_f)\big)\Big), \qquad n = 0, \dots, 6,$$

satisfies that $w^{(n)} - \tau(w_d)$ has weight at least $n+2$, so $w^{(7)} = \tau(w_d)$. The jets listed in Appendix C are $w^{(7)}$, and by the uniqueness above they are $\tau(w_0), \dots, \tau(w_8)$. Substituting them into (21.6), a direct calculation shows that the coefficients of the six Plücker coordinates in (21.6) at the monomials of weight at most 5 vanish except at 1, y, tx, and t^3, where they are given in Table 27.

TABLE 27. The coefficients of 1, y, tx, and t^3 in the six Plücker coordinates (21.6) on X near p.

	P_{012}	P_{015}	P_{024}	P_{036}	P_{123}	P_{126}
1	1	$-\frac{43}{14}$	-1	0	$-\frac{17}{40}$	$\frac{901}{160}$
y	0	$\frac{16000}{7889}$	$\frac{3200}{2093}$	0	$\frac{52}{161}$	1
tx	0	$-\frac{35804240}{183711143}$	$-\frac{450880}{3749207}$	$\frac{8880}{163009}$	$-\frac{97760}{3749207}$	0
t^3	0	$-\frac{21402982400}{11896582487251}$	$-\frac{145408000}{242787397699}$	$\frac{25600}{51999871}$	$-\frac{321817600}{2670661374689}$	0

The 4×6 matrix of Table 27 has rank 4, and a direct calculation shows that the kernel of the matrix is spanned by the coefficient vectors of ℓ_λ, $\lambda \in \mathbb{C}$, in the basis $(P_{012}, P_{015}, P_{024}, P_{036}, P_{123}, P_{126})$. Thus g_λ has weight at least 6 for any $\lambda \in \mathbb{C}$. A direct calculation from (21.6) and the jets listed in Appendix C shows

$$\tau(g_\lambda) = c_{y^2} y^2 + c_{txy} txy + c_{t^3y} t^3 y + c_{x^3} x^3 + c_{t^2x^2} t^2 x^2 + c_{t^4x} t^4 x + c_{t^6} t^6, \tag{21.7}$$

where

$$c_{y^2} = \frac{159840(211321\lambda+3172)}{5200504309}, \qquad c_{txy} = -\frac{864(9987403943\lambda+144320876)}{9315703372591},$$
$$c_{t^3y} = -\frac{184320(164823660111\lambda+3056315452)}{66358270362876292 57}, \qquad c_{x^3} = \frac{3978(608707\lambda-7076)}{210129477635},$$
$$c_{t^2x^2} = \frac{7488000(69342907139\lambda+980497652)}{15836239263939443041}, \qquad c_{t^4x} = \frac{2396160(1514746365466367\lambda+27602022724044)}{1128058080616419164306 6407},$$
$$c_{t^6} = \frac{197673615360000(314743\lambda+7076)}{7896406564314934150146 4849}.$$

Step 7. In this step we prove the properties of Φ and λ_* in the statement, and we show that the binary cubic attached to (21.7) at $\lambda = \lambda_*$ satisfies the condition $J = 0$ of (21.9) and has non-zero

discriminant. Here λ is the parameter of the pencil ℓ_λ, and $c_{y^2}, \dots, c_{t^6}$ are the coefficients of (21.7).

Let

$$\alpha_0 := -\frac{c_{x^3}}{c_{y^2}}, \quad \alpha_1 := \frac{c_{txy}^2}{4c_{y^2}^2} - \frac{c_{t^2x^2}}{c_{y^2}}, \quad \alpha_2 := \frac{c_{txy}c_{t^3y}}{2c_{y^2}^2} - \frac{c_{t^4x}}{c_{y^2}}, \quad \alpha_3 := \frac{c_{t^3y}^2}{4c_{y^2}^2} - \frac{c_{t^6}}{c_{y^2}}$$

in $\mathbb{Q}(\lambda)$. Completing the square in y in (21.7) implies

$$\frac{\tau(g_\lambda)}{c_{y^2}} = \left(y + \frac{c_{txy}tx + c_{t^3y}t^3}{2c_{y^2}}\right)^2 - \left(\alpha_0x^3 + \alpha_1t^2x^2 + \alpha_2t^4x + \alpha_3t^6\right). \tag{21.8}$$

Let

$$\operatorname{disc} := \alpha_1^2\alpha_2^2 - 4\alpha_0\alpha_2^3 - 4\alpha_1^3\alpha_3 - 27\alpha_0^2\alpha_3^2 + 18\alpha_0\alpha_1\alpha_2\alpha_3$$

be the discriminant of the binary cubic form $\alpha_0x^3 + \alpha_1x^2t^2 + \alpha_2xt^4 + \alpha_3t^6$ in the variables (x, t^2), and let

$$J := 27\alpha_0^2\alpha_3 - 9\alpha_0\alpha_1\alpha_2 + 2\alpha_1^3.$$

By **Step 8**, $J/(27\alpha_0^2)$ is the coefficient of t^6 in $\alpha_0x^3 + \alpha_1x^2t^2 + \alpha_2xt^4 + \alpha_3t^6$ after the substitution $x = x' - \frac{\alpha_1}{3\alpha_0}t^2$. Let

$$\Psi(\lambda) := 1785300708176201777\lambda^2 + 54485684902020328\lambda + 415712298600848.$$

A direct calculation shows

$$\begin{aligned} J &= \frac{7145409613482 01\Phi(\lambda)}{5523640076594481873333273454160289793500000(211321\lambda + 3172)^6}, \\ \operatorname{disc} &= -\frac{13082762544228458004 6848}{4748755266163073454980784197959389643964399646922833834975} \\ &\quad \cdot \frac{(421179\lambda - 3172)^2(9048127\lambda - 431636)^3\Psi(\lambda)}{(211321\lambda + 3172)^7}. \end{aligned} \tag{21.9}$$

Both identities are polynomial identities after multiplication by the denominators.

We show that Φ is irreducible over $\mathbb{Q}$. The leading coefficient of Φ is congruent to 1 modulo 19, so the reduction $\overline{\Phi} \in \mathbb{F}_{19}[\lambda]$ of Φ modulo 19 is monic of degree 6. The Frobenius remainder identities modulo 19 in Appendix C imply $\lambda^{19^6} \equiv \lambda$ modulo $\overline{\Phi}$, and the Bézout identities modulo 19 in Appendix C imply

$$\gcd(\overline{\Phi}, \lambda^{19^2} - \lambda) = \gcd(\overline{\Phi}, \lambda^{19^3} - \lambda) = 1.$$

For $n \geq 1$, the polynomial $\lambda^{19^n} - \lambda$ is the product of the monic irreducible polynomials over $\mathbb{F}_{19}$ whose degrees divide n [LN97, Theorem 3.20]. Thus any irreducible factor of $\overline{\Phi}$ has degree dividing 6, and degree not dividing 2 or 3. Hence $\overline{\Phi}$ is irreducible over $\mathbb{F}_{19}$. Since $\deg\overline{\Phi} = \deg\Phi$, the reduction criterion [Lan02, Chapter IV, Theorem 3.2] and the Gauss lemma [Lan02, Chapter IV, Theorem 2.1] imply that Φ is irreducible over $\mathbb{Q}$. In particular any root of Φ has degree six over $\mathbb{Q}$.

We show that Φ has exactly one root in $(1/102, 1/101)$ and that λ_* is a simple root of Φ. Let $\Phi_0 = \Phi, \Phi_1, \dots, \Phi_6$ be the Sturm chain in Appendix C: Φ_1 is a positive rational multiple of Φ', Φ_{i+1} is a positive rational multiple of the negative of the remainder of the Euclidean division of Φ_{i-1} by Φ_i for $1 \leq i \leq 5$, and $\Phi_6 = -1$. Multiplication by positive rational numbers does not change the numbers of sign variations. No Φ_i vanishes at $1/102$ or at $1/101$, and the sequences $(\Phi_i(1/102))_i$ and $(\Phi_i(1/101))_i$ have three and two sign variations respectively. By Sturm's theorem [BPR06, Theorem 2.50], Φ has exactly $3 - 2 = 1$ root in $(1/102, 1/101)$. Since Φ_6 is a non-zero constant, $\gcd(\Phi, \Phi') = 1$, so λ_* is a simple root.

We show that $c_{y^2}(\lambda_*)\alpha_0(\lambda_*)\operatorname{disc}(\lambda_*) \neq 0$ and $J(\lambda_*) = 0$. For $\lambda \in (1/102, 1/101)$ we have $211321\lambda + 3172 > 0$, $\Psi(\lambda) > 0$ since the coefficients of Ψ are positive, and

$$608707\lambda - 7076 < \tfrac{608707}{101} - 7076 < 0,$$
$$421179\lambda - 3172 > \tfrac{421179}{102} - 3172 > 0,$$
$$9048127\lambda - 431636 < \tfrac{9048127}{101} - 431636 < 0,$$

where the three right inequalities are equivalent to $608707 < 714676$, $421179 > 323544$, and $9048127 < 43595236$. Thus $c_{y^2}(\lambda_*) \neq 0$, $c_{x^3}(\lambda_*) \neq 0$, so $\alpha_0(\lambda_*) \neq 0$, and $\operatorname{disc}(\lambda_*) \neq 0$ by (21.9). Since $\Phi(\lambda_*) = 0$ and $211321\lambda_* + 3172 \neq 0$, (21.9) implies $J(\lambda_*) = 0$.

Step 8. In this step we prove (3). Recall from **Step 5** that near p the divisor D is defined by $g_{\lambda_*} = 0$, and from **Step 6** that $g_{\lambda_*} - \tau(g_{\lambda_*})$ has weight at least 7. In this step all functions of λ are evaluated at $\lambda = \lambda_*$.

We first reduce the weight-six part to $y^2 - x^3 - t^4x$. Let

$$u := \alpha_2 - \frac{\alpha_1^2}{3\alpha_0}, \qquad v := \alpha_3 - \frac{\alpha_1\alpha_2}{3\alpha_0} + \frac{2\alpha_1^3}{27\alpha_0^2} = \frac{J}{27\alpha_0^2} = 0.$$

Substituting $x = x' - \frac{\alpha_1}{3\alpha_0}t^2$ implies

$$\alpha_0x^3 + \alpha_1t^2x^2 + \alpha_2t^4x + \alpha_3t^6 = \alpha_0x'^3 + ut^4x' + vt^6 = \alpha_0x'^3 + ut^4x'.$$

The discriminant of a binary cubic form in (x, t^2) is invariant under the substitution $x = x' - \frac{\alpha_1}{3\alpha_0}t^2$, so $\operatorname{disc} = -4\alpha_0u^3 - 27\alpha_0^2v^2 = -4\alpha_0u^3$. By **Step 7**, $\operatorname{disc} \neq 0$, so $u \neq 0$. Let $\kappa, \nu \in \mathbb{C}$ be non-zero numbers with $\kappa^2 = u/\alpha_0$ and $\nu^2 = u\kappa$, so that $\alpha_0\kappa^3 = \nu^2$. Let

$$t'' := t, \qquad x'' := \frac{1}{\kappa}\Big(x + \frac{\alpha_1}{3\alpha_0}t^2\Big), \qquad y'' := \frac{1}{\nu}\Big(y + \frac{c_{txy}tx + c_{t^3y}t^3}{2c_{y^2}}\Big).$$

Each of t'', x'', y'' is a weighted homogeneous polynomial in (t, x, y) of weight $1, 2, 3$ respectively, and the linear part of $(t, x, y) \to (t'', x'', y'')$ is invertible, so (t'', x'', y'') are local coordinates of X at p, and a power series has weight at least 7 in (t, x, y) if and only if it has weight at least 7 in (t'', x'', y''). By (21.8),

$$\frac{g_{\lambda_*}}{c_{y^2}\nu^2} = h_0(t'', x'', y'') + \eta, \qquad h_0(t, x, y) := y^2 - x^3 - t^4x,$$

where η has weight at least 7 in (t'', x'', y'').

We compute the local algebra of h_0. The Jacobian ideal of h_0 is $I := (y, 3x^2 + t^4, t^3x)$. The ring $\mathbb{C}[t, x]/(3x^2 + t^4)$ is a free $\mathbb{C}[t]$-module with basis $1, x$, and multiplication by t^3x maps 1 to t^3x and x to $t^3x^2 = -\frac{1}{3}t^7$. Thus

$$\mathbb{C}[t, x, y]/I \cong \mathbb{C}[t, x]/(3x^2 + t^4, t^3x) \cong \mathbb{C}[t]/(t^7) \oplus \big(\mathbb{C}[t]/(t^3)\big)x,$$

with basis $1, t, t^2, t^3, t^4, t^5, t^6, x, tx, t^2x$. Since $t^7 \in I$ and $x^4 \equiv \frac{1}{9}t^8$ modulo I, the ideal I contains a power of (t, x, y), so $\mathbb{C}\{t, x, y\}/I\mathbb{C}\{t, x, y\} \cong \mathbb{C}[t, x, y]/I$. Thus the Milnor number of h_0 is 10, and h_0 is a non-degenerate quasihomogeneous polynomial of degree 6 for the weights $(1, 2, 3)$. The weights of the ten basis monomials are $0, 1, \dots, 6, 2, 3, 4$, all at most 6.

By the normal form theorem for semiquasihomogeneous functions [AGV85, Section 12.1, Theorem], $h_0 + \eta$ is equivalent to $h_0 + \sum_k c_ke_k$, where e_k runs through the basis monomials of the local algebra of h_0 of weight greater than 6. Since no such basis monomial exists, there exists a biholomorphic germ φ of $(\mathbb{C}^3, 0)$ with $(h_0 + \eta) \circ \varphi = h_0$. Since D is defined near p by $g_{\lambda_*} = c_{y^2}\nu^2(h_0 + \eta)$,

$$(D \ni p) \cong (\{y^2 = x^3 + t^4x\} \ni 0)$$

analytically, and the Milnor number of $(D \ni p)$ is 10.

We conclude the proof of (3). The constant term of g_{λ_*} vanishes, so $p \in D$. Since $c_{y^2}(\lambda_*)$ is non-zero, the section $\ell_{\lambda_*}|_X$ is non-zero. Since X is integral (**Step 4**), $D \geq 0$ is a Cartier

divisor with $\mathcal{O}_X(D) \cong \mathcal{O}_{\mathrm{Gr}(3,V)}(1)|_X \cong \mathcal{O}_X(-K_X)$, that is, $D \in |-K_X|$. The polynomial h_0 is quasihomogeneous of degree 6 for the weights $(1,2,3)$. The curve $\{y^2 = x^3 + t^4 x\} \subset \mathbb{P}(1,2,3)$ is smooth, since the partial derivatives $-4t^3x$, $-3x^2 - t^4$, and $2y$ have no common zero in $\mathbb{P}(1,2,3)$, and it does not contain the singular points $[0:1:0]$ and $[0:0:1]$ of $\mathbb{P}(1,2,3)$. On $\{t \neq 0\}$ the curve is the elliptic curve $y^2 = x^3 + x$, whose j-invariant is

$$j = 1728 \cdot \frac{4 \cdot 1^3}{4 \cdot 1^3 + 27 \cdot 0^2} = 1728.$$

Thus $(D \ni p)$ is a simple elliptic singularity of type $\tilde{E}_8$ with j-invariant 1728 [Sai74, Section 1]. This proves (3). □

22. Proofs of the main theorems

In this section, we prove the main theorems of the introduction. We first prove Theorems 1.2 and 1.3 and Corollary 1.4, and then Theorems 1.5, 1.7, and 1.8.

Proof of Theorem 1.2. Let F be a prime divisor over X, and let $h : X' \to X$ be a projective birational morphism from a smooth variety X' such that F is a prime divisor on X'. We show that $\beta_X(F) > 0$.

Step 1. In this step we reduce the positivity of $\beta_X(F)$ to lower bounds for $A_X(F)$.

Since X is smooth, $A_X(F) > 0$. Since X' is smooth, F is Cartier. Since h^*H is big and the big cone of X' is open, the volume of $h^*H - tF$ is positive for any sufficiently small $t \geq 0$. Hence $S_X(F) > 0$. Thus, if $c > 1$ and $k \geq 0$ are real numbers and $G \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on a higher model of X such that

$$A_X(F) \geq cS_X(F) + k \operatorname{mult}_F G,$$

then $\operatorname{mult}_F G \geq 0$ implies

$$\beta_X(F) \geq (c-1)S_X(F) > 0.$$

The center $\operatorname{center}_X F$ is a prime divisor, an integral curve, or a closed point of X. We treat the cases of a prime divisor, an integral curve, and a closed point in turn.

Step 2. In this step we assume that $D := \operatorname{center}_X F$ is a prime divisor on X.

Since X is normal, h is an isomorphism over the generic point of D. Thus F is the strict transform of D, and $\operatorname{ord}_F = \operatorname{ord}_D$. By Lemma 4.2, $D \sim aH$ for an integer $a \geq 1$, and

$$\beta_X(F) = A_X(D) - S_X(D) = 1 - \frac{1}{4a} \geq \frac{3}{4} > 0.$$

Step 3. In this step we assume that $C := \operatorname{center}_X F$ is an integral curve.

Since H is ample and Cartier, $H \cdot C$ is a positive integer. If $H \cdot C \geq 2$, then $A_X(F) > S_X(F)$ by Proposition 4.6. If $H \cdot C = 1$, then C is a line, because the morphism defined by $|H|$ is an embedding (Lemma 2.12(1)). Since $\operatorname{center}_X F = C$, Corollary 5.10 implies $A_X(F) > S_X(F)$.

Step 4. In this step we assume that $\operatorname{center}_X F = \{x\}$ for a closed point $x \in X$.

By Proposition 4.1, the incidence type of x is one of $\emptyset$, 1, $1+1$, $1+1+1$, 2, $2+1$, and 3. We treat the seven types in turn. By **Step 1**, in each type it suffices to prove a lower bound of the form $A_X(F) \geq cS_X(F) + k \operatorname{mult}_F G$ with $c > 1$ and $k \geq 0$, or to prove $\beta_X(F) > 0$ directly.

Type $\emptyset$. By Proposition 4.4, $\delta_x(X) \geq \frac{33}{32}$. Thus $A_X(F) \geq \frac{33}{32} S_X(F)$.

Type 1. By Proposition 6.1, $\delta_x(X) \geq \frac{1944}{1903}$. Thus $A_X(F) \geq \frac{1944}{1903} S_X(F)$.

Type $1+1$. By Theorem 6.5(1), there exists a line L through x such that, for the exceptional divisor E of $\mathrm{Bl}_L X \to X$ in Set-up 5.1,

$$A_X(F) \geq \frac{216}{215} S_X(F) + \frac{127}{215} \operatorname{mult}_F E.$$

Type $1+1+1$**.** Let E be the exceptional divisor of the blow-up $\mathrm{Bl}_x X \to X$ of the point x. By Theorem 6.18 with $p = x$,

$$A_X(F) \geq S_X(F) + \frac{3}{4}\,\mathrm{mult}_F E.$$

Since $\mathrm{center}_X F = \{x\}$ and $\mathfrak{m}_x \mathcal{O}_{\mathrm{Bl}_x X} = \mathcal{O}_{\mathrm{Bl}_x X}(-E)$, we have $\mathrm{mult}_F E = \mathrm{ord}_F(\mathfrak{m}_x) > 0$. Thus

$$\beta_X(F) \geq \frac{3}{4}\,\mathrm{mult}_F E > 0.$$

Type 2**.** Let L be the line supporting $\Sigma(X, x)$. By Theorem 7.2, L is special. Recall from Construction-Proposition 7.1 that $\hat{X} \to X$ is the blow-up of the unique non-reduced conic supported on L, and that P is the reduced exceptional divisor of $\hat{X} \to X$. By Theorem 7.2, whose statement includes $\mathrm{mult}_F P > 0$ for $\mathrm{center}_X F = \{x\}$,

$$\beta_X(F) \geq \frac{27}{95}\,\mathrm{mult}_F P > 0.$$

Type $2+1$**.** By Proposition 4.1, $\Sigma(X, x)$ consists of the reduced point $[L]$ of a non-special line L and a curvilinear double point at a special line M. Recall from Set-up 5.1, applied to L, that the exceptional divisor E of $\mathrm{Bl}_L X \to X$ is isomorphic to $\mathbb{F}_1$. Let n be the width of M in Proposition 5.3. By Proposition 5.3, $n \in \{2, 3, 4\}$.

If $n \in \{2, 3\}$, then Propositions 8.8 and 8.9 respectively give

$$A_X(F) \geq \frac{3360}{3323} S_X(F) + \frac{5798}{9969}\,\mathrm{mult}_F E.$$

Assume that $n = 4$. By Lemma 8.3, the configuration is that of Lemma 8.2. The lines through x are L and M. Moreover, L is non-special, M is special, $\Sigma(X, x)$ has length two at $[M]$, and M has width four in the link from L. Thus x satisfies the hypotheses of Construction-Proposition 8.10 with $p = x$. Let P be the exceptional divisor of the plt extraction at x with weights $(2, 3, 2)$ of Construction-Proposition 8.10 with $p = x$ (denoted E there). By Theorem 8.12 with $p = x$,

$$S_X(F) \leq \frac{71}{72} A_X(F) - \frac{95}{72}\,\mathrm{mult}_F P.$$

Thus

$$\beta_X(F) \geq \frac{1}{72} A_X(F) + \frac{95}{72}\,\mathrm{mult}_F P > 0.$$

Type 3**.** By Lemma 9.1, x has a normal form. We fix one. It is in case (A) or in case (B).

In case (A), let $E \cong \mathbb{P}(1, 2, 3)$ be the plt-type divisor over X of Construction-Proposition 9.2(A). Then X, x, and E are as in Set-up 9.3, and Theorem 9.9 gives

$$S_X(F) \leq \frac{47}{48} A_X(F) - \frac{5}{8}\,\mathrm{mult}_F E.$$

Thus

$$\beta_X(F) \geq \frac{1}{48} A_X(F) + \frac{5}{8}\,\mathrm{mult}_F E > 0.$$

In case (B), recall that E_0 is the plt-type divisor of Construction-Proposition 9.12, extracted by the weighted blow-up at x with weights $(1, 2, 3)$ with respect to the regular parameters (i, a, e), and that $E_0 \cong \mathbb{P}(1, 2, 3)$ and $A_X(E_0) = 6$. By Theorem 9.13,

$$S_X(F) \leq \frac{8}{9} A_X(F) - \frac{5}{12}\,\mathrm{mult}_F E_0.$$

Thus

$$\beta_X(F) \geq \frac{1}{9} A_X(F) + \frac{5}{12}\,\mathrm{mult}_F E_0 > 0.$$

Step 5. In this step we conclude the proof.

By **Steps 2–4**, $\beta_X(F) > 0$ for any prime divisor F over X. Since X is smooth, X is a Fano variety with klt singularities. By the valuative criterion of Fujita and Li [Fuj19, Theorems 1.4

and 1.6], [Li17], see also [BX19], in the form of Theorem 2.5(2), X is K-stable. By Theorem 2.5(4), X is uniformly K-stable, and by Theorem 2.5(3), $\delta(X) > 1$. □

Proof of Theorem 1.3. By Definition-Theorem 2.20, for any linear isomorphism $V_X \cong V$, we have $\Pi_X \in \Omega$ and $X \cong X_{\Pi_X}$, and the orbit $\mathrm{SL}(V) \cdot \Pi_X$ depends only on X. In particular, the condition $\Pi_X \in \Omega^{\mathrm{st}}$ depends only on X. Thus Theorems 17.5 and 16.12 apply to $\Pi = \Pi_X$, with $X_\Pi \cong X$. In particular, X is K-semistable by Theorem 17.5.

(1) By Theorem 17.5, the conditions (a), (b), (c), (d), and (e) are equivalent. By Theorem 19.1(1), $\Pi_X \in \Omega^{\mathrm{st}}$ if and only if any special line on X intersects another line on X, that is, (d) and (f) are equivalent. If (a)–(f) hold, then $\Pi_X \in \Omega^{\mathrm{st}}$, and $A_X(F) > S_X(F)$ for any prime divisor F over X by Theorem 17.5. Assume that (a)–(f) fail. If $\mathrm{SL}(V) \cdot \Pi_X$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, then $\delta(X) = 1$ by Theorem 17.5. If $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$, then the stabilizer of Π_X in $\mathrm{SL}(V)$ is infinite since $\Pi_X \notin \Omega^{\mathrm{st}}$. Since the stabilizer is a closed subgroup of $\mathrm{SL}(V)$, it is of positive dimension, and $\delta(X) = 1$ by Theorem 17.5.

(2) Since $H = -K_X$, Theorem 18.1 implies that (i), (ii), and (iii) are equivalent, and that for D and p as in (i), the line through p is unique, special, and intersects no other line on X. By Theorem 16.12(3), (iv) and (v) are equivalent. By Proposition 16.17(4) with $\Pi = \Pi_X$, (v) implies (ii). Conversely, if (ii) holds, then some special line on X intersects no other line on X, so $\Pi_X \notin \Omega^{\mathrm{st}}$ by Theorem 19.1(1). Thus (ii) and (v) are equivalent.

(3) By Theorem 17.5, X is K-polystable if and only if $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

Assume that X admits a faithful $\mathbb{G}_m$-action. Then X is K-polystable by Theorem 2.26, so $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ by Theorem 17.5. Assume that there exists no pair (D, p) as in (2)(i). Then (2)(i) fails, so (2)(v) fails by (2), that is, $\Pi_X \in \Omega^{\mathrm{st}}$. In particular, $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$.

Conversely, assume that $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$. If the stabilizer of Π_X in $\mathrm{SL}(V)$ is finite, then $\Pi_X \in \Omega^{\mathrm{st}}$, so (2)(v) fails, and there exists no pair (D, p) as in (2)(i) by (2). If the stabilizer of Π_X in $\mathrm{SL}(V)$ is of positive dimension, then X admits a faithful $\mathbb{G}_m$-action by Theorem 17.5.

Assume that $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Aut}(X)$ is infinite. By Theorem 16.12(2), there exists an exact sequence

$$1 \to \mu_7 \to \mathrm{Stab}_{\mathrm{SL}(V)}(\Pi_X) \to \mathrm{Aut}(X) \to 1.$$

Thus $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi_X)$ is infinite, hence of positive dimension, and Theorem 17.5 implies that X is K-polystable, not K-stable, and $\delta(X) = 1$. If $\mathrm{SL}(V) \cdot \Pi_X$ is not closed in $\mathbb{B}^{\mathrm{nd}}$, then X is K-semistable, not K-polystable, and $\delta(X) = 1$ by Theorem 17.5. □

Proof of Corollary 1.4. Let A be the ample generator of $\mathrm{Pic}(X)$, let r be the positive integer such that $-K_X \sim rA$, and let $d := A^3$. By the classification of smooth Fano threefolds of Picard number one [IP99, Section 12.2, Theorem 3.3.1, and Remarks 3.3.2] (see also [KPS18, Section 1.1, Tables 1 and 2]), $\mathrm{Pic}(X) = \mathbb{Z}[A]$ and one of the following holds:

(1) $r = 4$ and $X \cong \mathbb{P}^3$,
(2) $r = 3$ and X is isomorphic to a smooth quadric threefold Q^3,
(3) $r = 2$ and $1 \le d \le 5$, and $X \cong V_5$ if $d = 5$, or
(4) $r = 1$ and $d \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 22\}$.

Assume that $r = 2$ and $d \le 4$, or that $r = 1$ and $d \le 16$. Then $X \ncong \mathbb{P}^3, Q^3, V_5$ and $(-K_X)^3 = r^3 d \in \{8, 16, 24, 32\} \cup \{2, 4, \dots, 16\}$, so $(-K_X)^3 \ne 18, 22$. Hence X is uniformly K-stable by [AZ23, Theorem C]. Thus $\delta(X) > 1$ by Theorem 2.5(3), and X is K-semistable by Theorem 2.5(1).

Assume that X is isomorphic to $\mathbb{P}^3$, Q^3, or V_5. Then X is K-polystable by [ACC+23, Section 3.2] for $\mathbb{P}^3$ and Q^3 and by [ACC+23, Example 3.2] for V_5 (see also [CS09, Theorems 1.10 and 1.17] and [Ber16, Theorem 1.1]). In particular, X is K-semistable.

Assume that $r = 1$ and $d \in \{18, 22\}$. Then $-K_X \sim A$ and $\mathrm{Pic}(X) = \mathbb{Z}[-K_X]$, so X is a smooth prime Fano threefold of genus g with $2g - 2 = d$, that is, $g = 10$ or $g = 12$. If $g = 10$,

then X is K-stable by Theorem 1.2, so X is K-semistable by Theorem 2.5(1) and (2). If $g = 12$, then X is K-semistable by Theorem 1.3.

Thus any smooth Fano threefold of Picard number one is K-semistable, which answers the question of Xu and Zhuang [XZ26, Problem 10]. □

Proof of Theorem 1.5. For $M \in \mathrm{GL}_3$ and a ternary form c, write $c \circ M$ for the form $w \to c(Mw)$. The Aronhold invariant $\mathfrak{A}$ is an SL_3-invariant of degree four on ternary cubics, and $\mathfrak{A}(\lambda^3 c) = \lambda^{12}\mathfrak{A}(c)$ for $\lambda \in \mathbb{C}^*$. Since GL_3 is generated by SL_3 and the scalar matrices, $\mathfrak{A}(c \circ M) = \det(M)^4\mathfrak{A}(c)$ for any ternary cubic c and any $M \in \mathrm{GL}_3$. By the chain rule,

$$\sum_i a_i \partial_i (f \circ M) = \Big(\sum_i (Ma)_i \partial_i f\Big) \circ M$$

for $f \in \mathbb{C}[x, y, z]_4$ and $a \in \mathbb{C}^3$. Thus

$$\mathrm{Sc}(f \circ M) = \det(M)^4 \,\mathrm{Sc}(f) \circ M \quad \text{for any } f \in \mathbb{C}[x, y, z]_4 \text{ and } M \in \mathrm{GL}_3\,. \tag{22.1}$$

In particular, if f and f' are projectively equivalent, then so are $\mathrm{Sc}(f)$ and $\mathrm{Sc}(f')$. For a smooth prime Fano threefold X of genus 12, a basis of Π_X allows us to regard f_X and $\mathrm{Sc}(f_X)$ as elements of $\mathbb{C}[x, y, z]_4$, and by (22.1) if another basis differs from the first by $g \in \mathrm{PGL}_3$, it replaces $[f_X]$ and $[\mathrm{Sc}(f_X)]$ by $g \cdot [f_X]$ and $g \cdot [\mathrm{Sc}(f_X)]$. Hence the GIT-semistability, GIT-polystability, and GIT-stability of f_X and of Q_X do not depend on the choice of basis of Π_X, and they are the notions used in Theorem 19.1.

Let X be a smooth prime Fano threefold of genus 12. By Definition-Theorem 2.20, $\Pi_X \in \Omega$ and $X \cong X_{\Pi_X}$.

(1) By Theorem 16.11(1), the middle catalecticant $\mathrm{Cat}_2(f_X)$ is invertible. By Theorem 16.11(2), the scheme $\Sigma(X)$ is isomorphic to Q_X. By Theorem 19.1(1), f_X and Q_X are GIT-semistable.

(2) Since $X \cong X_{\Pi_X}$, by Theorem 17.5 with $\Pi = \Pi_X$, the threefold X is K-stable if and only if $\Pi_X \in \Omega^{\mathrm{st}}$, and X is K-polystable if and only if $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$. By Theorem 19.1(1), $\Pi_X \in \Omega^{\mathrm{st}}$ if and only if f_X is GIT-stable, if and only if Q_X is GIT-stable. By Theorem 19.1(2), $\mathrm{SL}(V) \cdot \Pi_X$ is closed in $\mathbb{B}^{\mathrm{nd}}$ if and only if f_X is GIT-polystable, if and only if Q_X is GIT-polystable. By Theorem 1.3(1) and (3), K-stability and K-polystability of X are equivalent to the conditions listed in Theorem 1.3(1) and Theorem 1.3(3), respectively. This proves the first two sentences of (2).

By Theorem 17.1, X is K-semistable, and by (1), f_X and Q_X are GIT-semistable. Thus X is K-semistable but not K-polystable if and only if X is not K-polystable, if and only if f_X, equivalently Q_X, is not GIT-polystable, if and only if f_X, equivalently Q_X, is GIT-semistable but not GIT-polystable. Since K-stability implies K-polystability and GIT-stability implies GIT-polystability, the first two sentences of (2) also imply that X is K-polystable but not K-stable if and only if f_X, equivalently Q_X, is GIT-polystable but not GIT-stable.

(3) Let $[f] \in U^{\mathrm{sm}}$. By Definition 2.27, the condition $(f^\perp)_2 = 0$ means that f is non-degenerate, equivalently $\mathrm{Cat}_2(f)$ is invertible. By Proposition 16.10(1), $\mathrm{VPS}(f, 6)$ is a smooth prime Fano threefold of genus 12 and $\mathrm{VSP}(f, 6) = \mathrm{VPS}(f, 6)$.

Let $M \in \mathrm{GL}_3$. Then $(f \circ M)^\perp$ is the image of $f^\perp$ under the automorphism of $\mathbb{C}[\partial]$ induced by M, so the automorphism of Hilb^6 of the dual plane induced by M maps $\mathrm{VPS}(f, 6)$ isomorphically onto $\mathrm{VPS}(f \circ M, 6)$. Hence sending $[f]$ to $\mathrm{VPS}(f, 6)$ induces a well-defined map from the set of PGL_3-orbits in U^{sm} to the set of isomorphism classes of smooth prime Fano threefolds of genus 12.

To see that the map is surjective, let X be a smooth prime Fano threefold of genus 12. By Proposition 16.10(3), $X \cong \mathrm{VSP}(f_X, 6)$. By (1), $\mathrm{Cat}_2(f_X)$ is invertible, so f_X is non-degenerate. By Proposition 16.10(2), $\mathrm{VSP}(f_X, 6) = \mathrm{VPS}(f_X, 6)$. Thus $\mathrm{VPS}(f_X, 6) \cong X$ is smooth, $[f_X] \in U^{\mathrm{sm}}$, and the orbit of $[f_X]$ maps to the class of X.

To see that the map is injective, let $[f], [f'] \in U^{\mathrm{sm}}$ with $\mathrm{VPS}(f, 6) \cong \mathrm{VPS}(f', 6)$, and let $X := \mathrm{VPS}(f, 6)$. By Definition 2.27, $[f], [f'] \in U^{\mathrm{cat}}$, so by Lemma 16.3(2), there exist $\Pi, \Pi' \in \mathbb{B}^{\mathrm{nd}}$

with bases in which f_Π and $f_{\Pi'}$ are non-zero multiples of f and f', and $X_\Pi \cong \mathrm{VPS}(f,6)$ and $X_{\Pi'} \cong \mathrm{VPS}(f',6)$. Since $[f],[f'] \in U^{\mathrm{sm}}$, the schemes X_Π and $X_{\Pi'}$ are smooth of dimension three by Proposition 16.10(1), so $\Pi,\Pi' \in \Omega$ by Proposition 16.5(1), and Π and Π' are the canonical Mukai nets of X_Π and $X_{\Pi'}$ by Theorem 16.12(1). Since $X_\Pi \cong X_{\Pi'}$, Definition-Theorem 2.20 implies $\Pi' \in \mathrm{SL}(V)\cdot\Pi$, and Lemma 16.3(3) implies that f and f' are projectively equivalent. Thus $[f'] \in \mathrm{PGL}_3 \cdot [f]$. Since the orbit of $[f_X]$ maps to the class of X, the inverse map sends X to the orbit of $[f_X]$.

Finally, let $[f] \in U^{\mathrm{sm}}$ and $X := \mathrm{VPS}(f,6)$. By Proposition 16.10(3) and (2), $\mathrm{VPS}(f_X,6) \cong X$, and $[f_X] \in U^{\mathrm{sm}}$ as shown above. Applying the previous paragraph to f and f_X, we see that f is projectively equivalent to f_X, and by (22.1), $\mathrm{Sc}(f)$ is projectively equivalent to $\mathrm{Sc}(f_X)$. In particular $\mathrm{Sc}(f) \neq 0$ by Theorem 16.11(1). Invertibility of the middle catalecticant and the GIT notions are invariant under projective equivalence, and $\Sigma(X) \cong (\mathrm{Sc}(f) = 0)$ by Theorem 16.11(4). Hence (1) and (2) hold for X with f and $\mathrm{Sc}(f)$ in place of f_X and $\mathrm{Sc}(f_X)$.

(4) Statement (4) follows from Theorem 19.1(3) and Proposition 16.17(1) and (3).

(5) Assume first that X admits a faithful $\mathbb{G}_m$-action. By Theorem 16.12(2) applied to $\Pi = \Pi_X$, the group $\mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi_X) \cong \mathrm{Aut}(X)$ contains a non-trivial torus. By Definition-Proposition 16.14, there exist $h \in \mathbb{C}\backslash\{1\}$ and $g \in \mathrm{PGL}(V)$ with $g\cdot\Pi_X = \Pi_h$. The automorphism of $\mathrm{Gr}(3,V)$ induced by g maps X_{Π_X} isomorphically onto X_{Π_h}, so X_{Π_h} is smooth, and $h \neq 2$ by Definition-Proposition 16.14. By Lemma 16.3(3), $f_X = f_{\Pi_X}$ is projectively equivalent to f_{Π_h}, which is F_h in the basis of Definition-Proposition 16.14.

Conversely, let $h \in \mathbb{C}\backslash\{1,2\}$. By Definition-Proposition 16.14, X_{Π_h} is a smooth prime Fano threefold of genus 12, the apolar quartic of Π_h is F_h, and the projective stabilizer of Π_h contains a copy of $\mathbb{G}_m$. By Proposition 16.5, $\Pi_h \in \Omega$. By Theorem 16.12(1), Π_h is the canonical Mukai net of X_{Π_h}, so $f_{X_{\Pi_h}}$ is projectively equivalent to F_h. By Theorem 16.12(2), $\mathrm{Aut}(X_{\Pi_h}) \cong \mathrm{Stab}_{\mathrm{PGL}(V)}(\Pi_h)$, so X_{Π_h} admits a faithful $\mathbb{G}_m$-action.

By Proposition 16.16, F_h and $F_{h'}$ are projectively equivalent only if $h = h'$. By Definition-Proposition 16.14, $X_{\Pi_6} \cong X^{\mathrm{MU}}$. This proves (5). □

Proof of Theorem 1.7. We first prove the following claim.

Claim 22.1. *Let $[f] \in U^{\mathrm{sm}}$. Then there exists $\Pi \in \Omega$ with a basis in which f_Π is a non-zero multiple of f, and $X_\Pi \cong \mathrm{VPS}(f,6)$. Moreover, $\mathrm{VPS}(f,6)$ is K-polystable if and only if $[f]$ is GIT-polystable.*

Proof. By Definition 2.27, $U^{\mathrm{sm}} \subset U^{\mathrm{cat}}$. By Lemma 16.3(2), there exists $\Pi \in \mathbb{B}^{\mathrm{nd}}$ with a basis in which f_Π is a non-zero multiple of f, and $\mathrm{VPS}(f,6) \cong X_\Pi$. By Proposition 16.10(1), X_Π is a smooth prime Fano threefold of genus 12, so $\Pi \in \Omega$ by Proposition 16.5. By Theorem 17.5, X_Π is K-polystable if and only if $\mathrm{SL}(V)\cdot\Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$, and by Proposition 20.11, $\mathrm{SL}(V)\cdot\Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ if and only if $[f_\Pi] = [f]$ is GIT-polystable. □

Proof of Theorem 1.7 continued. (1) The isomorphisms $\mathrm{SO}(q) \cong \mathrm{PSL}_2$ and $\mathrm{Harm}_4 \cong \mathrm{Sym}^8\mathbb{C}^2$ in the statement are proved in Subsection 20.3. By Theorem 20.14(1), Harm_4° is an $\mathrm{SO}(q)$-invariant open subset containing 0, $\mathrm{Harm}_4^\circ = \varpi^{-1}(\varpi(\mathrm{Harm}_4^\circ))$, and $\varpi(\mathrm{Harm}_4^\circ)$ is open. Since $\mathrm{SO}(q) \cong \mathrm{PSL}_2$ is linearly reductive and Harm_4 is affine, $[\mathrm{Harm}_4 / \mathrm{SO}(q)] \to \mathrm{Harm}_4 /\!\!/ \mathrm{SO}(q)$ is a good moduli space [Alp13, Theorem 13.2]. Since Harm_4° is saturated, the restriction $[\mathrm{Harm}_4^\circ / \mathrm{SO}(q)] \to \varpi(\mathrm{Harm}_4^\circ)$ is a good moduli space [Alp13, Remark 6.2], that is, $\varpi(\mathrm{Harm}_4^\circ) = \mathrm{Harm}_4^\circ /\!\!/ \mathrm{SO}(q)$.

(2) Statement (2) is Theorem 20.14(2).

(3) By (2), any X_b is a smooth prime Fano threefold of genus 12, so X_b is K-semistable by Theorem 17.1. Let X be a smooth K-polystable prime Fano threefold of genus 12. By Theorem 1.5(3), $[f_X] \in U^{\mathrm{sm}}$ and $\mathrm{VPS}(f_X,6) \cong X$. By Claim 22.1, $[f_X]$ is GIT-polystable. By Theorem 20.14(3), there exists $b \in \mathrm{Harm}_4^\circ$ whose $\mathrm{SO}(q)$-orbit is closed in Harm_4 such that $[f_b]$ is projectively equivalent to $[f_X]$. By Theorem 1.5(3), $X_b = \mathrm{VPS}(f_b,6) \cong \mathrm{VPS}(f_X,6) \cong X$.

(4) By Theorem 20.14(3), sending b to $[f_b]$ induces a surjective morphism $\rho : \varpi(\mathrm{Harm}_4^\circ) \to U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$. By Theorem 20.14(1), U^{sm} is a saturated invariant open subset of the GIT-semistable

locus, so $U^{\mathrm{sm}} \to U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$ is a good quotient [Alp13, Theorem 13.6 and Remark 6.2], and the closed points of $U^{\mathrm{sm}} /\!\!/ \mathrm{PGL}_3$ correspond bijectively to the PGL_3-orbits closed in U^{sm} [Alp13, Theorem 4.16(iv)]. Let $O \subset U^{\mathrm{sm}}$ be an orbit closed in U^{sm}. The unique closed orbit in the closure of O in the GIT-semistable locus has image in the quotient equal to the image of O, so it is contained in U^{sm} by saturation, and hence it is O. Thus the orbits closed in U^{sm} are exactly the GIT-polystable orbits in U^{sm}. By Theorem 1.5(3) and Claim 22.1, sending $[f]$ to $\mathrm{VPS}(f, 6)$ induces a bijection from the set of GIT-polystable orbits in U^{sm} onto the set of isomorphism classes of smooth K-polystable prime Fano threefolds of genus 12. This proves (4).

(5) By [KLPZ26, Theorem 2.25], with [ABHLX20, Corollary 1.2] for the existence of a separated good moduli space and [LXZ22, Theorem 1.3 and Remark 1.5] for properness and projectivity, the K-semistable $\mathbb{Q}$-Fano threefolds of anti-canonical volume 22 form an Artin stack $\mathcal{M}^{\mathrm{K}}_{3,22}$ of finite type with a projective good moduli space $\mathcal{M}^{\mathrm{K}}_{3,22} \to M^{\mathrm{K}}_{3,22}$, and the $\mathbb{C}$-points of $M^{\mathrm{K}}_{3,22}$ correspond bijectively to the K-polystable $\mathbb{Q}$-Fano threefolds of anti-canonical volume 22. For a K-semistable $\mathbb{Q}$-Fano threefold Y of anti-canonical volume 22, write $[Y] \in M^{\mathrm{K}}_{3,22}$ for the image of the point of $\mathcal{M}^{\mathrm{K}}_{3,22}$ defined by Y. If Y is K-polystable, then $[Y]$ is the point corresponding to Y. For a smooth projective morphism $\mathcal{X} \to B$ of relative dimension three with $\mathcal{X}$ integral, $\Omega_{\mathcal{X}/B}$ is locally free of rank three, so $\omega_{\mathcal{X}/B} = \wedge^3 \Omega_{\mathcal{X}/B}$ is invertible and commutes with base change [Har77, Example III.10.0.2 and Proposition II.8.10]. Thus a smooth projective morphism with integral total space whose fibers are K-semistable Fano threefolds of anti-canonical volume 22 is a family as in [KLPZ26, Theorem 2.25], and it induces a morphism from the base to $\mathcal{M}^{\mathrm{K}}_{3,22}$ and hence to $M^{\mathrm{K}}_{3,22}$. A smooth projective morphism $\mathcal{X} \to B$ with B smooth and irreducible and with connected fibers has $\mathcal{X}$ smooth [Har77, Proposition III.10.1(c)] and connected, since the morphism is open [Har77, Exercise III.9.1] and closed, so $\mathcal{X}$ is integral.

By Proposition 16.5, the universal isotropic scheme over Ω is a smooth projective morphism with fibers X_Π, and $\mathcal{O}_{\mathrm{Gr}(3,V)}(1)$ restricts to $\omega^{-1}_{X_\Pi}$ on each fiber (Proposition 16.5(2)). Since Ω is open in $\mathbb{B}$ and the fibers X_Π are connected, the universal isotropic scheme over Ω is integral. By Theorem 17.5, any X_Π with $\Pi \in \Omega$ is K-semistable. Hence we obtain a morphism $\kappa_\Omega : \Omega \to M^{\mathrm{K}}_{3,22}$ with $\kappa_\Omega(\Pi) = [X_\Pi]$. For $g \in \mathrm{SL}(V)$, the automorphism of $\mathrm{Gr}(3, V)$ induced by g maps X_Π isomorphically onto $X_{g\cdot\Pi}$, so κ_Ω is constant on $\mathrm{SL}(V)$-orbits. Since the fibers of κ_Ω over closed points are closed in Ω, κ_Ω is constant on the closure in Ω of any $\mathrm{SL}(V)$-orbit in Ω.

By (2), the threefolds X_b, $b \in \mathrm{Harm}^\circ_4$, form an $\mathrm{SO}(q)$-equivariant smooth projective family polarized by the relative anti-canonical bundle. Since Harm°_4 is open in Harm_4 and the fibers X_b are connected, the total space of the family is integral. By (3) the fibers are K-semistable. Hence the family induces an $\mathrm{SO}(q)$-invariant morphism $\mathrm{Harm}^\circ_4 \to M^{\mathrm{K}}_{3,22}$ sending b to $[X_b]$, that is, a morphism $[\mathrm{Harm}^\circ_4 / \mathrm{SO}(q)] \to M^{\mathrm{K}}_{3,22}$. By (1) and [Alp13, Theorem 6.6], the morphism factors uniquely through the good moduli space $[\mathrm{Harm}^\circ_4 / \mathrm{SO}(q)] \to \varpi(\mathrm{Harm}^\circ_4)$. Thus we obtain a morphism $\kappa : \varpi(\mathrm{Harm}^\circ_4) \to M^{\mathrm{K}}_{3,22}$ with $\kappa(\varpi(b)) = [X_b]$ for $b \in \mathrm{Harm}^\circ_4$.

Fix $b \in \mathrm{Harm}^\circ_4$. By Theorem 20.14(3), there exists a GIT-polystable $[f_{\mathrm{cl}}] \in U^{\mathrm{sm}}$, unique up to PGL_3, in the closure of $\mathrm{PGL}_3 \cdot [f_b]$. Let $X_{\mathrm{cl}} := \mathrm{VPS}(f_{\mathrm{cl}}, 6)$. By Claim 22.1, there exist $\Pi_b, \Pi_{\mathrm{cl}} \in \Omega$ with bases in which f_{Π_b} and $f_{\Pi_{\mathrm{cl}}}$ are non-zero multiples of f_b and f_{cl}, respectively, and with $X_{\Pi_b} \cong X_b$ and $X_{\Pi_{\mathrm{cl}}} \cong X_{\mathrm{cl}}$. Since $[f_{\mathrm{cl}}] \in U^{\mathrm{cat}}$, the point $[f_{\Pi_{\mathrm{cl}}}]$ belongs to the closure in U^{cat} of $\mathrm{PGL}_3 \cdot [f_{\Pi_b}]$. By Proposition 20.11 and since $\Pi_{\mathrm{cl}} \in \Omega$, Π_{cl} belongs to the closure of $\mathrm{SL}(V) \cdot \Pi_b$ in Ω. Therefore

$$\kappa(\varpi(b)) = [X_b] = \kappa_\Omega(\Pi_b) = \kappa_\Omega(\Pi_{\mathrm{cl}}) = [X_{\mathrm{cl}}].$$

By Claim 22.1, X_{cl} is K-polystable. Since Π_{cl} belongs to the closure of $\mathrm{SL}(V) \cdot \Pi_b$ in Ω, the threefold $X_{\mathrm{cl}} \cong X_{\Pi_{\mathrm{cl}}}$ is a degeneration of $X_b \cong X_{\Pi_b}$ in the family over Ω, and since X_{cl} is K-polystable and $[X_{\mathrm{cl}}] = [X_b]$, it is the K-polystable degeneration of X_b. Thus $\kappa(\varpi(b))$ is the point of $M^{\mathrm{K}}_{3,22}$ corresponding to $X_{\mathrm{cl}} = \mathrm{VPS}(f_{\mathrm{cl}}, 6)$, which is a smooth prime Fano threefold of genus 12 by Theorem 1.5(3). By (3), $X_{\mathrm{cl}} \cong X_{b'}$ for some $b' \in \mathrm{Harm}^\circ_4$ whose $\mathrm{SO}(q)$-orbit is closed, so $\kappa(\varpi(b))$ is the class of $X_{b'}$.

It remains to determine the image of κ on $\mathbb{C}$-points. Since ϖ is surjective on $\mathbb{C}$-points, the previous paragraph shows that any $\mathbb{C}$-point in the image of κ corresponds to a smooth K-polystable prime Fano threefold of genus 12. Conversely, let X be a smooth K-polystable prime Fano threefold of genus 12. By (3), $X \cong X_{b'}$ for some $b' \in \mathrm{Harm}_4^\circ$. Then $\kappa(\varpi(b')) = [X_{b'}]$ is the point corresponding to X, since X is K-polystable. This proves (5). □

Proof of Theorem 1.8. Statements (1), (2), and (3), except the last sentence of (3), are Construction-Proposition 21.1(1)(2)(3). In particular, $\det \mathrm{Cat}_2(f_1) = 216320 \neq 0$. By (1) and (2), $\Pi \in \mathbb{B}^{\mathrm{nd}}$ and $X = X_\Pi$ is smooth of dimension three, so $\Pi \in \Omega$ by Proposition 16.5. Since $D \in |-K_X|$ and the germ $(D \ni p)$ is analytically isomorphic to $(\{y^2 = x^3 + t^4 x\} \ni 0)$, we have $\Pi \in \Delta'$ by Definition 2.17.

(4) By (1), f_Π is a non-zero multiple of the smooth quartic f_1, so $[f_\Pi]$ is GIT-stable by Theorem 2.29. By Theorem 16.12(6), $\mathrm{SL}(V) \cdot \Pi$ is closed in $\mathbb{B}^{\mathrm{nd}}$ and $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$ is finite. Thus $\Pi \in \Omega^{\mathrm{st}}$. By Theorem 16.12(2), $\mathrm{Aut}(X)$ is the image of the finite group $\mathrm{Stab}_{\mathrm{SL}(V)}(\Pi)$, so $\mathrm{Aut}(X)$ is finite. By Theorem 17.5, $\delta(X) > 1$, and X is uniformly K-stable and K-stable. Since $\mathbb{G}_m$ is connected and $\mathrm{Aut}(X)$ is finite, any homomorphism of algebraic groups $\mathbb{G}_m \to \mathrm{Aut}(X)$ is trivial. Hence X admits no non-trivial $\mathbb{C}^*$-action, and in particular no faithful $\mathbb{G}_m$-action.

Finally, $\Pi \in \Delta'$ by (3), X admits no non-trivial $\mathbb{C}^*$-action by (4), and X is K-polystable since it is K-stable by (4). Thus X contradicts Conjecture 1.1. □

Appendix A. Tables for genus ten

We list the Zariski decompositions and the values of the integrals that the proofs in Sections 6–9 use.

A.1. **Three-line points.**

Remark A.1. The tables of this subsection are given in the notation of Set-up 6.6, Set-up 6.13, Lemma 6.14, and Definition 6.16, under condition (C). Let X be a smooth prime Fano threefold of genus 10 and $p \in X$ a closed point such that $\Sigma(X, p)$ consists of three distinct reduced points. By Lemma 6.7, X and p are isomorphic to a codimension two linear section and the point $[h]$ as in Set-up 6.6, and condition (C) holds. Thus the tables cover all cases of Theorem 6.18.

Lemma 6.14 divides the surfaces $\widetilde{E}$ into six configurations. This subsection gives the tables for the generic, the conic, and the edge configurations. The tables for the star, the path, and the cycle configurations follow them.

Remark A.2. Notation as in Definition 6.16. Fix a configuration and an irreducible curve $C \subset \widetilde{E}$ with $C^2 < 0$. The region of the points (t, y) with $0 \le t \le 4$, $y \ge 0$, and $P_t - yC$ big is divided into polygons, and the table for C has one row for each polygon. The first column lists the vertices of the polygon in order, the second column lists $N_{t,y}$, and the third column lists $(P_{t,y} \cdot C)$, where

$$P_{t,y} = P_t - yC - N_{t,y}.$$

On each polygon, the coefficients of $N_{t,y}$ and the number $(P_{t,y} \cdot C)$ are affine functions of (t, y). Each polygon is contained in one of the strips $0 \le t \le 1$, $1 \le t \le 3$, and $3 \le t \le 4$, and P_t is affine in t on each of them. The intersection matrix of the basis $(\eta, e_1, e_2, e_3, e_1', e_2', e_3')$ of the classes on $\widetilde{E}$ is the diagonal matrix with diagonal entries $1, -1, -1, -1, -1, -1, -1$. Thus on each polygon the integrands

$$P_{t,y}^2, \qquad (P_{t,y} \cdot C)^2, \qquad (P_{t,y} \cdot C)\,\mathrm{mult}_D\, N_{t,y}$$

of Definition 6.16 are linear combinations of products of two affine functions of (t, y), and the integrals of Definition 6.16 are the sums of the integrals over the polygons of the table, with the factors $\frac{1}{6}$, $\frac{1}{6}$, and $\frac{1}{3}$. The last row of the table for C lists S_C, θ_C, and $\kappa_{C,D}$ for the curves D of the configuration with $(C \cdot D) > 0$ and $\kappa_{C,D} \neq 0$.

In each row of the tables in this subsection, $N_{t,y}$ has non-negative coefficients, the support of $N_{t,y}$ has negative definite intersection matrix, and $P_{t,y}$ is orthogonal to every curve in the support. At every vertex of the polygon of the row, $P_{t,y}^2 \geq 0$, $(-K_{\widetilde{E}} \cdot P_{t,y}) \geq 0$, and $P_{t,y}$ has non-negative intersection with every curve in the list of Lemma 6.14 present in the configuration. On the polygon, $P_{t,y}$ is affine in (t, y), so $(P_{t,y} \cdot C') \geq 0$ at any point of the polygon for any curve C' of the list. Since $(-K_{\widetilde{E}})^2 = 3 > 0$ (Construction-Proposition 6.10(2)), the Hodge index theorem implies that the set of classes P with $P^2 \geq 0$ and $(-K_{\widetilde{E}} \cdot P) \geq 0$ is the closure of the connected component of $\{P : P^2 > 0\}$ containing $-K_{\widetilde{E}}$. Hence the set is convex and any two classes in the set have non-negative intersection, so the set contains $P_{t,y}$ for any (t, y) in the polygon. Since $-K_{\widetilde{E}}$ is nef (Construction-Proposition 6.10(2)), any irreducible curve of non-negative self-intersection belongs to the set, so $P_{t,y}$ has non-negative intersection with it. By Lemma 6.14, every irreducible curve of negative self-intersection is in the list. Hence $P_{t,y}$ is nef on the polygon, so that $P_{t,y} + N_{t,y}$ is the Zariski decomposition of $P_t - yC$ there. For each t the vertical sections of the polygons of the table for C cover the interval on which $P_t - yC$ is big. Each polygon is triangulated from the first vertex, and the integral of the product of two affine functions l, l' over a triangle with vertices x_1, x_2, x_3 and area V is

$$\frac{V}{12}\Big(\big(\sum_i l(x_i)\big)\big(\sum_i l'(x_i)\big) + \sum_i l(x_i)l'(x_i)\Big).$$

All values are exact rational numbers.

The generic configuration.

Remark A.3. By Lemma 6.14, in the generic configuration the irreducible curves $C \subset \widetilde{E}$ with $C^2 < 0$ are the fifteen curves e'_j, R_j, $L_{jj'}$, T_j, and $\mathcal{C}_j$, with the classes in Table 28. We give the tables for one curve in each orbit of the permutations of $\{1, 2, 3\}$ preserving the arrows (see the proof of Theorem 6.18): e'_1, R_1, L_{12}, T_1, and $\mathcal{C}_1$.

TABLE 28. The classes of the curves with negative self-intersection in the generic configuration, in the basis $(\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3)$.

curve	class	curve	class
e'_1	$(0,0,0,0,1,0,0)$	e'_2	$(0,0,0,0,0,1,0)$
e'_3	$(0,0,0,0,0,0,1)$	L_{12}	$(1,-1,-1,0,0,0,0)$
L_{13}	$(1,-1,0,-1,0,0,0)$	L_{23}	$(1,0,-1,-1,0,0,0)$
T_1	$(1,-1,0,0,-1,0,0)$	T_2	$(1,0,-1,0,0,-1,0)$
T_3	$(1,0,0,-1,0,0,-1)$	$\mathcal{C}_1$	$(2,-1,-1,-1,0,-1,-1)$
$\mathcal{C}_2$	$(2,-1,-1,-1,-1,0,-1)$	$\mathcal{C}_3$	$(2,-1,-1,-1,-1,-1,0)$
R_1	$(0,1,0,0,-1,0,0)$	R_2	$(0,0,1,0,0,-1,0)$
R_3	$(0,0,0,1,0,0,-1)$		

The conic configuration.

Remark A.4. By Lemma 6.14, in the conic configuration the irreducible curves $C \subset \widetilde{E}$ with $C^2 < 0$ are the thirteen curves e'_j, R_j, $L_{jj'}$, T_j, and $\mathcal{C}_0$, with the classes in Table 34. We give the tables for one curve in each orbit of the permutations of $\{1, 2, 3\}$ preserving the arrows (see the proof of Theorem 6.18): e'_1, R_1, L_{12}, T_1, and $\mathcal{C}_0$.

The edge configuration.

Remark A.5. In the edge configuration the arrow is $1 \to 2$. By Lemma 6.14, the irreducible curves $C \subset \widetilde{E}$ with $C^2 < 0$ are the twelve curves e'_j, R_j, L_{13}, L_{23}, T_2, T_3, $\mathcal{C}_1$, and D_{12}, with the classes in Table 40. A permutation σ of the indices maps D_{12} to $D_{\sigma(1)\sigma(2)}$, which belongs to Table 40 only if σ is the identity. Thus we give the tables for all twelve curves.

TABLE 29. The Zariski decompositions of $P_t - yC$ for $C = e_1'$ in the generic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(-t+y)T_1 + \frac{y}{2}R_1$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(3,1)$, $(3,\frac{4}{3})$, $(1,2)$	$(y-1)T_1 + (t+2y-5)\mathcal{C}_2 + (t+2y-5)\mathcal{C}_3 + \frac{y}{2}R_1 + (\frac{t}{2}+y-\frac{5}{2})R_2 + (\frac{t}{2}+y-\frac{5}{2})R_3$	$-\frac{3t}{2} - \frac{9y}{2} + \frac{21}{2}$
$(3,1)$, $(1,2)$, $(1,1)$	$(y-1)T_1 + \frac{y}{2}R_1$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(1,0)$, $(3,0)$, $(3,1)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(4,0)$, $(3,\frac{4}{3})$, $(3,1)$	$(t+y-4)T_1 + (2t+2y-8)\mathcal{C}_2 + (2t+2y-8)\mathcal{C}_3 + \frac{y}{2}R_1 + (t+y-4)R_2 + (t+y-4)R_3$	$-6t - \frac{9y}{2} + 24$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1$	$-t + \frac{y}{2} + 4$
$S_C = \frac{3}{4}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,T_1} = \frac{1}{12}$, $\kappa_{C,\mathcal{C}_2} = \frac{1}{72}$, $\kappa_{C,\mathcal{C}_3} = \frac{1}{72}$, $\kappa_{C,R_1} = \frac{3}{8}$.		

TABLE 30. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the generic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye_1'	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e_1'$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(-\frac{t}{2}+y+\frac{1}{2})e_1' + (2y-2)L_{12} + (2y-2)L_{13} + (\frac{t}{2}+y-\frac{5}{2})\mathcal{C}_1 + (y-1)R_2 + (y-1)R_3$	$-4y + 6$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e_1' + (2y-2)L_{12} + (2y-2)L_{13} + (y-1)R_2 + (y-1)R_3$	$\frac{t}{2} - 3y + \frac{7}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(t+y-4)e_1' + (2t+2y-8)L_{12} + (2t+2y-8)L_{13} + (t+y-4)\mathcal{C}_1 + (t+y-4)R_2 + (t+y-4)R_3$	$-6t - 4y + 24$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e_1'} = \frac{19}{72}$, $\kappa_{C,L_{12}} = \frac{1}{12}$, $\kappa_{C,L_{13}} = \frac{1}{12}$, $\kappa_{C,\mathcal{C}_1} = \frac{1}{72}$.		

TABLE 31. The Zariski decompositions of $P_t - yC$ for $C = L_{12}$ in the generic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye_1' + ye_2' + yR_1 + yR_2$	$t - y$
$(2,1)$, $(3,2)$, $(1,1)$	$(-t+y+1)e_1' + (-t+y+1)e_2' + (y-1)T_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t - 2y + 1$
$(1,0)$, $(2,1)$, $(1,1)$	$(-t+y+1)e_1' + (-t+y+1)e_2' + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t - y$
$(3,1)$, $(3,2)$, $(2,1)$	$(y-1)T_3 + \frac{y}{2}R_1 + \frac{y}{2}R_2$	$-y + 2$
$(1,0)$, $(3,0)$, $(3,1)$, $(2,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_2$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_3 + \frac{y}{2}R_1 + \frac{y}{2}R_2$	$-2t - y + 8$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_2$	$-t + 4$
$S_C = \frac{11}{18}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,T_3} = \frac{1}{18}$, $\kappa_{C,R_1} = \frac{1}{3}$, $\kappa_{C,R_2} = \frac{1}{3}$.		

TABLE 32. The Zariski decompositions of $P_t - yC$ for $C = T_1$ in the generic configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_1 + yR_1$	$t - y$
$(1,0)$, $(3,0)$, $(3,1)$	0	$y + 1$
$(1,0)$, $(3,1)$, $(1,1)$	$(-t + 2y + 1)e'_1 + (-\frac{t}{2} + y + \frac{1}{2})R_1$	$t - y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$-t + y + 4$
$S_C = \frac{17}{36}$, $\theta_C = \frac{11}{18}$, $\kappa_{C,e'_1} = \frac{7}{36}$.		

TABLE 33. The Zariski decompositions of $P_t - yC$ for $C = \mathcal{C}_1$ in the generic configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,\frac{1}{2})$	$ye'_1 + 2ye'_2 + 2ye'_3 + yR_1 + yR_2 + yR_3$	$2t - 4y$
$(1,0)$, $(2,1)$, $(1,\frac{1}{2})$	$(-t + y + 1)e'_1 + (-t + 2y + 1)e'_2 + (-t + 2y + 1)e'_3 + (-\frac{t}{2} + y + \frac{1}{2})R_1 + (-\frac{t}{2} + y + \frac{1}{2})R_2 + (-\frac{t}{2} + y + \frac{1}{2})R_3$	$2t - 4y$
$(3,1)$, $(3,\frac{4}{3})$, $(2,1)$	$(-t + 2y + 1)e'_2 + (-t + 2y + 1)e'_3 + (y - 1)T_1 + \frac{y}{2}R_1 + (-\frac{t}{2} + y + \frac{1}{2})R_2 + (-\frac{t}{2} + y + \frac{1}{2})R_3$	$\frac{3t}{2} - \frac{9y}{2} + \frac{3}{2}$
$(1,0)$, $(3,1)$, $(2,1)$	$(-t + 2y + 1)e'_2 + (-t + 2y + 1)e'_3 + \frac{y}{2}R_1 + (-\frac{t}{2} + y + \frac{1}{2})R_2 + (-\frac{t}{2} + y + \frac{1}{2})R_3$	$\frac{3t}{2} - \frac{7y}{2} + \frac{1}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_1$	$-\frac{t}{2} + \frac{y}{2} + \frac{5}{2}$
$(4,0)$, $(3,\frac{4}{3})$, $(3,1)$	$(2t + 2y - 8)e'_2 + (2t + 2y - 8)e'_3 + (t + y - 4)T_1 + \frac{y}{2}R_1 + (t + y - 4)R_2 + (t + y - 4)R_3$	$-6t - \frac{9y}{2} + 24$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1$	$-t + \frac{y}{2} + 4$
$S_C = \frac{91}{216}$, $\theta_C = \frac{47}{72}$, $\kappa_{C,e'_2} = \frac{23}{216}$, $\kappa_{C,e'_3} = \frac{23}{216}$, $\kappa_{C,T_1} = \frac{1}{216}$, $\kappa_{C,R_1} = \frac{47}{216}$.		

TABLE 34. The classes of the curves with negative self-intersection in the conic configuration, in the basis $(\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3)$.

curve	class	curve	class
e'_1	$(0,0,0,0,1,0,0)$	e'_2	$(0,0,0,0,0,1,0)$
e'_3	$(0,0,0,0,0,0,1)$	L_{12}	$(1,-1,-1,0,0,0,0)$
L_{13}	$(1,-1,0,-1,0,0,0)$	L_{23}	$(1,0,-1,-1,0,0,0)$
T_1	$(1,-1,0,0,-1,0,0)$	T_2	$(1,0,-1,0,0,-1,0)$
T_3	$(1,0,0,-1,0,0,-1)$	R_1	$(0,1,0,0,-1,0,0)$
R_2	$(0,0,1,0,0,-1,0)$	R_3	$(0,0,0,1,0,0,-1)$
$\mathcal{C}_0$	$(2,-1,-1,-1,-1,-1,-1)$		

TABLE 35. The Zariski decompositions of $P_t - yC$ for $C = e_1'$ in the conic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y}\cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(-t+y)T_1 + \frac{y}{2}R_1$	$t-\frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(2,1)$, $(1,2)$, $(1,1)$	$(y-1)T_1 + \frac{y}{2}R_1$	$\frac{t}{2}-\frac{y}{2}+\frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,2)$, $(2,1)$	$(y-1)T_1 + \frac{y}{2}R_1 + (\frac{t}{2}+\frac{y}{2}-\frac{3}{2})\mathcal{C}_0$	$-y+2$
$(1,0)$, $(3,0)$, $(2,1)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2}+\frac{y}{2}-\frac{1}{2}$
$(3,0)$, $(3,1)$, $(2,1)$	$\frac{y}{2}R_1 + (\frac{t}{2}+\frac{y}{2}-\frac{3}{2})\mathcal{C}_0$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_1 + \frac{y}{2}R_1 + \frac{y}{2}\mathcal{C}_0$	$-2t-y+8$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1 + \frac{y}{2}\mathcal{C}_0$	$-t+4$
$S_C = \frac{29}{36}$, $\theta_C = \frac{25}{72}$, $\kappa_{C,T_1} = \frac{1}{8}$, $\kappa_{C,R_1} = \frac{29}{72}$, $\kappa_{C,\mathcal{C}_0} = \frac{11}{72}$.		

TABLE 36. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the conic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y}\cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye_1'	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e_1'$	$\frac{t}{2}+y-\frac{1}{2}$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e_1' + (2y-2)L_{12} + (2y-2)L_{13} + (y-1)R_2 + (y-1)R_3$	$\frac{t}{2}-3y+\frac{7}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(2y-2)e_1' + (2y-2)L_{12} + (2y-2)L_{13} + (y-1)R_2 + (y-1)R_3 + (\frac{t}{2}+y-\frac{5}{2})\mathcal{C}_0$	$-4y+6$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(2t+2y-8)e_1' + (2t+2y-8)L_{12} + (2t+2y-8)L_{13} + (t+y-4)R_2 + (t+y-4)R_3 + (t+y-4)\mathcal{C}_0$	$-6t-4y+24$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e_1'} = \frac{5}{18}$, $\kappa_{C,L_{12}} = \frac{1}{12}$, $\kappa_{C,L_{13}} = \frac{1}{12}$.		

TABLE 37. The Zariski decompositions of $P_t - yC$ for $C = L_{12}$ in the conic configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y}\cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye_1' + ye_2' + yR_1 + yR_2$	$t-y$
$(2,1)$, $(3,2)$, $(1,1)$	$(-t+y+1)e_1' + (-t+y+1)e_2' + (y-1)T_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t-2y+1$
$(1,0)$, $(2,1)$, $(1,1)$	$(-t+y+1)e_1' + (-t+y+1)e_2' + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t-y$
$(3,1)$, $(3,2)$, $(2,1)$	$(y-1)T_3 + \frac{y}{2}R_1 + \frac{y}{2}R_2$	$-y+2$
$(1,0)$, $(3,0)$, $(3,1)$, $(2,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_2$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_3 + \frac{y}{2}R_1 + \frac{y}{2}R_2$	$-2t-y+8$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_2$	$-t+4$
$S_C = \frac{11}{18}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,T_3} = \frac{1}{18}$, $\kappa_{C,R_1} = \frac{1}{3}$, $\kappa_{C,R_2} = \frac{1}{3}$.		

TABLE 38. The Zariski decompositions of $P_t - yC$ for $C = T_1$ in the conic configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_1 + yR_1$	$t - y$
$(1,0)$, $(3,0)$, $(3,1)$	0	$y + 1$
$(1,0)$, $(3,1)$, $(1,1)$	$(-t + 2y + 1)e'_1 + (-\frac{t}{2} + y + \frac{1}{2})R_1$	$t - y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$-t + y + 4$
$S_C = \frac{17}{36}$, $\theta_C = \frac{11}{18}$, $\kappa_{C,e'_1} = \frac{7}{36}$.		

TABLE 39. The Zariski decompositions of $P_t - yC$ for $C = \mathcal{C}_0$ in the conic configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,\frac{1}{2})$	$2ye'_1 + 2ye'_2 + 2ye'_3 + yR_1 + yR_2 + yR_3$	$2t - 4y$
$(1,0)$, $(3,0)$, $(3,1)$	0	$-t + 2y + 3$
$(1,0)$, $(3,1)$, $(3,\frac{3}{2})$, $(1,\frac{1}{2})$	$(-t + 2y + 1)e'_1 + (-t + 2y + 1)e'_2 + (-t + 2y + 1)e'_3 + (-\frac{t}{2} + y + \frac{1}{2})R_1 + (-\frac{t}{2} + y + \frac{1}{2})R_2 + (-\frac{t}{2} + y + \frac{1}{2})R_3$	$2t - 4y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(2t + 2y - 8)e'_1 + (2t + 2y - 8)e'_2 + (2t + 2y - 8)e'_3 + (t + y - 4)R_1 + (t + y - 4)R_2 + (t + y - 4)R_3$	$-6t - 4y + 24$
$S_C = \frac{1}{2}$, $\theta_C = \frac{2}{3}$, $\kappa_{C,e'_1} = \frac{5}{36}$, $\kappa_{C,e'_2} = \frac{5}{36}$, $\kappa_{C,e'_3} = \frac{5}{36}$.		

TABLE 40. The classes of the curves with negative self-intersection in the edge configuration, in the basis $(\eta, e_1, e_2, e_3, e'_1, e'_2, e'_3)$.

curve	class	curve	class
e'_1	$(0,0,0,0,1,0,0)$	e'_2	$(0,0,0,0,0,1,0)$
e'_3	$(0,0,0,0,0,0,1)$	L_{13}	$(1,-1,0,-1,0,0,0)$
L_{23}	$(1,0,-1,-1,0,0,0)$	T_2	$(1,0,-1,0,0,-1,0)$
T_3	$(1,0,0,-1,0,0,-1)$	$\mathcal{C}_1$	$(2,-1,-1,-1,0,-1,-1)$
R_1	$(0,1,0,0,-1,0,0)$	R_2	$(0,0,1,0,0,-1,0)$
R_3	$(0,0,0,1,0,0,-1)$	D_{12}	$(1,-1,-1,0,-1,0,0)$

TABLE 41. The Zariski decompositions of $P_t - yC$ for $C = e'_1$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(-t + y)e'_2 + \frac{y}{2}R_1 + (-t + y)R_2 + (-t + y)D_{12}$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(2,2)$, $(1,2)$, $(1,1)$	$(-t + y)e'_2 + \frac{y}{2}R_1 + (-\frac{t}{2} + y - \frac{1}{2})R_2 + (y - 1)D_{12}$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,\frac{3}{2})$, $(3,2)$, $(2,2)$	$(t + 2y - 6)T_3 + \frac{y}{2}R_1 + (\frac{t}{2} + y - \frac{5}{2})R_2 + (t + 2y - 5)D_{12}$	$-\frac{t}{2} - \frac{3y}{2} + \frac{9}{2}$
$(1,0)$, $(3,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,\frac{3}{2})$, $(2,2)$, $(1,1)$	$\frac{y}{2}R_1 + (\frac{t}{6} + \frac{y}{3} - \frac{1}{2})R_2 + (\frac{t}{3} + \frac{2y}{3} - 1)D_{12}$	$\frac{t}{6} - \frac{y}{6} + \frac{1}{2}$
$(4,0)$, $(3,2)$, $(3,\frac{3}{2})$	$(3t + 2y - 12)T_3 + \frac{y}{2}R_1 + (t + y - 4)R_2 + (2t + 2y - 8)D_{12}$	$-3t - \frac{3y}{2} + 12$
$(3,0)$, $(4,0)$, $(3,\frac{3}{2})$	$\frac{y}{2}R_1 + \frac{y}{3}R_2 + \frac{2y}{3}D_{12}$	$-t - \frac{y}{6} + 4$
$S_C = \frac{7}{8}$, $\theta_C = \frac{5}{16}$, $\kappa_{C,R_1} = \frac{7}{16}$, $\kappa_{C,D_{12}} = \frac{55}{144}$.		

TABLE 42. The Zariski decompositions of $P_t - yC$ for $C = e'_2$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(-t+y)T_2 + \frac{y}{2}R_2$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{y}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(1,2)$	$(y-1)T_2 + (t+2y-5)\mathcal{C}_1 + (\frac{t}{2}+y-\frac{5}{2})R_1 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{y}{3}-1)D_{12}$	$-\frac{2t}{3} - \frac{8y}{3} + 6$
$(2,1)$, $(1,2)$, $(1,1)$	$(y-1)T_2 + \frac{y}{2}R_2$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,1)$, $(1,2)$, $(2,1)$	$(y-1)T_2 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{y}{3}-1)D_{12}$	$\frac{t}{3} - \frac{2y}{3} + 1$
$(1,0)$, $(3,0)$, $(2,1)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,1)$, $(2,1)$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{y}{3}-1)D_{12}$	$\frac{t}{3} + \frac{y}{3}$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(t+y-4)T_2 + (2t+2y-8)\mathcal{C}_1 + (t+y-4)R_1 + \frac{2y}{3}R_2 + \frac{y}{3}D_{12}$	$-4t - \frac{8y}{3} + 16$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{2y}{3}R_2 + \frac{y}{3}D_{12}$	$-t + \frac{y}{3} + 4$
$S_C = \frac{55}{72}$, $\theta_C = \frac{85}{216}$, $\kappa_{C,T_2} = \frac{5}{54}$, $\kappa_{C,\mathcal{C}_1} = \frac{1}{36}$, $\kappa_{C,R_2} = \frac{23}{54}$.		

TABLE 43. The Zariski decompositions of $P_t - yC$ for $C = e'_3$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(-t+y)T_3 + \frac{y}{2}R_3$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_3$	$\frac{y}{2}$
$(3,1)$, $(3,\frac{4}{3})$, $(1,2)$	$(t+3y-6)T_3 + (t+2y-5)\mathcal{C}_1 + (\frac{t}{2}+y-\frac{5}{2})R_1 + (\frac{t}{2}+y-\frac{5}{2})R_2 + \frac{y}{2}R_3 + (t+2y-5)D_{12}$	$-\frac{3t}{2} - \frac{9y}{2} + \frac{21}{2}$
$(3,1)$, $(1,2)$, $(1,1)$	$(y-1)T_3 + \frac{y}{2}R_3$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(1,0)$, $(3,0)$, $(3,1)$, $(1,1)$	$\frac{y}{2}R_3$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(4,0)$, $(3,\frac{4}{3})$, $(3,1)$	$(3t+3y-12)T_3 + (2t+2y-8)\mathcal{C}_1 + (t+y-4)R_1 + (t+y-4)R_2 + \frac{y}{2}R_3 + (2t+2y-8)D_{12}$	$-6t - \frac{9y}{2} + 24$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_3$	$-t + \frac{y}{2} + 4$
$S_C = \frac{3}{4}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,T_3} = \frac{7}{72}$, $\kappa_{C,\mathcal{C}_1} = \frac{1}{72}$, $\kappa_{C,R_3} = \frac{3}{8}$.		

TABLE 44. The Zariski decompositions of $P_t - yC$ for $C = L_{13}$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye'_1 + ye'_3 + yR_1 + yR_3$	$t - y$
$(2,1)$, $(3,2)$, $(1,1)$	$(-t+y+1)e'_1 + (-t+y+1)e'_3 + (y-1)T_2 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$t - 2y + 1$
$(1,0)$, $(2,1)$, $(1,1)$	$(-t+y+1)e'_1 + (-t+y+1)e'_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$t - y$
$(3,1)$, $(3,2)$, $(2,1)$	$(y-1)T_2 + \frac{y}{2}R_1 + \frac{y}{2}R_3$	$-y + 2$
$(1,0)$, $(3,0)$, $(3,1)$, $(2,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_3$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_2 + \frac{y}{2}R_1 + \frac{y}{2}R_3$	$-2t - y + 8$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_3$	$-t + 4$
$S_C = \frac{11}{18}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,T_2} = \frac{1}{18}$, $\kappa_{C,R_1} = \frac{1}{3}$, $\kappa_{C,R_3} = \frac{1}{3}$.		

TABLE 45. The Zariski decompositions of $P_t - yC$ for $C = L_{23}$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye'_2 + ye'_3 + yR_2 + yR_3$	$t - y$
$(1,0)$, $(2,1)$, $(1,1)$	$(-t+y+1)e'_2 + (-t+y+1)e'_3 + (-\frac{t}{2}+y+\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$t - y$
$(3,2)$, $(1,1)$, $(2,1)$	$(-t+2y)e'_2 + (-t+y+1)e'_3 + (-\frac{t}{2}+2y-\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3 + (y-1)D_{12}$	$t - 2y + 1$
$(3,\frac{3}{2})$, $(3,2)$, $(2,1)$	$(-t+2y)e'_2 + (-\frac{t}{2}+2y-\frac{1}{2})R_2 + \frac{y}{2}R_3 + (y-1)D_{12}$	$\frac{t}{2} - \frac{3y}{2} + \frac{3}{2}$
$(1,0)$, $(3,0)$, $(2,1)$	$\frac{y}{2}R_2 + \frac{y}{2}R_3$	1
$(3,0)$, $(3,\frac{3}{2})$, $(2,1)$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + \frac{y}{2}R_3 + (\frac{t}{3}+\frac{y}{3}-1)D_{12}$	$-\frac{t}{6} - \frac{y}{6} + \frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,\frac{3}{2})$	$(3t+2y-12)e'_2 + (2t+2y-8)R_2 + \frac{y}{2}R_3 + (t+y-4)D_{12}$	$-3t - \frac{3y}{2} + 12$
$(3,0)$, $(4,0)$, $(3,\frac{3}{2})$	$\frac{2y}{3}R_2 + \frac{y}{2}R_3 + \frac{y}{3}D_{12}$	$-t - \frac{y}{6} + 4$
$S_C = \frac{5}{8}$, $\theta_C = \frac{59}{144}$, $\kappa_{C,R_2} = \frac{59}{144}$, $\kappa_{C,R_3} = \frac{49}{144}$.		

TABLE 46. The Zariski decompositions of $P_t - yC$ for $C = T_2$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_2 + yR_2$	$t - y$
$(1,0)$, $(3,0)$, $(3,1)$	0	$y + 1$
$(1,0)$, $(3,1)$, $(1,1)$	$(-t+2y+1)e'_2 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t - y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$-t + y + 4$
$S_C = \frac{17}{36}$, $\theta_C = \frac{11}{18}$, $\kappa_{C,e'_2} = \frac{7}{36}$.		

TABLE 47. The Zariski decompositions of $P_t - yC$ for $C = T_3$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_3 + yR_3$	$t - y$
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$y + 1$
$(3,1)$, $(3,\frac{3}{2})$, $(1,1)$	$(-t+2y+1)e'_3 + (y-1)T_2 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12}$	$\frac{2t}{3} - \frac{8y}{3} + 2$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(-t+2y+1)e'_3 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12}$	$\frac{2t}{3} - \frac{5y}{3} + 1$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(-t+2y+1)e'_3 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$t - y$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12}$	$-\frac{t}{3} + \frac{y}{3} + 2$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(2t+2y-8)e'_3 + (t+y-4)T_2 + \frac{y}{3}R_2 + (t+y-4)R_3 + \frac{2y}{3}D_{12}$	$-4t - \frac{8y}{3} + 16$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{3}R_2 + \frac{2y}{3}D_{12}$	$-t + \frac{y}{3} + 4$
$S_C = \frac{37}{72}$, $\theta_C = \frac{55}{108}$, $\kappa_{C,e'_3} = \frac{13}{54}$, $\kappa_{C,T_2} = \frac{1}{72}$, $\kappa_{C,D_{12}} = \frac{5}{27}$.		

TABLE 48. The Zariski decompositions of $P_t - yC$ for $C = \mathcal{C}_1$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,\frac{1}{2})$	$ye'_1 + 2ye'_2 + 2ye'_3 + yR_1 + yR_2 + yR_3$	$2t - 4y$
$(1,0)$, $(2,1)$, $(1,\frac{1}{2})$	$(-t+y+1)e'_1 + (-t+2y+1)e'_2 + (-t+2y+1)e'_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$2t - 4y$
$(1,0)$, $(3,1)$, $(2,1)$	$(-t+2y+1)e'_2 + (-t+2y+1)e'_3 + \frac{y}{2}R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3$	$\frac{3t}{2} - \frac{7y}{2} + \frac{1}{2}$
$(3,1)$, $(3,\frac{4}{3})$, $(2,1)$	$(-t+3y)e'_2 + (-t+2y+1)e'_3 + \frac{y}{2}R_1 + (-\frac{t}{2}+2y-\frac{1}{2})R_2 + (-\frac{t}{2}+y+\frac{1}{2})R_3 + (y-1)D_{12}$	$\frac{3t}{2} - \frac{9y}{2} + \frac{3}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_1$	$-\frac{t}{2} + \frac{y}{2} + \frac{5}{2}$
$(4,0)$, $(3,\frac{4}{3})$, $(3,1)$	$(3t+3y-12)e'_2 + (2t+2y-8)e'_3 + \frac{y}{2}R_1 + (2t+2y-8)R_2 + (t+y-4)R_3 + (t+y-4)D_{12}$	$-6t - \frac{9y}{2} + 24$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_1$	$-t + \frac{y}{2} + 4$
$S_C = \frac{91}{216}$, $\theta_C = \frac{47}{72}$, $\kappa_{C,e'_2} = \frac{1}{9}$, $\kappa_{C,e'_3} = \frac{23}{216}$, $\kappa_{C,R_1} = \frac{47}{216}$.		

TABLE 49. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_1	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_1$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(-\frac{t}{2}+3y-\frac{3}{2})e'_1 + (2y-2)L_{13} + (\frac{t}{2}+y-\frac{5}{2})\mathcal{C}_1 + (y-1)R_2 + (y-1)R_3 + (2y-2)D_{12}$	$-4y + 6$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(-\frac{t}{2}+3y-\frac{3}{2})e'_1 + (2y-2)L_{13} + (y-1)R_2 + (y-1)R_3 + (2y-2)D_{12}$	$\frac{t}{2} - 3y + \frac{7}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(3t+3y-12)e'_1 + (2t+2y-8)L_{13} + (t+y-4)\mathcal{C}_1 + (t+y-4)R_2 + (t+y-4)R_3 + (2t+2y-8)D_{12}$	$-6t - 4y + 24$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e'_1} = \frac{25}{72}$, $\kappa_{C,L_{13}} = \frac{1}{12}$, $\kappa_{C,\mathcal{C}_1} = \frac{1}{72}$.		

TABLE 50. The Zariski decompositions of $P_t - yC$ for $C = R_2$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_2	y
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$2y$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_2$	$\frac{t}{2}+y-\frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_2 + (2y-2)L_{23} + (y-1)R_3 + (\frac{t}{4}+\frac{y}{2}-\frac{3}{4})D_{12}$	$\frac{t}{4}-\frac{3y}{2}+\frac{9}{4}$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_2 + (\frac{t}{4}+\frac{y}{2}-\frac{3}{4})D_{12}$	$\frac{t}{4}+\frac{y}{2}+\frac{1}{4}$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{4}+\frac{y}{2}-\frac{3}{4})D_{12}$	$-\frac{t}{4}+\frac{3y}{2}+\frac{3}{4}$
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)e'_2 + (2t+2y-8)L_{23} + (t+y-4)R_3 + \frac{y}{2}D_{12}$	$-3t-\frac{3y}{2}+12$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}D_{12}$	$\frac{3y}{2}$
$S_C = \frac{59}{72}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_2} = \frac{23}{72}$, $\kappa_{C,L_{23}} = \frac{13}{72}$, $\kappa_{C,D_{12}} = \frac{1}{4}$.		

TABLE 51. The Zariski decompositions of $P_t - yC$ for $C = R_3$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_3	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_3$	$\frac{t}{2}+y-\frac{1}{2}$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(-\frac{t}{2}+y+\frac{1}{2})e'_3 + (2y-2)L_{13} + (2y-2)L_{23} + (y-1)R_1 + (y-1)R_2$	$\frac{t}{2}-3y+\frac{7}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(-\frac{t}{2}+y+\frac{1}{2})e'_3 + (2y-2)L_{13} + (\frac{t}{2}+3y-\frac{9}{2})L_{23} + (y-1)R_1 + (\frac{t}{2}+2y-\frac{7}{2})R_2 + (\frac{t}{2}+y-\frac{5}{2})D_{12}$	$-4y+6$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(t+y-4)e'_3 + (2t+2y-8)L_{13} + (3t+3y-12)L_{23} + (t+y-4)R_1 + (2t+2y-8)R_2 + (t+y-4)D_{12}$	$-6t-4y+24$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e'_3} = \frac{19}{72}$, $\kappa_{C,L_{13}} = \frac{1}{12}$, $\kappa_{C,L_{23}} = \frac{7}{72}$.		

TABLE 52. The Zariski decompositions of $P_t - yC$ for $C = D_{12}$ in the edge configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_1 + ye'_2 + yR_1 + yR_2$	$t-y$
$(2,1)$, $(3,2)$, $(1,1)$	$(-t+2y+1)e'_1 + (-t+y+1)e'_2 + (y-1)T_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t-2y+1$
$(1,0)$, $(2,1)$, $(1,1)$	$(-t+2y+1)e'_1 + (-t+y+1)e'_2 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + (-\frac{t}{2}+y+\frac{1}{2})R_2$	$t-y$
$(3,1)$, $(3,2)$, $(2,1)$	$(-t+2y+1)e'_1 + (y-1)T_3 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + \frac{y}{2}R_2$	$\frac{t}{2}-\frac{3y}{2}+\frac{3}{2}$
$(1,0)$, $(3,1)$, $(2,1)$	$(-t+2y+1)e'_1 + (-\frac{t}{2}+y+\frac{1}{2})R_1 + \frac{y}{2}R_2$	$\frac{t}{2}-\frac{y}{2}+\frac{1}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_2$	$-\frac{t}{2}+\frac{3y}{2}+\frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,1)$	$(2t+2y-8)e'_1 + (t+y-4)T_3 + (t+y-4)R_1 + \frac{y}{2}R_2$	$-3t-\frac{3y}{2}+12$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_2$	$\frac{3y}{2}$
$S_C = \frac{25}{36}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_1} = \frac{31}{72}$, $\kappa_{C,T_3} = \frac{5}{72}$, $\kappa_{C,R_2} = \frac{3}{8}$.		

The star configuration.

Remark A.6. The star configuration of Lemma 6.14 has the arrows $2 \to 1$ and $3 \to 1$, and the curves with negative self-intersection in the star configuration are listed in Table 53. The configuration is invariant under the exchange of the indices 2 and 3. We give one curve in each orbit of the permutations of $\{1,2,3\}$ preserving the arrows (see the proof of Theorem 6.18), namely e_1', R_1, e_2', R_2, L_{23}, T_1, and D_{21}. Thus the data of the curves e_3', R_3, and D_{31} are the data of the curves e_2', R_2, and D_{21} with the indices 2 and 3 exchanged. The curves e_1' and R_1, which intersect at the point $R_1 \cap e_1'$, come first, in Tables 54 and 55.

TABLE 53. The classes of the curves with negative self-intersection in the star configuration, in the basis $(\eta, e_1, e_2, e_3, e_1', e_2', e_3')$.

curve	class	curve	class
e_1'	$(0,0,0,0,1,0,0)$	e_2'	$(0,0,0,0,0,1,0)$
e_3'	$(0,0,0,0,0,0,1)$	R_1	$(0,1,0,0,-1,0,0)$
R_2	$(0,0,1,0,0,-1,0)$	R_3	$(0,0,0,1,0,0,-1)$
L_{23}	$(1,0,-1,-1,0,0,0)$	T_1	$(1,-1,0,0,-1,0,0)$
D_{21}	$(1,-1,-1,0,0,-1,0)$	D_{31}	$(1,-1,0,-1,0,0,-1)$

TABLE 54. The Zariski decompositions of $P_t - yC$ for $C = e_1'$ in the star configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y-t)T_1 + \frac{y}{2}R_1$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(2,1)$, $(1,2)$, $(1,1)$	$(y-1)T_1 + \frac{y}{2}R_1$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,2)$, $(2,1)$	$(y-1)T_1 + (\frac{t}{2} + y - \frac{3}{2})R_1 + (\frac{t}{2} + \frac{y}{2} - \frac{3}{2})D_{21} + (\frac{t}{2} + \frac{y}{2} - \frac{3}{2})D_{31}$	$2 - y$
$(1,0)$, $(3,0)$, $(2,1)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,1)$, $(2,1)$	$(\frac{t}{2} + y - \frac{3}{2})R_1 + (\frac{t}{2} + \frac{y}{2} - \frac{3}{2})D_{21} + (\frac{t}{2} + \frac{y}{2} - \frac{3}{2})D_{31}$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_1 + yR_1 + \frac{y}{2}D_{21} + \frac{y}{2}D_{31}$	$8 - 2t - y$
$(3,0)$, $(4,0)$, $(3,1)$	$yR_1 + \frac{y}{2}D_{21} + \frac{y}{2}D_{31}$	$4 - t$
$S_C = \frac{29}{36}$, $\theta_C = \frac{25}{72}$, $\kappa_{C,R_1} = \frac{5}{9}$, $\kappa_{C,T_1} = \frac{1}{8}$.		

TABLE 55. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye_1'	y
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$2y$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e_1'$	$\frac{t}{2} + y - \frac{1}{2}$
$(2,\frac{1}{2})$, $(3,1)$, $(3,2)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e_1' + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{21} + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{31}$	1
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{21} + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{31}$	$y - \frac{t}{2} + \frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,1)$	$(t + y - 4)e_1' + \frac{y}{2}D_{21} + \frac{y}{2}D_{31}$	$4 - t$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}D_{21} + \frac{y}{2}D_{31}$	y
$S_C = \frac{65}{72}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,e_1'} = \frac{7}{18}$, $\kappa_{C,D_{21}} = \frac{7}{24}$, $\kappa_{C,D_{31}} = \frac{7}{24}$.		

TABLE 56. The Zariski decompositions of $P_t - yC$ for $C = e_2'$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y - t)e_1' + (y - t)R_1 + \frac{y}{2}R_2 + (y - t)D_{21}$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{y}{2}$
$(2,2)$, $(1,2)$, $(1,1)$	$(y - t)e_1' + (y - \frac{t}{2} - \frac{1}{2})R_1 + \frac{y}{2}R_2 + (y - 1)D_{21}$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,0)$, $(2,2)$, $(1,1)$	$(\frac{t}{6} + \frac{y}{3} - \frac{1}{2})R_1 + \frac{y}{2}R_2 + (\frac{t}{3} + \frac{2y}{3} - 1)D_{21}$	$\frac{t}{6} - \frac{y}{6} + \frac{1}{2}$
$(3,0)$, $(3,2)$, $(2,2)$	$(\frac{t}{2} + \frac{y}{2} - \frac{3}{2})R_1 + \frac{y}{2}R_2 + (\frac{t}{2} + \frac{3y}{4} - \frac{3}{2})D_{21} + (\frac{t}{2} + \frac{y}{4} - \frac{3}{2})D_{31}$	$1 - \frac{y}{4}$
$(1,0)$, $(3,0)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(4,0)$, $(3,2)$	$\frac{y}{2}R_1 + \frac{y}{2}R_2 + \frac{3y}{4}D_{21} + \frac{y}{4}D_{31}$	$4 - t - \frac{y}{4}$
$S_C = \frac{8}{9}$, $\theta_C = \frac{11}{36}$, $\kappa_{C,R_2} = \frac{4}{9}$, $\kappa_{C,D_{21}} = \frac{29}{72}$.		

TABLE 57. The Zariski decompositions of $P_t - yC$ for $C = R_2$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye_2'	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e_2'$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(3y - \frac{t}{2} - \frac{3}{2})e_2' + (2y - 2)L_{23} + (y - 1)R_1 + (y - 1)R_3 + (2y - 2)D_{21}$	$\frac{t}{2} - 3y + \frac{7}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(4y - 4)e_2' + (2y - 2)L_{23} + (\frac{t}{2} + 2y - \frac{7}{2})R_1 + (y - 1)R_3 + (\frac{t}{2} + 3y - \frac{9}{2})D_{21} + (\frac{t}{2} + y - \frac{5}{2})D_{31}$	$6 - 4y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(4t + 4y - 16)e_2' + (2t + 2y - 8)L_{23} + (2t + 2y - 8)R_1 + (t + y - 4)R_3 + (3t + 3y - 12)D_{21} + (t + y - 4)D_{31}$	$24 - 6t - 4y$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e_2'} = \frac{13}{36}$, $\kappa_{C,L_{23}} = \frac{1}{12}$.		

TABLE 58. The Zariski decompositions of $P_t - yC$ for $C = L_{23}$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye'_2 + ye'_3 + yR_2 + yR_3$	$t - y$
$(2,1)$, $(3,2)$, $(1,1)$	$(y-t+1)e'_2 + (y-t+1)e'_3 + (y-1)T_1 + (y-\frac{t}{2}+\frac{1}{2})R_2 + (y-\frac{t}{2}+\frac{1}{2})R_3$	$t - 2y + 1$
$(1,0)$, $(2,1)$, $(1,1)$	$(y-t+1)e'_2 + (y-t+1)e'_3 + (y-\frac{t}{2}+\frac{1}{2})R_2 + (y-\frac{t}{2}+\frac{1}{2})R_3$	$t - y$
$(3,1)$, $(3,2)$, $(2,1)$	$(y-1)T_1 + \frac{y}{2}R_2 + \frac{y}{2}R_3$	$2 - y$
$(1,0)$, $(3,0)$, $(3,1)$, $(2,1)$	$\frac{y}{2}R_2 + \frac{y}{2}R_3$	1
$(4,0)$, $(3,2)$, $(3,1)$	$(t+y-4)T_1 + \frac{y}{2}R_2 + \frac{y}{2}R_3$	$8 - 2t - y$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_2 + \frac{y}{2}R_3$	$4 - t$
$S_C = \frac{11}{18}$, $\theta_C = \frac{5}{12}$, $\kappa_{C,R_2} = \frac{1}{3}$, $\kappa_{C,R_3} = \frac{1}{3}$, $\kappa_{C,T_1} = \frac{1}{18}$.		

TABLE 59. The Zariski decompositions of $P_t - yC$ for $C = T_1$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_1 + yR_1$	$t - y$
$(1,0)$, $(3,0)$, $(3,1)$	0	$y + 1$
$(1,0)$, $(3,1)$, $(1,1)$	$(2y-t+1)e'_1 + (y-\frac{t}{2}+\frac{1}{2})R_1$	$t - y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$y - t + 4$
$S_C = \frac{17}{36}$, $\theta_C = \frac{11}{18}$, $\kappa_{C,e'_1} = \frac{7}{36}$.		

TABLE 60. The Zariski decompositions of $P_t - yC$ for $C = D_{21}$ in the star configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye'_1 + 2ye'_2 + yR_1 + yR_2$	$t - y$
$(1,0)$, $(2,1)$, $(1,1)$	$(y-t+1)e'_1 + (2y-t+1)e'_2 + (y-\frac{t}{2}+\frac{1}{2})R_1 + (y-\frac{t}{2}+\frac{1}{2})R_2$	$t - y$
$(3,\frac{3}{2})$, $(3,2)$, $(1,1)$, $(2,1)$	$(2y-t)e'_1 + (2y-t+1)e'_2 + (2y-\frac{t}{2}-\frac{1}{2})R_1 + (y-\frac{t}{2}+\frac{1}{2})R_2 + (y-1)D_{31}$	$t - 2y + 1$
$(1,0)$, $(\frac{7}{3},\frac{2}{3})$, $(2,1)$	$(2y-t+1)e'_2 + \frac{y}{2}R_1 + (y-\frac{t}{2}+\frac{1}{2})R_2$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(\frac{7}{3},\frac{2}{3})$, $(3,1)$, $(3,\frac{3}{2})$, $(2,1)$	$(2y-t+1)e'_2 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_1 + (y-\frac{t}{2}+\frac{1}{2})R_2 + (\frac{t}{3}+\frac{y}{3}-1)D_{31}$	$\frac{t}{3} - \frac{2y}{3} + 1$
$(1,0)$, $(3,0)$, $(\frac{7}{3},\frac{2}{3})$	$\frac{y}{2}R_1$	$\frac{3y}{2} - \frac{t}{2} + \frac{3}{2}$
$(3,0)$, $(3,1)$, $(\frac{7}{3},\frac{2}{3})$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_1 + (\frac{t}{3}+\frac{y}{3}-1)D_{31}$	$\frac{4y}{3} - \frac{2t}{3} + 2$
$(4,0)$, $(3,2)$, $(3,\frac{3}{2})$	$(3t+2y-12)e'_1 + (2t+2y-8)e'_2 + (2t+2y-8)R_1 + (t+y-4)R_2 + (t+y-4)D_{31}$	$16 - 4t - 2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(2t+2y-8)e'_2 + \frac{2y}{3}R_1 + (t+y-4)R_2 + \frac{y}{3}D_{31}$	$8 - 2t - \frac{2y}{3}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{2y}{3}R_1 + \frac{y}{3}D_{31}$	$\frac{4y}{3}$
$S_C = \frac{17}{24}$, $\theta_C = \frac{71}{162}$, $\kappa_{C,e'_2} = \frac{73}{162}$, $\kappa_{C,R_1} = \frac{301}{648}$.		

The path configuration.

Remark A.7. The path configuration of Lemma 6.14 has the arrows $1 \to 2$ and $2 \to 3$, and the curves with negative self-intersection in the path configuration are listed in Table 61. We give one curve in each orbit of the permutations of $\{1,2,3\}$ preserving the arrows (see the proof of Theorem 6.18). Only the identity preserves the arrows, so all ten curves e_1', e_2', e_3', R_1, R_2, R_3, L_{13}, T_3, D_{12}, and D_{23} are given.

TABLE 61. The classes of the curves with negative self-intersection in the path configuration, in the basis $(\eta, e_1, e_2, e_3, e_1', e_2', e_3')$.

curve	class	curve	class
e_1'	$(0,0,0,0,1,0,0)$	e_2'	$(0,0,0,0,0,1,0)$
e_3'	$(0,0,0,0,0,0,1)$	R_1	$(0,1,0,0,-1,0,0)$
R_2	$(0,0,1,0,0,-1,0)$	R_3	$(0,0,0,1,0,0,-1)$
L_{13}	$(1,-1,0,-1,0,0,0)$	T_3	$(1,0,0,-1,0,0,-1)$
D_{12}	$(1,-1,-1,0,-1,0,0)$	D_{23}	$(1,0,-1,-1,0,-1,0)$

TABLE 62. The Zariski decompositions of $P_t - yC$ for $C = e_1'$ in the path configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y-t)e_2' + \frac{y}{2}R_1 + (y-t)R_2 + (y-t)D_{12}$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(2,2)$, $(1,2)$, $(1,1)$	$(y-t)e_2' + \frac{y}{2}R_1 + (y-\frac{t}{2}-\frac{1}{2})R_2 + (y-1)D_{12}$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,\frac{3}{2})$, $(3,2)$, $(2,2)$	$(t+2y-6)T_3 + \frac{y}{2}R_1 + (\frac{t}{2}+y-\frac{5}{2})R_2 + (t+2y-5)D_{12}$	$\frac{9}{2} - \frac{t}{2} - \frac{3y}{2}$
$(1,0)$, $(3,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,\frac{3}{2})$, $(2,2)$, $(1,1)$	$\frac{y}{2}R_1 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12}$	$\frac{t}{6} - \frac{y}{6} + \frac{1}{2}$
$(4,0)$, $(3,2)$, $(3,\frac{3}{2})$	$(3t+2y-12)T_3 + \frac{y}{2}R_1 + (t+y-4)R_2 + (2t+2y-8)D_{12}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,\frac{3}{2})$	$\frac{y}{2}R_1 + \frac{y}{3}R_2 + \frac{2y}{3}D_{12}$	$4 - t - \frac{y}{6}$

$S_C = \frac{7}{8}$, $\theta_C = \frac{5}{16}$, $\kappa_{C,R_1} = \frac{7}{16}$, $\kappa_{C,D_{12}} = \frac{55}{144}$.

TABLE 63. The Zariski decompositions of $P_t - yC$ for $C = e'_2$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y-t)e'_3 + \frac{y}{2}R_2 + (y-t)R_3 + (y-t)D_{23}$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{y}{2}$
$(3,3)$, $(1,2)$, $(\frac{3}{2},\frac{3}{2})$	$(y-t)e'_3 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + (y-\frac{t}{2}-\frac{1}{2})R_3 + (\frac{t}{3}+\frac{y}{3}-1)D_{12} + (y-1)D_{23}$	$\frac{t}{3} - \frac{2y}{3} + 1$
$(\frac{3}{2},\frac{3}{2})$, $(1,2)$, $(1,1)$	$(y-t)e'_3 + \frac{y}{2}R_2 + (y-\frac{t}{2}-\frac{1}{2})R_3 + (y-1)D_{23}$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(1,0)$, $(3,0)$, $(1,1)$	$\frac{y}{2}R_2$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,3)$, $(\frac{3}{2},\frac{3}{2})$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_2 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_3 + (\frac{t}{3}+\frac{y}{3}-1)D_{12} + (\frac{t}{3}+\frac{2y}{3}-1)D_{23}$	$1 - \frac{y}{3}$
$(3,0)$, $(\frac{3}{2},\frac{3}{2})$, $(1,1)$	$\frac{y}{2}R_2 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_3 + (\frac{t}{3}+\frac{2y}{3}-1)D_{23}$	$\frac{t}{6} - \frac{y}{6} + \frac{1}{2}$
$(3,0)$, $(4,0)$, $(3,3)$	$\frac{2y}{3}R_2 + \frac{y}{3}R_3 + \frac{y}{3}D_{12} + \frac{2y}{3}D_{23}$	$4 - t - \frac{y}{3}$
$S_C = \frac{17}{18}$, $\theta_C = \frac{79}{288}$, $\kappa_{C,R_2} = \frac{157}{288}$, $\kappa_{C,D_{23}} = \frac{121}{288}$.		

TABLE 64. The Zariski decompositions of $P_t - yC$ for $C = e'_3$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y-t)T_3 + \frac{y}{2}R_3$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_3$	$\frac{y}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(1,2)$	$(t+3y-6)T_3 + (\frac{t}{2}+y-\frac{5}{2})R_2 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_3 + (t+2y-5)D_{12} + (\frac{t}{3}+\frac{y}{3}-1)D_{23}$	$6 - \frac{2t}{3} - \frac{8y}{3}$
$(2,1)$, $(1,2)$, $(1,1)$	$(y-1)T_3 + \frac{y}{2}R_3$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,1)$, $(1,2)$, $(2,1)$	$(y-1)T_3 + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_3 + (\frac{t}{3}+\frac{y}{3}-1)D_{23}$	$\frac{t}{3} - \frac{2y}{3} + 1$
$(1,0)$, $(3,0)$, $(2,1)$, $(1,1)$	$\frac{y}{2}R_3$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(3,1)$, $(2,1)$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_3 + (\frac{t}{3}+\frac{y}{3}-1)D_{23}$	$\frac{t}{3} + \frac{y}{3}$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(3t+3y-12)T_3 + (t+y-4)R_2 + \frac{2y}{3}R_3 + (2t+2y-8)D_{12} + \frac{y}{3}D_{23}$	$16 - 4t - \frac{8y}{3}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{2y}{3}R_3 + \frac{y}{3}D_{23}$	$\frac{y}{3} - t + 4$
$S_C = \frac{55}{72}$, $\theta_C = \frac{85}{216}$, $\kappa_{C,R_3} = \frac{23}{54}$, $\kappa_{C,T_3} = \frac{13}{108}$.		

TABLE 65. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_1	y
$(1,0)$, $(3,0)$, $(3,1)$	0	$2y$
$(1,0)$, $(3,1)$, $(1,1)$	$(y-\frac{t}{2}+\frac{1}{2})e'_1$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(2,\frac{3}{2})$, $(1,1)$	$(3y-\frac{t}{2}-\frac{3}{2})e'_1 + (2y-2)L_{13} + (y-1)R_2 + (y-1)R_3 + (2y-2)D_{12}$	$\frac{t}{2} - 3y + \frac{7}{2}$
$(3,1)$, $(3,\frac{3}{2})$, $(2,\frac{3}{2})$	$(3y-\frac{t}{2}-\frac{3}{2})e'_1 + (\frac{t}{2}+3y-\frac{9}{2})L_{13} + (y-1)R_2 + (\frac{t}{2}+2y-\frac{7}{2})R_3 + (2y-2)D_{12} + (\frac{t}{2}+y-\frac{5}{2})D_{23}$	$6 - 4y$
$(3,0)$, $(4,0)$, $(3,1)$	0	$2y$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(3t+3y-12)e'_1 + (3t+3y-12)L_{13} + (t+y-4)R_2 + (2t+2y-8)R_3 + (2t+2y-8)D_{12} + (t+y-4)D_{23}$	$24 - 6t - 4y$
$S_C = \frac{3}{4}$, $\theta_C = \frac{19}{36}$, $\kappa_{C,e'_1} = \frac{25}{72}$, $\kappa_{C,L_{13}} = \frac{7}{72}$.		

TABLE 66. The Zariski decompositions of $P_t - yC$ for $C = R_2$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_2	y
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$2y$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e'_2$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,1)$	$(3y - \frac{t}{2} - \frac{3}{2})e'_2 + (y-1)R_3 + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{12} + (2y-2)D_{23}$	$\frac{t}{4} - \frac{3y}{2} + \frac{9}{4}$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e'_2 + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{12}$	$\frac{t}{4} + \frac{y}{2} + \frac{1}{4}$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{12}$	$\frac{3y}{2} - \frac{t}{4} + \frac{3}{4}$
$(4,0)$, $(3,2)$, $(3,1)$	$(3t + 3y - 12)e'_2 + (t + y - 4)R_3 + \frac{y}{2}D_{12} + (2t + 2y - 8)D_{23}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}D_{12}$	$\frac{3y}{2}$
$S_C = \frac{59}{72}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_2} = \frac{1}{2}$, $\kappa_{C,D_{12}} = \frac{1}{4}$.		

TABLE 67. The Zariski decompositions of $P_t - yC$ for $C = R_3$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye'_3	y
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$2y$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e'_3$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e'_3 + (2y-2)L_{13} + (y-1)R_1 + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{23}$	$\frac{t}{4} - \frac{3y}{2} + \frac{9}{4}$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e'_3 + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{23}$	$\frac{t}{4} + \frac{y}{2} + \frac{1}{4}$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{23}$	$\frac{3y}{2} - \frac{t}{4} + \frac{3}{4}$
$(4,0)$, $(3,2)$, $(3,1)$	$(t + y - 4)e'_3 + (2t + 2y - 8)L_{13} + (t + y - 4)R_1 + \frac{y}{2}D_{23}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}D_{23}$	$\frac{3y}{2}$
$S_C = \frac{59}{72}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_3} = \frac{23}{72}$, $\kappa_{C,L_{13}} = \frac{13}{72}$, $\kappa_{C,D_{23}} = \frac{1}{4}$.		

TABLE 68. The Zariski decompositions of $P_t - yC$ for $C = L_{13}$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$ye'_1 + ye'_3 + yR_1 + yR_3$	$t - y$
$(1,0)$, $(2,1)$, $(1,1)$	$(y - t + 1)e'_1 + (y - t + 1)e'_3 + (y - \frac{t}{2} + \frac{1}{2})R_1 + (y - \frac{t}{2} + \frac{1}{2})R_3$	$t - y$
$(2,1)$, $(3,2)$, $(1,1)$	$(y - t + 1)e'_1 + (2y - t)e'_3 + (y - \frac{t}{2} + \frac{1}{2})R_1 + (2y - \frac{t}{2} - \frac{1}{2})R_3 + (y-1)D_{23}$	$t - 2y + 1$
$(3,\frac{3}{2})$, $(3,2)$, $(2,1)$	$(2y - t)e'_3 + \frac{y}{2}R_1 + (2y - \frac{t}{2} - \frac{1}{2})R_3 + (y-1)D_{23}$	$\frac{t}{2} - \frac{3y}{2} + \frac{3}{2}$
$(1,0)$, $(3,0)$, $(2,1)$	$\frac{y}{2}R_1 + \frac{y}{2}R_3$	1
$(3,0)$, $(3,\frac{3}{2})$, $(2,1)$	$\frac{y}{2}R_1 + (\frac{t}{6} + \frac{2y}{3} - \frac{1}{2})R_3 + (\frac{t}{3} + \frac{y}{3} - 1)D_{23}$	$\frac{3}{2} - \frac{t}{6} - \frac{y}{6}$
$(4,0)$, $(3,2)$, $(3,\frac{3}{2})$	$(3t + 2y - 12)e'_3 + \frac{y}{2}R_1 + (2t + 2y - 8)R_3 + (t + y - 4)D_{23}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,\frac{3}{2})$	$\frac{y}{2}R_1 + \frac{2y}{3}R_3 + \frac{y}{3}D_{23}$	$4 - t - \frac{y}{6}$
$S_C = \frac{5}{8}$, $\theta_C = \frac{59}{144}$, $\kappa_{C,R_1} = \frac{49}{144}$, $\kappa_{C,R_3} = \frac{59}{144}$.		

TABLE 69. The Zariski decompositions of $P_t - yC$ for $C = T_3$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_3 + \ yR_3$	$t - y$
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$y + 1$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(2y - t + 1)e'_3 + \ (\frac{t}{6} + \frac{y}{3} - \frac{1}{2})R_2 +$ $(y - \frac{t}{2} + \frac{1}{2})R_3 + \ (\frac{t}{3} + \frac{2y}{3} - 1)D_{12}$	$\frac{2t}{3} - \frac{5y}{3} + 1$
$(3,1)$, $(3,\frac{3}{2})$, $(1,1)$	$(3y - t)e'_3 + \ (\frac{t}{6} + \frac{y}{3} - \frac{1}{2})R_2 + \ (2y - \frac{t}{2} - \frac{1}{2})R_3 +$ $(\frac{t}{3} + \frac{2y}{3} - 1)D_{12} + \ (y - 1)D_{23}$	$\frac{2t}{3} - \frac{8y}{3} + 2$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(2y - t + 1)e'_3 + \ (y - \frac{t}{2} + \frac{1}{2})R_3$	$t - y$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{6} + \frac{y}{3} - \frac{1}{2})R_2 + \ (\frac{t}{3} + \frac{2y}{3} - 1)D_{12}$	$\frac{y}{3} - \frac{t}{3} + 2$
$(4,0)$, $(3,\frac{3}{2})$, $(3,1)$	$(3t + 3y - 12)e'_3 + \ \frac{y}{3}R_2 + \ (2t + 2y - 8)R_3 +$ $\frac{2y}{3}D_{12} + \ (t + y - 4)D_{23}$	$16 - 4t - \frac{8y}{3}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{3}R_2 + \ \frac{2y}{3}D_{12}$	$\frac{y}{3} - t + 4$

$S_C = \frac{37}{72}$, $\theta_C = \frac{55}{108}$, $\kappa_{C,e'_3} = \frac{55}{216}$, $\kappa_{C,D_{12}} = \frac{5}{27}$.

TABLE 70. The Zariski decompositions of $P_t - yC$ for $C = D_{12}$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_1 + \ ye'_2 + \ yR_1 + \ yR_2$	$t - y$
$(2,1)$, $(3,2)$, $(1,1)$	$(2y - t + 1)e'_1 + \ (y - t + 1)e'_2 + \ (y - 1)T_3 +$ $(y - \frac{t}{2} + \frac{1}{2})R_1 + \ (y - \frac{t}{2} + \frac{1}{2})R_2$	$t - 2y + 1$
$(1,0)$, $(2,1)$, $(1,1)$	$(2y - t + 1)e'_1 + \ (y - t + 1)e'_2 +$ $(y - \frac{t}{2} + \frac{1}{2})R_1 + \ (y - \frac{t}{2} + \frac{1}{2})R_2$	$t - y$
$(3,1)$, $(3,2)$, $(2,1)$	$(2y - t + 1)e'_1 + \ (y - 1)T_3 + \ (y - \frac{t}{2} + \frac{1}{2})R_1 +$ $\frac{y}{2}R_2$	$\frac{t}{2} - \frac{3y}{2} + \frac{3}{2}$
$(1,0)$, $(3,1)$, $(2,1)$	$(2y - t + 1)e'_1 + \ (y - \frac{t}{2} + \frac{1}{2})R_1 + \ \frac{y}{2}R_2$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_2$	$\frac{3y}{2} - \frac{t}{2} + \frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,1)$	$(2t + 2y - 8)e'_1 + \ (t + y - 4)T_3 +$ $(t + y - 4)R_1 + \ \frac{y}{2}R_2$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_2$	$\frac{3y}{2}$

$S_C = \frac{25}{36}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_1} = \frac{31}{72}$, $\kappa_{C,R_2} = \frac{3}{8}$, $\kappa_{C,T_3} = \frac{5}{72}$.

TABLE 71. The Zariski decompositions of $P_t - yC$ for $C = D_{23}$ in the path configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye'_2 + ye'_3 + yR_2 + yR_3$	$t - y$
$(1,0)$, $(2,1)$, $(1,1)$	$(2y - t + 1)e'_2 + (y - t + 1)e'_3 + (y - \frac{t}{2} + \frac{1}{2})R_2 + (y - \frac{t}{2} + \frac{1}{2})R_3$	$t - y$
$(3,2)$, $(1,1)$, $(2,1)$	$(3y - t)e'_2 + (y - t + 1)e'_3 + (2y - \frac{t}{2} - \frac{1}{2})R_2 + (y - \frac{t}{2} + \frac{1}{2})R_3 + (y - 1)D_{12}$	$t - 2y + 1$
$(1,0)$, $(3,1)$, $(2,1)$	$(2y - t + 1)e'_2 + (y - \frac{t}{2} + \frac{1}{2})R_2 + \frac{y}{2}R_3$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,1)$, $(3,2)$, $(2,1)$	$(3y - t)e'_2 + (2y - \frac{t}{2} - \frac{1}{2})R_2 + \frac{y}{2}R_3 + (y - 1)D_{12}$	$\frac{t}{2} - \frac{3y}{2} + \frac{3}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_3$	$\frac{3y}{2} - \frac{t}{2} + \frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,1)$	$(3t + 3y - 12)e'_2 + (2t + 2y - 8)R_2 + \frac{y}{2}R_3 + (t + y - 4)D_{12}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_3$	$\frac{3y}{2}$
$S_C = \frac{25}{36}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e'_2} = \frac{1}{2}$, $\kappa_{C,R_3} = \frac{3}{8}$.		

The cycle configuration.

Remark A.8. The cycle configuration of Lemma 6.14 has the arrows $1 \to 2$, $2 \to 3$, and $3 \to 1$, and the curves with negative self-intersection in the cycle configuration are listed in Table 72. The configuration, with the curves D_{12}, D_{23}, and D_{31}, is invariant under the cyclic permutation $\sigma = (123)$ of the indices. We give one curve in each orbit of the permutations of $\{1,2,3\}$ preserving the arrows (see the proof of Theorem 6.18), namely e_1', R_1, and D_{12}. Thus the data of the other six curves, e_2', e_3', R_2, R_3, D_{23}, and D_{31}, are the data of the curves e_1', R_1, and D_{12} with the indices permuted by σ or σ^2.

TABLE 72. The classes of the curves with negative self-intersection in the cycle configuration, in the basis $(\eta, e_1, e_2, e_3, e_1', e_2', e_3')$.

curve	class	curve	class
e_1'	$(0,0,0,0,1,0,0)$	e_2'	$(0,0,0,0,0,1,0)$
e_3'	$(0,0,0,0,0,0,1)$	R_1	$(0,1,0,0,-1,0,0)$
R_2	$(0,0,1,0,0,-1,0)$	R_3	$(0,0,0,1,0,0,-1)$
D_{12}	$(1,-1,-1,0,-1,0,0)$	D_{23}	$(1,0,-1,-1,0,-1,0)$
D_{31}	$(1,-1,0,-1,0,0,-1)$		

TABLE 73. The Zariski decompositions of $P_t - yC$ for $C = e_1'$ in the cycle configuration.

vertices (t,y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,1)$, $(1,2)$	$(y-t)e_2' + \frac{y}{2}R_1 + (y-t)R_2 + (y-t)D_{12}$	$t - \frac{y}{2}$
$(0,0)$, $(1,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{y}{2}$
$(\frac{3}{2},\frac{3}{2})$, $(1,2)$, $(1,1)$	$(y-t)e_2' + \frac{y}{2}R_1 + (y-\frac{t}{2}-\frac{1}{2})R_2 + (y-1)D_{12}$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,3)$, $(1,2)$, $(\frac{3}{2},\frac{3}{2})$	$(y-t)e_2' + (\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_1 + (y-\frac{t}{2}-\frac{1}{2})R_2 + (y-1)D_{12} + (\frac{t}{3}+\frac{y}{3}-1)D_{31}$	$\frac{t}{3} - \frac{2y}{3} + 1$
$(1,0)$, $(3,0)$, $(1,1)$	$\frac{y}{2}R_1$	$\frac{t}{2} + \frac{y}{2} - \frac{1}{2}$
$(3,0)$, $(\frac{3}{2},\frac{3}{2})$, $(1,1)$	$\frac{y}{2}R_1 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12}$	$\frac{t}{6} - \frac{y}{6} + \frac{1}{2}$
$(3,0)$, $(3,3)$, $(\frac{3}{2},\frac{3}{2})$	$(\frac{t}{6}+\frac{2y}{3}-\frac{1}{2})R_1 + (\frac{t}{6}+\frac{y}{3}-\frac{1}{2})R_2 + (\frac{t}{3}+\frac{2y}{3}-1)D_{12} + (\frac{t}{3}+\frac{y}{3}-1)D_{31}$	$1 - \frac{y}{3}$
$(3,0)$, $(4,0)$, $(3,3)$	$\frac{2y}{3}R_1 + \frac{y}{3}R_2 + \frac{2y}{3}D_{12} + \frac{y}{3}D_{31}$	$4 - t - \frac{y}{3}$

$S_C = \frac{17}{18}$, $\theta_C = \frac{79}{288}$, $\kappa_{C,R_1} = \frac{157}{288}$, $\kappa_{C,D_{12}} = \frac{121}{288}$.

TABLE 74. The Zariski decompositions of $P_t - yC$ for $C = R_1$ in the cycle configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	ye_1'	y
$(1,0)$, $(3,0)$, $(2,\frac{1}{2})$	0	$2y$
$(1,0)$, $(2,\frac{1}{2})$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e_1'$	$\frac{t}{2} + y - \frac{1}{2}$
$(3,1)$, $(3,2)$, $(1,1)$	$(3y - \frac{t}{2} - \frac{3}{2})e_1' + (y-1)R_2 + (2y-2)D_{12} + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{31}$	$\frac{t}{4} - \frac{3y}{2} + \frac{9}{4}$
$(2,\frac{1}{2})$, $(3,1)$, $(1,1)$	$(y - \frac{t}{2} + \frac{1}{2})e_1' + (\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{31}$	$\frac{t}{4} + \frac{y}{2} + \frac{1}{4}$
$(3,0)$, $(3,1)$, $(2,\frac{1}{2})$	$(\frac{t}{4} + \frac{y}{2} - \frac{3}{4})D_{31}$	$\frac{3y}{2} - \frac{t}{4} + \frac{3}{4}$
$(4,0)$, $(3,2)$, $(3,1)$	$(3t + 3y - 12)e_1' + (t + y - 4)R_2 + (2t + 2y - 8)D_{12} + \frac{y}{2}D_{31}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}D_{31}$	$\frac{3y}{2}$
$S_C = \frac{59}{72}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e_1'} = \frac{1}{2}$, $\kappa_{C,D_{31}} = \frac{1}{4}$.		

TABLE 75. The Zariski decompositions of $P_t - yC$ for $C = D_{12}$ in the cycle configuration.

vertices (t, y)	$N_{t,y}$	$(P_{t,y} \cdot C)$
$(0,0)$, $(1,0)$, $(1,1)$	$2ye_1' + ye_2' + yR_1 + yR_2$	$t - y$
$(1,0)$, $(2,1)$, $(1,1)$	$(2y - t + 1)e_1' + (y - t + 1)e_2' + (y - \frac{t}{2} + \frac{1}{2})R_1 + (y - \frac{t}{2} + \frac{1}{2})R_2$	$t - y$
$(3,2)$, $(1,1)$, $(2,1)$	$(3y - t)e_1' + (y - t + 1)e_2' + (2y - \frac{t}{2} - \frac{1}{2})R_1 + (y - \frac{t}{2} + \frac{1}{2})R_2 + (y-1)D_{31}$	$t - 2y + 1$
$(1,0)$, $(3,1)$, $(2,1)$	$(2y - t + 1)e_1' + (y - \frac{t}{2} + \frac{1}{2})R_1 + \frac{y}{2}R_2$	$\frac{t}{2} - \frac{y}{2} + \frac{1}{2}$
$(3,1)$, $(3,2)$, $(2,1)$	$(3y - t)e_1' + (2y - \frac{t}{2} - \frac{1}{2})R_1 + \frac{y}{2}R_2 + (y-1)D_{31}$	$\frac{t}{2} - \frac{3y}{2} + \frac{3}{2}$
$(1,0)$, $(3,0)$, $(3,1)$	$\frac{y}{2}R_2$	$\frac{3y}{2} - \frac{t}{2} + \frac{3}{2}$
$(4,0)$, $(3,2)$, $(3,1)$	$(3t + 3y - 12)e_1' + (2t + 2y - 8)R_1 + \frac{y}{2}R_2 + (t + y - 4)D_{31}$	$12 - 3t - \frac{3y}{2}$
$(3,0)$, $(4,0)$, $(3,1)$	$\frac{y}{2}R_2$	$\frac{3y}{2}$
$S_C = \frac{25}{36}$, $\theta_C = \frac{4}{9}$, $\kappa_{C,e_1'} = \frac{1}{2}$, $\kappa_{C,R_2} = \frac{3}{8}$.		

A.2. **Integral tables.** This subsection lists the integrals and the polygon and polytope data behind, successively, Proposition 6.2, Theorem 6.5, Propositions 8.8 and 8.9, and Theorem 8.12, and then, from Remark A.25 on, Lemmas 7.8, 7.9, 7.10, 7.16, and 9.5, Proposition 9.8, and Theorems 9.9 and 9.13. Only data and values are listed, and the arguments are in the main text. The data are arranged in groups of tables, and a letter introduced in a group is valid only within that group.

Remark A.9. Each integrand below is a piecewise polynomial function on a rational polygon in a plane with coordinates (w, y), (w, δ), or (a, u), or, for Theorem 8.12 and in several tables of Remark A.25 and after it, on a rational polytope in $\mathbb{R}^3$. It is built from maxima and minima of finitely many affine functions. For a real number x we write $(x)_+ := \max\{x, 0\}$. Cutting the region along the loci on which two of the affine functions are equal yields convex polygons, respectively polytopes, with rational vertices. On each polygon the lower and upper limits h_- and h_+ of the inner variable are single affine functions, and the integrand is a sum of products of two affine functions, for instance

$$\frac{h_+^2 - h_-^2}{2} = \frac{1}{2}(h_+ - h_-)(h_+ + h_-).$$

If φ and ψ are affine functions and T is a triangle with vertices P_1, P_2, P_3 and area $|T|$, then

$$\int_T \varphi\psi = \frac{|T|}{12}\Big(\sum_{i=1}^{3}\varphi(P_i)\psi(P_i) + \sum_{i=1}^{3}\varphi(P_i)\cdot\sum_{i=1}^{3}\psi(P_i)\Big).$$

We triangulate each polygon from one vertex and apply the formula on each triangle. Each polytope is cut into tetrahedra with a common vertex, or, for Theorem 8.12, into the three regions of Remark A.24, over which the integrals are iterated. A tetrahedron with vertices $0, x, y, z$ has volume $|\det(x, y, z)|/6$, and the integral of an affine function φ over a simplex is the volume times the mean of the values of φ at the vertices. All arithmetic is exact arithmetic in $\mathbb{Q}$, and no floating point number occurs. The limits $S = \lim_m S_m$ exist by [AZ22, Lemma 2.9]. Rounding, the finitely many boundary values of j, and the difference between the lattice point sums and the integrals change the normalized moments at level m by $O(1/m)$, and the tables list the limits.

Remark A.10. Notation is as in Set-up 5.1(1) and Construction-Proposition 5.4. In the first four groups of tables L is a non-special line, $m \geq 1$, and $0 \leq j \leq 5m/2$. For $m \leq j \leq 2m$ let $w := j/m - 1 \in [0, 1]$. Then $W_{m,j} \subset H^0(E, \mathcal{O}_E(js + (m + j)f))$, and after division by m the class $js + (m + j)f$ becomes $\alpha s + \beta f$ with

$$\alpha = 1 + w, \qquad \beta = 2 + w.$$

Exponents of monomials are also divided by m. The total of a flag is the sum of three integrals, over $0 \leq j \leq m$, over $m \leq j \leq 2m$, and over $2m \leq j \leq 5m/2$, and the tables list them from the first to the third. For $2m \leq j \leq 5m/2$, Construction-Proposition 5.4(3) shows $W_{m,j} = d_E^{j-2m} \cdot U_{5m-2j}$. Thus the moving part of the third integral is at most the corresponding integral for the unit model $(\alpha, \beta, w) = (2, 3, 1)$ multiplied by

$$\int_2^{5/2} (5 - 2t)^3 dt = \frac{1}{8},$$

and the tables list the product. By Lemma 5.2, $mN_m = 3m^4 + O(m^3)$, so the moment listed in the tables is one third of the total, plus the fixed contribution cJ described below. The fixed factor d_E^{j-2m} adds cJ to the moment of f_x, or of a point of f_x, at which $D|_E$, respectively $(D|_E - \mu f_x)|_{f_x}$, has coefficient c, where μ is the coefficient of f_x in $D|_E$ and $J = 1/72$ is the limit in Lemma 5.2.

Two lines, root off the negative section.

Remark A.11. In the group of tables for Proposition 6.2, z, u, d, ℓ, y, δ, n', and ω are the letters of **Steps 2–5** of the proof of Proposition 6.2. Let p be the point of $Z_U \cap f_x$ in Proposition 6.2, so that $p \notin s$, and let $q = s \cap f_x$. Let z be a coordinate on the base of the ruling with $f_x = (z = 0)$, and let u be a fiber coordinate on $E \backslash s$ with $p = (0,0)$. Over the fiber f_∞ over the point where z is ∞ we use $\zeta = 1/z$ and $u_\infty = u/z$, and $v_\infty = 1/u_\infty$ near s. Let $r := (\zeta, u_\infty) = (0,0)$ and $t := s \cap f_\infty$. For each of the seven pairs (n', ω) of **Step 3** of the proof of Proposition 6.2, the limit subscheme is the disjoint union of the subscheme defined by $(u, z^{n'})$ at p and a residual subscheme ω, which is empty for $n' = 3$ and otherwise one of the following:

- (r) the simple point r;
- (h) the subscheme defined by (u_∞, ζ^2) at r;
- (v) the subscheme defined by (ζ, u_∞^2) at r;
- (r+t) the simple points r and t;
- (t_v) the subscheme defined by (ζ, v_∞^2) at t.

A monomial $z^d u^\ell$ of $\mathcal{O}_E(\alpha s + \beta f)$ satisfies $0 \le \ell \le \alpha$ and $0 \le d \le \beta - \ell$, and d is the order along f_x. Let $y := \ell/m$ and $\delta := d/m$. For fixed (w, y) the variable δ runs from

$$\delta_-(w,y) = \max\big(0, n'(w-y)\big)$$

to $\delta_+(w,y)$, the minimum of $2 + w - y$ and the residual bound of ω:

$$\text{(r)} : 2, \quad \text{(h)} : 2 - w + y, \quad \text{(v)} : 2 - \tfrac{y}{2}, \quad \text{(r+t)} : 2 \text{ and } 3 + w - 2y, \quad (t_v) : \tfrac{5}{2} + \tfrac{w}{2} - \tfrac{3y}{2}.$$

The second integral is the integral over $0 \le w \le 1$ of the integral of δ over the polygon $\mathcal{P}(w)$ of **Step 5** of the proof of Proposition 6.2, that is, the integral of $\frac{1}{2}(\delta_+^2 - \delta_-^2)$ over $0 \le w \le 1$, $0 \le y \le 1 + w$, $\delta_- \le \delta_+$. The first integral equals $11/24$ for all seven pairs. The fixed factor adds nothing, since f_x is not a component of $D|_E$.

Table 76 lists the fiber moments of the seven pairs for the bound $S(W; f_x) \le 143/144$ of Proposition 6.2(1).

TABLE 76. Fiber moments of the seven pairs (n', ω) for Proposition 6.2.

n'	ω	$\delta_+(w,y)$	$m \le j \le 2m$	$2m \le j \le 5m/2$	moment
3	$\emptyset$	$2 + w - y$	13/6	17/48	143/144
2	(r)	$\min(2 + w - y, 2)$	2	5/16	133/144
2	(t_v)	$\min(2 + w - y, \frac{5}{2} + \frac{w}{2} - \frac{3y}{2})$	47/24	7/24	65/72
1	(h)	$\min(2 + w - y, 2 - w + y)$	11/6	13/48	41/48
1	(v)	$\min(2 + w - y, 2 - \frac{y}{2})$	11/6	13/48	41/48
1	(r+t)	$\min(2 + w - y, 2, 3 + w - 2y)$	47/24	5/16	131/144
1	(t_v)	$\min(2 + w - y, \frac{5}{2} + \frac{w}{2} - \frac{3y}{2})$	25/12	17/48	139/144

Remark A.12. Table 77 lists the four polygons of the pair $(2, (t_v))$ in Table 76. Their integrals sum to $47/24$, and, for the pair $(2, (t_v))$, the integral of $\frac{1}{2}\big(\delta_+(1,y)^2 - \delta_-(1,y)^2\big)$ over $0 \le y \le 2$, multiplied by $\frac{1}{8}$, is $\frac{7}{24}$.

TABLE 77. Polygons of the pair $(2, (t_v))$ for Proposition 6.2.

δ_-	δ_+	vertices (w,y)	integral
0	$w - y + 2$	$(0,0)$, $(1/2, 1/2)$, $(0,1)$	17/48
0	$w/2 - 3y/2 + 5/2$	$(1/2, 1/2)$, $(1,1)$, $(1,2)$, $(0,1)$	31/64
$2w - 2y$	$w - y + 2$	$(0,0)$, $(1,0)$, $(1/2, 1/2)$	29/48
$2w - 2y$	$w/2 - 3y/2 + 5/2$	$(1,0)$, $(1,1)$, $(1/2, 1/2)$	33/64

Remark A.13. For the point moments of Proposition 6.2(1), let δ be the normalized order along f_x. At q the row with normalized order δ along f_x is contained in the span of the u^ℓ with $\max(0, w-\delta) \le y \le \min(\alpha, \beta-\delta)$, and u^ℓ has order $\alpha - y$ at q. The second integrand, after integration in δ and y, is $w^2 + 2w + 5/6$. At a point of $f_x \backslash \{q\}$ the row is contained in the polynomials in u of degree $\min(\alpha, \beta-\delta)$, whose total order at a point is $\min(\alpha, \beta-\delta)^2/2$ after normalization. The coefficient of $(D|_E)|_{f_x}$ is at most 2 at q and at most 5 at any point. Table 78 lists the point moments for the bounds $S(W^{f_x}; q) \le 143/144$ and $S(W^{f_x}; a) \le 7/8$ of Proposition 6.2(1).

TABLE 78. Point moments on f_x for Proposition 6.2.

point	$0 \le j \le m$	$m \le j \le 2m$	$2m \le j \le 5m/2$	moving	fixed	moment
q	1/4	13/6	23/48	139/144	$2J = 1/36$	143/144
$a \in f_x \backslash \{q\}$	5/24	43/24	5/12	29/36	$5J = 5/72$	7/8

Two lines, roots on the negative sections.

Remark A.14. In the group of tables for Theorem 6.5, z, v, i, ℓ, d, y, and δ are the letters of the proof of Theorem 6.5. The data for Theorem 6.5 concern the situation of Theorem 6.5(2): the only point of $Z_U \cap f_x$ is $q = s \cap f_x$, and the marked component of Z_U at q has width $n \in \{2, 3, 4\}$. Let z be a coordinate on the base of the ruling with $f_x = (z = 0)$, let u be a fiber coordinate on $E \backslash s$, and let $v := 1/u$ near q. Over f_∞ we use $\zeta = 1/z$ and $u_\infty = u/z$, and $r := (\zeta, u_\infty) = (0, 0)$. Let

$$\mathfrak{r} := (\zeta, u_\infty), \qquad \mathfrak{r}' := (u_\infty, \zeta^2), \qquad \mathfrak{r}'' := (\zeta, u_\infty^2),$$

so that $\mathfrak{r}$ is the maximal ideal of r, and let

$$\mathfrak{c}_4 := (v - z, z^4), \quad \mathfrak{c}_3 := (v - z, z^3) \cap \mathfrak{r}, \quad \mathfrak{c}_2' := (v - z, z^2) \cap \mathfrak{r}', \quad \mathfrak{c}_2'' := (v - z, z^2) \cap \mathfrak{r}''.$$

After scaling u, the marked component is contained in $(v - z, z^2)$, and the torus action $(z, u) \to (\lambda^{-1} z, \lambda u)$ degenerates the marked component to $(v - z, z^n)$ and the residual part of Z_U to one of four limit models:

- $\mathfrak{c}_4$ $n = 4$ and no residual part;
- $\mathfrak{c}_3$ $n = 3$ and the simple point r;
- $\mathfrak{c}_2'$ $n = 2$ and the subscheme defined by $\mathfrak{r}'$;
- $\mathfrak{c}_2''$ $n = 2$ and the subscheme defined by $\mathfrak{r}''$.

A monomial $z^i v^\ell$ of $\mathcal{O}_E(\alpha s + \beta f)$ satisfies $0 \le \ell \le \alpha$ and $0 \le i \le \beta - \alpha + \ell$. Let $d := i + \ell$, so that the order along f_x is $i = d - \ell$, and let $y := \ell/m$ and $\delta := d/m$. For the four ideals $\mathfrak{c}$, the variable y runs from $\ell_-^{\mathfrak{c}}(w, \delta)$ to $\ell_+(w, \delta) = \min(1 + w, \delta)$, where

$$\begin{aligned}
\ell_-^{\mathfrak{c}_4}(w, \delta) &= \max\left(0, \tfrac{\delta - 1}{2}\right) + \max\left(0, \tfrac{4w - \delta}{3}\right), \\
\ell_-^{\mathfrak{c}_3}(w, \delta) &= \max\left(0, \tfrac{\delta - 1}{2}, \delta - 2\right) + \max\left(0, \tfrac{3w - \delta}{2}\right), \\
\ell_-^{\mathfrak{c}_2'}(w, \delta) &= \max\left(0, \tfrac{\delta - 1}{2}\right) + \max(0, 2w - \delta), \\
\ell_-^{\mathfrak{c}_2''}(w, \delta) &= \max\left(0, \tfrac{\delta - 1}{2}, \tfrac{2\delta - 3 + w}{3}\right) + \max(0, 2w - \delta)
\end{aligned}$$

are the limits of the counts of **Step 4** of the proof of Theorem 6.5, and $\mathfrak{c} = \mathfrak{c}_2'$ has the extra condition $\delta \le 3$. The fiber moment integrand is

$$\int_{\ell_-^{\mathfrak{c}}}^{\ell_+} (\delta - y) dy = (\ell_+ - \ell_-^{\mathfrak{c}}) \left(\delta - \frac{\ell_+ + \ell_-^{\mathfrak{c}}}{2}\right)$$

on $0 \le w \le 1$, $0 \le \delta \le 3 + 2w$, $\ell_-^{\mathfrak{c}} \le \ell_+$. The first integral equals 11/24 for all four models, and the third integral is computed at $w = 1$ over intervals of δ. Table 79 lists the polygons and the fiber moments of the four models for the bound $S(W; f_x) \le 533/576$ of Theorem 6.5(2).

TABLE 79. Polygons and fiber moments of the four limit models for Theorem 6.5.

$\ell_-^{\mathfrak{c}}$	ℓ_+	vertices (w,δ), or interval of δ at $w=1$	integral
model $\mathfrak{c}_4$			
0	δ	$(0,0)$, $(1/4,1)$, $(0,1)$	$1/32$
$4w/3-\delta/3$	δ	$(0,0)$, $(1,1)$, $(1/4,1)$	$1/32$
$\delta/2-1/2$	$w+1$	$(1,4)$, $(1,5)$, $(0,3)$, $(0,1)$, $(1/3,4/3)$	$3209/2592$
$\delta/2-1/2$	δ	$(0,1)$, $(1/4,1)$, $(1/3,4/3)$	$241/10368$
$4w/3+\delta/6-1/2$	$w+1$	$(1,2)$, $(1,4)$, $(1/3,4/3)$	$185/324$
$4w/3+\delta/6-1/2$	δ	$(1,1)$, $(1,2)$, $(1/3,4/3)$, $(1/4,1)$	$1253/10368$
$\delta/6+5/6$	δ	$\delta\in[1,2]$	$25/1728$
$\delta/6+5/6$	2	$\delta\in[2,4]$	$181/864$
$\delta/2-1/2$	2	$\delta\in[4,5]$	$5/64$
$m\le j\le 2m$: $129/64$,	$2m\le j\le 5m/2$: $29/96$,	moment $533/576$	
model $\mathfrak{c}_3$			
0	δ	$(0,0)$, $(1/3,1)$, $(0,1)$	$1/24$
$3w/2-\delta/2$	δ	$(0,0)$, $(1,1)$, $(1/3,1)$	$1/36$
$\delta/2-1/2$	$w+1$	$(1,3)$, $(0,3)$, $(0,1)$, $(1/2,3/2)$	$821/768$
$\delta/2-1/2$	δ	$(0,1)$, $(1/3,1)$, $(1/2,3/2)$	$113/2304$
$3w/2-1/2$	$w+1$	$(1,2)$, $(1,3)$, $(1/2,3/2)$	$181/768$
$3w/2-1/2$	δ	$(1,1)$, $(1,2)$, $(1/2,3/2)$, $(1/3,1)$	$305/2304$
$\delta-2$	$w+1$	$(1,3)$, $(1,4)$, $(0,3)$	$7/24$
1	δ	$\delta\in[1,2]$	$1/48$
1	2	$\delta\in[2,3]$	$1/8$
$\delta-2$	2	$\delta\in[3,4]$	$5/48$
$m\le j\le 2m$: $133/72$,	$2m\le j\le 5m/2$: $1/4$,	moment $23/27$	
model $\mathfrak{c}_2'$			
0	δ	$(0,0)$, $(1/2,1)$, $(0,1)$	$1/16$
$2w-\delta$	δ	$(0,0)$, $(1,1)$, $(1/2,1)$	$1/48$
$\delta/2-1/2$	$w+1$	$(1,2)$, $(1,3)$, $(0,3)$, $(0,1)$	$131/96$
$\delta/2-1/2$	δ	$(1,2)$, $(0,1)$, $(1/2,1)$	$11/64$
$2w-\delta/2-1/2$	δ	$(1,1)$, $(1,2)$, $(1/2,1)$	$19/192$
$3/2-\delta/2$	δ	$\delta\in[1,2]$	$3/64$
$\delta/2-1/2$	2	$\delta\in[2,3]$	$11/64$
$m\le j\le 2m$: $55/32$,	$2m\le j\le 5m/2$: $7/32$,	moment $115/144$	
model $\mathfrak{c}_2''$			
0	δ	$(0,0)$, $(1/2,1)$, $(0,1)$	$1/16$
$2w-\delta$	δ	$(0,0)$, $(1,1)$, $(1/2,1)$	$1/48$
$\delta/2-1/2$	$w+1$	$(2/3,5/3)$, $(0,3)$, $(0,1)$	$79/162$
$\delta/2-1/2$	δ	$(3/4,3/2)$, $(2/3,5/3)$, $(0,1)$, $(1/2,1)$	$5525/41472$
$2w-\delta/2-1/2$	δ	$(1,1)$, $(3/4,3/2)$, $(1/2,1)$	$61/1536$
$w/3+2\delta/3-1$	$w+1$	$(1,2)$, $(1,4)$, $(0,3)$, $(2/3,5/3)$	$8/9$
$w/3+2\delta/3-1$	δ	$(1,2)$, $(2/3,5/3)$, $(3/4,3/2)$	$491/13824$
$7w/3-\delta/3-1$	δ	$(1,1)$, $(1,2)$, $(3/4,3/2)$	$721/13824$
$4/3-\delta/3$	δ	$\delta\in[1,2]$	$1/27$
$2\delta/3-2/3$	2	$\delta\in[2,4]$	$5/27$
$m\le j\le 2m$: $991/576$,	$2m\le j\le 5m/2$: $2/9$,	moment $461/576$	

Remark A.15. For the point q in Theorem 6.5(2), the rows are contained in those of the ideal $\mathfrak{a}^{j-m}$, where $\mathfrak{a}:=(v-z,z^2)$, that is, of the model with $n=2$ and no residual part, and z^iv^ℓ has order ℓ at q. Thus y runs from

$$\ell_-^{\mathfrak{a}}(w,\delta)=\max\left(0,\frac{\delta-1}{2}\right)+\max(0,2w-\delta)\qquad\text{to}\qquad \ell_+(w,\delta)=\min(1+w,\delta),$$

and the integrand of the q-moment of the containing rows is $\big(\ell_+^2-(\ell_-^{\mathfrak{a}})^2\big)/2$ on $0\le w\le 1$, $0\le\delta\le 3+2w$, $\ell_-^{\mathfrak{a}}\le\ell_+$. The three integrals of the q-moment of the containing rows are $1/4$,

223/96, and 17/32. We use the functions $\rho(w)$ and $\kappa(w)$ of the proof of Theorem 6.5:

$$\rho(w) = \frac{3 + 4w - 3w^2}{2},$$

the limit of $m^{-2} \dim W_{m,j}$ for $j = m(1 + w)$ (Lemma 5.2), and the piecewise affine interpolation $\kappa(w)$ of $\frac{5}{4}w^2$ at $w = 0, 1/4, 1/2, 3/4, 1$:

$$\kappa(w) = \frac{5w}{16}, \quad \frac{15w}{16} - \frac{5}{32}, \quad \frac{25w}{16} - \frac{15}{32}, \quad \frac{35w}{16} - \frac{15}{16}$$

on $[0, 1/4]$, $[1/4, 1/2]$, $[1/2, 3/4]$, and $[3/4, 1]$, respectively. As in **Step 5** of the proof of Theorem 6.5, for an integer $c \geq 0$ the sum of the orders at q over a basis of the rows of $W_{m,j}$ is at most $c \dim W_{m,j}$ plus the sum of $\max(0, o - c)$ over a monomial basis of the containing rows, where o is the order at q. Taking $c = \lfloor m\kappa(w) \rfloor$, the bound without cutoff minus the bound with cutoff is, after normalization, the following saving. On an interval $[w_0, w_1]$ the saving of the cutoff $\kappa(w)$ is

$$\int_{w_0}^{w_1} \Big(\int_0^{3+2w} \int_{\ell_-^{\mathfrak{a}}}^{\ell_+} \min(\kappa(w), y) dy d\delta - \kappa(w)\rho(w) \Big) dw.$$

In the range $2m \leq j \leq 5m/2$ we work at unit scale, that is, with U_{5m-2j} in place of $W_{m,j}$ and all quantities divided by the powers of $5m - 2j$. There the cutoff is $5/4$, and the rank $\lim_{k \to +\infty} k^{-2} \dim U_k$ is $2 = \rho(1)$. Table 80 lists the polygons of the cutoff integrals for the bound on $S(W^{f_x}; q)$ in Theorem 6.5(2).

TABLE 80. Polygons and savings of the cutoff $\kappa(w)$ at q for Theorem 6.5.

	$\ell_-^{\mathfrak{a}}$	ℓ_+	vertices (w, δ), or interval of δ at $w = 1$	$\int \min(\kappa(w), y)$
$w \in [0, 1/4]$, $\kappa(w) = 5w/16$				
1	0	δ	$(0,0)$, $(1/4, 1/2)$, $(1/4, 1)$, $(0, 1)$	3235/786432
2	$2w - \delta$	δ	$(0,0)$, $(1/4, 1/4)$, $(1/4, 27/64)$	605/4194304
3	$2w - \delta$	δ	$(0,0)$, $(1/4, 27/64)$, $(1/4, 1/2)$	3925/25165824
4	$\delta/2 - 1/2$	$w + 1$	$(1/4, 5/4)$, $(1/4, 7/2)$, $(0, 3)$, $(0, 1)$	2255/196608
5	$\delta/2 - 1/2$	δ	$(1/4, 37/32)$, $(1/4, 5/4)$, $(0, 1)$	2755/4194304
6	$\delta/2 - 1/2$	δ	$(1/4, 1)$, $(1/4, 37/32)$, $(0, 1)$	2175/2097152
$\int \min(\kappa(w), y) = 442345/25165824$, $\int \kappa(w)\rho(w) = 1715/98304$, saving 3305/25165824				
$w \in [1/4, 1/2]$, $\kappa(w) = 15w/16 - 5/32$				
7	0	δ	$(1/2, 1)$, $(1/4, 1)$, $(1/4, 1/2)$	5885/786432
8	$2w - \delta$	δ	$(1/2, 1/2)$, $(1/2, 11/16)$, $(1/4, 27/64)$, $(1/4, 1/4)$	20195/12582912
9	$2w - \delta$	δ	$(1/2, 11/16)$, $(1/2, 1)$, $(1/4, 1/2)$, $(1/4, 27/64)$	151225/25165824
10	$\delta/2 - 1/2$	$w + 1$	$(1/2, 13/8)$, $(1/2, 4)$, $(1/4, 7/2)$, $(1/4, 5/4)$, $(5/14, 19/14)$	4550825/67436544
11	$\delta/2 - 1/2$	$w + 1$	$(1/2, 3/2)$, $(1/2, 13/8)$, $(5/14, 19/14)$	12149/4214784
12	$\delta/2 - 1/2$	δ	$(5/14, 19/14)$, $(1/4, 5/4)$, $(1/4, 37/32)$	912105/1438646272
13	$\delta/2 - 1/2$	δ	$(1/2, 1)$, $(1/2, 3/2)$, $(5/14, 19/14)$, $(1/4, 37/32)$, $(1/4, 1)$	43299287/2157969408
$\int \min(\kappa(w), y) = 2671645/25165824$, $\int \kappa(w)\rho(w) = 3285/32768$, saving 148765/25165824				

Remark A.16. The four savings for $m \leq j \leq 2m$ sum to

$$\frac{3305}{25165824} + \frac{148765}{25165824} + \frac{1104181}{25165824} + \frac{74328067}{478150656} = \frac{24549209}{119537664}.$$

TABLE 80. Polygons and savings of the cutoff $\kappa(w)$ at q for Theorem 6.5 (continued).

	$\ell_-^{\mathfrak{a}}$	ℓ_+	vertices (w,δ), or interval of δ at $w=1$	$\int \min(\kappa(w),y)$
$w \in [1/2,3/4]$, $\kappa(w) = 25w/16 - 15/32$				
14	$2w-\delta$	δ	$(3/4,3/4)$, $(3/4,51/64)$, $(1/2,11/16)$, $(1/2,1/2)$	7065/4194304
15	$2w-\delta$	δ	$(3/4,51/64)$, $(3/4,1)$, $(1/2,1)$, $(1/2,11/16)$	366353/25165824
16	$\delta/2-1/2$	$w+1$	$(3/4,77/32)$, $(3/4,9/2)$, $(1/2,4)$, $(1/2,13/8)$	654745/4194304
17	$\delta/2-1/2$	$w+1$	$(3/4,7/4)$, $(3/4,77/32)$, $(1/2,13/8)$, $(1/2,3/2)$	67655/1048576
18	$\delta/2-1/2$	δ	$(3/4,3/2)$, $(3/4,7/4)$, $(1/2,3/2)$, $(1/2,1)$	81307/1572864
19	$2w-\delta/2-1/2$	δ	$(3/4,1)$, $(3/4,3/2)$, $(1/2,1)$	23053/786432
$\int \min(\kappa(w),y) = 7999541/25165824$, $\int \kappa(w)\rho(w) = 26935/98304$, saving 1104181/25165824				
$w \in [3/4,1]$, $\kappa(w) = 35w/16 - 15/16$				
20	$2w-\delta$	δ	$(3/4,51/64)$, $(3/4,3/4)$, $(15/19,15/19)$	31725/1514143744
21	$2w-\delta$	δ	$(31/35,1)$, $(3/4,1)$, $(3/4,51/64)$, $(15/19,15/19)$	66570721309/20500709376000
22	$2w-\delta$	δ	$(1,1)$, $(31/35,1)$, $(15/19,15/19)$	37888/46433625
23	$\delta/2-1/2$	$w+1$	$(1,7/2)$, $(1,5)$, $(3/4,9/2)$, $(3/4,77/32)$	2425775/12582912
24	$\delta/2-1/2$	$w+1$	$(1,2)$, $(1,7/2)$, $(3/4,77/32)$, $(3/4,7/4)$	315695/1048576
25	$\delta/2-1/2$	δ	$(1,2)$, $(3/4,7/4)$, $(3/4,3/2)$	53525/1572864
26	$2w-\delta/2-1/2$	δ	$(1,5/4)$, $(1,2)$, $(3/4,3/2)$, $(3/4,1)$, $(31/35,1)$	4384610497/33718272000
27	$2w-\delta/2-1/2$	δ	$(1,1)$, $(1,5/4)$, $(31/35,1)$	186619/65856000
$\int \min(\kappa(w),y) = 105964289/159383552$, $\int \kappa(w)\rho(w) = 50075/98304$, saving 74328067/478150656				
$2m \le j \le 5m/2$, unit scale, cutoff 5/4				
	$3/2-\delta/2$	δ	$\delta \in [1,5/4]$	25/512
	$3/2-\delta/2$	δ	$\delta \in [5/4,2]$	387/512
	$\delta/2-1/2$	2	$\delta \in [2,7/2]$	63/32
	$\delta/2-1/2$	2	$\delta \in [7/2,5]$	45/64
$\int \min(5/4,y) = 445/128$, saving $\frac{1}{8}\left(\frac{445}{128} - \frac{5}{4}\cdot 2\right) = 125/1024$				

Table 81 lists the point moments of Theorem 6.5(2), with the fixed coefficient bounded by 5 at any point of f_x. In Table 81 the row "savings at q" lists the savings subtracted from the row q, and the entries moving, fixed, and moment of that row belong to the point q.

TABLE 81. Point moments on f_x for Theorem 6.5.

point	$0 \le j \le m$	$m \le j \le 2m$	$2m \le j \le 5m/2$	moving	fixed	moment
q	1/4	223/96	17/32			
savings at q	–	24549209/119537664	125/1024	331923623/358612992	5/72	356827303/358612992
$a \ne q$	5/24	43/24	5/12	29/36	5/72	7/8

Remark A.17. The bound at q is

$$\frac{1}{3}\left(\frac{1}{4} + \frac{223}{96} + \frac{17}{32} - \frac{24549209}{119537664} - \frac{125}{1024}\right) + \frac{5}{72} = \frac{356827303}{358612992},$$

and it is smaller than 215/216 because

$$215 \cdot 358612992 - 216 \cdot 356827303 = 27095832 > 0.$$

Mixed points of width two.

Remark A.18. In the group of tables for Proposition 8.8, z, u, v, ℓ, d, and δ are the letters of **Step 3** of the proof of Proposition 8.8, and ℓ is the order along f_x. Let p_0 be the point of $Z_U \cap f_x$ in Proposition 8.8, and let $q := s \cap f_x$. Let z be a coordinate on the base of the ruling with $f_x = (z = 0)$, let u be a fiber coordinate on $E\backslash s$, and let $v := 1/u$ near q. With u translated so that $p_0 = (0,0)$ if $p_0 \notin s$, the marked ideal is (z, u^2) if $p_0 \notin s$ and (z, v^2) if $p_0 \in s$. The torus action $(z,u) \to (\lambda^{-1}z, \lambda^B u)$, for a fixed integer $B > 0$, fixes the marked ideal and degenerates the residual part of length two to one of the models (h), (v), (r+t), and (t_v) introduced before Table 76. If $p_0 \in s$, then only (h) and (v) occur. The monomials are $z^\ell u^d$ with $0 \le d \le \alpha$ and $0 \le \ell \le \beta - d$ if $p_0 \notin s$, and $z^\ell v^d$ with $0 \le d \le \alpha$ and $0 \le \ell \le \beta - \alpha + d$ if $p_0 \in s$. The marked condition is $2\ell + d \ge 2(j - m)$. With $\delta := d/m$, the normalized order along f_x runs from

$$\ell_-(w,\delta) = \max\left(0, w - \frac{\delta}{2}\right)$$

to $\ell_+(w,\delta)$, the minimum of the following affine functions:

$$\begin{array}{lll}
p_0 \notin s,\ (\mathrm{h}): & 2 + w - \delta, & 2 - w + \delta;\\
p_0 \notin s,\ (\mathrm{v}): & 2 + w - \delta, & 2 - \delta/2;\\
p_0 \notin s,\ (\mathrm{r{+}t}): & 2 + w - \delta, & 2,\quad 3 + w - 2\delta;\\
p_0 \notin s,\ (\mathrm{t}_v): & 2 + w - \delta, & 5/2 + w/2 - 3\delta/2;\\
p_0 \in s,\ (\mathrm{h}): & 1 + \delta, & 3 - \delta;\\
p_0 \in s,\ (\mathrm{v}): & 1 + \delta, & 3/2 - w/2 + \delta/2.
\end{array}$$

The integrand is $(\ell_+^2 - \ell_-^2)/2$ on $0 \le w \le 1$, $0 \le \delta \le 1 + w$, $\ell_- \le \ell_+$. Table 82 lists the polygons of the six models for the bound $S(W; f_x) \le 35/36$ of Proposition 8.8.

TABLE 82. Polygons of the six models for Proposition 8.8.

model	ℓ_-	ℓ_+	vertices (w,δ)	integral
$p_0 \notin s$, (h)	0	$w - \delta + 2$	$(0,0)$, $(1,2)$, $(0,1)$	11/24
	$w - \delta/2$	$w - \delta + 2$	$(0,0)$, $(1,1)$, $(1,2)$	67/96
	$w - \delta/2$	$-w + \delta + 2$	$(0,0)$, $(1,0)$, $(1,1)$	61/96
$p_0 \notin s$, (v)	0	$w - \delta + 2$	$(0,0)$, $(1,2)$, $(0,1)$	11/24
	$w - \delta/2$	$2 - \delta/2$	$(0,0)$, $(1,0)$, $(1,2)$	4/3
$p_0 \notin s$, (r+t)	0	$w - \delta + 2$	$(0,0)$, $(1/2,1)$, $(0,1)$	55/192
	0	$w - 2\delta + 3$	$(1/2,1)$, $(1,2)$, $(0,1)$	19/192
	$w - \delta/2$	$w - \delta + 2$	$(0,0)$, $(1,1)$, $(1/2,1)$	5/12
	$w - \delta/2$	2	$(0,0)$, $(1,0)$, $(1,1)$	89/96
	$w - \delta/2$	$w - 2\delta + 3$	$(1,1)$, $(1,2)$, $(1/2,1)$	3/16
$p_0 \notin s$, (t_v)	0	$w - \delta + 2$	$(0,0)$, $(1/3,2/3)$, $(0,1)$	133/648
	0	$w/2 - 3\delta/2 + 5/2$	$(1/3,2/3)$, $(1,2)$, $(0,1)$	49/324
	$w - \delta/2$	$w - \delta + 2$	$(0,0)$, $(1,0)$, $(1/3,2/3)$	131/162
	$w - \delta/2$	$w/2 - 3\delta/2 + 5/2$	$(1,0)$, $(1,2)$, $(1/3,2/3)$	71/81
$p_0 \in s$, (h)	0	$\delta + 1$	$(0,0)$, $(1/2,1)$, $(0,1)$	17/48
	0	$3 - \delta$	$(1/2,1)$, $(1,2)$, $(0,1)$	17/48
	$w - \delta/2$	$\delta + 1$	$(0,0)$, $(1,0)$, $(1,1)$, $(1/2,1)$	47/64
	$w - \delta/2$	$3 - \delta$	$(1,1)$, $(1,2)$, $(1/2,1)$	67/192
$p_0 \in s$, (v)	0	$\delta + 1$	$(0,0)$, $(1/3,2/3)$, $(0,1)$	133/648
	0	$-w/2 + \delta/2 + 3/2$	$(1/3,2/3)$, $(1,2)$, $(0,1)$	193/324
	$w - \delta/2$	$\delta + 1$	$(0,0)$, $(1,0)$, $(1/3,2/3)$	73/324
	$w - \delta/2$	$-w/2 + \delta/2 + 3/2$	$(1,0)$, $(1,2)$, $(1/3,2/3)$	62/81

Remark A.19. Table 83 lists, for the six models, the terms of (8.11) in the proof of Proposition 8.8, where $\Psi(\alpha,\beta,\gamma)$ is the polygon integral of that proof. The integral over $0 \le j \le m$ equals 11/24 for all six models. The integrals over $m \le j \le 2m$ and over $2m \le j \le 5m/2$ are $\int_0^1 \Psi(1+w, 2+w, w)dw$ and $\Psi(2,3,1)/8$. The column moving is one third of the sum of the three integrals, and the last column adds the fixed term μJ, where $\mu \le 2$, and $\mu = 1$ if $p_0 \in s$.

TABLE 83. Fiber moments of the six models for Proposition 8.8.

model	$\int_0^1 \Psi(1+w,2+w,w)dw$	$\Psi(2,3,1)/8$	moving	with μJ
$p_0 \notin s$, (h)	43/24	1/4	5/6	31/36
$p_0 \notin s$, (v)	43/24	1/4	5/6	31/36
$p_0 \notin s$, (r+t)	23/12	7/24	8/9	11/12
$p_0 \notin s$, (t_v)	49/24	1/3	17/18	35/36
$p_0 \in s$, (h)	43/24	1/4	5/6	61/72
$p_0 \in s$, (v)	43/24	1/4	5/6	61/72

Remark A.20. For the point moments of Proposition 8.8, let $\xi := \ell/m$ be the normalized order along f_x. If $p_0 \notin s$, the row with normalized order ξ along f_x is contained in the span of the u^d with $\max(0, 2(w-\xi)) \le \delta \le \min(\alpha, \beta-\xi)$, and u^d has order $\alpha-\delta$ at q. At any point $a \neq q$ the row is contained in the polynomials of degree $\min(\alpha, \beta-\xi)$, as for Table 78. If $p_0 = q \in s$, the containing rows at q consist of the $z^\ell v^d$ with $0 \le \delta \le \alpha$ and $\max(0, w-\delta/2) \le \xi \le \beta-\alpha+\delta$, and $z^\ell v^d$ has order δ at q. The density in δ of the containing rows is

$$1+\delta-\max\Big(0, w-\frac{\delta}{2}\Big), \qquad 0 \le \delta \le 1+w,$$

and the area of the containing rows, $\int_0^{1+w} \big(1+\delta-\max(0,w-\delta/2)\big)d\delta$, exceeds $\rho(w) = \lim m^{-2} \dim W_{m,j}$ by w^2. The cutoff w^2 saves

$$w^2 \cdot w^2 - \int_0^{w^2} (w^2-\delta)\Big(1-w+\frac{3\delta}{2}\Big)d\delta = \frac{w^4}{2}+\frac{w^5}{2}-\frac{w^6}{4}, \qquad \int_0^1 \Big(\frac{w^4}{2}+\frac{w^5}{2}-\frac{w^6}{4}\Big)dw = \frac{31}{210}.$$

At unit scale for $2m \le j \le 5m/2$ the containing rows have area 3 and $\lim_{k\to+\infty} k^{-2} \dim U_k = 2$, the density is $3\delta/2$ on $[0,2]$, and the cutoff 1 saves $1-\int_0^1 (1-\delta)\frac{3\delta}{2}d\delta = \frac{3}{4}$, that is, $\frac{3}{4}\cdot\frac{1}{8} = \frac{3}{32}$ after multiplication by 1/8. The fixed coefficient of $(D|_E - \mu f_x)|_{f_x}$ is at most 5 at any point. Table 84 lists the point moments for the bound $S(W^{f_x}; a) \le 3323/3360$ of Proposition 8.8.

TABLE 84. Point moments on f_x for Proposition 8.8.

point	$0 \le j \le m$	$m \le j \le 2m$	$2m \le j \le 5m/2$	moving	fixed	moment
q, $p_0 \notin s$	1/4	2	5/12	8/9	5/72	23/24
$a \neq q$	5/24	43/24	5/12	29/36	5/72	7/8
$q = p_0 \in s$	1/4	9/4	1/2			
savings at $q = p_0 \in s$	–	31/210	3/32	9269/10080	5/72	3323/3360

Remark A.21. In Table 84, the moving entry of the point $q = p_0 \in s$ is $1-\frac{1}{3}\big(\frac{31}{210}+\frac{3}{32}\big) = 1-\frac{811}{10080}$.

Mixed points of width three.

Remark A.22. In the group of tables for Proposition 8.9, z, u, ℓ, d, and δ are the letters of **Step 3** of the proof of Proposition 8.9, and ℓ is the order along f_x. For Proposition 8.9, let $q := s \cap f_x$, let p_0 be the root of the component of length three of Z_U on f_x and p_1 the residual simple point. By Lemma 8.3, the case $p_0 \in s$ and the case $p_0 \notin s$, $p_1 \in s$ do not occur, so only the case $p_0 \notin s$, $p_1 \notin s$ is listed. After translation and scaling, the marked ideal is $(z-u^2, u^3)$, and the torus action $(z,u) \to (\lambda^{-2}z, \lambda^{-1}u)$ fixes it and, as $\lambda \to 0$, degenerates p_1 to the point $r = (\zeta, u_\infty) = (0,0)$ of the fiber f_∞ over the point where z is ∞, where $\zeta = 1/z$ and $u_\infty = u/z$. Group the monomials $z^\ell u^{d-2\ell}$ by d, and let $\delta := d/m$. The normalized order ℓ/m along f_x runs from

$$\ell_-(w,\delta) = \max\Big(0, \frac{\delta-1-w}{2}, \delta-2-w\Big) \quad \text{to} \quad \ell_+(w,\delta) = \min\Big(\frac{\delta}{2}, 2\Big) - \max(0, 3w-\delta),$$

and the integrand is $(\ell_+^2-\ell_-^2)/2$ on $0 \le w \le 1$, $\ell_- \le \ell_+$. Table 85 lists the six polygons and their integrals for the bound $S(W; f_x) \le 271/288$ of Proposition 8.9.

TABLE 85. Polygons of the case $p_0 \notin s$, $p_1 \notin s$ for Proposition 8.9.

ℓ_-	ℓ_+	vertices (w, δ)	integral
0	$\delta/2$	$(0,0)$, $(1/2,3/2)$, $(0,1)$	$19/768$
0	$-3w+3\delta/2$	$(0,0)$, $(1,2)$, $(1/2,3/2)$	$3/256$
$-w/2+\delta/2-1/2$	$\delta/2$	$(1,3)$, $(1,4)$, $(0,3)$, $(0,1)$, $(1/2,3/2)$	$913/768$
$-w/2+\delta/2-1/2$	$-3w+3\delta/2$	$(1,2)$, $(1,3)$, $(1/2,3/2)$	$59/768$
$-w+\delta-2$	$\delta/2$	$(1,4)$, $(0,4)$, $(0,3)$	$37/96$
$-w+\delta-2$	2	$(1,4)$, $(1,5)$, $(0,4)$	$7/24$

Remark A.23. The six integrals of Table 85 sum to $95/48$, and Table 86 lists the moment of f_x. **Step 4** of the proof of Proposition 8.9 proves the bounds for the points of f_x. Their values $23/24$ at q and $7/8$ elsewhere are the rows "q, $p_0 \notin s$" and "$a \neq q$" of Table 84.

TABLE 86. Fiber moment for Proposition 8.9.

curve	$0 \le j \le m$	$m \le j \le 2m$	$2m \le j \le 5m/2$	moving	fixed	moment
f_x	$11/24$	$95/48$	$29/96$	$263/288$	$\mu J \le 2J = 1/36$	$271/288$

Mixed points of width four.

Remark A.24. For the tables of Theorem 8.12, x, y, z, w denote the coordinates of Subsection 8.3. The variables $w = j/m - 1$ and $y = \ell/m$ of the tables above are not used there. We use the notation of Subsection 8.3: E has coarse space $\mathbb{P}(1,1,3)$ with coordinates $(x : z : w)$, $w = y^2$, $C_x = (x = 0)$, $C_z = (z = 0)$, $C_w = (w = 0)$, $\Delta_E = \frac{1}{2}C_w$, $o_x = [1 : 0 : 0]$, $o_z = [0 : 1 : 0]$, $o_w = [0 : 0 : 1]$, and $C_\theta = (w - \theta x^2 z = 0)$ for $\theta \in \mathbb{C}^*$. The auxiliary weight ω has $\omega(x) = 0$, $\omega(y) = 1$, $\omega(z) = 2$, and W' is the associated graded series of W for ω. A monomial $x^a y^b z^e$ has character $(u_1, u_2) = (a + b, b + 2e)$, and $\varphi_1 = x^2 z - y^2$ has character $(2, 2)$. Normalized characters and normalized φ_1-orders γ form the polytope

$$\mathcal{P} = \big\{(u_1, u_2, \gamma) : \gamma_-(u_1) \le \gamma \le \gamma_+(u_1, u_2),\ u_1 \le 3,\ u_2 \le 4\big\},$$

$$\gamma_-(u_1) = \max\Big(0, \frac{u_1 - 1}{2}\Big), \qquad \gamma_+(u_1, u_2) = \min\Big(\frac{u_1}{2}, \frac{u_2}{2}, 1 + u_1 - \frac{u_2}{2}, 3 - \frac{u_2}{2}\Big),$$

with vertices $(0,0,0)$, $(1,0,0)$, $(0,2,0)$, $(1,4,0)$, $(3,2,1)$, $(3,4,1)$, $(2,4,1)$, and $(3,3,3/2)$. Within a character the highest y-exponent is 2γ and the lowest is $2(\gamma - \gamma_-)$. In the order (γ, u_1, u_2), $\mathcal{P}$ is the union of the three regions

$$\begin{array}{lll} 0 \le \gamma \le 1, & 2\gamma \le u_1 \le 1 + \gamma, & 2\gamma \le u_2 \le 2 + 2u_1 - 2\gamma; \\ 0 \le \gamma \le 1, & 1 + \gamma \le u_1 \le 1 + 2\gamma, & 2\gamma \le u_2 \le 4; \\ 1 \le \gamma \le 3/2, & 2\gamma \le u_1 \le 3, & 2\gamma \le u_2 \le 6 - 2\gamma. \end{array}$$

Iterated integration over the three regions yields the first four integrals in Table 87. For γ_- write $\gamma_- = \frac{u_1 - 1}{2} + \frac{1}{2}(1 - u_1)_+$. The first term integrates to $43/48$. For $0 \le u_1 \le 1$ the slice of $\mathcal{P}$ has area $u_1 + u_1^2/2$, so the second term integrates to $\frac{1}{2}\int_0^1 (1 - u_1)(u_1 + u_1^2/2)du_1 = 5/48$. Table 87 lists the integrals over $\mathcal{P}$ and the means for Theorem 8.12.

Remark A.25. The tables below contain the chamber and polytope data for Lemmas 7.8, 7.9, 7.10, 7.16, and 9.5, Proposition 9.8, and Theorems 9.9 and 9.13. The integrals are computed as in Remark A.9. Here the integrands over the polygons are polynomials of degree at most two in two variables, integrated over the triangles of a subdivision directly, and each polytope is split into tetrahedra with a common vertex.

Chamber tables on $\mathbb{F}_2$.

TABLE 87. Integrals and means over the polytope $\mathcal{P}$ for Theorem 8.12.

function	$\int_{\mathcal{P}}$	mean $= \frac{1}{3}\int_{\mathcal{P}}$
1	3	1
u_1	$115/24$	$115/72$
u_2	$43/6$	$43/18$
γ	$13/8$	$13/24$
γ_-	1	$1/3$
$\gamma - \gamma_-$	$5/8$	$5/24$
$2u_1 + u_2$	$67/4$	$67/12 = S_X(E)$

Remark A.26. In Tables 88–92, $S = \mathbb{F}_2$ with negative section σ and fiber class f, and classes on blow-ups of S are total transforms. We set $a := k/m$. For a section in $R_{m,k}$ of order ℓ along the flag curve we set $u := \ell/m$, except in Table 89, where the flag curve is E_0 and the order along E_0 is $m(a-1)_+ + \ell$ (Remark A.29). The range is early for $0 \le a \le 1$ and middle for $1 \le a \le 2$. Each row describes a polygon in the (a,u)-plane by the vertices in boundary order, and on the interior of the polygon the residual divisor $D_{a,u}$ has the Zariski decomposition $D_{a,u} = D^+_{a,u} + D^-_{a,u}$ with the components of $D^-_{a,u}$ listed in the support column. The coefficients of $D^-_{a,u}$ are the solution of the linear equations

$$\big((D_{a,u} - D^-_{a,u}) \cdot \Gamma\big) = 0 \qquad \text{for the curves } \Gamma \text{ in the support column.}$$

The column w is the coefficient of the strict transform of σ in $D^-_{a,u}$, and d is the intersection number of $D^+_{a,u}$ with the flag curve. In the middle range the series has the fixed divisor $m(a-1)\sigma$. All integrals are taken with respect to $dadu$ over the union of the polygons of the table. Since $n_m = \frac{3}{2}m^3 + O(m^2)$, the normalization factor is $2/3$.

Tables for Lemma 7.8.

Remark A.27. Tables 88 and 89 belong to Lemma 7.8. We use the surfaces $\tilde{S}$ and $\tilde{S}'$, the classes $e_1, \dots, e_4$, and the curves $\tilde{\sigma}$, $c_1, \dots, c_4$, $e_1, \dots, e_4$ of Construction-Proposition 7.7 with Z_B reduced, and on $\tilde{S}'$ the curve $\tilde{\sigma}'$ of Construction-Proposition 7.7(3). In a row whose support contains e_3, the coefficient of e_3 in $D^-_{a,u}$ is w in the early range and $w - (a-1)$ in the middle range. In a row whose support contains c_1, c_2, both curves have coefficient $w - (2-a)$. If Z_B is not reduced, both tables hold on the surfaces of Construction-Proposition 7.7(2) and (3) after replacing $c_1 + c_2$ by $2C_1 + D_1$ in $D^-_{a,u}$.

For Table 88, the flag curve is the strict transform of a fiber C disjoint from $\operatorname{Supp} Z$, and

$$D_{a,u} = a\sigma + (a+1-u)f \quad (0 \le a \le 1),$$
$$D_{a,u} = \sigma + (a+1-u)f - (a-1)(e_1+e_2+e_3) \quad (1 \le a \le 2).$$

The restriction to C has the fixed part $m((a-1)_+ + w)$ at $C \cap \sigma$ and the free degree $md + O(1)$.

TABLE 88. Chambers along a fiber C disjoint from $\operatorname{Supp} Z$ for Lemma 7.8.

Ω_i	range	vertices in (a,u)	support of $D^-_{a,u}$	d	w
Ω_1	early	$(0,0),(1,0),(0,1)$	$\emptyset$	a	0
Ω_2	early	$(1,0),(1,2),(0,1)$	$\tilde{\sigma}, e_3$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$
Ω_3	middle	$(1,0),(2,0),(\frac{3}{2},\frac{3}{2})$	$\tilde{\sigma}$	$1 - \frac{u}{3}$	$\frac{u}{3}$
Ω_4	middle	$(1,0),(\frac{3}{2},\frac{3}{2}),(1,2)$	$\tilde{\sigma}, e_3$	$\frac{a-u+1}{2}$	$\frac{-a+u+1}{2}$
Ω_5	middle	$(2,0),(2,1),(\frac{3}{2},\frac{3}{2})$	$\tilde{\sigma}, c_1, c_2$	$5 - 2a - u$	$2a + u - 4$

Remark A.28. The integrals over Table 88 are

$$\int d = \frac{3}{2}, \qquad \int ud = \frac{11}{12}, \qquad \int \frac{d^2}{2} = \frac{1}{2}, \qquad \int d\big((a-1)_+ + w\big) = \frac{19}{24}.$$

Thus

$$S(R; \mathrm{ord}_C) = \frac{2}{3}\cdot\frac{11}{12} = \frac{11}{18}, \qquad S(R^C; C\cap\sigma) \le \frac{2}{3}\Big(\frac{1}{2}+\frac{19}{24}\Big) = \frac{31}{36},$$
$$S(R^C; q) \le \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3} \quad \text{for } q \in C\backslash\sigma.$$

Remark A.29. For Table 89, recall that E_0 is the exceptional curve of $\mathrm{Bl}_{q_0} S$. The strict transform of E_0 on $\tilde{S}'$ has class e_4 and is the flag curve, and

$$D_{a,u} = a\sigma + (a+1)f - ue_4 \quad (0 \le a \le 1),$$
$$D_{a,u} = \sigma + (a+1)f - (a-1)(e_1+e_2+e_3) - ue_4 \quad (1 \le a \le 2).$$

A section of order $m(a-1)_+ + \ell$ along E_0 gives the point (a,u) with $u := \ell/m$. The column w is the coefficient of $\tilde{\sigma}'$, the column z is the coefficient of c_4, and $d = u - w - z$. Let q_σ and q_f be the points of E_0 given by the tangent directions of σ and of the fiber through q_0. The restriction to E_0 has the fixed part $m((a-1)_+ + w)$ at q_σ, the fixed part mz at q_f, and the free degree $md + O(1)$.

TABLE 89. Chambers along E_0 for Lemma 7.8.

Ω'_i	range	vertices in (a,u)	support of $D^-_{a,u}$	d	w	z
Ω'_1	early	$(0,0),(1,0),(\frac12,\frac12)$	$\emptyset$	u	0	0
Ω'_2	early	$(0,0),(\frac12,\frac12),(0,1)$	c_4	a	0	$u-a$
Ω'_3	early	$(1,0),(1,1),(\frac12,\frac12)$	$\tilde{\sigma}', e_3$	$\frac{-a+2u+1}{3}$	$\frac{a+u-1}{3}$	0
Ω'_4	early	$(1,1),(1,3),(0,1),(\frac12,\frac12)$	$\tilde{\sigma}', c_4, e_3$	$\frac{2a-u+1}{3}$	$\frac{a+u-1}{3}$	$u-a$
Ω'_5	middle	$(1,0),(2,0),(\frac74,1),(\frac54,1)$	$\tilde{\sigma}'$	$\frac{3u}{4}$	$\frac{u}{4}$	0
Ω'_6	middle	$(\frac54,1),(\frac74,1),(\frac32,2)$	$\tilde{\sigma}', c_4$	$1-\frac{u}{4}$	$\frac{u}{4}$	$u-1$
Ω'_7	middle	$(1,0),(\frac54,1),(1,1)$	$\tilde{\sigma}', e_3$	$\frac{a+2u-1}{3}$	$\frac{-a+u+1}{3}$	0
Ω'_8	middle	$(2,0),(2,1),(\frac74,1)$	$\tilde{\sigma}', c_1, c_2$	$2-a+\frac{u}{2}$	$a+\frac{u}{2}-2$	0
Ω'_9	middle	$(1,3),(1,1),(\frac54,1),(\frac32,2)$	$\tilde{\sigma}', c_4, e_3$	$\frac{a-u+2}{3}$	$\frac{-a+u+1}{3}$	$u-1$
Ω'_{10}	middle	$(2,1),(2,2),(\frac32,2),(\frac74,1)$	$\tilde{\sigma}', c_1, c_2, c_4$	$3-a-\frac{u}{2}$	$a+\frac{u}{2}-2$	$u-1$
Ω'_{11}	middle	$(2,2),(1,3),(\frac32,2)$	$\tilde{\sigma}', c_1, c_2, c_4, e_3$	$4-a-u$	$a+u-3$	$u-1$

Remark A.30. The integrals over Table 89 are

$$\int d = \frac{3}{2}, \qquad \int \big((a-1)_+ + u\big)d = \frac{53}{24}, \qquad \int \frac{d^2}{2} = \frac{11}{32},$$
$$\int \Big(\frac{d^2}{2} + d\big((a-1)_+ + w\big)\Big) = \frac{31}{24}, \qquad \int \Big(\frac{d^2}{2} + dz\Big) = \frac{11}{12}.$$

Thus

$$S(R; \mathrm{ord}_{E_0}) = \frac{2}{3}\cdot\frac{53}{24} = \frac{53}{36} < \frac{3}{2} = A_{S,B/2}(\mathrm{ord}_{E_0}),$$

and

$$S(R^{E_0}; q_\sigma) \le \frac{2}{3}\cdot\frac{31}{24} = \frac{31}{36}, \qquad S(R^{E_0}; q_f) \le \frac{2}{3}\cdot\frac{11}{12} = \frac{11}{18},$$
$$S(R^{E_0}; q) \le \frac{2}{3}\cdot\frac{11}{32} = \frac{11}{48} \quad \text{for } q \in E_0\backslash\{q_\sigma, q_f\}.$$

Tables for Lemma 7.9.

Remark A.31. Table 90 belongs to Lemma 7.9. Let $\tilde{S} \to S$ be the composition of the three point blow-ups resolving Δ, with total exceptional classes e_1, e_2, e_3. The curves

$$\tilde{\sigma} = \sigma - e_1, \quad C_1 = f - e_1, \quad D_1 = e_1 - e_2, \quad D_2 = e_2 - e_3, \quad e_3$$

generate the cone of curves of $\tilde{S}$. We have $\tilde{\sigma}^2 = -3$, $D_1^2 = D_2^2 = -2$, $C_1^2 = e_3^2 = -1$, and

$$(\tilde{\sigma} \cdot D_1) = (C_1 \cdot D_1) = (D_1 \cdot D_2) = (D_2 \cdot e_3) = 1,$$

and the intersection numbers of the remaining pairs of distinct curves among them vanish. The flag curve is the strict transform of a fiber C not through q_0, and

$$D_{a,u} = a\sigma + (a+1-u)f \quad (0 \le a \le 1),$$
$$D_{a,u} = \sigma + (a+1-u)f - (a-1)(e_1+e_2+e_3) \quad (1 \le a \le 2).$$

The restriction to C has the fixed part $m((a-1)_+ + w)$ at $C \cap \sigma$ and the free degree $md + O(1)$.

TABLE 90. Chambers along a fiber C not through q_0 for Lemma 7.9.

Ω_i	range	vertices in (a,u)	support of $D^-_{a,u}$	d	w
Ω_1	early	$(0,0),(1,0),(0,1)$	$\emptyset$	a	0
Ω_2	early	$(1,0),(1,2),(0,1)$	$\tilde{\sigma}, D_1, D_2, e_3$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$
Ω_3	middle	$(1,0),(2,0),(\frac{4}{3},\frac{7}{3})$	$\tilde{\sigma}, D_1, D_2$	$1-\frac{3u}{7}$	$\frac{3u}{7}$
Ω_4	middle	$(2,0),(2,1),(\frac{4}{3},\frac{7}{3})$	$\tilde{\sigma}, C_1, D_1, D_2$	$5-2a-u$	$2a+u-4$
Ω_5	middle	$(1,0),(\frac{4}{3},\frac{7}{3}),(1,2)$	$\tilde{\sigma}, D_1, D_2, e_3$	$\frac{a-u+1}{2}$	$\frac{-a+u+1}{2}$

Remark A.32. The integrals over Table 90 are

$$\int d = \frac{3}{2}, \qquad \int ud = \frac{23}{24}, \qquad \int \frac{d^2}{2} = \frac{17}{36}, \qquad \int d\big((a-1)_+ + w\big) = \frac{61}{72}.$$

Thus

$$S(R; \mathrm{ord}_C) = \frac{2}{3} \cdot \frac{23}{24} = \frac{23}{36}, \qquad S(R^C; C \cap \sigma) \le \frac{2}{3}\Big(\frac{17}{36} + \frac{61}{72}\Big) = \frac{95}{108},$$
$$S(R^C; q) \le \frac{2}{3} \cdot \frac{17}{36} = \frac{17}{54} \quad \text{for } q \in C \backslash \sigma.$$

Tables for Lemma 7.10.

Remark A.33. Table 91 belongs to Lemma 7.10. Let $\tilde{S} \to S$ be the composition of the blow-up of q_0, the blow-up of the point of the exceptional curve given by the tangent direction of Z_0, and the blow-up of q_3, with total exceptional classes e_1, e_2, e_3. The curves

$$\tilde{\sigma} = \sigma - e_1, \quad C_1 = f - e_1, \quad c_3 = f - e_3, \quad D_1 = e_1 - e_2, \quad e_2, \quad e_3$$

generate the cone of curves of $\tilde{S}$. We have $\tilde{\sigma}^2 = -3$, $D_1^2 = -2$, $C_1^2 = c_3^2 = e_2^2 = e_3^2 = -1$, and

$$(\tilde{\sigma} \cdot c_3) = (\tilde{\sigma} \cdot D_1) = (C_1 \cdot D_1) = (c_3 \cdot e_3) = (D_1 \cdot e_2) = 1,$$

and the intersection numbers of the remaining pairs of distinct curves among them vanish, $(\tilde{\sigma} \cdot C_1) = 0$ in particular. The flag curve is the strict transform of a fiber C disjoint from Z, and

$$D_{a,u} = a\sigma + (a+1-u)f \quad (0 \le a \le 1),$$
$$D_{a,u} = \sigma + (a+1-u)f - (a-1)(e_1+e_2+e_3) \quad (1 \le a \le 2).$$

The restriction to C has the fixed part $m((a-1)_+ + w)$ at $C \cap \sigma$ and the free degree $md + O(1)$.

TABLE 91. Chambers along a fiber C disjoint from Z for Lemma 7.10.

Ω_i	range	vertices in (a,u)	support of $D^-_{a,u}$	d	w
Ω_1	early	$(0,0),(1,0),(0,1)$	$\emptyset$	a	0
Ω_2	early	$(1,0),(1,2),(0,1)$	$\tilde{\sigma}, D_1, e_2$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$
Ω_3	middle	$(1,0),(2,0),(\frac{4}{3},\frac{5}{3})$	$\tilde{\sigma}, D_1$	$1-\frac{2u}{5}$	$\frac{2u}{5}$
Ω_4	middle	$(2,0),(\frac{3}{2},2),(\frac{4}{3},\frac{5}{3})$	$\tilde{\sigma}, c_3, D_1$	$\frac{7-2a-2u}{3}$	$\frac{2a+2u-4}{3}$
Ω_5	middle	$(1,0),(\frac{4}{3},\frac{5}{3}),(1,2)$	$\tilde{\sigma}, D_1, e_2$	$\frac{a-u+1}{2}$	$\frac{-a+u+1}{2}$
Ω_6	middle	$(2,0),(2,1),(\frac{3}{2},2)$	$\tilde{\sigma}, C_1, c_3, D_1$	$5-2a-u$	$2a+u-4$
Ω_7	middle	$(1,2),(\frac{4}{3},\frac{5}{3}),(\frac{3}{2},2)$	$\tilde{\sigma}, c_3, D_1, e_2$	$2-u$	$u-1$

Remark A.34. The integrals over Table 91 are

$$\int d = \frac{3}{2}, \qquad \int ud = \frac{17}{18}, \qquad \int \frac{d^2}{2} = \frac{23}{48}, \qquad \int d\big((a-1)_+ + w\big) = \frac{5}{6}.$$

Thus

$$S(R;\mathrm{ord}_C) = \frac{2}{3}\cdot\frac{17}{18} = \frac{17}{27}, \qquad S(R^C; C\cap\sigma) \le \frac{2}{3}\Big(\frac{23}{48}+\frac{5}{6}\Big) = \frac{7}{8},$$
$$S(R^C;q) \le \frac{2}{3}\cdot\frac{23}{48} = \frac{23}{72} \quad \text{for } q \in C\backslash\sigma.$$

Tables for Lemma 7.16.

Remark A.35. Table 92 belongs to Lemma 7.16, which is used in configuration (I) of the proof of Theorem 7.2. There the base scheme on S is $Z_B \sqcup \{q_3\}$ with $q_3 \in \sigma\backslash B$, as in Construction-Proposition 7.7. The table treats the fiber C through q_3. Since $C \subset T_Z$, Lemma 7.8 does not cover C. We use $\tilde{S}$ and the curves of Construction-Proposition 7.7. The flag curve is c_3, the strict transform of C. Let $h_3 := (a-1-u)_+$. Then

$$D_{a,u} = a\sigma + (a+1-u)f \quad (0 \le a \le 1),$$
$$D_{a,u} = \sigma + (a+1-u)f - (a-1)(e_1+e_2) - h_3e_3 \quad (1 \le a \le 2).$$

When e_3 is in the support, the coefficient of e_3 in $D^-_{a,u}$ is w. The restriction to C has the fixed part $m((a-1)_+ + w + h_3)$ at $q_3 = C\cap\sigma$, where $(a-1)_+$ is denoted γ in the proof of Lemma 7.16, and the free degree $md + O(1)$.

TABLE 92. Chambers along the fiber C through q_3 in configuration (I) for Lemma 7.16.

range	vertices in (a,u)	support of $D^-_{a,u}$	d	w	h_3
early	$(0,0),(1,0),(0,1)$	$\emptyset$	a	0	0
early	$(1,0),(1,2),(0,1)$	$\tilde{\sigma}, e_3$	$\frac{a-u+1}{2}$	$\frac{a+u-1}{2}$	0
middle	$(1,0),(2,0),(2,1)$	$\emptyset$	$2-a+u$	0	$a-1-u$
middle	$(1,0),(2,1),(1,2)$	$\tilde{\sigma}, e_3$	$\frac{a-u+1}{2}$	$\frac{-a+u+1}{2}$	0

Remark A.36. The integrals over Table 92 are

$$\int d = \frac{3}{2}, \qquad \int ud = 1, \qquad \int \frac{d^2}{2} = \frac{1}{2}, \qquad \int d\big((a-1)_+ + w + h_3\big) = \frac{19}{24}.$$

Thus

$$S(R;\mathrm{ord}_C) = \frac{2}{3}, \qquad S(R^C;q_3) \le \frac{2}{3}\Big(\frac{1}{2}+\frac{19}{24}\Big) = \frac{31}{36},$$
$$S(R^C;q) \le \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3} \quad \text{for } q \in C\backslash\{q_3\}.$$

Tables for Lemma 9.5.

Remark A.37. Tables 93 and 94 belong to Lemma 9.5, which is used in Propositions 9.7 and 9.8. Recall from Set-up 9.3 that $E \cong \mathbb{P}(1,2,3)$ has coordinates $[i : a : e]$, that $q = [1 : 0 : 0]$, that $\lambda \neq 0$, that $f = e - a^2$ on the chart $i = 1$, and that $\nu_{1,3}$ and $\nu_{1,2}$ are the monomial valuations at q with weights $(1,3)$ and $(1,2)$ on (a,f). By Proposition 9.7, $\kappa \neq -1$. Recall from Set-up 9.4 that $\psi = ae - \lambda if$ and $z = \lambda f - a^3$. Since $\lambda \neq 0$, $\nu_{1,3}$ is also the monomial valuation with weights $(1,3)$ on (a,z). The degree-one generators $1, i, a, e, f, \psi, c, l, b, d, M, J$ of Lemma 9.5(1) have the second degrees $j = 0,1,2,3,4,5,6,6,7,8,9,10$. We use the coordinates (j,u,v) of Table 7: for the leading monomial of a product of the generators, j is the second degree, u is the exponent of a, and v is the exponent of z if $\kappa \neq 0$ and of f if $\kappa = 0$, each divided by the anti-canonical degree m.

(1) If $\kappa \neq 0$, let $\nu := \nu_{1,3}$, and let Δ_{gen} be the convex hull of the normalized exponents of the generators and of the normalized exponent $(6, 0, \frac{4}{3})$ of the element $\Phi \in W_{3,18}$ whose $\nu_{1,3}$-leading form is z^4.
(2) If $\kappa = 0$, let $\nu := \nu_{1,2}$, and let Δ_0 be the convex hull of the normalized exponents of the generators.

Let $\varphi : \tilde{E} \to E$ be the weighted blow-up of q defining ν, with exceptional curve C'. Let $p_* \in C'$ be the point where $z/a^3 = 0$ if $\kappa \neq 0$, respectively $f/a^2 = 0$ if $\kappa = 0$. Since $N_m = 3m^3 + O(m^2)$,

$$S(W;\nu) = \frac{1}{3}\Big(\int u + r\int v\Big), \qquad S(W^{C'};p_*) = \frac{1}{3}\int v,$$

where $r = 3$ on Δ_{gen} and $r = 2$ on Δ_0, and the integrals are taken over Δ_{gen}, respectively Δ_0, with respect to $djdudv$. The facets of Δ_{gen} and Δ_0 are

$$\begin{gathered}\Delta_{\mathrm{gen}}:\ u,v \geq 0,\ -9j+14u+45v \leq 6,\ -6j+8u+27v \leq 0,\\ -3j+5u+18v \leq 6,\ j-u-5v \leq 1,\ j-u-4v \leq 2;\\ \Delta_0:\ u,v \geq 0,\ -u+v \leq 1,\ -3j+5u+13v \leq 1,\ -2j+3u+8v \leq 0,\\ j-u-5v \leq 1,\ j-u-3v \leq 3.\end{gathered}$$

TABLE 93. Vertices of Δ_{gen} and of Δ_0 in (j,u,v) for Lemma 9.5.

vertex	Δ_{gen}	Δ_0
v_0	$(0,0,0)$	$(0,0,0)$
v_1	$(1,0,0)$	$(1,0,0)$
v_2	$(4,3,0)$	$(3,2,0)$
v_3	$(6,0,\frac{4}{3})$	$(4,0,1)$
v_4	$(6,0,1)$	$(6,0,1)$
v_5	$(9,3,1)$	$(9,3,1)$
v_6	$(10,0,2)$	$(10,1,2)$

Remark A.38. Thus

$$\begin{aligned}\kappa \neq 0:\ & S(W;\nu_{1,3}) = \frac{1}{3}\Big(\frac{23}{8} + 3\cdot\frac{55}{24}\Big) = \frac{13}{4}, \quad S(W^{C'};p_*) = \frac{55}{72},\\ \kappa = 0:\ & S(W;\nu_{1,2}) = \frac{1}{3}\Big(\frac{11}{4} + 2\cdot\frac{29}{12}\Big) = \frac{91}{36}, \quad S(W^{C'};p_*) = \frac{29}{36}.\end{aligned}$$

Tables for Proposition 9.8.

TABLE 94. A triangulation of Δ_{gen} and of Δ_0 by tetrahedra with vertex v_0, with volumes and linear moments, for Lemma 9.5.

polytope	tetrahedron	volume	$\int u$	$\int v$
Δ_{gen}	$v_0v_5v_3v_6$	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{13}{18}$
	$v_0v_5v_3v_2$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{7}{36}$
	$v_0v_4v_5v_6$	1	$\frac{3}{4}$	1
	$v_0v_4v_1v_2$	$\frac{1}{2}$	$\frac{3}{8}$	$\frac{1}{8}$
	$v_0v_4v_5v_2$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{4}$
	total	3	$\frac{23}{8}$	$\frac{55}{24}$
Δ_0	$v_0v_4v_5v_6$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
	$v_0v_3v_4v_6$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{1}{3}$
	$v_0v_2v_4v_1$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{12}$
	$v_0v_2v_4v_5$	$\frac{1}{2}$	$\frac{5}{8}$	$\frac{1}{4}$
	$v_0v_3v_2v_5$	$\frac{1}{6}$	$\frac{5}{24}$	$\frac{1}{12}$
	$v_0v_3v_5v_6$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
	total	3	$\frac{11}{4}$	$\frac{29}{12}$

Remark A.39. Table 95 belongs to Proposition 9.8, and uses the curves C_τ ($\tau \in \mathbb{C}^*$), C_a, C_e, and C_f through q of Proposition 9.8. For such a curve C, let w be the degree with $C \in |\mathcal{O}_E(w)|$, let $t := \nu_{1,3}(C)$, and let $s := S(W;C)$. The last column bounds $\limsup_m S_m(W^C;p)$ uniformly for $p \in C\backslash\{q\}$, by the expression

$$\frac{7w}{8} - \frac{13t}{12} + \Big(\frac{t^2}{3} - \frac{w^2}{6}\Big)s$$

together with $0 \le s \le \frac{23}{24}$ for all four curves and $s \ge \frac{1}{8}$ for C_τ. The expression uses $S(W;\nu_{1,3}) \ge \frac{13}{4}$. By Lemma 9.5(3) and (4), $S(W;\nu_{1,3}) = \frac{13}{4}$ if $\kappa \ne 0$, and $S(W;\nu_{1,3}) = \frac{1}{3}\big(\frac{11}{4} + 3\cdot\frac{29}{12}\big) = \frac{10}{3}$ if $\kappa = 0$, computed on Δ_0 of Table 94 with the weights $(1,3)$.

TABLE 95. Degrees, $\nu_{1,3}$-values, and bounds for the curves through q for Proposition 9.8.

curve	(w,t)	upper bound for $\limsup_m S_m(W^C;p)$
C_τ	$(5,3)$	$\frac{9}{8} - \frac{7}{6}s \le \frac{47}{48}$
C_a	$(2,1)$	$\frac{2}{3} - \frac{s}{3} \le \frac{2}{3}$
C_e	$(3,2)$	$\frac{11}{24} - \frac{s}{6} \le \frac{11}{24}$
C_f	$(4,3)$	$\frac{1}{4} + \frac{s}{3} \le \frac{41}{72}$

Tables for Theorem 9.9.

Remark A.40. Tables 96 and 97 belong to Theorem 9.9. Here $p_3 = [0:0:1]$, the coordinates $(\alpha_i, \alpha_a, \alpha_e)$ are the exponents of i, a, e of a leading monomial divided by the anti-canonical degree m, and $\nu_{p_3}(i^{n_i}a^{n_a}e^{n_e}) = (n_i + 2n_a)/3$. If $\kappa \ne -2$, the degree-one slice polytopes Δ_{n_0} of the semigroups generated by the twelve generators and $P_1, \dots, P_{n_0}, Q_0, \dots, Q_{n_0}$ increase to the polytope Δ_∞, whose vertices $v_3 = (2, 0, \frac{1}{2})$ and $v_5 = (2, 0, 2)$ are the limits of the normalized exponents of Q_r and P_m. If $\kappa = -2$, the twelve generators alone have the convex hull $\tilde{\Delta}$ of their

normalized exponents. In both cases, since $N_m = 3m^3 + O(m^2)$,

$$S(W;\nu_{p_3}) = \frac{1}{3}\int_{\Delta_\infty} \frac{\alpha_i + 2\alpha_a}{3} d\alpha_i d\alpha_a d\alpha_e \quad \text{if } \kappa \neq -2,$$
$$S(W;\nu_{p_3}) = \frac{1}{3}\int_{\tilde{\Delta}} \frac{\alpha_i + 2\alpha_a}{3} d\alpha_i d\alpha_a d\alpha_e \quad \text{if } \kappa = -2.$$

The facets are

$$\Delta_\infty : \ \alpha_i, \alpha_a, \alpha_e \geq 0,\ \alpha_a \leq 1,\ \alpha_a + \alpha_e \leq 3,\ \alpha_i + \alpha_a - 2\alpha_e \leq 1,$$
$$\alpha_i + \alpha_a \leq 2,\ \alpha_i + 2\alpha_a + \alpha_e \leq 4,\ 2\alpha_i + 3\alpha_a - 2\alpha_e \leq 3;$$
$$\tilde{\Delta} : \ \alpha_i, \alpha_a, \alpha_e \geq 0,\ \alpha_a + \alpha_e \leq 3,\ \alpha_i + \alpha_a - 2\alpha_e \leq 1,$$
$$\alpha_i + 2\alpha_a + \alpha_e \leq 4,\ \alpha_i + 3\alpha_a \leq 3.$$

TABLE 96. Vertices of Δ_∞ and of $\tilde{\Delta}$ in $(\alpha_i, \alpha_a, \alpha_e)$ for Theorem 9.9.

vertex	Δ_∞	vertex	$\tilde{\Delta}$
v_0	$(0,0,0)$	w_0	$(0,0,0)$
v_1	$(1,0,0)$	w_1	$(1,0,0)$
v_2	$(0,1,0)$	w_2	$(0,1,0)$
v_3	$(2,0,\frac{1}{2})$	w_3	$(3,0,1)$
v_4	$(1,1,1)$	w_4	$(0,1,2)$
v_5	$(2,0,2)$	w_5	$(0,0,3)$
v_6	$(0,1,2)$	w_6	$(1,0,3)$
v_7	$(0,0,3)$		
v_8	$(1,0,3)$		

TABLE 97. A triangulation of Δ_∞ and of $\tilde{\Delta}$ by tetrahedra with vertex v_0, respectively w_0, with volumes and linear moments, for Theorem 9.9.

polytope	tetrahedron	volume	$\int \alpha_i$	$\int \alpha_a$	$\int \alpha_e$
Δ_∞	$v_0v_4v_2v_3$	$\frac{1}{4}$	$\frac{3}{16}$	$\frac{1}{8}$	$\frac{3}{32}$
	$v_0v_1v_2v_3$	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{1}{48}$	$\frac{1}{96}$
	$v_0v_5v_4v_3$	$\frac{1}{2}$	$\frac{5}{8}$	$\frac{1}{8}$	$\frac{7}{16}$
	$v_0v_6v_4v_2$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$
	$v_0v_8v_6v_7$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$	1
	$v_0v_8v_5v_4$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{6}$	1
	$v_0v_8v_6v_4$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1
	total	3	$\frac{25}{12}$	$\frac{55}{48}$	$\frac{91}{24}$
$\tilde{\Delta}$	$w_0w_4w_2w_3$	1	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
	$w_0w_1w_2w_3$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{24}$	$\frac{1}{24}$
	$w_0w_6w_4w_5$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$	1
	$w_0w_6w_4w_3$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{1}{3}$	2
	total	3	$\frac{19}{8}$	1	$\frac{91}{24}$

Remark A.41. Thus, for $\kappa \neq -2$ and for $\kappa = -2$ respectively,

$$S(W;\nu_{p_3}) = \frac{1}{9}\Big(\frac{25}{12} + 2\cdot\frac{55}{48}\Big) = \frac{35}{72}, \qquad S(W;\nu_{p_3}) = \frac{1}{9}\Big(\frac{19}{8} + 2\cdot 1\Big) = \frac{35}{72}.$$

Tables for Theorem 9.13.

Remark A.42. Table 98 belongs to Theorem 9.13. Recall that $E_0 \cong \mathbb{P}(1,2,3)$ is the extracted divisor of Construction-Proposition 9.12, with coordinates $[i:a:e]$, and let $f := ie - a^2$. Let $\rho : \mathbb{G}_m \to \mathrm{Aut}(E_0)$ be the one-parameter subgroup with weights $(-1,0,1)$ on (i,a,e). Each bidegree part of the algebra B_0 of Construction-Proposition 9.12 on which ρ acts by one character is contained in the span $\mathcal{V}_{w,y,\epsilon}$ of the monomials $i^p a^\epsilon e^q f^r$ with fixed $\epsilon \in \{0,1\}$, fixed $w = p + r$ and $y = q + r$, and $r_- \le r \le r_+$ for integers $r_\pm$. Let $R := r_+ - r_-$. In the table and in the integrals below, w and y stand for the normalized real variables w/m and y/m. As $m \to +\infty$, $r_\pm/m$ tends to $\bar r_\pm(w,y)$, where

$$\bar r_-(w,y) = \max\Big\{0, \frac{3w+y-3}{4}, \frac{3w+2y-6}{2}\Big\}, \qquad \bar r_+(w,y) = \min\{w,y\},$$

and $R/m \to R_0 := \bar r_+ - \bar r_-$. The region $\{\bar r_- \le \bar r_+\}$ is contained in $[0,2]\times[0,3]$, and the six cells of Table 98 are obtained by fixing the term that realizes the maximum in $\bar r_-$ and the term that realizes the minimum in $\bar r_+$. Let $C_\zeta := (f - \zeta ie = 0)$ for $\zeta \in \mathbb{C}\backslash\{0,1\}$, let $C_0 = (f=0)$, $C_i = (i=0)$, $C_a = (a=0)$, and $C_e = (e=0)$, and let $z_i := [1:0:0]$ and $z_e := [0:0:1]$. Let π_i be the weighted blow-up of z_i with weights $(1,2)$ on (a,e), with exceptional divisor Q_i. Let q_e be the index-three cover of the chart $e=1$ of E_0 by the plane with coordinates (i,a), and let π_e be the weighted blow-up of the point over z_e with weights $(2,1)$ on (i,a), with exceptional divisor Q_e. For each column, the integrand is the limit, divided by m^2, of the largest total order along the curve, or of the largest total value of mult_{Q_i} or of $\mathrm{mult}_{Q_e} q_e^*$, over the bases of each $\mathcal{V}_{w,y,\epsilon}$, whose bounds are the lines of Table 10:

$$C_i:\ (w-\bar r_+)R_0 + \tfrac12 R_0^2, \qquad C_a:\ R_0^2, \qquad C_e:\ (y-\bar r_+)R_0 + \tfrac12 R_0^2, \qquad C_0:\ \tfrac12(\bar r_+^2 - \bar r_-^2),$$
$$C_\zeta:\ \tfrac12 R_0^2, \qquad \mathrm{mult}_{Q_i}:\ 2yR_0, \qquad \mathrm{mult}_{Q_e} q_e^*:\ 2wR_0.$$

The integrals are taken with respect to $dwdy$. The parameter ϵ takes two values and $N_m = 3m^3 + O(m^2)$, so the normalization factor is $2/3$.

TABLE 98. The six cells and their integrals for Theorem 9.13.

$(\bar r_-, \bar r_+)$	vertices in (w,y)	$\int R_0$	C_i	C_a	C_e	C_0	C_ζ	mult_{Q_i}	$\mathrm{mult}_{Q_e} q_e^*$
$(0,w)$	$(0,0),(\frac34,\frac34),(0,3)$	$\frac{9}{32}$	$\frac{27}{512}$	$\frac{27}{256}$	$\frac{135}{512}$	$\frac{27}{512}$	$\frac{27}{512}$	$\frac{81}{128}$	$\frac{27}{128}$
$(0,y)$	$(0,0),(1,0),(\frac34,\frac34)$	$\frac{3}{32}$	$\frac{21}{512}$	$\frac{9}{256}$	$\frac{9}{512}$	$\frac{9}{512}$	$\frac{9}{512}$	$\frac{9}{128}$	$\frac{15}{128}$
$(\frac{3w+y-3}{4}, w)$	$(\frac32,\frac32),(0,3),(\frac34,\frac34)$	$\frac{9}{16}$	$\frac{81}{512}$	$\frac{81}{256}$	$\frac{297}{512}$	$\frac{81}{256}$	$\frac{81}{512}$	$\frac{459}{256}$	$\frac{243}{256}$
$(\frac{3w+y-3}{4}, y)$	$(1,0),(2,1),(\frac32,\frac32),(\frac34,\frac34)$	$\frac{5}{16}$	$\frac{95}{512}$	$\frac{39}{256}$	$\frac{39}{512}$	$\frac{53}{256}$	$\frac{39}{512}$	$\frac{145}{256}$	$\frac{201}{256}$
$(\frac{3w+2y-6}{2}, w)$	$(2,2),(0,3),(\frac32,\frac32)$	$\frac{3}{16}$	$\frac{9}{256}$	$\frac{9}{128}$	$\frac{45}{256}$	$\frac{51}{256}$	$\frac{9}{256}$	$\frac{3}{4}$	$\frac{15}{32}$
$(\frac{3w+2y-6}{2}, y)$	$(2,1),(2,2),(\frac32,\frac32)$	$\frac{1}{16}$	$\frac{7}{256}$	$\frac{3}{128}$	$\frac{3}{256}$	$\frac{21}{256}$	$\frac{3}{256}$	$\frac{3}{16}$	$\frac{7}{32}$
total		$\frac{3}{2}$	$\frac{1}{2}$	$\frac{45}{64}$	$\frac{9}{8}$	$\frac{7}{8}$	$\frac{45}{128}$	4	$\frac{11}{4}$
times $\frac{2}{3}$		1	$\frac{1}{3}$	$\frac{15}{32}$	$\frac{3}{4}$	$\frac{7}{12}$	$\frac{15}{64}$	$\frac{8}{3}$	$\frac{11}{6}$

Remark A.43. The last row bounds, up to $O(1/m)$, the coefficients along C_i, C_a, C_e, C_0, and C_ζ, the value of mult_{Q_i}, and the value of $\mathrm{mult}_{Q_e} q_e^*$ of a normalized basis divisor of B_0 consisting of ρ-eigenvectors. Let t be the variable corresponding to r/m and integrate over $\{(w,y,t) : \bar r_-(w,y) \le t \le \bar r_+(w,y)\}$. Then $\int(w-t) = \frac12$ and $\int t = \frac78$ are the totals of the columns C_i and C_0 of Table 98, since $\int_{\bar r_-}^{\bar r_+}(w-t)dt = (w-\bar r_+)R_0 + \frac12 R_0^2$ and $\int_{\bar r_-}^{\bar r_+} t dt = \frac12(\bar r_+^2 - \bar r_-^2)$, which

checks the total of the column $\mathrm{mult}_{Q_e} q_e^*$:

$$\int 2wR_0 dwdy = 2\Big(\frac{1}{2} + \frac{7}{8}\Big) = \frac{11}{4}.$$

The curves C_ς, C_0, C_a, C_i, and C_e have log discrepancy 1, and Q_i and Q_e have log discrepancy 3, computed for Q_e on the cover q_e. Hence the ratios are $\frac{15}{64}$, $\frac{7}{12}$, $\frac{15}{32}$, $\frac{1}{3}$, $\frac{3}{4}$, $\frac{8}{9}$ at z_i, and $\frac{11}{18}$ at z_e, and the largest ratio is $\frac{8}{9}$, attained at z_i.

Appendix B. Tables for genus twelve

This section lists the lattice regions and the integrals used in Sections 13–15, and computes the Scorza differential at the quartics F_h used in Section 19 (Subsection B.2). Each integral in this section has an integrand which is a polynomial on each cell of a partition of a rational polygon or polytope into rational polygons or polytopes. The cells are cut out by the loci where two of the piecewise affine bounds coincide, and we cut each cell into simplices. After an affine change of variables, each simplex is evaluated exactly by

$$\int_{\{u_1,\dots,u_n\geq 0,\ u_1+\cdots+u_n\leq 1\}} u_1^{i_1}\cdots u_n^{i_n}\,du_1\cdots du_n = \frac{i_1!\cdots i_n!}{(i_1+\cdots+i_n+n)!} \qquad (n=1,2,3).$$

No numerical approximation enters. The regions are fixed rational polytopes, so the lattice sums differ from the integrals by $O(1/m)$, uniformly in the basis and the point. The limits of the normalized sums as $m\to+\infty$ exist by [AZ22, Lemma 2.9].

For the tables of Proposition 13.15 we normalize by $mN_m=\frac{11}{3}m^4+O(m^3)$. Thus a quantity whose rescaled values over the three ranges $0\leq j\leq m$, $m\leq j\leq 2m$, and $2m\leq j\leq 3m$ integrate to $I_{[0,1]}$, $I_{[1,2]}$, and $I_{[2,3]}$ has normalized limit

$$\frac{3}{11}\left(I_{[0,1]}+I_{[1,2]}+I_{[2,3]}\right). \tag{B.1}$$

B.1. **Integral tables.** We first treat Proposition 13.15 and Theorem 13.21.

Tables for Proposition 13.15.

Remark B.1. Notation as in Set-up 13.12 with $r=2$. Fix $\{i,j\}=\{1,2\}$, and write $E=E_i$, $s=s_i$, $f=f_{i,p}$, $Z=Z_{U_i}$, and $n=n_{ij}$. On $E\backslash s$ let z be a coordinate on the base $\mathbb{P}^1$ of the ruling with $f=(z=0)$, let u be a fiber coordinate of degree one, and let $\nu=1/u$ near s. Here u and ν are the coordinates u' and u of Set-up 13.4. For integers $b\geq a\geq 0$, the space $H^0(E,\mathcal{O}_E(as+bf))$ has the basis (13.42) of monomials z^ku^ℓ on $E\backslash s$, and the basis (13.43) of monomials $z^k\nu^\ell$ near s. For an integer $w\geq 0$ let

$$\mathcal{C}(a,b,w) := H^0(E,\mathcal{O}_E(as+bf)\otimes I^w_{Z_{U_i}}).$$

The tables integrate over the containing spaces of (13.41), namely

$$\begin{aligned}
&\mathcal{C}(j,m+j,0) && (0\leq j\leq m),\\
&\mathcal{C}(j,m+j,j-m) && (m\leq j\leq 2m),\\
&\mathcal{C}(2(3m-j),3(3m-j),3m-j) && (2m\leq j\leq 3m).
\end{aligned}$$

In the tables all exponents and degrees are divided by m. In the range $0\leq j\leq m$ we have $(a,b,w)=(t,1+t,0)$ with $t=j/m\in[0,1]$. In the range $m\leq j\leq 2m$ we write $w=j/m-1\in[0,1]$, so that $(a,b)=(1+w,2+w)$, and we write $\eta=\ell/m$. The range $2m\leq j\leq 3m$ is the cone over the value at $w=1$, scaled by $\theta=(3m-j)/m\in[0,1]$. Thus $I_{[2,3]}$ equals $\frac{1}{3}$ of the value at $w=1$ for ranks and $\frac{1}{4}$ of the value at $w=1$ for moments.

Off-section root.

Remark B.2. Assume that $p_{ij}\notin s$, and choose u with $p_{ij}=(z=u=0)$. Fix an integer $\mu>0$, and let ψ_τ be the automorphism $(z,u)\to(\tau^{-1}z,\tau^\mu u)$ of E for $\tau\in\mathbb{C}^*$. As $\tau\to 0$, the part of $\psi_\tau(Z)$ at p_{ij} tends to (u,z^n). Let $z_\infty=1/z$, $u_\infty=u/z$, and $\nu_\infty=1/u_\infty$. Let $f_\infty=(z_\infty=0)$ be the fiber over $z=+\infty$, and let

$$y_\infty=(z_\infty=u_\infty=0), \qquad q_\infty=s\cap f_\infty.$$

The limit of the remaining part of $\psi_\tau(Z)$ is one of the subschemes in the second column of Table 99. The limit of $\mathcal{C}(a,b,w)$ is spanned by monomials z^ku^ℓ. With $\eta=\ell/m\in[0,1+w]$, the monomial z^ku^ℓ is in the limit if and only if $k_{\min}\leq k/m\leq k_{\max}$, where

$$k_{\min}=\max(0,n(w-\eta)), \qquad k_{\max}=\min(2+w-\eta,\ \text{the bounds in the third column}).$$

The first moment along f in the range $m \le j \le 2m$ is

$$I_{[1,2]} = \int_0^1 \int_0^{1+w} \frac{\max(k_{\max}, k_{\min})^2 - k_{\min}^2}{2} d\eta dw, \tag{B.2}$$

and $I_{[2,3]}$ is $\frac{1}{4}$ of the inner integral at $w = 1$. In the range $0 \le j \le m$ the conditions from the base scheme are void, and

$$I_{[0,1]} = \int_0^1 \frac{(1+t)^3 - 1}{6} dt = \frac{11}{24}$$

for any n and any residual subscheme. The cells of (B.2) are cut out by the lines where two affine bounds coincide. For example, for the double point at q_∞ the extra cuts are $\eta = w$ and $\eta = 1 - w$, which cross at $(w, \eta) = (\frac{1}{2}, \frac{1}{2})$.

Applying (B.1) to $I_{[0,1]} = \frac{11}{24}$ and the last two columns of Table 99 yields the bounds in Table 19, the largest of which is 83/88.

TABLE 99. First moments along f at an off-section root p_{ij}, by length n and limit residual subscheme.

n	residual subscheme	extra bounds on k/m	$I_{[1,2]}$	$I_{[2,3]}$
3	none	none	13/6	17/24
2	y_∞	2	2	5/8
2	q_∞	$3 + w - 2\eta$	53/24	19/24
1	(u_∞, z_∞^2) at y_∞	$2 - w + \eta$	11/6	13/24
1	(z_∞, u_∞^2) at y_∞	$2 - \eta/2$	11/6	13/24
1	(z_∞, ν_∞^2) at q_∞	$5/2 + w/2 - 3\eta/2$	25/12	17/24
1	$y_\infty + q_\infty$	$2,\ 3 + w - 2\eta$	47/24	5/8

Roots on the negative section.

Remark B.3. Assume that $p_{ij} = s \cap f$ and $n \ge 2$. We use coordinates (z, ν) at p_{ij} in which the stalk of $I_{Z_{U_i}}$ at p_{ij} is $(\nu - z, z^n)$, and, for $n = 2$, the remaining point of Z is y_∞. The space $\mathcal{C}(a, b, w)$ is graded by $\deg z = \deg \nu = 1$ in the basis $z^k \nu^\ell$, and for $d \ge 0$ we write $\mathcal{C}(a, b, w)_d$ for the part of degree d. For $d \ge 0$ let

$$\ell_+ = \min(a, d),$$
$$\ell_- = \max\left(0, \left\lceil \tfrac{d-b+a}{2} \right\rceil\right) \quad (n = 3), \qquad \ell_- = \max\left(0, \left\lceil \tfrac{d-b+a}{2} \right\rceil, w + d - b\right) \quad (n = 2),$$
$$\kappa = \max\left(0, \left\lceil \tfrac{nw-d}{n-1} \right\rceil\right), \qquad Q = \ell_+ - \ell_- - \kappa.$$

By (13.50), $\mathcal{C}(a, b, w)_d$ is $\nu^{\ell_-} z^{d-\ell_+} (\nu - z)^\kappa \mathbb{C}[\nu, z]_Q$ if $Q \ge 0$, and zero otherwise. Thus ℓ_- and ℓ_+ are the lower and upper bounds of the exponent of ν in $\mathcal{C}(a, b, w)_d$, and κ is the power of $\nu - z$ dividing $\mathcal{C}(a, b, w)_d$. Let $C_1 = (\nu = z)$ and $C_t = (\nu = tz)$ for $t \ne 0, 1$. To leading order, the dimension of $\mathcal{C}(a, b, w)_d$, the maximal sum over a basis of the f-orders, the sum over a basis adapted to ord_f of the orders at p_{ij} on f of the leading coefficients along f, and the maximal sums over a basis of the coefficients of $s = (\nu = 0)$, of C_1, and of C_t are respectively

$$Q, \quad (d - \ell_+)Q + \tfrac{1}{2}Q^2, \quad \ell_+ Q - \tfrac{1}{2}Q^2, \quad \ell_- Q + \tfrac{1}{2}Q^2, \quad \kappa Q + \tfrac{1}{2}Q^2, \quad \tfrac{1}{2}Q^2. \tag{B.3}$$

In the range $m \le j \le 2m$ we keep the letters d, $\ell_\pm$, κ, and Q for the quantities divided by m. The rescaled bounds are

$$\ell_+ = \min(1 + w, d), \qquad \ell_- = \max\left(0, \tfrac{d-1}{2}, d - 2\right) \ (n = 2), \qquad \ell_- = \max\left(0, \tfrac{d-1}{2}\right) \ (n = 3),$$

with $\kappa = \max(0, \frac{nw-d}{n-1})$, and the functions (B.3) are integrated over $\{0 \le w \le 1,\ d \ge 0,\ Q \ge 0\}$. Table 100 lists the polygons in the (w,d)-plane on which ℓ_-, ℓ_+, and κ are affine. On each polygon the functions (B.3) are polynomials of degree at most two.

In the range $0 \le j \le m$ the conditions from the base scheme are void, and the integrals do not depend on n. For both n the rank integrals in Table 101 sum to

$$\frac{2}{3} + \frac{13}{6} + \frac{5}{6} = \frac{11}{3},$$

in accordance with $N_m = \frac{11}{3}m^3 + O(m^2)$. Applying (B.1) to the last five columns of Table 101 yields the five columns of Table 20 headed f, leading coeff. along f at q, s, C_1, and C_t. The column mult_q of Table 20 is the sum of the columns headed f and leading coeff. along f at q, since each element of $\mathcal{C}(a,b,w)_d$ has multiplicity d at p_{ij} and the second and third functions in (B.3) add up to dQ.

TABLE 100. The partition of the range $m \le j \le 2m$ for a root $p_{ij} \in s$ into polygons on which the bounds are affine.

n	vertices in the (w,d)-plane	ℓ_-	ℓ_+	κ
2	$(0,0),(\frac{1}{2},1),(0,1)$	0	d	0
2	$(0,0),(1,1),(\frac{1}{2},1)$	0	d	$2w-d$
2	$(1,2),(1,3),(0,3),(0,1)$	$\frac{d-1}{2}$	$1+w$	0
2	$(1,2),(0,1),(\frac{1}{2},1)$	$\frac{d-1}{2}$	d	0
2	$(1,1),(1,2),(\frac{1}{2},1)$	$\frac{d-1}{2}$	d	$2w-d$
2	$(1,3),(1,4),(0,3)$	$d-2$	$1+w$	0
3	$(0,0),(\frac{1}{3},1),(0,1)$	0	d	0
3	$(0,0),(1,1),(\frac{1}{3},1)$	0	d	$\frac{3w-d}{2}$
3	$(1,3),(1,5),(0,3),(0,1),(\frac{1}{2},\frac{3}{2})$	$\frac{d-1}{2}$	$1+w$	0
3	$(1,2),(1,3),(\frac{1}{2},\frac{3}{2})$	$\frac{d-1}{2}$	$1+w$	$\frac{3w-d}{2}$
3	$(\frac{1}{2},\frac{3}{2}),(0,1),(\frac{1}{3},1)$	$\frac{d-1}{2}$	d	0
3	$(1,1),(1,2),(\frac{1}{2},\frac{3}{2}),(\frac{1}{3},1)$	$\frac{d-1}{2}$	d	$\frac{3w-d}{2}$

TABLE 101. Integrals of the six functions (B.3) over the three ranges of j for a root $p_{ij} \in s$.

range	n	rank	f	p_{ij} on f	s	C_1	C_t
$0 \le j \le m$	$2,3$	$2/3$	$11/24$	$1/4$	$1/4$	$1/6$	$1/6$
$m \le j \le 2m$	2	$13/6$	$193/96$	$199/96$	2	$17/16$	$95/96$
$2m \le j \le 3m$	2	$5/6$	$31/48$	$41/48$	$19/24$	$19/48$	$1/3$
$m \le j \le 2m$	3	$13/6$	$20/9$	$161/72$	$197/96$	$17/16$	$253/288$
$2m \le j \le 3m$	3	$5/6$	$19/24$	$23/24$	$13/16$	$19/48$	$1/4$

Tables for Theorem 13.21.

Remark B.4. Notation as in Construction-Proposition 13.19. For $g \in \mathbb{C}[x,y,z] = \mathbb{C}[x,y,\zeta]$ let $\mathrm{in}_\zeta(g)$ be the sum of the terms of g of lowest degree in ζ. The generator column of Table 102 lists the fourteen generators of $\mathcal{R}$ in Construction-Proposition 13.19(1), except that the row $k = 4$

lists $T\zeta$ in place of Tz. Since $\zeta = z - xy$, we have $T\zeta = Tz - Txy$, so the fourteen entries of the column also generate $\mathcal{R}$. For the fourteen generators Tg listed in Table 102, the monomials $\mathrm{in}_\zeta(g)$ are $x^{i_1}y^{i_2}\zeta^{i_3}$ with $(i_1, i_2, i_3) = \mathbf{w}_0, \dots, \mathbf{w}_{13}$. For instance, $\mathrm{in}_\zeta(z^2\zeta) = x^2y^2\zeta$ and $\mathrm{in}_\zeta(z\zeta^2) = xy\zeta^2$. The polytope $\mathcal{P}$ is the convex hull of $\mathbf{w}_0, \dots, \mathbf{w}_{13}$, and $\mathrm{vol}(\mathcal{P}) = \frac{11}{3}$.

TABLE 102. The exponent vectors $\mathbf{w}_k$ of the monomials in_ζ of the generators of $\mathcal{R}$, with the heights h_k of the regular triangulation of $\mathcal{P}$.

k	$\mathbf{w}_k$	generator	h_k	k	$\mathbf{w}_k$	generator	h_k
0	$(0,0,0)$	T	982	7	$(1,1,1)$	$Tz\zeta$	2071
1	$(1,0,0)$	Tx	599	8	$(2,1,1)$	$Txz\zeta$	5004
2	$(0,1,0)$	Ty	1592	9	$(1,2,1)$	$Tyz\zeta$	6379
3	$(1,1,0)$	Txy	1049	10	$(2,0,1)$	$Tx^2\zeta$	4877
4	$(0,0,1)$	$T\zeta$	1894	11	$(1,3,0)$	Ty^2z	10286
5	$(1,0,1)$	$Tx\zeta$	2462	12	$(2,2,1)$	$Tz^2\zeta$	8323
6	$(1,2,0)$	Tyz	5611	13	$(1,1,2)$	$Tz\zeta^2$	5448

Remark B.5. Lift $\mathbf{w}_k$ to $(\mathbf{w}_k, h_k) \in \mathbb{R}^4$, where h_k is the height in Table 102. The lower faces of the convex hull of the lifted points project to the 22 tetrahedra

$$\begin{gathered}\{0,1,3,4\},\ \{0,2,3,4\},\ \{1,3,4,7\},\ \{1,3,7,8\},\ \{1,4,5,7\},\ \{1,5,7,10\},\\ \{1,7,8,10\},\ \{2,3,4,7\},\ \{2,3,6,9\},\ \{2,3,7,9\},\ \{2,4,7,13\},\ \{2,6,9,11\},\\ \{2,7,9,13\},\ \{3,6,9,12\},\ \{3,7,8,12\},\ \{3,7,9,12\},\ \{4,5,7,13\},\\ \{5,7,10,13\},\ \{6,9,11,12\},\ \{7,8,10,13\},\ \{7,8,12,13\},\ \{7,9,12,13\},\end{gathered} \tag{B.4}$$

given by the indices k of their vertices $\mathbf{w}_k$. The triangulation (B.4) is checked by the following finite tests with integer arithmetic.

(1) For each tetrahedron $\{k_0, k_1, k_2, k_3\}$ the 4×4 matrix with rows $(1, \mathbf{w}_{k_0}), \dots, (1, \mathbf{w}_{k_3})$ has determinant ± 1.
(2) For each tetrahedron, the affine function on $\mathbb{R}^3$ with values $h_{k_0}, \dots, h_{k_3}$ at $\mathbf{w}_{k_0}, \dots, \mathbf{w}_{k_3}$ is strictly smaller than h_k at $\mathbf{w}_k$ for the ten other indices k.
(3) Each triangular face of a tetrahedron in (B.4) is a face of exactly two of the tetrahedra, or is contained in one of the facets $i_1 = 0$, $i_2 = 0$, $i_3 = 0$, $i_1 - i_3 = 1$, $i_3 - i_2 = 1$, $i_1 + i_3 = 3$, $i_2 + i_3 = 3$, and $-2i_1 + i_2 + i_3 = 1$ of $\mathcal{P}$.

By (1), the 22 tetrahedra are unimodular. By (2), their interiors are pairwise disjoint. By (3), their union is a non-empty closed subset of $\mathcal{P}$ whose boundary is contained in $\partial\mathcal{P}$, so the union equals $\mathcal{P}$. The numbers of faces of the triangulation of dimensions 0, 1, 2, and 3 are

$$14, \qquad 46, \qquad 55, \qquad 22.$$

The relative interior of an r-dimensional unimodular simplex σ contains $\binom{m-1}{r}$ points of $m\sigma \cap \mathbb{Z}^3$. Decomposing $m\mathcal{P}$ into the relative interiors of the dilated faces, we obtain for $m \geq 1$

$$\#(m\mathcal{P} \cap \mathbb{Z}^3) = 14 + 46(m-1) + 55\binom{m-1}{2} + 22\binom{m-1}{3} = \frac{11}{3}m^3 + \frac{11}{2}m^2 + \frac{23}{6}m + 1 = N_m. \tag{B.5}$$

For $m = 1, 2, 3, 4$ both sides of (B.5) equal 14, 60, 161, and 339, and the value 14 at $m = 1$ is the number of generators of $\mathcal{R}$.

The rest of Subsection B.1 lists the integrals for Lemma 14.3, Lemma 14.5, and Theorem 15.6. Each entry is the integral of a polynomial over a rational polygon or polytope, given by vertices or by inequalities, evaluated as described at the beginning of this section.

Remark B.6. We list the integrals for Lemma 14.3. Notation as in Set-up 14.2. Write $a := k/m$, and call the ranges $0 \le a \le 1$, $1 \le a \le 2$, and $2 \le a \le 5/2$ early, middle, and tail. Uniformly in k,

$$\dim R_{m,k} = m^2\rho(a) + O(m), \qquad \rho(a) = \begin{cases} a, & 0 \le a \le 1, \\ a - (a-1)^2, & 1 \le a \le 2, \\ (5-2a)^2, & 2 \le a \le 5/2, \end{cases}$$

and $\int_0^{5/2} \rho(a)da = 11/6$. Thus $n_m = \frac{11}{6}m^3 + O(m^2)$, and an integral over a and one further scaled order becomes a normalized expected order after multiplication by $6/11$. Let e_1, e_2 be the total exceptional classes of the blow-up $\psi : S' \to S$ of S along Δ_B: the blow-up of $\operatorname{Supp}\Delta_B$ when Δ_B is reduced, or, when Δ_B is a double point, the blow-up of the point followed by the blow-up of the point of the exceptional curve determined by Δ_B.

For the refinement by B, let u be the scaled order along B of the free part. The free residual divisor on S' is

$$\begin{cases} (a-u)\sigma + (1-2u)f - (a-1-u)_+(e_1+e_2), & 0 \le a \le 2, \\ (10-4a-u)\sigma + (5-2a-2u)f - (5-2a-u)(e_1+e_2), & 2 \le a \le \frac{5}{2}, \end{cases}$$

with negative part $(a+u-2)_+[(\sigma-e_1)+(\sigma-e_2)]$ for $a \le 2$ and $u[(\sigma-e_1)+(\sigma-e_2)]$ in the tail. When Δ_B is a double point, $(\sigma-e_1)+(\sigma-e_2)$ is replaced by $2(\sigma-e_1)+(e_1-e_2)$, and the positive part equals the positive part in the case of two distinct points. In Table 103, n is the scaled degree of the free restriction to B, and φ_Δ (resp. φ_ι) is the scaled fixed coefficient of the restriction to B at each point of Δ_B (resp. $\iota(\Delta_B)$), counted with scheme multiplicity and with the factor $s_{R_\Delta}^{2k-4m}$ counted.

TABLE 103. Chambers of the refinement by B for Lemma 14.3.

a	u	n	$u+(a-2)_+$	φ_Δ	φ_ι
$[0,\frac12]$	$[0,a]$	$2a+1-4u$	u	0	0
$[\frac12,1]$	$[0,\frac12]$	$2a+1-4u$	u	0	0
$[1,\frac32]$	$[0,a-1]$	$3-2u$	u	$a-1-u$	0
$[1,\frac32]$	$[a-1,\frac12]$	$2a+1-4u$	u	0	0
$[\frac32,2]$	$[0,2-a]$	$3-2u$	u	$a-1-u$	0
$[\frac32,2]$	$[2-a,\frac12]$	$7-2a-4u$	u	$a-1-u$	$a+u-2$
$[2,\frac52]$	$[0,\frac{5-2a}{2}]$	$15-6a-4u$	$a-2+u$	$1-u$	$u+2(a-2)$

For each a, beyond the largest upper limit of u ($u=a$ for $a \le \frac12$, $u=\frac12$ for $\frac12 \le a \le 2$, and $u=\frac{5-2a}{2}$ for $2 \le a \le \frac52$), the coefficient of σ or of f in the residual divisor is negative, and the residual divisor has no sections. With the measure $dadu$ over the union of the chambers,

$$\int n = \frac{11}{6}, \quad \int (u+(a-2)_+)n = \frac{19}{48}, \quad \int \frac{n^2}{2} = \frac{47}{24}, \quad \int n\varphi_\Delta = \frac12, \quad \int n\varphi_\iota = \frac{5}{48}, \tag{B.6}$$

and the tail row contributes $\frac16$, $\frac{3}{64}$, $\frac{13}{96}$, $\frac{9}{64}$, and $\frac{13}{192}$ to the five integrals. For $z \in B \setminus \operatorname{Supp}\Delta_B$,

$$S(R;\operatorname{ord}_B) = \frac{6}{11}\cdot\frac{19}{48} = \frac{19}{88}, \qquad S(R^B;z) \le \frac{6}{11}\Big(\frac{47}{24} + \frac{5}{48}\operatorname{mult}_z \iota(\Delta_B)\Big) = \frac{94 + 5\operatorname{mult}_z \iota(\Delta_B)}{88}. \tag{B.7}$$

For the refinement by a fiber C of class f with $C \cap \Delta_B = \emptyset$, let u be the scaled order along C of the free part. The free residual divisor is $a\sigma + (1-u)f$ in the early range, $a\sigma + (1-u)f - (a-1)(e_1+e_2)$ with negative part $(a+u-2)_+[(\sigma-e_1)+(\sigma-e_2)]$ in the middle range, and $2(5-2a)\sigma + (5-2a-u)f - (5-2a)(e_1+e_2)$ with negative part $u[(\sigma-e_1)+(\sigma-e_2)]$ in the

tail. In Table 104, n is the scaled degree of the free restriction to C, and φ_R (resp. φ_B) is the scaled coefficient of $R_\Delta|_C$ (resp. $B|_C$) in the fixed part of the restriction, with the negative part counted.

TABLE 104. Chambers of the refinement by a fiber C of class f with $C \cap \Delta_B = \emptyset$ for Lemma 14.3.

a	u	n	φ_R	φ_B
$[0,1]$	$[0,1]$	a	0	0
$[1,2]$	$[0,2-a]$	a	0	0
$[1,2]$	$[2-a,1]$	$4-a-2u$	$a+u-2$	0
$[2,\frac{5}{2}]$	$[0,5-2a]$	$2(5-2a-u)$	$u+2(a-2)$	$a-2$

With the measure $dadu$,

$$\int n = \frac{11}{6}, \quad \int un = \frac{19}{24}, \quad \int \frac{n^2}{2} = 1, \quad \int n\varphi_R = \frac{5}{24}, \quad \int n\varphi_B = \frac{1}{48}, \tag{B.8}$$

so that, for any $z \in C$,

$$S(R;\mathrm{ord}_C) = \frac{19}{44}, \qquad S(R^C;z) \le \frac{6}{11} + \frac{5}{44}\,\mathrm{mult}_z(R_\Delta|_C) + \frac{1}{88}\,\mathrm{mult}_z(B|_C). \tag{B.9}$$

Let C be a fiber of class f with $C \cap \Delta_B \neq \emptyset$, and let $z \in C \backslash B$. The order along C is at most 1 in the early and middle ranges and at most $5-2a$ in the tail, and the free image after removing the order along C is contained in a complete linear system on C of scaled degree a, resp. $2(5-2a)$. Thus

$$\begin{aligned} S(R;\mathrm{ord}_C) &\le \frac{6}{11}\Big(\int_0^1 ada + \int_1^2 \big(a-(a-1)^2\big)da + \int_2^{5/2}(5-2a)^3da\Big) \\ &= \frac{6}{11}\Big(\frac{1}{2}+\frac{7}{6}+\frac{1}{8}\Big) = \frac{43}{44}, \\ S(R^C;z) &\le \frac{6}{11}\Big(\int_0^1 \frac{a^2}{2}da + \int_1^2 \frac{a^2}{2}da + \int_2^{5/2} 2(5-2a)^3da\Big) \\ &\quad + \frac{6}{11}\,\mathrm{mult}_z(R_\Delta|_C)\int_2^{5/2}(2a-4)(5-2a)^2da \\ &= \frac{19}{22} + \frac{\mathrm{mult}_z(R_\Delta|_C)}{44} \le \frac{10}{11}, \end{aligned} \tag{B.10}$$

where $\mathrm{mult}_z(R_\Delta|_C) \le 2$.

Remark B.7. We list the integrals for Lemma 14.5. Notation as in Set-up 14.4. Write $a := k/m$. The density ρ equals the density $\rho(a)$ of the series of Lemma 14.3, so the normalization factor is again $6/11$. Let $g : \tilde{S} \to S$ be the blow-up of S at q_σ and q_B, with exceptional curves e_1 over q_σ and e_2 over q_B. Let $C \neq C_{q_0}$ be a fiber, and let u be the scaled order along C of the free part. Let d be the scaled degree of the free restriction to C. Let ζ and b be the total scaled fixed coefficients of the restriction to C at $C \cap \sigma$ and at $C \cap B$, with the coefficients of e_1 and e_2 in the residual divisor on $\tilde{S}$ and the residual base multiplicities at q_σ and q_B counted. Let τ be the scaled order along C, so that $\tau = u$, except for $a \ge 2$ and $C \in \{C_\sigma, C_B\}$, where $\tau = u + 2a - 4$, since the factor $s_{C_\sigma+C_B}^{2k-4m}$ contains C_σ and C_B with scaled coefficient $2(a-2)$. Table 105 lists the integrals over the triangles of Table 23, with the measure $dadu$. Outside the triangles of Table 23

the residual divisor has volume zero. The totals give

$$S(R;\mathrm{ord}_C) = \frac{6}{11}\int \tau d, \qquad S(R^C;p) \le \frac{6}{11}\int \frac{d^2}{2},$$
$$S(R^C;C\cap\sigma) \le \frac{6}{11}\int \Big(\frac{d^2}{2}+d\zeta\Big), \quad S(R^C;C\cap B) \le \frac{6}{11}\int \Big(\frac{d^2}{2}+db\Big)$$

for $p \in C\backslash(\sigma\cup B)$.

TABLE 105. Integrals over the triangles of Table 23 for Lemma 14.5, where the rows marked "any" enter the totals of all three fibers.

fiber	vertices in (a,u)	$\int d$	$\int \tau d$	$\int \frac{d^2}{2}$	$\int d\zeta$	$\int db$
any	$(0,0),(1,0),(0,1)$	$\frac{1}{6}$	$\frac{1}{24}$	$\frac{1}{24}$	0	0
any	$(1,0),(1,2),(0,1)$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{12}$	$\frac{1}{12}$	0
$C\notin\{C_\sigma,C_B\}$	$(1,0),(2,0),(\frac{3}{2},\frac{3}{2})$	$\frac{5}{8}$	$\frac{9}{32}$	$\frac{17}{64}$	$\frac{13}{32}$	0
	$(1,0),(\frac{3}{2},\frac{3}{2}),(1,2)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{7}{96}$	$\frac{7}{48}$	0
	$(2,0),(2,2),(\frac{3}{2},\frac{3}{2})$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{7}{96}$	$\frac{5}{16}$	0
	$(2,2),(1,2),(\frac{3}{2},\frac{3}{2})$	$\frac{1}{24}$	$\frac{7}{96}$	$\frac{1}{192}$	$\frac{5}{96}$	0
	$(2,0),(\frac{5}{2},0),(2,2)$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{4}$	$\frac{1}{48}$
	total	$\frac{11}{6}$	$\frac{59}{48}$	$\frac{7}{12}$	$\frac{5}{4}$	$\frac{1}{48}$
C_σ	$(1,0),(2,0),(2,1)$	$\frac{1}{3}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{7}{24}$	0
	$(1,0),(2,1),(1,2)$	$\frac{2}{3}$	$\frac{7}{12}$	$\frac{1}{4}$	$\frac{5}{12}$	0
	$(2,1),(2,2),(1,2)$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{24}$	$\frac{5}{24}$	0
	$(2,0),(\frac{5}{2},0),(2,1)$	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{1}{48}$	$\frac{1}{8}$	$\frac{1}{96}$
	$(\frac{5}{2},0),(2,2),(2,1)$	$\frac{1}{12}$	$\frac{5}{48}$	$\frac{1}{48}$	$\frac{1}{8}$	$\frac{1}{96}$
	total	$\frac{11}{6}$	$\frac{17}{12}$	$\frac{7}{12}$	$\frac{5}{4}$	$\frac{1}{48}$
C_B	$(1,0),(2,0),(2,1)$	$\frac{5}{18}$	$\frac{7}{72}$	$\frac{19}{216}$	$\frac{43}{216}$	$\frac{5}{72}$
	$(1,0),(2,1),(2,3)$	$\frac{5}{9}$	$\frac{11}{18}$	$\frac{19}{108}$	$\frac{29}{54}$	0
	$(1,0),(2,3),(1,2)$	$\frac{1}{3}$	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	0
	$(2,0),(\frac{5}{2},0),(2,1)$	$\frac{1}{18}$	$\frac{1}{24}$	$\frac{1}{108}$	$\frac{17}{216}$	$\frac{1}{48}$
	$(\frac{5}{2},0),(2,3),(2,1)$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{1}{54}$	$\frac{5}{27}$	$\frac{1}{72}$
	total	$\frac{11}{6}$	$\frac{13}{8}$	$\frac{1}{2}$	$\frac{4}{3}$	$\frac{5}{48}$

Remark B.8. We list the integrals for Theorem 15.6. Notation as in Construction-Proposition 15.4. Let $\mathcal{Q}\subset\mathbb{R}^3$ be the polytope of the points (r,α,c) with

$$0\le r\le 2, \quad 0\le\alpha\le 2r, \quad 0\le c\le 2, \quad r+\alpha\le 3, \quad r+c\le 3, \quad r-\alpha+c\le 2,$$

the inequalities of Construction-Proposition 15.4(3) with $m=1$, and let $\mathcal{Q}'$ be its projection to the (r,α)-plane, the polygon with vertices $(0,0),(2,0),(2,1),(1,2)$. Then $\mathrm{vol}(\mathcal{Q})=\frac{11}{3}$, and $\langle\cdot\rangle$ denotes the mean with respect to the normalized Lebesgue measure on $\mathcal{Q}$. Let x,y,z have the auxiliary weights $0,1,2$. The monomial $T^m u^\alpha v^{2r-\alpha} h_C^c$ has exceptional degree $j=5r-2\alpha+6c$ and auxiliary weight $\ell=2r-\alpha+2c$. For fixed (m,j,ℓ), the integers c for which $(j-2\ell-2c, 2j-5\ell-2c, c)$

satisfies the inequalities of Construction-Proposition 15.4(3) form an interval $\{c_{\min},\dots,c_{\max}\}$. Let

$$\psi := \max\{0,1-r,1-\alpha\},\qquad \varphi := 3-r-\psi,\qquad \sigma := \min\left\{1-\frac{r}{2},\frac{3-r-\alpha}{4}\right\}$$

on $\mathcal{Q}'$, so that the fiber of $\mathcal{Q}$ over (r,α) is $0\le c\le\varphi$. At the point $(r,\alpha,c)\in\mathcal{Q}$ the scaled difference $c-c_{\min}$ equals $\min\{c,\sigma,3-r-c\}$. On $\mathcal{Q}'$ we have $0\le\sigma\le(3-r)/2$ and $\sigma\le\varphi$, and

$$\int_0^{\varphi}\min\{c,\sigma,3-r-c\}dc=\begin{cases}\sigma(3-r)-\frac{\sigma^2}{2}-\sigma\psi, & \psi\ge\sigma,\\ \sigma(3-r)-\sigma^2-\frac{\psi^2}{2}, & \psi\le\sigma.\end{cases}$$

Table 106 lists the six cells of $\mathcal{Q}'$ on which σ, ψ, and the case are given by one formula, and the integrals over them of the right-hand side above, with the measure $drd\alpha$. The cells overlap only along edges.

TABLE 106. The six cells for the integral of $c-c_{\min}$ over $\mathcal{Q}$ in Theorem 15.6.

vertices in (r,α)	σ	ψ	case	integral
$(1,0),(2,0),(2,1)$	$1-\frac{r}{2}$	$1-\alpha$	$\psi\ge\sigma$	$\frac{5}{96}$
$(2,1),(1,2),(1,1)$	$\frac{3-r-\alpha}{4}$	0	$\psi\le\sigma$	$\frac{13}{192}$
$(0,0),(\frac12,\frac12),(1,2)$	$\frac{3-r-\alpha}{4}$	$1-r$	$\psi\ge\sigma$	$\frac{47}{256}$
$(\frac12,\frac12),(1,1),(1,2)$	$\frac{3-r-\alpha}{4}$	$1-r$	$\psi\le\sigma$	$\frac{11}{96}$
$(0,0),(1,0),(2,1),(\frac12,\frac12)$	$\frac{3-r-\alpha}{4}$	$1-\alpha$	$\psi\ge\sigma$	$\frac{307}{768}$
$(2,1),(1,1),(\frac12,\frac12)$	$\frac{3-r-\alpha}{4}$	$1-\alpha$	$\psi\le\sigma$	$\frac{19}{192}$
total				$\frac{11}{12}$

Dividing the total by $\mathrm{vol}(\mathcal{Q})=11/3$, and using that c runs over all of the interval $\{c_{\min},\dots,c_{\max}\}$ in each eigenweight space,

$$\left\langle\frac{c_{\max}-c_{\min}}{2}\right\rangle=\langle c-c_{\min}\rangle=\frac14. \tag{B.11}$$

The exponents of x and z in the eigenweight space of (j,ℓ) are $\max\{0,j-3\ell\}=(\alpha-r)_+$ and $\max\{0,3\ell-j\}=(r-\alpha)_+$. Let ν_x be the valuation of the weighted blow-up of P_x with weights $(1,2)$ on (y,z) in the chart $x=1$, and let ν_z be the valuation of the weighted blow-up with weights $(2,1)$ on (x,y) on the index-five cover of the chart $z=1$. Both have log discrepancy 3 on the respective smooth chart. Table 107 lists, for each prime divisor and each of ν_x, ν_z, the scaled average order in one eigenweight space and its mean $\langle\cdot\rangle$, which is the limit of the bound for the coefficient, resp. the value, on any normalized eigenbasis divisor. In Table 107 the largest ratio of $\langle\cdot\rangle$ to the log discrepancy is $\frac{125}{44}/3=\frac{125}{132}$, attained by ν_x.

B.2. **The Scorza differential at the torus quartics.** This subsection proves Lemma B.10, the computational input to Theorem 19.1.

Notation B.9. For a ternary quartic f we write f_x,f_y,f_z for the partial derivatives of f, and $\mathrm{Hess}(f_x)$, $\mathrm{Hess}(f_y)$, $\mathrm{Hess}(f_z)$ for the Hessian matrices of the cubics f_x,f_y,f_z, which are symmetric 3×3 matrices of linear forms. For an alternating matrix M with rows and columns numbered $0,\dots,8$ and an ordered index set $I=(i_1,\dots,i_{2r})$, M_I denotes the submatrix of M on the rows and columns in I, and $\mathrm{Pf}(M_\emptyset)=1$. All Pfaffians below are computed by the expansion along the first row,

$$\mathrm{Pf}(M_I)=\sum_{j=2}^{2r}(-1)^j M_{i_1i_j}\,\mathrm{Pf}(M_{I\setminus\{i_1,i_j\}}). \tag{B.12}$$

TABLE 107. Coefficients of normalized eigenbasis divisors on E for Theorem 15.6.

divisor or valuation	average order in an eigenweight space	$\langle\cdot\rangle$
$(x=0)$	$(\alpha-r)_+ + \frac{c_{\max}-c_{\min}}{2}$	$\frac{135}{352}$
$(y=0)$	$\ell - 2(r-\alpha)_+ - (c_{\max}+c_{\min})$	$\frac{97}{176}$
$(z=0)$	$(r-\alpha)_+ + \frac{c_{\max}-c_{\min}}{2}$	$\frac{203}{352}$
$C=(D_2=0)$	$\frac{c_{\max}+c_{\min}}{2}$	$\frac{9}{11}$
$(xz=sy^2),\ s\in\mathbb{C}\backslash\{0,1\}$	$\frac{c_{\max}-c_{\min}}{2}$	$\frac{1}{4}$
ν_x	ℓ	$\frac{125}{44}$
ν_z	$2j-5\ell$	$\frac{27}{11}$

The torus $\mathbb{G}_m$ acts on $\mathbb{C}[x,y,z]$ through $(x,y,z)\to(x,sy,s^{-1}z)$, and the weight of a monomial $x^ay^bz^c$ is $b-c$.

Lemma B.10. *For a ternary quartic f let*

$$M(f) := \begin{pmatrix} 0 & \operatorname{Hess}(f_z) & -\operatorname{Hess}(f_y) \\ -\operatorname{Hess}(f_z) & 0 & \operatorname{Hess}(f_x) \\ \operatorname{Hess}(f_y) & -\operatorname{Hess}(f_x) & 0 \end{pmatrix},$$

an alternating 9×9 matrix of linear forms with rows and columns numbered $0,\dots,8$, and let $\operatorname{Sc}_0(f)\in\mathbb{C}[x,y,z]_4$ be $1/256$ times the Pfaffian of the submatrix of $M(f)$ on the rows and columns $0,\dots,7$. Then the following hold.

(1) $\operatorname{Sc}_0(f)$ is the Scorza quartic of Definition 2.27 for one choice of the Aronhold invariant $\mathfrak{A}$.

(2) For $f=c_0x^4+c_1x^2yz+c_2y^2z^2$ we have

$$\operatorname{Sc}_0(f) = \frac{c_1^4}{16}x^4 + \frac{c_1c_2(6c_0c_2-c_1^2)}{2}x^2yz + \frac{c_1^2c_2^2}{4}y^2z^2.$$

In particular, for $h\in\mathbb{C}\backslash\{1\}$ the quartic $F_h=\frac{h}{6}x^4+2x^2yz+y^2z^2$ of Definition 2.27 satisfies $\operatorname{Sc}_0(F_h)=x^4+(h-4)x^2yz+y^2z^2$.

(3) For $h\in\mathbb{C}\backslash\{1\}$, the differential of Sc_0 at F_h, as a linear endomorphism of $\mathbb{C}[x,y,z]_4$, preserves the weight spaces for the torus $(x,y,z)\to(x,sy,s^{-1}z)$, and the determinant of the differential is $-\frac{19683}{2}(h-2)^6$.

Proof. All Pfaffians in this proof are computed with (B.12) as exact polynomials with coefficients in $\mathbb{Q}[h]$, and in $\mathbb{Q}[h,\epsilon]$ for the perturbations $F_h+\epsilon m$ below. No numerical value of h is substituted.

Step 1. In this step we prove (2). For $f=c_0x^4+c_1x^2yz+c_2y^2z^2$ we have $f_x=4c_0x^3+2c_1xyz$, $f_y=c_1x^2z+2c_2yz^2$, and $f_z=c_1x^2y+2c_2y^2z$. Thus

$$\operatorname{Hess}(f_x)=\begin{pmatrix}24c_0x & 2c_1z & 2c_1y\\ 2c_1z & 0 & 2c_1x\\ 2c_1y & 2c_1x & 0\end{pmatrix},\quad \operatorname{Hess}(f_y)=\begin{pmatrix}2c_1z & 0 & 2c_1x\\ 0 & 0 & 4c_2z\\ 2c_1x & 4c_2z & 4c_2y\end{pmatrix},$$
$$\operatorname{Hess}(f_z)=\begin{pmatrix}2c_1y & 2c_1x & 0\\ 2c_1x & 4c_2z & 4c_2y\\ 0 & 4c_2y & 0\end{pmatrix}. \tag{B.13}$$

A direct calculation of the Pfaffian of $M(f)_{0,\dots,7}$ with (B.12) shows

$$\operatorname{Sc}_0(c_0x^4+c_1x^2yz+c_2y^2z^2) = \frac{c_1^4}{16}x^4 + \frac{c_1c_2(6c_0c_2-c_1^2)}{2}x^2yz + \frac{c_1^2c_2^2}{4}y^2z^2. \tag{B.14}$$

The quartic F_h corresponds to $(c_0, c_1, c_2) = (\frac{h}{6}, 2, 1)$, and (B.14) at $(c_0, c_1, c_2) = (\frac{h}{6}, 2, 1)$ is $x^4 + (h-4)x^2yz + y^2z^2$. This proves (2).

Step 2. In this step we prove (1). For a ternary cubic q define $M(q)$ by the formula defining $M(f)$, with q in place of f. The entries of $\mathrm{Hess}(q_x)$, $\mathrm{Hess}(q_y)$, $\mathrm{Hess}(q_z)$ are constants, so $M(q)$ is an alternating 9×9 matrix with entries in $\mathbb{C}$, linear in the coefficients of q. By [Ott24, Theorem 3.1], there exists $\lambda \in \mathbb{C}$ such that

$$\mathrm{Pf}(M(q)_{0,\dots,7}) = \lambda \mathfrak{A}(q)$$

for any ternary cubic q (see also [Ott24, Algorithm 5]). For $a = (a_1, a_2, a_3) \in \mathbb{C}^3$ write $D_a f := a_1 f_x + a_2 f_y + a_3 f_z$. The partial derivatives commute with D_a, and $\sum_i a_i \partial_i \ell = \ell(a)$ for a linear form ℓ. Thus $\mathrm{Hess}((D_a f)_x) = \mathrm{Hess}(f_x)(a)$, the matrix of linear forms $\mathrm{Hess}(f_x)$ evaluated at a, and likewise for y and z. Therefore $M(D_a f) = M(f)(a)$, and

$$256\,\mathrm{Sc}_0(f)(a) = \mathrm{Pf}(M(D_a f)_{0,\dots,7}) = \lambda \mathfrak{A}(D_a f).$$

If $\lambda = 0$, then $\mathrm{Sc}_0(f)(a) = 0$ for any quartic f and any $a \in \mathbb{C}^3$, which contradicts $\mathrm{Sc}_0(F_h) \neq 0$ from **Step 1**. Hence $\lambda \neq 0$, $\frac{\lambda}{256}\mathfrak{A}$ is an Aronhold invariant, and $\mathrm{Sc}_0(f)$ is the Scorza quartic of f for $\frac{\lambda}{256}\mathfrak{A}$. This proves (1).

Step 3. In this step we show that $(d\,\mathrm{Sc}_0)_{F_h}$ preserves the weight spaces and reduce the computation of $(d\,\mathrm{Sc}_0)_{F_h}$ to the weights $k \geq 0$. Here $(d\,\mathrm{Sc}_0)_{F_h}$ denotes the differential of Sc_0 at F_h, that is, $(d\,\mathrm{Sc}_0)_{F_h}(m)$ is the coefficient of ϵ in $\mathrm{Sc}_0(F_h + \epsilon m)$ for $m \in \mathbb{C}[x, y, z]_4$. The group SL_3 acts on $\mathbb{C}[x, y, z]_4$ by $(g \cdot r)(w) = r(g^{-1}w)$. For $g \in \mathrm{SL}_3$ and $a \in \mathbb{C}^3$ we have $D_a(g \cdot f) = g \cdot (D_{g^{-1}a} f)$, and $\mathfrak{A}$ is SL_3-invariant. Thus, by (1),

$$\mathrm{Sc}_0(g \cdot f) = g \cdot \mathrm{Sc}_0(f)$$

for any quartic f and any $g \in \mathrm{SL}_3$. If $g \cdot F_h = F_h$, then differentiating $\mathrm{Sc}_0(F_h + \epsilon g \cdot m) = g \cdot \mathrm{Sc}_0(F_h + \epsilon m)$ in ϵ implies that $(d\,\mathrm{Sc}_0)_{F_h}$ commutes with g.

The diagonal matrix $\mathrm{diag}(1, s, s^{-1})$ has determinant 1 and fixes F_h, and it acts on a monomial of weight k by multiplication by s^{-k}. Thus $(d\,\mathrm{Sc}_0)_{F_h}$ preserves the weight spaces, which proves the first part of (3).

The weight spaces of $\mathbb{C}[x, y, z]_4$ have the following ordered bases: (y^4) for $k = 4$, (xy^3) for $k = 3$, (x^2y^2, y^3z) for $k = 2$, (x^3y, xy^2z) for $k = 1$, (x^4, x^2yz, y^2z^2) for $k = 0$, and for $k < 0$ the basis of weight $-k$ with y and z exchanged, that is, (x^3z, xyz^2), (x^2z^2, yz^3), (xz^3), (z^4) for $k = -1, -2, -3, -4$. Concatenating the bases of weights $k = 4, 3, \dots, -4$, in decreasing order of k, we obtain an ordered basis

$$y^4,\ xy^3,\ x^2y^2,\ y^3z,\ x^3y,\ xy^2z,\ x^4,\ x^2yz,\ y^2z^2,\ x^3z,\ xyz^2,\ x^2z^2,\ yz^3,\ xz^3,\ z^4 \tag{B.15}$$

of $\mathbb{C}[x, y, z]_4$, and we use (B.15) on both sides of $(d\,\mathrm{Sc}_0)_{F_h}$.

Let $g \in \mathrm{SL}_3$ be $(x, y, z) \to (-x, z, y)$. The determinant of g is 1, $g^2 = \mathrm{id}_{\mathbb{C}^3}$, and $g \cdot F_h = F_h$, so $(d\,\mathrm{Sc}_0)_{F_h}$ commutes with g. The automorphism g sends $x^a y^b z^c$ to $(-1)^a x^a z^b y^c$. Since $a+b+c = 4$, the weight $k = b - c$ of $x^a y^b z^c$ satisfies $k \equiv a \pmod 2$. Thus g maps the basis of weight k to $(-1)^k$ times the basis of weight $-k$, element by element. The sign $(-1)^k$ appears on both sides of $(d\,\mathrm{Sc}_0)_{F_h}$ and cancels. Hence the block of $(d\,\mathrm{Sc}_0)_{F_h}$ of weight $-k$ is equal to the block of weight k.

Step 4. In this step we compute the blocks of weight $k \geq 0$. Let m be one of the six monomials of positive weight in (B.15). Since $M(f)$ is linear in f, we have $M(F_h + \epsilon m) = M(F_h) + \epsilon M(m)$.

A direct calculation of $\mathrm{Pf}((M(F_h)+\epsilon M(m))_{0,\dots,7})/256$ with (B.12) shows the exact identities

$$\mathrm{Sc}_0(F_h+\epsilon y^4)-\mathrm{Sc}_0(F_h)=-6\epsilon y^4, \tag{B.16}$$

$$\mathrm{Sc}_0(F_h+\epsilon xy^3)-\mathrm{Sc}_0(F_h)=-\tfrac{3}{2}(h-2)\epsilon xy^3, \tag{B.17}$$

$$\mathrm{Sc}_0(F_h+\epsilon x^2y^2)-\mathrm{Sc}_0(F_h)=-\epsilon y^2\big(-\epsilon y^2+(h-1)x^2-yz\big), \tag{B.18}$$

$$\mathrm{Sc}_0(F_h+\epsilon y^3z)-\mathrm{Sc}_0(F_h)=\tfrac{3}{4}\epsilon y^2\big(3\epsilon y^2+2(h-2)x^2\big), \tag{B.19}$$

$$\mathrm{Sc}_0(F_h+\epsilon x^3y)-\mathrm{Sc}_0(F_h)=-\tfrac{3}{4}\epsilon xy\big(4x^2-4yz-3\epsilon xy\big), \tag{B.20}$$

$$\begin{aligned}\mathrm{Sc}_0(F_h+\epsilon xy^2z)-\mathrm{Sc}_0(F_h)=\tfrac{\epsilon y}{16}\big(&\epsilon^3y^3+4\epsilon^2xy^2+4\epsilon(h+3)x^2y\\&-4\epsilon y^2z+8(h+2)x^3-32xyz\big).\end{aligned} \tag{B.21}$$

The coefficients of ϵ in (B.16)–(B.21) are

$$\begin{aligned}(d\,\mathrm{Sc}_0)_{F_h}(y^4)&=-6y^4, & (d\,\mathrm{Sc}_0)_{F_h}(xy^3)&=-\tfrac{3}{2}(h-2)xy^3,\\ (d\,\mathrm{Sc}_0)_{F_h}(x^2y^2)&=(1-h)x^2y^2+y^3z, & (d\,\mathrm{Sc}_0)_{F_h}(y^3z)&=\tfrac{3}{2}(h-2)x^2y^2,\\ (d\,\mathrm{Sc}_0)_{F_h}(x^3y)&=-3x^3y+3xy^2z, & (d\,\mathrm{Sc}_0)_{F_h}(xy^2z)&=\big(\tfrac{h}{2}+1\big)x^3y-2xy^2z.\end{aligned}$$

The six images above are the columns of the blocks of weights $4,3,2,1$ in Table 108.

For the weight 0, the quartic $F_h+\epsilon(e_0x^4+e_1x^2yz+e_2y^2z^2)$ with $e_0,e_1,e_2\in\mathbb{C}$ is of the form in (2) with $(c_0,c_1,c_2)=(\frac{h}{6}+\epsilon e_0,2+\epsilon e_1,1+\epsilon e_2)$. Thus the columns of the block of weight 0 are the partial derivatives of the right-hand side of (B.14) with respect to c_0, c_1, c_2 at $(\frac{h}{6},2,1)$:

$$\begin{aligned}(d\,\mathrm{Sc}_0)_{F_h}(x^4)&=6x^2yz,\\ (d\,\mathrm{Sc}_0)_{F_h}(x^2yz)&=2x^4+\big(\tfrac{h}{2}-6\big)x^2yz+y^2z^2,\\ (d\,\mathrm{Sc}_0)_{F_h}(y^2z^2)&=(2h-4)x^2yz+2y^2z^2.\end{aligned}$$

TABLE 108. The blocks of $(d\,\mathrm{Sc}_0)_{F_h}$ on the weight spaces, where the block of weight $-k<0$ is taken in the basis of weight k with y and z exchanged and the column of a basis monomial m lists the coefficients of $(d\,\mathrm{Sc}_0)_{F_h}(m)$.

weight $\pm k$	basis of weight k	block	determinant
±4	(y^4)	$\begin{pmatrix}-6\end{pmatrix}$	-6
±3	(xy^3)	$\begin{pmatrix}-\frac{3}{2}(h-2)\end{pmatrix}$	$-\frac{3}{2}(h-2)$
±2	(x^2y^2,y^3z)	$\begin{pmatrix}1-h & \frac{3}{2}(h-2)\\ 1 & 0\end{pmatrix}$	$-\frac{3}{2}(h-2)$
±1	(x^3y,xy^2z)	$\begin{pmatrix}-3 & \frac{h}{2}+1\\ 3 & -2\end{pmatrix}$	$-\frac{3}{2}(h-2)$
0	(x^4,x^2yz,y^2z^2)	$\begin{pmatrix}0 & 2 & 0\\ 6 & \frac{h}{2}-6 & 2h-4\\ 0 & 1 & 2\end{pmatrix}$	-24

Step 5. In this step we compute the determinant and finish the proof of (3). By **Steps 3 and 4**, the matrix of $(d\,\mathrm{Sc}_0)_{F_h}$ in the basis (B.15) is block diagonal with the blocks of Table 108, each block of weight $k\neq0$ appearing twice, once for k and once for $-k$. The determinants of the blocks are listed in the last column of Table 108. Therefore

$$\det(d\,\mathrm{Sc}_0)_{F_h}=(-6)^2\cdot\Big(-\frac{3}{2}(h-2)\Big)^6\cdot(-24)=-36\cdot24\cdot\frac{729}{64}(h-2)^6=-\frac{19683}{2}(h-2)^6. \tag{B.22}$$

This proves (3). □

Appendix C. Explicit identities for the counterexample

This section lists the polynomial data used in Section 21: the weighted jets of the chart coordinates at p, the univariate identities for the sextic Φ, and the unit ideal identities over $\mathbb{F}_3$ on eight standard charts of $\mathrm{Gr}(3,\mathbb{F}_3^7)$. All coefficients are exact rational numbers. Each identity below can be checked by polynomial multiplication, the relations between Φ_{i-1}, Φ_i, and Φ_{i+1} by division with remainder, and the sign table (C.12) by evaluating the polynomials.

Jets of the chart functions. Recall the affine chart (21.4) of $\mathrm{Gr}(3,V)$ at p, on which $P_{012}=1$, with the coordinates $w_0,\dots,w_{11}$ of (21.4), and the nine local equations $Lw+\mathcal{Q}(w)=0$ of X in (21.5). The local coordinates of X at p are

$$t=w_{11},\qquad x=w_{10},\qquad y=w_9,$$

with weights $\mathrm{wt}(t,x,y)=(1,2,3)$, and $w_0,\dots,w_8$ are convergent power series in t,x,y. For a convergent power series g in t,x,y, let $\tau(g)$ be the sum of the terms of g of weight at most 6. Tables 109, 110, and 111 list nine polynomials in t,x,y, and Section 21 shows that they are $\tau(w_0),\dots,\tau(w_8)$.

A monomial of weight at most 6 that does not appear in the table of $\tau(w_i)$ has coefficient zero in $\tau(w_i)$. Thus the monomials of $\tau(w_i)$ have weights 3 and 6 for $i\in\{0,4,8\}$, weights 2 and 5 for $i\in\{1,5,6\}$, and weights 1 and 4 for $i\in\{2,3,7\}$. Moreover,

$$w_9=y,\qquad w_{10}=x,\qquad w_{11}=t.$$

Substitute $\tau(w_0),\dots,\tau(w_8)$ for $w_0,\dots,w_8$ in the nine polynomials $Lw+\mathcal{Q}(w)$ of (21.5). Then the coefficient of any monomial of weight at most 6 is zero in each of the nine polynomials.

Table 109. The coefficients of the monomials of weights 3 and 6 in $\tau(w_0)$, $\tau(w_4)$, and $\tau(w_8)$.

monomial	$\tau(w_0)$	$\tau(w_4)$	$\tau(w_8)$
t^3	$-\frac{321817600}{2670661374689}$	$\frac{145408000}{242787397699}$	$-\frac{21402982400}{11896582487251}$
tx	$-\frac{97760}{3749207}$	$\frac{450880}{3749207}$	$-\frac{35804240}{183711143}$
y	$\frac{52}{161}$	$-\frac{3200}{2093}$	$\frac{16000}{7889}$
t^6	$-\frac{2536669593180569600000}{76206795810036094003135010 83}$	$\frac{1146152516845568000000}{69278905281850994548304 5553}$	$-\frac{1189322869697413120 0000}{48495233697295696183 81318871}$
t^4x	$-\frac{4484117618552832000000}{37540293502480834484302961}$	$\frac{2010721007500984320000}{34127539547709849531184 51}$	$-\frac{1870772823813586944 0000}{23889277683396894671 829157}$
t^3y	$\frac{2881263541592064000}{1612070833618793081303}$	$-\frac{16793367046717440000}{1905174621549482732449}$	$\frac{12542463575359488000}{1025863257757413779011}$
t^2x^2	$-\frac{556413478400000000}{52700927386552572743}$	$\frac{2719772519981056000}{5270092738655257 2743}$	$-\frac{1524746506268272640 00}{25823454419410760644 07}$
txy	$\frac{704533643955200}{2263105053744689}$	$-\frac{44747476213196800}{29420365698680957}$	$\frac{370609093268595200}{205942559890766699}$
y^2	$-\frac{9280000}{4173281}$	$\frac{7671296000}{705284489}$	$-\frac{34823680000}{2658379997}$
x^3	$\frac{5865600}{139856668721}$	$-\frac{27052800}{139856668721}$	$\frac{324864000}{139856668721}$

Univariate identities for Φ. Recall the integer sextic Φ of Construction-Proposition 21.1. The identities (C.4) and (C.6) below prove that Φ is irreducible over $\mathbb{Q}$, and the polynomials (C.7)–(C.11) prove that Φ has exactly one root in the open interval $(1/102,1/101)$, namely the

TABLE 110. The coefficients of the monomials of weights 2 and 5 in $\tau(w_1)$, $\tau(w_5)$, and $\tau(w_6)$.

monomial	$\tau(w_1)$	$\tau(w_5)$	$\tau(w_6)$
t^2	$\frac{25600}{51999871}$	$\frac{342080}{33090827}$	$-\frac{2633280}{363999097}$
x	$-\frac{4}{73}$	$\frac{80}{511}$	$-\frac{221}{1022}$
t^5	$\frac{1530475315200000}{6451330727355359079619}$	$\frac{20450976399360000}{4105392281044319414303}$	$-\frac{157428517109760000}{45159315091487513557333}$
t^3x	$\frac{97830006784000}{4539993474563940239}$	$\frac{2891090518016000}{2889086756540689243}$	$-\frac{19910373789696000}{31779954321947581673}$
t^2y	$-\frac{2083438592000}{2534457644579861}$	$-\frac{24227621324800}{1612836682914457}$	$\frac{14757573273600}{1364707962466079}$
tx^2	$-\frac{233409203200}{44614277321999}$	$\frac{248975353600}{28390903750363}$	$\frac{3069465734400}{312299941253993}$
xy	$\frac{5843200}{78075179}$	$-\frac{80422400}{546526253}$	$-\frac{5395200}{42040481}$

TABLE 111. The coefficients of the monomials of weights 1 and 4 in $\tau(w_2)$, $\tau(w_3)$, and $\tau(w_7)$.

monomial	$\tau(w_2)$	$\tau(w_3)$	$\tau(w_7)$
t	$-\frac{244}{2233}$	$\frac{26}{319}$	$-\frac{400}{2233}$
t^4	$-\frac{4234129408000}{277035716380614037}$	$\frac{1554388992000}{39576530911516291}$	$-\frac{14726922240000}{277035716380614037}$
t^2x	$-\frac{4467799219200}{1364707962466079}$	$\frac{1475760998400}{194958280352297}$	$-\frac{14820123801600}{1364707962466079}$
ty	$\frac{583456000}{10436302877}$	$-\frac{12547200}{114684647}$	$\frac{1713216000}{10436302877}$
x^2	$\frac{601600}{83297599}$	$-\frac{1560}{11899657}$	$-\frac{1277520}{83297599}$

root λ_* of Construction-Proposition 21.1, and that λ_* is a simple root. Section 21 deduces the irreducibility of Φ and the count of roots from (C.4), (C.6), (C.7)–(C.11), and (C.12), and uses them. The polynomials over $\mathbb{F}_{19}$ were computed by division with remainder and by the extended Euclidean algorithm in $\mathbb{F}_{19}[\lambda]$, and the polynomials over $\mathbb{Q}$ by division with remainder in $\mathbb{Q}[\lambda]$.

The leading coefficient of Φ is congruent to 1 modulo 19, and the reduction of Φ modulo 19 is

$$\overline{\Phi} = \lambda^6 - 7\lambda^5 - 7\lambda^4 - 6\lambda^3 + 3\lambda^2 - 2\lambda + 9 \in \mathbb{F}_{19}[\lambda]. \tag{C.1}$$

In (C.1)–(C.6) the coefficients are integers read modulo 19. Let

$$\begin{aligned} h_1 &= 5\lambda^5 - 8\lambda^4 - \lambda^2 + 9\lambda - 5 \\ h_2 &= 3\lambda^5 - 3\lambda^4 + 6\lambda^3 + 3\lambda^2 - 6\lambda - 4 \\ h_3 &= -6\lambda^5 - 8\lambda^4 - 3\lambda^3 + 9\lambda^2 - 8\lambda + 6 \\ h_4 &= -9\lambda^5 + 9\lambda^4 - 2\lambda^3 + 7\lambda^2 - \lambda - 5 \\ h_5 &= 7\lambda^5 - 9\lambda^4 - \lambda^3 + \lambda^2 + 5\lambda - 4 \\ h_6 &= \lambda, \end{aligned} \tag{C.2}$$

and let

$$
\begin{aligned}
g_0 &= \lambda^{13} + 7\lambda^{12} - \lambda^{11} - 9\lambda^{10} + 7\lambda^9 - \lambda^8 - 4\lambda^7 + 7\lambda^6 + 4\lambda^5 - \lambda^4 \\
&\quad - 9\lambda^3 - 9\lambda^2 + \lambda + 9 \\
g_1 &= 7\lambda^{19} - 7\lambda^{18} - 7\lambda^{17} + 9\lambda^{16} + 5\lambda^{15} - 7\lambda^{14} - 9\lambda^{12} + 6\lambda^{11} - 6\lambda^{10} + 8\lambda^9 \\
&\quad - 5\lambda^8 - 4\lambda^7 + 2\lambda^6 - 3\lambda^5 + 9\lambda^4 - 4\lambda^3 + 4\lambda^2 + 3\lambda - 9 \\
g_2 &= 8\lambda^{19} - 8\lambda^{18} - 8\lambda^{17} - 6\lambda^{16} + 3\lambda^{15} - 4\lambda^{14} - 9\lambda^{13} - \lambda^{12} - 6\lambda^{11} - \lambda^{10} \\
&\quad - 7\lambda^9 - 4\lambda^8 - 8\lambda^7 - \lambda^5 - 5\lambda^3 - 4\lambda^2 - 6\lambda + 3 \\
g_3 &= 3\lambda^{19} - 3\lambda^{18} - 3\lambda^{17} - 7\lambda^{16} - 6\lambda^{15} - 2\lambda^{14} - 7\lambda^{13} - 9\lambda^{12} - 2\lambda^{11} + 5\lambda^{10} \\
&\quad - 6\lambda^9 + 8\lambda^8 + \lambda^7 - \lambda^6 + 3\lambda^5 + 6\lambda^4 - 7\lambda^3 - 2\lambda^2 + 3\lambda + 2 \\
g_4 &= -5\lambda^{19} + 5\lambda^{18} + 5\lambda^{17} - \lambda^{16} - 9\lambda^{15} - 7\lambda^{14} + 8\lambda^{13} + 3\lambda^{12} - \lambda^{11} + 3\lambda^{10} \\
&\quad + 7\lambda^9 + 4\lambda^8 - 8\lambda^7 - 9\lambda^6 + 5\lambda^5 - 5\lambda^4 - 7\lambda^3 + \lambda^2 - \lambda + 4 \\
g_5 &= 6\lambda^{19} - 6\lambda^{18} - 6\lambda^{17} + 5\lambda^{16} + 7\lambda^{15} + \lambda^{14} + 8\lambda^{13} - 3\lambda^{12} + 3\lambda^{11} - \lambda^{10} \\
&\quad - 2\lambda^9 - 3\lambda^8 + 8\lambda^6 - 4\lambda^5 + 9\lambda^4 + 4\lambda^3 + \lambda^2 + \lambda.
\end{aligned} \tag{C.3}
$$

Then the following identities hold in $\mathbb{F}_{19}[\lambda]$:

$$
\lambda^{19} - h_1 = g_0\overline{\Phi}, \qquad h_i(h_1(\lambda)) - h_{i+1}(\lambda) = g_i\overline{\Phi} \quad \text{for } i = 1, \dots, 5. \tag{C.4}
$$

Since $h_i \in \mathbb{F}_{19}[\lambda]$, we have $h_i(\lambda)^{19} = h_i(\lambda^{19})$. Thus (C.4) and induction on i imply

$$
\lambda^{19^{i+1}} = \left(\lambda^{19^i}\right)^{19} \equiv h_i(\lambda)^{19} = h_i(\lambda^{19}) \equiv h_i(h_1(\lambda)) \equiv h_{i+1}(\lambda) \pmod{\overline{\Phi}}
$$

for $i = 1, \dots, 5$. Hence $\lambda^{19^i} \equiv h_i \pmod{\overline{\Phi}}$ for $i = 1, \dots, 6$. In particular, $\lambda^{19^6} \equiv \lambda \pmod{\overline{\Phi}}$, that is, $\overline{\Phi}$ divides $\lambda^{19^6} - \lambda$.

Let

$$
\begin{aligned}
u_2 &= 5\lambda^4 + 9\lambda^3 - 3\lambda^2 + 7\lambda - 3 \\
v_2 &= -8\lambda^5 + 7\lambda^4 + 6\lambda^3 + 9\lambda^2 + \lambda - 7 \\
u_3 &= -3\lambda^4 + 9\lambda^3 + 9\lambda^2 + 9\lambda - 5 \\
v_3 &= 9\lambda^5 - 7\lambda^4 - 7\lambda^3 - 6\lambda^2 + 9\lambda - 5.
\end{aligned} \tag{C.5}
$$

Then the following identities hold in $\mathbb{F}_{19}[\lambda]$:

$$
u_2\overline{\Phi} + v_2(h_2 - \lambda) = 1, \qquad u_3\overline{\Phi} + v_3(h_3 - \lambda) = 1. \tag{C.6}
$$

Since $h_i \equiv \lambda^{19^i} \pmod{\overline{\Phi}}$, (C.6) implies that $\overline{\Phi}$ and $\lambda^{19^i} - \lambda$ are coprime for $i = 2, 3$.

We now work over $\mathbb{Q}$. Let $\Phi_0 := \Phi$, and let $\Phi_1, \dots, \Phi_6$ be the following integer polynomials. In (C.7)–(C.11), a long integer is written in groups of ten digits, counted from the right and separated by thin spaces, and an integer that does not fit on one line continues on the next line.

$$
\begin{aligned}
\Phi_1 &= 4818703\,3962764740\,2014352550\,4661708633\,0808227377\lambda^5 \\
&\quad + 58432\,3716544680\,3466845887\,3910001668\,0866346820\lambda^4 \\
&\quad - 937\,3694688096\,6509544580\,9804391783\,6150247520\lambda^3 \\
&\quad - 10\,0555929854\,2767160191\,9504990051\,6781728640\lambda^2 \\
&\quad + 226716628\,7372477526\,1339303022\,2689813760\lambda \\
&\quad + 1909431\,9137040913\,3782208112\,9332388864
\end{aligned} \tag{C.7}
$$

$$\begin{aligned}\Phi_2 &= 77657\,1913826449\,8988765344\,6236668744\,1440701828\,2080123924\,0541505653\lambda^4\\ &\quad + 990\,0904156793\,8499918022\,1437899171\,7956476991\,9775570041\,0071448384\lambda^3\\ &\quad - 8\,8713975693\,0860113208\,5150673538\,8068398251\,0639602248\,4496307168\lambda^2\\ &\quad - 1144675942\,7709797770\,8242279933\,4303197872\,0856017283\,2456144384\lambda\\ &\quad - 3991056\,2634500332\,8576054358\,0385385775\,4139664804\,3689033472\end{aligned} \tag{C.8}$$

$$\begin{aligned}\Phi_3 &= 476\,9990285678\,1828211738\,7953452313\,6555019259\,9806938833\\ &\qquad 3157174389\,0005664421\,4595433182\,5259634144\,0089448627\lambda^3\\ &\quad + 4\,5099394162\,9329628865\,5033511739\,5712283479\,9528483947\\ &\qquad 6699172343\,2082408689\,3251060838\,4243289206\,6305523092\lambda^2\\ &\quad - 588513258\,2926194523\,0701866911\,5404911638\,3091457429\\ &\qquad 6822395358\,7971025699\,5738946153\,4268536143\,7053494256\lambda\\ &\quad - 2401533\,8579539946\,9798107924\,8875278115\,3562547750\\ &\qquad 5313026657\,3179664818\,0260131162\,4617958755\,3611852864\end{aligned} \tag{C.9}$$

$$\begin{aligned}\Phi_4 &= 17922\,1394868092\,8491101977\,7257449324\,4352920699\,7256129777\\ &\qquad 6213049307\,9954979893\,5593618953\,5114964497\,5798246783\,6397945223\lambda^2\\ &\quad + 459\,2865468734\,7706919390\,7641511908\,8543244022\,2721423074\\ &\qquad 3947960882\,3437997910\,6221844815\,9506040676\,2409659815\,2317987672\lambda\\ &\quad + 2\,8340915424\,6083327197\,2884421961\,0943801154\,6160881780\\ &\qquad 5081456821\,8108515412\,0715423615\,4203304337\,8793083755\,4885127152\end{aligned} \tag{C.10}$$

$$\begin{aligned}\Phi_5 &= -8\,5788804147\,2330237069\,4881012005\,4885304132\,2005261941\\ &\qquad 2198830280\,4222391403\,1196455835\,2299343195\,7188097511\lambda\\ &\quad - 1325566099\,4748031157\,2405973852\,4468393679\,9200273196\\ &\qquad 1560285388\,5080411774\,6969646499\,2478045834\,9932202252\end{aligned} \tag{C.11}$$

$$\Phi_6 = -1.$$

We have $\Phi' = 66\Phi_1$. For $i = 1, \dots, 5$, let $\operatorname{rem}(\Phi_{i-1}, \Phi_i)$ be the remainder of the division of Φ_{i-1} by Φ_i in $\mathbb{Q}[\lambda]$. Then for $i = 1, \dots, 5$, Φ_{i+1} is primitive in $\mathbb{Z}[\lambda]$, and

$$\Phi_{i+1} = -\gamma_i \operatorname{rem}(\Phi_{i-1}, \Phi_i)$$

for a positive rational number γ_i. The signs of $\Phi_0, \dots, \Phi_6$ at $\lambda = 1/102$ and at $\lambda = 1/101$ are

	Φ_0	Φ_1	Φ_2	Φ_3	Φ_4	Φ_5	Φ_6
$\lambda = 1/102$	+	−	−	+	+	−	−
$\lambda = 1/101$	−	−	−	+	+	−	−

(C.12)

They are the signs of the integers $102^{\deg \Phi_i}\Phi_i(1/102)$ and $101^{\deg \Phi_i}\Phi_i(1/101)$. No Φ_i vanishes at $1/102$ or at $1/101$, and the rows of (C.12) have three sign variations at $\lambda = 1/102$ and two sign variations at $\lambda = 1/101$.

Identities over $\mathbb{F}_3$. We now list the identities over $\mathbb{F}_3$ used in the proof of Construction-Proposition 21.1(2).

Set-up C.1. Let $e_0, \dots, e_6$ be the basis of $\mathbb{Z}^7$ and $e_{ij} := e_i^\vee \wedge e_j^\vee$, as in Construction-Proposition 21.1, and let

$$\begin{aligned}\omega_1 &= 6e_{02} - 2e_{05} + 10e_{14} - 47e_{23} + 6e_{35},\\ \omega_2 &= 6e_{03} - 2e_{06} + 6e_{12} + 8e_{15} - 37e_{24} + 6e_{36} + 6e_{45},\\ \omega_3 &= 6e_{13} - 2e_{16} + 10e_{25} - 47e_{34} + 6e_{46}\end{aligned}$$

be the forms of Construction-Proposition 21.1. Let $\omega_1' := 2\omega_1$, $\omega_2' := \omega_2$, and $\omega_3' := 2\omega_3$, and let $\overline{\omega}_k'$ be the reduction of ω_k' modulo 3. Then

$$\overline{\omega}_1' = -e_{05} - e_{14} - e_{23}, \qquad \overline{\omega}_2' = e_{06} - e_{15} - e_{24}, \qquad \overline{\omega}_3' = -e_{16} - e_{25} - e_{34}.$$

We regard an alternating form $\sum_{i<j} c_{ij}e_{ij}$ as the alternating 7×7 matrix whose (i,j) entry is c_{ij} for $i<j$, with rows and columns numbered $0,\dots,6$. Let $X_{\mathbb{F}_3}\subset \mathrm{Gr}(3,\mathbb{F}_3^7)$ be the scheme of three-dimensional subspaces isotropic for $\overline{\omega}'_1$, $\overline{\omega}'_2$, and $\overline{\omega}'_3$.

For an increasing triple $I=i_1i_2i_3$ of elements of $\{0,\dots,6\}$, let U_I be the 7×3 matrix over $\mathbb{F}_3[z_1,\dots,z_{12}]$ whose rows i_1, i_2, i_3 are $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, and whose other twelve entries are $z_1,\dots,z_{12}$, filled in row by row: the rows not in I are taken in increasing order, and each of them is filled from left to right. The column span of U_I defines an isomorphism from $\mathbb{A}^{12}_{\mathbb{F}_3}=\operatorname{Spec}\mathbb{F}_3[z_1,\dots,z_{12}]$ onto the chart $\{P_I\neq 0\}$ of $\mathrm{Gr}(3,\mathbb{F}_3^7)$. For $k=1,2,3$, let $\psi_{I,3k-2}$, $\psi_{I,3k-1}$, and $\psi_{I,3k}$ be the $(0,1)$, $(0,2)$, and $(1,2)$ entries of the 3×3 matrix $U_I^t\overline{\omega}'_kU_I$, with rows and columns numbered $0,1,2$. Thus $X_{\mathbb{F}_3}\cap\{P_I\neq 0\}$ is the closed subscheme of $\mathbb{A}^{12}_{\mathbb{F}_3}$ defined by $\psi_{I,1},\dots,\psi_{I,9}$. Let

$$J_I:=\left(\frac{\partial\psi_{I,r}}{\partial z_j}\right)_{1\le r\le 9,\ 1\le j\le 12}$$

be the 9×12 Jacobian matrix. For a sequence $\gamma=(\gamma_1,\dots,\gamma_9)$ of integers with $1\le\gamma_1<\cdots<\gamma_9\le 12$, let

$$\mu_{I,\gamma}:=\det J_I[\cdot,\gamma]$$

be the determinant of the 9×9 submatrix of J_I formed by the columns $\gamma_1,\dots,\gamma_9$ in increasing order.

Lemma C.2. *In the notation of Set-up C.1, the fourteen standard affine charts* $\{P_I\neq 0\}$ *with*

$$I\in\{012,456,013,356,246,256,245,014,024,124,125,135,145,045\}$$

cover $X_{\mathbb{F}_3}$. *The linear involution* σ *of* $\mathbb{Z}^7$ *with* $\sigma(e_i)=e_{6-i}$ *satisfies* $\sigma^*\omega'_1=-\omega'_3$, $\sigma^*\omega'_2=-\omega'_2$, *and* $\sigma^*\omega'_3=-\omega'_1$, *and the induced automorphism of* $\mathrm{Gr}(3,\mathbb{Z}^7)$ *exchanges the chart* 012 *with* 456, 013 *with* 356, 246 *with* 024, 256 *with* 014, 245 *with* 124, *and* 125 *with* 145.

Proof. A family of open subschemes covers $X_{\mathbb{F}_3}$ if, for every field K, the image of every K-point of $X_{\mathbb{F}_3}$ is contained in one of them, so it suffices to treat K-points. Let K be a field of characteristic 3, let $W\subset K^7$ be a three-dimensional subspace whose point $[W]$ belongs to $X_{\mathbb{F}_3}(K)$, and let v_1,v_2,v_3 be a basis of W. For $0\le i,j,k\le 6$, let P_{ijk} be the determinant of the 3×3 matrix formed by the rows i, j, k of the 7×3 matrix with columns v_1,v_2,v_3. Thus P_{ijk} is alternating in (i,j,k), $P_{ijk}=0$ if two of i,j,k coincide, and the P_{ijk} with $i<j<k$ are the Plücker coordinates of $[W]$ in the basis v_1,v_2,v_3. For an alternating form $\beta=\sum_{a<b}\beta_{ab}e_{ab}$ on K^7, the expansion of the determinant P_{iab} along the row i implies

$$\beta(v_2,v_3)v_1-\beta(v_1,v_3)v_2+\beta(v_1,v_2)v_3=\Bigl(\sum_{0\le a<b\le 6}\beta_{ab}P_{iab}\Bigr)_{0\le i\le 6}.$$

Since W is isotropic for $\overline{\omega}'_1$, $\overline{\omega}'_2$, and $\overline{\omega}'_3$, the left-hand side vanishes for $\beta=\overline{\omega}'_k$, $k=1,2,3$. Thus, for $0\le i\le 6$,

$$P_{i05}+P_{i14}+P_{i23}=0,\qquad P_{i06}-P_{i15}-P_{i24}=0,\qquad P_{i16}+P_{i25}+P_{i34}=0. \tag{C.13}$$

Writing the twenty-one relations (C.13) in increasing Plücker coordinates, with $i=0,\dots,6$ from top to bottom, we obtain

$$\begin{aligned}
P_{014}+P_{023}&=0, & P_{015}+P_{024}&=0, & P_{016}+P_{025}+P_{034}&=0,\\
P_{015}-P_{123}&=0, & P_{016}+P_{124}&=0, & P_{125}+P_{134}&=0,\\
P_{025}+P_{124}&=0, & P_{026}-P_{125}&=0, & P_{126}-P_{234}&=0,\\
P_{035}+P_{134}&=0, & P_{036}-P_{135}-P_{234}&=0, & P_{136}+P_{235}&=0,\\
P_{045}-P_{234}&=0, & P_{046}-P_{145}&=0, & P_{146}+P_{245}&=0,\\
P_{145}+P_{235}&=0, & P_{056}+P_{245}&=0, & P_{156}-P_{345}&=0,\\
P_{056}+P_{146}+P_{236}&=0, & P_{156}+P_{246}&=0, & P_{256}+P_{346}&=0.
\end{aligned}$$

Each equality in the following list of 21 eliminations is a relation of (C.13) that has two terms when written in increasing Plücker coordinates, or a chain of such relations, except the equalities for P_{034}, P_{036}, and P_{236}. Since $2=-1$ in $\mathbb{F}_3$, the relations $P_{016}+P_{025}+P_{034}=0$, $P_{016}=-P_{124}$, and $P_{025}=-P_{124}$ imply $P_{034}=2P_{124}=-P_{124}$. Substituting $P_{234}=P_{045}$ into $P_{036}=P_{135}+P_{234}$, we obtain $P_{036}=P_{135}+P_{045}$. Substituting $P_{056}=P_{146}=-P_{245}$ into $P_{056}+P_{146}+P_{236}=0$, we obtain $P_{236}=2P_{245}=-P_{245}$. The list is

$$\begin{aligned}
&P_{015}=P_{123}=-P_{024}, && P_{016}=P_{025}=P_{034}=-P_{124},\\
&P_{023}=-P_{014}, && P_{026}=P_{035}=-P_{134}=P_{125},\\
&P_{036}=P_{135}+P_{045}, && P_{046}=P_{136}=-P_{235}=P_{145},\\
&P_{056}=P_{146}=P_{236}=-P_{245}, && P_{126}=P_{234}=P_{045},\\
&P_{156}=P_{345}=-P_{246}, && P_{346}=-P_{256}.
\end{aligned}$$

In each chain, every term except the last is $\pm P_{ijk}$ with ijk not in

$$\{012,456,013,356,246,256,245,014,024,124,125,135,145,045\}.$$

Each of the 21 such increasing triples ijk occurs in exactly one such term. The last term of each chain is a linear combination of the fourteen coordinates P_I. If $P_I=0$ for the fourteen triples I, then all 35 Plücker coordinates of $[W]$ vanish, which is impossible since v_1,v_2,v_3 are linearly independent. Thus the fourteen charts cover $X_{\mathbb{F}_3}$.

We have $\sigma^* e_i^\vee = e_{6-i}^\vee$, hence $\sigma^* e_{ij} = e_{6-i,6-j}$, where $e_{ji}:=-e_{ij}$. Substituting into the five, seven, and five terms of ω_1, ω_2, and ω_3, we obtain

$$\begin{aligned}
\sigma^*\omega_1 &= 6e_{64}-2e_{61}+10e_{52}-47e_{43}+6e_{31}=-\omega_3,\\
\sigma^*\omega_2 &= 6e_{63}-2e_{60}+6e_{54}+8e_{51}-37e_{42}+6e_{30}+6e_{21}=-\omega_2,
\end{aligned}$$

and $\sigma^*\omega_3=\sigma^*\sigma^*(-\omega_1)=-\omega_1$ since $\sigma^2=\mathrm{id}_{\mathbb{Z}^7}$. Multiplying by 2, 1, and 2, we obtain $\sigma^*\omega_1'=-\omega_3'$, $\sigma^*\omega_2'=-\omega_2'$, and $\sigma^*\omega_3'=-\omega_1'$. Hence σ maps a three-dimensional subspace isotropic for $\omega_1',\omega_2',\omega_3'$ to a subspace isotropic for $\omega_1',\omega_2',\omega_3'$, so the induced automorphism of $\mathrm{Gr}(3,\mathbb{Z}^7)$ preserves the subscheme defined by $\omega_1',\omega_2',\omega_3'$. For a 7×3 matrix M, the row i of σM is the row $6-i$ of M. Hence, for $0\le i<j<k\le 6$,

$$P_{ijk}(\sigma M)=\det\big(\text{rows } 6-i,6-j,6-k \text{ of } M\big)=-P_{6-k,6-j,6-i}(M).$$

Thus the induced automorphism exchanges the charts of the triples ijk and $(6-k)(6-j)(6-i)$. For $ijk=012$, 013, 246, 256, 245, and 125, the triple $(6-k)(6-j)(6-i)$ is 456, 356, 024, 014, 124, and 145, respectively. □

The automorphism of $\mathrm{Gr}(3,\mathbb{Z}^7)$ induced by σ maps the chart 135 to itself and the chart 045 to the chart 126, which is not one of the fourteen charts of Lemma C.2. Thus, besides one chart from each pair exchanged in Lemma C.2, we treat the charts 135 and 045 directly.

Lemma C.3. *In the notation of Set-up C.1, let $I\in\{012,013,246,256,245,125,135,045\}$. Then there exist polynomials $a_{I,r}$, $1\le r\le 9$, and $b_{I,\gamma}$ in $\mathbb{F}_3[z_1,\dots,z_{12}]$, where γ runs over the sequences $1\le\gamma_1<\cdots<\gamma_9\le 12$, such that*

$$\sum_{r=1}^{9} a_{I,r}\psi_{I,r}+\sum_{\gamma} b_{I,\gamma}\mu_{I,\gamma}=1 \quad \text{in } \mathbb{F}_3[z_1,\dots,z_{12}]. \tag{C.14}$$

Proof. For each I, we list below the polynomials $\psi_{I,r}$, $1\le r\le 9$, and the non-zero coefficients $a_{I,r}$ and $b_{I,\gamma}$. The coefficients that are not listed are zero. Each identity (C.14) can be checked by expanding the 9×9 minors $\mu_{I,\gamma}$, multiplying, and reducing modulo 3 in $\mathbb{F}_3[z_1,\dots,z_{12}]$.
In the chart $I=012$, we write ψ_r and μ_γ for $\psi_{I,r}$ and $\mu_{I,\gamma}$. The equations are

$$\begin{aligned}
&\psi_1=z_4-z_8, && \psi_2=z_1-z_9, && \psi_3=z_2-z_6,\\
&\psi_4=z_7+z_{11}, && \psi_5=z_4+z_{12}, && \psi_6=z_5-z_9,
\end{aligned}$$

$$\psi_7 = -z_1z_5 + z_2z_4 + z_{10}, \qquad \psi_8 = -z_1z_6 + z_3z_4 + z_7,$$
$$\psi_9 = -z_2z_6 + z_3z_5 + z_8 - z_{12}$$

and the identity is

$$\mu_{(4,5,6,7,8,9,10,11,12)} = 1. \tag{C.15}$$

In the chart $I = 013$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$\psi_1 = z_4 - z_8, \qquad \psi_2 = -z_1 - z_9, \qquad \psi_3 = -z_2 - z_6,$$
$$\psi_4 = -z_1z_5 + z_2z_4 + z_7 + z_{11}, \qquad \psi_5 = -z_1z_6 + z_3z_4 + z_{12},$$
$$\psi_6 = -z_2z_6 + z_3z_5 - z_9, \qquad \psi_7 = -z_1z_8 + z_2z_7 + z_{10},$$
$$\psi_8 = -z_1z_9 + z_3z_7 + z_4, \qquad \psi_9 = -z_2z_9 + z_3z_8 + z_5 - z_{12}$$

the non-zero coefficients are

$$a_3 = z_2z_3^2 + z_3,$$
$$a_6 = z_3^2,$$

$$b_{(1,4,5,6,7,8,9,10,12)} = -1,$$

and the identity is

$$a_3\psi_3 + a_6\psi_6 + b_{(1,4,5,6,7,8,9,10,12)}\mu_{(1,4,5,6,7,8,9,10,12)} = 1. \tag{C.16}$$

In the chart $I = 246$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$\psi_1 = -z_1z_{11} + z_2z_{10} - z_4 - z_8, \qquad \psi_2 = -z_1z_{12} + z_3z_{10} - z_9,$$
$$\psi_3 = -z_2z_{12} + z_3z_{11} + z_6, \qquad \psi_4 = -z_4z_{11} + z_5z_{10} - 1,$$
$$\psi_5 = -z_4z_{12} + z_6z_{10} + z_1, \qquad \psi_6 = -z_5z_{12} + z_6z_{11} + z_2,$$
$$\psi_7 = -z_7 - z_{11}, \qquad \psi_8 = -z_4 - z_{12}, \qquad \psi_9 = -z_5 + z_9$$

the non-zero coefficients are

$$a_2 = z_{10},$$
$$a_4 = -1,$$
$$a_5 = z_{10}z_{12},$$
$$a_9 = z_{10},$$

$$b_{(1,2,3,5,7,8,9,11,12)} = -1,$$
$$b_{(1,2,3,6,8,9,10,11,12)} = -1,$$

and the identity is

$$a_2\psi_2 + a_4\psi_4 + a_5\psi_5 + a_9\psi_9 + b_{(1,2,3,5,7,8,9,11,12)}\mu_{(1,2,3,5,7,8,9,11,12)} + b_{(1,2,3,6,8,9,10,11,12)}\mu_{(1,2,3,6,8,9,10,11,12)} = 1. \tag{C.17}$$

In the chart $I = 256$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$\psi_1 = -z_4z_{11} + z_5z_{10} - z_1 - z_8, \qquad \psi_2 = -z_4z_{12} + z_6z_{10} - z_9,$$
$$\psi_3 = -z_5z_{12} + z_6z_{11} + z_3, \qquad \psi_4 = -z_4 - z_{11},$$
$$\psi_5 = z_1 - z_{12}, \qquad \psi_6 = z_2 + z_6,$$
$$\psi_7 = -z_7z_{11} + z_8z_{10} - 1, \qquad \psi_8 = -z_7z_{12} + z_9z_{10} - z_4,$$
$$\psi_9 = -z_8z_{12} + z_9z_{11} - z_5$$

the non-zero coefficients are

$$a_2 = -z_7^2z_{10}z_{11},$$
$$a_4 = -z_7^2z_{11} + z_7,$$

$$a_7 = z_7^2 z_{12} - 1,$$
$$a_8 = z_7^2 z_{11} + z_9 z_{10}^2 - z_7,$$
$$a_9 = -z_7^2 z_{10}^2 z_{12} + z_{10}^2,$$

$$b_{(2,3,4,5,6,8,10,11,12)} = -1,$$
$$b_{(2,3,5,6,7,8,10,11,12)} = z_7^2,$$
$$b_{(2,4,5,6,7,8,9,11,12)} = -z_7^2,$$

and the identity is

$$\text{(C.18)}\quad a_2\psi_2 + a_4\psi_4 + a_7\psi_7 + a_8\psi_8 + a_9\psi_9 + b_{(2,3,4,5,6,8,10,11,12)}\mu_{(2,3,4,5,6,8,10,11,12)} + b_{(2,3,5,6,7,8,10,11,12)}\mu_{(2,3,5,6,7,8,10,11,12)} + b_{(2,4,5,6,7,8,9,11,12)}\mu_{(2,4,5,6,7,8,9,11,12)} = 1.$$

In the chart $I = 245$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$\psi_1 = -z_4 - z_8, \qquad \psi_2 = -z_1 - z_9, \qquad \psi_3 = -z_2 + z_6,$$
$$\psi_4 = z_1 z_{11} - z_2 z_{10} - 1, \qquad \psi_5 = z_1 z_{12} - z_3 z_{10} - z_4,$$
$$\psi_6 = z_2 z_{12} - z_3 z_{11} - z_5, \qquad \psi_7 = -z_4 z_{11} + z_5 z_{10} - z_7,$$
$$\psi_8 = -z_4 z_{12} + z_6 z_{10} - 1, \qquad \psi_9 = -z_5 z_{12} + z_6 z_{11} + z_9$$

the non-zero coefficients are

$$a_1 = -z_1 z_{12}^3 + z_3 z_{10} z_{12}^2 + z_1 z_{11} z_{12} + z_6 z_{10} z_{12} + z_9 z_{11} z_{12} - z_{12},$$
$$a_2 = z_3 z_{10} z_{11} z_{12} - z_6 z_{10} z_{12}^2 - z_5 z_{10} z_{12} + z_6 z_{10} z_{11} - z_8 z_{11} z_{12} + z_1 z_{10} - z_9 z_{10} + z_{11},$$
$$a_3 = -z_1 z_{10} z_{12}^2 - z_9 z_{10} z_{12}^2 - z_{10},$$
$$a_4 = 1,$$
$$a_5 = -z_8 z_{12}^2,$$
$$a_6 = -z_9 z_{10} z_{12},$$
$$a_8 = z_8 z_{12} - z_9 z_{11} + 1,$$
$$a_9 = -z_9 z_{10},$$

$$b_{(1,2,5,6,7,8,9,11,12)} = -1,$$

and the identity is

$$\text{(C.19)}\quad a_1\psi_1 + a_2\psi_2 + a_3\psi_3 + a_4\psi_4 + a_5\psi_5 + a_6\psi_6 + a_8\psi_8 + a_9\psi_9 + b_{(1,2,5,6,7,8,9,11,12)}\mu_{(1,2,5,6,7,8,9,11,12)} = 1.$$

In the chart $I = 125$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$\psi_1 = z_4 - z_8, \qquad \psi_2 = -z_1 - z_9, \qquad \psi_3 = -z_2 - z_6,$$
$$\psi_4 = z_1 z_{11} - z_2 z_{10} + z_7, \qquad \psi_5 = z_1 z_{12} - z_3 z_{10} - 1,$$
$$\psi_6 = z_2 z_{12} - z_3 z_{11} - z_9, \qquad \psi_7 = -z_4 z_8 + z_5 z_7 - z_{11},$$
$$\psi_8 = -z_4 z_9 + z_6 z_7 - z_{12}, \qquad \psi_9 = -z_5 z_9 + z_6 z_8 - 1$$

the non-zero coefficients are

$$a_1 = -z_2^2 z_6 z_8 z_{12}^2 - z_2 z_3 z_6 z_8 z_{11} z_{12} - z_3^2 z_6 z_8 z_{11}^2 + z_1 z_2 z_6 z_8 z_{12} - z_2 z_3 z_4 z_8 z_{12} + z_2 z_3 z_5 z_7 z_{12} - z_2 z_3 z_6 z_8 z_{10} + z_2 z_3 z_8^2 z_{12} + z_3^2 z_4 z_8 z_{11} - z_3^2 z_5 z_7 z_{11} - z_3^2 z_8^2 z_{11} - z_3 z_5 z_8 z_9 z_{12} + z_3 z_6 z_8^2 z_{12} - z_3 z_5 z_{12}^2 - z_3 z_8^2 z_9 + z_5 z_9^3 - z_3 z_8 z_{12} - z_5 z_6 z_9 - z_6^2 z_8 + z_6 z_9 z_{12} - z_9^2,$$

$$
\begin{aligned}
a_2 &= -z_5^2z_6z_9z_{12}^2 + z_5z_6^2z_8z_{12}^2 - z_5^2z_9^2z_{12} - z_5z_6z_8z_9z_{12} - z_6^2z_8^2z_{12}\\
&\quad + z_3z_6z_{11}^2 + z_5z_6z_{12}^2 + z_6^2z_{11}z_{12} - z_3z_8z_{11} + z_5z_9z_{12} - z_6z_8z_{12} + z_{12},\\
a_3 &= z_1z_5^2z_9z_{12}^2 - z_1z_5z_6z_8z_{12}^2 - z_2z_5z_6z_7z_{12}^2 - z_2z_5z_8z_9z_{12}^2\\
&\quad - z_2z_6z_8^2z_{12}^2 + z_3z_5^2z_9z_{10}z_{12} + z_3z_5z_6z_7z_{11}z_{12} - z_3z_5z_6z_8z_{10}z_{12}\\
&\quad - z_3z_6z_8^2z_{11}z_{12} + z_5^2z_9^2z_{12}^2 + z_5z_6z_8z_9z_{12}^2 + z_6^2z_8^2z_{12}^2 + z_1z_5z_6z_7z_{12}\\
&\quad + z_1z_5z_8z_9z_{12} + z_1z_6z_8^2z_{12} - z_3z_5z_6z_7z_{10} - z_3z_5z_8z_9z_{10}\\
&\quad - z_3z_6z_8^2z_{10} + z_5z_6z_7z_9z_{12} - z_5z_8z_9^2z_{12} - z_3z_5z_{10}z_{12} - z_3z_6z_{10}z_{11}\\
&\quad - z_3z_8z_{11}z_{12} - z_6z_8z_{12}^2 - z_6z_9z_{11}z_{12} + z_3z_8z_{10} + z_6z_7z_{12} + z_8z_9z_{12}\\
&\quad - z_{12}^2,\\
a_4 &= z_3z_6z_{11} - z_3z_8,\\
a_5 &= -z_5^2z_9^2 + z_5z_6^2z_7 + z_5z_6z_{12} + z_6^2z_{11} + z_5z_9 - z_6z_8 + 1,\\
a_6 &= -z_3z_5z_6z_7z_{11} + z_3z_5z_8z_9z_{11} + z_5^2z_9^2z_{12} - z_5z_6^2z_7z_{12} - z_3z_5z_7z_8\\
&\quad + z_3z_8^3 + z_5z_6z_7z_9 - z_5z_6z_{12}^2 - z_5z_8z_9^2 - z_3z_5z_{10} - z_3z_8z_{11} - z_6z_8z_{12}\\
&\quad - z_6z_9z_{11} + z_6z_7 + z_8z_9 - z_{12},\\
a_7 &= z_3z_6z_8z_{12} + z_3z_8z_9 + z_6z_9^2 - z_6^2,\\
a_8 &= -z_3z_5z_8z_{12} + z_3z_8^2 + z_5z_9^2 - z_6z_8z_9 - z_5z_6 + z_6z_{12} - z_9,\\
a_9 &= z_3^2z_5z_{10}z_{11} + z_3^2z_8z_{11}^2 - z_3z_5z_6z_{10}z_{12} - z_3z_5z_9z_{11}z_{12}\\
&\quad - z_3z_6z_8z_{11}z_{12} + z_6^2z_8z_{12}^2 + z_3z_5z_9z_{10} - z_3z_6z_8z_{10} + z_3z_8^2z_{12}\\
&\quad - z_6z_8z_9z_{12} - z_3z_7z_8 + z_3z_{11}z_{12} + z_6z_7z_9 - z_3z_{10} + z_5z_9 - z_6z_8\\
&\quad + z_9z_{12} + 1,
\end{aligned}
$$

$$
\begin{aligned}
b_{(1,2,3,4,6,7,8,9,10)} &= z_3,\\
b_{(1,2,4,6,7,8,9,10,12)} &= z_{12},
\end{aligned}
$$

and the identity is

$$
\begin{aligned}
(\mathrm{C}.20)\quad & a_1\psi_1 + a_2\psi_2 + a_3\psi_3 + a_4\psi_4 + a_5\psi_5 + a_6\psi_6 + a_7\psi_7 + a_8\psi_8 + a_9\psi_9\\
&\quad + b_{(1,2,3,4,6,7,8,9,10)}\mu_{(1,2,3,4,6,7,8,9,10)} + b_{(1,2,4,6,7,8,9,10,12)}\mu_{(1,2,4,6,7,8,9,10,12)} = 1.
\end{aligned}
$$

In the chart $I = 135$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$
\begin{aligned}
\psi_1 &= -z_4 - z_8, & \psi_2 &= -z_1 - z_9,\\
\psi_3 &= -z_2 + z_6,\\
\psi_4 &= z_1z_{11} - z_2z_{10} - z_4z_8 + z_5z_7,\\
\psi_5 &= z_1z_{12} - z_3z_{10} - z_4z_9 + z_6z_7 - 1,\\
\psi_6 &= z_2z_{12} - z_3z_{11} - z_5z_9 + z_6z_8,\\
\psi_7 &= z_7 - z_{11}, & \psi_8 &= -z_4 - z_{12},\\
\psi_9 &= -z_5 - z_9
\end{aligned}
$$

the non-zero coefficients are

$$
\begin{aligned}
a_1 &= -z_3^2z_9z_{10}^2 - z_3z_6^2z_{10}^2 + z_6^3z_{10}z_{11} + z_6^2z_9z_{11}^2 - z_3z_9z_{10} - z_6^2z_{10}\\
&\quad - z_6z_9z_{11},\\
a_2 &= z_1z_3z_9z_{10}z_{11} - z_1z_6^2z_{10}z_{11} - z_3z_6z_9z_{10}^2 + z_3z_6z_{10}z_{11}z_{12}\\
&\quad + z_3z_9^2z_{10}z_{11} + z_3z_9z_{10}z_{12}^2 + z_3z_9z_{11}^2z_{12} - z_6^3z_{10}^2 - z_6^2z_9z_{10}z_{11}
\end{aligned}
$$

$$
\begin{aligned}
&+ z_6^2z_{10}z_{12}^2 + z_6z_9^2z_{10}z_{12} + z_6z_9z_{11}z_{12}^2 - z_9^3z_{11}z_{12} - z_3z_8z_{10} + z_6z_9z_{10}\\
&- z_6z_{11}z_{12} - z_{12},\\
a_3 = &-z_1z_3z_9z_{10}^2 + z_1z_6^2z_{10}^2 + z_3z_6z_8z_{10}^2 + z_3z_6z_{10}^2z_{12} - z_3z_6z_{10}z_{11}^2\\
&+ z_3z_9^2z_{10}^2 - z_6^2z_9z_{10}^2 + z_6^2z_{10}z_{11}z_{12} + z_6z_9z_{11}^2z_{12} + z_3z_{10}z_{11} - z_6z_8z_{10}\\
&- z_6z_{10}z_{12} - z_6z_{11}^2 - z_9z_{11}z_{12},\\
a_5 = &\,z_3z_6z_{10}z_{11} + z_3z_9z_{10}z_{12} + z_3z_9z_{11}^2 + z_6^2z_{10}z_{12} + z_6z_9^2z_{10}\\
&+ z_6z_9z_{11}z_{12} - z_9^3z_{11} + z_3z_{10} - z_6z_{11} - 1,\\
a_6 = &\,z_6^2z_{10}z_{11} + z_6z_9z_{11}^2 - z_9z_{11},\\
a_7 = &\,z_1z_3z_5z_9z_{10} - z_1z_3z_9^2z_{10} - z_1z_5z_6^2z_{10} + z_1z_6^2z_9z_{10} + z_2z_3z_6z_{10}z_{11}\\
&- z_3^2z_6z_{10}^2 + z_3^2z_7z_9z_{10} - z_3^2z_9z_{10}z_{11} + z_3z_5^2z_9z_{10} - z_3z_5z_9^2z_{10}\\
&- z_3z_6^2z_7z_{10} - z_3z_6^2z_{10}z_{11} + z_3z_6z_8z_9z_{10} - z_3z_6z_9z_{11}^2 - z_5^2z_6^2z_{10}\\
&+ z_5z_6^2z_9z_{10} - z_6^3z_8z_{10} + z_6^3z_{10}z_{12} - z_6^2z_9^2z_{10} - z_6^2z_9z_{11}z_{12} + z_6z_9^3z_{11}\\
&- z_2z_3z_{10} + z_2z_6z_{11} + z_3z_6z_{10} + z_6,\\
a_8 = &-z_1z_3z_6z_{10}z_{11} + z_1z_3z_8z_9z_{10} + z_1z_3z_9z_{10}z_{12} - z_1z_6^2z_8z_{10}\\
&- z_1z_6^2z_{10}z_{12} + z_3^2z_9z_{10}^2 + z_3z_6^2z_{10}^2 + z_3z_6z_9z_{10}z_{11} - z_3z_8z_9^2z_{10}\\
&+ z_3z_9^2z_{10}z_{12} - z_3z_9^2z_{11}^2 + z_6^3z_{10}z_{11} + z_6^2z_8z_9z_{10} - z_6^2z_9z_{11}^2 - z_6z_9^3z_{10}\\
&- z_6z_9^2z_{11}z_{12} + z_9^4z_{11} - z_1z_6z_{11} + z_6^2z_{10} + z_9,\\
a_9 = &\,z_1z_3z_6z_{10}^2 - z_2z_3z_9z_{10}^2 + z_2z_6^2z_{10}^2 - z_3z_4z_6z_{10}z_{11} + z_3z_4z_8z_9z_{10}\\
&+ z_3z_4z_9z_{10}z_{12} + z_3z_5z_6z_{10}^2 - z_3z_5z_9z_{10}z_{11} - z_3z_6z_9z_{10}^2\\
&- z_3z_6z_{10}z_{11}z_{12} + z_3z_8z_9z_{10}z_{12} + z_3z_9z_{10}z_{12}^2 - z_4z_6^2z_8z_{10}\\
&- z_4z_6^2z_{10}z_{12} + z_5z_6^2z_{10}z_{11} - z_6^3z_{10}^2 - z_6^2z_8z_{10}z_{12} - z_6^2z_9z_{10}z_{11}\\
&- z_6^2z_{10}z_{12}^2 - z_6z_9^2z_{11}^2 + z_1z_6z_{10} - z_3z_8z_{10} - z_3z_{10}z_{12} - z_4z_6z_{11}\\
&+ z_5z_6z_{10} + z_6z_9z_{10} - z_6z_{11}z_{12} + z_9^2z_{11},
\end{aligned}
$$

$$
\begin{aligned}
b_{(1,2,5,6,7,8,9,11,12)} &= -z_3z_9z_{10} + z_6^2z_{10},\\
b_{(1,3,5,6,7,8,9,11,12)} &= -z_3z_6z_{10} - z_6,\\
b_{(2,3,5,6,8,9,10,11,12)} &= z_3z_6z_{10} + z_6,\\
b_{(2,5,6,7,8,9,10,11,12)} &= -z_{10},\\
b_{(3,4,5,6,7,8,9,11,12)} &= z_3z_9z_{10} - z_6^2z_{10},
\end{aligned}
$$

and the identity is

$$
\begin{aligned}
a_1\psi_1 + a_2\psi_2 + a_3\psi_3 + a_5\psi_5 + a_6\psi_6 + a_7\psi_7 + a_8\psi_8 + a_9\psi_9\\
+ b_{(1,2,5,6,7,8,9,11,12)}\mu_{(1,2,5,6,7,8,9,11,12)} + b_{(1,3,5,6,7,8,9,11,12)}\mu_{(1,3,5,6,7,8,9,11,12)}\\
+ b_{(2,3,5,6,8,9,10,11,12)}\mu_{(2,3,5,6,8,9,10,11,12)} + b_{(2,5,6,7,8,9,10,11,12)}\mu_{(2,5,6,7,8,9,10,11,12)}\\
+ b_{(3,4,5,6,7,8,9,11,12)}\mu_{(3,4,5,6,7,8,9,11,12)} = 1.
\end{aligned}
\tag{C.21}
$$

In the chart $I = 045$, we write ψ_r, a_r, b_γ, and μ_γ for $\psi_{I,r}$, $a_{I,r}$, $b_{I,\gamma}$, and $\mu_{I,\gamma}$. The equations are

$$
\begin{aligned}
\psi_1 &= -z_4z_8 + z_5z_7 - z_1, & \psi_2 &= -z_4z_9 + z_6z_7 - 1,\\
\psi_3 &= -z_5z_9 + z_6z_8 + z_3, & \psi_4 &= -z_4 + z_{11},\\
\psi_5 &= -z_1 + z_{12}, & \psi_6 &= -z_2 + z_6,\\
\psi_7 &= -z_1z_{11} + z_2z_{10} - z_7, & \psi_8 &= -z_1z_{12} + z_3z_{10} - z_4,
\end{aligned}
$$

$$\psi_9 = -z_2z_{12} + z_3z_{11} - z_5 + z_9$$

the non-zero coefficients are

$$\begin{aligned}
a_1 = {} & z_3z_6z_9z_{11}^3z_{12}^2 - z_6^2z_9z_{11}^2z_{12}^3 - z_3z_5z_7z_9z_{11}z_{12}^2 + z_3z_8z_9z_{11}^2z_{12}^2 \\
& + z_5z_9^2z_{11}z_{12}^3 - z_6^2z_8z_{11}^2z_{12}^2 - z_3z_5z_7z_9z_{11}^2 - z_3z_6z_{11}^2z_{12}^2 + z_3z_8z_9z_{11}^3 \\
& - z_5z_8z_9z_{11}z_{12}^2 - z_6z_7z_8z_9z_{12}^2 + z_6z_7z_9^2z_{11}z_{12} + z_6z_8^2z_{11}z_{12}^2 \\
& - z_6z_8z_9z_{11}^2z_{12} - z_6z_9^3z_{10}z_{11} + z_6z_9^2z_{11}^3 + z_6z_9z_{12}^4 - z_7z_9^3z_{12}^2 \\
& - z_8z_9^2z_{11}z_{12}^2 - z_3z_9z_{11}^2z_{12} - z_5z_8z_9z_{11}^2 - z_6z_7^2z_9^2 + z_6z_7z_8z_9z_{11} \\
& + z_6z_8^2z_{11}^2 - z_6z_8z_{12}^3 + z_6z_9z_{11}z_{12}^2 + z_8z_9^2z_{11}^2 - z_5z_9z_{11}z_{12} \\
& - z_6z_8z_{11}z_{12} - z_6z_9^2z_{10} + z_6z_9z_{11}^2 - z_8z_9z_{12}^2 - z_9^2z_{11}z_{12} - z_6z_{12}^2 \\
& - z_7z_9^2 + z_8z_9z_{11} - z_6z_{11}, \\
a_2 = {} & -z_3z_5z_9z_{11}^3z_{12}^2 + z_5z_6z_9z_{11}^2z_{12}^3 + z_3^2z_6z_{10}z_{11}^2z_{12} + z_3z_5z_7z_8z_{11}z_{12}^2 \\
& - z_3z_6^2z_{10}z_{11}z_{12}^2 - z_3z_8^2z_{11}^2z_{12}^2 + z_5z_6z_8z_{11}^2z_{12}^2 - z_5z_8z_9z_{11}z_{12}^3 \\
& - z_3^2z_9z_{10}z_{11}^2 - z_3z_5z_6z_{10}z_{11}z_{12} + z_3z_5z_7z_8z_{11}^2 + z_3z_5z_7z_{12}^3 \\
& + z_3z_5z_9z_{10}z_{12}^2 + z_3z_5z_{11}^2z_{12}^2 + z_3z_6^2z_7z_{10}z_{12} + z_3z_6z_9z_{10}z_{11}z_{12} \\
& - z_3z_8^2z_{11}^3 - z_3z_8z_{11}z_{12}^3 + z_5z_7z_8z_9z_{12}^2 - z_5z_7z_9^2z_{11}z_{12} \\
& + z_5z_8z_9z_{11}^2z_{12} + z_5z_9^3z_{10}z_{11} - z_5z_9^2z_{11}^3 + z_7z_8z_9^2z_{12}^2 + z_8^2z_9z_{11}z_{12}^2 \\
& + z_3z_5z_7z_{11}z_{12} - z_3z_5z_9z_{10}z_{11} - z_3z_6z_7z_9z_{10} + z_3z_8z_{11}^2z_{12} \\
& + z_5z_6z_7z_{12}^2 + z_5z_7^2z_9^2 - z_5z_7z_8z_9z_{11} - z_5z_8z_{12}^3 + z_5z_9z_{11}z_{12}^2 \\
& - z_6z_8z_{11}z_{12}^2 - z_8^2z_9z_{11}^2 - z_8z_9z_{12}^3 + z_9^2z_{11}z_{12}^2 + z_3z_6z_{10}z_{12} - z_3z_{11}z_{12}^2 \\
& + z_5z_6z_7z_{11} + z_5z_9^2z_{10} - z_6z_8z_{11}^2 - z_8z_9z_{11}z_{12} - z_9^3z_{10} - z_3z_9z_{10} \\
& - z_3z_{11}^2 + z_6z_{11}z_{12} + z_9z_{12}^2 + z_5z_{11} - z_6z_7 + z_9z_{11} - 1, \\
a_3 = {} & z_3z_9z_{11}^4z_{12}^2 - z_6z_9z_{11}^3z_{12}^3 - z_3z_5z_7^2z_{11}z_{12}^2 + z_3z_7z_8z_{11}^2z_{12}^2 \\
& + z_5z_7z_9z_{11}z_{12}^3 - z_6z_8z_{11}^3z_{12}^2 - z_3z_5z_7^2z_{11}^2 + z_3z_7z_8z_{11}^3 - z_3z_{11}^3z_{12}^2 \\
& - z_5z_7z_8z_{11}z_{12}^2 + z_6z_{11}^2z_{12}^3 - z_7^2z_9^2z_{12}^2 + z_7z_8z_9z_{11}z_{12}^2 + z_7z_9^2z_{11}^2z_{12} \\
& + z_8^2z_{11}^2z_{12}^2 - z_8z_9z_{11}^3z_{12} - z_9^3z_{10}z_{11}^2 + z_9^2z_{11}^4 + z_9z_{11}z_{12}^4 - z_3z_7z_{11}^2z_{12} \\
& - z_5z_7z_8z_{11}^2 - z_7^2z_9^2z_{11} - z_7z_8z_9z_{11}^2 + z_7z_9z_{12}^3 + z_8^2z_{11}^3 - z_8z_{11}z_{12}^3 \\
& + z_9z_{11}^2z_{12}^2 - z_5z_7z_{11}z_{12} + z_7z_9z_{11}z_{12} - z_8z_{11}^2z_{12} - z_9^2z_{10}z_{11} - z_{11}z_{12}^2 \\
& - z_{11}^2, \\
a_4 = {} & z_3z_5z_9^2z_{11}^3z_{12}^2 - z_3z_6z_8z_9z_{11}^3z_{12}^2 - z_5z_6z_9^2z_{11}^2z_{12}^3 + z_6^2z_8z_9z_{11}^2z_{12}^3 \\
& - z_2z_3z_4z_6z_9z_{10}z_{12}^2 + z_2z_3z_6^2z_7z_{10}z_{12}^2 - z_2z_3z_6z_9z_{10}z_{11}z_{12}^2 \\
& + z_3^2z_4z_6z_9z_{10}z_{11}z_{12} - z_3^2z_6^2z_7z_{10}z_{11}z_{12} + z_3z_6^2z_9z_{10}z_{11}z_{12}^2 \\
& - z_5z_6z_8z_9z_{11}^2z_{12}^2 + z_6^2z_8^2z_{11}^2z_{12}^2 - z_1z_2z_8z_9z_{12}^3 - z_2z_3z_4z_6z_8z_{10}z_{12} \\
& + z_2z_3z_4z_9^2z_{10}z_{12} + z_2z_3z_5z_6z_7z_{10}z_{12} - z_2z_3z_6z_8z_{10}z_{11}z_{12} \\
& + z_2z_3z_7z_8z_{12}^3 + z_2z_3z_8z_9z_{10}z_{12}^2 + z_2z_3z_9^2z_{10}z_{11}z_{12} - z_2z_8z_9z_{12}^4 \\
& - z_3^2z_4z_9^2z_{10}z_{11} + z_3^2z_6z_7z_9z_{10}z_{11} - z_3^2z_7z_8z_{11}z_{12}^2 \\
& + z_3z_5z_6z_9z_{10}z_{11}z_{12} - z_3z_5z_7z_9z_{12}^3 - z_3z_5z_9^2z_{10}z_{12}^2 - z_3z_5z_9z_{11}^2z_{12}^2 \\
& - z_3z_6^2z_7z_9z_{10}z_{12} + z_3z_6z_8z_{11}^2z_{12}^2 - z_3z_6z_9^2z_{10}z_{11}z_{12} - z_3z_8z_9z_{11}z_{12}^3
\end{aligned}$$

$$
\begin{aligned}
&- z_5z_7z_8z_9^2z_{12}^2 + z_5z_7z_9^3z_{11}z_{12} + z_5z_8^2z_9z_{11}z_{12}^2 - z_5z_8z_9^2z_{11}^2z_{12}\\
&- z_5z_9^4z_{10}z_{11} + z_5z_9^3z_{11}^3 + z_6z_7z_8^2z_9z_{12}^2 - z_6z_7z_8z_9^2z_{11}z_{12}\\
&- z_6z_8^3z_{11}z_{12}^2 + z_6z_8^2z_9z_{11}^2z_{12} + z_6z_8z_9^3z_{10}z_{11} - z_6z_8z_9^2z_{11}^3\\
&- z_6z_8z_9z_{12}^4 - z_1z_2z_8^2z_{12}^2 - z_1z_2z_8z_9z_{11}z_{12} - z_1z_3z_9z_{12}^3 + z_2^2z_7z_8z_{12}^2\\
&+ z_2z_3z_4z_8z_9z_{10} - z_2z_3z_5z_7z_9z_{10} + z_2z_3z_7z_8z_{11}z_{12} - z_2z_3z_7z_9^2z_{10}\\
&- z_2z_3z_8z_9z_{10}z_{11} - z_2z_8^2z_{12}^3 - z_2z_8z_9z_{11}z_{12}^2 - z_3^2z_4z_6z_{10}z_{12}\\
&- z_3^2z_6z_{10}z_{11}z_{12} - z_3^2z_7z_8z_{11}^2 + z_3^2z_9z_{10}z_{12}^2 - z_3z_5z_7z_9z_{11}z_{12}\\
&+ z_3z_5z_9^2z_{10}z_{11} + z_3z_6z_7z_9^2z_{10} + z_3z_8^2z_{11}z_{12}^2 + z_3z_8z_9z_{11}^2z_{12}\\
&- z_5z_6z_7z_9z_{12}^2 - z_5z_7^2z_9^3 + z_5z_7z_8z_9^2z_{11} + z_5z_8^2z_9z_{11}^2 + z_5z_8z_9z_{12}^3\\
&+ z_5z_9^2z_{11}z_{12}^2 + z_6z_7^2z_8z_9^2 - z_6z_7z_8^2z_9z_{11} - z_6z_7z_9^2z_{12}^2 - z_6z_8^3z_{11}^2\\
&+ z_6z_8^2z_{12}^3 + z_8z_9^2z_{12}^3 - z_9^3z_{11}z_{12}^2 - z_1z_2z_8^2z_{11} - z_1z_3z_8z_{12}^2\\
&- z_1z_3z_9z_{11}z_{12} + z_2^2z_7z_8z_{11} - z_2z_8^2z_{11}z_{12} + z_2z_9z_{12}^3 + z_3^2z_4z_9z_{10}\\
&- z_3^2z_9z_{10}z_{11} - z_3z_6z_9z_{10}z_{12} + z_3z_8^2z_{11}^2 - z_3z_9z_{11}z_{12}^2 - z_5z_6z_7z_9z_{11}\\
&+ z_5z_8z_9z_{11}z_{12} - z_5z_9^3z_{10} + z_5z_9^2z_{11}^2 + z_6z_8^2z_{11}z_{12} + z_6z_8z_9^2z_{10}\\
&- z_6z_9z_{12}^3 + z_8^2z_9z_{12}^2 - z_8z_9^2z_{11}z_{12} + z_9^4z_{10} + z_9^3z_{11}^2 - z_1z_3z_8z_{11}\\
&+ z_2z_6z_9z_{10} + z_2z_8z_{12}^2 + z_2z_9z_{11}z_{12} - z_3z_4z_9z_{11} + z_3z_6z_7z_{11}\\
&+ z_3z_9^2z_{10} + z_4z_6z_9z_{12} + z_5z_9z_{12}^2 - z_6^2z_7z_{12} - z_6^2z_9z_{10} + z_6z_8z_{12}^2\\
&+ z_7z_8z_9^2 - z_8^2z_9z_{11} + z_9^2z_{12}^2 + z_2z_8z_{11} + z_4z_6z_8 - z_5z_6z_7 - z_5z_9z_{11}\\
&- z_6z_8z_{11} + z_3z_4 + z_3z_{11} - z_6z_{12} - z_9,\\
a_5 = &-z_3z_6z_9z_{11}^3z_{12}^2 + z_6^2z_9z_{11}^2z_{12}^3 + z_3z_5z_7z_9z_{11}z_{12}^2 + z_3z_5z_9z_{11}^3z_{12}\\
&- z_3z_6z_8z_{11}^3z_{12} + z_3z_7z_9^2z_{11}z_{12}^2 + z_5z_6z_7z_9z_{12}^3 + z_5z_9^2z_{11}z_{12}^3\\
&+ z_6^2z_8z_{11}^2z_{12}^2 - z_6z_7z_9^2z_{12}^3 - z_6z_8z_9z_{11}z_{12}^3 + z_3^2z_{11}^3z_{12} + z_3z_5z_7z_9z_{11}^2\\
&- z_3z_6z_{11}^2z_{12}^2 + z_3z_8z_9z_{11}^3 + z_3z_9z_{11}z_{12}^3 + z_5z_6z_7z_8z_{12}^2\\
&+ z_5z_6z_7z_9z_{11}z_{12} + z_5z_8z_9z_{11}z_{12}^2 + z_5z_9^2z_{11}^2z_{12} + z_6z_7z_8z_9z_{12}^2\\
&- z_6z_7z_9^2z_{11}z_{12} + z_6z_8^2z_{11}z_{12}^2 + z_6z_9^3z_{10}z_{11} - z_6z_9^2z_{11}^3 + z_6z_9z_{12}^4\\
&+ z_7z_9^3z_{12}^2 + z_8z_9^2z_{11}z_{12}^2 + z_3z_5z_7z_{12}^2 + z_3z_5z_{11}^2z_{12} + z_3z_7z_9z_{12}^2\\
&- z_3z_8z_{11}z_{12}^2 + z_3z_9z_{11}^2z_{12} + z_5z_6z_7z_8z_{11} + z_5z_8z_9z_{11}^2 + z_5z_9z_{12}^3\\
&+ z_6z_7^2z_9^2 - z_6z_7z_8z_9z_{11} + z_6z_8^2z_{11}^2 + z_6z_8z_{12}^3 - z_6z_9z_{11}z_{12}^2 - z_7z_9^3z_{11}\\
&- z_8z_9^2z_{11}^2 - z_9^2z_{12}^3 + z_3z_5z_7z_{11} + z_3z_7z_9z_{11} - z_3z_8z_{11}^2 + z_5z_9z_{11}z_{12}\\
&+ z_6z_8z_{11}z_{12} + z_6z_9^2z_{10} - z_6z_9z_{11}^2 + z_8z_9z_{12}^2 + z_9^2z_{11}z_{12} + z_7z_9^2\\
&- z_8z_9z_{11} + z_6z_{11} + z_9z_{12},\\
a_6 = &\, z_3z_6^2z_7z_{10}z_{11}z_{12}^2 - z_3z_6z_9z_{10}z_{11}^2z_{12}^2 + z_3z_9z_{11}^3z_{12}^3 - z_6z_9z_{11}^2z_{12}^4\\
&+ z_1z_5z_7z_9z_{12}^3 - z_1z_8z_9z_{11}z_{12}^3 + z_3z_5z_6z_7z_{10}z_{11}z_{12} - z_3z_5z_7^2z_{12}^3\\
&- z_3z_5z_7z_9z_{10}z_{12}^2 - z_3z_5z_9z_{10}z_{11}^2z_{12} - z_3z_6^2z_7^2z_{10}z_{12} + z_3z_7z_8z_{11}z_{12}^3\\
&- z_3z_7z_9^2z_{10}z_{12}^2 + z_3z_9^2z_{10}z_{11}^2z_{12} + z_5z_7z_9z_{12}^4 + z_6z_8z_{11}^2z_{12}^3 + z_7z_9^2z_{12}^4\\
&+ z_8z_9z_{11}z_{12}^4 - z_9^2z_{11}^2z_{12}^3 + z_1z_5z_7z_8z_{12}^2 + z_1z_5z_7z_9z_{11}z_{12}\\
&- z_1z_8^2z_{11}z_{12}^2 - z_1z_8z_9z_{11}^2z_{12} - z_2z_5z_7^2z_{12}^2 + z_2z_7z_8z_{11}z_{12}^2
\end{aligned}
$$

$$
\begin{aligned}
&- z_3^2z_{10}z_{11}^2z_{12} - z_3z_5z_7^2z_{11}z_{12} + z_3z_5z_7z_9z_{10}z_{11} + z_3z_6z_7^2z_9z_{10}\\
&- z_3z_6z_{10}z_{11}z_{12}^2 + z_3z_7z_8z_{11}^2z_{12} - z_3z_7z_9^2z_{10}z_{11} - z_3z_9z_{10}z_{12}^3\\
&- z_5z_6z_7^2z_{12}^2 + z_5z_7z_8z_{12}^3 + z_5z_7z_9z_{11}z_{12}^2 + z_6z_7z_8z_{11}z_{12}^2 - z_7z_8z_9z_{12}^3\\
&+ z_7z_9^2z_{11}z_{12}^2 - z_8^2z_{11}z_{12}^3 - z_9^3z_{10}z_{11}z_{12} + z_9^2z_{11}^3z_{12} - z_9z_{12}^5\\
&+ z_1z_5z_7z_8z_{11} - z_1z_8^2z_{11}^2 - z_2z_5z_7^2z_{11} + z_2z_7z_8^2z_{11}^2 - z_3z_5z_{10}z_{11}z_{12}\\
&- z_3z_9z_{10}z_{11}z_{12} - z_5z_6z_7^2z_{11} + z_5z_7z_8z_{11}z_{12} + z_6z_7z_8z_{11}^2 - z_7^2z_9^2z_{12}\\
&+ z_7z_8z_9z_{11}z_{12} + z_7z_9^3z_{10} - z_8^2z_{11}^2z_{12} - z_8z_{12}^4 - z_9z_{11}z_{12}^3 - z_5z_7z_{12}^2\\
&- z_6^2z_7z_{10} + z_6z_9z_{10}z_{11} - z_7z_9z_{12}^2 + z_8z_{11}z_{12}^2 - z_9^2z_{10}z_{12} + z_9z_{11}^2z_{12}\\
&- z_5z_7z_{11} - z_7z_9z_{11} + z_8z_{11}^2 + z_{11}z_{12},\\
a_7 = &-z_3z_5z_9z_{11}^2z_{12} + z_3z_6z_8z_{11}^2z_{12} - z_3z_7z_9^2z_{12}^2 - z_3z_8z_9z_{11}z_{12}^2\\
&- z_3^2z_{11}^2z_{12} - z_3z_6z_{11}z_{12}^2 + z_3z_8z_9z_{11}^2 - z_3z_9z_{12}^3 + z_9^3z_{11}z_{12}\\
&- z_3z_5z_{11}z_{12} - z_3z_9z_{11}z_{12} + z_7z_9^3 + z_9^2z_{12},\\
a_8 = &\,z_5z_9^2z_{11}z_{12}^2 + z_6z_7z_9^2z_{12}^2 - z_3z_9z_{11}z_{12}^2 - z_5z_9^2z_{11}^2 + z_6z_9z_{12}^3 - z_9^3z_{11}^2\\
&- z_5z_9z_{12}^2 + z_9^2z_{12}^2 - z_9^2z_{11} + z_6z_{12} - z_9,\\
a_9 = &-z_3z_9z_{11}^3z_{12}^2 + z_6z_9z_{11}^2z_{12}^3 - z_6z_8z_{11}^2z_{12}^2 - z_7z_9^2z_{12}^3 + z_8z_9z_{11}z_{12}^3\\
&+ z_9^2z_{11}^2z_{12}^2 + z_7z_8z_9z_{12}^2 - z_7z_9^2z_{11}z_{12} - z_8z_9z_{11}^2z_{12} - z_9^3z_{10}z_{11} - z_9^2z_{11}^3\\
&+ z_9z_{12}^4 + z_7^2z_9^2 - z_7z_8z_9z_{11} + z_8z_{12}^3 - z_9z_{11}z_{12}^2 - z_9^2z_{10} - z_{11},
\end{aligned}
$$

$$
\begin{aligned}
b_{(1,2,4,5,6,9,10,11,12)} &= z_{12}^2 + z_{11},\\
b_{(2,3,5,6,8,9,10,11,12)} &= z_3z_6z_{12} - z_3z_9,\\
b_{(2,4,5,6,7,8,9,11,12)} &= -z_6,\\
b_{(3,5,6,7,8,9,10,11,12)} &= -1,
\end{aligned}
$$

and the identity is

$$
\begin{aligned}
(\mathrm{C.22})\quad & a_1\psi_1 + a_2\psi_2 + a_3\psi_3 + a_4\psi_4 + a_5\psi_5 + a_6\psi_6 + a_7\psi_7 + a_8\psi_8 + a_9\psi_9\\
&+ b_{(1,2,4,5,6,9,10,11,12)}\mu_{(1,2,4,5,6,9,10,11,12)} + b_{(2,3,5,6,8,9,10,11,12)}\mu_{(2,3,5,6,8,9,10,11,12)}\\
&+ b_{(2,4,5,6,7,8,9,11,12)}\mu_{(2,4,5,6,7,8,9,11,12)} + b_{(3,5,6,7,8,9,10,11,12)}\mu_{(3,5,6,7,8,9,10,11,12)} = 1.
\end{aligned}
$$

□

Let $\mathcal{R}$ be the rank-drop locus of **Step 2** of the proof of Construction-Proposition 21.1, and $\mathcal{R}_{\mathbb{F}_3} := \mathcal{R} \times_{\mathbb{Z}_{(3)}} \mathbb{F}_3$. On the chart $\{P_I \neq 0\}$, $\mathcal{R}_{\mathbb{F}_3}$ is defined by the $\psi_{I,r}$ and the $\mu_{I,\gamma}$, so by (C.14) we have $\mathcal{R}_{\mathbb{F}_3} \cap \{P_I \neq 0\} = \emptyset$ for the eight triples I of Lemma C.3. By Lemma C.2, σ^* maps $\omega_1', \omega_2', \omega_3'$ to $-\omega_3', -\omega_2', -\omega_1'$, so the automorphism of $\mathrm{Gr}(3, \mathbb{Z}_{(3)}^7)$ induced by σ maps $X_{\mathbb{Z}_{(3)}}$ onto itself. Since $\mathcal{R}$ is defined by the rank of the relative derivative ds of the section s of the bundle $\mathcal{E}$ of that step, and on $X_{\mathbb{Z}_{(3)}}$ the derivative ds does not depend on the trivialization of $\mathcal{E}$, the automorphism induced by σ also maps $\mathcal{R}$ onto itself. It maps the chart of each of the six triples $012, 013, 246, 256, 245, 125$ onto the chart of the triple paired with it in Lemma C.2. Hence $\mathcal{R}_{\mathbb{F}_3} \cap \{P_I \neq 0\} = \emptyset$ for the fourteen triples I of Lemma C.2. By Lemma C.2, the fourteen charts $\{P_I \neq 0\}$ cover $X_{\mathbb{F}_3} \supset \mathcal{R}_{\mathbb{F}_3}$, so $\mathcal{R}_{\mathbb{F}_3} = \emptyset$.

References

[AZ22] H. Abban and Z. Zhuang, *K-stability of Fano varieties via admissible flags*, Forum Math. Pi **10** (2022), Paper No. e15, 43 pp.

[AZ23] H. Abban and Z. Zhuang, *Seshadri constants and K-stability of Fano manifolds*, Duke Math. J. **172** (2023), no. 6, 1109–1144.

[Alp13] J. Alper, *Good moduli spaces for Artin stacks*, Ann. Inst. Fourier (Grenoble) **63** (2013), no. 6, 2349–2402.

[ABHLX20] J. Alper, H. Blum, D. Halpern-Leistner, and C. Xu, *Reductivity of the automorphism group of K-polystable Fano varieties*, Invent. Math. **222** (2020), no. 3, 995–1032.

[ACC+23] C. Araujo, A.-M. Castravet, I. A. Cheltsov, K. Fujita, A.-S. Kaloghiros, J. Martinez-Garcia, C. Shramov, H. Süß, and N. Viswanathan, *The Calabi problem for Fano threefolds*, London Math. Soc. Lecture Note Ser. **485**, Cambridge Univ. Press, Cambridge (2023).

[AGV85] V. I. Arnold, S. M. Gusein-Zade, and A. N. Varchenko, *Singularities of differentiable maps. Vol. I: The classification of critical points, caustics and wave fronts*, Monographs in Mathematics **82**, Birkhäuser Boston, Boston, MA (1985).

[Arz08] I. V. Arzhantsev, *Invariant ideals and Matsushima's criterion*, Comm. Algebra **36** (2008), no. 12, 4368–4374.

[ADL24] K. Ascher, K. DeVleming, and Y. Liu, *Wall crossing for K-moduli spaces of plane curves*, Proc. Lond. Math. Soc. (3) **128** (2024), no. 6, Paper No. e12615.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley, Reading, MA (1969).

[GBP22] E. R. García Barroso and A. Płoski, *A note on the plane curve singularities in positive characteristic*, arXiv:2207.14523.

[BPR06] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in real algebraic geometry*, 2nd ed., Algorithms and Computation in Mathematics **10**, Springer-Verlag, Berlin (2006).

[BGM19] M. Bate, H. Geranios, and B. Martin, *Orbit closures and invariants*, Math. Z. **293** (2019), no. 3-4, 1121–1159.

[BHMR17] M. Bate, S. Herpel, B. Martin, and G. Röhrle, *Cocharacter-closure and the rational Hilbert–Mumford theorem*, Math. Z. **287** (2017), no. 1-2, 39–72.

[Bau09] T. Bauer, *A simple proof for the existence of Zariski decompositions on surfaces*, J. Algebraic Geom. **18** (2009), no. 4, 789–793.

[BKM26a] A. Bayer, A. G. Kuznetsov, and E. Macrì, *Mukai bundles on Fano threefolds*, Compos. Math. **162** (2026), no. 1, 59–99.

[BKM26b] A. Bayer, A. G. Kuznetsov, and E. Macrì, *Mukai models of Fano varieties*, J. Reine Angew. Math. **836** (2026), 111–162.

[BR15] M. Beck and S. Robins, *Computing the continuous discretely. Integer-point enumeration in polyhedra*, 2nd ed., Undergraduate Texts in Mathematics, Springer, New York (2015).

[Ber16] R. J. Berman, *K-polystability of $\mathbb{Q}$-Fano varieties admitting Kähler–Einstein metrics*, Invent. Math. **203** (2016), no. 3, 973–1025.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[BJ20] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. **365** (2020), 107062, 57 pp.

[BL21] H. Blum and Y. Liu, *The normalized volume of a singularity is lower semicontinuous*, J. Eur. Math. Soc. (JEMS) **23** (2021), no. 4, 1225–1256.

[BLX22] H. Blum, Y. Liu, and C. Xu, *Openness of K-semistability for Fano varieties*, Duke Math. J. **171** (2022), no. 13, 2753–2797.

[BX19] H. Blum and C. Xu, *Uniqueness of K-polystable degenerations of Fano varieties*, Ann. of Math. (2) **190** (2019), no. 2, 609–656.

[BO95] A. Bondal and D. Orlov, *Semiorthogonal decomposition for algebraic varieties*, arXiv:alg-geom/9506012.

[Bor91] A. Borel, *Linear algebraic groups*, 2nd ed., Grad. Texts in Math. **126**, Springer-Verlag, New York (1991).

[BLR90] S. Bosch, W. Lütkebohmert, and M. Raynaud, *Néron Models*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **21**, Springer-Verlag, Berlin (1990).

[BC11] S. Boucksom and H. Chen, *Okounkov bodies of filtered linear series*, Compos. Math. **147** (2011), no. 4, 1205–1229.

[BFJ09] S. Boucksom, C. Favre, and M. Jonsson, *Differentiability of volumes of divisors and a problem of Teissier*, J. Algebraic Geom. **18** (2009), no. 2, 279–308.

[BHJ17] S. Boucksom, T. Hisamoto, and M. Jonsson, *Uniform K-stability, Duistermaat–Heckman measures and singularities of pairs*, Ann. Inst. Fourier (Grenoble) **67** (2017), no. 2, 743–841.

[Bou02] N. Bourbaki, *Lie Groups and Lie Algebras. Chapters 4–6*, Elements of Mathematics, Springer-Verlag, Berlin (2002).

[BH98] W. Bruns and J. Herzog, *Cohen–Macaulay rings*, rev. ed., Cambridge Studies in Advanced Mathematics **39**, Cambridge University Press, Cambridge (1998).

[BE77] D. A. Buchsbaum and D. Eisenbud, *Algebra structures for finite free resolutions, and some structure theorems for ideals of codimension 3*, Amer. J. Math. **99** (1977), no. 3, 447–485.

[Cas00] E. Casas-Alvero, *Singularities of Plane Curves*, London Mathematical Society Lecture Note Series **276**, Cambridge University Press, Cambridge (2000).

[Che17] I. A. Cheltsov, *Worst singularities of plane curves of given degree*, J. Geom. Anal. **27** (2017), no. 3, 2302–2338.

[CDGF+25] I. A. Cheltsov, T. Duarte Guerreiro, K. Fujita, I. Krylov, and J. Martinez-Garcia, *K-stability of Casagrande–Druel varieties*, J. Reine Angew. Math. **818** (2025), 53–113.

[CRZ19] I. A. Cheltsov, Y. A. Rubinstein, and K. Zhang, *Basis log canonical thresholds, local intersection estimates, and asymptotically log del Pezzo surfaces*, Selecta Math. (N.S.) **25** (2019), no. 2, Paper No. 34.

[CS09] I. A. Cheltsov and C. A. Shramov, *Extremal metrics on del Pezzo threefolds*, Proc. Steklov Inst. Math. **264** (2009), no. 1, 30–44.

[CS23] I. A. Cheltsov and C. A. Shramov, *Kähler–Einstein Fano threefolds of degree* 22, J. Algebraic Geom. **32** (2023), no. 3, 385–428.

[CDS15] X. Chen, S. Donaldson, and S. Sun, *Kähler–Einstein metrics on Fano manifolds. I, II, III*, J. Amer. Math. Soc. **28** (2015), no. 1, 183–197, 199–234, 235–278.

[CLS11] D. A. Cox, J. B. Little, and H. K. Schenck, *Toric Varieties*, Graduate Studies in Mathematics **124**, American Mathematical Society, Providence, RI (2011).

[DK01] J.-P. Demailly and J. Kollár, *Semi-continuity of complex singularity exponents and Kähler–Einstein metrics on Fano orbifolds*, Ann. Sci. École Norm. Sup. (4) **34** (2001), no. 4, 525–556.

[DK25] E. Denisova and A.-S. Kaloghiros, *On K-stability of one-nodal prime Fano threefolds of genus* 12, Taiwanese J. Math. **29** (2025), no. 6, 1297–1330.

[DKK15] S. Dinew, G. Kapustka, and M. Kapustka, *Remarks on Mukai threefolds admitting* $\mathbb{C}^*$ *action*, arXiv:1506.03286.

[DK93] I. Dolgachev and V. Kanev, *Polar covariants of plane cubics and quartics*, Adv. Math. **98** (1993), no. 2, 216–301.

[Dol12] I. V. Dolgachev, *Classical algebraic geometry: a modern view*, Cambridge University Press, Cambridge (2012).

[Don02] S. K. Donaldson, *Scalar curvature and stability of toric varieties*, J. Differential Geom. **62** (2002), no. 2, 289–349.

[Don07] S. K. Donaldson, *A note on the* α*-invariant of the Mukai–Umemura 3-fold*, arXiv:0711.4357.

[Don08] S. K. Donaldson, *Kähler geometry on toric manifolds, and some other manifolds with large symmetry*, in: *Handbook of Geometric Analysis. No. 1*, Adv. Lect. Math. (ALM) **7**, Int. Press, Somerville, MA (2008), 29–75.

[Don18] S. K. Donaldson, *Stability of algebraic varieties and Kähler geometry*, in: Algebraic geometry: Salt Lake City 2015, Proc. Sympos. Pure Math. **97.1**, Amer. Math. Soc., Providence, RI (2018), pp. 199–221.

[DW16] W. Donovan and M. Wemyss, *Noncommutative deformations and flops*, Duke Math. J. **165** (2016), no. 8, 1397–1474.

[DFK25] A. Dubouloz, K. Fujita, and T. Kishimoto, *Another view on smooth prime Fano threefolds of degree* 22 *with infinite automorphism groups*, arXiv:2512.08409.

[Eis95] D. Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, Graduate Texts in Mathematics **150**, Springer-Verlag, New York (1995).

[FR06] J. Fine and J. Ross, *A note on positivity of the CM line bundle*, Int. Math. Res. Not. **2006**, Art. ID 95875, 14 pp.

[Fuj19] K. Fujita, *A valuative criterion for uniform K-stability of* $\mathbb{Q}$*-Fano varieties*, J. Reine Angew. Math. **751** (2019), 309–338.

[Fuj23] K. Fujita, *On Fano threefolds of degree* 22 *after Cheltsov and Shramov*, in: *Birational Geometry, Kähler–Einstein Metrics and Degenerations*, Springer Proc. Math. Stat. **409**, Springer, Cham (2023), 249–262.

[Fuj26a] K. Fujita, *Elementary links from prime Fano threefolds along two lines*, arXiv:2604.08097.

[Fuj26b] K. Fujita, *On the coupled stability thresholds of graded linear series*, J. Math. Sci. Univ. Tokyo **33** (2026), 161–281.

[FO18] K. Fujita and Y. Odaka, *On the K-stability of Fano varieties and anticanonical divisors*, Tohoku Math. J. (2) **70** (2018), no. 4, 511–521.

[Fuj79] T. Fujita, *On Zariski problem*, Proc. Japan Acad. Ser. A Math. Sci. **55** (1979), no. 3, 106–110.

[Ful98] W. Fulton, *Intersection theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge **2**, Springer-Verlag, Berlin (1998).

[Ful08] W. Fulton, *Algebraic curves. An introduction to algebraic geometry*, 3rd ed., 2008, available at the author's homepage (first edition W. A. Benjamin, New York, 1969).

[FH91] W. Fulton and J. Harris, *Representation Theory. A First Course*, Graduate Texts in Mathematics **129**, Springer-Verlag, New York (1991).

[FT92] M. Furushima and M. Tada, *Non-normal del Pezzo surfaces and Fano threefolds of the first kind*, J. Reine Angew. Math. **429** (1992), 183–190.

[GLT14] F. Greer, Z. Li, and Z. Tian, *Picard groups on moduli of K3 surfaces with Mukai models*, arXiv:1402.2330.

[GLS07] G.-M. Greuel, C. Lossen, and E. Shustin, *Introduction to Singularities and Deformations*, Springer Monographs in Mathematics, Springer, Berlin (2007).

[Gro57] A. Grothendieck, *Sur la classification des fibrés holomorphes sur la sphère de Riemann*, Amer. J. Math. **79** (1957), no. 1, 121–138.

[Gro61] A. Grothendieck, *Éléments de géométrie algébrique. II. Étude globale élémentaire de quelques classes de morphismes*, Inst. Hautes Études Sci. Publ. Math. **8** (1961), 5–222.

[Har92] J. Harris, *Algebraic Geometry: A First Course*, Graduate Texts in Mathematics **133**, Springer-Verlag, New York (1992).

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York–Heidelberg (1977).

[Har10] R. Hartshorne, *Deformation theory*, Graduate Texts in Mathematics **257**, Springer, New York (2010).

[HS06] C. Huneke and I. Swanson, *Integral Closure of Ideals, Rings, and Modules*, London Mathematical Society Lecture Note Series **336**, Cambridge University Press, Cambridge (2006).

[HL10] D. Hyeon and Y. Lee, *Log minimal model program for the moduli of stable curves of genus three*, Math. Res. Lett. **17** (2010), no. 4, 625–636.

[IK99] A. Iarrobino and V. Kanev, *Power sums, Gorenstein algebras, and determinantal loci*, Lecture Notes in Mathematics **1721**, Springer-Verlag, Berlin (1999).

[Isk77] V. A. Iskovskikh, *Fano threefolds. I*, Izv. Akad. Nauk SSSR Ser. Mat. **41** (1977), no. 3, 516–562.

[Isk78] V. A. Iskovskikh, *Fano threefolds. II*, Izv. Akad. Nauk SSSR Ser. Mat. **42** (1978), no. 3, 506–549.

[Isk89] V. A. Iskovskikh, *Double projection from a line on Fano threefolds of the first kind*, Mat. Sb. **180** (1989), no. 2, 260–278; English transl. in Math. USSR-Sb. **66** (1990), no. 1, 265–284.

[IP99] V. A. Iskovskikh and Yu. G. Prokhorov, *Fano varieties*, in: Algebraic Geometry V, Encyclopaedia Math. Sci. **47**, Springer, Berlin (1999), 1–247.

[JR11] P. Jahnke and I. Radloff, *Terminal Fano threefolds and their smoothings*, Math. Z. **269** (2011), no. 3–4, 1129–1136.

[JM12] M. Jonsson and M. Mustaţă, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.

[KLPZ26] A.-S. Kaloghiros, Y. Liu, A. Petracci, and J. Zhao, *The boundary of K-moduli of prime Fano threefolds of genus twelve*, arXiv:2603.29827.

[Kan12] A. Kanazawa, *Pfaffian Calabi–Yau threefolds and mirror symmetry*, Commun. Number Theory Phys. **6** (2012), no. 3, 661–696.

[Kap11] M. Kapustka, *Some degenerations of G_2 and Calabi–Yau varieties*, arXiv:1103.4623.

[KK12] K. Kaveh and A. G. Khovanskii, *Newton–Okounkov bodies, semigroups of integral points, graded algebras and intersection theory*, Ann. of Math. (2) **176** (2012), no. 2, 925–978.

[Kaw07] M. Kawakita, *Inversion of adjunction on log canonicity*, Invent. Math. **167** (2007), no. 1, 129–133.

[KN79] G. Kempf and L. Ness, *The length of vectors in representation spaces*, in: *Algebraic Geometry (Copenhagen, 1978)*, Lecture Notes in Math. **732**, Springer, Berlin (1979), 233–243.

[KPZ14] T. Kishimoto, Yu. G. Prokhorov, and M. Zaidenberg, *Affine cones over Fano threefolds and additive group actions*, Osaka J. Math. **51** (2014), 1093–1112.

[Kle05] S. L. Kleiman, *The Picard scheme*, in: Fundamental Algebraic Geometry, Math. Surveys Monogr. **123**, Amer. Math. Soc., Providence, RI (2005), 235–321.

[KM76] F. F. Knudsen and D. Mumford, *The projectivity of the moduli space of stable curves. I. Preliminaries on "det" and "Div"*, Math. Scand. **39** (1976), no. 1, 19–55.

[Kol89] J. Kollár, *Flops*, Nagoya Math. J. **113** (1989), 15–36.

[Kol96] J. Kollár, *Rational curves on algebraic varieties*, Ergeb. Math. Grenzgeb. (3) **32**, Springer-Verlag, Berlin (1996).

[Kol13] J. Kollár, *Singularities of the minimal model program*, with a collaboration of S. Kovács, Cambridge Tracts in Math. **200**, Cambridge Univ. Press, Cambridge (2013).

[Kol23] J. Kollár, *Families of varieties of general type*, with the collaboration of K. Altmann and S. J. Kovács, Cambridge Tracts in Mathematics **231**, Cambridge University Press, Cambridge (2023).

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Kol+92] J. Kollár et al., *Flips and abundance for algebraic threefolds*, Astérisque **211**, Société Mathématique de France, Paris (1992).

[KP18] A. G. Kuznetsov and Yu. G. Prokhorov, *Prime Fano threefolds of genus* 12 *with a* $\mathbb{G}_m$*-action and their automorphisms*, Épijournal Géom. Algébrique **2** (2018), Art. 3, 14 pp.

[KP20] A. G. Kuznetsov and Yu. G. Prokhorov, *Rationality of Mukai varieties over non-closed fields*, arXiv:2003.10761.

[KP23] A. G. Kuznetsov and Yu. G. Prokhorov, *Rationality of Fano threefolds over non-closed fields*, Amer. J. Math. **145** (2023), no. 2, 335–411.

[KP25] A. G. Kuznetsov and Yu. G. Prokhorov, 1*-nodal Fano threefolds with Picard number* 1, Izv. Math. **89** (2025), no. 3, 495–594.

[KPS18] A. G. Kuznetsov, Yu. G. Prokhorov, and C. A. Shramov, *Hilbert schemes of lines and conics and automorphism groups of Fano threefolds*, Jpn. J. Math. **13** (2018), no. 1, 109–185.

[Lan02] S. Lang, *Algebra*, rev. 3rd ed., Graduate Texts in Mathematics **211**, Springer-Verlag, New York (2002).

[Lat23] R. Laterveer, *Algebraic cycles and Fano threefolds of genus 10*, Bull. Soc. Math. France **151** (2023), no. 4, 565–593.

[Laz06] R. Laza, *Deformations of singularities and variation of GIT quotients*, Ph.D. thesis, Columbia University, 2006; arXiv:math/0607003.

[Laz04a] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **48**, Springer-Verlag, Berlin (2004).

[Laz04b] R. Lazarsfeld, *Positivity in algebraic geometry. II. Positivity for vector bundles, and multiplier ideals*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **49**, Springer-Verlag, Berlin (2004).

[Li17] C. Li, *K-semistability is equivariant volume minimization*, Duke Math. J. **166** (2017), no. 16, 3147–3218.

[LX14] C. Li and C. Xu, *Special test configuration and K-stability of Fano varieties*, Ann. of Math. (2) **180** (2014), no. 1, 197–232.

[LN97] R. Lidl and H. Niederreiter, *Finite fields*, 2nd ed., Encyclopedia of Mathematics and its Applications **20**, Cambridge University Press, Cambridge (1997).

[LX19] Y. Liu and C. Xu, *K-stability of cubic threefolds*, Duke Math. J. **168** (2019), no. 11, 2029–2073.

[LXZ22] Y. Liu, C. Xu, and Z. Zhuang, *Finite generation for valuations computing stability thresholds and applications to K-stability*, Ann. of Math. (2) **196** (2022), no. 2, 507–566.

[MS01] N. Manolache and F.-O. Schreyer, *Moduli of* $(1,7)$*-polarized abelian surfaces via syzygies*, Math. Nachr. **226** (2001), 177–203.

[Mat89] H. Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1989).

[Mil17] J. S. Milne, *Algebraic groups. The theory of group schemes of finite type over a field*, Cambridge Stud. Adv. Math. **170**, Cambridge University Press, Cambridge (2017).

[Mil68] J. Milnor, *Singular points of complex hypersurfaces*, Annals of Mathematics Studies, no. 61, Princeton University Press, Princeton, NJ (1968).

[MM81] S. Mori and S. Mukai, *Classification of Fano* 3*-folds with* $B_2 \geq 2$, Manuscripta Math. **36** (1981/82), 147–162.

[Muk88] S. Mukai, *Curves, K3 surfaces and Fano* 3*-folds of genus* ≤ 10, in: *Algebraic Geometry and Commutative Algebra, Vol. I*, Kinokuniya, Tokyo (1988), 357–377.

[Muk89] S. Mukai, *Biregular classification of Fano* 3*-folds and Fano manifolds of coindex* 3, Proc. Nat. Acad. Sci. U.S.A. **86** (1989), no. 9, 3000–3002.

[Muk92] S. Mukai, *Fano* 3*-folds*, Complex projective geometry (Trieste, 1989/Bergen, 1989), London Math. Soc. Lecture Note Ser. **179**, Cambridge Univ. Press, Cambridge (1992), pp. 255–263.

[Muk04] S. Mukai, *Plane quartics and Fano threefolds of genus twelve*, in: The Fano Conference, Univ. Torino, Turin (2004), 563–572.

[MU83] S. Mukai and H. Umemura, *Minimal rational threefolds*, in: Algebraic Geometry (Tokyo/Kyoto, 1982), Lecture Notes in Math. **1016**, Springer, Berlin (1983), 490–518.

[Mum74] D. Mumford, *Abelian varieties*, 2nd ed., Tata Inst. Fund. Res. Stud. Math. **5**, Oxford Univ. Press, London (1974).

[MFK94] D. Mumford, J. Fogarty, and F. Kirwan, *Geometric invariant theory*, 3rd ed., Ergeb. Math. Grenzgeb. (2) **34**, Springer-Verlag, Berlin (1994).

[Nam97] Y. Namikawa, *Smoothing Fano 3-folds*, J. Algebraic Geom. **6** (1997), no. 2, 307–324.

[OSS16] Y. Odaka, C. Spotti, and S. Sun, *Compact moduli spaces of del Pezzo surfaces and Kähler–Einstein metrics*, J. Differential Geom. **102** (2016), no. 1, 127–172.

[Ott24] G. Ottaviani, *Effective methods for plane quartics, their theta characteristics and the Scorza map*, arXiv:2403.19599.

[PT06] S. T. Paul and G. Tian, *CM stability and the generalized Futaki invariant I*, arXiv:math/0605278.

[Pro90] Yu. G. Prokhorov, *Automorphism groups of Fano 3-folds*, Russian Math. Surveys **45** (1990), no. 3, 222–223.

[Pro01] Yu. G. Prokhorov, *Lectures on complements on log surfaces*, MSJ Memoirs **10**, Mathematical Society of Japan, Tokyo (2001).

[Pro16] Yu. G. Prokhorov, *Singular Fano threefolds of genus* 12, Sb. Math. **207** (2016), no. 7, 983–1009.

[PCS19] V. V. Przyjalkowski, I. A. Cheltsov, and C. A. Shramov, *Fano threefolds with infinite automorphism groups*, Izv. Math. **83** (2019), no. 4, 860–907.

[RR85] S. Ramanan and A. Ramanathan, *Projective normality of flag varieties and Schubert varieties*, Invent. Math. **79** (1985), no. 2, 217–224.

[Rei83] M. Reid, *Minimal models of canonical* 3*-folds*, Algebraic varieties and analytic varieties (Tokyo, 1981), Adv. Stud. Pure Math. **1**, North-Holland, Amsterdam (1983), pp. 131–180.

[Sai74] K. Saito, *Einfach-elliptische Singularitäten*, Invent. Math. **23** (1974), 289–325.

[Sch01] F.-O. Schreyer, *Geometry and algebra of prime Fano* 3*-folds of genus* 12, Compositio Math. **127** (2001), no. 3, 297–319.

[Ser79] J.-P. Serre, *Local fields*, Graduate Texts in Mathematics **67**, Springer-Verlag, New York (1979).

[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.

[Sta] The Stacks Project Authors, *The Stacks Project*, https://stacks.math.columbia.edu.

[Tak89] K. Takeuchi, *Some birational maps of Fano* 3*-folds*, Compos. Math. **71** (1989), no. 3, 265–283.

[Tia97] G. Tian, *Kähler–Einstein metrics with positive scalar curvature*, Invent. Math. **130** (1997), no. 1, 1–37.

[Tia15] G. Tian, *K-stability and Kähler–Einstein metrics*, Comm. Pure Appl. Math. **68** (2015), no. 7, 1085–1156.

[Vir26] N. Virin, *Cylinders in Fano threefolds of genus* 9 *and* 10, arXiv:2605.30875.

[Voi02] C. Voisin, *Hodge Theory and Complex Algebraic Geometry. I*, Cambridge Stud. Adv. Math. **76**, Cambridge University Press, Cambridge (2002).

[Wey03] J. Weyman, *Cohomology of Vector Bundles and Syzygies*, Cambridge Tracts in Mathematics **149**, Cambridge University Press, Cambridge (2003).

[XZ26] C. Xu and Z. Zhuang, *Open problems in K-stability of Fano varieties*, arXiv:2601.15576.

[Zar62] O. Zariski, *The theorem of Riemann–Roch for high multiples of an effective divisor on an algebraic surface*, Ann. of Math. (2) **76** (1962), 560–615.

[ZS60] O. Zariski and P. Samuel, *Commutative algebra. Vol. II*, The University Series in Higher Mathematics, D. Van Nostrand Co., Princeton, NJ–Toronto–London–New York (1960).

[Zhu20] Z. Zhuang, *Birational superrigidity and K-stability of Fano complete intersections of index 1*, Duke Math. J. **169** (2020), no. 12, 2205–2229.